T972

Datasheet

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Revision History

Issue 05 (2021-10-25)

This is the 05 release.

Compared to the previous version, the following contents are changed.

Section	Change Description
6.7.3.9	Added a note.

Issue 04 (2021-02-20)

This is the 04 release.

Compared to the previous version, the following contents are changed.

Section	Change Description
1 and 2	Deleted RealVideo related description.

Issue 03 (2020-11-05)

This is the 03 release.

Compared to the previous version, the following contents are changed.

Section	Change Description
4.4	Added thermal resistance specification of FCCSP package.
5	Added dimension figure of FCCSP package.

Issue 02 (2020-07-20)

This is the 02 release.

Compared to the previous version, the following contents are changed.

Section	Change Description
2	Changed eMMC specification to HS400.
4.6.2	Added description of eMMC HS400.

Issue 01 (2020-02-18)

This is the 01 release.

Compared to last version, the following topics are added.

- 4.6 and its child topics.

Compared to last version, the following contents are changed.

Section	Change Description
1 and 2	Added HDCP 2.3 description.
4.2	Changed the parameter values of VDD_EE.

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Section	Change Description
4.5.2	Added a note explaining “OD 5V”.
• 6.12.1 • 6.7.4	Optimized some descriptions about CPU.
9.10.12	Updated register description.

Issue 0.2 (2019-05-15)

This is the preliminary 0.2 release.

Compared to last version, the following contents are changed.

Section	Change Description
4.5.3	Updated the specifications of DC Specifications - LPDDR4 mode.
4.10	Updated maximum current of VDDCPU and VDD_EE.
2	Updated specifications of Video/Picture CODEC.
4.4	Added thermal resistance specifications.
3.3	Updated the pin description of the LVDS part.
Whole document	Modified some editing issues.
6.5	Modified descriptions of Vulkan and OpenCL.

Issue 0.1 (2018-12-06)

This is the initial release.

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1 General Description

T972 is an advanced application processor designed for worldwide UHD TV applications. It integrates a powerful CPU/GPU subsystem, a best-in-class HDR image processing pipeline, a secured 8K/4K video CODEC engine with all major peripherals to form the ultimate cost-effective smart TV chip.

The main system CPU is a quad-core ARM Cortex-A55 CPU with shared L3 cache to improve system performance. In addition, the Cortex-A55 CPU includes the NEON SIMD co-processor to improve software media processing capability.

The graphic subsystem consists of two graphic engines and a flexible video/graphic output pipeline. The ARM Mali-G31 MP2 GPU handles all OpenGL ES 3.2, Vulkan 1.1 and OpenCL 2.0 graphic programs, while the 2.5D graphics processor handles additional scaling, alpha, rotation and color space conversion operations. Together, the CPU and GPU handle all operating system, network, user-interface and game related tasks.

Amlogic Video Engine (AVE-10) is a subsystem which uses dedicated hardware video decoders and encoders to offloads the Cortex-A55 CPUs from all video CODEC processing. AVE-10 is capable of decoding 4K2K resolution video within Trusted Video Path (TVP) for secured DRM applications. It supports all major video formats including MVC, MPEG-1/2/4, VC-1/WMV, AVS+, AVS2, MJPEG, H.264, H.265-10, VP9-10 and also JPEG.

The video/graphics output pipeline includes HDR10+, HDR10, HLG and Technicolor Prime HDR processing, BT.2020/ BT.2100 processing, motion compensated and motion adaptive de-interlacer, flexible programmable super scaler, local dimming and many picture enhancement filters before passing the enhanced image to the video output ports. The 8-lane V-by-one and dual-channel LVDS interface are available for UHD/FHD TV panel, and 12-lane P2P interface with internal flexible timing control module Optional for UHD TCON-less panels including CEDS, CHPI, CMPI and iSP.

3 HDMI 2.1 receiver ports plus two sets of CVBS composite analog input ports are available. The HDMI ports support HDCP 1.4/2.2/2.3 and can receive up to 4K2K HDR video.

The processor integrates the ATV demodulators which fully support worldwide analog TV standards including NTSC, PAL, and SECAM. DTV broadcasting streams can be received by the internal DTMB demodulator or the transport stream (TS) interface. The built-in three demux can process the TV streams from the serial transport stream input interface, which can connect to external tuner/demodulator. DVB Common Descrambler 1.0 is supported in addition to DES, Triple DES (TDES/3DES) and AES streaming crypto formats. An integrated ISO7816 controller is included for interfacing to external smart card.

It is optimized for low power far-field voice application. The powerful main CPU can enable top of the line audio front end and wake word algorithms. It also has built-in Voice Activity Detection (VAD) module for ultra-low power operations during system standby and full digital MIC interface including PDM, TDM and I2S up to 8 channels are available.

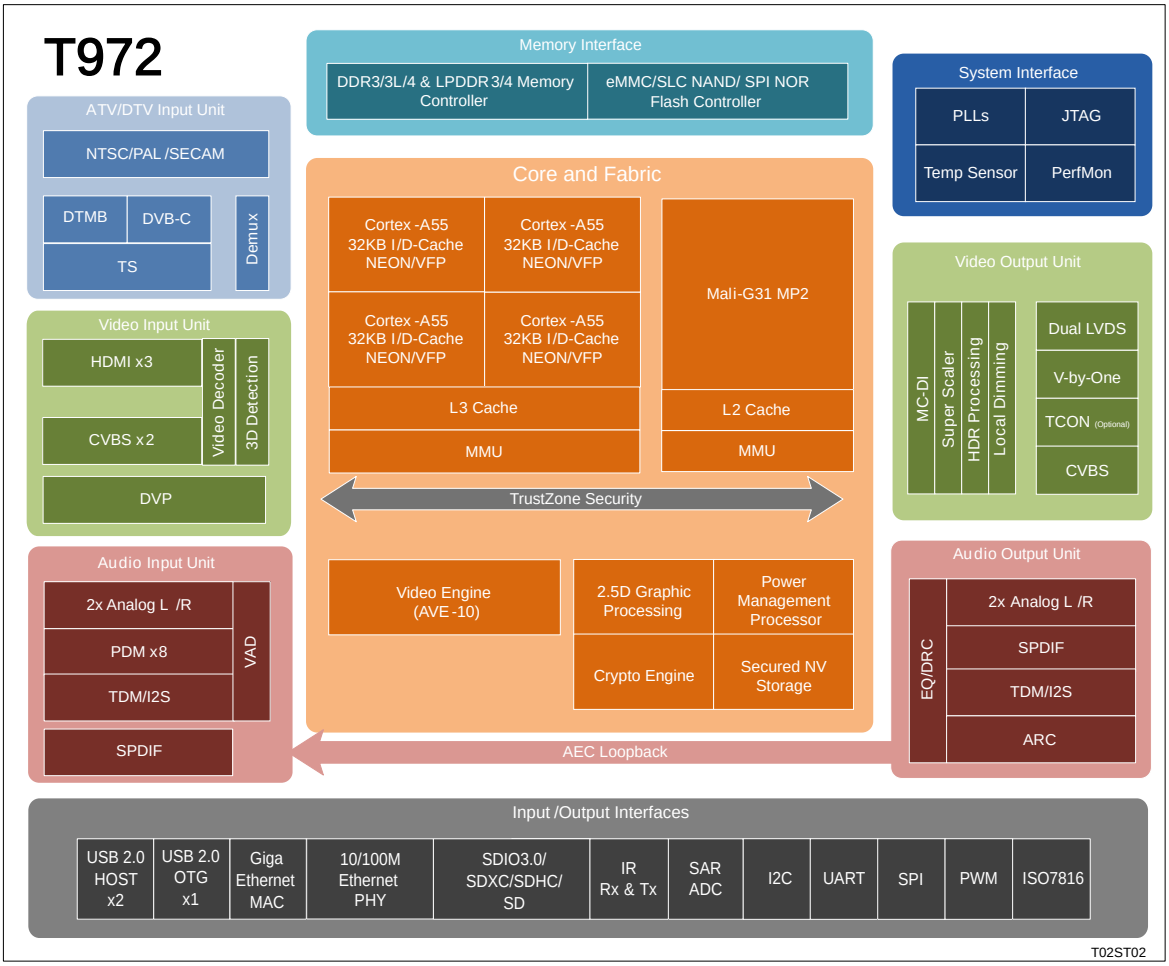
The SoC integrates rich advanced network and peripheral interfaces, including a 10/100/1000M Ethernet MAC with RGMII, 10/100M Ethernet PHY, USB 2.0 high-speed port, SDIO 3.0 controller, eMMC 5.0 controller, SLC NAND controller and multiple SDIO/SD card controllers, UART, I2C, high-speed SPI PWMs and a built-in IR blaster. The flexible and programmable QoS-based switch fabric and memory controller tie all the processing cores and peripherals together and connects to the DRAM memory bus.

Standard development environment utilizing SecureOS, Linux and GNU/GCC Android tool chain is supported. Please contact your AMLOGIC sales representative for more information.

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2 Features Summary

Figure 2-1 Features Summary



CPU Sub-system

- Quad core ARM Cortex-A55 CPU
- ARMv8.2 architecture with Neon extensions
- Unified system L3 cache
- Advanced TrustZone security system
- Application based traffic optimization using internal QoS-based switching fabrics
- CoreSight debugger support

3D Graphics Processing Unit

- ARM Mali-G31 MP2 GPU
- 4-wide warps, dual texture pipe, 2x 4-wide execution engines (EE)
- Concurrent multi-core processing
- OpenGL ES 3.2, Vulkan 1.1 and OpenCL 2.0 support

2.5D Graphics Processor

- Fast bitblt engine with dual inputs and single output

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- Programmable raster operations (ROP)
- Programmable polyphase scaling filter
- Supports multiple video formats 4:2:0, 4:2:2 and 4:4:4 and multiple pixel formats (8/16/24/32 bits graphics layer)
- Fast color space conversion
- Advanced anti-flickering filter

Crypto Engine

- AES block cipher with 128/256 bits keys, standard 16 bytes block size and streaming ECB, CBC and CTR modes
- DES/3DES block cipher with ECB and CBC modes supporting 64 bits key for DES and 192 bits key for 3DES
- Hardware key-ladder operation and DVB-CSA for transport stream encryption
- Built-in hardware True Random Number Generator (TRNG) and SHA-1/SHA-2 engine

Video/Picture CODEC

- Amlogic Video Engine (AVE-10) with dedicated hardware decoders up to 4Kx2K@75fps
- Video/Picture Decoding
  - VP9 Profile 2-10 up to 8Kx4K@24fps or 4Kx2K@60fps
  - H.265 HEVC MP-10@L5.1 up to 8Kx4K@24fps or 4Kx2K@60fps
  - AVS2-P2 Profile up to 4Kx2K@60fps
  - H.264 AVC HP@L5.1 up to 4Kx2K@30fps
  - H.264 MVC up to 1080P@60fps
  - MPEG-4 ASP@L5 up to 1080P@60fps (ISO-14496)
  - WMV/VC-1 SP/MP/AP up to 1080P@60fps
  - AVS-P16(AVS+) /AVS-P2 JiZhun Profile up to 1080P@60fps
  - MPEG-2 MP/HL up to 1080P@60fps (ISO-13818)
  - MPEG-1 MP/HL up to 1080P@60fps (ISO-11172)
  - Multiple language and multiple format sub-title video support
  - MJPEG and JPEG unlimited pixel resolution decoding (ISO/IEC-10918)
  - Supports JPEG thumbnail, scaling, rotation and transition effects
  - Supports *.mkv,*.wmv,*.mpg, *.mpeg, *.dat, *.avi, *.mov, *.iso, *.mp4 and *.jpg file formats

9th Generation Advanced Amlogic TruLife Image Engine

- Supports HDR10/10+, HLG, Technicolor Prime HDR
- Motion compensated noise reduction and 3D digital noise reduction for random noise
- Block noise, mosquito noise, spatial noise, contour noise reduction
- Motion compensated and motion adaptive de-interlacer
- Edge interpolation with low angle protection and processing
- 3:2/2:2 pulldown and Video on Film (VOF) detection and processing
- Smart sharpness with SuperScaler technology including de-contouring, de-ring, LTI, CTI, de-jaggy, peaking
- Local contrast and dynamic non-Linear contrast for detail enhancement
- 3D LUTs with 17x17x17 nodes, provide 4913 different control points, which is competent for matching calibrated displays to a target colorspace

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- High precision HSL color space based color management with low saturation protection, independent luma/hue/saturation adjustment to achieve blue/green extension, fresh tone correction, and wider gamut for video
- Video mixer: 2 video planes and 2 graphics planes
- Independent HDR re-mapping of video and graphic layer
- Local dimming control for high nits backlights

LCD Panel Output

- 8-lane V-By-One output with 1, 2, 4 regions supported, up to 4Kx2K 60Hz resolution
- Dual-channel LVDS output supporting up to 1920x1080 60Hz resolution
- Built-in (1-port 6-pair)/(2-port 3-pair) mini-LVDS output with programmable HD/FHD timing controller Optional up to 1920x1080Hz 60Hz resolution
- 12-lane CEDS/CHPI/CMPI/iSP output with programmable UHD timing controller Optional for UHD TCON-less panel, up to 4Kx2K 60Hz resolution
- Three independent Gamma table for LCD panel tuning
- Dithering logic for mapping to different LCD panel color depth

Video Input/output Interface

- 3x HDMI 2.1 receiver ports with Dynamic HDR, ARC, HDCP 1.4/2.2/2.3, 4Kx2K@60 max resolution input
- 2x CVBS 480i/576i standard definition inputs
- Supports CVBS (PAL/NTSC) bypass output
- ITU 601/656 parallel camera input supporting 8-bit RGB565, CCIR656, CCIR601, YUV422, YCbCr422

Audio Processing and Input/Output

- Supports MP3, AAC, WMA, RM, FLAC, Dolby Audio Optional, Ogg and programmable with 7.1/5.1 down-mixing
- Low-power VAD and internal AEC loopback path
- 3 built-in TDM/PCM/I2S ports with TDM/PCM mode up to 384kHz x32bits x 8ch or 96kHz x 32bits x 32ch and I2S mode up to 384kHz x 32bits x 8ch
- Digital microphone PDM voice input with programmable CIC, LPF; HPF, support up to 8 DMICs
- Built-in serial digital audio SPDIF/IEC958 output
- 2 L/R analog input channels and 2 L/R output channels
- Supports concurrent dual audio stereo channel output with combination of I2S+PCM
- Supports Audio EQ/DRC for audio speaker

TV Demodulator

- Standard compliant NTSC, NTSC-J, PAL-BG, PAL-DK1, PAL-I, PAL-DK, PAL-M, PAL-N, SECAM-DK2, SECAM-DK3, SECAM-L ATV demodulators
- Worldwide analog TV audio standard: BTSC, A2, EIA-J and NICAM
- Supports Teletext, close caption, V-chip
- DTMB/DVB-C DTV demodulators
- Build-in VIF demodulator supports low IF interface from tuner module

DTV Broadcasting Interface

- 3x Transport stream (TS) input interface with built-in demux processor for connecting to external digital TV tuner/demodulator

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- Built-in PWM, I2C and SPI interfaces to control tuner and demodulator
- Integrated ISO 7816 smart card controller

Memory and Storage Interface

- 32-bit DRAM memory interface with dual ranks and max 4GB total address space
- Compatible with JEDEC standard DDR3-2133 /DDR3L-2133 /DDR4-2666 /LPDDR3-2133 /LPDDR4-3200 SDRAM
- SDSC/SDHC/SDXC card and SDIO interface with 1-bit and 4-bit data bus width supporting spec version 2.x/3.x/4.x DS/HS modes up to UHS-I SDR104
- eMMC memory interface with 1/4/8-bit data bus width fully supporting spec version 5.0 HS400
- SLC NAND Flash controller
- Built-in 4K bits OTP memory for secured key storage

Network Interface

- IEEE 802.3 10/100/1000M Ethernet MAC with RGMII interface
- 10/100M Ethernet PHY interface
- WiFi/IEEE802.11 supporting via USB or SDIO
- Bluetooth supporting via USB or UART
- Network interface optimized for mixed WIFI and BT traffic

Integrated I/O Controllers and Interfaces

- Triple USB 2.0 high-speed USB I/O, two USB Hosts and one USB OTG
- Multiple UARTs, I2Cs and PWMs
- SPI interface
- Programmable remote control input circuitry and IR-blaster output
- Built-in 10bit SAR ADC with 4 input channels
- General Purpose IOs with built-in pull up and pull down

System, Peripherals and Misc. Interfaces

- Integrated general purpose timers, counters, DMA controllers
- 24 MHz crystal input
- Embedded debug interface using ICE/JTAG

Power Management

- Multiple internal power domains controlled by software
- Multiple sleep modes for CPU, system, DRAM, etc.
- Multiple internal PLLs to adjust the operating frequencies
- Multi-voltage I/O design for 1.8V and 3.3V

Security

- Trustzone based Trusted Execution Environment (TEE)
- Secured boot, encrypted hardware self-setup OTP, encrypted DRAM with memory integrity checker, hardware key ladder and internal control buses and storage
- Separated secure/non-secure Entropy true RNG
- Pre-region/ID memory security control and electric fence
- Hardware based Trusted Video Path (TVP), and secured contents (needs SecureOS software)
- Secured IO and secured clock

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Package

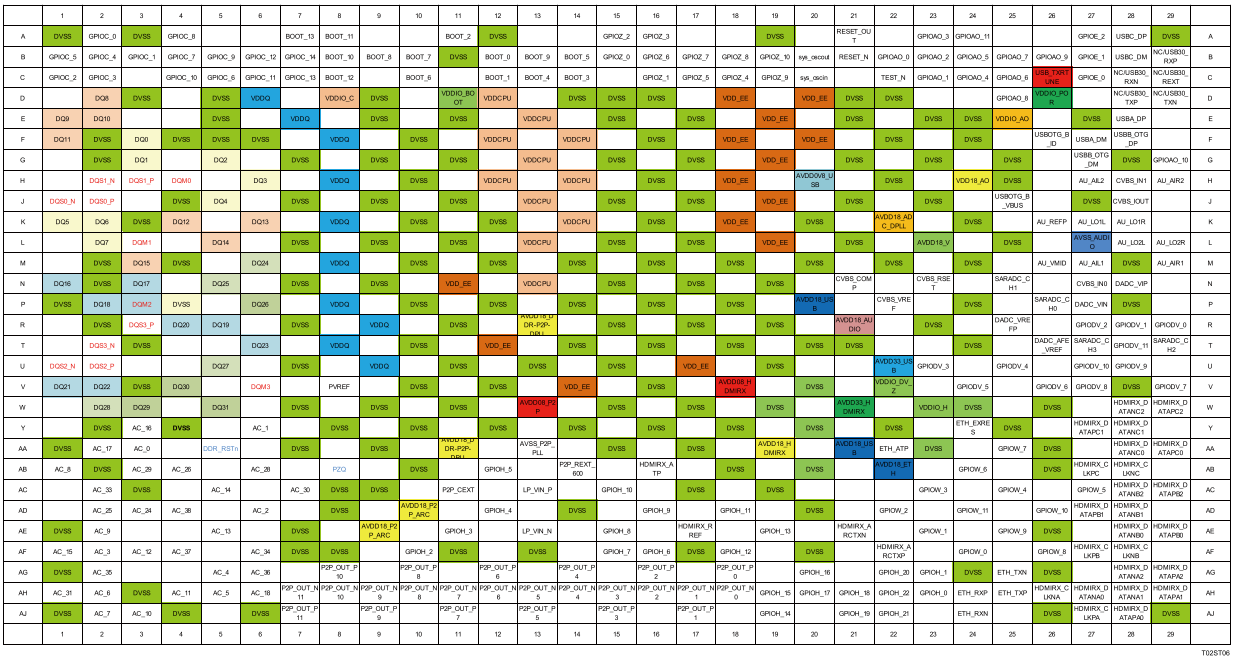
- FCBGA or FCCSP, 19 mm x 19 mm, 0.65 ball pitch, RoHS compliant

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3 Pinout Specification

3.1 Pinout Diagram

Figure 3-1 Pinout Diagram (top view)



3.2 Pin Order

BALL #	NET NAME
A1	DVSS
A2	GPIOC_0
A3	DVSS
A4	GPIOC_8
A7	BOOT_13
A8	BOOT_11
A11	BOOT_2
A12	DVSS
A15	GPIOZ_2
A16	GPIOZ_3
A19	DVSS
A21	RESET_OUT
A23	GPIOAO_3
A24	GPIOAO_11
A27	GPIOE_2

BALL #	NET NAME
A28	USBC_DP
A29	DVSS
B1	GPIOC_5
B2	GPIOC_4
B3	GPIOC_1
B4	GPIOC_7
B5	GPIOC_9
B6	GPIOC_12
B7	GPIOC_14
B8	BOOT_10
B9	BOOT_8
B10	BOOT_7
B11	DVSS
B12	BOOT_0
B13	BOOT_9

BALL #	NET NAME
B14	BOOT_5
B15	GPIOZ_0
B16	GPIOZ_6
B17	GPIOZ_7
B18	GPIOZ_8
B19	GPIOZ_10
B20	SYS_OSCOUT
B21	RESET_N
B22	GPIOAO_0
B23	GPIOAO_2
B24	GPIOAO_5
B25	GPIOAO_7
B26	GPIOAO_9
B27	GPIOE_1
B28	USBC_DM

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BALL #	NET NAME
B29	NC
C1	GPIOC_2
C2	GPIOC_3
C4	GPIOC_10
C5	GPIOC_6
C6	GPIOC_11
C7	GPIOC_13
C8	BOOT_12
C10	BOOT_6
C12	BOOT_1
C13	BOOT_4
C14	BOOT_3
C16	GPIOZ_1
C17	GPIOZ_5
C18	GPIOZ_4
C19	GPIOZ_9
C20	SYS_OSCIN
C22	TEST_N
C23	GPIOAO_1
C24	GPIOAO_4
C25	GPIOAO_6
C26	USB_TXRTUNE
C27	GPIOE_0
C28	NC
C29	NC
D2	DQ8
D3	DVSS
D5	DVSS
D6	VDDQ
D8	VDDIO_C
D9	DVSS
D11	VDDIO_BOOT
D12	VDDCPU
D14	DVSS
D15	DVSS
D16	DVSS

BALL #	NET NAME
D18	VDD_EE
D20	VDD_EE
D21	DVSS
D22	DVSS
D25	GPIOAO_8
D26	VDDIO_POR
D28	NC
D29	NC
E1	DQ9
E2	DQ10
E5	DVSS
E7	VDDQ
E9	DVSS
E11	DVSS
E13	VDDCPU
E15	DVSS
E17	DVSS
E19	VDD_EE
E21	DVSS
E23	DVSS
E24	DVSS
E25	VDDIO_AO
E27	DVSS
E28	USBA_DP
F1	DQ11
F2	DVSS
F3	DQ0
F4	DVSS
F5	DVSS
F6	DVSS
F8	VDDQ
F10	DVSS
F12	VDDCPU
F14	VDDCPU
F16	DVSS
F18	VDD_EE
F20	VDD_EE
F22	DVSS

BALL #	NET NAME
F24	DVSS
F26	USBOTG_B_ID
F27	USBA_DM
F28	USBB_OTG_DP
G2	DVSS
G3	DQ1
G5	DQ2
G7	DVSS
G9	DVSS
G11	DVSS
G13	VDDCPU
G15	DVSS
G17	DVSS
G19	VDD_EE
G21	DVSS
G23	DVSS
G25	DVSS
G27	USBB_OTG_DM
G28	DVSS
G29	GPIOAO_10
H2	DQS1_N
H3	DQS1_P
H4	DQM0
H6	DQ3
H8	VDDQ
H10	DVSS
H12	VDDCPU
H14	VDDCPU
H16	DVSS
H18	VDD_EE
H20	AVDD0V8_USB
H22	DVSS
H24	VDD18_AO
H25	DVSS
H27	AU_AIL2
H28	CVBS_IN1
H29	AU_AIR2

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BALL #	NET NAME
J1	DQS0_N
J2	DQS0_P
J4	DVSS
J5	DQ4
J7	DVSS
J9	DVSS
J11	DVSS
J13	VDDCPU
J15	DVSS
J17	DVSS
J19	VDD_EE
J21	DVSS
J23	DVSS
J25	USBOTG_B_VBUS
J27	DVSS
J28	CVBS_IOUT
K1	DQ5
K2	DQ6
K3	DVSS
K4	DQ12
K6	DQ13
K8	VDDQ
K10	DVSS
K12	DVSS
K14	VDDCPU
K16	DVSS
K18	VDD_EE
K20	DVSS
K22	AVDD18_ADC_DPLL
K24	DVSS
K26	AU_REFP
K27	AU_LO1L
K28	AU_LO1R
L2	DQ7
L3	DQM1
L5	DQ14
L7	DVSS
L9	DVSS

BALL #	NET NAME
L11	DVSS
L13	VDDCPU
L15	DVSS
L17	DVSS
L19	VDD_EE
L21	DVSS
L23	AVDD18_V
L25	DVSS
L27	AVSS_AUDIO
L28	AU_LO2L
L29	AU_LO2R
M2	DVSS
M3	DQ15
M4	DVSS
M6	DQ24
M8	VDDQ
M10	DVSS
M12	DVSS
M14	DVSS
M16	DVSS
M18	DVSS
M20	DVSS
M22	DVSS
M24	DVSS
M26	AU_VMID
M27	AU_AIL1
M28	DVSS
M29	AU_AIR1
N1	DQ16
N2	DVSS
N3	DQ17
N5	DQ25
N7	DVSS
N9	DVSS
N11	VDD_EE
N13	VDDCPU
N15	DVSS
N17	DVSS

BALL #	NET NAME
N19	DVSS
N21	CVBS_COMP
N23	CVBS_RSET
N25	SARADC_CH1
N27	CVBS_IN0
N28	DADC_VIP
P1	DVSS
P2	DQ18
P3	DQM2
P4	DVSS
P6	DQ26
P8	VDDQ
P10	DVSS
P12	DVSS
P14	DVSS
P16	DVSS
P18	DVSS
P20	AVDD18_USB
P22	CVBS_VREF
P24	DVSS
P26	SARADC_CH0
P27	DADC_VIN
P28	DVSS
R2	DVSS
R3	DQS3_P
R4	DQ20
R5	DQ19
R7	DVSS
R9	VDDQ
R11	DVSS
R13	AVDD18_DDR-P2P-DPLL
R15	DVSS
R17	DVSS
R19	DVSS
R21	AVDD18_AUDIO
R23	DVSS
R25	DADC_VREFP

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BALL #	NET NAME
R27	GPIODV_2
R28	GPIODV_1
R29	GPIODV_0
T2	DQS3_N
T3	DVSS
T6	DQ23
T8	VDDQ
T10	DVSS
T12	VDD_EE
T14	DVSS
T16	DVSS
T18	DVSS
T20	DVSS
T21	DVSS
T24	DVSS
T26	DADC_AFE_VREF
T27	SARADC_CH3
T28	GPIODV_11
T29	SARADC_CH2
U1	DQS2_N
U2	DQS2_P
U5	DQ27
U7	DVSS
U9	VDDQ
U11	DVSS
U13	DVSS
U15	DVSS
U17	VDD_EE
U19	DVSS
U22	AVDD33_USB
U23	GPIODV_3
U25	GPIODV_4
U27	GPIODV_10
U28	GPIODV_9
V1	DQ21
V2	DQ22
V3	DVSS

BALL #	NET NAME
V4	DQ30
V6	DQM3
V8	PVREF
V10	DVSS
V12	DVSS
V14	VDD_EE
V16	DVSS
V18	AVDD08_HDMIRX
V20	DVSS
V22	VDDIO_DV_Z
V24	GPIODV_5
V26	GPIODV_6
V27	GPIODV_8
V28	DVSS
V29	GPIODV_7
W2	DQ28
W3	DQ29
W5	DQ31
W7	DVSS
W9	DVSS
W11	DVSS
W13	AVDD08_P2P
W15	DVSS
W17	DVSS
W19	DVSS
W21	AVDD33_HDMIRX
W23	VDDIO_H
W24	DVSS
W26	DVSS
W28	HDMIRX_DATANC2
W29	HDMIRX_DATAPC2
Y2	DVSS
Y3	AC_16
Y4	DVSS
Y6	AC_1
Y8	DVSS
Y10	DVSS

BALL #	NET NAME
Y12	DVSS
Y14	DVSS
Y16	DVSS
Y18	DVSS
Y20	DVSS
Y22	DVSS
Y24	ETH_EXRES
Y25	DVSS
Y27	HDMIRX_DATAPC1
Y28	HDMIRX_DATANC1
AA1	DVSS
AA2	AC_17
AA3	AC_0
AA5	DDR_RSTn
AA7	DVSS
AA9	DVSS
AA11	AVDD18_DDR-P2P-DPLL
AA13	AVSS_P2P_PLL
AA15	DVSS
AA17	DVSS
AA19	AVDD18_HDMIRX
AA21	AVDD18_USB
AA22	ETH_ATP
AA23	DVSS
AA25	GPIOW_7
AA26	DVSS
AA28	HDMIRX_DATANC0
AA29	HDMIRX_DATAPC0
AB1	AC_8
AB2	DVSS
AB3	AC_29
AB4	AC_26
AB6	AC_28
AB8	PZQ
AB10	DVSS
AB12	GPIOH_5

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BALL #	NET NAME
AB14	P2P_REXT_600
AB16	HDMIRX_ATP
AB18	DVSS
AB20	DVSS
AB22	AVDD18_ETH
AB24	GPIOW_6
AB26	DVSS
AB27	HDMIRX_CLKPC
AB28	HDMIRX_CLKNC
AC2	AC_33
AC3	DVSS
AC5	AC_14
AC7	AC_30
AC8	DVSS
AC9	DVSS
AC11	P2P_CEXT
AC13	LP_VIN_P
AC15	GPIOH_10
AC17	DVSS
AC19	DVSS
AC23	GPIOW_3
AC25	GPIOW_4
AC27	GPIOW_5
AC28	HDMIRX_DATANB2
AC29	HDMIRX_DATAPB2
AD2	AC_25
AD3	AC_24
AD4	AC_38
AD6	AC_2
AD8	DVSS
AD10	AVDD18_P2P_ARC
AD12	GPIOH_4
AD14	DVSS
AD16	GPIOH_9
AD18	GPIOH_11
AD20	DVSS

BALL #	NET NAME
AD22	GPIOW_2
AD24	GPIOW_11
AD26	GPIOW_10
AD27	HDMIRX_DATAPB1
AD28	HDMIRX_DATANB1
AE1	DVSS
AE2	AC_9
AE5	AC_13
AE7	DVSS
AE9	AVDD18_P2P_ARC
AE11	GPIOH_3
AE13	LP_VIN_N
AE15	GPIOH_8
AE17	HDMIRX_RREF
AE19	GPIOH_13
AE21	HDMIRX_ARCTXN
AE23	GPIOW_1
AE25	GPIOW_9
AE26	DVSS
AE28	HDMIRX_DATANB0
AE29	HDMIRX_DATAPB0
AF1	AC_15
AF2	AC_3
AF3	AC_12
AF4	AC_37
AF6	AC_34
AF7	DVSS
AF8	DVSS
AF10	GPIOH_2
AF11	DVSS
AF13	DVSS
AF15	GPIOH_7
AF16	GPIOH_6
AF17	DVSS
AF18	GPIOH_12
AF20	DVSS

BALL #	NET NAME
AF22	HDMIRX_ARCTXP
AF24	GPIOW_0
AF26	GPIOW_8
AF27	HDMIRX_CLKPB
AF28	HDMIRX_CLKNB
AG1	DVSS
AG2	AC_35
AG5	AC_4
AG6	AC_36
AG8	P2P_OUT_P10
AG10	P2P_OUT_P8
AG12	P2P_OUT_P6
AG14	P2P_OUT_P4
AG16	P2P_OUT_P2
AG18	P2P_OUT_P0
AG20	GPIOH_16
AG22	GPIOH_20
AG23	GPIOH_1
AG24	DVSS
AG25	ETH_TXN
AG26	DVSS
AG28	HDMIRX_DATANA2
AG29	HDMIRX_DATAPA2
AH1	AC_31
AH2	AC_6
AH3	DVSS
AH4	AC_11
AH5	AC_5
AH6	AC_18
AH7	P2P_OUT_N11
AH8	P2P_OUT_N10
AH9	P2P_OUT_N9
AH10	P2P_OUT_N8
AH11	P2P_OUT_N7
AH12	P2P_OUT_N6
AH13	P2P_OUT_N5

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BALL #	NET NAME
AH14	P2P_OUT_N4
AH15	P2P_OUT_N3
AH16	P2P_OUT_N2
AH17	P2P_OUT_N1
AH18	P2P_OUT_N0
AH19	GPIOH_15
AH20	GPIOH_17
AH21	GPIOH_18
AH22	GPIOH_22
AH23	GPIOH_0
AH24	ETH_RXP
AH25	ETH_TXP
AH26	HDMIRX_CLKNA

BALL #	NET NAME
AH27	HDMIRX_DATANA0
AH28	HDMIRX_DATANA1
AH29	HDMIRX_DATAPA1
AJ1	DVSS
AJ2	AC_7
AJ3	AC_10
AJ4	DVSS
AJ6	DVSS
AJ7	P2P_OUT_P11
AJ9	P2P_OUT_P9
AJ11	P2P_OUT_P7
AJ13	P2P_OUT_P5
AJ15	P2P_OUT_P3

BALL #	NET NAME
AJ17	P2P_OUT_P1
AJ19	GPIOH_14
AJ21	GPIOH_19
AJ22	GPIOH_21
AJ24	ETH_RXN
AJ26	DVSS
AJ27	HDMIRX_CLKPA
AJ28	HDMIRX_DATAPA0
AJ29	DVSS

3.3 Pin Description

The pin assignments are described in the following table.

Table 3-1 Pin Name Assignments

Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
GPIOC- refer to Table 3-2 for functional multiplex information.					
GPIOC_0	DIO	Up	General purpose input/output bank C signal 0	VDDIO_C	NC
GPIOC_1	DIO	Up	General purpose input/output bank C signal 1	VDDIO_C	NC
GPIOC_2	DIO	Up	General purpose input/output bank C signal 2	VDDIO_C	NC
GPIOC_3	DIO	Up	General purpose input/output bank C signal 3	VDDIO_C	NC
GPIOC_4	DIO	Up	General purpose input/output bank C signal 4	VDDIO_C	NC
GPIOC_5	DIO	Up	General purpose input/output bank C signal 5	VDDIO_C	NC
GPIOC_6	DIO	Up	General purpose input/output bank C signal 6	VDDIO_C	NC
GPIOC_7	DIO	Up	General purpose input/output bank C signal 7	VDDIO_C	NC
GPIOC_8	DIO	Up	General purpose input/output bank C signal 8	VDDIO_C	NC
GPIOC_9	DIO	Up	General purpose input/output bank C signal 9	VDDIO_C	NC
GPIOC_10	DIO	Up	General purpose input/output bank C signal 10	VDDIO_C	NC
GPIOC_11	DIO	Down	General purpose input/output bank C signal 11	VDDIO_C	NC
GPIOC_12	DIO	Up	General purpose input/output bank C signal 12	VDDIO_C	NC
GPIOC_13	DIO	Down	General purpose input/output bank C signal 13	VDDIO_C	NC
GPIOC_14	DIO	Up	General purpose input/output bank C signal 14	VDDIO_C	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
VDDIO_C	P	-	Power supply for GPIO bank C	-	1.8V or 3.3V
GPIOZ - refer to Table 3-3 for functional multiplex information.					
GPIOZ_0	DIO	Down	General purpose input/output bank Z signal 0	VDDIO_DV_Z	NC
GPIOZ_1	DIO	Up	General purpose input/output bank Z signal 1	VDDIO_DV_Z	NC
GPIOZ_2	DIO	Up	General purpose input/output bank Z signal 2	VDDIO_DV_Z	NC
GPIOZ_3	DIO	Up	General purpose input/output bank Z signal 3	VDDIO_DV_Z	NC
GPIOZ_4	DIO	Down	General purpose input/output bank Z signal 4	VDDIO_DV_Z	NC
GPIOZ_5	DIO	Down	General purpose input/output bank Z signal 5	VDDIO_DV_Z	NC
GPIOZ_6	DIO	Down	General purpose input/output bank Z signal 6	VDDIO_DV_Z	NC
GPIOZ_7	DIO	Down	General purpose input/output bank Z signal 7	VDDIO_DV_Z	NC
GPIOZ_8	DIO	Down	General purpose input/output bank Z signal 8	VDDIO_DV_Z	NC
GPIOZ_9	DIO	Up	General purpose input/output bank Z signal 9	VDDIO_DV_Z	NC
GPIOZ_10	DIO	Up	General purpose input/output bank Z signal 10	VDDIO_DV_Z	NC
VDDIO_DV_Z	P	-	Power supply for GPIO bank DV and bank Z	-	1.8V or 3.3V
BOOT- refer to Table 3-4 for functional multiplex information.					
BOOT_0	DIO	Up	General purpose input/output bank BOOT signal 0	VDDIO_BOOT	NC
BOOT_1	DIO	Up	General purpose input/output bank BOOT signal 1	VDDIO_BOOT	NC
BOOT_2	DIO	Up	General purpose input/output bank BOOT signal 2	VDDIO_BOOT	NC
BOOT_3	DIO	Up	General purpose input/output bank BOOT signal 3	VDDIO_BOOT	NC
BOOT_4	DIO	Up	General purpose input/output bank BOOT signal 4	VDDIO_BOOT	NC
BOOT_5	DIO	Up	General purpose input/output bank BOOT signal 5	VDDIO_BOOT	NC
BOOT_6	DIO	Up	General purpose input/output bank BOOT signal 6	VDDIO_BOOT	NC
BOOT_7	DIO	Up	General purpose input/output bank BOOT signal 7	VDDIO_BOOT	NC
BOOT_8	DIO	Up	General purpose input/output bank BOOT signal 8	VDDIO_BOOT	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
BOOT_9	DIO	Down	General purpose input/output bank BOOT signal 9	VDDIO_BOOT	NC
BOOT_10	DIO	Up	General purpose input/output bank BOOT signal 10	VDDIO_BOOT	NC
BOOT_11	DIO	Down	General purpose input/output bank BOOT signal 11	VDDIO_BOOT	NC
BOOT_12	DIO	Up	General purpose input/output bank BOOT signal 12	VDDIO_BOOT	NC
BOOT_13	DIO	Up	General purpose input/output bank BOOT signal 13	VDDIO_BOOT	NC
VDDIO_BOOT	P	-	Power supply for GPIO bank BOOT	-	1.8V or 3.3V
GPIOH- refer to Table 3-5 for functional multiplex information.					
GPIOH_0	DIO	Down	General purpose input/output bank H signal 0	VDDIO_H	NC
GPIOH_1	DIO	Down	General purpose input/output bank H signal 1	VDDIO_H	NC
GPIOH_2	DIO	Down	General purpose input/output bank H signal 2	VDDIO_H	NC
GPIOH_3	DIO	Down	General purpose input/output bank H signal 3	VDDIO_H	NC
GPIOH_4	DIO	Down	General purpose input/output bank H signal 4	VDDIO_H	NC
GPIOH_5	DIO	Down	General purpose input/output bank H signal 5	VDDIO_H	NC
GPIOH_6	DIO	Down	General purpose input/output bank H signal 6	VDDIO_H	NC
GPIOH_7	DIO	Down	General purpose input/output bank H signal 7	VDDIO_H	NC
GPIOH_8	DIO	Down	General purpose input/output bank H signal 8	VDDIO_H	NC
GPIOH_9	DIO	Down	General purpose input/output bank H signal 9	VDDIO_H	NC
GPIOH_10	DIO	Down	General purpose input/output bank H signal 10	VDDIO_H	NC
GPIOH_11	DIO	Down	General purpose input/output bank H signal 11	VDDIO_H	NC
GPIOH_12	DIO	Down	General purpose input/output bank H signal 12	VDDIO_H	NC
GPIOH_13	DIO	Down	General purpose input/output bank H signal 13	VDDIO_H	NC
GPIOH_14	DIO	Down	General purpose input/output bank H signal 14	VDDIO_H	NC
GPIOH_15	DIO	Down	General purpose input/output bank H signal 15	VDDIO_H	NC
GPIOH_16	OD5V	-	General purpose input/output bank H signal 16	VDD18_AO	NC
GPIOH_17	DIO	Z	General purpose input/output bank H signal 17	VDDIO_H	NC
GPIOH_18	DIO	Z	General purpose input/output bank H signal 18	VDDIO_H	NC
GPIOH_19	DIO	Z	General purpose input/output bank H signal 19	VDDIO_H	NC
GPIOH_20	DIO	Z	General purpose input/output bank H signal 20	VDDIO_H	NC
GPIOH_21	DIO	Z	General purpose input/output bank H signal 21	VDDIO_H	NC
GPIOH_22	DIO	Z	General purpose input/output bank H signal 22	VDDIO_H	NC
VDDIO_H	P	-	Power supply for GPIO bank H	-	1.8V or 3.3V
GPIOAO- refer to Table 3-6 for functional multiplex information.					

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
GPIOAO_0	DIO	Up	General purpose input/output bank AO signal 0	VDDIO_AO	NC
GPIOAO_1	DIO	Up	General purpose input/output bank AO signal 1	VDDIO_AO	NC
GPIOAO_2	DIO	Up	General purpose input/output bank AO signal 2	VDDIO_AO	NC
GPIOAO_3	DIO	Down	General purpose input/output bank AO signal 3	VDDIO_AO	NC
GPIOAO_4	DIO	Down	General purpose input/output bank AO signal 4	VDDIO_AO	NC
GPIOAO_5	DIO	Up	General purpose input/output bank AO signal 5	VDDIO_AO	NC
GPIOAO_6	DIO	Down	General purpose input/output bank AO signal 6	VDDIO_AO	NC
GPIOAO_7	DIO	Down	General purpose input/output bank AO signal 7	VDDIO_AO	NC
GPIOAO_8	DIO	Down	General purpose input/output bank AO signal 8	VDDIO_AO	NC
GPIOAO_9	DIO	Down	General purpose input/output bank AO signal 9	VDDIO_AO	NC
GPIOAO_10	DIO	Z	General purpose input/output bank AO signal 10	VDDIO_AO	NC
GPIOAO_11	DIO	Down	General purpose input/output bank AO signal 11	VDDIO_AO	NC
TEST_N	DIO	Up	SOC test pin and general purpose input/output bank AO signal 12. Should be pulled up during normal power-on.	VDDIO_AO	NC
RESET_N	Input	Down	System reset input	VDDIO_AO	To RESET_N
VDDIO_AO	P	-	Power supply for GPIO bank AO	VDDIO_AO	1.8V or 3.3V
GPIOE- refer to Table 3-7 for functional multiplex information.					
GPIOE_0	DIO	Z	General purpose input/output bank E signal 0	VDD18_AO	NC
GPIOE_1	DIO	Z	General purpose input/output bank E signal 1	VDD18_AO	NC
GPIOE_2	DIO	Z	General purpose input/output bank E signal 2	VDD18_AO	NC
VDD18_AO	P	-	Power supply for GPIO bank E, SARADC, SYS_SOC, EFUSE, and IOVREF	-	VDD18_AO
GPIODV- refer to Table 3-8 for functional multiplex information.					
GPIODV_0	DIO	Up	General purpose input/output bank DV signal 0	VDDIO_DV_Z	NC
GPIODV_1	DIO	Up	General purpose input/output bank DV signal 1	VDDIO_DV_Z	NC
GPIODV_2	DIO	Up	General purpose input/output bank DV signal 2	VDDIO_DV_Z	NC
GPIODV_3	DIO	Down	General purpose input/output bank DV signal 3	VDDIO_DV_Z	NC
GPIODV_4	DIO	Down	General purpose input/output bank DV signal 4	VDDIO_DV_Z	NC
GPIODV_5	DIO	Down	General purpose input/output bank DV signal 5	VDDIO_DV_Z	NC
GPIODV_6	DIO	Up	General purpose input/output bank DV signal 6	VDDIO_DV_Z	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
GPIODV_7	DIO	Up	General purpose input/output bank DV signal 7	VDDIO_DV_Z	NC
GPIODV_8	DIO	Up	General purpose input/output bank DV signal 8	VDDIO_DV_Z	NC
GPIODV_9	DIO	Up	General purpose input/output bank DV signal 9	VDDIO_DV_Z	NC
GPIODV_10	DIO	Up	General purpose input/output bank DV signal 10	VDDIO_DV_Z	NC
GPIODV_11	DIO	Up	General purpose input/output bank DV signal 11	VDDIO_DV_Z	NC
VDDIO_DV_Z	P	-	Power supply for GPIO bank DV and bank Z	-	1.8V or 3.3V
GPIOW- refer to Table 3-9 for functional multiplex information.					
GPIOW_0	OD5V	-	General purpose input/output bank W signal 0	VDD18_AO	NC
GPIOW_1	OD5V	-	General purpose input/output bank W signal 1	VDD18_AO	NC
GPIOW_2	OD5V	-	General purpose input/output bank W signal 2	VDD18_AO	NC
GPIOW_3	OD5V	-	General purpose input/output bank W signal 3	VDD18_AO	NC
GPIOW_4	OD5V	-	General purpose input/output bank W signal 4	VDD18_AO	NC
GPIOW_5	OD5V	-	General purpose input/output bank W signal 5	VDD18_AO	NC
GPIOW_6	OD5V	-	General purpose input/output bank W signal 6	VDD18_AO	NC
GPIOW_7	OD5V	-	General purpose input/output bank W signal 7	VDD18_AO	NC
GPIOW_8	OD5V	-	General purpose input/output bank W signal 8	VDD18_AO	NC
GPIOW_9	OD5V	-	General purpose input/output bank W signal 9	VDD18_AO	NC
GPIOW_10	OD5V	-	General purpose input/output bank W signal 10	VDD18_AO	NC
GPIOW_11	OD5V	-	General purpose input/output bank W signal 11	VDD18_AO	NC
SARADC					
SARADC_CH0	AI	-	ADC channel 0 input	VDD18_AO	NC
SARADC_CH1	AI	-	ADC channel 1 input	VDD18_AO	NC
SARADC_CH2	AI	-	ADC channel 2 input	VDD18_AO	NC
SARADC_CH3	AI	-	ADC channel 3 input	VDD18_AO	NC
VDD18_AO	P	-	Power supply for AO 1.8V	-	To 1.8V
CVBS OUT					
CVBS_COMP	A	-	CVBS external compensation capacitor connection	AVDD18_V	NC
CVBS_RSET	A	-	CVBS output strength setting resistor	AVDD18_V	NC
CVBS_VREF	A	-	CVBS reference voltage filter cap	AVDD18_V	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
CVBS_IOUT	AO	-	CVBS output	AVDD18_V	NC
AVDD18_V	P	-	Power Supply for analog 1.8V	-	To 1.8V
CVBS IN					
CVBS_IN0	AI	-	Analog CVBS input port 0	AVDD18_V	NC
CVBS_IN1	AI	-	Analog CVBS input port 1	AVDD18_V	NC
DADC_ AFE_VREF	A	-	CVBS input reference voltage	AVDD18_V	NC
DADC_ VREFP	A	-	CVBS ADC internal positive reference voltage	AVDD18_V	NC
AVDD18_ ADC_DPLL	A	-	Power supply for ADC PLL	-	1.8V
Demod					
DADC_VIN	AI	-	Demod ADC Low IF_N input	AVDD18_V	NC
DADC_VIP	AI	-	Demod ADC Low IF_P input	AVDD18_V	NC
LVDS/V-by-One/P2P-TCON					
P2P_OUT_ N0	AO	-	LVDS A/V-by-one/P2P data0 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P0	AO	-	LVDS A/V-by-one/P2P data0 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N1	AO	-	LVDS A/V-by-one/P2P data1 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P1	AO	-	LVDS A/V-by-one/P2P data1 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N2	AO	-	LVDS A/V-by-one/P2P data2 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P2	AO	-	LVDS A/V-by-one/P2P data2 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N3	AO	-	LVDS A CLK, V-by-one/P2P data3 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P3	AO	-	LVDS A CLK, V-by-one/P2P data3 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N4	AO	-	LVDS A data3, V-by-one/P2P data4 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P4	AO	-	LVDS A data3, V-by-one/P2P data4 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N5	AO	-	LVDS A data4, V-by-one/P2P data5 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P5	AO	-	LVDS A data4, V-by-one/P2P data5 positive output	AVDD18_ P2P_ARC	NC
P2P_OUT_ N6	AO	-	LVDS B data0, V-by-one/P2P data6 negative output	AVDD18_ P2P_ARC	NC
P2P_OUT_ P6	AO	-	LVDS B data0, V-by-one/P2P data6 positive output	AVDD18_ P2P_ARC	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
P2P_OUT_N7	AO	-	LVDS B data1, V-by-one/P2P data7 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P7	AO	-	LVDS B data1, V-by-one/P2P data7 positive output	AVDD18_P2P_ARC	NC
P2P_OUT_N8	AO	-	LVDS port B data2, P2P data8 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P8	AO	-	LVDS port B data2, P2P data8 positive output	AVDD18_P2P_ARC	NC
P2P_OUT_N9	AO	-	LVDS port B CLK, P2P data9 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P9	AO	-	LVDS port B CLK, P2P data9 positive output	AVDD18_P2P_ARC	NC
P2P_OUT_N10	AO	-	LVDS port B data3, P2P data10 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P10	AO	-	LVDS port B data3, P2P data10 positive output	AVDD18_P2P_ARC	NC
P2P_OUT_N11	AO	-	LVDS port B data4, P2P data11 negative output	AVDD18_P2P_ARC	NC
P2P_OUT_P11	AO	-	LVDS port B data4, P2P data11 positive output	AVDD18_P2P_ARC	NC
P2P_REXT_600	AO	-	LVDS and V-by-one reference resistor connection	AVDD18_P2P_ARC	NC
P2P_CEXT	AO	-	LVDS and P2P port test pin	AVDD18_P2P_ARC	NC
LP_VIN_P	AO	-	LVDS and P2P port test output	AVDD18_P2P_ARC	NC
LP_VIN_N	AO	-	LVDS and P2P port test output	AVDD18_P2P_ARC	NC
AVDD18_P2P_ARC	AP	-	Analog power supply 1.8V for V-by-one	-	1.8V
AVSS_P2P_PLL	AP	-	Analog ground for P2P PLL	-	GND
AVDD08_P2P	AP	-	Power supply for P2P analog 0.8V	-	0.8V
HDMI RX					
HDMIRX_ARCTXN	AO	-	Audio Return Channel output	AVDD18_P2P_ARC	NC
HDMIRX_ARCTXP	AO	-	Audio Return Channel output	AVDD18_P2P_ARC	NC
HDMIRX_RREF	A	-	HDMIRX Reference resistor connection	AVDD18_HDMI	NC
HDMIRX_CLKNA	AI	-	HDMIRX Port A TMDS clock negative input	AVDD33_HDMI	NC
HDMIRX_CLKPA	AI	-	HDMIRX Port A TMDS clock positive input	AVDD33_HDMI	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
HDMIRX_DATANA0	AI	-	HDMIRX Port A TMDS data0 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPA0	AI	-	HDMIRX Port A TMDS data0 positive input	AVDD33_HDMI	NC
HDMIRX_DATANA1	AI	-	HDMIRX Port A TMDS data1 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPA1	AI	-	HDMIRX Port A TMDS data1 positive input	AVDD33_HDMI	NC
HDMIRX_DATANA2	AI	-	HDMIRX Port A TMDS data2 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPA2	AI	-	HDMIRX Port A TMDS data2 positive input	AVDD33_HDMI	NC
HDMIRX_CLKNB	AI	-	HDMIRX Port B TMDS clock negative input	AVDD33_HDMI	NC
HDMIRX_CLKPB	AI	-	HDMIRX Port B TMDS clock positive input	AVDD33_HDMI	NC
HDMIRX_DATANB0	AI	-	HDMIRX Port B TMDS data0 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPB0	AI	-	HDMIRX Port B TMDS data0 positive input	AVDD33_HDMI	NC
HDMIRX_DATANB1	AI	-	HDMIRX Port B TMDS data1 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPB1	AI	-	HDMIRX Port B TMDS data1 positive input	AVDD33_HDMI	NC
HDMIRX_DATANB2	AI	-	HDMIRX Port B TMDS data2 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPB2	AI	-	HDMIRX Port B TMDS data2 positive input	AVDD33_HDMI	NC
HDMIRX_CLKNC	AI	-	HDMIRX Port C TMDS clock negative input	AVDD33_HDMI	NC
HDMIRX_CLKPC	AI	-	HDMIRX Port C TMDS clock positive input	AVDD33_HDMI	NC
HDMIRX_DATANC0	AI	-	HDMIRX Port C TMDS data0 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPC0	AI	-	HDMIRX Port C TMDS data0 positive input	AVDD33_HDMI	NC
HDMIRX_DATANC1	AI	-	HDMIRX Port C TMDS data1 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPC1	AI	-	HDMIRX Port C TMDS data1 positive input	AVDD33_HDMI	NC
HDMIRX_DATANC2	AI	-	HDMIRX Port C TMDS data2 negative input	AVDD33_HDMI	NC
HDMIRX_DATAPC2	AI	-	HDMIRX Port C TMDS data2 positive input	AVDD33_HDMI	NC
HDMIRX_ATP	-	-	HDMIRX test port	AVDD33_HDMI	NC
AVDD33_HDMIRX	AP	-	Power supply for HDMI RX analog 3.3V	-	To 3.3V

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
AVDD18_ HDMIRX	AP	-	Power supply for HDMI RX analog 1.8V	-	To 1.8V
AVDD08_ HDMIRX	AP	-	Power supply for HDMI RX analog 0.8V	-	To 0.8V
Audio					
AU_AIL1	AI	-	Audio line in, left channel of port 1	AVDD18_ AUDIO	NC
AU_AIR1	AI	-	Audio line in, right channel of port 1	AVDD18_ AUDIO	NC
AU_AIL2	AI	-	Audio line in, left channel of port 2	AVDD18_ AUDIO	NC
AU_AIR2	AI	-	Audio line in, right channel of port 2	AVDD18_ AUDIO	NC
AU_LO1L	AO	-	Audio line out, left channel of port 1	AVDD18_ AUDIO	NC
AU_LO1R	AO	-	Audio line out, right channel of port 1	AVDD18_ AUDIO	NC
AU_LO2L	AO	-	Audio line out, left channel of port 2	AVDD18_ AUDIO	NC
AU_LO2R	AO	-	Audio line out, right channel of port 2	AVDD18_ AUDIO	NC
AU_REFP	A	-	Audio ADC/DAC positive reference voltage	AVDD18_ AUDIO	NC
AU_VMID	A	-	MID voltage of audio ADC/DAC	AVDD18_ AUDIO	NC
AVDD18_ AUDIO	AP	-	Analog 1.8V for Audio CODEC	-	To 1.8V
AVSS_ Audio	AP	-	Analog ground for Audio CODEC	-	To VSS
DRAM- refer to Table 3-10 for functional multiplex information.					
AC_0	DO	-	DDR PHY address/command/control signal bit 0	VDDQ	NC
AC_1	DO	-	DDR PHY address/command/control signal bit 1	VDDQ	NC
AC_2	DO	-	DDR PHY address/command/control signal bit 2	VDDQ	NC
AC_3	DO	-	DDR PHY address/command/control signal bit 3	VDDQ	NC
AC_4	DO	-	DDR PHY address/command/control signal bit 4	VDDQ	NC
AC_5	DO	-	DDR PHY address/command/control signal bit 5	VDDQ	NC
AC_6	DO	-	DDR PHY address/command/control signal bit 6	VDDQ	NC
AC_7	DO	-	DDR PHY address/command/control signal bit 7	VDDQ	NC
AC_8	DO	-	DDR PHY address/command/control signal bit 8	VDDQ	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
AC_9	DO	-	DDR PHY address/command/control signal bit 9	VDDQ	NC
AC_10	DO	-	DDR PHY address/command/control signal bit 10	VDDQ	NC
AC_11	DO	-	DDR PHY address/command/control signal bit 11	VDDQ	NC
AC_12	DO	-	DDR PHY address/command/control signal bit 12	VDDQ	NC
AC_13	DO	-	DDR PHY address/command/control signal bit 13	VDDQ	NC
AC_14	DO	-	DDR PHY address/command/control signal bit 14	VDDQ	NC
AC_15	DO	-	DDR PHY address/command/control signal bit 15	VDDQ	NC
AC_16	DO	-	DDR PHY address/command/control signal bit 16	VDDQ	NC
AC_17	DO	-	DDR PHY address/command/control signal bit 17	VDDQ	NC
AC_18	DO	-	DDR PHY address/command/control signal bit 18	VDDQ	NC
AC_24	DO	-	DDR PHY address/command/control signal bit 24	VDDQ	NC
AC_25	DO	-	DDR PHY address/command/control signal bit 25	VDDQ	NC
AC_26	DO	-	DDR PHY address/command/control signal bit 26	VDDQ	NC
AC_28	DO	-	DDR PHY address/command/control signal bit 28	VDDQ	NC
AC_29	DO	-	DDR PHY address/command/control signal bit 29	VDDQ	NC
AC_30	DO	-	DDR PHY address/command/control signal bit 30	VDDQ	NC
AC_31	DO	-	DDR PHY address/command/control signal bit 31	VDDQ	NC
AC_33	DO	-	DDR PHY address/command/control signal bit 33	VDDQ	NC
AC_34	DO	-	DDR PHY address/command/control signal bit 34	VDDQ	NC
AC_35	DO	-	DDR PHY address/command/control signal bit 35	VDDQ	NC
AC_36	DO	-	DDR PHY address/command/control signal bit 36	VDDQ	NC
AC_37	DO	-	DDR PHY address/command/control signal bit 37	VDDQ	NC
AC_38	DO	-	DDR PHY address/command/control signal bit 38	VDDQ	NC
DDR_RSTn	DO	-	DDR3/DDR4/LPDDR4 RSTn	VDDQ	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
DQ0	DIO	-	DRAM data bus bit 0	VDDQ	To DRAM
DQ1	DIO	-	DRAM data bus bit 1	VDDQ	To DRAM
DQ2	DIO	-	DRAM data bus bit 2	VDDQ	To DRAM
DQ3	DIO	-	DRAM data bus bit 3	VDDQ	To DRAM
DQ4	DIO	-	DRAM data bus bit 4	VDDQ	To DRAM
DQ5	DIO	-	DRAM data bus bit 5	VDDQ	To DRAM
DQ6	DIO	-	DRAM data bus bit 6	VDDQ	To DRAM
DQ7	DIO	-	DRAM data bus bit 7	VDDQ	To DRAM
DQ8	DIO	-	DRAM data bus bit 8	VDDQ	To DRAM
DQ9	DIO	-	DRAM data bus bit 9	VDDQ	To DRAM
DQ10	DIO	-	DRAM data bus bit 10	VDDQ	To DRAM
DQ11	DIO	-	DRAM data bus bit 11	VDDQ	To DRAM
DQ12	DIO	-	DRAM data bus bit 12	VDDQ	To DRAM
DQ13	DIO	-	DRAM data bus bit 13	VDDQ	To DRAM
DQ14	DIO	-	DRAM data bus bit 14	VDDQ	To DRAM
DQ15	DIO	-	DRAM data bus bit 15	VDDQ	To DRAM
DQ16	DIO	-	DRAM data bus bit 16	VDDQ	NC
DQ17	DIO	-	DRAM data bus bit 17	VDDQ	NC
DQ18	DIO	-	DRAM data bus bit 18	VDDQ	NC
DQ19	DIO	-	DRAM data bus bit 19	VDDQ	NC
DQ20	DIO	-	DRAM data bus bit 20	VDDQ	NC
DQ21	DIO	-	DRAM data bus bit 21	VDDQ	NC
DQ22	DIO	-	DRAM data bus bit 22	VDDQ	NC
DQ23	DIO	-	DRAM data bus bit 23	VDDQ	NC
DQ24	DIO	-	DRAM data bus bit 24	VDDQ	NC
DQ25	DIO	-	DRAM data bus bit 25	VDDQ	NC
DQ26	DIO	-	DRAM data bus bit 26	VDDQ	NC
DQ27	DIO	-	DRAM data bus bit 27	VDDQ	NC
DQ28	DIO	-	DRAM data bus bit 28	VDDQ	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
DQ29	DIO	-	DRAM data bus bit 29	VDDQ	NC
DQ30	DIO	-	DRAM data bus bit 30	VDDQ	NC
DQ31	DIO	-	DRAM data bus bit 31	VDDQ	NC
DQM0	DIO	-	DRAM data mask 0	VDDQ	To DRAM
DQM1	DIO	-	DRAM data mask 1	VDDQ	To DRAM
DQM2	DIO	-	DRAM data mask 2	VDDQ	NC
DQM3	DIO	-	DRAM data mask 3	VDDQ	NC
DQS0_P	DIO	-	DRAM data strobe 0	VDDQ	To DRAM
DQS0_N	DIO	-	DRAM data strobe 0 complementary	VDDQ	To DRAM
DQS1_P	DIO	-	DRAM data strobe 1	VDDQ	To DRAM
DQS1_N	DIO	-	DRAM data strobe 1 complementary	VDDQ	To DRAM
DQS2_P	DIO	-	DRAM data strobe 2	VDDQ	NC
DQS2_N	DIO	-	DRAM data strobe 2 complementary	VDDQ	NC
DQS3_P	DIO	-	DRAM data strobe 3	VDDQ	NC
DQS3_N	DIO	-	DRAM data strobe 3 complementary	VDDQ	NC
PZQ	A	-	DRAM reference pin for ZQ calibration, to GND by 240ohm	VDDQ	-
PVREF	A	-	DRAM reference voltage	VDDQ	NC
USB					
USBA_DP	AIO	-	USB 2.0 Port A host positive data signal	AVDD33_ USB	NC
USBA_DM	AIO	-	USB 2.0 Port A host negative data signal	AVDD33_ USB	NC
USBB_ OTG_DP	AIO	-	USB 2.0 Port B OTG positive data signal	AVDD33_ USB	NC
USBB_ OTG_DM	AIO	-	USB 2.0 Port B OTG negative data signal	AVDD33_ USB	NC
USBOTG_ B_ID	AI	-	USB 2.0 Port B ID detect signal, internal pull up to 1.8V	AVDD18_ USB	NC
USBOTG_ B_VBUS	AI	-	USB 2.0 Port B host cable power detection (1.8V input tolerance)	AVDD18_ USB	NC
USBC_DP	AIO	-	USB 2.0 Port C host positive data signal	AVDD33_ USB	NC
USBC_DM	AIO	-	USB 2.0 Port C host negative data signal	AVDD33_ USB	NC
USB_ TXRTUNE	AIO	-	USB 2.0 Port host output strength setting resistor	AVDD33_ USB	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
AVDD33_ USB	AP	-	Power supply for USB 2.0(A&B&C Channel) analog 3.3V	-	To 3.3V
AVDD18_ USB	AP	-	Power supply for USB 2.0(A&B&C Channel) analog 1.8V	-	To 1.8V
AVDD0V8_ USB	AP	-	Power supply for USB analog 0.8V	-	To 0.8V
Ethernet					
ETH_ATP	-	-	Ethernet test port	AVDD18_ ETH	NC
ETH_ EXRES	A	-	Ethernet PHY external resistor connection	AVDD18_ ETH	NC
ETH_RXN	AIO	-	Ethernet PHY receive date negative input	AVDD18_ ETH	NC
ETH_RXP	AIO	-	Ethernet PHY receive data positive input	AVDD18_ ETH	NC
ETH_TXN	AIO	-	Ethernet PHY transmit data negative output	AVDD18_ ETH	NC
ETH_TXP	AIO	-	Ethernet PHY transmit data positive output	AVDD18_ ETH	NC
AVDD18_ ETH	AP	-	Analog 1.8V power supply for Ethernet	-	To 1.8V
System Clock & PLL					
SYS_ OSCIN	AI	-	24MHz crystal oscillator input	VDD18_AO	-
SYS_ OSCOUT	AO	-	24MHz crystal oscillator output	VDD18_AO	-
AVDD18_ DDR-P2P-DPLL	AP	-	Power supply for DPLL analog 1.8V	-	-
Power On Reset					
RESET_ OUT	AO	Down	Power on reset output	VDDIO_ POR	NC
VDDIO_ POR	AP	-	Power supply for POR	-	NC
Digital Power					
VDDCPU	P	-	Power supply for CPU (Cortex A55)	-	To VDDCP-U
VDDQ	P	-	DDR IO Power supply for DDR PHY	-	To VDDQ
VDD_EE	P	-	Power supply for GPU and core logic	-	To VDD_ EE
VDD18_AO	P	-	1.8V Power supply for Always On Domain	-	To VDD18_ AO
VDDIO_ BOOT	P	-	Power supply for GPIO bank BOOT	-	NC

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Net Name	Type	Default Pull Up/DN	Description	Power Domain	If Unused
VDDIO_C	P	-	Power supply for GPIO bank C	-	1.8V or 3.3V
VDDIO_H	P	-	Power supply for GPIO bank H	-	NC
VDDIO_AO	P	-	Power supply for GPIO bank AO	-	NC
VDDIO_DV_Z	P	-	Power supply for GPIO bank DV and bank Z	-	NC
Digital Ground					
DVSS	P	-	Digital power ground	-	To GND

Abbreviations:

- DI = Digital input pin
- DO = Digital output pin
- DIO = Digital input/output pin
- OD5V = 5V input tolerant open drain (OD) output pin, need external pull up
- A = Analog setting or filtering pin
- AI = Analog input pin
- AO = Analog output pin
- AIO = Analog input/output pin
- P = Power pin
- AP = Analog power pin
- NC = No connection
- Up = Pull-Up
- Down = Pull-down
- Z = High-Z

3.4 Pin Multiplexing Tables

Multiple usage pins are used to conserve pin consumption for different features. The device can be used in many different applications but each application will not utilize all the onchip features. As a result, some of the features share the same pin. Most of the multiple usage pins can be used as a GPIO pin as well.

Table 3-2 GPIOC_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIOC_0	SDCARD_D0		TDMB_D0	TSout_CLK	SPI_MISO_B	
GPIOC_1	SDCARD_D1		TDMB_D1	TSout_SOP	SPI_MOSI_B	
GPIOC_2	SDCARD_D2		TDMB_D2	TSout_VALID	SPI_CLK_B	
GPIOC_3	SDCARD_D3		TDMB_D3	TSout_D0	SPI_SS0_B	
GPIOC_4	SDCARD_CLK		TDMB_SCLK	TSout_D1	SPI_SS1_B	
GPIOC_5	SDCARD_CMD		TDMB_FS	TSout_D2	SPI_SS2_B	

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Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIOC_6	UART_EE_A_TX	PDM_DCLK	TDMB_DIN0	TSout_D3	DTV_IF_AGC	ATV_IF_AGC
GPIOC_7	UART_EE_A_RX	PDM_DIN0	TDMB_DIN1	TSout_D4	DTV_IF_AGC	DTV_RF_AGC
GPIOC_8	UART_EE_A_CTS	PDM_DIN1	TDMB_DIN2	TSout_D5	I2C_EE_M0_SCL	
GPIOC_9	UART_EE_A_RTS	PDM_DIN2	TDMB_DIN3	TSout_D6	I2C_EE_M0_SDA	
GPIOC_10	PWM_B	PDM_DIN3	MCLK_1	TSout_D7		
GPIOC_11	SPI_MISO_A	TDMB_D1	TDMB_DIN1			
GPIOC_12	SPI_MOSI_A	TDMB_D0	TDMB_DIN0	WIFI_BEACON		
GPIOC_13	SPI_CLK_A	TDMB_SCLK	TDMB_SLV_SCLK	I2C_EE_M3_SCL		
GPIOC_14	SPI_SS0_A	TDMB_FS	TDMB_SLV_FS	I2C_EE_M3_SDA		

Table 3-3 GPIOZ_x Multi-Function Pin

Pin Name	Fun1	Func2	Func3	Func4
GPIOZ_0	MCLK_0	PDM_DIN1		
GPIOZ_1	TDMA_SCLK	TDMA_SLV_SCLK	I2C_EE_M1_SDA	
GPIOZ_2	TDMA_FS	TDMA_SLV_FS	I2C_EE_M1_SCL	
GPIOZ_3	TDMA_D0			PWM_A
GPIOZ_4	TDMA_D1	PDM_DIN2	WIFI_BEACON	PWM_B
GPIOZ_5	TDMA_D2		PWM_VS	PWM_C
GPIOZ_6	TDMA_D3		PWM_VS	PWM_D
GPIOZ_7	SPDIF_OUT	TDMA_DIN0	PDM_DCLK	
GPIOZ_8	MCLK_1	TDMA_DIN1	PDM_DIN0	
GPIOZ_9	I2C_EE_M2_SCL	TDMA_DIN2	PDM_DIN1	PWM_E
GPIOZ_10	I2C_EE_M2_SDA	TDMA_DIN3	PDM_DIN2	PWM_F

Table 3-4 BOOT_x Multi-Function Pin

Pin Name	Func1	Func2	Func3
BOOT_0	EMMC_D0		
BOOT_1	EMMC_D1		
BOOT_2	EMMC_D2		
BOOT_3	EMMC_D3		SPIF_HOLD
BOOT_4	EMMC_D4		SPIF_MO
BOOT_5	EMMC_D5		SPIF_MI

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Pin Name	Func1	Func2	Func3
BOOT_6	EMMC_D6		SPIF_CLK
BOOT_7	EMMC_D7		SPIF_WP
BOOT_8	EMMC_CLK	NAND_WEN_CLK	
BOOT_9		NAND_ALE	
BOOT_10	EMMC_CMD	NAND_REN_WR	
BOOT_11	EMMC_DS	NAND_CLE	
BOOT_12		NAND_CE0	
BOOT_13			SPIF_CS

Table 3-5 GPIOH_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIOH_0	TCON_0	SPI_SS1_A	HSYNC	ETH_ACT_LED	SPDIF_OUT	I2C_EE_M3_SCL
GPIOH_1	TCON_1	SPI_SS2_A	VSYNC	ETH_LINK_LED		I2C_EE_M3_SDA
GPIOH_2	TCON_2	ETH_MDIO	BT656_A_CLK	TCON_SFC		
GPIOH_3	TCON_3	ETH_MDC	BT656_A_VS			
GPIOH_4	TCON_4	ETH_RGMII_RX_CLK	BT656_A_HS	TSin_D0_B	MCLK_0	
GPIOH_5	TCON_5	ETH_RX_DV	BT656_A_DIN0	TSin_CLK_B	TDMA_SCLK	
GPIOH_6	TCON_6	ETH_RXD0	BT656_A_DIN1	TSin_SOP_B	TDMA_FS	
GPIOH_7	TCON_7	ETH_RXD1	BT656_A_DIN2	TSin_VALID_B	TDMA_D0	
GPIOH_8	TCON_8	ETH_RXD2_RGMII	BT656_A_DIN3	TSin_D0_C	TDMA_D1	
GPIOH_9	TCON_9	ETH_RXD3_RGMII	BT656_A_DIN4	TSin_CLK_C	TDMA_D2	
GPIOH_10	TCON_10	ETH_RGMII_TX_CLK	BT656_A_DIN5	TSin_SOP_C	TDMA_D3	
GPIOH_11	TCON_11	ETH_TXEN	BT656_A_DIN6	TSin_VALID_C	TDMA_DIN0	
GPIOH_12	TCON_12	ETH_TXD0	BT656_A_DIN7		TDMA_DIN1	
GPIOH_13	TCON_13	ETH_TXD1	CLK12_24	PWM_C	TDMA_DIN2	
GPIOH_14	TCON_14	ETH_TXD2_RGMII	3D_SYNC_OUT		TDMA_DIN3	
GPIOH_15	TCON_15	ETH_TXD3_RGMII	VX1_HTPDN			
GPIOH_16	TCON_LOCK	TCON_SFC	VX1_LOCKN			
GPIOH_17	TCON_SPI_MO	SPI_MISO_A		UART_EE_B_TX	GEN_CLK_EE	

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Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIOH_18	TCON_SPI_MI	SPI_MOSI_A		UART_EE_B_RX		
GPIOH_19	TCON_SPI_CLK	SPI_CLK_A	I2C_EE_M2_SCL	UART_EE_B_CTS	I2C_EE_M3_SCL	
GPIOH_20	TCON_SPI_SS	SPI_SS0_A	I2C_EE_M2_SDA	UART_EE_B_RTS	I2C_EE_M3_SDA	
GPIOH_21	I2C_EE_M1_SCL				GEN_CLK_EE	
GPIOH_22	I2C_EE_M1_SDA					

Table 3-6 GPIOAO_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4	Func5
GPIOAO_0	UART_AO_A_TX				
GPIOAO_1	UART_AO_A_RX				
GPIOAO_2	I2C_AO_M0_SCL	UART_AO_B_TX	I2C_AO_S0_SCL		
GPIOAO_3	I2C_AO_M0_SDA	UART_AO_B_RX	I2C_AO_S0_SDA		
GPIOAO_4	IR_REMOTE_OUT	CLK_32K_IN	PWMAO_C	PWMAO_C_HIZ	
GPIOAO_5	IR_REMOTE_INPUT		PWMAO_D		
GPIOAO_6	JTAG_A_CLK	SPDIF_OUT	PWMAO_C		
GPIOAO_7	JTAG_A_TMS	SPDIF_IN	PWMAO_C_HIZ		
GPIOAO_8	JTAG_A_TDI		UART_AO_B_TX		
GPIOAO_9	JTAG_A_TDO	IR_REMOTE_OUT	UART_AO_B_RX		
GPIOAO_10	AO_CEC_A	AO_CEC_B	PWMAO_D		
GPIOAO_11		PWMAO_A_HIZ	PWMAO_A	GEN_CLK_EE	GEN_CLK_AO

Table 3-7 GPIOE_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4
GPIOE_0	UART_AO_A_CTS	UART_AO_B_CTS	PWMAO_B	I2C_AO_M0_SCL
GPIOE_1	UART_AO_A_RTS	UART_AO_B_RTS	PWMAO_D	I2C_AO_M0_SDA
GPIOE_2	CLK12_24	CLK25_EE	PWM_A	

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Table 3-8 GPIODV_x Multi-Function Pin

Pin Name	Func1	Func2	Func3	Func4	Func5	Func6
GPIODV_0	I2C_EE_M0_SDA	TCON_SLV_SDA	TSin_D7_A			
GPIODV_1	I2C_EE_M0_SCL	TCON_SLV_SCK	TSin_D6_A			
GPIODV_2	DTV_IF_AGC	ATV_IF_AGC	TSin_D5_A	TCON_AGING	PWM_C	
GPIODV_3	DTV_IF_AGC	DTV_RF_AGC	TSin_D4_A		PWM_D	
GPIODV_4	SPDIF_OUT	TCON_FLICKER	TSin_D3_A	ISO7816_CLK		
GPIODV_5	SPDIF_IN		TSin_D2_A	ISO7816_DATA		
GPIODV_6	TDMC_D3	TDMC_DIN3	TSin_D1_A	PWM_E		
GPIODV_7	TDMC_SCLK	TDMC_SLV_SCLK	TSin_D0_A	SPI_MISO_B	UART_EE_C_TX	
GPIODV_8	TDMC_FS	TDMC_SLV_FS	TSin_CLK_A	SPI_MOSI_B	UART_EE_C_RX	
GPIODV_9	TDMC_D0	TDMC_DIN0	TSin_SOP_A	SPI_CLK_B	UART_EE_C_CTS	I2C_EE_M3_SCL
GPIODV_10	TDMC_D1	TDMC_DIN1	TSin_VALID_A	SPI_SS0_B	UART_EE_C_RTS	I2C_EE_M3_SDA
GPIODV_11	TDMC_D2	TDMC_DIN2	SPDIF_OUT	SPI_SS1_B	PWM_F	

Table 3-9 GPIOW_x Multi-Function Pin

Pin Name	Func1	Func2
GPIOW_0	HDMIRX_HPD_A	
GPIOW_1	HDMIRX_5VDET_A	
GPIOW_2	HDMIRX_SDA_A	UART_AO_A_TX
GPIOW_3	HDMIRX_SCL_A	UART_AO_A_RX
GPIOW_4	HDMIRX_HPD_C	
GPIOW_5	HDMIRX_5VDET_C	
GPIOW_6	HDMIRX_SDA_C	UART_AO_A_TX
GPIOW_7	HDMIRX_SCL_C	UART_AO_A_RX
GPIOW_8	HDMIRX_HPD_B	
GPIOW_9	HDMIRX_5VDET_B	
GPIOW_10	HDMIRX_SDA_B	UART_AO_A_TX
GPIOW_11	HDMIRX_SCL_B	UART_AO_A_RX

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Table 3-10 DDR AC Multi-Function Pin

Pin Name	LPDDR3	LPDDR4	DDR3	DDR4
AC_0	CKEA0	CKEA0	CKE0	CKE0
AC_1	CKEA1	CKEA1	CKE1	CKE1
AC_2	CSA0	CSA0	CS_N0	CS_N0
AC_3	CSA1	CSA1	NC	C0
AC_4	CLKA_T	CLKA_T	BA2	ACT_N
AC_5	CLKA_C	CLKA_C	RAS_N	WE_N
AC_6	NC	NC	A7	BA0
AC_7	NC	NC	A5	BG0
AC_8	CAA2	CAA1	A10	A10
AC_9	CAA7	CAA0	A14	A1
AC_10	CAA1	CAA4	BA0	A8
AC_11	CAA4	CAA5	CAS_N	A6
AC_12	CAA5	CAA2	A9	A2
AC_13	CAA6	CAA3	A13	A11
AC_14	CAA0	NC	WE_N	A4
AC_15	CAA3	NC	A11	A13
AC_16	CAA9	NC	CLK0_T	CLK0_T
AC_17	CAA8	NC	CLK0_C	CLK0_C
AC_18	ODTA	NC	NC	NC
AC_24	NC	CLKB_T	A6	A7
AC_25	NC	CLKB_C	A4	A5
AC_26	NC	NC	A1	A12
AC_28	NC	CAB2	A8	BA1
AC_29	NC	CAB0	BA1	RAS_N
AC_30	NC	CAB5	A15	A0
AC_31	NC	CAB4	A2	A9
AC_33	NC	CAB3	A12	CAS_N
AC_34	NC	NC	A3	BG1
AC_35	NC	NC	A0	A3
AC_36	NC	NC	ODT0	ODT0
AC_37	NC	NC	ODT1	ODT1
AC_38	NC	NC	CS_N1	CS_N1
DDR_RSTN	NC	DDR_RSTN	DDR_RSTN	DDR_RSTN
PVREF	PVREF	PVREF	PVREF	PVREF
PZQ	PZQ	PZQ	PZQ	PZQ

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3.5 Signal Description

Table 3-11 SD Card Interface Signal Description

Signal Name	Type	Description
SDCARD_D0	DIO	SD Card data bus bit 0 signal
SDCARD_D1	DIO	SD Card data bus bit 1 signal
SDCARD_D2	DIO	SD Card data bus bit 2 signal
SDCARD_D3	DIO	SD Card data bus bit 3 signal
SDCARD_CLK	DO	SD Card clock signal
SDCARD_CMD	DIO	SD Card command signal

Table 3-12 Clock Interface Signal Description

Signal Name	Type	Description
CLK_32K_IN	DI	32KHz clock input
CLK12_24	DO	12MHz/24MHZ clock output
CLK25_EE	DO	25MHz clock output

Table 3-13 UART Interface Signal Description

Signal Name	Type	Description
UART_AO_A_TX	DO	UART Port A data output in AO domain
UART_AO_A_RX	DI	UART Port A data input in AO domain
UART_AO_A_CTS	DI	UART Port A Clear To Send Signal in AO domain
UART_AO_A_RTS	DO	UART Port A Ready To Send Signal in AO domain
UART_AO_B_TX	DO	UART Port B data output in AO domain
UART_AO_B_RX	DI	UART Port B data input in AO domain
UART_AO_B_CTS	DI	UART Port B Clear To Send Signal in AO domain
UART_AO_B_RTS	DO	UART Port B Ready To Send Signal in AO domain
UART_EE_A_TX	DO	UART Port A data output in EE domain
UART_EE_A_RX	DI	UART Port A data input in EE domain
UART_EE_A_CTS	DI	UART Port A Clear To Send Signal in EE domain
UART_EE_A_RTS	DO	UART Port A Ready To Send Signal in EE domain
UART_EE_B_TX	DO	UART Port B data output in EE domain
UART_EE_B_RX	DI	UART Port B data input in EE domain
UART_EE_B_CTS	DI	UART Port B Clear To Send Signal in EE domain
UART_EE_B_RTS	DO	UART Port B Ready To Send Signal in EE domain

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Signal Name	Type	Description
UART_EE_C_TX	DO	UART Port C data output in EE domain
UART_EE_C_RX	DI	UART Port C data input in EE domain
UART_EE_C_CTS	DI	UART Port C Clear To Send Signal in EE domain
UART_EE_C_RTS	DO	UART Port C Ready To Send Signal in EE domain

Table 3-14 ISO7816 Interface Signal Description

Signal Name	Type	Description
ISO7816_DATA	DIO	ISO7816 data signal
ISO7816_CLK	DO	ISO7816 clock signal

Table 3-15 TS In Interface Signal Description

Signal Name	Type	Description
TSin_D0_A	DI	TS input port A data bus bit 0 (LSB)/ port A TS Serial data input
TSin_D1_A	DI	TS input port A data bus bit 1
TSin_D2_A	DI	TS input port A data bus bit 2
TSin_D3_A	DI	TS input port A data bus bit 3
TSin_D4_A	DI	TS input port A data bus bit 4
TSin_D5_A	DI	TS input port A data bus bit 5
TSin_D6_A	DI	TS input port A data bus bit 6
TSin_D7_A	DI	TS input port A data bus bit 7(MSB)
TSin_CLK_A	DI	TS input port A clock
TSin_SOP_A	DI	TS input port A start of stream signal
TSin_VALID_A	DI	TS input port A data valid signal
TSin_D0_B	DI	TS input port B data bus bit 0 (LSB) /port B TS Serial data input
TSin_CLK_B	DI	TS input port B clock
TSin_SOP_B	DI	TS input port B start of stream signal
TSin_VALID_B	DI	TS input port B data valid signal
TSin_D0_C	DI	TS input port C data bus bit 0 (LSB) /port C TS Serial data input
TSin_CLK_C	DI	TS input port C clock
TSin_SOP_C	DI	TS input port C start of stream signal
TSin_VALID_C	DI	TS input port C data valid signal

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Table 3-16 TS Out Interface Signal Description

Signal Name	Type	Description
TSout_D0	DO	TS output data bus bit 0 (LSB)
TSout_D1	DO	TS output data bus bit 1
TSout_D2	DO	TS output data bus bit 2
TSout_D3	DO	TS output data bus bit 3
TSout_D4	DO	TS output data bus bit 4
TSout_D5	DO	TS output data bus bit 5
TSout_D6	DO	TS output data bus bit 6
TSout_D7	DO	TS output data bus bit 7 (MSB)
TSout_CLK	DO	TS output clock
TSout_SOP	DO	TS output start of stream signal
TSout_VALID	DO	TS output date valid signal

Table 3-17 PWM Interface Signal Description

Signal Name	Type	Description
PWM_A	DO	PWM channel A output signal
PWM_B	DO	PWM channel B output signal
PWM_C	DO	PWM channel C output signal
PWM_D	DO	PWM channel D output signal
PWM_E	DO	PWM channel E output signal
PWM_F	DO	PWM channel F output signal
PWM_VS	DO	PWM VSYNC output signal
PWMAO_A	DO	PWM A output signal in Always On domain
PWMAO_B	DO	PWM B output signal in Always On domain
PWMAO_C	DO	PWM C output signal in Always On domain
PWMAO_D	DO	PWM D output signal in Always On domain
PWMAO_A_HIZ	DO	Extended HiZ function of PWMAO_A
PWMAO_C_HIZ	DO	Extended HiZ function of PWMAO_C

Table 3-18 I2C Interface Signal Description

Signal Name	Type	Description
I2C_AO_M0_SCL	DO	I2C bus port 0 clock output, Master mode, in AO domain
I2C_AO_M0_SDA	DIO	I2C bus port 0 data input/output, Master mode, in AO domain
I2C_AO_S0_SCL	DI	I2C bus port 0 clock input, Slave mode, in AO domain
I2C_AO_S0_SDA	DIO	I2C bus port 0 data input/output, Slave mode, in AO domain

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Signal Name	Type	Description
I2C_EE_M0_SCL	DO	I2C bus port 0 clock output, Master mode, in EE domain
I2C_EE_M0_SDA	DIO	I2C bus port 0 data input/output, Master mode, in EE domain
I2C_EE_M1_SCL	DO	I2C bus port 1 clock output, Master mode, in EE domain
I2C_EE_M1_SDA	DIO	I2C bus port 1 data input/output, Master mode, in EE domain
I2C_EE_M2_SCL	DO	I2C bus port 2 clock output, Master mode, in EE domain
I2C_EE_M2_SDA	DIO	I2C bus port 2 data input/output, Master mode, in EE domain
I2C_EE_M3_SCL	DO	I2C bus port 3 clock output, Master mode, in EE domain
I2C_EE_M3_SDA	DIO	I2C bus port 3 data input/output, Master mode, in EE domain

Table 3-19 TCON Interface Signal Description

Signal Name	Type	Description
TCON_0	DO	TCON control signal output
TCON_1	DO	TCON control signal output
TCON_2	DO	TCON control signal output
TCON_3	DO	TCON control signal output
TCON_4	DO	TCON control signal output
TCON_5	DO	TCON control signal output
TCON_6	DO	TCON control signal output
TCON_7	DO	TCON control signal output
TCON_8	DO	TCON control signal output
TCON_9	DO	TCON control signal output
TCON_10	DO	TCON control signal output
TCON_11	DO	TCON control signal output
TCON_12	DO	TCON control signal output
TCON_13	DO	TCON control signal output
TCON_14	DO	TCON control signal output
TCON_15	DO	TCON control signal output
TCON_LOCK	DIO	Feedback signal from source driver
TCON_SPI_MO	DIO	TCON SPI master output
TCON_SPI_MI	DI	TCON SPI master input
TCON_SPI_CLK	DO	TCON SPI Serial Clock
TCON_SPI_SS	DO	TCON SPI Slave Select
TCON_FLICKER	DI	Test flicker input signal
TCON_AGING	DI	TCON AGING mode input
TCON_SFC	DO	Dedicated USIT signal

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Table 3-20 eMMC Interface Signal Description

Signal Name	Type	Description
EMMC_D0	DIO	eMMC/NAND data bus bit 0 signal
EMMC_D1	DIO	eMMC/NAND data bus bit 1 signal
EMMC_D2	DIO	eMMC/NAND data bus bit 2 signal
EMMC_D3	DIO	eMMC/NAND data bus bit 3 signal
EMMC_D4	DIO	eMMC/NAND data bus bit 4 signal
EMMC_D5	DIO	eMMC/NAND data bus bit 5 signal
EMMC_D6	DIO	eMMC/NAND data bus bit 6 signal
EMMC_D7	DIO	eMMC/NAND data bus bit 7 signal
EMMC_CLK	DO	eMMC clock signal
EMMC_CMD	DIO	eMMC command signal
EMMC_DS	DI	eMMC data strobe

Table 3-21 NAND Signal Description

Signal Name	Type	Description
NAND_ALE	DO	NAND address latch enable
NAND_CE0	DO	NAND chip enable 0
NAND_CLE	DO	NAND command latch enable
NAND_REN_WR	DO	NAND read enable or write/read
NAND_WEN_CLK	DO	NAND write enable or clock

Table 3-22 HDMI Interface Signal Description

Signal Name	Type	Description
HDMIRX_HPD_A	DO	HDMI RX port A hot plug in signal output
HDMIRX_5VDET_A	DI	HDMI RX port A 5V power detection
HDMIRX_SDA_A	DIO	HDMI RX port A DDC_I2C interface data signal
HDMIRX_SCL_A	DI	HDMI RX port A DDC_I2C interface clock signal
HDMIRX_HPD_B	DO	HDMI RX port B hot plug in signal output
HDMIRX_5VDET_B	DI	HDMI RX port B 5V power detection
HDMIRX_SDA_B	DIO	HDMI RX port B DDC_I2C interface data signal
HDMIRX_SCL_B	DI	HDMI RX port B DDC_I2C interface clock signal
HDMIRX_HPD_C	DO	HDMI RX port C hot plug in signal output
HDMIRX_5VDET_C	DI	HDMI RX port C 5V power detection
HDMIRX_SDA_C	DIO	HDMI RX port C DDC_I2C interface data signal

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Signal Name	Type	Description
HDMIRX_SCL_C	DI	HDMI RX port C DDC_I2C interface clock signal
AO_CEC_A	DIO	Customer Electronics Control signal in AO domain
AO_CEC_B	DIO	2nd pin of Customer Electronics Control signal in AO domain

Table 3-23 SPIF Interface Signal Description

Signal Name	Type	Description
SPIF_CS	DO	SPIF chip select
SPIF_CLK	DO	SPIF Serial Clock
SPIF_MO	DIO	SPIF 1bit mode Output, 2/4 bit mode data I/O 0
SPIF_MI	DIO	SPIF 1bit mode Input, 2/4 bit mode data I/O 1
SPIF_WP	DIO	SPIF Write protection output, 4 bit mode data I/O 2
SPIF_HOLD	DIO	SPIF bus hold output, 4 bit mode data I/O 3

Table 3-24 SPDIF Interface Signal Description

Signal Name	Type	Description
SPDIF_IN	DI	SPDIF input signal
SPDIF_OUT	DO	SPDIF output signal

Table 3-25 SPI Interface Signal Description

Signal Name	Type	Description
SPI_MOSI_A	DIO	SPI port A master output, slave input
SPI_MISO_A	DIO	SPI port A master input, slave output
SPI_CLK_A	DIO	SPI clock, port A
SPI_SS0_A	DIO	SPI slave select 0, port A
SPI_SS1_A	DIO	SPI slave select 1, port A
SPI_SS2_A	DIO	SPI slave select 2, port A
SPI_MOSI_B	DIO	SPI port B master output, slave input
SPI_MISO_B	DIO	SPI port B master input, slave output
SPI_CLK_B	DIO	SPI clock, port B
SPI_SS0_B	DIO	SPI slave select 0, port B
SPI_SS1_B	DIO	SPI slave select 1, port B
SPI_SS2_B	DIO	SPI slave select 2, port B

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Table 3-26 Remote Interface Signal Description

Signal Name	Type	Description
IR_REMOTE_INPUT	DI	IR remote control input
IR_REMOTE_OUT	DO	IR remote control output

Table 3-27 Time Division Multiplexing Signal Description

Signal Name	Type	Description
MCLK_0	DO	Master clock output 0, for I2S master mode
MCLK_1	DO	Master clock output 1, for I2S master mode
TDMA_DIN0	DI	Data input 0 of TDM port A
TDMA_DIN1	DI	Data input 1 of TDM port A
TDMA_DIN2	DI	Data input 2 of TDM port A
TDMA_DIN3	DI	Data input 3 of TDM port A
TDMA_D0	DIO	Data input/output 0 of TDM port A
TDMA_D1	DIO	Data input/output 1 of TDM port A
TDMA_D2	DIO	Data input/output 2 of TDM port A
TDMA_D3	DIO	Data input/output 3 of TDM port A
TDMA_SCLK	DO	Bit clock output of TDM port A
TDMA_FS	DO	Frame sync output of TDM port A (Word clock of I2S)
TDMA_SLV_SCLK	DI	Bit clock input of TDM port A
TDMA_SLV_FS	DI	Frame sync input of TDM port A (Word clock of I2S)
TDMB_DIN0	DI	Data input 0 of TDM port B
TDMB_DIN1	DI	Data input 1 of TDM port B
TDMB_DIN2	DI	Data input 2 of TDM port B
TDMB_DIN3	DI	Data input 3 of TDM port B
TDMB_D0	DIO	Data input/output 0 of TDM port B
TDMB_D1	DIO	Data input/output 1 of TDM port B
TDMB_D2	DIO	Data input/output 2 of TDM port B
TDMB_D3	DIO	Data input/output 3 of TDM port B
TDMB_SCLK	DO	Bit clock output of TDM port B
TDMB_FS	DO	Frame sync output of TDM port B (Word clock of I2S)
TDMB_SLV_SCLK	DI	Bit clock input of TDM port B
TDMB_SLV_FS	DI	Frame sync input of TDM port B (Word clock of I2S)
TDMC_DIN0	DI	Data input 0 of TDM port C
TDMC_DIN1	DI	Data input 1 of TDM port C
TDMC_DIN2	DI	Data input 2 of TDM port C

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Signal Name	Type	Description
TDMC_DIN3	DI	Data input 3 of TDM port C
TDMC_D0	DIO	Data input/output 0 of TDM port C
TDMC_D1	DIO	Data input/output 1 of TDM port C
TDMC_D2	DIO	Data input/output 2 of TDM port C
TDMC_D3	DIO	Data input/output 3 of TDM port C
TDMC_SCLK	DO	Bit clock output of TDM port C
TDMC_FS	DO	Frame sync output of TDM port C (Word clock of I2S)
TDMC_SLV_SCLK	DI	Bit clock input of TDM port C
TDMC_SLV_FS	DI	Frame sync input of TDM port C (Word clock of I2S)

Table 3-28 PDM Signal Description

Signal Name	Type	Description
PDM_DIN0	DI	PDM input data 0 signal
PDM_DIN1	DI	PDM input data 1 signal
PDM_DIN2	DI	PDM input data 2 signal
PDM_DIN3	DI	PDM input data 3 signal
PDM_DCLK	DO	PDM output clock signal

Table 3-29 Tuner/Demod Interface Signal Description

Signal Name	Type	Description
DTV_IF_AGC	DO	DTV IF AGC output for external demodulator
DTV_RF_AGC	DO	DTV RF AGC output for external tuner
ATV_IF_AGC	DO	ATV IF AGC output for external demodulator

Table 3-30 3D Interface Signal Description

Signal Name	Type	Description
3D_SYNC_OUT	DO	3D sync signal output

Table 3-31 JTAG Interface Signal Description

Signal Name	Type	Description
JTAG_A_TDO	DO	JTAG data output channel A
JTAG_A_TDI	DI	JTAG data input channel A
JTAG_A_TMS	DI	JTAG Test mode select input channel A
JTAG_A_CLK	DI	JTAG Test clock input channel A

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Table 3-32 BT656 Interface Signal Description

Signal Name	Type	Description
BT656_A_DIN0	DI	BT656 input data bus bit 0
BT656_A_DIN1	DI	BT656 input data bus bit 1
BT656_A_DIN2	DI	BT656 input data bus bit 2
BT656_A_DIN3	DI	BT656 input data bus bit 3
BT656_A_DIN4	DI	BT656 input data bus bit 4
BT656_A_DIN5	DI	BT656 input data bus bit 5
BT656_A_DIN6	DI	BT656 input data bus bit 6
BT656_A_DIN7	DI	BT656 input data bus bit 7
BT656_A_CLK	DI	BT656 input Clock
BT656_A_HS	DI	BT656 input HSYNC Signal
BT656_A_VS	DI	BT656 input VSYNC Signal

Table 3-33 Ethernet Interface Signal Description

Signal Name	Type	Description
ETH_LINK_LED	DO	Ethernet link LED indicator
ETH_ACT_LED	DO	Ethernet active LED indicator
ETH_RGMII_RX_CLK	DI	Ethernet RGMII interface receive clock input
ETH_RGMII_TX_CLK	DO	Ethernet RGMII transmit clock
ETH_TXEN	DO	Ethernet RMII/RGMII Interface transmit enable
ETH_TXD3_RGMII	DO	Ethernet RGMII interface transmit data 3
ETH_TXD2_RGMII	DO	Ethernet RGMII interface transmit data 2
ETH_TXD1	DO	Ethernet RMII/RGMII interface transmit data 1
ETH_TXD0	DO	Ethernet RMII/RGMII interface transmit data 0
ETH_RX_DV	DI	Ethernet RMII/RGMII interface receive data valid signal
ETH_RXD3_RGMII	DI	Ethernet RGMII interface receive data 3
ETH_RXD2_RGMII	DI	Ethernet RGMII interface receive data 2
ETH_RXD1	DI	Ethernet RMII/RGMII interface receive data 1
ETH_RXD0	DI	Ethernet RMII/RGMII interface receive data 0
ETH_MDIO	DIO	Ethernet SMI interface management data input/output
ETH_MDC	DO	Ethernet SMI interface management clock

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Table 3-34 Other Signal Description

Signal Name	Type	Description
WIFI_BEACON	DI	General clock sync input, to sync clock of multi devices
GEN_CLK_EE	DO	General clock output for EE domain clock, for debug
GEN_CLK_AO	DO	General clock output for AO domain clock, for debug
HSYNC	DO	Horizontal synchronizing signal
VSYNC	DO	Vertical synchronizing signal
VX1_HTPDN	DI	Connection status of V-by-one receiver
VX1_LOCKN	DI	V-by-one receiver PLL lock state

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4 Electrical Characteristics

4.1 Absolute Maximum Ratings

The table below gives the absolute maximum ratings. Exposure to stresses beyond those listed in this table may result in permanent device damage, unreliability or both.

Table 4-1 Absolute Maximum Ratings

Characteristic	Value	Unit
VDDCPU Supply Voltage	1.15	V
VDD_EE Supply Voltage	1.02	V
VDDQ Supply Voltage	1.7	V
1.8V Supply Voltage	1.98	V
3.3V Supply Voltage	3.63	V
Input voltage, V_I	-0.3 ~VDDIO+0.3	V
Junction Temperature	125	°C

4.2 Recommended Operating Conditions

Table 4-2 Recommended Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDDCPU	Voltage for Cortex A55 CPU	0.70[1]	-	1.03[2]	V
VDD_EE and other 0.8V domain	Voltage for GPU & core logic	0.77	0.84	0.9	V
VDDQ	DDR3/DDR3L/DDR4/LPDDR3/LPDDR4 IO Supply Voltage	1.06	-	1.575	V
AVDD18	1.8V AVDD for HDMI, USB, AUDIO, VX1/LVDS/P2P, ETHERNET phy, DDRPLL, P2P_PLL and DPLL	1.74	1.80	1.89	V
VDD18_AO	1.8V VDD for XTAL, CVBS, SARADC, efuse and IOVREF	1.74	1.80	1.89	V
AVDD33	3.3V AVDD for USB and HDMIRX	3.15	3.3	3.45	V
VDDIO	LV mode	1.71	1.80	1.89	V
	HV mode	3.0[3]	3.3	3.45	V
T_J	Operating Junction Temperature	0	-	105[4]	°C
T_A	Operating Ambient Temperature	0	-	70	°C

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Note

1. Minimal VDDCPU/VDD_EE voltage is for sleep mode while system runs at very low speed. Higher clock will need higher voltage. Considering the power supply may have 2% deviation, the minimal voltage in actual application should not be designed to lower than the min spec value plus 0.02V.
2. Likewise, maximum VDDCPU voltage in actual application should not be higher than (max spec -0.02V). Voltage of VDDCPU will affect CPU speed. Use lower voltage when CPU runs on lower speed to save power. Recommend to use +/-1.5% or higher precision DCDC.
3. GPIO cannot work if VDDIO voltage is out of the spec of LV / HV mode. GPIO output at HV mode will be weaker & max operating speed will be lower if VDDIO are design to 3.0V. Do not design VDDIO to lower than 3.0V in HV mode.
4. For operating temperature, good heat sink may be needed to guarantee T_J < max spec.

4.3 Ripple Voltage Specifications

Table 4-3 Ripple Voltage Specifications

Power	Max Ripple	Unit	Test State
VDDCPU	40	+/-mV	Run APK StabilityTest
VDD_EE and other 0.8V domain	40	+/-mV	Run APK Basemark ES 2.0 Taiji
DDR3 VDDQ	60	+/-mV	Kernel boot
DDR3L VDDQ	60	+/-mV	Kernel boot
LPDDR3 VDDQ	40	+/-mV	Kernel boot
DDR4 VDDQ	40	+/-mV	Kernel boot
LPDDR4 VDDQ	40	+/-mV	Kernel boot
AVDD18	30	+/-mV	Kernel boot
VDD18_AO	30	+/-mV	Kernel boot
AVDD33	50	+/-mV	WIFI SCAN
VDDIO LV	60	+/-mV	Kernel boot
VDDIO HV	60	+/-mV	WIFI SCAN

Note

The ripple specifications are for reference only. Customers should perform pressure / performance / reliability tests (high / low temperature tests, wet / heat tests, functional tests, etc.) on the product to confirm system stability.

4.4 Thermal Resistance

Simulation condition: JEDEC 2s2p board 101.5 mm x 114.5 mm, natural convection, ambient temperature 25°C.

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Symbol	Parameter	Value.	Unit
Θ_{JA}	Package junction-to-ambient thermal resistance in nature convection	FCBGA: 15.30 FCCSP: 15.1	°C/Watt
Θ_{JB}	Package junction-to-board thermal resistance in nature convection	FCBGA:7.14 FCCSP: 7.2	°C/Watt
Θ_{JC}	Package junction-to-case thermal resistance in nature convection	FCBGA: 0.18 FCCSP: 4.4	°C/Watt

Note

Due to the thinness of the SoC, DRAM or capacitors placed close to SoC may prevent heatsink touching SoC top side. A special convex shape heatsink is recommended.

4.5 DC Electrical Characteristics

4.5.1 Normal GPIO Specifications

Table 4-4 Normal GPIO Specifications (For DIO)

Symbol	Parameter	Min.	Typ.	Max.	Unit
$V_{iH(VDDIO=3.3V)}$	High-level input voltage	IOVREF +0.37	-	VDDIO+0.3	V
$V_{iL(VDDIO=3.3V)}$	Low-level input voltage	-0.3	-	IOVREF-0.23	V
$V_{iH(VDDIO=1.8V)}$	High-level input voltage	IOVREF/2 +0.3	-	VDDIO+0.3	V
$V_{iL(VDDIO=1.8V)}$	Low-level input voltage	-0.3	-	IOVREF/2-0.3	V
$R_{PU/PD}$	Built-in pull up/down resistor	-	60K		ohm
IoL/IoH(DS=0)[4]	GPIO driving capability	0.5	-	-	mA
IoL/IoH(DS=1)	GPIO driving capability	2.5	-	-	mA
IoL/IoH(DS=2)	GPIO driving capability	3	-	-	mA
IoL/IoH(DS=3)	GPIO driving capability	4[1]	-	-	mA
VOH	Output high level with IoL/ IoH loading	VDDIO-0.5	-	-	V
VOL	Output low level with IoL/ IoH loading	-	-	0.4	V

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Note

1. With Minimal IoL/IoH driving capability loading, IO is garanteed to meet Vol < 0.4V or VOH > (VDDIO-0.5V) spec.
2. Maximal GPIO loading is 6mA for application such as driving LED, which does not care about Vol/Voh spec. Please set DS=3 for such application.
3. VDD18_AO supplies power to IOVREF.
4. Do not use this setting, it's too weak for most applications.

4.5.2 Open Drain GPIO Specifications (For DIO_OD)

Table 4-5 Open Drain GPIO Specifications (For DIO_OD)

Symbol	Parameter	Min.	Typ.	Max.	Unit
$V_{iH(}$OD5V)	High-level input voltage	1.5	-	5.5	V
$V_{iL(}$OD5V)	Low-level input voltage	-0.3	-	0.8	V
$R_{PU/PD}$	No built-in pull up/down resistor on OD IO	-	-	-	ohm
IoL	OD IO driving low capability	4[1]	-	6[2]	mA
VOL	Output low level with min Io loading	-	-	0.4	V

Note

1. With Minimal IoL driving capability loading, IO is garanteed to meet Vol < 0.4V spec.
2. Maximal GPIO loading is 6mA for application such as driving LED, which does not care about Vol spec.
3. The ViL/ViH of OD PAD is irrelevant to VDDIO voltage.
4. "OD 5V" means that in applications such as I2C, use a resistor greater than 1Kohm to pull it up to 5V. Do not connect the pad directly to the 5V power supply.

4.5.3 DDR3/DDR3L/DDR4/LPDDR3/LPDDR4 SDRAM Specifications

Table 4-6 Recommended Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDDQ	IO supply voltage (DDR3)	1.425	1.50	1.575	V
VDDQ	IO supply voltage (DDR3L)	1.283	1.35	1.45	V
VDDQ	IO supply voltage (DDR4)	1.14	1.20	1.26	V
VDDQ	IO supply voltage (LPDDR3)	1.14	1.2	1.30	V
VDDQ	IO supply voltage (LPDDR4)	1.06	1.1	1.17	V
Vref	Input reference supply voltage	0.49 x VDDQ	0.5 x VDDQ	0.51 x VDDQ	V

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Note

The minimal VDDQ voltage in sleep mode is defined by memory.

Table 4-7 DC specifications - DDR3/DDR3L mode

Symbol	Parameter	Min.	Typ.	Max.	Unit
VIH	DC input voltage high	Vref + 0.100	-	-	V
VIL	DC input voltage low	-	-	Vref-0.100	V
VOH	DC output logic high	0.8*VDDQ	-	-	V
VOL	DC output logic low	-	-	0.2 x VDDQ	V
RTT	Input termination resistance to VDDQ/2	100 54 36	120 60 40	140 66 44	ohm

Table 4-8 DC specifications - DDR4 mode

Symbol	Parameter	Min.	Typ.	Max.	Unit
VdlVW_total	Rx Mask voltage-p-p total	-	-	136	mv
VOH	DC output logic high	0.9 x VDDQ	-	-	V
VOL	DC output logic low	-	-	0.1 x VDDQ	V
RTT	Input termination resistance to VDDQ	200	240	280	ohm
		100	120	140	
		67	80	93	
		50	60	70	
		42	48	56	
		34	40	46	
		28	34	40	

Table 4-9 DC Specifications - LPDDR3 mode

Symbol	Parameter	Min.	Typ.	Max.	Unit
VIH	DC input voltage high	Vref + 0.100	-	-	V
VIL	DC input voltage low	-	-	Vref-0.100	V
VOH	DC output logic high	0.9*VDDQ	-	-	V
VOL	DC output logic low	-	-	0.1*VDDQ	V
RTT	Input termination resistance to VDDQ	100 200	120 240	140 280	ohm

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Table 4-10 DC Specifications - LPDDR4 mode

Symbol	Parameter	Min.	Typ.	Max.	Unit
VOH	DC output logic high	0.9*VDDQ	-	-	V
VOL	DC output logic low	-	-	0.1*VDDQ	V
RTT	Input termination resistance to VDDQ	216	240	264	ohm
		108	120	132	
		72	80	88	
		54	60	66	
		43.2	48	52.8	
		36	40	44	

4.6 Timing Information

4.6.1 I2C Timing Specification

The I2C master interface Fast/Standard mode timing specifications are shown below.

Figure 4-1 I2C Interface Timing, FS mode

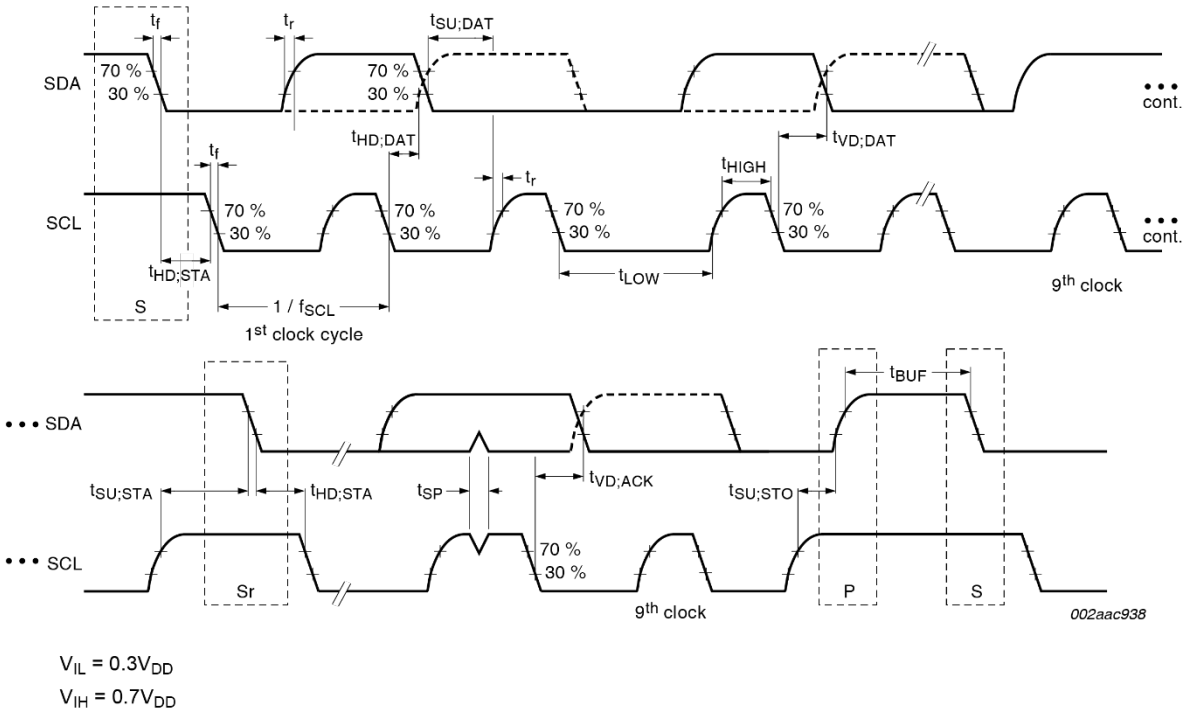


Table 4-11 I2C Interface Timing Specification, SF mode

Symbol	Parameter	Standard-mode		Fast-mode		Unit
		Min.	Max	Min	Max	
tR	Rise time of SDA and SCL signals	-	1000	-	300	ns
tF	Fall time of SDA and SCL signals	-	300	-	300	ns

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Symbol	Parameter	Standard-mode		Fast-mode		Unit
		Min.	Max	Min	Max	
fSCL	SCL clock frequency	-	100	-	400	KHz
tLOW	LOW period of the SCL clock	4.7	-	1.3	-	μs
tHIGH	HIGH period of the SCL clock	4	-	0.6	-	μs
tSu;STA	Setup time for START	4.7	-	0.6	-	μs
tSu;DAT	Setup time for SDA	250	-	100	-	ns
tSu;STO	Setup time for STOP	4	-	0.6	-	μs
tHd;STA	Hold time for START	4	-	0.6	-	μs
tHd;DAT	Hold time for SDA	0	3.45	0	0.9	μs
tBuf	Bus free time between stop and start	4.7	-	1.3	-	μs

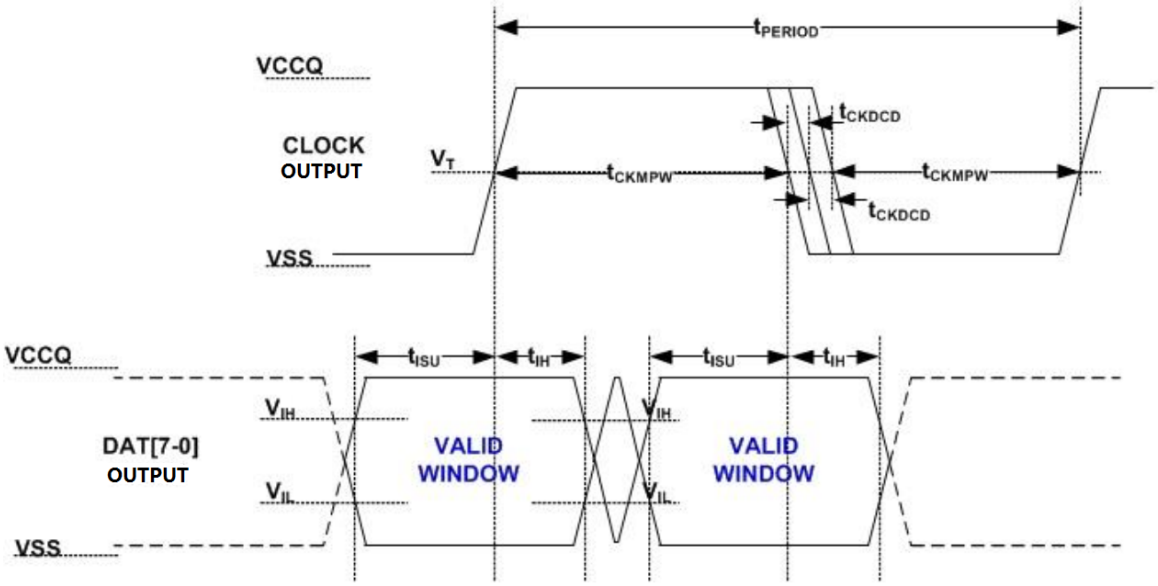
Note

Driving strength of OD pin is not adjustable.

4.6.2 EMMC/SDIO Timing Specification

Timing specification for EMMC and SD are shown as below.

Figure 4-2 EMMC HS400 Data Output Timing



NOTE V_T = 50% of V_{CCQ}, indicates clock reference point for timing measurements.

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Table 4-12 HS400 Output Timing Specification

Symbol	Parameter	Min	Max	Unit
tPERIOD	Cycle time data transfer mode	5	-	ns
SR	Slew rate	1.125	-	V/ns
tCKDCD	Duty cycle distortion	0	0.3	ns
tCKMPW	Minimum pulse width	2.2	-	ns
tISU	input set-up time	1.4	-	ns
tIH	input hold time	0.8	-	ns
tISUddr	input set-up time	0.4	-	ns
tIHddr	input hold time	0.4	-	ns

Figure 4-3 EMMC HS200 Data Output Timing

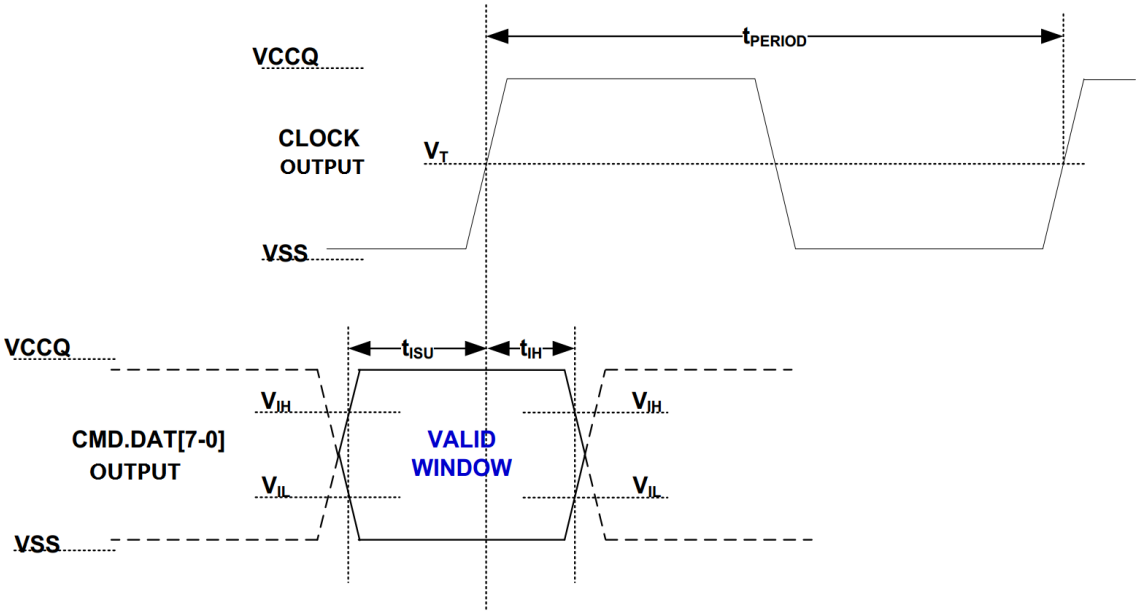
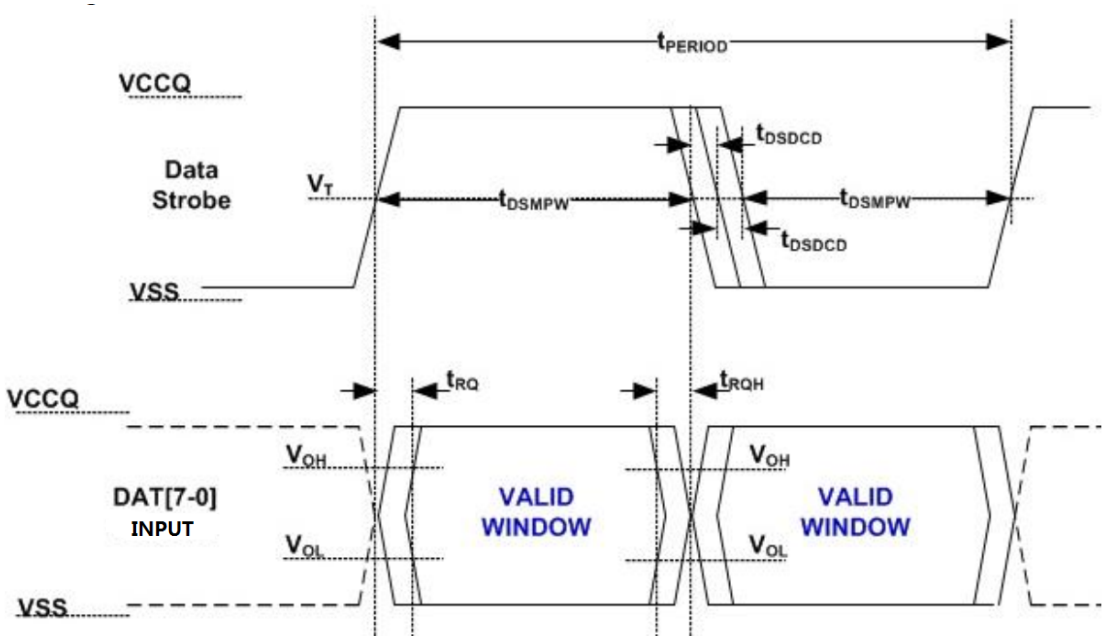


Table 4-13 HS200 Output Timing Specification

Symbol	Parameter	Min	Max	Unit
tPERIOD	Cycle time data transfer mode	5	-	ns
tISU	output set-up time	1.4	-	ns
tIH	output hold time	0.8	-	ns

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Figure 4-4 EMMC HS400 Data Input Timing



NOTE V_T = 50% of V_{CCQ}, indicates clock reference point for timing measurements.

Table 4-14 HS400 Data Input Timing Specification

Symbol	Parameter	Min	Max	Unit
tPERIOD	Cycle time data transfer mode	5	-	ns
SR	Slew rate	1.125	-	V/ns
tCKDCD	Duty cycle distortion	0	0.2	ns
tCKMPW	Minimum pulse width	2	-	ns
tRQ	Input skew	-	0.4	ns
tRQH	input hold skew	-	0.4	ns

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Figure 4-5 EMMC HS200 Data Input Timing

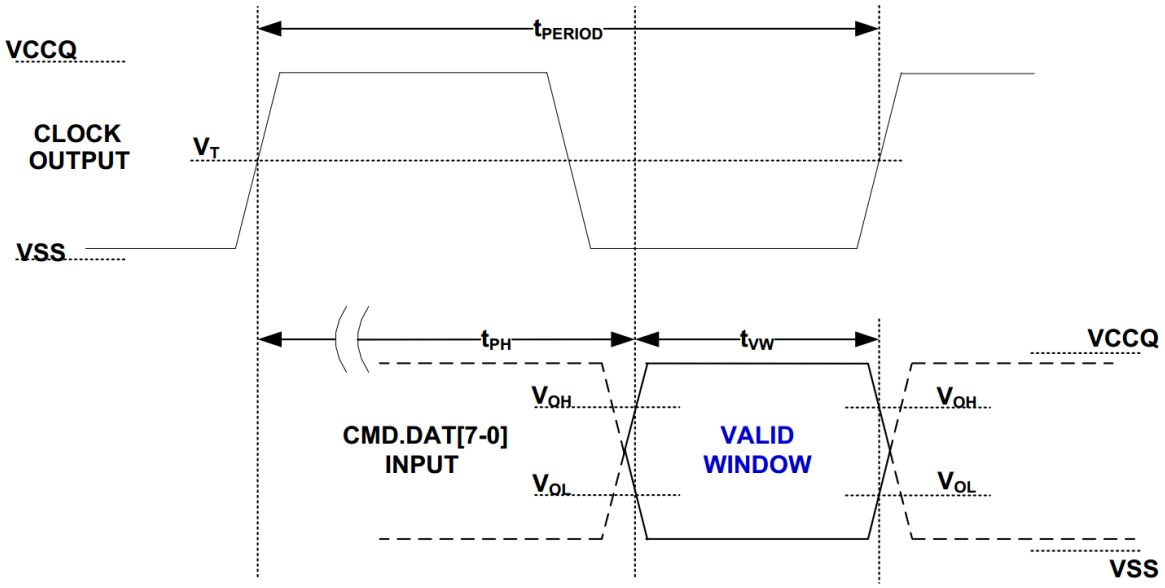
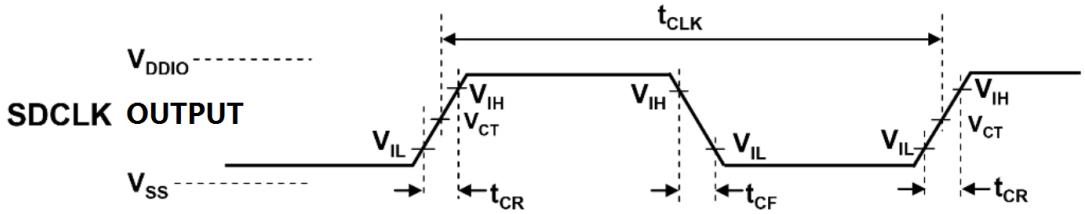


Table 4-15 HS200 Input Timing Specification

Symbol	Parameter	Min	Max	Unit
tPH	Device output momentary phase from CLK input to CMD or DAT line output. Does not include a long-term temperature drift.	0	2	UI
ΔTPH	Delay variation due to temperature change after tuning. Total allowable shift of output valid window (TVW) from last system Tuning procedure ΔTPH is 2600ps for ΔT from -25 °C to 125 °C during operation.	-350(ΔT=-20deg.C)	1550(ΔT=90deg.C)	ps
tVW	Valid Data Simple window	0.575	-	UI

Figure 4-6 SDIO(SDR104) Clock Signal Timing



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Table 4-16 SDIO(SDR104) Clock Timing Specification

Symbol	Parameter (SDR104 Mode)	Min	Max	Unit
tCLK	clock period Data Transfer Mode(PP)	4.8	-	ns
Duty	Clock Duty	30	70	%
tCR	clock rise time	-	0.96	ns
tCF	clock fall time	-	0.96	ns

Figure 4-7 SDIO(SDR104) Output Timing Diagram

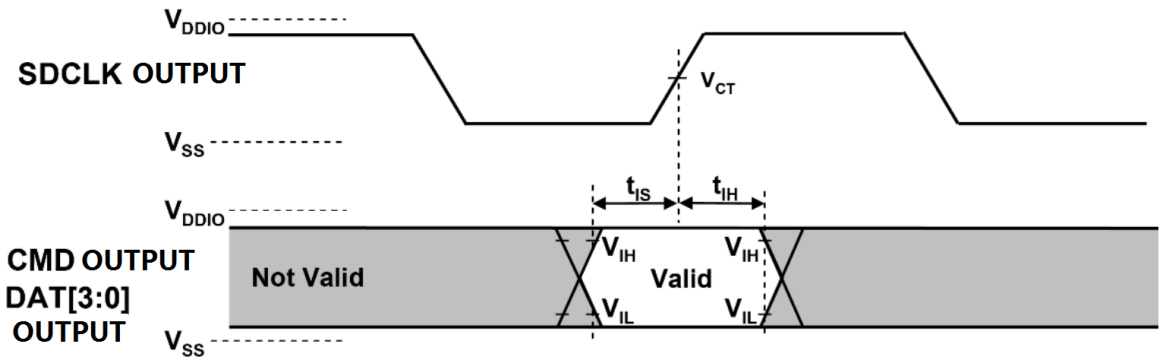


Table 4-17 SDIO(SDR104) Timing Specification

Inputs CMD, DAT (referenced to CLK)				
Symbol	**Parameter**	**Min**	**Max**	**Unit**
tIS	input set-up time	1.4	-	ns
tIH	input hold time	0.8	-	ns

Note

SD card interface uses SDIO protocol.

4.6.3 NAND Timing Specification

Nand timing specifications are shown as below.

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Figure 4-8 Async Waveform for Command/Address/Data Output Timing

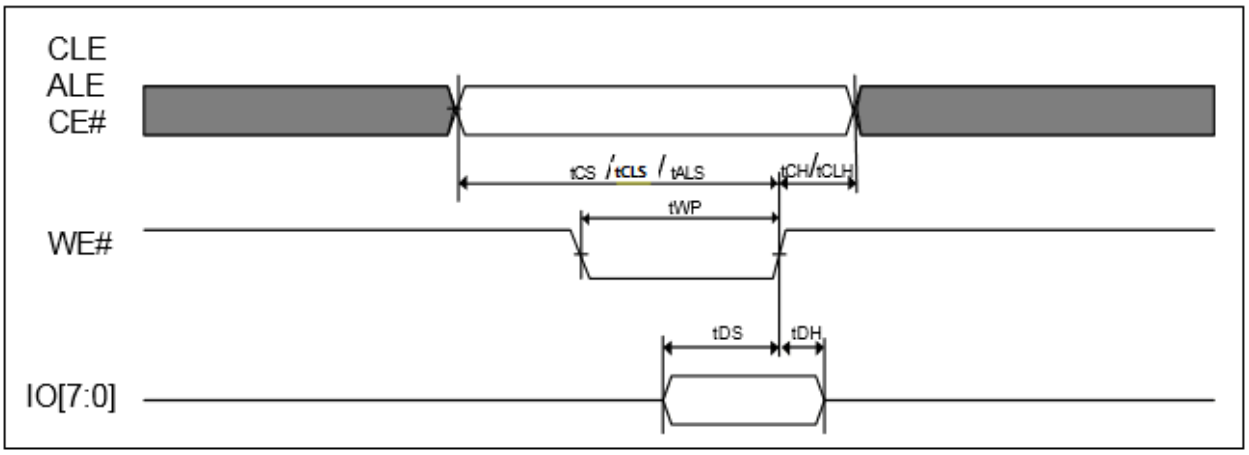
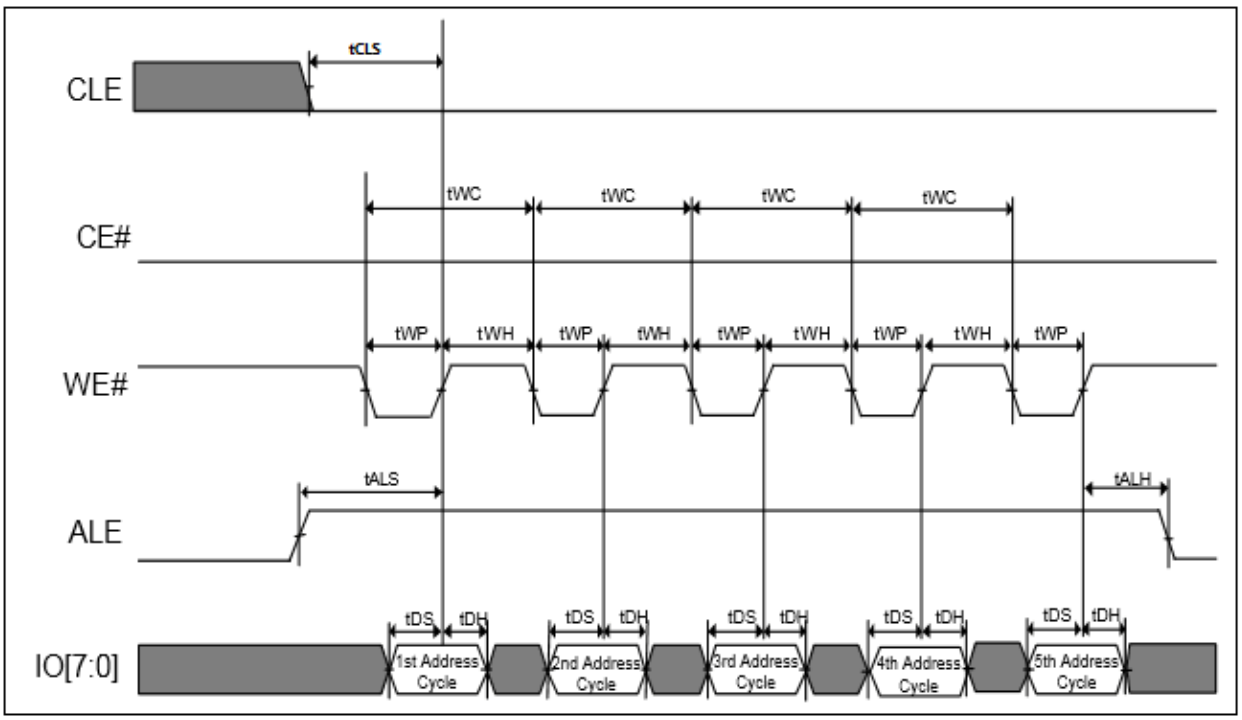


Figure 4-9 Async Waveform for Address Output Cycle



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Figure 4-10 Async Waveform for Sequential Data Read Cycle(After Read)-EOD Mode

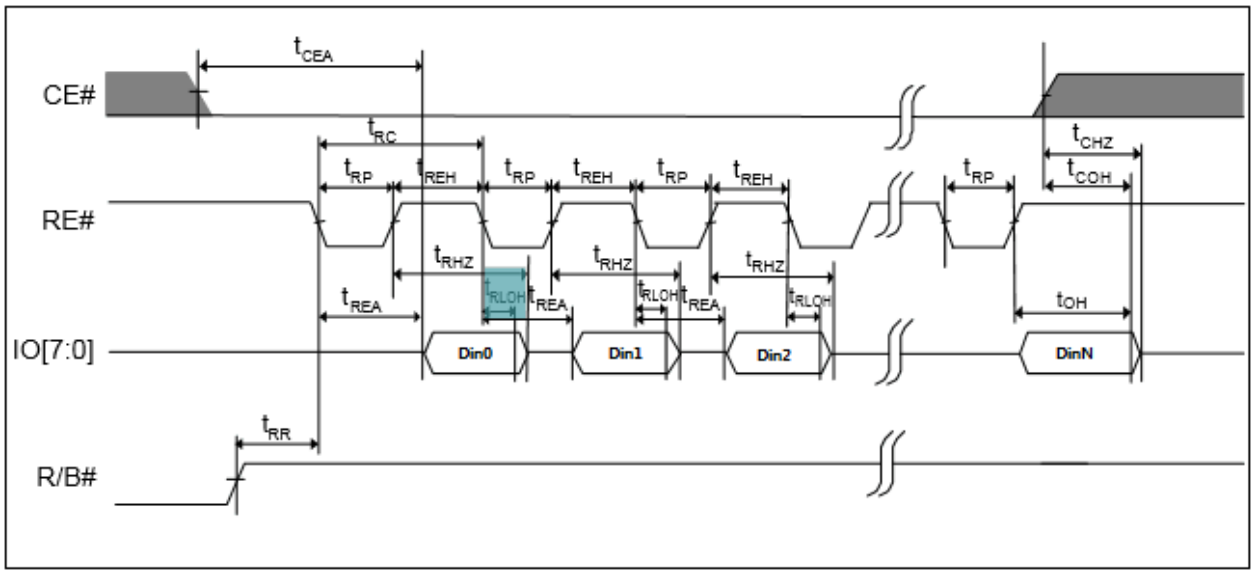


Table 4-18 Nand Timing Specifications

Symbol	Parameter (Asynchronous) (mode 5)	Min	Max	Unit
tCLS	CLE setup time	10	-	ns
tCLH	CLE hold time	5	-	ns
tALS	ALE setup	10	-	ns
tALH	ALE hold	5	-	ns
tDS	Data setup time	7	-	ns
tDH	Data hold time	5	-	ns
tWC	WE# cycle time	20	-	ns
tWP	WE# pulse width	10	-	ns
tWH	WE# high lold time	7	-	ns
tREA	RE# access time	-	16	ns
tOH	Data output hold time	15	-	ns
tRLOH	RE#-low to data hold time (EDO)	5	-	ns
tRP	RE# pulse width	10	-	ns
tREH	RE# high hold time	7	-	ns
tRC	RE# cycle time	20	-	ns

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4.6.4 SPICC Timing Specification

Figure 4-11 SPICC Timing Diagram

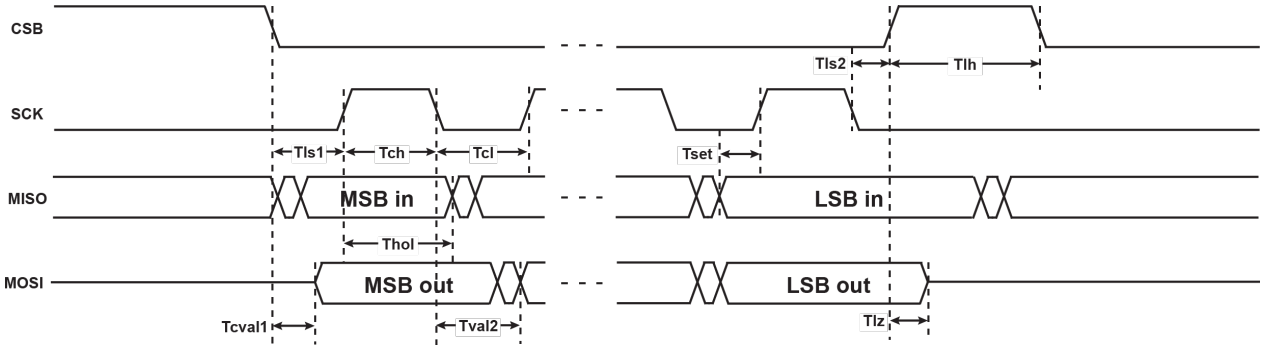
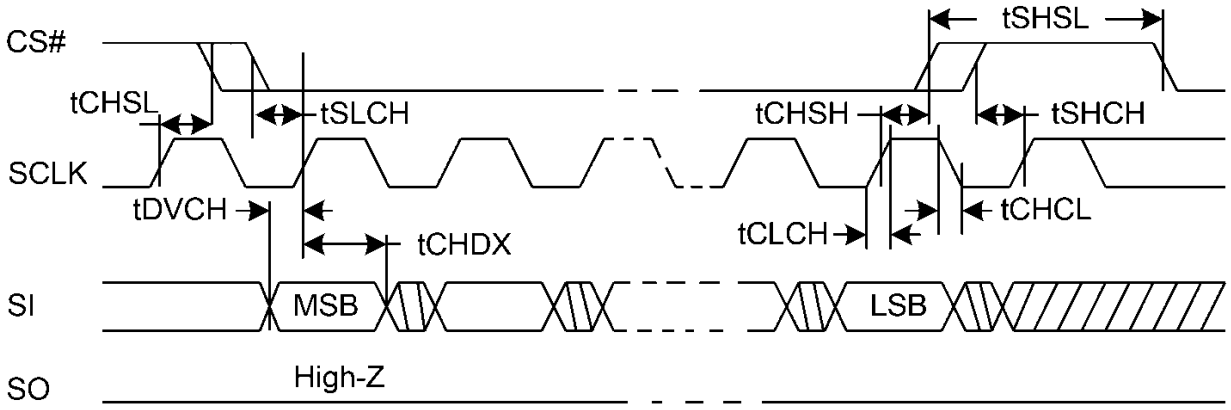


Table 4-19 SPICC Master Timing Specification

Symbol	Description	Min.	Max.	Unit
fCLK	Clock Frequency	1	80	MHz
TCH	Clock high time	5		ns
TCL	Clock low time	5		ns
TLS1	CS fall to First Rising CLK Edge	50		ns
TSET	Data input Setup Time	4		ns
THOL	Data input Hold Time	4		ns
TLH	Minimum idling time between transfers (minimum ss high time)	5		ns

4.6.5 SPIFC Timing Specification

Figure 4-12 SPIFC Serial Input Timing Diagram



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Figure 4-13 SPIFC Out Timing Diagram

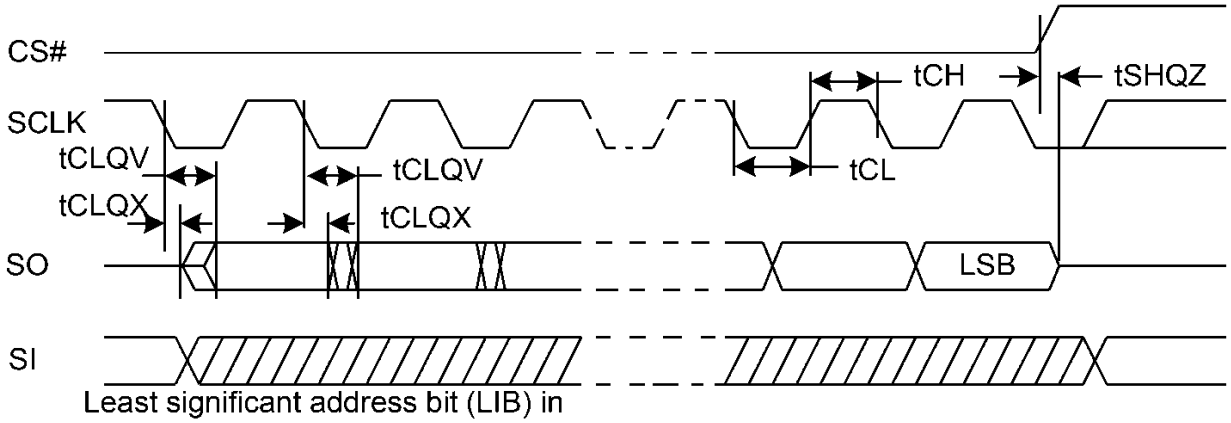


Table 4-20 SPIFC Master Timing Specification

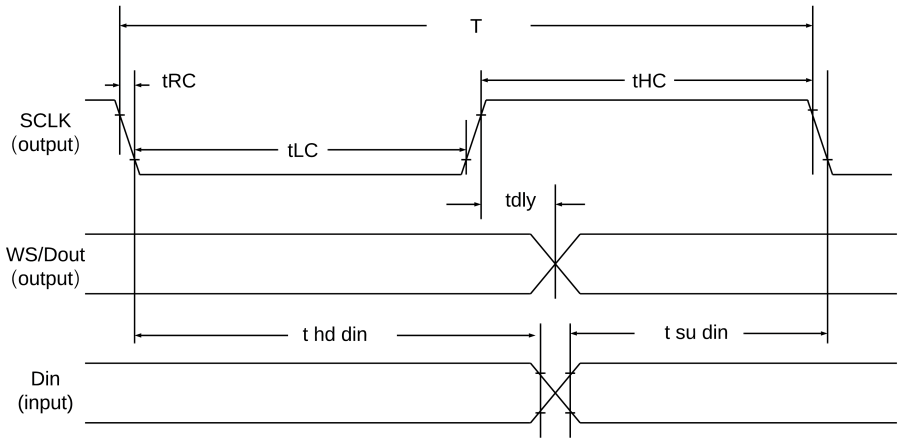
Symbol	Parameter (Clock 41.7MHz)	Min	Max	Unit
fRSCLK	Clock Frequency for READ instructions	-	50	Mhz
tCH	Clock High Time	8	-	ns
tCL	Clock Low Time	8	-	ns
tCLCH	Clock Rise Time (peak to peak)	0.1	-	V/ns
tCHCL	Clock Fall Time (peak to peak)	0.1	-	V/ns
tSLCH	CS# Active Setup Time (relative to SCLK)	4	-	ns
tCHSH	CS# Active Hold Time (relative to SCLK)	4	-	ns
tDVCH	Data In Setup Time	2	-	ns
tCHDX	Data In Hold Time	3	-	ns
tSHQZ	Output Disable Time (relative to CS#)	-	8	ns
tCLQV	Clock Low to Output Valid	-	6	ns
tCLQX	Output Hold Time	1	-	ns

4.6.6 Audio Timing Specification

There are two modes for the audio I2S/TDM interface: Master mode and Slave mode, as shown below.

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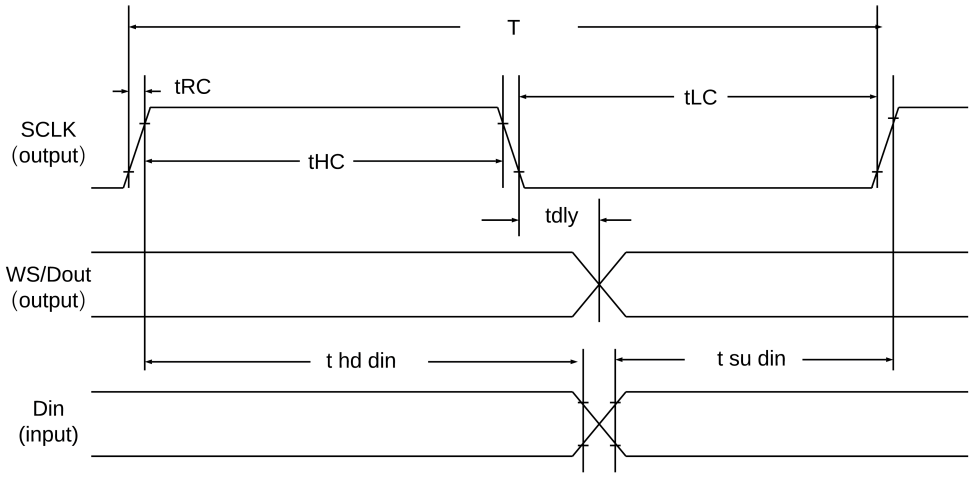
Figure 4-14 Output Data of SCLK Rising Edge TDM/I2S Timing, Master Mode



Note

Input data was sampled on SCLK rising/falling edge via software setting.

Figure 4-15 Output Data of SCLK Falling Edge TDM/I2S Timing, Master Mode



Note

Input data was sampled on SCLK rising/falling edge via software setting.

Table 4-21 Audio I2S/TDM Timing Specification, Transmitter, Master Mode

Symbol	Parameter	Min	Typ	Max	Unit
T	Clock period	10			ns
tHC	High level of SCLK	0.35			T
tLC	Low level of SCLK	0.35			T
tRC	Edge time of SCLK			0.15	T
tdly	Delay from SCLK to WS	-2	3	5	

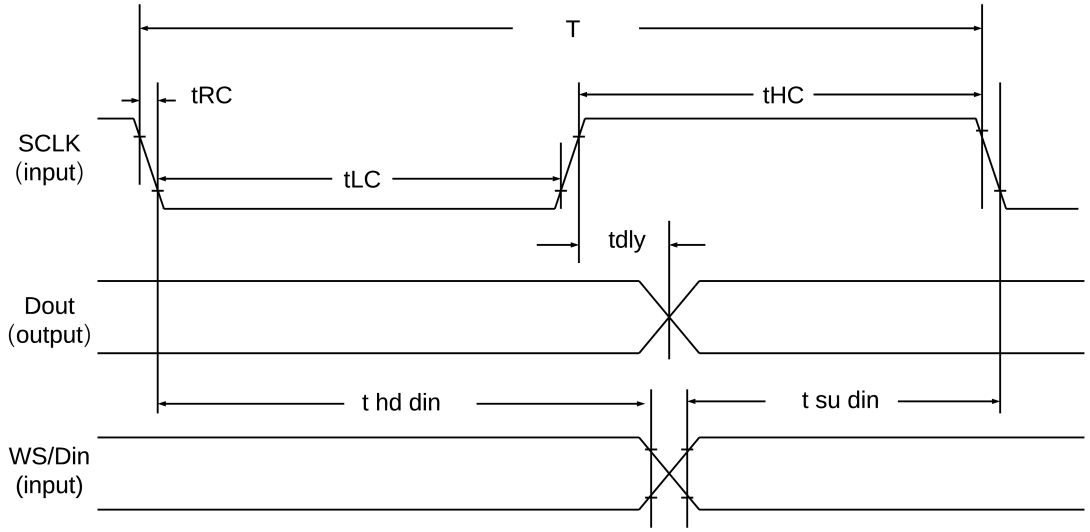
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Symbol	Parameter	Min	Typ	Max	Unit
tsuin	Setup time of Din	4			ns
thdin	Hold time of Din	4			ns

Note

Measure Point refers to VIH, ViL parameters of Normal GPIO Specifications.

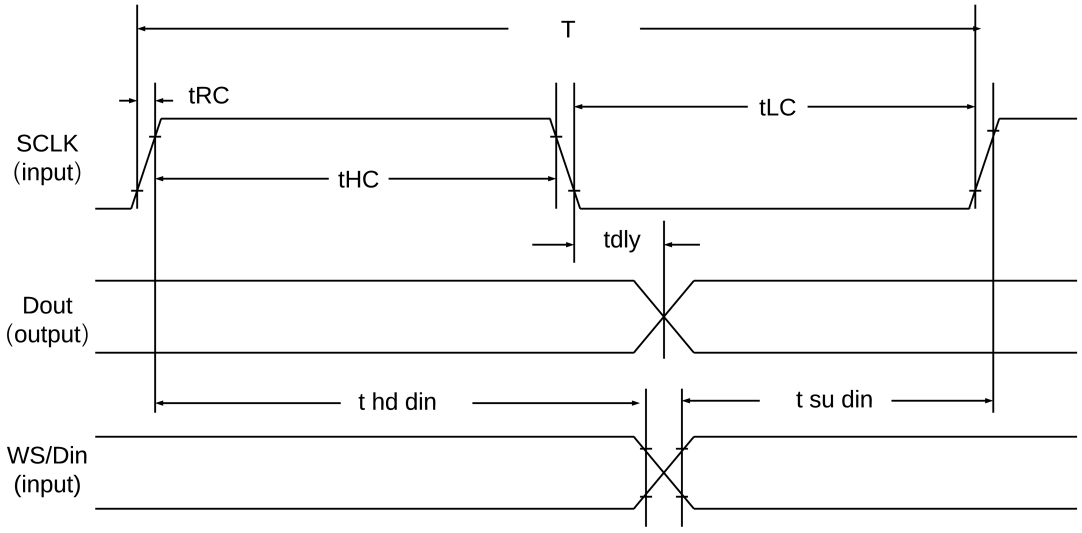
Figure 4-16 Output Data of SCLK Rising Edge TDM/I2S Timing, Slave Mode



Note

Input data was sampled on SCLK rising/falling edge via software setting.

Figure 4-17 Output Data of SCLK Falling Edge TDM/I2S Timing, Slave Mode



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Note

Input data was sampled on SCLK rising/falling edge via software setting.

Table 4-22 Audio I2S/TDM Timing Specification, Transmitter, Slave Mode

Symbol	Parameter	Min	Typ	Max	unit
T(out)	Clock period	40			ns
T(in)	Clock period	10			ns
tHC	High level of SCLK	0.35			T
tLC	Low level of SCLK	0.35			T
tRC	Edge time of SCLK			0.15	T
tsu din	Setup time of WS/Din	4			ns
thd din	Hold time of WS/Din	4			ns
tdly	Delay between SCLK and Dout	2	12	15	ns

Note

Measure Point refers to VIH, ViL parameter of Normal GPIO Specifications.

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4.6.7 PDM Timing Specification

Figure 4-18 PDM Timing Diagram

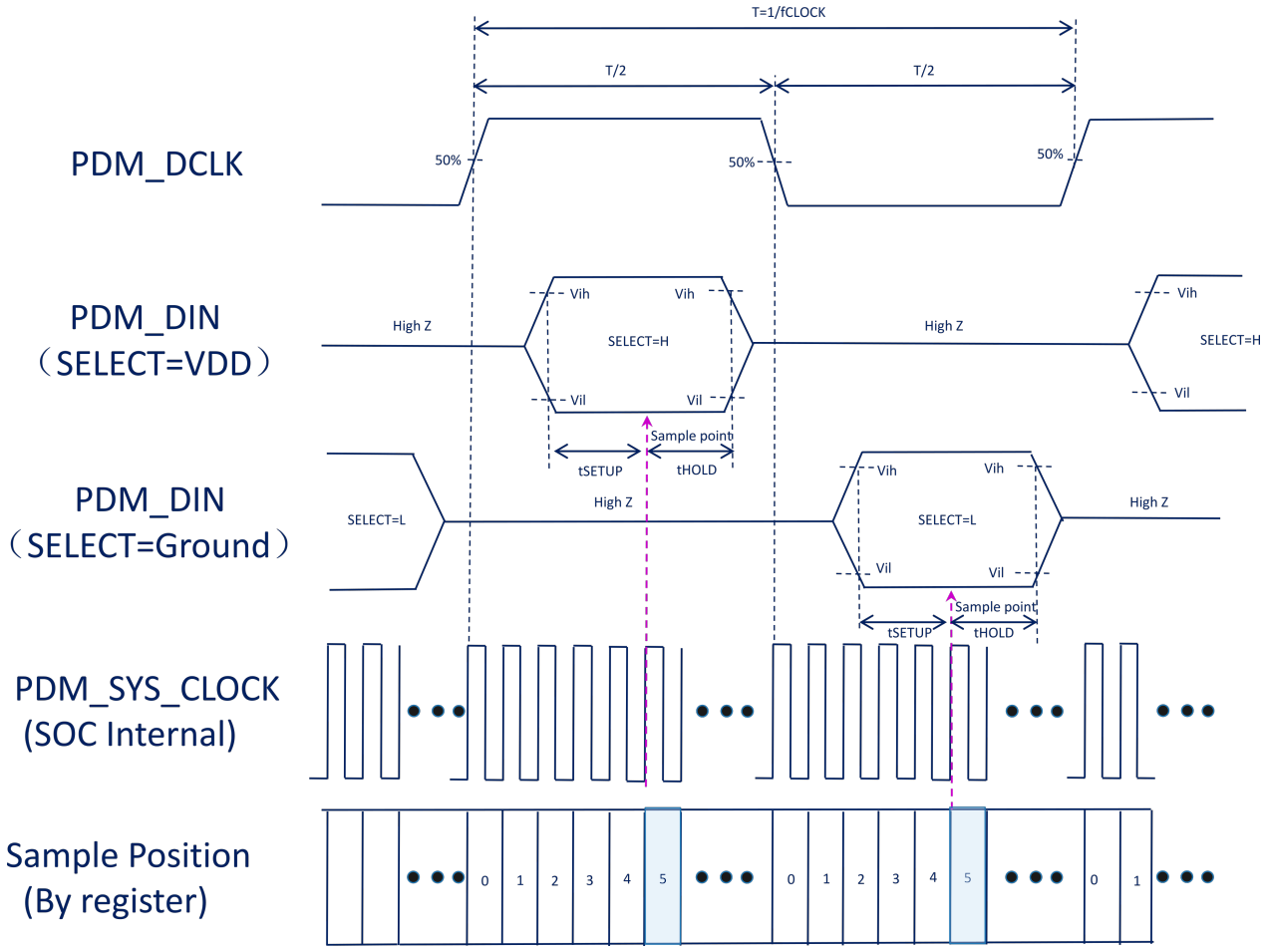


Table 4-23 PDM Timing Specification

Parameter	Symbol	Min.	Typ.	Max.	Unit
PDM clock period	tDCLK	200			ns
PDM clock duty cycle	tHIGH/tLOW	48%		52%	tDCLK
PDM Data setup time	tSETUP	20			ns
PDM Data hold time	tHOLD	20			ns
Sys clock period	tSYSCLK	5	7.5		ns

Note

1. Default PDM_SYS_CLOCK=133MHz.
2. For Sample position, please refer to PDM register PDM_CHAN_CTRL, PDM_CHAN_CTRL1.

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4.6.8 UART Timing Specification

Figure 4-19 UART Timing Diagram

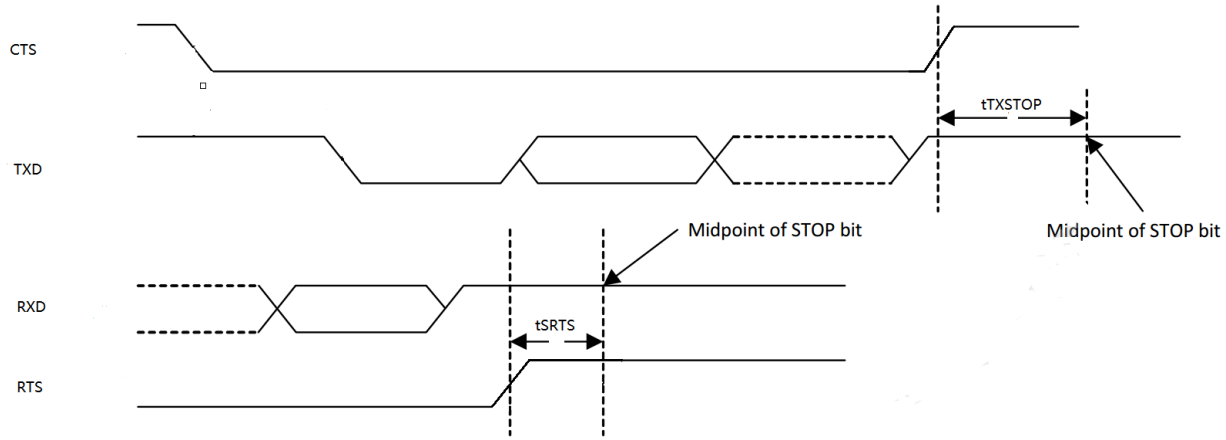


Table 4-24 UART Timing Specification

Parameter	Symbole	Min.	Max.	Unit
Delay time, CTS high before midpoint of stop bit	tTXSTOP	-	0.5	Bit Periods
Delay time, midpoint of stop bit to RTS hgih	tSRTS	-	0.5	Bit Periods

4.7 Recommended Oscillator Electrical Characteristics

The processor requires the 24MHz oscillator for generating the main clock source.

Table 4-25 24MHz Oscillator Specification

Symbol	Description	Min.	Typ.	Max.	Unit	Note
F_o	Nominal Frequency		24		MHz	
$\Delta f/f_o$	Frequency Tolerance	-30		30	ppm	At 25°C
		-50		50	ppm	At -20~85 °C
C_L	Load Capacitance	7.5	12	12.5	pF	
ESR	Equivalent Series Resistance			100	oHm	

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Note

- 10ppm Tolerance is preferred if 24MHz XTAL is also driving WIFI module.
- For user external clock source□please connect input clock output to SYS_OSCIN, let SYS_OSCOUT float.
- The threshold of Xin inverter is around 0.9V (Xin range: -0.3V to +2.1V). Therefore, follow the suggestions below for input clock.
  - Without DC blocking capacitor, use a higher Vpp output TCXO. The high voltage should be higher than 1.35V (VSWING >1.35V, 0V to >1.35V).
  - With DC blocking capacitor, re-bias the middle voltage at 0.9V, VSWING >2*0.45V.

4.8 Power On Config

3 Boot pins are used as power on config (POC) pins, to set the booting sequence.

POC setting is latched at the rising edge of reset signal.

3 POC pins are all pull high internal, CPU will try to boot from nand/eMMC first, if fails then try to boot from SD CARD, if still fails then try to boot from USB (PC).

External 4.7K ohm pull down resistors can be used to change the POC setting. The resistors should be placed on right location, avoid stubs on high speed signals.

The Power on configuration is listed below.

Table 4-26 Power On Configuration Pin Table

POC	Boot Pin	Name	Pull Low	Pull High
POC0	Boot [4]	SPI NAND First	SPI NAND boot first	Default sequence
POC1	Boot [5]	USB First	USB boot first	Default sequence
POC2	Boot [6]	SPI NOR First	SPI NOR first	Default sequence

Table 4-27 Booting Sequence Diagram

No.	POC_0 (SPI_ NAND)	POC_1 (USB_ BOOT)	POC_2 (SPI_ NOR, eMMC/ NAND)	1 st Boot device	2 nd Boot device	3 rd Boot device	4 th Boot device
1	0	0	0	USB (short delay)	SPI_NOR	NAND/eMMC	SD Card
2	0	0	1	USB (short delay)	NAND/ eMMC	SD Card	-
3	0	1	0	SPI_NOR	NAND/ eMMC	SD Card	USB
4	0	1	1	SPI_NAND	NAND/ eMMC	USB	-
5	1	0	0	USB (short delay)	SPI_NOR	NAND/eMMC	SD Card
6	1	0	1	USB (short delay)	NAND/ eMMC	SD Card	-

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No.	POC_0 (SPI_ NAND)	POC_1 (USB_ BOOT)	POC_2 (SPI_ NOR, eMMC/ NAND)	1 st Boot device	2 nd Boot device	3 rd Boot device	4 th Boot device
7	1	1	0	SPI_NOR	NAND/ eMMC	SD Card	USB
8	1	1	1	NAND/ eMMC	SD Card	USB	-

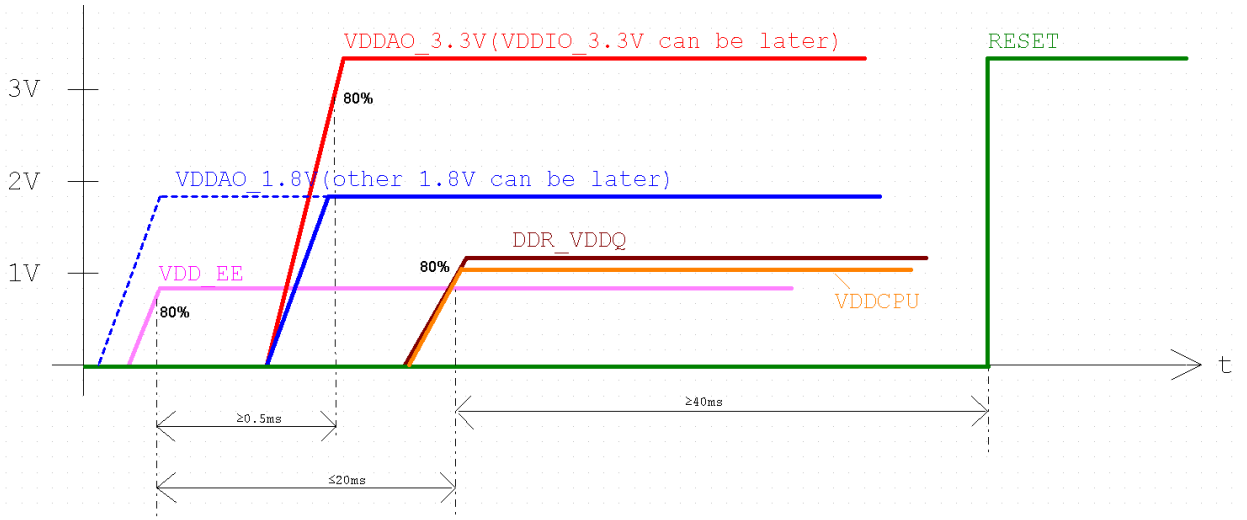
Note

If GPIOC is not working as SDCARD port, do not pull CARD_DET(GPIOC_10) low when system is booting up, to avoid romcode trying to boot from SDCARD.

4.9 Recommended Power On Sequence

The example of power on sequence is shown in the following figure.

Figure 4-20 Power On Sequence



Note

1. All test values refer to 80% of typical power voltage.
2. VDDAO_3.3V & VDDIO_3.3V should ramp up > 0.5ms later than VDD_EE.
3. All power sources should get stable within 20ms (except for DDR_VDDQ).
4. No sequence requirement between VDDCPU & DDR_VDDQ and other power source.
5. VDDAO_1.8V should ramps up no later than VDDAO_3.3V & VDDIO_3.3V, and VDDAO_ 3.3V & VDDIO_3.3V should never be 2.5V higher than VDDAO_1.8V.
6. In some designs, VDDCPU & VDD_EE are merged to VCC_CORE, the power on sequence should be same as VDD_EE.
7. RESET_n should keep low for at least 40ms after power up (except DDR_VDDQ).

Please refer to reference schematics.

4.10 Power Consumption

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Note

The specs listed here are tested typical value. Enough margin in circuit needs to be reserved.

Symbol	Maximum Current (mA)	Note
VDDCPU	2500	-
VDD_EE	2000	-
VDDQ	1600	Total power of controller and DDR3 memory

Symbol	Typical Current (mA)	Maximum Current (mA)	Note
AVDD08_P2P	TBD	200	
AVDD08_HDMIRX	80	110	
AVDD0V8_USB	25.8	39	Per USB channel at high speed full loading
VDD18_AO	2.3	-	Including VDD18_EFUSE Max 100 when programing EFUSE
AVDD18_DDR-P2P-DPLL	40	52	
AVDD18_ETH	33.2	50	-
AVDD18_AUDIO	12.3	-	-
AVDD18_V	56	110	
AVDD18_ADC_DPLL	5	7	-
AVDD18_HDMIRX	35	40	At 6Gbps mode
AVDD18_P2P_HRX-EARC	128	158	
AVDD18_USB	12.4	15	Per USB channel at high speed full loading
AVDD33_USB	5	29	Per USB channel at full speed full loading
AVDD33_HDMIRX	56	90	Per connected channel
VDDIO	-	-	[1]

Note

VDDIO=1.8V, DS=3, output 200MHz clock:

- IO pad itself consumes about 1.4mA.
- Driving a 55ohm trace with length of 50mm and width of 0.1mm will consumes about 2.8mA additional current (low impedance trace consumes more power)
- Base #2, add 5pF cap will consumes about 1.8mA additional current, total about 6mA
- When VDDIO=3.3V, GPIO consumes about 70% higher current, about 13mA
- Internal & external pull down resistor consumes additional current

4.11 Storage and Baking Conditions

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The processor is moisture-sensitive device of MSL level 3, defined by IPC/JEDEC J-STD-020. Please follow the storage and backing guidelines.

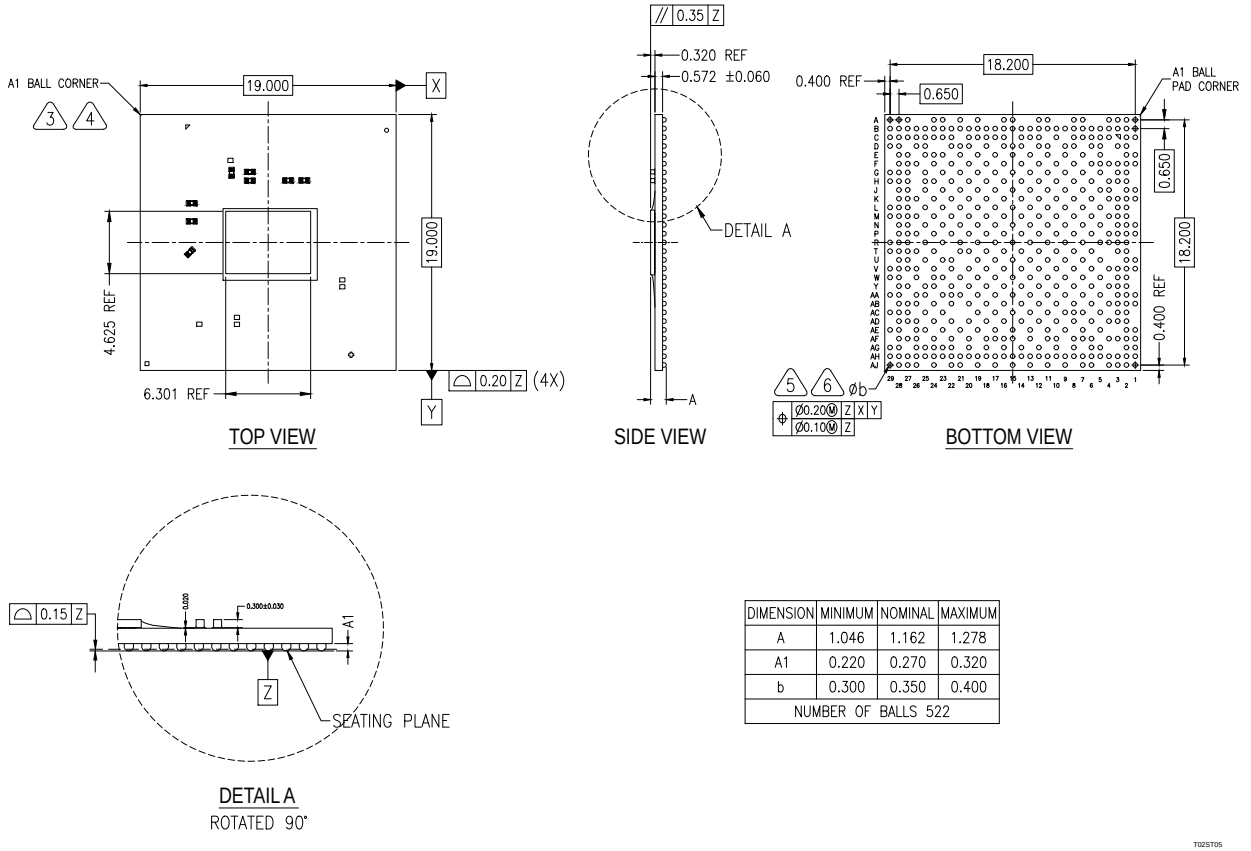
1. Calculated shelf life in sealed bag: 12 months at <40°C and <90% relative humidity (RH).
2. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must comply with either of the following principles.
   a. Mounted with 168 hours of factory conditions ≤30°C/60% RH
   b. Stored per J-STD-033
3. If the humidity indicator card reads >10%, devices should be baked before mounting.
4. If baking is required, see IPC/JEDEC J-STB-033 for baking process.

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5 Mechanical Dimensions

The processor comes in a 29x29 ball matrix FCBGA or FCCSP RoHS package. The mechanical dimensions are given in millimeters as the following figure.

Figure 5-1 Dimensions (FCBGA)



DIMENSION	MINIMUM	NOMINAL	MAXIMUM
A	1.046	1.162	1.278
A1	0.220	0.270	0.320
b	0.300	0.350	0.400
NUMBER OF BALLS 522			

Figure 5-2 Dimensions (FCCSP)

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6 System

This chapter describes the system architecture of the chip.

6.1 Memory Map

Memory map of the chip is listed in the following table.

Table 6-1 Memory Map

START	END	REGION (NORMAL)
FFFF0000	FFFFFFFF	A55_ROM
FFFF0000	FFFEFFFF	RESERVED
FFFA0000	FFFEFFFF	AHB SRAM
FFE80000	FFF9FFFF	RESERVED
FFE40000	FFE7FFFF	DVALIN
FFE0E000	FFE3FFFF	RESERVED
FFE0D000	FFE0DFFF	HRX_HDCP22
FFE0B000	FFE0CFFF	RESERVED
FFE09000	FFE0AFFF	USBCTRL
FFE07000	FFE08FFF	EMMCC
FFE05000	FFE06FFF	EMMCB
FFE03000	FFE04FFF	RESERVED
FFE02000	FFE02FFF	BT656
FFE01000	FFE01FFF	RESERVED
FFD27000	FFE00FFF	RESERVED
FFD26000	FFD26FFF	ASYNC_FIFO3
FFD25000	FFD25FFF	SC
FFD24000	FFD24FFF	UART0
FFD23000	FFD23FFF	UART1
FFD22000	FFD22FFF	UART2
FFD21000	FFD21FFF	RESERVED
FFD20000	FFD20FFF	RESERVED
FFD1F000	FFD1FFFF	I2C_M0
FFD1E000	FFD1EFFF	I2C_M1
FFD1D000	FFD1DFFF	I2C_M2
FFD1C000	FFD1CFFF	I2C_M3
FFD1B000	FFD1BFFF	PWM_AB
FFD1A000	FFD1AFFF	PWM_CD
FFD19000	FFD19FFF	PWM_EF
FFD18000	FFD18FFF	MSR_CLK
FFD17000	FFD17FFF	RESERVED

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START	END	REGION (NORMAL)
FFD16000	FFD16FFF	RESERVED
FFD15000	FFD15FFF	SPICC_1
FFD14000	FFD14FFF	SPIFC
FFD13000	FFD13FFF	SPICC_0
FFD12000	FFD12FFF	RESERVED
FFD11000	FFD11FFF	RESERVED
FFD10000	FFD10FFF	RESERVED
FFD0F000	FFD0FFFF	ISA
FFD0E000	FFD0EFFF	PARSER
FFD0D000	FFD0DFFF	RESERVED
FFD0C000	FFD0CFFF	SANA
FFD0B000	FFD0BFFF	STREAM
FFD0A000	FFD0AFFF	ASYNC_FIFO
FFD09000	FFD09FFF	ASYNC_FIFO2
FFD08000	FFD08FFF	ASSIST
FFD07000	FFD07FFF	RESERVED
FFD06000	FFD06FFF	STB
FFD05000	FFD05FFF	AIFIFO
FFD04000	FFD04FFF	RESERVED
FFD03000	FFD03FFF	RESERVED
FFD02000	FFD02FFF	RESERVED
FFD01000	FFD01FFF	RESET
FFD00000	FFD00FFF	RESERVED
FFC08000	FFCFFFFF	RESERVED
FFC00000	FFC07FFF	GIC
FFB00000	FFBFFFFF	GPV
FF950000	FFAFFFFF	RESERVED
FF940000	FF94FFFF	GE2D
FF900000	FF93FFFF	VPU
FF80B000	FF8FFFFF	RESERVED
FF80A000	FF80AFFF	AO_MAILBOX
FF809000	FF809FFF	SAR_ADC
FF808000	FF808FFF	IR_DEC
FF807000	FF807FFF	PWM_AB
FF806000	FF806FFF	I2C_S
FF805000	FF805FFF	I2C_M
FF804000	FF804FFF	UART2
FF803000	FF803FFF	UART

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START	END	REGION (NORMAL)
FF802000	FF802FFF	PWM_CD
FF801000	FF801FFF	RESERVED
FF800000	FF800FFF	RTI
FF670000	FF7FFFFF	RESERVED
FF660000	FF66FFFF	TCON
FF65A000	FF65FFFF	RESERVED
FF658000	FF659FFF	USBPHY22
FF656000	FF657FFF	ATV_DMD
FF654000	FF655FFF	TVFE
FF654000	FF653FFF	RESERVED
FF650000	FF653FFF	DEMOD
FF64E000	FF64FFFF	RESET_SEC
FF64C000	FF64DFFF	ETH_PHY
FF64A000	FF64BFFF	ADEC
FF640000	FF649FFF	RESERVED
FF63E000	FF63FFFF	DMA
FF63C000	FF63DFFF	HIU
FF63A000	FF63BFFF	USBPHY21
FF638000	FF639FFF	DMC
FF636000	FF637FFF	USBPHY20
FF635400	FF635FFF	RESERVED
FF635000	FF6353FF	TS_SAR
FF634C00	FF634FFF	TS_DDR
FF634800	FF634BFF	TS_PLL
FF634400	FF6347FF	PERIPHS_REG
FF634000	FF6343FF	RESERVED
FF632000	FF633FFF	ACODEC
FF630000	FF631FFF	EFUSE
FF620000	FF62FFFF	DOS
FF610000	FF61FFFF	HDMIRX
FF600000	FF60FFFF	AUDIO
FF500000	FF5FFFFF	USB0
FF400000	FF4FFFFF	USB1
FF3F0000	FF3FFFFF	ETH
FF000000	FF3EFFFF	RESERVED
FE000000	FEFFFFFF	DDR_CTRL
FC000000	FDFFFFFF	RESERVED
FA000000	FBFFFFFF	RESERVED

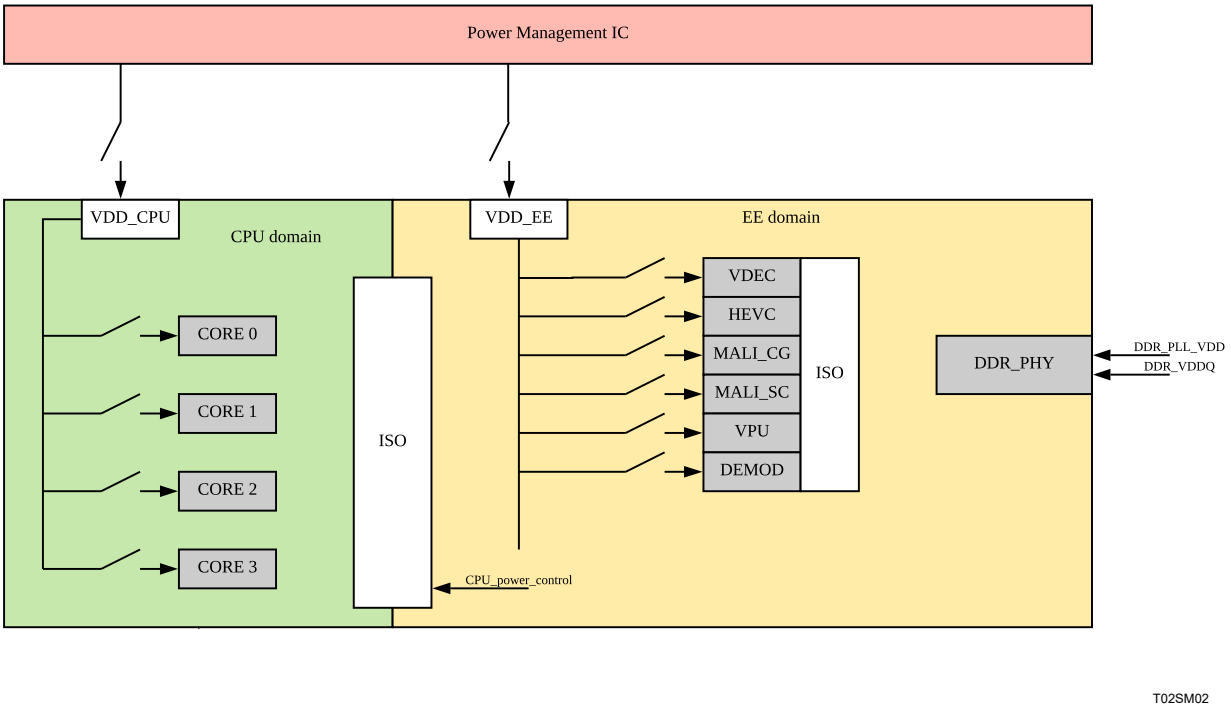
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START	END	REGION (NORMAL)
F6000000	F9FFFFFF	FLASH
F5800000	F5FFFFFF	A55_DBG
0	F57FFFFF	DDR

6.2 Power Domain

The following figure shows the power domain.

Figure 6-1 Power Domain



6.2.1 Top Level Power Domains

The power supplies for the different domains must follow a specific power supply order:

- The A55 can't be powered without the EE domain.
- The EE domain can't be powered on without the AO domain.

Please read the following table from left to right and then right to left, that's the essential power up/down sequence for the entire chip.

	EE Domain		A55 domain				
	EE Logic	Mali and DOS, DEMOD and VPU	L2 Cache	CPU0	CPU1	CPU2	CPU3
STATE 0 All off	The EE domain is OFF		The A55 domain must be off if the EE domain is OFF				
STATE 1 EE only (for example, audio applications or simple video	On	On or off as needed	Off	Off	Off	Off	Off

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	EE Domain		A55 domain				
applications that don't need the A55)							
STATE 2 Any single CPU	On	On or off as needed	On	On	Off	Off	Off
STATE 3 Any 2 CPU's	On	On or off as needed	On	On	On	Off	Off
STATE 4 Any 3 CPUs	On	On or off as needed	On	On	On	On	Off
STATE 5 All 4 CPUs	On	On or off as needed	On	On	On	On	On

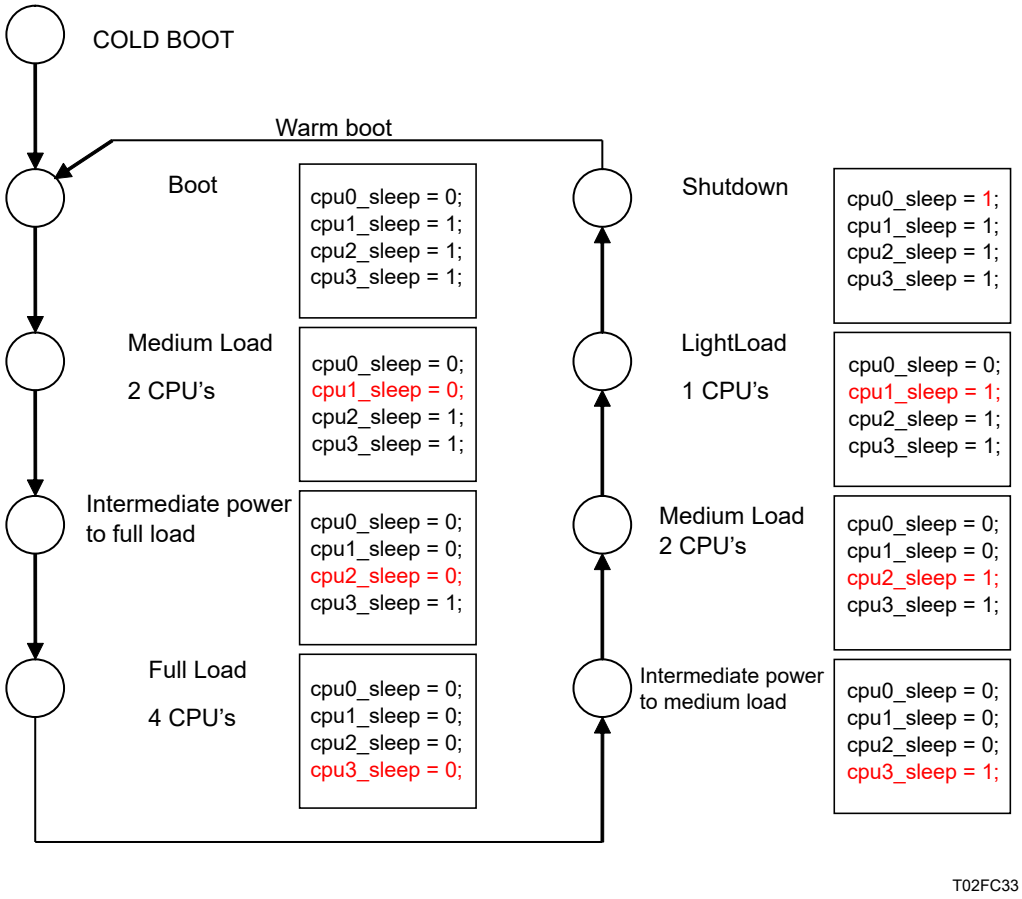
6.2.2 A55 Power Modes

The A55 domain is the last to power up and the first to power down. The A55 domain itself consists of a quad (4) CPUs (L1/L2 inside), an L3 cache controller and an SCU. The A55 CPU boots with the SCU/L3 powered and CPU0 powered. After CPU0 boots, subsequent CPU's can be enabled and disabled independently of one another using the control bits described below. The most likely scenario would be for the CPU0 to be used for low load conditions, CPU0 and CPU1 to be used for medium load conditions and CPU0, 1, 2 and CPU3 to all be used for heavy load conditions. The flow-diagram below illustrates the transition to each legal state.

A55	CPU0	CPU1	CPU2	CPU3	A55_pwr
Domain Power Sleep bit AO_RTI_PWR_SYS_CPU_CNTL1 (1 = power off)	Bit[2]	Bit[4]	Bit[6]	Bit[8]	
Domain Power Acknowledge bit AO_RTI_PWR_SYS_CPU_CNTL1 AO_RTI_PWR_	Bit[16]	Bit[17]	Bit[18]	Bit[19]	
Output signal isolation bit AO_RTI_PWR_SYS_CPU_CNTL0 (1 = isolated)	Bit[0]	Bit[1]	Bit[2]	Bit[3]	Bit[12]

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Figure 6-2 Power Seuquence of A55



6.2.3 EE Top Level Power Modes

The EE domain consits of severl sub-domains: DOS, VPU_HDMI, Mali and DEMOD.

6.2.3.1 DOS

DOS Power Mode contains two major components: VDEC and HEVC.

These should be powered up in sequence so as to reduce the surge currents. The module consists of both logic and memories. The logic and memories should be powered up/down with the clocks disabled. Each power domain in the DOS module is controlled by a single control bit. There is also an "acknowledge" bit that can be used to check if the power up/down condition is complete. The typical shutdown/power up time is typically less than 200 ns.

Table 6-2 DOS Power Control Summary

	Register	VDEC	HEVC
Domain Power Sleep (1 = power off)	AO_RTI_GEN_PWR_ SLEEP0	Bit[3:2]	Bit[7:6]
Domain Power Acknowledge	AO_RTI_GEN_PWR_ ACK0	Bit[1]	Bit[3]
Input Signal isolation (1 = isolated)	AO_RTI_GEN_PWR_ ISO0	Bit[6]	Bit[10]

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	Register	VDEC	HEVC
Output signal isolation (1 = isolated)	AO_RTI_GEN_PWR_ISO0	Bit[7]	Bit[11]
Memory Power Down/Up 0 = powered on 1 = powered off		DOS_MEM_PD_VDEC	DOS_MEM_PD_HEVC
Reset_n		DOS_SW_RESET0	DOS_SW_RESET3

6.2.3.2 VPU_HDMI

The VPU power mode contains multipule video paths and the HDMI-TX module.

Table 6-3 VPU_HDMI Power Control Summary

	Register	Bit
Domain Power Sleep Bit 1 = power off	AO_RTI_GEN_PWR_SLEEP0	Bit[8]
Domain Power Acknowledge Status 1 = module powered off	AO_RTI_GEN_PWR_ACK0	Bit[9:8]
Input Signal Isolation Bit 1 = Set isolation	AO_RTI_GEN_PWR_SLEEP0	Bit[9]
Memory Power Down/Up 0x00000000 = powered on 0xFFFFFFFF = powered off	HHI_VPU_MEM_PD_REG0	Bit[31:0]
	HHI_VPU_MEM_PD_REG1	Bit[31:0]
	HHI_VPU_MEM_PD_REG2	Bit[31:0]
	HHI_MEM_PD_REG0	Bit[15:8]

6.2.3.3 DEMOD

The Demod domain consits of ISDBT, ATSC, QAM, and DTMB modules.

Table 6-4 DEMOD Power Control Summary

	Register	Bit
Domain Power Sleep bit 1 = power off	AO_RTI_GEN_PWR_SLEEP0	Bit[10]
Domain Power Acknowledge status 1 = module powered off	AO_RTI_GEN_PWR_ACK0	Bit[10]
Input Signal isolation bit 1 = Set isolation	AO_RTI_GEN_PWR_ISO0	Bit[15:14]
Memory Power Down/Up 0x00000000 = powered on 0xFFFFFFFF = powered off	HHI_DEMOD_MEM_PD_REG0	Bit[31:0]

6.2.3.4 Mali

Mali power mode consits of three sub-domains: SC0, CG and GL.

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The power on/down sequence is as follows:

- GL is always powered on.
- Power on sequence: CG -> SC0
- Power down sequence: SC0 -> CG

6.2.4 Register Description

This section describes power/isolation/memory power down register summary.

Table 6-5 AO_RTI_PWR_CNTL_REG0 0x04

Bit(s)	R/W	Default	Description
31-24	R	0	Unused
23-18	R/W	0	Mem_pd[7:0]; [1:0] ahb_sram_32k;
17-16	R/W	0	Unused
15	R/W	0	Unused
14	R/W	1	Set to 1 to enable the crystal clock throughout the AO domain. Set to 0 to disable the crystal clock (and therefore save power)
13	R/W	0	Reserved
12-10	R/W	0	ALT_32K_INPUT_SEL: 000 = XTAL generate 32k 001 = gpio pad
9	R/W	1	EE_RESET_N: This bit is OR'd with the RESET_N signal from the pad to generate a new signal called reset_n_to_ee. Reset_n_to_ee is the general reset for the entire Everything Else domain. Therefore, if EE_RESET_N is set low, the entire Everything Else domain is reset.
8	R/W	0	RTC_OSCIN_SEL: Set this bit to 1 to mux the RTC 32khz crystal in place of the clk81 signal that comes from the Everything Else domain to drive the Always On domain.
7	R/W	0	Unused
6	R/W	0	Unused
5	R/W	0	Unused
4	R/W	0	Ee_iso_in_en
3	R/W	0	Ee_iso_out_en
2	R/W	0	Unused
1	R/W	0	Reset_iso_en
0	R/W	0	Unused

Table 6-6 AO_RTI_PWR_SYS_CPU_CNTL0 0x38

Bit(s)	R/W	Default	Description
31-30	R	-	A53_pwrctli3 (readback)
29-28	R	-	A53_pwrctli2 (readback)
27-26	R	-	A53_pwrctli1 (readback)
25-24	R	-	A53_pwrctli0 (readback)

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Bit(s)	R/W	Default	Description
23-22	R/W	0x3	A53_pwrctli3
21-20	R/W	0x3	A53_pwrctli2
19-18	R/W	0x3	A53_pwrctli1
17-16	R/W	0	A53_pwrctli0
15-0	R/W	0	unused

Table 6-7 AO_RTI_PWR_SYS_CPU_CNTL1 0x39

Bit(s)	R/W	Default	Description
31-24	R	0	unused
23-16	R	-	A53 sleep status
15-10	R/W	0	Unused
9-8	R/W	3	CPU3_SLEEP
7-6	R/W	3	CPU2_SLEEP
5-4	R/W	3	CPU1_SLEEP
3-2	R/W	0	CPU0_SLEEP
1	R/W	0	Unused
0	R/W	0	Unused

Table 6-8 AO_RTI_GEN_PWR_SLEEP0 0x3a

Bit(s)	R/W	Default	Description
31-8	R/W	0	unused
7	R/W	1	DOS_HEVC _D1_PWR_OFF
6	R/W	1	DOS_HEVC_PWR_OFF
5	R/W	1	DOS_VDEC2_D1_PWR_OFF
4	R/W	1	DOS_VDEC2_PWR_OFF
3	R/W	1	DOS_VDEC1_D1_PWR_OFF
2	R/W	1	DOS_VDEC1_PWR_OFF
1	R/W	1	DOS_HCODEC_D1_PWR_OFF
0	R/W	1	DOS_HCODEC_PWR_OFF

Table 6-9 AO_RTI_GEN_PWR_ISO0 0x3b

Bit(s)	R/W	Default	Description
31-12	R/W	0	unused
11	R/W	1	DOS_HEVC_OUT_EN
10	R/W	1	DOS_HEVC_IN_EN

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Bit(s)	R/W	Default	Description
9	R/W	1	DOS_VDEC2_ISO_OUT_EN
8	R/W	1	DOS_VDEC2_ISO_IN_EN
7	R/W	1	DOS_VDEC1_ISO_OUT_EN
6	R/W	1	DOS_VDEC1_ISO_IN_EN
5	R/W	1	DOS_HCODEC_ISO_OUT_EN
4	R/W	1	DOS_HCODEC_ISO_IN_EN
3	R/W	1	GPU_ISO_OUT_EN
2	R/W	1	GPU_ISO_IN_EN
1	R/W	1	Unused
0	R/W	1	Unused

Table 6-10 AO_RTI_GEN_PWR_ACK0 0x3c

Bit(s)	R/W	Default	Description
31-10	R	0	Reserved
9	R/W	1	VPU_ACK
8	R/W	1	HDMI_ACK
7	R/W	1	MALI_PWRUP_ACK[3]
6	R/W	1	MALI_PWRUP_ACK[3]
5	R/W	1	MALI_PWRUP_ACK[3]
4	R/W	1	MALI_PWRUP_ACK[3]
3	R/W	1	DOS_HEVC
2	R/W	1	DOS_VDEC2
1	R/W	1	DOS_VDEC1
0	R/W	1	DOS_HCODEC

Table 6-11 AO_RTI_GEN_PWR_SYS_CPU_MEM_PD0 0x3d

Bit(s)	R/W	Default	Description
31-0	R/W	0xFF-F00000	sys_cpu_pwr_mem_pd[31:0]

Table 6-12 AO_RTI_GEN_PWR_SYS_CPU_MEM_PD1 0x3e

Bit(s)	R/W	Default	Description
31-0	R/W	0	sys_cpu_pwr_mem_pd[63:32]

6.3 System Booting

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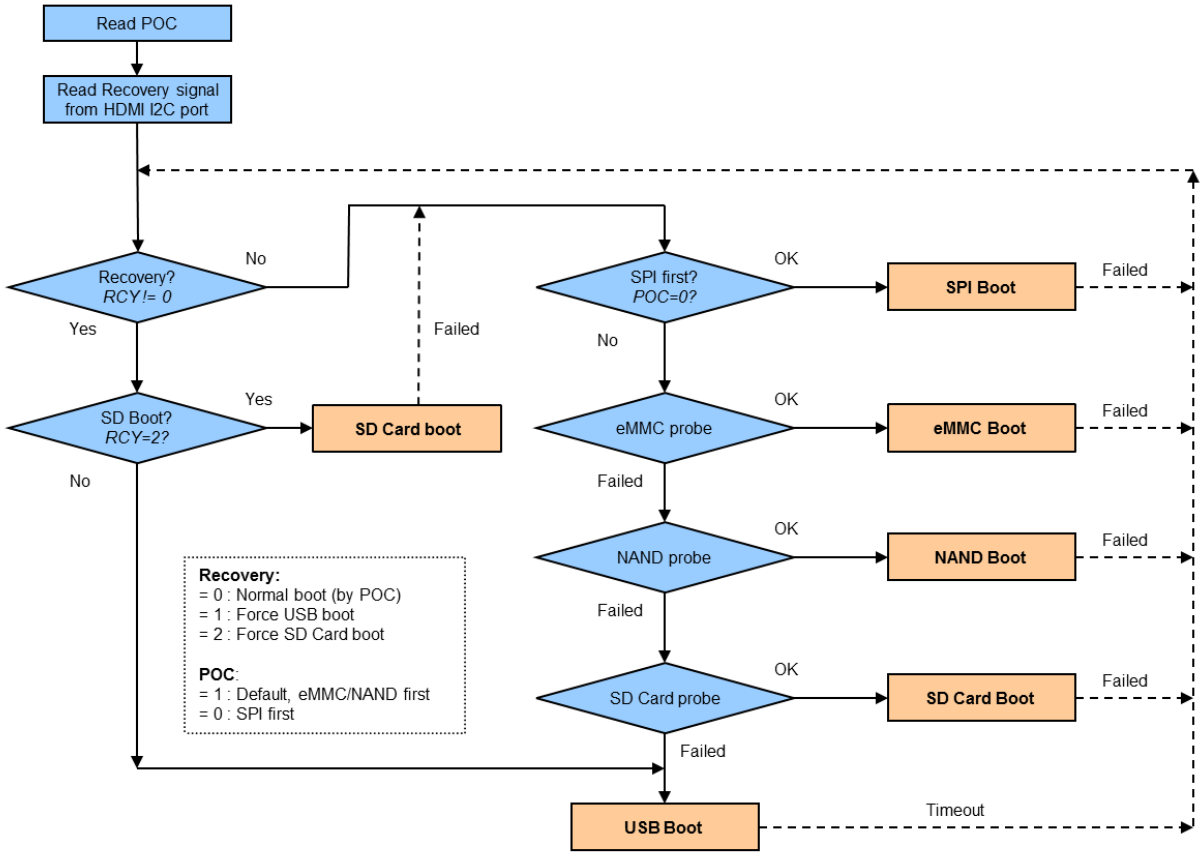
6.3.1 Overview

The part describes the power-on mode configurations, which include two portions: Cortex-M3 for security control and A55 for others.

6.3.2 Power-on Flow Chart

The following figure illustrates power on sequence of the chip.

Figure 6-3 Power-on Flow Chart



6.4 CPU

The Cortex ™-A55 MP subsystem of the chip is a high-performance, low-power, ARM macrocell with an L1, L2 and L3 cache subsystem that provide full virtual memory capabilities. The Cortex-A55 processor implements the ARMv8.2 architecture and runs 32-bit ARM instructions and 64 bit ARMv8 instructions. The developers can follow the ARM official reference documents for programming details.

The Cortex-A55 processor features are:

- In-order pipeline with dynamic branch prediction
- ARM, Thumb, and ThumbEE instruction set support
- TrustZone security extensions
- Harvard level 1 memory system with a Memory Management Unit (MMU)
- 128-bit AXI master interface
- ARM CoreSight debug architecture
- Trace support through an Embedded Trace Macrocell (ETMv4) interface

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- Intelligent Energy Manager (IEM) support with
  - Asynchronous AXI wrappers
  - Two voltage domains
- Media Processing Engine (MPE) with NEON technology
- Supports FPU

6.5 GPU

The Mali-G31 MP2 GPU is a hardware accelerator for 2D and 3D graphics system which compatible with the following graphics standards: OpenGL ES 3.2, Vulkan 1.1 and OpenCL 1.2. The developers can follow the ARM and Khronos official reference documents for programming details.

6.6 NPU

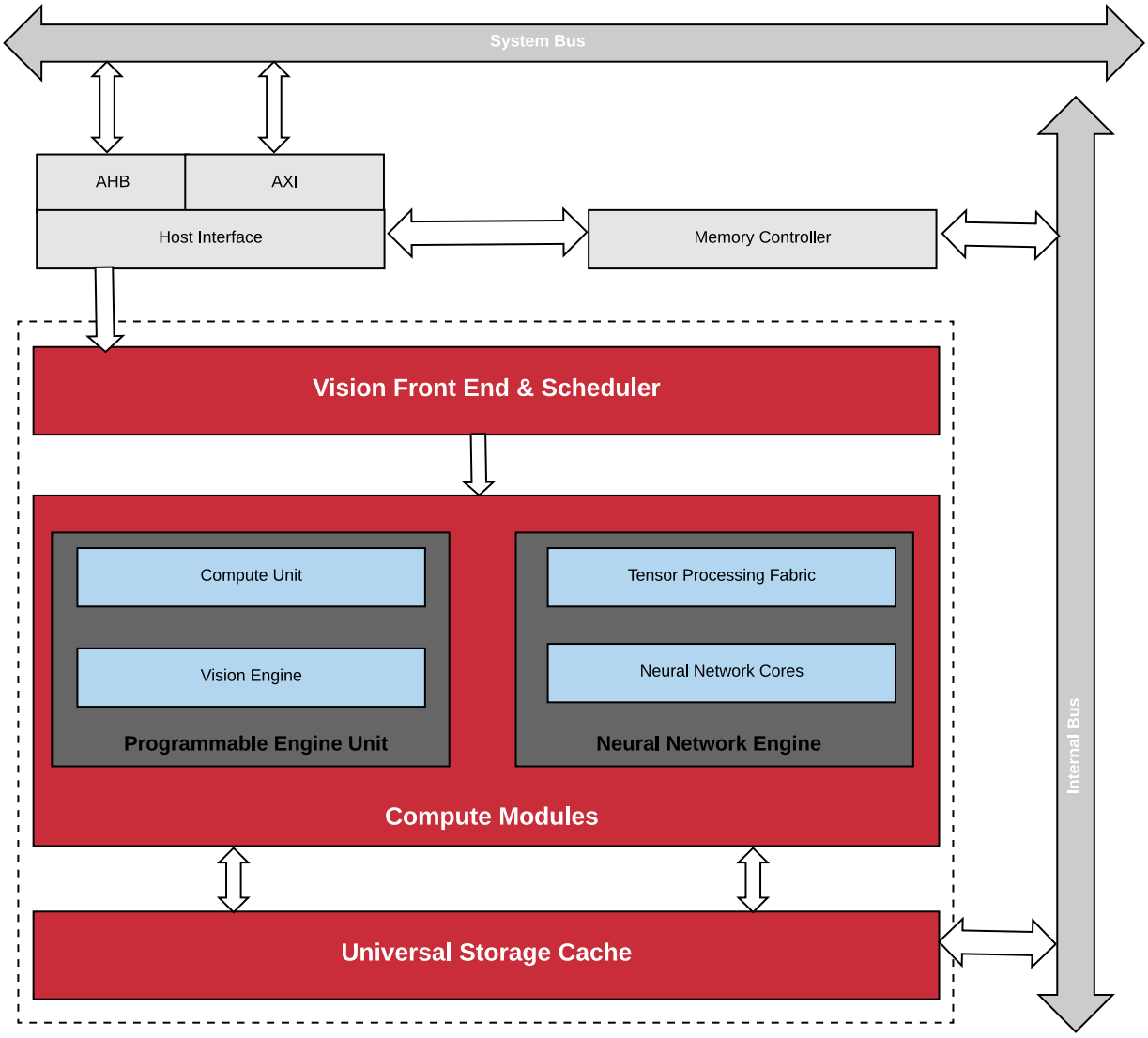
6.6.1 Overview

NPU is the process unit which is dedicated to neural network. It is designed to accelerate the neural network arithmetic in field of AI (artificial intelligence) such as machine vision and natural language processing.

The following figure shows the hardware structure of NPU.

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Figure 6-4 NPU Structure



NPU supports the following features.

- **Host interface**
  - 32 bit AHB interface used for configuration and only support single mode access
  - 128 bit AXI interface used to fetch data memory
- **Neural Network**
  - Support integer 8, integer 16 convolution operation
  - 768MAC (multiply-add units) per cycle (int 8)
  - 192MAC per cycle (int 16)
  - Support Liner, MIMO, Fully Connected, Fully Convolution
  - Unlimited network size (bound by system resource)
  - Inference Engine: Tensorflow, Tensorflow lite, Darknet, Onnxc Caffe, Caffe2, Pytorch, Keras backend, OpenCL, OpenVX
  - Support network sparse coefficient decompression
  - Support max, average pooling

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- Max pooling support 2x2, 3x3, stride ≤ min (input width, input height)
- Local average pooling size ≤ 11x11
- Support unpooling
- Support batch normalize, l2 normalize, local response normalize, Instance normalize
- Support region proposal
- Support permute, reshape, concat, depth to space, space to depth, flatten, reorg, squeeze and split
- Support prior_box layer
- Support non-max Suppression
- Support ROI pooling
- Convolution size NXN, N ≤ 11*stride, stride ≤ min(input width, input height)
- Support dilate convolution, N ≤ 11*stride, stride ≤ min(input width, input height), dilation < 1024
- Support de-convolution, N ≤ 11*stride, stride ≤ min(input width, input height)
- Support Element_wise add, div, floor, max, mul, scale, sub
- Support ReLU, Leaky ReLU, Softmax, PreLU, Sigmod, Tanh
- Support LSTM, RNN
- Support channel shuffle
- Support dequantize, dropout
- Include embedded lookup table
- Support hashtable lookup
- Support lsh projection
- Support svdf
- Support reserve

6.6.2 Register Description

Table 6-13 AQHiClockControl 0x0000

Bit(s)	R/W	Name	Description
0	R/W	CLK3D_ DIS	Disable 3D clock. Software core clock disable signal for 3D modules(clk_3d) clock. When set to 1, this clock is frozen.
1	R/W	CLK2D_ DIS	Disable 2D clock. Software clock disable signal. For this core both bits CLK3D_ DIS and CLK2D__DIS should be controlled by software. The AXI interface clock is the only block not stalled at that point.
8:2	R/W	FSCAL- E_VAL	Core clock frequency scale value.
9	R/W	FSCAL- E_CMD_ LOAD	Core clock frequency scale value enable. When writing a 1 to this bit, it updates the frequency scale factor with the value FSCALE_VAL[6:0]. The bit must be set back to 0 after that. If this bit is set and FSCALE_VAL=0 (an invalid combination), the HREADYOUT output signal will get stuck to 0.
10	R/W	DISA- BLE_ RAM_ CLOCK_ GATING	Disables clock gating for rams.
13	R/W	DISA- BLE_	Disables ram power optimization.

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Bit(s)	R/W	Name	Description
		RAM_ POW- ER_OP- TIMIZA- TION	
16	R	IDLE3_D	3D pipe is idle.
19	R/W	ISO- LATE_ GPU	Isolate GPU bit. Used for power on/off, isolation only for multi-core GPUs.
23:20	R/W	MULTI_ PIPE_ REG_ SELECT	Determines which HI/MC to use while reading registers.

Table 6-14 AQHildle 0x0001

Bit(s)	R/W	Name	Description
0	R	IDLE_FE	FE is idle.
1	R	IDLE_ SH	SH is idle.
31	R	AXI_LP	AXI is in low power mode.

Table 6-15 AQAxiStatus 0x0003

Bit(s)	R/W	Name	Description
9	R	DET_ RD_ERR	Detect read error
8	R	DET_ WR_ ERR	Detect write error
7:4	R	RD_ ERR_ID	Read Error ID
3:0	R	WR_ ERR_ID	Write Error ID

Table 6-16 AQIntrAcknowledge 0x0004

Bit(s)	R/W	Name	Description
31:0	R	INTR_ VEC	For each interrupt event, 0=Clear, 1=InterruptActive Bit 31 is AXI_BUS_ERROR, 0 = No Error.

Table 6-17 AQIntrEnbl 0x0005

Bit(s)	R/W	Name	Description
31:0	R/W	INTR_ ENBL_ VEC	0: Disable interrupt; 1: Enable interrupt

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Table 6-18 GCChipRev 0x0009

Bit(s)	R/W	Name	Description
31:0	R	REV	Revision

Table 6-19 GCChipDate 0x000A

Bit(s)	R/W	Name	Description
31:0	R	DATE	Date.

Table 6-20 gcTotalCycles 0x001E

Bit(s)	R/W	Name	Description
31:0	R/W	CYCLES	

Table 6-21 gcregHIChipPatchRev 0x0026

Bit(s)	R/W	Name	Description
7:0	R	PATCH_ REV	

Table 6-22 gcProductId 0x002A

Bit(s)	R/W	Name	Description
3:0	R	GRADE_ LEVEL	0: None-no extra letter on the product name 1:N-Nano 2:L-Lite 3:UL-UltraLite
23:4	R	NUM	Product Number
27:24	R	TYPE	VIP (OpenVX/VIP only core)
31:28	R		this value is 0 and Convolution core count is provided in gcEcoId bits 31:24.

Table 6-23 gcEcoId 0x003A

Bit(s)	R/W	Name	Description
7:0	R	ID	ECO Id.
31:24	R	CONV_ COUNT	For example: 0x00: No convolution cores. 0x01: 1 CC, (= .5 NN). 0x02: 2 CC, S variant (Single = 1NN). 0x03: 3 CC, S+ variant (= 1.5 NN). 0x04: 4 CC, D variant (Dual = 2NN). 0x06: 6 CC, D+ variant (= 3 NN). 0x08: 8 CC, Q variant (Quad = 4 NN). 0x10: 16 CC, O variant (Octo = 8 NN). 0x20: 32 CC, H variant (Hex = 16 NN).

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Table 6-24 gcModulePowerControls 0x0040

Bit(s)	R/W	Name	Description
0	R/W	ENABLE_MODULE_CLOCK_GATING	Enables module level clock gating.
1	R/W	DISABLE_STALL_MODULE_CLOCK_GATING	Disables module level clock gating for stall condition.
2	R/W	DISABLE_STARVE_MODULE_CLOCK_GATING	Disables module level clock gating for starve/idle condition.
7:4	R/W	TURN_ON_COUNTER	Number of clock cycles to wait after turning on the clock.
31:16	R/W	TURN_OFF_COUNTER	Counter value for clock gating the module if the module is idle for this amount of clock cycles.

Table 6-25 gcregMMUAHBControl 0x00E2

Bit(s)	R/W	Name	Description
0	W	MMU	Enable the MMU. For security reasons, once the MMU is enabled it cannot be disabled anymore. writeOnce

Table 6-26 gcregMMUAHBTableArrayBaseAddressLow 0x00E3

Bit(s)	R/W	Name	Description
31:0	R/W	ADDRESS	32 bit Address for MMUTableArrayBaseLow 32bit.

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Table 6-27 gcregMMUAHBTableArrayBaseAddressHigh 0x00E4

Bit(s)	R/W	Name	Description
7:0	R/W	MASTER_TLB	Upper 8-bits of the master TLB address to form a true 40-bit address.
8	R/W	MASTER_TLB_SECURE	Bit that defines whether the master TLB address is secure or not
9	R/W	MASTER_TLB_SHAREABLE	Bit that defines whether the master TLB address is shareable or not.

Table 6-28 gcregMMUAHBTableArraySize 0x00E5

Bit(s)	R/W	Name	Description
15:0	R/W	SIZE	16 bit MMUTableArraySize.

Table 6-29 gcregMMUAHBSafeNonSecureAddress 0x00E6

Bit(s)	R/W	Name	Description
31:0	R/W	ADDRESS	A 64-byte address that will acts as a 'safe' zone. Any address that would cause an exception is routed to this safe zone. Reads will happen and writes will go to this address, but with a write-enable of 0. This register can only be programmed once after a reset - any attempt to write to this register after the initial write-after-reset will be ignored. This is in NonSecure memory.

Table 6-30 gcregMMUAHBSafeSecureAddress 0x00E7

Bit(s)	R/W	Name	Description
31:0	R/W	ADDRESS	Description: A 64-byte address that will act as a 'safe' zone. Any address that would cause an exception is routed to this safe zone. Reads will happen and writes will go to this address, but with a write-enable of 0. This register can only be programmed once after a reset - any attempt to write to this register after the initial write-after-reset will be ignored. This is in Secure memory.

Table 6-31 gcregCmdBufferAHBCtrl 0x00E9

Bit(s)	R/W	Name	Description
15:0	W	PREFETCH	Number of 64-bit words to fetch from the command buffer.
16	W	ENABLE	Enable the command parser. 0 => DISABLE 1 => ENABLE

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Table 6-32 gcregHiAHBControl 0x00EA

Bit(s)	R/W	Name	Description
0	R/W	SOFT_ RESET	Soft resets the IP. 1 => ENABLE 0 => DISABLE
1	R/W	DEBUG_ MODE	Enable debug mode if disabled debug registers return 0xFFFF FFFF. 1 => ENABLE 0 => DISABLE

Table 6-33 gcregAxiAHBConfig 0x00EB

Bit(s)	R/W	Name	Description
3:0	R/W	AWID	
7:4	R/W	ARID	
11:8	R/W	AWC- ACHE	Set AWCACHE[3:0] value
15:12	R/W	ARC- ACHE	Set ARCACHE[3:0] value
17:16	R/W	AXDO- MAIN_ SHARE- D	Configure AxDOMAIN walue for shareable request
19:18	R/W	AXDO- MAIN_ NON_ SHARE- D	Configure AxDOMAIN walue for non-shareable request
23:20	R/W	AXC- ACHE_ OVER- RIDE_ SHARE- D	Configure AxCACHE walue for shareable request

Table 6-34 AQMemoryDebug 0x0105

Bit(s)	R/W	Name	Description
7:0	R/W	MAX_ OUT- STAND- ING_ READS	Limits the total number of outstanding read requests.

Table 6-35 AQRegisterTimingControl 0x010B

Bit(s)	R/W	Name	Description
7:0	R/W	FOR_ RF1P	For 1 port RAM
15:8	R/W	FOR_ RF2P	For 2 port RAM
17:16	R/W	FAST_ RTC	RTC for fast RAM

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Bit(s)	R/W	Name	Description
19:18	R/W	FAST_ WTC	WTC for fast RAM
20	R/W	POW-ER_ DOWN	Power down memory
21	R/W	DEEP_ SLEEP	Deep sleep. Bit to allow SOC to manage sleep for embedded memories.
22	R/W	LIGHT_ SLEEP	Light sleep. Bit to allow SOC to manage sleep for embedded memories.

Table 6-36 AQCmdBufferAddr 0x0195

Bit(s)	R/W	Name	Description
30:0	W	AD-DRESS	Base address for the command buffer. The address must be 64-bit aligned and it is always physical. You can write all bits (the virtual bit is legacy). This register cannot be read. To check the value of the current fetch address use AQFEDe-bugCurCmdAdr. This WRITE ONLY register it has no set reset value.
31	W	TYPE	0 => SYSTEM 1 => VIRTUAL_SYSTEM

Table 6-37 AQFEDebugCurCmdAdr 0x0199

Bit(s)	R/W	Name	Description
31:3	R	CUR_ CMD_ ADR	This is the command decoder address. It has no reset value. READ ONLY

6.7 Clock

6.7.1 Overview

The clock and reset unit is an APB slave module that is designed for generating all of the internal and system clocks, resets of chip. external 24MHz crystal is used; the PLLs are shown in the following table.

MPLL (FIXPLL)/SYS_PLL/HDMI_PLL/GP0_PLL/GP1_PLL/DDR_PLL/HIFI_PLL/ETHPHY_PLL/USB-PHY_PLL/TCON_PLL/ADC_PLL

Table 6-38 PLL Description

PLL Type	Integer/Fractional	Max Frequency	Spread Spectrum
SYS_PLL	Fractional	6G	Yes
GP0_PLL	Fractional	6G	Yes
GP1_PLL		6G	Yes
HIFI_PLL	Fractional	6G	Yes
ETHPHY_PLL	Fractional	500M(fixed)	No
USBPHY_PLL	Fractional	480M(fixed)	No

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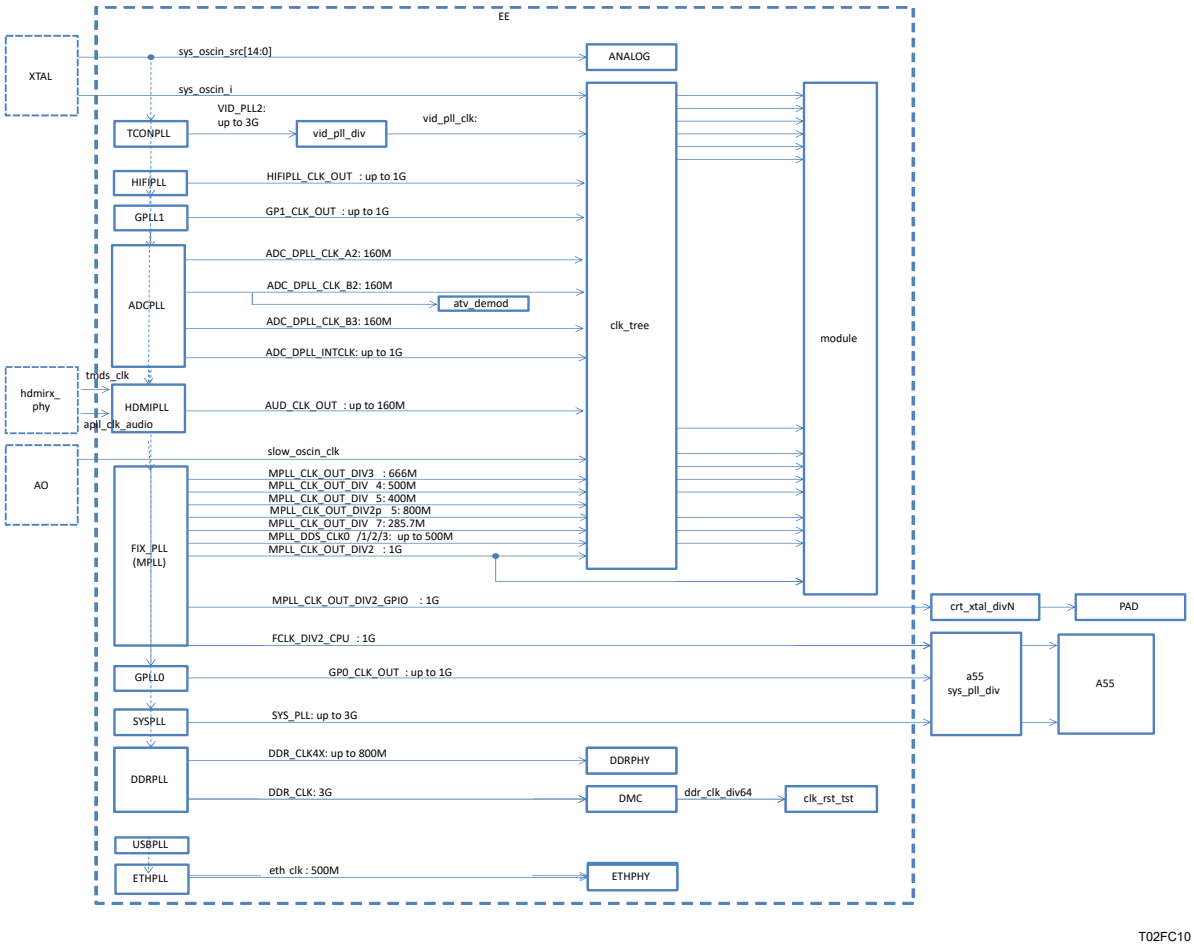
PLL Type		Integer/Fractional	Max Frequency	Spread Spectrum
MPLL	FCLK_DIV2_CPU		1G(fixed)	
	FCLK_DIV3_CPU		666M(fixed)	
	MPLL_CLK_OUT_ DIV2_GPIO		1G(fixed)	
	MPLL_CLK_OUT_DIV2		1G(fixed)	
	MPLL_CLK_OUT_ DIV2p5		800M(fixed)	
	MPLL_CLK_OUT_DIV3		666M(fixed)	
	MPLL_CLK_OUT_DIV4		500M(fixed)	
	MPLL_CLK_OUT_DIV5		400M(fixed)	
	MPLL_CLK_OUT_DIV7		285.7M(fixed)	
	MPLL_DDS_CLK0	Fractional	500M	No
	MPLL_DDS_CLK1	Fractional	500M	No
	MPLL_DDS_CLK2	Fractional	500M	Yes
	MPLL_DDS_CLK3	Fractional	500M	Yes
HDMI_ PLL	APLL_CLK_AUDIO	Fractional	3G	Yes
	APLL_CLK_OUT_DIV	Fractional	3G	Yes
TCON_ PLL	tcon_clk_ph0		6G	Yes
	tcon_clk_out2		6G	Yes
ADC_ PLL	ADC_DPLL_CLK_A2		64M	
	ADC_DPLL_CLK_B2		64M	
	ADC_DPLL_CLK_B3		64M	
	ADC_DPLL_INTCLK		1G	
DDR_ PLL	DDR_CLK4X_OUT	Fractional	4.8G	Yes
	DDR_CLK_OUT	Fractional	4.8G	

6.7.2 Clock Trees

The following figure shows the clock connections of the chip. In this part, we will discuss A55 clock tree, AO clock tree, HDMI clock tree and EE clock tree in details.

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Figure 6-5 Clock Connections



6.7.2.1 24MHz Crystal Oscillator Layout

Oscillator from XTAL pad will connect all PLLs and some analog modules:

Table 6-39 Xtal & Module Connections

Xtal Number	Modules
1	DDR_PLL
3	HDMIRX_TOP_PLL
4	PLL_TOP
5	TCON_PLL
6	USBPLL0
7	USBPLL1
8	USBPLL2
10	ETHPLL
12	ADC_DPLL
13	HDMIRX_AUDIO_PLL

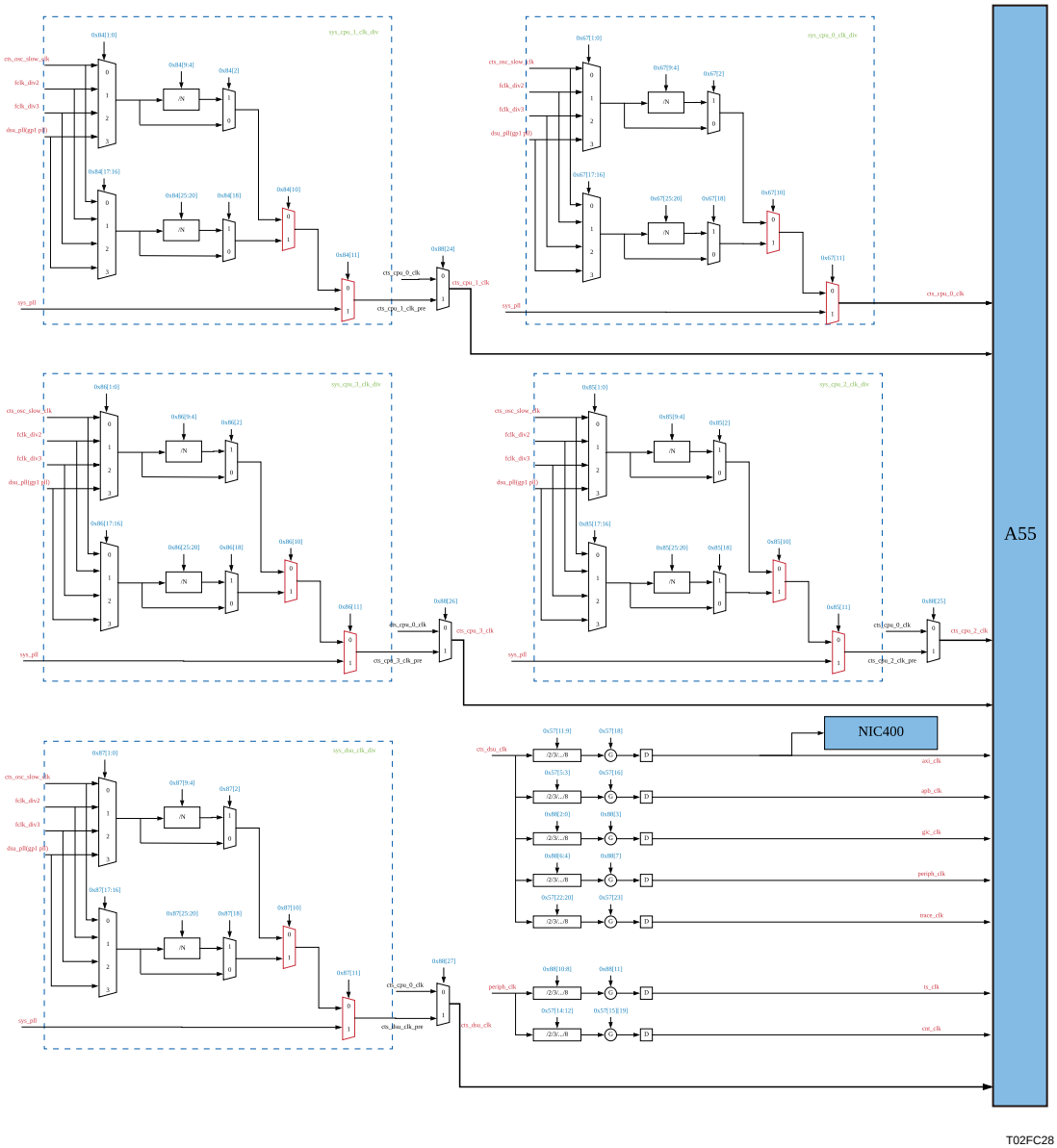
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6.7.2.2 A55 Clock Tree

A55 has 5 clock source, as shown in the following figure, among which,

1. cts_osc_slow_clk is for low power and debug clock.
2. fclk_div2 and fclk_div3 are for frequencies lower than 1G.
3. sys_pll and gp1_clk are for frequencies higher than 1G.

Figure 6-6 Mutil Phase PLLS of A55



To avoid glitch when change frequencies, there are 2 specially designed dynamic muxes, labeled by red in the above figure. When frequencies changes, the dynamic muxes will first stop the first frequency, then start the second so there will be no mixing of 2 different frequencies thus generate no frequency glitch.

It is possible to do the following switch without glitch:

- Between any 2 frequencies lower than 1GHz;
- From a frequency lower than 1GHz to a frequency higher than 1GHz

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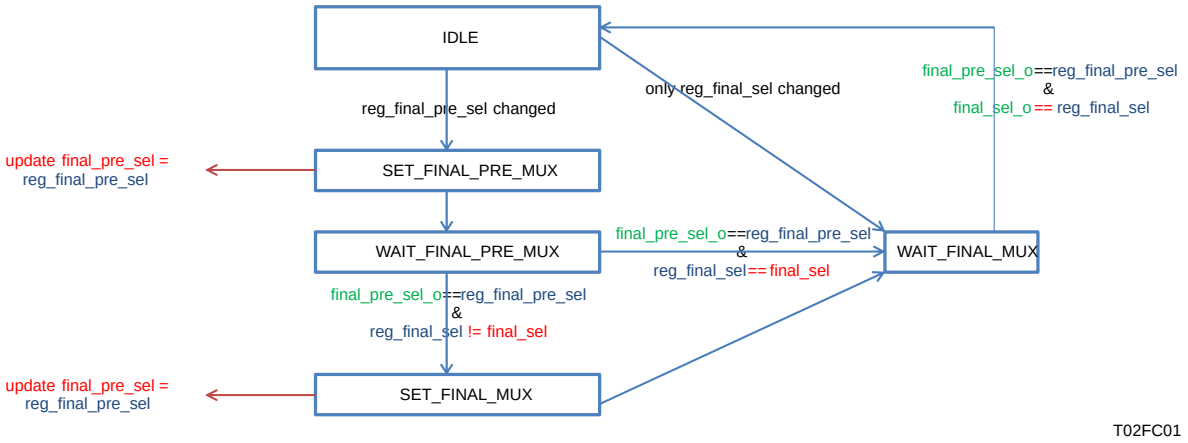
If the user want to switch between 2 frequencies both higher than 1GHz, it is strongly recommended to change to frequencies lower than 1GHz first.

The diagram of specially designed dynamic mux is shown in the following diagram.

Note

When use these 2 dynamic muxes, both control bits of these muxes have to be configured at the same time, otherwise they will not function correctly.

Figure 6-7 Dynamic Mux

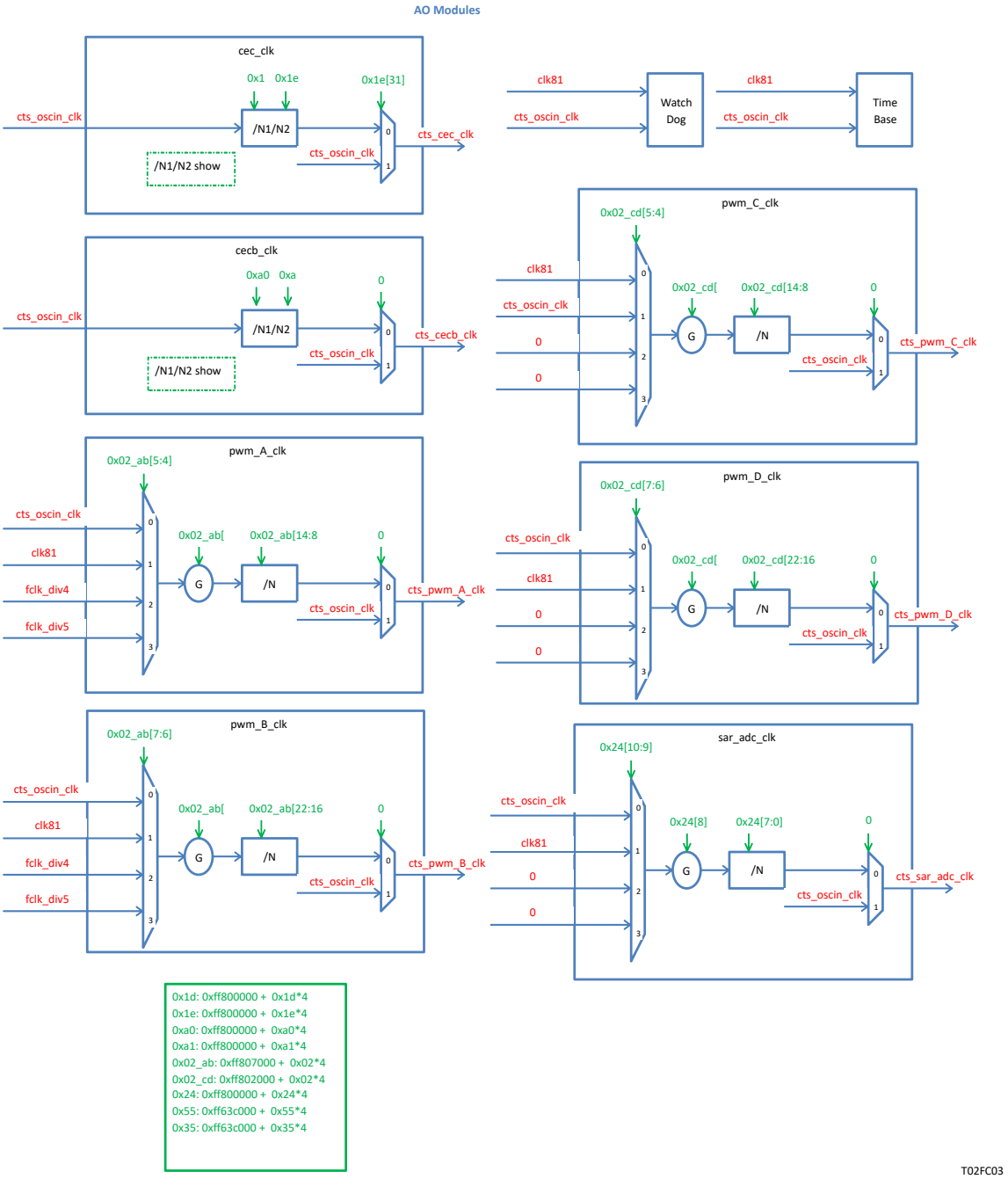


6.7.2.3 AO Clock Tree

The following figure shows the clock source of AO modules.

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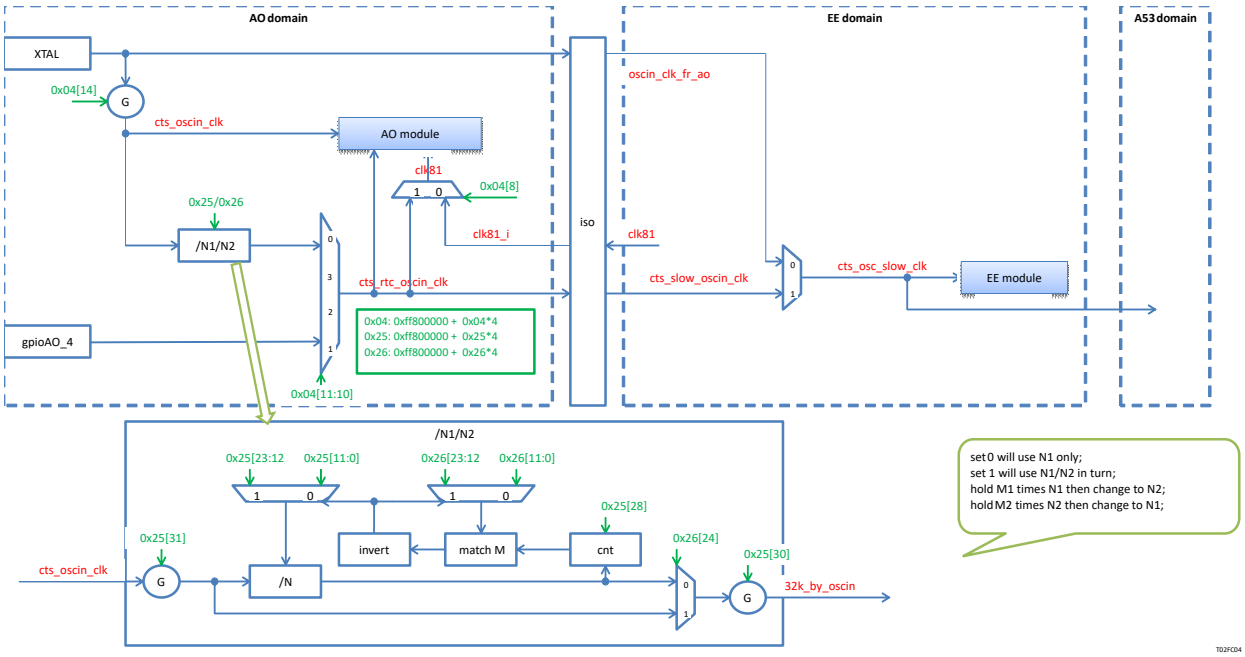
Figure 6-8 AO Clock Sources



For low power mode, 32KHz clock is needed, and it is generated in AO domain as shown in the following diagram.

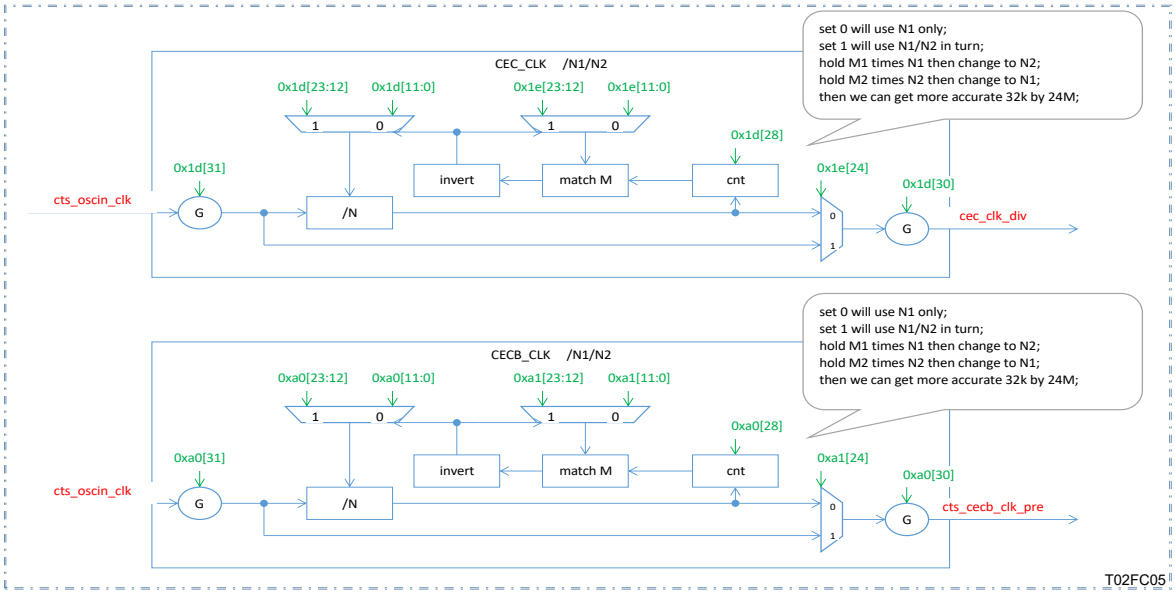
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Figure 6-9 How to generate 32KHz Clock



To generate the exact 32768Hz frequency, please check the following diagram.

Figure 6-10 How to generate 32786Hz Clock

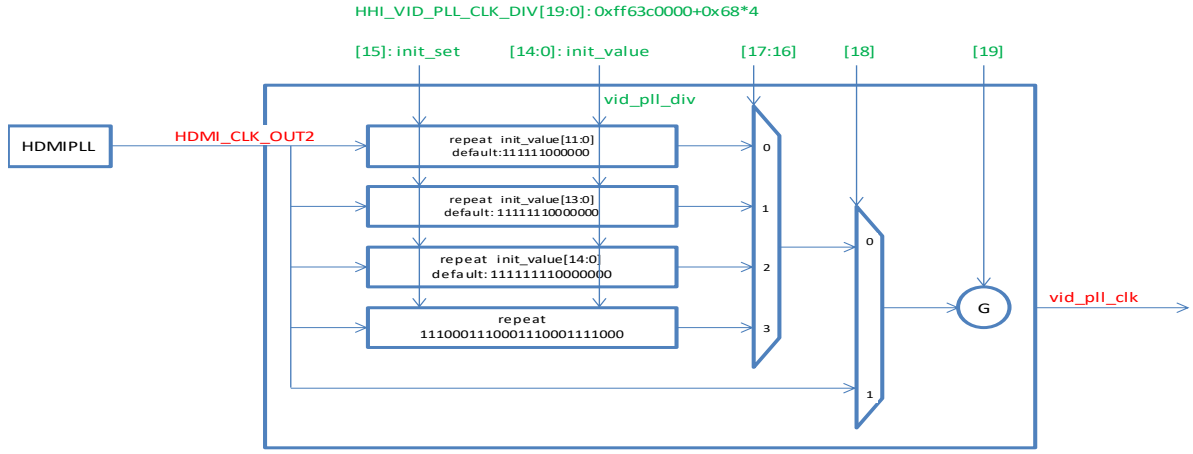


6.7.2.4 HDMI Clock Tree

The HDMI_PLL goes through vid_pll_div to generate new clock. The HDMI clock tree is shown in the following figure.

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Figure 6-11 HDMI Clock Tree

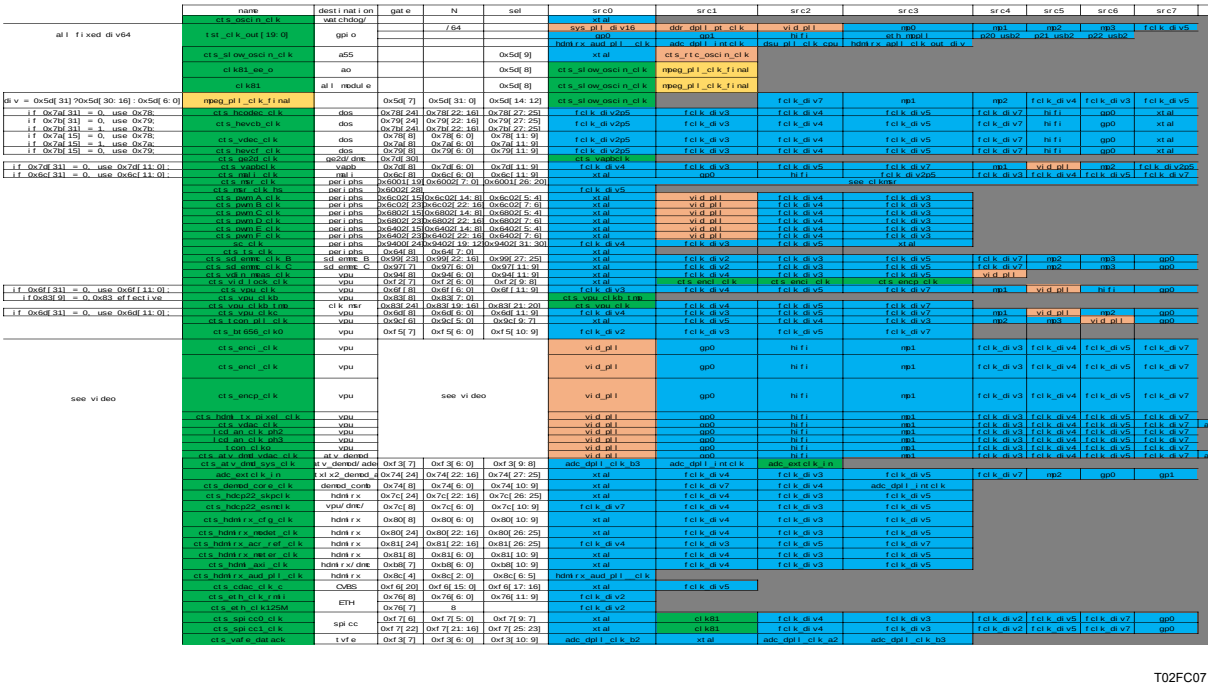


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6.7.2.5 EE Clock Tree

All EE clocks except video clock are listed in the following table.

Figure 6-12 EE Clock

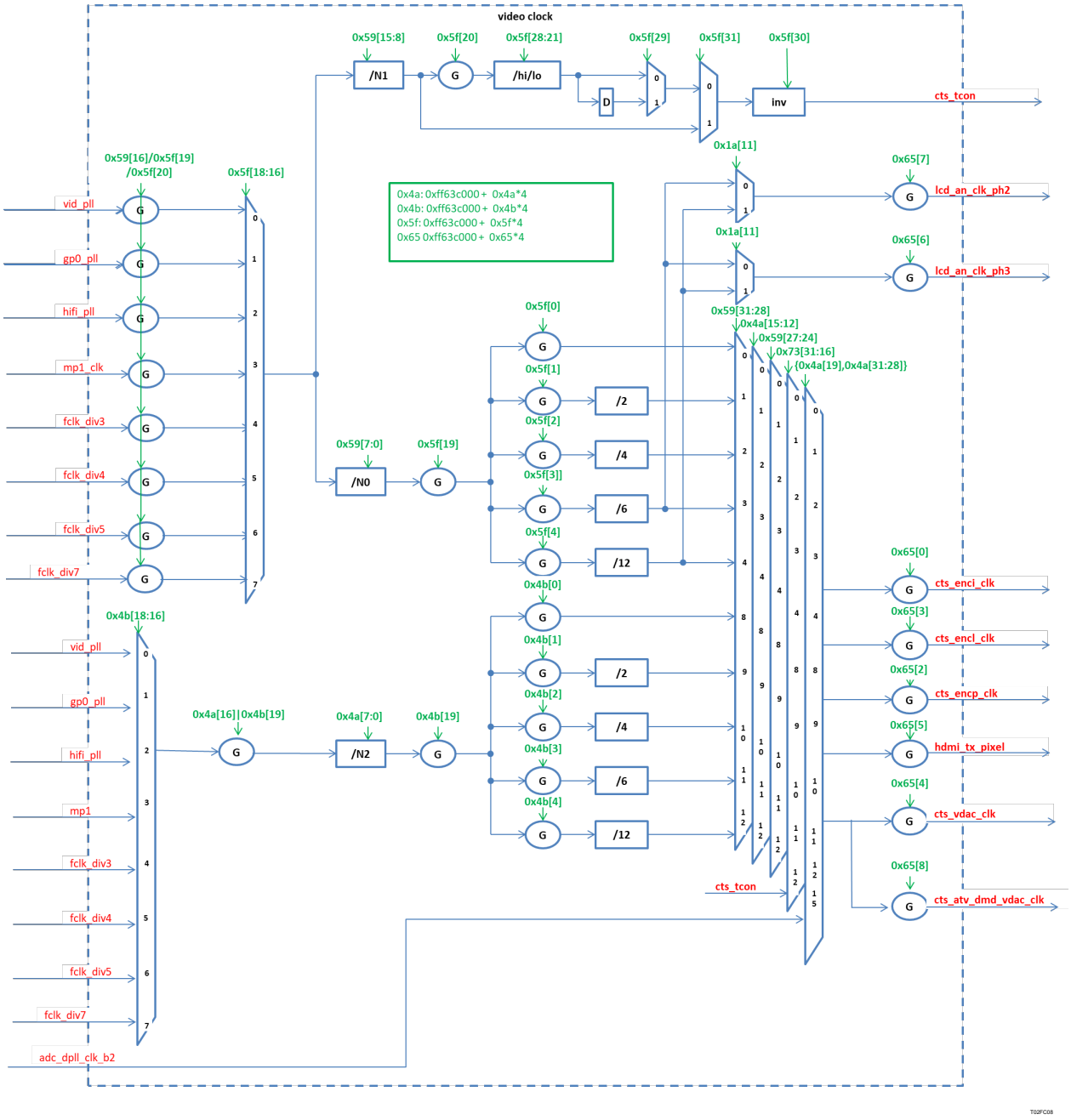


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The video clock tree is shown in the following figure.

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Figure 6-13 Video Clock Tree



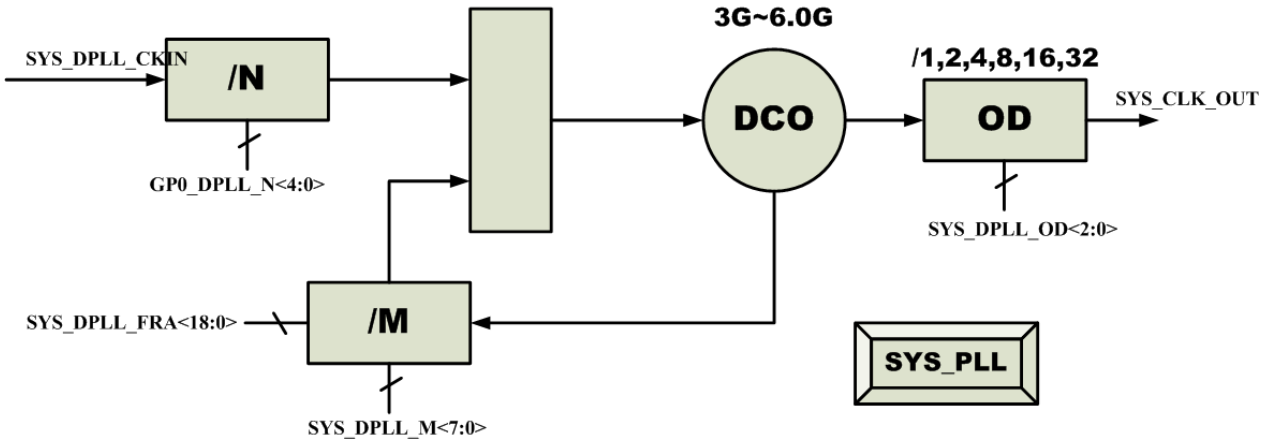
6.7.3 Frequency Calculation

6.7.3.1 SYS_PLL

SYS_PLL diagram is shown in the following figure.

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Figure 6-14 SYS_PLL



DCO frequency is calculated with the following equation:

$$f_{DCO} = f_{REF} \cdot (M + frac)/N$$

OD control table is as following.

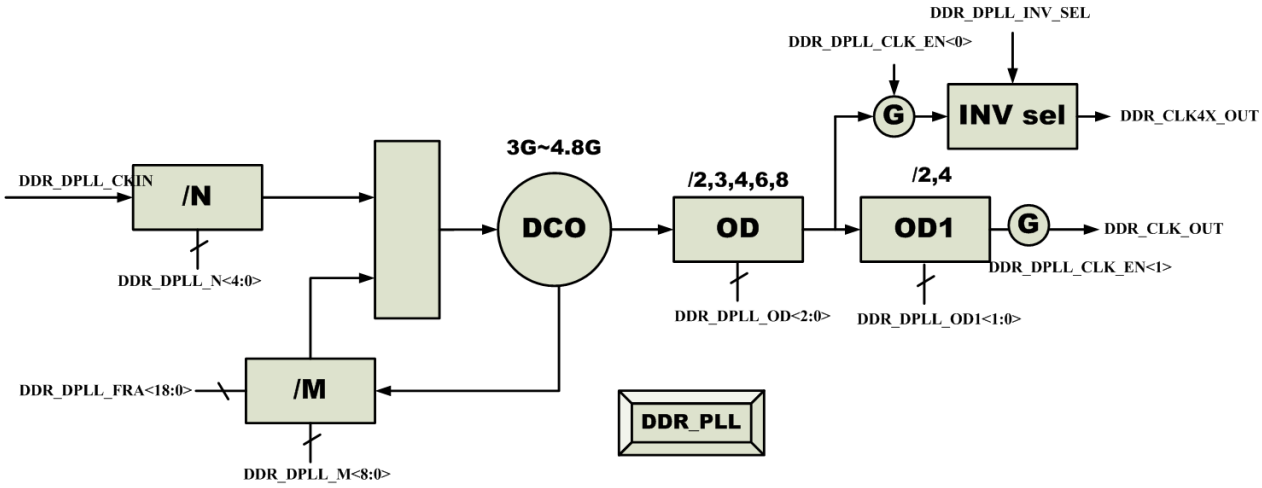
Table 6-40 SYS_PLL OD Control

Block	Register	Function
OD	SYS_DPLL_OD<2:0>	000:/1
		001:/2
		010:/4
		011:/8
		100:/16
		101:/32

6.7.3.2 DDR_PLL

DDR_PLL diagram is shown in the following figure.

Figure 6-15 DDR_PLL



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OD control table is as following.

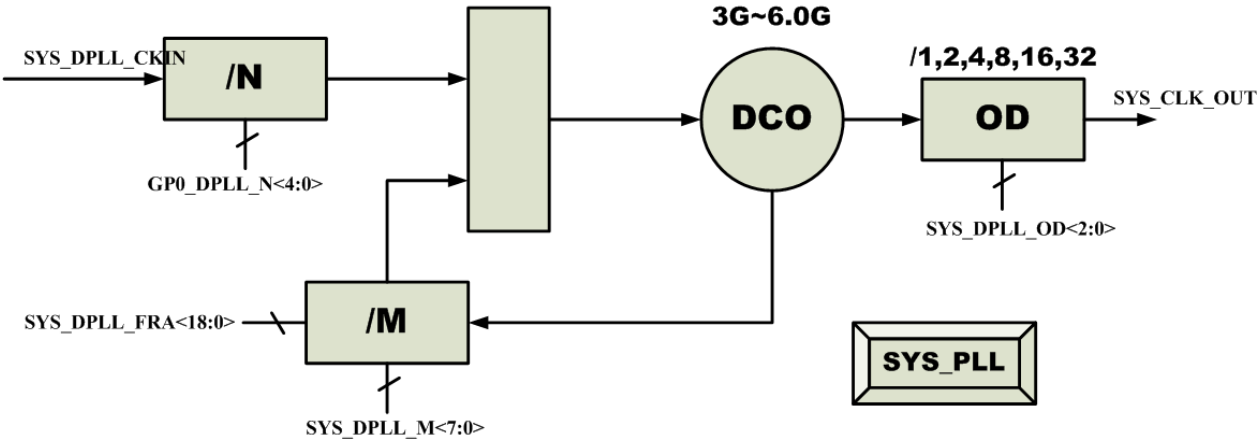
Table 6-41 DDR_PLL OD Control

Block	Register	Function
OD	DDR_DPLL_OD<2:0>	000: /2
		001:/3
		010:/4
		011:/6
		100:/8
OD1	DDR_DPLL_OD1<1:0>	*0:/2
		*1:/4

6.7.3.3 GP0_PLL

GP0_PLL diagram is shown in the following figure.

Figure 6-16 GP0_PLL



DCO frequency is calculated with the following equation:

$$f_{DCO} = f_{REF} \cdot (M + frac)/N$$

OD control table is as following.

Table 6-42 GP0_PLL OD Control

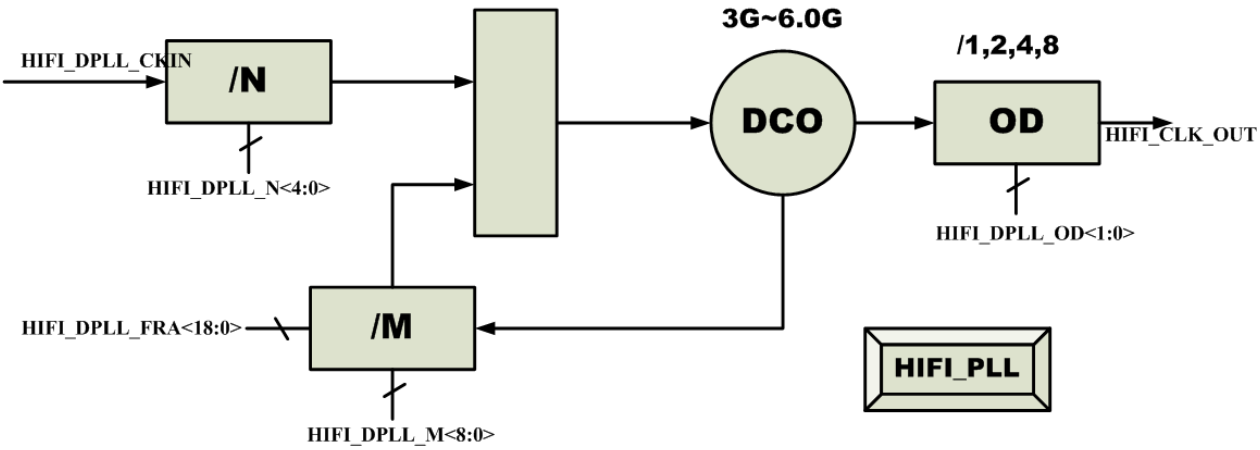
Block	Register	Function
OD	GP0_DPLL_OD<2:0>	000:/1
		001:/2
		010:/4
		011:/8
		100:/16
		101:/32

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6.7.3.4 HIFI PLL

HIFI PLL diagram is shown in the following figure.

Figure 6-17 HIFI PLL



DCO frequency is calculated with the following equation:

$$f_{DCO} = f_{REF} \cdot (M + frac)/N$$

OD control table is as following.

Table 6-43 HIFI PLL OD Control

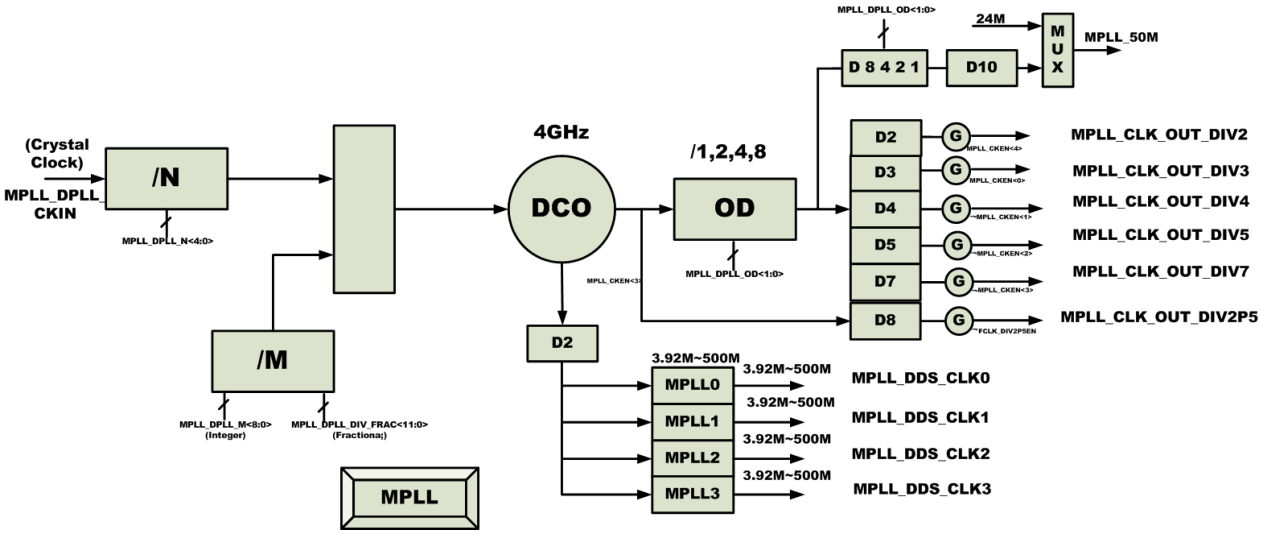
Block	Register	Function
OD	HIFI_DPLL_OD<1:0>	00:/1
		01:/2
		10:/4
		10:/8

6.7.3.5 MPLL

MPLL diagram is shown in the following figure.

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Figure 6-18 MPLL



DCO frequency is calculated with the following equation:

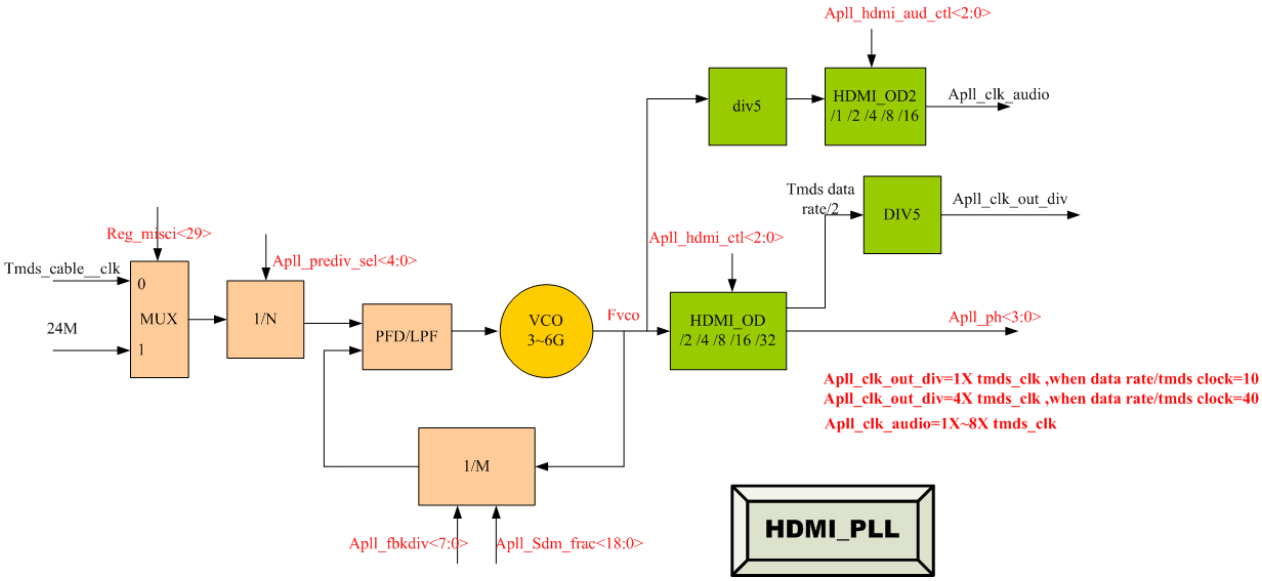
$$Fout_2G=Fref*M*OD_FB/N$$

$$MPLL_CLK_OUT2=f(N2_integer, SDM_IN) = \left(\frac{2Ghz}{(N2_integer + \frac{SDM_IN}{16384})}\right)$$

6.7.3.6 HDMI PLL

HDMI PLL diagram is shown in the following figure.

Figure 6-19 HDMI PLL



OD control table is as following.

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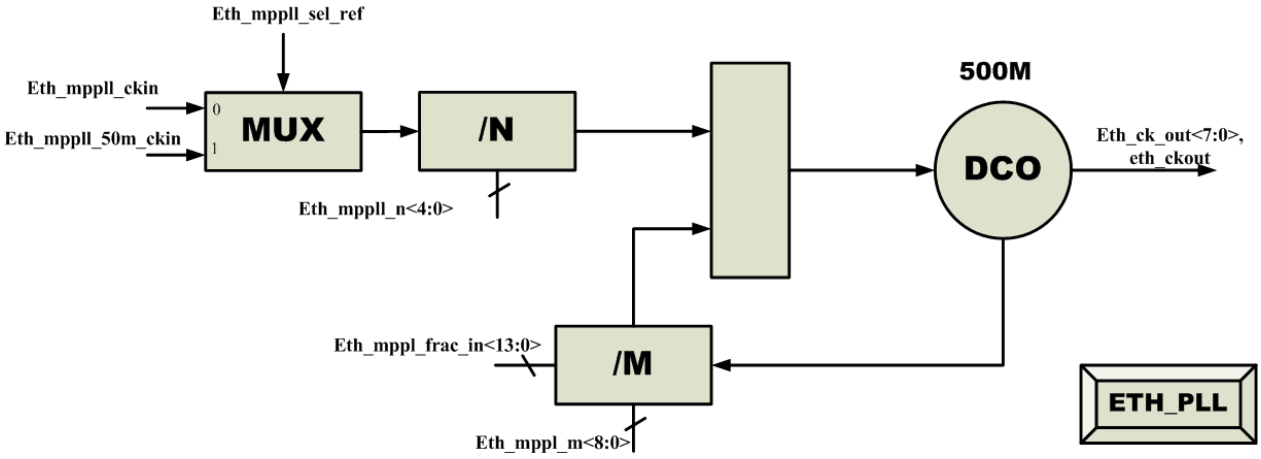
Table 6-44 HDMI PLL OD Control

Block	Register	Function
HDMI_OD	Apll_hdmi_ctrl<2:0>	001: /2
		010:/4
		011:/8
		100:/16
		101:/32
HDMI_OD2	Apll_hdmi_aud_ctrl<2:0>	000: /1
		001:/2
		010:/4
		011:/8
		100:/16

6.7.3.7 ETH PLL

ETH PLL diagram is shown in the following figure.

Figure 6-20 ETH PLL



DCO frequency is 500M.

The ETH PLL fractional value weight table is shown below.

Table 6-45 ETH PLL Fractional Value Weight Table

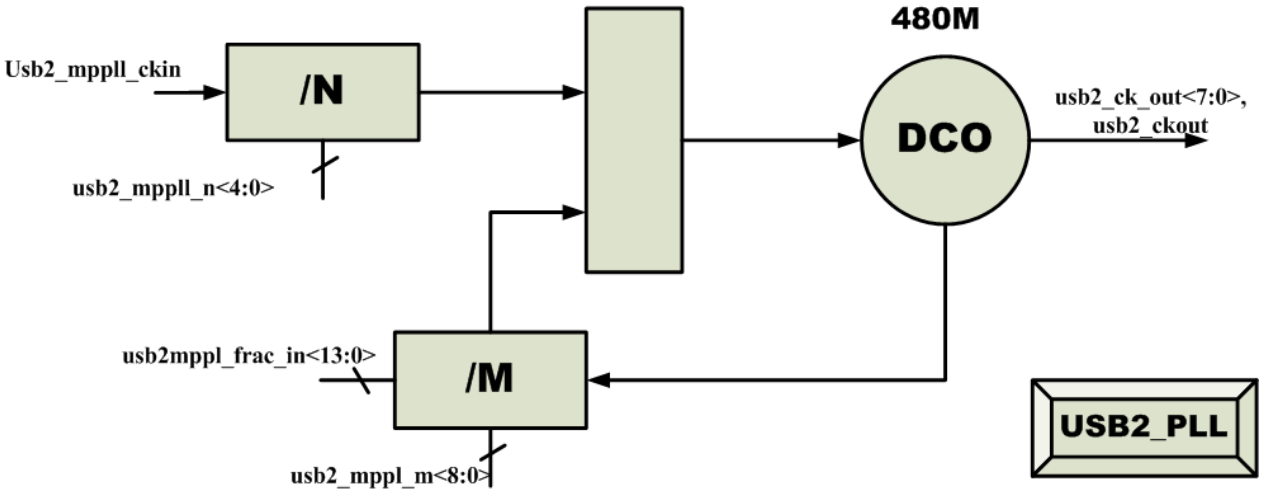
	Bit13	Bit12	Bit11	Bit10	..	..	..	..	..	..	..	Bit0
Weig-ht	1/2	1/2^2	1/2^3	1/2^4	..	..	..	..	..	..	..	1/ 2^14

6.7.3.8 USB PLL

USB PLL diagram is shown in the following figure.

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Figure 6-21 USB PLL



DCO frequency is 480M.

The USB PLL fractional value weight table is shown below.

Table 6-46 USB PLL Fractional Value Weight Table

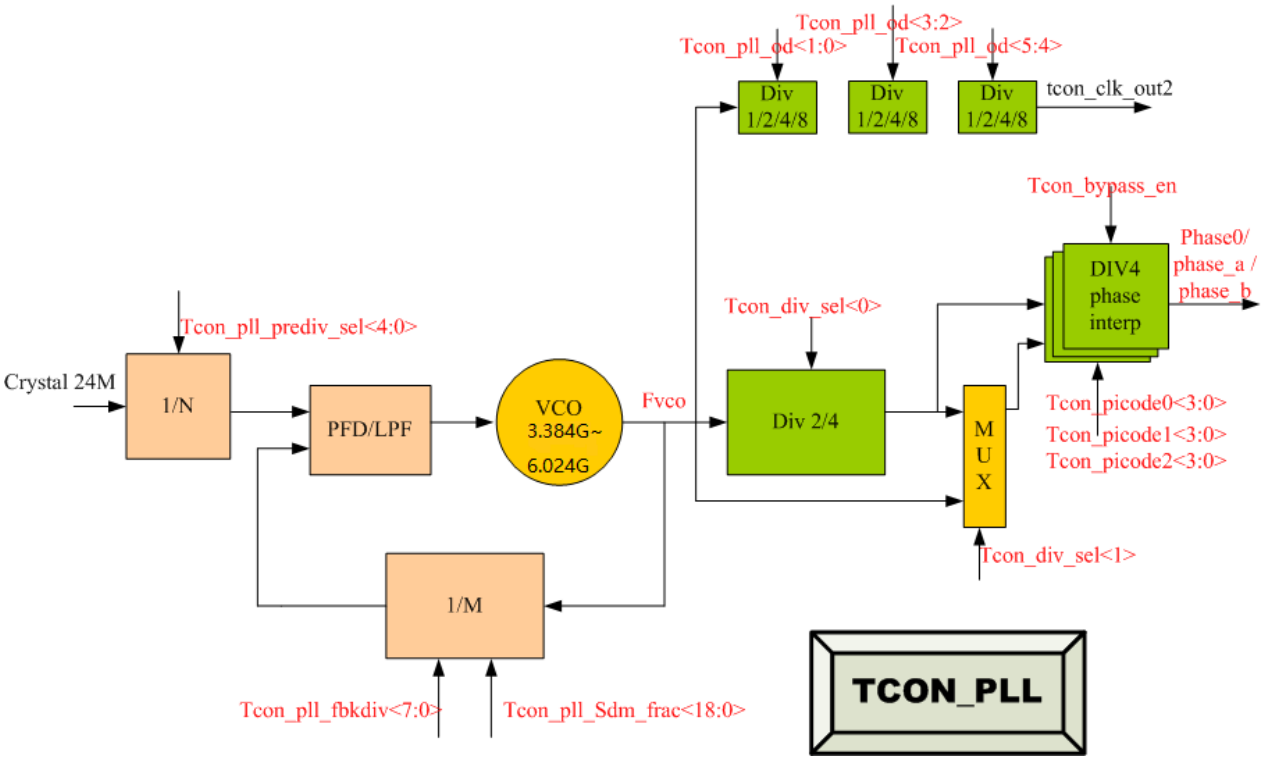
	Bit13	Bit12	Bit11	Bit10	..	..	..	..	..	..	..	Bit0
Weig-ht	1/2	1/2^2	1/2^3	1/2^4	..	..	..	..	..	..	..	1/ 2^14

6.7.3.9 TCON PLL

TCON PLL diagram is shown in the following figure.

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Figure 6-22 TCON PLL



Note

Pixel clock frequency must be less than 600MHz for Vbyone UHD.

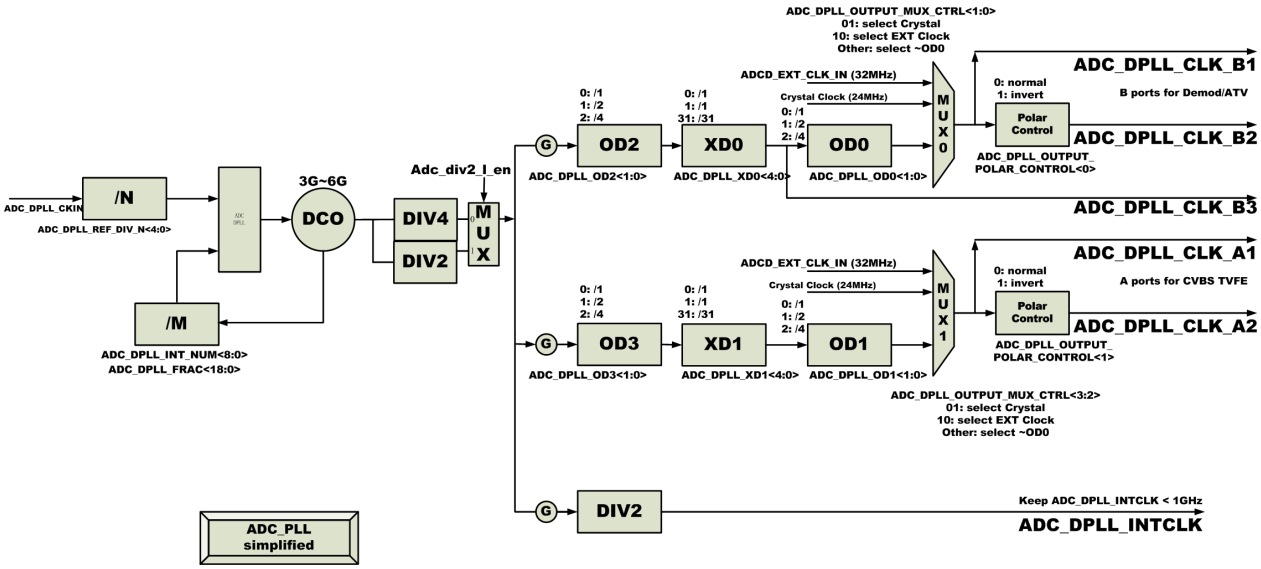
Wire	Register	Function
Tcon_pll_od<5:0>	tcon_pll_ctrl0<24:19>	00:/1
		01:/2
		10:/4
		11:/8

6.7.3.10 ADC PLL

ADC PLL diagram is shown in the following figure.

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Figure 6-23 ADC PLL



ADC PLL DCO set same ase SYS PLL, output OD same as TXLX ADC PLL.

$$f_{DCO} = f_{REF} \cdot (NUM + frac)/N$$

Take ADC_DPLL_CLK_B2=25M for example

N=1 NUM=200 Frac=0 -> Fdco=4800M

Adc_div2_l_en=0, Fdco/4=1200M

OD2=4 XD0=12 ADC_DPLL_OUTPUT_MUX_CTRL<1:0>=11, ADC_DPLL_CLK_B2=1200/4/12=25M

6.7.4 Clock Gating

Modules and sub-modules within the chip can be disabled by shutting off the clockR. The control for these clocks comes from six CBUS registers that collectively make up a 64-bit register that controls the MPEG_DOMAIN and a 32-bit register that controls the OTHER_ DOMAIN. The table below indicates the Bits associated with either the MPEG_DOMAIN and OTHER_DOMAIN gated clock enables. The table is organized by function rather than by bit order because it makes it easier to determine how to turn on/off a particular function within the chip. If a bit is set high, the clock is enabled. If a bit is set low, the clock is turned off and the module is disabled.

Table 6-47 AO Domain Clock Gating

Address	Bit(s)	Module Description
0xff80004C	10	dspb
	9	dspa
	8	saradc
	7	ir_out
	6	uart2
	5	prod_i2c
	4	uart2
	3	i2c_s

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Address	Bit(s)	Module Description
	2	i2c_m
	1	ir_in
	0	ahb
0xff800050	3	rti
	2	ahb_sram
	1	n205
	0	mailbox

Table 6-48 EE Domain Clock Gating (clk81)

Address	Bit(s)	Module Description
0xff63c140 0xff63c0c0	31	dsp
	30	spi
	29	tcon
	28	acodec
	27	adec
	26	emmc_c
	25	emmc_b
	24	-
	23	ASSIST_MISC
	22	atv_demod_top
	21	hdmirx_top
	20	
	19	HIU Registers
	18	tvfe
	17	-
	16	AML async fifo
	15	stream
	14	spicc_1
	13	uart0
	12	-
	11	smartcard
	10	sana
	9	i2c
	8	spicc_0
	7	PERIPHS module top level (there are separate register bits for internal blocks)
	6	PL310 (AXI Matrix) to CBUS
	5	ISA module

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Address	Bit(s)	Module Description
	4	eth_phy
	3-2	-
	1	u_dos_top()
	0	dmc/am2axi_arb
0xff63c144 0xff63c0c4	31-30	
	29	AHB ARB0
	28-27	-
	26	USB General
	25	U_parser_top()
	24	-
	23	RESET
	22-21	-
	20	General 2D Graphics Engine
	19-17	-
	16	uart1
	15-14	
	13	ADC
	12	
	11	aififo
	10-5	
	4	Set top box demux module u_stb_top.clk
	3	Ethernet core logic
	2-1	
	0	audio
0xff63c148 0xff63c0c8	31	-
	30	gic
	29	-
	28	demod_comb
	27	no use
	26	-
	25	VPU Interrupt
	24-23	-
	22	ts
	21-16	-
	15	uart2
	14	-
	13	hdmirx_top
	12	-

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Address	Bit(s)	Module Description
	11	MMC PCLK
	10-9	-
	8	USB1 to DDR bridge
	7	-
	6	bt656
	5-3	-
	2	AHB control bus
	1	AHB data bus
	0	-
0xff63c150 0xff63c0cc	31-27	-
	26	VCLK2_OTHER
	25	VCLK2_VENCL
	24	VCLK2_VENCL MMC Clock All
	23	VCLK2_ENCL
	22	VCLK2_ENCT
	21	Random Number Generator
	20	ENC480P
	19-11	-
	10	DAC_CLK
	9	VCLK2_ENCP
	8	VCLK2_ENCI
	7	VCLK2_OTHER
	6	VCLK2_VENCT
	5	VCLK2_VENCT
	4	VCLK2_VENCP
	3	VCLK2_VENCP
	2	VCLK2_VENCI
	1	VCLK2_VENCI
	0	-

6.7.5 Clock Measure

The chip contains a module that can measure the frequency of internal clock. The frequency measurement is a simple counter measurement in which the counter (driven by MCLK) is enabled for a programmable amount of time. The sources are listed in the following table.

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Table 6-49 Clock Measure Source

No.	Source
255~159	1'b0
158	1'b0
157	1'b0
156	1'b0
155	1'b0
154	1'b0
153	1'b0
152	1'b0
151	1'b0
150	1'b0
149	1'b0
148	1'b0
147	1'b0
146	1'b0
145	1'b0
144	ts_pll_clk
143	mainclk
142	demod_ts_clk
141	ts_ddr_clk
140	audio_toacodec_bclk
139	AU_ADC_CLK_G128X
138	dsu_pll_clk_cpu
137	atv_dmd_i2s_sclk
136	sys_pll_clk
135	tvfe_sample_clk
134	adc_extclk_in
133	atv_dmd_mono_clk_32
132	audio_toacodec_mclk
131	ts_sar_clk
130	AU_DAC2_CLK_G128X
129	lvds_fifo_clk
128	cts_tcon_pll_clk
127	hdmirx_vid_clk
126	sar_ring_osc_clk
125	cts_hdmi_axi_clk
124	cts_demod_core_clk

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No.	Source
123	mod_audio_pdm_dclk_o
122	audio_spdifin_mst_clk
121	audio_spdifout_mst_clk
120	audio_spdifout_b_mst_clk
119	audio_pdm_sysclk
118	audio_resamplea_clk
117	audio_resampleb_clk
116	audio_tdmin_a_sclk
115	audio_tdmin_b_sclk
114	audio_tdmin_c_sclk
113	audio_tdmin_lb_sclk
112	audio_tdmout_a_sclk
111	audio_tdmout_b_sclk
110	audio_tdmout_c_sclk
109	o_vad_clk
108	acodec_i2sout_bclk
107	AU_DAC1_CLK_G128X
106	ephy_test_clk
105	am_ring_osc_clk_out_ee[9]
104	am_ring_osc_clk_out_ee[8]
103	am_ring_osc_clk_out_ee[7]
102	am_ring_osc_clk_out_ee[6]
101	am_ring_osc_clk_out_ee[5]
100	am_ring_osc_clk_out_ee[4]
99	am_ring_osc_clk_out_ee[3]
98	cts_ts_clk
97	cts_vpu_clkb_tmp
96	cts_vpu_clkb
95	eth_phy_plltxclk
94	eth_phy_rxclk
93	sys_cpu_ring_osc_clk[3]
92	sys_cpu_ring_osc_clk[2]
91	hdmirx_audmeas_clk
90	am_ring_osc_clk_out_ee[11]
89	am_ring_osc_clk_out_ee[10]

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No.	Source
88	cts_hdmirx_meter_clk
87	1'b0
86	cts_hdmirx_modet_clk
85	cts_hdmirx_acr_ref_clk
84	co_tx_clk
83	co_rx_clk
82	cts_ge2d_clk
81	cts_vapbclk
80	rng_ring_osc_clk[3]
79	rng_ring_osc_clk[2]
78	rng_ring_osc_clk[1]
77	rng_ring_osc_clk[0]
76	hdmirx_aud_clk
75	cts_hevcf_clk
74	hdmirx_aud_pll_clk
73	cts_pwm_C_clk
72	cts_pwm_D_clk
71	cts_pwm_E_clk
70	cts_pwm_F_clk
69	cts_hdcp22_skpclk
68	cts_hdcp22_esmclk
67	hdmirx_apll_clk_audio
66	cts_vid_lock_clk
65	cts_spicc_0_clk
64	cts_spicc_1_clk
63	hdmirx_tmds_clk
62	cts_hevcb_clk
61	gpio_clk_msr
60	cts_hdmirx_aud_pll_clk
59	cts_hcodec_clk
58	cts_vafe_datack
57	cts_atv_dmd_vdac_clk
56	cts_atv_dmd_sys_clk
55	vid_pll_div_clk_out
54	cts_vpu_clkc

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No.	Source
53	ddr_4xclk
52	cts_sd_emmc_clk_B
51	cts_sd_emmc_clk_C
50	mp3_clk_out
49	mp2_clk_out
48	mp1_clk_out
47	ddr_dpll_pt_clk
46	cts_vpu_clk
45	cts_pwm_A_clk
44	cts_pwm_B_clk
43	fclk_div5
42	mp0_clk_out
41	mac_eth_rx_clk_rmii
40	cts_hdmirx_cfg_clk
39	cts_bt656_clk0
38	cts_vdin_meas_clk
37	cts_cdac_clk_c
36	cts_hdmi_tx_pixel_clk
35	cts_mali_clk
34	eth_mppll_50m_ckout
33	sys_cpu_ring_osc_clk[1]
32	cts_vdec_clk
31	mpll_clk_test_out
30	hdmirx_cable_clk
29	hdmirx_apll_clk_out_div
28	cts_sar_adc_clk
27	co_clkin_to_mac
26	sc_clk_int
25	cts_eth_clk_rmii
24	cts_eth_clk125Mhz
23	mpll_clk_50m
22	mac_eth_phy_ref_clk
21	lcd_an_clk_ph3
20	rtc_osc_clk_out
19	lcd_an_clk_ph2

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No.	Source
18	sys_cpu_clk_div16
17	sys_pll_div16
16	cts_FEC_CLK_2
15	cts_FEC_CLK_1
14	cts_FEC_CLK_0
13	mod_tcon_clko
12	hifi_pll_clk
11	mac_eth_tx_clk
10	cts_vdac_clk
9	cts_encl_clk
8	cts_encp_clk
7	clk81
6	cts_enci_clk
5	gp1_pll_clk
4	gp0_pll_clk
3	sys_cpu_ring_osc_clk[0]
2	am_ring_osc_clk_out_ee[2]
1	am_ring_osc_clk_out_ee[1]
0	am_ring_osc_clk_out_ee[0]

6.7.6 Register Description

HIU(base:32'hFF63C000)

Each register final address = BASE + address * 4

Table 6-50 HHI_MIPI_CNTL0 [31:0] 0x00

Bit(s)	R/W	Default	Description
31:16	R/W	0	REGISTER CONTROL B<15:10>:common ref gen control B9:Vref select 0=VR 1=Vbg B8:Iref select 0=I_rout 1=Ibg B7:enable Ibg B6/5:unused B<4:0>:VR trimming control 10mV/step Work mode:1010,0100,1000,0111
15:0	R/W	0	REGISTER CONTROL B3:VR gen from Ibg enable, B2:VR gen by avdd18 enable Other bits unused set 0 Work mode:0000,0000,0000,1000

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Table 6-51 HHI_MIPI_CNTL1 [31:0] 0x01

Bit(s)	R/W	Default	Description
19	R/W	0	dsi_edp_aux_tx_en AUX TX ENABLE(useless)
18	R/W	0	dsi_prbs_clk_in CLOCK FROM CORE FOR TEST.(max frequency 1GHZ)
17	R/W	0	dsi_edp_aux_tx AUX TX INPUT(useless)
16	R/W	0	dsi_vbg_en Bandgap Enable signal
15:0	R/W	0	REGISTER CONTROL B<15:12>:unused B<11:8>:test output select b7:RX output inv select B6:unused B<5:4>:LPTX slew-rate select B3:LRTX leak-current enable B2:input inv select B1:input select 0=prbs 1=normal input B0:prbs7 enable Work mode:0000,0000,0010,1110

Table 6-52 HHI_MIPI_CNTL2 [31:0] 0x02

Bit(s)	R/W	Default	Description
31:16	R/W	0	REGISTER CONTROL B<15:9>:CH cell power up B<8:3>:CH control B2/1:unused B0:LPULPS enable Work mode:0010,0110,1000,0000
15:0	R/W	0	REGISTER CONTROL B<15:11>: CH<0:4> power up B10:RX Ibias enable B9:unused B<8:0>:LPRX reference voltage control Work mode:1111,1100,0101,1010

Table 6-53 HHI_MIPI_STS [31:0] 0x03

Bit(s)	R/W	Default	Description
8	R	0	AUX RX OUTPUT(useless)
7:0	R	0	LPRX and LPCD receiver primitive state data <LPCD_OUTN_L, LPCD_OUTN_L, LPCD_OUTP_H, LPCD_OUTP_L, LP_OUTN_H, LP_OUTN_L, LP_OUTP_H, LP_OUTP_L>

Table 6-54 HHI_CHECK_CLK_RESULT 0x04

Bit(s)	R/W	Default	Description
31	R	0	cpu_0_clk_result
30	R	0	cpu_1_clk_result
29	R	0	cpu_2_clk_result
28	R	0	cpu_3_clk_result
27	R	0	cpu_dsu_clk_result
2	R	0	encp_check_result

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Bit(s)	R/W	Default	Description
1	R	0	hevcb_clk_check_result
0	R	0	mali_clk_check_result

Table 6-55 SCR System Clock Reference 0x0B

Bit(s)	R/W	Default	Description
31-0	R/W	0	System clock reference high: bits 31:16

Table 6-56 TIMEOUT_VALUE: Program timer 0x0F

Bit(s)	R/W	Default	Description
15-12	R	0	Unused
11-0	R/W	0	Program timer

Note

↑Increased by 1 every 900 cycles. Triggers timer interrupt to CPU when it expires.

Table 6-57 HHI_GP0_PLL_CNTL0 0x10

Bit(s)	R/W	Default	Description
[31]	R	0	gp0_dpll_lock
[30]	R	0	gp0_dpll_lock_a
[29]	R/W	1	gp0_dpll_reset
[28]	R/W	0	gp0_dpll_en
[18:16]	R/W	0	gp0_dpll_od
[14:10]	R/W	0	gp0_dpll_N
[8:0]	R/W	0	gp0_dpll_M

Table 6-58 HHI_GP0_PLL_CNTL1 0x11

Bit(s)	R/W	Default	Description
[18:0]	R/W	0	gp0_dpll_frac

Table 6-59 HHI_GP0_PLL_CNTL2 0x12

Bit(s)	R/W	Default	Description
[22:20]	R/W	0	gp0_dpll_fref_sel
[17:16]	R/W	0	gp0_dpll_os_ssc
[15:12]	R/W	0	gp0_dpll_ssc_str_m
[8]	R/W	0	gp0_dpll_ssc_en

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Bit(s)	R/W	Default	Description
[7:4]	R/W	0	gp0_dpll_ssc_dep_sel
[1:0]	R/W	0	gp0_dpll_ss_mode

Table 6-60 HHI_GP0_PLL_CNTL3 0x13

Bit(s)	R/W	Default	Description
31	R/W	0	gp0_dpll_afc_bypass
30	R/W	0	gp0_dpll_afc_clk_sel
29	R/W	0	gp0_dpll_code_new
28	R/W	0	gp0_dpll_dco_m_en
27	R/W	0	gp0_dpll_dco_sdm_en
26	R/W	0	
25	R/W	0	gp0_dpll_div_mode
24	R/W	0	gp0_dpll_fast_lock
23	R/W	0	gp0_dpll_fb_pre_div
22	R/W	0	gp0_dpll_filter_mode
21	R/W	0	gp0_dpll_fix_en
20	R/W	0	gp0_dpll_freq_shift_en
19	R/W	0	gp0_dpll_load
18	R/W	0	gp0_dpll_load_en
17	R/W	0	gp0_dpll_lock_f
16	R/W	0	gp0_dpll_pulse_width_en
15	R/W	0	gp0_dpll_sdmnc_en
14	R/W	0	gp0_dpll_sdmnc_mode
13	R/W	0	gp0_dpll_sdmnc_range
12	R/W	0	gp0_dpll_tdc_en
11	R/W	0	gp0_dpll_tdc_mode_sel
10	R/W	0	gp0_dpll_wait_en

Table 6-61 HHI_GP0_PLL_CNTL4 0x14

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	gp0_dpll_alpha
[27:24]	R/W	0	gp0_dpll_rou
[22:20]	R/W	0	
[18:16]	R/W	0	gp0_dpll_lambda0
[13:12]	R/W	0	gp0_dpll_acq_gain

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Bit(s)	R/W	Default	Description
[11:8]	R/W	0	gp0_dpll_filter_pvt2
[7:4]	R/W	0	gp0_dpll_filter_pvt1
[1:0]	R/W	0	gp0_dpll_pfd_gain

Table 6-62 HHI_GP0_PLL_CNTL5 0x15

Bit(s)	R/W	Default	Description
[30:28]	R/W	0	gp0_dpll_adj_vco_ldo
[27:24]	R/W	0	gp0_dpll_lm_w
[21:16]	R/W	0	gp0_dpll_lm_s
[15:0]	R/W	0	gp0_dpll_reve

Table 6-63 HHI_GP0_PLL_CNTL6 0x16

Bit(s)	R/W	Default	Description
[31:30]	R/W	0	gp0_dpll_afc_hold_t
[29:28]	R/W	0	gp0_dpll_lkw_sel
[27:26]	R/W	0	gp0_dpll_dco_sdm_clk_sel
[25:24]	R/W	0	gp0_dpll_afc_in
[23:22]	R/W	0	gp0_dpll_afc_nt
[21:20]	R/W	0	gp0_dpll_vc_in
[19:18]	R/W	0	gp0_dpll_lock_long
[17:16]	R/W	0	gp0_dpll_freq_shift_v
[14:12]	R/W	0	gp0_dpll_data_sel
[10:8]	R/W	0	gp0_dpll_sdmnc_ulms
[6:0]	R/W	0	gp0_dpll_sdmnc_power

Table 6-64 HHI_GP0_PLL_STS 0x17

Bit(s)	R/W	Default	Description
[31]	R	0	gp0_dpll_lock
[30]	R	0	gp0_dpll_lock_a
[29]	R	0	gp0_dpll_afc_done
[22:16]	R	0	gp0_dpll_sdmnc_monitor
[9:0]	R	0	gp0_dpll_out_rsv

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Table 6-65 HHI_GP1_PLL_CNTL0 0x18

Bit(s)	R/W	Default	Description
[31]	R	0	gp1_dpll_lock
[30]	R	0	gp1_dpll_lock_a
[29]	R/W	1	gp1_dpll_reset
[28]	R/W	0	gp1_dpll_en
[18:16]	R/W	0	gp1_dpll_od
[14:10]	R/W	0	gp1_dpll_N
[8:0]	R/W	0	gp1_dpll_M

Table 6-66 HHI_GP1_PLL_CNTL1 0x19

Bit(s)	R/W	Default	Description
[18:0]	R/W	0	gp1_dpll_frac

Table 6-67 HHI_GP1_PLL_CNTL2 0x1a

Bit(s)	R/W	Default	Description
[22:20]	R/W	0	gp1_dpll_fref_sel
[17:16]	R/W	0	gp1_dpll_os_ssc
[15:12]	R/W	0	gp1_dpll_ssc_str_m
[8]	R/W	0	gp1_dpll_ssc_en
[7:4]	R/W	0	gp1_dpll_ssc_dep_sel
[1:0]	R/W	0	gp1_dpll_ss_mode

Table 6-68 HHI_GP1_PLL_CNTL3 0x1b

Bit(s)	R/W	Default	Description
31	R/W	0	gp1_dpll_afc_bypass
30	R/W	0	gp1_dpll_afc_clk_sel
29	R/W	0	gp1_dpll_code_new
28	R/W	0	gp1_dpll_dco_m_en
27	R/W	0	gp1_dpll_dco_sdm_en
25	R/W	0	gp1_dpll_div_mode
24	R/W	0	gp1_dpll_fast_lock
23	R/W	0	gp1_dpll_fb_pre_div
22	R/W	0	gp1_dpll_filter_mode
21	R/W	0	gp1_dpll_fix_en
20	R/W	0	gp1_dpll_freq_shift_en

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Bit(s)	R/W	Default	Description
19	R/W	0	gp1_dpll_load
18	R/W	0	gp1_dpll_load_en
17	R/W	0	gp1_dpll_lock_f
16	R/W	0	gp1_dpll_pulse_width_en
15	R/W	0	gp1_dpll_sdmnc_en
14	R/W	0	gp1_dpll_sdmnc_mode
13	R/W	0	gp1_dpll_sdmnc_range
12	R/W	0	gp1_dpll_tdc_en
11	R/W	0	gp1_dpll_tdc_mode_sel
10	R/W	0	gp1_dpll_wait_en

Table 6-69 HHI_GP1_PLL_CNTL4 0x1c

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	gp1_dpll_alpha
[27:24]	R/W	0	gp1_dpll_rou
[18:16]	R/W	0	gp1_dpll_lambda0
[13:12]	R/W	0	gp1_dpll_acq_gain
[11:8]	R/W	0	gp1_dpll_filter_pvt2
[7:4]	R/W	0	gp1_dpll_filter_pvt1
[1:0]	R/W	0	gp1_dpll_pfd_gain

Table 6-70 HHI_GP1_PLL_CNTL5 0x1d

Bit(s)	R/W	Default	Description
[30:28]	R/W	0	gp1_dpll_adj_vco_ldo
[27:24]	R/W	0	gp1_dpll_lm_w
[21:16]	R/W	0	gp1_dpll_lm_s
[15:0]	R/W	0	gp1_dpll_reve

Table 6-71 HHI_GP1_PLL_CNTL6 0x1e

Bit(s)	R/W	Default	Description
[31:30]	R/W	0	gp1_dpll_afc_hold_t
[29:28]	R/W	0	gp1_dpll_lkw_sel
[27:26]	R/W	0	gp1_dpll_dco_sdm_clk_sel
[25:24]	R/W	0	gp1_dpll_afc_in
[23:22]	R/W	0	gp1_dpll_afc_nt

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Bit(s)	R/W	Default	Description
[21:20]	R/W	0	gp1_dpll_vc_in
[19:18]	R/W	0	gp1_dpll_lock_long
[17:16]	R/W	0	gp1_dpll_freq_shift_v
[14:12]	R/W	0	gp1_dpll_data_sel
[10:8]	R/W	0	gp1_dpll_sdmnc_ulms
[6:0]	R/W	0	gp1_dpll_sdmnc_power

Table 6-72 HHI_GP1_PLL_STS 0x1f

Bit(s)	R/W	Default	Description
[31]	R	0	gp1_dpll_lock
[30]	R	0	gp1_dpll_lock_a
[29]	R	0	gp1_dpll_afc_done
[22:16]	R	0	gp1_dpll_sdmnc_monitor
[9:0]	R	0	gp1_dpll_out_rsv

Table 6-73 HHI_TCON_PLL_CNTL0 0x20

Bit(s)	R/W	Default	Description
[31]	R	0	tcon_pll_lock
[30]	R	0	tcon_pll_lock_a
[29]	R/W	1	tcon_pll_reset
[28]	R/W	0	tcon_pll_en
[27]	R/W	0	tcon_pll_vco_div_sel
[26]	R/W	0	tcon_pll_afc_start
[24:19]	R/W	0	tcon_pll_od
[18:17]	R/W	0	tcon_pll_gate_ctrl
[16]	R/W	0	tcon_pll_bypass_en
[14:10]	R/W	0	tcon_pll_prediv_sel
[7:0]	R/W	0	tcon_pll_fbkdiv

Table 6-74 HHI_TCON_PLL_CNTL1 0x21

Bit(s)	R/W	Default	Description
[27:24]	R/W	0	tcon_picode0
[23]	R/W	0	tcon_pll_load_en
[22:21]	R/W	0	tcon_div_sel

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Bit(s)	R/W	Default	Description
[20]	R/W	0	tcon_pll_sdm_en
[18:0]	R/W	0	tcon_pll_sdm_frac

Table 6-75 HHI_TCON_PLL_CNTL2 0x22

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	tcon_pll_ssc_dep_sel
[26:24]	R/W	0	tcon_pll_ssc_fref_sel
[23:22]	R/W	0	tcon_pll_ssc_mode
[21:20]	R/W	0	tcon_pll_ssc_offset
[19:16]	R/W	0	tcon_pll_str_m
[15:0]	R/W	0	tcon_pll_reserve

Table 6-76 HHI_TCON_PLL_CNTL3 0x23

Bit(s)	R/W	Default	Description
31	R/W	0	tcon_pll_afc_bypass_en
29:28:00	R/W	0	tcon_pll_afc_hold_t
26:20:00	R/W	0	tcon_pll_in
19	R/W	0	tcon_pll_nt
18:17	R/W	0	tcon_pll_div
16	R/W	0	tcon_pll_lpf_en
15:12	R/W	0	tcon_pll_icap
11:08	R/W	0	tcon_pll_ires
5:04	R/W	0	tcon_pll_cpi

Table 6-77 HHI_TCON_PLL_STS 0x24

Bit(s)	R/W	Default	Description
[31]	R	0	tcon_pll_lock
[30]	R	0	tcon_pll_afc_don
[6:0]	R	0	tcon_pll_afc_out

Table 6-78 HHI_DADC_CNTL 0x27

Bit(s)	R/W	Default	Description
[26:24]	R/W	0	dadc_ml
[22:21]	R/W	0	dadc_vcm_clk_adj
[20]	R/W	0	dadc_stop_adj

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Bit(s)	R/W	Default	Description
[18:16]	R/W	0	dadc_mh
[13:12]	R/W	0	dadc_m2a
[6:3]	R/W	0	dadc_vddasel
[2]	R/W	0	dadc_en_dem
[0]	R/W	0	dadc_input_sel

Table 6-79 HHI_DADC_CNTL2 0x28

Bit(s)	R/W	Default	Description
[15]	R/W	0	dadc_pt_rst
[14:12]	R/W	0	dadc_ns_gain_sel
[10]	R/W	0	dadc_force_tm
[7:0]	R/W	0	dadc_ctrl_rsv2

Table 6-80 HHI_DADC_RDBK0_I 0x29

Bit(s)	R/W	Default	Description
[2:0]	R	0	dadc_tm_adj_flag

Table 6-81 HHI_DADC_CNTL3 0x2a

Bit(s)	R/W	Default	Description
[29]	R/W	0	dadc_fb_en
[28:27]	R/W	0	dadc_i_adj2
[25]	R/W	0	dadc_en_ext_clk
[23:16]	R/W	0	dadc_ctrl_rsv
[12:10]	R/W	0	dadc_tm_adj
[9:7]	R/W	0	dadc_i_adj
[6:2]	R/W	0	dadc_clk_div
[1]	R/W	0	dadc_ldo_en
[0]	R/W	0	dadc_en_adc

Table 6-82 HHI_DADC_CNTL 0x2e

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	afe_detect_rsv
[24:22]	R/W	0	afe_ch2_dc_level_adj
[21:19]	R/W	0	afe_ch2_comp_level_adj
[18]	R/W	0	afe_ch2_en_dc_bias

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Bit(s)	R/W	Default	Description
[17]	R/W	0	afe_ch2_detect_mode_select
[16]	R/W	0	afe_ch2_comp_hys_adj
[15]	R/W	0	afe_ch2_en_detect
[9:7]	R/W	0	afe_ch1_dc_level_adj
[6:4]	R/W	0	afe_ch1_comp_level_adj
[3]	R/W	0	afe_ch1_dc_bias
[2]	R/W	0	afe_ch1_detect_mode_select
[1]	R/W	0	afe_ch1_hys_adj

Table 6-83 HHI_XTAL_DIVN_CNTL 0x2f

Bit(s)	R/W	Default	Description
12	R/W	0	crt_clk25_en
11	R/W	0	crt_clk24_en
10	R/W	0	clk24_div2_en
7:0	R/W	0	clk25_div

Table 6-84 HHI_GCLK2_MPEG0 0x30

Bit(s)	R/W	Default	Description
31-0	R/W	All 0	Bits [31:0] of the composite MPEG clock gating register

Table 6-85 HHI_GCLK2_MPEG1 0x31

Bit(s)	R/W	Default	Description
31-0	R/W	All 0	Bits [63:32] of the composite MPEG clock gating register

Table 6-86 HHI_GCLK2_MPEG2 0x32

Bit(s)	R/W	Default	Description
31-0	R/W	All 0	Bits [63:32] of the composite MPEG clock gating register

Table 6-87 HHI_GCLK2_OTHER 0x34

Bit(s)	R/W	Default	Description
31-0	R/W	All 0	Bits [31:0] of the composite Other clock gating register

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Table 6-88 HHI_HIFI_PLL_CNTL0 0x36

Bit(s)	R/W	Default	Description
[31]	R	0	hifi_dpll_lock
[30]	R	0	hifi_dpll_lock_a
[29]	R/W	1	hifi_dpll_reset
[28]	R/W	0	hifi_dpll_en
[17:16]	R/W	0	hifi_dpll_od
[14:10]	R/W	0	hifi_dpll_N
[8:0]	R/W	0	hifi_dpll_M

Table 6-89 HHI_HIFI_PLL_CNTL1 0x37

Bit(s)	R/W	Default	Description
[18:0]	R	0	hifi_dpll_frac

Table 6-90 HHI_HIFI_PLL_CNTL2 0x38

Bit(s)	R/W	Default	Description
[22:20]	R/W	0	hifi_dpll_fref_sel
[17:16]	R/W	0	hifi_dpll_os_ssc
[15:12]	R/W	0	hifi_dpll_ssc_str_m
[8]	R/W	0	hifi_dpll_ssc_en
[7:4]	R/W	0	hifi_dpll_ssc_dep_sel
[1:0]	R/W	0	hifi_dpll_ss_mode

Table 6-91 HHI_HIFI_PLL_CNTL3 0x39

Bit(s)	R/W	Default	Description
31	R/W	0	hifi_dpll_afc_bypass
30	R/W	0	hifi_dpll_afc_clk_sel
29	R/W	0	hifi_dpll_code_new
28	R/W	0	hifi_dpll_dco_m_en
27	R/W	0	hifi_dpll_dco_sdm_en
26	R/W	0	hifi_dpll_div2
25	R/W	0	hifi_dpll_div_mode
24	R/W	0	hifi_dpll_fast_lock
23	R/W	0	hifi_dpll_fb_pre_div
22	R/W	0	hifi_dpll_filter_mode
21	R/W	0	hifi_dpll_fix_en

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Bit(s)	R/W	Default	Description
20	R/W	0	hifi_dpll_freq_shift_en
19	R/W	0	hifi_dpll_load
18	R/W	0	hifi_dpll_load_en
17	R/W	0	hifi_dpll_lock_f
16	R/W	0	hifi_dpll_pulse_width_en
15	R/W	0	hifi_dpll_sdmnc_en
14	R/W	0	hifi_dpll_sdmnc_mode
13	R/W	0	hifi_dpll_sdmnc_range
12	R/W	0	hifi_dpll_tdc_en
11	R/W	0	hifi_dpll_tdc_mode_sel
10	R/W	0	hifi_dpll_wait_en

Table 6-92 HHI_HIFI_PLL_CNTL4 0x3a

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	hifi_dpll_alpha
[27:24]	R/W	0	hifi_dpll_rou
[22:20]	R/W	0	-
[18:16]	R/W	0	hifi_dpll_lambda0
[13:12]	R/W	0	hifi_dpll_acq_gain
[11:8]	R/W	0	hifi_dpll_filter_pvt2
[7:4]	R/W	0	hifi_dpll_filter_pvt1
[1:0]	R/W	0	hifi_dpll_pfd_gain

Table 6-93 HHI_HIFI_PLL_CNTL5 0x3b

Bit(s)	R/W	Default	Description
[30:28]	R/W	0	hifi_dpll_adj_vco_ldo
[27:24]	R/W	0	hifi_dpll_lm_w
[21:16]	R/W	0	hifi_dpll_lm_s
[15:0]	R/W	0	hifi_dpll_reve

Table 6-94 HHI_HIFI_PLL_CNTL6 0x3c

Bit(s)	R/W	Default	Description
[31:30]	R/W	0	hifi_dpll_afc_hold_t
[29:28]	R/W	0	hifi_dpll_lkw_sel
[27:26]	R/W	0	hifi_dpll_dco_sdm_clk_sel

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Bit(s)	R/W	Default	Description
[25:24]	R/W	0	hifi_dpll_afc_in
[23:22]	R/W	0	hifi_dpll_afc_nt
[21:20]	R/W	0	hifi_dpll_vc_in
[19:18]	R/W	0	hifi_dpll_lock_long
[17:16]	R/W	0	hifi_dpll_freq_shift_v
[14:12]	R/W	0	hifi_dpll_data_sel
[10:8]	R/W	0	hifi_dpll_sdmnc_ulms
[6:0]	R/W	0	hifi_dpll_sdmnc_power

Table 6-95 HHI_HIFI_PLL_STS 0x3d

Bit(s)	R/W	Default	Description
[31]	R	0	hifi_dpll_lock
[30]	R	0	hifi_dpll_lock_a
[29]	R	0	hifi_dpll_afc_done
[22:16]	R	0	hifi_dpll_sdmnc_monitor
[9:0]	R	0	hifi_dpll_out_rsv

Table 6-96 HHI_TIMER90K 0x3f

Bit(s)	R/W	Default	Description
31-16	R/W	0	Unused
15-0	R/W	0x384	90khz divider

Table 6-97 HHI_MEM_PD_REG0 0x40

Bit(s)	R/W	Default	Description
31~16	R/W	0xFFFF-FFFF	Reserved
19~18	R/W	0x3	Dsp and tvfe memory PD
17~16	R/W	0x3	Atv_dmd and ddr memory PD
3~2	R/W	0x3	Ethernet memory PD
1~0	R/W	0x3	Adec memory PD

Table 6-98 HHI_VPU_MEM_PD_REG0 0x41

Bit(s)	R/W	Default	Description
31~30	R/W	0x3	sharp
29~28	R/W	0x3	Deinterlacer – di_post: 11 = power down. 00 = normal operation

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Bit(s)	R/W	Default	Description
27~26	R/W	0x3	Deinterlacer – di_pre
25~24	R/W	0x3	Vi_di_scaler
23~22	R/W	0x3	afbc_dec1
21~20	R/W	0x3	Srscl super scaler
19~18	R/W	0x3	Vdin1 memory
17~16	R/W	0x3	Vdin0 memory
15~14	R/W	0x3	Osd_scaler memory
13~12	R/W	0x3	Scaler memory
11~10	R/W	0x3	Vpp output fifo
9~8	R/W	0x3	Color management module
7~6	R/W	0x3	Vd2 memory
5~4	R/W	0x3	Vd1 memory
3~2	R/W	0x3	Osd2 memory
1~0	R/W	0x3	Osd1 memory

Table 6-99 HHI_VPU_MEM_PD_REG1 0x42

Bit(s)	R/W	Default	Description
31~30	R/W	0x3	Vd2_osd2_scaler
29~28	R/W	0x3	tv decoder cvd2 / ldim_stts
27~26	R/W	0x3	Ls_stts
25~24	R/W	0x3	CVBS inci interface
23~22	R/W	0x3	Panel encl top
21~20	R/W	0x3	Hdmi encp interface
19~18	R/W	0x3	Vd2_scale
17~16	R/W	0x3	Afbc dec0
15~14	R/W	0x3	Vpu arb
13~12	R/W	0x3	dolby1b
11~10	R/W	0x3	dolby1a
9~8	R/W	0x3	dolby0
7~6	R/W	0x3	dolby core3
5~4	R/W	0x3	vkstone
3~2	R/W	0x3	VIU2_OFIFO
1~0	R/W	0x3	VIU2 OSD1

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Table 6-100 HHI_DEMOD_MEM_PD_REG 0x43

Bit(s)	R/W	Default	Description
31-0	R/W	Ffffffff	demod

Table 6-101 HHI_MEM_PD_REG0 0x45

Bit(s)	R/W	Default	Description
23~12	R/W	0xffffffff	Audio memory PD
7~0	R/W	0xffffffff	Audio memory PD

Table 6-102 HHI_VIID_CLK_DIV 0x4a

Bit(s)	R/W	Default	Description
31-28	R/W	0	DAC0_CLK_SEL
27-24	R/W	0	DAC1_CLK_SEL
23-20	R/W	0	DAC2_CLK_SEL
19	R/W	0	Select adc_pll_clk_b2 to be cts_clk_vdac
18	R/W	0	Unused
17	R/W	0	V2_cntl_clk_div_reset
16	R/W	0	V2_cntl_clk_div_en
15-12	R/W	0	Encl_clk_sel
14-8	R/W	0	Unused
7-0	R/W	0	V2_cntl_xd0

Table 6-103 HHI_VIID_CLK_CNTL 0x4b

Bit(s)	R/W	Default	Description
31-20	R/W	0	Unused
19	R/W	0	V2_cntl_clk_en0
18-16	R/W	0	V2_cntl_clk_in_sel
15	R/W	0	V2_cntl_soft_reset
14-5	R/W	0	Unused
4	R/W	0	V2_cntl_div12_en
3	R/W	0	V2_cntl_div6_en
2	R/W	0	V2_cntl_div4_en
1	R/W	0	V2_cntl_div2_en
0	R/W	0	V2_cntl_div1_en

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Table 6-104 HHI_VPU_MEM_PD_REG4 0x4c

Bit(s)	R/W	Default	Description
31-7	R/W	ffffffff	unused
6-4	R/W	3	afbce
3-2	R/W	3	Axiwr0/axird2/axird0
1-0	R/W	3	Axiwr1/axi_rd1

Table 6-105 HHI_VPU_MEM_PD_REG2 0x4d

Bit(s)	R/W	Default	Description
31-28	R/W	3	Unused
27-24	R/W	f	Vi_dipre
23-22	R/W	3	Viu2
21-20	R/W	3	Lut3d
19-18	R/W	3	ds
17-16	R/W	3	Vd2_ofifo
15-14	R/W	3	Prime_dolby_ram
13-12	R/W	3	Osd_bld34
11-10	R/W	3	Vd1_scaler
9-8	R/W	3	Mem_pd_mail_afbcd
7-6	R/W	3	Mem_pd_vi_osd4
5-4	R/W	3	Mem_pd_vi_osd3
3-2	R/W	3	Mem_pd_tcon
1-0	R/W	3	vpp water mark

Table 6-106 HHI_VPU_MEM_PD_REG3 0x4e

Bit(s)	R/W	Default	Description
31-0	R/W	ffffffff	TCON

Table 6-107 HHI_GCLK_LOCK 0x4F

wr	rd
SP/SCP sec only	all access

Bit(s)	R/W	Default	Description
31	R/W	0	enable error return if write gclk after lock
8	R/W	0	lock gclk2_other
7	R/W	0	lock gclk2_mpeg2

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Bit(s)	R/W	Default	Description
6	R/W	0	lock gclk2_mpeg1
5	R/W	0	lock gclk2_mpeg0
4	R/W	0	lock gclk_sp_mpeg
3	R/W	0	lock gclk1_other
2	R/W	0	lock gclk1_mpeg2
1	R/W	0	lock gclk1_mpeg1
0	R/W	0	lock gclk1_mpeg0

Table 6-108 HHI_GCLK_MPEG0 0x50

Bit(s)	R/W	Default	Description
31-0	R/W	All 1	Bits [31:0] of the composite MPEG clock gating register

Table 6-109 HHI_GCLK_MPEG1 0x51

Bit(s)	R/W	Default	Description
31-0	R/W	All 1	Bits [63:32] of the composite MPEG clock gating register

Table 6-110 HHI_GCLK_MPEG2 0x52

Bit(s)	R/W	Default	Description
31-0	R/W	All 1	Bits [63:32] of the composite MPEG clock gating register

Table 6-111 HHI_GCLK_OTHER 0x54

Bit(s)	R/W	Default	Description
31-0	R/W	All 1	Bits [31:0] of the composite Other clock gating register

Table 6-112 HHI_GCLK_SP_MPEG 0x55

wr	rd
SP sec only	SP sec only

Bit(s)	R/W	Default	Description
31-0	R/W	0	bits[31:0] of secure module clock gating.

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Table 6-113 HHI_SYS_CPU_CLK_CNTL1 0x57

Bit(s)	R/W	Default	Description
31-26	R/W	0	Reserved
25	RW	0	Soft_reset_periph
24	R/W	0	Sys_pll_div16_en
23	R/W	0	A55_trace_clk _DIS: Set to 1 to manually disable the A55_trace_clk when changing the mux selection. Typically this bit is set to 0 since the clock muxes can switch without glithces. This is a “just in case” bit
22:20	R/W	5	A55_trace_clk: 0 A55 clock divided by 2 1 A55 clock divided by 4 2 A55 clock divided by 6 3 A55 clock divided by 8 4 A55 clock divided by 10 5 A55 clock divided by 12 6 A55 clock divided by 14 7 A55 clock divided by 16
19	R/W	0	Timestamp CNTCLKEN_dis
18	R/W	0	AXI_CLK_DIS: Set to 1 to manually disable the AXI clock when changing the mux selection. Typically this bit is set to 0 since the clock muxes can switch without glithces. This is a “just in case” bit
16	R/W	0	APB_CLK_DIS: Set to 1 to manually disable the APB clock when changing the mux selection. Typically this bit is set to 0 since the clock muxes can switch without glithces. This is a “just in case” bit
15	R/W	0	Timestamp CNTCLKEN
14~12	R/W	0	Timestamp cntclk mux(not used)
11~9	R/W	1	AXI_CLK _MUX: 0 A53 clock divided by 2 1 A53 clock divided by 3 2 A53 clock divided by 4 3 A53 clock divided by 5 4 A53 clock divided by 6 5 A53 clock divided by 7 6 A53 clock divided by 8
5~3	R/W	4	APB_CLK _MUX: 0 A53 clock divided by 2 1 A53 clock divided by 3 2 A53 clock divided by 4 3 A53 clock divided by 5 4 A53 clock divided by 6 5 A53 clock divided by 7 6 A53 clock divided by 8
2	R/W	0	Soft_reset_cpu
1	R/W	0	Sys_cpu_clk_div16_en
0	R/W	0	Pclk_en_dbg

Table 6-114 HHI_SYS_CPU_RESET_CNTL 0x58

wr	rd
SP/SCP/AP sec only	all access

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Bit(s)	R/W	Default	Description
16	R/W	0	Cpu_axi_reset
14	R/W	0	nPERIPHRESET
13	R/W	0	nGICRESET
11	R/W	0	nPERSETDBG
9	R/W	0	nSRESET
8	R/W	0	nSPORSET
7-4	R/W	0	nCORERESET[3:0]
3-0	R/W	0	Cpu_soft_reset[3:0]

Table 6-115 HHI_VID_CLK_DIV 0x59

Bit(s)	R/W	Default	Description
31-28	R/W	0	ENCI_CLK_SEL
27-24	R/W	0	ENCP_CLK_SEL
23-20	R/W	0	ENCT_CLK_SEL
19-18	R/W	0	UNUSED
17	R/W	0	CLK_DIV_RESET
16	R/W	0	CLK_DIV_EN
15-8	R/W	1	XD1
7-0	R/W	1	XD0

Table 6-116 HHI_MPEG_CLK_CNTL 0x5d

Bit(s)	R/W	Default	Description
31	R/W	0	NEW_DIV_EN: If this bit is set to 1, then bits[30:16] make up the clk81 divider. If this bit is 0, then bits[6:0] dictate the divider value. This is a new feature that allows clk81 to be divided down to a very slow frequency.
30~16	R/W	0	NEW_DIV: New divider value if bit[31] = 1
15	R/W	0	Production clock enable
14-12	R.W	6	MPEG_CLK_SEL (See clock document)
11-10	R/W	0	unused
9	R.W	0	RTC Oscillator Enable: Set this bit to 1 to connect the RTC 32khz oscillator output as the XTAL input for the divider above
8	R/W	0	Divider Mux: 0 = the ARC clock and the MPEG system clock are connected to the 27Mhz crystal. 1 = the ARC clock and the MPEG system clock are connected to the MPEG PLL divider
7	R/W	1	PLL Mux: 0 = all circuits associated with the MPEG PLL are connected to 27Mhz. 1 = all circuits associated with the MPEG PLL are connected to the MPEG PLL
6-0	R/W	0	PLL Output divider. The MPEG System clock equals the video PLL clock frequency divided by (N+1). Note: N must be odd (1,3,5,….) so that the MPEG clock is divided by an even number to generate a 50% duty cycle.

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Table 6-117 HHI_VID_CLK_CNTL 0x5f

Bit(s)	R/W	Default	Description
31-21	R/W	0	TCON_CLK0_CTRL
20	R/W	0	CLK_EN1
19	R/W	0	CLK_EN0
18-16	R/W	0	CLK_IN_SEL
15	R/W	0	SOFT_RESET
14	R/W	0	PH23_ENABLE
13	R/W	0	DIV12_PH23
12-5	R/W	0	UNUSED
4	R/W	0	DIV12_EN
3	R/W	0	DIV6_EN
2	R/W	0	DIV4_EN
1	R/W	0	DIV2_EN
0	R/W	0	DIV1_EN

Table 6-118 HHI_TS_CLK_CNTL 0x64

Bit(s)	R/W	Default	Description
8	R/W	0	ts_clk clk_en
7:0	R/W	0	ts_clk clk_div

Table 6-119 HHI_VID_CLK_CNTL2 0x65

Bit(s)	R/W	Default	Description
31-16	R	0	
15-9	R/W	0	Reserved
8	R/W	0	Atv demod vdac gated clock control
7	R/W	1	LCD_AN_CLK_PHY2 gated clock control. 1 = enable
6	R/W	1	LCD_AN_CLK_PH3 gated clock control
5	R/W	1	HDMI_TX_PIXEL_CLK gated clock control
4	R/W	1	VDAC_clk gated clock control
3	R/W	1	ENCL gated clock control
2	R/W	1	ENCP gated clock control
1	R/W	1	ENCT gated clock control
0	R/W	1	ENCI gated clock control

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Table 6-120 HHI_SYS_CPU_CLK_CNTL0 0x67

Bit(s)	R/W	Default	Description
31	R	0	Cpu0:Final_mux_sel
30	R	0	Cpu0:Final_dyn_mux_sel
29	R	0	Cpu0:Busy_cnt
28	R	0	Cpu0:busy
26	R/W	0	Cpu0:Dyn_enable
25-20	R/W	0	Cpu0:Mux1_divn_tcnt
18	R/W	0	Cpu0:Postmux1
17-16	R/W	0	Cpu0:Premux1
15	R/W	0	Cpu0:Manual_mux_mode
14	R/W	0	Cpu0:Manual_mode_post
13	R/W	0	Cpu0:Manual_mode_pre
12	R/W	0	Cpu0:Force_update_t
11	R/W	0	Cpu0:Final_mux_sel
10	R/W	0	Cpu0:Final_dyn_mux_sel
9-4	R/W	0	Cpu0:mux0_divn_tcnt
3	R/W	0	Rev
2	R/W	0	Cpu0:Postmux0
1-0	R/W	0	Cpu0:Premux0

Table 6-121 HHI_VID_PLL_CLK_DIV 0x68

Bit(s)	R/W	Default	Description
31~24	R	0	RESERVED
23~20	R/W	0	Reserved
19	R/W	0	CLK_FINAL_EN
18	R/W	0	CLK_DIV1
17~16	R/W	0	CLK_SEL
15	R/W	0	SET_PRESET
14-0	R/W	0	SHIFT_PRESET

Table 6-122 HHI_MALI_CLK_CNTL 0x6c

Bit(s)	R/W	Default	Description
31	R/W	0	0: mali_clk_cntl[14:0]; 1: mali_clk_cntl[30:16];
27~25	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_mali _clk 0:oscin; 1:gp0pll;

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Bit(s)	R/W	Default	Description
			2:hifi_pll; 3:fclk_div2p5; 4:fclk_div3; 5:fclk_div4; 6:fclk_div5; 7:fclk_div7;
24	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_ mali _clk
23	R/W	0	Reserved
22~16	R/W	0	CLK_DIV: See the Clock Tree document for information related to cts_ mali _clk
15~12			
11~9	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_mali _clk 0:oscin; 1:gp0pll; 2:mpll2; 3:mpll1; 4:fclk_div7; 5:fclk_div4; 6:fclk_div3; 7:fclk_div5;
8	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_ mali _clk
7	R/W	0	Reserved
6-0	R/W	0	CLK_DIV: See the Clock Tree document for information related to cts_ mali _clk

Table 6-123 HHI_VPU_CLKC_CNTL 0x6D

Bit(s)	R/W	Default	Description
31	R/W	0	Final mux sel
30-29	R/W	0	Reserved
27~25	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_vpu_clkc 0:fclk_div4; 1:fclk_div3; 2:fclk_div5; 3:fclk_div7; 4:mpll1; 5:vid_pll; 6:mpll2; 7:gp0pll;
24	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_vpu_clkc
23	R/W	0	Reserved
22~16	R/W	0	CLK_DIV: See the Clock Tree document for information related to cts_vpu_clkc
15~12			
11~9	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_vpu_clkc 0:fclk_div4; 1:fclk_div3; 2:fclk_div5; 3:fclk_div7; 4:mpll1; 5:vid_pll; 6:mpll2; 7:gp0pll;
8	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_vpu_clkc

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Bit(s)	R/W	Default	Description
7	R/W	0	Reserved
6-0	R/W	0	CLK_DIV: See the Clock Tree document for information related to cts_vpu_clkc

Table 6-124 HHI_VPU_CLK_CNTL 0x6f

Bit(s)	R/W	Default	Description
31	R/W	0	Final mux sel
30-29	R/W	0	Reserved
27~25	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_vpu_clk 0:fclk_div3; 1:fclk_div4; 2:fclk_div5; 3:fclk_div7; 4:mpll1; 5:vid_pll; 6:hifi_pll; 7:gp0_pll;
24	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_vpu_clk
23	R/W	0	Reserved
22~16	R/W	0	CLK_DIV: See the Clock Tree document for information related to cts_vpu_clk
15~12			
11~9	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_vpu_clk 0:fclk_div3; 1:fclk_div4; 2:fclk_div5; 3:fclk_div7; 4:mpll1; 5:vid_pll; 6:hifi_pll; 7:gp0_pll;
8	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_vpu_clk
7	R/W	0	Reserved
6-0	R/W	0	CLK_DIV: See the Clock Tree document for information related to cts_vpu_clk

Table 6-125 HHI_HDMI_CLK_CNTL 0x73

Bit(s)	R/W	Default	Description
31-20	R/W	0	Reserved
19~16	R/W	0	crt_hdmi_pixel_clk_sel
15~11	R/W	0	Reserved
10~9	R/W	0	
8	R/W	0	
7	R/W	0	Reserved
6-0	R/W	0	

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Table 6-126 HHI_AUDPLL_CLK_OUT_CNTL 0x74

Bit(s)	R/W	Default	Description
27-25	R/W	0	Clk_sel:See the Clock Tree document for information related to adc_extclk_in
24	R/W	0	Clk_en:See the Clock Tree document for information related to adc_extclk_in
22-16	R/W	0	Clk_div:See the Clock Tree document for information related to adc_extclk_in
10-9	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_demod_ core_clk
8	R/W	0	Clk_en:See the Clock Tree document for information related to cts_demod_ core_clk
6-0	R/W	0	Clk_div:See the Clock Tree document for information related to cts_demod_ core_clk

Table 6-127 HHI_ETH_CLK_CNTL 0x76

Bit(s)	R/W	Default	Description
17	R/W	0	eth_mac_speed; 0:div 20; 1: div2;
16	R/W	0	invert clk_rmii_pad_i
13	R/W	0	invert clk_rmii_pad_o
12	R/W	0	clk_rmii_pad_o; 0: rmii_clk; 1: rmii_div;
11:9	R/W	0	rmii_clk; clk_sel; 0: fclk_div2; 7: clk_rmii_pad_i;
8	R/W	0	rmii_clk; clk_en
7	R/W	0	eth_clk125M; clk_en;
6:0	R/W	0	rmii_clk; clk_div

Table 6-128 HHI_VDEC_CLK_CNTL 0x78

Bit(s)	R/W	Default	Description
31-28	R/W	0	Reserved
27~25	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_hcodec_ clk 0:fclk_div2p5; 1:fclk_div3; 2:fclk_div4; 3:fclk_div5; 4:fclk_div7; 5:hifi_pll; 6:gp0pll; 7:oscin;
24	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_hcodec_ clk
23	R/W	0	Reserved
22~16	R/W	0	CLK_DIV: See the Clock Tree document for information related to ctshcodec_ clk

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Bit(s)	R/W	Default	Description
15~12	R/W	0	Reserved
11~9	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_vdec_clk 0:fclk_div2p5; 1:fclk_div3; 2:fclk_div4; 3:fclk_div5; 4:fclk_div7; 5:hifi_pll; 6:gp0pll; 7:oscin;
8	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_vdec_clk
7	R/W	0	Reserved
6-0	R/W	0	CLK_DIV: See the Clock Tree document for information related to cts_vdec_clk

Table 6-129 HHI_VDEC2_CLK_CNTL 0x79

Bit(s)	R/W	Default	Description
31-28	R/W	0	Reserved
27~25	R/W	0	HEVCB_CLK_SEL: See the Clock Tree document for information related to cts_hevcb_clk 0:fclk_div2p5; 1:fclk_div3; 2:fclk_div4; 3:fclk_div5; 4:fclk_div7; 5:hifi_pll; 6:gp0pll; 7:oscin;
24	R/W	0	HEVCB_CLK_EN: See the Clock Tree document for information related to cts_hevcb_clk
23	R/W	0	Reserved
22~16	R/W	0	HEVCB_CLK_DIV: See the Clock Tree document for information related to cts_hevcb_clk
15~12			
11~9	R/W	0	HEVCF_CLK_SEL: See the Clock Tree document for information related to cts_hevcf_clk 0:fclk_div2p5; 1:fclk_div3; 2:fclk_div4; 3:fclk_div5; 4:fclk_div7; 5:hifi_pll; 6:gp0pll; 7:oscin;
8	R/W	0	HEVCF_CLK_EN: See the Clock Tree document for information related to cts_hevcf_clk
7	R/W	0	Reserved
6-0	R/W	0	HEVCF_CLK_DIV: See the Clock Tree document for information related to cts_hevcf_clk

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Table 6-130 HHI_VDEC3_CLK_CNTL 0x7a

Bit(s)	R/W	Default	Description
31	R/W	0	cts_hcodec_clk; 0: use VDEC_CLK_CNTL; 1: use VDEC3_CLK_CNTL;
27~25	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_hcodec_clk 0:fclk_div2p5; 1:fclk_div3; 2:fclk_div4; 3:fclk_div5; 4:fclk_div7; 5:hifi_pll; 6:gp0pll; 7:oscin;
24	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_hcodec_clk
23	R/W	0	Reserved
22~16	R/W	0	CLK_DIV: See the Clock Tree document for information related to ctshcodec_clk
15	R/W	0	cts_vdec_clk; 0: use VDEC_CLK_CNTL; 1: use VDEC3_CLK_CNTL;
11~9	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_vdec_clk 0:fclk_div2p5; 1:fclk_div3; 2:fclk_div4; 3:fclk_div5; 4:fclk_div7; 5:hifi_pll; 6:gp0pll; 7:oscin;
8	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_vdec_clk
7	R/W	0	Reserved
6-0	R/W	0	CLK_DIV: See the Clock Tree document for information related to cts_vdec_clk

Table 6-131 HHI_VDEC4_CLK_CNTL 0x7b

Bit(s)	R/W	Default	Description
31	R/W	0	HEVCB_CLK; 0: use VDEC2_CLK_CNTL; 1: use VDEC4_CLK_CNTL;
27~25	R/W	0	HEVCB_CLK_SEL: See the Clock Tree document for information related to cts_hevcb_clk 0:fclk_div2p5; 1:fclk_div3; 2:fclk_div4; 3:fclk_div5; 4:fclk_div7; 5:hifi_pll; 6:gp0pll; 7:oscin;
24	R/W	0	HEVCB_CLK_EN: See the Clock Tree document for information related to cts_hevcb_clk
23	R/W	0	Reserved

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Bit(s)	R/W	Default	Description
22~16	R/W	0	HEVCB_CLK_DIV: See the Clock Tree document for information related to cts_hevcb_clk
15	R/W	0	HEVCF_CLK; 0: use VDEC2_CLK_CNTL; 1: use VDEC4_CLK_CNTL;
11~9	R/W	0	HEVCF_CLK_SEL: See the Clock Tree document for information related to cts_hevcf_clk 0:fclk_div2p5; 1:fclk_div3; 2:fclk_div4; 3:fclk_div5; 4:fclk_div7; 5:hifi_pll; 6:gp0pll; 7:oscin;
8	R/W	0	HEVCF_CLK_EN: See the Clock Tree document for information related to cts_hevcf_clk
7	R/W	0	Reserved
6-0	R/W	0	HEVCF_CLK_DIV: See the Clock Tree document for information related to cts_hevcf_clk

Table 6-132 HHI_HDCP22_CLK_CNTL 0x7c

Bit(s)	R/W	Default	Description
31-0	R/W	0	Reserved
26-25	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_hdcp22_skpclk
24	R/W	0	Clk_en:See the Clock Tree document for information related to cts_hdcp22_skpclk
22-16	R/W	0	Clk_div:See the Clock Tree document for information related to cts_hdcp22_skpclk
10-9	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_hdcp22_esmclk
8	R/W	0	Clk_en:See the Clock Tree document for information related to cts_hdcp22_esmclk
6-0	R/W	0	Clk_div:See the Clock Tree document for information related to cts_hdcp22_esmclk

Table 6-133 HHI_VAPBCLK_CNTL 0x7d

Bit(s)	R/W	Default	Description
31	R/W	0	Final_mux_sel
30	R/W	0	Enable
29-28	R/W	0	Reserved
27-25	R/W	0	Mux1_sel: 0:fclk_div4 1:fclk_div3 2:fclk_div5 3:fclk_div7 4:mp1_clk_out

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Bit(s)	R/W	Default	Description
			5:vid_pll_clk 6:mp2_clk_out 7:fclk_div2p5
24	R/W	0	Mux1_en
23	R/W	0	Reserved
22-16	R/W	0	Mux1_div
15-12	R/W	0	Reserved
11-9	R/W	0	Mux0_sel,as mux1_sel
8	R/W	0	Mux0_en
7	R/W	0	Reserved
6-0	R/W	0	Mux0_div

Table 6-134 HHI_HDMIRX_CLK_CNTL 0x80

Bit(s)	R/W	Default	Description
31-0	R/W	0	Reserved
26-25	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_hdmirx_modet_clk
24	R/W	0	Clk_en:See the Clock Tree document for information related to cts_hdmirx_modet_clk
22-16	R/W	0	Clk_div:See the Clock Tree document for information related to cts_hdmirx_modet_clk
10-9	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_hdmirx_cfg_clk
8	R/W	0	Clk_en:See the Clock Tree document for information related to cts_hdmirx_cfg_clk
6-0	R/W	0	Clk_div:See the Clock Tree document for information related to cts_hdmirx_cfg_clk

Table 6-135 HHI_HDMIRX_AUD_CLK_CNTL 0x81

Bit(s)	R/W	Default	Description
31-0	R/W	0	Reserved
26-25	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_hdmirx_acr_ref_clk
24	R/W	0	Clk_en:See the Clock Tree document for information related to cts_hdmirx_acr_ref_clk
22-16	R/W	0	Clk_div:See the Clock Tree document for information related to cts_hdmirx_acr_ref_clk

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Table 6-136 HHI_VPU_CLKB_CNTL 0x83

Bit(s)	R/W	Default	Description
31-25	R/W	0	Reserved
24	R/W	0	Cts_vpu_clkb_tmp en
21-20	R/W	0	Cts_vpu_clkb_tmp sel 0:cts_vpu_clk 1:fclk_div4 2:fclk_div5 3:fclk_div7
19-16	R/W	0	Cts_vpu_clkb_tmp div
9	R/W	0	Vpu_clkb_latch_en
8	R/W	0	Vpu_clkb_en
7-0	R/W	0	Vpu_clkb_div

Table 6-137 HHI_SYS_CPU_CLK_CNTL2 0x84

Bit(s)	R/W	Default	Description
31	R	0	Cpu1:Final_mux_sel
30	R	0	Cpu1:Final_dyn_mux_sel
29	R	0	Cpu1:Busy_cnt
28	R	0	Cpu1:busy
26	R/W	0	Cpu1:Dyn_enable
25-20	R/W	0	Cpu1:Mux1_divn_tcnt
18	R/W	0	Cpu1:Postmux1
17-16	R/W	0	Cpu1:Premux1
15	R/W	0	Cpu1:Manual_mux_mode
14	R/W	0	Cpu1:Manual_mode_post
13	R/W	0	Cpu1:Manual_mode_pre
12	R/W	0	Cpu1:Force_update_t
11	R/W	0	Cpu1:Final_mux_sel
10	R/W	0	Cpu1:Final_dyn_mux_sel
9-4	R/W	0	Cpu1:mux0_divn_tcnt
3	R/W	0	Rev
2	R/W	0	Cpu1:Postmux0
1-0	R/W	0	Cpu1:Premux0

Table 6-138 HHI_SYS_CPU_CLK_CNTL3 0x85

Bit(s)	R/W	Default	Description
31	R	0	Cpu2:Final_mux_sel
30	R	0	Cpu2:Final_dyn_mux_sel
29	R	0	Cpu2:Busy_cnt

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Bit(s)	R/W	Default	Description
28	R	0	Cpu2:busy
26	R/W	0	Cpu2:Dyn_enable
25-20	R/W	0	Cpu2:Mux1_divn_tcnt
18	R/W	0	Cpu2:Postmux1
17-16	R/W	0	Cpu2:Premux1
15	R/W	0	Cpu2:Manual_mux_mode
14	R/W	0	Cpu2:Manual_mode_post
13	R/W	0	Cpu2:Manual_mode_pre
12	R/W	0	Cpu2:Force_update_t
11	R/W	0	Cpu2:Final_mux_sel
10	R/W	0	Cpu2:Final_dyn_mux_sel
9-4	R/W	0	Cpu2:mux0_divn_tcnt
3	R/W	0	Rev
2	R/W	0	Cpu2:Postmux0
1-0	R/W	0	Cpu2:Premux0

Table 6-139 HHI_SYS_CPU_CLK_CNTL4 0x86

Bit(s)	R/W	Default	Description
31	R	0	Cpu3:Final_mux_sel
30	R	0	Cpu3:Final_dyn_mux_sel
29	R	0	Cpu3:Busy_cnt
28	R	0	Cpu3:busy
26	R/W	0	Cpu3:Dyn_enable
25-20	R/W	0	Cpu3:Mux1_divn_tcnt
18	R/W	0	Cpu3:Postmux1
17-16	R/W	0	Cpu3:Premux1
15	R/W	0	Cpu3:Manual_mux_mode
14	R/W	0	Cpu3:Manual_mode_post
13	R/W	0	Cpu3:Manual_mode_pre
12	R/W	0	Cpu3:Force_update_t
11	R/W	0	Cpu3:Final_mux_sel
10	R/W	0	Cpu3:Final_dyn_mux_sel
9-4	R/W	0	Cpu3:mux0_divn_tcnt
3	R/W	0	Rev

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Bit(s)	R/W	Default	Description
2	R/W	0	Cpu3:Postmux0
1-0	R/W	0	Cpu3:Premux0

Table 6-140 HHI_SYS_CPU_CLK_CNTL5 0x87

Bit(s)	R/W	Default	Description
31	R	0	dsu:Final_mux_sel
30	R	0	dsu:Final_dyn_mux_sel
29	R	0	dsu:Busy_cnt
28	R	0	dsu:busy
26	R/W	0	dsu:Dyn_enable
25-20	R/W	0	dsu:Mux1_divn_tcnt
18	R/W	0	dsu:Postmux1
17-16	R/W	0	dsu:Premux1
15	R/W	0	dsu:Manual_mux_mode
14	R/W	0	dsu:Manual_mode_post
13	R/W	0	dsu:Manual_mode_pre
12	R/W	0	dsu:Force_update_t
11	R/W	0	dsu:Final_mux_sel
10	R/W	0	dsu:Final_dyn_mux_sel
9-4	R/W	0	dsu:mux0_divn_tcnt
3	R/W	0	Rev
2	R/W	0	dsu:Postmux0
1-0	R/W	0	dsu:Premux0

Table 6-141 HHI_SYS_CPU_CLK_CNTL6 0x88

Bit(s)	R/W	Default	Description
27-24	R/W	0	Cpu_clk_sync_mux_sel
23	R/W	0	Sys_pll_mux_sel
22-20	R/W	0	Sys_clk_mux_sel
11	R/W	0	Tsclk_dis
10-8	R/W	0	Tsclk_mux
7	R/W	0	Periphclk_dis
6-4	R/W	0	Periphclk_mux
3	R/W	0	Gicclk_dis
2-0	R/W	0	Gciclk_mux

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Table 6-142 HHI_GEN_CLK_CNTL 0x8a

Bit(s)	R/W	Default	Description
31-17	R/W	0	Reserved
16~12	R/W	0	CLK_SEL: 0:cts_oscin_clk 1:rtc_oscin_i 2:sys_cpu_clk_div16 3:ddr_dpll_pt_clk 4:vid_pll_clk 5:gp0_pll_clk 6:gp1_pll_clk 7:hifi_pll_clk 8:dsu_pll_clk_cpu 9:eth_mppll_test 10:hdmirx_aud_pll_clk 11:hdmirx_apll_clk_out_div 12:cts_msr_clk 13:adc_dpll_intclk 14:AU_ADC_CLK_TO_GPIO 15:AU_DAC2_CLK_TOGPIO 16:AU_DAC1_CLK_TOGPIO 17:sys_cpu_clk_div16; 18:sys_pll_div16 19:fclk_div2 20:fclk_div2p5 21:fclk_div3; 22:fclk_div4; 23:fclk_div5; 24:fclk_div7; 25:mpll0_clk; 26:mpll1_clk; 27:mpll2_clk; 28:mpll3_clk; 29:adc_dpll_clk_a2 30:adc_dpll_clk_b2 31:adc_dpll_clk_b3
11	R/W	0	CLK_EN: See the Clock Tree document for information related to gen_clk_out
10~0	R/W	0	CLK_DIV: See the Clock Tree document for information related to gen_clk_out

Table 6-143 HHI_AUDPLL_CLK_OUT_CNTL 0x8c

Bit(s)	R/W	Default	Description
6-5	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_hdmirx_aud_pll_clk
4	R/W	0	Clk_en:See the Clock Tree document for information related to cts_hdmirx_aud_pll_clk
2-0	R/W	0	Clk_div:See the Clock Tree document for information related to cts_hdmirx_aud_pll_clk

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Table 6-144 HHI_HDMIRX_METER_CLK_CNTL 0x8d

Bit(s)	R/W	Default	Description
10-9	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_hdmirx_meter_clk
8	R/W	0	Clk_en:See the Clock Tree document for information related to cts_hdmirx_meter_clk
6-0	R/W	0	Clk_div:See the Clock Tree document for information related to cts_hdmirx_meter_clk

Table 6-145 HHI_DIF_CSI_PHY_CNTL10 0x8e

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch5_chdig_ctl
15:0	R/W	0	Ch4_chdig_ctl

Table 6-146 HHI_DIF_CSI_PHY_CNTL11 0x8f

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch7_chdig_ctl
15:0	R/W	0	Ch6_chdig_ctl

Table 6-147 HHI_DIF_CSI_PHY_CNTL12 0x90

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch9_chdig_ctl
15:0	R/W	0	Ch8_chdig_ctl

Table 6-148 HHI_DIF_CSI_PHY_CNTL13 0x91

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch11_chdig_ctl
15:0	R/W	0	Ch10_chdig_ctl

Table 6-149 HHI_DIF_CSI_PHY_CNTL14 0x92

Bit(s)	R/W	Default	Description
31:16	R/W	0	t12nm_p2p_common1
15:0	R/W	0	t12nm_p2p_common0

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Table 6-150 HHI_DIF_CSI_PHY_CNTL15 0x93

Bit(s)	R/W	Default	Description
31:16	R/W	0	t12nm_p2p_vinlp
15:0	R/W	0	t12nm_p2p_pictl

Table 6-151 HHI_VDIN_MEAS_CLK_CNTL 0x94 Clock control for cts_vdin_meas_clk

Bit(s)	R/W	Default	Description
31-12	R/W	0	unused
11-9	R/W	0	CLK_SEL: See the Clock Tree document for information related to cts_vdin_meas_clk 0:oscin; 1:fclk_div4; 2:fclk_div3; 3:fclk_div5; 4:vid_pll; 5:0 6:0 7:0
8	R/W	0	CLK_EN: See the Clock Tree document for information related to cts_vdin_meas_clk
6-0	R/W	48	CLK_DIV: See the Clock Tree document for information related to cts_vdin_meas_clk

Table 6-152 HHI_MIPIDSI_PHY_CLK_CNTL 0x95

Bit(s)	R/W	Default	Description
14:12	R/W	0	mipi_dsi_phy_clk clk_sel; 0:vid_pll; 1:gp0_pll; 2:hifi_pll; 3:mpll1; 4:fclk_div2; 5:fclk_div2p5; 6:fclk_div3; 7:fclk_div7;
8	R/W	0	mipi_dsi_phy_clk clk_en
6:0	R/W	0	mipi_dsi_phy_clk clk_div

Table 6-153 HHI_NAND_CLK_CNTL 0x97

Bit(s)	R/W	Default	Description
31-12	R/W	0	unused
11-9	R/W	0	CLK_SEL: 0:cts_oscin_clk 1:fclk_div2 2:fclk_div3 3:fclk_div5 4:fclk_div7 5:mp2_clk_out 6:mp3_clk_out 7:gp0_pll_clk

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Bit(s)	R/W	Default	Description
8			Reserved
7	R/W	1	CLK_EN:
6-0	R/W	0	CLK_DIV

Table 6-154 HHI_SD_EMMC_CLK_CNTL 0x99

Bit(s)	R/W	Default	Description
31-28	R/W	0	unused
27-25	R/W	0	Sd_emmc_B_CLK_SEL: 0:cts_oscin_clk 1:fclk_div2 2:fclk_div3 3:fclk_div5 4:fclk_div7 5:mp2_clk_out 6:mp3_clk_out 7:gp0_pll_clk
24			Reserved
23	R/W	1	Sd_emmc_B_CLK_EN:
22-16	R/W	0	Sd_emmc_B_CLK_DIV
15-12	R/W	0	reseverd

Table 6-155 HHI_WAVE420L_CLK_CNTL 0x9A

Bit(s)	R/W	Default	Description
27:25	R/W	0	wave420l_cclk clk_sel; 0:oscin; 1:fclk_div4; 2:fclk_div3; 3:fclk_div5; 4:fclk_div7; 5:mpll2; 6:mpll3; 7:gp0pll;
24	R/W	0	wave420l_cclk clk_en
22:16	R/W	0	wave420l_cclk clk_div
11:9	R/W	0	wave420l_bclk clk_sel; 0:oscin; 1:fclk_div4; 2:fclk_div3; 3:fclk_div5; 4:fclk_div7; 5:mpll2; 6:mpll3; 7:gp0pll;
8	R/W	0	wave420l_bclk clk_en
6:0	R/W	0	wave420l_bclk clk_div

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Table 6-156 HHI_WAVE420L_CLK_CNTL2 0x9B

Bit(s)	R/W	Default	Description
11:9	R/W	0	wave420l_aclk clk_sel; 0:oscin; 1:fclk_div4; 2:fclk_div3; 3:fclk_div5; 4:fclk_div7; 5:mpll2; 6:mpll3; 7:gp0pll;
8	R/W	0	wave420l_aclk clk_en
6:0	R/W	0	wave420l_aclk clk_div

Table 6-157 HHI_LVDS_TX_PHY_CNTL0 0x9a

Bit(s)	R/W	Default	Description
27-16	R/W	0	Cntl_ser_en
13	R/W	0	Cntl_prbs_en
12	R/W	0	Cntl_prbs_err_en
11-10	R/W	0	Cntl_mode_set_high
9-8	R/W	0	Cntl_mode_set_low
7-6	R/W	0	Fifo_clk_sel
4	R/W	0	Mode_port_rev
3	R/W	0	Mode_bit_rev
2	R/W	0	Mode_inv_p_n
0	R/W	0	Soft_reset

Table 6-158 HHI_LVDS_TX_PHY_CNTL1 0x9b

Bit(s)	R/W	Default	Description
31-24	R/W	0	Phy_control1
23-0	R	0	Phy_control_out

Table 6-159 HHI_TCON_CLK_CNTL 0x9c

Bit(s)	R/W	Default	Description
9-7	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_tcon_pll_clk
6	R/W	0	Clk_en:See the Clock Tree document for information related to cts_tcon_pll_clk
5-0	R/W	0	Clk_div:See the Clock Tree document for information related to cts_tcon_pll_clk

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Table 6-160 HHI_MPLL_CNTL0 0x9E

Bit(s)	R/W	Default	Description
[21:20]	R/W	0	mpll_sys_dpll_exldo
[18:16]	R/W	0	mpll_test_c
[15:12]	R/W	0	mpll_dds_reve
[11:10]	R/W	0	mpll_dds_vc_vdd
[9:8]	R/W	0	mpll_dds_vr_fb1
[7:6]	R/W	0	mpll_dds_vr_fb2
[1]	R/W	0	mpll_dds_enldo
[0]	R/W	0	mpll_dds_ldo_rupsel

Table 6-161 HHI_MPLL_CNTL1 0x9F

Bit(s)	R/W	Default	Description
[31]	R/W	0	mpll_dds0_en
[30]	R/W	0	mpll_dds0_sdm_en
[29]	R/W	0	mpll_dds0_ss_en
[28:20]	R/W	0	mpll_dds0_n_in
[13:0]	R/W	0	mpll_dds0_sdm_in

Table 6-162 HHI_MPLL_CNTL2 0xA0

Bit(s)	R/W	Default	Description
[31]	R/W	0	mpll_dds0_ir_bypass
[30]	R/W	0	mpll_dds0_load
[29]	R/W	0	mpll_dds0_lp_en
[28]	R/W	0	mpll_dds0_modesel
[20:16]	R/W	0	mpll_dds0_f_set
[15:12]	R/W	0	mpll_dds0_ir_byin
[10:8]	R/W	0	mpll_dds0_p_set
[7:4]	R/W	0	mpll_dds0_vref_cf
[3:0]	R/W	0	mpll_dds0_vref_cs

Table 6-163 HHI_MPLL_CNTL3 0xA1

Bit(s)	R/W	Default	Description
[31]	R/W	0	mpll_dds1_en
[30]	R/W	0	mpll_dds1_sdm_en
[29]	R/W	0	mpll_dds1_ss_en

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Bit(s)	R/W	Default	Description
[28:20]	R/W	0	mpll_dds1_n_in
[13:0]	R/W	0	mpll_dds1_sdm_in

Table 6-164 HHI_MPLL_CNTL4 0xA2

Bit(s)	R/W	Default	Description
[31]	R/W	0	mpll_dds1_ir_bypass
[30]	R/W	0	mpll_dds1_load
[29]	R/W	0	mpll_dds1_lp_en
[28]	R/W	0	mpll_dds1_modesel
[20:16]	R/W	0	mpll_dds1_f_set
[15:12]	R/W	0	mpll_dds1_ir_byin
[10:8]	R/W	0	mpll_dds1_p_set
[7:4]	R/W	0	mpll_dds1_vref_cf
[3:0]	R/W	0	mpll_dds1_vref_cs

Table 6-165 HHI_MPLL_CNTL5 0xA3

Bit(s)	R/W	Default	Description
[31]	R/W	0	mpll_dds2_en
[30]	R/W	0	mpll_dds2_sdm_en
[29]	R/W	0	mpll_dds2_ss_en
[28:20]	R/W	0	mpll_dds2_n_in
[13:0]	R/W	0	mpll_dds2_sdm_in

Table 6-166 HHI_MPLL_CNTL6 0xA4

Bit(s)	R/W	Default	Description
[31]	R/W	0	mpll_dds2_ir_bypass
[30]	R/W	0	mpll_dds2_load
[29]	R/W	0	mpll_dds2_lp_en
[28]	R/W	0	mpll_dds2_modesel
[20:16]	R/W	0	mpll_dds2_f_set
[15:12]	R/W	0	mpll_dds2_ir_byin
[10:8]	R/W	0	mpll_dds2_p_set
[7:4]	R/W	0	mpll_dds2_vref_cf
[3:0]	R/W	0	mpll_dds2_vref_cs

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Table 6-167 HHI_MPLL_CNTL7 0xA5

Bit(s)	R/W	Default	Description
[31]	R/W	0	mpll_dds3_en
[30]	R/W	0	mpll_dds3_sdm_en
[29]	R/W	0	mpll_dds3_ss_en
[28:20]	R/W	0	mpll_dds3_n_in
[13:0]	R/W	0	mpll_dds3_sdm_in

Table 6-168 HHI_MPLL_CNTL8 0xA6

Bit(s)	R/W	Default	Description
[31]	R/W	0	mpll_dds3_ir_bypass
[30]	R/W	0	mpll_dds3_load
[29]	R/W	0	mpll_dds3_lp_en
[28]	R/W	0	mpll_dds3_modesel
[20:16]	R/W	0	mpll_dds3_f_set
[15:12]	R/W	0	mpll_dds3_ir_byin
[10:8]	R/W	0	mpll_dds3_p_set
[7:4]	R/W	0	mpll_dds3_vref_cf
[3:0]	R/W	0	mpll_dds3_vref_cs

Table 6-169 HHI_MPLL_STS 0xA7

Bit(s)	R/W	Default	Description
[28]	R	0	mpll_dds0_ir_done
[27:24]	R	0	mpll_dds0_ir_out
[20]	R	0	mpll_dds1_ir_done
[19:16]	R	0	mpll_dds1_ir_out
[12]	R	0	mpll_dds2_ir_done
[11:8]	R	0	mpll_dds2_ir_out
[4]	R	0	mpll_dds3_ir_done
[3:0]	R	0	mpll_dds3_ir_out

Table 6-170 HHI_FIX_PLL_CNTL0 0xA8

Bit(s)	R/W	Default	Description
[31]	R	0	fix_dpll_lock
[30]	R	0	fix_dpll_lock_a
[29]	R/W	1	fix_dpll_reset

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Bit(s)	R/W	Default	Description
[28]	R/W	0	fix_dpll_en
[17:16]	R/W	0	fix_dpll_od
[14:10]	R/W	0	fix_dpll_N
[8:0]	R/W	0	fix_dpll_M

Table 6-171 HHI_FIX_PLL_CNTL1 0xA9

Bit(s)	R/W	Default	Description
[25]	R/W	0	fix_dpll_div2p5_en
[24:20]	R/W	0	fixpll_dpll_clken
[18:0]	R/W	0	fix_dpll_frac

Table 6-172 HHI_FIX_PLL_CNTL2 0xAA

Bit(s)	R/W	Default	Description
[22:20]	R/W	0	fix_dpll_fref_sel
[17:16]	R/W	0	fix_dpll_os_ssc
[15:12]	R/W	0	fix_dpll_ssc_str_m
[8]	R/W	0	fix_dpll_ssc_en
[7:4]	R/W	0	fix_dpll_ssc_dep_sel
[1:0]	R/W	0	fix_dpll_ss_mode

Table 6-173 HHI_FIX_PLL_CNTL3 0xAB

Bit(s)	R/W	Default	Description
31	R/W	0	fix_dpll_afc_bypass
30	R/W	0	fix_dpll_afc_clk_sel
29	R/W	0	fix_dpll_code_new
28	R/W	0	fix_dpll_dco_m_en
27	R/W	0	fix_dpll_dco_sdm_en
26	R/W	0	fix_dpll_od_en
25	R/W	0	fix_dpll_div_mode
24	R/W	0	fix_dpll_fast_lock
23	R/W	0	fix_dpll_fb_pre_div
22	R/W	0	fix_dpll_filter_mode
21	R/W	0	fix_dpll_fix_en
20	R/W	0	fix_dpll_freq_shift_en
19	R/W	0	fix_dpll_load

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Bit(s)	R/W	Default	Description
18	R/W	0	fix_dpll_load_en
17	R/W	0	fix_dpll_lock_f
16	R/W	0	fix_dpll_pulse_width_en
15	R/W	0	fix_dpll_sdmnc_en
14	R/W	0	fix_dpll_sdmnc_mode
13	R/W	0	fix_dpll_sdmnc_range
12	R/W	0	fix_dpll_tdc_en
11	R/W	0	fix_dpll_tdc_mode_sel
10	R/W	0	fix_dpll_wait_en
7	R/W	0	fix_dpll_ssclk_sel
6	R/W	0	fix_dpll_clk_irsel
5	R/W	0	fix_dpll_clk50m_en

Table 6-174 HHI_FIX_PLL_CNTL4 0xAC

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	fix_dpll_alpha
[27:24]	R/W	0	fix_dpll_rou
[18:16]	R/W	0	fix_dpll_lambda0
[13:12]	R/W	0	fix_dpll_acq_gain
[11:8]	R/W	0	fix_dpll_filter_pvt2
[7:4]	R/W	0	fix_dpll_filter_pvt1
[1:0]	R/W	0	fix_dpll_pfd_gain

Table 6-175 HHI_FIX_PLL_CNTL5 0xAD

Bit(s)	R/W	Default	Description
[30:28]	R/W	0	fix_dpll_adj_vco_ldo
[27:24]	R/W	0	fix_dpll_lm_w
[21:16]	R/W	0	fix_dpll_lm_s
[15:0]	R/W	0	fix_dpll_reve

Table 6-176 HHI_FIX_PLL_CNTL6 0xAE

Bit(s)	R/W	Default	Description
[31:30]	R/W	0	fix_dpll_afc_hold_t
[29:28]	R/W	0	fix_dpll_lkw_sel
[27:26]	R/W	0	fix_dpll_dco_sdm_clk_sel

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Bit(s)	R/W	Default	Description
[25:24]	R/W	0	fix_dpll_afc_in
[23:22]	R/W	0	fix_dpll_afc_nt
[21:20]	R/W	0	fix_dpll_vc_in
[19:18]	R/W	0	fix_dpll_lock_long
[17:16]	R/W	0	fix_dpll_freq_shift_v
[14:12]	R/W	0	fix_dpll_data_sel
[10:8]	R/W	0	fix_dpll_sdmnc_ulms
[6:0]	R/W	0	fix_dpll_sdmnc_power

Table 6-177 HHI_FIX_PLL_STS 0xAF

Bit(s)	R/W	Default	Description
[31]	R	0	fix_dpll_lock
[30]	R	0	fix_dpll_lock_a
[29]	R	0	fix_dpll_afc_done
[22:16]	R	0	fix_dpll_sdmnc_monitor
[9:0]	R	0	fix_dpll_out_rsv

Table 6-178 HHI_ADC_PLL_CNTL0 0xb0

Bit(s)	R/W	Default	Description
[31]	R	0	adc_dpll_lock
[30]	R	0	adc_dpll_lock_a
[29]	R/W	1	adc_dpll_reset
[28]	R/W	0	adc_dpll_en
[24]	R/W	0	adc_div2_l_en
[23:22]	R/W	0	adc_dpll_od3
[21:20]	R/W	0	adc_dpll_od2
[19:18]	R/W	0	adc_dpll_od1
[17:16]	R/W	0	adc_dpll_od0
[14:10]	R/W	0	adc_dpll_ref_div_n
[8:0]	R/W	0	adc_dpll_int_num

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Table 6-179 HHI_ADC_PLL_CNTL1 0xb1

Bit(s)	R/W	Default	Description
[31:27]	R/W	0	adc_dpll_xd1
[26:22]	R/W	0	adc_dpll_xd0
[18:0]	R/W	0	adc_dpll_frac

Table 6-180 HHI_ADC_PLL_CNTL2 0xb2

Bit(s)	R/W	Default	Description
[31:29]	R/W	0	adc_dpll_gate_ctrl
[28:27]	R/W	0	adc_dpll_output_polar_control
[26:23]	R/W	0	adc_dpll_output_mux_ctrl
[22:20]	R/W	0	adc_dpll_fref_sel
[17:16]	R/W	0	adc_dpll_os_ssc
[15:12]	R/W	0	adc_dpll_ssc_str_m
[8]	R/W	0	adc_dpll_ssc_en
[7:4]	R/W	0	adc_dpll_ssc_dep_sel
[1:0]	R/W	0	adc_dpll_ss_mode

Table 6-181 HHI_ADC_PLL_CNTL3 0xb3

Bit(s)	R/W	Default	Description
31	R/W	0	adc_dpll_afc_bypass
30	R/W	0	adc_dpll_afc_clk_sel
29	R/W	0	adc_dpll_code_new
28	R/W	0	adc_dpll_dco_m_en
27	R/W	0	adc_dpll_dco_sdm_en
25	R/W	0	adc_dpll_div_mode
24	R/W	0	adc_dpll_fast_lock
23	R/W	0	adc_dpll_fb_pre_div
22	R/W	0	adc_dpll_filter_mode
21	R/W	0	adc_dpll_fix_en
20	R/W	0	adc_dpll_freq_shift_en
19	R/W	0	adc_dpll_load
18	R/W	0	adc_dpll_load_en
17	R/W	0	adc_dpll_lock_f
16	R/W	0	adc_dpll_pulse_width_en
15	R/W	0	adc_dpll_sdmnc_en

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Bit(s)	R/W	Default	Description
14	R/W	0	adc_dpll_sdmnc_mode
13	R/W	0	adc_dpll_sdmnc_range
12	R/W	0	adc_dpll_tdc_en
11	R/W	0	adc_dpll_tdc_mode_sel
10	R/W	0	adc_dpll_wait_en

Table 6-182 HHI_ADC_PLL_CNTL4 0xb4

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	adc_dpll_alpha
[27:24]	R/W	0	adc_dpll_rou
[18:16]	R/W	0	adc_dpll_lambda0
[13:12]	R/W	0	adc_dpll_acq_gain
[11:8]	R/W	0	adc_dpll_filter_pvt2
[7:4]	R/W	0	adc_dpll_filter_pvt1
[1:0]	R/W	0	adc_dpll_pfd_gain

Table 6-183 HHI_ADC_PLL_CNTL5 0xb5

Bit(s)	R/W	Default	Description
[30:28]	R/W	0	adc_dpll_adj_vco_ldo
[27:24]	R/W	0	adc_dpll_lm_w
[21:16]	R/W	0	adc_dpll_lm_s
[15:0]	R/W	0	adc_dpll_reve

Table 6-184 HHI_ADC_PLL_CNTL6 0xb6

Bit(s)	R/W	Default	Description
[31:30]	R/W	0	adc_dpll_afc_hold_t
[29:28]	R/W	0	adc_dpll_lkw_sel
[27:26]	R/W	0	adc_dpll_dco_sdm_clk_sel
[25:24]	R/W	0	adc_dpll_afc_in
[23:22]	R/W	0	adc_dpll_afc_nt
[21:20]	R/W	0	adc_dpll_vc_in
[19:18]	R/W	0	adc_dpll_lock_long
[17:16]	R/W	0	adc_dpll_freq_shift_v
[14:12]	R/W	0	adc_dpll_data_sel

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Bit(s)	R/W	Default	Description
[10:8]	R/W	0	adc_dpll_sdmnc_ulms
[6:0]	R/W	0	adc_dpll_sdmnc_power

Table 6-185 HHI_ADC_PLL_STS 0xb7

Bit(s)	R/W	Default	Description
[31]	R	0	adc_dpll_lock
[30]	R	0	adc_dpll_lock_a
[29]	R	0	adc_dpll_afc_done
[22:16]	R	0	adc_dpll_sdmnc_monitor
[9:0]	R	0	adc_dpll_out_rsv

Table 6-186 HHI_HDMIRX_AXI_CLK_CNTL 0xb8

Bit(s)	R/W	Default	Description
10-9	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_hdmirx_axi_clk
8	R/W	0	Clk_en:See the Clock Tree document for information related to cts_hdmirx_axi_clk
6-0	R/W	0	Clk_div:See the Clock Tree document for information related to cts_hdmirx_axi_clk

Table 6-187 HHI_VDAC_CNTL0 0xBB

Bit(s)	R/W	Default	Description
27	R/W	0	CDAC_BIAS_C
26	R/W	0	CDAC_EXT_VREF_EN
25	R/W	0	CDAC_DRIVER_ADJ
24	R/W	0	CDAC_CLK_PHASE_SEL
23~21	R/W	0	CDAC_RL_ADJ
20~16	R/W	0	CDAC_VREF_ADJ
15~8	R/W	0	CDAC_CTRL_RESV2
7~0	R/W	0	CDAC_CTRL_RESV1

Table 6-188 HHI_VDAC_CNTL1 0xBC

Bit(s)	R/W	Default	Description
23~16	R	0	CDAC_DIG_OUT_RESV
11~4	R/W	0	Cdac_ctrl_resv3

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Bit(s)	R/W	Default	Description
3	R/W	0	Cdac_pwd
2~0	R/W	0	CDAC_GSW

Table 6-189 HHI_SYS_PLL_CNTL0 0xBD

Bit(s)	R/W	Default	Description
[31]	R	0	sys_dpll_lock
[30]	R	0	sys_dpll_lock_a
[29]	R/W	1	sys_dpll_reset
[28]	R/W	0	sys_dpll_en
[18:16]	R/W	0	sys_dpll_od
[14:10]	R/W	0	sys_dpll_N
[8:0]	R/W	0	sys_dpll_M

Table 6-190 HHI_SYS_PLL_CNTL1 0xBE

Bit(s)	R/W	Default	Description
[18:0]	R/W	0	sys_dpll_frac

Table 6-191 HHI_SYS_PLL_CNTL2 0xBF

Bit(s)	R/W	Default	Description
[22:20]	R/W	0x0	sys_dpll_fref_sel
[17:16]	R/W	0x0	sys_dpll_os_ssc
[15:12]	R/W	0x0	sys_dpll_ssc_str_m
[8]	R/W	0x0	sys_dpll_ssc_en
[7:4]	R/W	0x0	sys_dpll_ssc_dep_sel
[1:0]	R/W	0x0	sys_dpll_ss_mode

Table 6-192 HHI_SYS_PLL_CNTL3 0xC0

Bit(s)	R/W	Default	Description
31	R/W	0x0	sys_dpll_afc_bypass
30	R/W	0x0	sys_dpll_afc_clk_sel
29	R/W	0x0	sys_dpll_code_new
28	R/W	0x0	sys_dpll_dco_m_en
27	R/W	0x0	sys_dpll_dco_sdm_en
26	R/W	0x0	
25	R/W	0x0	sys_dpll_div_mode

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Bit(s)	R/W	Default	Description
24	R/W	0x0	sys_dpll_fast_lock
23	R/W	0x0	sys_dpll_fb_pre_div
22	R/W	0x0	sys_dpll_filter_mode
21	R/W	0x0	sys_dpll_fix_en
20	R/W	0x0	sys_dpll_freq_shift_en
19	R/W	0x0	sys_dpll_load
18	R/W	0x0	sys_dpll_load_en
17	R/W	0x0	sys_dpll_lock_f
16	R/W	0x0	sys_dpll_pulse_width_en
15	R/W	0x0	sys_dpll_sdmnc_en
14	R/W	0x0	sys_dpll_sdmnc_mode
13	R/W	0x0	sys_dpll_sdmnc_range
12	R/W	0x0	sys_dpll_tdc_en
11	R/W	0x0	sys_dpll_tdc_mode_sel
10	R/W	0x0	sys_dpll_wait_en

Table 6-193 HHI_SYS_PLL_CNTL4 0xC1

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	sys_dpll_alpha
[27:24]	R/W	0	sys_dpll_rou
[22:20]	R/W	0	
[18:16]	R/W	0	sys_dpll_lambda0
[13:12]	R/W	0	sys_dpll_acq_gain
[11:8]	R/W	0	sys_dpll_filter_pvt2
[7:4]	R/W	0	sys_dpll_filter_pvt1
[1:0]	R/W	0	sys_dpll_pfd_gain

Table 6-194 HHI_SYS_PLL_CNTL5 0xC2

Bit(s)	R/W	Default	Description
[30:28]	R/W	0	sys_dpll_adj_vco_ldo
[27:24]	R/W	0	sys_dpll_lm_w
[21:16]	R/W	0	sys_dpll_lm_s
[15:0]	R/W	0	sys_dpll_reve

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Table 6-195 HHI_SYS_PLL_CNTL6 0xC3

Bit(s)	R/W	Default	Description
[31:30]	R/W	0	sys_dpll_afc_hold_t
[29:28]	R/W	0	sys_dpll_lkw_sel
[27:26]	R/W	0	sys_dpll_dco_sdm_clk_sel
[25:24]	R/W	0	sys_dpll_afc_in
[23:22]	R/W	0	sys_dpll_afc_nt
[21:20]	R/W	0	sys_dpll_vc_in
[19:18]	R/W	0	sys_dpll_lock_long
[17:16]	R/W	0	sys_dpll_freq_shift_v
[14:12]	R/W	0	sys_dpll_data_sel
[10:8]	R/W	0	sys_dpll_sdmnc_ulms
[6:0]	R/W	0	sys_dpll_sdmnc_power

Table 6-196 HHI_SYS_PLL_STS 0xC4

Bit(s)	R/W	Default	Description
[31]	R	0	sys_dpll_lock
[30]	R	0	sys_dpll_lock_a
[29]	R	0	sys_dpll_afc_done
[22:16]	R	0	sys_dpll_sdmnc_monitor
[9:0]	R	0	sys_dpll_out_rsv

Table 6-197 HHI_HDMI_PLL_VLOCK_CNTL 0xd1

Bit(s)	R/W	Default	Description
[1]	R/W	0	Vlcok_adj_en_to_pll sel: 1:from vlock 0:from hi_hdmi_pll_vlock_cntl[2]
[0]	R/W	0	Tcon framc fbkdiv sel: 1:tcon_pll_framc and tcon_pll_fbkdiv from vlock 0:tcon_pll_framc and tcon_pll_fbkdiv from hiu register

Table 6-198 HHI_HDMIRX_APLL_PLL_CNTL0 0xd2

Bit(s)	R/W	Default	Description
[31]	R	0	hdmirx_apll_lock
[30]	R	0	hdmimrx_apll_afc_done
[29]	R/W	0	apll_reset
[28]	R/W	0	apll_en
[27]	R/W	0	apll_vco_div_sel

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Bit(s)	R/W	Default	Description
[26]	R/W	0	apll_prediv_sel
[14:10]	R/W	0	apll_afc_start
[7:0]	R/W	0	apll_fbkdiv

Table 6-199 HHI_HDMIRX_APLL_PLL_CNTL1 0xd3

Bit(s)	R/W	Default	Description
[20]	R/W	0	apll_sdm_en
[18:0]	R/W	0	apll_sdm_frac

Table 6-200 HHI_HDMIRX_APLL_PLL_CNTL2 0xd4

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	apll_ssc_dep_sel
[26:24]	R/W	0	apll_ssc_fref_sel
[23:22]	R/W	0	apll_ssc_mode
[21:20]	R/W	0	apll_ssc_offset
[19:16]	R/W	0	apll_str_m
[15:0]	R/W	0	apll_reserve

Table 6-201 HHI_HDMIRX_APLL_PLL_CNTL3 0xd5

Bit(s)	R/W	Default	Description
[31]	R/W	0	apll_afc_bypass_en
[29:28]	R/W	0	apll_afc_hold_t
[26:20]	R/W	0	apll_afc_in
[19]	R/W	0	apll_afc_nt
[18:17]	R/W	0	apll_afc_div
[16]	R/W	0	apll_bias_lpf_en
[15:12]	R/W	0	apll_cp_icap
[11:8]	R/W	0	apll_cp_ires
[5:4]	R/W	0	apll_cpi
[2:0]	R/W	0	apll_hdmi_aud_cntl

Table 6-202 HHI_HDMIRX_APLL_PLL_CNTL4 0xd6

Bit(s)	R/W	Default	Description
[27:24]	R/W	0	apll_hdmi_cntl
[23]	R/W	0	apll_vctrl_mon_en

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Bit(s)	R/W	Default	Description
[21:20]	R/W	0	apll_lpf_cap
[19:16]	R/W	0	apll_lpf_capadj
[13:12]	R/W	0	apll_lpf_res
[11]	R/W	0	apll_lpf_sf
[10]	R/W	0	apll_lvr_od_en
[9]	R/W	0	apll_refclk_mon_en
[8]	R/W	0	apll_fbkclk_mon_en

Table 6-203 HHI_HDMIRX_PHY_MISC0 0xd7

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_misci[31:0]

Table 6-204 HHI_HDMIRX_PHY_MISC1 0xd8

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_misci[63:32]

Table 6-205 HHI_DIF_CSI_PHY_CNTL1 0xd9

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch1_reg
15:0	R/W	0	Ch0_reg

Table 6-206 HHI_DIF_CSI_PHY_CNTL2 0xda

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch3_reg
15:0	R/W	0	Ch2_reg

Table 6-207 HHI_DIF_CSI_PHY_CNTL3 0xdb

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch5_reg
15:0	R/W	0	Ch4_reg

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Table 6-208 HHI_DIF_CSI_PHY_CNTL16 0xde

Bit(s)	R/W	Default	Description
31	R/W	0	P2p_rb_in
23:12	R/W	0	ckdi
11:0	R/W	0	ck12i

Table 6-209 HHI_TCON_PLL_CNTL4 0xdf

Bit(s)	R/W	Default	Description
[31:28]	R/W	0	tcon_picode1
[27:24]	R/W	0	tcon_picode2
[23]	R/W	0	tcon_pll_vctrl_mon_en
[21:20]	R/W	0	tcon_pll_lpf_cap
[19:16]	R/W	0	tcon_pll_lpf_capadj
[13:12]	R/W	0	tcon_pll_lpf_res
[11]	R/W	0	tcon_pll_lpf_sf
[10]	R/W	0	tcon_pll_lvr_od_en
[9]	R/W	0	tcon_pll_refclk_mon_en
[8]	R/W	0	tcon_pll_fbkclk_mon_en

Table 6-210 HHI_HDMIRX_PHY_MISC2 0xe0

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_misci[95:64]

Table 6-211 HHI_HDMIRX_PHY_MISC3 0xe1

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_misci[127:96]

Table 6-212 HHI_HDMIRX_PHY_DCHA0 0xe2

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_dcha[31:0]

Table 6-213 HHI_HDMIRX_PHY_DCHA1 0xe3

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_dcha[63:32]

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Table 6-214 HHI_HDMIRX_PHY_DCHA2 0xe4

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_dcha[95:64]

Table 6-215 HHI_HDMIRX_PHY_DCHD0 0xe5

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_dchd[31:0]

Table 6-216 HHI_HDMIRX_PHY_DCHD1 0xe6

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_dchd[63:32]

Table 6-217 HHI_HDMIRX_PHY_DCHD2 0xe7

Bit(s)	R/W	Default	Description
[31:0]	R/W	0	Reg_dchd[95:64]

Table 6-218 HHI_HDMIRX_ARC_CNTL 0xe8

Bit(s)	R/W	Default	Description
[5]	R/W	0	Arc debug single when [4:2] == 3'b000
[4:2]	R/W	0	Audio_spdif source sel: 3'b001: audio_spdif_a 3'b010:audio_spdif_b 3'b100:hdmir_aud_spdif 3'b000:hi_hdmirx_arc_cntl[5]
[1]	R/W	0	arc_cmdc_valid
[0]	R/W	0	arc_dmac_valid

Table 6-219 HHI_DIF_CSI_PHY_CNTL4 0xe9

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch7_reg
15:0	R/W	0	Ch6_reg

Table 6-220 HHI_DIF_CSI_PHY_CNTL6 0xea

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch9_reg
15:0	R/W	0	Ch8_reg

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Table 6-221 HHI_DIF_CSI_PHY_CNTL7 0xeb

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch11_reg
15:0	R/W	0	Ch10_reg

Table 6-222 HHI_DIF_CSI_PHY_CNTL8 0xec

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch1_chdig_ctl
15:0	R/W	0	Ch0_chdig_ctl

Table 6-223 HHI_DIF_CSI_PHY_CNTL9 0xed

Bit(s)	R/W	Default	Description
31:16	R/W	0	Ch3_chdig_ctl
15:0	R/W	0	Ch2_chdig_ctl

Table 6-224 HHI_HDMIRX_PHY_MISC_STAT 0xee

Bit(s)	R/W	Default	Description
[10:1]	R	0	sta_rtermenb
[0]	R	0	sta_rcallock

Table 6-225 HHI_HDMIRX_PHY_DCHD_STAT 0xef

Bit(s)	R/W	Default	Description
[23:16]	R	0	sta_dchd2
[15:8]	R	0	sta_dchd1
[7:0]	R	0	sta_dchd0

Table 6-226 HHI_VID_LOCK_CLK_CNTL 0xF2

Bit(s)	R/W	Default	Description
31~10	R/W	0	reserved
9-8	R/W	0	Clk_sel: 0:cts_oscin_clk 1:cts_encl_clk 2:cts_enci_clk 3:cts_encp_clk
7	R/W	0	Clk_en
6-0	R/W	0	Clk_div

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Table 6-227 HHI_ATV_DMD_SYS_CLK_CNTL 0xf3

Bit(s)	R/W	Default	Description
25-24	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_vafe_datack
23	R/W	0	Clk_en:See the Clock Tree document for information related to cts_vafe_datack
22-16	R/W	0	Clk_div:See the Clock Tree document for information related to cts_vafe_datack
9-8	R/W	0	Clk_sel:See the Clock Tree document for information related to cts_atv_dmd_ sys_clk
7	R/W	0	Clk_en:See the Clock Tree document for information related to cts_atv_dmd_ sys_clk
6-0	R/W	0	Clk_div:See the Clock Tree document for information related to cts_atv_dmd_ sys_clk

Table 6-228 HHI_AXI_PIPEL_CNTL 0xF4

Bit(s)	R/W	Default	Description
31	R	-	MALI DMC Pipeline control: CHAN_IDLE
30	R/W	1	MALI DMC Pipeline control: REQ_EN
29	R/W	1	MALI DMC Pipeline control: AUTO_GCLK_EN
28	R/W	0	MALI DMC Pipeline control: DISABLE_GCLK
27	R	-	DOS DMC Pipeline1 control: CHAN_IDLE
26	R/W	1	DOS DMC Pipeline1 control: REQ_EN
25	R/W	1	DOS DMC Pipeline1 control: AUTO_GCLK_EN
24	R/W	0	DOS DMC Pipeline1 control: DISABLE_GCLK
23	R	-	DOS DMC Pipeline0 control: CHAN_IDLE
22	R/W	1	DOS DMC Pipeline0 control: REQ_EN
21	R/W	1	DOS DMC Pipeline0 control: AUTO_GCLK_EN
20	R/W	0	DOS DMC Pipeline0 control: DISABLE_GCLK
19	R	-	GE2D DMC Pipeline control: CHAN_IDLE
18	R/W	1	GE2D DMC Pipeline control: REQ_EN
17	R/W	1	GE2D DMC Pipeline control: AUTO_GCLK_EN
16	R/W	0	GE2D DMC Pipeline control: DISABLE_GCLK
15	R	-	VPU2 DMC Pipeline control: CHAN_IDLE
14	R/W	1	VPU2 DMC Pipeline control: REQ_EN
13	R/W	1	VPU2 DMC Pipeline control: AUTO_GCLK_EN
12	R/W	0	VPU2 DMC Pipeline control: DISABLE_GCLK
11	R	-	VPU1 DMC Pipeline control: CHAN_IDLE
10	R/W	1	VPU1 DMC Pipeline control: REQ_EN
9	R/W	1	VPU1 DMC Pipeline control: AUTO_GCLK_EN
8	R/W	0	VPU1 DMC Pipeline control: DISABLE_GCLK

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Bit(s)	R/W	Default	Description
7	R	-	VPU0 DMC Pipeline control: CHAN_IDLE
6	R/W	1	VPU0 DMC Pipeline control: REQ_EN
5	R/W	1	VPU0 DMC Pipeline control: AUTO_GCLK_EN
4	R/W	0	VPU0 DMC Pipeline control: DISABLE_GCLK
3	R	-	HCODEC DMC PipelinePipeline control: CHAN_IDLE
2	R/W	1	HCODEC DMC PipelinePipeline control: REQ_EN
1	R/W	1	HCODEC DMC PipelinePipeline control: AUTO_GCLK_EN
0	R/W	0	HCODEC DMC PipelinePipeline control: DISABLE_GCLK

Table 6-229 HHI_BT656_CLK_CNTL 0xF5

Bit(s)	R/W	Default	Description
10-9	R/W	0	Bt656_1_clk_sel 0:fclk_div2 1:fclk_div3 2:fclk_div5 3:fclk_div7
8	R/W	0	Reserved
7	R/W	0	Bt656_1_clk_en
6-0	R/W	0	Bt656_1_clk_div

Table 6-230 HHI_CDAC_CLK_CNTL 0xF6

Bit(s)	R/W	Default	Description
20	R/W	0	vdac_clk_c clk_en
17:16	R/W	0	vdac_clk_c clk_sel 0:osc; 1:fclk_div5;
15:0	R/W	0	vdac_clk_c: clk_div

Table 6-231 HHI_SPICC_CLK_CNTL 0xF7

Bit(s)	R/W	Default	Description
25:23	R/W	0	spicc_1_clk clk_sel 0:osc; 1:clk81; 2:fclk_div4; 3:fclk_div3; 4:fclk_div2; 5:fclk_div5; 6:fclk_div7; 7:gp0_pll;
22	R/W	0	spicc_1_clk clk_en
21:16	R/W	0	spicc_1_clk clk_div
9:7	R/W	0	spicc_0_clk clk_sel

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Bit(s)	R/W	Default	Description
			0:osc; 1:clk81; 2:fclk_div4; 3:fclk_div3; 4:fclk_div2; 5:fclk_div5; 6:fclk_div7; 7:gp0_pll;
6	R/W	0	spicc_0_clk clk_en
5:0	R/W	0	spicc_0_clk clk_div

Table 6-232 HHI_HDMIRX_AUD_PLL_CNTL 0xF8

Bit(s)	R/W	Default	Description
30	R/W	0	aud_dds_en
29	R/W	0	aud_dds_reset
15	R/W	0	aud_dds_update_sel
14	R/W	0	aud_dds_dsel_in
13	R/W	0	aud_dds_ext_clkdiv2_en
12	R/W	0	aud_dds_lrg_lvrod_en
11:8	R/W	0	aud_dds_rc_filter
7:6	R/W	0	aud_dds_ldo_ad
5	R/W	0	aud_dds_sin_cos
4:0	R/W	0	aud_dds_data_mux

Table 6-233 HHI_HDMIRX_AUD_PLL_CNTL2 0xF9

Bit(s)	R/W	Default	Description
11:0	R/W	0	aud_dds_dac_cntrl

Table 6-234 HHI_HDMIRX_AUD_PLL_CNTL3 0xFA

Bit(s)	R/W	Default	Description
31:16	R/W	0	aud_dds_tmds_flag
10:8	R/W	0	aud_dds_ss_adder_sel
7	R/W	0	aud_dds_ss_amp
6:4	R/W	0	aud_dds_ss_en
3	R/W	0	aud_dds_ss_speed
2:0	R/W	0	aud_dds_digi_reve

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Table 6-235 HHI_AXI_PIPEL_CNTL 0xFb

Bit(s)	R/W	Default	Description
6	R/W	1	HDMIRX DMC Pipeline1 control: REQ_EN
5	R/W	1	HDMIRX DMC Pipeline1 control: AUTO_GCLK_EN
4	R/W	0	HDMIRX DMC Pipeline1 control: DISABLE_GCLK
3	R	-	HDMIRX DMC Pipeline0 control: CHAN_IDLE
2	R/W	1	HDMIRX DMC Pipeline0 control: REQ_EN
1	R/W	1	HDMIRX DMC Pipeline0 control: AUTO_GCLK_EN
0	R/W	0	HDMIRX DMC Pipeline0 control: DISABLE_GCLK

Table 6-236 HHI_HDMIRX_AUD_PLL_CNTL_I 0xfe

Bit(s)	R/W	Default	Description
[31]	R	0	aud_pll_lock
[7:0]	R	0	aud_pll_out_rsv

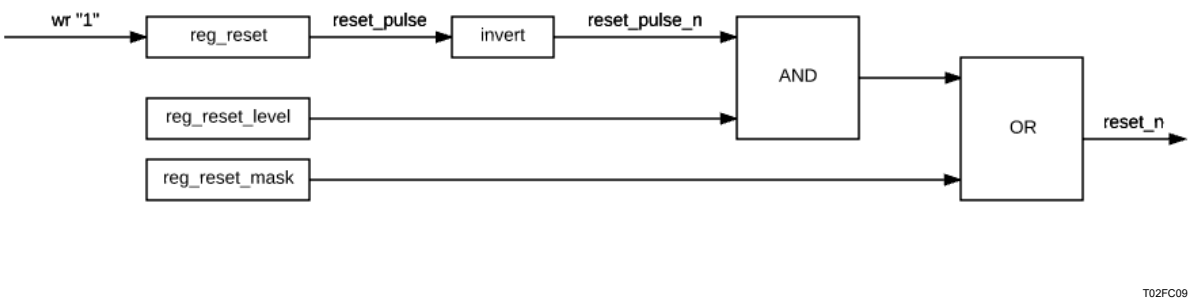
6.8 Reset

6.8.1 Overview

The following figure shows the reset design.

- Each reset includes 3 control registers: reg_reset, reg_reset_level, and reg_reset_mask.
- When write “1” to reg_reset, it will generate a signal: reset_pulse.
- Most of module’s reset is negative, so reset_pulse will invert to reset_pulse_n.
- The usage of reg_reset_level is hold reset_pulse_n to “0”.
- The usage of reg_reset_mask is hold reset_pulse_n to “1”.
- AO module has special soft reset control.
- A55 has special soft reset control.

Figure 6-24 Reset Design



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6.8.2 Register Description

6.8.2.1 EE Reset

Base_address: 0xFFD0_1000

Each register final address = base_address + address * 4

Table 6-237 RESET0_REGISTER: 0xFFD01004

Bit(s)	R/W	Default	Description
31-27	R/W	0	Reserved
26	R/W	0	vcbus
25	R/W	0	ahb_data
24	R/W	0	ahb_cntl
23	R/W	0	cbus_capb3
22	R/W	0	Reserved
21	R/W	0	dos_capb3
20	R/W	0	Dvalin_capb3
19	R/W	0	-
18	R/W	0	hdmirx_apb
17	R/W	0	decode_capb3
16	R/W	0	gic
15	R/W	0	-
14	R/W	0	-
13	R/W	0	vcbus
12	R/W	0	-
11	R/W	0	assist
10	R/W	0	venc
9	R/W	0	hdmirx
8	R/W	0	demod
7	R/W	0	vid_pll_div
6	R/W	0	afifo
5	R/W	0	viu
4	R/W	0	Reserved
3	R/W	0	Reserved
2	R/W	0	dos
1	R/W	0	Reserved
0	R/W	0	hiu

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Table 6-238 RESET1_REGISTER: 0xFFD01008

Bit(s)	R/W	Default	Description
31-30	R/W	0	Reserved
29	R/W	0	Audio codec
28:19	R/W	0	Reserved
18	R/W	0	u2phy22
17	R/W	0	u2phy21
16	R/W	0	u2phy20
15	R/W	0	Reserved
14	R/W	0	sd_emmcC
13	R/W	0	sd_emmcB
12	R/W	0	Reserved
11	R/W	0	eth
10	R/W	0	isa
9	R/W	0	avt_dmd
8	R/W	0	parser
7	R/W	0	tvfe_top
6	R/W	0	ahb_sram
5	R/W	0	bt656
4	R/W	0	aml_tcon
3	R/W	0	ddr
2	R/W	0	usb
1	R/W	0	demux
0	R/W	0	Reserved

Table 6-239 RESET2_REGISTER: 0xFFD0100C

Bit(s)	R/W	Default	Description
14	R/W	0	Dvalin
13	R/W	0	Reserved
12	R/W	0	Reserved
11	R/W	0	Reserved
10	R/W	0	parser_top
9	R/W	0	parser_ctl
8	R/W	0	parser_fetch
7	R/W	0	parser_reg
6	R/W	0	ge2d
5	R/W	0	Reserved
4	R/W	0	Reserved

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Bit(s)	R/W	Default	Description
3	R/W	0	adec_top_wrap
2	R/W	0	Reserved
1	R/W	0	audio
0	R/W	0	Reserved

Table 6-240 RESET3_REGISTER: 0xFFD01010

Bit(s)	R/W	Default	Description
15	R/W	0	demux_2
14	R/W	0	demux_1
13	R/W	0	demux_0
12	R/W	0	demux_s2p_1
11	R/W	0	demux_s2p_0
10	R/W	0	demux_des_pl
9	R/W	0	demux_top
8-1	R/W	0	Reserved
0	R/W	0	demux_s2p_2

Table 6-241 RESET4_REGISTER: 0xFFD01014

Bit(s)	R/W	Default	Description
15	R/W	0	i2c_m2
14	R/W	0	i2c_m1
13	R/W	0	vencl
12	R/W	0	vdi6
11-10	R/W	0	Reserved
9	R/W	0	vdac
8	R/W	0	Reserved
7	R/W	0	vencp
6	R/W	0	venci
5	R/W	0	rdma
4-0	R/W	0	Reserved

Table 6-242 RESET6_REGISTER: 0xFFD0101c

Bit(s)	R/W	Default	Description
15	R/W	0	async3
14	R/W	0	i2c_m3
13	R/W	0	spifc

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Bit(s)	R/W	Default	Description
12	R/W	0	Async2
11	R/W	0	async
10	R/W	0	uart1_2
9	R/W	0	uart0
8	R/W	0	ts_cpu
7	R/W	0	stream
6	R/W	0	spicc1
5	R/W	0	ts_pll
4	R/W	0	i2c_m0
3	R/W	0	sana_top
2	R/W	0	sc
1	R/W	0	spicc0
0	R/W	0	gen

Table 6-243 RESET7_REGISTER: 0xFFD01020

Bit(s)	R/W	Default	Description
10	R/W	0	ge2d_dmc_pipl
9	R/W	0	dmc_vpu_pipl
8	R/W	0	Reserved
7	R/W	0	vpu_vid_lock
6	R/W	0	Dvalin_dmc_pipl
5	R/W	0	device_mmc_arb
4	R/W	0	ts_gpu
3	R/W	0	usb_ddr3
2	R/W	0	Reserved
1	R/W	0	usb_ddr1
0	R/W	0	Reserved

The Bits of the following registers correspond to the RESET[n] REGISTERs above. If a bit is set in these registers, then when the watchdog timer fires, that particular module will NOT be RESET.

RESET0_MASK 0xFFD01040

RESET1_MASK 0xFFD01044

RESET2_MASK 0xFFD01048

RESET3_MASK 0xFFD0104C

RESET4_MASK 0xFFD01050

RESET5_MASK 0xFFD01054

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RESET6_MASK 0xFFD01058

RESET7_MASK 0xFFD0105C

The bits of the following registers correspond to the RESET[n] REGISTERs above. The default value of these registers is 0xFFFFFFFF. Setting any bit to 0, forces the corresponding RESET LOW. These register allow the software to “Hold” a reset LOW (in a reset condition).

RESET0_LEVEL 0xFFD0_1080

RESET1_LEVEL 0xFFD0_1084

RESET2_LEVEL 0xFFD0_1088

RESET3_LEVEL 0xFFD0_108C

RESET4_LEVEL 0xFFD0_1090

RESET5_LEVEL 0xFFD0_1094

RESET6_LEVEL 0xFFD0_1098

RESET7_LEVEL 0xFFD0_109C

6.8.2.2 AO Soft Rest

Base_address: 0xFF80_0040

Each register final address = base_address + address * 4

Table 6-244 AO Soft Rest 0xFF800044

Bit(s)	R/W	Default	Description
31-24	R/W	0	Reserved
23	R/W	0	ir_out
22	R/W	0	uart2
21	R/W	0	Reserved
20	R/W	0	sar_adc
19	R/W	0	i2c_s
18	R/W	0	i2c_m
17	R/W	0	uart
16	R/W	0	ir_in
15-0	R/W	0	Reserved

6.8.2.3 SEC Reset

Table 6-245 reset0 0xFF64E000

Bit(s)	R/W	Default	Description
6	R/W	0	ao_mailbox
5	R/W	0	ao
4	R/W	0	dma
3	R/W	0	Reserved

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Bit(s)	R/W	Default	Description
2	R/W	0	am_ring_osc
1	R/W	0	Reserved
0	R/W	0	ao_cpu

Table 6-246 reset1 0xFF64E004

Bit(s)	R/W	Default	Description
20	R/W	0	perphreset
19	R/W	0	gicreset
18	R/W	0	Reserved
17	R/W	0	sreset
16	R/W	0	sporeset
15	R/W	0	sys_cpu3
14	R/W	0	sys_cpu2
13	R/W	0	sys_cpu1
12	R/W	0	sys_cpu0
11	R/W	0	sys_core3
10	R/W	0	sys_core2
9	R/W	0	sys_core1
8	R/W	0	sys_core0
7	R/W	0	sys_pll_div
6	R/W	0	presetn_sys_cpu_capb3
5	R/W	0	rom_boot
4	R/W	0	syscpu_axi
3	R/W	0	
2	R/W	0	syscpu_preset
1	R/W	0	syscpu_mbist
0	R/W	0	

The bits of the following registers correspond to the RESET[n]_SEC REGISTERs above. The default value of these registers is 0xFFFFFFFF. Setting any bit to 0, forces the corresponding RESET LOW. These register allow the software to “Hold” a reset LOW (in a reset condition).

RESET0_sec_LEVEL 0xFF64_E040

RESET1_sec_LEVEL 0xFF64_E044

RESET2_sec_LEVEL 0xFF64_E048

The bits of the following registers correspond to the RESET[n]_SEC REGISTERs above. The default value of these registers is 0xFFFFFFFF. Setting any bit to 0, forces the corresponding RESET LOW. These register allow the software to “Hold” a reset LOW (in a reset condition).

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RESET0_sec_MASK 0xFF64_E080

RESET1_sec_MASK 0xFF64_E084

RESET2_sec_MASK 0xFF64_E088

6.8.2.4 A55 Soft Reset

Table 6-247 a55_soft_reset 0xFF63C160

Bit(s)	R/W	Default	Description
16	R/W	0	cpu_axi
15	R/W	0	Reserved
14	R/W	0	periph_reset
13	R/W	0	gic
12	R/W	0	atreset
11	R/W	0	presetdbg
10	R/W	0	Reserved
9	R/W	0	sreset
8	R/W	0	sporeset
7	R/W	0	sys_core3
6	R/W	0	sys_core2
5	R/W	0	sys_core1
4	R/W	0	sys_core0
3	R/W	0	sys_cpu3_por
2	R/W	0	sys_cpu2_por
1	R/W	0	sys_cpu1_por
0	R/W	0	sys_cpu0_por

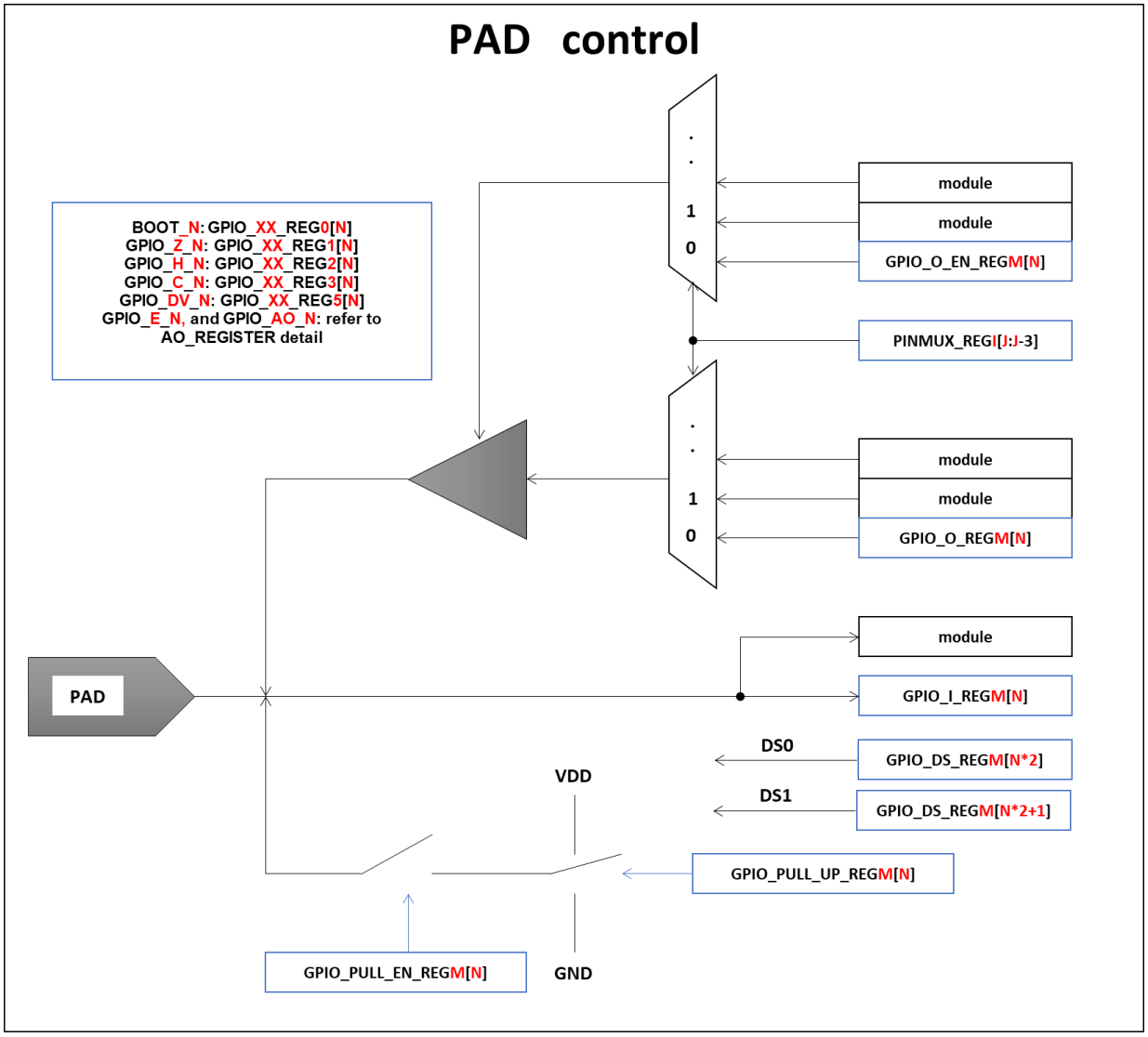
6.9 GPIO

6.9.1 Overview

The SOC has a number of multi-function digital I/O pads that can be multiplexed to a number of internal resources (e.g. PWM generators, SDIO controllers). When a digital I/O is not being used for any specific purpose, it is converted to a general purpose GPIO pin. A GPIO pin can be statically set to high/low logical levels. The structure of a GPIO is given below.

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Figure 6-25 GPIO Structure



T02FC11

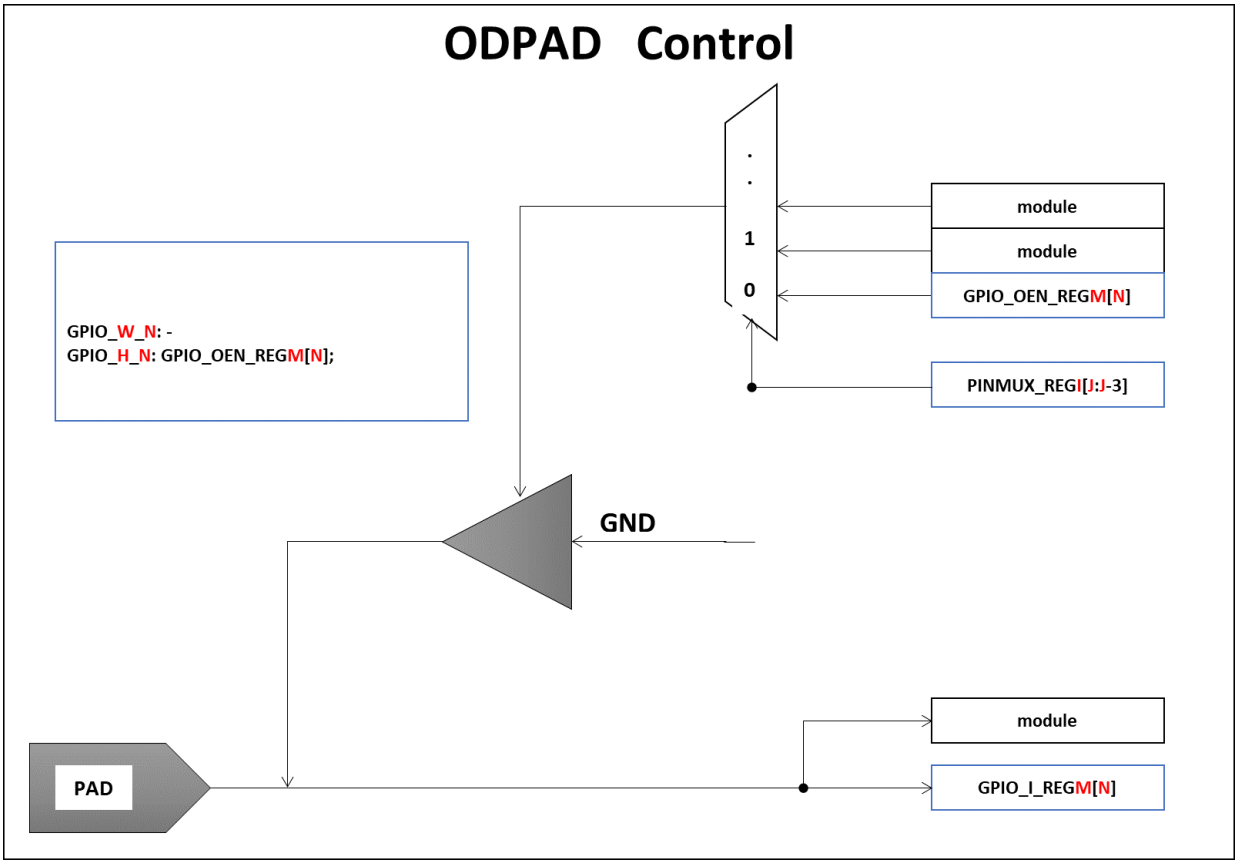
Each GPIO pad has 6 related registers, among which:

- GPIO_O_REG is used to control the output of the pad;
- GPIO_I_REG is used to store the input value of the pad;
- GPIO_O_EN_REG is used to enable GPIO output function;
- PINMUX_REG is used to define the function;
- GPIO_PULL_EN_REG is used to enable the pull-up function of GPIO pad;
- GPIO_PULL_UP_REG is used to control GPIO PAD to be whether pull-up or pull-down;
- For AO GPIO pads, refer to AO GPIO registers.

There are OD pins that can not be used as output, with different structure, shown as following.

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Figure 6-26 OD Pad Structure



T02FC12

6.9.2 GPIO Multiplex Function

The GPIO multiplex functions are shown in the sections below, where the RegNN[MM] corresponds to CBUS registers defined in the following table. The base address is 0xff634400 for PERIPHS_PIN_MUX_X, PREG_PAD_GPIO_EN_N, PREG_PAD_GPIO_O, PREG_PAD_GPIO_I, PAD_PULL_UP_EN_REG, PAD_PULL_UP_REG; and 0xff800000 for AO_RTI_PIN_MUX_REG, the final address is calculated as: final address = base address + offset*4.

Table 6-248 Pin Mux Registers

Pin Mux Registers	Offset	Reset by Watchdog
AO_GPIO_I	0x0a	Yes
AO_GPIO_O	0x0d	Yes
AO_GPIO_O_EN_N	0x09	Yes
AO_RTI_PULL_UP_REG	0x0b	Yes
AO_RTI_PULL_UP_EN_REG	0x0c	Yes
AO_PAD_DS_A	0x07	Yes
AO_PAD_DS_B	0x08	Yes
AO_RTI_PINMUX_REG0	0x05	Yes
AO_RTI_PINMUX_REG1	0x06	Yes

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Pin Mux Registers	Offset	Reset by Watchdog
PERIPHS_PIN_MUX_0	0xb0	Yes
PERIPHS_PIN_MUX_1	0xb1	Yes
PERIPHS_PIN_MUX_2	0xb2	Yes
PERIPHS_PIN_MUX_3	0xb3	Yes
PERIPHS_PIN_MUX_4	0xb4	Yes
PERIPHS_PIN_MUX_5	0xb5	Yes
PERIPHS_PIN_MUX_6	0xb6	Yes
PERIPHS_PIN_MUX_7	0xb7	Yes
PERIPHS_PIN_MUX_8	0xb8	Yes
PERIPHS_PIN_MUX_9	0xb9	Yes
PERIPHS_PIN_MUX_A	0xba	Yes
PERIPHS_PIN_MUX_B	0xbb	Yes
PERIPHS_PIN_MUX_C	0xbc	Yes
PERIPHS_PIN_MUX_D	0xbd	Yes
PERIPHS_PIN_MUX_E	0xbe	Yes
PERIPHS_PIN_MUX_F	0xbf	Yes
PAD_DS_REG0A	0xd0	Yes
PAD_DS_REG1A	0xd1	Yes
PAD_DS_REG2A	0xd2	Yes
PAD_DS_REG2B	0xd3	Yes
PAD_DS_REG3A	0xd4	Yes
PAD_DS_REG4A	0xd5	Yes
PAD_DS_REG5A	0xd6	Yes
PAD_PULL_UP_REG0	0x3a	Yes
PAD_PULL_UP_REG1	0x3b	Yes
PAD_PULL_UP_REG2	0x3c	Yes
PAD_PULL_UP_REG3	0x3d	Yes
PAD_PULL_UP_REG4	0x3e	Yes
PAD_PULL_UP_REG5	0x3f	Yes
PAD_PULL_UP_EN_REG0	0x48	Yes
PAD_PULL_UP_EN_REG1	0x49	Yes
PAD_PULL_UP_EN_REG2	0x4a	Yes
PAD_PULL_UP_EN_REG3	0x4b	Yes
PAD_PULL_UP_EN_REG4	0x4c	Yes
PAD_PULL_UP_EN_REG5	0x4d	Yes

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Pin Mux Registers	Offset	Reset by Watchdog
PREG_PAD_GPIO0_EN_N	0x10	Yes
PREG_PAD_GPIO0_O	0x11	Yes
PREG_PAD_GPIO0_I	0x12	Yes
PREG_PAD_GPIO1_EN_N	0x13	Yes
PREG_PAD_GPIO1_O	0x14	Yes
PREG_PAD_GPIO1_I	0x15	Yes
PREG_PAD_GPIO2_EN_N	0x16	Yes
PREG_PAD_GPIO2_O	0x17	Yes
PREG_PAD_GPIO2_I	0x18	Yes
PREG_PAD_GPIO3_EN_N	0x19	Yes
PREG_PAD_GPIO3_O	0x1a	Yes
PREG_PAD_GPIO3_I	0x1b	Yes
PREG_PAD_GPIO4_EN_N	0x1c	Yes
PREG_PAD_GPIO4_O	0x1d	Yes
PREG_PAD_GPIO4_I	0x1e	Yes
PREG_PAD_GPIO5_EN_N	0x20	Yes
PREG_PAD_GPIO5_O	0x21	Yes
PREG_PAD_GPIO5_I	0x22	Yes
AO_GPIO_TEST_N	0xd7	No

Table 6-249 GPIOAO_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5
GPIO-AO_0	AO_RTI_PIN-MUX_REG0	[3:0]	UART_AO_A_TX				
GPIO-AO_1		[7:4]	UART_AO_A_RX				
GPIO-AO_2		[11:-8]	I2C_AO_M0_SCL	UART_AO_B_TX	I2C_AO_S0_SCL		
GPIO-AO_3		[15:-12]	I2C_AO_M0_SDA	UART_AO_B_RX	I2C_AO_S0_SDA		
GPIO-AO_4		[19:-16]	IR_REMOTE_OUT	CLK_32K_IN	PWMAO_C	PWMAO_C_HIZ	
GPIO-AO_5		[23:-20]	IR_REMOTE_INPUT		PWMAO_D		
GPIO-AO_6		[27:-24]	JTAG_A_CLK	SPDIF_OUT	PWMAO_C		
GPIO-AO_7		[31:-28]	JTAG_A_TMS	SPDIF_IN	PWMAO_C_HIZ		

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Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5
GPIO-AO_8	AO_RTI_PIN-MUX_REG1	[3:0]	JTAG_A_TDI		UART_AO_B_TX		
GPIO-AO_9		[7:4]	JTAG_A_TDO	IR_RE-MOTE_OUT	UART_AO_B_RX		
GPIO-AO_10		[11:-8]	AO_CEC_A	AO_CEC_B	PWMAO_D		
GPIO-AO_11		[15:-12]		PWMAO_A_HIZ	PWMAO_A	GEN_CLK_EE	GEN_CLK_AO

Table 6-250 GPIOE_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4
GPIO-E_0	AO_RTI_PINMUX_REG1	[19:1-6]	UART_AO_A_CTS	UART_AO_B_CTS	PWMAO_B	I2C_AO_M0_SCL
GPIO-E_1		[23:2-0]	UART_AO_A_RTS	UART_AO_B_RTS	PWMAO_D	I2C_AO_M0_SDA
GPIO-E_2		[27:2-4]	CLK12_24	CLK25_EE	PWM_A	

Table 6-251 BOOT_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3
BOOT_0	PERIPHS_PIN_MUX_0	[3:0]	EMMC_D0		
BOOT_1		[7:4]	EMMC_D1		
BOOT_2		[11:8]	EMMC_D2		
BOOT_3		[15:12]	EMMC_D3		SPIF_HOLD
BOOT_4		[19:16]	EMMC_D4		SPIF_MO
BOOT_5		[23:20]	EMMC_D5		SPIF_MI
BOOT_6		[27:24]	EMMC_D6		SPIF_CLK
BOOT_7		[31:28]	EMMC_D7		SPIF_WP
BOOT_8	PERIPHS_PIN_MUX_1	[3:0]	EMMC_CLK	NAND_WEN_CLK	
BOOT_9		[7:4]		NAND_ALE	
BOOT_10		[11:8]	EMMC_CMD	NAND_REN_WR	
BOOT_11		[15:12]	EMMC_DS	NAND_CLE	
BOOT_12		[19:16]		NAND_CE0	
BOOT_13		[23:20]			SPIF_CS

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Table 6-252 GPIODV_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIO-D-V_0	PERIPHS_PIN_MUX_C	[3:0]	I2C_EE_M0_SDA	TCON_SLV_SDA	TSin_D7_A			LVDS_PROD_TEST_O_22	DEBUG_BUS_OUT0
GPIO-D-V_1		[7:4]	I2C_EE_M0_SCL	TCON_SLV_SCK	TSin_D6_A			LVDS_PROD_TEST_O_23	DEBUG_BUS_OUT1
GPIO-D-V_2		[11:8]	DTV_IF_AGC	ATV_IF_AGC	TSin_D5_A	TCON_AGING	PWM_C	LVDS_PROD_TEST_O_24	DEBUG_BUS_OUT2
GPIO-D-V_3		[15:12]	DTV_IF_AGC	DTV_RF_AGC	TSin_D4_A		PWM_D		DEBUG_BUS_OUT3
GPIO-D-V_4		[19:16]	SPDIF_OUT	TCON_FLICKER	TSin_D3_A	ISO7816_CLK			DEBUG_BUS_OUT4
GPIO-D-V_5		[23:20]	SPDIF_IN		TSin_D2_A	ISO7816_DATA			DEBUG_BUS_OUT5
GPIO-D-V_6		[27:24]	TDMC_D3	TDMC_DIN3	TSin_D1_A	PWM_E			DEBUG_BUS_OUT6
GPIO-D-V_7		[31:28]	TDMC_SCLK	TDMC_SLV_SCLK	TSin_D0_A	SPI_MISO_B	UART_EE_C_TX		DEBUG_BUS_OUT7
GPIO-D-V_8	PERIPHS_PIN_MUX_D	[3:0]	TDMC_FS	TDMC_SLV_FS	TSin_CLK_A	SPI_MOSI_B	UART_EE_C_RX		DEBUG_BUS_OUT8
GPIO-D-V_9		[7:4]	TDMC_D0	TDMC_DIN0	TSin_SOP_A	SPI_CLK_B	UART_EE_C_CTS	I2C_EE_M3_SCL	DEBUG_BUS_OUT9
GPIO-D-V_10		[11:8]	TDMC_D1	TDMC_DIN1	TSin_VALID_A	SPI_SS0_B	UART_EE_C_RTS	I2C_EE_M3_SDA	DEBUG_BUS_OUT10
GPIO-D-V_11		[15:12]	TDMC_D2	TDMC_DIN2	SPDIF_OUT	SPI_SS1_B	PWM_F		DEBUG_BUS_OUT11

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Table 6-253 GPIOW_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Fun2
GPIOW_0	PERIPHS_PIN_MUX_A	[3:0]	HDMIRX_HPD_A	
GPIOW_1		[7:4]	HDMIRX_5VDET_A	
GPIOW_2		[11:8]	HDMIRX_SDA_A	UART_AO_A_TX
GPIOW_3		[15:12]	HDMIRX_SCL_A	UART_AO_A_RX
GPIOW_4		[19:16]	HDMIRX_HPD_C	
GPIOW_5		[23:20]	HDMIRX_5VDET_C	
GPIOW_6		[27:24]	HDMIRX_SDA_C	UART_AO_A_TX
GPIOW_7		[31:28]	HDMIRX_SCL_C	UART_AO_A_RX
GPIOW_8	PERIPHS_PIN_MUX_B	[3:0]	HDMIRX_HPD_B	
GPIOW_9		[7:4]	HDMIRX_5VDET_B	
GPIOW_10		[11:8]	HDMIRX_SDA_B	UART_AO_A_TX
GPIOW_11		[15:12]	HDMIRX_SCL_B	UART_AO_A_RX

Table 6-254 GPIOH_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIO-H_0	PERIPHS_PIN_MUX_7	[3:0]	TCON_0	SPI_SS1_A	HSYNC	ETH_ACT_LED	SPDIF_OUT	I2C_EE_M3_SCL	ETH_DEBUG_IN0
GPIO-H_1		[7:4]	TCON_1	SPI_SS2_A	VSYNC	ETH_LINK_LED		I2C_EE_M3_SDA	ETH_DEBUG_IN1
GPIO-H_2		[11:8]	TCON_2	ETH_MDIO	BT656_A_CLK	TCON_SFC			ETH_DEBUG_IN2
GPIO-H_3		[15:12]	TCON_3	ETH_MDC	BT656_A_VS				ETH_DEBUG_IN3
GPIO-H_4		[19:16]	TCON_4	ETH_RGMII_RX_CLK	BT656_A_HS	TSin_D0_B	MCLK_0		ETH_DEBUG_IN4
GPIO-H_5		[23:20]	TCON_5	ETH_RX_DV	BT656_A_DIN0	TSin_CLK_B	TDMA_SCLK		ETH_DEBUG_IN5
GPIO-H_6		[27:24]	TCON_6	ETH_RXD0	BT656_A_DIN1	TSin_SOP_B	TDMA_FS		ETH_DEBUG_IN6
GPIO-H_7		[31:28]	TCON_7	ETH_RXD1	BT656_A_DIN2	TSin_VALID_B	TDMA_D0		ETH_DEBUG_IN7

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Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIO-H_8	PERIPHS_PIN_MUX_8	[3:0]	TCON_8	ETH_RXD2_RGMII	BT656_A_DIN3	TSin_D0_C	TDMA_D1		ETH_DEBUG_IN8
GPIO-H_9		[7:4]	TCON_9	ETH_RXD3_RGMII	BT656_A_DIN4	TSin_CLK_C	TDMA_D2		ETH_DEBUG_IN9
GPIO-H_10		[11:8]	TCON_10	ETH_RGMII_TX_CLK	BT656_A_DIN5	TSin_SOP_C	TDMA_D3		ETH_DEBUG_IN10
GPIO-H_11		[15:12]	TCON_11	ETH_TXEN	BT656_A_DIN6	TSin_VALID_C	TDMA_DIN0		ETH_DEBUG_IN11
GPIO-H_12		[19:16]	TCON_12	ETH_TXD0	BT656_A_DIN7		TDMA_DIN1		ETH_DEBUG_IN12
GPIO-H_13		[23:20]	TCON_13	ETH_TXD1	CLK12_24	PWM_C	TDMA_DIN2		ETH_DEBUG_IN13
GPIO-H_14		[27:24]	TCON_14	ETH_TXD2_RGMII	3D_SYNC_OUT		TDMA_DIN3		
GPIO-H_15		[31:28]	TCON_15	ETH_TXD3_RGMII	VX1_HTPDN				
GPIO-H_16	PERIPHS_PIN_MUX_9	[3:0]	TCON_LOCK	TCON_SFC	VX1_LOCKN				
GPIO-H_17		[7:4]	TCON_SPI_MO	SPI_MISO_A		UART_EE_B_TX	GEN_CLK_EE		DEBUG_BUS_OUT12
GPIO-H_18		[11:8]	TCON_SPI_MI	SPI_MOSI_A		UART_EE_B_RX			DEBUG_BUS_OUT13
GPIO-H_19		[15:12]	TCON_SPI_CLK	SPI_CLK_A	I2C_EE_M2_SCL	UART_EE_B_CTS	I2C_EE_M3_SCL		DEBUG_BUS_OUT14
GPIO-H_20		[19:16]	TCON_SPI_SS	SPI_SS0_A	I2C_EE_M2_SDA	UART_EE_B_RTS	I2C_EE_M3_SDA		DEBUG_BUS_OUT15
GPIO-H_21		[23:20]	I2C_EE_M1_SCL				GEN_CLK_EE		DEBUG_BUS_OUT16

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Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIO-H_22		[27:-24]	I2C_EE_M1_SDA						DEBUG_BUS_OUT17

Table 6-255 GPIOC_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIOC_0	PERIPHS_PIN_MUX_4	[3:0]	SDCARD_D0	JTAG_B_TDO	TDMB_D0	TSout_CLK	SPI_MISO_B		
GPIOC_1		[7:4]	SDCARD_D1	JTAG_B_TDI	TDMB_D1	TSout_SOP	SPI_MOSI_B		
GPIOC_2		[11:8]	SDCARD_D2	UART_AO_A_RX	TDMB_D2	TSout_VALID	SPI_CLK_B	LVDS_PROD_TEST_O_0	TST_CLK_OUT0
GPIOC_3		[15:12]	SDCARD_D3	UART_AO_A_TX	TDMB_D3	TSout_D0	SPI_SS0_B	LVDS_PROD_TEST_O_1	TST_CLK_OUT1
GPIOC_4		[19:16]	SDCARD_CLK	JTAG_B_CLK	TDMB_SCLK	TSout_D1	SPI_SS1_B	LVDS_PROD_TEST_O_2	TST_CLK_OUT2
GPIOC_5		[23:20]	SDCARD_CMD	JTAG_B_TMS	TDMB_FS	TSout_D2	SPI_SS2_B	LVDS_PROD_TEST_O_3	TST_CLK_OUT3
GPIOC_6		[27:24]	UART_EE_A_TX	PDM_DCLK	TDMB_DIN0	TSout_D3	DTV_IF_AGC	ATV_IF_AGC	TST_CLK_OUT4
GPIOC_7		[31:28]	UART_EE_A_RX	PDM_DIN0	TDMB_DIN1	TSout_D4	DTV_IF_AGC	DTV_RF_AGC	TST_CLK_OUT5
GPIOC_8	PERIPHS_PIN_MUX_5	[3:0]	UART_EE_A_CTS	PDM_DIN1	TDMB_DIN2	TSout_D5	I2C_EE_M0_SCL	LVDS_PROD_TEST_O_4	TST_CLK_OUT6
GPIOC_9		[7:4]	UART_EE_A_RTS	PDM_DIN2	TDMB_DIN3	TSout_D6	I2C_EE_M0_SDA	LVDS_PROD_TEST_O_5	TST_CLK_OUT7
GPIOC_10		[11:8]	PWM_B	PDM_DIN3	MCLK_1	TSout_D7		LVDS_PROD_TEST_O_6	TST_CLK_OUT8
GPIOC_11		[15:12]	SPI_MISO_A	TDMB_D1	TDMB_DIN1			LVDS_PROD_TEST_O_7	TST_CLK_OUT9

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Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIOC_12		[19:16]	SPI_MOSI_A	TDMB_D0	TDMB_DIN0	WIFI_BEACON		LVDS_PROD_TEST_O_8	TST_CLK_OUT10
GPIOC_13		[23:20]	SPI_CLK_A	TDMB_SCLK	TDMB_SLV_SCLK	I2C_EE_M3_SCL		LVDS_PROD_TEST_O_9	TST_CLK_OUT11
GPIOC_14		[27:24]	SPI_SS0_A	TDMB_FS	TDMB_SLV_FS	I2C_EE_M3_SDA		LVDS_PROD_TEST_O_10	TST_CLK_OUT12

Table 6-256 GPIOZ_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func6	Func7
GPIOZ_0	PERIPHS_PIN_MUX_2	[3:0]	MCLK_0				LVDS_PROD_TEST_O_11	TST_CLK_OUT13
GPIOZ_1		[7:4]	TDMA_SCLK	TDMA_SLV_SCLK	I2C_EE_M1_SDA		LVDS_PROD_TEST_O_12	TST_CLK_OUT14
GPIOZ_2		[11:8]	TDMA_FS	TDMA_SLV_FS	I2C_EE_M1_SCL		LVDS_PROD_TEST_O_13	TST_CLK_OUT15
GPIOZ_3		[15:12]	TDMA_D0			PWM_A	LVDS_PROD_TEST_O_14	TST_CLK_OUT16
GPIOZ_4		[19:16]	TDMA_D1		WIFI_BEACON	PWM_B	LVDS_PROD_TEST_O_15	TST_CLK_OUT17
GPIOZ_5		[23:20]	TDMA_D2		PWM_VS	PWM_C	LVDS_PROD_TEST_O_16	TST_CLK_OUT18
GPIOZ_6		[27:24]	TDMA_D3		PWM_VS	PWM_D	LVDS_PROD_TEST_O_17	TST_CLK_OUT19
GPIOZ_7		[31:28]	SPDIF_OUT	TDMA_DIN0	PDM_DCLK		LVDS_PROD_TEST_O_18	
GPIOZ_8	PERIPHS_PIN_MUX_3	[3:0]	MCLK_1	TDMA_DIN1	PDM_DIN0		LVDS_PROD_TEST_O_19	

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Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func6	Func7
GPIOZ_9		[7:4]	I2C_EE_M2_SCL	TDMA_DIN2	PDM_DIN1	PWM_E	LVDS_PROD_TEST_O_20	
GPIOZ_10		[11:8]	I2C_EE_M2_SDA	TDMA_DIN3	PDM_DIN2	PWM_F	LVDS_PROD_TEST_O_21	

6.9.3 Register Description

The following table shows the mapping information of GPIO I/O Registers and Nets.

Register	Offset	Net Name
PREG_PAD_GPIO0_EN_N	0x10	BOOT
PREG_PAD_GPIO0_O	0x11	
PREG_PAD_GPIO0_I	0x12	
PAD_PULL_UP_REG0	0x3a	
PAD_PULL_UP_EN_REG0	0x48	
PAD_DS_REG0A	0xd0	
PERIPHS_PIN_MUX_0	0xb0	
PERIPHS_PIN_MUX_1	0xb1	
PREG_PAD_GPIO3_EN_N	0x19	GPIOC
PREG_PAD_GPIO3_O	0x1a	
PREG_PAD_GPIO3_I	0x1b	
PAD_PULL_UP_REG3	0x3d	
PAD_PULL_UP_EN_REG3	0x4b	
PERIPHS_PIN_MUX_4	0xb4	
PERIPHS_PIN_MUX_5	0xb5	
PREG_PAD_GPIO2_EN_N	0x16	GPIOH
PREG_PAD_GPIO2_O	0x17	
PREG_PAD_GPIO2_I	0x18	
PAD_PULL_UP_REG2	0x3c	
PAD_PULL_UP_EN_REG2	0x4a	
PAD_DS_REG2A	0xd2	
PAD_DS_REG2B	0xd3	
PAD_DS_REG3A	0xd4	
PERIPHS_PIN_MUX_7	0xb7	

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Register	Offset	Net Name
PERIPHS_PIN_MUX_8	0xb8	
PERIPHS_PIN_MUX_9	0xb9	
PREG_PAD_GPIO4_EN_N	0x1c	GPIOW
PREG_PAD_GPIO4_O	0x1d	
PREG_PAD_GPIO4_I	0x1e	
PERIPHS_PIN_MUX_A	0xba	
PERIPHS_PIN_MUX_B	0xbb	
PREG_PAD_GPIO5_EN_N	0x20	GPIODV
PREG_PAD_GPIO5_O	0x21	
PREG_PAD_GPIO5_I	0x22	
PAD_PULL_UP_REG5	0x3f	
PAD_PULL_UP_EN_REG5	0x4d	
PAD_DS_REG5A	0xd6	
PERIPHS_PIN_MUX_C	0xbc	
PERIPHS_PIN_MUX_D	0xbd	
PAD_PULL_UP_REG1	0x3b	GPIOZ
PAD_PULL_UP_EN_REG1	0x49	
PAD_DS_REG1A	0xd1	
PERIPHS_PIN_MUX_2	0xb2	
PERIPHS_PIN_MUX_3	0xb3	
PREG_PAD_GPIO1_EN_N	0x13	
PREG_PAD_GPIO1_O	0x14	
PREG_PAD_GPIO1_I	0x15	

6.9.3.1 Pin MUX Registers

Table 6-257 PERIPHS_PIN_MUX_0 0xb0

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for BOOT_0
[7:4]	R/W	0	pin mux select bits for BOOT_1
[11:8]	R/W	0	pin mux select bits for BOOT_2
[15:12]	R/W	0	pin mux select bits for BOOT_3
[19:16]	R/W	0	pin mux select bits for BOOT_4
[23:20]	R/W	0	pin mux select bits for BOOT_5

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Bit(s)	R/W	Default	Description
[27:24]	R/W	0	pin mux select bits for BOOT_6
[31:28]	R/W	0	pin mux select bits for BOOT_7

Table 6-258 PERIPHS_PIN_MUX_1 0xb1

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for BOOT_8
[7:4]	R/W	0	pin mux select bits for BOOT_9
[11:8]	R/W	0	pin mux select bits for BOOT_10
[15:12]	R/W	0	pin mux select bits for BOOT_11
[19:16]	R/W	0	pin mux select bits for BOOT_12
[23:20]	R/W	0	pin mux select bits for BOOT_13

Table 6-259 PERIPHS_PIN_MUX_2 0xb2

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOZ_0
[7:4]	R/W	0	pin mux select bits for GPIOZ_1
[11:8]	R/W	0	pin mux select bits for GPIOZ_2
[15:12]	R/W	0	pin mux select bits for GPIOZ_3
[19:16]	R/W	0	pin mux select bits for GPIOZ_4
[23:20]	R/W	0	pin mux select bits for GPIOZ_5
[27:24]	R/W	0	pin mux select bits for GPIOZ_6
[31:28]	R/W	0	pin mux select bits for GPIOZ_7

Table 6-260 PERIPHS_PIN_MUX_3 0xb3

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOZ_8
[7:4]	R/W	0	pin mux select bits for GPIOZ_9
[11:8]	R/W	0	pin mux select bits for GPIOZ_10

Table 6-261 PERIPHS_PIN_MUX_4 0xb4

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOC_0
[7:4]	R/W	0	pin mux select bits for GPIOC_1
[11:8]	R/W	0	pin mux select bits for GPIOC_2
[15:12]	R/W	0	pin mux select bits for GPIOC_3

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Bit(s)	R/W	Default	Description
[19:16]	R/W	0	pin mux select bits for GPIOC_4
[23:20]	R/W	0	pin mux select bits for GPIOC_5
[27:24]	R/W	0	pin mux select bits for GPIOC_6
[31:28]	R/W	0	pin mux select bits for GPIOC_7

Table 6-262 PERIPHS_PIN_MUX_5 0xb5

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOC_8
[7:4]	R/W	0	pin mux select bits for GPIOC_9
[11:8]	R/W	0	pin mux select bits for GPIOC_10
[15:12]	R/W	0	pin mux select bits for GPIOC_11
[19:16]	R/W	0	pin mux select bits for GPIOC_12
[23:20]	R/W	0	pin mux select bits for GPIOC_13
[27:24]	R/W	0	pin mux select bits for GPIOC_14

PERIPHS_PIN_MUX_6; 0xb6

Reserved.

Table 6-263 PERIPHS_PIN_MUX_7 0xb7

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOH_0
[7:4]	R/W	0	pin mux select bits for GPIOH_1
[11:8]	R/W	0	pin mux select bits for GPIOH_2
[15:12]	R/W	0	pin mux select bits for GPIOH_3
[19:16]	R/W	0	pin mux select bits for GPIOH_4
[23:20]	R/W	0	pin mux select bits for GPIOH_5
[27:24]	R/W	0	pin mux select bits for GPIOH_6
[31:28]	R/W	0	pin mux select bits for GPIOH_7

Table 6-264 PERIPHS_PIN_MUX_8 0xb8

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOH_8
[7:4]	R/W	0	pin mux select bits for GPIOH_9
[11:8]	R/W	0	pin mux select bits for GPIOH_10
[15:12]	R/W	0	pin mux select bits for GPIOH_11
[19:16]	R/W	0	pin mux select bits for GPIOH_12

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Bit(s)	R/W	Default	Description
[23:20]	R/W	0	pin mux select bits for GPIOH_13
[27:24]	R/W	0	pin mux select bits for GPIOH_14
[31:28]	R/W	0	pin mux select bits for GPIOH_15

Table 6-265 PERIPHS_PIN_MUX_9 0xb9

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOH_16
[7:4]	R/W	0	pin mux select bits for GPIOH_17
[11:8]	R/W	0	pin mux select bits for GPIOH_18
[15:12]	R/W	0	pin mux select bits for GPIOH_19
[19:16]	R/W	0	pin mux select bits for GPIOH_20
[23:20]	R/W	0	pin mux select bits for GPIOH_21
[27:24]	R/W	0	pin mux select bits for GPIOH_22

Table 6-266 PERIPHS_PIN_MUX_A 0xba

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOW_0
[7:4]	R/W	0	pin mux select bits for GPIOW_1
[11:8]	R/W	0	pin mux select bits for GPIOW_2
[15:12]	R/W	0	pin mux select bits for GPIOW_3
[19:16]	R/W	0	pin mux select bits for GPIOW_4
[23:20]	R/W	0	pin mux select bits for GPIOW_5
[27:24]	R/W	0	pin mux select bits for GPIOW_6
[31:28]	R/W	0	pin mux select bits for GPIOW_7

Table 6-267 PERIPHS_PIN_MUX_B 0xbb

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOW_8
[7:4]	R/W	0	pin mux select bits for GPIOW_9
[11:8]	R/W	0	pin mux select bits for GPIOW_10
[15:12]	R/W	0	pin mux select bits for GPIOW_11

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Table 6-268 PERIPHS_PIN_MUX_C 0xbc

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIODV_0
[7:4]	R/W	0	pin mux select bits for GPIODV_1
[11:8]	R/W	0	pin mux select bits for GPIODV_2
[15:12]	R/W	0	pin mux select bits for GPIODV_3
[19:16]	R/W	0	pin mux select bits for GPIODV_4
[23:20]	R/W	0	pin mux select bits for GPIODV_5
[27:24]	R/W	0	pin mux select bits for GPIODV_6
[31:28]	R/W	0	pin mux select bits for GPIODV_7

Table 6-269 PERIPHS_PIN_MUX_D 0xbd

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIODV_8
[7:4]	R/W	0	pin mux select bits for GPIODV_9
[11:8]	R/W	0	pin mux select bits for GPIODV_10
[15:12]	R/W	0	pin mux select bits for GPIODV_11

PERIPHS_PIN_MUX_E 0xbe

Reserved.

PERIPHS_PIN_MUX_F 0xbf

Reserved.

6.9.3.2 Pad Pull-up/down Direction

The I/O pads contain both a pull-up and a pull-down. If a bit is set to 1 in the registers below, then the pull-up is enabled. If a bit is set to 0, then the pull-down is enabled.

Note

There are separate pull-up “enables” that must also be set to 1 for the pull-up/down direction to function. If an “enable” is set to 0, then the pull-up/down feature is disabled, and the bits below are ignored (on a per pad basis).

Table 6-270 PAD_PULL_UP_REG0 0x3a

Bit(s)	R/W	Default	Description
31~14	R/W	0x35ff	Unused
13~0	R/W		Boot[13:0] 1 = pull up. 0 = pull down

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Table 6-271 PAD_PULL_UP_REG1 0x3b

Bit(s)	R/W	Default	Description
31~11	R/W	0x60e	Unused
10~0	R/W		gpioZ[10:0] 1 = pull up. 0 = pull down

Table 6-272 PAD_PULL_UP_REG2 0x3c

Bit(s)	R/W	Default	Description
31~23	R/W	0x0000-00	Reserved
22~17	R/W		gpioH[22:17] 1 = pull up. 0 = pull down
16	R/W		Reserved
15~0	R/W		gpioH[15:0] 1 = pull up. 0 = pull down

Table 6-273 PAD_PULL_UP_REG3 0x3d

Bit(s)	R/W	Default	Description
31~15	R/W	0x57ff	Reserved
14~0	R/W		gpioC[14:0] 1 = pull up. 0 = pull down

PAD_PULL_UP_REG4 0x3e

Reserved

Table 6-274 PAD_PULL_UP_REG5 0x3f

Bit(s)	R/W	Default	Description
31~12	R/W	0xfc7	Unused
11~0	R/W		gpioDV[11:0] 1 = pull up. 0 = pull down

6.9.3.3 Pad Pull-up/down Enables

Each I/O pad has a selectable pull-up or pull-down resistor. For the pull-up direction (up or down) to be operational, the appropriate bit below must be set in order to enable the pull-up/down function.

Table 6-275 PAD_PULL_UP_EN_REG0 0x48

Bit(s)	R/W	Default	Description
31~14	R/W	0x3fff	Unused
13~0	R/W		boot[13:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

Table 6-276 PAD_PULL_UP_EN_REG1 0x49

Bit(s)	R/W	Default	Description
31~11	R/W	0x7ff	Unused

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Bit(s)	R/W	Default	Description
10~0	R/W		gpioZ[10:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

Table 6-277 PAD_PULL_UP_EN_REG2 0x4a

Bit(s)	R/W	Default	Description
31~23	R/W	0x00ffff	Reserved
22~17	R/W		gpioH[22:17] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down
16	R/W		Reserved
15~0	R/W		gpioH[15:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

Table 6-278 PAD_PULL_UP_EN_REG3 0x4b

Bit(s)	R/W	Default	Description
31~15	R/W	0x7fff	reserved
14~0	R/W		gpioC[14:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

PAD_PULL_UP_EN_REG4 0x4c

Reserved

Table 6-279 PAD_PULL_UP_EN_REG5 0x4d

Bit(s)	R/W	Default	Description
31~12	R/W	0xfff	reserved
11~0	R/W		gpioDV[11:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

6.9.3.4 PREG_PAD_GPIO_I Registers

Table 6-280 PREG_PAD_GPIO0_I 0x12

Bit(s)	R/W	Default	Description
0	R/W	0	BOOT_0 select
1	R/W	0	BOOT_1 select
2	R/W	0	BOOT_2 select
3	R/W	0	BOOT_3 select
4	R/W	0	BOOT_4 select
5	R/W	0	BOOT_5 select
6	R/W	0	BOOT_6 select
7	R/W	0	BOOT_7 select

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Bit(s)	R/W	Default	Description
8	R/W	0	BOOT_8 select
9	R/W	0	BOOT_9 select
10	R/W	0	BOOT_10 select
11	R/W	0	BOOT_11 select
12	R/W	0	BOOT_12 select
13	R/W	0	BOOT_13 select

Table 6-281 PREG_PAD_GPIO1_I 0x15

Bit(s)	R/W	Default	Description
0	R/W	0	GPIOZ_0 select
1	R/W	0	GPIOZ_1 select
2	R/W	0	GPIOZ_2 select
3	R/W	0	GPIOZ_3 select
4	R/W	0	GPIOZ_4 select
5	R/W	0	GPIOZ_5 select
6	R/W	0	GPIOZ_6 select
7	R/W	0	GPIOZ_7 select
8	R/W	0	GPIOZ_8 select
9	R/W	0	GPIOZ_9 select
10	R/W	0	GPIOZ_10 select

Table 6-282 PREG_PAD_GPIO2_I 0x18

Bit(s)	R/W	Default	Description
0	R/W	0	GPIOH_0 select
1	R/W	0	GPIOH_1 select
2	R/W	0	GPIOH_2 select
3	R/W	0	GPIOH_3 select
4	R/W	0	GPIOH_4 select
5	R/W	0	GPIOH_5 select
6	R/W	0	GPIOH_6 select
7	R/W	0	GPIOH_7 select
8	R/W	0	GPIOH_8 select
9	R/W	0	GPIOH_9 select
10	R/W	0	GPIOH_10 select
11	R/W	0	GPIOH_11 select

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Bit(s)	R/W	Default	Description
12	R/W	0	GPIOH_12 select
13	R/W	0	GPIOH_13 select
14	R/W	0	GPIOH_14 select
15	R/W	0	GPIOH_15 select
16	R/W	0	GPIOH_16 select
17	R/W	0	GPIOH_17 select
18	R/W	0	GPIOH_18 select
19	R/W	0	GPIOH_19 select
20	R/W	0	GPIOH_20 select
21	R/W	0	GPIOH_21 select
22	R/W	0	GPIOH_22 select

Table 6-283 PREG_PAD_GPIO3_I 0x1b

Bit(s)	R/W	Default	Description
0	R/W	0	GPIOC_0 select
1	R/W	0	GPIOC_1 select
2	R/W	0	GPIOC_2 select
3	R/W	0	GPIOC_3 select
4	R/W	0	GPIOC_4 select
5	R/W	0	GPIOC_5 select
6	R/W	0	GPIOC_6 select
7	R/W	0	GPIOC_7 select
8	R/W	0	GPIOC_8 select
9	R/W	0	GPIOC_9 select
10	R/W	0	GPIOC_10 select
11	R/W	0	GPIOC_11 select
12	R/W	0	GPIOC_12 select
13	R/W	0	GPIOC_13 select
14	R/W	0	GPIOC_14 select

Table 6-284 PREG_PAD_GPIO4_I 0x1e

Bit(s)	R/W	Default	Description
0	R/W	0	GPIOW_0 select
1	R/W	0	GPIOW_1 select
2	R/W	0	GPIOW_2 select

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Bit(s)	R/W	Default	Description
3	R/W	0	GPIOW_3 select
4	R/W	0	GPIOW_4 select
5	R/W	0	GPIOW_5 select
6	R/W	0	GPIOW_6 select
7	R/W	0	GPIOW_7 select
8	R/W	0	GPIOW_8 select
9	R/W	0	GPIOW_9 select
10	R/W	0	GPIOW_10 select
11	R/W	0	GPIOW_11 select

Table 6-285 PREG_PAD_GPIO5_I 0x22

Bit(s)	R/W	Default	Description
0	R/W	0	GPIODV_0 select
1	R/W	0	GPIODV_1 select
2	R/W	0	GPIODV_2 select
3	R/W	0	GPIODV_3 select
4	R/W	0	GPIODV_4 select
5	R/W	0	GPIODV_5 select
6	R/W	0	GPIODV_6 select
7	R/W	0	GPIODV_7 select
8	R/W	0	GPIODV_8 select
9	R/W	0	GPIODV_9 select
10	R/W	0	GPIODV_10 select
11	R/W	0	GPIODV_11 select

6.9.3.5 PREG_PAD_GPIO_O Registers

Table 6-286 PREG_PAD_GPIO0_O 0x11

Bit(s)	R/W	Default	Description
0	R/W	1	BOOT_0 select
1	R/W	1	BOOT_1 select
2	R/W	1	BOOT_2 select
3	R/W	1	BOOT_3 select
4	R/W	1	BOOT_4 select
5	R/W	1	BOOT_5 select

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Bit(s)	R/W	Default	Description
6	R/W	1	BOOT_6 select
7	R/W	1	BOOT_7 select
8	R/W	1	BOOT_8 select
9	R/W	1	BOOT_9 select
10	R/W	1	BOOT_10 select
11	R/W	1	BOOT_11 select
12	R/W	1	BOOT_12 select
13	R/W	1	BOOT_13 select

Table 6-287 PREG_PAD_GPIO1_O 0x14

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOZ_0 select
1	R/W	1	GPIOZ_1 select
2	R/W	1	GPIOZ_2 select
3	R/W	1	GPIOZ_3 select
4	R/W	1	GPIOZ_4 select
5	R/W	1	GPIOZ_5 select
6	R/W	1	GPIOZ_6 select
7	R/W	1	GPIOZ_7 select
8	R/W	1	GPIOZ_8 select
9	R/W	1	GPIOZ_9 select
10	R/W	1	GPIOZ_10 select

Table 6-288 PREG_PAD_GPIO2_O 0x17

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOH_0 select
1	R/W	1	GPIOH_1 select
2	R/W	1	GPIOH_2 select
3	R/W	1	GPIOH_3 select
4	R/W	1	GPIOH_4 select
5	R/W	1	GPIOH_5 select
6	R/W	1	GPIOH_6 select
7	R/W	1	GPIOH_7 select
8	R/W	1	GPIOH_8 select
9	R/W	1	GPIOH_9 select

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Bit(s)	R/W	Default	Description
10	R/W	1	GPIOH_10 select
11	R/W	1	GPIOH_11 select
12	R/W	1	GPIOH_12 select
13	R/W	1	GPIOH_13 select
14	R/W	1	GPIOH_14 select
15	R/W	1	GPIOH_15 select
16	R/W	1	NA
17	R/W	1	GPIOH_17 select
18	R/W	1	GPIOH_18 select
19	R/W	1	GPIOH_19 select
20	R/W	1	GPIOH_20 select
21	R/W	1	GPIOH_21 select
22	R/W	1	GPIOH_22 select

Table 6-289 PREG_PAD_GPIO3_O 0x1a

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOC_0 select
1	R/W	1	GPIOC_1 select
2	R/W	1	GPIOC_2 select
3	R/W	1	GPIOC_3 select
4	R/W	1	GPIOC_4 select
5	R/W	1	GPIOC_5 select
6	R/W	1	GPIOC_6 select
7	R/W	1	GPIOC_7 select
8	R/W	1	GPIOC_8 select
9	R/W	1	GPIOC_9 select
10	R/W	1	GPIOC_10 select
11	R/W	1	GPIOC_11 select
12	R/W	1	GPIOC_12 select
13	R/W	1	GPIOC_13 select
14	R/W	1	GPIOC_14 select

PREG_PAD_GPIO4_O 0x1d

Reserved.

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Table 6-290 PREG_PAD_GPIO5_O 0x21

Bit(s)	R/W	Default	Description
0	R/W	1	GPIODV_0 select
1	R/W	1	GPIODV_1 select
2	R/W	1	GPIODV_2 select
3	R/W	1	GPIODV_3 select
4	R/W	1	GPIODV_4 select
5	R/W	1	GPIODV_5 select
6	R/W	1	GPIODV_6 select
7	R/W	1	GPIODV_7 select
8	R/W	1	GPIODV_8 select
9	R/W	1	GPIODV_9 select
10	R/W	1	GPIODV_10 select
11	R/W	1	GPIODV_11 select

6.9.3.6 PREG PAD_GPIO_EN_N Registers

Table 6-291 PREG_PAD_GPIO0_EN_N 0x10

Bit(s)	R/W	Default	Description
0	R/W	1	BOOT_0 select
1	R/W	1	BOOT_1 select
2	R/W	1	BOOT_2 select
3	R/W	1	BOOT_3 select
4	R/W	1	BOOT_4 select
5	R/W	1	BOOT_5 select
6	R/W	1	BOOT_6 select
7	R/W	1	BOOT_7 select
8	R/W	1	BOOT_8 select
9	R/W	1	BOOT_9 select
10	R/W	1	BOOT_10 select
11	R/W	1	BOOT_11 select
12	R/W	1	BOOT_12 select
13	R/W	1	BOOT_13 select

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Table 6-292 PREG_PAD_GPIO1_EN_N 0x13

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOZ_0 select
1	R/W	1	GPIOZ_1 select
2	R/W	1	GPIOZ_2 select
3	R/W	1	GPIOZ_3 select
4	R/W	1	GPIOZ_4 select
5	R/W	1	GPIOZ_5 select
6	R/W	1	GPIOZ_6 select
7	R/W	1	GPIOZ_7 select
8	R/W	1	GPIOZ_8 select
9	R/W	1	GPIOZ_9 select
10	R/W	1	GPIOZ_10 select

Table 6-293 PREG_PAD_GPIO2_EN_N 0x16

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOH_0 select
1	R/W	1	GPIOH_1 select
2	R/W	1	GPIOH_2 select
3	R/W	1	GPIOH_3 select
4	R/W	1	GPIOH_4 select
5	R/W	1	GPIOH_5 select
6	R/W	1	GPIOH_6 select
7	R/W	1	GPIOH_7 select
8	R/W	1	GPIOH_8 select
9	R/W	1	GPIOH_9 select
10	R/W	1	GPIOH_10 select
11	R/W	1	GPIOH_11 select
12	R/W	1	GPIOH_12 select
13	R/W	1	GPIOH_13 select
14	R/W	1	GPIOH_14 select
15	R/W	1	GPIOH_15 select
16	R/W	1	GPIOH_16 select
17	R/W	1	GPIOH_17 select
18	R/W	1	GPIOH_18 select
19	R/W	1	GPIOH_19 select

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Bit(s)	R/W	Default	Description
20	R/W	1	GPIOH_20 select
21	R/W	1	GPIOH_21 select
22	R/W	1	GPIOH_22 select

Table 6-294 PREG_PAD_GPIO3_EN_N 0x19

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOC_0 select
1	R/W	1	GPIOC_1 select
2	R/W	1	GPIOC_2 select
3	R/W	1	GPIOC_3 select
4	R/W	1	GPIOC_4 select
5	R/W	1	GPIOC_5 select
6	R/W	1	GPIOC_6 select
7	R/W	1	GPIOC_7 select
8	R/W	1	GPIOC_8 select
9	R/W	1	GPIOC_9 select
10	R/W	1	GPIOC_10 select
11	R/W	1	GPIOC_11 select
12	R/W	1	GPIOC_12 select
13	R/W	1	GPIOC_13 select
14	R/W	1	GPIOC_14 select

Table 6-295 PREG_PAD_GPIO4_EN_N 0x1c

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOW_0 select
1	R/W	1	GPIOW_1 select
2	R/W	1	GPIOW_2 select
3	R/W	1	GPIOW_3 select
4	R/W	1	GPIOW_4 select
5	R/W	1	GPIOW_5 select
6	R/W	1	GPIOW_6 select
7	R/W	1	GPIOW_7 select
8	R/W	1	GPIOW_8 select
9	R/W	1	GPIOW_9 select

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Bit(s)	R/W	Default	Description
10	R/W	1	GPIOW_10 select
11	R/W	1	GPIOW_11 select

Table 6-296 PREG_PAD_GPIO5_EN_N 0x20

Bit(s)	R/W	Default	Description
0	R/W	1	GPIODV_0 select
1	R/W	1	GPIODV_1 select
2	R/W	1	GPIODV_2 select
3	R/W	1	GPIODV_3 select
4	R/W	1	GPIODV_4 select
5	R/W	1	GPIODV_5 select
6	R/W	1	GPIODV_6 select
7	R/W	1	GPIODV_7 select
8	R/W	1	GPIODV_8 select
9	R/W	1	GPIODV_9 select
10	R/W	1	GPIODV_10 select
11	R/W	1	GPIODV_11 select

6.9.3.7 PAD_DS Registers

Table 6-297 PAD_DS_REG0A 0xd0

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	BOOT_0 select
[3:2]	R/W		BOOT_1 select
[5:4]	R/W		BOOT_2 select
[7:6]	R/W		BOOT_3 select
[9:8]	R/W		BOOT_4 select
[11:10]	R/W		BOOT_5 select
[13:12]	R/W		BOOT_6 select
[15:14]	R/W		BOOT_7 select
[17:16]	R/W		BOOT_8 select
[19:18]	R/W		BOOT_9 select
[21:20]	R/W		BOOT_10 select
[23:22]	R/W		BOOT_11 select
[25:24]	R/W		BOOT_12 select

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Bit(s)	R/W	Default	Description
[27:26]	R/W		BOOT_13 select

Table 6-298 PAD_DS_REG1A 0xd1

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	GPIOZ_0 select
[3:2]	R/W		GPIOZ_1 select
[5:4]	R/W		GPIOZ_2 select
[7:6]	R/W		GPIOZ_3 select
[9:8]	R/W		GPIOZ_4 select
[11:10]	R/W		GPIOZ_5 select
[13:12]	R/W		GPIOZ_6 select
[15:14]	R/W		GPIOZ_7 select
[17:16]	R/W		GPIOZ_8 select
[19:18]	R/W		GPIOZ_9 select
[21:20]	R/W		GPIOZ_10 select

Table 6-299 PAD_DS_REG2A 0xd2

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	GPIOH_0 select
[3:2]	R/W		GPIOH_1 select
[5:4]	R/W		GPIOH_2 select
[7:6]	R/W		GPIOH_3 select
[9:8]	R/W		GPIOH_4 select
[11:10]	R/W		GPIOH_5 select
[13:12]	R/W		GPIOH_6 select
[15:14]	R/W		GPIOH_7 select
[17:16]	R/W		GPIOH_8 select
[19:18]	R/W		GPIOH_9 select
[21:20]	R/W		GPIOH_10 select
[23:22]	R/W		GPIOH_11 select
[25:24]	R/W		GPIOH_12 select
[27:26]	R/W		GPIOH_13 select
[29:28]	R/W		GPIOH_14 select
[31:30]	R/W		GPIOH_15 select

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Table 6-300 PAD_DS_REG2B 0xd3

Bit(s)	R/W	Default	Description
[3:2]	R/W	2'b10	GPIOH_17
[5:4]	R/W		GPIOH_18
[7:6]	R/W		GPIOH_19
[9:8]	R/W		GPIOH_20
[11:10]	R/W		GPIOH_21
[13:12]	R/W		GPIOH_22

Table 6-301 PAD_DS_REG3A 0xd4

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	GPIOC_0
[3:2]	R/W		GPIOC_1
[5:4]	R/W		GPIOC_2
[7:6]	R/W		GPIOC_3
[9:8]	R/W		GPIOC_4
[11:10]	R/W		GPIOC_5
[13:12]	R/W		GPIOC_6
[15:14]	R/W		GPIOC_7
[17:16]	R/W		GPIOC_8
[19:18]	R/W		GPIOC_9
[21:20]	R/W		GPIOC_10
[23:22]	R/W		GPIOC_11
[25:24]	R/W		GPIOC_12
[27:26]	R/W		GPIOC_13
[29:28]	R/W		GPIOC_14

PAD_DS_REG4A 0xd5

Reserved.

Table 6-302 PAD_DS_REG5A 0xd6

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	GPIODV_0
[3:2]	R/W		GPIODV_1
[5:4]	R/W		GPIODV_2
[7:6]	R/W		GPIODV_3
[9:8]	R/W		GPIODV_4

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Bit(s)	R/W	Default	Description
[11:10]	R/W		GPIODV_5
[13:12]	R/W		GPIODV_6
[15:14]	R/W		GPIODV_7
[17:16]	R/W		GPIODV_8
[19:18]	R/W		GPIODV_9
[21:20]	R/W		GPIODV_10
[23:22]	R/W		GPIODV_11

6.9.3.8 GPIOAO Registers

Always On GPIO Input levels

Table 6-303 AO_GPIO_I 0x0a

Bit(s)	R/W	Default	Description
31:19	R	0	reserved
18:16	R	0	INPUT_LEVELS: These bits correspond to the INPUT levels on the gpioE[2:0] pins
15:12	R	0	reserved
11-0	R	0	INPUT_LEVELS: These bits correspond to the INPUT levels on the gpioAO[11:0] pins

Table 6-304 AO_GPIO_O 0x0d

Bit(s)	R/W	Default	Description
31	R/W	1	reserved
18-16	R/W	0xFFF	OUTPUT_LEVEL: These bits correspond to the output levels on the gpioE[2:0] pins when in GPIO mode.
11-0	R/W	0xFFF	OUTPUT_LEVEL: These bits correspond to the output levels on the gpioAO [11:0] pins when in GPIO mode.

Table 6-305 AO_RTI_PIN_MUX_REG0 0x05

Bit(s)	R/W	Default	Description
31-0	R/W	0	Pin mux settings

Table 6-306 AO_RTI_PIN_MUX_REG1 0x06

Bit(s)	R/W	Default	Description
31-0	R/W	0	Pin mux settings

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Table 6-307 AO_GPIO_O_EN_N 0x09

Bit(s)	R/W	Default	Description
30-19	R	0	Reserved
18-16	R/W	0x7	OUTPUT_ENABLE: These bits correspond to the output levels on the gpioE[2:0] pins when in GPIO mode. A '0' sets the gpioE pin to be an output.
15-12	R	0	Reserved
11-0	R/W	0xfff	OUTPUT_ENABLE: These bits correspond to the output levels on the gpioAO [11:0] pins when in GPIO mode. A '0' sets the gpioAO pin to be an output.

Table 6-308 AO_RTI_PULL_UP_REG 0x0b

Bit(s)	R/W	Default	Description
31	R/W	1	Reserved.
18-16	R/W	0x3dff	gpioE[2:0] pull-up/down. 1 = pull-up 0 = pull-down
11-0	R/W	0x9df	gpioAO[11:0] pull up/down. 1 = pull-up 0 = pull-down

Table 6-309 AO_RTI_PULL_UP_EN_REG 0x0c

Bit(s)	R/W	Default	Description
31	R/W	1	Reserved.
18-16	R/W	0x3dff	gpioE[2:0] pull-up/down enable. 1 = pull-up/down enable, 0 = pull-up/down disable
11-0	R/W	0x9df	gpioAO[11:0] pull-up/down enable. 1 = pull-up/down enable, 0 = pull-up/down disable

Pad ds0/ds1

Table 6-310 AO_PAD_DS_A 0x07

Bit(s)	R/W	Default	Description
23:22	R/W	2'b10	gpioAO_11{ds1 ds0}
21:20	R/W		gpioAO_10{ds1 ds0}
19:18	R/W		gpioAO_9{ds1 ds0}
17:16	R/W		gpioAO_8{ds1 ds0}
15:14	R/W		gpioAO_7{ds1 ds0}
13:12	R/W		gpioAO_6{ds1 ds0}
11:10	R/W		gpioAO_5{ds1 ds0}
9:8	R/W		gpioAO_4{ds1 ds0}
7:6	R/W		gpioAO_3{ds1 ds0}
5:4	R/W		gpioAO_2{ds1 ds0}
3:2	R/W		gpioAO_1{ds1 ds0}
1:0	R/W		gpioAO_0 {ds1 ds0}

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Pad ds0/ds1

Table 6-311 AO_PAD_DS_B 0x08

Bit(s)	R/W	Default	Description
31:30	R/W	2'b10	reserved
5:4	R/W		gpioE_2{ds1 ds0}
3:2	R/W		gpioE_1{ds1 ds0}
1:0	R/W		gpioE_0{ds1 ds0}

Table 6-312 AO_PINMUX_LOCK 0x17

Bit(s)	R/W	Default	Description
31	R/W	0	1: generate error when write to each locked register;
7	R/W	0	lock AO gpio_o
6	R/W	0	lock AO gpio pull en
5	R/W	0	lock AO gpio pull up/down
4	R/W	0	lock AO gpio_oen_n
1	R/W	0	lock ao pin_mux_reg1
0	R/W	0	lock ao pin_mux_reg0

Always On GPIO controls.

Caution

This register is NOT reset during a watchdog event.

Table 6-313 AO_GPIO_TEST_N 0xd7

Bit(s)	R/W	Default	Description
31	R/W	1	TEST_N_OUTPUT_LEVEL: This bit controls the output level of the test_n pin when TEST_N_GPIO_EN is set to 1.
30	R	0	TEST_N input level
29-12	R/W	0	reserved
11-8	R/W	0	testn_sel
7-6	R/W	0	reserved
5	R/W	1	test_n_pull_up_en
4	R/W	1	test_n_pull_up
3-2	R/W	2’b10	test_n_ds
1-0	R/W	2’b10	reset_n_ds

6.10 Interrupt Control

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6.10.1 Overview

Generic Interrupt Controller (GIC) is a centralized resource that supports and manages interrupts in a system. For more details about GIC, please refer to the ARM GIC Architecture Specification V2.0.

6.10.2 Interrupt Source

There is a set of interrupt sources in the chip. All of the interrupts are connected to the integrated GIC in Cortex-A55 while the AO-CPU see a sub-set of the interrupts. The control Bits of AO-CPU interrupt are listed in the following table.

Table 6-314 EE Interrupt Source

A55 GIC Bit	Interrupt Source	Description
255	1'b0	
254	1'b0	
253	1'b0	
252	1'b0	
251	1'b0	
250	1'b0	
249	1'b0	
248	1'b0	
247	m_i2c_2_irq	
246	m_i2c_1_irq	
245	mbox_irq_send5	
244	mbox_irq_send4	
243	mbox_irq_send3	
242	mbox_irq_receiv2	
241	mbox_irq_receiv1	
240	mbox_irq_receiv0	
239	ao_gpio_irq1	
238	ao_gpio_irq0	
237	ao_timerB_irq	
236	ao_timerA_irq	
235	cecb_irq	
234	ao_watchdog_irq	
233	ao_m_i2c_to_irq	
232	sar_adc_irq	
231	ao_cec_irq	
230	ao_ir_blaster_irq	
229	ao_uart2_irq	
228	ao_ir_dec_irq	

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A55 GIC Bit	Interrupt Source	Description
227	ao_i2c_m_irq	
226	ao_i2c_s_irq	
225	ao_uart_irq	
224	1'b0	
223	SD_EMMC_C	
222	SD_EMMC_B	
221	1'b0	
220	1'b0	
219	1'b0	
218	1'b0	
217	dma_irq[5]	
216	dma_irq[4]	
215	dma_irq[3]	
214	dma_irq[2]	
213	dma_irq[1]	
212	dma_irq[0]	
211	eth_phy_irq8	
210	eth_phy_irq7	
209	eth_phy_irq6	
208	eth_phy_irq5	
207	eth_phy_irq4	
206	eth_phy_irq3	
205	eth_phy_irq2	
204	eth_phy_irq1	
203	eth_phy_irq0	
202	mali_irq_pp3	reserved
201	mali_irq_ppmmu2	reserved
200	mali_irq_pp2	reserved
199	mali_irq_ppmmu1	reserved
198	mali_irq_pp1	reserved
197	mali_irq_ppmmu0	reserved
196	mali_irq_pp0	reserved
195	mali_irq_pmu	
194	mali_irq_pp	
193	mali_irq_gpmmu	
192	mali_irq_gp	

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A55 GIC Bit	Interrupt Source	Description
191	AUDIO_IRQ11	
190	AUDIO_IRQ10	
189	AUDIO_IRQ9	reserved
188	AUDIO_IRQ8	toram
187	AUDIO_IRQ7	power_detect
186	AUDIO_IRQ6	frddr_C
185	AUDIO_IRQ5	frddr_B
184	AUDIO_IRQ4	frddr_A
183	AUDIO_IRQ3	spdifin
182	AUDIO_IRQ2	toddr_C
181	AUDIO_IRQ1	toddr_B
180	AUDIO_IRQ0	toddr_A
179	1'b0	
178	ge2d	
177	cusad	
176	1'b0	
175	viu1_wm_int	
174	demod	
173	A55irq[4]	EXTERRIRQ_a
172	A55irq[3]	|CTIIRQ[3:0]
171	A55irq[2]	|VCPUMNTIRQ_a[3:0]
170	A55irq[1]	|COMMIRQ_a[3:0]
169	A55irq[0]	|PMUIRQ_a[3:0]
168	mbox_rec7	
167	mbox_sed6	
166	A55IRQ[1]	
165	A55IRQ[0]	
164	viu1_line	
163	asssit_mbox_irq3	
162	asssit_mbox_irq2	
161	asssit_mbox_irq1	
160	asssit_mbox_irq0	
127	m_i2c_3_TO	
126	m_i2c_2_TO	
125	uart2_irq	
124	m_i2c_1_TO	

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A55 GIC Bit	Interrupt Source	Description
123	m_i2c_0_TO	
122	spicc_1_int	
121	rdma_done_int	
120	tcon_intr_out	
119	lc_curve_int	
118	vid1_wr_irq	
117	vdin1_vsync_int	
116	vdin1_hsync_int	
115	vdin0_vsync_int	
114	vdin0_hsync_int	
113	spicc_0_int	
112	spi_int	
111	vid0_wr_irq	
110	venc_vx1_int	
109	viu1_dolby_int	
108	viu1_mail_afbc	
107	uart1_irq	
106	1'b0	
105	1'b0	
104	vpu_crash_int	
103	gpio_irq[7]	
102	gpio_irq[6]	
101	gpio_irq[5]	
100	gpio_irq[4]	
99	gpio_irq[3]	
98	gpio_irq[2]	
97	gpio_irq[1]	
96	gpio_irq[0]	
95	TimerI	TimerI
94	TimerH	TimerH
93	TimerG	TimerG
92	TimerF	TimerF
91	viu2_line	
90	1'b0	
89	-	
88	viu2_vsync	

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A55 GIC Bit	Interrupt Source	Description
87	viu2_hsync	
86	dmc_test_irq	
85	demux_int_2	
84	dmc_irq	
83	dmc_sec_irq	
82	audio_irq[3]	
81	audio_irq[2]	
80	audio_irq[1]	
79	audio_irq[0]	
78	di_pre	Reserved for Deinterlacer
77	dos_mbox_slow_irq[2]	DOS Mailbox 2
76	dos_mbox_slow_irq[1]	DOS Mailbox 1
75	dos_mbox_slow_irq[0]	DOS Mailbox 0
74	htx_hdcp22_intr	
73	-	
72	di_post	
71	m_i2c_3_irq	I2C Master #3
70	ts_sar	
69	smartcard_irq	
68	ts_ddr	
67	ts_pll	
66	nand_irq	
65	viff_empty_int_cpu	
64	parser_int_cpu	
63	U2d_interrupt	USB
62	U3h_interrupt	USB
61	Timer D	Timer D
60	bus_mon1_fast_irq	
59	bus_mon0_fast_irq	
58	uart0_irq	
57	async_fifo2_flush_irq	
56	async_fifo2_fill_irq	
55	demux_int	
54	encif_irq	
53	m_i2c_0_irq	

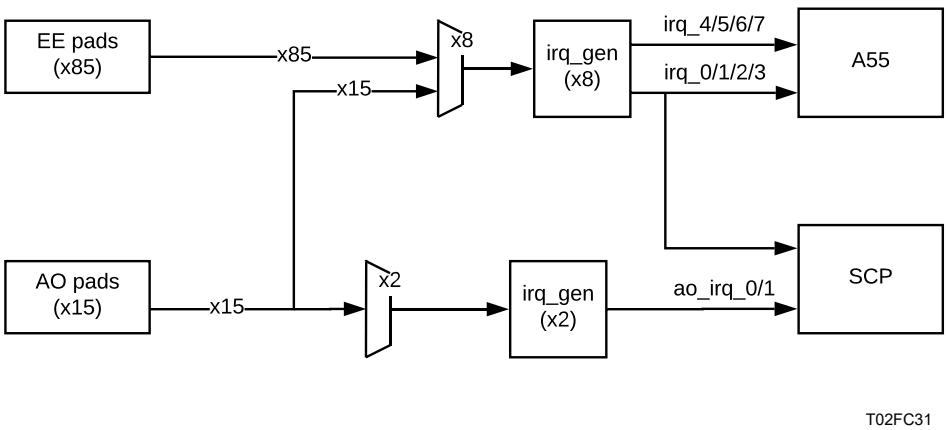
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A55 GIC Bit	Interrupt Source	Description
52	bt656	
51	async_fifo_flush_irq	
50	async_fifo_fill_irq	
49	async_fifo3_flush_irq	
48	usb_iddig_irq	
47	async_fifo3_fill_irq	unused
46	eth_lip_intro_o	
45	cvbs_det_irq1_out	
44	cvbs_det_irq0_out	
43	Timer B	Timer B
42	Timer A	Timer A
41	eth_phy_irq_or	
40	eth_gmac_int	
39	1'b0	
38	Timer C	Timer C
37	demux_int_1	
36	eth_pmt_intr_o	
35	viu1_vsync_int	VSYNC
34	viu1_hsync_int	HSYNC
33	1'b0	
32	ee_wd_irq	Watchdog Timer

6.10.3 GPIO Interrupt

There are 10 interrupts that are routed to all CPUs (A55/SCP). The tables below outline the interrupts associated with each interrupt group.

Figure 6-27 GPIO Interrupt



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There are 8 independent filtered GPIO interrupt modules that can be programmed to use any of the GPIOs in the chip as an interrupt source (listed in the table below). For example, to select gpioZ_15 as the source for GPIO IRQ #0, then CBUS 0x2609 bits[7:0] = 27 (according to the table below).

Input Mux Location **CBUS registers 0x2621 and 0x2622**	**Description**
[255:102]	0
[101:99]	gpioE[2:0]
[98:87]	gpioDV[11:0]
[86:75]	gpioW[11:0]
[74:60]	gpioC[14:0]
[59:46]	boot[13:0]
[45:23]	gpioH[22:0]
[22:12]	gpioZ[10:0]
[11:0]	gpioAO[11:0]

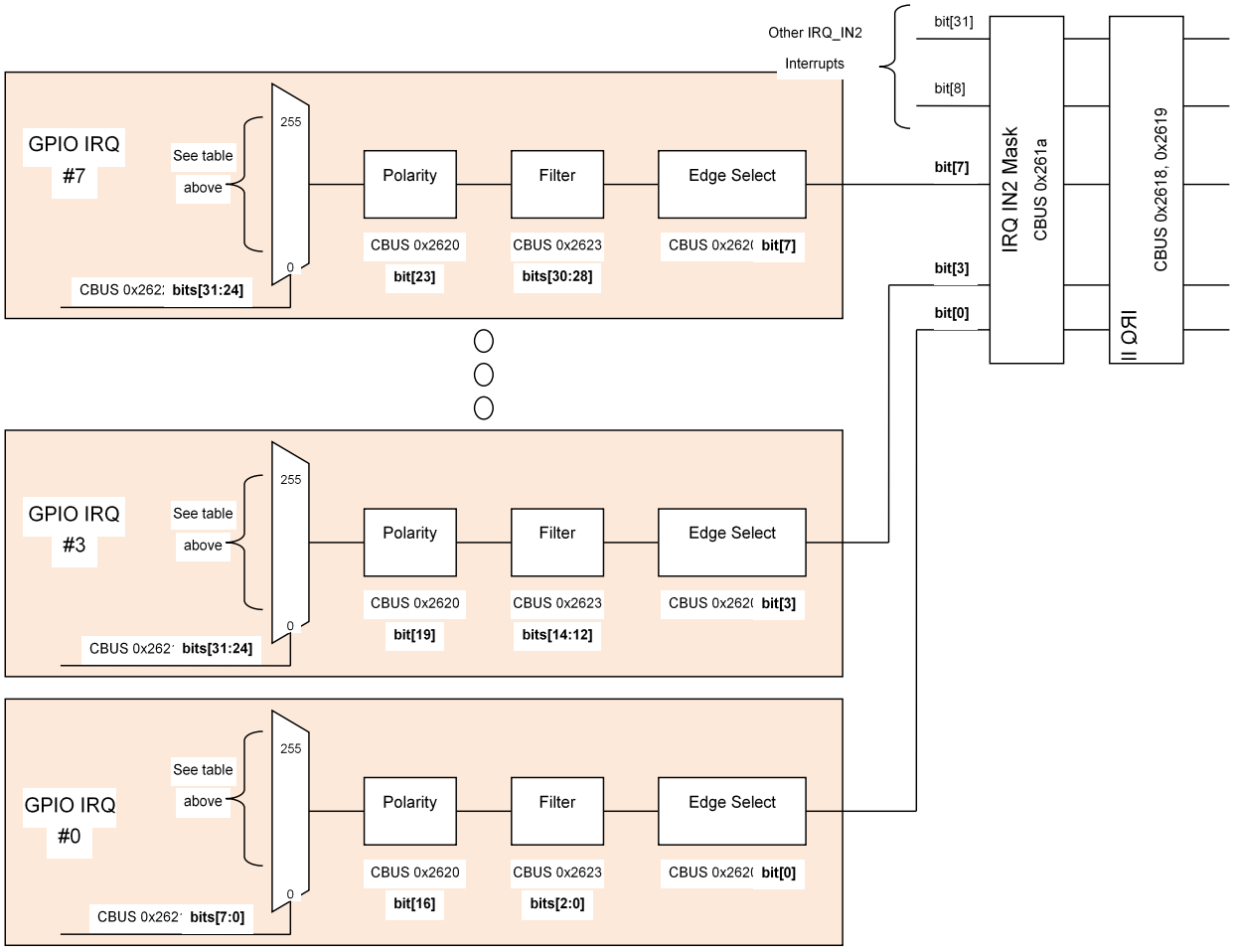
The diagram below illustrates the path a GPIO takes to become an interrupt. The eight GPIO interrupts respond to the MASK, STATUS and STATUS/CLEAR registers just like any other interrupt in the chip. The difference for the GPIO interrupts is that they can be filtered and conditioned.

Note

The input for the GPIO interrupt module (the input into the 256:1 mux) comes directly from the I/O pad of the chip. Therefore if a pad (say gpioA_11) is configured as a UART TX pin, then in theory, the UART TX pin can be a GPIO interrupt since the TX pin will drive gpioA_11 which in turn can drive the GPIO interrupt module.

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Figure 6-28 GPIO Interrupt Path



T02FC32

6.10.4 Register Description

Interrupt masking and FIQ select

Table 6-315 AO_IRQ_GPIO_REG 0x21

Bit(s)	R/W	Default	Description
31-20	R	0	Unused
19	R/W	0	GPIO_1_IRQ_EDGE: If this bit is 1, then GPIO_1_IRQ generates an interrupt on the rising (or falling edge if GPIO_1_POL is set) of the selected interrupt.
18	R/W	0	GPIO_0_IRQ_EDGE: If this bit is 1, then GPIO_0_IRQ generates an interrupt on the rising (or falling edge if GPIO_0_POL is set) of the selected interrupt.
17	R/W	0	GPIO_1_POL: This bit controls the polarity of the GPIO muxed into the GPIO_1 interrupt filter module
16	R/W	0	GPIO_0_POL: This bit controls the polarity of the GPIO muxed into the GPIO_0 interrupt filter module

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Bit(s)	R/W	Default	Description
15	R/W	0	GPIO_1_FILTER_USE_CLK: Setting this bit to 1 connects the input filter to the system clock (the clock used by the media CPU). If this bit is 0, then the filter uses a 125nS clock to filter the GPIO_1 input
14-12	R/W	0	GPIO_1_FILTER_SEL: 0 = no filter, 1 = max filter (about 2.6uS)
11	R/W	0	GPIO_0_FILTER_USE_CLK
10-8	R/W	0	GPIO_0_FILTER_SEL: 0 = no filter, 1 = max filter (about 2.6uS)
7-4	R/W	0	GPIO_1_INPUT_SEL: These bits select which gpioAO[11:0] pin is connected to the GPIO_1 interrupt filter module.
3-0	R/W	0	GPIO_0_INPUT_SEL: These bits select which gpioAO[11:0] pin is connected to the GPIO_1 interrupt filter module.

This register controls the polarity of the GPIO interrupts and whether or not the interrupts are level or edge triggered. There are 8 GPIO interrupts. These 8 GPIO interrupts can be assigned to any one of up to 256 pins on the chip.

Table 6-316 GPIO Interrupt EDGE and Polarity: 0x3c20

Bit(s)	R/W	Default	Description
31-24	R	0	unused
23			GPIO_POLARITY_PATH_7: If a bit in this field is 1, then the GPIO signal for GPIO interrupt path 7 is inverted.
22			GPIO_POLARITY_PATH_6:
21			GPIO_POLARITY_PATH_5:
20			GPIO_POLARITY_PATH_4:
19			GPIO_POLARITY_PATH_3:
18			GPIO_POLARITY_PATH_2:
17			GPIO_POLARITY_PATH_1:
16	R/W	0	GPIO_POLARITY_PATH_0:
15-8	R	0	Unused
7	R/W		GPIO_EDGE_SEL_PATH_7: If a bit is set to 1, then the GPIO interrupt for GPIO path 7 is configured to be an edge generated interrupt. If the polarity (above) is 0, then the interrupt is generated on the rising edge. If the polarity is 1, then the interrupt is generated on the falling edge of the GPIO. If a bit is this field is 0, then the GPIO is a level interrupt.
6	R/W		GPIO_EDGE_SEL_PATH_6
5	R/W		GPIO_EDGE_SEL_PATH_5
4	R/W		GPIO_EDGE_SEL_PATH_4
3	R/W		GPIO_EDGE_SEL_PATH_3
2	R/W		GPIO_EDGE_SEL_PATH_2
1	R/W		GPIO_EDGE_SEL_PATH_1
0	R/W	0	GPIO_EDGE_SEL_PATH_0

GPIO 0 ~ 3 Pin Select: 0x3c21

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Each GPIO interrupt can select from any number of up to 256 GPIO pins on the chip. The bits below control the pin selection for GPIO interrupts 0 ~3.

	R/W	Default	Description
31-24	R/W	0	GPIO_PIN_SEL3: This value select which of up to 256 pins on the chip can be mapped to GPIO interrupt 3
23-16	R/W	0	GPIO_PIN_SEL2: This value select which of up to 256 pins on the chip can be mapped to GPIO interrupt 2
15-8	R/W	0	GPIO_PIN_SEL1: This value select which of up to 256 pins on the chip can be mapped to GPIO interrupt 1
7-0	R/W	0	GPIO_PIN_SEL0: This value select which of up to 256 pins on the chip can be mapped to GPIO interrupt 0

GPIO 4 ~ 7 Pin Select: 0x3c22

Bit(s)	R/W	Default	Description
31-24	R/W	0	GPIO_PIN_SEL7: This value select which of up to 256 pins on the chip can be mapped to GPIO interrupt 7
23-16	R/W	0	GPIO_PIN_SEL6: This value select which of up to 256 pins on the chip can be mapped to GPIO interrupt 6
15-8	R/W	0	GPIO_PIN_SEL5: This value select which of up to 256 pins on the chip can be mapped to GPIO interrupt 5
7-0	R/W	0	GPIO_PIN_SEL4: This value select which of up to 256 pins on the chip can be mapped to GPIO interrupt 4

GPIO Filter Select (interrupts 0~7): 0x3c23

Bit(s)	R/W	Default	Description
31	R/W	0	unused
30-28	R/W	0	FILTER_SEL7: (see FILTER_SEL0)
27	R/W	0	Unused
26-24	R/W	0	FILTER_SEL6: (see FILTER_SEL0)
23	R/W	0	Unused
22-20	R/W	0	FILTER_SEL5: (see FILTER_SEL0)
19	R/W	0	Unused
18-16	R/W	0	FILTER_SEL4: (see FILTER_SEL0)
15	R/W	0	Unused
14-12	R/W	0	FILTER_SEL3: (see FILTER_SEL0)
11	R/W	0	Unused
10-8	R/W	0	FILTER_SEL2: (see FILTER_SEL0)
7	R/W	0	Unused
6-4	R/W	0	FILTER_SEL1: (see FILTER_SEL0)

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Bit(s)	R/W	Default	Description
3	R/W	0	unused
2-0	R/W	0	FILTER_SEL0: This value sets the filter selection for GPIO interrupt 0. A value of 0 = no filtering. A value of 7 corresponds to 7 x 3 x (111nS) of filtering.

6.11 Timer

6.11.1 Overview

The SOC contains 15 general purpose timers and 2 watchdog timers.

6.11.2 General-Purpose Timer

The SOC contains a number of general-purpose timers that can be used as general counters or interrupt generators. Each counter (except TIMER E) can be configured as a periodic counter (for generating periodic interrupts) or a simple count-down and stop counter. Additionally, the timers have a programmable count rate ranging from 1uS to 1mS. The table below outlines the general-purpose timers available in the chip.

Timer	Timebase Options	Counter size	Comment
Timer A	1uS, 10uS, 100uS, 1mS	16-bits	The 16-bit counter allows the timer to generate interrupts as infrequent as every 65.535 Seconds
Timer B	1uS, 10uS, 100uS, 1mS	16-bits	
Timer C	1uS, 10uS, 100uS, 1mS	16-bits	
Timer D	1uS, 10uS, 100uS, 1mS	16-bits	
Timer E	System clock, 1uS, 10uS, 100uS, 1mS	64-bits	Doesn't generate an interrupt. This is a count up counter that counts from 0 to 0xFFFFFFFF. The counter can be written at any time to reset the value to 0.
Timer F	1uS, 10uS, 100uS, 1mS	16-bits	
Timer G	1uS, 10uS, 100uS, 1mS	16-bits	
Timer H	1uS, 10uS, 100uS, 1mS	16-bits	
Timer I	1uS, 10uS, 100uS, 1mS	16-bits	
Timer A-AO	System clock, 1uS, 10uS, 100uS	32-bits	Used in the Always On domain to generate interrupts for the AO-CPU
Timer B-AO	System clock, 1uS, 10uS, 100uS	32-bits	Used in the Always On domain to generate interrupts for the AO-CPU
Timer C-AO	System clock, 1uS, 10uS, 100uS	32-bits	Used in the Always On domain to generate interrupts for the AO-CPU
Timer E-AO	System clock	64-bits	This Always On counter doesn't generate an interrupt. Instead it simply counts up from 0 to 0xFFFFFFFF. The counter can be written at any time to reset the value to 0.

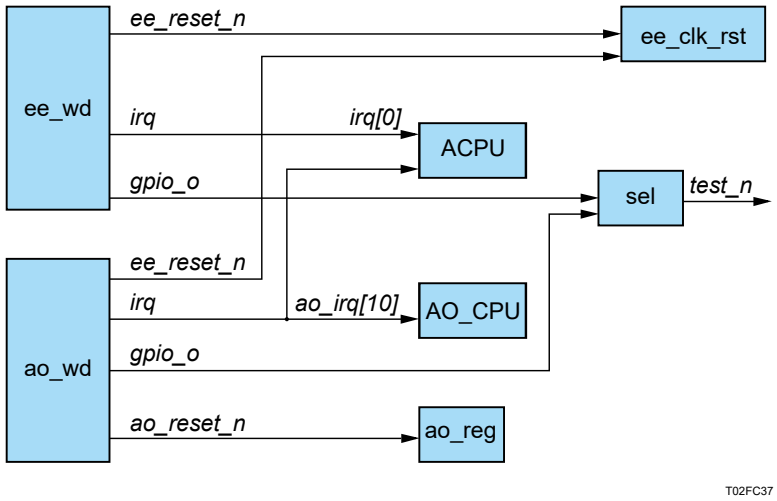
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Timer	Timebase Options	Counter size	Comment
Timer F-AO	System clock	64-bits	This Always On counter doesn't generate an interrupt. Instead it simply counts up from 0 to 0xFFFFFFFF. The counter can be written at any time to reset the value to 0.
Timer G-AO	System clock	64-bits	This Always On counter doesn't generate an interrupt. Instead it simply counts up from 0 to 0xFFFFFFFF. The counter can be written at any time to reset the value to 0.

6.11.3 Watchdog Timer

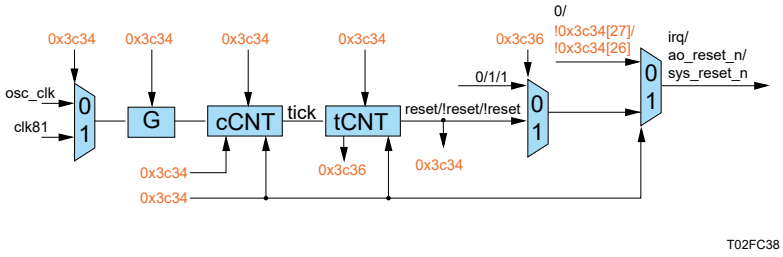
There are also two watchdog timers, one in AO and the other in EE domain, illustrated as following:

Figure 6-29 Watch Timer



AO domain watchdog timer and EE domain watchdog timer have the same design, as illustrated in the following figure:

Figure 6-30 EE domain Watchdog Timer Degisn



The AO Domain watchdog timer is driven from the system clock (typically 157Mhz). It is a 16-bit counter that is periodically reset by either the AO CPU or the System CPU (A55). This AO-watchdog timer can be used to generate an interrupt of the AO domain. Additionally, the AO-watchdog timer can be used to "enable" a delay generator that can toggle a GPIO pin (currently the TEST_n I/O pad). The "delay generator" allows an interrupt to first be acknowledged by the AO-CPU before the TEST_N pad is toggled. The "delay generator" is programmable from 1 to 65535 system clocks (typically 417uS).

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It should be noted that the AO watchdog timer can also be used to reset the AO domain but this feature is only used when operating in a suspend mode (only the AO-domain is powered). As long as the system periodically resets the AO-watchdog timer the WD_GPIO_CNT (delay generator) will not be enabled and the I/O pad will not toggle.

Note

The maximum delay between two AO-watchdog periodic resets is about 100mS (assuming a 157Mhz system clock).

The EE Domain watchdog timer is driven by the 24Mhz crystal clock and can be used to generate an interrupt to the system CPU (the A55) or optionally, the watchdog timer can completely reset the chip (causing a cold boot). There are a few registers that are not affected by watchdog timer. These registers are only reset by the external RESET_n I/O pad and can be used to store information related to a possible watchdog event. As long as the system CPU periodically resets the EE-watchdog timer, it will never timeout and cause an interrupt or system reset.

Note

The maximum delay between two EE-watchdog periodic resets is about 8.3 Seconds. This time is independent of the system clocks and is driven by the external 24Mhz crystal.

6.11.4 Register Description

Each register final address = 0xffd00000 + address * 4

Table 6-317 ISA_TIMER_MUX 0x3c50

Bit(s)	R/W	Default	Description
31-20	R	0	unused
19	R/W	1	TIMERD_ EN: Set to 1 to enable Timer D
18	R/W	1	TIMERC_ EN: Set to 1 to enable Timer C
17	R/W	1	TIMERB_ EN: Set to 1 to enable Timer B
16	R/W	1	TIMERA_EN: Set to 1 to enable Timer A
15	R/W	0	TIMERD_MODE: If this bit is set to 1, then timerD is a periodic . 0 = one-shot timer (M1 default)
14	R/W	0	TIMERC_MODE: If this bit is set to 1, then timerC is a periodic . 0 = one-shot timer (M1 default)
13	R/W	1	TIMERB_MODE: If this bit is set to 1, then timerB is a periodic (M1 Default). 0 = one-shot timer
12	R/W	1	TIMERA_MODE: If this bit is set to 1, then timerA is a periodic (M1 Default). 0 = one-shot timer
11	R	0	unused
10-8	R/W	0x1	TIMER E input clock selection: 000: System clock 001: 1uS Timebase resolution 010: 10uS Timebase resolution 011: 100uS Timebase resolution 100: 1mS timebase NOTE: The mux selection for Timer E is different from timer A, B, C and D
7-6	R/W	0x0	TIMER D input clock selection: See TIMER A below
5-4	R/W	0x0	TIMER C input clock selection: See TIMER A below

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Bit(s)	R/W	Default	Description
2-3	R/W	0x0	TIMER B input clock selection: See TIMER A below
1-0	R/W	0x0	TIMER A Input clock selection: These bits select the input timebase for the counters for TimerA 00: 1uS Timebase resolution 01: 10uS Timebase resolution 10: 100uS Timebase resolution 11: 1mS Timebase resolution

Note

Timer A is a 16 bit count DOWN counter driven by the clock selected in register 0x01000530. TIMER A will count down from some value to zero, generate an interrupt and then re-load the original start count value. This timer can be used to generate a periodic interrupt (e.g. interrupt every 22 uS).

Table 6-318 ISA_TIMERA 0x3c51

Bit(s)	R/W	Default	Description
31-16	R	-	Current Count value
15-0	R/W	0x0	Starting count value. Write this value to start TIMER A.

Note

Timer B is just like Timer A.

Table 6-319 ISA_TIMERB 0x3c52

Bit(s)	R/W	Default	Description
31-16	R	-	Current Count value
15-0	R/W	0x0	Starting count value. Write this value to start TIMER B

Note

Timer C is just like Timer A.

Table 6-320 ISA_TIMERC 0x3c53

Bit(s)	R/W	Default	Description
31-16	R	0	unused
15-0	R/W	0x0	Starting count value. Write this value to start TIMER C

Note

Timer D is identical to Timer A.

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Table 6-321 ISA_TIMERD 0x3c54

Bit(s)	R/W	Default	Description
31-16	R	0	unused
15-0	R/W	0x0	Starting count value. Write this value to start TIMER C

Note

Timer E is simply a 64-bit counter that increments at a rate set by register 0x3c50. To reset the counter to zero, simply write this register with any value. The value below is a read-only value that reflects the current count of the internal counter. This register can be used by software to simply provide a polling delay loop based on a programmable timebase.

Table 6-322 ISA_TIMERE 0x3c62

Bit(s)	R/W	Default	Description
31-0	R	-	Current value of Timer E. Write this register with any value to clear the counter.

Table 6-323 ISA_TIMERE_HI 0x3c63

Bit(s)	R/W	Default	Description
31-0	R	0	Counter[63:32], start by write TIMERE, not write TIMERE_HI.

Table 6-324 ISA_TIMER_MUX1 0x3c64

Bit(s)	R/W	Default	Description
31-20	R	0	unused
19	R/W	1	TIMERD_ EN: Set to 1 to enable Timer D
18	R/W	1	TIMERC_ EN: Set to 1 to enable Timer C
17	R/W	1	TIMERB_ EN: Set to 1 to enable Timer B
16	R/W	1	TIMERA_EN: Set to 1 to enable Timer A
15	R/W	0	TIMERD_MODE: If this bit is set to 1, then timerD is a periodic . 0 = one-shot timer (M1 default)
14	R/W	0	TIMERC_MODE: If this bit is set to 1, then timerC is a periodic . 0 = one-shot timer (M1 default)
13	R/W	1	TIMERB_MODE: If this bit is set to 1, then timerB is a periodic (M1 Default). 0 = one-shot timer
12	R/W	1	TIMERA_MODE: If this bit is set to 1, then timerA is a periodic (M1 Default). 0 = one-shot timer
11	R	0	unused
10-8	R/W	0x1	TIMER E input clock selection: 000: System clock 001: 1uS Timebase resolution 010: 10uS Timebase resolution 011: 100uS Timebase resolution 100: 1mS timebase NOTE: The mux selection for Timer E is different from timer A, B, C and D
7-6	R/W	0x0	TIMER D input clock selection: See TIMER A below
5-4	R/W	0x0	TIMER C input clock selection: See TIMER A below

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Bit(s)	R/W	Default	Description
2-3	R/W	0x0	TIMER B input clock selection: See TIMER A below
1-0	R/W	0x0	TIMER A Input clock selection: These bits select the input timebase for the counters for TimerA 00: 1uS Timebase resolution 01: 10uS Timebase resolution 10: 100uS Timebase resolution 11: 1mS Timebase resolution

Note

↓Timer F is a 16 bit count DOWN counter driven by the clock selected in register 0x01000530. TIMER A will count down from some value to zero, generate an interrupt and then re-load the original start count value. This timer can be used to generate a periodic interrupt (e.g. interrupt every 22 uS).

Table 6-325 ISA_TIMERF 0x3c65

Bit(s)	R/W	Default	Description
31-16	R	-	Current Count value
15-0	R/W	0x0	Starting count value. Write this value to start TIMER F.

Note

Timer G is just like Timer F.

Table 6-326 ISA_TIMERG 0x3c66

Bit(s)	R/W	Default	Description
31-16	R	-	Current Count value
15-0	R/W	0x0	Starting count value. Write this value to start TIMER G

Note

Timer H is just like Timer F.

Table 6-327 ISA_TIMER_H 0x3c67

Bit(s)	R/W	Default	Description
31-16	R	0	unused
15-0	R/W	0x0	Starting count value. Write this value to start TIMER H

Note

Timer I is just like Timer F.

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Table 6-328 ISA_TIMER_I 0x3c68

Bit(s)	R/W	Default	Description
31-16	R	0	unused
15-0	R/W	0x0	Starting count value. Write this value to start TIMER I

Table 6-329 WATCHDOG_CNTL 0x3c34

Bit(s)	R/W	Default	Description
31	R	0	Watchdog_reset
30-28	R/W	0	Reserved
27	R/W	0	Ao_reset_n_now, if watchdog_en =0, output ao_reset_n = ! ao_reset_n_now
26	R/W	0	Sys_reset_n_now, if watch_dog_en = 0, output sys_reset_n = !sys_reset_n_ now.
25	R/W	0	Clk_div_en: 0: no tick; 1: generate tick;
24	R/W	0	Clk_en: 0: no clk; 1: clk work;
23	R/W	0	Interrupt_en: 0: no irq out; 1: irq = watchdog_reset;
22	R/W	0	Ao_reset_n_en: 0: output ao_reset_n = 1; 1: output ao_reset_n = ! watchdog_ reset;
21	R/W	0	Sys_reset_n_en 0: output sys_reset_n = 1; 1: output sys_reset_n = ! watchdog_ reset;
20	R/W	0	Reserved
19	R/W	0	Clk_sel: 0:osc_clk; 1:clk_81
18	R/W	0	Watchdog_en 0: no watchdog reset 1: gen watchdog reset
17-0	R/W	0	Clk_div_tcnt, when clk_div_en is 1, generate a tick each tcnt clock

Table 6-330 WATCHDOG_CNTL1: 0x3c35

Bit(s)	R/W	Default	Description
31-18	R	0	Reserved
17	R/W	0	Gpio_pulse 0:level reset 1:pulse reset
16	R/W	0	Gpio_polarity 0: 1 is reset; 1: 0 is reset.
15-0	R/W	0	Gpio_pulse_tcnt If gpio_pulse is 1, level reset will hold tcnt clock.

Table 6-331 WATCHDOG_TCNT: 0x3c36

Bit(s)	R/W	Default	Description
31-16	R	0	The cnt of tick.
15-0	R/W	5000	If watchdog_en is 1, when tick cnt reached “5000”, generate watchdog_reset.

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Table 6-332 WATCHDOG_RESET: 0x3c37

Bit(s)	R/W	Default	Description
31-0	W	0	When write any value(include 0), watchdog module will be reset.

Below registers have the same base address: 0xFF800000, register address: 0xFF800000 + offset * 4, same for secure and non-secure access

Table 6-333 AO_TIMER_CTRL 0xf0

Bit(s)	R/W	Default	Description
31-7	R/W	0	Unused
6-5	R/W	0	TIMERE_CLK_MUX: 00 = Timer E clock = Media CPU clock 01 = Timer E clock = 1uS ticks 10 = Timer E clock = 10uS ticks 11 = Timer E clock = 100uS ticks
4	R/W	0	TIMER_E_EN
3	R/W	0	TIMER_A_EN
2	R/W	0	TIMER_A_MODE: 1 = periodic, 0 = one-shot
1-0	R/W	0	TIMERA_CLK_MUX: 00 = TimerA clock = Media CPU clock 01 = Timer A clock = 1uS ticks 10 = Timer A clock = 10uS ticks 11 = Timer A clock = 100uS ticks

Table 6-334 AO_TIMER_SEC_SCP_CTRL 0xf1

Bit(s)	R/W	Default	Description
31-7	R/W	0	Unused
6-5	R/W	0	TIMER F_CLK_MUX: 00 = Timer F clock = Media CPU clock 01 = Timer F clock = 1uS ticks 10 = Timer F clock = 10uS ticks 11 = Timer F clock = 100uS ticks
4	R/W	0	TIMER_F_EN
3	R/W	0	TIMER_B_EN
2	R/W	0	TIMER_B_MODE: 1 = periodic, 0 = one-shot
1-0	R/W	0	TIMERB_CLK_MUX: 00 = TimerB clock = Media CPU clock 01 = Timer B clock = 1uS ticks 10 = Timer B clock = 10uS ticks 11 = Timer B clock = 100uS ticks

Table 6-335 AO_TIMER_SEC_SP_CTRL 0xf2

Bit(s)	R/W	Default	Description
31-7	R/W	0	Unused
6-5	R/W	0	TIMER G_CLK_MUX: 00 = Timer G clock = Media CPU clock 01 = Timer G clock = 1uS ticks 10 = Timer G clock = 10uS ticks 11 = Timer G clock = 100uS ticks
4	R/W	0	TIMER_G_EN
3	R/W	0	TIMER_C_EN

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Bit(s)	R/W	Default	Description
2	R/W	0	TIMER_C_MODE: 1 = periodic, 0 = one-shot
1-0	R/W	0	TIMERC_CLK_MUX: 00 = Timer C clock = Media CPU clock 01 = Timer C clock = 1uS ticks 10 = Timer C clock = 10uS ticks 11 = Timer C clock = 100uS ticks

Note

Timer A starts at a non-zero value and decrements to 0. When timer A reaches a count of 0 it will re-load with the TIMER_A_TCNT value.

Table 6-336 AO_TIMERA_REG 0xf3

Bit(s)	R/W	Default	Description
31-0	R/W	0	timer A set value; if mode = 0, will clr timer to 0;

Table 6-337 AO_TIMERA_CUR_REG 0xf4

Bit(s)	R/W	Default	Description
31-0	R	0	TIMER A current count.

Table 6-338 AO_TIMERB_REG 0xf5

Bit(s)	R/W	Default	Description
31-0	R/W	0	timer B set value; if mode = 0, will clr timer to 0;

Table 6-339 AO_TIMERB_CUR_REG 0xf6

Bit(s)	R/W	Default	Description
31-0	R	0	TIMER B current count.

Table 6-340 AO_TIMERC_REG 0xf7

Bit(s)	R/W	Default	Description
31-0	R/W	0	timer C set value; if mode = 0, will clr timer to 0;

Table 6-341 AO_TIMERC_CUR_REG 0xf8

Bit(s)	R/W	Default	Description
31-0	R	0	TIMER C current count.

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Note

If this register is written (with any value), then Timer E is reset to 0. Immediately after being cleared, timer E will start incrementing at a clock rate equal to the clock used for the Media CPU.

Table 6-342 AO_TIMERE_REG 0xf9

Bit(s)	R/W	Default	Description
31-0	R/W	0	TIMER E current Count[31:0]

Table 6-343 AO_TIMERE_HI_REG 0xfa

Bit(s)	R/W	Default	Description
31-0	R/W	0	TIMER E current Count[63:32], latched by read timerE[31:0];

Note

If this register is written (with any value), then Timer F is reset to 0. Immediately after being cleared, timer F will start incrementing at a clock rate equal to the clock used for the Media CPU.

Table 6-344 AO_TIMERF_REG 0xfb

Bit(s)	R/W	Default	Description
31-0	R/W	0	TIMER F current Count[31:0]

Table 6-345 AO_TIMERF_HI_REG 0xfc

Bit(s)	R/W	Default	Description
31-0	R/W	0	TIMER F current Count[63:32], latched by read timerF[31:0];

Note

If this register is written (with any value), then Timer G is reset to 0. Immediately after being cleared, timer G will start incrementing at a clock rate equal to the clock used for the Media CPU.

Table 6-346 AO_TIMERG_REG 0xfd

Bit(s)	R/W	Default	Description
31-0	R/W	0	TIMER G current Count[31:0]

Table 6-347 AO_TIMERG_HI_REG 0xfe

Bit(s)	R/W	Default	Description
31-0	R/W	0	TIMER G current Count[63:32], latched by read timerG[31:0];

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Table 6-348 AO_WATCHDOG_CNTL 0x48

Bit(s)	R/W	Default	Description
31-28	R	0	unused
27	R/W	0	Ao_reset_n_now
26	R/W	0	Sys_reset_n_now
25	R/W	0	Clk_div_en
24	R/W	1	Clk_en
23	R/W	0	Interrupt_en
22	R/W	0	Ao_reset_n_en
21	R/W	0	Sys_reset_n_en
20	R/W	0	Noused
19	R/W	0	Clk_sel
18	R/W	0	Watchdog_en
17-0	R/W	23999	Clk_div_tcnt

Table 6-349 AO_WATCHDOG_CNTL1 0x49

Bit(s)	R/W	Default	Description
31-17	R	0	unused
17	R/W	0	Gpio_pulse
16	R/W	0	Gpio_polarity
15-0	R/W	0	Gpio_pulse_tcnt

Table 6-350 AO_WATCHDOG_TCNT 0x4a

Bit(s)	R/W	Default	Description
31-16	R	0	Watchdog count read
15-0	R/W	0	Watchdog_count set

Note

Write any value can reset watchdog.

Table 6-351 AO_WATCHDOG_RESET 0x4b

Bit(s)	R/W	Default	Description
31-0	W	0	Watchdog soft reset

Note

Write any value can reset watchdog.

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Table 6-352 AO_RTI_STICKY_REG0~3 0x4c~0x4f

Bit(s)	R/W	Default	Description
31-0	W	0	

6.12 Crypto

6.12.1 Overview

The crypto engine is one encrypt/decrypt function accelerator. Crypto engine supports 4 different modes, i.e. AP (Application Processor) secure, AP non secure, SCP (System Control Processor) secure and SCP non secure. The crypto engine has special internal DMA controller to transfer data.

It has the following features.

- AES block cipher with 128/256 bits keys, standard 16 bytes block size and streaming ECB, CBC and CTR modes
- DES/3DES block cipher with ECB and CBC modes supporting 64 bits key for DES and 192 bits key for 3DES
- Hardware key-ladder operation and DVB-CSA for transport stream encryption
- Built-in hardware True Random Number Generator (TRNG) and SHA-1/SHA-2 engine

6.12.2 Key Ladder

The Key Ladder is a series of TDES / AES crypto processes that iterates on different user supplied and OTP keys. The key ladder module uses a single AES / TDES crypto module and iterates by using internal storage to hold temporary states.

6.12.3 RNG

Functionality the Random Number Generator (RNG) contains two main modules, True Random Number Generator (TRNG) and Deterministic Random Number Generator (DRNG).

- True Random Number Generator (TRNG): is realized by using metastability and jitter for random bit generation based on four free running ring oscillator
- Deterministic Random Number Generator (DRNG): which has 32-bit random number generator, is mainly designed to increase the throughput and do post-processing of the Digital TRNG, which will need hundreds cycles to collect entropy.

6.12.4 EFUSE

The EFUSE consists of a 4k bit One Time Programmable (OTP) memory that is broken up into 32, 128-bit blocks. Data is always read/written in 128-bit blocks using the APB bus (software) or by the Key-ladder which is integrated with EFUSE block.

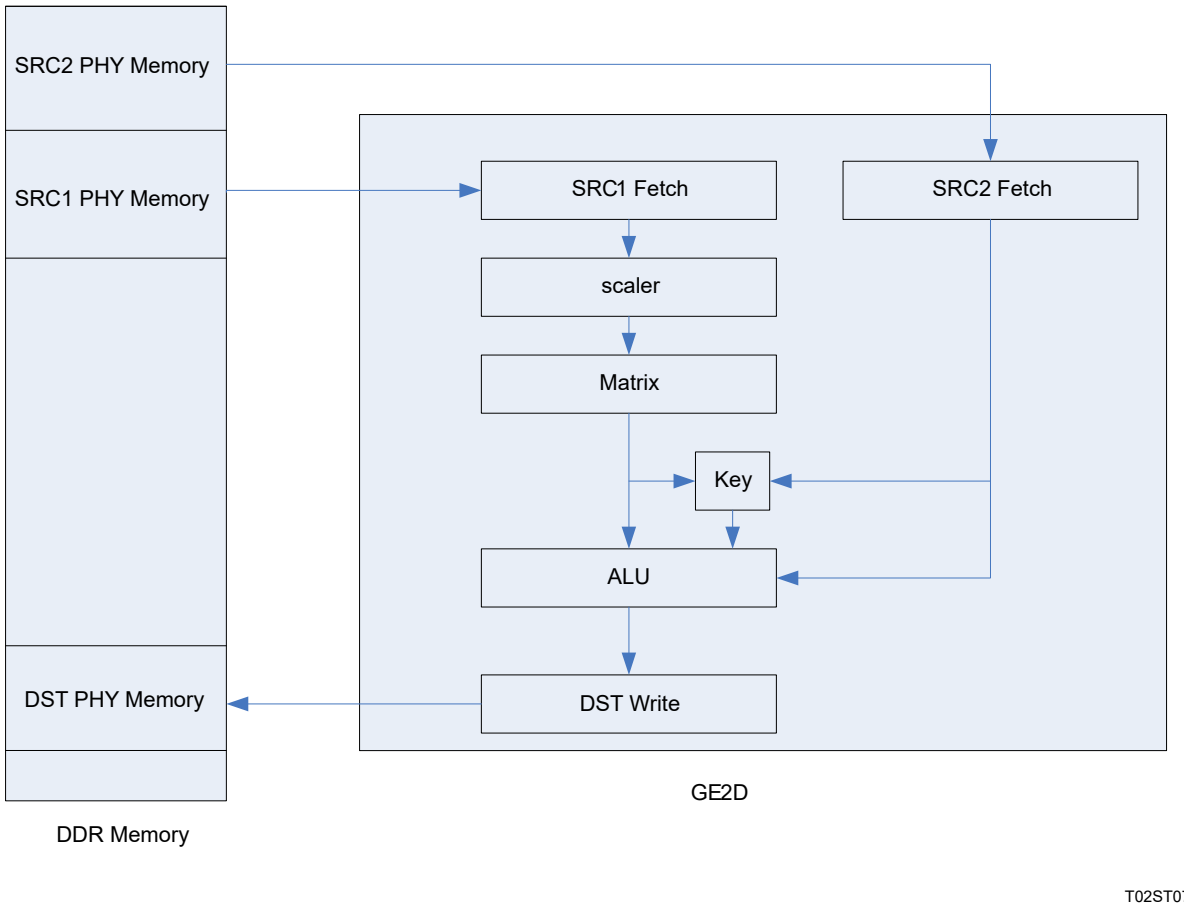
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7 GE2D

7.1 Overview

The basic structure of GE2D is shown in figure below.

Figure 7-1 GE2D Structure



7.2 Register Description

Table 7-1 GE2D_GEN_CTRL0 0x8A0

Bit(s)	R/W	Default	Description
31	R/W	0	dst_bytemask_only: Applicable only if dst_bitmask_en=1.
30	R/W	0	dst_bitmask_en: destination bitmask enable. 0: disable; 1: enable.
29	R/W	0	src2_key_en: source2 key enable. 0: disable; 1: enable.
28	R/W	0	src2_key_mode: source2 key mode. 0: mask data when match; 1: mask data when unmatch.

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Bit(s)	R/W	Default	Description
27	R/W	0	src1_key_en: source1 key enable. 0: disable; 1: enable.
26	R/W	0	src1_key_mode: source1 key mode. 0: mask data when match; 1: mask data when unmatch.
25-24	R/W	0	dst1_8b_mode_sel: Destination 1's 8-bit mode component selection. 0: Select Y or R; 1: Select Cb or G; 2: Select Cr or B; 3: Select Alpha.
23	R/W	0	dst_clip_mode: 0: Write inside clip window; 1: Write outside clip window.
22-17	R	0	Unused
16-15	R/W	0	src2_8b_mode_sel: Applicable only when src2_format=0, define the property of the 8-bit output data. 0: The 8-bit output data is for Y or R; 1: The 8-bit output data is for Cb or G; 2: The 8-bit output data is for Cr or B; 3: The 8-bit output data is for Alpha.
14	R/W	0	src2_fill_mode: When the display window is outside the boundary of the clipping window, this field defines how to fill the area outside the clipping window. 0: Fill with the pixels at the boundary of the clipping window; 1: Fill with the default color defined by register GE2D_SRC2_DEF_COLOR.
13-12	R/W	0	src2_pic_struct: Define how source2 reads the picture stored in DDR memory. 0: Read all lines; 1: Reserved; 2: Read even lines only; 3: Read odd lines only.
11	R/W	0	src2_X_yc_ratio: Source2 x direction yc ratio. 0: 1:1; 1: 2:1.
9-7	R	0	Unused
6-5	R/W	0	src1_8b_mode_sel: Applicable only when src1_sep_en =0, src1_format=0 and src1_lut_en=0, define the property of the 8-bit output data. 0: the 8-bit output data is for Y or R; 1: the 8-bit output data is for Cb or G; 2: the 8-bit output data is for Cr or B; 3: the 8-bit output data is for Alpha.
4	R/W	0	src1_fill_mode: When the display window is outside the boundary of the clipping window, this field defines how to fill the area outside the clipping window. 0: Fill with the pixels at the boundary of the clipping window; 1: Fill with the default color defined by register GE2D_SRC1_DEF_COLOR.
3	R/W	0	src1_lut_en: Applicable only when src1_sep_en =0 and src1_format= 0, define whether to enable 8-bit input data to look up a 32-bit pixel for output. 0: Disable; 1: Enable.
2-1	R/W	0	src1_pic_struct: Define how source1 reads the picture stored in DDR memory. 0: Read all lines; 1: Reserved; 2: Read even lines only; 3: Read odd lines only.

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Table 7-2 GE2D_GEN_CTRL1 0x8A1

Bit(s)	R/W	Default	Description
31	R/W	0	soft_rst: If true, reset GE2D.
30	R/W	0	dst write response counter reset
29	R/W	0	disable adding dst write response count to busy bit
28-27	R	0	Unused
26	R/W	0	Color_conversion_mode[1] in alu. Mode[1:0] 3 : color_out = color; 2 : color_out = (color != 255) ? color : color + 1; 1 : color_out = (color < 128) ? color : color + 1; 0 : color_out = (color == 0) ? color : color + 1.
25-24	R/W	0	interrupt_ctrl: If bit[0] true, generate interrupt when one command done; if bit[1] true, generate interrupt when ge2d change from busy to not busy.
23-22	R/W	0x3	src2_burst_size_ctrl: Source2 DDR request burst size control. (Note: data to source2 are stored together in one DDR memory block.) 0: Burst size = 24 x 64-bit; 1: Burst size = 32 x 64-bit; 2: Burst size = 48 x 64-bit; 3: Burst size = 64 x 64-bit.
21-16	R/W	0x3F	src1_burst_size_ctrl: Source1 DDR request burst size control. Bit[21:20] control Y burst size, bit[19:18] control Cb burst size, bit[17:16] control Cr burst size. Each 2-bit is decoded as below: 0: Burst size = 24 x 64-bit; 1: Burst size = 32 x 64-bit; 2: Burst size = 48 x 64-bit; 3: Burst size = 64 x 64-bit.
15-14	R/W	0	dst1_pic_struct: Define how destination 1 write the picture to DDR memory. 0: Write all lines (whole frame); 1: Reserved; 2: Write even lines only (top); 3: Write odd lines only (bottom).
13-12	R/W	0	src_rd_ctrl: Bit[13] if true, force read src1, bit[12] if true, force read src2.
11	R/W	0	dst2_urgent_en: Destination 2 DDR request urgent enable.
10	R/W	0	src1_urgent_en: Source1 DDR request urgent enable.
9	R/W	0	src2_urgent_en: Source2 DDR request urgent enable.
8	R/W	0	dst1_urgent_en: Destination 1 DDR request urgent enable.
7-0	R/W	0	src1_gb_alpha: Source1 global alpha.

Table 7-3 GE2D_GEN_CTRL2 0x8A2

Bit(s)	R/W	Default	Description
31	R/W	0	alpha_conversion_mode[0] in alu. Mode[1:0]: 2,3 : alpha_out = (alpha!=255) ? alpha : alpha + 1; 1 : alpha_out = (alpha < 128) ? alpha : alpha + 1; 0 : alpha_out = (alpha == 0) ? alpha : alpha + 1.
30	R/W	0	Color_conversion_mode[0] in alu. Mode[1:0] 3 : color_out = color; 2 : color_out = (color != 255) ? color : color + 1; 1 : color_out = (color < 128) ? color : color + 1;

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Bit(s)	R/W	Default	Description
			0 : color_out = (color == 0) ? color : color + 1.
29	R/W	0	src1_gb_alpha_en in alu. As = src1_gb_alpha_en ? Asr * Ag: Asr
28	R/W	0	dst1_COLOR_round_MODE. 1 = Truncate the full bit color components to required output bit width; 0 = Round (+ 0.5) the full bit color components to required output bit width.
27	R/W	0	src2_COLOR_EXPAND_MODE. 1 = Expand the color components to 8-bit by padding LSBs with MSBs. E.g. If the input is 5'b11000, the output is expanded to 8'b11000110; 0 = Expand the color components to 8-bit by padding LSBs with 0.
26	R/W	0	src2_ALPHA_EXPAND_MODE. 1 = Expand alpha value to 8-bit by padding LSBs with MSBs. E.g. If the input is 5'b11000, the output is expanded to 8'b11000110; 0 = If input alpha value is all 1, then expand the value to 8-bit by padding LSBs with 1; otherwise, pad LSBs with 0.
25	R/W	0	src1_COLOR_EXPAND_MODE. 1 = Expand the color components to 8-bit by padding LSBs with MSBs. E.g. If the input is 5'b11000, the output is expanded to 8'b11000110; 0 = Expand the color components to 8-bit by padding LSBs with 0.
24	R/W	0	src1_ALPHA_EXPAND_MODE. 1 = Expand alpha value to 8-bit by padding LSBs with MSBs. E.g. If the input is 5'b11000, the output is expanded to 8'b11000110; 0 = If input alpha value is all 1, then expand the value to 8-bit by padding LSBs with 1; otherwise, pad LSBs with 0.
23	R/W	0	dst_little_endian: define the endianess of SRC2 input data. 1 = Little endian; 0 = Big endian.
22-19	R/W	0	dst1_color_map: Applicable to 16-bit, 24-bit and 32-bit pixel, defines the bit-field allocation of the pixel data. For whether to truncate or round full 8-bit to output, refer to ge2d_gen_ctrl2.dst1_color_round_mode. For 16-bit mode (dst1_format=1): 0 = Unused; 1 = 6:5:5 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:5] is Cb[7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3]; 2 = 8:4:4 format. Bit[15:8] is Y or R, bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]]; 3 = 6:4:4:2 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:6] is Cb[7:4] or G[7:4], bit [5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[7:6]; 4 = 4:4:4:4 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:8] is Cb[7:4] or G[7:4], bit [7:4] is Cr[7:4] or B[7:4], bit[3:0] is Alpha[7:4]; 5 = 5:6:5 format. Bit[15:11] is Y[7:3] or R[7:3], bit[10:5] is Cb[7:2] or G[7:2], bit [4:0] is Cr[7:3] or B[7:3]; 6 = 4:4:4:4 format. Bit[15:12] is Alpha[7:4], bit[11:8] is Y[7:4] or R[7:4], bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]; 7 = 1:5:5:5 format. Bit[15] is Alpha[7], bit[14:10] is Y[7:3] or R[7:3], bit[9:5] is Cb [7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3]; 8 = 4:6:4:2 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:6] is Cb[7:2] or G[7:2], bit [5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[1:0]. 9 = CbCr format. Bit[15:8] is Cb, bit[7:0] is Cr; 10 = CrCb format. Bit[15:8] is Cr, bit[7:0] is Cb. For 24-bit mode (dst1_format=2): 0 = RGB 8:8:8 mode. Bit[23:16] is Y or R, bit[15:8] is Cb or G, bit[7:0] is Cr or B; 1 = RGBA 5:6:5:8 mode. Bit[23:19] is Y[7:3] or R[7:3], bit[18:13] is Cb[7:2] or G [7:2], bit[12:8] is Cr[7:3] or B[7:3], bit[7:0] is Alpha; 2 = ARGB 8:5:6:5 mode. Bit[23:16] is Alpha, bit[15:11] is Y[7:3] or R[7:3], bit [10:5] is Cb[7:2] or G[7:2], bit[4:0] is Cr[7:3] or B[7:3]; 3 = RGBA 6:6:6:6 mode. Bit[23:18] is Y[7:2] or R[7:2], bit[17:12] is Cb[7:2] or G [7:2], bit[11:6] is Cr[7:2] or B[7:2], bit[5:0] is Alpha[7:2];

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Bit(s)	R/W	Default	Description
			4 = ARGB 6:6:6:6 mode. Bit[23:18] is Alpha[7:2];, bit[17:12] is Y[7:2] or R[7:2], bit[11:6] is Cb[7:2] or G[7:2], bit[5:0] is Cr[7:2] or B[7:2]; 5 = BGR 8:8:8 mode. Bit[23:16] is Cr or B, bit[15:8] is Cb or G, bit[7:0] is Y or R. For 32-bit mode (dst1_format=3): 0 = RGBA 8:8:8:8 format. Bit[31:24] is Y or R, bit[23:16] is Cb or G; bit[15:8] is Cr or B; bit[7:0] is Alpha; 1 = ARGB 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Y or R; bit[15:8] is Cb or G; bit[7:0] is Cr or B; 2 = ABGR 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Cr or B; bit[15:8] is Cb or G; bit[7:0] is Y or R; 3 = BGRA 8:8:8:8 format. Bit[31:24] is Cr or B, bit[23:16] is Cb or G; bit[15:8] is Y or R; bit[7:0] is Alpha.
18	R/W	0	ALU_MULT_MODE: 1: mult result rounding else truncation
17-16	R/W	0	dst1_format: define output pixel byte-width. 0: Output pixel is 1-byte (8-bit) color component; 1: Output pixel is 2-byte (16-bit), refer to GE2D_GEN_CTRL2.dst1_COLOR_map for further pixel color mapping; 2: Output pixel is 3-byte (24-bit), refer to GE2D_GEN_CTRL2.dst1_COLOR_MAP for further pixel color mapping; 3: Output pixel is 4-byte (32-bit), refer to GE2D_GEN_CTRL2.dst1_COLOR_MAP for further pixel color mapping.
15	R/W	0	src2_little_endian: define the endianess of SRC2 input data. 1 = Little endian; 0 = Big endian.
14-11	R/W	0	src2_color_map: Applicable to 16-bit, 24-bit and 32-bit pixel, defines the bit-field allocation of the pixel data. For expanding the bit-fields to full 8-bit, refer to ge2d_gen_ctrl2.src2_color_expand_mode and ge2d_gen_ctrl2.src2_alpha_expand_mode. For 16-bit mode (src2_format=1): 0 = Unused; 1 = 6:5:5 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:5] is Cb[7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3], for Alpha value refer to src2_def_color_alpha 0x8af[7:0]; 2 = 8:4:4 format. Bit[15:8] is Y or R, bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]], for Alpha value refer to src2_def_color_alpha 0x8af[7:0]; 3 = 6:4:4:2 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:6] is Cb[7:4] or G[7:4], bit [5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[7:6]; 4 = 4:4:4:4 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:8] is Cb[7:4] or G[7:4], bit [7:4] is Cr[7:4] or B[7:4], bit[3:0] is Alpha[7:4]; 5 = 5:6:5 format. Bit[15:11] is Y[7:3] or R[7:3], bit[10:5] is Cb[7:2] or G[7:2], bit [4:0] is Cr[7:3] or B[7:3], for Alpha value refer to src2_def_color_alpha 0x8af [7:0]; 6 = 4:4:4:4 format. Bit[15:12] is Alpha[7:4], bit[11:8] is Y[7:4] or R[7:4], bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]; 7 = 1:5:5:5 format. Bit[15] is Alpha[7], bit[14:10] is Y[7:3] or R[7:3], bit[9:5] is Cb [7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3]; 8 = 4:6:4:2 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:6] is Cb[7:2] or G[7:2], bit [5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[1:0]. For 24-bit mode (src2_format=2): 0 = RGB 8:8:8 mode. Bit[23:16] is Y or R, bit[15:8] is Cb or G, bit[7:0] is Cr or B, for Alpha value refer to src2_def_color_alpha 0x8af[7:0]; 1 = RGBA 5:6:5:8 mode. Bit[23:19] is Y[7:3] or R[7:3], bit[18:13] is Cb[7:2] or G [7:2], bit[12:8] is Cr[7:3] or B[7:3], bit[7:0] is Alpha; 2 = ARGB 8:5:6:5 mode. Bit[23:16] is Alpha, bit[15:11] is Y[7:3] or R[7:3], bit [10:5] is Cb[7:2] or G[7:2], bit[4:0] is Cr[7:3] or B[7:3]; 3 = RGBA 6:6:6:6 mode. Bit[23:18] is Y[7:2] or R[7:2], bit[17:12] is Cb[7:2] or G [7:2], bit[11:6] is Cr[7:2] or B[7:2], bit[5:0] is Alpha[7:2]; 4 = ARGB 6:6:6:6 mode. Bit[23:18] is Alpha[7:2];, bit[17:12] is Y[7:2] or R[7:2], bit[11:6] is Cb[7:2] or G[7:2], bit[5:0] is Cr[7:2] or B[7:2]; 5 = BGR 8:8:8 mode. Bit[23:16] is Cr or B, bit[15:8] is Cb or G, bit[7:0] is Y or R, for Alpha value refer to src2_def_color_alpha 0x8af[7:0].

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Bit(s)	R/W	Default	Description
			For 32-bit mode (src2_format=3): 0 = RGBA 8:8:8:8 format. Bit[31:24] is Y or R, bit[23:16] is Cb or G; bit[15:8] is Cr or B; bit[7:0] is Alpha; 1 = ARGB 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Y or R; bit[15:8] is Cb or G; bit[7:0] is Cr or B; 2 = ABGR 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Cr or B; bit[15:8] is Cb or G; bit[7:0] is Y or R; 3 = BGRA 8:8:8:8 format. Bit[31:24] is Cr or B, bit[23:16] is Cb or G; bit[15:8] is Y or R; bit[7:0] is Alpha.
10	R/W	0	alpha_conversion_mode[1] in alu. Mode[1:0]: 2,3 : alpha_out = (alpha!=255) ? alpha : alpha + 1; 1 : alpha_out = (alpha < 128) ? alpha : alpha + 1; 0 : alpha_out = (alpha == 0) ? alpha : alpha + 1.
9-8	R/W	0	src2_format: define input pixel byte-width. 0: Input pixel is 1-byte (8-bit) color component; 1: Input pixel is 2-byte (16-bit), refer to GE2D_GEN_CTRL2.SRC2_COLOR_MAP for further pixel color mapping; 2: Input pixel is 3-byte (24-bit), refer to GE2D_GEN_CTRL2.SRC2_COLOR_MAP for further pixel color mapping; 3: Input pixel is 4-byte (32-bit), refer to GE2D_GEN_CTRL2.SRC2_COLOR_MAP for further pixel color mapping.
7	R/W	0	src1_little_endian: define the endianess of SRC1 input data. 1 = Little endian; 0 = Big endian.
6-3	R/W	0	src1_color_map: Note: If SRC1_DEEPCOLOR=0, the SRC1_COLOR_MAP's definitions is as below. If SRC1_DEEPCOLOR=1, please refer to SRC1_DEEPCOLOR entry for new meaning. Applicable to 16-bit, 24-bit and 32-bit pixel, defines the bit-field allocation of the pixel data. For expanding the bit-fields to full 8-bit, refer to ge2d_gen_ctrl2.src1_color_expand_mode and ge2d_gen_ctrl2.src1_alpha_expand_mode. For 16-bit mode (src1_format=1): 0 = 4:2:2 format (Y0Cb0Y1Cr0); 1 = 6:5:5 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:5] is Cb[7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3], for Alpha value refer to src1_def_color_alpha 0x8a6[7:0]; 2 = 8:4:4 format. Bit[15:8] is Y or R, bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]], for Alpha value refer to src1_def_color_alpha 0x8a6[7:0]; 3 = 6:4:4:2 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:6] is Cb[7:4] or G[7:4], bit [5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[7:6]; 4 = 4:4:4:4 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:8] is Cb[7:4] or G[7:4], bit [7:4] is Cr[7:4] or B[7:4], bit[3:0] is Alpha[7:4]; 5 = 5:6:5 format. Bit[15:11] is Y[7:3] or R[7:3], bit[10:5] is Cb[7:2] or G[7:2], bit [4:0] is Cr[7:3] or B[7:3], for Alpha value refer to src1_def_color_alpha 0x8a6 [7:0]; 6 = 4:4:4:4 format. Bit[15:12] is Alpha[7:4], bit[11:8] is Y[7:4] or R[7:4], bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]; 7 = 1:5:5:5 format. Bit[15] is Alpha[7], bit[14:10] is Y[7:3] or R[7:3], bit[9:5] is Cb [7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3]; 8 = 4:6:4:2 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:6] is Cb[7:2] or G[7:2], bit [5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[1:0]. For 24-bit mode (src1_format=2): 0 = RGB 8:8:8 mode. Bit[23:16] is Y or R, bit[15:8] is Cb or G, bit[7:0] is Cr or B, for Alpha value refer to src1_def_color_alpha 0x8a6[7:0]; 1 = RGBA 5:6:5:8 mode. Bit[23:19] is Y[7:3] or R[7:3], bit[18:13] is Cb[7:2] or G [7:2], bit[12:8] is Cr[7:3] or B[7:3], bit[7:0] is Alpha; 2 = ARGB 8:5:6:5 mode. Bit[23:16] is Alpha, bit[15:11] is Y[7:3] or R[7:3], bit [10:5] is Cb[7:2] or G[7:2], bit[4:0] is Cr[7:3] or B[7:3];

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Bit(s)	R/W	Default	Description
			3 = RGBA 6:6:6:6 mode. Bit[23:18] is Y[7:2] or R[7:2], bit[17:12] is Cb[7:2] or G[7:2], bit[11:6] is Cr[7:2] or B[7:2], bit[5:0] is Alpha[7:2]; 4 = ARGB 6:6:6:6 mode. Bit[23:18] is Alpha[7:2];, bit[17:12] is Y[7:2] or R[7:2], bit[11:6] is Cb[7:2] or G[7:2], bit[5:0] is Cr[7:2] or B[7:2]; 5 = BGR 8:8:8 mode. Bit[23:16] is Cr or B, bit[15:8] is Cb or G, bit[7:0] is Y or R, for Alpha value refer to src1_def_color_alpha 0x8a6[7:0]. 14=NV12 format. 8-bit Y and 16-bit CbCr; 15=NV21 format. 8-bit Y and 16-bit CrCb; For 32-bit mode (src1_format=3): 0 = RGBA 8:8:8:8 format. Bit[31:24] is Y or R, bit[23:16] is Cb or G; bit[15:8] is Cr or B; bit[7:0] is Alpha; 1 = ARGB 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Y or R; bit[15:8] is Cb or G; bit[7:0] is Cr or B; 2 = ABGR 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Cr or B; bit[15:8] is Cb or G; bit[7:0] is Y or R; 3 = BGRA 8:8:8:8 format. Bit[31:24] is Cr or B, bit[23:16] is Cb or G; bit[15:8] is Y or R; bit[7:0] is Alpha.
2	R/W	0	src1_DEEPCOLOR: 1 = Enable deepcolor formats support, the formats are defined by SRC1_FORMAT and SRC1_COLOR_MAP; 0 = Disable deepcolor. The supported deepcolor formats are as below: src1_deep_color=1, src1_format=2'b01, src1_color_map=4'b0000 or 4'b0001: 10-bit 422 in one canvas – 10-bit Y + 10-bit C = 20-bit per pixel in canvas. If src1_color_map=4'b0000, the sequence is Y0Cb0, Y1Cr0, ... If src1_color_map=4'b0001, the sequence is Y0Cr0, Y1Cb0, ... src1_deep_color=1, src1_format=2'b01, src1_color_map=4'b1000 or 4'b1001: 12-bit 422 in one canvas – 12-bit Y + 12-bit C = 24-bit per pixel in canvas. If src1_color_map=4'b1000, the sequence is Y0Cb0, Y1Cr0, ... If src1_color_map=4'b1001, the sequence is Y0Cr0, Y1Cb0, ... src1_deep_color=1, src1_format=2'b10: 10-bit 444 in one canvas – 10-bit Y + 10-bit Cb + 10-bit Cr + 2-bit stuffing = 32-bit per pixel in canvas. If src1_color_map=4'b0000, the sequence is Y0Cb0Cr0, Y1Cb1Cr1, ... If src1_color_map=other value, the sequence is Cr0Cb0Y0, Cr1Cb1Y1, ...
1-0	R/W	0	src1_format: define input pixel byte-width. Note: If SRC1_DEEPCOLOR=0, the SRC1_FORMAT's definitions is as below. If SRC1_DEEPCOLOR=1, please refer to SRC1_DEEPCOLOR entry for new meaning. 0: Input pixel is 1-byte (8-bit), it is either an 8-bit color component or 8-bit address to look up a 32-bit pixel, refer to GE2D_GEN_CTRL0. src1_lut_en 1: Input pixel is 2-byte (16-bit), refer to GE2D_GEN_CTRL2.SRC1_COLOR_MAP for further pixel color mapping; 2: Input pixel is 3-byte (24-bit), refer to GE2D_GEN_CTRL2.SRC1_COLOR_MAP for further pixel color mapping; 3: Input pixel is 4-byte (32-bit), refer to GE2D_GEN_CTRL2.SRC1_COLOR_MAP for further pixel color mapping.

Table 7-4 GE2D_CMD_CTRL 0x8A3

Bit(s)	R/W	Default	Description
31-10	R	0	unused
9	R/W	0	src2_fill_color_en: if true, all src2 data use default color.
8	R/W	0	src1_fill_color_en: if true, all src1 data use default color.
7	R/W	0	dst_xy_swap: if true, dst x/y swap.

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Bit(s)	R/W	Default	Description
6	R/W	0	dst_x_rev: if true, dst x direction reversely read.
5	R/W	0	dst_y_rev: if true, dst y direction reversely read.
4	R/W	0	src2_x_rev: if true, src2 x direction reversely read.
3	R/W	0	src2_y_rev: if true, src2 y direction reversely read.
2	R/W	0	src1_x_rev: if true, src1 x direction reversely read.
1	R/W	0	src1_y_rev: if true, src1 y direction reversely read.
0	R/W	0	cbus_cmd_wr: If true, generate a pulse to validate a GE2D command, the command is described by the rest of the field of this register.

Table 7-5 GE2D_STATUS0 0x8A4

Bit(s)	R/W	Default	Description
31-29	R	0	unused
28-17	R	0	dst write response counter, for debug only.
16-7	R	0	dp_status: ge2d_dp status, for debug only.
6	R	0	r1cmd_rdy: read src2 cmd ready.
5	R	0	r2cmd_rdy: read src2 cmd ready.
4	R	0	pdpcmd_v: pre dpcmd ready.
3	R	0	dpcmd_rdy: GE2D dpcmd ready.
2	R	0	buf_cmd_v: GE2D buffer command valid.
1	R	0	curr_cmd_v: GE2D current command valid.
0	R	0	ge2d_busy: GE2D busy.

Table 7-6 GE2D_STATUS1 0x8A5

Bit(s)	R/W	Default	Description
31-30	R	0	unused
29-16	R	0	ge2d_dst1_status, for debug only.
15	R	1	ge2d_rd_src2 core.fifo_empty.
14	R	0	ge2d_rd_src2 core.fifo_overflow.
13-12	R	0	ge2d_rd_src2 core.req_st. Same as req_st_y.
11	R	0	ge2d_rd_src2 cmd_if.cmd_err, true if cmd_format=1.
10	R	0	ge2d_rd_src2 cmd_if.cmd_st, 0=IDLE state, 1=BUSY state.
9	R	1	ge2d_rd_src1 luma_core(chroma_core).fifo_empty.
8	R	0	ge2d_rd_src1 luma_core(chroma_core).fifo_overflow.
7-6	R	0	ge2d_rd_src1 chroma_core.req_st_cr. Same as req_st_y.
5-4	R	0	ge2d_rd_src1 chroma_core.req_st_cb. Same as req_st_y.

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Bit(s)	R/W	Default	Description
3-2	R	0	ge2d_rd_src1 luma_core.req_st_y. 0: IDLE; 1: WAIT_FIFO_ROOM; 2: REQUEST; 3: WAIT_FINISH.
1	R	0	ge2d_rd_src1 cmd_if.stat_read_window_err, 1=reading/clipping window setting exceed limit.
0	R	0	ge2d_rd_src1 cmd_if.cmd_st, 0=IDLE state, 1=BUSY state.

Table 7-7 GE2D_SRC1_DEF_COLOR 0x8A6

Bit(s)	R/W	Default	Description
31-24	R/W	0	Default Y or R.
23-16	R/W	0x80	Default Cb or G.
15-8	R/W	0x80	Default Cr or B.
7-0	R/W	0	Default Alpha.

Table 7-8 GE2D_SRC1_CLIPX_START_END 0x8A7

Bit(s)	R/W	Default	Description
31	R/W	0	SRC1 clip x start extra, if true, one more data is read for chroma.
30-29	R	0	Unused.
28-16	R/W	0	SRC1 clip x start.
15	R/W	0	SRC1 clip x end extra, if true, one more data is read for chroma.
14-13	R	0	Unused.
12-0	R/W	0x1FFF	SRC1 clip x end.

Table 7-9 GE2D_SRC1_CLIPY_START_END 0x8A8

Bit(s)	R/W	Default	Description
30-29	R	0	Unused.
28-16	R/W	0	SRC1 clip y start.
14-13	R	0	Unused.
12-0	R/W	0x1FFF	SRC1 clip y end.

Table 7-10 GE2D_SRC1_CANVAS 0x8A9

Bit(s)	R/W	Default	Description
31-24	R/W	0	SRC1 canvas address0, for Y only or Y/Cb/Cr stored together.
7-0	R	0	Unused.

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Table 7-11 GE2D_SRC1_X_START_END 0x8AA

Bit(s)	R/W	Default	Description
31	R/W	0	SRC1 x start extra bit1, if true, one more chroma data is read for x even start chroma data when y/c ratio = 2 or x even/odd start chroma extra data when y/c ratio = 1
30	R/W	0	SRC1 x start extra bit0, if true, one more chroma data is read for x odd start chroma data when y/c ratio = 2
29-16	R/W	0	SRC1 x start, signed data
15	R/W	0	SRC1 x end extra bit1, if true, one more chroma data is read for x odd end chroma data when y/c ratio = 2 or x even/odd end chroma extra data when y/c ratio = 1
14	R/W	0	SRC1 x end extra bit0, if true, one more chroma data is read for x even end chroma data when y/c ratio = 2
13-0	R/W	0	SRC1 x end, signed data.

Table 7-12 GE2D_SRC1_Y_START_END 0x8AB

Bit(s)	R/W	Default	Description
31	R/W	0	SRC1 y start extra bit1, if true, one more chroma line is read for y even start chroma data when y/c ratio = 2 or x even/odd start chroma extra data when y/c ratio = 1
30	R/W	0	SRC1 y start extra bit0, if true, one more chroma line is read for y odd start chroma data when y/c ratio = 2
29-16	R/W	0	SRC1 y start, signed data
15	R/W	0	SRC1 y end extra bit1, if true, one more chroma line is read for y odd end chroma data when y/c ratio = 2 or y even/odd end chroma extra data when y/c ratio = 1
14	R/W	0	SRC1 y end extra bit0, if true, one more chroma line is read for y even end chroma data when y/c ratio = 2
13-0	R/W	0	SRC1 y end, signed data.

Table 7-13 GE2D_SRC1_LUT_ADDR 0x8AC

Bit(s)	R/W	Default	Description
31-9	R	0	Unused.
8	R/W	1	0 = Write LUT, 1 = Read LUT.
7-0	R/W	0	lut_addr: The initial read or write address of the look-up table

Table 7-14 GE2D_SRC1_LUT_DAT 0x8AD

Bit(s)	R/W	Default	Description
31-24	R/W	0	Current LUT entry's Y or R
23-16	R/W	0	Current LUT entry's Cb or G
15-8	R/W	0	Current LUT entry's Cr or B
7-0	R/W	0	Current LUT entry's Alpha.

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Table 7-15 GE2D_SRC1_FMT_CTRL 0x8AE

Bit(s)	R/W	Default	Description
31-20	R	0	Unused.
19	R/W	0	src1_chfmt_rpt_pix: if true, horizontal formatter using repeat to get the pixel, otherwise using interpolation.
18	R/W	0	src1_chfmt_en: horizontal formatter enable.
17	R/W	0	src1_cvfmt_rpt_pix: if true, vertical formatter using repeat to get the pixel, otherwise using interpolation.
16	R/W	0	src1_cvfmt_en: vertical formatter enable.
15-8	R/W	0	src1_x_chr_phase: X direction chroma phase, Bit[15:12] for x direction even start/end chroma phase when y/c ratio = 2 or start/end even/odd chroma phase when y/c ratio = 1; Bit[11:8] for x direction odd start/end chroma phase only when y/c ration = 2.
7-0	R/W	0	src1_y_chr_phase:Y direction chroma phase. Bit[7:4] for y direction even start/end chroma phase when y/c ratio = 2 or start/end even/odd chroma phase when y/c ratio = 1; Bit[3:0] for y direction odd start/end chroma phase only when y/c ration = 2.

Table 7-16 GE2D_SRC2_DEF_COLOR 0x8AF

Bit(s)	R/W	Default	Description
31-24	R/W	0	Default Y or R.
23-16	R/W	0x80	Default Cb or G.
15-8	R/W	0x80	Default Cr or B.
7-0	R/W	0	Default Alpha.

Table 7-17 GE2D_SRC2_CLIPX_START_END 0x8B0

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	SRC2 clip x start.
15-13	R	0	Unused.
12-0	R/W	0x1FFF	SRC2 clip x end.

Table 7-18 GE2D_SRC2_CLIPY_START_END 0x8B1

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	SRC2 clip y start.
15-13	R	0	Unused.
12-0	R/W	0x1FFF	SRC2 clip y end.

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Table 7-19 GE2D_SRC2_X_START_END 0x8B2

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	SRC2 x start.
15-13	R	0	Unused.
12-0	R/W	0	SRC2 x end.

Table 7-20 GE2D_SRC2_Y_START_END 0x8B3

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	SRC2 y start.
15-13	R	0	Unused.
12-0	R/W	0	SRC2 y end.

Table 7-21 GE2D_DST_CLIPX_START_END 0x8B4

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	DST clip x start.
15-13	R	0	Unused.
12-0	R/W	0x1FFF	DST clip x end.

Table 7-22 GE2D_DST_CLIPY_START_END 0x8B5

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	DST clip y start.
15-13	R	0	Unused.
12-0	R/W	0x1FFF	DST clip y end.

Table 7-23 GE2D_DST_X_START_END 0x8B6

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	DST x start.
15-13	R	0	Unused.
12-0	R/W	0	DST x end.

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Table 7-24 GE2D_DST_Y_START_END 0x8B7

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	DST y start.
15-13	R	0	Unused.
12-0	R/W	0	DST y end.

Table 7-25 GE2D_SRC2_DST_CANVAS 0x8B8

Bit(s)	R/W	Default	Description
31-24	R	0	Unused.
23-16	R/W	0	DST2 canvas address.
15-8	R/W	0	SRC2 canvas address.
7-0	R/W	0	DST1 canvas address.

Table 7-26 GE2D_VSC_START_PHASE_STEP 0x8B9

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-0	R/W	0x0100-0000	5.24 format.

Table 7-27 GE2D_VSC_PHASE_SLOPE 0x8BA

Bit(s)	R/W	Default	Description
31-25	R	0	Unused.
24-0	R/W	0	Signed data.

Table 7-28 GE2D_VSC_INI_CTRL 0x8BB

Bit(s)	R/W	Default	Description
31	R	0	Unused.
30-29	R/W	0	vertical repeat line0 number.
28-24	R	0	Unused.
23-0	R/W	0	vertical scaler initial phase.

Table 7-29 GE2D_HSC_START_PHASE_STEP 0x8BC

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-0	R/W	0x0100-0000	5.24 format.

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Table 7-30 GE2D_HSC_PHASE_SLOPE 0x8BD

Bit(s)	R/W	Default	Description
31-25	R	0	Unused.
24-0	R/W	0	Signed data.

Table 7-31 GE2D_HSC_INI_CTRL 0x8BE

Bit(s)	R/W	Default	Description
31	R	0	Unused.
30-29	R/W	0	horizontal repeat line0 number.
28-24	R	0	Unused.
23-0	R/W	0	horizontal scaler initial phase.

Table 7-32 GE2D_HSC_ADV_CTRL 0x8BF

Bit(s)	R/W	Default	Description
31-24	R/W	0	advance number in this round, if horizontal scaler is working on dividing mode.
23-0	R/W	0	horizontal scaler advance phase in this round, if horizontal scaler is working on dividing mode.

Table 7-33 GE2D_SC_MISC_CTRL 0x8C0

Bit(s)	R/W	Default	Description
31	R	0	Unused.
30	R/W	0	vsc_nearest_en: vertical nearest mode enable, must set vt_bank_length = 4.
29	R/W	0	hsc_nearest_en: horizontal nearest mode enable, must set hz_bank_length = 4.
28	R/W	0	hsc_div_en: horizontal scaler dividing mode enable.
27-15	R/W	0	hsc_div_length: horizontal dividing length, if bit 25 is enable.
14	R/W	0	pre horizontal scaler enable.
13	R/W	0	pre vertical scale enable.
12	R/W	0	vertical scale enable.
11	R/W	0	horizontal scaler enable.
10	R	0	Unused.
9	R/W	0	HSc_rpt_ctrl: if true, treat horizontal repeat line number(GE2D_HSC_INI_CTRL bit 30:29) as repeating line, otherwise using treat horizontal repeat line number as minus line number.
8	R/W	0	VSc_rpt_ctrl: if true, treat vertical repeat line number(GE2D_VSC_INI_CTRL bit 30:29) as repeating line, otherwise using treat vertical repeat line number as minus line number.
7	R/W	0	vsc_phase0_always_en: if true, always use phase0 in vertical scaler.
6-4	R/W	2	vsc_bank_length: vertical scaler bank length.

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Bit(s)	R/W	Default	Description
3	R/W	0	hsc_phase0_always_en: if true, always use phase0 in horizontal scaler.
2-0	R/W	2	hsc_bank_length: horizontal scaler bank length.

Table 7-34 GE2D_VSC_NRND_POINT 0x8C1

Bit(s)	R/W	Default	Description
31-14	R	0	Unused.
13-0	R	0	vertical scaler next round integer pixel pointer, signed data.

Table 7-35 GE2D_VSC_NRND_PHASE 0x8C2

Bit(s)	R/W	Default	Description
31-24	R	0	Unused.
23-0	R	0	vertical scaler next round phase.

Table 7-36 GE2D_HSC_NRND_POINT 0x8C3

Bit(s)	R/W	Default	Description
31-14	R	0	Unused.
13-0	R	0	horizontal scaler next round integer pixel pointer, signed data.

Table 7-37 GE2D_HSC_NRND_PHASE 0x8C4

Bit(s)	R/W	Default	Description
31-24	R	0	Unused.
23-0	R	0	horizontal scaler next round phase.

Table 7-38 GE2D_MATRIX_PRE_OFFSET 0x8C5

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-20	R/W	0	pre_offset0.
19	R	0	Unused.
18-10	R/W	0	pre_offset1.
9	R	0	Unused.
8-0	R/W	0	pre_offset2.

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Table 7-39 GE2D_MATRIX_COEF00_01 0x8C6

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	coef00.
15-13	R	0	Unused.
12-0	R/W	0	coef01.

Table 7-40 GE2D_MATRIX_COEF02_10 0x8C7

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	Coef02.
15-13	R	0	Unused.
12-0	R/W	0	Coef10.

Table 7-41 GE2D_MATRIX_COEF11_12 0x8C8

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	Coef11.
15-13	R	0	Unused.
12-0	R/W	0	Coef12.

Table 7-42 GE2D_MATRIX_COEF20_21 0x8C9

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	Coef20.
15-13	R	0	Unused.
12-0	R/W	0	Coef21.

Table 7-43 GE2D_MATRIX_COEF22_CTRL 0x8CA

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-16	R/W	0	coef22.
15-8	R	0	Unused.
7	R/W	0	input y/cb/cr saturation enable.
6-1	R	0	Unused.
0	R/W	0	conversion matrix enable.

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Table 7-44 GE2D_MATRIX_OFFSET 0x8CB

Bit(s)	R/W	Default	Description
31-29	R	0	Unused.
28-20	R/W	0	offset0.
19	R	0	Unused.
18-10	R/W	0	offset1.
9	R	0	Unused.
8-0	R/W	0	offset2.

Table 7-45 GE2D_ALU_OP_CTRL 0x8CC

Bit(s)	R/W	Default	Description
31-27	R	0	Unused.
26-25	R/W	0	SRC1 color multiplier alpha selection. if 00, Cs = Csr if 01, Cs = Csr * Asr * Ag (if source is not premultiplied) if 10, Cs = Csr * Ag (if source is premultipied).
24	R/W	0	SRC2 color multiplier alpha selection. if 0, no multiplier, Cd = Cdr, otherwise, Cd = Cdr * Ad.
23	R	0	Unused.
22-12	R/W	0x010	ALU color operation. Bit[22:20] Blending Mode Parameter. 3'b000: ADD Cs*Fs + Cd*Fd 3'b001: SUBTRACT Cs*Fs - Cd*Fd 3'b010: REVERSE SUBTRACT Cd*Fd - Cs*Fs 3'b011: MIN min(Cs*Fs, Cd*Fd) 3'b100: MAX max(Cs*Fs, Cd*Fd) 3'b101: LOGIC OP Cs op Cd .... reserved Bit[19:16] Source Color Blending Factor CFs. 4'b0000: ZERO 0 4'b0001: ONE 1 4'b0010: SRC_COLOR Cs(RGBs) 4'b0011: ONE_MINUS_SRC_COLOR 1 - Cs(RGBs) 4'b0100: DST_COLOR Cd(RGBd) 4'b0101: ONE_MINUS_DST_COLOR 1 - Cd(RGBd) 4'b0110: SRC_ALPHA As 4'b0111: ONE_MINUS_SRC_ALPHA 1 – As 4'b1000: DST_ALPHA Ad 4'b1001: ONE_MINUS_DST_ALPHA 1 – Ad 4'b1010: CONST_COLOR Cc(RGBc) 4'b1011: ONE_MINUS_CONST_COLOR 1 - Cc(RGBc) 4'b1100: CONST_ALPHA Ac 4'b1101: ONE_MINUS_CONST_ALPHA 1 – Ac 4'b1110: SRC_ALPHA_SATURATE min(As,1-Ad) .... reserved Bit[15:12] dest Color Blending Factor CFd, when bit[22:20]!= LOGIC OP. 4'b0000: ZERO 0 4'b0001: ONE 1 4'b0010: SRC_COLOR Cs(RGBs) 4'b0011: ONE_MINUS_SRC_COLOR 1 - Cs(RGBs) 4'b0100: DST_COLOR Cd(RGBd) 4'b0101: ONE_MINUS_DST_COLOR 1 - Cd(RGBd) 4'b0110: SRC_ALPHA As 4'b0111: ONE_MINUS_SRC_ALPHA 1 – As 4'b1000: DST_ALPHA Ad 4'b1001: ONE_MINUS_DST_ALPHA 1 – Ad 4'b1010: CONST_COLOR Cc(RGBc) 4'b1011: ONE_MINUS_CONST_COLOR 1 - Cc(RGBc) 4'b1100: CONST_ALPHA Ac 4'b1101: ONE_MINUS_CONST_ALPHA 1 – Ac 4'b1110: SRC_ALPHA_SATURATE min(As,1-Ad) .... reserved Bit[15:12] logic operations, when bit[22:20]== LOGIC OP. 4'b0000: CLEAR 0 4'b0001: COPY s 4'b0010: NOOP d 4'b0011: SET 1 4'b0100: COPY_INVERT ~s 4'b0101: INVERT ~d 4'b0110: AND_REVERSE s & ~d 4'b0111: OR_REVERSE s | ~d 4'b1000: AND s & d 4'b1001: OR s | d 4'b1010: NAND ~(s & d) 4'b1011: NOR ~(s | d) 4'b1100: XOR s ^ d 4'b1101: EQUIV ~(s ^ d) 4'b1110: AND_INVERTED ~s & d 4'b1111: OR_INVERTED ~s | d

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Bit(s)	R/W	Default	Description
11	R	0	Unused.
10-0	R/W	0x010	ALU alpha operation. Bit[10:8] Blending Equation Math Operation. 3'b000: ADD As*Fs + Ad*Fd 3'b001: SUBTRACT As*Fs - Ad*Fd 3'b010: REVERSE SUBTRACT Ad*Fd - As*Fs 3'b011: MIN min(As*Fs, Ad*Fd) 3'b100: MAX max(As*Fs, Ad*Fd) 3'b101: LOGIC OP As op Ad .... reserved Bit[7:4] Source alpha Blending Factor AFs. 4'b0000 0 4'b0001 1 4'b0010 As 4'b0011 1 - As 4'b0100 Ad 4'b0101 1 – Ad 4'b0110 Ac 4'b0111 1 – Ac .... reserved Bit[3:0] Destination alpha Blending Factor AFd, when bit[10:8] != LOGIC OP. 4'b0000 0 4'b0001 1 4'b0010 As 4'b0011 1 - As 4'b0100 Ad 4'b0101 1 - Ad 4'b0110 Ac 4'b0111 1 - Ac .... reserved Bit[3:0] logic operations, when bit[10:8] == LOGIC OP. 4'b0000: CLEAR 0 4'b0001: COPY s 4'b0010: NOOP d 4'b0011: SET 1 4'b0100: COPY_INVERT ~s 4'b0101: INVERT ~d 4'b0110: AND_REVERSE s & ~d 4'b0111: OR_REVERSE s | ~d 4'b1000: AND s & d 4'b1001: OR s | d 4'b1010: NAND ~(s & d) 4'b1011: NOR ~(s | d) 4'b1100: XOR s ^ d 4'b1101: EQUIV ~(s ^ d) 4'b1110: AND_INVERTED ~s & d 4'b1111: OR_INVERTED ~s | d

Table 7-46 GE2D_ALU_CONST_COLOR 0x8CD

Bit(s)	R/W	Default	Description
31-0	R/W	0x0080-8000	RGBA or YCbCrA.

Table 7-47 GE2D_SRC1_KEY 0x8CE

Bit(s)	R/W	Default	Description
31-0	R/W	0	SRC1 Key.

Table 7-48 GE2D_SRC1_KEY_MASK 0x8CF

Bit(s)	R/W	Default	Description
31-0	R/W	0	SRC1 Key Mask.

Table 7-49 GE2D_SRC2_KEY 0x8D0

Bit(s)	R/W	Default	Description
31-0	R/W	0	SRC2 Key.

Table 7-50 GE2D_SRC2_KEY_MASK 0x8D1

Bit(s)	R/W	Default	Description
31-0	R/W	0	Destination Bit Mask.

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Table 7-51 GE2D_DP_ONOFF_CTRL 0x8D3

Bit(s)	R/W	Default	Description
31	R/W	0	DP onoff mode. 0: on_counter means how many pixels will output before ge2d turns off; 1: on_counter means how many clocks will ge2d turn on before ge2d turns off.
30-16	R/W	0	DP on counter.
15	R/W	0	0: vd_format doesnt have onoff mode, 1: vd format has onoff mode.
14-0	R/W	0	DP off counter.

Because there are many coefficients used in the vertical filter and horizontal filters, indirect access the coefficients of vertical filter and horizontal filter is used. For vertical filter, there are 33x4 coefficients. For horizontal filter, there are 33x4 coefficients

Table 7-52 GE2D_SCALE_COEF_IDX 0x8D4

Bit(s)	R/W	Default	Description
31-16	R	0	Unused.
15	R/W	0	index increment, if bit9 == 1 then (0: index increase 1, 1: index increase 2) else (index increase 2).
14	R/W	0	1: read coef through cbus enable, just for debug purpose in case when we wanna check the coef in ram in correct or not.
13-10	R	0	Unused.
9	R/W	0	if true, use 9bit resolution coef, other use 8bit resolution coef.
8	R/W	0	type of index, 0: vertical coef; 1: horizontal coef.
7	R	0	Unused.
6-0	R/W	0	coef index.

Table 7-53 GE2D_SCALE_COEF 0x8D5

Bit(s)	R/W	Default	Description
31-0	R/W	0	coefficients for vertical filter and horizontal filter.

Table 7-54 GE2D_SRC_OUTSIDE_ALPHA 0x8D6

Bit(s)	R/W	Default	Description
31-25	R	0	Unused.

24	R/W	0	src2 alpha fill mode: together with GE2D_GEN_CTRL0[14](fill_mode), define what alpha values are used.
			0: repeat innner alpha, 1: fill src2 outside alpha.
			
			for the area outside the clipping window. As below:
			fill_mode=0, alpha_fill_mode=0 : use inner alpha, (or default_alpha if src data have no alpha values);
			fill_mode=0, alpha_fill_mode=1 : use outside_alpha;
			fill_mode=1, alpha_fill_mode=0 : use default_alpha;
			fill_mode=1, alpha_fill_mode=1 : use outside_alpha.

23-16	R/W	16	src2 outside alpha.
15-9	R	0	Unused.

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Bit(s)	R/W	Default	Description
8	R/W	1	src1 alpha fill mode, refer to src2 alpha fill mode above.
7-0	R/W	0	src1 outside alpha.

Table 7-55 GE2D_ANTIFLICK_CTRL0 0x8D8

Bit(s)	R/W	Default	Description
31	R/W	0	antiflick enable
24	R/W	0	1: alpha value for the first line use repeated alpha, 0: use bit 23:16 as the first line alpha
23-16	R/W	0	register value for the first line alpha when bit 24 is 1.
8	R/W	0	1: alpha value for the last line use repeated alpha, 0: use bit 7:0 as the last line alpha
7-0	R/W	0	register value for the last line alpha when bit 8 is 1.

Table 7-56 GE2D_ANTIFLICK_CTRL1 0x8D9

Bit(s)	R/W	Default	Description
25	R/W	0	rgb_sel, 1: antiflick RGBA, 0: antiflick YCbCrA
24	R/W	0	cbcr_en, 1: also filter cbcr in case of antiflicking YCbCrA, 0: no filter on cbcr in case of antiflicking YCbCrA
23-16	R/W	0	R mult coef for converting RGB to Y
15- 8	R/W	0	G mult coef for converting RGB to Y
7-0	R/W	0	B mult coef for converting RGB to Y

Y = (R * y_r + G * y_g + B * y_b) / 256

Table 7-57 GE2D_ANTIFLICK_COLOR_FILT0 0x8DA

Bit(s)	R/W	Default	Description
31-24	R/W	0	Y threhold1, when 0<Y<=th1, use filter0.
23-16	R/W	0	color antiflick filter0 n3
15- 8	R/W	0	color antiflick filter0 n2
7-0	R/W	0	color antiflick filter0 n1

Y = (line_up * n1 + line_center * n2 + line_dn * n3) / 128

Table 7-58 GE2D_ANTIFLICK_COLOR_FILT1 0x8DB

Bit(s)	R/W	Default	Description
31-24	R/W	0	Y threhold2, when th1<Y<=th2, use filter1.
23-16	R/W	0	color antiflick filter1 n3
15- 8	R/W	0	color antiflick filter1 n2
7-0	R/W	0	color antiflick filter1 n1

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Table 7-59 GE2D_ANTIFLICK_COLOR_FILT2 0x8DC

Bit(s)	R/W	Default	Description
31-24	R/W	0	Y threhold3, when th2<Y<=th3, use filter2 ; Y>th3, use filter3.
23-16	R/W	0	color antiflick filter2 n3
15- 8	R/W	0	color antiflick filter2 n2
7-0	R/W	0	color antiflick filter2 n1

Table 7-60 GE2D_ANTIFLICK_COLOR_FILT3 0x8DD

Bit(s)	R/W	Default	Description
23-16	R/W	0	color antiflick filter3 n3
15- 8	R/W	0	color antiflick filter3 n2
7-0	R/W	0	color antiflick filter3 n1

Table 7-61 GE2D_ANTIFLICK_ALPHA_FILT0 0x8DE

Bit(s)	R/W	Default	Description
31-24	R/W	0	Alpha threhold1, when 0< Alpha<=th1, use filter0.
23-16	R/W	0	Alpha antiflick filter0 n3
15- 8	R/W	0	Alpha antiflick filter0 n2
7-0	R/W	0	Alpha antiflick filter0 n1

Alpha = (line_up * n1 + line_center * n2 + line_dn * n3) / 128

Table 7-62 GE2D_ANTIFLICK_ALPHA_FILT1 0x8DF

Bit(s)	R/W	Default	Description
31-24	R/W	0	Alpha threhold2, when th1< Alpha<=th2, use filter1.
23-16	R/W	0	Alpha antiflick filter1 n3
15- 8	R/W	0	Alpha antiflick filter1 n2
7-0	R/W	0	Alpha antiflick filter1 n1

Table 7-63 GE2D_ANTIFLICK_ALPHA_FILT2 0x8E0

Bit(s)	R/W	Default	Description
31-24	R/W	0	Alpha threhold3, when th2< Alpha<=th3, use filter2; Alpha>th3, use filter3.
23-16	R/W	0	Alpha antiflick filter2 n3
15- 8	R/W	0	Alpha antiflick filter2 n2
7-0	R/W	0	Alpha antiflick filter2 n1

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Table 7-64 GE2D_ANTIFLICK_ALPHA_FILT3 0x8E1

Bit(s)	R/W	Default	Description
23-16	R/W	0	Alpha antiflick filter3 n3
15- 8	R/W	0	Alpha antiflick filter3 n2
7-0	R/W	0	Alpha antiflick filter3 n1

Table 7-65 GE2D_SRC1_RANGE_MAP_Y_CTRL 0x8E3

Bit(s)	R/W	Default	Description
30-22	R/W	0	din_offset (signed data)
21-14	R/W	0	map_coef (unsigned data)
13- 10	R/W	0	map_sr (unsigned data)
9-1	R/W	0	dout_offset (signed data)
0	R/W	0	enable

dout = clipto_0_255(((din + din_offset) * map_coef + ((1 << (map_sr - 1))) >> map_sr + dout_offset)

Table 7-66 GE2D_SRC1_RANGE_MAP_CB_CTRL 0x8E4

Bit(s)	R/W	Default	Description
30-22	R/W	0	din_offset (signed data)
21-14	R/W	0	map_coef (unsigned data)
13- 10	R/W	0	map_sr (unsigned data)
9-1	R/W	0	dout_offset (signed data)
0	R/W	0	enable

dout = clipto_0_255(((din + din_offset) * map_coef + ((1 << (map_sr - 1))) >> map_sr + dout_offset)

Table 7-67 GE2D_SRC1_RANGE_MAP_CR_CTRL 0x8E5

Bit(s)	R/W	Default	Description
30-22	R/W	0	din_offset (signed data)
21-14	R/W	0	map_coef (unsigned data)
13- 10	R/W	0	map_sr (unsigned data)
9-1	R/W	0	dout_offset (signed data)
0	R/W	0	enable

dout = clipto_0_255(((din + din_offset) * map_coef + ((1 << (map_sr - 1))) >> map_sr + dout_offset)

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Table 7-68 GE2D_ARB_BURST_NUM 0x8E6

Bit(s)	R/W	Default	Description
21-16	R/W	0x3f	Src1 prearbitor burst number
13-8	R/W	0x3f	Src2 prearbitor burst number
5-0	R/W	0x3f	dst prearbitor burst number

Table 7-69 GE2D_TID_TOKEN 0x8E7

Bit(s)	R/W	Default	Description
21-16	R/W	0x3f	Src1 ID. High 4bit are thread ID, low 2bits are the token
13-8	R/W	0x3f	Src2 ID. High 4bit are thread ID, low 2bits are the token
5-0	R/W	0x3f	dst ID. High 4bit are thread ID, low 2bits are the token

Table 7-70 GE2D_GEN_CTRL3 0x8E8

Bit(s)	R/W	Default	Description
31-28	R/W	0	dst2_bytemask_val: 1-bit mask for each byte (8-bit). Applicable only if both dst_ bitmask_en=1 and dst_bytemask_only=1.
27-26	R/W	0	dst2_pic_struct: Define how destination 2 write the picture to DDR memory. 0: Write all lines (whole frame); 1: Reserved; 2: Write even lines only (top); 3: Write odd lines only (bottom).
25-24	R/W	0	dst2_8b_mode_sel: Destination 8-bit mode component selection. 0: Select Y or R; 1: Select Cb or G; 2: Select Cr or B; 3: Select Alpha.
23	R	0	Unused
22-19	R/W	0	dst2_color_map: Applicable to 16-bit, 24-bit and 32-bit pixel, defines the bit-field allocation of the pixel data. For whether to truncate or round full 8-bit to output, refer to ge2d_gen_ctrl3.dst2_color_round_mode. For 16-bit mode (dst2_format=1): 0 = Unused; 1 = 6:5:5 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:5] is Cb[7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3]; 2 = 8:4:4 format. Bit[15:8] is Y or R, bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]]; 3 = 6:4:4:2 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:6] is Cb[7:4] or G[7:4], bit [5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[7:6]; 4 = 4:4:4:4 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:8] is Cb[7:4] or G[7:4], bit [7:4] is Cr[7:4] or B[7:4], bit[3:0] is Alpha[7:4]; 5 = 5:6:5 format. Bit[15:11] is Y[7:3] or R[7:3], bit[10:5] is Cb[7:2] or G[7:2], bit [4:0] is Cr[7:3] or B[7:3]; 6 = 4:4:4:4 format. Bit[15:12] is Alpha[7:4], bit[11:8] is Y[7:4] or R[7:4], bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]; 7 = 1:5:5:5 format. Bit[15] is Alpha[7], bit[14:10] is Y[7:3] or R[7:3], bit[9:5] is Cb [7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3]; 8 = 4:6:4:2 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:6] is Cb[7:2] or G[7:2], bit [5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[1:0]; 9 = CbCr format. Bit[15:8] is Cb, bit[7:0] is Cr; 10 = CrCb format. Bit[15:8] is Cr, bit[7:0] is Cb. For 24-bit mode (dst2_format=2):

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Bit(s)	R/W	Default	Description
			0 = RGB 8:8:8 mode. Bit[23:16] is Y or R, bit[15:8] is Cb or G, bit[7:0] is Cr or B; 1 = RGBA 5:6:5:8 mode. Bit[23:19] is Y[7:3] or R[7:3], bit[18:13] is Cb[7:2] or G [7:2], bit[12:8] is Cr[7:3] or B[7:3], bit[7:0] is Alpha; 2 = ARGB 8:5:6:5 mode. Bit[23:16] is Alpha, bit[15:11] is Y[7:3] or R[7:3], bit [10:5] is Cb[7:2] or G[7:2], bit[4:0] is Cr[7:3] or B[7:3]; 3 = RGBA 6:6:6:6 mode. Bit[23:18] is Y[7:2] or R[7:2], bit[17:12] is Cb[7:2] or G [7:2], bit[11:6] is Cr[7:2] or B[7:2], bit[5:0] is Alpha[7:2]; 4 = ARGB 6:6:6:6 mode. Bit[23:18] is Alpha[7:2];, bit[17:12] is Y[7:2] or R[7:2], bit [11:6] is Cb[7:2] or G[7:2], bit[5:0] is Cr[7:2] or B[7:2]; 5 = BGR 8:8:8 mode. Bit[23:16] is Cr or B, bit[15:8] is Cb or G, bit[7:0] is Y or R. For 32-bit mode (dst2_format=3): 0 = RGBA 8:8:8:8 format. Bit[31:24] is Y or R, bit[23:16] is Cb or G; bit[15:8] is Cr or B; bit[7:0] is Alpha; 1 = ARGB 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Y or R; bit[15:8] is Cb or G; bit[7:0] is Cr or B; 2 = ABGR 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Cr or B; bit[15:8] is Cb or G; bit[7:0] is Y or R; 3 = BGRA 8:8:8:8 format. Bit[31:24] is Cr or B, bit[23:16] is Cb or G; bit[15:8] is Y or R; bit[7:0] is Alpha.
18	R	0	Unused
17-16	R/W	0	dst2_format: define DST2 output pixel byte-width. 0: Output pixel is 1-byte (8-bit) color component; 1: Output pixel is 2-byte (16-bit), refer to GE2D_GEN_CTRL3.dst2_COLOR_ map for further pixel color mapping; 2: Output pixel is 3-byte (24-bit), refer to GE2D_GEN_CTRL3.dst2_COLOR_ MAP for further pixel color mapping; 3: Output pixel is 4-byte (32-bit), refer to GE2D_GEN_CTRL3.dst2_COLOR_ MAP for further pixel color mapping.
15	R	0	Unused
14	R/W	0	dst2_COLOR_round_MODE. 1 = Truncate the full bit color components to required output bit width; 0 = Round (+ 0.5) the full bit color components to required output bit width.
13-12	R/W	0	dst2_x_discard_mode: Define how DST2 discard X direction data before writing to DDR. Note: x is post reverse/rotation. 0: No discard; 1: Reserved; 2: discard even x; 3: discard odd x.
11-10	R/W	0	dst2_y_discard_mode: Define how DST2 discard Y direction data before writing to DDR. Note: y is post reverse/rotation. 0: No discard; 1: Reserved; 2: discard even y; 3: discard odd y.
9	R	0	Unused
8	R/W	0	dst2_enable: 0: Disable destination 2; 1: Enable destination 2.
7-6	R	0	Unused
5-4	R/W	0	dst1_x_discard_mode: Define how DST1 discard X direction data before writing to DDR. Note: x is post reverse/rotation. 0: No discard; 1: Reserved; 2: discard even x; 3: discard odd x.

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Bit(s)	R/W	Default	Description
3-2	R/W	0	dst1_y_discard_mode: Define how DST1 discard Y direction data before writing to DDR. Note: y is post reverse/rotation. 0: No discard; 1: Reserved; 2: discard even y; 3: discard odd y.
1	R	0	Unused
0	R/W	1	dst1_enable: 0: Disable destination 1; 1: Enable destination 1.

Table 7-71 GE2D_STATUS2 0x8E9

Bit(s)	R/W	Default	Description
13-12	R	0	ge2d_dst1.ctrl_status
11	R	0	ge2d_dst1.map_srdy
10	R	0	ge2d_dst1.map_d1_srdy
9	R	0	ge2d_dst1.s_v
8	R	0	ge2d_dst1.ofifo_dout_srdy
7-1	R	0	ge2d_dst1.ofifo_cnt
0	R	0	ge2d_dst1.dst_busy

Table 7-72 GE2D_GEN_CTRL4 0x8EA

Bit(s)	R/W	Default	Description
0	R/W	0	dis_dp_hang_bugfix.

Table 7-73 GE2D_DST1_BADDR_CTRL 0x8f1

Bit(s)	R/W	Default	Description
31-0	R/W	0	DST1 base address in 64bits.

Table 7-74 GE2D_DST1_STRIDE_CTRL 0x8f2

Bit(s)	R/W	Default	Description
19-0	R/W	0	DST1 stride size in 64bits.

Table 7-75 GE2D_SRC1_BADDR_CTRL 0x8f3

Bit(s)	R/W	Default	Description
31-0	R/W	0	SRC1 base address in 64bits.

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Table 7-76 GE2D_SRC1_STRIDE_CTRL 0x8f4

Bit(s)	R/W	Default	Description
19-0	R/W	0	SRC1 stride size in 64bits.

Table 7-77 GE2D_SRC2_BADDR_CTRL 0x8f5

Bit(s)	R/W	Default	Description
31-0	R/W	0	SRC2 base address in 64bits.

Table 7-78 GE2D_SRC2_STRIDE_CTRL 0x8f6

Bit(s)	R/W	Default	Description
19-0	R/W	0	SRC2 stride size in 64bits.

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8 Video Path

The working flow of the video path module is as following:

- For TS input, it will go to DEMUX, which will separate the data into video data, audio data or control data, and will write video data into DDR, the VDEC module will read this data from DDR and decode it, and then write it back to DDR, which will be sent to VPU module and display.
- For HDMIRX and ATV input, the data will send to VDIN first and then write to DDR, and VPU will read the data from DDR, then display.
- For DVP input, DVP module will write it directly to DDR, and VPU module read it and display.

The display format supports either LVDS, VX1, TCON or CVBS.

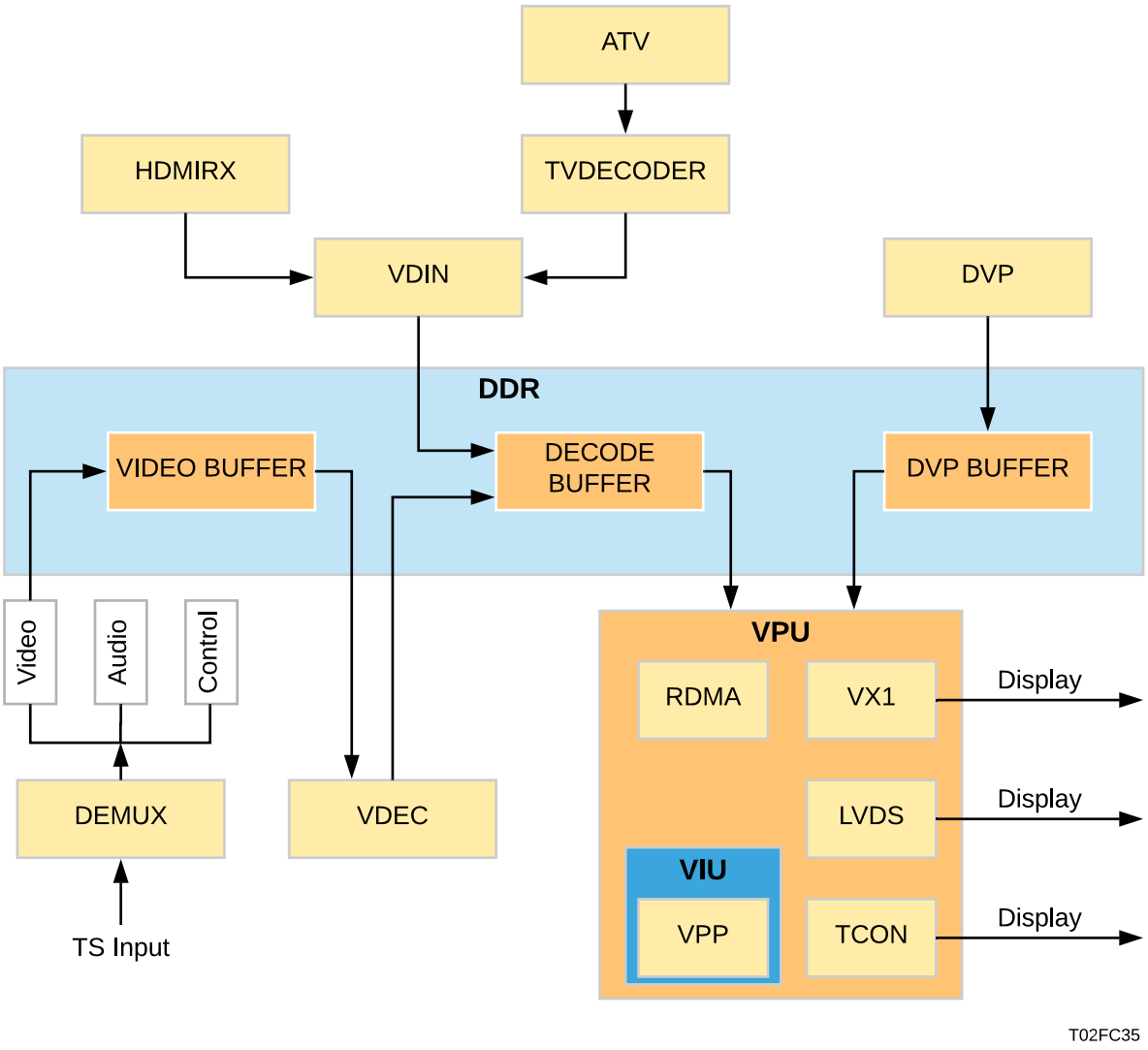
This section describe the video path from the following aspects:

- Video Input
- Video Output
  - RDMA
  - VPP(VIU/VPP)
  - CVBS

The data path of the video path module is shown in the figure below:

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Figure 8-1 Data Path of Video Path



8.1 Video Input

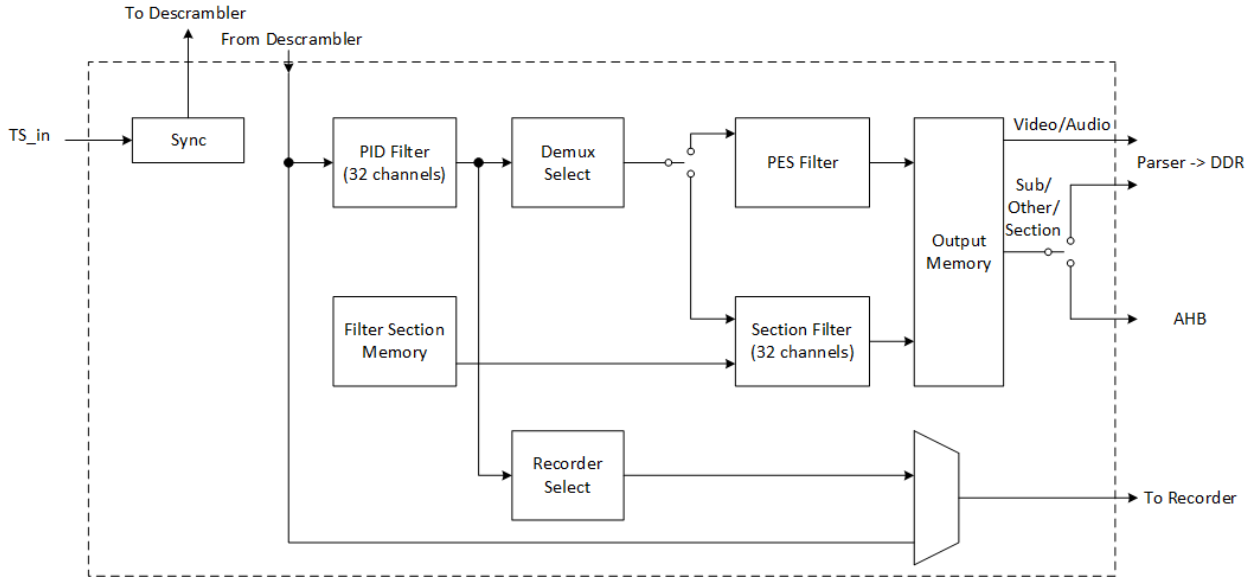
8.1.1 Overview

This part describes the video input module which uses DEMUX to handle input video signals, and store the demuxed signal in DDR, which can be read directly.

Demux submodule is designed for connecting to external digital TV tuner/demodular and de-mux the input signal. It decomposes the input signal from TS/DDR into video signal, audio signal, and clock signal, the video signal will be first written in to DDR, then decoded by the video decoder(VDEC). Below shows the diagram of Demux submodule.

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Figure 8-2 Demux Submodule



The chip intergrades 3 demux controllers, with the following features:

- Supports PID filter up to 32 channels
- Supports session filter up to 32 channels
- Supports up to 96 channels filter when use 3 demux as one group
- Supports AES/DES/DVB-CSA 1.0 crypto
- Supports inputs from both external TS inputs (serial or parallel, depending on pinmux) and DDR memory (file playback path)
- Supports input loopback to TS-out pins
- Supports separate selection for playback and recorder

8.1.2 HDMIRX Registers

8.1.2.1 Register Access

Accessing HDMIRX Top-Level and RX IP registers is by directly accessing memory addresses. Due to the difference between the data width of normal registers and EDID RAM maps – Normal registers is 32-bit wide and EDID RAM is 8-bit wide, below list the correct ways to access HDMIRX EDID RAM, Top-level registers.

HDMIRX Top-Level Register Access

- To access Top-level normal registers, use type uint32_t
- To access Top-Level EDID RAM, use type uint8_t:

```
# define HDMIRX_TOP_BASE_OFFSET0xff618000
# define HDMIRX_TOP_EDID_ADDR_S0x1000
# define HDMIRX_TOP_EDID_ADDR_E0x11ff

void hdmirx_wr_only_TOP (uint32_t addr, uint32_t data)
{
    if ((addr >= HDMIRX_TOP_EDID_ADDR_S) && (addr <= HDMIRX_TOP_EDID_ADDR_E)) {
        *((volatile uint8_t *) (HDMIRX_TOP_BASE_OFFSET+addr)) = (uint8_t)(data&0xff);
    } else {
        *((volatile uint32_t *) (HDMIRX_TOP_BASE_OFFSET+addr)) = (data);
    }
}
```


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```
uint32_t hdmirx_rd_TOP (uint32_t addr)
{
    uint32_t    data;
    if ((addr >= HDMIRX_TOP_EDID_ADDR_S) && (addr <= HDMIRX_TOP_EDID_ADDR_E)) {
        data = (uint32_t)(*((volatile uint8_t *) (HDMIRX_TOP_BASE_OFFSET+addr)));
    } else {
        data = *((volatile uint32_t *) (HDMIRX_TOP_BASE_OFFSET+addr));
    }
    return (data);
```

HRX_HDCP2.2 IP Register Access

Table 8-1 HRX_HDCP2.2 IP register base address

Absolute Address	Address Mnemonic	Description
0xffe0d000	HRX_ELP_ESM_HPI_REG_BASE	Address base to HRX_HDCP2.2 IP register access.

To write to an HDCP2.2 IP register:

```
*((volatile uint32_t *) (HRX_ELP_ESM_HPI_REG_BASE+addr)) = data;
```

To read from an HDCP2.2 IP register:

```
data = *((volatile uint32_t *) (HRX_ELP_ESM_HPI_REG_BASE+addr));
```

HDMIRX_PHY Register Access

Below lists the HDMIRX_PHY control and status registers:

Table 8-2 HDMIRX_PHY Registers in HIU Module

Absolute Address	Name
0xff63c000 + (0x0d2 << 2)	HHI_HDMIRX_APLL_CNTL0
0xff63c000 + (0x0d3 << 2)	HHI_HDMIRX_APLL_CNTL1
0xff63c000 + (0x0d4 << 2)	HHI_HDMIRX_APLL_CNTL2
0xff63c000 + (0x0d5 << 2)	HHI_HDMIRX_APLL_CNTL3
0xff63c000 + (0x0d6 << 2)	HHI_HDMIRX_APLL_CNTL4
0xff63c000 + (0x0d7 << 2)	HHI_HDMIRX_PHY_MISC0
0xff63c000 + (0x0d8 << 2)	HHI_HDMIRX_PHY_MISC1
0xff63c000 + (0x0e0 << 2)	HHI_HDMIRX_PHY_MISC2
0xff63c000 + (0x0e1 << 2)	HHI_HDMIRX_PHY_MISC3
0xff63c000 + (0x0e2 << 2)	HHI_HDMIRX_PHY_DCHA_CNTL0
0xff63c000 + (0x0e3 << 2)	HHI_HDMIRX_PHY_DCHA_CNTL1
0xff63c000 + (0x0e4 << 2)	HHI_HDMIRX_PHY_DCHA_CNTL2
0xff63c000 + (0x0e5 << 2)	HHI_HDMIRX_PHY_DCHD_CNTL0
0xff63c000 + (0x0e6 << 2)	HHI_HDMIRX_PHY_DCHD_CNTL1
0xff63c000 + (0x0e7 << 2)	HHI_HDMIRX_PHY_DCHD_CNTL2
0xff63c000 + (0x0e8 << 2)	HHI_HDMIRX_ARC_CNTL

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Absolute Address	Name
0xff63c000 + (0x0ee << 2)	HHI_HDMIRX_PHY_MISC_STAT
0xff63c000 + (0x0ef << 2)	HHI_HDMIRX_PHY_DCHD_STAT

8.1.2.2 Top-Level Register Description

Table 8-3 HDMIRX Top-Level Register Summary

Addr	Name	RW	Function
0x000<<2	HDMIRX_TOP_SW_RESET	RW	Software reset sub-modules.
0x001<<2	HDMIRX_TOP_CLK_CNTL	RW	Clock gating and inversion.
0x002<<2	HDMIRX_TOP_HPD_PWR5V	RW	HPD control and PWR5V status.
0x003<<2	HDMIRX_TOP_PORT_SEL	RW	RX input port selection.
0x004<<2	HDMIRX_TOP_EDID_GEN_ CNTL	RW	EDID control.
0x005<<2	HDMIRX_TOP_EDID_ADDR_ CEC	RW	EDID's CEC bytes auto-replace.
0x006<<2	HDMIRX_TOP_EDID_DATA_ CEC_PORT01	RW	EDID's CEC bytes auto-replace.
0x007<<2	HDMIRX_TOP_EDID_DATA_ CEC_PORT2	RW	EDID's CEC bytes auto-replace.
0x008<<2	HDMIRX_TOP_EDID_GEN_ STAT_A	RW	PortA EDID status.
0x009<<2	HDMIRX_TOP_INTR_MASKN	RW	Interrupt mask.
0x00A<<2	HDMIRX_TOP_INTR_STAT	RW	Interrupt status.
0x00B<<2	HDMIRX_TOP_INTR_STAT_ CLR	RW	Interrupt clear.
0x00C<<2	HDMIRX_TOP_VID_CNTL	RW	Video processing control.
0x00D<<2	HDMIRX_TOP_VID_STAT	R	Video processing status.
0x00E<<2	HDMIRX_TOP_ACR_CNTL_ STAT	RW	ACR control and status.
0x00F<<2	HDMIRX_TOP_ACR_ AUDFIFO	RW	ACR control.
0x011<<2	HDMIRX_TOP_METER_ HDMI_CNTL	RW	tmds_clk meter control.
0x012<<2	HDMIRX_TOP_METER_ HDMI_STAT	R	tmds_clk meter status.
0x013<<2	HDMIRX_TOP_VID_CNTL2	RW	Video processing control.
0x014<<2	HDMIRX_TOP_MEM_PD	RW	Memory power down control.
0x015<<2	HDMIRX_TOP_EDID_RAM_ OVR0	RW	EDID RAM override byte0 address.
0x016<<2	HDMIRX_TOP_EDID_RAM_ OVR0_DATA	RW	EDID RAM override byte0 data.

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Addr	Name	RW	Function
0x017<<2	HDMIRX_TOP_EDID_RAM_OVR1	RW	EDID RAM override byte1 address.
0x018<<2	HDMIRX_TOP_EDID_RAM_OVR1_DATA	RW	EDID RAM override byte1 data.
0x019<<2	HDMIRX_TOP_EDID_RAM_OVR2	RW	EDID RAM override byte2 address.
0x01A<<2	HDMIRX_TOP_EDID_RAM_OVR2_DATA	RW	EDID RAM override byte2 data.
0x01B<<2	HDMIRX_TOP_EDID_RAM_OVR3	RW	EDID RAM override byte3 address.
0x01C<<2	HDMIRX_TOP_EDID_RAM_OVR3_DATA	RW	EDID RAM override byte3 data.
0x01D<<2	HDMIRX_TOP_EDID_RAM_OVR4	RW	EDID RAM override byte4 address.
0x01E<<2	HDMIRX_TOP_EDID_RAM_OVR4_DATA	RW	EDID RAM override byte4 data.
0x01F<<2	HDMIRX_TOP_EDID_RAM_OVR5	RW	EDID RAM override byte5 address.
0x020<<2	HDMIRX_TOP_EDID_RAM_OVR5_DATA	RW	EDID RAM override byte5 data.
0x021<<2	HDMIRX_TOP_EDID_RAM_OVR6	RW	EDID RAM override byte6 address.
0x022<<2	HDMIRX_TOP_EDID_RAM_OVR6_DATA	RW	EDID RAM override byte6 data.
0x023<<2	HDMIRX_TOP_EDID_RAM_OVR7	RW	EDID RAM override byte7 address.
0x024<<2	HDMIRX_TOP_EDID_RAM_OVR7_DATA	RW	EDID RAM override byte7 data.
0x025<<2	HDMIRX_TOP_EDID_GEN_STAT_B	RW	PortB EDID status.
0x026<<2	HDMIRX_TOP_EDID_GEN_STAT_C	RW	PortC EDID status.
0x028<<2	HDMIRX_TOP_CHAN_SWITCH_0	RW	Data channel input control.
0x029<<2	HDMIRX_TOP_TMDS_ALIGN_CNTL0	RW	Data channel alignment control.
0x02A<<2	HDMIRX_TOP_TMDS_ALIGN_CNTL1	RW	Data channel alignment control.
0x02B<<2	HDMIRX_TOP_TMDS_ALIGN_STAT	R	Data channel alignment status.
0x02C<<2	HDMIRX_TOP_INFILTER_HDCP	RW	HDCP and SCDC input glitch filter control.
0x02D<<2	HDMIRX_TOP_INFILTER_I2C_0	RW	DDC SCL/SDA input glitch filter control for port 0.
0x02E<<2	HDMIRX_TOP_INFILTER_I2C_1	RW	DDC SCL/SDA input glitch filter control for port 1.
0x02F<<2	HDMIRX_TOP_INFILTER_I2C_2	RW	DDC SCL/SDA input glitch filter control for port 2.

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Addr	Name	RW	Function
0x033<<2	HDMIRX_TOP_PRBS_GEN	RW	PRBS generator control.
0x034<<2	HDMIRX_TOP_PRBS_ANA_0	RW	PRBS analyzer control.
0x035<<2	HDMIRX_TOP_PRBS_ANA_1	RW	PRBS analyzer control.
0x036<<2	HDMIRX_TOP_PRBS_ANA_ STAT	R	PRBS analyzer status.
0x037<<2	HDMIRX_TOP_PRBS_ANA_ BER_CH0	R	PRBS analyzer BER count for data channel 0.
0x038<<2	HDMIRX_TOP_PRBS_ANA_ BER_CH1	R	PRBS analyzer BER count for data channel 1.
0x039<<2	HDMIRX_TOP_PRBS_ANA_ BER_CH2	R	PRBS analyzer BER count for data channel 2.
0x03A<<2	HDMIRX_TOP_METER_CA-BLE_CNTL	RW	cable_clk meter control.
0x03B<<2	HDMIRX_TOP_METER_CA-BLE_STAT	R	cable_clk meter status.
0x03C<<2	HDMIRX_TOP_CHAN_ SWITCH_1	RW	Data channel input control.
0x040<<2	HDMIRX_TOP_AUDPLL_ LOCK_FILTER	RW	Aud_pll lock input filter.
0x041<<2	HDMIRX_TOP_CHAN01_ ERRCNT	R	Character error counter for data channel 0 and 1.
0x042<<2	HDMIRX_TOP_CHAN2_ ERRCNT	R	Character error counter for data channel 2.
0x043<<2	HDMIRX_TOP_ACR_CNTL2	RW	ACR control.
0x044<<2	HDMIRX_TOP_ACR_N_STAT	R	Read back N value to AUD_PLL.
0x045<<2	HDMIRX_TOP_ACR_CTS_ STAT	R	Read back CTS value to AUD_PLL.
0x046<<2	HDMIRX_TOP_AUDMEAS_ CTRL	RW	Audio clock measure control.
0x047<<2	HDMIRX_TOP_AUDMEAS_ CYCLES_M1	RW	Audio clock measure period length.
0x048<<2	HDMIRX_TOP_AUDMEAS_ INTR_MASKN	RW	Audio clock measure interrupt mask.
0x049<<2	HDMIRX_TOP_AUDMEAS_ INTR_STAT	RW	Audio clock measure interrupt status.
0x04A<<2	HDMIRX_TOP_AUDMEAS_ REF_CYCLES_STAT0	R	Audio clock measure result.
0x04B<<2	HDMIRX_TOP_AUDMEAS_ REF_CYCLES_STAT1	R	Audio clock measure result.
0x060<<2	HDMIRX_TOP_HDCP22_ BSOD	RW	HDCP22 BSOD control.
0x061<<2	HDMIRX_TOP_SKP_CNTL_ STAT	RW	SKP interface control for HDCP2.2.
0x062<<2	HDMIRX_TOP_NONCE_0	RW	Nonce[31:0] for HDCP2.2.
0x063<<2	HDMIRX_TOP_NONCE_1	RW	Nonce[63:32] for HDCP2.2.

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Addr	Name	RW	Function
0x064<<2	HDMIRX_TOP_NONCE_2	RW	Nonce[95:64] for HDCP2.2.
0x065<<2	HDMIRX_TOP_NONCE_3	RW	Nonce[127:96] for HDCP2.2.
0x066<<2	HDMIRX_TOP_PKF_0	RW	PKF[31:0] for HDCP2.2.
0x067<<2	HDMIRX_TOP_PKF_1	RW	PKF[63:32] for HDCP2.2.
0x068<<2	HDMIRX_TOP_PKF_2	RW	PKF[95:64] for HDCP2.2.
0x069<<2	HDMIRX_TOP_PKF_3	RW	PKF[127:96] for HDCP2.2.
0x06A<<2	HDMIRX_TOP_DUK_0	RW	DUK [31:0] for HDCP2.2.
0x06B<<2	HDMIRX_TOP_DUK_1	RW	DUK [63:32] for HDCP2.2.
0x06C<<2	HDMIRX_TOP_DUK_2	RW	DUK [95:64] for HDCP2.2.
0x06D<<2	HDMIRX_TOP_DUK_3	RW	DUK [127:96] for HDCP2.2.
0x06E<<2	HDMIRX_TOP_NSEC_ SCRATCH	RW	Scratch register for non-secure access.
0x06F<<2	HDMIRX_TOP_SEC_ SCRATCH	RW	Scratch register for secure access.
0x070<<2	HDMIRX_TOP_EMP_DDR_ START_A	RW	DDR pointer A to store EMP.
0x071<<2	HDMIRX_TOP_EMP_DDR_ START_B	RW	DDR pointer B to store EMP.
0x072<<2	HDMIRX_TOP_EMP_DDR_ FILTER	RW	Packet filter to go to DDR.
0x073<<2	HDMIRX_TOP_EMP_CNTL_0	RW	EMP to DDR control.
0x074<<2	HDMIRX_TOP_EMP_CNTL_1	RW	EMP to DDR control.
0x075<<2	HDMIRX_TOP_EMP_CNTMAX	RW	Maximum number of packets to store in DDR per frame.
0x076<<2	HDMIRX_TOP_EMP_ERR_ STAT	RW	EMP error status.
0x077<<2	HDMIRX_TOP_EMP_RCV_ CNT_CUR	R	Current number of packet received.
0x078<<2	HDMIRX_TOP_EMP_RCV_ CNT_BUF	R	Buffered number of packet received.
0x079<<2	HDMIRX_TOP_EMP_DDR_ ADDR_CUR	R	Current DDR pointer for EMP.
0x07A<<2	HDMIRX_TOP_EMP_DDR_ PTR_S_BUF	R	Buffered DDR pointer for EMP.
0x07B<<2	HDMIRX_TOP_EMP_STAT_0	R	EMP status.
0x07C<<2	HDMIRX_TOP_EMP_STAT_1	R	EMP status.
0x080<<2	HDMIRX_TOP_AXI_CNTL_0	RW	AXI control.
0x081<<2	HDMIRX_TOP_AXI_ASYNC_ HOLD_ESM	RW	AXI async control for ESM.
0x082<<2	HDMIRX_TOP_AXI_ASYNC_ HOLD_EMP	RW	AXI async control for EMP.
0x083<<2	HDMIRX_TOP_AXI_STAT_0	R	AXI async status.

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Addr	Name	RW	Function
0x084<<2	HDMIRX_TOP_MISC_STAT0	R	MISC status – aud_fmt, PHY sqofclk
0x1000-0x11ff	HDMIRX_TOP_EDID_ADDR_S-HDMIRX_TOP_EDID_ADDR_E	RW	For Host to program EDID RAM.

Table 8-4 HDMIRX_TOP_SW_RESET

Bit	R/W	Default	Description
31:16	R	0	Reserved
15	RW	0	sw_reset_audmeas: to reset audio clock measure module. 0=Release from reset; 1=Apply reset.
14	RW	0	sw_reset_axiarb: to reset AXI arbitor. 0=Release from reset; 1=Apply reset.
13	RW	0	sw_reset_emp: to reset EMP module. 0=Release from reset; 1=Apply reset.
12	RW	1	sw_reset_hdcp22: to reset HRX_HDCP22 IP. 0=Release from reset; 1=Apply reset.
11	RW	0	sw_reset_flt: to reset DDC input glitch filter. 0=Release from reset; 1=Apply reset.
10	RW	0	sw_reset_skp: to reset interface to HRX_HDCP22 SKP ports. 0=Release from reset; 1=Apply reset.
9	RW	0	sw_reset_align: to reset channel_align module. 0=Release from reset; 1=Apply reset.
8	RW	0	sw_reset_bist: to reset BIST module. 0=Release from reset; 1=Apply reset.
7	RW	0	sw_reset_chan: to reset channel_switch module. 0=Release from reset; 1=Apply reset.
6	RW	0	sw_reset_cable_meter: to reset cable_clk meter. 0=Release from reset; 1=Apply reset.
5	RW	0	sw_reset_hdmi_meter: to reset hdmi_clk meter. 0=Release from reset; 1=Apply reset.
4	RW	0	sw_reset_acr: to reset ACR module. 0=Release from reset; 1=Apply reset.
3	RW	0	sw_reset_vid: to reset video processing path. 0=Release from reset; 1=Apply reset.
2	RW	0	sw_reset_intr: to reset interrupt block. 0=Release from reset; 1=Apply reset.

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Bit	R/W	Default	Description
1	RW	0	sw_reset_edid: to reset EDID interface. 0=Release from reset; 1=Apply reset.
0	RW	1	sw_reset_core: To reset RX Controller IP. 0=Release from reset; 1=Apply reset.

Table 8-5 HDMIRX_TOP_CLK_CNTL

Bit	R/W	Default	Description
31	RW	0	free_clk_en: 0= Enable clock gating for power saving; 1= Disable clock gating, enable free-run clock.
30:18	R	0	Reserved
17	RW	0	audfifo_rd_en: 1=Enable audio data FIFO; 0=Disable audio data FIFO.
16	RW	0	pktfifo_rd_en: 1=Enable packet FIFO; 0=Disable packet FIFO.
15:12	RW	0	Reserved
11	RW	0	clk_en_hdcp22_tmdsclk: 1= Enable hdcp22_tmdsclk.
10	RW	0	clk_en_skp: 1= Enable hdcp22_skpclk.
9	RW	0	clk_en_esm: 1= Enable hdcp22_esmclk.
8	RW	0	tmds_ch2_clk_inv: 1= Invert tmds_ch2_clk.
7	RW	0	tmds_ch1_clk_inv: 1= Invert tmds_ch1_clk.
6	RW	0	tmds_ch0_clk_inv: 1= Invert tmds_ch0_clk.
5	RW	0	pll4x_cfg
4	RW	0	force_pll4x: 1=Force pll4x_en value to be pll4x_cfg. 0=Use auto detect.
3	RW	0	phy_clk_inv: 1= Invert phy_clk.
2:0	R	0	Reserved

Table 8-6 HDMIRX_TOP_HPD_PWR5V

Bit	R/W	Default	Description
31:23	R	0	Reserved
22	R	0	stat_5v_portC: Status of 5v power for port C.
21	R	0	stat_5v_portB: Status of 5v power for port B.

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Bit	R/W	Default	Description
20	R	0	stat_5v_portA: Status of 5v power for port A.
19	R	0	Reserved
18	R	0	stat_hpd_portC: Status of IP's own HPD value for port C.
17	R	0	stat_hpd_portB: Status of IP's own HPD value for port B.
16	R	0	stat_hpd_portA: Status of IP's own HPD value for port A.
15:6	R	0	Reserved
5	RW	0	invert_hpd: 1=Invert the HPD value before final output; 0=No invert.
4	RW	1	force_hpd: 1=Select hpd_config as HPD value; 0=Select IP's own value as HPD.
3	R	0	Reserved
2:0	RW	0	hpd_config: HPD values when force_hpd=1, bit0 for port A, bit1 for port B, bit2 for port C.

Table 8-7 HDMIRX_TOP_PORT_SEL

Bit	R/W	Default	Description
31:5	R	0	Reserved
4	RW	0	multi_edid: 1=Enable all ports available for EDID transaction; 0=Only the selected port can do EDID.
2:0	RW	0	port_sel: HDMI port select for EDID. 0001=Select port A; 0010=Select port B; 0100=Select port C

Table 8-8 HDMIRX_TOP_EDID_GEN_CNTL

Bit	R/W	Default	Description
31:16	RW	0	Reserved
15:13	RW	0	i2c_filter_enable[2:0]. One bit for each port. 1=Enable glitch filter on SCL and SDA; 0=No glitch filter, same as M6TV.
12	R	0	Reserved
11	RW	0	auto_cec_enable: Control whether to auto-replace the port-related CEC information within EDID info. 1=Auto-replace data byte with edid_byte0_cec_portA/B/C/D, if the byte's location matches edid_addr0_cec, and port matches A/B/C/D; and auto-replace data byte with edid_byte1_cec_portA/B/C/D, if the byte's location matches edid_addr1_cec, and port matches A/B/C/D; 0=Send EDID data exactly as filled in RAM.
10	RW	0	scl_stretch_trigger_config: 1=Manual trigger for clock stretch, applicable when force_scl_stretch_trigger=1. 0=No trigger.

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Bit	R/W	Default	Description
9	RW	0	force_scl_stretch_trigger: Select the source to trigger start clock-stretch. 1=Manual trigger of scl_stretch_trigger_config=1; 0=Auto trigger on every 2 EDID blocks sent.
8	RW	0	scl_stretch_enable: 1=Enable I2C clock-stretch feature to suspend I2C when Host need time to fill new EDID data into RAM during I2C transaction; 0=Disable I2C clock-stretch feature.
7:0	RW	0	clk_divide_m1: Divide system clock down for EDID I/F clock, no need to divide down, but can divide down if for power saving purpose.

Table 8-9 HDMIRX_TOP_EDID_ADDR_CEC

Bit	R/W	Default	Description
31:16	RW	0x99	edid_addr1_cec: byte1 address for auto-replacing EDID's CEC info.
15:0	RW	0x98	edid_addr0_cec: byte0 address for auto-replacing EDID's CEC info.

Table 8-10 HDMIRX_TOP_EDID_DATA_CEC_PORT01

Bit	R/W	Default	Description
31:24	RW	0	edid_byte1_cec_portB: byte1 data for auto-replacing EDID's CEC info.
23:16	RW	0	edid_byte0_cec_portB: byte0 data for auto-replacing EDID's CEC info.
15:8	RW	0	edid_byte1_cec_portA: byte1 data for auto-replacing EDID's CEC info.
7:0	RW	0	edid_byte0_cec_portA: byte0 data for auto-replacing EDID's CEC info.

Table 8-11 Register HDMIRX_TOP_EDID_DATA_CEC_PORT2

Bit	R/W	Default	Description
15:8	RW	0	edid_byte1_cec_portC: byte1 data for auto-replacing EDID's CEC info.
7:0	RW	0	edid_byte0_cec_portC: byte0 data for auto-replacing EDID's CEC info.

The following table shows EDID status for portA.

Table 8-12 HDMIRX_TOP_EDID_GEN_STAT_A

Bit	R/W	Default	Description
31:25	R	0	Reserved
24:22	R	0	i2c_state: Status of I2C transaction. 0=IDLE; 1=SEND_BYTE; 2=GET_ACK; 3=READ_BYTE; 4=PUT_ACK.
21:20	R	0	slave_state: Status of I2C slave. 0=IDLE; 1=OFFSET, read offset address;

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Bit	R/W	Default	Description
			2=SEGMENT, read segment pointer; 3=SEND, sending data.
19:17	R	0	Reserved
16	RW	0	edid_addr_intr_stat_clr: This bit read back 1 after 2-block EDID data were sent, and there are further block to be sent. Write this bit 1 to clear this bit.
15:0	R	0	edid_addr: Status of current EDID address.

Interrupt MASKN, one bit per interrupt source.

- 0= Disable interrupt source;
- 1= Enable interrupt source.

Table 8-13 HDMIRX_TOP_INTR_MASKN

Bit	R/W	Default	Description
31:30	R	0	Reserved
29	RW	0	hdmirx_sqofclk_fall
28	RW	0	hdmirx_sqofclk_rise
27	RW	0	de_rise_irq: DE rise edge.
26	RW	0	last_emp_done: For EMP-to-DDR mode, this bit means last EMP packet received; For TMDS-to-DDR mode, this bit means last TMDS character received.
25	RW	0	emp_field_done: EMP receive done for last field. Note: EMP field end at DE rise.
24	RW	0	tmds_align_stable_chg: tmds alignment stable/unstable status change.
23	RW	0	meter_stable_chg_cable: cable_clk measure stable/unstable status change
22	RW	0	hdcp22_rnd_err_pls: Error generating random number to HRX_HDCP22
21	RW	0	HRX_HDCP22 SKP port signal nonce_rfrsh rise
20	RW	0	Reserved.
19	RW	0	edid_addr_intr for the port C
18	RW	0	edid_addr_intr for the port B
17	RW	0	edid_addr_intr for the port A
16	RW	0	hdcp_enc_state_fall
15	RW	0	hdcp_enc_state_rise
14	RW	0	hdcp_auth_start_fall
13	RW	0	hdcp_auth_start_rise
12	RW	0	meter_stable_chg_hdmi: tmds_clk measure stable/unstable status change
11	RW	0	vid_colour_depth_chg: video data color_depth change
10	RW	0	vid_fmt_chg: video data format change
9	RW	0	Reserved.

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Bit	R/W	Default	Description
8	RW	0	Port C 5v fall
7	RW	0	Port B 5v fall
6	RW	0	Port A 5v fall
5	RW	0	Reserved.
4	RW	0	Port C 5v rise
3	RW	0	Port B 5v rise
2	RW	0	Port A 5v rise
1	RW	0	edid_addr_intr for the selected port
0	RW	0	RX Controller IP interrupt.

In the following table, the interrupt status, for each bit of bit[29:1], write 1 to manually set the interrupt bit, read back the interrupt status.

Table 8-14 HDMIRX_TOP_INTR_STAT

Bit	R/W	Default	Description
31	R	0	Shadowing HRX_HDCP2.2 IP interrupt status flag.
29	RW	0	hdmirx_sqofclk_fall
28	RW	0	hdmirx_sqofclk_rise
27	RW	0	de_rise_irq: DE rise edge.
26	RW	0	last_emp_done: For EMP-to-DDR mode, this bit means last EMP packet received; For TMDS-to-DDR mode, this bit means last TMDS character received.
25	RW	0	emp_field_done: EMP receive done for last field. Note: EMP field end at DE rise.
24	RW	0	tmds_align_stable_chg: tmds alignment stable/unstable status change.
23	RW	0	meter_stable_chg_cable: cable_clk measure stable/unstable status change
22	RW	0	hdcp22_rnd_err_pls: Error generating random number to HRX_HDCP22
21	RW	0	HRX_HDCP22 SKP port signal nonce_rfrsh rise
20	RW	0	Reserved.
19	RW	0	edid_addr_intr for the port C
18	RW	0	edid_addr_intr for the port B
17	RW	0	edid_addr_intr for the port A
16	RW	0	hdcp_enc_state_fall
15	RW	0	hdcp_enc_state_rise
14	RW	0	hdcp_auth_start_fall
13	RW	0	hdcp_auth_start_rise
12	RW	0	meter_stable_chg_hdmi: tmds_clk measure stable/unstable status change
11	RW	0	vid_colour_depth_chg: video data color_depth change
10	RW	0	vid_fmt_chg: video data format change

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Bit	R/W	Default	Description
9	RW	0	Reserved.
8	RW	0	Port C 5v fall
7	RW	0	Port B 5v fall
6	RW	0	Port A 5v fall
5	RW	0	Reserved.
4	RW	0	Port C 5v rise
3	RW	0	Port B 5v rise
2	RW	0	Port A 5v rise
1	RW	0	edid_addr_intr for the selected port
0	RW	0	Reserved.

Interrupt status clear. For each bit[29:1], write 1 to clear the interrupt bit.

Table 8-15 HDMIRX_TOP_INTR_STAT_CLR

Bit	R/W	Default	Description
31:30	R	0	Reserved
29	RW	0	hdmirx_sqofclk_fall
28	RW	0	hdmirx_sqofclk_rise
27	RW	0	de_rise_irq: DE rise edge.
26	RW	0	last_emp_done: For EMP-to-DDR mode, this bit means last EMP packet received; For TMDS-to-DDR mode, this bit means last TMDS character received.
25	RW	0	emp_field_done: EMP receive done for last field. Note: EMP field end at DE rise.
24	RW	0	tmds_align_stable_chg: tmds alignment stable/unstable status change.
23	RW	0	meter_stable_chg_cable: cable_clk measure stable/unstable status change
22	W	0	hdcp22_rnd_err_pls: Error generating random number to HRX_HDCP22
21	W	0	HRX_HDCP22 SKP port signal nonce_rfrsh rise
20	W	0	Reserved.
19	W	0	edid_addr_intr for the port C
18	W	0	edid_addr_intr for the port B
17	W	0	edid_addr_intr for the port A
16	W	0	hdcp_enc_state_fall
15	W	0	hdcp_enc_state_rise
14	W	0	hdcp_auth_start_fall
13	W	0	hdcp_auth_start_rise
12	W	0	meter_stable_chg_hdmi: tmds_clk measure stable/unstable status change
11	W	0	vid_colour_depth_chg: video data color_depth change
10	W	0	vid_fmt_chg: video data format change

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Bit	R/W	Default	Description
9	W	0	Reserved.
8	W	0	Port C 5v fall
7	W	0	Port B 5v fall
6	W	0	Port A 5v fall
5	W	0	Reserved.
4	W	0	Port C 5v rise
3	W	0	Port B 5v rise
2	W	0	Port A 5v rise
1	W	0	edid_addr_intr for the selected port
0	W	0	Reserved.

Table 8-16 HDMIRX_TOP_VID_CNTL

Bit	R/W	Default	Description
31:29	RW	0	Reserved
28	RW	0	vs_timing: 0=input VS rising edge to generate output VS pulse; 1=input VS falling edge to generate output VS pulse.
27	RW	0	hs_timing: 0=input HS rising edge to generate output HS pulse; 1=input HS falling edge to generate output HS pulse.
26:24	RW	0	data_map: Re-map component data to different position. Reserved for simulation purpose only.
23:20	RW	0	Reserved
19	RW	0	mode_422: 1=Average neighbor pixels to interpolate to 444; 0=Repeat previous pixel to interpolate to 444.
18	RW	0	Reserved
17	RW	0	dbv422_mode: 1=For DolbyVision 422 mode, pass through data to downstream video path without converting to 444. The data map is as follows: vid_comp2[9:0]: {Y[11:4], 2'h0} vid_comp1[9:0]: {C[11:4], 2'h0} vid_comp0[9:0]: {C[3:0], Y[3:0], 2'h0}.
16:0	RW	0	Reserved

Table 8-17 HDMIRX_TOP_VID_STAT

Bit	R/W	Default	Description
31:24	R	0	Reserved
23:20	R	0	vid_fmt: input video data format. 0=RGB; 1=YUV422; 2=YUV444; 3=YUV420.

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Bit	R/W	Default	Description
19:16	R	0	vid_colour_depth: input video data color_depth. 0-3=Reserved; 4=24-bit per pixel; 5=30-bit; 6=36-bit; 7=48-bit; 8-15=Reserved.
15:0	R	0	window_start_pix: Applicable to 420 mode only. The measure input data line's position from HS rise to DE rise, minus cntl_ last420l_rdwin_auto.

Table 8-18 HDMIRX_TOP_ACR_CNTL_STAT

Bit	R/W	Default	Description
31:29	R	0	afif_status: Applicable only to digital ACR. Audio FIFO level status: 0=IDLE; 4=EMPTY; 5=LOW; 1=BELOW_NOMINAL; 3=NOMINAL; 2=ABOVE_NOMINAL; 6=HIGH; 7=FULL.
28:27	R	0	Reserved
26:16	R	0	afif_level: Applicable only to digital ACR. IP's Audio FIFO level.
15:13	RW	0	n_cts_update_width: 000=hdmirx_n_cts_update pulse width is 1 tmds_clk cycle; 001=2-cycle; 011=3-cycle; 111=4-cycle.
12	RW	0	n_cts_update_on_pll4x: 1=aud_pll4x_en change will result in a hdmirx_n_cts_update pulse; 0=No hdmirx_n_cts_update pulse on aud_pll4x_en change.
11	RW	0	n_cts_update_man: Write 1 to manually generate an hdmirx_n_cts_update pulse. This bit self-clears.
10	RW	0	pll_tmode_cntl: Set to 1 for production testing audio ACR PLL. Normal working mode set to 0.
9	RW	0	force_afif_status: Applicable only to digital ACR. 1=Use afif_status_cfg as fifo status; 0=Use detected afif_status.
8	RW	0	afif_status_auto: Applicable only to digital ACR. 1=Enable audio FIFO status auto-exit EMPTY/FULL, if FIFO level is back to LipSync; 0=Once enters EMPTY/FULL, never exits.
7: 6	RW	0	afif_nom_threshold: Applicable only to digital ACR. Audio FIFO nominal level : 0=s_total/4; 1=s_total/8; 2=s_total/16; 3=s_total/32.
5:4	RW	0	afif_crit_threshold: Applicable only to digital ACR. Audio FIFO critical level:

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Bit	R/W	Default	Description
			0=s_total/4; 1=s_total/8; 2=s_total/16; 3=s_total/32.
3	RW	0	afif_status_clr: Applicable only to digital ACR. 1=Clear audio FIFO status to IDLE.
2	RW	0	dig_acr_en: 1=Enable digital ACR; 0=Disable.
1	RW	0	audmeas_clk_sel: Select which audio clock source to be measured. 0=Measure audio ACR PLL; 1=Measure digital ACR output.
0	RW	0	aud_clk_sel: Select audio clock source. 0=Use audio ACR PLL; 1=Use digital ACR output.

Table 8-19 HDMIRX_TOP_ACR_AUDFIFO

Bit	R/W	Default	Description
30:3	R	0	Reserved
2:0	RW	0	afif_status_cfg: Applicable only to digital ACR. When force_afif_status=1, afif_status_cfg is used as fifo status.

Table 8-20 HDMIRX_TOP_METER_HDMI_CNTL

Bit	R/W	Default	Description
31:28	RW	0x1	meas_tolerance
27:24	RW	0	Reserved
23:0	RW	0x80	ref_cycles

Table 8-21 HDMIRX_TOP_METER_HDMI_STAT

Bit	R/W	Default	Description
31	R	0	meas_stable
30:24	R	0	Reserved
23:0	R	0	meas_cycles. Freq(tmds_clk)=Freq(ref_clk)*meas_cycles/ref_cycles

Table 8-22 HDMIRX_TOP_VID_CNTL2

Bit	R/W	Default	Description
31:21	RW	0	Reserved
20	RW	0	last420l_rdwin_mode: Control when to start read from FIFO for outputting 420 even or last line. 0=End Of Active Video by detecting no DE cntl_last420l_rdwin_auto + window_ start_pix clock cycles after go_line. 1=End of Active Video after cntl_last420l_rdwin_manual active line passed.

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Bit	R/W	Default	Description
19:8	RW	0	last420l_rdwin_manual
7:0	RW	0xa	last420l_rdwin_auto

Table 8-23 THDMIRX_TOP_MEM_PD

Bit	R/W	Default	Description
31:14	RW	0	Reserved
15:14	RW	0x3	mem_pd_emp
13:12	RW	0x3	mem_pd_hrx_hdcp22
11:10	RW	0x3	mem_pd_ksv
9:8	RW	0x3	mem_pd_packet
7:6	RW	0x3	mem_pd_audio
5:4	RW	0x3	Reserved
3:2	RW	0x3	mem_pd_vid_420
1:0	RW	0x3	mem_pd_edid

Table 8-24 HDMIRX_TOP_EDID_RAM_OVR0

Bit	R/W	Default	Description
31:19	R	0	Reserved
18:16	RW	0	edid_ram_ovr0_en: For each bit, 1=Enable override the data read from EDID RAM address edid_ram_ovr0_addr, with the data defined in HDMIRX_EDID_RAM_OVR0_DATA. Bit[0] for portA, bit[1] for portB, bit[2] for portC
15:0	RW	0	edid_ram_ovr0_addr

Table 8-25 HDMIRX_TOP_EDID_RAM_OVR0_DATA

Bit	R/W	Default	Description
23:16	RW	0	edid_ram_ovr0_data for portC
15:8	RW	0	edid_ram_ovr0_data for portB
7:0	RW	0	edid_ram_ovr0_data for portA

Table 8-26 HDMIRX_TOP_EDID_RAM_OVR1

Bit	R/W	Default	Description
31:19	R	0	Reserved
18:16	RW	0	edid_ram_ovr1_en: For each bit, 1=Enable override the data read from EDID RAM address edid_ram_ovr1_addr, with the data defined in HDMIRX_EDID_RAM_OVR1_DATA. Bit[0] for portA, bit[1] for portB, bit[2] for portC
15:0	RW	0	edid_ram_ovr1_addr

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Table 8-27 HDMIRX_TOP_EDID_RAM_OVR1_DATA

Bit	R/W	Default	Description
23:16	RW	0	edid_ram_ovr1_data for portC
15:8	RW	0	edid_ram_ovr1_data for portB
7:0	RW	0	edid_ram_ovr1_data for portA

Table 8-28 HDMIRX_TOP_EDID_RAM_OVR2

Bit	R/W	Default	Description
31:19	R	0	Reserved
18:16	RW	0	edid_ram_ovr2_en: For each bit, 1=Enable override the data read from EDID RAM address edid_ram_ovr2_addr, with the data defined in HDMIRX_EDID_RAM_OVR2_DATA. Bit[0] for portA, bit[1] for portB, bit[2] for portC
15:0	RW	0	edid_ram_ovr2_addr

Table 8-29 HDMIRX_TOP_EDID_RAM_OVR2_DATA

Bit	R/W	Default	Description
23:16	RW	0	edid_ram_ovr2_data for portC
15:8	RW	0	edid_ram_ovr2_data for portB
7:0	RW	0	edid_ram_ovr2_data for portA

Table 8-30 HDMIRX_TOP_EDID_RAM_OVR3

Bit	R/W	Default	Description
31:19	R	0	Reserved
18:16	RW	0	edid_ram_ovr3_en: For each bit, 1=Enable override the data read from EDID RAM address edid_ram_ovr3_addr, with the data defined in HDMIRX_EDID_RAM_OVR3_DATA. Bit[0] for portA, bit[1] for portB, bit[2] for portC
15:0	RW	0	edid_ram_ovr3_addr

Table 8-31 HDMIRX_TOP_EDID_RAM_OVR3_DATA

Bit	R/W	Default	Description
23:16	RW	0	edid_ram_ovr3_data for portC
15:8	RW	0	edid_ram_ovr3_data for portB
7:0	RW	0	edid_ram_ovr3_data for portA

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Table 8-32 HDMIRX_TOP_EDID_RAM_OVR4

Bit	R/W	Default	Description
31:19	R	0	Reserved
18:16	RW	0	edid_ram_ovr4_en: For each bit, 1=Enable override the data read from EDID RAM address edid_ram_ovr4_addr, with the data defined in HDMIRX_EDID_RAM_OVR4_DATA. Bit[0] for portA, bit[1] for portB, bit[2] for portC
15:0	RW	0	edid_ram_ovr4_addr

Table 8-33 HDMIRX_TOP_EDID_RAM_OVR4_DATA

Bit	R/W	Default	Description
23:16	RW	0	edid_ram_ovr4_data for portC
15:8	RW	0	edid_ram_ovr4_data for portB
7:0	RW	0	edid_ram_ovr4_data for portA

Table 8-34 HDMIRX_TOP_EDID_RAM_OVR5

Bit	R/W	Default	Description
31:19	R	0	Reserved
18:16	RW	0	edid_ram_ovr5_en: For each bit, 1=Enable override the data read from EDID RAM address edid_ram_ovr5_addr, with the data defined in HDMIRX_EDID_RAM_OVR5_DATA. Bit[0] for portA, bit[1] for portB, bit[2] for portC
15:0	RW	0	edid_ram_ovr5_addr

Table 8-35 HDMIRX_TOP_EDID_RAM_OVR5_DATA

Bit	R/W	Default	Description
23:16	RW	0	edid_ram_ovr5_data for portC
15:8	RW	0	edid_ram_ovr5_data for portB
7:0	RW	0	edid_ram_ovr5_data for portA

Table 8-36 HDMIRX_TOP_EDID_RAM_OVR6

Bit	R/W	Default	Description
31:19	R	0	Reserved
18:16	RW	0	edid_ram_ovr6_en: For each bit, 1=Enable override the data read from EDID RAM address edid_ram_ovr6_addr, with the data defined in HDMIRX_EDID_RAM_OVR6_DATA. Bit[0] for portA, bit[1] for portB, bit[2] for portC
15:0	RW	0	edid_ram_ovr6_addr

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Table 8-37 HDMIRX_TOP_EDID_RAM_OVR6_DATA

Bit	R/W	Default	Description
23:16	RW	0	edid_ram_ovr6_data for portC
15:8	RW	0	edid_ram_ovr6_data for portB
7:0	RW	0	edid_ram_ovr6_data for portA

Table 8-38 HDMIRX_TOP_EDID_RAM_OVR7

Bit	R/W	Default	Description
31:19	R	0	Reserved
18:16	RW	0	edid_ram_ovr7_en: For each bit, 1=Enable override the data read from EDID RAM address edid_ram_ovr7_addr, with the data defined in HDMIRX_EDID_RAM_OVR7_DATA. Bit[0] for portA, bit[1] for portB, bit[2] for portC
15:0	RW	0	edid_ram_ovr7_addr

Table 8-39 HDMIRX_TOP_EDID_RAM_OVR7_DATA

Bit	R/W	Default	Description
23:16	RW	0	edid_ram_ovr7_data for portC
15:8	RW	0	edid_ram_ovr7_data for portB
7:0	RW	0	edid_ram_ovr7_data for portA

EDID status for portB.

Table 8-40 HDMIRX_TOP_EDID_GEN_STAT_B

Bit	R/W	Default	Description
31:25	R	0	Reserved
24:22	R	0	i2c_state: Status of I2C transaction. 0=IDLE; 1=SEND_BYTE; 2=GET_ACK; 3=READ_BYTE; 4=PUT_ACK.
21:20	R	0	slave_state: Status of I2C slave. 0=IDLE; 1=OFFSET, read offset address; 2=SEGMENT, read segment pointer; 3=SEND, sending data.
19:17	R	0	Reserved
16	RW	0	edid_addr_intr_stat_clr: This bit read back 1 after 2-block EDID data were sent, and there are further block to be sent. Write this bit 1 to clear this bit.
15:0	R	0	edid_addr: Status of current EDID address.

EDID status for portC.

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Table 8-41 HDMIRX_TOP_EDID_GEN_STAT_C

Bit	R/W	Default	Description
31:25	R	0	Reserved
24:22	R	0	i2c_state: Status of I2C transaction. 0=IDLE; 1=SEND_BYTE; 2=GET_ACK; 3=READ_BYTE; 4=PUT_ACK.
21:20	R	0	slave_state: Status of I2C slave. 0=IDLE; 1=OFFSET, read offset address; 2=SEGMENT, read segment pointer; 3=SEND, sending data.
19:17	R	0	Reserved
16	RW	0	edid_addr_intr_stat_clr: This bit read back 1 after 2-block EDID data were sent, and there are further block to be sent. Write this bit 1 to clear this bit.
15:0	R	0	edid_addr: Status of current EDID address.

Table 8-42 HDMIRX_TOP_CHAN_SWITCH_0

Bit	R/W	Default	Description
29:28	RW	0	source_chan2: 0=select chan0 input; 1=select chan1 input; 2=select chan2 input.
27:26	RW	0	source_chan1: 0=select chan0 input; 1=select chan1 input; 2=select chan2 input.
25:24	RW	0	source_chan0: 0=select chan0 input; 1=select chan1 input; 2=select chan2 input.
22:20	RW	0	skew_chan2: 0=Delay data by 1 clock cycle; 1=Delay by 2 cycles; ...; 7=Delay by 8.
18:16	RW	0	skew_chan1: 0=Delay data by 1 clock cycle; 1=Delay by 2 cycles; ...; 7=Delay by 8.
14:12	RW	0	skew_chan0: 0=Delay data by 1 clock cycle; 1=Delay by 2 cycles; ...; 7=Delay by 8.
10	RW	0	bitswap_chan2: 1=Reverse input data endian.
9	RW	0	bitswap_chan1: 1=Reverse input data endian.

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Bit	R/W	Default	Description
8	RW	0	bitswap_chan0: 1=Reverse input data endian.
6	RW	0	polarity_chan2: 1=Invert input data polarity.
5	RW	0	polarity_chan1: 1=Invert input data polarity.
4	RW	0	polarity_chan0: 1=Invert input data polarity.
0	RW	0	enable: 1=enable data channels.

Table 8-43 HDMIRX_TOP_TMDS_ALIGN_CNTL0

Bit	R/W	Default	Description
31	RW	0	force_valid_cfg: Refer to cntl_valid_mode
29:28	RW	0	glitch_filter_en_2:Glitch filter enable for CONTROL character detect. 1=Enable filter. Otherwise disable.
27:26	RW	0	glitch_filter_en_1:Glitch filter enable for CONTROL character detect. 1=Enable filter. Otherwise disable.
25:24	RW	0	glitch_filter_en_0:Glitch filter enable for CONTROL character detect. 1=Enable filter. Otherwise disable.
23:22	RW	0	tolerance_2: Refer to tolerance_0.
21:19	RW	0	win_score_bound_2: Refer to win_score_bound_0.
18:16	RW	0	win_score_delay_2: Refer to win_score_delay_0.
15:14	RW	0	tolerance_1: Refer to tolerance_0.
13:11	RW	0	win_score_bound_1: Refer to win_score_bound_0.
10:8	RW	0	win_score_delay_1: Refer to win_score_delay_0.
7:6	RW	0	tolerance_0: Hamming distance for character match.
5:3	RW	0	win_score_bound_0: 0=Score if CONTROL Character delay boundary for once; 1=Score if CONTROL Character delay boundary three times; 2=Score if CONTROL Character delay boundary 7 times; 3=Score if CONTROL Character delay boundary 15 times; 4=Score if CONTROL Character delay boundary 31 times; 5=Score if CONTROL Character delay boundary 63 times; 6=Score if CONTROL Character delay boundary 127 times; 7=Score if CONTROL Character delay boundary 255 times.
2:0	RW	0	win_score_delay_0: 0=Score if CONTROL Character delay match for once; 1=Score if CONTROL Character delay match three times; 2=Score if CONTROL Character delay match 7 times; 3=Score if CONTROL Character delay match 15 times; 4=Score if CONTROL Character delay match 31 times; 5=Score if CONTROL Character delay match 63 times; 6=Score if CONTROL Character delay match 127 times; 7=Score if CONTROL Character delay match 255 times.

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Table 8-44 HDMIRX_TOP_TMDS_ALIGN_CNTL1

Bit	R/W	Default	Description
31:30	RW	0	cntl_valid_mode
29	RW	0	forced_delay_adjustment_enable_2
28	RW	0	forced_bound_adjustment_enable_2
27	RW	0	forced_delay_adjustment_enable_1
26	RW	0	forced_bound_adjustment_enable_1
25	RW	0	forced_delay_adjustment_enable_0
24	RW	0	forced_bound_adjustment_enable_0
23:20	RW	0	forced_delay_adjustment_2
19:16	RW	0	forced_bound_adjustment_2
15:12	RW	0	forced_delay_adjustment_1
11:8	RW	0	forced_bound_adjustment_1
7:4	RW	0	forced_delay_adjustment_0
3:0	RW	0	forced_bound_adjustment_0

Table 8-45 HDMIRX_TOP_TMDS_ALIGN_STAT

Bit	R/W	Default	Description
29	R	0	data_ch2_delay_stability
28	R	0	data_ch2_bound_stability
27	R	0	data_ch1_delay_stability
26	R	0	data_ch1_bound_stability
25	R	0	data_ch0_delay_stability
24	R	0	data_ch0_bound_stability
23:20	R	0	data_ch2_delay_status
19:16	R	0	data_ch2_bound_status
15:12	R	0	data_ch1_delay_status
11:8	R	0	data_ch1_bound_status
7:4	R	0	data_ch0_delay_status
3:0	R	0	data_ch0_bound_status

Table 8-46 HDMIRX_TOP_INFILTER_HDCP

Bit	R/W	Default	Description
31:29	RW	0	For HDCP/SCDC SDA infilter: filter internal clock divider. Refer to SCL infilter.
28:16	RW	0	For HDCP/SCDC SDA infilter: sampling clock divider. Refer to SCL infilter.

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Bit	R/W	Default	Description
15: 13	RW	0	For HDCP/SCDC SCL infilter: filter internal clock divider. 0=No divide; 1=Divide by 2; 2=Divide by 3; … 7=Divide by 8.
12:0	RW	0	For HDCP/SCDC SCL infilter: sampling clock divider. 0=No divide; 1=Divide the filter sampling clock by 2; 2=Divide the filter sampling clock by 3; … 8191=Divide the filter sampling clock by 8192;

Table 8-47 HDMIRX_TOP_INFILTER_I2C_0/1/2

Bit	R/W	Default	Description
31:29	RW	0	For EDID SDA infilter: filter internal clock divider. Refer to HDCP infilter.
28:16	RW	0	For EDID SDA infilter: sampling clock divider. Refer to HDCP infilter.
15: 13	RW	0	For EDID SCL infilter: filter internal clock divider. Refer to HDCP infilter.
12:0	RW	0	For EDID SCL infilter: sampling clock divider. Refer to HDCP infilter.

Table 8-48 HDMIRX_TOP_PRBS_GEN

Bit	R/W	Default	Description
8	RW	0	bist_loopback: 0=Loopback PRBS data to PHY; 1=Loopback original PHY data to PHY.
7:5	RW	3	decoup_thresh_prbs: Read start threshold for decouple RIFO at PRBS generator.
4:3	RW	0	prbs_mode: 0=PRBS11; 1=PRBS15; 2=PRBS7; 3=PRBS31.
2:1	RW	0	prbs_width: 0=output 0; 1=output 8-bit pattern; 2=output 1-bit pattern; 3=output 10-bit pattern.
0	RW	0	prbs_enable

Table 8-49 HDMIRX_TOP_PRBS_ANA_0

Bit	R/W	Default	Description
23:22	RW	0	chan2_prbs_mode: 0=PRBS11; 1=PRBS15; 2=PRBS7; 3=PRBS31.
21:20	RW	0	chan2_prbs_width: 0=idle; 1=8-bit pattern; 2=reserved; 3=10-bit pattern.
19	RW	0	chan2_clr_BER: 1=Clear BER result.
18	RW	0	chan2_freeze_BER. 1:Freeze BER result so it can be sample.
17	RW	0	chan2_inverse: 1=Reverse bit order/endian.

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Bit	R/W	Default	Description
15:14	RW	0	chan1_prbs_mode: 0=PRBS11; 1=PRBS15; 2=PRBS7; 3=PRBS31.
13:12	RW	0	chan1_prbs_width: 0=idle; 1=8-bit pattern; 2=reserved; 3=10-bit pattern.
11	RW	0	chan1_clr_BER: 1=Clear BER result.
10	RW	0	chan1_freeze_BER. 1:Freeze BER result so it can be sample.
9	RW	0	chan1_inverse: 1=Reverse bit order/endian.
7:6	RW	0	chan0_prbs_mode: 0=PRBS11; 1=PRBS15; 2=PRBS7; 3=PRBS31.
5:4	RW	0	chan0_prbs_width: 0=idle; 1=8-bit pattern; 2=reserved; 3=10-bit pattern.
3	RW	0	chan0_clr_BER: 1=Clear BER result.
2	RW	0	chan0_freeze_BER. 1:Freeze BER result so it can be sample.
1	RW	0	chan0_inverse: 1=Reverse bit order/endian.

Table 8-50 HDMIRX_TOP_PRBS_ANA_1

Bit	R/W	Default	Description
19:8	RW	0	time_window: Define the number of clocks after which the lock signal goes high if error number is smaller than error_threshold.
7:0	RW	0	err_threshold: Allowed number of errors.

Table 8-51 HDMIRX_TOP_PRBS_ANA_STAT

Bit	R/W	Default	Description
5	R	0	chan2_pattern_not_ok
4	R	0	chan2_lock
3	R	0	chan1_pattern_not_ok
2	R	0	chan1_lock
1	R	0	chan0_pattern_not_ok
0	R	0	chan0_lock

Table 8-52 HDMIRX_TOP_PRBS_ANA_BER_CH0

Bit	R/W	Default	Description
19:0	R	0	chan0_BER_result

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Table 8-53 HDMIRX_TOP_PRBS_ANA_BER_CH1

Bit	R/W	Default	Description
19:0	R	0	chan1_BER_result

Table 8-54 HDMIRX_TOP_PRBS_ANA_BER_CH2

Bit	R/W	Default	Description
19:0	R	0	chan2_BER_result

Table 8-55 HDMIRX_TOP_METER_CABLE_CNTL

Bit	R/W	Default	Description
31:28	RW	0x1	meas_tolerance: If difference between the consecutive measure results is greater than meas_tolerance, then stat_meas_stable=0; Otherwise stat_meas_stable=1.
23:0	RW	0x80	ref_cycles: Number of meter_clk cycles as one measure duration.

Table 8-56 HDMIRX_TOP_METER_CABLE_STAT

Bit	R/W	Default	Description
31	R	0	meas_stable: 1=If difference between the consecutive measure results is greater than meas_tolerance.
23:0	R	0	meas_cycles: Number of cable_clk cycles counted during one measure duration. Freq(cable_clk)=Freq(ref_clk)*meas_cycles/ref_cycles

Table 8-57 HDMIRX_TOP_CHAN_SWITCH_1

Bit	R/W	Default	Description
4	RW	0	valid_always: 0=use tmdsvalid signal from FIFO to gate data receiving; 1=Regardless tmdsvalid signal level, always receive data form PHY.
3:0	RW	7	decoup_thresh_chan: Read start threshold for decouple FIFO at channel input.

Table 8-58 HDMIRX_TOP_AUDPLL_LOCK_FILTER

Bit	R/W	Default	Description
7:0	RW	0	audpll_lock_filter value N: filter out glitch <= N

Table 8-59 HDMIRX_TOP_CHAN01_ERRCNT

Bit	R/W	Default	Description
30:16	R	0	erdet_ch1: shadow IP's periodical error counter for channel 1.
14:0	R	0	erdet_ch0: shadow IP's periodical error counter for channel 0.

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Table 8-60 HDMIRX_TOP_CHAN2_ERRCNT

Bit	R/W	Default	Description
14:0	R	0	erdet_ch0: shadow IP's periodical error counter for channel 2.

Table 8-61 HDMIRX_TOP_ACR_CNTL2

Bit	R/W	Default	Description
31:5	R	0	Reserved
4	RW	0	n_cts_auto_mode: 0=Auto update on every ACR packet, regardless N&CTS; value change or not; 1=Auto update on ACR packet's N or CTS value change, so may not update on every ACR.
3	RW	0	n_cts_update_inv: 1=Invert hdmirx_n_cts_update output polarity.
2:0	RW	0	n_cts_update_del: 0=hdmirx_n_cts_update pulse is aligned with N&CTS value change; 1=hdmirx_n_cts_update pulse is 1 cycle behind N&CTS value change; ... 7=hdmirx_n_cts_update pulse is 7 cycle behind N&CTS value change.

Table 8-62 HDMIRX_TOP_ACR_N_STAT

Bit	R/W	Default	Description
19:0	R	0	Records N value to AUD_PLL.

Table 8-63 HDMIRX_TOP_ACR_CTS_STAT

Bit	R/W	Default	Description
19:0	R	0	Records CTS value to AUD_PLL.

Table 8-64 HDMIRX_TOP_AUDMEAS_CTRL

Bit	R/W	Default	Description
31:16	RW	0	meas_tolerance. Clk is considered stable, if the difference between two consecutive measurements is less than meas_tolerance. Applicable meas_mode=0, periodic measure.
2	RW	0	meas_mode. 0=Periodic measure; 1=Accumulate time stamp.
1	RW	0	enable. 0=Disable; 1=Enable.
0	RW	0	sw_reset. 0=No reset; 1=Reset.

Table 8-65 HDMIRX_TOP_AUDMEAS_CYCLES_M1

Bit	R/W	Default	Description
31:0	RW	0	Set (meas_clk_cycles_m1+1) of meas_clk cycles as one measure period, to count the number of ref_clk cycles.

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Table 8-66 HDMIRX_TOP_AUDMEAS_INTR_MASKN

Bit	R/W	Default	Description
3:0	RW	0	Interrupt src mask, one bit per interrupt src. 0=disable the src to interrupt; 1=enable the src to interrupt. Bit[3]: meas_stable fall Bit[2]: meas_stable rise Bit[1]: ref_counter overflow Bit[0]: one measure period completes

Table 8-67 HDMIRX_TOP_AUDMEAS_INTR_STAT

Bit	R/W	Default	Description
3:0	RW	0	Read back interrupt status, one bit per interrupt src. For each bit, write 1to clear the interrupt. Bit[3]: meas_stable fall Bit[2]: meas_stable rise Bit[1]: ref_counter overflow Bit[0]: one measure period completes

Table 8-68 HDMIRX_TOP_AUDMEAS_REF_CYCLES_STAT0

Bit	R/W	Default	Description
31:0	R	0	ref_clk_cycles_m1[31:0]. ref_clk_cycles_m1[47:0]+1 is the number of ref_clk cycles during one measure period (meas_mode=0), or during N measure period (meas_mode=1).

Table 8-69 HDMIRX_TOP_AUDMEAS_REF_CYCLES_STAT1

Bit	R/W	Default	Description
17	R	0	meas_stable. 1=meas_clk is stable; 0=not stable. Applicable only to meas_mode=0.
16	R	0	ref_clk_cycles_ovl. 1=ref_clk_cycles_m1 wrapped over.
15:0	R	0	ref_clk_cycles_m1[47:32].

Table 8-70 HDMIRX_TOP_HDCP22_BSOD

Bit	R/W	Default	Description
25	RW	0	hdcp22_no_bsod. 1=Do not replace content with BSOD, prior to authentication; 0=Replace with BSOD prior to authentication. Original behaviour.
24	RW	0	hdcp22_bsod_override. 1=Override BSOD value with hdcp22_bsod_val. 0=Original BSOD value fixed by FW.
23:0	RW	0x8000-80	hdcp22_bsod_val

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Table 8-71 HDMIRX_TOP_SKP_CNTL_STAT

Bit	R/W	Default	Description
31	R	0	Status of nonce_vld signal.
30:3	R	0	Reserved
2	RW	0	DUK_vld:Set to 1 once DUK is written.
1	RW	0	PKF_vld:Set to 1 once PKF is written.
0	RW	0	nonce_hw_en: 1=Use HW nonce; 0=Use SW nonce from reg HDMIRX_TOP_NONCE_0/1/2/3.

Table 8-72 HDMIRX_TOP_NONCE_0

Bit	R/W	Default	Description
31:0	W	0	nonce[31:0]

Table 8-73 HDMIRX_TOP_NONCE_1

Bit	R/W	Default	Description
31:0	W	0	nonce[63:32]

Table 8-74 HDMIRX_TOP_NONCE_2

Bit	R/W	Default	Description
31:0	W	0	nonce[95:64]

Table 8-75 HDMIRX_TOP_NONCE_3

Bit	R/W	Default	Description
31:0	W	0	nonce[127:96]

Table 8-76 HDMIRX_TOP_PKF_0

Bit	R/W	Default	Description
31:0	W	0	PKF[31:0]

Table 8-77 HDMIRX_TOP_PKF_1

Bit	R/W	Default	Description
31:0	W	0	PKF[63:32]

Table 8-78 HDMIRX_TOP_PKF_2

Bit	R/W	Default	Description
31:0	W	0	PKF[95:64]

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Table 8-79 HDMIRX_TOP_PKF_3

Bit	R/W	Default	Description
31:0	W	0	PKF[127:96]

Table 8-80 HDMIRX_TOP_DUK_0

Bit	R/W	Default	Description
31:0	W	0	DUK[31:0]

Table 8-81 HDMIRX_TOP_ DUK_1

Bit	R/W	Default	Description
31:0	W	0	DUK[63:32]

Table 8-82 HDMIRX_TOP_ DUK_2

Bit	R/W	Default	Description
31:0	W	0	DUK[95:64]

Table 8-83 HDMIRX_TOP_ DUK_3

Bit	R/W	Default	Description
31:0	W	0	DUK[127:96]

Table 8-84 HDMIRX_TOP_NSEC_SCRATCH

Bit	R/W	Default	Description
31:0	RW	0	Scratch register that can be used for either secure or non-secure reg access.

Table 8-85 HDMIRX_TOP_SEC_SCRATCH

Bit	R/W	Default	Description
31:0	RW	0	Scratch register that can be used for secure reg access only.

Table 8-86 HDMIRX_TOP_EMP_DDR_START_A

Bit	R/W	Default	Description
31:0	RW	0	DDR pointer A for storing EMP packets or TMDS. 16-byte aligned, so low 4-bit is ignored. For EMP-to-DDR, the pointer is alternated between A and B, frame by frame. For TMDS-to-DDR, the pointer is fixed at A.

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Table 8-87 HDMIRX_TOP_EMP_DDR_START_B

Bit	R/W	Default	Description
31:0	RW	0	DDR pointer B for storing EMP packets. 16-byte aligned, so low 4-bit is ignored. For EMP-to-DDR, the pointer is alternated between A and B, frame by frame. Not applicable for TMDS-to-DDR.

The following register is applicable only to EMP-to-DDR mode.

Table 8-88 HDMIRX_TOP_EMP_DDR_FILTER

Bit	R/W	Default	Description
24	RW	0	ddr_store_other. 1=Enable store other packets that is neither NULL nor any that has already been described in bit[22:0].
23	RW	0	Reserved.
22	RW	0	ddr_store_drm. 1=Enable store Dynamic Range and Mastering InfoFrame.
21	RW	0	ddr_store_ntscvbi. 1=Enable store NTSC VBI InfoFrame.
20	RW	0	ddr_store_mpegs. 1=Enable store MPEG Source InfoFrame.
19	RW	0	ddr_store_aif. 1=Enable store Audio InfoFrame.
18	RW	0	ddr_store_spd. 1=Enable store Source Product Description InfoFrame.
17	RW	0	ddr_store_avi. 1=Enable store Auxiliary Video InfoFrame.
16	RW	0	ddr_store_vsi. 1=Enable store Vendor Specific InfoFrame.
15	RW	0	ddr_store_emp. 1=Enable store Extended Metadata packet.
14	RW	0	ddr_store_obm. 1=Enable store One bit Multi-stream audio sample packet.
13	RW	0	ddr_store_mas. 1=Enable store Multi-Stream Audio sample packet.
12	RW	0	ddr_store_amp. 1=Enable store Audio Metadata packet.
11	RW	0	ddr_store_ob3d. 1=Enable store One Bit 3D audio sample packet.
10	RW	0	ddr_store_auds3d. 1=Enable store 3D Audio Sample packet.
9	RW	0	ddr_store_gmd. 1=Enable store Gamut MetaData packet.
8	RW	0	ddr_store_hbr. 1=Enable store High-BitRate audio stream packet.
7	RW	0	ddr_store_dst. 1=Enable store DST audio packet.
6	RW	0	ddr_store_oba. 1=Enable store One Bit Audio sample packet.
5	RW	0	ddr_store_isrc2. 1=Enable store ISRC2 packet.
4	RW	0	ddr_store_isrc1. 1=Enable store ISRC1 packet.
3	RW	0	ddr_store_acp. 1=Enable store Audio Content Protection packet.
2	RW	0	ddr_store_gcp. 1=Enable store General Control packet.
1	RW	0	ddr_store_auds. 1=Enable store Audio Sample packet.
0	RW	0	ddr_store_acr. 1=Enable store Audo Clock Regeneration packet.

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Table 8-89 HDMIRX_TOP_EMP_CNTL_0

Bit	R/W	Default	Description
23:16	RW	0	hs_beat_rate. 0=Break-up DDR write burst on every Hsync rise; N=Break-up DDR write burst on the (N+1)th Hsync rise. Appliable only for EMP-to-DDR mode.
14	RW	0	buffer_info_mode. 0=Buffer received-packet-count and DDR-pointer-in-use on EMP_FIELD_ DONE; 1=Buffer on the last EMP packet. Appliable only for EMP-to-DDR mode.
13	RW	0	reset_on_de. 1=Reset on DE rise. Appliable only for EMP-to-DDR mode.
12	RW	0	brst_end_on_last_emp. 1=Break-up DDR write burst at the last EMP packet. 0=Do not break-up. The burst will end at 16x128-bit, or at programmable Hsync interval or at DE rise. Note: Set this bit to 0 for TMDS-to-DDR mode.
11:2	RW	0	de_rise_delay. 0=Generate DE rise on actual DE rise; N=Generate DE rise (N+1) tmds_clk cycles after actual DE rise. Applicable only for EMP-to-DDR mode.
1:0	RW	0	Endian for DDR storage. 0=Little-endian. 3=Big-endian per 8-byte.

Table 8-90 HDMIRX_TOP_EMP_CNTL_1

Bit	R/W	Default	Description
1	RW	0	ddr_mode. 0=Store packet, EMP-to-DDR; 1=Store tmds character from PHY, TMDS-to-DDR.
0	RW	0	ddr_en. 1=Enable storing either packet or tmds to DDR.

Table 8-91 HDMIRX_TOP_EMP_CNTMAX

Bit	R/W	Default	Description
31:0	RW	0	For EMP-to-DDR, maximum number of packets to store per frame; For TMDS-to-DDR, maximum number of 3x10-b tmds characters to store.

Table 8-92 HDMIRX_TOP_EMP_ERR_STAT

Bit	R/W	Default	Description
9	R	0	packet_num_exceeds_max
8	R	0	axi_overdue
7	R	0	bfifo_underflow
6	R	0	bfifo_overflow
5	R	0	axi_bresp error

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Bit	R/W	Default	Description
4	R	0	Extra axi_bresp
3	R	0	wfifo_underflow
2	R	0	wfifo_overflow
1	R	0	awfifo_underflow
0	R	0	awfifo_overflow

Table 8-93 HDMIRX_TOP_EMP_RCV_CNT_CUR

Bit	R/W	Default	Description
31:0	R	0	packet_rcv_cnt_cur. Current received packet count. Appliable only for EMP-to-DDR mode..

Table 8-94 HDMIRX_TOP_EMP_RCV_CNT_BUF

Bit	R/W	Default	Description
31:0	R	0	packet_rcv_cnt_buf. Received packet count, buffered on either LAST_EMP_ DONE or EMP_FIELD_DONE. Appliable only for EMP-to-DDR mode.

Table 8-95 HDMIRX_TOP_EMP_DDR_ADDR_CUR

Bit	R/W	Default	Description
31:0	R	0	Current DDR pointer. Low 4-bit always 0.

Table 8-96 HDMIRX_TOP_EMP_DDR_PTR_S_BUF

Bit	R/W	Default	Description
31:0	R	0	DDR start pointer buffered on either LAST_EMP_DONE or EMP_FIELD_DONE. Appliable only for EMP-to-DDR mode.

Table 8-97 HDMIRX_TOP_EMP_STAT_0

Bit	R/W	Default	Description
31	R	0	odd_field
26:18	R	0	wfifo_count
17:16	R	0	axi_w_state. 0=IDLE; 1=WAIT_READY_L; 2=WAIT_READY_H.
10:2	R	0	awfifo_cnt
1:0	R	0	axi_aw_state. 0=IDLE; 1=WAIT_READY; 2=WAIT_W.

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Table 8-98 HDMIRX_TOP_EMP_STAT_1

Bit	R/W	Default	Description
15:0	R	0	aw_pending_cnt

Table 8-99 HDMIRX_TOP_AXI_CNTL_0

Bit	R/W	Default	Description
31:24	RW	0x80	axi_async_waiting_limit_emp
18	RW	0	axi_async_disable_clk_emp
17	RW	1	axi_async_auto_gclk_en_emp
16	RW	1	axi_async_req_en_emp
15:8	RW	0x80	axi_async_waiting_limit_esm
6:5	RW	0	axi_rd_urgent
4:3	RW	0	axi_wr_urgent
2	RW	0	axi_async_disable_clk_esm
1	RW	1	axi_async_auto_gclk_en_esm
0	RW	1	axi_async_req_en_esm

Table 8-100 HDMIRX_TOP_AXI_ASYNC_HOLD_ESM

Bit	R/W	Default	Description
31:24	RW	0x18	axi_wr_hold_num_esm
23:16	RW	0x10	axi_wr_rel_num_esm
15:8	RW	0x18	axi_rd_hold_num_esm
7:0	RW	0x10	axi_rd_rel_num_esm

Table 8-101 HDMIRX_TOP_AXI_ASYNC_HOLD_EMP

Bit	R/W	Default	Description
31:24	RW	0x18	axi_wr_hold_num_emp
23:16	RW	0x10	axi_wr_rel_num_emp
15:8	RW	0x18	axi_rd_hold_num_emp
7:0	RW	0x10	axi_rd_rel_num_emp

Table 8-102 HDMIRX_TOP_AXI_STAT_0

Bit	R/W	Default	Description
1	R	1	axi_async_chan_idle_emp
0	R	1	axi_async_chan_idle_esm

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Table 8-103 HDMIRX_TOP_MISC_STAT0

Bit	R/W	Default	Description
3:1	R	0	aud_fmt. 0=Audio Sample; 1=One-bit audio; 2=DST audio; 3=HBR audio; 4=MAS audio; 5=OBM audio.
0	R	0	PHY.sqofclk status

HDMIRX_TOP_EDID_ADDR_S-HDMIRX_TOP_EDID_ADDR_E (0x1000-0x11ff)

0x1000 for EDID byte 0, 0x1001 for EDID byte1, and so on. Store up to 512-byte EDID data.

Table 8-104 HDMIRX_TOP_EDID_ADDR

Bit	R/W	Default	Description
7:0	RW	0	EDID byte.

8.1.2.3 Controller Register Updates

The following are the additional updates added to the original Controller. Please read the following in combination with the original databook of the Controller.

Table 8-105 HDCP_CTRL 0xc0

Bits	Name	Memory Access	Description
27	nonhdc-p_del_en	R/W	0: Original behaviour 1: Balance path delay between non-HDCP and Value After Reset: 0x0
26	hdcp14_del_en	R/W	0: Original behaviour 1: Balance path delay between HDCP14 and HDCP22. Value After Reset: 0x0

Table 8-106 PDEC_CTRL 0x300

Bits	Name	Memory Access	Description
30	pfifo_store_emp_if	R/W	Enable packet FIFO to store Extended Metadata Packet. When pfifo_store_filter_en is enabled this bit controls packet insertion: 0: Packet not stored in the packet FIFO 1: Packet is stored in the packet FIFO Value After Reset: 0x0 Note: If use DDR to store EMP, then do not enable this bit to save packet FIFO space.
1	emp_rcv_intr_each	R/W	Select whether to trigger interrupt on every EMP packet received or only the last EMP received. 0: Only last EMP received triggers emp_rcv interrupt 1: Every EMP received triggers emp_rcv interrupt Value After Reset: 0x0

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Table 8-107 PDEC_ERR_FILTER 0x33c

Bits	Name	Memory Access	Description
12	bchsp_ emp_ err_filter	R/W	BCH error filter for sub-packet of Extended Metadata Packet enable Value After Reset: 0x0

Table 8-108 PDEC_STS 0x360

Bits	Name	Memory Access	Description
9	emp_rcv	R	New Extended Metadata Packet received 0: No change 1: packet received Value After Reset: 0x0

Table 8-109 HDMI20_CONTROL 0x800

Bits	Name	Memory Access	Description
3	ignore_ sscp_ charerr	R/W	To support some TX that sends out SSCP even when not scrambling: 0: Original behaviour 1: During TMDS character error detection, treat SSCP character as normal TMDS character. Note: If scramble is turned on, this bit will not take effect, revert to original IP behaviour. Value After Reset: 0x0
2	ignore_ sscp_ tmds	R/W	To support some TX that sends out SSCP even when not scrambling: 0: Original behaviour 1: During TMDS decoding, treat SSCP character as normal TMDS character Note: If scramble is turned on, this bit will not take effect, revert to original IP behaviour. Value After Reset: 0x0

Table 8-110 HDCP22_STATUS 0x8fc

Bits	Name	Memory Access	Description
20	hdcp22_ authenti- cation_ fail	R	Value After Reset: 0
19	hdcp22_ authenti- cated	R	Value After Reset: 0
18	hdcp22_ authenti- cation_ lost	R	Value After Reset: 0
17	hdcp22_ not_ capable	R	Value After Reset: 0
16	hdcp22_ capable	R	Value After Reset: 0

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Table 8-111 PDEC_IEN_CLR 0xf78

Bits	Name	Memory Access	Description
9	emp_ rcv_ ENCLR	W	Extended Metadata Packet received interrupt – Enable clear 0: No action 1: Disable the interrupt. The enable status bit changes to 0. Value After Reset: 0x0

Table 8-112 PDEC_IEN_SET 0xf7c

Bits	Name	Memory Access	Description
9	emp_ rcv_ ENSET	W	Extended Metadata Packet received interrupt – Enable set 0: No action 1: Enable the interrupt. The enable status bit changes to 1. Value After Reset: 0x0

Table 8-113 PDEC_ISTS 0xf80

Bits	Name	Memory Access	Description
9	emp_ rcv_ISTS	R	Extended Metadata Packet received interrupt flag status 0: inactive 1: active Value After Reset: 0x0

Table 8-114 PDEC_IEN 0xf84

Bits	Name	Memory Access	Description
9	emp_ rcv_IEN	R	Extended Metadata Packet received interrupt enable status 0: disabled 1: enabled Value After Reset: 0x0

Table 8-115 PDEC_ICLR 0xf88

Bits	Name	Memory Access	Description
9	emp_ rcv_ ICLR	W	Extended Metadata Packet received interrupt flag clear 0: No action 1: action Value After Reset: 0x0

Table 8-116 PDEC_ISET 0xf8c

Bits	Name	Memory Access	Description
9	emp_ rcv_ISET	W	Extended Metadata Packet received interrupt set 0: No action 1: action Value After Reset: 0x0

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8.1.2.4 Operating Procedures

8.1.2.4.1 EMP-To-DDR (Store Extended Metadata Packet to DDR)

EMPs are transmitted during the video blanking periods. DE rise edge signals the end of video blanking period, where the packets should all have been received, and the content of the packets should apply to the first active video pixel following the video blank.

Enable emp_field_done interrupt by setting HDMIRX_TOP_INTR_MASKN bit[25]=1.

Program the following DDR start address pointers A and B, the DDR start pointer will alternate between A and B frame-by-frame. The address must be at least 16-byte aligned, meaning the low 4-bit will be ignored.

- HDMIRX_TOP_EMP_DDR_START_A
- HDMIRX_TOP_EMP_DDR_START_B

Program HDMIRX_TOP_EMP_DDR_FILTER, to select the packet type(s) to be stored in DDR.

Program HDMIRX_TOP_EMP_CNTMAX to limit the number packets to be stored in DDR. E.g., if this register is set to 5, only the first 5 type-selected packets are allow to go to DDR.

Program HDMIRX_TOP_EMP_CNTL_0 appropriately, for detail refer to register description section 2.88. Recommended values for the following bit fields:

- [23:16] hs_beat_rate=0xf
- [14] buffer_info_mode=0
- [13] reset_on_de=1
- [12] burst_end_on_last_emp=1
- [11:2] de_rise_delay=0
- [1:0] Endian = 0

Write HDMIRX_TOP_EMP_CNTL_1 bit[1] ddr_mode=0 first, to select the function mode to be EMP-to-DDR. Then write HDMIRX_TOP_EMP_CNTL_1 bit[0] ddr_en=1, to enable the function.

On emp_field_done interrupt, read the following registers to determine where and how many packets in DDR: HDMIRX_TOP_EMP_DDR_PTR_S_BUF – The DDR start pointer to read the stored packets HDMIRX_TOP_EMP_RCV_CNT_BUF – The number of packets stored in DDR

The format of the packet data in DDR memory (little-endian) is as below:

```
---------------------------------- 1st received Packet
Addr_S+0:Packet Header HB0 (E.g: EMP=0x7F, DRM=0x87)
Addr_S+1:Packet Header HB1 (E.g: EMP=0x80/0x00/0x40, DRM=InfoFrame_version)
Addr_S+2:Packet Header HB2 (E.g: EMP=Sequence_index, DRM=InfoFrame_length)
Addr_S+3:Packet Header PB0 (E.g: EMP=Payload byte0, DRM=Checksum byte)
Addr_S+4:Packet Header PB1 (E.g: EMP=Payload byte1, DRM=Byte1)
Addr_S+5:Packet Header PB2 (E.g: EMP=Payload byte2, DRM=Byte2)
……
Addr_S+30:Packet Header PB27 (E.g: EMP=Payload byte27, DRM=Byte27)
Addr_S+31:0x00
---------------------------------- 2nd received Packet
Addr_S+32:Packet Header HB0
……
Addr_S+63:0x00
---------------------------------- 3rd received Packet
……
……
---------------------------------- N-th/Last received Packet
Addr_S+32*(N-1):Packet Header HB0
……
Addr_S+32*N-1:0x00
----------------------------------
```


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Note

* Addr_S is the start address of a DDR location to store the packet data, must be 16-byte aligned. If use LPDDR4 which is most efficient at 64-byte aligned access, then it is better that the Addr_S is 64-byte aligned, for memory bandwidth efficiency.

* N is the total number of packets that were received in a Frame. Need 32*N byte memory space.

* All types of packets are supported.

8.1.2.4.2 TMDS-To-DDR (Store TMDS characters to DDR)

This function stores a pre-defined number of 3-channel TMDS characters from PHY, into DDR. It is for debugging only, and can not co-exist with EMP-to-DDR function. If EMP-to-DDR is enabled, then you cannot do TMDS-to-DDR, and vice-versa.

If previously EMP-to-DDR is enabled, disable it by write HDMIRX_TOP_EMP_CNTL_1 bit[0] ddr_en= 0. Wait until HDMIRX_TOP_EMP_STAT_0 bit[30:0] reads back 0, and HDMIRX_TOP_EMP_STAT_1 reads back 0.

Check no error in HDMIRX_TOP_EMP_ERR_STAT.

Enable last_emp_done interrupt by setting HDMIRX_TOP_INTR_MASKN bit[26]=1.

Program HDMIRX_TOP_EMP_DDR_START_A as the start address to store TMDS.

Program HDMIRX_TOP_EMP_CNTMAX to limit the number 3x10-bit TMDS to be stored in DDR. E. g., if this register is set to 8-1=7, then 8 of 3x10-bit TMDS characters will be stored to DDR.

Note

(HDMIRX_TOP_EMP_CNTMAX + 1) must be integer multipler of 8.

Program HDMIRX_TOP_EMP_CNTL_0 appropriately, for detail refer to register description section 2.88. Recommended values for the following bit fields:

- [23:16] hs_beat_rate. N/A for TMDS-to-DDR.
- [14] buffer_info_mode. N/A for TMDS-to-DDR.
- [13] reset_on_de. N/A for TMDS-to-DDR.
- [12] burst_end_on_last_emp=0. Note: Must be 0 for TMDS-to-DDR.
- [11:2] de_rise_delay. N/A for TMDS-to-DDR.
- [1:0] Endian = 0

Write HDMIRX_TOP_EMP_CNTL_1 bit[1] ddr_mode=1 first, to select the function mode to be TMDS-to-DDR. Then write HDMIRX_TOP_EMP_CNTL_1 bit[0] ddr_en=1, to enable recording TMDS.

On last_emp_done interrupt, write HDMIRX_TOP_EMP_CNTL_1 bit[0] ddr_en=0. Dump DDR from address HDMIRX_TOP_EMP_DDR_START_A, of (HDMIRX_TOP_EMP_CNTMAX + 1)*4 bytes.

The format of TMDS data in DDR memory (little-endian) is as below:

```
-------------------------------- 1st 3x10-bit TMDS
{Addr_S+3,+2,+1,+0}:{2'h0, tmds_ch2[9:0], tmds_ch1[9:0], tmds_ch0[9:0]}
-------------------------------- 2nd 3x10-bit TMDS
{Addr_S+7,+6,+5,+4}:{2'h0, tmds_ch2[9:0], tmds_ch1[9:0], tmds_ch0[9:0]}
-------------------------------- 3rd 3x10-bit TMDS
……
……
-------------------------------- N-th/Last 3x10-bit TMDS
Addr_S+32*(N-1):Packet Header HB0
……
Addr_S+32*N-1:0x00
{Addr_S+(N-1)*4+3 ,+(N-1)*4+2, ,+(N-1)*4+1,+(N-1)*4}:
```


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{2'h0, tmds_ch2[9:0], tmds_ch1[9:0], tmds_ch0[9:0]}

Note

* *Addr_S* is the start address of a DDR location to store the packet data, must be 16-byte aligned. If use LPDDR4 which is most efficient at 64-byte aligned access, then it is better that the Addr_S is 64-byte aligned, for memory bandwidth efficiency.

* *N* is the total number of 3x10-bit TMDS that were stored in DDR. Need 4**N* byte memory space.

8.1.2.4.3 Audo Clock Measure

Periodic measure (meas_mode=0)

This function measures the frequency of either Audio PLL clk or internal digital ACR clk.

Select either Audio PLL clk or digital ACR clk to measure: HDMIRX_TOP_ACR_CNTL_STAT bit[1] – 0=Audio PLL clk; 1=digital ACR clk.

Program ref_clk to 24MHz xtal clock for measuring audio clk. HHI_HDMIRX_METER_CLK_CNTL

- [6:0] clk_div = 0
- [8] clk_en = 1
- [10:9] clk_sel = 0, to select xtal clock 24MHz.

Initialization: HDMIRX_TOP_AUDMEAS_CTRL

- [0] sw_reset = 1
- [1] enable = 1
- [2] meas_mode = 0
- [31:16] meas_tolerance = appropriate value, e.g. 2 for Aud PLL, 500 for ACR.

HDMIRX_TOP_AUDMEAS_CYCLES_M1 = meas_cycles_m1 [31:0]. Set to appropriate value, e. g. 65535.

Start measure: HDMIRX_TOP_AUDMEAS_CTRL [0] sw_reset = 0

One measure period is T(audio clk) * (meas_cycles_m1+1). From initialization, wait about two periods, because the very first period maybe an incomplete measure.

Read back the result periodically.

- HDMIRX_TOP_AUDMEAS_REF_CYCLES_STAT0 – ref_cycles_m1[31:0]
- HDMIRX_TOP_AUDMEAS_REF_CYCLES_STAT1
- [15:0] ref_cycles_m1[47:32]
- [16] ref_cycles_overflow. N/A for periodic measure.
- [17] meas_stable.
- [31:24] timestamp_num. N/A for periodic measure.

The audio clock frequency is:

Freq(audio_clk) = Freq(ref_clk) * (meas_cycles_m1+1) / (ref_cycles_m1+1)

If meas_stable=0, it means that, between the two measure periods, the difference of ref_cycles_m1 exceeds meas_tolerance.

ABS(last ref_cycles_m1 - current ref_cycles_m1) > meas_tolerance.

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Timestamp (meas_mode=1)

This function generates audio clock timestamps.

Select either Audio PLL clk or digital ACR clk to measure:

HDMIRX_TOP_ACR_CNTL_STAT bit[1] – 0=Audio PLL clk; 1=digital ACR clk.

Program ref_clk to 24MHz xtal clock for measuring audio clk.

HHI_HDMIRX_METER_CLK_CNTL

- [6:0] clk_div = 0
- [8] clk_en = 1
- [10:9] clk_sel = 0, to select xtal clock 24MHz.

Initialization: HDMIRX_TOP_AUDMEAS_CTRL

- [0] sw_reset = 1
- [1] enable = 1
- [2] meas_mode = 1
- [31:16] meas_tolerance. N/A for timestamp mode.

HDMIRX_TOP_AUDMEAS_CYCLES_M1 = meas_cycles_m1 [31:0]. Set to appropriate value, e. g. 65535.

Start measure: HDMIRX_TOP_AUDMEAS_CTRL

[0] sw_reset = 0

One measure period is T(audio clk) * (meas_cycles_m1+1). From initialization, wait about two periods, because the very first period maybe an incomplete measure.

Read back the result periodically.

- HDMIRX_TOP_AUDMEAS_REF_CYCLES_STAT0 – ref_cycles_m1[31:0]
- HDMIRX_TOP_AUDMEAS_REF_CYCLES_STAT1
- [15:0] ref_cycles_m1[47:32]
- [16] ref_cycles_overflow
- [17] meas_stable. N/A for timestamp mode.
- [31:24] timestamp_num

timestamp_num and ref_cycles_m1 reflects the time elapsed since the beginning of measurement. timestamp_num shows the number of measure periods elapsed since measure begins; (ref_cycles_ m1+1) shows the total number of ref_clk cycles since measure begins.

To assist timestamp checking, AUDMEAS interrupt can be enabled to generate one interrupt per measure period. Refer to reg HDMIRX_TOP_AUDMEAS_INTR_MASKN/STAT.

8.1.3 Demux Register Description

Table 8-117 STB_RECORDER2_CNTL 0xee

Bit(s)	R/W	Default	Description
31:2	RW	0	reserved
1:0	RW	1	stb_recorder2_sel

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Table 8-118 STB_S2P2_CONFIG 0xef

Bit(s)	R/W	Default	Description
31:12	RW	0	reserved
11	RW	0	s2p2_disable
10:7	RW	7	s2p2_clk_div
6	RW	0	invert fec_error for S2P2
5	RW	0	invert fec_data for S2P2
4	RW	0	invert fec_sync for S2P2
3	RW	0	invert fec_valid for S2P2
2	RW	0	invert fec_clk for S2P2
1:0	RW	0	fec_s_sel for S2P2 00 - select TS0, 01 – select TS1, 10 – select TS2, 11 - TS3

Table 8-119 STB_TOP_CONFIG 0xf0

Bit(s)	R/W	Default	Description
30:28	RW	0	ciplus_o_sel
27:26	RW	0	ciplus_i_sel
25	RW	0	use FAIL from TS2
24	RW	0	use FAIL from TS1
23	RW	0	use FAIL from TS0
22	RW	0	invert fec_error for S2P1
21	RW	0	invert fec_data for S2P1
20	RW	0	invert fec_sync for S2P1
19	RW	0	invert fec_valid for S2P1
18	RW	0	invert fec_clk for S2P1
17:16	RW	0	fec_s_sel for S2P1 00 - select TS0, 01 – select TS1, 10 – select TS2, 11 - TS3
15	RW	0	enable_des_pl_clk
14:13	RW	0	reserved
13	RW	0	use FAIL for TS3
12:10	RW	0	ts_out_select, 0-TS0, 1-TS1, 2-TS2, 3-TS3, 4-S2P3, 5-S2P1, 6-S2P0, 7-File
9:8	RW	0	des_i_sel 00 – select_fec_0, 01 – select_fec_1, 10 – select_fec_2, 11 - reserved
7	RW	0	enable_des_pl
6	RW	0	invert fec_error for S2P0
5	RW	0	invert fec_data for S2P0
4	RW	0	invert fec_sync for S2P0
3	RW	0	invert fec_valid for S2P0

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Bit(s)	R/W	Default	Description
2	RW	0	invert fec_clk for S2P0
1:0	RW	0	fec_s_sel for S2P0 00 - select TS0, 01 – select TS1, 10 – select TS2, 11 - TS3

Table 8-120 TS_TOP_CONFIG 0xf1

Bit(s)	R/W	Default	Description
31:28	RW	7	s2p1_clk_div
27:24	RW	7	s2p0_clk_div
23	RW	0	s2p1_disable
22	RW	0	s2p0_disable
21	RW	0	Reserved
20	RW	0	TS_OUT_error_INVERT
19	RW	0	TS_OUT_data_INVERT
18	RW	0	TS_OUT_sync_INVERT
17	RW	0	TS_OUT_valid_INVERT
16	RW	0	TS_OUT_clk_INVERT
15:8	RW	187	TS_package_length_sub_1 (default : 187)
7:0	RW	0x47	fec_sync_byte (default : 0x47)

Table 8-121 TS_FILE_CONFIG 0xf2

Bit(s)	R/W	Default	Description
25:24	RW	3	transport_scrambling_control_odd_2 // should be 3
23:16	RW	0	file_m2ts_skip_bytes
15:8	RW	0	des_out_dly
7:6	RW	3	transport_scrambling_control_odd // should be 3
5	RW	0	ts_hiu_enable
4:0	RW	4	fec_clk_div

Table 8-122 TS_PL_PID_INDEX 0xf3

Bit(s)	R/W	Default	Description
19:14	R	0	des_2 ts pl state – Read Only
13:8	R	0	des ts pl state – Read Only
3:0	RW	0	PID index to 8 PID to get key-set auto increse after TS_PL_PID_DATA read/ write

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Table 8-123 TS_PL_PID_DATA 0xf4

Bit(s)	R/W	Default	Description
29	RW	0	PID #INDEX +1 match disble
28:16	RW	0	PID #INDEX+1
13	RW	0	PID #INDEX match disble
12:0	RW	0	PID #INDEX

Table 8-124 COMM_DESC_KEY0 0xf5

Bit(s)	R/W	Default	Description
31:0	RW	0	Common descrambler key (key Bits[63:32])

Table 8-125 COMM_DESC_KEY1 0xf6

Bit(s)	R/W	Default	Description
31:0	RW	0	Common descrambler key (key Bits[31:0])

Table 8-126 COMM_DESC_KEY_RW 0xf7

Bit(s)	R/W	Default	Description
7	RW	0	Key endian;
6	RW	0	Write key ladder cw [127:64] to key;
5	RW	0	Write key ladder cw [63:0] to key;
4	RW	0	0: write to descramble 1; 1: write to descramble 2;
3:0	RW	0	The address of key

Table 8-127 CIPLUS_KEY0 0xf8

Bit(s)	R/W	Default	Description
31:0	RW	0	CI+ Register defines Bits[31:0] of the key

Table 8-128 CIPLUS_KEY1 0xf9

Bit(s)	R/W	Default	Description
31:0	RW	0	CI+ Register defines Bits[63:32] of the key

Table 8-129 CIPLUS_KEY2 0xfa

Bit(s)	R/W	Default	Description
31:0	RW	0	CI+ Register defines Bits[95:64] of the key

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Table 8-130 CIPLUS_KEY3 0xfb

Bit(s)	R/W	Default	Description
31:0	RW	0	CI+ Register defines Bits[127:96] of the key

Table 8-131 CIPLUS_KEY_WR 0xfc

Bit(s)	R/W	Default	Description
5	RW	0	write AES IV B value
4	RW	0	write AES IV A value
3	RW	0	write AES B key
2	RW	0	write AES A key
1	RW	0	write DES B key
0	RW	0	write DES A key

Table 8-132 CIPLUS_CONFIG 0xfd

Bit(s)	R/W	Default	Description
15:8	RW	0	TS out delay. This controls the rate at which the CIplus module drives TS out
3	RW	0	General enable for the ciplus module
2	RW	0	AES CBC disable (default should be 0 to enable AES CBC)
1	RW	0	AES Enable
0	RW	0	DES Eanble

Table 8-133 CIPLUS_ENDIAN 0xfe

Bit(s)	R/W	Default	Description
31:28	RW	0	AES IV endian
27:24	RW	0	AES message out endian
23:20	RW	0	AES message in endian
19:16	RW	0	AES key endian
15:11	RW	0	unused
10:8	RW	0	DES message out endian
6:4	RW	0	DES message in endian
2:0	RW	0	DES key endian

Table 8-134 COMM_DESC_2_CTL 0xff

Bit(s)	R/W	Default	Description
15:8	RW	0	des_out_dly_2
7	RW	0	reserved

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Bit(s)	R/W	Default	Description
6	RW	0	enable_des_pl_clk_2
5	RW	0	enable_des_pl_2
4:2	RW	0	use_des_2 Bit[2] – demux0, Bit[3] – demux1, Bit[4] – demux2
1:0	RW	0	des_i_sel_2 00 – select_fec_0, 01 – select_fec_1, 10 – select_fec_2, 11 - reserved

8.1.4 Demux Core Registers

demux core 0 Final address = 0xc1105800 + offset * 4

demux core 1 Final address = 0xc1105940 + offset * 4

demux core 2 Final address = 0xc1105a80 + offset * 4

Table 8-135 STB_VERSION_O 0x00

Bit(s)	R/W	Default	Description
31:0	R	0x30003	The version of stb

Table 8-136 STB_TEST_REG_O 0x01

Bit(s)	R/W	Default	Description
31:0	RW	0xfe015-aa5	Test register.

Table 8-137 FEC_INPUT_CONTROL_O 0x02

Bit(s)	R/W	Default	Description
15	RW	0	fec_core_select 1 - select descramble output
14:12	RW	0	fec_select 0-TS0, 1-TS1, 2-TS2, 3,4-Reserved, 5-S2P1, 6-S2P0, 7-File
11	RW	0	FEC_CLK
10	RW	0	SOP
9	RW	0	D_VALID
8	RW	0	D_FAIL
7:0	RW	0	D_DATA 7:0

Table 8-138 FEC_INPUT_DATA_O 0x03

Bit(s)	R/W	Default	Description
11	R	0	FEC_CLK
20	R	0	SOP
9	R	0	VALID

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Bit(s)	R/W	Default	Description
8	R	0	FAIL
7:0	R	0	FEC DATAIN

Table 8-139 DEMUX_CONTROL_O 0x04

Bit(s)	R/W	Default	Description
31	RW	0	enable_free_clk_fec_data_valid
30	RW	0	enable_free_clk_stb_reg
29	RW	0	always_use_pes_package_length
28	RW	0	disable_pre_incomplete_section_fix
27	RW	0	pointer_field_multi_pre_en
26	RW	0	ignore_pre_incomplete_section
25	RW	0	video2_enable
24:22	RW	0	video2_type
21	RW	0	do_not_trust_pes_package_length
20 (bit4)	RW	0	Bypass use recoder path
19 (bit3)	RW	0	clear_PID_continuity_counter_valid
18 (bit2)	RW	0	Disable Splicing
17 (bit1)	RW	0	Insert PES_STRONG_SYNC in Audio PES
16 (bit0)	RW	0	Insert PES_STRONG_SYNC in Video PES
15	RW	0	do not trust section length
14	RW	0	om cmd push even zero
13	RW	0	set_buff_ready_even_not_busy
12	RW	0	SUB, OTHER PES interrupt at beginning of PES
11	RW	0	discard_av_package – for ts_recorder use only
10	RW	0	ts_recorder_select 0:after PID filter 1:before PID filter
9	RW	0	ts_recorder_enable
8	RW	0	(table_id == 0xff) means section_end
7	RW	0	do not send uncomplete section
6	RW	0	do not discard duplicate package
5	RW	0	search SOP when trasport_error_indicator
4	RW	0	stb demux enable
3	RW	0	do not reset state machine on SOP
2	RW	0	search SOP when error happened (when ignore_fail_n_sop, will have this case)
1	RW	0	do not use SOP input (check FEC sync byte instead)
0	RW	0	ignore fec_error bit when non sop (check error on SOP only)

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Table 8-140 FEC_SYNC_BYTE_O 0x05

Bit(s)	R/W	Default	Description
15:8	RW	187	demux package length - 1 (default : 187)
7:0	RW	0	default is 0x47

Table 8-141 FM_WR_DATA_O 0x06

Bit(s)	R/W	Default	Description
31:16	RW	0	filter memory write data hi[31:16]
15:0	RW	0	filter memory write data low [15:0]

Table 8-142 FM_WR_ADDR_O 0x07

Bit(s)	R/W	Default	Description
31:24	RW	0	advanced setting hi
23:16	RW	0	advanced setting low
15	R	0	filter memory write data request
7:0	R	0	filter memory write addr

Table 8-143 MAX_FM_COMP_ADDR_O 0x08

Bit(s)	R/W	Default	Description
13:8	R	0	demux state – read only
7:4	RW	0	maxnum section filter compare address
3:0	RW	0	maxnum PID filter compare address

Table 8-144 TS_HEAD_0_O 0x09

Bit(s)	R/W	Default	Description
15	RW	0	transport_error_indicator
14	RW	0	payload_unit_start_indicator
13	RW	0	transport_priority
12:0	RW	0	PID

Table 8-145 TS_HEAD_1_O 0x0a

Bit(s)	R/W	Default	Description
7:6	R	0	transport_scrambling_control
5:4	R	0	adaptation_field_control
3:0	R	0	continuity_counter

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Table 8-146 OM_CMD_STATUS_O 0x0b

Bit(s)	R/W	Default	Description
15:12	R	0	om_cmd_count (read only)
11:9	R	0	overflow_count // bit 11:9 – om_cmd_wr_ptr (read only)
8:6	R	0	om_overwrite_count // bit 8:6 – om_cmd_rd_ptr (read only)
5:3	R	0	type_stb_om_w_rd (read only)
2	R	0	unit_start_stb_om_w_rd (read only)
1	R	0	om_cmd_overflow (read only)
0	R	0	om_cmd_pending (read)
0	R	0	om_cmd_read_finished (write)

Table 8-147 OM_CMD_DATA_O 0x0c

Bit(s)	R/W	Default	Description
15:9	R	0	count_stb_om_w_rd (read only)
8:0	R	0	start_stb_om_wa_rd (read only)

Table 8-148 OM_CMD_DATA2_O 0x0d

Bit(s)	R/W	Default	Description
11:0	R	0	offset for section data

Table 8-149 SEC_BUFF_01_START_O 0x0e

Bit(s)	R/W	Default	Description
31:16	RW	0	base address for section buffer group 0 (*0x400 to get real address)
15:0	RW	0	base address for section buffer group 1 (*0x400 to get real address)

Table 8-150 SEC_BUFF_23_START_O 0x0f

Bit(s)	R/W	Default	Description
31:16	RW	0	base address for section buffer group 2 (*0x400 to get real address)
15:0	RW	0	base address for section buffer group 3 (*0x400 to get real address)

Table 8-151 SEC_BUFF_SIZE_O 0x10

Bit(s)	R/W	Default	Description
3:0	RW	0	section buffer size for group 0 (bitused, for example, 10 means 1K)
7:4	RW	0	section buffer size for group 1

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Bit(s)	R/W	Default	Description
11:8	RW	0	section buffer size for group 2
15:12	RW	0	section buffer size for group 3

Table 8-152 SEC_BUFF_BUSY_O 0x11

Bit(s)	R/W	Default	Description
31:0	R	0	Section buffer busy status for buff 31:0 (Read Only)

Table 8-153 SEC_BUFF_READY_O 0x12

Bit(s)	R/W	Default	Description
31:0	RW	0	section buffer write status for buff 31:0 – Read

Table 8-154 SEC_BUFF_NUMBER_O 0x13

Bit(s)	R/W	Default	Description
4:0	RW	0	SEC_BUFFER_INDEX RW
12:8	RW	0	SEC_BUFFER_NUMBER for the INDEX buffer Read_Only
14	RW	0	output_section_buffer_valid
15	RW	0	section_reset_busy (Read Only)

Table 8-155 ASSIGN_PID_NUMBER_O 0x14

Bit(s)	R/W	Default	Description
9:5	RW	0	BYPASS PID number
4:0	RW	0	PCR PID number

Table 8-156 VIDEO_STREAM_ID_O 0x15

Bit(s)	R/W	Default	Description
31:16	RW	0	for video2
15:0	RW	0	stream_id filter Bit(s) enable

Table 8-157 AUDIO_STREAM_ID_O 0x16

Bit(s)	R/W	Default	Description
15:0	RW	0	For audio

Table 8-158 SUB_STREAM_ID_O 0x17

Bit(s)	R/W	Default	Description
15:0	RW	0	For sub

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Table 8-159 OTHER_STREAM_ID_O 0x18

Bit(s)	R/W	Default	Description
15:0	RW	0	For other

Table 8-160 PCR90K_CTL_O 0x19

Bit(s)	R/W	Default	Description
12	RW	0	PCR_EN
11:0	RW	0	PCR90K_DIV

Table 8-161 PCR_DEMUX_O 0x1a

Bit(s)	R/W	Default	Description
31:0	RW	0	PCR

Table 8-162 VIDEO_PTS_DEMUX_O 0x1b

Bit(s)	R/W	Default	Description
31:0	RW	0	VPTS

Table 8-163 VIDEO_DTS_DEMUX_O 0x1c

Bit(s)	R/W	Default	Description
31:0	RW	0	VDTS

Table 8-164 AUDIO_PTS_DEMUX_O 0x1d

Bit(s)	R/W	Default	Description
31:0	RW	0	APTS

Table 8-165 SUB_PTS_DEMUX_O 0x1e

Bit(s)	R/W	Default	Description
31:0	RW	0	SPTS

Table 8-166 STB_PTS_DTS_STATUS_O 0x1f

Bit(s)	R/W	Default	Description
15	R	0	SUB_PTS[32]
14	R	0	AUDIO_PTS[32]
13	R	0	VIDEO_DTS[32]
12	R	0	VIDEO_PTS[32]

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Bit(s)	R/W	Default	Description
3	R	0	sub_pts_ready
2	R	0	audio_pts_ready
1	R	0	video_dts_ready
0	R	0	video_pts_ready

Table 8-167 STB_DEBUG_INDEX_O 0x20

Bit(s)	R/W	Default	Description
3	RW	0	pes_ctr_byte[7:0], pes_flag_byte[7:0]
2	RW	0	pes_package_bytes_left[15:0]
1	RW	0	stream_id[7:0], pes_header_bytes_left[7:0]
0	RW	0	adaptation_field_length[7:0], adaption_field_byte_1[7:0]

Table 8-168 STB_DEBUG_DATAOUT_O 0x21

Bit(s)	R/W	Default	Description
15:0	R	0	Debug data out[15:0]

Table 8-169 STB_MOM_CTL_O 0x22

Bit(s)	R/W	Default	Description
31	RW	0	no_match_record_en
30:16	RW	0	reserved
15:9	RW	0	MAX OM DMA COUNT (default: 0x40)
8:0	RW	0	LAST ADDR OF OM ADDR (default: 127)

Table 8-170 STB_INT_STATUS_O 0x23

Bit(s)	R/W	Default	Description
12	R	0	INPUT_TIME_OUT
11	R	0	PCR_ready
10	R	0	audio_splicing_point
9	R	0	video_splicing_point
8	R	0	other_PES_int
7	R	0	sub_PES_int
6	R	0	discontinuity
5	R	0	duplicated_pack_found
4	R	0	New PDTS ready
3	R	0	om_cmd_buffer ready for access

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Bit(s)	R/W	Default	Description
2	R	0	section buffer ready
1	R	0	transport_error_indicator
0	R	0	TS ERROR PIN

Table 8-171 DEMUX_ENDIAN_O 0x24

Bit(s)	R/W	Default	Description
23:21	RW	0	demux om write endian control for OTHER_PES_PACKET
20:18	RW	0	demux om write endian control for SCR_ONLY_PACKET
17:15	RW	0	demux om write endian control for SUB_PACKET
14:12	RW	0	demux om write endian control for AUDIO_PACKET
11:9	RW	0	demux om write endian control for VIDEO_PACKET
8:6	RW	0	demux om write endian control for else
5:3	RW	0	demux om write endian control for bypass
2:0	RW	0	demux om write endian control for section

Table 8-172 TS_HIU_CTL_O 0x25

Bit(s)	R/W	Default	Description
15:8	RW	0	last_burst_threshold
7	RW	0	use hi_bsf interface
6:2	RW	0	fec_clk_div
1	RW	0	ts_source_sel
0	RW	0	Hiu TS generate enable

Table 8-173 SEC_BUFF_BASE_O 0x26

Bit(s)	R/W	Default	Description
15:0	RW	0	base address for section buffer start (*0x10000 to get real base)

Table 8-174 DEMUX_MEM_REQ_EN_O 0x27

Bit(s)	R/W	Default	Description
11	RW	0	mask bit for OTHER_PES_AHB_DMA_EN
10	RW	0	mask bit for SUB_AHB_DMA_EN
9	RW	0	mask bit for BYPASS_AHB_DMA_EN
8	RW	0	mask bit for SECTION_AHB_DMA_EN
7	RW	0	mask bit for recoder stream
6:0	RW	0	mask bit for each type

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Table 8-175 VIDEO_PDTS_WR_PTR_O 0x28

Bit(s)	R/W	Default	Description
31:0	RW	0	vb_wr_ptr for video PDTS

Table 8-176 AUDIO_PDTS_WR_PTR_O 0x29

Bit(s)	R/W	Default	Description
31:0	RW	0	ab_wr_ptr for video PDTS

Table 8-177 SUB_WR_PTR_O 0x2a

Bit(s)	R/W	Default	Description
20:0	RW	0	SB_WRITE_PTR (sb_wr_ptr << 3 == byte write position)

Table 8-178 SB_START_O 0x2b

Bit(s)	R/W	Default	Description
19:0	RW	0	SB_START (sb_start << 12 == byte address);

Table 8-179 SB_LAST_ADDR_O 0x2c

Bit(s)	R/W	Default	Description
20:0	RW	0	SB_SIZE (sb_size << 3 == byte size, 16M maximun)

Table 8-180 SB_PES_WR_PTR_O 0x2d

Bit(s)	R/W	Default	Description
31:0	RW	0	sb_wr_ptr for sub PES

Table 8-181 OTHER_WR_PTR_O 0x2e

Bit(s)	R/W	Default	Description
31:21	RW	0	ob_wr_ptr for other PES
20:0	RW	0	OB_WRITE_PTR (ob_wr_ptr << 3 == byte write position)

Table 8-182 OB_START_O 0x2f

Bit(s)	R/W	Default	Description
19:0	RW	0	OB_START (ob_start << 12 == byte address);

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Table 8-183 OB_LAST_ADDR_O 0x30

Bit(s)	R/W	Default	Description
20:0	RW	0	OB_SIZE (ob_size << 3 == byte size, 16M maximun)

Table 8-184 OB_PES_WR_PTR_O 0x31

Bit(s)	R/W	Default	Description
31:0	RW	0	ob_wr_ptr for sub PES

Table 8-185 STB_INT_MASK_O 0x32

Bit(s)	R/W	Default	Description
9	RW	0	splicing_point
8	RW	0	other_PES_int
7	RW	0	sub_PES_int
6	RW	0	discontinuity
5	RW	0	duplicated_pack_found
4	RW	0	New PDTS ready
3	RW	0	om_cmd_buffer ready for access
2	RW	0	section buffer ready
1	RW	0	transport_error_indicator
0	RW	0	TS ERROR PIN

Table 8-186 VIDEO_SPLICING_CTL_O 0x33

Bit(s)	R/W	Default	Description
15	RW	0	splicing VIDEO PID change enable
14:10	RW	0	VIDEO PID FILTER ADDRESS
9	RW	0	PES splicing active (Read Only)
8	RW	0	splicing active (Read Only)
7:0	RW	0	splicing countdown (Read Only)

Table 8-187 AUDIO_SPLICING_CTL_O 0x34

Bit(s)	R/W	Default	Description
15	RW	0	splicing AUDIO PID change enable
14:10	RW	0	AUDIO PID FILTER ADDRESS
9	RW	0	PES splicing active (Read Only)
8	RW	0	splicing active (Read Only)
7:0	RW	0	splicing countdown (Read Only)

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Table 8-188 TS_PACKAGE_BYTE_COUNT_O 0x35

Bit(s)	R/W	Default	Description
23:16	RW	0	M2TS_SKIP_BYTES
15:8	RW	0	LAST TS PACKAGE BYTE COUNT (Read Only)
7:0	RW	0	PACKAGE BYTE COUNT (Read Only)

Table 8-189 PES_STRONG_SYNC_O 0x36

Bit(s)	R/W	Default	Description
15:0	RW	0	2 bytes strong sync add to PES

Table 8-190 OM_DATA_RD_ADDR_O 0x37

Bit(s)	R/W	Default	Description
15	RW	0	stb_om_ren
14:11	RW	0	reserved
10:0	RW	0	OM_DATA_RD_ADDR

Table 8-191 OM_DATA_RD_O 0x38

Bit(s)	R/W	Default	Description
15:0	RW	0	OM_DATA_RD

Table 8-192 SECTION_AUTO_STOP_3_O 0x39

Bit(s)	R/W	Default	Description
31	RW	0	Auto stop count 31 Wr_en
30:28	RW	0	Auto stop count 31
27	RW	0	Auto stop count 30 Wr_en
26:24	RW	0	Auto stop count 30
23	RW	0	Auto stop count 29 Wr_en
22:20	RW	0	Auto stop count 29
19	RW	0	Auto stop count 28 Wr_en
18:16	RW	0	Auto stop count 28
15	RW	0	Auto stop count 27 Wr_en
14:12	RW	0	Auto stop count 27
11	RW	0	Auto stop count 26 Wr_en
10:8	RW	0	Auto stop count 26
7	RW	0	Auto stop count 25 Wr_en

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Bit(s)	R/W	Default	Description
6:4	RW	0	Auto stop count 25
3	RW	0	Auto stop count 24 Wr_en
2:0	RW	0	Auto stop count 24

Table 8-193 SECTION_AUTO_STOP_2_O 0x3a

Bit(s)	R/W	Default	Description
31	RW	0	Auto stop count 23 Wr_en
30:28	RW	0	Auto stop count 23
27	RW	0	Auto stop count 22 Wr_en
26:24	RW	0	Auto stop count 22
23	RW	0	Auto stop count 21 Wr_en
22:20	RW	0	Auto stop count 21
19	RW	0	Auto stop count 20 Wr_en
18:16	RW	0	Auto stop count 20
15	RW	0	Auto stop count 19 Wr_en
14:12	RW	0	Auto stop count 19
11	RW	0	Auto stop count 18 Wr_en
10:8	RW	0	Auto stop count 18
7	RW	0	Auto stop count 17 Wr_en
6:4	RW	0	Auto stop count 17
3	RW	0	Auto stop count 16 Wr_en
2:0	RW	0	Auto stop count 16

Table 8-194 SECTION_AUTO_STOP_1_O 0x3b

Bit(s)	R/W	Default	Description
31	RW	0	Auto stop count 15 Wr_en
30:28	RW	0	Auto stop count 15
27	RW	0	Auto stop count 14 Wr_en
26:24	RW	0	Auto stop count 14
23	RW	0	Auto stop count 13 Wr_en
22:20	RW	0	Auto stop count 13
19	RW	0	Auto stop count 12 Wr_en
18:16	RW	0	Auto stop count 12
15	RW	0	Auto stop count 11 Wr_en
14:12	RW	0	Auto stop count 11

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Bit(s)	R/W	Default	Description
11	RW	0	Auto stop count 10 Wr_en
10:8	RW	0	Auto stop count 10
7	RW	0	Auto stop count 9 Wr_en
6:4	RW	0	Auto stop count 9
3	RW	0	Auto stop count 8 Wr_en
2:0	RW	0	Auto stop count 8

Table 8-195 SECTION_AUTO_STOP_0_O 0x3c

Bit(s)	R/W	Default	Description
31	RW	0	Auto stop count 7 Wr_en
30:28	RW	0	Auto stop count 7
27	RW	0	Auto stop count 6 Wr_en
26:24	RW	0	Auto stop count 6
23	RW	0	Auto stop count 5 Wr_en
22:20	RW	0	Auto stop count 5
19	RW	0	Auto stop count 4 Wr_en
18:16	RW	0	Auto stop count 4
15	RW	0	Auto stop count 3 Wr_en
14:12	RW	0	Auto stop count 3
11	RW	0	Auto stop count 2 Wr_en
10:8	RW	0	Auto stop count 2
7	RW	0	Auto stop count 1 Wr_en
6:4	RW	0	Auto stop count 1
3	RW	0	Auto stop count 0 Wr_en
2:0	RW	0	Auto stop count 0

Table 8-196 DEMUX_CHANNEL_RESET_O 0x3d

Bit(s)	R/W	Default	Description
31:0	R	0	Bit 31:0 reset channel status - Each Bit reset each channel

Table 8-197 DEMUX_SCRAMBLING_STATE_O 0x3e

Bit(s)	R/W	Default	Description
31:0	R	0	Scrambling state of each channel

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Table 8-198 DEMUX_CHANNEL_ACTIVITY_O 0x3f

Bit(s)	R/W	Default	Description
31:0	R	0	Channel activity of each channel

Table 8-199 DEMUX_STAMP_CTL_O 0x40

Bit(s)	R/W	Default	Description
4	RW	0	video_stamp_use_dts
3	RW	0	audio_stamp_sync_1_en
2	RW	0	audio_stamp_insert_en
1	RW	0	video_stamp_sync_1_en
0	RW	0	video_stamp_insert_en

Table 8-200 DEMUX_VIDEO_STAMP_SYNC_0_O 0x41

Bit(s)	R/W	Default	Description
31:0	RW	0	Video stamp sync [63:32]

Table 8-201 DEMUX_VIDEO_STAMP_SYNC_1_O 0x42

Bit(s)	R/W	Default	Description
31:0	RW	0	Video stamp sync [31:0]

Table 8-202 DEMUX_AUDIO_STAMP_SYNC_0_O 0x43

Bit(s)	R/W	Default	Description
31:0	RW	0	Aideo stamp sync [63:32]

Table 8-203 DEMUX_AUDIO_STAMP_SYNC_1_O 0x44

Bit(s)	R/W	Default	Description
31:0	RW	0	Aideo stamp sync [31:0]

Table 8-204 DEMUX_SECTION_RESET_O 0x45

Bit(s)	R/W	Default	Description
31:0	R	0	Write : Bit[4:0] secter filter number for reset Read : select according to output_ section_buffer_valid: per bit per section buffer valid status or section_buffer_ ignore

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Table 8-205 DEMUX_INPUT_TIMEOUT_C_O 0x46

Bit(s)	R/W	Default	Description
31:0	RW	0	channel_reset_timeout_disable

Table 8-206 DEMUX_INPUT_TIMEOUT_O 0x47

Bit(s)	R/W	Default	Description
31	RW	0	no_match_reset_timeout_disable
30:0	RW	0	input_time_out_int_cnt (0 – means disable) Wr-setting, Rd-count

Table 8-207 DEMUX_PACKET_COUNT_O 0x48

Bit(s)	R/W	Default	Description
31:0	RW	0	channel_packet_count_disable

Table 8-208 DEMUX_PACKET_COUNT_C_O 0x49

Bit(s)	R/W	Default	Description
31	RW	0	no_match_packet_count_disable
30:0	RW	0	input_packet_count

Table 8-209 DEMUX_CHAN_RECORD_EN_O 0x4a

Bit(s)	R/W	Default	Description
31:0	RW	0xffffffff	channel_record_enable

Table 8-210 DEMUX_CHAN_PROCESS_EN_O 0x4b

Bit(s)	R/W	Default	Description
31:0	RW	0xffffffff	channel_process_enable

Table 8-211 DEMUX_SMALL_SEC_CTL_O 0x4c

Bit(s)	R/W	Default	Description
31:24	RW	0	small_sec_size ((n+1) * 256 Bytes)
23:16	RW	0	small_sec_rd_ptr
15:8	RW	0	small_sec_wr_ptr
7:2	RW	0	reserved
1	RW	0	small_sec_wr_ptr_wr_enable
0	RW	0	small_section_enable

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8.1.5 DTMB Registers

Table 8-212 ctrl_sw_rst 0x01

Bit(s)	R/W	Default	Description
1	R/W	0x0	SW reset but not include register
0	R/W	0x0	SW reset include register

Table 8-213 testbus 0x02

Bit(s)	R/W	Default	Description
16	R/W	0x0	testbus_en
15:0	R/W	0x1000	testbus_addr

Table 8-214 tb 0x03

Bit(s)	R/W	Default	Description
13	R/W	0x0	tb_self_test
12	R/W	0x0	tb_capture_stop
11:8	R/W	0x0	tb_dc_mk
4:0	R/W	0x0	tb_act_width

Table 8-215 tb_v 0x04

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	

Table 8-216 tb_addr_begin 0x05

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	

Table 8-217 tb_addr_end 0x06

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	

Table 8-218 ctrl_enable 0x07

Bit(s)	R/W	Default	Description
27:0	R/W	0xffffff	ctrl_snr_loss_enable

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Table 8-219 ctrl_loop 0x08

Bit(s)	R/W	Default	Description
5	R/W	0x0	sfo track disable ,high active
4	R/W	0x1	icfo feedback enable to nf
3	R/W	0x1	icfo feedback enable
2	R/W	0x1	fcfo feedback enable
1	R/W	0x1	sfo feedback enable
0	R/W	0x1	sfo fron pnphaseDet feedback enable

Table 8-220 ctrl_fsm 0x09

Bit(s)	R/W	Default	Description
12:9	R/W	0x3	ctrl_reset_state
8	R/W	0x0	ctrl_fsm_v
4:0	R/W	0x0	ctrl_fsm_state

Table 8-221 ctrl_agc 0x0a

Bit(s)	R/W	Default	Description
3	R/W	0x0	sfo_strong0_bypass, 1, bypass strong0 shift for sfo input
2	R/W	0x0	ts_cfo_bypass, 1, bypass strong0 shift for sfo input
1	R/W	0x0	ctrl_agc_bypass
0	R/W	0x1	ctrl_fast_agc

Table 8-222 ctrl_ts_sfo_cfo 0x0b

Bit(s)	R/W	Default	Description
31:24	R/W	0x20	ctrl_cfo_q, 8,-16US, fcfo lock thresh, 8.94*2/25e6
23:20	R/W	0x8	ctrl_sfo_q, 4.-16US, sfo lock thresh, 2.3e-6 * 25/(2*7.56)
19:12	R/W	0x3	ctrl_pnphase_q, 7.6TC, quality thresh of pnphase det
9:0	R/W	0x6	ctrl_ts_q, 10.4TC, quality thresh of Timingsync

Table 8-223 ctrl_fec 0x0c

Bit(s)	R/W	Default	Description
31:24	R/W	0x1	ctrl_che_to_th
23:16	R/W	0x8	ctrl_sfo_to_th
15:8	R/W	0x1	ctrl_pnphase_to_th
7:0	R/W	0x1	ctrl_ts_to_th

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Table 8-224 ctrl_intlv_time 0x0d

Bit(s)	R/W	Default	Description
23:12	R/W	0x280	ctrl_intlv240_time
11:0	R/W	0x500	ctrl_intlv720_time

Table 8-225 ctrl_dagc_cci 0x0e

Bit(s)	R/W	Default	Description
10	R/W	0x0	cci_restart_en, 1: restart cci after icfo ready
9	R/W	0x1	fec_inzero_check, "0" not check fec input zero; "1" check, if zero fec unlock
8	R/W	0x0	new_sync2, 2, timing adjust according TS3
7	R/W	0x0	new_sync1, 1, use correlation before FE when PN595, only one case must use it
5	R/W	0x0	fe_bypass
4	R/W	0x0	cci_bypass
3:2	R/W	0x1	cci_dagc_mode
1:0	R/W	0x1	front_dagc_mode, 1, use target_h and target_l before TPS, use target_her and ler after it; 2, use her and ler always; 0, use soft gain before TPS, and target_her after TPS; 3, use soft gain always

Table 8-226 ctrl_tps 0x0f

Bit(s)	R/W	Default	Description
31:24	R/W	0x1	ctrl_fec_to_th
23:16	R/W	0x1	ctrl_tps_to_th
11	R/W	0x0	tps_known
10	R/W	0x0	freq_reverse_known
9	R/W	0x0	tps_carrier_mode
8:7	R/W	0x0	constell
6:5	R/W	0x0	code_rate
4	R/W	0x0	intlv_mode
3	R/W	0x0	qam4_nr
2	R/W	0x0	freq_reverse
1:0	R/W	0x2	sfo_gain

Table 8-227 tps_bit 0x10

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	tps_32bits

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Table 8-228 fsm_watchdog 0x11

Bit(s)	R/W	Default	Description
16	R/W	0x1	fsm_watchdog_vld, watchdog enable, 1 is on, 0 is off
15:0	R/W	0xffff	fsm_watchdog_thr, atch dog timeout threshold , {this, 16'h0}

Table 8-229 fcfo_all_th 0x12

Bit(s)	R/W	Default	Description
19:0	R/W	0x7ffff	

Table 8-230 sfo_all_th 0x13

Bit(s)	R/W	Default	Description
24:0	R/W	0xffffff	

Table 8-231 che_snr_th 0x14

Bit(s)	R/W	Default	Description
13:0	R/W	0x2	

Table 8-232 che_path 0x15

Bit(s)	R/W	Default	Description
27:16	R	0x0	che_path_dist
11:0	R	0x0	che_path_dist_pn_rmv

Table 8-233 che_strong_path 0x16

Bit(s)	R/W	Default	Description
27:16	R	0x0	che_strong_path1_loc
11:0	R	0x0	che_strong_path0_loc

Table 8-234 che_timing_offset 0x17

Bit(s)	R/W	Default	Description
10:0	R	0x0	che_timing_offset

Table 8-235 com_phase 0x18

Bit(s)	R/W	Default	Description
26:16	R	0x0	che_com_phs_q
10:0	R	0x0	che_com_phs_i

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Table 8-236 fec_loss_ofdm_th 0x19

Bit(s)	R/W	Default	Description
19:0	R/W	0x1c00	fec loss time out th

Table 8-237 fe_corr_status 0x1a

Bit(s)	R/W	Default	Description
28:16	R	0x0	sync_corr_quality
12:0	R	0x0	sync_corr_max_pos

Table 8-238 cci_flg 0xc7

Bit(s)	R/W	Default	Description
25:9	R	0x0	front_sfo_adjust_accu
8	R	0x0	che_m_cci_ready
7:1	R	0x0	che_cci_flg_cnt

Table 8-239 testbus_out 0xc8

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-240 tbus_dc_addr 0xc9

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-241 sync_ts 0xcb

Bit(s)	R/W	Default	Description
24:15	R	0x0	sync_ts_q
14:2	R	0x0	sync_ts_pos
1:0	R	0x0	sync_ts_idx

Table 8-242 sync_pnphase 0xcd

Bit(s)	R/W	Default	Description
28:17	R	0x0	sync_pnphase_sfo
16:10	R	0x0	sync_pnphase_max_q
9:2	R	0x0	sync_pnphase
1:0	R	0x0	sync_pnphase_max_q_idx

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Table 8-243 ctrl_ddc_icfo 0xd2

Bit(s)	R/W	Default	Description
19:0	R	0x0	

Table 8-244 ctrl_ddc_fcfo 0xd3

Bit(s)	R/W	Default	Description
31:17	R	0x0	ctrl_ddc_fcfo
16:0	R	0x0	ctrl_src_sfo

Table 8-245 ctrl_fsm_state0 0xd4

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-246 ctrl_fsm_state1 0xd5

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-247 ctrl_fsm_state2 0xd6

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-248 ctrl_fsm_state3 0xd7

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-249 ctrl_ts2 0xd8

Bit(s)	R/W	Default	Description
31:24	R	0x0	sync_fe_workcnt
23:16	R	0x0	ctrl_sfo_workcnt
15:8	R	0x0	ctrl_pnphase_workcnt
7:0	R	0x0	ctrl_ts2_workcnt

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Table 8-250 front_agc 0xd9

Bit(s)	R/W	Default	Description
31:22	R	0x0	front_agc_power
21:10	R	0x0	front_agc_rf_gain
10:0	R	0x0	front_agc_if_gain

Table 8-251 front_dagc 0xda

Bit(s)	R/W	Default	Description
19:8	R	0x0	front_dagc_gain
7:0	R	0x0	front_dagc_power

Table 8-252 ctrl_icfo_all 0xe0

Bit(s)	R/W	Default	Description
19:0	R	0x0	

Table 8-253 ctrl_fcfo_all 0xe1

Bit(s)	R/W	Default	Description
19:0	R	0x0	

Table 8-254 ctrl_sfo_all 0xe2

Bit(s)	R/W	Default	Description
24:0	R	0x0	

Table 8-255 fec_lock_snr 0xe3

Bit(s)	R/W	Default	Description
30	R	0x0	demap_do_bad
29:16	R	0x0	che_snr_avg
14	R	0x0	fec_lock
13:0	R	0x0	che_snr

Table 8-256 che_seg_factor 0xe4

Bit(s)	R/W	Default	Description
29:16	R	0x0	
13:0	R	0x0	

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Table 8-257 ctrl_che_workcnt 0xe5

Bit(s)	R/W	Default	Description
23	R	0x0	ctrl_carrier_mode
22	R	0x0	ctrl_freq_reverse
21	R	0x0	ctrl_qam4_nr
20	R	0x0	ctrl_intlv_mode
19:18	R	0x0	ctrl_code_rate
17:16	R	0x0	ctrl_constell
15:8	R	0x0	ctrl_fec_workcnt
7:0	R	0x0	ctrl_che_workcnt

Table 8-258 che_obs_state1 0xe6

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-259 che_obs_state2 0xe7

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-260 che_obs_state3 0xe8

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-261 che_obs_state4 0xe9

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-262 che_obs_state5 0xea

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-263 sync_cci_nf1 0xee

Bit(s)	R/W	Default	Description
29:20	R	0x0	sync_cci_nf1_a1
19:10	R	0x0	sync_cci_nf1_a2
9:0	R	0x0	sync_cci_nf1_b1

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Table 8-264 sync_cci_nf2 0xef

Bit(s)	R/W	Default	Description
29:20	R	0x0	sync_cci_nf2_a1
19:10	R	0x0	sync_cci_nf2_a2
9:0	R	0x0	sync_cci_nf2_b1
s			

Table 8-265 ync_cci_nf2_position 0xf0

Bit(s)	R/W	Default	Description
23	R	0x0	sync_cci_nf1_det
22	R	0x0	sync_cci_nf2_det
21:11	R	0x0	sync_cci_nf1_position
10:0	R	0x0	sync_cci_nf2_position

Table 8-266 ctrl_sys_ofdm_cnt 0xf1

Bit(s)	R/W	Default	Description
29:28	R	0x0	che_working_state
26:8	R	0x0	front_mobi_det_power_var
7:0	R	0x0	ctrl_sys_ofdm_cnt

Table 8-267 ctrl_tps_q_final 0xf2

Bit(s)	R/W	Default	Description
13:7	R	0x0	tps_success_cnt
6:0	R	0x0	tps_q_final

Table 8-268 front_dc 0xf3

Bit(s)	R/W	Default	Description
19:10	R	0x0	front_dc_i
9:0	R	0x0	front_dc_q

Table 8-269 che_debug 0xf6

Bit(s)	R/W	Default	Description
19:0	R	0x0	

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Table 8-270 fsm_watchdog_mnt 0xf7

Bit(s)	R/W	Default	Description
31:8	R	0x0	front_ddc_accum_phase
7:0	R	0x0	fsm_watchdog_mnt

Table 8-271 fsm_watchdog_cnt 0xf8

Bit(s)	R/W	Default	Description
31:0	R	0x0	

Table 8-272 ctrl_totps_ready_cnt 0xff

Bit(s)	R/W	Default	Description
31:16	R	0x0	

Table 8-273 afifo_adc 0x20

Bit(s)	R/W	Default	Description
16	R/W	0x0	adc_IQ, obsolete, not connected to any net
15:10	R/W	0x0	adc_sample, obsolete, not connected to any net
9	R/W	0x0	afifo_bypass, set to 1 to bypass afifo
8	R/W	0x0	afifo_data_format, 1, 2's complement, 0, offset binary
7:0	R/W	0x29	afifo_nco_rate

Table 8-274 agc_config1 0x21

Bit(s)	R/W	Default	Description
30:28	R/W	0x0	agc_a_filter_coef1, alpha filter1 coef, right shift (this+1),
26:24	R/W	0x0	agc_a_filter_coef2, alpha filter2 coef, right shift (this+1)
21:16	R/W	0x1	agc_gain_step1, initial gain and combine with gain_step to derive gain_delta in sigma-delta output
13:8	R/W	0x1	agc_gain_step2, initial gain and combine with gain_step to derive gain_delta in sigma-delta output
7:4	R/W	0x2	agc_cal_intv, accumation interval (64+2^this)
2:0	R/W	0x2	agc_a_filter_coef0, alpha filter0 coef, right shift (this+1)

Table 8-275 agc_config2 0x22

Bit(s)	R/W	Default	Description
30:24	R/W	0x7	agc_sd_rate, sigma-delta accum interval (1<
21:16	R/W	0x20	agc_signal_clip_thr, thr to determine if there is power clipping
12:8	R/W	0xa	agc_clip_ratio, gain applied to filtered output when clipping

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Bit(s)	R/W	Default	Description
5	R/W	0x0	agc_iq_exchange
4	R/W	0x1	agc_imp_en, impulsive noise detection enable
3:0	R/W	0x6	agc_imp_thresh, threshold for impulsive noise detection

Table 8-276 agc_config3 0x23

Bit(s)	R/W	Default	Description
31	R/W	0x0	agc_iffb_set, IF gain set trigger signal
30	R/W	0x1	agc_if_only, when set, only IF agc control in function
29	R/W	0x0	agc_ifgain_freeze, when set, IF gain is freezed
28	R/W	0x0	agc_gain_step_if, 1, shift right 1bit from gain step1
27	R/W	0x0	agc_rffb_set, RF gain set trigger signal
26	R/W	0x0	agc_tuning_slope, when set, only RF agc control in function
25	R/W	0x1	agc_rfgain_freeze, when set, RF gain is freezed
24	R/W	0x0	agc_gain_step_rf, 1, shift right 1bit from gain step1
23:12	R/W	0x190	agc_iffb_value, IF gain setting value
11:0	R/W	0x190	agc_rffb_value, RF gain setting value

Table 8-277 agc_config4 0x24

Bit(s)	R/W	Default	Description
31:24	R/W	0x7f	agc_iffb_gain_sat, IF gain saturation value
23:16	R/W	0x80	agc_iffb_gain_sat_i
15:8	R/W	0x7f	agc_rffb_gain_sat, RF gain saturation value
7:0	R/W	0x80	agc_rffb_gain_sat_i

Table 8-278 ddc_bypass 0x25

Bit(s)	R/W	Default	Description
28	R/W	0x0	ddc_bypass, set to 1 to bypass
24:0	R/W	0x62c1-a5	ddc_phase, u8.0

Table 8-279 dc_hold 0x28

Bit(s)	R/W	Default	Description
26:24	R/W	0x0	dc_len
20	R/W	0x0	dc_bypass
19:16	R/W	0xa	channel_portable_th

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Bit(s)	R/W	Default	Description
15:12	R/W	0x1	channel_static_th
10:8	R/W	0x0	mobi_det_observe_len
6:4	R/W	0x6	mobi_det_accu_len
3:1	R/W	0x3	dc_alpha
0	R/W	0x0	dc_hold

Table 8-280 dagc_target_power 0x29

Bit(s)	R/W	Default	Description
31:24	R/W	0x14	dagc_target_power_her
23:16	R/W	0xa	dagc_target_power_ler
15:8	R/W	0x10	dagc_target_power_h
7:0	R/W	0xe	dagc_target_power_l

Table 8-281 acf_bypass 0x2a

Bit(s)	R/W	Default	Description
24	R/W	0x0	acf_bypass, set to 1 to bypass
22:12	R/W	0x274	coef66, 11.0TC, coef66 to coef0 is half number of the 133-tap FIR coefficents
10:0	R/W	0x217	coef65

Table 8-282 coef_set1 0x2b

Bit(s)	R/W	Default	Description
22:12	R/W	0x131	coef64
10:0	R/W	0x36	coef63

Table 8-283 coef_set2 0x2c

Bit(s)	R/W	Default	Description
9:0	R/W	0x396	coef62

Table 8-284 coef_set3 0x2d

Bit(s)	R/W	Default	Description
21:12	R/W	0x37f	coef61
9:0	R/W	0x3cc	coef60

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Table 8-285 coef_set4 0x2e

Bit(s)	R/W	Default	Description
9:0	R/W	0x29	coef59

Table 8-286 coef_set5 0x2f

Bit(s)	R/W	Default	Description
20:12	R/W	0x4f	coef58
8:0	R/W	0x31	coef57

Table 8-287 coef_set6 0x30

Bit(s)	R/W	Default	Description
23:16	R/W	0xf3	coef56
15:8	R/W	0xcb	coef55
7:0	R/W	0xd4	coef54

Table 8-288 coef_set7 0x31

Bit(s)	R/W	Default	Description
6:0	R/W	0x7e	coef53

Table 8-289 coef_set8 0x32

Bit(s)	R/W	Default	Description
30:24	R/W	0x23	coef52
22:16	R/W	0x27	coef51
14:8	R/W	0xb	coef50
6:0	R/W	0x6a	coef49

Table 8-290 coef_set9 0x33

Bit(s)	R/W	Default	Description
30:24	R/W	0x5f	coef48
22:16	R/W	0x70	coef47
14:8	R/W	0xc	coef46
6:0	R/W	0x1b	coef45

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Table 8-291 coef_set10 0x34

Bit(s)	R/W	Default	Description
21:16	R/W	0x13	coef44
13:8	R/W	0x3c	coef43
5:0	R/W	0x2b	coef42

Table 8-292 coef_set11 0x35

Bit(s)	R/W	Default	Description
29:24	R/W	0x2d	coef41
21:16	R/W	0x3e	coef40
13:8	R/W	0xf	coef39
5:0	R/W	0x12	coef38

Table 8-293 coef_set12 0x36

Bit(s)	R/W	Default	Description
29:24	R/W	0x6	coef37
21:16	R/W	0x36	coef36
13:8	R/W	0x30	coef35
5:0	R/W	0x38	coef34

Table 8-294 coef_set13 0x37

Bit(s)	R/W	Default	Description
29:24	R/W	0x6	coef33
21:16	R/W	0xe	coef32
13:8	R/W	0xa	coef31
5:0	R/W	0x3e	coef30

Table 8-295 coef_set14 0x38

Bit(s)	R/W	Default	Description
20:16	R/W	0x15	coef29
12:8	R/W	0x16	coef28
4:0	R/W	0x1f	coef27

Table 8-296 coef_set15 0x39

Bit(s)	R/W	Default	Description
28:24	R/W	0x8	coef26
20:16	R/W	0x9	coef25

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Bit(s)	R/W	Default	Description
12:8	R/W	0x3	coef24
4:0	R/W	0x1b	coef23

Table 8-297 coef_set16 0x3a

Bit(s)	R/W	Default	Description
28:24	R/W	0x18	coef22
20:16	R/W	0x1c	coef21
12:8	R/W	0x3	coef20
4:0	R/W	0x7	coef19

Table 8-298 coef_set17 0x3b

Bit(s)	R/W	Default	Description
28:24	R/W	0x5	coef18
20:16	R/W	0x1f	coef17
12:8	R/W	0x1a	coef16
4:0	R/W	0x1b	coef15

Table 8-299 coef_set18 0x3c

Bit(s)	R/W	Default	Description
27:24	R/W	0x0	coef14
23:20	R/W	0x4	coef13
19:16	R/W	0x5	coef12
15:12	R/W	0x1	coef11
11:8	R/W	0xd	coef10
7:4	R/W	0xc	coef09
3:0	R/W	0xe	coef08

Table 8-300 coef_set19 0x3d

Bit(s)	R/W	Default	Description
31:28	R/W	0x2	coef07
27:24	R/W	0x4	coef06
23:20	R/W	0x2	coef05
19:16	R/W	0xf	coef04
15:12	R/W	0xd	coef03
11:8	R/W	0xe	coef02
7:4	R/W	0x1	coef01
3:0	R/W	0x2	coef00

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Table 8-301 src_config1 0x3e

Bit(s)	R/W	Default	Description
28	R/W	0x1	src_ted_disable, 1, disable src timing error detection loop
27:24	R/W	0x3	src_tim_shr, timing loop counter
23:0	R/W	0x1-a747d	src_norm_inrate, input rate, set to ratio=Fsamp/(2*Fsymb)*1048576

Table 8-302 src_config2 0x3f

Bit(s)	R/W	Default	Description
28:24	R/W	0x11	src_time_mu2, timing loop coef2
20:16	R/W	0x4	src_time_mu1, timing loop coef1
15:12	R/W	0xa	src_time_err_thr, timing error threshold
10:8	R/W	0x2	src_ted_beta, timing error filter coeff (1>>this)
6:4	R/W	0x6	src_seg_len
3:0	R/W	0x4	src_stable_timeout

Table 8-303 sfifo_out_len 0x40

Bit(s)	R/W	Default	Description
3:0	R/W	0x7	sfifo depth

Table 8-304 dagc_gain 0x41

Bit(s)	R/W	Default	Description
29:24	R/W	0x4	dagc_gain_step, 6.0 US, dagc gain step
23:28	R/W	0x1	dagc_gain_step_er, 6.0US, dagc gain step when CHE
17:6	R/W	0x0	dagc_gain_ctrl
5:3	R/W	0x3	dagc_bw
2:1	R/W	0x1	dagc_power_alpha
0	R/W	0x0	dagc_bypass

Table 8-305 src_config3 0x42

Bit(s)	R/W	Default	Description
15:12	R/W	0xf	src_tim_acc_gap
8	R/W	0x0	src_sweep_onoff
4:0	R/W	0x0	src_sweep_speed

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Table 8-306 src_config4 0x43

Bit(s)	R/W	Default	Description
23:0	R/W	0x10ee6	src_sweep_range

Table 8-307 st_config 0x44

Bit(s)	R/W	Default	Description
28:24	R/W	0x18	st_len
22:20	R/W	0x3	st_dist, auto-correlation differential distance
19:12	R/W	0x0	st_Q_thrsh
10:8	R/W	0x0	st_alpha
6:4	R/W	0x5	st_dc_len
0	R/W	0x0	st_enable

Table 8-308 st_freq 0x45

Bit(s)	R/W	Default	Description
31:28	R/W	0xc	agc_warmup_intarget_thr
27:24	R/W	0x2	agc_fast_intarget_thr
23:20	R/W	0x4	agc_slow_intarget_thr
19:1	R/W	0x0	st_freq_i
0	R/W	0x0	st_freq_v

Table 8-309 agc_config5 0x46

Bit(s)	R/W	Default	Description
29:24	R/W	0x1a	agc_target
16:20	R/W	0x1f	agc_warmup_intarget_len
12:8	R/W	0xf	agc_fast_intarget_len
4:0	R/W	0xf	agc_slow_intarget_len

Table 8-310 ts_cfo_pn_value 0x57

Bit(s)	R/W	Default	Description
31:16	R/W	0x25b	ts_cfo_pn0_value, 16.-6TC, icfo feedback for pn420, 1800*2/1e6/fs
15:0	R/W	0x243	ts_cfo_pn1_value, 16.-6TC, icfo feedback for pn595, 1728*2/1e6/fs

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Table 8-311 ts_cfo_err_limit 0x58

Bit(s)	R/W	Default	Description
31:16	R/W	0x218	ts_cfo_pn2_value, 16.-6TC, icfo feedback for pn945, 1600*2/1e6/fs
15:0	R/W	0x86	ts_cfo_err_limit, 16.-6TC, error limit to icfo, if >ts_cfo_err_limit, output is zero, 400*2/1e6/fs

Table 8-312 ts_cfo_pn_modify 0x59

Bit(s)	R/W	Default	Description
31:16	R/W	0x136e	ts_cfo_pn0_modify, 16.-6TC, for PN420, 7.56/255/fs
15:0	R/W	0x121	ts_cfo_pn1_modify, 16.-6TC, for PN595, 7.56/4375/fs

Table 8-313 ts_gain 0x5a

Bit(s)	R/W	Default	Description
31:16	R/W	0x9b2	ts_cfo_pn2_modify, 16.-6TC, for PN945, 7.56/511/fs
15:12	R/W	0x5	ts_cfo_cut,
10:9	R/W	0x0	ts_fixpn, fix pnmode
8	R/W	0x0	ts_fixpn_en, 1 is enable fixpn, ts only search the ts_fixpn
6:4	R/W	0x2	ts_sat_shift, cut 5-ts_sat_shift to 6 bit
1:0	R/W	0x3	ts_gain, gain = 2^(ts_gain+1)

Table 8-314 fe_config 0x5b

Bit(s)	R/W	Default	Description
31:16	R/W	0x4d6a	fe_modify, 16.-7TC, factor to get the fe feedback , = 2*7.56/fs/256
11:8	R/W	0xa	fe_cut
6:4	R/W	0x2	fe_sat_shift
3:0	R/W	0x5	fe_lock_len, 2 ^ 5 =32

Table 8-315 pnphase_offset 0x5c

Bit(s)	R/W	Default	Description
11:8	R/W	0xc	pnphase_offset0, 4.2TC, -238*4200/1e6
7:4	R/W	0x0	pnphase_offset1, 4.2TC
3:0	R/W	0x4	pnphase_offset2, 4.2TC, 238*4200/1e6

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Table 8-316 pnphase_config 0x5d

Bit(s)	R/W	Default	Description
31:16	R/W	0xd3b	pnphase_modify, 16.-15TC, 1e-6*fs/(2*7.56)
11:8	R/W	0xa	pnphase_cut
7:4	R/W	0x3	pnphase_sat_shift, for PN420, fftout(16.6) * (2^(pnphase_sat_shift-2))->7.6
1:0	R/W	0x2	pnphase_gain, 2^(2+pnphase_gain)

Table 8-317 sfo_sfo_pn0_modify 0x5e

Bit(s)	R/W	Default	Description
31:16	R/W	0x33a	sfo_sfo_pn0_modify, 0.5*256 / 4200 * 25 /(2*7.56)
15:0	R/W	0x4b8	sfo_cfo_pn0_modify, 7.56/4200/25

Table 8-318 sfo_sfo_pn1_modify 0x5f

Bit(s)	R/W	Default	Description
31:16	R/W	0x631	sfo_sfo_pn1_modify, 0.5*512 / 4375 * 25 /(2*7.56)
15:0	R/W	0x488	sfo_cfo_pn1_modify, 7.56/4375/25

Table 8-319 sfo_sfo_pn2_modify 0x60

Bit(s)	R/W	Default	Description
31:16	R/W	0x5bc	sfo_sfo_pn2_modify, 0.5*512 / 4725 * 25 /(2*7.56)
15:0	R/W	0x432	sfo_cfo_pn2_modify, 7.56/4725/25

Table 8-320 sfo_config 0x61

Bit(s)	R/W	Default	Description
27:26	R/W	0x0	sfo_cci_th_sft
25:24	R/W	0x2	sfo_cci_blank_rang
23:20	R/W	0x6	sfo_cci_th, 4.4US, cci thresh in sfo
19:16	R/W	0x4	sfo_sfo_cut, RTL setting == C setting +3
15:12	R/W	0x4	sfo_cfo_cut
9:8	R/W	0x0	sfo_dist, sfo distance, 0 is 0.5*Nfft, 1 is 0.25*Nfft, 2 is 0.125*Nfft, 3 is 0.0625*Nfft
5:4	R/W	0x2	sfo_gain, 2^(3+sfo_gain)
3:0	R/W	0x1	sfo_sat_shift, for PN420, fftout(16.6) * (2^sfo_sat_shift)->8.5

Table 8-321 ts_cfo_modify_mask 0x62

Bit(s)	R/W	Default	Description
4:0	R/W	0x0	ts cfo modify mask bit ,high active ,default all postion is working

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Table 8-322 track_cfo_max 0x6d

Bit(s)	R/W	Default	Description
20:17	R/W	0x4	ctrl_fe_to_th
16	R/W	0x1	track_max_en, 1 is enable adjust max limit
15:8	R/W	0x7	track_sfo_max, 8.-15 TC, max sfo adjust feedback, 0.5e-6 * 25/(2*7.56)
7:0	R/W	0x1	track_cfo_max, 8.-16 TC, max cfo adjust feedback, 0.8*(2/fs/1e6)

Table 8-323 cci_dagc_config1 0x6e

Bit(s)	R/W	Default	Description
29:24	R/W	0x4	cci_dagc_gain_step, 6.0 US, dagc gain step
23:18	R/W	0x1	cci_dagc_gain_step_er, 6.0US, dagc gain step when CHE
17:6	R/W	0x0	cci_dagc_gain_ctrl
5:3	R/W	0x3	cci_dagc_bw, avglen=2^(9+cci_dagc_bw), algo is 3
2:1	R/W	0x1	cci_dagc_power_alpha
0	R/W	0x0	cci_dagc_bypass

Table 8-324 cci_dagc_config2 0x6f

Bit(s)	R/W	Default	Description
31:24	R/W	0x14	cci_dagc_target_power_her, u8.0, higher target when CHE
23:16	R/W	0xa	cci_dagc_target_power_ler, lower target
15:8	R/W	0x10	cci_dagc_target_power_h, high target when SYNC
7:0	R/W	0xe	cci_dagc_target_power_l, low target

Table 8-325 cci_rp 0x70

Bit(s)	R/W	Default	Description
24:12	R/W	0xbbf	cci_rp_n, related to notch bw and doesn't change
9:0	R/W	0x376	cci_rpsq_n, related to notch bw and doesn't change

Table 8-326 cci_det_thres 0x71

Bit(s)	R/W	Default	Description
31:16	R/W	0x61a8	nf_modify, parameter for nf filter coefficient recalculate
10:8	R/W	0x5	cci_det_thres, power compared to this value (1<
4:0	R/W	0x4	cci_avr_times, averaging number of times among fft symbols

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Table 8-327 cci_notch1_config1 0x72

Bit(s)	R/W	Default	Description
12	R/W	0x0	cci_notch1_en
9:0	R/W	0x0	cci_notch1_a1, a1 of iir filter, can be set to remove known cci

Table 8-328 cci_notch1_config2 0x73

Bit(s)	R/W	Default	Description
12	R/W	0x0	cci_notch1_a2, a2 of iir filter
9:0	R/W	0x0	cci_notch1_b1, b1 of iir filter

Table 8-329 cci_notch2_config1 0x74

Bit(s)	R/W	Default	Description
31:16	R/W	0x0	cci_mpthres, minimum power threshold to prevent false cci detection
12	R/W	0x0	cci_notch2_en
9:0	R/W	0x0	cci_notch2_a1

Table 8-330 cci_notch2_config2 0x75

Bit(s)	R/W	Default	Description
12	R/W	0x0	cci_notch2_a2
9:0	R/W	0x0	cci_notch2_b1

Table 8-331 timing_est_thr0 0x78

Bit(s)	R/W	Default	Description
29:16	R/W	0x14	tm_thr_ma0
8:0	R/W	0x134	tm_thr_avg0

Table 8-332 timing_est_thr1 0x79

Bit(s)	R/W	Default	Description
29:16	R/W	0x0c	tm_thr_ma1
8:0	R/W	0x12c	tm_thr_avg1

Table 8-333 timing_est_thr2 0x7a

Bit(s)	R/W	Default	Description
29:16	R/W	0x0a	tm_thr_ma2
8:0	R/W	0x12a	tm_thr_avg2

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Table 8-334 timing_est_thr3 0x7b

Bit(s)	R/W	Default	Description
29:16	R/W	0x13c	tm_thr_ma3
8:0	R/W	0xb4	tm_thr_avg3

Table 8-335 timing_est_thr4 0x7c

Bit(s)	R/W	Default	Description
29:16	R/W	0x13c	tm_thr_ma4
8:0	R/W	0xb4	tm_thr_avg4

Table 8-336 timing_snr_min0 0x7d

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	cci_mask_range0
22:12	R/W	0x04	tm_snr_min0
10:0	R/W	0x19	tm_snr_min1

Table 8-337 timing_snr_min1 0x7e

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	cci_mask_range1
22:12	R/W	0x28	tm_snr_min2
10:0	R/W	0x4f	tm_snr_min3

Table 8-338 0x7f

Bit(s)	R/W	Default	Description
Bit(s)	R/W	Default	Description
31:24	R/W	0x0d	qstep_sc_lmt_04q_dot4
23:12	R/W	0x166	qstep_sc_04q_dot4
10:0	R/W	0x080	qstep_sc_16q_dot4

Table 8-339 0x80

Bit(s)	R/W	Default	Description
31:24	R/W	0x16	qstep_sc_lmt_16q_dot4
23:12	R/W	0x029	qstep_sc_64q_dot4
10:0	R/W	0x200	qstep_sc_04q_dot6

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Table 8-340 0x81

Bit(s)	R/W	Default	Description
31:24	R/W	0x1f	qstep_sc_lmt_64q_dot4
23:12	R/W	0x0b3	qstep_sc_16q_dot6
10:0	R/W	0x033	qstep_sc_64q_dot6

Table 8-341 0x82

Bit(s)	R/W	Default	Description
31:24	R/W	0x18	qstep_sc_lmt_04q_dot6
23:12	R/W	0x200	qstep_sc_04q_dot8
10:0	R/W	0x0cd	qstep_sc_16q_dot8

Table 8-342 0x83

Bit(s)	R/W	Default	Description
31:24	R/W	0x25	qstep_sc_lmt_16q_dot6
23:12	R/W	0x066	qstep_sc_64q_dot8
10:0	R/W	0x066	qstep_sc_32q_dot8

Table 8-343 0x84

Bit(s)	R/W	Default	Description
31:24	R/W	0x2e	qstep_sc_lmt_64q_dot6
23:16	R/W	0x21	qstep_sc_lmt_04q_dot8
15:8	R/W	0x3c	qstep_sc_lmt_16q_dot8
7:0	R/W	0x72	qstep_sc_lmt_64q_dot8

Table 8-344 0x85

Bit(s)	R/W	Default	Description
28:16	R/W	0xa	cci_mask_pos0
12:0	R/W	0x7d0	cci_mask_pos1

Table 8-345 m_cci_thr_config1 0x86

Bit(s)	R/W	Default	Description
31:28	R/W	0x9	dist0
25:16	R/W	0x20	m_cci_thr_mc3, 10.9US, cci thresh for mc 64QAM
9:0	R/W	0x20	m_cci_thr_mc2, 10.9US, cci thresh for mc 32QAM

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Table 8-346 m_cci_thr_config2 0x87

Bit(s)	R/W	Default	Description
31:28	R/W	0x4	dist1
25:16	R/W	0x20	m_cci_thr_mc1, 10.9US, cci thresh for mc 16QAM
9:0	R/W	0x3f	m_cci_thr_mc0, 10.9US, cci thresh for mc 4QAM or 4QAM_NR

Table 8-347 m_cci_thr_config3 0x88

Bit(s)	R/W	Default	Description
31:28	R/W	0x3	dist2
25:16	R/W	0x20	m_cci_thr_sc3, 10.9US, cci thresh for sc 64QAM
9:0	R/W	0x20	m_cci_thr_sc2, 10.9US, cci thresh for sc 64QAM

Table 8-348 m_cci_thr_config4 0x89

Bit(s)	R/W	Default	Description
31:28	R/W	0x2	dist3
25:16	R/W	0x20	m_cci_thr_sc1, 10.9US, cci thresh for sc 16QAM
9:0	R/W	0x20	m_cci_thr_sc0, 10.9US, cci thresh for sc 4QAM or 4QAM_NR

Table 8-349 m_cci_thr_config5 0x8a

Bit(s)	R/W	Default	Description
25:16	R/W	0x1f6	m_cci_thr_ma, 10.9US, cci thresh for move average
15:12	R/W	0x2	ccidet_masize, 1:2^1,2:2^2,3:2^3,4:2^4,5:2^5,2^(masize+1) +1 as ma length
8	R/W	0x0	m_cci_bypass, 1 is bypass
4	R/W	0x1	tune_mask, 1, force blank tune in SC
1:0	R/W	0x0	ccicnt_out_sel

Table 8-350 ibdfe_config0 0x8b

Bit(s)	R/W	Default	Description
31:28	R/W	0x5	ibdfe_err_iter
24	R/W	0x0	ibdfe_blank_y
20:16	R/W	0x1f	gd_len
12	R/W	0x1	eq_soft_slicer_en, 1 is enable soft slicer for data remove
11:8	R/W	0x4	eq_dist_thr_tps, H good dist thresh for tps
7:4	R/W	0x7	ibdfe_eqout_iter, ibdfe iteration times for eqout
3:0	R/W	0x4	ibdfe_fdbk_iter, ibdfe iteration times for Hrebuild

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Table 8-351 ibdfe_config1 0x8c

Bit(s)	R/W	Default	Description
27	R/W	0x1	sc_cci_switch_en
26	R/W	0x1	mc_cci_switch_en
25:16	R/W	0x15f	cci_switch_num
8:0	R/W	0xeb	ibdfe_dmsg_start_cnt

Table 8-352 te_hreb_snr 0x8d

Bit(s)	R/W	Default	Description
31:21	R/W	0x583	qstep_mc_04q_dot8
20:0	R/W	0x13b7b	te_hreb_snr

Table 8-353 timing_est_thr_max0 0x8e

Bit(s)	R/W	Default	Description
31:21	R/W	0x240	qstep_mc_16q_dot8
20:12	R/W	0x90	tm_thr_max0
8:0	R/W	0x1ba	tm_thr_max1

Table 8-354 timing_est_thr_max1 0x8f

Bit(s)	R/W	Default	Description
28:20	R/W	0x1ba	tm_thr_max2
16:8	R/W	0x1ea	tm_thr_max3
6:4	R/W	0x3	dist, 3.3US, MA compare dist
2:0	R/W	0x1	h_valid, h_valid for h exceed MA thresh

Table 8-355 timing_limit 0x90

Bit(s)	R/W	Default	Description
30:28	R/W	0x4	ma_size
24:20	R/W	0x1f	timing_limit, 5.5US, max timing adjust
15:8	R/W	0x50	strong_loc_thd, 8.8US, the min distance between first and second strong path
6:4	R/W	0x4	coh_thd, 3.3US, 2^coh_thd is coh length in timingest
2:0	R/W	0x3	ncoh_thd,3.3US, 2^ncoh_thd is non-coh length in timingest

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Table 8-356 tps_config 0x91

Bit(s)	R/W	Default	Description
25	R/W	0x1	chi_path_dist_sel
24	R/W	0x1	dcm_path_dist_sel
23:16	R/W	0xc0	chi_power_thr, 8.1US, hinit power thresh to h dc remove
12:8	R/W	0x0	tps_pre_num
4:0	R/W	0x0	tps_pst_num

Table 8-357 ccidet_config 0x92

Bit(s)	R/W	Default	Description
30:28	R/W	0x0	ccidect_sat_sft
24:20	R/W	0x2	ccidet_mc_mask_rng, 5.5US, mc mask range
16:12	R/W	0x0	ccidet_sc_mask_rng, 5.5US, sc mask range
10:8	R/W	0x5	ccidet_malpha, 3.3US, 2^-ccidet_malpha is alpha filter coeff
6:0	R/W	0x10	ccidet_dly, ccidet will give out a result after ccidet_dly symbols

Table 8-358 qstep_set 0x93

Bit(s)	R/W	Default	Description
28	R/W	0x0	demap_mode
25	R/W	0x0	qstep_set_val, 1, use qstep_set in demap
24:12	R/W	0x0	qstep_set, 11.6us for mc, 13.11us for sc
9:0	R/W	0x20	factor_stable_thres, 10.2US, use segment if factor_min

Table 8-359 seg_config 0x94

Bit(s)	R/W	Default	Description
25	R/W	0x0	seg_bypass, 1 is bypass
24	R/W	0x0	seg_read_val
23:12	R/W	0x0	seg_read_addr
11:8	R/W	0x1	noise_input_shift, 4.4TC, select 16bit [16+noise_input_shift:1+noise_input_shift] noise error to noisePow avg calc
6:4	R/W	0x5	seg_alpha, 3.3US, 2^-seg_alpha is noise alpha filter in MC segment calculation
2:0	R/W	0x0	seg_num_1seg_log2, 3.3US, 2^[seg_num_1seg_log2+2] is one seg subcarrier number

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Table 8-360 td_stepsize 0x95

Bit(s)	R/W	Default	Description
31:21	R/W	0x40b	qstep_mc_04q_dot4
20:16	R/W	0x0	td_stepsize_thr01, For SC mode, 5.5US, downsample 5, 0 is 0, other is 2^-td_ stepsize_thr01, td stepsize in stage1
12:8	R/W	0x8	td_stepsize_thr02, td stepsize in stage2
4:0	R/W	0xa	td_stepsize_thr03, td stepsize in stage3

Table 8-361 fd_stepsize 0x96

Bit(s)	R/W	Default	Description
31:21	R/W	0x160	qstep_mc_16q_dot4
20:16	R/W	0x0	fd_stepsize_thr01, 5.5US, fd stepsize in stage1
12:8	R/W	0x7	fd_stepsize_thr02, fd stepsize in stage2
4:0	R/W	0xa	fd_stepsize_thr03, fd stepsize in stage3

Table 8-362 fd_td_coeff 0x97

Bit(s)	R/W	Default	Description
30:26	R/W	0x0	fd_coeff_addr
25	R/W	0x0	td_coeff_done
24	R/W	0x0	fd_coeff_rst
23	R/W	0x0	fd_coeff_done
22	R/W	0x0	fd_coeff_init
21	R/W	0x0	td_coeff_rst
20	R/W	0x0	td_coeff_init
19:16	R/W	0x0	td_coeff_addr
6:0	R/W	0x0	td_coeff_frz, td taps enable, bit 0 is td eariliest symbol , bit 10 is current symbol, bit 11~13 is future symbol

Table 8-363 td_leaksize_config1 0x98

Bit(s)	R/W	Default	Description
31:21	R/W	0x92	qstep_mc_64q_dot4
20:16	R/W	0x0	td_leaksize_thr01, 5.5US, downsample 5, 0 is 0, other is 2^-td_leak_thr01, td leaksize in stage1
12:8	R/W	0x0	td_leaksize_thr02, td leaksize in stage2
4:0	R/W	0x0	td_leaksize_thr03, td leaksize in stage2

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Table 8-364 fd_leaksize_config1 0x99

Bit(s)	R/W	Default	Description
31:21	R/W	0x4c7	qstep_mc_04q_dot6
20:16	R/W	0x0	fd_leaksize_thr01, 5.5US, fd leaksize in stage1
12:8	R/W	0x0	fd_leaksize_thr02, fd leaksize in stage2
4:0	R/W	0x0	fd_leaksize_thr03, fd leaksize in stage3

Table 8-365 td_leaksize_config2 0x9a

Bit(s)	R/W	Default	Description
31:21	R/W	0x1c0	qstep_mc_16q_dot6
20:16	R/W	0x0	td_leaksize_thr11, 5.5US, downsample 3, 0 is 0, other is 2^-td_leak_thr01, td leaksize in stage1
12:8	R/W	0x0	td_leaksize_thr12, td leaksize in stage2
4:0	R/W	0x0	td_leaksize_thr13, td leaksize in stage2

Table 8-366 fd_leaksize_config2 0x9b

Bit(s)	R/W	Default	Description
31:21	R/W	0xd3	qstep_mc_64q_dot6
20:16	R/W	0x0	fd_leaksize_thr11, 5.5US, fd leaksize in stage1
12:8	R/W	0x0	fd_leaksize_thr12, fd leaksize in stage2
4:0	R/W	0x0	fd_leaksize_thr13, fd leaksize in stage3

Table 8-367 href_v_mask 0x9c

Bit(s)	R/W	Default	Description
28:16	R/W	0x5	h_ref_v_mask_pos0
12:0	R/W	0xeba	h_ref_v_mask_pos1

Table 8-368 ibdfe_config2 0x9d

Bit(s)	R/W	Default	Description
26:24	R/W	0x4	ibdfe_dmsg_alp
12:0	R/W	0x2	ibdfe_cci_just_thr

Table 8-369 ibdfe_config3 0x9e

Bit(s)	R/W	Default	Description
25:16	R/W	0x3ff	ibdfe_rou_rat_0, 10.0US, rou_ratio for 4QAM or 4QAM_NR in IBDFE
9:0	R/W	0x39a	ibdfe_rou_rat_1, 10.0US, rou_ratio for 16QAM

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Table 8-370 ibdfe_config4 0x9f

Bit(s)	R/W	Default	Description
25:16	R/W	0x333	ibdfe_rou_rat_2, 10.0US, rou_ratio for 32QAM
9:0	R/W	0x2cd	ibdfe_rou_rat_3, 10.0US, rou_ratio for 64QAM

Table 8-371 td_coeff 0xa0

Bit(s)	R/W	Default	Description
26:24	R/W	0x0	td_stepsize_adj, 2^td_stepsize_adj for td tps training
23:0	R/W	0x0	td_coeff

Table 8-372 fd_coeff_frz 0xa1

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	

Table 8-373 fd_coeff 0xa2

Bit(s)	R/W	Default	Description
26:24	R/W	0x0	fd_stepsize_adj, 2^fd_stepsize_adj for fd tps training
23:0	R/W	0x0	fd_coeff

Table 8-374 fd_leaksize 0xa3

Bit(s)	R/W	Default	Description
31:20	R/W	0x400	dcm_pow_avg_init
17:0	R/W	0x0	fd_leaksize

Table 8-375 protect_gd_tps 0xa4

Bit(s)	R/W	Default	Description
28:20	R/W	0x1ea	tm_thr_max4
18:16	R/W	0x2	coh_thd_tps, 3.3US, 2^coh_thd_tps, coh length for tps timing est
14:12	R/W	0x2	ncoh_thd_tps, 3.3US, 2^ncoh_thd_tps, non-coh length for tps timing est
9:0	R/W	0x215	pow_norm, 10.5US, 1/power normalized

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Table 8-376 tps_timingest_config 0xa5

Bit(s)	R/W	Default	Description
31:24	R/W	0x32	qstep_sc_lmt_32q_dot8
23:16	R/W	0x3	tm_switch_num
9:0	R/W	0x28b	path_dist_num, 10'h28a

Table 8-377 ibdfe_config5 0xa6

Bit(s)	R/W	Default	Description
27:16	R/W	0xa	eq_init_snr, 12.12US, init snr for mceq
11:0	R/W	0x64	ibdfe_init_snr, 12.12US, init snr for ibdfe, 100 is 20dB

Table 8-378 ibdfe_config6 0xa7

Bit(s)	R/W	Default	Description
31:28	R/W	0x9	ibdfe_threshold0, 4.3US, ibdfe soft slicer constellation half dist for 4QAM
27:24	R/W	0x4	ibdfe_threshold1, 4.3US, ibdfe soft slicer constellation half dist for 16QAM
23:20	R/W	0x3	ibdfe_threshold2, 4.3US, ibdfe soft slicer constellation half dist for 32QAM
19:16	R/W	0x2	ibdfe_threshold3, 4.3US, ibdfe soft slicer constellation half dist for 64QAM
15:12	R/W	0x2	ibdfe_const_thr0, 4.3US, ibdfe constlelation max value for 4QAM
11:8	R/W	0x8	ibdfe_const_thr1, 16QAM
7:4	R/W	0xc	ibdfe_const_thr2, 32QAM
3:0	R/W	0xc	ibdfe_const_thr3, 64QAM

Table 8-379 ibdfe_config7 0xa8

Bit(s)	R/W	Default	Description
31:24	R/W	0xc0	ibdfe_pick_thr0, 8.0US, 4QAM, ibdfe soft slicer, if eqout-demap < ibdfe_pick_ thr0*ibdfe_threshold, use decision, otherwise, use eqout
23:16	R/W	0x9a	ibdfe_pick_thr1, 16QAM
15:8	R/W	0xa8	ibdfe_pick_thr2, 32QAM
7:0	R/W	0xcd	ibdfe_pick_thr3, 64QAM

Table 8-380 dcm_sc_mc_gd_len 0xa9

Bit(s)	R/W	Default	Description
31:16	R/W	0x8	eq_dsnr_slc2drm, 16.13US, delta snr for drm
13:8	R/W	0x0	dcm_sc_gd_len, 6.6US sc path protect length in DCM
5:0	R/W	0x14	dcm_mc_gd_len, sc path protect length in DCM

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Table 8-381 eqmc_pick_thr 0xaa

Bit(s)	R/W	Default	Description
31:24	R/W	0xc0	eqmc_pick_thr0, 8.0US, 4QAM, MCEQ, soft slicer, if eqout-demap < ibdfe_pick_thr0*ibdfe_threshold, use decision, otherwise, use eqout
23:16	R/W	0xc0	eqmc_pick_thr1, 16QAM
15:8	R/W	0xc0	eqmc_pick_thr2, 32QAM
7:0	R/W	0xc0	eqmc_pick_thr3, 64QAM

Table 8-382 eqmc_threshold 0xab

Bit(s)	R/W	Default	Description
31:28	R/W	0x9	eqmc_threshold0, 4.3us, MCEQ, soft slicer, constellation half dist for 4QAM
27:24	R/W	0x4	eqmc_threshold1, 16QAM
23:20	R/W	0x3	eqmc_threshold2, 32QAM
19:16	R/W	0x2	eqmc_threshold3, 64QAM
15:12	R/W	0x9	eqmc_const_thr0, 4.3US, MCEQ, soft slicer, constlelation max value for 4QAM
11:8	R/W	0xc	eqmc_const_thr1, 16QAM
7:4	R/W	0xf	eqmc_const_thr2, 32QAM
3:0	R/W	0xe	eqmc_const_thr3, 64QAM

Table 8-383 power_thr 0xac

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	pow_thr0
23:16	R/W	0x8f	pow_thr1
15:8	R/W	0x50	pow_thr2
7:0	R/W	0x93	pow_thr3

Table 8-384 eqsc_pick_thr 0xad

Bit(s)	R/W	Default	Description
31:24	R/W	0x80	eqsc_pick_thr0, no use
23:16	R/W	0x9a	eqsc_pick_thr1
15:8	R/W	0xb3	eqsc_pick_thr2
7:0	R/W	0xcd	eqsc_pick_thr3

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Table 8-385 eqsc_threshold 0xae

Bit(s)	R/W	Default	Description
31:28	R/W	0x9	eqsc_threshold0
27:24	R/W	0x4	eqsc_threshold1
23:20	R/W	0x3	eqsc_threshold2
19:16	R/W	0x2	eqsc_threshold3
15:12	R/W	0x9	eqsc_const_thr0
11:8	R/W	0xc	eqsc_const_thr1
7:4	R/W	0xf	eqsc_const_thr2
3:0	R/W	0xe	eqsc_const_thr3

Table 8-386 td_stepsize_thr1 0xaf

Bit(s)	R/W	Default	Description
31:21	R/W	0x120	qstep_mc_32q_dot8
20:16	R/W	0x0	td_stepsize_thr11, For MC mode, 5.5US, downsample 3, 0 is 0, other is 2^-td_stepsize_thr01, td stepsize in stage1
12:8	R/W	0x8	td_stepsize_thr12, td stepsize in stage2
4:0	R/W	0xa	td_stepsize_thr13, td stepsize in stage3

Table 8-387 fd_stepsize_thr1 0xb0

Bit(s)	R/W	Default	Description
31:21	R/W	0x10d	qstep_mc_64q_dot8
20:16	R/W	0x0	fd_stepsize_thr11, 5.5US, fd stepsize in stage1
12:8	R/W	0x7	fd_stepsize_thr12, fd stepsize in stage2
4:0	R/W	0x8	fd_stepsize_thr13, fd stepsize in stage3

Table 8-388 tdfd_switch_sym1 0xb1

Bit(s)	R/W	Default	Description
31:16	R/W	0x12c	tdfd_switch_sym01, 16.16US, downsample 5, tdfd switch to stage3 after tdfd_switch_sym01
15:0	R/W	0x14	tdfd_switch_sym00, 16.16US, downsample 5, tdfd switch to stage2 after tdfd_switch_sym00

Table 8-389 tdfd_switch_sym2 0xb2

Bit(s)	R/W	Default	Description
31:16	R/W	0x12c	tdfd_switch_sym11, 16.16US, downsample 3, same above
15:0	R/W	0x14	tdfd_switch_sym10, 16.16US, downsample 3, same above

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Table 8-390 eq_config 0xb3

Bit(s)	R/W	Default	Description
30	R/W	0x0	snr_avg_freeze
29	R/W	0x0	snr_imp_freeze
28	R/W	0x0	dcm_mc_bypass, if 1, bypass mc H limit in above
27	R/W	0x0	dcm_sc_bypass, if 1, bypass sc H limit in above
26	R/W	0x0	eq_imp_setzero_en, if 1, set eqout and csi to zero if impulse
25	R/W	0x1	eq_cmp_en, if 1, enable commonphase compensation
24:23	R/W	0x0	dcm_alpha, default is 2
22:20	R/W	0x0	eqsnr_avg_alp, 3.3US, 2^-eqsnr_avg_alp is alpha filter coeff for average snr to demap
18:16	R/W	0x2	eqsnr_imp_alp, 3.3US, 2^-eqsnr_imp_alp is alpha filter coeff for average snr to impulse detection
15:12	R/W	0x7	dcm_mc_h_limit, 4.4US, for mc, blank H if H_pow>H_powAvg*2^dcm_mc_h_limit
11:8	R/W	0x7	dcm_sc_h_limit, 4.4US, for sc, blank H if H_pow>H_powAvg*2^dcm_sc_h_limit
5:0	R/W	0x19	eq_dsnr_h2drm, 6.3US, snr_esti+eq_dsnr_h2drm used in H_drm estimation

Table 8-391 eqsc_snr_imp_thr1 0xb4

Bit(s)	R/W	Default	Description
29:24	R/W	0x1c	tps_amp0
23:12	R/W	0x20	eqsc_snr_imp_thr0, 12.10US, for SC,4QAM, snr
11:0	R/W	0x7e	eqsc_snr_imp_thr1, 16QAM

Table 8-392 eqsc_snr_imp_thr2 0xb5

Bit(s)	R/W	Default	Description
29:24	R/W	0x18	tps_amp1
23:12	R/W	0xfc	eqsc_snr_imp_thr2, 32QAM
11:0	R/W	0xfc	eqsc_snr_imp_thr3, 64QAM

Table 8-393 eqmc_snr_imp_thr1 0xb6

Bit(s)	R/W	Default	Description
29:24	R/W	0x12	tps_amp2
23:12	R/W	0x20	eqmc_snr_imp_thr0, 12.10US, for MC,4QAM, snr
11:0	R/W	0x7e	eqmc_snr_imp_thr1, 16QAM

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Table 8-394 eqmc_snr_imp_thr2 0xb7

Bit(s)	R/W	Default	Description
29:24	R/W	0x1	tps_minus_stepsize, tps minus step size configuration
23:12	R/W	0xfc	eqmc_snr_imp_thr2, 32QAM
11:0	R/W	0xfc	eqmc_snr_imp_thr3, 64QAM

Table 8-395 eqsc_snr_drop_thr 0xb8

Bit(s)	R/W	Default	Description
31:24	R/W	0x1f	eqsc_snr_drop_thr0, 8.8US, for SC, 4QAM, , snr
23:16	R/W	0x1f	eqsc_snr_drop_thr1, 16QAM
15:8	R/W	0x1f	eqsc_snr_drop_thr2, 32QAM
7:0	R/W	0x1f	eqsc_snr_drop_thr3, 64QAM

Table 8-396 eqmc_snr_drop_thr 0xb9

Bit(s)	R/W	Default	Description
31:24	R/W	0x5	eqmc_snr_drop_thr0, 8.8US, for MC,4QAM, snr
23:16	R/W	0x5	eqmc_snr_drop_thr1, 16QAM
15:8	R/W	0x5	eqmc_snr_drop_thr2, 32QAM
7:0	R/W	0x5	eqmc_snr_drop_thr3, 64QAM

Table 8-397 m_cci_thr 0xba

Bit(s)	R/W	Default	Description
29:20	R/W	0x1f6	m_cci_thr_ma_tps, 10.9US, for cci ma thresh in tps decoder status, (10^(24/10)) *2
17:8	R/W	0x1f6	m_cci_thr_tps
4:0	R/W	0x2	ccidet_mask_rng_tps

Table 8-398 tps_mc 0xbb

Bit(s)	R/W	Default	Description
30:28	R/W	0x2	tps_mc_alpha, 3.3US, 2^-tps_mc_alpha is used to input alpha filter
26:20	R/W	0x9	tps_mc_q_thr, 7.4US, MC quality thresh
18:12	R/W	0x8	tps_mc_suc_limit, 7.7US, MC successful times
9:0	R/W	0x3e8	tps_mc_run_tim_limit, 10.10US, MC timeout

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Table 8-399 tps_sc 0xbc

Bit(s)	R/W	Default	Description
30:28	R/W	0x2	tps_sc_alpha, 3.3US, 2^-tps_sc_alpha is used to input alpha filter
26:20	R/W	0x14	tps_sc_q_thr, 7.4US, SC quality thresh
18:12	R/W	0x8	tps_sc_suc_limit, 7.7US, SC successful times
9:0	R/W	0x3e8	tps_sc_run_tim_limit, 10.10US, SC timeout

Table 8-400 che_set_fsm 0xbd

Bit(s)	R/W	Default	Description
28	R/W	0x0	che_set_fsm_en, set che fsm state enable
26:24	R/W	0x0	che_set_fsm_st, set che fsm state value
11:0	R/W	0x65	che_open_loop_len, 10.10U, open loop length

Table 8-401 zero_num_thr 0xbe

Bit(s)	R/W	Default	Description
31	R/W	0x0	snr_obs_freeze
30:28	R/W	0x7	snr_obs_alp
27:16	R/W	0xec4	zero_num_thr, max zero number in one frame for demap out
15:0	R/W	0x40	null_frame_thr, max null frame for demap out

Table 8-402 timing_ready 0xbf

Bit(s)	R/W	Default	Description
16	R	0x0	timing_ready, first timing offset ready
10:0	R	0x0	timing_offset, timing offset value

Table 8-403 timing_est_protect 0xc0

Bit(s)	R/W	Default	Description
28:24	R/W	0x1e	tm_protect_gd0
20:16	R/W	0x19	tm_protect_gd1
12:8	R/W	0x1a	tm_protect_gd2
4:0	R/W	0x1a	tm_protect_gd3

Table 8-404 timing_est 0xc1

Bit(s)	R/W	Default	Description
30	R/W	0x1	tm_mc_h_mode
29	R/W	0x1	tm_sc_h_mode

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Bit(s)	R/W	Default	Description
28	R/W	0x1	tm_path_dist_sel
24:20	R/W	0x4	tm_gd_max_pow
15:12	R/W	0x5	tm_fb_delay
11:8	R/W	0x5	tm_snr_adj_sft
4:0	R/W	0x1a	tm_protect_gd4

Table 8-405 0xc2

Bit(s)	R/W	Default	Description
28	R/W	0x1	tps_suc_limit_sel
25:16	R/W	0x20	cci_thr_err
15:0	R/W	0x8	dsnr_href

Table 8-406 0xc3

Bit(s)	R/W	Default	Description
28:0	R/W	0x2560-20	sc_pow

Table 8-407 0xc4

Bit(s)	R/W	Default	Description
25:16	R/W	0x1f6	cci_thr_ma_err
9:0	R/W	0x20	cci_thr_min_err

Table 8-408 0xc5

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	h_ref_v_mask_range0
22:12	R/W	0x480	pn_amp0
10:0	R/W	0x478	pn_amp1

Table 8-409 0xc6

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	h_ref_v_mask_range1
22:12	R/W	0x4bd	pn_amp2
10:0	R/W	0x494	pn_amp3

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Table 8-410 fec_cfg 0x47

Bit(s)	R/W	Default	Description
31	R/W	0x0	fifo_base
30:28	R/W	0x0	tbus_cfg
27:22	R/W	0x0	fec_reg_reset
21	R/W	0x0	time_cfg
20	R/W	0x1	ts_fast
19	R/W	0x0	ts_clk_neg
18	R/W	0x0	bch_off
17:0	R/W	0x33221	ldpc_it_cfg

Table 8-411 fec_debug_cfg 0x48

Bit(s)	R/W	Default	Description
31:25	R/W	0x0	debug_spare,[0]?debug_q:time_di_q [1]time_deint,waiting for job [2] bch debug mode& fec_lock [3]&fifo_overflow: fec_cnt/fec_lock=0
24	R/W	0x1	bad_to_zero
23:21	R/W	0x3	fec_lost_cfg
20:18	R/W	0x4	fec_lock_cfg
17	R/W	0x0	debug_on
16	R/W	0x0	debug_mode
15:0	R/W	0x0	debug_len, 0x16800000

Table 8-412 data_ddr_adr 0x49

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	

Table 8-413 debug_ddr_adr 0x4a

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	

Table 8-414 fec_sim_cfg1 0x4b

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	

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Table 8-415 fec_sim_cfg2 0x4c

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	

Table 8-416 fec_cfg1 0x4d

Bit(s)	R/W	Default	Description
24	R/W	0x0	ts_sync_sel, 0x2ffbe0
23:20	R/W	0x2	ldpc_tracks_first
19:16	R/W	0xf	ldpc_tracks
15	R/W	0x1	ldpc_alpha_en
14:10	R/W	0x1e	ldpc_alpha2
9:5	R/W	0x1f	ldpc_alpha1
4:0	R/W	0x0	ldpc_alpha0

Table 8-417 fec_spct_cfg 0x4e

Bit(s)	R/W	Default	Description
15:12	R/W	0xf	freq_rev_unlock_cfg, 0xf604
11:8	R/W	0x6	freq_sch_cfg
7:0	R/W	0x04	freq_rev_lock_cfg

Table 8-418 testbus_out 0xc8

Bit(s)	R/W	Default	Description
31:0	R		

Table 8-419 tbus_dc_addr 0xc9

Bit(s)	R/W	Default	Description
31:0	R		

Table 8-420 fec_time_sts 0xdb

Bit(s)	R/W	Default	Description
31:0	R		

Table 8-421 fec_ldpc_sts 0xdc

Bit(s)	R/W	Default	Description
31:0	R		

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Table 8-422 fec_ldpc_it_avg 0xdd

Bit(s)	R/W	Default	Description
29	R		fec_freq_rev
28:16	R		fec_ldpc_per_rpt
15:0	R		fec_ldpc_it_avg

Table 8-423 fec_ldpc_unc_acc 0xde

Bit(s)	R/W	Default	Description
31:0	R		

Table 8-424 fec_bch_acc 0xdf

Bit(s)	R/W	Default	Description
31:0	R		

8.1.6 CEC Registers

8.1.6.1 AO_CEC_A Registers

Register Access

Accessing AO_CEC_A registers is by indirectly method through an AO module register.

```
#define      AO_CEC_RW_REG           (0xff800000 + (0x041 << 2))

void aocec_poll_reg_busy (uint32_t reg_busy)
{
    while (((Rd(AO_CEC_RW_REG)) & (1 << 23)) != (reg_busy << 23)) ;
} /* aocec_poll_reg_busy */

void aocec_wr_only_reg (uint32_t addr, uint32_t data)
{
    uint32_t     data32;

    aocec_poll_reg_busy(0);

    data32  = 0;
    data32 |= (1     << 16); // [16]      cec_reg_wr
    data32 |= (data << 8);  // [15:8]    cec_reg_wrdata
    data32 |= (addr << 0);  // [7:0]     cec_reg_addr
    Wr(AO_CEC_RW_REG,    data32);
} /* aocec_wr_only_reg */

uint32_t aocec_rd_reg (uint32_t addr)
{
    uint32_t     data32;

    aocec_poll_reg_busy(0);

    data32  = 0;
    data32 |= (0     << 16); // [16]      cec_reg_wr
    data32 |= (0     << 8);  // [15:8]    cec_reg_wrdata
    data32 |= (addr << 0);  // [7:0]     cec_reg_addr
    Wr(AO_CEC_RW_REG,    data32);

// Wait for some time.

    aocec_poll_reg_busy(0);
```


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```
    data32 = ((Rd(AO_CEC_RW_REG)) >> 24) & 0xff;
    return (data32);
} /* aocec_rd_reg */
```

Register Description

Addr	Name	RW	Function
0x00	TX_MSG_HEADER	RW	TX message Header byte.
0x01	TX_MSG_OPCODE	RW	TX message Op-code.
0x02	TX_MSG_OP1	RW	TX message 1st operand.
0x03	TX_MSG_OP2	RW	TX message 2nd operand.
0x04	TX_MSG_OP3	RW	TX message 3rd operand.
0x05	TX_MSG_OP4	RW	TX message 4th operand.
0x06	TX_MSG_OP5	RW	TX message 5th operand.
0x07	TX_MSG_OP6	RW	TX message 6th operand.
0x08	TX_MSG_OP7	RW	TX message 7th operand.
0x09	TX_MSG_OP8	RW	TX message 8th operand.
0x0A	TX_MSG_OP9	RW	TX message 9th operand.
0x0B	TX_MSG_OP10	RW	TX message 10th operand.
0x0C	TX_MSG_OP11	RW	TX message 11th operand.
0x0D	TX_MSG_OP12	RW	TX message 12th operand.
0x0E	TX_MSG_OP13	RW	TX message 13th operand.
0x0F	TX_MSG_OP14	RW	TX message 14th operand.
0x10	TX_MSG_LENGTH	RW	TX message length.
0x11	TX_MSG_CMD	RW	TX message command.
0x12-0x13	Reserved	RW	
0x14	RX_MSG_CMD	RW	Command for reading RX msg.
0x15	RX_CLEAR_BUF	RW	Clear RX msg buffer.
0x16	LOGICAL_ADDR0	RW	Device logical address .
0x17-0x1A	Reserved	RW	
0x1B	CLOCK_DIV_H	RW	MSB of the clock divider.
0x1C	CLOCK_DIV_L	RW	LSB of the clock divider.
0x1D-0x1F	Reserved	RW	
0x20	QUIESCENT_25MS_BIT7_0	RW	CEC bit timing parameter.
0x21	QUIESCENT_25MS_BIT11_8	RW	CEC bit timing parameter.
0x22	STARTBITMINL2H_3MS5_BIT7_0	RW	CEC bit timing parameter.
0x23	STARTBITMINL2H_3MS5_BIT8	RW	CEC bit timing parameter.

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Addr	Name	RW	Function
0x24	STARTBITMAXL2H_3MS9_BIT7_0	RW	CEC bit timing parameter.
0x25	STARTBITMAXL2H_3MS9_BIT8	RW	CEC bit timing parameter.
0x26	STARTBITMINH_0MS6_BIT7_0	RW	CEC bit timing parameter.
0x27	STARTBITMINH_0MS6_BIT8	RW	CEC bit timing parameter.
0x28	STARTBITMAXH_1MS0_BIT7_0	RW	CEC bit timing parameter.
0x29	STARTBITMAXH_1MS0_BIT8	RW	CEC bit timing parameter.
0x2A	STARTBITMINTOTAL_4MS3_BIT7_0	RW	CEC bit timing parameter.
0x2B	STARTBITMINTOTAL_4MS3_BIT9_8	RW	CEC bit timing parameter.
0x2C	STARTBITMAXTOTAL_4MS7_BIT7_0	RW	CEC bit timing parameter.
0x2D	STARTBITMAXTOTAL_4MS7_BIT9_8	RW	CEC bit timing parameter.
0x2E	LOGIC1MINL2H_0MS4_BIT7_0	RW	CEC bit timing parameter.
0x2F	LOGIC1MINL2H_0MS4_BIT8	RW	CEC bit timing parameter.
0x30	LOGIC1MAXL2H_0MS8_BIT7_0	RW	CEC bit timing parameter.
0x31	LOGIC1MAXL2H_0MS8_BIT8	RW	CEC bit timing parameter.
0x32	LOGIC0MINL2H_1MS3_BIT7_0	RW	CEC bit timing parameter.
0x33	LOGIC0MINL2H_1MS3_BIT8	RW	CEC bit timing parameter.
0x34	LOGIC0MAXL2H_1MS7_BIT7_0	RW	CEC bit timing parameter.
0x35	LOGIC0MAXL2H_1MS7_BIT8	RW	CEC bit timing parameter.
0x36	LOGICMINTOTAL_2MS05_BIT7_0	RW	CEC bit timing parameter.
0x37	LOGICMINTOTAL_2MS05_BIT9_8	RW	CEC bit timing parameter.
0x38	LOGICMAXHIGH_2MS8_BIT7_0	RW	CEC bit timing parameter.
0x39	LOGICMAXHIGH_2MS8_BIT8	RW	CEC bit timing parameter.
0x3A	LOGICERRLOW_3MS4_BIT7_0	RW	CEC bit timing parameter.
0x3B	LOGICERRLOW_3MS4_BIT8	RW	CEC bit timing parameter.

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Addr	Name	RW	Function
0x3C	NOMSMPPOINT_1MS05	RW	CEC bit timing parameter.
0x3D	Reserved	RW	
0x3E	DELCNTR_LOGICERR	RW	CEC bit timing parameter.
0x3F	Reserved	RW	
0x40	TXTIME_17MS_BIT7_0	RW	CEC bit timing parameter.
0x41	TXTIME_17MS_BIT15_8	RW	CEC bit timing parameter.
0x42	TXTIME_2BIT_BIT7_0	RW	CEC bit timing parameter.
0x43	TXTIME_2BIT_BIT15_8	RW	CEC bit timing parameter.
0x44	TXTIME_4BIT_BIT7_0	RW	CEC bit timing parameter.
0x45	TXTIME_4BIT_BIT15_8	RW	CEC bit timing parameter.
0x46	STARTBITNOML2H_3MS7_ BIT7_0	RW	CEC bit timing parameter.
0x47	STARTBITNOML2H_3MS7_ BIT8	RW	CEC bit timing parameter.
0x48	STARTBITNOMH_0MS8_ BIT7_0	RW	CEC bit timing parameter.
0x49	STARTBITNOMH_0MS8_ BIT8	RW	CEC bit timing parameter.
0x4A	LOGIC1NOML2H_0MS6_ BIT7_0	RW	CEC bit timing parameter.
0x4B	LOGIC1NOML2H_0MS6_ BIT8	RW	CEC bit timing parameter.
0x4C	LOGIC0NOML2H_1MS5_ BIT7_0	RW	CEC bit timing parameter.
0x4D	LOGIC0NOML2H_1MS5_ BIT8	RW	CEC bit timing parameter.
0x4E	LOGIC1NOMH_1MS8_ BIT7_0	RW	CEC bit timing parameter.
0x4F	LOGIC1NOMH_1MS8_BIT8	RW	CEC bit timing parameter.
0x50	LOGIC0NOMH_0MS9_ BIT7_0	RW	CEC bit timing parameter.
0x51	LOGIC0NOMH_0MS9_BIT8	RW	CEC bit timing parameter.
0x52	LOGICERRLOW_3MS6_ BIT7_0	RW	CEC bit timing parameter.
0x53	LOGICERRLOW_3MS6_ BIT8	RW	CEC bit timing parameter.
0x54	CHKCONTENTION_0MS1	RW	CEC bit timing parameter.
0x55	Reserved	RW	
0x56	PREPARENXTBIT_0MS05_ BIT7_0	RW	CEC bit timing parameter.
0x57	PREPARENXTBIT_0MS05_ BIT8	RW	CEC bit timing parameter.

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Addr	Name	RW	Function
0x58	NOMSMPACKPOINT_ 0MS45	RW	CEC bit timing parameter.
0x59	Reserved	RW	
0x5A	ACK0NOML2H_1MS5_ BIT7_0	RW	CEC bit timing parameter.
0x5B	ACK0NOML2H_1MS5_BIT8	RW	CEC bit timing parameter.
0x5C-0x5F	Reserved	RW	
0x60	BUGFIX_DISABLE_0	RW	Bug fix control.
0x61	BUGFIX_DISABLE_1	RW	Bug fix control.
0x62-0x7F	Reserved	RW	
0x80	RX_MSG_HEADER	R	RX message Header byte.
0x81	RX_MSG_OPCODE	R	RX message Op-code.
0x82	RX_MSG_OP1	R	RX message 1st operand.
0x83	RX_MSG_OP2	R	RX message 2nd operand.
0x84	RX_MSG_OP3	R	RX message 3rd operand.
0x85	RX_MSG_OP4	R	RX message 4th operand.
0x86	RX_MSG_OP5	R	RX message 5th operand.
0x87	RX_MSG_OP6	R	RX message 6th operand.
0x88	RX_MSG_OP7	R	RX message 7th operand.
0x89	RX_MSG_OP8	R	RX message 8th operand.
0x8A	RX_MSG_OP9	R	RX message 9th operand.
0x8B	RX_MSG_OP10	R	RX message 10th operand.
0x8C	RX_MSG_OP11	R	RX message 11th operand.
0x8D	RX_MSG_OP12	R	RX message 12th operand.
0x8E	RX_MSG_OP13	R	RX message 13th operand.
0x8F	RX_MSG_OP14	R	RX message 14th operand.
0x90	RX_MSG_LENGTH	R	Length in bytes of RX message.
0x91	RX_MSG_STATUS	R	Status of the received message.
0x92	RX_NUM_MSG	R	Number of received messages.
0x93	TX_MSG_STATUS	R	Status of the transmitted message.
0x94-0x9F	Reserved	R	
0xA0	STAT_0_0	R	Internal status register.
0xA1	STAT_0_1	R	Internal status register.
0xA2	STAT_0_2	R	Internal status register.
0xA3	STAT_0_3	R	Internal status register.
0xA4	STAT_1_0	R	Internal status register.

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Addr	Name	RW	Function
0xA5	STAT_1_1	R	Internal status register.
0xA6	STAT_1_2	R	Internal status register.

Table 8-425 TX_MSG_HEADER

Bit	R/W	Default	Description
7:4	RW	0	Initiator Address.
3:0	RW	0	Follower Address.

Table 8-426 TX_MSG_OPCODE

Bit	R/W	Default	Description
7:0	RW	0	Op-code

Table 8-427 TX_MSG_OP1 – TX_MSG_OP14

Bit	R/W	Default	Description
7:0	RW	0	1st – 14th operand.

Table 8-428 TX_MSG_LENTH

Bit	R/W	Default	Description
3:0	RW	0	Tx msg length in bytes, excluding Header byte.

Table 8-429 TX_MSG_CMD

Bit	R/W	Default	Description
1:0	RW	0	Command for transmitting a message. 0=TX_NO_OP. No transaction. 1=TX_REQ_CURRENT. Transmit message in buffer. 2=TX_ABORT. Abort transmit. 3=Same as TX_REQ_CURRENT.

Table 8-430 RX_MSG_CMD

Bit	R/W	Default	Description
1:0	RW	0	Command for reading a received message. 0=RX_NO_OP. No transaction. 1=RX_ACK_CURRENT. Read message in buffer. 2=RX_DISABLE. Disable receiving message. 3=Same as RX_ACK_CURRENT.

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Table 8-431 RX_CLEAR_BUF

Bit	R/W	Default	Description
0	RW	0	Clear message in Rx buffer.

Table 8-432 LOGICAL_ADDR0

Bit	R/W	Default	Description
4	RW	0	1=Device logical address valid. 0=Not valid.
3:0	RW	0	Device logical address

Table 8-433 LOGICAL_ADDR_HI

Bit	R/W	Default	Description
7	RW	1	1=Enable receiving message destined to logical address 15.
6	RW	0	1=Enable receiving message destined to logical address 14.
5	RW	0	1=Enable receiving message destined to logical address 13.
4	RW	0	1=Enable receiving message destined to logical address 12.
3	RW	0	1=Enable receiving message destined to logical address 11.
2	RW	0	1=Enable receiving message destined to logical address 10.
1	RW	0	1=Enable receiving message destined to logical address 9.
0	RW	0	1=Enable receiving message destined to logical address 8.

Note

↓If CEC works in original behaviour, meaning the internal beat rate is once per 0.01ms, then for example: Input clock is 24MHz, then set {CLOCK_DIV_H, CLOCK_DIV_L} = 240-1 = 239, so that the internal beat rate is once per 0.01ms.

If CEC works under input clock = 32768Hz, no clock division, CLOK_DIV_H/L=0.

Table 8-434 CLOCK_DIV_H – CLOCK_DIV_L

Bit	R/W	Default	Description
7:0	RW	0	MSB/LSB of the clock divider.

Table 8-435 QUIESCENT_25MS_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x33	Quiescent_25ms[7:0]

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Table 8-436 QUIESCENT_25MS_BIT11_8

Bit	R/W	Default	Description
3:0	RW	0x3	Quiescent_25ms[11:8]

Table 8-437 STARTBITMINL2H_3MS5_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x70	StartBitMinL2H_3ms5[7:0]

Table 8-438 STARTBITMINL2H_3MS5_BIT8

Bit	R/W	Default	Description
0	RW	0x0	StartBitMinL2H_3ms5[8]

Table 8-439 STARTBITMAXL2H_3MS9_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x7c	StartBitMaxL2H_3ms9[7:0]

Table 8-440 STARTBITMAXL2H_3MS9_BIT8

Bit	R/W	Default	Description
0	RW	0x0	StartBitMaxL2H_3ms9[8]

Table 8-441 STARTBITMINH_0MS6_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x11	StartBitMinH_0ms6[7:0]

Table 8-442 STARTBITMINH_0MS6_BIT8

Bit	R/W	Default	Description
0	RW	0x0	StartBitMinH_0ms6[8]

Table 8-443 STARTBITMAXH_1MS0_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x1d	StartBitMaxH_1ms0[7:0]

Table 8-444 STARTBITMAXH_1MS0_BIT8

Bit	R/W	Default	Description
0	RW	0x0	StartBitMaxH_1ms0[8]

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Table 8-445 STARTBITMINTOTAL_4MS3_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x8a	StartBitMinTotal_4ms3[7:0]

Table 8-446 STARTBITMINTOTAL_4MS3_BIT9_8

Bit	R/W	Default	Description
1:0	RW	0x0	StartBitMinTotal_4ms3[9:8]

Table 8-447 STARTBITMAXTOTAL_4MS7_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x96	StartBitMaxTotal_4ms7[7:0]

Table 8-448 STARTBITMAXTOTAL_4MS7_BIT9_8

Bit	R/W	Default	Description
1:0	RW	0x0	StartBitMaxTotal_4ms7[9:8]

Table 8-449 LOGIC1MINL2H_0MS4_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x0a	Logic1MinL2H_0ms4[7:0]

Table 8-450 LOGIC1MINL2H_0MS4_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Logic1MinL2H_0ms4[8]

Table 8-451 LOGIC1MAXL2H_0MS8_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x16	Logic1MaxL2H_0ms8[7:0]

Table 8-452 LOGIC1MAXL2H_0MS8_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Logic1MaxL2H_0ms8[8]

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Table 8-453 LOGIC0MINL2H_1MS3_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x27	Logic0MinL2H_1ms3[7:0]

Table 8-454 LOGIC0MINL2H_1MS3_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Logic0MinL2H_1ms3[8]

Table 8-455 LOGIC0MAXL2H_1MS7_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x34	Logic0MaxL2H_1ms7[7:0]

Table 8-456 LOGIC0MAXL2H_1MS7_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Logic0MaxL2H_1ms7[8]

Table 8-457 LOGICMINTOTAL_2MS05_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x40	LogicMinTotal_2ms05[7:0]

Table 8-458 LOGICMINTOTAL_2MS05_BIT9_8

Bit	R/W	Default	Description
1:0	RW	0x0	LogicMinTotal_2ms05[9:8]

Table 8-459 LOGICMAXHIGH_2MS8_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x59	LogicMaxHigh_2ms8[7:0]

Table 8-460 LOGICMAXHIGH_2MS8_BIT8

Bit	R/W	Default	Description
0	RW	0x0	LogicMaxHigh_2ms8[8]

Table 8-461 LOGICERRLOW_3MS4_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x6c	LogicErrLow_3ms4[7:0]

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Table 8-462 LOGICERRLOW_3MS4_BIT8

Bit	R/W	Default	Description
0	RW	0x0	LogicErrLow_3ms4[8]

Table 8-463 NOMSMPPOINT_1MS05

Bit	R/W	Default	Description
7:0	RW	0x1f	NomSmpPoint_1ms05[7:0]

Table 8-464 DELCNTR_LOGICERR

Bit	R/W	Default	Description
6:4	RW	0x3	DelCntr_LogicErrHigh[2:0]
2:0	RW	0x5	DelCntr_LogicErrLow[2:0]

Table 8-465 TXTIME_17MS_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x2c	TxTime_17ms[7:0]

Table 8-466 TXTIME_17MS_BIT15_8

Bit	R/W	Default	Description
7:0	RW	0x02	TxTime_17ms[15:8]

Table 8-467 TXTIME_2BIT_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0xa2	TxTime_2Bit[7:0]

Table 8-468 TXTIME_2BIT_BIT15_8

Bit	R/W	Default	Description
7:0	RW	0x0	TxTime_2Bit[15:8]

Table 8-469 TXTIME_4BIT_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x3f	TxTime_4Bit[7:0]

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Table 8-470 TXTIME_4BIT_BIT15_8

Bit	R/W	Default	Description
7:0	RW	0x01	TxTime_4Bit[15:8]

Table 8-471 STARTBITNOML2H_3MS7_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x78	StartBitNomL2H_3ms7[7:0]

Table 8-472 STARTBITNOML2H_3MS7_BIT8

Bit	R/W	Default	Description
0	RW	0x0	StartBitNomL2H_3ms7[8]

Table 8-473 STARTBITNOMH_0MS8_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x19	StartBitNomH_0ms8[7:0]

Table 8-474 STARTBITNOMH_0MS8_BIT8

Bit	R/W	Default	Description
0	RW	0x0	StartBitNomH_0ms8[8]

Table 8-475 LOGIC1NOML2H_0MS6_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x13	Logic1NomL2H_0ms6[7:0]

Table 8-476 LOGIC1NOML2H_0MS6_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Logic1NomL2H_0ms6[8]

Table 8-477 LOGIC0NOML2H_1MS5_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x30	Logic0NomL2H_1ms5[7:0]

Table 8-478 LOGIC0NOML2H_1MS5_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Logic0NomL2H_1ms5[8]

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Table 8-479 LOGIC1NOMH_1MS8_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x3a	Logic1NomH_1ms8[7:0]

Table 8-480 LOGIC1NOMH_1MS8_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Logic1NomH_1ms8[8]

Table 8-481 LOGIC0NOMH_0MS9_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x1d	Logic0NomH_0ms9[7:0]

Table 8-482 LOGIC0NOMH_0MS9_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Logic0NomH_0ms9[8]

Table 8-483 LOGICERRLOW_3MS6_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x75	LogicErrLow_3ms6[7:0]

Table 8-484 LOGICERRLOW_3MS6_BIT8

Bit	R/W	Default	Description
0	RW	0x0	LogicErrLow_3ms6[8]

Table 8-485 CHKCONTENTION_0MS1

Bit	R/W	Default	Description
3:0	RW	0x3	ChkContention_0ms1[7:0]

Table 8-486 PREPARENXTBIT_0MS05_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x05	PrepareNxtBit_0ms05[7:0]

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Table 8-487 PREPARENXTBIT_0MS05_BIT8

Bit	R/W	Default	Description
0	RW	0x0	PrepareNxtBit_0ms05[8]

Table 8-488 NOMSMPACKPOINT_0MS45

Bit	R/W	Default	Description
7:0	RW	0x0f	NomSmpAckPoint_0ms45[7:0]

Table 8-489 ACK0NOML2H_1MS5_BIT7_0

Bit	R/W	Default	Description
7:0	RW	0x2c	Ack0NomL2H_1ms5[7:0]

Table 8-490 ACK0NOML2H_1MS5_BIT8

Bit	R/W	Default	Description
0	RW	0x0	Ack0NomL2H_1ms5[8]

Table 8-491 BUGFIX_DISABLE_0

Bit	R/W	Default	Description
7	RW	0	Bug H: In special case it send a msg with initiator_addr=follower_addr, it should not consider itself a follower, as it should know itself being the sender. 0=Apply the fix. 1=Disable the fix. Original behaviour.
6	RW	0	Bug G: At start bit, cec_rx_monitor's should also check cec_data_out when deciding whether the current transactionit is initiated by itself or not. 0=Apply the fix. 1=Disable the fix. Original behaviour.
5	RW	0	Bug G: cec_tx_frame_composer only update broadcast_msg during IDLE, would cause it to be incorrect if the follower_adr is updated outside the IDLE period. 0=Apply the fix. 1=Disable the fix. Original behaviour.
4	RW	1	Bug E: Line error handling. Before it knows it is a follower, it does not detect line error. The probably is correct behaviour. But I make it an option to detect line error, even before it knows to be a follower. However, this bit is default to be 1, as it is likely not to be a bug. 0=Apply the fix. 1=Disable the fix. Original behaviour.
3	RW	0	Bug D: cec_tx_signal_free_time_ctrl use (bit_start&last_bit) to start counting for observing free time rule. But bit_start is too wide, (it basically is the inversion of the cec_data_in with the exception of start_bit). And last_bit straddles from the previous bit's sampling point to current bit's sampling point. If the previous CEC bit is logic0, then bit_start for this previous CEC bit is extra wide that it also makes (bit_start&last_bit) condition to true, it shifts the free time counter start point, thus making free time in-accurate. 0=Apply the fix. 1=Disable the fix. Original behaviour.
2	RW	0	Bug C: cec_tx_timing_ctrl's status signal msg_tx. If previous transmit has failed, the status should be TX_ERROR. However, a transaction on CEC line will

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Bit	R/W	Default	Description
			change this status to TX_DONE, even if the transaction does not involve our CEC. So if we don't read this status fast enough, we may not safely know whether the last transmit was successful or not. 0=Apply the fix. 1=Disable the fix. Original behaviour.
1	RW	0	Bug B: cec_rx_ser2par's couldn't differentiate the two scenarios as below: 1. a new message following previous message that it is not a follower; 2. subsequent message bytes within a message that it is a follower. So for scenario 1, it treats it as subsequent bytes of the previous message, so it will not update follower address field, therefore won't recognize that itself is to be the follower, so it will not ACK to the sender.. 0=Apply the fix. 1=Disable the fix. Original behaviour.
0	RW	0	Bug A: cec_rx_monitor's should also check cec_data_out when deciding whether the current transactionit is initiated by itself or not. 0=Apply the fix. 1=Disable the fix. Original behaviour.

Table 8-492 BUGFIX_DISABLE_1

Bit	R/W	Default	Description
7:0	RW	0x0	Reserved.

Table 8-493 RX_MSG_HEADER

Bit	R/W	Default	Description
7:4	R	0x0	RX msg Initiator address.
3:0	R	0x0	RX msg Follower address.

Table 8-494 RX_MSG_OPCODE

Bit	R/W	Default	Description
7:0	R	0x0	RX msg Op-code.

Table 8-495 RX_MSG_OP1 - RX_MSG_OP14

Bit	R/W	Default	Description
7:0	R	0x0	RX msg 1st – 14th operand.

Table 8-496 RX_MSG_LENGTH

Bit	R/W	Default	Description
3:0	R	0x0	Length in bytes of Rx msg. Excluding Header byte.

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Table 8-497 RX_MSG_STATUS

Bit	R/W	Default	Description
1:0	R	0x0	Status of the received message. 0=RX_IDLE. No transaction. 2=RX_DONE. Message has been received successfully. 1,3=Reserved.

Table 8-498 RX_NUM_MSG

Bit	R/W	Default	Description
3:0	R	0x0	Number of received messages in Rx buffer.

Table 8-499 TX_MSG_STATUS

Bit	R/W	Default	Description
1:0	R	0x0	Status of the transmitted message. 0=TX_IDLE. No transaction. 1=TX_BUSY. Transmitter is busy. 2=TX_DONE. Message has been successfully transmitted. 3=TX_ERROR. Message has been transmitted with error.

Table 8-500 STAT_0_0

Bit	R/W	Default	Description
7	R	0x0	rx_msg_status[0]. rx_msg_status[1:0]: 0=RX_IDLE. No transaction. 2=RX_DONE. Message has been received succesfully. 1,3=Reserved.
6:4	R	0x0	cec_ifc_rx_state. 0=IDLE. Idle state when reset. 1=WAIT4DATA_VALID. Send Rx buffer write enable. 2=SEND_WRITE_BUF.Send Rx buffer write enable. 3=WAIT4NXT_BYTE. Wait for the next byte to be received. 4=WAIT4SEND_NXT_BYTE. Wait for Tx_frame_composer to request for the next byte. 5=SEND_NXT_BYTE. Send the next byte of the message. 6=WAIT4END_OF_MSG. Wait for the whole message to be received. 7=WAIT4BUF_NOT_FULL.Wait for the Rx buffer to become not full.
3:2	R	0x0	rx_msg_rd_state. 0= IDLE_RD. Idle state when reset and when buffer empty. 1=READ_CUR_MSG. The host is reading the current message. 2=WAIT4NXT_READ. Wait for the next read. 3=Reserved.
1:0	R	0x0	rx_msg_wr_state. 0= IDLE_WR. Idle state when reset and when buffer empty. 1=WRITE_CUR_BYTE. Write to current buffer. 2=WAIT4NXT_WRITE. Wait for the next write. 3=WAIT4NXT_MSG. Current message was received, wait for next message.

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Table 8-501 STAT_0_1

Bit	R/W	Default	Description
7	R	0x0	cec_ifc_tx_state[0]. cec_ifc_tx_state[2:0]: 0=IDLE. Idle state when reset. 1=SEND_ACK_GET_MSG. Send acknowledge to get_msg signal. 2=GET_HEADER. Get message header. 3=WAIT4SEND_NXT_BYTE. Wait for Tx_frame_composer to request for the next byte. 4=SEND_NXT_BYTE. Send the next byte of the message. 5=SEND_DIN_ENB. Send din_enb signal. 6=WAIT4SEND_LAST_BYTE. Send next FIFO read enable. 7=RETRANSMIT_MSG. Wait for retransmit_message low.
6:5	R	0x0	msg_status_tx_cec. 0=TX_IDLE. No transaction. 1=Reserved. 3=TX_DONE. Message has been successfully transmitted. 4=TX_ERROR. Message has been transmitted with error.
4	R	0x0	tx_msg_buf_state. 0=IDLE. Idle state when reset and when buffer empty. 1=TX_ON_GOING. The data is kept in the buffer during the tansmition.
3:1	R	0x0	host_ifc_rx_state. 0=IDLE. Idle state when reset. 1=WAIT4_HOST_ACK. A message was written in the Rx buffer and the host has to acknowledge reading it. 2=WAIT4_HOST_NO_OP. Wait for RX_NO_OP from the host before reading the next message. 3=WAIT4NOT_RX_DONE. Wait for msg_stat_rx_buf to be not RX_DONE. 4=WAIT4ACK_RX_DIS. Wait for CEC interface to acknowledge the recive disable.
0	R	0x0	rx_msg_status[1].

Table 8-502 STAT_0_2

Bit	R/W	Default	Description
7	R	0x0	frame_compose_state[0]. frame_compose_state[3:0]: 0=IDLE. Idle state when reset. 1=SEND_REQUEST. Send request to the arbiter for transimition of a new frame. 2=SEND_START_BIT. If Tx granted tell TX_timing controller to send Start Bit. 3=WAIT4SEND_FIRST_BIT. Wait for signal send_nxt_bit from the TX_timing controller. 4=SEND_DIN_7. Send bit 7 of din bus. 5=SEND_DIN_6. Send bit 6 of din bus. 6=SEND_DIN_5. Send bit 5 of din bus. 7=SEND_DIN_4. Send bit 4 of din bus. 8=SEND_DIN_3. Send bit 3 of din bus. 9=SEND_DIN_2. Send bit 2 of din bus. 10=SEND_DIN_1. Send bit 1 of din bus. 11=SEND_DIN_0. Send bit 0 of din bus. 12=EOM_TRANSMITION. End of Message is transmitted. 13=WAIT4ACK_SEND. Wait for ACK to be sent.
6:4	R	0x0	host_ifc_tx_state. 0=IDLE. Idle state when reset. 1=SEND_SAMP_DATA. Assert samp_data signal. 2=SEND_GET_MSG . Send get_msg to CEC interface. 3=WAIT4RD_NXT_BYTE_1. Increase rd_addr every read_nxt_byte pulse and go to WAIT4RD_NXT_BYTE_0. 4=WAIT4RD_NXT_BYTE_0. If rd_addr_r = msg_length go to WAIT4TX_DONE. 5=WAIT4TX_DONE . Wait for TX_DONE and go to IDLE. 6=SEND_ABORT. Send abort to CEC interface.

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Bit	R/W	Default	Description
			7=CHECK_TX_NO_OP. Wait for TX_NO_OP command from the host.
3:2	R	0x0	tx_msg_status. 0=TX_IDLE. No transaction. 1=TX_BUSY. Transmitter is busy. 2=TX_DONE. Message has been successfully transmitted. 3=TX_ERROR. Message has been transmitted with error.
1:0	R	0x0	cec_ifc_tx_state[2:1].

Table 8-503 STAT_0_3

Bit	R/W	Default	Description
7:6	R	0x0	signal_free_time_state. 0=IDLE. Idle state when reset. 1=FREE_TIME_MEASURE. Start measuring the Signal Free Time. 2=FREE_TIME_MEASURE_END. Wait for end_free_time_req signal.
5:3	R	0x0	tx_arbiter_state. 0=IDLE. Idle state when reset. 1=CHECK_SIGNAL_FREE_TIME. Send request to the arbiter for transimition of a new frame. 2=GRANT_TRANSMITION. If Tx granted tell TX_timing controller to send Start Bit. 3=ABORT_TRANSMITION. If there is a contention on CEC bus abort transmition. 4=END_ARBITRATION. Current device got control of the CEC bus as an initiator.
2:0	R	0x0	frame_compose_state[3:1].

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Table 8-504 STAT_1_0

Bit	R/W	Default	Description
7:6	R	0x0	ack_send_state[1:0]. ack_send_state[2:0]: 0=IDLE_ACK. Idle state when reset. 1=SEND_ACK_LOW. Put the CEC bus on low impedance for a '0', ACK transmition. 2=SEND_ACK_HIGH. Put the CEC bus on high impedance for a '0', ACK transmition. 3=CHECK_LINE_ERROR. Check send_line_error signal again. 4=SEND_LINE_ERROR. Put the CEC bus on low impedance for 3.6 ms to signalize that a CEC line error occured.
5:2	R	0x0	tx_timing_ctrl_state. 0=IDLE. Idle state when reset. 1=WAIT_FOR_GRANT. Wait for grant from the arbiter. 2=SEND_START_BIT_LOW. If Tx granted send Start Bit low. 3=SEND_START_BIT_HIGH. Start Bit high. 4=SEND_1_LOW. Put the CEC bus on low impedance for a '1' transmition. 5=SEND_0_LOW. Put the CEC bus on low impedance for a '0' transmition. 6=SEND_1_HIGH. Put the CEC bus on high impedance for a '1' transmition. 7=SEND_0_HIGH. Put the CEC bus on high impedance for a '0' transmition. 8=CHECK_ACK. Check acknowledge. 9=DATA_BLOCK_OK. Header/Data block was acknowledged, receive the next one. 10=REQUEST_RETRANSMIT. A valid frame was lost and therefore it should be re-transmitted. 11=WAIT4TX_NO_OP. Wait for host to send tx_cmd = TX_NO_OP.
1:0	R	0x0	msg_tx. 0=TX_IDLE. No transaction. 1=Reserved. 2=TX_DONE. Message has been successfully transmitted. 3=TX_ERROR. Message has been transmitted with error.

Table 8-505 STAT_1_1

Bit	R/W	Default	Description
7:6	R	0x0	rx_ser2par_state[1:0]. rx_ser2par_state[3:0]: 0=IDLE. Idle state when reset. 1=RECEIVE_EIGTH_BIT. Receive the eigth data bit. 2=RECEIVE_SEVENTH_BIT. Receive the seventh data bit. 3=RECEIVE_SIXTH_BIT. Receive the sixth data bit. 4=RECEIVE_FIFTH_BIT. Receive the fifth data bit. 5=RECEIVE_FOURTH_BIT. Receive the fourth data bit. 6=RECEIVE_THIRD_BIT. Receive the third data bit. 7=RECEIVE_SECOND_BIT. Receive the second data bit. 8=RECEIVE_FIRST_BIT. Receive the first data bit. 9=RECEIVE_EOM_BIT. Receive the EOM bit. 10=RECEIVE_ACK_BIT. Receive the ACK bit.
5:4	R	0x0	msg_rx. 0=RX_IDLE. No transaction. 1=RX_BUSY. Receiver is busy. 2=RX_DONE. Message has been received succesfully. 3=RX_ERROR. Message has been received with error.

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Bit	R/W	Default	Description
3:1	R	0x0	rx_monitor_state. 0=IDLE. Idle state when reset. 1=CHECK_START_BIT_LOW. Check the start bit low period. 2=CHECK_START_BIT_HIGH. Check the start bit high period. 3=CHECK_BIT_LOW. Check data bit low period. 4=CHECK_BIT_HIGH. Check data bit high period. 5=DELAY_OUTPUT_SIGNAL. The del_cntr counter creates a desired delay for an output signal.
0	R	0x0	ack_send_state[2].

Table 8-506 STAT_1_2

Bit	R/W	Default	Description
1:0	R	0x0	rx_ser2par_state[3:2].

Table 8-507 AO_CEC_GEN_CNTL 0X40

Bit(s)	R/W	Default	Description
31-15	R/W	0	Reserved
14-12	R/W	0	Filter_del. For glitch-filtering CEC line, ignore signal change pulse width < filter_del * T(filter_tick) * 3.
11-10	R/W	0	Reserved
9-8	R/W	0	Filter_tick_sel: Select which periodical pulse for glitch-filtering CEC line signal. 0=Use T(xtal)*3 = 125ns; 1=Use once-per-1us pulse; 2=Use once-per-10us pulse; 3=Use once-per-100us pulse.
7-5	R/W	0	Reserved
4	R/W	0	Together with AO_CECB_GEN_CNTL bit[4], to select whether CEC_A/B to share same pin mux. {AO_CEC_GEN_CNTL[4], AO_CECB_GEN_CNTL[4]}: {0,0} = A and B separate, original behaviour. Use this if pin-mu selects either A-only or B-only. {1,0} = Use this if pin-mux selects A, but also wants to use B through the same pin. {0,1} = Use this if pin-mux selects B, but also wants to use A through the same pin. {1,1} = This value is not allowed.
3	R/W	0	Reserved
2-1	R/W	0	CLK_CNTL. 0=Disable clock; 1=Enable gated clock (normal working mode); 2= Enable free-run clock (Debug mode).
0	R/W	0	SW_RESET. 1=Apply reset; 0=No reset.

CEC Bit Timing Parameters

The following table lists the approprieate values to apply to the timing parameters for two scenarios, respectively.

- Scenario 1, original behaviour, when the internal beat rate is once per 0.01ms;
- Scenario 2, when the input clock = internal beat rate = 32768Hz.

The timing parameter values defaultly apply to scenario 2 – 32768Hz.

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Table 8-508 CEC Bit Timing Parameter Setting

Name	Scen1 100KHz	Scen2 32768Hz	Description
Quiescent_25ms[11:0]	0x9C4	0x333	No more activity, >=25 ms
StartBitMinL2H_3ms5[8:0]	0x15E	0x070	Earliest time for Start Bit L->H, 3.5 ms
StartBitMaxL2H_3ms9[8:0]	0x186	0x07C	Latest time for Start Bit L->H, 3.9 ms
StartBitMinH_0ms6[8:0]	0x03C	0x011	Min Start Bit High pulse width, 0.6 ms
StartBitMaxH_1ms0[8:0]	0x064	0x01D	Max start Bit High pulse width, 1.0 ms
StartBitMinTotal_4ms3[9:0]	0x1AE	0x08A	Min Start Bit total time, 4.3 ms
StartBitMaxTotal_4ms7 [9:0]	0x1D6	0x096	Max Start Bit total time, 4.7 ms
Logic1MinL2H_0ms4[8:0]	0x028	0x00A	Earliest time for logic1 L->H, T1=0.4 ms
Logic1MaxL2H_0ms8[8:0]	0x050	0x016	Latest time for logic1 L->H, T2=0.8 ms
Logic0MinL2H_1ms3[8:0]	0x082	0x027	Earliest time for logic0 L->H, T5=1.3 ms
Logic0MaxL2H_1ms7[8:0]	0x0AA	0x034	Latest time for logic0 L->H, T6=1.7 ms
LogicMinTotal_2ms05[9:0]	0x0CD	0x040	Earliest time for next bit, T7=2.05 ms
LogicMaxHigh_2ms8[8:0]	0x118	0x059	Go to IDLE if High for >=2.8 ms
LogicErrLow_3ms4[8:0]	0x154	0x06C	Assert Low time error when active, >=3.4 ms
NomSmpPoint_1ms05[7:0]	0x69	0x1F	Nominal sample time, 1.05 ms
DelCntr_LogicErrLow[2:0]	0x5	0x5	Initial value for del_cntr on LogicErrLow.
DelCntr_LogicErrHigh[2:0]	0x3	0x3	Initial value for del_cntr on LogicErrHigh.
TxTime_17ms[15:0]	0x06A4	0x022C	17 ms
TxTime_2Bit[15:0]	0x01F0	0x00A2	Time for 7-5 data bit, 4.96 ms
TxTime_4Bit[15:0]	0x03D0	0x013F	Time for 7-3 data bit, 9.76 ms
StartBitNomL2H_3ms7 [8:0]	0x172	0x078	Nominal time for Start Bit L->H, 3.7 ms
StartBitNomH_0ms8[8:0]	0x050	0x019	Nominal Start Bit High pulse width, 0.8 ms
Logic1NomL2H_0ms6[8:0]	0x03C	0x013	Nominal time for logic1 L->H, 0.6 ms
Logic0NomL2H_1ms5[8:0]	0x096	0x030	Nominal time for logic0 L->H, 1.5 ms
Logic1NomH_1ms8[8:0]	0x0B4	0x03A	Nominal Logic1 High pulse width, 1.8 ms
Logic0NomH_0ms9[8:0]	0x05A	0x01D	Nominal Logic0 High pulse width, 0.9 ms
LogicErrLow_3ms6[8:0]	0x168	0x075	Assert Low time error when active, >=3.6 ms
ChkContention_0ms1[3:0]	0xA	0x3	Check for bus contention 0.1ms=100us after H
PrepareNxtBit_0ms05[8:0]	0x005	0x005	Prepare next bit, 0.05ms before end of prev H
NomSmpAckPoint_0ms45 [7:0]	0x2D	0x0F	Nominal sample time 1.05ms=0.6(Logic1NomL2H_ 0ms6)+0.45
Ack0NomL2H_1ms5[8:0]	0x096	0x02C	Nominal time for Ack0 L->H, 1.5 ms

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8.1.6.2 AO_CEC_B Registers

Register Access

Accessing AO_CEC_B registers is by indirectly method through an AO module register.

```
#define      AO_CECB_RW_REG                  (0xff800000 + (0x0a3 << 2))

void aocecb_wr_only_reg (uint32_t addr, uint32_t data)
{
    uint32_t     data32;

    data32  = 0;
    data32 |= (1     << 16); // [16]      cec_reg_wr
    data32 |= (data << 8);  // [15:8]    cec_reg_wrdata
    data32 |= (addr << 0);  // [7:0]     cec_reg_addr
    Wr(AO_CECB_RW_REG,  data32);
} /* aocecb_wr_only_reg */

uint32_t aocecb_rd_reg (,uint32_t addr)
{
    uint32_t     data32;

    data32  = 0;
    data32 |= (0     << 16); // [16]      cec_reg_wr
    data32 |= (0     << 8);  // [15:8]    cec_reg_wrdata
    data32 |= (addr << 0);  // [7:0]     cec_reg_addr
    Wr(AO_CECB_RW_REG,  data32);
    data32 = ((Rd(AO_CECB_RW_REG)) >> 24) & 0xff;

    return (data32);
} /* aocecb_rd_reg */
```

Register Description

Table 8-509 AO_CEC_B Registers

Addr	Name	RW	Function
0x00	CEC_CTRL	RW	CEC control.
0x01	CEC_CTRL2	RW	CEC control.
0x02	CEC_INTR_MASK	RW	Interrupt mask.
0x05	CEC_LADD_LOW	RW	Logical address.
0x06	CEC_LADD_HIGH	RW	Logical address.
0x07	CEC_TX_CNT	RW	Tx byte count.
0x08	CEC_RX_CNT	R	Rx byte count.
0x09	CEC_STAT0	R	Internal status.
0x10	CEC_TX_DATA00	RW	TX byte 0.
0x11	CEC_TX_DATA01	RW	TX byte 1.
0x12	CEC_TX_DATA02	RW	TX byte 2.
0x13	CEC_TX_DATA03	RW	TX byte 3.
0x14	CEC_TX_DATA04	RW	TX byte 4.
0x15	CEC_TX_DATA05	RW	TX byte 5.
0x16	CEC_TX_DATA06	RW	TX byte 6.
0x17	CEC_TX_DATA07	RW	TX byte 7.
0x18	CEC_TX_DATA08	RW	TX byte 8.

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Addr	Name	RW	Function
0x19	CEC_TX_DATA09	RW	TX byte 9.
0x1A	CEC_TX_DATA10	RW	TX byte 10.
0x1B	CEC_TX_DATA11	RW	TX byte 11.
0x1C	CEC_TX_DATA12	RW	TX byte 12.
0x1D	CEC_TX_DATA13	RW	TX byte 13.
0x1E	CEC_TX_DATA14	RW	TX byte 14.
0x1F	CEC_TX_DATA15	RW	TX byte 15.
0x20	CEC_RX_DATA00	R	RX byte 0.
0x21	CEC_RX_DATA01	R	RX byte 1.
0x22	CEC_RX_DATA02	R	RX byte 2.
0x23	CEC_RX_DATA03	R	RX byte 3.
0x24	CEC_RX_DATA04	R	RX byte 4.
0x25	CEC_RX_DATA05	R	RX byte 5.
0x26	CEC_RX_DATA06	R	RX byte 6.
0x27	CEC_RX_DATA07	R	RX byte 7.
0x28	CEC_RX_DATA08	R	RX byte 8.
0x29	CEC_RX_DATA09	R	RX byte 9.
0x2A	CEC_RX_DATA10	R	RX byte 10.
0x2B	CEC_RX_DATA11	R	RX byte 11.
0x2C	CEC_RX_DATA12	R	RX byte 12.
0x2D	CEC_RX_DATA13	R	RX byte 13.
0x2E	CEC_RX_DATA14	R	RX byte 14.
0x2F	CEC_RX_DATA15	R	RX byte 15.
0x30	CEC_LOCK_BUF	RW	CEC lock register.
0x31	CEC_WAKEUPCTRL	RW	Wakeup status.

The AO_CECB_GEN_CNTL register supports multiple addresses, please refer to the original databook for details.

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Table 8-510 CEC_CTRL2

Bit	R/W	Default	Description
6	RW	0	curb_err_init. Bug fix: Do not signal initiator error, when it's myself who pulled down the line when functioning as a follower.
5	RW	0	en_chk_sbitlow. Bug fix – “Receive messages are ignored and not acknowledge” – to resolve bus contention during Start bit, when Tx is sending while Rx is receiving,
4:0	RW	0	rise_del_max. To support CEC line slow rise delay. 0=Support rise delay up to 1*30.5us; 1=Support rise delay up to 2*30.5us; … N=Support rise delay up to (N+1)*30.5us.

Table 8-511 CEC_STAT0

Bit	R/W	Default	Description
6:4	RW	0	p2s_state. 0=TIDLE; 1=SEND_SBIT; Send Start bit. Apply to initiator only. 2=SEND_DBIT; Send Data bit. Apply to initiator only. 3=SEND_EOM; Send End-Of-Message bit. Apply to initiator only. 4=SEND_ACK; ACK bit period. Apply to initiator only. 5=FBACK_ACK; Send ACK bit. Apply to follower only. 6=FBACK_RX_ERR. Signal line error. Apply to follower only.
3:0	R	0	s2p_status. 0x0=IDLE; 0x1=SBITLOWER; Start Bit's Low impedance period. 0x2=SBH; Start Bit's High impedance period. 0x5=L1LOWER; Data bit (logic 0 or 1), the period before one can decide if it's logic 0 or 1. 0x6=SMP1; Detected the data bit is logic 1. 0x7=SMP0; Detected the data bit is logic 0. 0x8=L0H; Logic 0 bit's High impedance period. 0x9=ERRLMIN; Detecting line error. 0xE=ERRLMAX. Line error detected.

8.2 Video Output

8.2.1 Overview

This section describes VPU sub-module, including RDMA sub-module, VIU sub-module, CVBS sub-module.

8.2.2 VPU

VPU is display process unit, the main function is to receive data from decoder/ddr/hdmirx etc, then process the source data in order to get the high-quality video picture, and finally send out the video to the screen by CVBS etc.

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8.2.3 Register Description

8.2.3.1 VPU Registers

Table 8-512 VPU_CRC_CTRL 0x2701

Bit(s)	R/W	Default	Description
10	R/W	0	vpu axi read channel 2 bist enable
9	R/W	0	vpu axi read channel 1 bist enable
8	R/W	0	vpu axi read channel 0 bist enable
4	R/W	0	vpu axi write channel 1 crc start
3	R/W	0	vpu axi write channel 0 crc start
2	R/W	0	vpu axi read channel 2 crc start
1	R/W	0	vpu axi read channel 1 crc start
0	R/W	0	vpu axi read channel 0 crc start

Table 8-513 VPP_RO_CRC0 0x2702

Bit(s)	R/W	Default	Description
31-0	R	0	vpu axi read 0 crc

Table 8-514 VPP_RO_CRC1 0x2703

Bit(s)	R/W	Default	Description
31-0	R	0	vpu axi read 1 crc

Table 8-515 VPP_RO_CRC2 0x2704

Bit(s)	R/W	Default	Description
31-0	R	0	vpu axi read 2 crc

Table 8-516 VPP_RO_CRC3 0x2705

Bit(s)	R/W	Default	Description
31-0	R	0	vpu axi write 0 cmd crc

Table 8-517 VPP_RO_CRC4 0x2706

Bit(s)	R/W	Default	Description
31-0	R	0	vpu axi write 0 data crc

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Table 8-518 VPP_RO_CRC5 0x2707

Bit(s)	R/W	Default	Description
31-0	R	0	vpu axi write 1 cmd crc

Table 8-519 VPP_RO_CRC6 0x2708

Bit(s)	R/W	Default	Description
31-0	R	0	vpu axi write 1 data crc

Table 8-520 VPP_INTF_CTRL 0x270a

Bit(s)	R/W	Default	Description
18	R/W	1	vpu axi write 1 enable
17	R/W	0	vpu axi write 1 gate clock auto
16	R/W	0	vpu axi write 1 disable clock
14	R/W	1	vpu axi write 0 enable
13	R/W	0	vpu axi write 0 gate clock auto
12	R/W	0	vpu axi write 0 disable clock
10	R/W	1	vpu axi read 2 enable
9	R/W	0	vpu axi read 2 gate clock auto
8	R/W	0	vpu axi read 2 disable clock
6	R/W	1	vpu axi read 1 enable
5	R/W	0	vpu axi read 1 gate clock auto
4	R/W	0	vpu axi read 1 disable clock
2	R/W	1	vpu axi read 0 enable
1	R/W	0	vpu axi read 0 gate clock auto
0	R/W	0	vpu axi read 0 disable clock

Table 8-521 VPP_APB_PROT_CTRL 0x270b

Bit(s)	R/W	Default	Description
21-2	R/W	0	hold number
1	R/W	0	protect enable
0	R/W	0	soft reset

Table 8-522 VPP_ENC_ERROR 0x270c

Bit(s)	R/W	Default	Description
31-24	R	0	RO enci error count
23-16	R	0	RO encp error count

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Bit(s)	R/W	Default	Description
15-8	R	0	RO encl error count
2	R/W	1	encl afifo error count enable
1	R/W	1	encp afifo error count enable
0	R/W	1	enci afifo error count enable

Table 8-523 VPU_VDIN_PRE_ARB_CTRL 0x2714

Bit(s)	R/W	Default	Description
31	R/W	0	VDIN_PREARB_SOFT_RESET: 1=SW reset vdin_mmc_pre_arb.
30-18	R	0	Reserved.
17	R/W	1	VDIN_ACG_EN: 1=Enable auto-clock gating of vdin_mmc_pre_arb.
16	R/W	0	VDIN_DISABLE_CLK: 1=Disalbe the clock to vdin_mmc_pre_arb.
15-0	R/W	0xffff	VDIN_REQ_EN: vdin_mmc_pre_arb request enable.

Table 8-524 VPU_VDISP_PRE_ARB_CTRL 0x2715

Bit(s)	R/W	Default	Description
31	R/W	0	VDISP_PREARB_SOFT_RESET: 1=SW reset vdisp_mmc_pre_arb.
30-18	R	0	Reserved.
17	R/W	1	VDISP_ACG_EN: 1=Enable auto-clock gating of vdisp_mmc_pre_arb.
16	R/W	0	VDISP_DISABLE_CLK: 1=Disalbe the clock to vdisp_mmc_pre_arb.
15-0	R/W	0xffff	VDISP_REQ_EN: vdisp_mmc_pre_arb request enable.

Table 8-525 VPU_VPUARB2_PRE_ARB_CTRL 0x2716

Bit(s)	R/W	Default	Description
31	R/W	0	VPUARB2_PREARB_SOFT_RESET: 1=SW reset vpuarb2_mmc_pre_arb.
30-18	R	0	Reserved.
17	R/W	1	VPUARB2_ACG_EN: 1=Enable auto-clock gating of vpuarb2_mmc_pre_arb.
16	R/W	0	VPUARB2_DISABLE_CLK: 1=Disalbe the clock to vpuarb2_mmc_pre_arb.
15-0	R/W	0xffff	VPUARB2_REQ_EN: vpuarb2_mmc_pre_arb request enable.

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Table 8-526 VPU_VIU_VENC_MUX_CTRL 0x271a

Bit(s)	R/W	Default	Description
20	R/W	0	Viu_sync_mode_en 1:Viu_sync_mode_en enable// ENCL/P/I vsync send to both viu1 and viu2 Bit[1:0] select data fbettwen viu1 and ENCL/P/I Bit[5:4] select ctrl singal(vsync) bettwen ENCL/P/I and viu1. Bit[3:2] select data and vsync bettwen viu2 and ENCL/P/I 0:Viu_sync_mode_en disable// Bit[1:0] select data and vsync fbettwen viu1 and ENCL/P/I Bit[3:2] select data and vsync bettwen viu2 and ENCL/P/I Bit[5:4] unuse
17-16	W	0	RASP DPI CLOCK SEL : 00 : cph1 01 : cph2 10/11 : cph3
15:6	W	0	reserved
5-4			VIU1_SEL_VENC_CTRL: Select which one of the encl/P/T/L control singal that Viu1 connects to when viu_sync_model enable.unuse when viu_sync_model disable. 0: ENCL 1: ENCI 2: ENCP 3: ENCT
3-2	R/W	0	VIU2_SEL_VENC: Select which one of the encl/P/T/L that Viu2 connects to (both vsync and data). 0: ENCL 1: ENCI 2: ENCP 3: ENCT
1-0	R/W	0	VIU1_SEL_VENC: Select which one of the encl/P/T/L that Viu1 connects to. Only select data when viu_sync_model enable,select both data and vsync when viu_sync_model disable, 0: ENCL 1: ENCI 2: ENCP 3: ENCT

Table 8-527 VPU_HDMI_SETTING 0x271b

Bit(s)	R/W	Default	Description
15-12	R/W	0	RD_RATE: Read rate to the async FIFO between VENC and HDMI. 0: One read every rd_clk 1: One read every 2 rd_clk 2: One read every 3 rd_clk … 15: One read every 16 rd_clk
11-8	R/W	0	WR_RATE: Write rate to the async FIFO between VENC and HDMI. 0: One write every wr_clk 1: One write every 2 wr_clk 2: One write every 3 wr_clk … 15: One write every 16 wr_clk
7-5	R/W	0	DATA_COMP_MAP: Input data is CrYCr(BRG), map the output data to desired format: 0: output CrYCb (BRG) 1: output YCbCr (RGB) 2: output YCrCb (RBG) 3: output CbCrY (GBR) 4: output CbYCr (GRB) 5: output CrCbY (BGR) 6,7: Reserved

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Bit(s)	R/W	Default	Description
4	R/W	0	INV_DVI_CLK: If true, invert the polarity of clock output to external DVI interface. (NOT internal HDMI).
3	R/W	0	INV_VSYNC: If true, invert the polarity of VSYNC input from VENC
2	R/W	0	INV_HSYNC: If true, invert the polarity of HSYNC input from VENC
1-0	R/W	0	SRC_SEL: Select which HDMI source from between ENCI and ENCP. 2'b00: Disable HDMI source 2'b01: Select ENCI data to HDMI 2'b10: Select ENCP data to HDMI

Table 8-528 ENCI_INFO_READ 0x271c

Bit(s)	R/W	Default	Description
31-29	R	0	Current ENCI field status.
28-25	R	0	Reserved
24-16	R	0	Current ENCI line counter status.
15-11	R	0	Reserved
10-0	R	0	Current ENCI pixel counter status.

Table 8-529 ENCP_INFO_READ 0x271d

Bit(s)	R/W	Default	Description
31-29	R	0	Current ENCP field status.
28-16	R	0	Current ENCP line counter status.
15-13	R	0	Reserved
12-0	R	0	Current ENCP pixel counter status.

Table 8-530 ENCT_INFO_READ 0x271e

Bit(s)	R/W	Default	Description
31-29	R	0	Current ENCT field status.
28-16	R	0	Current ENCT line counter status.
15-13	R	0	Reserved
12-0	R	0	Current ENCT pixel counter status.

Table 8-531 ENCL_INFO_READ 0x271f

Bit(s)	R/W	Default	Description
31-29	R	0	Current ENCL field status.
28-16	R	0	Current ENCL line counter status.
15-13	R	0	Reserved
12-0	R	0	Current ENCL pixel counter status.

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Table 8-532 VPU_SW_RESET 0x2720

Bit(s)	R/W	Default	Description
3	R/W	0	vpuarb2_mmc_arb_rst_n
2	R/W	0	vdisp_mmc_arb_rst_n
1	R/W	0	vdin_mmc_arb_rst_n
0	R/W	0	viu_rst_n

Table 8-533 VPU_D2D3_MMC_CTRL 0x2721

Bit(s)	R/W	Default	Description
30	R/W	0	d2d3_depr_req_sel, 0:vdisp_pre_arb, 1: vpuarb2_pre_arb
27-22	R/W	0x3f	d2d3_depr_brst_num
21-16	R/W	0x2d	d2d3_depr_id
14	R/W	0x0	d2d3_depw_req_sel, 0: vdin_pre_arb, 1: vdisp_pre_arb
11-6	R/W	0x3f	d2d3_depw_brst_num
5-0	R/W	0x2e	d2d3_depw_id

Table 8-534 VPU_CONT_MMC_CTRL 0x2722

Bit(s)	R/W	Default	Description
30	R/W	0x0	mtn_contrd_req_sel, 0:vdisp_post_arb, 1: vpuarb2_pre_arb
27-22	R/W	0x3f	mtn_contrd_brst_num
21-16	R/W	0x2b	mtn_contrd_id
14	R/W	0x0	mtn_contwr_req_sel, 0: vdisp_post_arb, 1: vpuarb2_pre_arb
11-6	R/W	0x3f	mtn_contwr_brst_num
5-0	R/W	0x2c	mtn_contwr_id

Table 8-535 VPU_CLK_GATE 0x2723

Bit(s)	R/W	Default	Description
30-18	R/W	0x0	Reserved
17	R/W	0x1	Clk_b control register no latch
16	R/W	0x1	Clk_vib enable
15	R/W	0x1	Gvapbclk enable
14	R/W	0x1	Clk mpeg vlock enable
13	R/W	0x1	Reserved
12	R/W	0x1	Venc_dac_process_clk enable
11	R/W	0x1	Venc_i_top enable
10	R/W	0x1	Venci_int enable

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Bit(s)	R/W	Default	Description
9:8	R/W	0x11	Clk_vib latch sync source select 00: di2ldim_go_field, 01: post_frame_rst;10: pre_frame_rst, 11:viu_go_field
7	R/W	0x1	Reserved
6	R/W	0x1	Vpu_misc_clk enable
5	R/W	0x1	Venc_l_top enable
4	R/W	0x1	Venc_l_int enable
3	R/W	0x1	Venc_p_int enable
2	R/W	0x1	Reserved
1	R/W	0x1	Vi_top clock enable
0	R/W	0x1	Venc_p_top enable

Table 8-536 VPU_HDMI_DATA_OVR 0x2727

Bit(s)	R/W	Default	Description
31	R/W	0	DATA_OVR_EN: Control if override HDMI input data with DATA_OVR[29:0], for display e.g. black or blue screen. 0: No override 1: Enable override
30	R	0	Reserved
29-0	R/W	0	DATA_OVR: programmable pixel data value for override.

Table 8-537 VPU_VPU_PWM_V0 0x2730

Bit(s)	R/W	Default	Description
31	R/W	0	reg_vpu_pwm_inv, 1: invert the pwm signal, active low
30-29	R/W	0	reg_vpu_pwm_src_sel, 00: encl, enct, encp
28-16	R/W	0	reg_vpu_pwm_v_end0
15-14	R/W	0	reg_vpu_pwm_setting_latch_mode
13	R/W	1	reg_vpu_pwm_vs_inv
12-0	R/W	0	reg_vpu_pwm_v_start0

Table 8-538 VPU_VPU_PWM_V1 0x2731

Bit(s)	R/W	Default	Description
28-16	R/W	0	reg_vpu_pwm_v_end1
12-0	R/W	0	reg_vpu_pwm_v_start1

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Table 8-539 VPU_VPU_PWM_V2 0x2732

Bit(s)	R/W	Default	Description
28-16	R/W	0	reg_vpu_pwm_v_end2
12-0	R/W	0	reg_vpu_pwm_v_start2

Table 8-540 VPU_VPU_PWM_V3 0x2733

Bit(s)	R/W	Default	Description
28-16	R/W	0	reg_vpu_pwm_v_end3
12-0	R/W	0	reg_vpu_pwm_v_start3

Table 8-541 VPU_VPU_PWM_H0 0x2734

Bit(s)	R/W	Default	Description
28-16	R/W	0	reg_vpu_pwm_h_end0
12-0	R/W	0	reg_vpu_pwm_h_start0

Table 8-542 VPU_VPU_PWM_H1 0x2735

Bit(s)	R/W	Default	Description
28-16	R/W	0	reg_vpu_pwm_h_end1
12-0	R/W	0	reg_vpu_pwm_h_start1

Table 8-543 VPU_VPU_PWM_H2 0x2736

Bit(s)	R/W	Default	Description
28-16	R/W	0	reg_vpu_pwm_h_end2
12-0	R/W	0	reg_vpu_pwm_h_start2

Table 8-544 VPU_VPU_PWM_H3 0x2737

Bit(s)	R/W	Default	Description
28-16	R/W	0	reg_vpu_pwm_h_end3
12-0	R/W	0	reg_vpu_pwm_h_start3

Table 8-545 VPU_VPU_3D_SYNC1 0x2738

Bit(s)	R/W	Default	Description
31	R/W	0	reg_3dsync_enable, 1: enable 3d sync output
30	R/W	0	3dsync setting vsync latch
29	R/W	0	3dsync go high field polarity: 1, go high while field[0]=1

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Bit(s)	R/W	Default	Description
28-16	R/W	0	reg_3dsync_v_end0
15	R/W	0	3dsync out inv
14	R/W	0	3dsync vbo out inv
13	R/W	1	Vbo 3d en, to v by one, 3d enable
12-0	R/W	0	reg_3dsync_v_start0

Table 8-546 VPU_VPU_3D_SYNC2 0x2739

Bit(s)	R/W	Default	Description
31	R/W	0	Reg_3dsync_field_bit_sel: 1. Keep 3dsync not changed for two fields, i.e. L-L-R-R-L-L-R-R (11001100) 0. Change 3dsync every field i.e. L-R-L-R (1010)
28-16	R/W	0	reg_3dsync_h_end
12-0	R/W	0	reg_3dsync_h_start

Table 8-547 VPU_HDMI_FMT_CTRL 0x2743

Bit(s)	R/W	Default	Description
12 bit to 10 bit dither control register. 10 bit to 8 bit see VPU_HDMI_DITH_CNTL			
21-19	R/W	0	rame count offset for B
18-16	R/W	0	frame count offset for G
15	R/W	0	hcnt hold when de valid
14	R/W	0	RGB frame count seperate
13	R/W	0	dith4x4 : frame random enable
12	R/W	0	dith4x4 enable
11	R/W	0	tunnel enable for DOLBY
10	R/W	0	rounding enable
9-6	R/W	0	Cntl_hdmi_dith10 :
5	R/W	0	Cntl_hdmi_dith_md:
4	R/W	0	Cntl_hdmi_dith_en: dither 10-b to 8-b enable
3-2	R/W	0	Cntl_chroma_dnsmp: Chroma down sample mode when convert to 422 or 420. 0 = use pixel 0; 1 = use pixel 1; 2 = use average;
1-0	R/W	0	Cntl_hdmi_vid_fmt: Control whether to convert ENCP's 444 data to 422 or 420 0 = No convertion; 1 = Convert to 422; 2 = Convert to 420;

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Table 8-548 VPU_VDIN_ASYNC_HOLD_CTRL 0x2744

Bit(s)	R/W	Default	Description
31-24	R/W	‘h18	Wr_hold_num
23-16	R/W	‘h10	Wr_rel_num
15-8	R/W	‘h18	Rd_hold_num
7-0	R/W	‘h10	Rd_rel_num

Table 8-549 VPU_VDISP_ASYNC_HOLD_CTRL 0x2745

Bit(s)	R/W	Default	Description
31-24	R/W	‘h18	Wr_hold_num
23-16	R/W	‘h10	Wr_rel_num
15-8	R/W	‘h18	Rd_hold_num
7-0	R/W	‘h10	Rd_rel_num

Table 8-550 VPU_VPUARB2_ASYNC_HOLD_CTRL 0x2746

Bit(s)	R/W	Default	Description
31-24	R/W	‘h18	Wr_hold_num
23-16	R/W	‘h10	Wr_rel_num
15-8	R/W	‘h18	Rd_hold_num
7-0	R/W	‘h10	Rd_rel_num

Table 8-551 VPU_ARB_URG_CTRL 0x2747

Bit(s)	R/W	Default	Description
11	R/W	0	Rdma_ddr_reg_busy to vpuarb2_urg_ctrl
10	R/W	0	Rdma_ddr_reg_busy to vdisp_urg_ctrl
9	R/W	0	Rdma_ddr_reg_busy to vdin_urg_ctrl
8	R/W	0	Vdin1_lff_urg_ctrl to vpuarb2_urg_ctrl
7	R/W	0	Vdin0_lff_urg_ctrl to vpuarb2_urg_ctrl
6	R/W	0	Vpp_off_urg_ctrl to vpuarb2_urg_ctrl
5	R/W	0	Vdin1_lff_urg_ctrl to vdisp_urg_ctrl
4	R/W	0	Vdin0_lff_urg_ctrl to vdisp_urg_ctrl
3	R/W	0	Vpp_off_urg_ctrl to vdisp_urg_ctrl
2	R/W	0	Vdin1_lff_urg_ctrl to vdin_urg_ctrl
1	R/W	0	Vdin0_lff_urg_ctrl to vdin_urg_ctrl
0	R/W	0	Vpp_off_urg_ctrl to vdin_urg_ctrl

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Table 8-552 VPU_VENCL_DITH_EN 0x2749

R/W	Default	Description
R/W	0	dith_en //dith bettwen vpp and encl_int
R/W	0	dith hsize

Table 8-553 VPU_422TO444_RST 0x274a

Bit(s)	R/W	Default	Description
3	R/W	0	Change output of viu_12bit422to10bit444_vd1 to 10bits 1:cut high 2 bits 0:cut low 2 bits
2	R/W	0	Change output of viu_12bit422to10bit444_encp to 10bits 1:cut high 2 bits 0:cut low 2 bits
1	R/W	0	Soft rst enable of viu_12bit422to10bit444_vd1 module active high
0	R/W	0	Soft rst enable of viu_12bit422to10bit444_encp module active high

Table 8-554 VPU_422TO444_CTRL0 0x274b

Bit(s)	R/W	Default	Description
31	R/W	0	bypass_mode : active high if this bit set high ,viu_12bit422to10bit444_encp work in bypass mode, dout = din
30	/	/	reversed
29	R/W	0	clip10_mode: active high if this bit set high ,viu_12bit422to10bit444_encp work clip10_mode ,output high 10bits value of din and clip data to 10bit,then expand to 12bits according clip_ lend.
28	R/W	0	Clip8_mode: active high if this bit set high ,viu_12bit422to10bit444_encp work clip8_mode ,output high 8bits value of din and clip data to 8bit,then expand to 12bits according clip_lend.
27	R/W	0	clip_lend :active high make output data to 12bits in clip10_mode/clip8_mode/scramble 1;expand bits in high bits 0: ;expand bits low bits
26	R/W	0	scramble_mode :active high viu_12bit422to10bit444_encp module get luma and chroma from 422 format data erery pixel ,then reorginize data to 8bits according to reg_tunnel value(this model can change data from 422 to 444??). Luma = din[35:24] Chroma =odd_pixel?din[23:12]:din[11:0]
25	R/W	0	go_field_en 1:rst odd_pixel to 0 when go_filed come
24	R/W	0	go_line_en 1:rst rst odd_pixel to 0 when go_line come
23	R/W	0	oft_rst_en 1:soft rst rst odd_pixel to 0
22	R/W	0	de_sel : input data de singal 1:chose encp require singal as de 0:close this module
17:15	R/W	0	reg_tunnel_sel_b1 : select high 4 bits for B from {luma,chroma} ,only work in scramble mode: 0:bit[3:0] 1:bit[7:3] 2:bit[11:8] 3:bit[15:12]

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Bit(s)	R/W	Default	Description
			4:bit[19:16] 5:bit[23:20] Default:bit[23:20]
14:12	R/W	0	reg_tunnel_sel_g1 : select high 4 bits for G from {luma,chroma} ,only work in scramble mode: 0:bit[3:0] 1:bit[7:3] 2:bit[11:8] 3:bit[15:12] 4:bit[19:16] 5:bit[23:20] Default:bit[23:20]
11:9	R/W	0	reg_tunnel_sel_r1 : select high 4 bits for R from {luma,chroma} ,only work in scramble mode: 0:bit[3:0] 1:bit[7:3] 2:bit[11:8] 3:bit[15:12] 4:bit[19:16] 5:bit[23:20] Default:bit[23:20]
8:6	R/W	0	reg_tunnel_sel_B0 : select low 4 bits for B from {luma,chroma} ,only work in scramble mode: 0:bit[3:0] 1:bit[7:3] 2:bit[11:8] 3:bit[15:12] 4:bit[19:16] 5:bit[23:20] Default:bit[23:20]
5:3	R/W	0	reg_tunnel_sel_g0 : select low 4 bits for G from {luma,chroma} ,only work in scramble mode: 0:bit[3:0] 1:bit[7:3] 2:bit[11:8] 3:bit[15:12] 4:bit[19:16] 5:bit[23:20] Default:bit[23:20]
2:0	R/W	0	reg_tunnel_sel_r0 : select low 4 bits for R from {luma,chroma} ,only work in scramble mode: 0:bit[3:0] 1:bit[7:3] 2:bit[11:8] 3:bit[15:12] 4:bit[19:16] 5:bit[23:20] Default:bit[23:20]

VPU_422TO444_CTRL1 0x274c Same as VPU_422TO444_CTRL0,viu_12bit422to10bit444_vd1

Table 8-555 VPU_VIU2VDIN_HDN_CTRL 0x2780

Bit(s)	R/W	Default	Description
20	R/W	0	software reset
19-18	R/W	0	reg_viu2vdin_dn_ratio: down-scale ratio: 0->no scale; 1-> 1/2; 2->1/4; 3->reserved
17-16	R/W	0	reg_viu2vdin_flt_mode: filter mode; 0->no filter; 1->[0 2 2 0]/4; 2->[1 1 1 1]/4; 3->[1 3 3 1]/8

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Bit(s)	R/W	Default	Description
15-14	R/W	0	reserved
13-0	R/W	0	reg_viu2vdin_hsize: source horizontal size

Table 8-556 VPU_VIU_ASYNC_MASK 0x2781

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:24	R/W	0	VDIN_STTs_data => VPP
23:20	R/W	0	Nr/di -> ldim
19-18	R/W	0	reg_viu2vdin_dn_ratio: down-scale ratio: 0->no scale; 1-> 1/2; 2->1/4; 3->reserved
11:8	R/W	0	Vpp afbc-> di pre inp
7:4	R/W	0	Mask vd1-> di post input
3:0	R/W	0	Mask di post => VPP_VD1_IN

Table 8-557 VPU_VDIN_MISC_CTRL 0x2782

Bit(s)	R/W	Default	Description
31:26	R/W	0	reserved
25-24	R/W	0	afbce gate clock ctrl
23	R/W	0	afbce_go_field & afbce_go_line select : 0:vdin 1:di
22	R/W	0	vdin_go_field & vdin_go_line select : 0:vdin0 1:vdin1
21	R/W	0	vdin0 output to afbc enc
20	R/W	0	vdin0 output to write mif
19	R/W	0	vdin0 output to mif enable
18	R/W	0	vdin1 output to afbc enc
17	R/W	0	vdin1 output to write mif
16	R/W	0	vdin1 output to mif enable
15	R/W	0	vdin line int enable
14-11	R/W	0	vdin arbiter gate clock ctrl
10	R/W	0	ldim stts data select : 0: vdin0 1:vdin1
9-8	R/W	0	vdin1 write mif fix disable
7-6	R/W	0	vdin0 write mif fix disable
4	R/W	0	Vdin1_wr_rst_n
3	R/W	0	Vdin0_wr_rst_n
2	R/W	0	Nrin_mux_wr_rst_n
1	R/W	0	Vdin1_rst_n
0	R/W	0	Vdin0_rst_n

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Table 8-558 VPU_VIU_VDIN_IF_MUX_CTRL 0x2783

Bit(s)	R/W	Default	Description
28-24	R/W	0	//Select VIU to “VDIN0/VDIN1 source 9” data path , must clear it first before changing the path sel //5’b00000= Disable VIU to VDIN path //5’b00001= Enble VIU of ENC_I domain to VDIN //5’b00010= Enble VIU of ENC_P domain to VDIN //5’b00100= Enble VIU of ENC_T domain to VDIN //5’b01000= Enble VIU WriteBack 1 domain to VDIN //5’b10000= Enble VIU WriteBack 2 domain to VDIN
20-16	R/W	0	//Select VIU Clock to “VDIN0/VDIN1 source 9” clk path , must clear it first before changing the path sel //5’b00000= Disable VIU to VDIN clock //5’b00001= Enble clock VIU of ENC_I domain to VDIN //5’b00010= Enble clock VIU of ENC_P domain to VDIN //5’b00100= Enble clock VIU of ENC_T domain to VDIN //5’b01000= Enble clock VIU WriteBack 1 domain to VDIN //5’b10000= Enble clock VIU WriteBack 2 domain to VDIN
12-8	R/W	0	//Select VIU to “VDIN0/VDIN1 source 7” data path , must clear it first before changing the path sel //5’b00000= Disable VIU to VDIN clock //5’b00001= Enble clock VIU of ENC_I domain to VDIN //5’b00010= Enble clock VIU of ENC_P domain to VDIN //5’b00100= Enble clock VIU of ENC_T domain to VDIN //5’b01000= Enble clock VIU WriteBack 1 domain to VDIN //5’b10000= Enble clock VIU WriteBack 2 domain to VDIN
4-0	R/W	0	//Select Clock to “VDIN0/VDIN1 source 7” , must clear it first before changing the path sel //5’b00000= Disable VIU to VDIN clock //5’b00001= Enble clock VIU of ENC_I domain to VDIN //5’b00010= Enble clock VIU of ENC_P domain to VDIN //5’b00100= Enble clock VIU of ENC_T domain to VDIN //5’b01000= Enble clock VIU WriteBack 1 domain to VDIN //5’b10000= Enble clock VIU WriteBack 2 domain to VDIN

Table 8-559 VPU_VIU2VDIN1_HDN_CTRL 0x2784

Bit(s)	R/W	Default	Description
20	R/W	0	software reset
19-18	R/W	0	reg_viu2vdin2_dn_ratio: down-scale ratio: 0->no scale; 1-> 1/2; 2->1/4; 3->reserved
17-16	R/W	0	reg_viu2vdin2_flt_mode: filter mode; 0->no filter; 1->[0 2 2 0]/4; 2->[1 1 1 1]/4; 3->[1 3 3 1]/8
15-14	R/W	0	reserved
13-0	R/W	0	reg_viu2vdin2_hsize: source horizontal size

Table 8-560 VPU_VENCX_CLK_CTRL 0x2785

Bit(s)	R/W	Default	Description
2	R/W	0	Enci_afifo_clk: 0: cts_vpu_clk_tm 1: cts_vpu_clkc_tm
1	R/W	0	Encl_afifo_clk: 0: cts_vpu_clk_tm 1: cts_vpu_clkc_tm
0	R/W	0	Encp_afifo_clk: 0: cts_vpu_clk_tm 1: cts_vpu_clkc_tm

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Table 8-561 VPU_RDARB_MODE_L1C1 0x2790

Bit(s)	R/W	Default	Description
21:16	R/W	0	rdarb_sel : uns, default = 0 , rdarb_sel [0]==0 slave dc0 connect master port0 rdarb_sel[0]==1 slave dc0 connect master port1 rdarb_sel [1]==0 slave dc1 connect master port0 rdarb_sel[1]==1 slave dc1 connect master port1 rdarb_sel [2]==0 slave dc2 connect master port0 rdarb_sel[2]==1 slave dc2 connect master port1 rdarb_sel [3]==0 slave dc3 connect master port0 rdarb_sel[3]==1 slave dc3 connect master port1 rdarb_sel [4]==0 slave dc4 connect master port0 rdarb_sel[4]==1 slave dc4 connect master port1 rdarb_sel [5]==0 slave dc5 connect master port0 rdarb_sel[5]==1 slave dc5 connect master port1
9:8	R/W	0	rdarb_arb_mode : uns, default = 0 , rdarb_arb_mode [0] master port0 arb way, rdarb_arb_mode [1] master port1 arb way,
3:0	R/W	0	rdarb_gate_clk_ctrl : uns, default = 0 , rdarb_gate_clk_ctrl [1:0] master port0 clk gate control rdarb_gate_clk_ctrl [3:2] master port1 clk gate control

Table 8-562 VPU_RDARB_REQEN_SLV_L1C1 0x2791

Bit(s)	R/W	Default	Description
11:0	R/W	0xfff	rdarb_dc_req_en : unsigned , default = 12'hfff rdarb_dc_req_en [0]: the slv0 req to mst port0 enable, rdarb_dc_req_en [1]: the slv1 req to mst port0 enable, rdarb_dc_req_en [2]: the slv2 req to mst port0 enable, rdarb_dc_req_en [3]: the slv3 req to mst port0 enable, rdarb_dc_req_en [4]: the slv4 req to mst port0 enable, rdarb_dc_req_en [5]: the slv5 req to mst port0 enable, rdarb_dc_req_en [6]: the slv0 req to mst port1 enable, rdarb_dc_req_en [7]: the slv1 req to mst port1 enable, rdarb_dc_req_en [8]: the slv2 req to mst port1 enable, rdarb_dc_req_en [9]: the slv3 req to mst port1 enable, rdarb_dc_req_en [10]: the slv4 req to mst port1 enable, rdarb_dc_req_en [11]: the slv5 req to mst port1 enable,

Table 8-563 VPU_RDARB_WEIGH0_SLV_L1C1 0x2792

Bit(s)	R/W	Default	Description
29:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [0*6+:6]: the slv0 req weigh number rddc_weigh_sxn [1*6+:6]: the slv1 req weigh number rddc_weigh_sxn [2*6+:6]: the slv2 req weigh number rddc_weigh_sxn [3*6+:6]: the slv3 req weigh number rddc_weigh_sxn [4*6+:6]: the slv4 req weigh number

Table 8-564 VPU_RDARB_WEIGH1_SLV_L1C1 0x2793

Bit(s)	R/W	Default	Description
5:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [5*6+:6]: the slv5 req weigh number

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Table 8-565 VPU_WRARB_MODE_L1C1 0x2794

Bit(s)	R/W	Default	Description
21:16	R/W	0	wrarb_sel : uns, default = 0 , wrarb_sel [0]==0 slave dc0 connect master port0 wrarb_sel[0]==1 slave dc0 connect master port1 wrarb_sel [1]==0 slave dc1 connect master port0 wrarb_sel[1]==1 slave dc1 connect master port1 wrarb_sel [2]==0 slave dc2 connect master port0 wrarb_sel[2]==1 slave dc2 connect master port1 wrarb_sel [3]==0 slave dc3 connect master port0 wrarb_sel[3]==1 slave dc3 connect master port1 wrarb_sel [4]==0 slave dc4 connect master port0 wrarb_sel[4]==1 slave dc4 connect master port1 wrarb_sel [5]==0 slave dc5 connect master port0 wrarb_sel[5]==1 slave dc5 connect master port1
9:8	R/W	0	wrarb_arb_mode : uns, default = 0 , wrarb_arb_mode [0] master port0 arb way, wrarb_arb_mode [1] master port1 arb way,
3:0	R/W	0	wrarb_gate_clk_ctrl : uns, default = 0 , wrarb_gate_clk_ctrl [1:0] master port0 clk gate control wrarb_gate_clk_ctrl [3:2] master port1 clk gate control

Table 8-566 VPU_WRARB_REQEN_SLV_L1C1 0x2795

Bit(s)	R/W	Default	Description
11:0	R/W	0	wrarb_dc_req_en : unsigned , default = 0 wrarb_dc_req_en [0]: the slv0 req to mst port0 enable, wrarb_dc_req_en [1]: the slv1 req to mst port0 enable, wrarb_dc_req_en [2]: the slv2 req to mst port0 enable, wrarb_dc_req_en [3]: the slv3 req to mst port0 enable, wrarb_dc_req_en [4]: the slv4 req to mst port0 enable, wrarb_dc_req_en [5]: the slv5 req to mst port0 enable, wrarb_dc_req_en [0]: the slv0 req to mst port1 enable, wrarb_dc_req_en [1]: the slv1 req to mst port1 enable, wrarb_dc_req_en [2]: the slv2 req to mst port1 enable, wrarb_dc_req_en [3]: the slv3 req to mst port1 enable, wrarb_dc_req_en [4]: the slv4 req to mst port1 enable, wrarb_dc_req_en [5]: the slv5 req to mst port1 enable,

Table 8-567 VPU_WRARB_WEIGH0_SLV_L1C1 0x2796

Bit(s)	R/W	Default	Description
29:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 wrdc_weigh_sxn [0*6+:6]: the slv0 req weigh number wrdc_weigh_sxn [1*6+:6]: the slv1 req weigh number wrdc_weigh_sxn [2*6+:6]: the slv2 req weigh number wrdc_weigh_sxn [3*6+:6]: the slv3 req weigh number wrdc_weigh_sxn [4*6+:6]: the slv4 req weigh number

Table 8-568 VPU_WRARB_WEIGH1_SLV_L1C1 0x2797

Bit(s)	R/W	Default	Description
5:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 wrdc_weigh_sxn [5*6+:6]: the slv5 req weigh number

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Table 8-569 VPU_RDWR_ARB_STATUS_L1C1 0x2798

Bit(s)	R/W	Default	Description
3:2	R/W	0	wrarb_arb_busy : unsigned , default = 0
1:0	R/W	0	rdarb_arb_busy : unsigned , default = 0

Table 8-570 VPU_RDARB_MODE_L1C2 0x2799

Bit(s)	R/W	Default	Description
20:16	R/W	0	rdarb_sel : uns, default = 0 , rdarb_sel [0]==0 slave dc0 connect master port0 rdarb_sel[0]==1 slave dc0 connect master port1 rdarb_sel [1]==0 slave dc1 connect master port0 rdarb_sel[1]==1 slave dc1 connect master port1 rdarb_sel [2]==0 slave dc2 connect master port0 rdarb_sel[2]==1 slave dc2 connect master port1 rdarb_sel [3]==0 slave dc3 connect master port0 rdarb_sel[3]==1 slave dc3 connect master port1 rdarb_sel [4]==0 slave dc4 connect master port0 rdarb_sel[4]==1 slave dc4 connect master port1
9:8	R/W	0	rdarb_arb_mode : uns, default = 0 , rdarb_arb_mode [0] master port0 arb way, rdarb_arb_mode [1] master port1 arb way,
3:0	R/W	0	rdarb_gate_clk_ctrl : uns, default = 0 , rdarb_gate_clk_ctrl [1:0] master port0 clk gate control rdarb_gate_clk_ctrl [3:2] master port0 clk gate control

Table 8-571 VPU_RDARB_REQEN_SLV_L1C2 0x279a

Bit(s)	R/W	Default	Description
9:0	R/W	0	rdarb_dc_req_en : unsigned , default = 0 rdarb_dc_req_en [0]: the slv0 req to mst port0 enable, rdarb_dc_req_en [1]: the slv1 req to mst port0 enable, rdarb_dc_req_en [2]: the slv2 req to mst port0 enable, rdarb_dc_req_en [3]: the slv3 req to mst port0 enable, rdarb_dc_req_en [4]: the slv4 req to mst port0 enable, rdarb_dc_req_en [5]: the slv0 req to mst port1 enable, rdarb_dc_req_en [6]: the slv1 req to mst port1 enable, rdarb_dc_req_en [7]: the slv2 req to mst port1 enable, rdarb_dc_req_en [8]: the slv3 req to mst port1 enable, rdarb_dc_req_en [9]: the slv4 req to mst port1 enable,

Table 8-572 VPU_RDARB_WEIGH0_SLV_L1C2 0x279b

Bit(s)	R/W	Default	Description
29:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [0*6+:6]: the slv0 req weigh number rddc_weigh_sxn [1*6+:6]: the slv1 req weigh number rddc_weigh_sxn [2*6+:6]: the slv2 req weigh number rddc_weigh_sxn [3*6+:6]: the slv3 req weigh number rddc_weigh_sxn [4*6+:6]: the slv4 req weigh number

Table 8-573 VPU_RDWR_ARB_STATUS_L1C2 0x279c

Bit(s)	R/W	Default	Description
1:0	R/W	0	rdarb_arb_busy : unsigned , default = 0

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Table 8-574 VPU_RDARB_MODE_L2C1 0x279d

Bit(s)	R/W	Default	Description
27:16	R/W	0	rdarb_sel : uns, default = 0 , rdarb_sel [0]==0 slave dc0 connect master port0 rdarb_sel[0]==1 slave dc0 connect master port1 rdarb_sel [1]==0 slave dc1 connect master port0 rdarb_sel[1]==1 slave dc1 connect master port1 rdarb_sel [2]==0 slave dc2 connect master port0 rdarb_sel[2]==1 slave dc2 connect master port1 rdarb_sel [3]==0 slave dc3 connect master port0 rdarb_sel[3]==1 slave dc3 connect master port1 rdarb_sel [4]==0 slave dc4 connect master port0 rdarb_sel[4]==1 slave dc4 connect master port1 rdarb_sel [5]==0 slave dc5 connect master port0 rdarb_sel[5]==1 slave dc5 connect master port1
10:8	R/W	0	rdarb_arb_mode : uns, default = 0 , rdarb_arb_mode [0] master port0 arb way, rdarb_arb_mode [1] master port1 arb way,
5:0	R/W	0	rdarb_gate_clk_ctrl : uns, default = 0 , rdarb_gate_clk_ctrl [1:0] master port0 clk gate control rdarb_gate_clk_ctrl [3:2] master port1 clk gate control rdarb_gate_clk_ctrl [5:4] master port2 clk gate control

Table 8-575 VPU_RDARB_REQEN_SLV_L2C1 0x279e

Bit(s)	R/W	Default	Description
17:0	R/W	0	rdarb_dc_req_en : unsigned , default = 0 rdarb_dc_req_en [0]: the slv0 req to mst port0 enable, rdarb_dc_req_en [1]: the slv1 req to mst port0 enable, rdarb_dc_req_en [2]: the slv2 req to mst port0 enable, rdarb_dc_req_en [3]: the slv3 req to mst port0 enable, rdarb_dc_req_en [4]: the slv4 req to mst port0 enable, rdarb_dc_req_en [5]: the slv5 req to mst port0 enable, rdarb_dc_req_en [0]: the slv0 req to mst port1 enable, rdarb_dc_req_en [1]: the slv1 req to mst port1 enable, rdarb_dc_req_en [2]: the slv2 req to mst port1 enable, rdarb_dc_req_en [3]: the slv3 req to mst port1 enable, rdarb_dc_req_en [4]: the slv4 req to mst port1 enable, rdarb_dc_req_en [5]: the slv5 req to mst port1 enable,

Table 8-576 VPU_RDARB_WEIGH0_SLV_L2C1 0x279f

Bit(s)	R/W	Default	Description
29:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [0*6+:6]: the slv0 req weigh number rddc_weigh_sxn [1*6+:6]: the slv1 req weigh number rddc_weigh_sxn [2*6+:6]: the slv2 req weigh number rddc_weigh_sxn [3*6+:6]: the slv3 req weigh number rddc_weigh_sxn [4*6+:6]: the slv4 req weigh number

Table 8-577 VPU_RDARB_WEIGH1_SLV_L2C1 0x27a0

Bit(s)	R/W	Default	Description
5:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [5*6+:6]: the slv5 req weigh number

Table 8-578 VPU_RDWR_ARB_STATUS_L2C1 0x27a1

Bit(s)	R/W	Default	Description
3:2	R/W	0	wrarb_arb_busy : unsigned , default = 0
1:0	R/W	0	rdarb_arb_busy : unsigned , default = 0

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Table 8-579 VPU_WRARB_MODE_L2C1 0x27a2

Bit(s)	R/W	Default	Description
19:16	R/W	0	wrarb_sel : uns, default = 0 , wrarb_sel [0]==0 slave dc0 connect master port0 wrarb_sel[0]==1 slave dc0 connect master port1 wrarb_sel [1]==0 slave dc1 connect master port0 wrarb_sel[1]==1 slave dc1 connect master port1 wrarb_sel [2]==0 slave dc2 connect master port0 wrarb_sel[2]==1 slave dc2 connect master port1 wrarb_sel [3]==0 slave dc3 connect master port0 wrarb_sel[3]==1 slave dc3 connect master port1
9:8	R/W	0	wrarb_arb_mode : uns, default = 0 , wrarb_arb_mode [0] master port0 arb way, wrarb_arb_mode [1] master port1 arb way,
3:0	R/W	0	wrarb_gate_clk_ctrl : uns, default = 0 , wrarb_gate_clk_ctrl [1:0] master port0 clk gate control wrarb_gate_clk_ctrl [3:2] master port0 clk gate control

Table 8-580 VPU_WRARB_REQEN_SLV_L2C1 0x27a3

Bit(s)	R/W	Default	Description
7:0	R/W	0	wrarb_dc_req_en : unsigned , default = 0 wrarb_dc_req_en [0]: the slv0 req to mst port0 enable, wrarb_dc_req_en [1]: the slv1 req to mst port0 enable, wrarb_dc_req_en [2]: the slv2 req to mst port0 enable, wrarb_dc_req_en [3]: the slv3 req to mst port0 enable, wrarb_dc_req_en [0]: the slv0 req to mst port1 enable, wrarb_dc_req_en [1]: the slv1 req to mst port1 enable, wrarb_dc_req_en [2]: the slv2 req to mst port1 enable, wrarb_dc_req_en [3]: the slv3 req to mst port1 enable,

Table 8-581 VPU_WRARB_WEIGH0_SLV_L2C1 0x27a4

Bit(s)	R/W	Default	Description
23:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 wrdc_weigh_sxn [0*6+:6]: the slv0 req weigh number wrdc_weigh_sxn [1*6+:6]: the slv1 req weigh number wrdc_weigh_sxn [2*6+:6]: the slv2 req weigh number wrdc_weigh_sxn [3*6+:6]: the slv3 req weigh number

Table 8-582 VPU_ASYNC_RD_MODE0 0x27a5

Bit(s)	R/W	Default	Description
18	R/W	0	req_en : unsigned , default = 0 async enable
17:16	R/W	0	clk_gate_ctrl : unsigned , default = 0 async clock gate control
15:12	R/W	4	auto_arugt_weight : unsigned , default = 4
10:9	R/W	0	arugt_sel : unsigned , default = 0 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 00 : use the input arguent
8	R/W	0	arguent_cfg : unsigned , default = 0 register arguent control bit
7:4	R/W	4	rd_hold_num : unsigned , default = 4 hold the read command threshold
3:0	R/W	0	rd_rel_num : unsigned , default = 0 release the read command threshold

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Table 8-583 VPU_ASYNC_RD_MODE1 0x27a6

Bit(s)	R/W	Default	Description
18	R/W	0	req_en : unsigned , default = 0 async enable
17:16	R/W	0	clk_gate_ctrl : unsigned , default = 0 async clock gate control
15:12	R/W	4	auto_arugt_weight : unsigned , default = 4
10:9	R/W	0	arugt_sel : unsigned , default = 0 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 00 : use the input arguent
8	R/W	0	arguent_cfg : unsigned , default = 0 register arguent control bit
7:4	R/W	4	rd_hold_num : unsigned , default = 4 hold the read command threshold
3:0	R/W	0	rd_rel_num : unsigned , default = 0 release the read command threshold

Table 8-584 VPU_ASYNC_RD_MODE2 0x27a7

Bit(s)	R/W	Default	Description
18	R/W	0	req_en : unsigned , default = 0 async enable
17:16	R/W	0	clk_gate_ctrl : unsigned , default = 0 async clock gate control
15:12	R/W	4	auto_arugt_weight : unsigned , default = 4
10:9	R/W	0	arugt_sel : unsigned , default = 0 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 00 : use the input arguent
8	R/W	0	arguent_cfg : unsigned , default = 0 register arguent control bit
7:4	R/W	4	rd_hold_num : unsigned , default = 4 hold the read command threshold
3:0	R/W	0	rd_rel_num : unsigned , default = 0 release the read command threshold

Table 8-585 VPU_ASYNC_RD_MODE3 0x27a8

Bit(s)	R/W	Default	Description
18	R/W	0	req_en : unsigned , default = 0 async enable
17:16	R/W	0	clk_gate_ctrl : unsigned , default = 0 async clock gate control
15:12	R/W	4	auto_arugt_weight : unsigned , default = 4
10:9	R/W	0	arugt_sel : unsigned , default = 0 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 00 : use the input arguent
8	R/W	0	arguent_cfg : unsigned , default = 0 register arguent control bit
7:4	R/W	4	rd_hold_num : unsigned , default = 4 hold the read command threshold
3:0	R/W	0	rd_rel_num : unsigned , default = 0 release the read command threshold

Table 8-586 VPU_ASYNC_RD_MODE4 0x27a9

Bit(s)	R/W	Default	Description
18	R/W	0	req_en : unsigned , default = 0 async enable
17:16	R/W	0	clk_gate_ctrl : unsigned , default = 0 async clock gate control

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Bit(s)	R/W	Default	Description
15:12	R/W	4	auto_arugt_weight : unsigned , default = 4
10:9	R/W	0	arugt_sel : unsigned , default = 0 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 00 : use the input arguent
8	R/W	0	arguent_cfg : unsigned , default = 0 register arguent control bit
7:4	R/W	4	rd_hold_num : unsigned , default = 4 hold the read command threshold
3:0	R/W	0	rd_rel_num : unsigned , default = 0 release the read command threshold

Table 8-587 VPU_ASYNC_WR_MODE0 0x27aa

Bit(s)	R/W	Default	Description
18	R/W	0	req_en : unsigned , default = 0 async enable
17:16	R/W	0	clk_gate_ctrl : unsigned , default = 0 async clock gate control
15:12	R/W	4	auto_arugt_weight : unsigned , default = 4
10:9	R/W	0	arugt_sel : unsigned , default = 0 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 00 : use the input arguent
8	R/W	0	arguent_cfg : unsigned , default = 0 register arguent control bit
7:4	R/W	4	wr_hold_num : unsigned , default = 4 hold the read command threshold
3:0	R/W	0	wr_rel_num : unsigned , default = 0 release the write command threshold

Table 8-588 VPU_ASYNC_WR_MODE1 0x27ab

Bit(s)	R/W	Default	Description
18	R/W	0	req_en : unsigned , default = 0 async enable
17:16	R/W	0	clk_gate_ctrl : unsigned , default = 0 async clock gate control
15:12	R/W	4	auto_arugt_weight : unsigned , default = 4
10:9	R/W	0	arugt_sel : unsigned , default = 0 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 00 : use the input arguent
8	R/W	0	arguent_cfg : unsigned , default = 0 register arguent control bit
7:4	R/W	4	wr_hold_num : unsigned , default = 4 hold the read command threshold
3:0	R/W	0	wr_rel_num : unsigned , default = 0 release the write command threshold

Table 8-589 VPU_ASYNC_WR_MODE2 0x27ac

Bit(s)	R/W	Default	Description
18	R/W	0	req_en : unsigned , default = 0 async enable
17:16	R/W	0	clk_gate_ctrl : unsigned , default = 0 async clock gate control
15:12	R/W	4	auto_arugt_weight : unsigned , default = 4
10:9	R/W	0	arugt_sel : unsigned , default = 0 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 00 : use the input arguent
8	R/W	0	arguent_cfg : unsigned , default = 0 register arguent control bit

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Bit(s)	R/W	Default	Description
7:4	R/W	4	wr_hold_num : unsigned , default = 4 hold the read command threshold
3:0	R/W	0	wr_rel_num : unsigned , default = 0 release the write command threshold

Table 8-590 VPU_ASYNC_STAT 0x27ad

Bit(s)	R/W	Default	Description
18	R/W	0x0	axiwr2_chan_idle : unsigned , RO, axi write channel2 idle state
17	R/W	0x0	axiwr1_chan_idle : unsigned , RO, axi write channel1 idle state
16	R/W	0x0	axiwr0_chan_idle : unsigned , RO, axi write channel0 idle state
4	R/W	0x0	axird4_chan_idle : unsigned , RO, axi read channel4 idle state
3	R/W	0x0	axird3_chan_idle : unsigned , RO, axi read channel3 idle state
2	R/W	0x0	axird2_chan_idle : unsigned , RO, axi read channel2 idle state
1	R/W	0x0	axird1_chan_idle : unsigned , RO, axi read channel1 idle state
0	R/W	0x0	axird0_chan_idle : unsigned , RO, axi read channel0 idle state

Table 8-591 VPU_HDMI_DITH_01_04 0x27f0

Bit(s)	R/W	Default	Description
31-0	R/W	0x8214_1428	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-592 VPU_HDMI_DITH_01_15 0x27f1

Bit(s)	R/W	Default	Description
31-0	R/W	0x4128_2841	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-593 VPU_HDMI_DITH_01_26 0x27f2

Bit(s)	R/W	Default	Description
31-0	R/W	0x2841_4182	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-594 VPU_HDMI_DITH_01_37 0x27f3

Bit(s)	R/W	Default	Description
31-0	R/W	0x1482_8214	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

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Table 8-595 VPU_HDMI_DITH_10_04 0x27f4

Bit(s)	R/W	Default	Description
31-0	R/W	0x9669_9696	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-596 VPU_HDMI_DITH_10_15 0x27f5

Bit(s)	R/W	Default	Description
31-0	R/W	0x3c3c_6969	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-597 VPU_HDMI_DITH_10_26 0x27f6

Bit(s)	R/W	Default	Description
31-0	R/W	0x6996_9696	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-598 VPU_HDMI_DITH_10_37 0x27f7

Bit(s)	R/W	Default	Description
31-0	R/W	0xc3c3_6969	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-599 VPU_HDMI_DITH_11_04 0x27f8

Bit(s)	R/W	Default	Description
31-0	R/W	0x7deb_ebd7	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-600 VPU_HDMI_DITH_11_15 0x27f9

Bit(s)	R/W	Default	Description
31-0	R/W	0xbed7_d7be	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-601 VPU_HDMI_DITH_11_26 0x27fa

Bit(s)	R/W	Default	Description
31-0	R/W	0xd7be_be7d	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

Table 8-602 VPU_HDMI_DITH_11_37 0x27fb

Bit(s)	R/W	Default	Description
31-0	R/W	0xeb7d_7deb	dith lut VPU_HDMI_DITH_CNTL[0] : 1 : lut for 10b to 8b 0 : lut for 12b to 10b

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Table 8-603 VPU_HDMI_DITH_CNTL 0x27fc

Bit(s)	R/W	Default	Description
10b to 8b dither control register. 12b to 10b see VPU_HDMI_FMT_CTRL			
21-19	R/W	0	rame count offset for B
18-16	R/W	0	frame count offset for G
15	R/W	0	hcnt hold when de valid
14	R/W	0	RGB frame count seperate
13	R/W	0	dith4x4 : frame random enable
12	R/W	0	dith4x4 enable
11	R/W	0	tunnel enable for DOLBY
10	R/W	0	rounding enable
9-6	R/W	0	Cntl_hdmi_dith10 :
5	R/W	0	Cntl_hdmi_dith_md:
4	R/W	0	Cntl_hdmi_dith_en: dither 10-b to 8-b enable
3	R/W	0	hsync invert
2	R/W	0	vsync invert
0	R/W	0	dither lut sel : 1 : sel 10b to 8b 0: sel 12b to 10b

Table 8-604 VPU_VENCL_DITH_CTRL 0x27e0

Bit(s)	R/W	Default	Description
19-17	R/W	0	dither 2x2 : frame number sel
16	R/W	0	dither 2x2 : frame number random
15-14	R/W	0	Reserved
13-11	R/W	7	G frame number offset
10-8	R/W	3	B frame number offset
7	R/W	0	Reserved
6	R/W	1	dither 4x4 : G/B frame number = B frame number + offset g/b
5	R/W	0	dither 4x4 : frame number random
4	R/W	1	dither 4x4 : enable
3	R/W	0	Reserved
2	R/W	0	dither md
1	R/W	0	rounding enable
0	R/W	1	dither enable

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Table 8-605 VPU_VENCL_DITH_LUT_1 0x27e1

Bit(s)	R/W	Default	Description
31-0	R/W	0x8214_1428	dith lut

Table 8-606 VPU_VENCL_DITH_LUT_2 0x27e2

Bit(s)	R/W	Default	Description
31-0	R/W	0x4128_2841	dith lut

Table 8-607 VPU_VENCL_DITH_LUT_3 0x27e3

Bit(s)	R/W	Default	Description
31-0	R/W	0x2841_4182	dith lut

Table 8-608 VPU_VENCL_DITH_LUT_4 0x27e4

Bit(s)	R/W	Default	Description
31-0	R/W	0x1482_8214	dith lut

Table 8-609 VPU_VENCL_DITH_LUT_5 0x27e5

Bit(s)	R/W	Default	Description
31-0	R/W	0x9669_9696	dith lut

Table 8-610 VPU_VENCL_DITH_LUT_6 0x27e6

Bit(s)	R/W	Default	Description
31-0	R/W	0x3c3c_6969	dith lut

Table 8-611 VPU_VENCL_DITH_LUT_7 0x27e7

Bit(s)	R/W	Default	Description
31-0	R/W	0x6996_9696	dith lut

Table 8-612 VPU_VENCL_DITH_LUT_8 0x27e8

Bit(s)	R/W	Default	Description
31-0	R/W	0xc3c3_6969	dith lut

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Table 8-613 VPU_VENCL_DITH_LUT_9 0x27e9

Bit(s)	R/W	Default	Description
31-0	R/W	0x7deb_ebd7	dith lut

Table 8-614 VPU_VENCL_DITH_LUT_10 0x27ea

Bit(s)	R/W	Default	Description
31-0	R/W	0xbed7_d7be	dith lut

Table 8-615 VPU_VENCL_DITH_LUT_11 0x27eb

Bit(s)	R/W	Default	Description
31-0	R/W	0xd7be_be7d	dith lut

Table 8-616 VPU_VENCL_DITH_LUT_12 0x27ec

Bit(s)	R/W	Default	Description
31-0	R/W	0xeb7d_7deb	dith lut

8.2.3.2 VPU Video Lock Registers

Table 8-617 VPU_VLOCK_CTRL 0x3000

Bit(s)	R/W	Default	Description
31	R/W	0x0	Vid_lock_en: 1: enable video lock module
30	R/W	0x0	Reg_adj_enc: enable video lock to adjust encoder
29	R/W	0x0	Adj_pll: enable video lock to adjust PLL
28	R/W	0x0	Mpeg_vs: set this to 1, then 0, this is software controlled mpeg vsync
27-26	R/W	0x0	Output goes to which module: 0: encl, 1: encp, 2:enci
25-20	R/W	0x0	Output vsync width extend: make sure the vsync width is extended big enough for vpu_vid_lock_clk to sample
19	R/W	0x0	m frac right shift 1 : right shift 2 bit 0 : no shift
18-16	R/W	0x0	Input Vsync source select: 0: unuse, 1: fromhdmi rx , 2:from tv-decoder, 3: from dvin, 4: from dvin, 5: from 2nd bt656
15			Output vsync invert: 1, invert
14			Input vsync invert: 1, invert
13-8	R/W		Input vsync width extend: make sure the vsync width is extended big enough for vpu_vid_lock_clk to sample
7			Force loop1 err enable: 1. Force error of loop1

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Bit(s)	R/W	Default	Description
6			Force loop0 err enable: 1. Force error of loop0
5			Overwrite accum0 enable
4			Loop0 adjust capture enable
3			Loop0 adjust pll enable
2			Overwrite accum1 enable
1			Loop1 adjust capture enable
0			Loop0 adjust pll enable

Table 8-618 VPU_VLOCK_MISC_CTRL 0x3001

Bit(s)	R/W	Default	Description
28	R/W	0x0	for interlace : get average of every two input vsync
26-24	R/W	0x0	Adj_capt_pxgrouprs, make sure the pixel number in one line of encoder is multiples of 2^pxgrouprs
23-16	R/W	0x0	Ifrm_cnt_mod: (output vsync freq)/(input vsync_freq * ifrm_cnt_mod) must be integer
15-8	R/W	0x0	Output vsync frequency
7-0	R/W	0x0	Input vsync frequency

Table 8-619 VPU_VLOCK_LOOP0_ACCUM_LMT 0x3002

Bit(s)	R/W	Default	Description
26-0	R/W	0x0	LOOP0 accumulator limit

Table 8-620 VPU_VLOCK_LOOP0_CTRL0 0x3003

Bit(s)	R/W	Default	Description
31-24	R/W	0x0	Loop0 errclip rate
23-20	R/W	0x0	Loop0_adj_pll_rs, right shift of loop0 adjust pll portion
19-12	R/W	0x0	Loop0_adj_pll_gain, u1.7
11-8	R/W	0x0	Loop0_adj_capt_rs, right shift of loop0 adjust capture portion
7-0	R/W	0x0	Loop0_adj_capt_gain

Table 8-621 VPU_VLOCK_LOOP0_CTRL1 0x3004

Bit(s)	R/W	Default	Description
23-20	R/W	0x0	Loop1_adj_pll_rs
19-12	R/W	0x0	Loop1_adj_pll_gain
11-8	R/W	0x0	Loop1_adj_capt_rs
7-0	R/W	0x0	Loop1_adj_capt_gain

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Table 8-622 VPU_VLOCK_LOOP1_IMISSYNC_MAX 0x3005

Bit(s)	R/W	Default	Description
27-0	R/W	0x0	Loop1 imissync max, input signal is missed after input vsync counter is larger than this max threshold

Table 8-623 VPU_VLOCK_LOOP1_IMISSYNC_MIN 0x3006

Bit(s)	R/W	Default	Description
27-0	R/W	0x0	Loop1 imissync min, input signal is missed after input vsync counter is less than this max threshold

Table 8-624 VPU_VLOCK_OVERWRITE_ACCUM0 0x3007

Bit(s)	R/W	Default	Description
27-0	R/W	0x0	Overwrite value of accum0

Table 8-625 VPU_VLOCK_OVERWRITE_ACCUM1 0x3008

Bit(s)	R/W	Default	Description
27-0	R/W	0x0	Overwrite value of accum1

Table 8-626 VPU_VLOCK_OUTPUT0_CAPT_LMT 0x3009

Bit(s)	R/W	Default	Description
26-0	R/W	0x0	Output0 capture limit

Table 8-627 VPU_VLOCK_OUTPUT0_PLL_LMT 0x300a

Bit(s)	R/W	Default	Description
26-0	R/W	0x0	Output0 pll limit

Table 8-628 VPU_VLOCK_OUTPUT1_CAPT_LMT 0x300b

Bit(s)	R/W	Default	Description
26-0	R/W	0x0	Output1 capture limit

Table 8-629 VPU_VLOCK_OUTPUT1_PLL_LMT 0x300c

Bit(s)	R/W	Default	Description
26-0	R/W	0x0	Output1 pll limit

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Table 8-630 VPU_VLOCK_LOOP1_PHSDIF_TARGET 0x300d

Bit(s)	R/W	Default	Description
27-0	R/W	0x0	Loop1 phase difference target, (input vsync - output vsync) phase distance target

Table 8-631 VPU_VLOCK_RO_LOOP0_ACCUM 0x300e

Bit(s)	R/W	Default	Description
27-0	R	0x0	Read only, loop0 accum result

Table 8-632 VPU_VLOCK_RO_LOOP1_ACCUM 0x300f

Bit(s)	R/W	Default	Description
27-0	R	0x0	Read only, loop1 accum result

Table 8-633 VPU_VLOCK_OROW_OCOL_MAX 0x3010

Bit(s)	R/W	Default	Description
29-16	R/W	0x0	Ocol_max
13-0	R/W	0x0	Orow_max

Table 8-634 VPU_VLOCK_RO_VS_I_D 0x3011

Bit(s)	R/W	Default	Description
27-0	R	0x0	Read only, input vsync counter

Table 8-635 VPU_VLOCK_RO_VS_O_D 0x3012

Bit(s)	R/W	Default	Description
27-0	R	0x0	Read only, output vsync counter

Table 8-636 VPU_VLOCK_RO_LINE_PIX_ADJ 0x3013

Bit(s)	R/W	Default	Description
29-16	R	0x0	Read only, encoder line adjust number
13-0	R	0x0	Read only, encoder pix adjust number

Table 8-637 VPU_VLOCK_RO_OUTPUT_00_01 0x3014

Bit(s)	R/W	Default	Description
31-16	R	0x0	Read only, accum0 output 00
15-0	R	0x0	Read only, accum0 output 01

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Table 8-638 VPU_VLOCK_RO_OUTPUT_10_11 0x3015

Bit(s)	R/W	Default	Description
31-16	R	0x0	Read only, accum1 output 10
15-0	R	0x0	Read only, accum1 output 11

Table 8-639 VPU_VLOCK_MX4096 0x3016

Bit(s)	R/W	Default	Description
20-0	R/W	0x0	Mx4096

Table 8-640 VPU_VLOCK_STBDET_WIN0_WIN1 0x3017

Bit(s)	R/W	Default	Description
15-8	R/W	0x0	Verr_stbdet_win1
7-0	R/W	0x0	Verr_stbdet_win0

Table 8-641 VPU_VLOCK_STBDET_CLP 0x3018

Bit(s)	R/W	Default	Description
15-8	R	0x0	Read only, ro_verr_clp_win1, verr_clp number in win0
7-0	R	0x0	Read only, ro_verr_clp_win0, verr clp number in win1

Table 8-642 VPU_VLOCK_STBDET_ABS_WIN0 0x3019

Bit(s)	R/W	Default	Description
23-0	R	0x0	Read only, ro_verr_abs_win0

Table 8-643 VPU_VLOCK_STBDET_ABS_WIN1 0x301a

Bit(s)	R/W	Default	Description
23-0	R	0x0	Read only, ro_verr_abs_win1

Table 8-644 VPU_VLOCK_STBDET_SGN_WIN0 0x301b

Bit(s)	R/W	Default	Description
23-0	R	0x0	Read only, ro_verr_sgn_win0

Table 8-645 VPU_VLOCK_STBDET_SGN_WIN1 0x301c

Bit(s)	R/W	Default	Description
23-0	R	0x0	Read only, ro_verr_sgn_win1

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Table 8-646 VPU_VLOCK_ADJ_EN_SYNC_CTRL 0x301d

Bit(s)	R/W	Default	Description
31-24	R/W	0x0	PLL adjust enable signal sync ctrl, adj_en_for_pll_end, end counter of adj_en_pll signal fall to 0
23-16	R/W	0x0	PLL adjust enable signal sync ctrl, adj_en_for_pll_start, start counter of adj_en_ pll signal go to 1, start must be larger than end
15-8	R/W	0x0	Adj_en_sync_latch_cnt, this is a delay to latch the adj_en signal
7-0	R/W	0x0	Adj_en_ext_cnt, extend the adj_en signal from vid_lock clock domain to pll sample domain, make sure it's wide enough

Table 8-647 VPU_VLOCK_GCLK_EN 0x301e

Bit(s)	R/W	Default	Description
2	R/W	0x0	Ref clock enable
1	R/W	0x0	Vsout clk enable
0	R/W	0x0	Vsin clk enable

Table 8-648 VPU_VLOCK_LOOP1_ACCUM_LMT 0x301f

Bit(s)	R/W	Default	Description
26-0	R/W	0x0	LOOP1 accumulator limit

Table 8-649 VPU_VLOCK_RO_M_INT_FRAC 0x3020

Bit(s)	R/W	Default	Description
29-16	R	0x0	Read only, m_int to PLL
13-0	R	0x0	Read only, m_frac to PLL

Table 8-650 VPU_VLOCK_RO_PH_DIS 0x3021

Bit(s)	R/W	Default	Description
27-0	R	0x0	phase distance RO

Table 8-651 VPU_VLOCK_RO_PH_ERR 0x3022

Bit(s)	R/W	Default	Description
27-0	R	0x0	phase error RO

Table 8-652 VPU_VLOCK_LOCK_TH 0x3023

Bit(s)	R/W	Default	Description
31-16	R/W	0x0	phase error threshold

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Table 8-653 VPU_VLOCK_LOCK_TH1 0x302b

Bit(s)	R/W	Default	Description
21-0	R/W	0x0	frequency error threshold

Table 8-654 VPU_VLOCK_RO_LCK_FRM 0x3024

Bit(s)	R/W	Default	Description
31-18	R/W	0x0	reserved
17	R/W	0x0	phase lock flag
16	R/W	0x0	frequency lock flag
15-0	R/W	0x0	ro_lock_frame_count

Table 8-655 VPU_VLOCK_WIN0_TH 0x3025

Bit(s)	R/W	Default	Description
23-0	R/W	0x0	reg_win0_th

Table 8-656 VPU_VLOCK_WIN0_RATIO 0x3026

Bit(s)	R/W	Default	Description
7-0	R/W	0x0	reg_win0_ratio

Table 8-657 VPU_VLOCK_WIN0_FILTER_CNTL 0x3027

Bit(s)	R/W	Default	Description
20-0	R/W	0x0	reg_win0_lock_filter_cntl

Table 8-658 VPU_VLOCK_WIN1_TH 0x3028

Bit(s)	R/W	Default	Description
23-0	R/W	0x0	reg_win0_th

Table 8-659 VPU_VLOCK_WIN1_RATIO 0x3029

Bit(s)	R/W	Default	Description
7-0	R/W	0x0	reg_win0_ratio

Table 8-660 VPU_VLOCK_WIN1_FILTER_CNTL 0x302a

Bit(s)	R/W	Default	Description
20-0	R/W	0x0	reg_win0_lock_filter_cntl

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8.2.3.3 VIU Top-Level Registers

Table 8-661 VIU_SW_RESET 0x1A01

Bit(s)	R/W	Default	Description
31	R/W	0	Osd1 afbcd reset
30	R/W	0	hist_spl reset
29	R/W	0	Ldim stts reset
8	R/W	0	Vd2 Dos afbcd reset
7	R/W	0	vpp_reset
6	R/W	0	di_dsr1to2_reset
5	R/W	0	vd2_fmt_reset
4	R/W	0	vd2_reset
3	R/W	0	vd1_fmt_reset
2	R/W	0	vd1_reset
1	R/W	0	osd2_reset
0	R/W	0	osd1_reset

Table 8-662 VIU_SW_RESET0 0x1A02

Bit(s)	R/W	Default	Description
2	R/W	0	Vd1 Dos afbcd reset

Table 8-663 VIU_MISC_CTRL0 0x1a06

Bit(s)	R/W	Default	Description
17	R/W	0	Vdin0_wr_out_ctrl : 0: nr_inp to vdin 1: vdin wr dout
16	R/W	0	Afbc_inp_sel : 0: mif to INP 1: afbc to INP
16	R/W	0	di_mif0_en: vd1(afbc) to di post(if0) enable
8	R/W	0	vsync_int_ctrl : default = 0
6:5	R/W	2	0: close mif data 1: mif to tfbf_downscale: 2: mif to nr , 3: mif to tfbf_downscale & NR
4	R/W		Vpp_di_mif0_sel 0:buf0_data from di_mif0 1: buf0_data from din_post_din
0	R/W	0	scan_reg : default = 0

Table 8-664 VIU_MISC_CTRL1 0x1A07

Bit(s)	R/W	Default	Description
27-22	R/W	0	afbc gate clk ctrl
15:14	R/W	0	Mali afbcd clock gate control
12	R/W	0	Osd1 axi bus select 1 : select mali afbcd 0 : normal osd1

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Bit(s)	R/W	Default	Description
11:8	R/W	0	di_mad_en: di post to vpp enable
7-2	R/W	0	Afbcd2 Clock gate control
1	R/W	0	1 : connect dos afbcd2 to vpp vd1, 0 : connect mif to vpp vd1
0	R/W	0	1 : Dos afbcd2 output to di ; 0 : dos afbcd2 output to vpp

Table 8-665 VIUB_SW_RESET 0x2001

Bit(s)	R/W	Default	Description
31	R/W	0	mcvecwr_mif_rst_n
30	R/W	0	reserved
29	R/W	0	reserved
28	R/W	0	di_cont_rd_mif_rst_n
27	R/W	0	di_cont_wr_mif_rst_n
26	R/W	0	reserved
25	R/W	0	reserved
24	R/W	0	vdin1_wr_rst_n
23	R/W	0	vdin0_wr_rst_n
22	R/W	0	nrin_mux_rst_n
21	R/W	0	vdin1_rst_n
20	R/W	0	vdin0_rst_n
19	R/W	0	di_mad_rst_n
18	R/W	0	di_mtn_rd_mif_rst_n
17	R/W	0	di_mtn_wr_mif_rst_n
16	R/W	0	di_chan2_mif_rst_n
15	R/W	0	dein_wr_mif_rst_n
14	R/W	0	di_nr_wr_mif_rst_n
13	R/W	0	di_mem_fmt_rst_n
12	R/W	0	di_mem_rst_n
11	R/W	0	di_inp_fmt_rst_n
10	R/W	0	di_inp_rst_n
9	R/W	0	di_if1_fmt_rst_n
8	R/W	0	di_if1_rst_n
7-0	R/W	0	RESERVED

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Table 8-666 VIUB_SW_RESET0 0x2002

Bit(s)	R/W	Default	Description
3	R/W	0	di_axi_arb_rst_n
2	R/W	0	mcinford_mif_rst_n
1	R/W	0	mcinfowr_mif_rst_n
0	R/W	0	mcvecrd_mif_rst_n

Table 8-667 VIUB_MISC_CTRL0 0x2006

Bit(s)	R/W	Default	Description
17	R/W	0	input2pre enable: 1->di inp data from vdin0 0->di inp data from inp_afbc(see bit16)
16	R/W	0	AFBC_INP_SEL: 1->di inp_afbc data from afbc 0->di inp_afbc data from inp mif(ddr)
6:5	R/W	0	0:mask input 1:buf_data ->nr_dscale 2: buf_data->NR 3: buf_data->nr_dscale &NR
4	R/W	0	0:post_din to buf 1: di_mif0 -> buf
3-2	R/W	0	Fix_disable:dein_wr_mif
1-0	R/W	0	Fix_disable: di_nr_wr_mif

Table 8-668 VIUB_GCLK_CTRL0 0x2007

Bit(s)	R/W	Default	Description
31:16	R/W	0	Reserved
15	R/W	0	Di_gate_all,for old di
14	R/W	1	Di_no_clk_gate,for old di
13	R/W	0	reserved
12	R/W	0	Di_post clock enable ,from div clock
11	R/W	0	Mcdi clock enable,from div clock
10	R/W	0	Div clock enable,di slow clock including di&MCDI
9	R/W	0	Mad post clock enable,from mad clock
8	R/W	0	Mad pre clock enable,from mad clock
7:1	R/W		reserved
0	R/W	1	Def=1 di_top_wrap clk enable

Table 8-669 VIUB_GCLK_CTRL1 0x2008

Bit(s)	R/W	Default	Description
31:28	R/W	0	Reserved
27:26	R/W	0	Mcdi pre mif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
25:24	R/W	0	Mtn mif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock

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Bit(s)	R/W	Default	Description
23:22	R/W	0	Nr wrmif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
21:20	R/W	0	Chan rdmif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
19:18	R/W	0	Mem rdmif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
17:16	R/W	0	Inp rdmif clock gate clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
15:14	R/W	0	reserved
13:12	R/W	0	Mcdi post clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
11-10	R/W	0	Mtnrd post mif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
9-8	R/W	0	De wrmif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
7-6	R/W	0	if2 rdmif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
5-4	R/W	0	If1 rdmif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
3-2	R/W	0	If0 rdmif clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
1-0	R/W	0	Mif-sub-arb clock gate 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock

Table 8-670 VIUB_GCLK_CTRL2 0x2009

Bit(s)	R/W	Default	Description
31-14	R/W	0	reserved
13-12	R/W	0	mcdi clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
11:10	R/W	0	Nr&dnr blend clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
9-8	R/W	0	Dnr clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
7-6	R/W	0	nning clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
5-4	R/W	0	Mtn det clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
3-2	R/W	0	pd clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
1-0	R/W	0	Nr clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock

Table 8-671 VIUB_GCLK_CTRL3 0x200a

Bit(s)	R/W	Default	Description
31-6	R/W	0	Reserved
5-4	R/W	0	Di blend clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
3-2	R/W	0	Ei clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock
1-0	R/W	0	Ei_0 clock gate ctrl 2'b00: gate clock ,2'b01: close clock,2'b1x: always clock

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Table 8-672 DI_DBG_CTRL 0x200B

Bit(s)	R/W	Default	Description
31:7	R/W	0	Reserved
6	R/W	0	go_field_sel : 1: post_frame_rst 0: pre_frame_rst
5:0	R/W	0	Debug_sel : 0:diwr_out_mif 1: mtn1_din 2:mtn1_dout 3:nrdm_bld 4:dnr_dm 5:dnr_din 6:ds_dout 7:ds_din 8:deband_din 9:me_mtn 10:mc_info_mif 11:me_din_pipe 12:buf1_dout 13:mem2post 14:cont_buf 15:nr_wr_out 16:mtn_din 17:me_din 18:nr_din 19:dnr_out 20:nr_out 21:contp_mif 22:chan2 23:di_inp_mif 24:di_mem_mif 25:contp2_mif 26:contp_mif 27:chan2_mif 28:di_inp 29:di_mem 30:mcinfo_pipe 31:mad_dout 32:deint_pipe 33:mtnp_mif 34:blend_1_dout 35:blend_buf1_din

Table 8-673 DI_DBG_CTRL1 0x200C

Bit(s)	R/W	Default	Description
31:16	R/W	0	Rrdy_to_srdy_max_num
15:0	R/W	0	Srdy_to_rrdy_max_num

Table 8-674 DI_DBG_SRDY_INF 0x200D

Bit(s)	R/W	Default	Description
31:16	RO	0	wait_rrdy_bmax_dbg_cnt
15:0	RO	0	Srdy_to_rrdy_dbg_cnt

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Table 8-675 DI_DBG_RRDY_INF 0x200E

Bit(s)	R/W	Default	Description
31:16	RO	0	Wait_srdy_bmax_dbg_cnt
15:0	RO	0	Rrdy_to_srdy_dbg_cnt

Table 8-676 VIU_MISC_CTRL1 0x3107

Bit(s)	R/W	Default	Description
15:14	R/W	0x0	mali_afbcd_gclk_ctrl : mali_afbcd clock gate control[5:4]
12	R/W	0x0	osd1_afbcd_axi_mux : 0 : use the osd mif as input; 1 : use afbcd as input
11:8	R/W	0x0	mali_afbcd_gclk_ctrl : mali_afbcd clock gate control[3:0]
7:2	R/W	0x0	vd2_afbcd_gclk_ctrl : vd2_afbcd clock gate control
1	R/W	0x0	vpp_vd2_din_sel : 0: vpp vd2 sel the mif input; 1: vpp vd2 sel the dos afbcd
0	R/W	0x0	vd2_afbcd_out_sel : 0: vd2_afbcd output to vpp; 1 : vd2_afbcd output to di inp

8.2.3.4 DI_AXI_ARB Registers

Table 8-677 DI_RDARB_MODE_L1C1 0x2050

Bit(s)	R/W	Default	Description
21:16	R/W	0	rdarb_sel : uns, default = 0 , rdarb_sel [0]==0 slave dc0 connect master port0 rdarb_sel[0]==1 slave dc0 connect master port1 rdarb_sel [1]==0 slave dc1 connect master port0 rdarb_sel[1]==1 slave dc1 connect master port1 rdarb_sel [2]==0 slave dc2 connect master port0 rdarb_sel[2]==1 slave dc2 connect master port1 rdarb_sel [3]==0 slave dc3 connect master port0 rdarb_sel[3]==1 slave dc3 connect master port1 rdarb_sel [4]==0 slave dc4 connect master port0 rdarb_sel [4]==1 slave dc4 connect master port1 rdarb_sel [5]==0 slave dc5 connect master port0 rdarb_sel[5]==1 slave dc5 connect master port1
9:8	R/W	0	rdarb_arb_mode : uns, default = 0 , rdarb_arb_mode [0] master port0 arb way, rdarb_arb_mode [1] master port1 arb way,
3:0	R/W	0	rdarb_gate_clk_ctrl : uns, default = 0 , rdarb_gate_clk_ctrl [1:0] master port0 clk gate control rdarb_gate_clk_ctrl [3:2] master port1 clk gate control

Table 8-678 DI_RDARB_REQEN_SLV_L1C1 0x2051

Bit(s)	R/W	Default	Description
11:0	R/W	0xfff	rdarb_dc_req_en : unsigned , default = 12'hfff rdarb_dc_req_en [0]: the slv0 req to mst port0 enable, rdarb_dc_req_en [1]: the slv1 req to mst port0 enable, rdarb_dc_req_en [2]: the slv2 req to mst port0 enable, rdarb_dc_req_en [3]: the slv3 req to mst port0 enable, rdarb_dc_req_en [4]: the slv4 req to mst port0 enable, rdarb_dc_req_en [5]: the slv5 req to mst port0 enable, rdarb_dc_req_en [6]: the slv0 req to mst port1 enable, rdarb_dc_req_en [7]: the slv1 req to mst port1 enable, rdarb_dc_req_en [8]: the slv2 req to mst port1 enable, rdarb_dc_req_en [9]: the slv3 req to mst port1 enable, rdarb_dc_req_en [10]: the slv4 req to mst port1 enable, rdarb_dc_req_en [11]: the slv5 req to mst port1 enable,

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Table 8-679 DI_RDARB_WEIGH0_SLV_L1C1 0x2052

Bit(s)	R/W	Default	Description
29:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [0*6+:6]: the slv0 req weigh number rddc_weigh_sxn [1*6+:6]: the slv1 req weigh number rddc_weigh_sxn [2*6+:6]: the slv2 req weigh number rddc_weigh_sxn [3*6+:6]: the slv3 req weigh number rddc_weigh_sxn [4*6+:6]: the slv4 req weigh number

Table 8-680 DI_RDARB_WEIGH1_SLV_L1C1 0x2053

Bit(s)	R/W	Default	Description
5:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [5*6+:6]: the slv5 req weigh number

Table 8-681 DI_WRARB_MODE_L1C1 0x2054

Bit(s)	R/W	Default	Description
21:16	R/W	0	wrarb_sel : uns, default = 0 , wrarb_sel [0]==0 slave dc0 connect master port0 wrarb_sel[0]==1 slave dc0 connect master port1 wrarb_sel [1]==0 slave dc1 connect master port0 wrarb_sel[1]==1 slave dc1 connect master port1 wrarb_sel [2]==0 slave dc2 connect master port0 wrarb_sel[2]==1 slave dc2 connect master port1 wrarb_sel [3]==0 slave dc3 connect master port0 wrarb_sel[3]==1 slave dc3 connect master port1 wrarb_sel [4]==0 slave dc4 connect master port0 wrarb_sel[4]==1 slave dc4 connect master port1 wrarb_sel [5]==0 slave dc5 connect master port0 wrarb_sel[5]==1 slave dc5 connect master port1
9:8	R/W	0	wrarb_arb_mode : uns, default = 0 , wrarb_arb_mode [0] master port0 arb way, wrarb_arb_mode [1] master port1 arb way,
3:0	R/W	0	wrarb_gate_clk_ctrl : uns, default = 0 , wrarb_gate_clk_ctrl [1:0] master port0 clk gate control wrarb_gate_clk_ctrl [3:2] master port1 clk gate control

Table 8-682 DI_WRARB_REQEN_SLV_L1C1 0x2055

Bit(s)	R/W	Default	Description
11:0	R/W	0	wrarb_dc_req_en : unsigned , default = 0 wrarb_dc_req_en [0]: the slv0 req to mst port0 enable, wrarb_dc_req_en [1]: the slv1 req to mst port0 enable, wrarb_dc_req_en [2]: the slv2 req to mst port0 enable, wrarb_dc_req_en [3]: the slv3 req to mst port0 enable, wrarb_dc_req_en [4]: the slv4 req to mst port0 enable, wrarb_dc_req_en [5]: the slv5 req to mst port0 enable, wrarb_dc_req_en [0]: the slv0 req to mst port1 enable, wrarb_dc_req_en [1]: the slv1 req to mst port1 enable, wrarb_dc_req_en [2]: the slv2 req to mst port1 enable, wrarb_dc_req_en [3]: the slv3 req to mst port1 enable, wrarb_dc_req_en [4]: the slv4 req to mst port1 enable, wrarb_dc_req_en [5]: the slv5 req to mst port1 enable,

Table 8-683 DI_WRARB_WEIGH0_SLV_L1C1 0x2056

Bit(s)	R/W	Default	Description
29:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 wrdc_weigh_sxn [0*6+:6]: the slv0 req weigh number wrdc_weigh_sxn [1*6+:6]: the slv1 req weigh number wrdc_weigh_sxn [2*6+:6]: the slv2 req weigh number wrdc_weigh_sxn [3*6+:6]: the slv3 req weigh number wrdc_weigh_sxn [4*6+:6]: the slv4 req weigh number

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Table 8-684 DI_WRARB_WEIGH1_SLV_L1C1 0x2057

Bit(s)	R/W	Default	Description
5:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 wrdc_weigh_sxn [5*6+:6]: the slv5 req weigh number

Table 8-685 DI_RDWR_ARB_STATUS_L1C1 0x2058

Bit(s)	R/W	Default	Description
3:2	R/W	0	wrarb_arb_busy : unsigned , default = 0
1:0	R/W	0	rdarb_arb_busy : unsigned , default = 0

Table 8-686 DI_ARB_DBG_CTRL_L1C1 0x2059

Bit(s)	R/W	Default	Description
31:0	R/W	8	det_cmd_ctrl : unsigned , default = 8

Table 8-687 DI_ARB_DBG_STAT_L1C1 0x205a

Bit(s)	R/W	Default	Description
31:0	R/W	0	det_dbg_stat : unsigned , default = 0

Table 8-688 DI_RDARB_UGT_L1C1 0x205b

Bit(s)	R/W	Default	Description
15:0	R/W	0x0	rdarb_ugt_basic : unsigned , default = {8{2'b1}};

Table 8-689 DI_RDARB_LIMT0_L1C1 0x205c

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	rdarb_req_limt_num : unsigned , default = {2{16'h3f3f}};

Table 8-690 DI_WRARB_UGT_L1C1 0x205d

Bit(s)	R/W	Default	Description
11:0	R/W	0	wrarb_ugt_basic : unsigned , default = 0

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Table 8-691 DI_SUB_RDARB_MODE 0x37c0

Bit(s)	R/W	Default	Description
23:16	R/W	0	rdarb_sel : uns, default = 0 , rdarb_sel [0]==0 slave dc0 connect master port0 rdarb_sel[0]==1 slave dc0 connect master port1 rdarb_sel [1]==0 slave dc1 connect master port0 rdarb_sel[1]==1 slave dc1 connect master port1 rdarb_sel [2]==0 slave dc2 connect master port0 rdarb_sel[2]==1 slave dc2 connect master port1 rdarb_sel [3]==0 slave dc3 connect master port0 rdarb_sel[3]==1 slave dc3 connect master port1 rdarb_sel [4]==0 slave dc4 connect master port0 rdarb_sel [4]==1 slave dc4 connect master port1 rdarb_sel [5]==0 slave dc5 connect master port0 rdarb_sel[5]==1 slave dc5 connect master port1 rdarb_sel [6]==0 slave dc5 connect master port0 rdarb_sel[6]==1 slave dc6 connect master port1 rdarb_sel [7]==0 slave dc5 connect master port0 rdarb_sel[7]==1 slave dc7 connect master port1
9:8	R/W	0	rdarb_arb_mode : uns, default = 0 , rdarb_arb_mode [0] master port0 arb way, rdarb_arb_mode [1] master port1 arb way,
3:0	R/W	0	rdarb_gate_clk_ctrl : uns, default = 0 , rdarb_gate_clk_ctrl [1:0] master port0 clk gate control rdarb_gate_clk_ctrl [3:2] master port1 clk gate control

Table 8-692 DI_SUB_RDARB_REQEN_SLV 0x37c1

Bit(s)	R/W	Default	Description
15:0	R/W	0xffff	rdarb_dc_req_en : uns, default = 16'hffff , slv0~slv7 enable to mst.

Table 8-693 DI_SUB_RDARB_WEIGH0_SLV 0x37c2

Bit(s)	R/W	Default	Description
29:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [0*6+:6]: the slv0 req weigh number rddc_weigh_sxn [1*6+:6]: the slv1 req weigh number rddc_weigh_sxn [2*6+:6]: the slv2 req weigh number rddc_weigh_sxn [3*6+:6]: the slv3 req weigh number rddc_weigh_sxn [4*6+:6]: the slv4 req weigh number

Table 8-694 DI_SUB_RDARB_WEIGH1_SLV 0x37c3

Bit(s)	R/W	Default	Description
17:0	R/W	0	rddc_weigh_sxn : unsigned , default = 0 rddc_weigh_sxn [0*6+:6]: the slv6 req weigh number rddc_weigh_sxn [1*6+:6]: the slv7 req weigh number rddc_weigh_sxn [2*6+:6]: the slv8 req weigh number

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Table 8-695 DI_SUB_RDARB_UGT 0x37c4

Bit(s)	R/W	Default	Description
15:0	R/W	0x0	rdarb_ugt_basic : unsigned , default = {8'{2'h1}} rdarb_ugt_basic [0*1+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [1*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [2*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [3*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [4*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [5*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [6*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [7*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen

Table 8-696 DI_SUB_RDARB_LIMT0 0x37c5

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	rdarb_req_limt_num : unsigned , default = {2{16'h3f3f}}

Table 8-697 DI_SUB_WRARB_MODE 0x37c6

Bit(s)	R/W	Default	Description
21:16	R/W	0	wrarb_sel : uns, default = 0 , wrarb_sel [0]==0 slave dc0 connect master port0 wrarb_sel[0]==1 slave dc0 connect master port1 wrarb_sel [1]==0 slave dc1 connect master port0 wrarb_sel[1]==1 slave dc1 connect master port1 wrarb_sel [2]==0 slave dc2 connect master port0 wrarb_sel[2]==1 slave dc2 connect master port1 wrarb_sel [3]==0 slave dc3 connect master port0 wrarb_sel[3]==1 slave dc3 connect master port1 wrarb_sel [4]==0 slave dc4 connect master port0 wrarb_sel[4]==1 slave dc3 connect master port1 wrarb_sel [5]==0 slave dc5 connect master port0 wrarb_sel[5]==1 slave dc3 connect master port1
8	R/W	0	wrarb_arb_mode : uns, default = 0 , wrarb_arb_mode[0] master port0 arb way,
1:0	R/W	0	wrarb_gate_clk_ctrl : uns, default = 0 , wrarb_gate_clk_ctrl[1:0] master port0 clk gate control

Table 8-698 DI_SUB_WRARB_REQEN_SLV 0x37c7

Bit(s)	R/W	Default	Description
5:0	R/W	0	wrarb_dc_req_en : unsigned , default = 0 wrarb_dc_req_en [0]: the slv0 req to mst port0 enable, wrarb_dc_req_en [1]: the slv1 req to mst port0 enable, wrarb_dc_req_en [2]: the slv2 req to mst port0 enable, wrarb_dc_req_en [0]: the slv0 req to mst port1 enable, wrarb_dc_req_en [1]: the slv1 req to mst port1 enable, wrarb_dc_req_en [2]: the slv2 req to mst port1 enable,

Table 8-699 DI_SUB_WRARB_WEIGH0_SLV 0x37c8

Bit(s)	R/W	Default	Description
29:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 wrdc_weigh_sxn [0*6+:6]: the slv0 req weigh number wrdc_weigh_sxn [1*6+:6]: the slv1 req weigh number wrdc_weigh_sxn [2*6+:6]: the slv2 req weigh number wrdc_weigh_sxn [3*6+:6]: the slv3 req weigh number wrdc_weigh_sxn [4*6+:6]: the slv4 req weigh number

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Table 8-700 DI_SUB_WRARB_WEIGH1_SLV 0x37c9

Bit(s)	R/W	Default	Description
5:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 the slv5 req weigh number

Table 8-701 DI_SUB_WRARB_UGT 0x37ca

Bit(s)	R/W	Default	Description
11:0	R/W	0x0	rdarb_ugt_basic : unsigned , default = {8'{2'h1}} rdarb_ugt_basic [0*1+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [1*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [2*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [3*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [4*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen rdarb_ugt_basic [5*2+:2]: 00 : use auto fifo arugt generate the output arugt. 01 : use the register bit control 10 : use the input arguen

Table 8-702 DI_SUB_RDWR_ARB_STATUS 0x37cb

Bit(s)	R/W	Default	Description
2	R.O	0	ro_wrarb_arb_busy : unsigned , default = 0
1	R/W	0x0	reserve :
0	R.O	0	ro_rdarb_arb_busy : unsigned , default = 0

Table 8-703 DI_SUB_ARB_DBG_CTRL 0x37cc

Bit(s)	R/W	Default	Description
31:0	R.O	0	ro_det_cmd_ctrl : unsigned , default = 0

Table 8-704 DI_SUB_ARB_DBG_STAT 0x37cd

Bit(s)	R/W	Default	Description
31:0	R.O	0	ro_det_dbg_stat : unsigned , default = 0

Table 8-705 CONTRD_CTRL1 0x37d0

Bit(s)	R/W	Default	Description
25:24	R/W	0	reg_sync_sel : unsigned , default = 0
23:16	R/W	0	reg_canvas_id : unsigned , default = 0
14:12	R/W	1	reg_cmd_intr_len : unsigned , default = 1
11:10	R/W	1	reg_cmd_req_size : unsigned , default = 1
9:8	R/W	2	reg_burst_len : unsigned , default = 2
7	R/W	0	reg_swap_64bit : unsigned , default = 0
6	R/W	0	reg_little_endian : unsigned , default = 0

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Bit(s)	R/W	Default	Description
5	R/W	0	reg_y_rev : unsigned , default = 0
4	R/W	0	reg_x_rev : unsigned , default = 0
2:0	R/W	1	reg_pack_mode : unsigned , default = 1

Table 8-706 CONTRD_CTRL2 0x37d1

Bit(s)	R/W	Default	Description
31:30	R/W	0	reg_sw_rst : unsigned , default = 0
23:18	R/W	0	reg_gclk_ctrl : unsigned , default = 0
16	R/W	0	urgent
15	R/W	0	auto_urgent_en
14	R/W	0	urgent_wr
7:4	R/W	0	up_th
3:0	R/W	0	dn_th

Table 8-707 CONTRD_SCOPE_X 0x37d2

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_x_end : unsigned , default = 0
12:0	R/W	0	reg_x_start : unsigned , default = 0

Table 8-708 CONTRD_SCOPE_Y 0x37d3

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_y_end : unsigned , default = 0
12:0	R/W	0	reg_y_start : unsigned , default = 0

Table 8-709 CONTRD_RO_STAT 0x37d4

Bit(s)	R/W	Default	Description
15:0	R.O	0	ro_reg_status : unsigned , default = 0

Table 8-710 CONT2RD_CTRL1 0x37d5

Bit(s)	R/W	Default	Description
25:24	R/W	0	reg_sync_sel : unsigned , default = 0
23:16	R/W	0	reg_canvas_id : unsigned , default = 0
14:12	R/W	1	reg_cmd_intr_len : unsigned , default = 1
11:10	R/W	1	reg_cmd_req_size : unsigned , default = 1

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Bit(s)	R/W	Default	Description
9:8	R/W	2	reg_burst_len : unsigned , default = 2
7	R/W	0	reg_swap_64bit : unsigned , default = 0
6	R/W	0	reg_little_endian : unsigned , default = 0
5	R/W	0	reg_y_rev : unsigned , default = 0
4	R/W	0	reg_x_rev : unsigned , default = 0
2:0	R/W	1	reg_pack_mode : unsigned , default = 1

Table 8-711 CONT2RD_CTRL2 0x37d6

Bit(s)	R/W	Default	Description
31:30	R/W	0	reg_sw_rst : unsigned , default = 0
23:18	R/W	0	reg_gclk_ctrl : unsigned , default = 0
16	R/W	0	urgent
15	R/W	0	auto_urgent_en
14	R/W	0	urgent_wr
7:4	R/W	0	up_th
3:0	R/W	0	dn_th

Table 8-712 CONT2RD_SCOPE_X 0x37d7

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_x_end : unsigned , default = 0
12:0	R/W	0	reg_x_start : unsigned , default = 0

Table 8-713 CONT2RD_SCOPE_Y 0x37d8

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_y_end : unsigned , default = 0
12:0	R/W	0	reg_y_start : unsigned , default = 0

Table 8-714 CONT2RD_RO_STAT 0x37d9

Bit(s)	R/W	Default	Description
15:0	R.O	0	ro_reg_status : unsigned , default = 0

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Table 8-715 MTNRD_CTRL1 0x37da

Bit(s)	R/W	Default	Description
25:24	R/W	0	reg_sync_sel : unsigned , default = 0
23:16	R/W	0	reg_canvas_id : unsigned , default = 0
14:12	R/W	1	reg_cmd_intr_len : unsigned , default = 1
11:10	R/W	1	reg_cmd_req_size : unsigned , default = 1
9:8	R/W	2	reg_burst_len : unsigned , default = 2
7	R/W	0	reg_swap_64bit : unsigned , default = 0
6	R/W	0	reg_little_endian : unsigned , default = 0
5	R/W	0	reg_y_rev : unsigned , default = 0
4	R/W	0	reg_x_rev : unsigned , default = 0
2:0	R/W	1	reg_pack_mode : unsigned , default = 1

Table 8-716 MTNRD_CTRL2 0x37db

Bit(s)	R/W	Default	Description
31:30	R/W	0	reg_sw_rst : unsigned , default = 0
23:18	R/W	0	reg_gclk_ctrl : unsigned , default = 0
16	R/W	0	urgent
15	R/W	0	auto_urgent_en
14	R/W	0	urgent_wr
7:4	R/W	0	up_th
3:0	R/W	0	dn_th

Table 8-717 MTNRD_SCOPE_X 0x37dc

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_x_end : unsigned , default = 0
12:0	R/W	0	reg_x_start : unsigned , default = 0

Table 8-718 MTNRD_SCOPE_Y 0x37dd

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_y_end : unsigned , default = 0
12:0	R/W	0	reg_y_start : unsigned , default = 0

Table 8-719 MTNRD_RO_STAT 0x37de

Bit(s)	R/W	Default	Description
15:0	R.O	0	ro_reg_status : unsigned , default = 0

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Table 8-720 MCVECRD_CTRL1 0x37df

Bit(s)	R/W	Default	Description
25:24	R/W	0	reg_sync_sel : unsigned , default = 0
23:16	R/W	0	reg_canvas_id : unsigned , default = 0
14:12	R/W	1	reg_cmd_intr_len : unsigned , default = 1
11:10	R/W	1	reg_cmd_req_size : unsigned , default = 1
9:8	R/W	2	reg_burst_len : unsigned , default = 2
7	R/W	0	reg_swap_64bit : unsigned , default = 0
6	R/W	0	reg_little_endian : unsigned , default = 0
5	R/W	0	reg_y_rev : unsigned , default = 0
4	R/W	0	reg_x_rev : unsigned , default = 0
2:0	R/W	1	reg_pack_mode : unsigned , default = 1

Table 8-721 MCVECRD_CTRL2 0x37e0

Bit(s)	R/W	Default	Description
31:30	R/W	0	reg_sw_rst : unsigned , default = 0
23:18	R/W	0	reg_gclk_ctrl : unsigned , default = 0
16	R/W	0	urgent
15	R/W	0	auto_urgent_en
14	R/W	0	urgent_wr
7:4	R/W	0	up_th
3:0	R/W	0	dn_th

Table 8-722 MCVECRD_SCOPE_X 0x37e1

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_x_end : unsigned , default = 0
12:0	R/W	0	reg_x_start : unsigned , default = 0

Table 8-723 MCVECRD_SCOPE_Y 0x37e2

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_y_end : unsigned , default = 0
12:0	R/W	0	reg_y_start : unsigned , default = 0

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Table 8-724 MCVECRD_RO_STAT 0x37e3

Bit(s)	R/W	Default	Description
15:0	R.O	0	ro_reg_status : unsigned , default = 0

Table 8-725 MCINFRD_CTRL1 0x37e4

Bit(s)	R/W	Default	Description
25:24	R/W	0	reg_sync_sel : unsigned , default = 0
23:16	R/W	0	reg_canvas_id : unsigned , default = 0
14:12	R/W	1	reg_cmd_intr_len : unsigned , default = 1
11:10	R/W	1	reg_cmd_req_size : unsigned , default = 1
9:8	R/W	2	reg_burst_len : unsigned , default = 2
7	R/W	0	reg_swap_64bit : unsigned , default = 0
6	R/W	0	reg_little_endian : unsigned , default = 0
5	R/W	0	reg_y_rev : unsigned , default = 0
4	R/W	0	reg_x_rev : unsigned , default = 0
2:0	R/W	1	reg_pack_mode : unsigned , default = 1

Table 8-726 MCINFRD_CTRL2 0x37e5

Bit(s)	R/W	Default	Description
31:30	R/W	0	reg_sw_rst : unsigned , default = 0
23:18	R/W	0	reg_gclk_ctrl : unsigned , default = 0
16	R/W	0	urgent
15	R/W	0	auto_urgent_en
14	R/W	0	urgent_wr
7:4	R/W	0	up_th
3:0	R/W	0	dn_th

Table 8-727 MCINFRD_SCOPE_X 0x37e6

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_x_end : unsigned , default = 0
12:0	R/W	0	reg_x_start : unsigned , default = 0

Table 8-728 MCINFRD_SCOPE_Y 0x37e7

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_y_end : unsigned , default = 0
12:0	R/W	0	reg_y_start : unsigned , default = 0

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Table 8-729 MCINFRD_RO_STAT 0x37e8

Bit(s)	R/W	Default	Description
15:0	R.O	0	ro_reg_status : unsigned , default = 0

Table 8-730 CONTWR_X 0x37e9

Bit(s)	R/W	Default	Description
31:30	R/W	2	burst_len : unsigned , default = 2
29	R/W	0	rev_x : unsigned , default = 0
28:16	R/W	0	start_x : unsigned , default = 0
12:0	R/W	2	end_x : unsigned , default = 2cf

Table 8-731 CONTWR_Y 0x37ea

Bit(s)	R/W	Default	Description
31:30	R/W	0	canvas_id : unsigned , default = 0
29	R/W	0	rev_y : unsigned , default = 0
28:16	R/W	0	start_y : unsigned , default = 0
12:0	R/W	0	end_y : unsigned , default = 0x1df

Table 8-732 CONTWR_CTRL 0x37eb

Bit(s)	R/W	Default	Description
31:16	R/W	0	urgent_ctrl : unsigned , default = 0
15	R/W	0	force_wvalid : unsigned , default = 0
14	R/W	0	canvas_syncen : unsigned , default = 0
13	R/W	1	canvas_wr : unsigned , default = 1
12	R/W	0	req_en : unsigned , default = 0
10	R/W	0	clr_wrrsp : unsigned , default = 0
8	R/W	0	urgent : unsigned , default = 0
7:0	R/W	0	canvas_index : unsigned , default = 0

Table 8-733 CONTWR_CAN_SIZE 0x37ec

Bit(s)	R/W	Default	Description
30:29	R/W	0	reg_rst : unsigned , default = 0
28:16	R/W	0	hsizem1 : unsigned , default = 0x2cf
14	R/W	0	reg_reset : unsigned , default = 0

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Bit(s)	R/W	Default	Description
13	R/W	0	little_endian : unsigned , default = 0
12:0	R/W	0	vsizem1 : unsigned , default = 0x1df

Table 8-734 MTNWR_X 0x37ed

Bit(s)	R/W	Default	Description
31:30	R/W	2	burst_len : unsigned , default = 2
29	R/W	0	rev_x : unsigned , default = 0
28:16	R/W	0	start_x : unsigned , default = 0
12:0	R/W	2	end_x : unsigned , default = 2cf

Table 8-735 MTNWR_Y 0x37ee

Bit(s)	R/W	Default	Description
31:30	R/W	0	canvas_id : unsigned , default = 0
29	R/W	0	rev_y : unsigned , default = 0
28:16	R/W	0	start_y : unsigned , default = 0
12:0	R/W	0	end_y : unsigned , default = 0x1df

Table 8-736 MTNWR_CTRL 0x37ef

Bit(s)	R/W	Default	Description
31:16	R/W	0	urgent_ctrl : unsigned , default = 0
15	R/W	0	force_wvalid : unsigned , default = 0
14	R/W	0	canvas_syncen : unsigned , default = 0
13	R/W	1	canvas_wr : unsigned , default = 1
12	R/W	0	req_en : unsigned , default = 0
10	R/W	0	clr_wrrsp : unsigned , default = 0
8	R/W	0	urgent : unsigned , default = 0
7:0	R/W	0	canvas_index : unsigned , default = 0

Table 8-737 MTNWR_CAN_SIZE 0x37f0

Bit(s)	R/W	Default	Description
30:29	R/W	0	reg_rst : unsigned , default = 0
28:16	R/W	0	hsizem1 : unsigned , default = 0x2cf
14	R/W	0	reg_reset : unsigned , default = 0
13	R/W	0	little_endian : unsigned , default = 0
12:0	R/W	0	vsizem1 : unsigned , default = 0x1df

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Table 8-738 MCVECWR_X 0x37f1

Bit(s)	R/W	Default	Description
31:30	R/W	2	burst_len : unsigned , default = 2
29	R/W	0	rev_x : unsigned , default = 0
28:16	R/W	0	start_x : unsigned , default = 0
12:0	R/W	2	end_x : unsigned , default = 2cf

Table 8-739 MCVECWR_Y 0x37f2

Bit(s)	R/W	Default	Description
31:30	R/W	0	canvas_id : unsigned , default = 0
29	R/W	0	rev_y : unsigned , default = 0
28:16	R/W	0	start_y : unsigned , default = 0
12:0	R/W	0	end_y : unsigned , default = 0x1df

Table 8-740 MCVECWR_CTRL 0x37f3

Bit(s)	R/W	Default	Description
31:16	R/W	0	urgent_ctrl : unsigned , default = 0
15	R/W	0	force_wvalid : unsigned , default = 0
14	R/W	0	canvas_syncen : unsigned , default = 0
13	R/W	1	canvas_wr : unsigned , default = 1
12	R/W	0	req_en : unsigned , default = 0
10	R/W	0	clr_wrrsp : unsigned , default = 0
8	R/W	0	urgent : unsigned , default = 0
7:0	R/W	0	canvas_index : unsigned , default = 0

Table 8-741 MCVECWR_CAN_SIZE 0x37f4

Bit(s)	R/W	Default	Description
30:29	R/W	0	reg_rst : unsigned , default = 0
28:16	R/W	0	hsizem1 : unsigned , default = 0x2cf
14	R/W	0	reg_reset : unsigned , default = 0
13	R/W	0	little_endian : unsigned , default = 0
12:0	R/W	0	vsizem1 : unsigned , default = 0x1df

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Table 8-742 MCINFWR_X 0x37f5

Bit(s)	R/W	Default	Description
31:30	R/W	2	burst_len : unsigned , default = 2
29	R/W	0	rev_x : unsigned , default = 0
28:16	R/W	0	start_x : unsigned , default = 0
12:0	R/W	2	end_x : unsigned , default = 2cf

Table 8-743 MCINFWR_Y 0x37f6

Bit(s)	R/W	Default	Description
31:30	R/W	0	canvas_id : unsigned , default = 0
29	R/W	0	rev_y : unsigned , default = 0
28:16	R/W	0	start_y : unsigned , default = 0
12:0	R/W	0	end_y : unsigned , default = 0x1df

Table 8-744 MCINFWR_CTRL 0x37f7

Bit(s)	R/W	Default	Description
31:16	R/W	0	urgent_ctrl : unsigned , default = 0
15	R/W	0	force_wvalid : unsigned , default = 0
14	R/W	0	canvas_syncen : unsigned , default = 0
13	R/W	1	canvas_wr : unsigned , default = 1
12	R/W	0	req_en : unsigned , default = 0
10	R/W	0	clr_wrrsp : unsigned , default = 0
8	R/W	0	urgent : unsigned , default = 0
7:0	R/W	0	canvas_index : unsigned , default = 0

Table 8-745 MCINFWR_CAN_SIZE 0x37f8

Bit(s)	R/W	Default	Description
30:29	R/W	0	reg_rst : unsigned , default = 0
28:16	R/W	0	hsizem1 : unsigned , default = 0x2cf
14	R/W	0	reg_reset : unsigned , default = 0
13	R/W	0	little_endian : unsigned , default = 0
12:0	R/W	0	vsizem1 : unsigned , default = 0x1df

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8.2.3.5 DI_RMEM_IF0 Registers

Table 8-746 DI_IF0_GEN_REG 0x2030

Bit(s)	R/W	Default	Description
31	R/W	0	ENABLE_FREE_CLK. 0: Gated clock for power saving 1: Free-running clock to drive logic
30	R/W	0	SW_RESET: Write 1 to this bit to generate a pulse to reset everything except registers.
29	R/W	0	RESET_ON_GO_FIELD: Define whether to reset state machines on go_field pulse. 0: No reset on go_field 1: go_field reset everything except registers
28	R/W	0	URGENT_CHROMA: Set urgent level for chroma fifo request from DDR. 0: Non urgent 1: Urgent
27	R/W	0	URGENT_LUMA: Set urgent level for luma fifo request from DDR. 0: Non urgent 1: Urgent
26	R/W	0	Chroma_end_at_last line: For chroma line, similar to luma_end_at_last_line, as below. Not used if data are stored together in one canvas.
25	R/W	0	Luma_end_at_last_line: Control whether continue outputting luma line past last line. 0: Repeat the last line or dummy pixels, after past the last line 1: Stop outputting data, once past the last line.
24-19	R/W	4	Hold_lines: After go_field, the number of lines to hold before the module is enabled.
18	R/W	0	LAST_LINE: This bit controls whether we simply repeat the last line or we push dummy pixels. ‘1’ tells the state-machines to repeat the last line using the dummy pixels defined in the register below. ‘0’ indicates that the state-machine should re-read the last line of real data.
17	R	0	Busy status of the state-machines. ‘1’ = busy, ‘0’ = idle
16	R/W	0	DEMUX_MODE: 0 = 4:2:2, 1 = RGB (24-bit). This value is used to control the demuxing logic when the picture is stored together. When a picture is stored together, the data is read into a single FIFO (the Y FIFO) and must be demultiplexed into the “drain” outputs. In the case of 4:2:2 the data is assumed to be stored in memory in 16-bit chunks: <YCb><YCr><YCb><YCr>,… the Y, Cb and Cr 8-bit values are pulled from the single Y-FIFO and sent out in pairs. This value is only valid when the picture is stored together. If the picture is separated into different canvases, then this bit field is ignored.
15-14	R/W	0	BYTES_PER_PIXEL: This value is used to determine how many bytes are associated with each pixel. 0: This value should be used if the image is stored separately (e.g. RGB or Y, Cb, Cr). 1: This value should be used if the data is 4:2:2 data stored together. In this case each pixel , YCb or YCr, is 16-bits (two bytes). 2: This value should be used if the RGB (24-bit) data is stored together. 3: reserved for future use (alpha RGB).
13-12	R/W	0	DDR_BURST_SIZE_CR: This value is used to control the DDR burst request size for the Cr FIFO. 0: Maximum burst = 24 64-bit values 1: Maximum burst = 32 64-bit values 2: Maximum burst = 48 64-bit values 3: Maximum burst = 64 64-bit values
11-10	R/W	0	DDR_BURST_SIZE_CB: This value is used to control the DDR burst request size for the Cb FIFO.

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Bit(s)	R/W	Default	Description
			0: Maximum burst = 24 64-bit values 1: Maximum burst = 32 64-bit values 2: Maximum burst = 48 64-bit values 3: Maximum burst = 64 64-bit values
9-8	R/W	0	DDR_BURST_SIZE_Y: This value is used to control the DDR burst request size for the Y FIFO. 0: Maximum burst = 24 64-bit values 1: Maximum burst = 32 64-bit values 2: Maximum burst = 48 64-bit values 3: Maximum burst = 64 64-bit values
7	R/W	0	MANUAL_START_FRAME: non-latching bit that can be used to simulate the go_ field signal for simulation.
6	R/W	0	CHRO_RPT_LASTL_CTRL: This bit controls whether to allow VPP's chroma-repeat request. 0: Chroma-repeat pulses from VPP are ignored 1: Chroma-repeat pulses from VPP are used.
5	R/W	0	Unused
4	R/W	0	LITTLE_ENDIAN: This bit defines the endianess of the memory data . 0: Pixel data are stored big-endian in memory 1: Chroma-repeat pulses from VPP are used.
3	R/W	0	Chroma_hz_avg: For chroma line output control, similar to luma_hz_avg, as below. Not used if data are stored together in one canvas.
2	R/W	0	Luma_hz_avg: Enable output half amount of data per line to save bandwidth. 0: Output every pixel per line 1: Output half line, each data averaged between every 2 pixels Note: For 4:2:2 mode data stored together in one canvas, only do averaging over luma data.
1	R/W	0	SEPARATE_EN: Set this bit to 1 if the image is in separate canvas locations.
0	R/W	0	ENABLE: This bit is set to 1 to enable the FIFOs and other logic. This bit can be set to 0 to cleanup and put the logic into an IDLE state.

Table 8-747 DI_IF0_CANVAS0 – Picture 0 0x2031

Bit(s)	R/W	Default	Description
31-24	R/W	0	unused
23-16	R/W	0	CANVAS0_ADDR2: Canvas table address for picture 0 for component 2 (Cr FIFO). This value is ignored when the picture is stored together
15-8	R/W	0	CANVAS0_ADDR1: Canvas table address for picture 0 for component 1 (Cb FIFO). This value is ignored when the picture is stored together
7-0	R/W	0	CANVAS0_ADDR0: Canvas table address for picture 0 for component 0 (Y FIFO).

Table 8-748 DI_IF0_LUMA_X0 – Picture 0 0x2032

Bit(s)	R/W	Default	Description
31	R/W	0	Unused
30-16	R/W	0	LUMA_X_END0: Picture 0, luma X end value
15	R/W	0	Unused
14-0	R/W	0	LUMA_X_START0: Picture 0, luma X start value

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Table 8-749 DI_IF0_LUMA_Y0 – Picture 0 0x2033

Bit(s)	R/W	Default	Description
31-29	R/W	0	Unused
28-16	R/W	0	LUMA_Y_END0: Picture 0, luma Y end value
15-13	R/W	0	Unused
12-0	R/W	0	LUMA_Y_START0: Picture 0, luma Y start value

Table 8-750 DI_IF0_CHROMA_X0 – Picture 0 0x2034

Bit(s)	R/W	Default	Description
31	R/W	0	Unused
30-16	R/W	0	CHROMA_X_END0: Picture 0, chroma X end value. This value is only used when the picture is not stored together.
15	R/W	0	Unused
14-0	R/W	0	CHROMA _X_START0: Picture 0, chroma X start value. This value is only used when the picture is not stored together.

Table 8-751 DI_IF0_ CHROMA_Y0 – Picture 0 0x2035

Bit(s)	R/W	Default	Description
31-29	R/W	0	Unused
28-16	R/W	0	CHROMA _Y_END0: Picture 0, chroma Y end value. This value is only used when the picture is not stored together.
15-13	R/W	0	Unused
12-0	R/W	0	CHROMA _Y_START0: Picture 0, chroma Y start value. This value is only used when the picture is not stored together.

Table 8-752 DI_IF0_REPEAT_LOOP – Pictures 0 and 1 0x2036

Bit(s)	R/W	Default	Description
31-24	R/W	0	CHROMA_RPT_LOOP1: Repeat loop for Picture 1. Bits[6:4] = start loop pointer, bits [2:0] = end loop pointer. Bits [7] and [3] are ignored.
23-16	R/W	0	LUMA_RPT_LOOP1: Repeat loop for Picture 1. Bits[6:4] = start loop pointer, bits [2:0] = end loop pointer. Bits [7] and [3] are ignored.
15-8	R/W	0	CHROMA_RPT_LOOP0: Repeat loop for Picture 0. Bits[6:4] = start loop pointer, bits [2:0] = end loop pointer. Bits [7] and [3] are ignored.
7-0	R/W	0	LUMA_RPT_LOOP0: Repeat loop for Picture 0. Bits[6:4] = start loop pointer, bits [2:0] = end loop pointer. Bits [7] and [3] are ignored.

Table 8-753 DI_IF0_LUMA0_RPT_PAT – Picture 0 LUMA repeat pattern 0x2037

Bit(s)	R/W	Default	Description
31-0	R/W	0	Luma repeat/skip pattern for picture 0

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Bits	Pattern Index	Pattern description
31-28	7	Repeat/skip pattern: Bit[3] = 0 indicates repeat. Bit[3] = 1 indicates either skip, or output this line and then skip. How to interpret this bit depends on the value of the previous pattern's Bit[3]. If previous Bit[3]=0, then skip; If previous Bit[3]=1, then output this line and then skip. Bits[2:0] indicate the skip / repeat count. Blow is an example of consecutive patterns, the start line is line 0: {0010} Repeat this line (line 0) two more times for a total of three line reads. Proceed to next line (line 1). {0000} Don't repeat this line (line 1). This line will be read just once. Proceed to next line (line 2). {1000} Skip one line (line 2) to get to the next line (line 3). The skip implies that the next line (line 3) should be read at least once. {1011} Read this line (line 3) once, and then skip the next four lines to get to the next line (line 8). The skip implies that the next line (line 8) should be read at least once. {0100} Repeat this line (line 8) four more times for a total of five line read. Proceed to next line (line 9). {1001} Skip two lines to get to the next line (line 11). The skip implies that the next line (line 11) should be read at least once.
27-24	6	See pattern definition above.
23-20	5	See pattern definition above.
19-16	4	See pattern definition above.
15-12	3	See pattern definition above.
11-8	2	See pattern definition above.
7-4	1	See pattern definition above.
3-0	0	See pattern definition above.

Table 8-754 DI_IF0_CHROMA0_RPT_PAT – Picture 0 CHROMA repeat pattern 0x2038

Bit(s)	R/W	Default	Description
31-0	R/W	0	Chroma repeat/skip pattern for picture 0. See picture 0 luma pattern for description. This value is only used when the picture is not stored together.

Table 8-755 DI_IF0_DUMMY_PIXEL 0x2039

Bit(s)	R/W	Default	Description
31-24	R/W	0	Y or R dummy pixel value
23-16	R/W	0	Cb or G dummy pixel value
15-8	R/W	0	Cr or B dummy pixel value
7-0	R/W	0	unused

Table 8-756 DI_IF0_LUMA_FIFO_SIZE 0x203A

Bit(s)	R/W	Default	Description
8-0	W/R	128	fifo size

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Table 8-757 DI_IF0_RANGE_MAP_Y 0x203B

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-758 DI_IF0_RANGE_MAP_CB 0x203C

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-759 DI_IF0_RANGE_MAP_CR 0x203D

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Output data range conversion function:

Y [n] = clip (Round (((Y [n] +DIN_OFFSET) * RANGE_MAP_COEF) / (1<< RANGE_MAP_SR)) +DOUT_OFFSET);

To perform VC-1 range reduction, set the following:

DIN_OFFSET = 0x180 = -128;

RANGE_MAP_COEF = RANGE_MAPY + 9

RANGE_MAP_SR = 3

DOUT_OFFSET = 0x080 = 128

To get the equivalent function:

Y[n] = clip((((Y[n]-128) * (RANGE_MAPY + 9) + 4) >> 3) +128);

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Table 8-760

Bit(s)	R/W	Default	Description
31-23	R/W	0	DIN_OFFSET
22-15	R/W	0	RANGE_MAP_COEF
14	R/W	0	unused
13-10	R/W	0	RANGE_MAP_SR
9-1	R/W	0	DOUT_OFFSET
0	R/W	0	RANGE_MAP_EN

Table 8-761 DI_IF0_GEN_REG2 0x203E

Bit(s)	R/W	Default	Description
31-30	R/W	0	unused
29	R/W	0	chroma line read sel
28	R/W	0	luma line read sel
25:24	R/W	0	shift_pat_cr
17:16	R/W	0	shift_pat_cb
9-8	R/W	0	shift_pat_y
6	R/W	0	hold_line[6]
3	R/W	0	y_rev
2	R/W	0	x_rev
1-0	R/W	0	COLOR_MAP: Define color map for NV12 or NV21 mode. Only applicable when VD1_IF0_GEN_REG.SEPARATE_EN = 1. 0: NOT NV12 or NV21; 1: NV12 (CbCr); 2: NV21 (CrCb).

Table 8-762 DI_IF0_FMT_CTRL 0x203F

Bit(s)	R/W	Default	Description
31	R/W	0	gate_clk_en. 0=No clock gating, free-running; 1=Enable clock gating for power saving.
30	R/W	0	soft_rst. If true, reset formatters.
29	R/W	0	unused
28	R/W	0	if true, horizontal formatter use repeating to generete pixel, otherwise use bilinear interpolation
27-24	R/W	0	horizontal formatter initial phase
23	R/W	0	horizontal formatter repeat pixel 0 enable
22-21	R/W	0	horizontal Y/C ratio, 00: 1:1, 01: 2:1, 10: 4:1
20	R/W	0	horizontal formatter enable
19	R/W	0	if true, always use phase0 while vertical formater, meaning always repeat data, no interpolation

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Bit(s)	R/W	Default	Description
18	R/W	0	if true, disable vertical formatter chroma repeat last line
17	R/W	0	veritcal formatter dont need repeat line on phase0, 1: enable, 0: disable
16	R/W	0	veritcal formatter repeat line 0 enable
15-12	R/W	0	vertical formatter skip line num at the beginning
11-8	R/W	0	vertical formatter initial phase
7-1	R/W	0	vertical formatter phase step (3.4)
0	R/W	0	vertical formatter enable

Table 8-763 DI_IF0_FMT_W 0x2040

Bit(s)	R/W	Default	Description
27-16	R/W	0	horizontal formatter width
11-0	R/W	0	vertical formatter width

8.2.3.6 VIUB Registers (slow clock)

DI_IF2_GEN_REG 0x2010

Same as DI_IF1_GEN_REG

DI_IF2_CANVAS0 0x2011

Same as DI_IF1_CANVAS0

DI_IF2_LUMA_X0 0x2012

Same as DI_IF1_LUMA_X0

DI_IF2_LUMA_Y0 0x2013

Same as DI_IF1_LUMA_Y0

DI_IF2_CHROMA_X0 0x2014

Same as DI_IF1_CHROMA_X0

DI_IF2_CHROMA_Y0 0x2015

Same as DI_IF1_CHROMA_Y0

DI_IF2_RPT_LOOP 0x2016

Same as DI_IF1_RPT_LOOP

DI_IF2_LUMA0_RPT_PAT 0x2017

Same as DI_IF1_LUMA0_RPT_PAT

DI_IF2_CHROMA0_RPT_PAT 0x2018

Same as DI_IF1_CHROMA0_RPT_PAT

DI_IF2_DUMMY_PIXEL 0x2019

Same as DI_IF1_DUMMY_PIXEL

DI_IF2_LUMA_FIFO_SIZE 0x201a

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Same as DI_IF1_LUMA_FIFO_SIZE

DI_IF2_RANGE_MAP_Y 0x201b

Same as DI_IF1_RANGE_MAP_Y

DI_IF2_RANGE_MAP_CB 0x201c

Same as DI_IF1_RANGE_MAP_CB

DI_IF2_RANGE_MAP_CR 0x201d

Same as DI_IF1_RANGE_MAP_CR

DI_IF2_GEN_REG2 0x201e

Same as DI_IF1_GEN_REG2

DI_IF2_FMT_CTRL 0x201f

Same as DI_IF1_FMT_CTRL

DI_IF2_FMT_W 0x2020

Same as DI_IF1_FMT_W

DI_IF2_URGENT_CTRL 0x2021

Same as DI_IF1_URGENT_CTRL

Table 8-764 DI_IF0_GEN_REG3 0x2042

Bit(s)	R/W	Default	Description
11-10	R/W	0	cntl_dbg_mode
9-8	R/W	0	cntl_bits_mode : 0->8bit 1->10bit 422 2->10bit 444
6-4	R/W	3	cntl_blk_len
2-1	R/W	1	cntl_burst_len
0	R/W	1	cntl_64bit_rev

Table 8-765 DI_IF1_GEN_REG3 0x20a7

Bit(s)	R/W	Default	Description
11-10	R/W	0	cntl_dbg_mode
9-8	R/W	0	cntl_bits_mode : 0->8bit 1->10bit 422 2->10bit 444
6-4	R/W	3	cntl_blk_len
2-1	R/W	1	cntl_burst_len
0	R/W	1	cntl_64bit_rev

Table 8-766 DI_IF2_GEN_REG3 0x2022

Bit(s)	R/W	Default	Description
11-10	R/W	0	cntl_dbg_mode
9-8	R/W	0	cntl_bits_mode : 0->8bit 1->10bit 422 2->10bit 444
6-4	R/W	3	cntl_blk_len

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Bit(s)	R/W	Default	Description
2-1	R/W	1	cntl_burst_len
0	R/W	1	cntl_64bit_rev

Table 8-767 DI_INP_GEN_REG3 0x20a8

Bit(s)	R/W	Default	Description
11-10	R/W	0	cntl_dbg_mode
9-8	R/W	0	cntl_bits_mode : 0->8bit 1->10bit 422 2->10bit 444
6-4	R/W	3	cntl_blk_len
2-1	R/W	1	cntl_burst_len
0	R/W	1	cntl_64bit_rev

Table 8-768 DI_MEM_GEN_REG3 0x20a9

Bit(s)	R/W	Default	Description
11-10	R/W	0	cntl_dbg_mode
9-8	R/W	0	cntl_bits_mode : 0->8bit 1->10bit 422 2->10bit 444
6-4	R/W	3	cntl_blk_len
2-1	R/W	1	cntl_burst_len
0	R/W	1	cntl_64bit_rev

Table 8-769 DI_CHAN2_GEN_REG3 0x20aa

Bit(s)	R/W	Default	Description
11-10	R/W	0	cntl_dbg_mode
9-8	R/W	0	cntl_bits_mode : 0->8bit 1->10bit 422 2->10bit 444
6-4	R/W	3	cntl_blk_len
2-1	R/W	1	cntl_burst_len
0	R/W	1	cntl_64bit_rev

8.2.3.7 De-Interlace Registers

De-Interlace mif Registers

Table 8-770 DI_IF1_GEN_REG 0x17E8

Bit(s)	R/W	Default	Description
31	W/R	0	enable free clk
30	W/R	0	sw reset : pulse bit
29	W/R	0	reset on go field

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Bit(s)	R/W	Default	Description
28	W/R	0	urgent chroma
27	W/R	0	urgent luma
26	W/R	0	chroma end at last line : 0 = read last line or push dummy after last line; 1 = stop read after last line
25	W/R	0	luma end at last line : 0 = read last line or push dummy after last line; 1 = stop read after last line
24-19	W/R	4	hold line[5:0], see GEN_REG2[6]
18	W/R	1	last line mode: 0 = read last line; 1 = push fixed value
16	W/R	0	demux mode: 0 = 4:2:2 demux; 1 = RGB demuxing from a single FIFO
15-14	W/R	0	bytes per pixel : 0= 1byte per pixel; 1 = 2 bytes per pixel; 2 = 3bytes per pixel
13-12	W/R	0	burst size cr: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
11-10	W/R	0	burst size cb: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
9-8	W/R	0	burst size y: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
7	W/R	0	start frame manual : pulse bit
6	W/R	0	chroma repeat last1
5	W/R	0	Reserved
4	W/R	0	little endian: 0=Pixels are big-endian in memory; 1=Pixel are little-endian in memory
3	W/R	0	chroma hz avg: 0= output pixel by pixel per line; 1= output half line ,average between every 2 pixels
2	W/R	0	luma_hz_avg: 0= output pixel by pixel per line; 1= output half line ,average between every 2 pixels
1	W/R	0	separate_en: Set to 1 to use 3 separate FIFO’s
0	W/R	0	enable

Table 8-771 DI_IF1_GEN_REG2 0x1790

Bit(s)	R/W	Default	Description
31-30	R/W	0	unused
29	R/W	0	chroma line read sel
28	R/W	0	luma line read sel
25-14	W/R	0	shift pat cr
17-16	W/R	0	shift pat cb
9-8	W/R	0	shift pat y
6	W/R	0	hold_lines[6]
3	W/R	0	y_rev: X read direction: 0=default ,normal read; 1=reverse read
2	W/R	0	x_rev: Y read direction: 0=default ,normal read; 1=reverse read
1-0	W/R	0	color map: 0=default color map as defined by “bytes per pixel”; 1=NV12(CbCr); 2=NV21(CrCb)

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Table 8-772 DI_IF1_CANVAS0 0x17E9

Bit(s)	R/W	Default	Description
31	W/R	0	canvas addr syncen
23-16	W/R	0	canvas addr2
15-8	W/R	0	canvas addr1
7-0	W/R	0	canvas addr0

Table 8-773 DI_IF1_LUMA_X0 0x17EA

Bit(s)	R/W	Default	Description
30-16	W/R	0	luma_x_end
14-0	W/R	0	luma_x_start

Table 8-774 DI_IF1_LUMA_Y0 0x17EB

Bit(s)	R/W	Default	Description
28-16	W/R	0	luma_y_end
12-0	W/R	0	luma_y_start

Table 8-775 DI_IF1_CHROMA_X0 0x17EC

Bit(s)	R/W	Default	Description
30-16	W/R	0	chroma_x_end
14-0	W/R	0	chroma_x_start

Table 8-776 DI_IF1_CHROMA_Y0 0x17ED

Bit(s)	R/W	Default	Description
28-16	W/R	0	chroma_y_end
12-0	W/R	0	chroma_y_start

Table 8-777 DI_IF1_RPT_LOOP 0x17EE

Bit(s)	R/W	Default	Description
15-8	W/R	0	chroma repeat loop
7-0	W/R	0	luma repeat loop

Table 8-778 DI_IF1_LUMA0_RPT_PAT 0x17EF

Bit(s)	R/W	Default	Description
31-0	W/R	0	luma repeat pattern

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Table 8-779 DI_IF1_CHROMA0_RPT_PAT 0x17F0

Bit(s)	R/W	Default	Description
7-0	W/R	0	chroma repeat loop

Table 8-780 DI_IF1_DUMMY_PIXEL 0x17F1

Bit(s)	R/W	Default	Description
31-0	W/R	0x8080-00	dummy pixel

Table 8-781 DI_IF1_LUMA_FIFO_SIZE 0x17F2

Bit(s)	R/W	Default	Description
8-0	W/R	128	fifo size

Table 8-782 DI_IF1_RANGE_MAP_Y 0x17FC

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-783 DI_IF1_RANGE_MAP_CB 0x17FD

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-784 DI_IF1_RANGE_MAP_CR 0x17FE

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

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Table 8-785 DI_IF1_URGENT_CTRL 0x17A3

Bit(s)	R/W	Default	Description
13-16	W/R	0	urgent_ctrl_luma: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold
15-0	W/R	0	urgent_ctrl_chroma: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold

Table 8-786 DI_IF1_FMT_CTRL 0x17F3

Bit(s)	R/W	Default	Description
31	R/W	0	gate_clk_en. 0=No clock gating, free-running; 1=Enable clock gating for power saving.
30	R/W	0	soft_rst. If true, reset formatters.
29	R/W	0	unused
28	R/W	0	if true, horizontal formatter use repeating to generete pixel, otherwise use bilinear interpolation
27-24	R/W	0	horizontal formatter initial phase
23	R/W	0	horizontal formatter repeat pixel 0 enable
22-21	R/W	0	horizontal Y/C ratio, 00: 1:1, 01: 2:1, 10: 4:1
20	R/W	0	horizontal formatter enable
19	R/W	0	if true, always use phase0 while vertical formater, meaning always repeat data, no interpolation
18	R/W	0	if true, disable vertical formatter chroma repeat last line
17	R/W	0	veritcal formatter dont need repeat line on phase0, 1: enable, 0: disable
16	R/W	0	veritcal formatter repeat line 0 enable
15-12	R/W	0	vertical formatter skip line num at the beginning
11-8	R/W	0	vertical formatter initial phase
7-1	R/W	0	vertical formatter phase step (3.4)
0	R/W	0	vertical formatter enable

Table 8-787 DI_IF1_FMT_W 0x17F4

Bit(s)	R/W	Default	Description
27-16	R/W	0	horizontal formatter width
12-0	R/W	0	vertical formatter width

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Table 8-788 DI_INP_GEN_REG 0x17CE

Bit(s)	R/W	Default	Description
31	W/R	0	enable free clk
30	W/R	0	sw reset : pulse bit
29	W/R	0	reset on go field
28	W/R	0	urgent chroma
27	W/R	0	urgent luma
26	W/R	0	chroma end at last line : 0 = read last line or push dummy after last line; 1 = stop read after last line
25	W/R	0	luma end at last line : 0 = read last line or push dummy after last line; 1 = stop read after last line
24-19	W/R	4	hold line[5:0], see GEN_REG2[6]
18	W/R	1	last line mode: 0 = read last line; 1 = push fixed value
16	W/R	0	demux mode: 0 = 4:2:2 demux; 1 = RGB demuxing from a single FIFO
15-14	W/R	0	bytes per pixel : 0= 1byte per pixel; 1 = 2 bytes per pixel; 2 = 3bytes per pixel
13-12	W/R	0	burst size cr: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
11-10	W/R	0	burst size cb: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
9-8	W/R	0	burst size y: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
7	W/R	0	start frame manual : pulse bit
6	W/R	0	chroma repeat last1
5	W/R	0	Reserved
4	W/R	0	little endian: 0=Pixels are big-endian in memory; 1=Pixel are little-endian in memory
3	W/R	0	chroma hz avg: 0= output pixel by pixel per line; 1= output half line ,average between every 2 pixels
2	W/R	0	luma_hz_avg: 0= output pixel by pixel per line; 1= output half line ,average between every 2 pixels
1	W/R	0	separate_en: Set to 1 to use 3 separate FIFO’s
0	W/R	0	enable

Table 8-789 DI_INP_GEN_REG2 0x1791

Bit(s)	R/W	Default	Description
29	W/R	0	chroma line read sel
28	W/R	0	luma line read sel
25-14	W/R	0	shift pat cr
17-16	W/R	0	shift pat cb
9-8	W/R	0	shift pat y
6	W/R	0	hold_lines[6]
3	W/R	0	y_rev: X read direction: 0=default ,normal read; 1=reverse read

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Bit(s)	R/W	Default	Description
2	W/R	0	x_rev: Y read direction: 0=default ,normal read; 1=reverse read
1-0	W/R	0	color map: 0=default color map as defined by “bytes per pixel”; 1=NV12(CbCr); 2=NV21(CrCb)

Table 8-790 DI_INP_CANVAS0 0x17CF

Bit(s)	R/W	Default	Description
31	W/R	0	canvas addr syncen
23-16	W/R	0	canvas addr2
15-8	W/R	0	canvas addr1
7-0	W/R	0	canvas addr0

Table 8-791 DI_INP_LUMA_X0 0x17D0

Bit(s)	R/W	Default	Description
30-16	W/R	0	luma_x_end
14-0	W/R	0	luma_x_start

Table 8-792 DI_INP_LUMA_Y0 0x17D1

Bit(s)	R/W	Default	Description
28-16	W/R	0	luma_y_end
12-0	W/R	0	luma_y_start

Table 8-793 DI_INP_CHROMA_X0 0x17D2

Bit(s)	R/W	Default	Description
30-16	W/R	0	chroma_x_end
14-0	W/R	0	chroma_x_start

Table 8-794 DI_INP_CHROMA_Y0 0x17D3

Bit(s)	R/W	Default	Description
28-16	W/R	0	chroma_y_end
12-0	W/R	0	chroma_y_start

Table 8-795 DI_INP_RPT_LOOP 0x17D4

Bit(s)	R/W	Default	Description
15-8	W/R	0	chroma repeat loop
7-0	W/R	0	luma repeat loop

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Table 8-796 DI_INP_LUMA0_RPT_PAT 0x17D5

Bit(s)	R/W	Default	Description
31-0	W/R	0	luma repeat pattern

Table 8-797 DI_INP_CHROMA0_RPT_PAT 0x17D6

Bit(s)	R/W	Default	Description
7-0	W/R	0	chroma repeat loop

Table 8-798 DI_INP_DUMMY_PIXEL 0x17D7

Bit(s)	R/W	Default	Description
31-0	W/R	0x8080-00	dummy pixel

Table 8-799 DI_INP_LUMA_FIFO_SIZE 0x17D8

Bit(s)	R/W	Default	Description
8-0	W/R	128	fifo size

Table 8-800 DI_INP_RANGE_MAP_Y 0x17BA

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-801 DI_INP_RANGE_MAP_CB 0x17BB

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

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Table 8-802 DI_INP_RANGE_MAP_CR 0x17BC

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-803 DI_INP_URGENT_CTRL 0x17A4

Bit(s)	R/W	Default	Description
13-16	W/R	0	urgent_ctrl_luma: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold
15-0	W/R	0	urgent_ctrl_chroma: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold

Table 8-804 DI_INP_FMT_CTRL 0x17D9

Bit(s)	R/W	Default	Description
31	R/W	0	gate_clk_en. 0=No clock gating, free-running; 1=Enable clock gating for power saving.
30	R/W	0	soft_rst. If true, reset formatters.
29	R/W	0	unused
28	R/W	0	if true, horizontal formatter use repeating to generete pixel, otherwise use bilinear interpolation
27-24	R/W	0	horizontal formatter initial phase
23	R/W	0	horizontal formatter repeat pixel 0 enable
22-21	R/W	0	horizontal Y/C ratio, 00: 1:1, 01: 2:1, 10: 4:1
20	R/W	0	horizontal formatter enable
19	R/W	0	if true, always use phase0 while vertical formater, meaning always repeat data, no interpolation
18	R/W	0	if true, disable vertical formatter chroma repeat last line
17	R/W	0	veritcal formatter dont need repeat line on phase0, 1: enable, 0: disable
16	R/W	0	veritcal formatter repeat line 0 enable
15-12	R/W	0	vertical formatter skip line num at the beginning
11-8	R/W	0	vertical formatter initial phase
7-1	R/W	0	vertical formatter phase step (3.4)
0	R/W	0	vertical formatter enable

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Table 8-805 DI_INP_FMT_W 0x17DA

Bit(s)	R/W	Default	Description
27-16	R/W	0	horizontal formatter width
12-0	R/W	0	vertical formatter width

Table 8-806 DI_MEM_GEN_REG 0x17DB

Bit(s)	R/W	Default	Description
31	W/R	0	enable free clk
30	W/R	0	sw reset : pulse bit
29	W/R	0	reset on go field
28	W/R	0	urgent chroma
27	W/R	0	urgent luma
26	W/R	0	chroma end at last line : 0 = read last line or push dummy after last line; 1 = stop read after last line
25	W/R	0	luma end at last line : 0 = read last line or push dummy after last line; 1 = stop read after last line
24-19	W/R	4	hold line[5:0], see GEN_REG2[6]
18	W/R	1	last line mode: 0 = read last line; 1 = push fixed value
16	W/R	0	demux mode: 0 = 4:2:2 demux; 1 = RGB demuxing from a single FIFO
15-14	W/R	0	bytes per pixel : 0= 1byte per pixel; 1 = 2 bytes per pixel; 2 = 3bytes per pixel
13-12	W/R	0	burst size cr: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
11-10	W/R	0	burst size cb: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
9-8	W/R	0	burst size y: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
7	W/R	0	start frame manual : pulse bit
6	W/R	0	chroma repeat last1
5	W/R	0	Reserved
4	W/R	0	little endian: 0=Pixels are big-endian in memory; 1=Pixel are little-endian in memory
3	W/R	0	chroma hz avg: 0= output pixel by pixel per line; 1= output half line ,average between every 2 pixels
2	W/R	0	luma_hz_avg: 0= output pixel by pixel per line; 1= output half line ,average between every 2 pixels
1	W/R	0	separate_en: Set to 1 to use 3 separate FIFO’s
0	W/R	0	enable

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Table 8-807 DI_MEM_GEN_REG2 0x1792

Bit(s)	R/W	Default	Description
25-14	W/R	0	shift pat cr
17-16	W/R	0	shift pat cb
9-8	W/R	0	shift pat y
6	W/R	0	hold_lines[6]
3	W/R	0	y_rev: X read direction: 0=default ,normal read; 1=reverse read
2	W/R	0	x_rev: Y read direction: 0=default ,normal read; 1=reverse read
1-0	W/R	0	color map: 0=default color map as defined by “bytes per pixel”; 1=NV12(CbCr); 2=NV21(CrCb)

Table 8-808 DI_MEM_CANVAS0 0x17DC

Bit(s)	R/W	Default	Description
31	W/R	0	canvas addr syncen
23-16	W/R	0	canvas addr2
15-8	W/R	0	canvas addr1
7-0	W/R	0	canvas addr0

Table 8-809 DI_MEM_LUMA_X0 0x17DD

Bit(s)	R/W	Default	Description
30-16	W/R	0	luma_x_end
14-0	W/R	0	luma_x_start

Table 8-810 DI_MEM_LUMA_Y0 0x17DE

Bit(s)	R/W	Default	Description
28-16	W/R	0	luma_y_end
12-0	W/R	0	luma_y_start

Table 8-811 DI_MEM_CHROMA_X0 0x17DF

Bit(s)	R/W	Default	Description
30-16	W/R	0	chroma_x_end
14-0	W/R	0	chroma_x_start

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Table 8-812 DI_MEM_CHROMA_Y0 0x17E0

Bit(s)	R/W	Default	Description
28-16	W/R	0	chroma_y_end
12-0	W/R	0	chroma_y_start

Table 8-813 DI_MEM_RPT_LOOP 0x17E1

Bit(s)	R/W	Default	Description
15-8	W/R	0	chroma repeat loop
7-0	W/R	0	luma repeat loop

Table 8-814 DI_MEM_LUMA0_RPT_PAT 0x17E2

Bit(s)	R/W	Default	Description
31-0	W/R	0	luma repeat pattern

Table 8-815 DI_MEM_CHROMA0_RPT_PAT 0x17E3

Bit(s)	R/W	Default	Description
7-0	W/R	0	chroma repeat loop

Table 8-816 DI_MEM_DUMMY_PIXEL 0x17E4

Bit(s)	R/W	Default	Description
31-0	W/R	0x8080-00	dummy pixel

Table 8-817 DI_MEM_LUMA_FIFO_SIZE 0x17E5

Bit(s)	R/W	Default	Description
8-0	W/R	128	fifo size

Table 8-818 DI_MEM_RANGE_MAP_Y 0x17BD

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

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Table 8-819 DI_MEM_RANGE_MAP_CB 0x17BE

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-820 DI_MEM_RANGE_MAP_CR 0x17BF

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-821 DI_MEM_URGENT_CTRL 0x17A5

Bit(s)	R/W	Default	Description
13-16	W/R	0	urgent_ctrl_luma: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold
15-0	W/R	0	urgent_ctrl_chroma: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold

Table 8-822 DI_MEM_FMT_CTRL 0x17E6

Bit(s)	R/W	Default	Description
31	R/W	0	gate_clk_en. 0=No clock gating, free-running; 1=Enable clock gating for power saving.
30	R/W	0	soft_rst. If true, reset formatters.
29	R/W	0	unused
28	R/W	0	if true, horizontal formatter use repeating to generete pixel, otherwise use bilinear interpolation
27-24	R/W	0	horizontal formatter initial phase
23	R/W	0	horizontal formatter repeat pixel 0 enable
22-21	R/W	0	horizontal Y/C ratio, 00: 1:1, 01: 2:1, 10: 4:1
20	R/W	0	horizontal formatter enable

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Bit(s)	R/W	Default	Description
19	R/W	0	if true, always use phase0 while vertical formater, meaning always repeat data, no interpolation
18	R/W	0	if true, disable vertical formatter chroma repeat last line
17	R/W	0	veritcal formatter dont need repeat line on phase0, 1: enable, 0: disable
16	R/W	0	veritcal formatter repeat line 0 enable
15-12	R/W	0	vertical formatter skip line num at the beginning
11-8	R/W	0	vertical formatter initial phase
7-1	R/W	0	vertical formatter phase step (3.4)
0	R/W	0	vertical formatter enable

Table 8-823 DI_MEM_FMT_W 0x17E7

Bit(s)	R/W	Default	Description
27-16	R/W	0	horizontal formatter width
12-0	R/W	0	vertical formatter width

Table 8-824 DI_CHAN2_GEN_REG 0x17F5

Bit(s)	R/W	Default	Description
31	W/R	0	enable free clk
30	W/R	0	sw reset : pulse bit
29	W/R	0	reset on go field
28	W/R	0	urgent chroma
27	W/R	0	urgent luma
26	W/R	0	chroma end at last line : 0 = read last line or push dummy after last line; 1 = stop read after last line
25	W/R	0	luma end at last line : 0 = read last line or push dummy after last line; 1 = stop read after last line
24-19	W/R	4	hold line[5:0], see GEN_REG2[6]
18	W/R	1	last line mode: 0 = read last line; 1 = push fixed value
16	W/R	0	demux mode: 0 = 4:2:2 demux; 1 = RGB demuxing from a single FIFO
15-14	W/R	0	bytes per pixel : 0= 1byte per pixel; 1 = 2 bytes per pixel; 2 = 3bytes per pixel
13-12	W/R	0	burst size cr: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
11-10	W/R	0	burst size cb: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
9-8	W/R	0	burst size y: 0 = 24x64; 1 = 32x64; 2 = 48x64; 3 = 64x64
7	W/R	0	start frame manual : pulse bit
6	W/R	0	chroma repeat last1
5	W/R	0	Reserved

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Bit(s)	R/W	Default	Description
4	W/R	0	little endian: 0=Pixels are big-endian in memory; 1=Pixel are little-endian in memory
3	W/R	0	chroma hz avg: 0= output pixel by pixel per line; 1= output half line ,average between every 2 pixels
2	W/R	0	luma_hz_avg: 0= output pixel by pixel per line; 1= output half line ,average between every 2 pixels
1	W/R	0	separate_en: Set to 1 to use 3 separate FIFO's
0	W/R	0	enable

Table 8-825 DI_CHAN2_GEN_REG2 0x17B7

Bit(s)	R/W	Default	Description
25-14	W/R	0	shift pat cr
17-16	W/R	0	shift pat cb
9-8	W/R	0	shift pat y
6	W/R	0	hold_lines[6]
3	W/R	0	y_rev: X read direction: 0=default ,normal read; 1=reverse read
2	W/R	0	x_rev: Y read direction: 0=default ,normal read; 1=reverse read
1-0	W/R	0	color map: 0=default color map as defined by "bytes per pixel"; 1=NV12(CbCr); 2=NV21(CrCb)

Table 8-826 DI_CHAN2_CANVAS0 0x17F6

Bit(s)	R/W	Default	Description
31	W/R	0	canvas addr syncen
23-16	W/R	0	canvas addr2
15-8	W/R	0	canvas addr1
7-0	W/R	0	canvas addr0

Table 8-827 DI_CHAN2_LUMA_X0 0x17F7

Bit(s)	R/W	Default	Description
30-16	W/R	0	luma_x_end
14-0	W/R	0	luma_x_start

Table 8-828 DI_CHAN2_LUMA_Y0 0x17F8

Bit(s)	R/W	Default	Description
28-16	W/R	0	luma_y_end
12-0	W/R	0	luma_y_start

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Table 8-829 DI_CHAN2_CHROMA_X0 0x17F9

Bit(s)	R/W	Default	Description
30-16	W/R	0	chroma_x_end
14-0	W/R	0	chroma_x_start

Table 8-830 DI_CHAN2_CHROMA_Y0 0x17FA

Bit(s)	R/W	Default	Description
28-16	W/R	0	chroma_y_end
12-0	W/R	0	chroma_y_start

Table 8-831 DI_CHAN2_RPT_LOOP 0x17FB

Bit(s)	R/W	Default	Description
15-8	W/R	0	chroma repeat loop
7-0	W/R	0	luma repeat loop

Table 8-832 DI_CHAN2_LUMA0_RPT_PAT 0x17B0

Bit(s)	R/W	Default	Description
31-0	W/R	0	luma repeat pattern

Table 8-833 DI_CHAN2_CHROMA0_RPT_PAT 0x17B1

Bit(s)	R/W	Default	Description
7-0	W/R	0	chroma repeat loop

Table 8-834 DI_CHAN2_DUMMY_PIXEL 0x17B2

Bit(s)	R/W	Default	Description
31-0	W/R	0x8080-00	dummy pixel

Table 8-835 DI_CHAN2_LUMA_FIFO_SIZE 0x17B3

Bit(s)	R/W	Default	Description
8-0	W/R	128	fifo size

Table 8-836 DI_CHAN2_RANGE_MAP_Y 0x17B4

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef

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Bit(s)	R/W	Default	Description
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-837 DI_CHAN2_RANGE_MAP_CB 0x17B5

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-838 DI_CHAN2_RANGE_MAP_CR 0x17B6

Bit(s)	R/W	Default	Description
31-23	W/R	0	din offset
22-15	W/R	0	range map coef
13-10	W/R	0	range map din offset mult
9-1	W/R	0	dout offset
0	W/R	0	range map enable

Table 8-839 DI_CHAN2_URGENT_CTRL 0x17A6

Bit(s)	R/W	Default	Description
13-16	W/R	0	urgent_ctrl_luma: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold
15-0	W/R	0	urgent_ctrl_chroma: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold

Table 8-840 DI_CHAN2_FMT_CTRL 0x17B8

Bit(s)	R/W	Default	Description
31	R/W	0	gate_clk_en. 0=No clock gating, free-running; 1=Enable clock gating for power saving.
30	R/W	0	soft_rst. If true, reset formatters.
29	R/W	0	unused

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Bit(s)	R/W	Default	Description
28	R/W	0	if true, horizontal formatter use repeating to generete pixel, otherwise use bilinear interpolation
27-24	R/W	0	horizontal formatter initial phase
23	R/W	0	horizontal formatter repeat pixel 0 enable
22-21	R/W	0	horizontal Y/C ratio, 00: 1:1, 01: 2:1, 10: 4:1
20	R/W	0	horizontal formatter enable
19	R/W	0	if true, always use phase0 while vertical formater, meaning always repeat data, no interpolation
18	R/W	0	if true, disable vertical formatter chroma repeat last line
17	R/W	0	veritcal formatter dont need repeat line on phase0, 1: enable, 0: disable
16	R/W	0	veritcal formatter repeat line 0 enable
15-12	R/W	0	vertical formatter skip line num at the beginning
11-8	R/W	0	vertical formatter initial phase
7-1	R/W	0	vertical formatter phase step (3.4)
0	R/W	0	vertical formatter enable

Table 8-841 DI_CHAN2_FMT_W 0x17B9

Bit(s)	R/W	Default	Description
27-16	R/W	0	horizontal formatter width
12-0	R/W	0	vertical formatter width

Table 8-842 DI_NRWR_CTRL 0x17C2

Bit(s)	R/W	Default	Description
31	R/W	0	Pending_ddr_wrrsp_nrwr
30	R/W	0	Nrwr_reg_swap
29-26	R/W	0	Nrwr_burst_lim
25	R/W	0	Nrwr_canvas_syncen
24	R/W	0	Nrwr_no_clk_gate
23-22	R/W	0	Nrwr_rgb_mode , 0: 4:2:2 to one canvas; 1: 4:4:4 to one canvas; 2: Y to luma canvas, CbCr to chroma canvas, for NV12/21; 3: Reserved.
21-20	R/W	0	Nrwr_hconv_mode
19-18	R/W	0	Nrwr_vconv_mode
17	R/W	0	Nrwr_swap_cbcr
16	R/W	0	Nrwr_urgent
15-8	R/W	0	Nrwr_canvas_index_chroma
7-0	R/W	0	Nrwr_canvas_index_luma

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Table 8-843 DI_NRWR_X 0x17C0

Bit(s)	R/W	Default	Description
31	R/W	0	Nrwr_little_endian
30	R/W	0	Nrwr_rev_x
29-16	R/W	0	Nrwr_start_x
15-14	R/W	0	nrwr_words_lim[3:2]
13-0			Nrwr_end_x

Table 8-844 DI_NRWR_Y 0x17C1

Bit(s)	R/W	Default	Description
31-30	R/W	1	Nrwr_words_lim[1:0]
29	R/W	0	Nrwr_rev_y
28-16	R/W	0	Nrwr_start_y
15	R/W	0	Nrwr_ext_en
14	R/W	1	Nrwr bit10 mode
12-0	R/W	0	Nrwr_end_y

Table 8-845 DI_DIWR_CTRL 0x17C8

Bit(s)	R/W	Default	Description
31	R/W	0	Pending_ddr_wrrsp_Diwr
30	R/W	0	Diwr_reg_swap
29-26	R/W	0	Diwr_burst_lim
25	R/W	0	Diwr_canvas_syncen
24	R/W	0	Diwr_no_clk_gate
23-22	R/W	0	Diwr_rgb_mode , 0: 4:2:2 to one canvas; 1: 4:4:4 to one canvas; 2: Y to luma canvas, CbCr to chroma canvas, for NV12/21; 3: Reserved.
21-20	R/W	0	Diwr_hconv_mode
19-18	R/W	0	Diwr_vconv_mode
17	R/W	0	Diwr_swap_cbcr
16	R/W	0	Diwr_urgent
15-8	R/W	0	Diwr_canvas_index_chroma
7-0	R/W	0	Diwr_canvas_index_luma

Table 8-846 DI_DIWR_X 0x17C6

Bit(s)	R/W	Default	Description
31	R/W	0	Diwr_little_endian
30	R/W	0	Diwr_rev_x

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Bit(s)	R/W	Default	Description
29-16	R/W	0	Diwr_start_x
15-14	R/W	0	Diwr_words_lim[3:2]
13-0			Diwr_end_x

Table 8-847 DI_DIWR_Y 0x17C7

Bit(s)	R/W	Default	Description
31-30	R/W	1	Diwr_words_lim[1:0]
29	R/W	0	Diwr_rev_y
28-16	R/W	0	Diwr_start_y
15	R/W	0	Diwr_ext_en
14	R/W	1	Diwr bit10 mode
12-0	R/W	0	Diwr_end_y

Table 8-848 DI_CONTWR_X 0x17A0

Bit(s)	R/W	Default	Description
30	R/W	0	rev_x
29-16	R/W	0	start_x
13-0	R/W	0	end_x

Table 8-849 DI_CONTWR_Y 0x17A1

Bit(s)	R/W	Default	Description
29	R/W	0	rev_x
28-16	R/W	0	start_x
12-0	R/W	0	end_x

Table 8-850 DI_CONTWR_CTRL 0x17A2

Bit(s)	R/W	Default	Description
31	R/W	0	write : clear wrrsp; read : Pending_ddr_wrrsp
11	R/W	0	canvas sync_enable
10	R/W	0	bits per pixel
8	R/W	0	urgent
7-0	R/W	0	canvas_index

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Table 8-851 DI_MTNWR_X 0x17C3

Bit(s)	R/W	Default	Description
30	R/W	0	rev_x
29-16	R/W	0	start_x
13-0	R/W	0	end_x

Table 8-852 DI_MTNWR_Y 0x17C4

Bit(s)	R/W	Default	Description
29	R/W	0	rev_x
28-16	R/W	0	start_x
12-0	R/W	0	end_x

Table 8-853 DI_MTNWR_CTRL 0x17C5

Bit(s)	R/W	Default	Description
31	R/W	0	write : clear wrrsp; read : Pending_ddr_wrrsp
11	R/W	0	canvas sync_enable
10	R/W	0	bits per pixel
8	R/W	0	urgent
7-0	R/W	0	canvas_index

Table 8-854 DI_MTNRD_X 0x17CB

Bit(s)	R/W	Default	Description
29-16	R/W	0	start_x
13-0	R/W	0	end_x

Table 8-855 DI_MTNRD_Y 0x17CC

Bit(s)	R/W	Default	Description
28-16	R/W	0	start_x
12-0	R/W	0	end_x

Table 8-856 DI_MTNRD_CTRL 0x17CD

Bit(s)	R/W	Default	Description
23	R/W	0	canvas sync_enable
22	R/W	0	bits per pixel
18	R/W	0	x flip
17	R/W	0	y flip

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Bit(s)	R/W	Default	Description
16	R/W	0	urgent
15-8	R/W	0	canvas_index

Table 8-857 DI_CONTRD_X 0x17A3

Bit(s)	R/W	Default	Description
29-16	R/W	0	start_x
13-0	R/W	0	end_x

Table 8-858 DI_CONTRD_Y 0x17A4

Bit(s)	R/W	Default	Description
28-16	R/W	0	start_x
12-0	R/W	0	end_x

Table 8-859 DI_CONTRD2_X 0x17A5

Bit(s)	R/W	Default	Description
29-16	R/W	0	start_x
13-0	R/W	0	end_x

Table 8-860 DI_CONTRD2_Y 0x17A6

Bit(s)	R/W	Default	Description
28-16	R/W	0	start_x
12-0	R/W	0	end_x

Table 8-861 DI_CONTRD_CTRL 0x17A7

Bit(s)	R/W	Default	Description
24	R/W	0	canvas sync_enable
23	R/W	0	contp bits per pixel
22	R/W	0	contp2 bits per pixel
18	R/W	0	contp x flip
17	R/W	0	contp y flip
18	R/W	0	contp2 x flip
17	R/W	0	contp2 y flip
16	R/W	0	urgent
15-8	R/W	0	contp2 canvas_index
7-0	R/W	0	contp canvas_index

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Table 8-862 MCDI_MCVECWR_X 0x2f92

Bit(s)	R/W	Default	Description
29	R/W	0	rev_x
28-16	R/W	0	start_x
12-0	R/W	0	end_x

Table 8-863 MCDI_MCVECWR_Y 0x2f93

Bit(s)	R/W	Default	Description
29	R/W	0	rev_y
28-16	R/W	0	start_y
12-0	R/W	0	end_y

Table 8-864 MCDI_MCVECWR_CTRL 0x2f94

Bit(s)	R/W	Default	Description
31-16	R/W	0	urgent_ctrl: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold
14	R/W	0	canvas sync enable
13	R/W	0	canvas_wr
12	R/W	0	req_en
11	R/W	0	fix disable
10	R/W	0	clear wrrsp
9	R/W	0	no clk gate
8	R/W	0	urgent
7-0	R/W	0	canvas index

Table 8-865 MCDI_MCVECRD_X 0x2f95

Bit(s)	R/W	Default	Description
30	R/W	0	x flip
28-16	R/W	0	start_x
12-0	R/W	0	end_x

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Table 8-866 MCDI_MCVECRD_Y 0x2f96

Bit(s)	R/W	Default	Description
30	R/W	0	y flip
28-16	R/W	0	start_y
12-0	R/W	0	end_y

Table 8-867 MCDI_MCVECRD_CTRL 0x2f97

Bit(s)	R/W	Default	Description
31-16	R/W	0	urgent_ctrl: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold
10	R/W	0	canvas sync enable
9	R/W	0	req_en
8	R/W	0	urgent
7-0	R/W	0	canvas index

Table 8-868 MCDI_MCINFOWR_X 0x2f98

Bit(s)	R/W	Default	Description
29	R/W	0	rev_x
28-16	R/W	0	start_x
12-0	R/W	0	end_x

Table 8-869 MCDI_MCINFOWR_Y 0x2f99

Bit(s)	R/W	Default	Description
29	R/W	0	rev_y
28-16	R/W	0	start_y
12-0	R/W	0	end_y

Table 8-870 MCDI_MCINFOWR_CTRL 0x2f9a

Bit(s)	R/W	Default	Description
31-16	R/W	0	urgent_ctrl: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold
14	R/W	0	canvas sync enable
13	R/W	0	canvas_wr
12	R/W	0	req_en

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Bit(s)	R/W	Default	Description
11	R/W	0	fix disable
10	R/W	0	clear wrrsp
9	R/W	0	no clk gate
8	R/W	0	urgent
7-0	R/W	0	canvas index

Table 8-871 MCDI_MCINFORD_X 0x2f9b

Bit(s)	R/W	Default	Description
30	R/W	0	x flip
28-16	R/W	0	start_x
12-0	R/W	0	end_x

Table 8-872 MCDI_MCINFORD_Y 0x2f9c

Bit(s)	R/W	Default	Description
30	R/W	0	y flip
28-16	R/W	0	start_y
12-0	R/W	0	end_y

Table 8-873 MCDI_MCINFORD_CTRL 0x2f9d

Bit(s)	R/W	Default	Description
31-16	R/W	0	urgent_ctrl: bit 15: auto urgent_en bit 14: urgent_wr bit 7-4: up_threshold bit 3-0: down_threshold
10	R/W	0	canvas sync enable
9	R/W	0	req_en
8	R/W	0	urgent
7-0	R/W	0	canvas index

Table 8-874 MCDI_MCVECWR_CANVAS_SIZE 0x2f65

Bit(s)	R/W	Default	Description
28-16	R/W	0	hsizem1
12-0	R.W	0	vsizem1

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Table 8-875 MCDI_MCVECRD_CANVAS_SIZE 0x2f66

Bit(s)	R/W	Default	Description
28-16	R/W	0	hsizem1
12-0	R.W	0	vsizem1

Table 8-876 MCDI_MCINFOWR_CANVAS_SIZE 0x2f67

Bit(s)	R/W	Default	Description
28-16	R/W	0	hsizem1
12-0	R.W	0	vsizem1

Table 8-877 MCDI_MCINFORD_CANVAS_SIZE 0x2f68

Bit(s)	R/W	Default	Description
28-16	R/W	0	hsizem1
12-0	R.W	0	vsizem1

Di mad registers

Table 8-878 DI_PRE_GL_CTRL 0x20ab

Bit(s)	R/W	Default	Description
31	W/R	0	pre count enable
30	W/R	0	pre count reset
29:16	W/R	0x20	total line number for pre count
15	W	0	pre mif manual start
13:0	W/R	0xc	the line number of pre frame reset

Table 8-879 DI_PRE_GL_THD 0x20ac

Bit(s)	R/W	Default	Description
21:16	W/R	10	DI PRE hold line number
15:0	W/R	1920	H total pixel number for pre count

Table 8-880 DI_POST_GL_CTRL 0x20ad

Bit(s)	R/W	Default	Description
31	W/R	0	post count enable
30	W/R	0	post count reset
29:16	W/R	0x20	total line number for post count
15	W	0	post mif manual start
13:0	W/R	0xc	the line number of post frame reset

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Table 8-881 DI_POST_GL_THD 0x20ae

Bit(s)	R/W	Default	Description
21:16	W/R	10	DI POST hold line number
15:0	W/R	1920	H total pixel number for post count

Table 8-882 DI_PRE_CTRL 0x1700

Bit(s)	R/W	Default	Description
31	W/R		cbus_pre_frame_rst
30	W/R		cbus_pre_soft_rst
29	W/R		pre_field_num
28	W/R		di mem mif bypass : 1:bypass
27:26	W/R		mode_444c422
25	W/R		di_cont_read_en
24:23	W/R		mode_422c444
22	W/R		mtn_after_nr
21:16	W/R		pre_hold_fifo_lines
21	W/R		pre field num for nr
20	W/R		pre field num for pulldown
19	W/R		pre field num for mcdi
18	W/R		pd_mtn_swap
17	W/R		reg_me_autoen
16	W/R		reg_me_en
15	W/R		nr_wr_by
14	W/R		use_vdin_go_line
13	W/R		di_prevdin_en
12	W/R		di_pre_viu_link
11	W/R		di_chan3_enable
10	W/R		di_mcinfo_rd_mif en
9	W/R		di_buf2_en
8	W/R		di_chan2_en
7	W/R		prenr_hist_en
6	W/R		chan2_hist_en
5	W/R		hist_check_en
4	W/R		check_after_nr
3	W/R		check222p_en

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Bit(s)	R/W	Default	Description
2	W/R		check322p_en
1	W/R		mtn_en
0	W/R		nr_en

Table 8-883 DI_POST_CTRL 0x1701

Bit(s)	R/W	Default	Description
31	W/R		cbus_post_frame_rst
30	W/R		cbus_post_soft_rst
29	W/R		post_field_num
21:16	W/R		post_hold_fifo_lines
14	W/R		mc vector invert 1:invert 0:no invert
13	W/R		prepost_link
12	W/R		di_post_viu_link
11	W/R		di_post_repeat
10	W/R		di_post_drop_1st
9	W/R		mif0_to_vpp_en
8	W/R		di_vpp_out_en
7	W/R		di_wr_bk_en
6	W/R		di_mux_en
5	W/R		di_blend_en
4	W/R		di_mtnp_read_en
3	W/R		di_mcvec_read_en
2	W/R		di_ei_en
1	W/R		di_buf1_en
0	W/R		di_buf0_en

Table 8-884 DI_POST_SIZE 0x1702

Bit(s)	R/W	Default	Description
31:29	W/R	0	diwr_field_mode
28:16	W/R	0	vsize1post
12:0	W/R	0	hsize1post

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Table 8-885 DI_PRE_SIZE 0x1703

Bit(s)	R/W	Default	Description
31:29	W/R	0	nrwr_field_mode
28:16	W/R	0	vsize1pre
12:0	W/R	0	hsize1pre

Table 8-886 DI_EI_CTRL0 0x1704

Bit(s)	R/W	Default	Description
23:16	W/R		ei0_filter[2:+] abs_diff_left>filter && ...right>filter && ...top>filter && ...bot>filter -> filter
15:8	W/R		ei0_threshold[2:+]
3	W/R		ei0_vertical
2	W/R		ei0_bpscf2
1	W/R		ei0_bpsfar1

Table 8-887 DI_EI_CTRL1 0x1705

Bit(s)	R/W	Default	Description
31:24	W/R		ei0_diff
23:16	W/R		ei0_angle45
15:8	W/R		ei0_peak
7:0	W/R		ei0_cross

Table 8-888 DI_EI_CTRL2 0x1706

Bit(s)	R/W	Default	Description
31:24	W/R		ei0_close2
23:16	W/R		ei0_close1
15:8	W/R		ei0_far2
7:0	W/R		ei0_far1

Table 8-889 DI_NR_CTRL0 0x1707

Bit(s)	R/W	Default	Description
26	W/R		nr_cue_en
25	W/R		nr2_en

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Table 8-890 DI_NR_CTRL1 0x1708

Bit(s)	R/W	Default	Description
31:30	W/R		mot_p1txtcore_mode
29:24	W/R		mot_p1txtcore_clmt
21:16	W/R		mot_p1txtcore_ylmt
15:8	W/R		mot_p1txtcore_crate
7:0	W/R		mot_p1txtcore_yrate

Table 8-891 DI_NR_CTRL2 0x1709

Bit(s)	R/W	Default	Description
29:24	W/R		mot_curtxtcore_clmt
21:16	W/R		mot_curtxtcore_ylmt
15:8	W/R		mot_curtxtcore_crate
7:0	W/R		mot_curtxtcore_yrate

Table 8-892 DI_CANVAS_URGENT0 0x170a

Bit(s)	R/W	Default	Description
26	R/W	0	di write mif bvalid_sel: 1. Bvalid_singnal from bus, 0: bytes_wr handshakes
25	R/W	0	di write mif burst last sel: 1. All kind of burst last signal include ext_data_last. 0. Used the normal burst last signal
24:16	W/R	0	Di write mif urgent ctrl
9	R/W	0	nr write mif bvalid_sel: 1. Bvalid_singnal from bus, 0: bytes_wr handshakes
8	R/W	0	nr write mif burst last sel: 1. All kind of burst last signal include ext_data_last. 0. Used the normal burst last signal
7:0	W/R	0	Nr write mif urgent ctrl

Table 8-893 DI_MTN_CTRL 0x170b

Bit(s)	R/W	Default	Description
31	W/R	0	reg_mtn_1_en if 0x170b[0]=1 (DI_MTN_CTRL)
30	W/R	0	reg_mtn_init if 0x170b[0]=1 (DI_MTN_CTRL)
29	W/R	0	reg_di2nr_txt_en if 0x170b[0]=1 (DI_MTN_CTRL)
28	W/R	0	reg_di2nr_txt_mode if 0x170b[0]=1 (DI_MTN_CTRL)
27:24	W/R	0	reg_mtn_def if 0x170b[0]=1 (DI_MTN_CTRL)
23: 0	W/R	0	reserved

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Table 8-894 DI_MTN_CTRL1 0x170c

Bit(s)	R/W	Default	Description
17	W/R		Invert NR field
16	W/R		Invert pulldown field
15	W/R		Invert mcdi field
14	W/R		Swap line0 and line2 of mtn input data
13	W/R		me enable
12	W/R		me autoenable
11:8	W/R		mtn_paramtnthd
7:0	W/R		mtn_paraflthd

Table 8-895 DI_BLEND_CTRL 0x170d

Bit(s)	R/W	Default	Description
31	W/R		blend_1_en
30	W/R		blend_mtn_lpf
28	W/R		post_mb_en
27	W/R		blend_mtn3p_max
26	W/R		blend_mtn3p_min
25	W/R		blend_mtn3p_ave
24	W/R		blend_mtn3p_maxtb
23	W/R		blend_mtn_flt_en
22	W/R		blend_data_flt_en
21:20	W/R		blend_top_mode 00: mtn, 01: weave mode, 10: bob mode, 11 : blend mode
19	W/R		blend_reg3_enable
18	W/R		blend_reg2_enable
17	W/R		blend_reg1_enable
16	W/R		blend_reg0_enable
15:14	W/R		blend_reg3_mode
13:12	W/R		blend_reg2_mode
11:10	W/R		blend_reg1_mode
9:8	W/R		blend_reg0_mode
7:0	W/R		kdeint

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Table 8-896 DI_ARB_CTRL 0x170f

Bit(s)	R/W	Default	Description
31:26	W/R	0x20	Di_arb_thd1
25:20	W/R	0x20	Di_arb_thd0
19	W/R	0	Di_arb_tid_mode
18	W/R	0	Di_arb_arb_mode
17	W/R	0	Di_arb_acg_en
16	W/R	0	Di_arb_disable_clk
15:0	W/R	0	Di_arb_req_en

Table 8-897 DI_BLEND_REG0_X 0x1710

Bit(s)	R/W	Default	Description
28:16	W/R		blend_reg0_startx
12:0	W/R		blend_reg0_endx

Table 8-898 DI_BLEND_REG0_Y 0x1711

Bit(s)	R/W	Default	Description

Table 8-899 DI_BLEND_REG1_X 0x1712

Bit(s)	R/W	Default	Description

Table 8-900 DI_BLEND_REG1_Y 0x1713

Bit(s)	R/W	Default	Description

Table 8-901 DI_BLEND_REG2_X 0x1714

Bit(s)	R/W	Default	Description

Table 8-902 DI_BLEND_REG2_Y 0x1715

Bit(s)	R/W	Default	Description

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Table 8-903 DI_BLEND_REG3_X 0x1716

Bit(s)	R/W	Default	Description

Table 8-904 DI_BLEND_REG3_Y 0x1717

Bit(s)	R/W	Default	Description

Table 8-905 DI_EI_CTRL4 0x171a

Bit(s)	R/W	Default	Description
29	W/R	0	reg_ei_caldrt_amblike2_biasvertical
28:24	W/R	21	reg_ei_caldrt_addxla2list_drtmax
23	W/R	0	N/A
22:20	W/R	1	reg_ei_caldrt_addxla2list_signm0th
19	W/R	1	reg_ei_caldrt_addxla2list_mode
18:16	W/R	3	reg_ei_signm_sad_cor_rate
15:12	W/R	3	reg_ei_signm_sadi_cor_rate
11:6	W/R	2	reg_ei_signm_sadi_cor_ofst
5:0	W/R	4	reg_ei_signm_sad_ofst

Table 8-906 DI_EI_CTRL5 0x171b

Bit(s)	R/W	Default	Description
30:28	W/R	5	reg_ei_caldrt_cnflcctchk_frcverthrd
27	W/R	0	N/A
26:24	W/R	2	reg_ei_caldrt_cnflctchk_mg
23:22	W/R	1	reg_ei_caldrt_cnflctchk_ws
21	W/R	1	reg_ei_caldrt_cnflctchk_en
20	W/R	1	reg_ei_caldrt_verfrc_final_en
19	W/R	0	reg_ei_caldrt_verfrc_retimflt_en
18:16	W/R	3	reg_ei_caldrt_verftc_eithratemth
15	W/R	0	reg_ei_caldrt_verfrc_retiming_en
14:12	W/R	2	reg_ei_caldrt_verfrc_bothratemth
11:9	W/R	0	reg_ei_caldrt_ver_thrd
8:4	W/R	4	reg_ei_caldrt_addxla2list_drtmin
3:0	W/R	15	reg_ei_caldrt_addxla2list_drtlimit

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Table 8-907 DI_EI_CTRL6 0x171c

Bit(s)	R/W	Default	Description
31:24	W/R	80	reg_ei_caldrt_abext_sad12thhig
23:16	W/R	35	reg_ei_caldrt_abext_sad00thlow
15:8	W/R	28	reg_ei_caldrt_abext_sad12thlow
6:4	W/R	1	reg_ei_caldrt_abext_ratemth
2:0	W/R	5	reg_ei_caldrt_abext_drtthrd

Table 8-908 DI_EI_CTRL7 0x171d

Bit(s)	R/W	Default	Description
29	W/R	1	reg_ei_caldrt_xlanopeak_codien
28:24	W/R	15	reg_ei_caldrt_xlanopeak_drtmax
23	W/R	1	reg_ei_caldrt_xlanopeak_en
22:20	W/R	3	reg_ei_caldrt_abext_monotrnd_alpha
19:18	W/R	1	reg_ei_caldrt_abext_mononum12_thrd
17:16	W/R	1	reg_ei_caldrt_abext_mononum00_thrd
15:12	W/R	6	reg_ei_caldrt_abext_sad00rate
11:8	W/R	6	reg_ei_caldrt_abext_sad12rate
7:0	W/R	80	reg_ei_caldrt_abext_sad00thhig

Table 8-909 DI_EI_CTRL8 0x171e

Bit(s)	R/W	Default	Description
30:28	W/R	2	reg_ei_assign_headtail_magin
26:24	W/R	3	reg_ei_retime_lastcurpncnfltchk_mode
22:21	W/R	0	reg_ei_retime_lastcurpncnfltchk_drtth
13:11	W/R	3	reg_ei_caldrt_amblike2_drtmg
10:8	W/R	1	reg_ei_caldrt_amblike2_valmg
7:4	W/R	10	reg_ei_caldrt_amblike2_alpha
3:0	W/R	4	reg_ei_caldrt_amblike2_drtth

Table 8-910 DI_EI_CTRL9 0x171f

Bit(s)	R/W	Default	Description
31:28	W/R	7	reg_ei_caldrt_hcnfcheck_frcvert_xla_th3
27	W/R	1	reg_ei_caldrt_hcnfcheck_frcvert_xla_en
26:24	W/R	4	reg_ei_caldrt_conf_drtth
23:20	W/R	11	reg_ei_caldrt_conf_absdrtth

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Bit(s)	R/W	Default	Description
19:18	W/R	2	reg_ei_caldrt_abcheck_mode1
17:16	W/R	1	reg_ei_caldrt_abcheck_mode0
15:12	W/R	11	reg_ei_caldrt_abcheck_drtth1
11:8	W/R	11	reg_ei_caldrt_abcheck_drtth0
6:4	W/R	3	reg_ei_caldrt_abpnchk1_th
1	W/R	1	reg_ei_caldrt_abpnchk1_en
0	W/R	1	reg_ei_caldrt_abpnchk0_en

Table 8-911 DI_EI_CTRL10 0x1793

Bit(s)	R/W	Default	Description
31:28	W/R	0	reg_ei_caldrt_hstrrgchk_drtth
27:24	W/R	8	reg_ei_caldrt_hstrrgchk_frcverthrd
23:20	W/R	4	reg_ei_caldrt_hstrrgchk_mg
19	W/R	0	reg_ei_caldrt_hstrrgchk_1sidnul
18	W/R	0	reg_ei_caldrt_hstrrgchk_excpcnf
17:16	W/R	2	reg_ei_caldrt_hstrrgchk_ws
15	W/R	1	reg_ei_caldrt_hstrrgchk_en
14:13	W/R	2	reg_ei_caldrt_hpncheck_mode
12	W/R	0	reg_ei_caldrt_hpncheck_mute
11:9	W/R	3	reg_ei_caldrt_hcnfcheck_mg2
8:6	W/R	2	reg_ei_caldrt_hcnfcheck_mg1
5:4	W/R	2	reg_ei_caldrt_hcnfcheck_mode
3:0	W/R	9	reg_ei_caldrt_hcnfcheck_frcvert_xla_th5

Table 8-912 DI_EI_CTRL11 0x179e

Bit(s)	R/W	Default	Description
30:29	W/R	2	reg_ei_amb_detect_mode
28:24	W/R	8	reg_ei_amb_detect_winth
23:21	W/R	3	reg_ei_amb_decide_rppth
20:19	W/R	1	reg_ei_retime_lastmappncnfltchk_drtth
18:16	W/R	2	reg_ei_retime_lastmappncnfltchk_mode
15:14	W/R	2	reg_ei_retime_lastmapvertfrcchk_mode
13:12	W/R	3	reg_ei_retime_lastvertfrcchk_mode
11:8	W/R	0	reg_ei_retime_lastpnchk_drtth
6	W/R	1	reg_ei_retime_lastpnchk_en

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Bit(s)	R/W	Default	Description
5:4	W/R	3	reg_ei_retime_mode
3	W/R	1	reg_ei_retime_last_en
2	W/R		reg_ei_retime_ab_en
1	W/R	1	reg_ei_caldrt_hstrvertfrcchk_en
0	W/R	0	reg_ei_caldrt_hstrrgchk_mode

Table 8-913 DI_EI_CTRL12 0x179f

Bit(s)	R/W	Default	Description
31:28	W/R	13	reg_ei_drtdelay2_lmt
27:26	W/R	2	reg_ei_drtdelay2_notver_lrwin
25:24	W/R	3	reg_ei_drtdelay_mode
23	W/R	0	reg_ei_drtdelay2_mode
22:20	W/R	0	reg_ei_assign_xla_signm0th
19	W/R	1	reg_ei_assign_pkbiasvert_en
18	W/R	1	reg_ei_assign_xla_en
17:16	W/R	0	reg_ei_assign_xla_mode
15:12	W/R	2	reg_ei_assign_nlfilter_magin
11:8	W/R	5	reg_ei_localsearch_maxrange
7:4	W/R	0	reg_ei_xla_drtth
3:0	W/R	3	reg_ei_flatmsad_thrd

Table 8-914 DI_EI_CTRL13 0x17a8

Bit(s)	R/W	Default	Description
27:24	W/R	15	reg_ei_int_drt2x_chrdrt_limit
23:20	W/R	0	reg_ei_int_drt16x_core
19:16	W/R	2	reg_ei_int_drtdelay2_notver_cancv
15:8	W/R	20	reg_ei_int_drtdelay2_notver_sadth
7:0	W/R	20	reg_ei_int_drtdelay2_vlddrt_sadth

Table 8-915 DI_EI_DRT_CTRL 0x2028

Bit(s)	R/W	Default	Description
31	W/R	0	reg_rectg_en: Low angle enable.
30	W/R	0	reg_recbld_en: New and old drt blend enable.
29:28	W/R	2	reg_rectg_ws : window sideto calculate the reference direction: 0:1x1; 1:1x3; 2:1x5; 3:1x7

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Bit(s)	R/W	Default	Description
27			reserved
26:24	W/R	2	reg_abq_margin : top and bottom curve trend quantilization margin of noise for direction assignments.
23			reserved
22:20	W/R	3	reg_trend_mg : the Margin of the top/bot trend.
19:16	W/R	1	reg_int_d16xc1 : Coring to drtf.
15:14			reserved
13:8	W/R	40	reg_int_chlmt1: Limit to drtf(16x) for chroma angle
7			reserved
6:4	W/R	5	reg_nscheck_thrd:check whether the pixels id noise or not.
3			reserved
2:0	W/R	7	reg_horsl_ws: window side to check the existent number of low angle drt, if the number < the value, drt=raw drt.

Table 8-916 DI_EI_DRT_PIXTH 0x2029

Bit(s)	R/W	Default	Description
31:24	W/R	22	reg_min_pix: the threshold of min pix of photos, < threshold the pix do not participate in the monotonic trend calculation.
23:16	W/R	203	reg_max_pix:the threshold of max pix of photos, >threshold the pix do not participate in the monotonic trend calculation.
15:8	W/R	50	reg_dmaxmin_thrdma: the max pixel and min pixel difference is larger than the value the trend existent.
7:0	W/R	30	reg_dmaxmin_thrdmi: the max pixel and min pixel difference is smaller than the value the trend non-existent.

Table 8-917 DI_EI_DRT_CORRPIXTH 0x202a

Bit(s)	R/W	Default	Description
31:24	W/R	40	reg_newcorrpix_maxthrd:the new low angle drt sad threshold.
23:16	W/R	60	reg_corrpix_diffthrd: the top and bottom pixel difference is larger than the value, the case may be ultra-low angle.
15:8	W/R	10	reg_corrpix_minthrd: the difference of top and bottom pixel is smaller than the value, the drt may be raw drt.
7:0	W/R	20	reg_corrpix_maxthrd: the difference of top and bottom pixel is larger than the value, the drt may be raw drt.

Table 8-918 DI_EI_DRT_RECTG_WAVE 0x202b

Bit(s)	R/W	Default	Description
31:29			reserved
28:24	W/R	0	reg_max_pixwave: the wave of the max pix threshold, prevent min pix close to reg_max_pix caused the number between max and min pix zeros.

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Bit(s)	R/W	Default	Description
23:21			reserved
20:16	W/R	15	reg_pix_wave: the wave of the max and min pix, the max pixel smaller than the value or the min pixel larger than the value, may be the ultra-low angle case.
15:14			reserved
13:8	W/R	40	reg_maxdrt_thrd: the threshold of the low angle max drt.
7:0	W/R	20	reg_wave_thrd:in bilateral cases tow pixel difference is smaller than the value, the trend between the tow pixel not change.

Table 8-919 DI_EI_DRT_PIX_DIFFTH 0x202c

Bit(s)	R/W	Default	Description
31:24	W/R	32	reg_newraw_thrd: the old drt and new drt transition threshold.
23:16	W/R	10	reg_tb_max_thrd: the threshold of top and bottom max or min pixel.
15:8	W/R	20	reg_diffpix_thrd: Max-Min<the value,the trend is non-existent.
7:6			reserved
5:0	W/R	5	reg_bilt_trendnumt: in bilateral cases the difference between the top and bottom pixel number of the monotonic trend smaller than the value is low angle.

Table 8-920 DI_EI_DRT_UNBITREND_TH 0x202d

Bit(s)	R/W	Default	Description
31:29			reserved
28:24	W/R	10	reg_trend_numb:in bilateral cases the pixel number of the monotonic trend larger than the value is low angle.
23:21			reserved
20:16	W/R	4	reg_bilt_trendnum:in bilateral cases the pixel number of the trend larger than the value is low angle.
15:13			reserved
12:8	W/R	7	reg_unil_trendnumt: in unilateral cases the difference between the top and bottom pixel number of the monotonic trend smaller than the value is low angle.
7:5			reserved
4:0	W/R	10	reg_trend_num: in unilateral cases the pixel number of the trend larger than the value is ultra-low angle.

Table 8-921 DI_EI_XWIN0 0x1798

Bit(s)	R/W	Default	Description
27:16	W/R		ei_xend0
11:0	W/R		ei_xstart0

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Table 8-922 DI_EI_XWIN1 0x1799

Bit(s)	R/W	Default	Description

Table 8-923 DI_MC_REG0_X 0x1720

Bit(s)	R/W	Default	Description
27:16	W/R		mc_reg0_start_x
11:0	W/R		mc_reg0_end_x

Table 8-924 DI_MC_REG0_Y 0x1721

Bit(s)	R/W	Default	Description

Table 8-925 DI_MC_REG1_X 0x1722

Bit(s)	R/W	Default	Description

Table 8-926 DI_MC_REG1_Y 0x1723

Bit(s)	R/W	Default	Description

Table 8-927 DI_MC_REG2_X 0x1724

Bit(s)	R/W	Default	Description

Table 8-928 DI_MC_REG2_Y 0x1725

Bit(s)	R/W	Default	Description

Table 8-929 DI_MC_REG3_X 0x1726

Bit(s)	R/W	Default	Description

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Table 8-930 DI_MC_REG3_Y 0x1727

Bit(s)	R/W	Default	Description

Table 8-931 DI_MC_REG4_X 0x1728

Bit(s)	R/W	Default	Description

Table 8-932 DI_MC_REG4_Y 0x1729

Bit(s)	R/W	Default	Description

Table 8-933 DI_MC_32LVL0 0x172a

Bit(s)	R/W	Default	Description
31:24	W/R		mc_reg2_32lvl
23:16	W/R		mc_reg1_32lvl
15:8	W/R		mc_reg0_32lvl
7:0	W/R		field_32lvl

Table 8-934 DI_MC_32LVL1 0x172b

Bit(s)	R/W	Default	Description
15:8	W/R		mc_reg3_32lvl
7:0	W/R		mc_reg4_32lvl

Table 8-935 DI_MC_22LVL0 0x172c

Bit(s)	R/W	Default	Description
31:16	W/R		mc_reg0_22lvl
15:0	W/R		field_22lvl

Table 8-936 DI_MC_22LVL1 0x172d

Bit(s)	R/W	Default	Description
31:16	W/R		mc_reg2_22lvl
15:0	W/R		mc_reg1_22lvl

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Table 8-937 DI_MC_22LVL2 0x172e

Bit(s)	R/W	Default	Description
31:16	W/R		mc_reg4_22lvl
15:0	W/R		mc_reg3_22lvl

Table 8-938 DI_MC_CTRL 0x172f

Bit(s)	R/W	Default	Description
4	W/R		mc_reg4_en
3	W/R		mc_reg3_en
2	W/R		mc_reg2_en
1	W/R		mc_reg1_en
0	W/R		mc_reg0_en

Table 8-939 DI_INTR_CTRL 0x1730

Bit(s)	R/W	Default	Description
31	W/R	0	When Deint_irq_mode is 0: When predi_int or postdi_int is generated, the interrupt flag will be erected; Deint_irq_mode==1: When the interrupt source of any module inside DI is generated and no Mask operation is performed, the interrupt flag will be erected.
30:26	W/R	0	reserved
25	w/R	0	NrDownscale_int_mask
24	w/R	0	Det3d_int_mask
23	w/R	0	Mcinfowr_int_mask
22	w/R	0	Mcvecwr_int_mask
21	w/R	0	Medi_int_mask
20	w/R	0	Contwr_int_mask
19	w/R	0	Hist_int_mask
18	w/R	0	Diwr_int_mask
17	w/R	0	Mtn_wr_int_mask
16	w/R	0	Nrwr_int_mask
15:10	W/R	0	reserved
9	R		Nrdownscale_done
8	R		Det3d_done
7	R		Mcinfowr_done (not valid in GX)
6	R		Mcvecwr_done (not valid in GX)
5	R		Medi_done(not valid in GX)
4	R		Contwr_done

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Bit(s)	R/W	Default	Description
3	R		Hist_done
2	R		diwr_done
1	R		Mtnwr_done
0	R		Nrwr_done

Table 8-940 DI_INFO_ADDR 0x1731 Addr_0

Bit(s)	R/W	Default	Description
31:0	R		Field_32p , sum of difference between n-2 and n

Table 8-941 Addr_1

Bit(s)	R/W	Default	Description
31:24	R		Field_32max, maximum difference between n-2 and n
23:0	R		Field_32num, numbers of pixels difference > threshold

Table 8-942 Addr_2

Bit(s)	R/W	Default	Description
31:0	R		Field_22p, sum of difference between temporal and vertical difference

Table 8-943 Addr_3

Bit(s)	R/W	Default	Description
15:0	R		Field_22max , maximum difference between temporal and verticaldifference

Table 8-944 Addr_4

Bit(s)	R/W	Default	Description
23:0	R		Field_22num, pixel sum which difference > threshold

Table 8-945 Addr_5

Bit(s)	R/W	Default	Description
31:0	R		Luma sum

Table 8-946 Addr_6

Bit(s)	R/W	Default	Description
31:0	R		Difference of 32, sum in area 0

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Table 8-947 Addr_7

Bit(s)	R/W	Default	Description
31:0	R		Difference of 32, sum in area 1

Table 8-948 Addr_8

Bit(s)	R/W	Default	Description
31:0	R		Difference of 32, sum in area 2

Table 8-949 Addr_9

Bit(s)	R/W	Default	Description
31:0	R		Difference of 32, sum in area 3

Table 8-950 Addr_10

Bit(s)	R/W	Default	Description
31:0	R		Difference of 32, sum in area 4

Table 8-951 Addr_11

Bit(s)	R/W	Default	Description
31:0	R		Difference of 22, sum in area 0

Table 8-952 Addr_12

Bit(s)	R/W	Default	Description
31:0	R		Difference of 22, sum in area 1

Table 8-953 Addr_13

Bit(s)	R/W	Default	Description
31:0	R		Difference of 22, sum in area 2

Table 8-954 Addr_14

Bit(s)	R/W	Default	Description
31:0	R		Difference of 22, sum in area 3

Table 8-955 Addr_15

Bit(s)	R/W	Default	Description
31:0	R		Difference of 22, sum in area 4

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Table 8-956 Addr_16

Bit(s)	R/W	Default	Description
31:0	R		luma, sum in area 0

Table 8-957 Addr_17

Bit(s)	R/W	Default	Description
31:0	R		luma, sum in area 1

Table 8-958 Addr_18

Bit(s)	R/W	Default	Description
31:0	R		luma, sum in area 2

Table 8-959 Addr_19

Bit(s)	R/W	Default	Description
31:0	R		luma, sum in area 3

Table 8-960 Addr_20

Bit(s)	R/W	Default	Description
31:0	R		luma, sum in area 4

Table 8-961 Addr_21

Bit(s)	R/W	Default	Description
31:24	R		Field_32max, maximum difference between n-2 and n in area0
23:0	R		Field_32num, numbers of pixels difference > threshold in area0

Table 8-962 Addr_22

Bit(s)	R/W	Default	Description
31:24	R		Field_32max, maximum difference between n-2 and n in area1
23:0	R		Field_32num, numbers of pixels difference > threshold in area1

Table 8-963 Addr_23

Bit(s)	R/W	Default	Description
31:24	R		Field_32max, maximum difference between n-2 and n in area2
23:0	R		Field_32num, numbers of pixels difference > threshold in area2

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Table 8-964 Addr_24

Bit(s)	R/W	Default	Description
31:24	R		Field_32max, maximum difference between n-2 and n in area3
23:0	R		Field_32num, numbers of pixels difference > threshold in area3

Table 8-965 Addr_25

Bit(s)	R/W	Default	Description
31:24	R		Field_32max, maximum difference between n-2 and n in area4
23:0	R		Field_32num, numbers of pixels difference > threshold in area4

Table 8-966 Addr_26

Bit(s)	R/W	Default	Description
31:20	R		Field_22max/16, in area 0
19:0	R		Field_22 num/16, in area 0

Table 8-967 Addr_27

Bit(s)	R/W	Default	Description
31:20	R		Field_22max/16, in area 1
19:0	R		Field_22 num/16, in area 1

Table 8-968 Addr_28

Bit(s)	R/W	Default	Description
31:20	R		Field_22max/16, in area 2
19:0	R		Field_22 num/16, in area 2

Table 8-969 Addr_29

Bit(s)	R/W	Default	Description
31:20	R		Field_22max/16, in area 3
19:0	R		Field_22 num/16, in area 3

Table 8-970 Addr_30

Bit(s)	R/W	Default	Description
31:20	R		Field_22max/16, in area 4
19:0	R		Field_22 num/16, in area 4

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Table 8-971 DI_INFO_DATA 0x1732

Bit(s)	R/W	Default	Description

Table 8-972 DI_PRE_HOLD 0x1733

Bit(s)	R/W	Default	Description
31	R/W	0	cntl_pre_hold_enable
27:16	R/W	0	cntl_pre_hold_count
11:0	R/W	0	cntl_pre_pass_count

Table 8-973 DI_MTN_1_CTRL1 0x1740

Bit(s)	R/W	Default	Description
31	W/R		reg_mtn_1_en if 0x170b[0]=0 (DI_MTN_CTRL)
30	W/R		reg_mtn_init if 0x170b[0]=0 (DI_MTN_CTRL)
29	W/R		reg_di2nr_txt_en if 0x170b[0]=0 (DI_MTN_CTRL)
28	W/R		reg_di2nr_txt_mode if 0x170b[0]=0 (DI_MTN_CTRL)
27:24	W/R		reg_mtn_def if 0x170b[0]=0 (DI_MTN_CTRL)
23:16	W/R	32	reg_DI_cmb_adp_YCrate
15: 8	W/R	32	reg_DI_cmb_adp_2Crate
7: 0	W/R	21	reg_DI_cmb_adp_2Yrate

Table 8-974 DI_MTN_1_CTRL2 0x1741

Bit(s)	R/W	Default	Description
31:24	W/R	26	reg_DI_m1b_core_Ykinter
23:16	W/R	26	reg_DI_m1b_core_Ckinter
15:8	W/R	58	reg_DI_m1b_core_Ykintra
7:0	W/R	98	reg_DI_m1b_core_Ckintra

Table 8-975 DI_MTN_1_CTRL3 0x1742

Bit(s)	R/W	Default	Description
31:24	W/R	21	reg_DI_m1b_thr_2Yrate
23:16	W/R	32	reg_DI_m1b_thr_2Crate
15: 8	W/R	10	reg_DI_m1b_core_mxcmbY
7: 0	W/R	10	reg_DI_m1b_core_mxcmbC

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Table 8-976 DI_MTN_1_CTRL4 0x1743

Bit(s)	R/W	Default	Description
31:24	W/R	1	reg_DI_m1b_coreY
23:16	W/R	0	reg_DI_m1b_coreC
15: 8	W/R	8	reg_DI_m1b_thrd_min
7: 0	W/R	128	reg_DI_m1b_thrd_max

Table 8-977 DI_MTN_1_CTRL5 0x1744

Bit(s)	R/W	Default	Description
31:27	W/R	7	reg_DI_m1b_pp_extnd_num
27:24	W/R	4	reg_DI_m1b_pp_errod_num
21:20	W/R	0	Re_di2nr_txt_mode 0:average of top/bot 1:max; 2: |a+c-2b|;
15: 8	W/R	13	reg_DI_mot_core_Ykinter
7: 0	W/R	13	reg_DI_mot_core_Ckinter

Table 8-978 DI_MTN_1_CTRL6 0x17a9

Bit(s)	R/W	Default	Description
31:24	W/R	13	reg_DI_mot_core_Ykintra
23:16	W/R	90	reg_DI_mot_core_Ckintra
15: 8	W/R	21	reg_DI_mot_cor_2Yrate
7: 0	W/R	32	reg_DI_mot_cor_2Crate

Table 8-979 DI_MTN_1_CTRL7 0x17aa

Bit(s)	R/W	Default	Description
31:24	W/R	10	reg_DI_mot_core_mxcmbY
23:16	W/R	10	reg_DI_mot_core_mxcmbC
15: 8	W/R	2	reg_DI_mot_coreY
7: 0	W/R	1	reg_DI_mot_coreC

Table 8-980 DI_MTN_1_CTRL8 0x17ab

Bit(s)	R/W	Default	Description
31:24	W/R	26	reg_DI_fmot_core_Ykinter
23:16	W/R	26	reg_DI_fmot_core_Ckinter
15: 8	W/R	38	reg_DI_fmot_core_Ykintra
7: 0	W/R	98	reg_DI_fmot_core_Ckintra

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Table 8-981 DI_MTN_1_CTRL9 0x17ac

Bit(s)	R/W	Default	Description
31:24	W/R	13	reg_DI_fmot_cor_2Yrate
23:16	W/R	32	reg_DI_fmot_cor_2Crate
15: 8	W/R	3	reg_DI_fmot_coreY
7: 0	W/R	2	reg_DI_fmot_coreC

Table 8-982 DI_MTN_1_CTRL10 0x17ad

Bit(s)	R/W	Default	Description
27:24	W/R	2	reg_DI_m1b_suremot_num_fld0
19:16	W/R	2	reg_DI_m1b_surestl_num_fld0
11: 8	W/R	6	reg_DI_m1b_suremot_num_fld1
3: 0	W/R	6	reg_DI_m1b_surestl_num_fld1

Table 8-983 DI_MTN_1_CTRL11 0x17ae

Bit(s)	R/W	Default	Description
27:24	W/R	5	reg_DI_m1b_suremot_evn_th
20:16	W/R	8	reg_DI_m1b_suremot_odd_th
11: 8	W/R	3	reg_DI_m1b_surestl_evn_th
6	W/R	0	reg_DI_m1b_suremot_fast_en
5	W/R	0	reg_DI_m1b_surestl_fast_en
4: 0	W/R	4	reg_DI_m1b_surestl_odd_th

Table 8-984 DI_MTN_1_CTRL12 0x17af

Bit(s)	R/W	Default	Description
31:24	W/R	64	reg_DI_mot_norm_gain
17:16	W/R	2	reg_DI_mot_alpha_lpf
15: 8	W/R	10	reg_DI_m1b_surestl_thrd
4: 0	W/R	4	reg_DI_mot_surestl_gain

Table 8-985 DI_CRC_CHK0 0x17cb

Bit(s)	R/W	Default	Description
0	W	0	di pre crc check start (pulse)

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Table 8-986 DI_CRC_CHK1 0x17cc

Bit(s)	R/W	Default	Description
0	W	0	di post crc check start (pulse)

Table 8-987 DI_RO_CRC_NRWR 0x17c3

Bit(s)	R/W	Default	Description
31-0	W	0	nrwr crc output (lock by pre frame rst)

Table 8-988 DI_RO_CRC_MTNWR 0x17c4

Bit(s)	R/W	Default	Description
31-0	W	0	mtnwr crc output (lock by pre frame rst)

Table 8-989 DI_RO_CRC_DEINT 0x17c5

Bit(s)	R/W	Default	Description
31-0	W	0	di post crc output (lock by pre frame rst)

8.2.3.8 NR2 Registers

Table 8-990 DET3D_MOTN_CFG 0x1734

Bit(s)	R/W	Default	Description
16	R/W	0	reg_det3d_intr_en : Det3d interrupt enable
9:8	R/W	0	reg_Det3D_Motion_Mode : U2 Different mode for Motion Calculation of Luma and Chroma: 0 : MotY, 1: (2*MotY + (MotU + MotV))/4; 2: Max(MotY, MotU, MotV); 3:Max(MotY, (MotU+MotV)/2)
7:4	R/W	0	reg_Det3D_Motion_Core_Rate : U4 K Rate to Edge (HV) details for coring of Motion Calculations, normalized to 32
3:0	R/W	0	reg_Det3D_Motion_Core_Thrd : U4 2X: static coring value for Motion Detection.

Table 8-991 DET3D_CB_CFG 0x1735

Bit(s)	R/W	Default	Description
7:4	R/W	0	reg_Det3D_ChessBd_HV_ofst : U4, Noise immune offset for Horizotnal or vertical combing detection.
3:0	R/W	0	reg_Det3D_ChessBd_NHV_ofst : U4, Noise immune offset for NON-Horizotnal or vertical combing detection.

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Table 8-992 DET3D_SPLT_CFG 0x1736

Bit(s)	R/W	Default	Description
7:4	R/W	0x0	reg_Det3D_SplitValid_ratio : U4, Ratio between max_value and the avg_value of the edge mapping for split line valid detection. The smaller of this value, the easier of the split line detected.
3:0	R/W	0x0	reg_Det3D_AvgIdx_ratio : U4, Ratio to the avg_value of the edge mapping for split line position estimation. The smaller of this value, the more samples will be added to the estimation.

Table 8-993 DET3D_HV_MUTE 0x1737

Bit(s)	R/W	Default	Description
23:20	R/W	0x0	reg_Det3D_Edge_Ver_Mute : U4 X2: Horizontal pixels to be mute from H/V Edge calculation Top and Bottom border part.
19:16	R/W	0x0	reg_Det3D_Edge_Hor_Mute : U4 X2: Horizontal pixels to be mute from H/V Edge calculation Left and right border part.
15:12	R/W	0x0	reg_Det3D_ChessBd_Ver_Mute : U4 X2: Horizontal pixels to be mute from ChessBoard statistics calculation in middle part
11:8	R/W	0x0	reg_Det3D_ChessBd_Hor_Mute : U4 X2: Horizontal pixels to be mute from ChessBoard statistics calculation in middle part
7:4	R/W	0x0	reg_Det3D_STA8X8_Ver_Mute : U4 1X: Vertical pixels to be mute from 8x8 statistics calculation in each block.
3:0	R/W	0x0	reg_Det3D_STA8X8_Hor_Mute : U4 1X: Horizontal pixels to be mute from 8x8 statistics calculation in each block.

Table 8-994 DET3D_MAT_STA_P1M1 0x1738

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_Det3D_STA8X8_P1_K0_R8 : U8 SAD to SAI ratio to decide P1, normalized to 256 (0.8)
23:16	R/W	0x0	reg_Det3D_STA8X8_P1_K1_R7 : U8 SAD to ENG ratio to decide P1, normalized to 128 (0.5)
15:8	R/W	0x0	reg_Det3D_STA8X8_M1_K0_R6 : U8 SAD to SAI ratio to decide M1, normalized to 64 (1.1)
7:0	R/W	0x0	reg_Det3D_STA8X8_M1_K1_R6 : U8 SAD to ENG ratio to decide M1, normalized to 64 (0.8)

Table 8-995 DET3D_MAT_STA_P1TH 0x1739

Bit(s)	R/W	Default	Description
23:16	R/W	0x0	reg_Det3D_STAYUV_P1_TH_L4 : U8 SAD to ENG Thrd offset to decide P1, X16 (100)
15:8	R/W	0x0	reg_Det3D_STAEDG_P1_TH_L4 : U8 SAD to ENG Thrd offset to decide P1, X16 (80)
7:0	R/W	0x0	reg_Det3D_STAMOT_P1_TH_L4 : U8 SAD to ENG Thrd offset to decide P1, X16 (48)

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Table 8-996 DET3D_MAT_STA_M1TH 0x173a

Bit(s)	R/W	Default	Description
23:16	R/W	0x0	reg_Det3D_STAYUV_M1_TH_L4 : U8 SAD to ENG Thrd offset to decide M1, X16 (100)
15:8	R/W	0x0	reg_Det3D_STAEDG_M1_TH_L4 : U8 SAD to ENG Thrd offset to decide M1, X16 (80)
7:0	R/W	0x0	reg_Det3D_STAMOT_M1_TH_L4 : U8 SAD to ENG Thrd offset to decide M1, X16 (64)

Table 8-997 DET3D_MAT_STA_RSFT 0x173b

Bit(s)	R/W	Default	Description
5:4	R/W	0x0	reg_Det3D_STAYUV_RSHFT : U2 YUV statistics SAD and SAI calculation result right shift bits to accommodate the 12bits clipping: 0 : mainly for images <=720x480: 1: mainly for images <=1366x768: 2: mainly for images <=1920X1080: 2; 3: other higher resolutions
3:2	R/W	0x0	reg_Det3D_STAEDG_RSHFT : U2 Horizontal and Vertical Edge Statistics SAD and SAI calculation result right shift bits to accommodate the 12bits clipping: 0 : mainly for images <=720x480: 1: mainly for images <=1366x768: 2: mainly for images <=1920X1080: 2; 3: other higher resolutions
1:0	R/W	0x0	reg_Det3D_STAMOT_RSHFT : U2 Motion SAD and SAI calculation result right shift bits to accommodate the 12bits clipping: 0 : mainly for images <=720x480: 1: mainly for images <=1366x768: 2: mainly for images <=1920X1080: 2; 3: other higher resolutions

Table 8-998 DET3D_MAT_SYMTC_TH 0x173c

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_Det3D_STALUM_symtc_Th : U8 threshold to decide if the Luma statistics is TB or LR symmetric.
23:16	R/W	0x0	reg_Det3D_STACHR_symtc_Th : U8 threshold to decide if the Chroma (UV) statistics is TB or LR symmetric.
15:8	R/W	0x0	reg_Det3D_STAEDG_symtc_Th : U8 threshold to decide if the Horizontal and Vertical Edge statistics is TB or LR symmetric.
7:0	R/W	0x0	reg_Det3D_STAMOT_symtc_Th : U8 threshold to decide if the Motion statistics is TB or LR symmetric.

Table 8-999 DET3D_RO_DET_CB_HOR 0x173d

Bit(s)	R/W	Default	Description
31:16	R.O	0x0	RO_Det3D_ChessBd_NHor_value : U16 X64: number of Pixels of Horizontally Surely NOT matching Chessboard pattern.
15:0	R.O	0x0	RO_Det3D_ChessBd_Hor_value : U16 X64: number of Pixels of Horizontally Surely matching Chessboard pattern.

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Table 8-1000 DET3D_RO_DET_CB_VER 0x173e

Bit(s)	R/W	Default	Description
31:16	R.O	0x0	RO_Det3D_ChessBd_NVer_value : U16 X64: number of Pixels of Vertically Surely NOT matching Chessboard pattern.
15:0	R.O	0x0	RO_Det3D_ChessBd_Ver_value : U16 X64: number of Pixels of Vertically Surely matching Chessboard pattern.

Table 8-1001 DET3D_RO_SPLT_HT 0x173f

Bit(s)	R/W	Default	Description
24	R.O	0x0	RO_Det3D_Split_HT_valid : U1 horizontal LR split border detected valid signal for top half picture
20:16	R.O	0x0	RO_Det3D_Split_HT_pxnum : U5 number of pixels included for the LR split position estimation for top half picture
9:0	R.O	0x0	RO_Det3D_Split_HT_idxX4 : S10 X4: horizontal pixel shifts of LR split position to the (ColMax/2) for top half picture

NR2 REG DEFINE BEGIN////

Table 8-1002 NR2_MET_NM_CTRL 0x1745

Bit(s)	R/W	Default	Description
28	R/W	0x0	reg_NM_reset : Reset to the status of the Loop filter.
27:24	R/W	0x0	reg_NM_calc_length : Length mode of the Noise measurement sample number for statistics. 0 : 256 samples; 1: 512 samples; 2: 1024 samples; ¡X: 2^(8+x) samples
23:20	R/W	0x0	reg_NM_inc_step : Loop filter input gain increase step.
19:16	R/W	0x0	reg_NM_dec_step : Loop filter input gain decrease step.
15:8	R/W	0x0	reg_NM_YHPmot_thrd : Luma channel HP portion motion for condition of pixels included in Luma Noise measurement.
7:0	R/W	0x0	reg_NM_CHPmot_thrd : Chroma channel HP portion motion for condition of pixels included in Chroma Noise measurement.

Table 8-1003 NR2_MET_NM_YCTRL 0x1746

Bit(s)	R/W	Default	Description
31:28	R/W	0x0	reg_NM_YPLL_target : Target rate of NM_Ynoise_thrd to mean of the Luma Noise
27:24	R/W	0x0	reg_NM_YLPmot_thrd : Luma channel LP portion motion for condition of pixels included in Luma Noise measurement.
23:16	R/W	0x0	reg_NM_YHPmot_thrd_min : Minimum threshold for Luma channel HP portion motion to decide whether the pixel will be included in Luma noise measurement.
15:8	R/W	0x0	reg_NM_YHPmot_thrd_max : Maximum threshold for Luma channel HP portion motion to decide whether the pixel will be included in Luma noise measurement.
7:0	R/W	0x0	reg_NM_Ylock_rate : Rate to decide whether the Luma noise measurement is lock or not.

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Table 8-1004 NR2_MET_NM_CCTRL 0x1747

Bit(s)	R/W	Default	Description
31:28	R/W	0x0	reg_NM_CPLL_target : Target rate of NM_Cnoise_thrd to mean of the Chroma Noise
27:24	R/W	0x0	reg_NM_CLPmot_thrd : Chroma channel LP portion motion for condition of pixels included in Chroma Noise measurement.
23:16	R/W	0x0	reg_NM_CHPmot_thrd_min : Minimum threshold for Chroma channel HP portion motion to decide whether the pixel will be included in Chroma noise measurement.
15:8	R/W	0x0	reg_NM_CHPmot_thrd_max : Maximum threshold for Chroma channel HP portion motion to decide whether the pixel will be included in Chroma noise measurement.
7:0	R/W	0x0	reg_NM_Clock_rate : Rate to decide whether the Chroma noise measurement is lock or not;

Table 8-1005 NR2_MET_NM_TNR 0x1748

Bit(s)	R/W	Default	Description
25	R.O	0x0	ro_NM_TNR_Ylock : Read-only register to tell ifLuma channel noise measurement is locked or not.
24	R.O	0x0	ro_NM_TNR_Clock : Read-only register to tell if Chroma channel noise measurement is locked or not.
23:12	R.O	0x0	ro_NM_TNR_Ylevel : Read-only register to give Luma channel noise level. It was 16x of pixel difference in 8 bits of YHPmot.
11:0	R.O	0x0	ro_NM_TNR_Clevel : Read-only register to give Chroma channel noise level. It was 16x of pixel difference in 8 bits of CHPmot.

Table 8-1006 NR2_MET_NMFRM_TNR_YLEV 0x1749

Bit(s)	R/W	Default	Description
28:0	R.O	0x0	ro_NMFrm_TNR_Ylevel : Frame based Read-only register to give Luma channel noise level within one frame/field.

Table 8-1007 NR2_MET_NMFRM_TNR_YCNT 0x174a

Bit(s)	R/W	Default	Description
23:0	R.O	0x0	ro_NMFrm_TNR_Ycount : Number ofLuma channel pixels included in Frame/Field based noise level measurement.

Table 8-1008 NR2_MET_NMFRM_TNR_CLEV 0x174b

Bit(s)	R/W	Default	Description
28:0	R.O	0x0	ro_NMFrm_TNR_Clevel : Frame based Read-only register to give Chroma channel noise level within one frame/field.

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Table 8-1009 NR2_MET_NMFRM_TNR_CCNT 0x174c

Bit(s)	R/W	Default	Description
23:0	R.O	0x0	ro_NMFrm_TNR_Ccount : Number of Chroma channel pixels included in Frame/ Field based noise level measurement.

Table 8-1010 NR2_3DEN_MODE 0x174d

Bit(s)	R/W	Default	Description
6:4	R/W	0x0	Blend_3dnr_en_r :
2:0	R/W	0x0	Blend_3dnr_en_l :

Table 8-1011 NR2_IIR_CTRL 0x174e

Bit(s)	R/W	Default	Description
15:14	R/W	0x0	reg_LP_IIR_8bit_mode : LP IIR membitwidth mode:0: 10bits will be store in memory;1: 9bits will be store in memory; 2 : 8bits will be store in memory;3: 7bits will be store in memory;
13:12	R/W	0x0	reg_LP_IIR_mute_mode : Mode for the LP IIR mute,
11:8	R/W	0x0	reg_LP_IIR_mute_thrd : Threshold of LP IIR mute to avoid ghost:
7:6	R/W	0x0	reg_HP_IIR_8bit_mode : IIR membitwidth mode:0: 10bits will be store in memory;1: 9bits will be store in memory; 2 : 8bits will be store in memory;3: 7bits will be store in memory;
5:4	R/W	0x0	reg_HP_IIR_mute_mode : Mode for theLP IIR mute
3:0	R/W	0x0	reg_HP_IIR_mute_thrd : Threshold of HP IIR mute to avoid ghost

Table 8-1012 NR2_SW_EN 0x174f

Bit(s)	R/W	Default	Description
17:8	R/W	0x0	Clk_gate_ctrl :
7	R/W	0x0	Cfr_enable :
5	R/W	0x0	Det3d_en :
4	R/W	0x0	Nr2_proc_en :
0	R/W	0x0	Nr2_sw_en :

Table 8-1013 NR2_FRM_SIZE 0x1750

Bit(s)	R/W	Default	Description
27:16	R/W	0x0	Frm_heigh : Frame/field height
11: 0	R/W	0x0	Frm_width : Frame/field width

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Table 8-1014 NR2_SNR_SAD_CFG 0x1751

Bit(s)	R/W	Default	Description
12	R/W	0x1	reg_MATNR_SNR_SAD_CenRPL : U1, Enable signal for Current pixel position SAD to be replaced by SAD_min.0: do not replace Current pixel position SAD by SAD_min;1: do replacements
11:8	R/W	0x3	reg_MATNR_SNR_SAD_coring : Coring value of the intra-frame SAD. sum = (sum - reg_MATNR_SNR_SAD_coring);sum = (sum<0) ? 0: (sum>255)? 255: sum;
6:5	R/W	0x1	reg_MATNR_SNR_SAD_WinMod : Unsigned, Intra-frame SAD matching window mode:0: 1x1; 1: [1 1 1] 2: [1 2 1]; 3: [1 2 2 2 1];
4:0	R/W	0x1	Sad_coef_num : Sad coeffient

Table 8-1015 NR2_MATNR_SNR_OS 0x1752

Bit(s)	R/W	Default	Description
7:4	R/W	0x8	reg_MATNR_SNR_COS : SNR Filter overshoot control margin for UV channel (X2 to u10 scale)
3:0	R/W	0xd	reg_MATNR_SNR_YOS : SNR Filter overshoot control margin for luma channel (X2 to u10 scale)

Table 8-1016 NR2_MATNR_SNR_NRM_CFG 0x1753

Bit(s)	R/W	Default	Description
23:16	R/W	0x40	reg_MATNR_SNR_NRM_ofst : Edge based SNR boosting normalization offset to SAD_max ;
15:8	R/W	0xff	reg_MATNR_SNR_NRM_max : Edge based SNR boosting normalization Max value
7:0	R/W	0x0	reg_MATNR_SNR_NRM_min : Edge based SNR boosting normalization Min value

Table 8-1017 NR2_MATNR_SNR_NRM_GAIN 0x1754

Bit(s)	R/W	Default	Description
15:8	R/W	0x0	reg_MATNR_SNR_NRM_Cgain : Edge based SNR boosting normalization Gain for Chrm channel (norm 32 as 1)
7:0	R/W	0x20	reg_MATNR_SNR_NRM_Ygain : Edge based SNR boosting normalization Gain for Luma channel (norm 32 as 1)

Table 8-1018 NR2_MATNR_SNR_LPF_CFG 0x1755

Bit(s)	R/W	Default	Description
23:16	R/W	0xc	reg_MATNR_SNRLPF_SADmaxTH : U8, Threshold to SADmax to use TNRLPF to replace SNRLPF. i.e.if (SAD_max <reg_MATNR_SNRLPF_SADmaxTH) SNRLPF_yuv[k] = TNRLPF_yuv[k];
13:11	R/W	0x2	reg_MATNR_SNRLPF_Cmode : LPF based SNR filtering mode on CHRM channel: 0 : gradient LPF [1 1]/2, 1: gradient LPF [2 1 1]/4; 2: gradient LPF [3 3 2]/8; 3: gradient LPF [5 4 4 3]/16; 4 : TNRLPF; 5 : CurLPF3x3_yuv[]; 6: CurLPF3o3_yuv [] 7: CurLPF3x5_yuv[]

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Bit(s)	R/W	Default	Description
10: 8	R/W	0x2	reg_MATNR_SNRLPF_Ymode : LPF based SNR filtering mode on LUMA channel: 0 : gradient LPF //Bit [1 1]/2, 1: gradient LPF [2 1 1]/4; 2: gradient LPF [3 3 2]/8;3: gradient LPF [5 4 4 3]/16; 4 : TNRLPF; 5 : CurLPF3x3_yuv[]; 6: CurLPF3o3_yuv[] 7: CurLPF3x5_yuv[]
7:4	R/W	0x6	reg_MATNR_SNRLPF_SADmin3TH : Offset threshold to SAD_min to Discard SAD_min3 corresponding pixel in LPF SNR filtering. (X8 to u8 scale)
3:0	R/W	0x4	reg_MATNR_SNRLPF_SADmin2TH : Offset threshold to SAD_min to Discard SAD_min2 corresponding pixel in LPF SNR filtering. (X8 to u8 scale)

Table 8-1019 NR2_MATNR_SNR_USF_GAIN 0x1756

Bit(s)	R/W	Default	Description
15:8	R/W	0x0	reg_MATNR_SNR_USF_Cgain : Un-sharp (HP) compensate back Chrm portion gain, (norm 64 as 1)
7:0	R/W	0x0	reg_MATNR_SNR_USF_Ygain : Un-sharp (HP) compensate back Luma portion gain, (norm 64 as 1)

Table 8-1020 NR2_MATNR_SNR_EDGE2B 0x1757

Bit(s)	R/W	Default	Description
15:8	R/W	0x80	reg_MATNR_SNR_Edge2Beta_ofst : U8, Offset for Beta based on Edge.
7:0	R/W	0x10	reg_MATNR_SNR_Edge2Beta_gain : U8. Gain to SAD_min for Beta based on Edge. (norm 16 as 1)

Table 8-1021 NR2_MATNR_BETA_EGAIN 0x1758

Bit(s)	R/W	Default	Description
15:8	R/W	0x20	reg_MATNR_CBeta_Egain : U8, Gain to Edge based Beta for Chrm channel. (normalized to 32 as 1)
7:0	R/W	0x20	reg_MATNR_YBeta_Egain : U8, Gain to Edge based Beta for Luma channel. (normalized to 32 as 1)

Table 8-1022 NR2_MATNR_BETA_BRT 0x1759

Bit(s)	R/W	Default	Description
31:28	R/W	0x0	reg_MATNR_beta_BRT_limt_hi : U4, Beta adjustment based on Brightness high side Limit. (X16 to u8 scale)
27:24	R/W	0x0	reg_MATNR_beta_BRT_slop_hi : U4, Beta adjustment based on Brightness high side slope. Normalized to 16 as 1
23:16	R/W	0xa0	reg_MATNR_beta_BRT_thrd_hi : U8, Beta adjustment based on Brightness high threshold.(u8 scale)
15:12	R/W	0x6	reg_MATNR_beta_BRT_limt_lo : U4, Beta adjustment based on Brightness low side Limit. (X16 to u8 scale)

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Bit(s)	R/W	Default	Description
11:8	R/W	0x6	reg_MATNR_beta_BRT_slop_lo : U4, Beta adjustment based on Brightness low side slope. Normalized to 16 as 1
7:0	R/W	0x64	reg_MATNR_beta_BRT_thrd_lo : U8, Beta adjustment based on Brightness low threshold.(u8 scale)

Table 8-1023 NR2_MATNR_XBETA_CFG 0x175a

Bit(s)	R/W	Default	Description
19:18	R/W	0x0	reg_MATNR_CBeta_use_mode : U2, Beta options (mux) from beta_motion and beta_edge for Chrm channel;
17:16	R/W	0x0	reg_MATNR_YBeta_use_mode : U2, Beta options (mux) from beta_motion and beta_edge for Luma channel;
15: 8	R/W	0x0	reg_MATNR_CBeta_Ofst : U8, Offset to Beta for Chrm channel.(after beta_edge and beta_motion mux)
7: 0	R/W	0x0	reg_MATNR_YBeta_Ofst : U8, Offset to Beta for Luma channel.(after beta_edge and beta_motion mux)

Table 8-1024 NR2_MATNR_YBETA_SCL 0x175b

Bit(s)	R/W	Default	Description
31:24	R/W	0x3c	reg_MATNR_YBeta_scale_min : U8, Final step Beta scale low limit for Luma channel;
23:16	R/W	0xff	reg_MATNR_YBeta_scale_max : U8, Final step Beta scale high limit for Luma channe;
15: 8	R/W	0x20	reg_MATNR_YBeta_scale_gain : U8, Final step Beta scale Gain for Luma channel (normalized 32 to 1);
7 : 0	R/W	0x0	reg_MATNR_YBeta_scale_ofst : S8, Final step Beta scale offset for Luma channel ;

Table 8-1025 NR2_MATNR_CBETA_SCL 0x175c

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_MATNR_CBeta_scale_min : Final step Beta scale low limit for Chrm channel.Similar to Y
23:16	R/W	0xff	reg_MATNR_CBeta_scale_max : U8, Final step Beta scale high limit for Chrm channel.Similar to Y
15: 8	R/W	0x20	reg_MATNR_CBeta_scale_gain : U8, Final step Beta scale Gain for Chrm channel Similar to Y
7: 0	R/W	0x0	reg_MATNR_CBeta_scale_ofst : S8, Final step Beta scale offset for Chrm channel Similar to Y

Table 8-1026 NR2_SNR_MASK 0x175d

Bit(s)	R/W	Default	Description
20:0	R/W	0x0	SAD_MSK : Valid signal in the 3x7 SAD surface

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Table 8-1027 NR2_SAD2NORM_LUT0 0x175e

Bit(s)	R/W	Default	Description
31:24	R/W	0x72	reg_MATNR_SAD2Norm_LUT_3 : SAD convert normal LUT node 3
23:16	R/W	0x92	reg_MATNR_SAD2Norm_LUT_2 : SAD convert normal LUT node 2
15: 8	R/W	0xab	reg_MATNR_SAD2Norm_LUT_1 : SAD convert normal LUT node 1
7: 0	R/W	0xcd	reg_MATNR_SAD2Norm_LUT_0 : SAD convert normal LUT node 0

Table 8-1028 NR2_SAD2NORM_LUT1 0x175f

Bit(s)	R/W	Default	Description
31:24	R/W	0x1c	reg_MATNR_SAD2Norm_LUT_7 : SAD convert normal LUT node 7
23:16	R/W	0x23	reg_MATNR_SAD2Norm_LUT_6 : SAD convert normal LUT node 6
15: 8	R/W	0x31	reg_MATNR_SAD2Norm_LUT_5 : SAD convert normal LUT node 5
7: 0	R/W	0x4f	reg_MATNR_SAD2Norm_LUT_4 : SAD convert normal LUT node 4

Table 8-1029 NR2_SAD2NORM_LUT2 0x1760

Bit(s)	R/W	Default	Description
31:24	R/W	0xf	reg_MATNR_SAD2Norm_LUT_11 : SAD convert normal LUT node 11
23:16	R/W	0x11	reg_MATNR_SAD2Norm_LUT_10 : SAD convert normal LUT node 10
15: 8	R/W	0x13	reg_MATNR_SAD2Norm_LUT_9 : SAD convert normal LUT node 9
7: 0	R/W	0x17	reg_MATNR_SAD2Norm_LUT_8 : SAD convert normal LUT node 8

Table 8-1030 NR2_SAD2NORM_LUT3 0x1761

Bit(s)	R/W	Default	Description
31:24	R/W	0x8	reg_MATNR_SAD2Norm_LUT_15 : SAD convert normal LUT node 15
23:16	R/W	0x9	reg_MATNR_SAD2Norm_LUT_14 : SAD convert normal LUT node 14
15:8	R/W	0xa	reg_MATNR_SAD2Norm_LUT_13 : SAD convert normal LUT node 13
7:0	R/W	0xc	reg_MATNR_SAD2Norm_LUT_12 : SAD convert normal LUT node 12

Table 8-1031 NR2_EDGE2BETA_LUT0 0x1762

Bit(s)	R/W	Default	Description
31:24	R/W	0x80	reg_MATNR_Edge2Beta_LUT_3 : Edge convert beta LUT node 3
23:16	R/W	0xa0	reg_MATNR_Edge2Beta_LUT_2 : Edge convert beta LUT node 2
15: 8	R/W	0xe0	reg_MATNR_Edge2Beta_LUT_1 : Edge convert beta LUT node 1
7: 0	R/W	0xff	reg_MATNR_Edge2Beta_LUT_0 : Edge convert beta LUT node 0

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Table 8-1032 NR2_EDGE2BETA_LUT1 0x1763

Bit(s)	R/W	Default	Description
31:24	R/W	0x4	reg_MATNR_Edge2Beta_LUT_7 : Edge convert beta LUT node 7
23:16	R/W	0x10	reg_MATNR_Edge2Beta_LUT_6 : Edge convert beta LUT node 6
15: 8	R/W	0x20	reg_MATNR_Edge2Beta_LUT_5 : Edge convert beta LUT node 5
7: 0	R/W	0x50	reg_MATNR_Edge2Beta_LUT_4 : Edge convert beta LUT node 4

Table 8-1033 NR2_EDGE2BETA_LUT2 0x1a64

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_MATNR_Edge2Beta_LUT_11 : Edge convert beta LUT node 11
23:16	R/W	0x0	reg_MATNR_Edge2Beta_LUT_10 : Edge convert beta LUT node 10
15: 8	R/W	0x0	reg_MATNR_Edge2Beta_LUT_9 : Edge convert beta LUT node 9
7: 0	R/W	0x2	reg_MATNR_Edge2Beta_LUT_8 : Edge convert beta LUT node 8

Table 8-1034 NR2_EDGE2BETA_LUT3 0x1765

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_MATNR_Edge2Beta_LUT_15 : Edge convert beta LUT node 15
23:16	R/W	0x0	reg_MATNR_Edge2Beta_LUT_14 : Edge convert beta LUT node 14
15: 8	R/W	0x0	reg_MATNR_Edge2Beta_LUT_13 : Edge convert beta LUT node 13
7: 0	R/W	0x0	reg_MATNR_Edge2Beta_LUT_12 : Edge convert beta LUT node 12

Table 8-1035 NR2_MOTION2BETA_LUT0 0x1766

Bit(s)	R/W	Default	Description
31:24	R/W	0x20	reg_MATNR_Mot2Beta_LUT_3 : Motion convert beta LUT node 3
23:16	R/W	0x10	reg_MATNR_Mot2Beta_LUT_2 : Motion convert beta LUT node 2
15: 8	R/W	0x4	reg_MATNR_Mot2Beta_LUT_1 : Motion convert beta LUT node 1
7: 0	R/W	0x0	reg_MATNR_Mot2Beta_LUT_0 : Motion convert beta LUT node 0

Table 8-1036 NR2_MOTION2BETA_LUT1 0x1767

Bit(s)	R/W	Default	Description
31:24	R/W	0xc4	reg_MATNR_Mot2Beta_LUT_7 : Motion convert beta LUT node 7
23:16	R/W	0x80	reg_MATNR_Mot2Beta_LUT_6 : Motion convert beta LUT node 6
15: 8	R/W	0x40	reg_MATNR_Mot2Beta_LUT_5 : Motion convert beta LUT node 5
7: 0	R/W	0x30	reg_MATNR_Mot2Beta_LUT_4 : Motion convert beta LUT node 4

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Table 8-1037 NR2_MOTION2BETA_LUT2 0x1768

Bit(s)	R/W	Default	Description
31:24	R/W	0xff	reg_MATNR_Mot2Beta_LUT_11 : Motion convert beta LUT node 11
23:16	R/W	0xff	reg_MATNR_Mot2Beta_LUT_10 : Motion convert beta LUT node 10
15: 8	R/W	0xf0	reg_MATNR_Mot2Beta_LUT_9 : Motion convert beta LUT node 9
7: 0	R/W	0xe0	reg_MATNR_Mot2Beta_LUT_8 : Motion convert beta LUT node 8

Table 8-1038 NR2_MOTION2BETA_LUT3 0x1769

Bit(s)	R/W	Default	Description
31:24	R/W	0xff	reg_MATNR_Mot2Beta_LUT_15 : Motion convert beta LUT node 15
23:16	R/W	0xff	reg_MATNR_Mot2Beta_LUT_14 : Motion convert beta LUT node 14
15: 8	R/W	0xff	reg_MATNR_Mot2Beta_LUT_13 : Motion convert beta LUT node 13
7: 0	R/W	0xff	reg_MATNR_Mot2Beta_LUT_12 : Motion convert beta LUT node 12

Table 8-1039 NR2_MATNR_MTN_CRTL 0x176a

Bit(s)	R/W	Default	Description
25:24	R/W	0x0	reg_MATNR_Vmtn_use_mode : Motion_yuvV channel motion selection mode:0: Vmot;1:Ymot/2 + (Umot+Vmot)/4; 2:Ymot/2 + max(Umot,Vmot)/2; 3: max(Ymot, Umot, Vmot)
21:20	R/W	0x0	reg_MATNR_Umtn_use_mode : Motion_yuvU channel motion selection mode:0: Umot;1:Ymot/2 + (Umot+Vmot)/4; 2:Ymot/2 + max(Umot,Vmot)/2; 3: max(Ymot, Umot, Vmot)
17:16	R/W	0x0	reg_MATNR_Ymtn_use_mode : Motion_yuvLuma channel motion selection mode:0: Ymot, 1: Ymot/2 + (Umot+Vmot)/4; 2: Ymot/2 + max(Umot,Vmot)/2; 3: max(Ymot,Umot, Vmot)
13:12	R/W	0x1	reg_MATNR_mtn_txt_mode : Texture detection mode for adaptive coring of HP motion
9: 8	R/W	0x1	reg_MATNR_mtn_cor_mode : Coring selection mode based on texture detection;
6: 4	R/W	0x8	reg_MATNR_mtn_hpf_mode : video mode of current and previous frame/field for MotHPF_yuv[k] calculation:
2: 0	R/W	0x6	reg_MATNR_mtn_lpf_mode : LPF video mode of current and previous frame/ field for MotLPF_yuv[k] calculation:

Table 8-1040 NR2_MATNR_MTN_CRTL2 0x176b

Bit(s)	R/W	Default	Description
18:16	R/W	0x6	reg_MATNR_iir_BS_Ymode : IIR TNR filter Band split filter mode for Luma LPF result generation (Cur and Prev);
15: 8	R/W	0x40	reg_MATNR_mtnb_alpLP_Cgain : Scale of motion_brthp_uv to motion_brtlp_uv, normalized to 32 as 1
7: 0	R/W	0x40	reg_MATNR_mtnb_alpLP_Ygain : Scale of motion_brthp_y to motion_brtlp_y, normalized to 32 as 1

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Table 8-1041 NR2_MATNR_MTN_COR 0x176c

Bit(s)	R/W	Default	Description
15:12	R/W	0x3	reg_MATNR_mtn_cor_Cofst : Coring Offset for Chroma Motion.
11: 8	R/W	0x3	reg_MATNR_mtn_cor_Cgain : Gain to texture based coring for Chroma Motion. Normalized to 16 as 1
7: 4	R/W	0x3	reg_MATNR_mtn_cor_Yofst : Coring Offset for Luma Motion.
3: 0	R/W	0x3	reg_MATNR_mtn_cor_Ygain : Gain to texture based coring for Luma Motion. Normalized to 16 as 1

Table 8-1042 NR2_MATNR_MTN_GAIN 0x176d

Bit(s)	R/W	Default	Description
31:24	R/W	0x40	reg_MATNR_mtn_hp_Cgain : Gain to MotHPF_yuv[k] Chrm channel for motion calculation, normalized to 64 as 1
23:16	R/W	0x40	reg_MATNR_mtn_hp_Ygain : Gain to MotHPF_yuv[k] Luma channel for motion calculation, normalized to 64 as 1
15: 8	R/W	0x40	reg_MATNR_mtn_lp_Cgain : Gain to MotLPF_yuv[k] Chrm channel for motion calculation, normalized to 32 as 1
7: 0	R/W	0x40	reg_MATNR_mtn_lp_Ygain : Gain to MotLPF_yuv[k] Luma channel for motion calculation, normalized to 32 as 1

Table 8-1043 NR2_MATNR_DEGHOST 0x176e

Bit(s)	R/W	Default	Description
30:28	R/W	0	reg_matnr_deghost_mode : // unsigned , default = 0 0:old_deghost; 1:soft_ denoise & strong_deghost; 2:strong_denoise & soft_deghost; 3:strong_denoise & strong_deghost
24:20	R/W	4	reg_matnr_deghost_ygain : // unsigned , default = 4
16:12	R/W	4	reg_matnr_deghost_cgain : // unsigned , default = 4
8	R/W	1	reg_matnr_deghost_en : // unsigned , default = 1 0: disable; 1: enable Enable signal for DeGhost function:0: disable; 1: enable
7: 4	R/W	3	reg_matnr_deghost_cos : // unsigned , default = 3 DeGhost Overshoot margin for UV channel, (X2 to u10 scale)
3: 0	R/W	3	reg_matnr_deghost_yos : // unsigned , default = 3 DeGhost Overshoot margin for Luma channel, (X2 to u10 scale)

Table 8-1044 NR2_MATNR_ALPHALP_LUT0 0x176f

Bit(s)	R/W	Default	Description
31:24	R/W	0x40	reg_MATNR_AlphaLP_LUT_3 : Matnr low-pass filter alpha LUT node 3
23:16	R/W	0x80	reg_MATNR_AlphaLP_LUT_2 : Matnr low-pass filter alpha LUT node 2
15: 8	R/W	0x80	reg_MATNR_AlphaLP_LUT_1 : Matnr low-pass filter alpha LUT node 1
7: 0	R/W	0x80	reg_MATNR_AlphaLP_LUT_0 : Matnr low-pass filter alpha LUT node 0

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Table 8-1045 NR2_MATNR_ALPHALP_LUT1 0x1770

Bit(s)	R/W	Default	Description
31:24	R/W	0xff	reg_MATNR_AlphaLP_LUT_7 : Matnr low-pass filter alpha LUT node 7
23:16	R/W	0x80	reg_MATNR_AlphaLP_LUT_6 : Matnr low-pass filter alpha LUT node 6
15: 8	R/W	0x50	reg_MATNR_AlphaLP_LUT_5 : Matnr low-pass filter alpha LUT node 5
7: 0	R/W	0x40	reg_MATNR_AlphaLP_LUT_4 : Matnr low-pass filter alpha LUT node 4

Table 8-1046 NR2_MATNR_ALPHALP_LUT2 0x1771

Bit(s)	R/W	Default	Description
31:24	R/W	0xff	reg_MATNR_AlphaLP_LUT_11 : Matnr low-pass filter alpha LUT node 11
23:16	R/W	0xff	reg_MATNR_AlphaLP_LUT_10 : Matnr low-pass filter alpha LUT node 10
15: 8	R/W	0xff	reg_MATNR_AlphaLP_LUT_9 : Matnr low-pass filter alpha LUT node 9
7: 0	R/W	0xff	reg_MATNR_AlphaLP_LUT_8 : Matnr low-pass filter alpha LUT node 8

Table 8-1047 NR2_MATNR_ALPHALP_LUT3 0x1772

Bit(s)	R/W	Default	Description
31:24	R/W	0xff	reg_MATNR_AlphaLP_LUT_15 : Matnr low-pass filter alpha LUT node 15
23:16	R/W	0xff	reg_MATNR_AlphaLP_LUT_14 : Matnr low-pass filter alpha LUT node 14
15: 8	R/W	0xff	reg_MATNR_AlphaLP_LUT_13 : Matnr low-pass filter alpha LUT node 13
7: 0	R/W	0xff	reg_MATNR_AlphaLP_LUT_12 : Matnr low-pass filter alpha LUT node 12

Table 8-1048 NR2_MATNR_ALPHAHP_LUT0 0x1773

Bit(s)	R/W	Default	Description
31:24	R/W	0x40	reg_MATNR_AlphaHP_LUT_3 : Matnr high-pass filter alpha LUT node 3
23:16	R/W	0x80	reg_MATNR_AlphaHP_LUT_2 : Matnr high-pass filter alpha LUT node 2
15: 8	R/W	0x80	reg_MATNR_AlphaHP_LUT_1 : Matnr high-pass filter alpha LUT node 1
7: 0	R/W	0x80	reg_MATNR_AlphaHP_LUT_0 : Matnr high-pass filter alpha LUT node 0

Table 8-1049 NR2_MATNR_ALPHAHP_LUT1 0x1774

Bit(s)	R/W	Default	Description
31:24	R/W	0xff	reg_MATNR_AlphaHP_LUT_7 : Matnr high-pass filter alpha LUT node 7
23:16	R/W	0x80	reg_MATNR_AlphaHP_LUT_6 : Matnr high-pass filter alpha LUT node 6
15: 8	R/W	0x50	reg_MATNR_AlphaHP_LUT_5 : Matnr high-pass filter alpha LUT node 5
7: 0	R/W	0x40	reg_MATNR_AlphaHP_LUT_4 : Matnr high-pass filter alpha LUT node 4

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Table 8-1050 NR2_MATNR_ALPHAHP_LUT2 0x1775

Bit(s)	R/W	Default	Description
31:24	R/W	0xff	reg_MATNR_AlphaHP_LUT_11 : Matnr high-pass filter alpha LUT node 11
23:16	R/W	0xff	reg_MATNR_AlphaHP_LUT_10 : Matnr high-pass filter alpha LUT node 10
15: 8	R/W	0xff	reg_MATNR_AlphaHP_LUT_9 : Matnr high-pass filter alpha LUT node 9
7: 0	R/W	0xff	reg_MATNR_AlphaHP_LUT_8 : Matnr high-pass filter alpha LUT node 8

Table 8-1051 NR2_MATNR_ALPHAHP_LUT3 0x1776

Bit(s)	R/W	Default	Description
31:24	R/W	0xff	reg_MATNR_AlphaHP_LUT_15 : Matnr high-pass filter alpha LUT node 15
23:16	R/W	0xff	reg_MATNR_AlphaHP_LUT_14 : Matnr high-pass filter alpha LUT node 14
15: 8	R/W	0xff	reg_MATNR_AlphaHP_LUT_13 : Matnr high-pass filter alpha LUT node 13
7: 0	R/W	0xff	reg_MATNR_AlphaHP_LUT_12 : Matnr high-pass filter alpha LUT node 12

Table 8-1052 NR2_MATNR_MTNB_BRT 0x1777

Bit(s)	R/W	Default	Description
31:28	R/W	0x0	reg_MATNR_mtnb_BRT_limt_hi : Motion adjustment based on Brightness high side Limit. (X16 to u8 scale)
27:24	R/W	0x0	reg_MATNR_mtnb_BRT_slop_hi : Motion adjustment based on Brightness high side slope. Normalized to 16 as 1
23:16	R/W	0xa0	reg_MATNR_mtnb_BRT_thrd_hi : Motion adjustment based on Brightness high threshold.(u8 scale)
15:12	R/W	0x6	reg_MATNR_mtnb_BRT_limt_lo : Motion adjustment based on Brightness low side Limit. (X16 to u8 scale)
11: 8	R/W	0x6	reg_MATNR_mtnb_BRT_slop_lo : Motion adjustment based on Brightness low side slope. Normalized to 16 as 1
7: 0	R/W	0x64	reg_MATNR_mtnb_BRT_thrd_lo : Motion adjustment based on Brightness low threshold.(u8 scale)

Table 8-1053 NR2_CUE_MODE 0x1778

Bit(s)	R/W	Default	Description
31:20	R/W	0x0	Reserved
19	R/W	0x0	Cue2_isabv_org_invert1
18	R/W	0x1	Cue2_valid_condition
17:16	R/W	0x1	Cue2_orgline_flt_sel
15:12	R/W	0x4	Cue2_orgline_flt_alph
11	R/W	0x0	Cue2_isabv_org_invert
10	R/W	0x0	Cue2_iscur_org_invert
9	R/W	0x0	Cue_enable_r : Cue right half frame enable

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Bit(s)	R/W	Default	Description
8	R/W	0x0	Cue_enable_l : Cue left half frame enable
6:4	R/W	0x0	reg_CUE_CON_RPLC_mode : U3, CUE pixel chroma replace mode;
3:0	R/W	0x0	reg_CUE_CHRM_FLT_mode : U4, CUE improvement filter mode,

Table 8-1054 NR2_CUE_CON_MOT_TH 0x1779

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_CUE_CON_Cmot_thrd2 : U8, Motion Detection threshold of up/down two rows, Chroma channel in Chroma Up-sampling Error (CUE) Detection (tighter).
23:16	R/W	0x0	reg_CUE_CON_Ymot_thrd2 : U8, Motion Detection threshold of up/mid/down three rows, Luma channel in Chroma Up-sampling Error (CUE) Detection (tighter).
15: 8	R/W	0x0	reg_CUE_CON_Cmot_thrd : U8, Motion Detection threshold of up/down two rows, Chroma channel in Chroma Up-sampling Error (CUE) Detection.
7: 0	R/W	0x0	reg_CUE_CON_Ymot_thrd : U8, Motion Detection threshold of up/mid/down three rows, Luma channel in Chroma Up-sampling Error (CUE) Detection.

Table 8-1055 NR2_CUE_CON_DIF0 0x177a

Bit(s)	R/W	Default	Description
15:8	R/W	0x0	reg_CUE_CON_difP1_thrd : U8, P1 field Intra-Field top/below line chroma difference threshold,
7:0	R/W	0x0	reg_CUE_CON_difCur_thrd : U8, Current Field/Frame Intra-Field up/down line chroma difference threshold,

Table 8-1056 NR2_CUE_CON_DIF1 0x177b

Bit(s)	R/W	Default	Description
19:16	R/W	0x0	reg_CUE_CON_rate0 : U4, The Krate to decide CUE by relationship between CUE_diflG and CUE_difEG
15: 8	R/W	0x0	reg_CUE_CON_difEG_thrd : U8, Theshold to the difference between current Field/Frame middle line to down line color channel(CUE_difEG).
7: 0	R/W	0x0	reg_CUE_CON_diflG_thrd : U8, Threshold to the difference between P1 field top line to current Field/Frame down line color channel (CUE_diflG).

Table 8-1057 NR2_CUE_CON_DIF2 0x177c

Bit(s)	R/W	Default	Description
19:16	R/W	0x0	reg_CUE_CON_rate1 : U4, The Krate to decide CUE by relationship between CUE_difnC and CUE_difEC
15: 8	R/W	0x0	reg_CUE_CON_difEC_thrd : U8, Theshold to the difference between current Field/Frame middle line to up line color channel(CUE_difEC).
7: 0	R/W	0x0	reg_CUE_CON_difnC_thrd : U8, Threshold to the difference between P1 field bot line to current Field/Frame up line color channel (CUE_difnC).

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Table 8-1058 NR2_CUE_CON_DIF3 0x177d

Bit(s)	R/W	Default	Description
19:16	R/W	0x0	reg_CUE_CON_rate2 : U4, The Krate to decide CUE by relationship between CUE_difP1 and CUE_difEP1
15: 8	R/W	0x0	reg_CUE_CON_difEP1_thrd : U8, Inter-Field top/below line to current field/frame middle line chroma difference (CUE_difEP1) threshold.
7: 0	R/W	0x0	reg_CUE_CON_difP1_thrd2 : U8, P1 field Intra-Field top/below line chroma difference threshold (tighter),

Table 8-1059 NR2_CUE_PRG_DIF 0x177e

Bit(s)	R/W	Default	Description
20	R/W	0x0	reg_CUE_PRG_Enable : Enable bit for progressive video CUE detection.If interlace input video,
19:16	R/W	0x0	reg_CUE_PRG_rate : U3, The Krate to decide CUE by relationship between CUE_difCur and (CUE_difEC+CUE_difEG)
15: 8	R/W	0x0	reg_CUE_PRG_difCEG_thrd : U8, Current Frame Intra-Field up-mid and mid-down line chroma difference threshold for progressive video CUE detection,
7: 0	R/W	0x0	reg_CUE_PRG_difCur_thrd : U8, Current Frame Intra-Field up/down line chroma difference threshold,

Table 8-1060 NR2_CONV_MODE 0x177f

Bit(s)	R/W	Default	Description
3:2	R/W	0x0	Conv_c444_mode : The format convert mode about 422 to 444 when data read out line buffer
1:0	R/W	0x0	Conv_c422_mode : the format convert mode about 444 to 422 when data write to line buffer

Table 8-1061 DET 3D REG DEFINE BEGIN //// 8 'h80~8'h8f DET3D_RO_SPLT_HB 0x1780

Bit(s)	R/W	Default	Description
24	R.O	0x0	RO_Det3D_Split_HB_valid : U1 horizontal LR split border detected valid signal for top half picture
20:16	R.O	0x0	RO_Det3D_Split_HB_pxnum : U5 number of pixels included for the LR split position estimation for top half picture
9: 0	R.O	0x0	RO_Det3D_Split_HB_idxX4 : S10 X4: horizontal pixel shifts of LR split position to the (ColMax/2) for top half picture

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Table 8-1062 DET3D_RO_SPLT_VL 0x1781

Bit(s)	R/W	Default	Description
24	R.O	0x0	RO_Det3D_Split_VL_valid : U1 horizontal LR split border detected valid signal for top half picture
20:16	R.O	0x0	RO_Det3D_Split_VL_pxnum : U5 number of pixels included for the LR split position estimation for top half picture
9: 0	R.O	0x0	RO_Det3D_Split_VL_idxX4 : S10 X4: horizontal pixel shifts of LR split position to the (ColMax/2) for top half picture

Table 8-1063 DET3D_RO_SPLT_VR 0x1782

Bit(s)	R/W	Default	Description
24	R.O	0x0	RO_Det3D_Split_VR_valid : U1 horizontal LR split border detected valid signal for top half picture
20:16	R.O	0x0	RO_Det3D_Split_VR_pxnum : U5 number of pixels included for the LR split position estimation for top half picture
9: 0	R.O	0x0	RO_Det3D_Split_VR_idxX4 : S10 X4: horizontal pixel shifts of LR split position to the (ColMax/2) for top half picture

Table 8-1064 DET3D_RO_MAT_LUMA_LR 0x1783

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Luma_LR_score : S2*8 LUMA statistics left right decision score for each band (8bands vertically), it can be -1/0/1:-1: most likely not LR symmetric 0: not sure 1: most likely LR symmetric
7:0	R.O	0x0	RO_Luma_LR_symtc : U1*8 Luma statistics left right pure symmetric for each band (8bands vertically), it can be 0/1: 0: not sure 1: most likely LR is pure symmetric
4:0	R.O	0x0	RO_Luma_LR_sum : S5 Total score of 8x8 Luma statistics for LR like decision, the larger this score, the more confidence that this is a LR 3D video. It is sum of RO_Luma_LR_score[0~7]

Table 8-1065 DET3D_RO_MAT_LUMA_TB 0x1784

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Luma_TB_score : S2*8 LUMA statistics Top/Bottom decision score for each band (8bands Horizontally),
7:0	R.O	0x0	RO_Luma_TB_symtc : Luma statistics Top/Bottompure symmetric for each band (8bands Horizontally),
4:0	R.O	0x0	RO_Luma_TB_sum : Total score of 8x8 Luma statistics for TB like decision,

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Table 8-1066 DET3D_RO_MAT_CHRU_LR 0x1785

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_ChrU_LR_score : S2*8 LUMA statistics left right decision score for each band (8bands vertically),
7:0	R.O	0x0	RO_ChrU_LR_symtc : CHRU statistics left right pure symmetric for each band (8bands vertically),
4:0	R.O	0x0	RO_ChrU_LR_sum : Total score of 8x8 ChrU statistics for LR like decision,

Table 8-1067 DET3D_RO_MAT_CHRU_TB 0x1786

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_ChrU_TB_score : S2*8 CHRU statistics Top/Bottom decision score for each band (8bands Horizontally)
7:0	R.O	0x0	RO_ChrU_TB_symtc : CHRU statistics Top/Bottompure symmetric for each band (8bands Horizontally)
4:0	R.O	0x0	RO_ChrU_TB_sum : Total score of 8x8 ChrU statistics for TB like decision

Table 8-1068 DET3D_RO_MAT_CHRV_LR 0x1787

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_ChrV_LR_score : S2*8 CHRUstatistics left right decision score for each band (8bands vertically)
7:0	R.O	0x0	RO_ChrV_LR_symtc : CHRV statistics left right pure symmetric for each band (8bands vertically)
4:0	R.O	0x0	RO_ChrV_LR_sum : Total score of 8x8 ChrV statistics for LR like decision

Table 8-1069 DET3D_RO_MAT_CHRV_TB 0x1788

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_ChrV_TB_score : CHRV statistics Top/Bottom decision score for each band (8bands Horizontally)
7:0	R.O	0x0	RO_ChrV_TB_symtc : CHRV statistics Top/Bottompure symmetric for each band (8bands Horizontally)
4:0	R.O	0x0	RO_ChrV_TB_sum : Total score of 8x8 ChrV statistics for TB like decision

Table 8-1070 DET3D_RO_MAT_HEDG_LR 0x1789

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Hedg_LR_score : Horizontal Edge statistics left right decision score for each band (8bands vertically)
7:0	R.O	0x0	RO_Hedg_LR_symtc : Horizontal Edge statistics left right pure symmetric for each band (8bands vertically)
4:0	R.O	0x0	RO_Hedg_LR_sum : Total score of 8x8 Hedg statistics for LR like decision

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Table 8-1071 DET3D_RO_MAT_HEDG_TB 0x178a

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Hedg_TB_score : Horizontal Edge statistics Top/Bottom decision score for each band (8bands Horizontally)
7:0	R.O	0x0	RO_Hedg_TB_symtc : Horizontal Edge statistics Top/Bottompure symmetric for each band (8bands Horizontally)
4:0	R.O	0x0	RO_Hedg_TB_sum : Total score of 8x8 Hedg statistics for TB like decision

Table 8-1072 DET3D_RO_MAT_VEDG_LR 0x178b

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Vedg_LR_score : Vertical Edge statistics left right decision score for each band (8bands vertically)
7:0	R.O	0x0	RO_Vedg_LR_symtc : Vertical Edge statistics left right pure symmetric for each band (8bands vertically)
4:0	R.O	0x0	RO_Vedg_LR_sum : Total score of 8x8 Vedg statistics for LR like decision

Table 8-1073 DET3D_RO_MAT_VEDG_TB 0x178c

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Vedg_TB_score : Vertical Edge statistics Top/Bottom decision score for each band (8bands Horizontally)
7:0	R.O	0x0	RO_Vedg_TB_symtc : Vertical Edge statistics Top/Bottompure symmetric for each band (8bands Horizontally)
4:0	R.O	0x0	RO_Vedg_TB_sum : Total score of 8x8 Vedg statistics for TB like decision

Table 8-1074 DET3D_RO_MAT_MOTN_LR 0x178d

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Motn_LR_score : Motion statistics left right decision score for each band (8bands vertically)
7:0	R.O	0x0	RO_Motn_LR_symtc : Motion statistics left right pure symmetric for each band (8bands vertically)
4:0	R.O	0x0	RO_Motn_LR_sum : Total score of 8x8 Motion statistics for LR like decision

Table 8-1075 DET3D_RO_MAT_MOTN_TB 0x178e

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Motn_TB_score : Motion statistics Top/Bottom decision score for each band (8bands Horizontally)
7:0	R.O	0x0	RO_Motn_TB_symtc : Motion statistics Top/Bottompure symmetric for each band (8bands Horizontally)
4:0	R.O	0x0	RO_Motn_TB_sum : Total score of 8x8 Motion statistics for TB like decision

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Table 8-1076 DET3D_RO_FRM_MOTN 0x178f

Bit(s)	R/W	Default	Description
15:0	R.O	0x0	RO_Det3D_Frame_Motion : U16 frame based motion value sum for still image decision in FW. mat ram read enter addr

DET3D_RAMRD_ADDR_PORT 0x179a

DET3D_RAMRD_DATA_PORT 0x179b

Table 8-1077 NR2_CFR_PARA_CFG0 0x179c

Bit(s)	R/W	Default	Description
8	R/W	0x0	reg_CFR_CurDif_luma_mode : Current Field Top/Bot line Luma difference calculation mode
7:6	R/W	0x0	reg_MACFR_frm_phase : U2 This will be a field based phase register that need to be set by FW phase to phase: this will be calculated based on dbdr_phase of the specific line of this frame. u1 : dbdr_phase=1, center line is DB in current line; dbdr_phase=2, center line is Dr in current line;
5:4	R/W	0x0	reg_CFR_CurDif_tran_mode : U2 Current Field Top/Bot line Luma/Chroma transition level calculation mode,
3:2	R/W	0x0	reg_CFR_alpha_mode : U2 Alpha selection mode for CFR block from curAlp and motAlp i.e. 0: motAlp; 1: (motAlp+curAlp)/2; 2: min(motAlp,curAlp); 3: max(motAlp,curAlp);
1:0	R/W	0x0	reg_CFR_Motion_Luma_mode : U2 LumaMotion Calculation mode for MA-CFR. 0: top/bot Lumma motion; 1: middle Luma Motion 2: top/bot + middle motion; 3: max(top/tot motion, middle motion)

Table 8-1078 NR2_CFR_PARA_CFG1 0x179d

Bit(s)	R/W	Default	Description
23:16	R/W	0x0	reg_CFR_alpha_gain : gain to map muxed curAlp and motAlp to alpha that will be used for final blending.
15: 8	R/W	0x0	reg_CFR_Motion_ofst : Offset to Motion to calculate the motAlp, e,g:motAlp= reg_CFR_Motion_ofst- Motion;This register can be seen as the level of motion that we consider it at moving.
7: 0	R/W	0x0	reg_CFR_CurDif_gain : gain to CurDif to map to alpha, normalized to 32;

Table 8-1079 NR3_MODE 0x2ff0

Bit(s)	R/W	Default	Description
5	R/W	0x0	reg_3dnr_nr3_vtxt_mode : ;
4	R/W	0x0	reg_3dnr_nr3_cbyy_ignor_coop : ; // u1: ignore coop condition for cbyy motion decision
3	R/W	0x0	reg_3dnr_nr3_ybyc_ignor_cnoop : ; // u1: ignore cnoop condition for ybyc motion decision
2:0	R/W	0x3	reg_3dnr_nr3_suremot_txt_mode : ; // u3: 0: cur, 1:p2; 2: (cur+p2)/2; 3/up: min (cur,p2)

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Table 8-1080 NR3_COOP_PARA 0x2ff1

Bit(s)	R/W	Default	Description
21:20	R/W	0x2	reg_3dnr_nr3_coop_mode : ; // u2 0 original pixel 1: [1 2 1]/4 lpf; 2: [1 2 2 2 1]/8; 3: 3x3 lpf
19:16	R/W	0x8	reg_3dnr_nr3_coop_ratio : ; // u4 cur and p2 color oop decision ratio: (avg1< (MAX(sat0,sat2)*ratio/8 + ofst));
15:8	R/W	0x0	reg_3dnr_nr3_coop_ofset : ; // s8 cur and p2 color oop decision ofst: (avg1< (MAX(sat0,sat2)*ratio/8 + ofst));
7:0	R/W	0x0	reg_3dnr_nr3_coop_sat_thrd : ; // u8 cur and p2 color oop decision min(sat0, sat1) threshold;

Table 8-1081 NR3_CNOOP_GAIN 0x2ff2

Bit(s)	R/W	Default	Description
23:20	R/W	0x8	reg_3dnr_nr3_cnoop_ratio0 : ; // u4 cur and p2 color noop decision ratio0: (avg1<(MAX(sat0,sat2)*ratio0/8 + ofst0));
19:16	R/W	0x8	reg_3dnr_nr3_cnoop_ratio1 : ; // u4 cur and p2 color noop decision ratio1: (dif1< (MIN(sat0,sat2)*ratio1/8 + ofst1));
15:8	R/W	0x19	reg_3dnr_nr3_cnoop_ofset0 : ; // s8 cur and p2 color noop decision ofset0: (avg1<(MAX(sat0,sat2)*ratio0/8 + ofst0));
7:0	R/W	0x0	reg_3dnr_nr3_cnoop_ofset1 : ; // s8 cur and p2 color noop decision ofset1: (dif1< (MIN(sat0,sat2)*ratio1/8 + ofst1));

Table 8-1082 NR3_YMOT_PARA 0x2ff3

Bit(s)	R/W	Default	Description
19	R/W	0x1	reg_3dnr_nr3_ymot_only_en : ; // u1: enable signal for ignor chroma motion: (ytxt & coop)
18	R/W	0x1	reg_3dnr_nr3_ymot_only_cmtmode : ; // u1: 0: cmot=ymot; 1: cmot = MIN(ymot, cmot)
17:16	R/W	0x0	reg_3dnr_nr3_ymot_only_txtmode : ; // u2: 0, min(txt0,txt2); 1, max(txt0,txt2);2, (txt0+txt2)/2; 3: sat(txt0, txt2)
15:8	R/W	0xa	reg_3dnr_nr3_ymot_only_txtthrd : ; // u8: threshold to luma texture to decide use ymot only
7:0	R/W	0x1e	reg_3dnr_nr3_ymot_only_motthrd : ; // u8: threshold to luma motion to decide use ymot only

Table 8-1083 NR3_CMOT_PARA 0x2ff4

Bit(s)	R/W	Default	Description
19	R/W	0x1	reg_3dnr_nr3_cmot_only_en : ; // u1: enable signal for ignor luma motion: (ctxt & coop)
18	R/W	0x0	reg_3dnr_nr3_cmot_only_ymtmode : ; // u1: 0: ymot=cmot+ymot/4; 1: ymot = MIN(ymot, cmot)
17:16	R/W	0x0	reg_3dnr_nr3_cmot_only_txtmode : ; // u2: 0, min(txt0,txt2); 1, max(txt0,txt2);2, (txt0+txt2)/2; 3: sat(txt0, txt2)

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Bit(s)	R/W	Default	Description
15:8	R/W	0x14	reg_3dnr_nr3_cmot_only_txtthrd : ; // u8: threshold to chroma texture to decide use cmot only
7:0	R/W	0xf	reg_3dnr_nr3_cmot_only_motthrd : ; // u8: threshold to chroma motion to decide use cmot only

Table 8-1084 NR3_SUREMOT_YGAIN 0x2ff5

Bit(s)	R/W	Default	Description
31:24	R/W	0x10	reg_3dnr_nr3_suremot_dec_yrate : ; // u8: (norm 16)lpfMot>(dec_rate*txt +ofst) then force lpfMot*frc_gain+frc_ofset
23:16	R/W	0xc	reg_3dnr_nr3_suremot_dec_yofst : ; // u8: lpfMot>(dec_rate*txt +ofst) then force lpfMot*frc_gain+frc_ofset
15:8	R/W	0x40	reg_3dnr_nr3_suremot_frc_ygain : ; // u8: (norm 8)lpfMot>(dec_rate*txt +ofst) then force lpfMot*frc_gain+frc_ofset
7:0	R/W	0x14	reg_3dnr_nr3_suremot_frc_yofst : ; // u8: lpfMot>(dec_rate*txt +ofst) then force lpfMot*frc_gain+frc_ofset

Table 8-1085 NR3_SUREMOT_CGAIN 0x2ff6

Bit(s)	R/W	Default	Description
31:24	R/W	0x22	reg_3dnr_nr3_suremot_dec_crate : ; // u8: (norm 16)lpfMot>(dec_rate*txt +ofst) then force lpfMot*frc_gain+frc_ofset
23:16	R/W	0x26	reg_3dnr_nr3_suremot_dec_cofst : ; // u8: lpfMot>(dec_rate*txt +ofst) then force lpfMot*frc_gain+frc_ofset
15:8	R/W	0x40	reg_3dnr_nr3_suremot_frc_cgain : ; // u8: (norm 8)lpfMot>(dec_rate*txt +ofst) then force lpfMot*frc_gain+frc_ofset
7:0	R/W	0x14	reg_3dnr_nr3_suremot_frc_cofst : ; // u8: lpfMot>(dec_rate*txt +ofst) then force lpfMot*frc_gain+frc_ofset

8.2.3.9 LBUF Registers

Table 8-1086 LBUF_TOP_CTRL 0x2fff

Bit(s)	R/W	Default	Description
25:20	R/W	6'd0	gate clk control of line buf . LBUF_TOP_CTRL[25:24] is the clk control of current line linebuffer. LBUF_TOP_CTRL[23:22] is the clk control of previous one line.linebuffer, LBUF_TOP_CTRL[21:20] is the clk control of previous two line linebuffer.
17	R/W	1'b1	lbuf_fmt444_mode; format of data store in linebuf ,high mean store 444 data into linebuf
16	R/W	1'b1	lbuf_line5_mode;Store 5 line or 3 lines in linebuf ,high means 5 lines
12:0	R/W	13'd342	pre_lbuf_size: size of linebuf

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8.2.3.10 DI_SCALE Registers

Table 8-1087 DI_SCO_FIFO_CTRL 0x374e

Bit(s)	R/W	Default	Description
28:0	R/W	0x0	sco_fifo_ctrl

Table 8-1088 DI_SC_TOP_CTRL 0x374f

Bit(s)	R/W	Default	Description
31	R.O	0	prog_interlace : no use // unsigned , default = 0x0
30	R/W	0	path_sel : 1 : di pre scaler for nr inp 0: di post scaler // unsigned , default = 0x0
29	R/W	0	go_field_sel : 1 : di pre go field 0: di post go filed // unsigned , default = 0x0
28	R/W	0	sw_resets : // unsigned , default = 0x0
27	R/W	0	pps_dummy_data_mode : 1: use low 8 bits 0: use high 8 bits // unsigned , default = 0x0
26	R/W	0	field_inv : field reverse// unsigned , default = 0x0
25	R/W	0	reg_field : no use// unsigned , default = 0x0
5:4	R/W	0	hdr_gclk_ctrl : 01 : no cbus clock for hdr other: free cbus clock for hdr // unsigned , default = 0x0
1	R/W	0	reg_gclk_ctrl : 1 : free clock for di scaler register 0: auto gate for di scalercbus // unsigned , default = 0x0

Table 8-1089 DI_SC_DUMMY_DATA 0x3750

Bit(s)	R/W	Default	Description
29:20	R/W	0	VD1_SC_Y : // unsigned , default = 0x10,dummy data used in the VD1 scaler,according VPP_DOLBY_CTRL[17] 1:set 8bit value 2:set 10bit value
19:10	R/W	0	VD1_SC_CB : // unsigned , default = 0x80,dummy data used in the VD1 scaler, according VPP_DOLBY_CTRL[17] 1:set 8bit value 2:set 10bit value
9 :0	R/W	0	VD1_SC_CR : // unsigned , default = 0x80,dummy data used in the VD1 scaler, according VPP_DOLBY_CTRL[17] 1:set 8bit value 2:set 10bit value

Table 8-1090 DI_SC_LINE_IN_LENGTH 0x3751

Bit(s)	R/W	Default	Description
13:0	R/W	14	line_in_length : // unsigned , default = 14'd1920,VD1 scaler input hsize

Table 8-1091 DI_SC_PIC_IN_HEIGHT 0x3752

Bit(s)	R/W	Default	Description
12:0	R/W	0x1fff	line_in_height : // unsigned , default = 13'h1fff,VD1 scaler input vsize

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Table 8-1092 DI_SC_COEF_IDX 0x3753

Bit(s)	R/W	Default	Description
15	R/W	0	index_inc : // unsigned , default = 0x0 ,index increment, if bit9 = 1 then (0: index increase 1, 1: index increase 2) else (index increase 2)
14	R/W	0	rd_cbus_coef_en : // unsigned , default = 0x0 ,1: read coef through cbus enable, just for debug purpose in case when we wanna check the coef in ram in correct or not
13	R/W	0	vf_sep_coef_en : // unsigned , default = 0x0 ,if true, vertical separated coef enable
9	R/W	0	high_reso_en : // unsigned , default = 0x0 ,if true, use 9bit resolution coef, other use 8bit resolution coef
8:7	R/W	0	type_index : // unsigned , default = 0x0 ,type of index, 00: vertical coef, 01: vertical chroma coef: 10: horizontal coef, 11: resevered
6:0	R/W	0	coef_index : // unsigned , default = 0x0 ,coef index

Table 8-1093 DI_SC_COEF 0x3754

Bit(s)	R/W	Default	Description
31:24	R/W	0	coef0 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
23:16	R/W	0	coef1 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
15:8	R/W	0	coef2 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
7 :0	R/W	0	coef3 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter

Table 8-1094 DI_VSC_REGION12_STARTP 0x3755

Bit(s)	R/W	Default	Description
28:16	R/W	0	region1_startp : //unsigned , default = 0 ,region1 startp
12:0	R/W	0	region2_startp : //unsigned , default = 0 ,region2 startp

Table 8-1095 DI_VSC_REGION34_STARTP 0x3756

Bit(s)	R/W	Default	Description
28:16	R/W	0	region3_startp : //unsigned , default = 0x0438,region3 startp
12:0	R/W	0	region4_startp : //unsigned , default = 0x0438,region4 startp

Table 8-1096 DI_VSC_REGION4_ENDP 0x3757

Bit(s)	R/W	Default	Description
12:0	R/W	13	region4_endp : //unsigned , default = 13'd1079 ,region4 endp

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Table 8-1097 DI_VSC_START_PHASE_STEP 0x3758

Bit(s)	R/W	Default	Description
27:24	R/W	1	integer_part : //unsigned , default = 1,vertical start phase step, (source/dest)* (2^24),integer part of step
23:0	R/W	0	fraction_part : //unsigned , default = 0,vertical start phase step, (source/dest)* (2^24),fraction part of step

Table 8-1098 DI_VSC_REGION0_PHASE_SLOPE 0x3759

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,vertical scaler region0 phase slope, region0 phase slope

Table 8-1099 DI_VSC_REGION1_PHASE_SLOPE 0x375a

Bit(s)	R/W	Default	Description
24:0	R/W	0	region1_phase_slope : //signed , default = 0,region1 phase slope

Table 8-1100 DI_VSC_REGION3_PHASE_SLOPE 0x375b

Bit(s)	R/W	Default	Description
24:0	R/W	0	region3_phase_slope : //signed , default = 0,region3 phase slope

Table 8-1101 DI_VSC_REGION4_PHASE_SLOPE 0x375c

Bit(s)	R/W	Default	Description
24:0	R/W	0	region4_phase_slope : //signed , default = 0,region4 phase slope

Table 8-1102 DI_VSC_PHASE_CTRL 0x375d

Bit(s)	R/W	Default	Description
18:17	R/W	0	vsc_double_line_mode : //unsigned , default = 0, double line mode, input/output line width of vscaler becomes 2X, so only 2 line buffer in this case, use for 3D line by line interleave scaling bit1 true, double the input width and half input height, bit0 true, change line buffer 2 lines instead of 4 lines
16	R.O	0	prog_interlace : //unsigned , default = 0,0: progressive output, 1: interlace output
15	R/W	0	vsc_bot_l0_out_en : //unsigned , default = 0,vertical scaler output line0 in advance or not for bottom field
14:13	R/W	1	vsc_bot_rpt_l0_num : //unsigned , default = 1,vertical scaler initial repeat line0 number for bottom field
11:8	R/W	4	vsc_bot_ini_rcv_num : //unsigned , default = 4,vertical scaler initial receiving number for bottom field
7	R/W	0	vsc_top_l0_out_en : //unsigned , default = 0,vertical scaler output line0 in advance or not for top field

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Bit(s)	R/W	Default	Description
6:5	R/W	1	vsc_top_rpt_l0_num : //unsigned , default = 1,vertical scaler initial repeat line0 number for top field
3:0	R/W	4	vsc_top_ini_rcv_num : //unsigned , default = 4,vertical scaler initial receiving number for top field

Table 8-1103 DI_VSC_INI_PHASE 0x375e

Bit(s)	R/W	Default	Description
31:16	R/W	0	vertical scaler field initial phase for bottom field
15:0	R/W	0	vertical scaler field initial phase for top field

Table 8-1104 DI_HSC_REGION12_STARTP 0x3760

Bit(s)	R/W	Default	Description
28:16	R/W	0	region1_startp : //unsigned , default = 0,region1 startp
12:0	R/W	0	region2_startp : //unsigned , default = 0,region2 startp

Table 8-1105 DI_HSC_REGION34_STARTP 0x3761

Bit(s)	R/W	Default	Description
28:16	R/W	0	region3 : startp //unsigned , default = 0x780,region3 startp
12:0	R/W	0	region4 : startp //unsigned , default = 0x780,region4 startp

Table 8-1106 DI_HSC_REGION4_ENDP 0x3762

Bit(s)	R/W	Default	Description
12:0	R/W	13	region4 : startp //unsigned , default = 13'd1919,region4 startp

Table 8-1107 DI_HSC_START_PHASE_STEP 0x3763

Bit(s)	R/W	Default	Description
27:24	R/W	1	integer_part : //unsigned , default = 1,integer part of step
23:0	R/W	0	fraction_part : //unsigned , default = 0,fraction part of step

Table 8-1108 DI_HSC_REGION0_PHASE_SLOPE 0x3764

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region0 phase slope

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Table 8-1109 DI_HSC_REGION1_PHASE_SLOPE 0x3765

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region1 phase slope

Table 8-1110 DI_HSC_REGION3_PHASE_SLOPE 0x3766

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region3 phase slope

Table 8-1111 DI_HSC_REGION4_PHASE_SLOPE 0x3767

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region4 phase slope

Table 8-1112 DI_HSC_PHASE_CTRL 0x3768

Bit(s)	R/W	Default	Description
22:21	R/W	1	hsc_rpt_p0_num0 : //unsigned , default = 1 ,horizontal scaler initial repeat pixel0 number0
19:16	R/W	4	hsc_ini_rcv_num0 : //unsigned , default = 4 ,horizontal scaler initial receiving number0
15:0	R/W	0	hsc_ini_phase0 : //unsigned , default = 0 ,horizontal scaler top field initial phase0

Table 8-1113 DI_SC_MISC 0x3769

Bit(s)	R/W	Default	Description
22	R/W	0	hsc_len_div2_en : //unsigned , default = 0 ,if true, divide VSC line length 2 as the HSC input length, othwise VSC length length is the same as the VSC line length just for special usage, more flexibilityi
21	R/W	0	lbuf_mode : //unsigned , default = 0 ,if true, prevsc uses lin buffer, otherwise prevsc does not use line buffer, it should be same as prevsc_en
20	R/W	0	prehsc_en : //unsigned , default = 0 ,prehsc_en
19	R/W	0	prevsc_en : //unsigned , default = 0 ,prevsc_en
18	R/W	0	vsc_en : //unsigned , default = 0 ,vsc_en
17	R/W	0	hsc_en : //unsigned , default = 0 ,hsc_en
16	R/W	0	sc_top_en : //unsigned , default = 0 ,scale_top_en
15	R/W	0	sc_vd_en : //unsigned , default = 0 ,video1 scale out enable
12	R/W	1	hsc_nonlinear_4region_en : //unsigned , default = 1 ,if true, region0,region4 are nonlinear regions, otherwise they are not scaling regions, for horizontal scaler
10:8	R/W	0	hsc_bank_length : //unsigned , default = 0 ,horizontal scaler bank length
5	R/W	4	vsc_phase_field_mode : //unsigned , default = 4 ,vertical scaler phase field mode, if true, disable the opposite parity line output, more bandwith needed if output 1080i

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Bit(s)	R/W	Default	Description
4	R/W	0	vsc_nonlinear_4region_en : //unsigned , default = 0 ,if true, region0,region4 are nonlinear regions, otherwise they are not scaling regions, for vertical scaler
2:0	R/W	4	vsc_bank_length : //unsigned , default = 4 ,vertical scaler bank length

Table 8-1114 DI_HSC_PHASE_CTRL1 0x376a

Bit(s)	R/W	Default	Description
22:21	R/W	1	hsc_rpt_p0_num0 : //unsigned , default = 1 ,horizontal scaler initial repeat pixel0 number0
19:16	R/W	4	hsc_ini_rcv_num0 : //unsigned , default = 4 ,horizontal scaler initial receiving number0
15:0	R/W	0	hsc_ini_phase0 : //unsigned , default = 0 ,horizontal scaler top field initial phase0

Table 8-1115 VPP_SC_MISC 0x1D19

Bit(s)	R/W	Default	Description
22	R/W	0	hsc_len_div2_en : //unsigned , default = 0 ,if true, divide VSC line length 2 as the HSC input length, othwise VSC length length is the same as the VSC line length just for special usage, more flexibilityi
21	R/W	0	lbuf_mode : //unsigned , default = 0 ,if true, prevsc uses lin buffer, otherwise prevsc does not use line buffer, it should be same as prevsc_en
20	R/W	0	prehsc_en : //unsigned , default = 0 ,prehsc_en
19	R/W	0	prevsc_en : //unsigned , default = 0 ,prevsc_en
18	R/W	0	vsc_en : //unsigned , default = 0 ,vsc_en
17	R/W	0	hsc_en : //unsigned , default = 0 ,hsc_en
16	R/W	0	sc_top_en : //unsigned , default = 0 ,scale_top_en
15	R/W	0	sc_vd_en : //unsigned , default = 0 ,video1 scale out enable
12	R/W	1	hsc_nonlinear_4region_en : //unsigned , default = 1 ,if true, region0,region4 are nonlinear regions, otherwise they are not scaling regions, for horizontal scaler
10:8	R/W	0	hsc_bank_length : //unsigned , default = 0 ,horizontal scaler bank length
5	R/W	4	vsc_phase_field_mode : //unsigned , default = 4 ,vertical scaler phase field mode, if true, disable the opposite parity line output, more bandwith needed if output 1080i
4	R/W	0	vsc_nonlinear_4region_en : //unsigned , default = 0 ,if true, region0,region4 are nonlinear regions, otherwise they are not scaling regions, for vertical scaler
2:0	R/W	4	vsc_bank_length : //unsigned , default = 4 ,vertical scaler bank length

Table 8-1116 DI_HSC_INI_PAT_CTRL 0x376b

Bit(s)	R/W	Default	Description
31:24	R/W	0	prehsc_pattern : //unsigned , default = 0, prehsc pattern, each patten 1 bit, from lsb -> msb
22:20	R/W	0	prehsc_pat_star : //unsigned , default = 0, prehsc pattern start

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Bit(s)	R/W	Default	Description
18:16	R/W	0	prehsc_pat_end : //unsigned , default = 0, prehsc pattern end
15:8	R/W	0	hsc_pattern : //unsigned , default = 0, hsc pattern, each patten 1 bit, from lsb -> msb
6:4	R/W	0	hsc_pat_start : //unsigned , default = 0, hsc pattern start
2:0	R/W	0	hsc_pat_end : //unsigned , default = 0, hsc pattern end

Table 8-1117 DI_SC_GCLK_CTRL 0x376c

Bit(s)	R/W	Default	Description
15:0	R/W	0	vpp_sc_gclk_ctrl : //unsigned , default = 0,

Table 8-1118 DI_SC_HOLD_LINE 0x376d

Bit(s)	R/W	Default	Description
31:0	R/W	0	sc_hold_line : //unsigned , default = 0,

Table 8-1119 DI_HDR_IN_HSIZE 0x376e

Bit(s)	R/W	Default	Description
12:0	R/W	0	hdr input h size

Table 8-1120 DI_HDR_IN_VSIZE 0x376f

Bit(s)	R/W	Default	Description
12:0	R/W	0	hdr input v size

Table 8-1121 DI_HDR2_CTRL 0x3770

Bit(s)	R/W	Default	Description
20:18	R/W	0	reg_din_swap : // unsigned , default = 0
17	R/W	0	reg_out_fmt : // unsigned , default = 0
16	R/W	0	reg_only_mat : // unsigned , default = 0
13	R/W	0	reg_VDIN0_HDR2_top_en : // unsigned , default = 0
12	R/W	1	reg_cgain_mode : // unsigned , default = 1
7: 6	R/W	1	reg_gmut_mode : // unsigned , default = 1
5	R/W	0	reg_in_shift : // unsigned , default = 0
4	R/W	1	reg_in_fmt : // unsigned , default = 1
3	R/W	1	reg_eo_enable : // unsigned , default = 1
2	R/W	1	reg_oe_enable : // unsigned , default = 1

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Bit(s)	R/W	Default	Description
1	R/W	1	reg_ogain_enable : // unsigned , default = 1
0	R/W	1	reg_cgain_enable : // unsigned , default = 1

Table 8-1122 DI_HDR2_CLK_GATE 0x3771

Bit(s)	R/W	Default	Description
31:30	R/W	0	clk_tm : gate clock ctrl (main clock) // unsigned , default = 0
29:28	R/W	0	output : matrix clock gate ctrl // unsigned , default = 0
25:24	R/W	0	input : matrix clock gate ctrl // unsigned , default = 0
23:22	R/W	0	hdr : top cbus clock gate ctrl // unsigned , default = 0
21:20	R/W	0	eotf : cbus clock gate ctrl // unsigned , default = 0
19:18	R/W	0	oetf : cbus clock gate ctrl // unsigned , default = 0
17:16	R/W	0	gamma : mult cbus clock gate ctrl // unsigned , default = 0
15:14	R/W	0	adaptive : cbus scaler clock gate ctrl // unsigned , default = 0
13:12	R/W	0	cgain : cbus clock gate ctrl // unsigned , default = 0
11:10	R/W	0	eotf : clock gate ctrl // unsigned , default = 0
9:8	R/W	0	oetf : clock gate ctrl // unsigned , default = 0
7:6	R/W	0	gamma : mult clock gate ctrl // unsigned , default = 0
5:4	R/W	0	adaptive : scaler clock gate ctrl // unsigned , default = 0
3:2	R/W	0	uv : gain clock gate ctrl // unsigned , default = 0
1:0	R/W	0	cgain : clock gate ctrl // unsigned , default = 0

Table 8-1123 DI_HDR2_MATRIXI_COEF00_01 0x3772

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-1124 DI_HDR2_MATRIXI_COEF02_10 0x3773

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-1125 DI_HDR2_MATRIXI_COEF11_12 0x3774

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

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Table 8-1126 DI_HDR2_MATRIXI_COEF20_21 0x3775

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-1127 DI_HDR2_MATRIXI_COEF22 0x3776

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-1128 DI_HDR2_MATRIXI_COEF30_31 0x3777

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-1129 DI_HDR2_MATRIXI_COEF32_40 0x3778

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-1130 DI_HDR2_MATRIXI_COEF41_42 0x3779

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-1131 DI_HDR2_MATRIXI_OFFSET0_1 0x377A

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-1132 DI_HDR2_MATRIXI_OFFSET2 0x377B

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

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Table 8-1133 DI_HDR2_MATRIXI_PRE_OFFSET0_1 0x377C

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-1134 DI_HDR2_MATRIXI_PRE_OFFSET2 0x377D

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-1135 DI_HDR2_MATRIXO_COEF00_01 0x377E

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-1136 DI_HDR2_MATRIXO_COEF02_10 0x377F

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-1137 DI_HDR2_MATRIXO_COEF11_12 0x3780

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-1138 DI_HDR2_MATRIXO_COEF20_21 0x3781

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-1139 DI_HDR2_MATRIXO_COEF22 0x3782

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

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Table 8-1140 DI_HDR2_MATRIXO_COEF30_31 0x3783

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-1141 DI_HDR2_MATRIXO_COEF32_40 0x3784

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-1142 DI_HDR2_MATRIXO_COEF41_42 0x3785

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-1143 DI_HDR2_MATRIXO_OFFSET0_1 0x3786

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-1144 DI_HDR2_MATRIXO_OFFSET2 0x3787

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-1145 DI_HDR2_MATRIXO_PRE_OFFSET0_1 0x3788

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-1146 DI_HDR2_MATRIXO_PRE_OFFSET2 0x3789

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

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Table 8-1147 DI_HDR2_MATRIXI_CLIP 0x378A

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-1148 DI_HDR2_MATRIXO_CLIP 0x378B

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-1149 DI_HDR2_CGAIN_OFFT 0x378C

Bit(s)	R/W	Default	Description
26:16	R/W	0	reg_cgain_oft2 : // signed , default = 0
10:0	R/W	0	reg_cgain_oft1 : // signed , default = 0

Table 8-1150 DI_EOTF_LUT_ADDR_PORT 0x378E

Bit(s)	R/W	Default	Description
7:0	R/W	0	eotf_lut_addr : // unsigned , default = 0

Table 8-1151 DI_EOTF_LUT_DATA_PORT 0x378F

Bit(s)	R/W	Default	Description
19:0	R/W	0	eotf_lut_data : // unsigned , default = 0

Table 8-1152 DI_OETF_LUT_ADDR_PORT 0x3790

Bit(s)	R/W	Default	Description
7:0	R/W	0	oetf_lut_addr : // unsigned , default = 0

Table 8-1153 DI_OETF_LUT_DATA_PORT 0x3791

Bit(s)	R/W	Default	Description
11:0	R/W	0	oetf_lut_data : // unsigned , default = 0

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Table 8-1154 DI_CGAIN_LUT_ADDR_PORT 0x3792

Bit(s)	R/W	Default	Description
7:0	R/W	0	cgain_lut_addr : // unsigned , default = 0

Table 8-1155 DI_CGAIN_LUT_DATA_PORT 0x3793

Bit(s)	R/W	Default	Description
11:0	R/W	0	cgain_lut_data : // unsigned , default = 0

Table 8-1156 DI_HDR2_CGAIN_COEF0 0x3794

Bit(s)	R/W	Default	Description
27:16	R/W	0	reg_cgain_coef1 : // unsigned , default = 0
11:0	R/W	0	reg_cgain_coef0 : //unsigned , default = 0

Table 8-1157 DI_HDR2_CGAIN_COEF1 0x3795

Bit(s)	R/W	Default	Description
11:0	R/W	0	reg_cgain_coef2 : // unsigned , default = 0

Table 8-1158 DI_OGAIN_LUT_ADDR_PORT 0x3796

Bit(s)	R/W	Default	Description
7:0	R/W	0	ogain_lut_addr : // unsigned , default = 0

Table 8-1159 DI_OGAIN_LUT_DATA_PORT 0x3797

Bit(s)	R/W	Default	Description
11:0	R/W	0	ogain_lut_data : // unsigned , default = 0

Table 8-1160 DI_HDR2_ADPS_CTRL 0x3798

Bit(s)	R/W	Default	Description
6	R/W	1	reg_adpscl_bypass2 : // unsigned , default = 1
5	R/W	1	reg_adpscl_bypass1 : // unsigned , default = 1
4	R/W	1	reg_adpscl_bypass0 : // unsigned , default = 1
1:0	R/W	1	reg_adpscl_mode : // unsigned , default = 1

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Table 8-1161 DI_HDR2_ADPS_ALPHA0 0x3799

Bit(s)	R/W	Default	Description
29:16	R/W	0x1000	reg_adpscl_alpha1 : // unsigned , default = 0x1000
13:0	R/W	0x1000	reg_adpscl_alpha0 : // unsigned , default = 0x1000

Table 8-1162 DI_HDR2_ADPS_ALPHA1 0x379A

Bit(s)	R/W	Default	Description
27:24	R/W	0xc	reg_adpscl_shift0 : // unsigned , default = 0xc
23:20	R/W	0xc	reg_adpscl_shift1 : // unsigned , default = 0xc
19:16	R/W	0xc	reg_adpscl_shift2 : // unsigned , default = 0xc
13:0	R/W	0x1000	reg_adpscl_alpha2 : // unsigned , default = 0x1000

Table 8-1163 DI_HDR2_ADPS_BETA0 0x379B

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta0_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta0 : // unsigned , default = 0xfc000

Table 8-1164 DI_HDR2_ADPS_BETA1 0x379C

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta1_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta1 : // unsigned , default = 0xfc000

Table 8-1165 DI_HDR2_ADPS_BETA2 0x379D

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta2_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta2 : // unsigned , default = 0xfc000

Table 8-1166 DI_HDR2_ADPS_COEF0 0x379E

Bit(s)	R/W	Default	Description
27:16	R/W	460	reg_adpscl_ys_coef1 : // unsigned , default = 460
11:0	R/W	1188	reg_adpscl_ys_coef0 : // unsigned , default = 1188

Table 8-1167 DI_HDR2_ADPS_COEF1 0x379F

Bit(s)	R/W	Default	Description
11:0	R/W	104	reg_adpscl_ys_coef2 : // unsigned , default = 104

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Table 8-1168 DI_HDR2_GMUT_CTRL 0x37A0

Bit(s)	R/W	Default	Description
3:0	R/W	14	reg_gmut_shift : // unsigned , default = 14

Table 8-1169 DI_HDR2_GMUT_COEF0 0x37A1

Bit(s)	R/W	Default	Description
31:16	R/W	674	reg_gmut_coef01 : // unsigned , default = 674
15:0	R/W	1285	reg_gmut_coef00 : // unsigned , default = 1285

Table 8-1170 DI_HDR2_GMUT_COEF1 0x37A2

Bit(s)	R/W	Default	Description
31:16	R/W	142	reg_gmut_coef10 : // unsigned , default = 142
15:0	R/W	89	reg_gmut_coef02 : // unsigned , default = 89

Table 8-1171 DI_HDR2_GMUT_COEF2 0x37A3

Bit(s)	R/W	Default	Description
31:16	R/W	23	reg_gmut_coef12 : // unsigned , default = 23
15:0	R/W	1883	reg_gmut_coef11 : // unsigned , default = 1883

Table 8-1172 DI_HDR2_GMUT_COEF3 0x37A4

Bit(s)	R/W	Default	Description
31:16	R/W	180	reg_gmut_coef21 : // unsigned , default = 180
15:0	R/W	34	reg_gmut_coef20 : // unsigned , default = 34

Table 8-1173 DI_HDR2_GMUT_COEF4 0x37A5

Bit(s)	R/W	Default	Description
15:0	R/W	1834	reg_gmut_coef22 : // unsigned , default = 1834

Table 8-1174 DI_HDR2_PIPE_CTRL1 0x37A6

Bit(s)	R/W	Default	Description
31:24	R/W	4	vblank_num_oetf : // unsigned , default = 4
23:16	R/W	4	hblank_num_oetf : // unsigned , default = 4
15:8	R/W	10	vblank_num_eotf : // unsigned , default = 10
7:0	R/W	10	hblank_num_eotf : // unsigned , default = 10

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Table 8-1175 DI_HDR2_PIPE_CTRL2 0x37A7

Bit(s)	R/W	Default	Description
31:24	R/W	10	vblank_num_cgain : // unsigned , default = 10
23:16	R/W	10	hblank_num_cgain : // unsigned , default = 10
15:8	R/W	11	vblank_num_gmut : // unsigned , default = 11
7:0	R/W	11	hblank_num_gmut : // unsigned , default = 11

Table 8-1176 DI_HDR2_PIPE_CTRL3 0x37A8

Bit(s)	R/W	Default	Description
31:24	R/W	22	vblank_num_adps : // unsigned , default = 22
23:16	R/W	2	hblank_num_adps : // unsigned , default = 2
15:8	R/W	4	vblank_num_uv : // unsigned , default = 4
7:0	R/W	4	hblank_num_uv : // unsigned , default = 4

Table 8-1177 DI_HDR2_PROC_WIN1 0x37A9

Bit(s)	R/W	Default	Description
28:16	R/W	0	proc_win_h_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_h_st : // unsigned , default = 0

Table 8-1178 DI_HDR2_PROC_WIN2 0x37AA

Bit(s)	R/W	Default	Description
31	R/W	0	proc_win_gmut_en : // unsigned , default = 0
30	R/W	0	proc_win_adps_en : // unsigned , default = 0
29	R/W	0	proc_win_cgain_en : // unsigned , default = 0
28:16	R/W	0	proc_win_v_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_v_st : // unsigned , default = 0

Table 8-1179 DI_HDR2_MATRIXI_EN_CTRL 0x37AB

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

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Table 8-1180 DI_HDR2_MATRIXO_EN_CTRL 0x37AC

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

8.2.3.11 NR_SCALE Registers

Table 8-1181 NRDSWR_X 0x37f9

Bit(s)	R/W	Default	Description
31:30	R/W	2	burst_len : unsigned , default = 2
29	R/W	0	rev_x : unsigned , default = 0
28:16	R/W	0	start_x : unsigned , default = 0
12:0	R/W	2	end_x : unsigned , default = 2cf

Table 8-1182 NRDSWR_Y 0x37fa

Bit(s)	R/W	Default	Description
31:30	R/W	0	canvas_id : unsigned , default = 0
29	R/W	0	rev_y : unsigned , default = 0
28:16	R/W	0	start_y : unsigned , default = 0
12:0	R/W	0	end_y : unsigned , default = 0x1df

Table 8-1183 NRDSWR_CTRL 0x37fb

Bit(s)	R/W	Default	Description
31:16	R/W	0	urgent_ctrl : unsigned , default = 0
15	R/W	0	force_wvalid : unsigned , default = 0
14	R/W	0	canvas_syncen : unsigned , default = 0
13	R/W	1	canvas_wr : unsigned , default = 1
12	R/W	0	req_en : unsigned , default = 0
10	R/W	0	clr_wrrsp : unsigned , default = 0
8	R/W	0	urgent : unsigned , default = 0
7:0	R/W	0	canvas_index : unsigned , default = 0

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Table 8-1184 NRDSWR_CAN_SIZE 0x37fc

Bit(s)	R/W	Default	Description
30:29	R/W	0	reg_rst : unsigned , default = 0
28:16	R/W	0	hsizem1 : unsigned , default = 0x2cf
14	R/W	0	reg_reset : unsigned , default = 0
13	R/W	0	little_endian : unsigned , default = 0
12:0	R/W	0	vsizem1 : unsigned , default = 0x1df

Table 8-1185 NR_DS_BUF_SIZE 0x3740

Bit(s)	R/W	Default	Description
31:24	R.O	96	dsbuf_rowmax : // unsigned , default = 96
23:16	R/W	128	dsbuf_colmax : // unsigned , default = 128
15: 8	R.O	128	dsbuf_orow : // unsigned , default = 128
7: 0	R/W	128	dsbuf_ocol : // unsigned , default = 128

Table 8-1186 NR_DS_CTRL 0x3741

Bit(s)	R/W	Default	Description
30	R/W	0	Nr_ds_enable
29:24	R/W	8	reg_h_step : // unsigned , default = 8 rand lut0
21:16	R/W	8	reg_v_step : // unsigned , default = 8 rand lut0
14:12	R/W	4	reg_haa_sel : // unsigned , default = 4
10: 8	R/W	4	reg_vaa_sel : // unsigned , default = 4
6: 4	R/W	1	reg_use_hphase : // unsigned , default = 1
0	R/W	0	reg_yuv_bldmode : // unsigned , default = 0

Table 8-1187 NR_DS_OFFSET 0x3742

Bit(s)	R/W	Default	Description
25:16	R/W	0	reg_h_ofst : // signed , default = 0
9: 0	R/W	0	reg_v_ofst : // signed , default = 0

Table 8-1188 NR_DS_BLD_COEF 0x3743

Bit(s)	R/W	Default	Description
23:16	R/W	128	reg_yuv_bldcoef2 : // unsigned , default = 128
15: 8	R/W	64	reg_yuv_bldcoef1 : // unsigned , default = 64
7: 0	R/W	64	reg_yuv_bldcoef0 : // unsigned , default = 64

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8.2.3.12 MCDI Registers

Table 8-1189 MCDI_HV_SIZEIN 0x2f00

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-16	R/W	1024	reg_mcdi_hsize image horizontal size (number of cols) default=1024
15-13	R/W		reserved
12-0	R/W	1024	reg_mcdi_vsize image vertical size (number of rows) default=1024

Table 8-1190 MCDI_HV_BLKSIZEIN 0x2f01

Bit(s)	R/W	Default	Description
31	R/W	0	reg_mcdi_vrev default = 0
30	R/W	0	reg_mcdi_hrev default = 0
29-28	R/W		reserved
27-16	R/W	1024	reg_mcdi_blkhsize image horizontal blk size (number of cols) default=1024
15-13	R/W		reserved
11-0	R/W	1024	reg_mcdi_blkvsize image vertical blk size (number of rows) default=1024

Table 8-1191 MCDI_BLKTOTAL 0x2f02

Bit(s)	R/W	Default	Description
31-24	R/W		reserved
23-0	R/W	0	reg_mcdi_blktotal

Table 8-1192 MCDI_MOTINEN 0x2f03

Bit(s)	R/W	Default	Description
31-2	R/W		reserved
1	R/W	1	reg_mcdi_motionrefen. enable motion refinement of MA, default = 1
0	R/W	1	reg_mcdi_motionparadoxen. enable motion paradox detection, default = 1

Table 8-1193 MCDI_CTRL_MODE 0x2f04

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28	R/W	0	mc info read enable
27-26	R/W	2	reg_mcdi_lmvlocken 0:disable, 1: use max Lmv, 2: use no-zero Lmv, lmv lock enable mode, default = 2
25	R/W	1	reg_mcdi_reldetrptchken 0-unable; 1: enableenable repeat pattern check (not repeat mv detection) in rel det part, default = 1

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Bit(s)	R/W	Default	Description
24	R/W	1	reg_mcdi_reldetgmvpd22chken 0-unable; 1: enable enable pull-down 22 mode check in gmv lock mode for rel det, default = 1
23	R/W	1	reg_mcdi_pd22chken 0-unable; 1: enable enable pull-down 22 mode check (lock) function, default = 1
22	R/W	1	reg_mcdi_reldetlpfen 0-unable; 1: enable enable det value lpf, default = 1
21	R/W	1	reg_mcdi_reldetlmvpd22chken 0-unable; 1: enable enable pull-down 22 mode check in lmv lock mode for rel det, default = 1
20	R/W	1	reg_mcdi_reldetlmvdifchken 0-unable; 1: enable enable lmv dif check in lmv lock mode for rel det, default = 1
19	R/W	1	reg_mcdi_reldetgmvdifchken 0-unable; 1: enable enable lmv dif check in lmv lock mode for rel det, default = 1
18	R/W	1	reg_mcdi_reldetpd22chken 0-unable; 1: enable enable pull-down 22 mode check for rel det refinement, default = 1
17	R/W	1	reg_mcdi_reldetfrqchken 0-unable; 1: enable enable mv frequency check in rel det, default = 1
16	R/W		reg_mcdi_qmeen 0-unable; 1: enable enable quarter motion estimation, defautl = 1
15	R/W	1	reg_mcdi_refrptmven 0-unable; 1: enable use repeat mv in refinement, default = 1
14	R/W	1	reg_mcdi_refgmven 0-unable; 1: enable use gmv in refinement, default = 1
13	R/W	1	reg_mcdi_reflmven 0-unable; 1: enable use lmvs in refinement, default = 1
12	R/W	1	reg_mcdi_refnmven 0-unable; 1: enable use neighoring mvs in refinement, default = 1
11	R/W		reserved
10	R/W	1	reg_mcdi_referrfrqchken 0-unable; 1: enable enable mv frquency check while finding min err in ref, default = 1
9	R/W	1	reg_mcdi_refen 0-unable; 1: enable enable mv refinement, default = 1
8	R/W	1	reg_mcdi_horlineen 0-unable; 1: enable enable horizontal lines detection by sad map, default = 1
7	R/W	1	reg_mcdi_highvertfrqdeten 0-unable; 1: enable enable high vertical frequency pattern detection, default = 1
6	R/W	1	reg_mcdi_gmvlocken 0-unable; 1: enable enable gmv lock mode, default = 1
5	R/W	1	reg_mcdi_rptmven 0-unable; 1: enable enable repeat pattern detection, default = 1
4	R/W	1	reg_mcdi_gmven 0-unable; 1: enable enable global motion estimation, default = 1
3	R/W	1	reg_mcdi_lmven0-unable; 1: enable enable line mv estimation for hme, default = 1
2	R/W	1	reg_mcdi_chkedgeen0-unable; 1: enable enable check edge function, default = 1
1	R/W	1	reg_mcdi_txtdeten 0-unable; 1: enable enable texture detection, default = 1

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Table 8-1194 MCDI_UNI_MVDST 0x2f05

Bit(s)	R/W	Default	Description
31-20	R/W		reserved
19-17	R/W	1	reg_mcdi_unimvdstabsseg0 segment0 for uni-mv abs, default = 1
16-12	R/W	15	reg_mcdi_unimvdstabsseg1 segment1 for uni-mv abs, default = 15
11-8	R/W	2	reg_mcdi_unimvdstabsdifgain0 2/2, gain0 of uni-mv abs dif for segment0, normalized 2 to '1', default = 2
7-5	R/W	2	reg_mcdi_unimvdstabsdifgain1 2/2, gain1 of uni-mv abs dif for segment1, normalized 2 to '1', default = 2
4-2	R/W	2	reg_mcdi_unimvdstabsdifgain2 2/2, gain2 of uni-mv abs dif beyond segment1, normalized 2 to '1', default = 2
1-0	R/W	0	reg_mcdi_unimvdstsgnshft shift for neighboring distance of uni-mv, default = 0

Table 8-1195 MCDI_BI_MVDST 0x2f06

Bit(s)	R/W	Default	Description
31-20	R/W		reserved
19-17	R/W	1	reg_mcdi_bimvdstabsseg0 segment0 for bi-mv abs, default = 1
16-12	R/W	9	reg_mcdi_bimvdstabsseg1 segment1 for bi-mv abs, default = 9
11-8	R/W	6	reg_mcdi_bimvdstabsdifgain0 6/2, gain0 of bi-mv abs dif for segment0, normalized 2 to '1', default = 6
7-5	R/W	3	reg_mcdi_bimvdstabsdifgain1 3/2, gain1 of bi-mvabs dif for segment1, normalized 2 to '1', default = 3
4-2	R/W	2	reg_mcdi_bimvdstabsdifgain2 2/2, gain2 of bi-mvabs dif beyond segment1, normalized 2 to '1', default = 2
1-0	R/W	0	reg_mcdi_bimvdstsgnshft shift for neighboring distance of bi-mv, default = 0

Table 8-1196 MCDI_SAD_GAIN 0x2f07

Bit(s)	R/W	Default	Description
31-19	R/W		reserved
18-17	R/W	3	reg_mcdi_unisadcorepxlgain uni-sad core pixels gain, default = 3
16	R/W	0	reg_mcdi_unisadcorepxlnormen enable uni-sad core pixels normalization, default = 0
15-11	R/W		reserved
10-9	R/W	3	reg_mcdi_bisadcorepxlgain bi-sad core pixels gain, default = 3
8	R/W	1	reg_mcdi_bisadcorepxlnormen enable bi-sad core pixels normalization, default = 1
7-3	R/W		reserved
2-1	R/W	3	reg_mcdi_biqsadcorepxlgain bi-qsad core pixels gain, default = 3
0	R/W	1	reg_mcdi_biqsadcorepxlnormen enable bi-qsad core pixels normalization, default = 1

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Table 8-1197 MCDI_TXT_THD 0x2f08

Bit(s)	R/W	Default	Description
31-24	R/W		reserved
23-16	R/W	24	reg_mcdi_txtminmaxdifthd, min max dif threshold (>=) for texture detection, default = 24
15-8	R/W	9	reg_mcdi_txtmeandifthd, mean dif threshold (<) for texture detection, default = 9
7-3	R/W		reserved
2-0	R/W	2	reg_mcdi_txtdetthd, texture detecting threshold, 0~4, default = 2

Table 8-1198 MCDI_FLT_MODESEL 0x2f09

Bit(s)	R/W	Default	Description
31	R/W		reserved
30-28	R/W	1	reg_mcdi_flthorlineselmode mode for horizontal line detecting flat calculation, default = 1, same as below
27	R/W		reserved
26-24	R/W	4	reg_mcdi_fltgmvselmode mode for gmv flat calculation, default = 4, same as below
23	R/W		reserved
22-20	R/W	2	reg_mcdi_fltsadselmode mode for sad flat calculation, default = 2, same as below
19	R/W		reserved
18-16	R/W	3	reg_mcdi_fltbadwselmode mode for badw flat calculation, default = 3, same as below
15	R/W		reserved
14-12	R/W	4	reg_mcdi_fltrptmvselmode mode for repeat mv flat calculation, default = 4, same as below
11	R/W		reserved
10-8	R/W	4	reg_mcdi_fltbadrelselmode mode for bad rel flat calculation, default = 4, same as below
7	R/W		reserved
6-4	R/W	2	reg_mcdi_fltcolcfdselmode mode for col cfd flat calculation, default = 2, same as below
3	R/W		reserved
2-0	R/W	2	reg_mcdi_fltpd22chkselmode mode for pd22 check flat calculation, default = 2, 0:cur dif h, 1: cur dif v, 2: pre dif h, 3: pre dif v, 4: cur flt, 5: pre flt, 6: cur+pre, 7: max all(cur,pre)

Table 8-1199 MCDI_CHK_EDGE_THD 0x2f0a

Bit(s)	R/W	Default	Description
23-28	R/W		reserved.
27-24	R/W	1	reg_mcdi_chkedgedifsadthd. thd (<=) for sad dif check, 0~8, default = 1
23-16	R/W		reserved.

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Bit(s)	R/W	Default	Description
15-12	R/W	15	reg_mcdi_chkedgemaxedgethd. max drt of edge, default = 15
11-8	R/W	2	reg_mcdi_chkedgeminedgethd. min drt of edge, default = 2
7	R/W		reserved.
6-0	R/W	14	reg_mcdi_chkedgevdifthd. thd for vertical dif in check edge, default = 14

Table 8-1200 MCDI_CHK_EDGE_GAIN_OFFST 0x2f0b

Bit(s)	R/W	Default	Description
31-24	R/W		reserved.
23-20	R/W	4	reg_mcdi_chkedgedifthd1. thd1 for edge dif check (<=), default = 4
19-16	R/W	15	reg_mcdi_chkedgedifthd0. thd0 for edge dif check (>=), default = 15
-15	R/W		reserved.
14-10	R/W	24	reg_mcdi_chkedgechklen. total check length for edge check, 1~24 (>0), default = 24
9-8	R/W	1	reg_mcdi_chkedgeedgesel. final edge select mode, 0: original start edge, 1: lpf start edge, 2: orignal start+end edge, 3: lpf start+end edge, default = 1
7-3	R/W	4	reg_mcdi_chkedgesaddstgain. distance gain for sad calc while getting edges, default = 4
2	R/W		reg_mcdi_chkedgechkmode. edge used in check mode, 0- original edge, 1: lpf edge, defautl = 1
1	R/W		reg_mcdi_chkedgestartedge. edge mode for start edge, 0- original edge, 1: lpf edge, defautl = 0
0	R/W	0	reg_mcdi_chkedgeedgelpf. edge lpf mode, 0-[0,2,4,2,0], 1:[1,2,2,2,1], default = 0

Table 8-1201 MCDI_LMV_RT0x2f0c

Bit(s)	R/W	Default	Description
31-15	R/W		reserved
14-12	R/W		reg_mcdi_lmvvalidmode valid mode for lmv calc., 100b: use char det, 010b: use flt, 001b: use hori flg
11-10	R/W	1	reg_mcdi_lmvgainmvmode four modes of mv selection for lmv weight caluclua-tion, default = 1
// // lst(x-1	R/W		x,x+1); 1- cur(x-4,x-3), lst(x,x+1); 2: cur(x-5,x-4,x-3), lst(x-1,x,x+1,x+2,x+3); 3: cur(x-6,x-5,x-4,x-3), lst(x-1,x,x+1,x+2);
9	R/W	0	reg_mcdi_lmvinitmode initial lmvs at first row of input field, 0- intial value = 0; 1: inital = 32 (invalid), default = 0
8	R/W		reserved
7-4	R/W	5	reg_mcdi_lmvrt0 ratio of max mv, default = 5
3-0	R/W	5	reg_mcdi_lmvrt1 ratio of second max mv, default = 5

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Table 8-1202 MCDI_LMV_GAINTHD 0x2f0d

Bit(s)	R/W	Default	Description
31-24	R/W	96	reg_mcdi_lmvvxmaxgain max gain of lmv weight, default = 96
23	R/W		reserved
22-20	R/W	1	reg_mcdi_lmvdifthd0 dif threshold 0 (<) for small lmv, default = 1
19-17	R/W	2	reg_mcdi_lmvdifthd1 dif threshold 1 (<) for median lmv, default = 2
16-14	R/W	3	reg_mcdi_lmvdifthd2 dif threshold 2 (<) for large lmv, default = 3
13-8	R/W	20	reg_mcdi_lmvnumlmt least/limit number of (total number - max0), default = 20
7-0	R/W	9	reg_mcdi_lmvfltthd flt cnt thd (<) for lmv, default = 9

Table 8-1203 MCDI_RPTMV_THD0 0x2f0e

Bit(s)	R/W	Default	Description
31-25	R/W	64	reg_mcdi_rptmvslpthd2 slope thd (>=) between i and i+3/i-3 (i+4/i-4), default = 64
24-20	R/W	4	reg_mcdi_rptmvslpthd1 slope thd (>=) between i and i+2/i-2, default = 4
19-10	R/W	300	reg_mcdi_rptmvampthd2 amplitude thd (>=) between max and min, when count cycles, default = 300
9-0	R/W	400	reg_mcdi_rptmvampthd1 amplitude thd (>=) between average of max and min, default = 400

Table 8-1204 MCDI_RPTMV_THD1 0x2f0f

Bit(s)	R/W	Default	Description
31-28	R/W		reserved
27-25	R/W	2	reg_mcdi_rptmvcyccntthd thd (>=) of total cycles count, default = 2
24-21	R/W	3	reg_mcdi_rptmvcycdifthd dif thd (<) of cycles length, default = 3
20-18	R/W	1	reg_mcdi_rptmvcycvldthd thd (>) of valid cycles number, default = 1
17-15	R/W	2	reg_mcdi_rptmvhalfcycminthd min length thd (>=) of half cycle, default = 2
14-11	R/W	5	reg_mcdi_rptmvhalfcycdifthd neighboring half cycle length dif thd (<), default = 5
10-8	R/W	2	reg_mcdi_rptmvminmaxcntthd least number of valid max and min, default = 2
7-5	R/W	2	reg_mcdi_rptmvcycminthd min length thd (>=) of cycles, default = 2
4-0	R/W	17	reg_mcdi_rptmvcycmaxthd max length thd (<) of cycles, default = 17

Table 8-1205 MCDI_RPTMV_THD2 0x2f10

Bit(s)	R/W	Default	Description
31-24	R/W		reserved
23-16	R/W	8	reg_mcdi_rptmvhdifthd0 higher hdif thd (>=) (vertical edge) for rpt detection, default = 8

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Bit(s)	R/W	Default	Description
15-8	R/W	4	reg_mcdi_rptmvhdifthd1 hdif thd (>=) (slope edge) for rpt detection, default = 4
7-0	R/W	1	reg_mcdi_rptmvvdifthd vdif thd (>=) (slope edge) for rpt detection, default = 1

Table 8-1206 MCDI_RPTMV_SAD 0x2f11

Bit(s)	R/W	Default	Description
31-26	R/W		reserved
25-16	R/W	336	reg_mcdi_rptmvsaddifthdgain 7x3x(16/16), gain for sad dif thd in rpt mv detection, 0~672, normalized 16 as '1', default = 336
15-10	R/W		reserved
9-0	R/W	16	reg_mcdi_rptmvsaddifthdoffst offset for sad dif thd in rpt mv detection, -512~511, default = 16

Table 8-1207 MCDI_RPTMV_FLG 0x2f12

Bit(s)	R/W	Default	Description
31-18	R/W		reserved
17-16	R/W	2	reg_mcdi_rptmvmode select mode of mvs for repeat motion estimation, 0: hmv, 1: qmv/2, 2 or 3: qmv/4, default = 2
15-8	R/W	64	reg_mcdi_rptmvflgcntthd thd (>=) of min count number for rptmv of whole field, for rptmv estimation, default = 64
7-5	R/W		reserved
4-0	R/W		reg_mcdi_rptmvflgcntrt 4/32, ratio for repeat mv flag count, normalized 32 as '1', set 31 to 32,

Table 8-1208 MCDI_RPTMV_GAIN 0x2f13

Bit(s)	R/W	Default	Description
31-24	R/W	96	reg_mcdi_rptmvlftgain up repeat mv gain for hme, default = 96
23-16	R/W	32	reg_mcdi_rptmvuplftgain up left repeat mv gain for hme, default = 32
15-8	R/W	64	reg_mcdi_rptmvupgain up repeat mv gain for hme, default = 64
7-0	R/W	32	reg_mcdi_rptmvuprightgain up right repeat mv gain for hme, default = 32

Table 8-1209 MCDI_GMV_RT0x2f14

Bit(s)	R/W	Default	Description
31	R/W		reserved
30-24	R/W	32	reg_mcdi_gmvmtnrt0 ratio 0 for motion senario, set 127 to 128, normalized 128 as '1', default =32
23	R/W		reserved
22-16	R/W	56	reg_mcdi_gmvmtnrt1 ratio 1 for motion senario, set 127 to 128, normalized 128 as '1', default = 56

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Bit(s)	R/W	Default	Description
15	R/W		reserved
14-8	R/W	56	reg_mcdi_gmvstlrt0 ratio 0 for still senario, set 127 to 128, normalized 128 as '1', default = 56
7	R/W		reserved
6-0	R/W	80	reg_mcdi_gmvstlrt1 ratio 1 for still senario, set 127 to 128, normalized 128 as '1', default = 80

Table 8-1210 MCDI_GMV_GAIN 0x2f15

Bit(s)	R/W	Default	Description
31-25	R/W	100	reg_mcdi_gmvzeromvlockrt0 ratio 0 for locking zero mv, set 127 to 128, normalized 128 as '1', default = 100
24-18	R/W	112	reg_mcdi_gmvzeromvlockrt1 ratio 1 for locking zero mv, set 127 to 128, normalized 128 as '1', default = 112
17-16	R/W	3	reg_mcdi_gmvvalidmode valid mode for gmv calc., 10b: use flt, 01b: use hori flg, default = 3
15-8	R/W	0	reg_mcdi_gmvvxgain gmv's vx gain when gmv locked for hme, default = 0
7-0	R/W	3	reg_mcdi_gmvfltthd flat thd (<) for gmv calc. default = 3

Table 8-1211 MCDI_HOR_SADOFST 0x2f16

Bit(s)	R/W	Default	Description
31-25	R/W		reserved
24-16	R/W	21	reg_mcdi_horsaddifthdgain 21*1/8, gain/divisor for sad dif threshold in hor line detection, normalized 8 as '1', default = 21
15-8	R/W	0	reg_mcdi_horsaddifthdoffst offset for sad dif threshold in hor line detection, -128~127, default = 0
7-0	R/W	24	reg_mcdi_horvdifthd threshold (>=) of vertical dif of next block for horizontal line detection, default = 24

Table 8-1212 MCDI_REF_MV_NUM 0x2f17

Bit(s)	R/W	Default	Description
31-2	R/W		reserved
1-0	R/W	0	reg_mcdi_refmcmode. motion compensated mode used in refinement, 0: pre, 1: next, 2: (pre+next)/2, default = 0

Table 8-1213 MCDI_REF_BADW_THD_GAIN0x2f18

Bit(s)	R/W	Default	Description
31-28	R/W		reserved
27-24	R/W	6	reg_mcdi_refbadwcnt2gain. gain for badwv count num==3, default = 6
23-20	R/W	3	reg_mcdi_refbadwcnt1gain. gain for badwv count num==2, default = 3
19-16	R/W	1	reg_mcdi_refbadwcnt0gain. gain for badwv count num==1, default = 1

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Bit(s)	R/W	Default	Description
15-12	R/W	4	reg_mcdi_refbadwthd3. threshold 3 for detect badweave with largest average luma, default = 4
11-8	R/W	3	reg_mcdi_refbadwthd2. threshold 2 for detect badweave with third smallest average luma, default = 3
7-4	R/W	2	reg_mcdi_refbadwthd1. threshold 1 for detect badweave with second smallest average luma, default = 2
3-0	R/W	1	reg_mcdi_refbadwthd0. threshold 0 for detect badweave with smallest average luma, default = 1

Table 8-1214 MCDI_REF_BADW_SUM_GAIN0x2f19

Bit(s)	R/W	Default	Description
31-13	R/W		reserved
12-8	R/W	8	reg_mcdi_refbadwsumgain0. sum gain for r channel, 0~16, default = 8
7-5	R/W		reserved
4	R/W	0	reg_mcdi_refbadwcalcmode. mode for badw calculation, 0-sum, 1:max, default = 0
3-0	R/W		reserved

Table 8-1215 MCDI_REF_BS_THD_GAIN 0x2f1a

Bit(s)	R/W	Default	Description
31-28	R/W	2	reg_mcdi_refbsudgain1. up & down block stregth gain1, normalized to 8 as '1', default = 2
27-24	R/W	4	reg_mcdi_refbsudgain0. up & down block stregth gain0, normalized to 8 as '1', default = 4
23-19	R/W		reserved
18-16	R/W	0	reg_mcdi_refbslftgain. left block strength gain, default = 0
15-13	R/W		reserved
12-8	R/W	16	reg_mcdi_refbsthd1. threshold 1 for detect block stregth in refinment, default = 16
7-5	R/W		reserved
4-0	R/W	8	reg_mcdi_refbsthd0. threshold 0 for detect block stregth in refinment, default = 8

Table 8-1216 MCDI_REF_ERR_GAIN0 0x2f1b

Bit(s)	R/W	Default	Description
31	R/W		reserved
30-24	R/W	48	reg_mcdi_referrnbrdstgain. neighoring mv distances gain for err calc. in ref, normalized to 8 as '1', default = 48
23-20	R/W		reserved
19-16	R/W	4	reg_mcdi_referrbsgain. bs gain for err calc. in ref, normalized to 8 as '1', default = 4

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Bit(s)	R/W	Default	Description
15	R/W		reserved
14-8	R/W	64	reg_mcdi_referrbadwgain. badw gain for err calc. in ref, normalized to 8 as '1', default = 64
7-4	R/W		reserved
3-0	R/W	4	reg_mcdi_referrsadgain. sad gain for err calc. in ref, normalized to 8 as '1', default = 4

Table 8-1217 MCDI_REF_ERR_GAIN1 0x2f1c

Bit(s)	R/W	Default	Description
31-20	R/W		reserved
19-16	R/W	4	reg_mcdi_referrchkedgegain. check edge gain for err calc. in ref, normalized to 8 as '1', default = 4
15-12	R/W		reserved
11-8	R/W	0	reg_mcdi_referrlmvgain. (locked) lmv gain for err calc. in ref, normalized to 8 as '1', default = 0
7-4	R/W		reserved
3-0	R/W	0	reg_mcdi_referrgmvgain. (locked) gmv gain for err calc. in ref, normalized to 8 as '1', default = 0

Table 8-1218 MCDI_REF_ERR_FRQ_CHK 0x2f1d

Bit(s)	R/W	Default	Description
31-28	R/W		reserved
27-24	R/W	10	reg_mcdi_referrfrqgain. gain for mv frquency, normalized to 4 as '1', default = 10
23-21	R/W		reserved
20-16	R/W	31	reg_mcdi_referrfrqmax. max gain for mv frquency check, default = 31
15	R/W		reserved
14-12	R/W	3	reg_mcdi_ref_errfrqmvdifthd2. mv dif threshold 2 (<) for mv frquency check, default = 3
11	R/W		reserved
10-8	R/W	2	reg_mcdi_ref_errfrqmvdifthd1. mv dif threshold 1 (<) for mv frquency check, default = 2
7	R/W		reserved
6-4	R/W	1	reg_mcdi_ref_errfrqmvdifthd0. mv dif threshold 0 (<) for mv frquency check, default = 1
3-0	R/W		reserved

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Table 8-1219 MCDI_QME_LPF_MSK 0x2f1e

Bit(s)	R/W	Default	Description
31-28	R/W		reserved
27-24	R/W	7	reg_mcdi_qmechkedgelpfmsk0. lpf mask0 for chk edge in qme, 0~8, msk1 = (8-msk0), normalized to 8 as '1', default = 7
23-20	R/W		reserved
19-16	R/W	7	reg_mcdi_qmebslpfmsk0. lpf mask0 for bs in qme, 0~8, msk1 = (8-msk0), normalized to 8 as '1', default = 7
15-12	R/W		reserved
11-8	R/W	7	reg_mcdi_qmebadwlpfmsk0. lpf mask0 for badw in qme, 0~8, msk1 = (8-msk0), normalized to 8 as '1', default = 7
7-4	R/W		reserved
3-0	R/W	7	reg_mcdi_qmesadlpfmsk0. lpf mask0 for sad in qme, 0~8, msk1 = (8-msk0), normalized to 8 as '1', default = 7

Table 8-1220 MCDI_REL_DIF_THD_02 0x2f1f

Bit(s)	R/W	Default	Description
31-24	R/W		reserved.
23-16	R/W	9	reg_mcdi_reldifthd2. thd (<) for (hdif+vdif), default = 9
15-8	R/W	5	reg_mcdi_reldifthd1. thd (<) for (vdif), default = 5
7-0	R/W	48	reg_mcdi_reldifthd0. thd (>=) for (hdif-vdif), default = 48

Table 8-1221 MCDI_REL_DIF_THD_34 0x2f20

Bit(s)	R/W	Default	Description
31-16	R/W		reserved.
15-8	R/W	255	reg_mcdi_reldifthd4. thd (<) for (hdif), default = 255
7-0	R/W	48	reg_mcdi_reldifthd3. thd (>=) for (vdif-hdif), default = 48

Table 8-1222 MCDI_REL_BADW_GAIN_OFFST_01 0x2f21

Bit(s)	R/W	Default	Description
31-24	R/W	0	reg_mcdi_relbadwoffst1. offset for badw adj, for flat block, -128~127, default = 0
23-16	R/W	128	reg_mcdi_relbadwgain1. gain for badw adj, for flat block, default = 128
15-8	R/W	0	reg_mcdi_relbadwoffst0. offset for badw adj, for vertical block, -128~127, default = 0
7-0	R/W	160	reg_mcdi_relbadwgain0. gain for badw adj, for vertical block, default = 160

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Table 8-1223 MCDI_REL_BADW_GAIN_OFFST_23 0x2f22

Bit(s)	R/W	Default	Description
31-24	R/W	0	reg_mcdi_relbadwoffst3. offset for badw adj, for other block, -128~127, default = 0
23-16	R/W	48	reg_mcdi_relbadwgain3. gain for badw adj, for other block, default = 48
15-8	R/W	0	reg_mcdi_relbadwoffst2. offset for badw adj, for horizontal block, -128~127, default = 0
7-0	R/W	48	reg_mcdi_relbadwgain2. gain for badw adj, for horizontal block, default = 48

Table 8-1224 MCDI_REL_BADW_THD_GAIN_OFFST 0x2f23

Bit(s)	R/W	Default	Description
31-23	R/W		reserved.
22-16	R/W	0	reg_mcdi_relbadwoffst. offset for badw thd adj, -64~63, default = 0
15-8	R/W		reserved.
7-0	R/W	16	reg_mcdi_relbadwthdgain. gain0 for badw thd adj, normalized to 16 as '1', default = 16

Table 8-1225 MCDI_REL_BADW_THD_MIN_MAX 0x2f24

Bit(s)	R/W	Default	Description
31-18	R/W		reserved.
17-8	R/W	256	reg_mcdi_relbadwthdmax. max for badw thd adj, default = 256
7-0	R/W	16	reg_mcdi_relbadwthdmin. min for badw thd adj, default = 16

Table 8-1226 MCDI_REL_SAD_GAIN_OFFST_01 0x2f25

Bit(s)	R/W	Default	Description
31-24	R/W	0	reg_mcdi_relsadoffst1. offset for sad adj, for flat block, -128~127, default = 0
23-20	R/W		reserved.
19-16	R/W	8	reg_mcdi_relsadgain1. gain for sad adj, for flat block, normalized to 8 as '1', default = 8
15-8	R/W	0	reg_mcdi_relsadoffst0. offset for sad adj, for vertical block, -128~127, default = 0
7-4	R/W		reserved.
3-0	R/W	6	reg_mcdi_relsadgain0. gain for sad adj, for vertical block, normalized to 8 as '1', default = 6

Table 8-1227 MCDI_REL_SAD_GAIN_OFFST_23 0x2f26

Bit(s)	R/W	Default	Description
31-24	R/W	0	reg_mcdi_relsadoffst3. offset for sad adj, for other block, -128~127, default = 0
23-20	R/W		reserved.

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Bit(s)	R/W	Default	Description
19-16	R/W	8	reg_mcdi_relsadgain3. gain for sad adj, for other block, normalized to 8 as '1', default = 8
15-8	R/W	0	reg_mcdi_relsadoffst2. offset for sad adj, for horizontal block, -128~127, default = 0
7-4	R/W		reserved.
3-0	R/W	12	reg_mcdi_relsadgain2. gain for sad adj, for horizontal block, normalized to 8 as '1', default = 12

Table 8-1228 MCDI_REL_SAD_THD_GAIN_OFFST 0x2f27

Bit(s)	R/W	Default	Description
31-24	R/W		reserved.
23-16	R/W	0	reg_mcdi_relsadoffst. offset for sad thd adj, -128~127, default = 0
15-10	R/W		reserved.
9-0	R/W	42	reg_mcdi_relsadthdgain. gain for sad thd adj, 21*2/16, normalized to 16 as '1', default = 42

Table 8-1229 MCDI_REL_SAD_THD_MIN_MAX 0x2f28

Bit(s)	R/W	Default	Description
31-27	R/W		reserved.
26-16	R/W	672	reg_mcdi_relsadthdmax. max for sad thd adj, 21*32, default = 672
15-9	R/W		reserved.
8-0	R/W	42	reg_mcdi_relsadthdmin. min for sad thd adj, 21*2, default = 42

Table 8-1230 MCDI_REL_DET_GAIN_00 0x2f29

Bit(s)	R/W	Default	Description
31-21	R/W		reserved.
20-16	R/W	8	reg_mcdi_reldetbsgain0. gain0 (gmv locked) for bs, for det. calc. normalized to 16 as '1', default = 8
15-14	R/W		reserved.
13-8	R/W	12	reg_mcdi_reldetbadwgain0. gain0 (gmv locked) for badw, for det. calc. normalized to 16 as '1', default = 12
7-5	R/W		reserved.
4-0	R/W	8	reg_mcdi_reldetsadgain0. gain0 (gmv locked) for qsad, for det. calc. normalized to 16 as '1', default = 8

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Table 8-1231 MCDI_REL_DET_GAIN_01 0x2f2a

Bit(s)	R/W	Default	Description
31-14	R/W		reserved.
12-8	R/W	2	reg_mcdi_reldetchkedgegain0. gain0 (gmv locked) for chk_edge, for det. calc. normalized to 16 as '1', default = 2
7	R/W		reserved.
6-0	R/W	24	reg_mcdi_reldetnbrdstgain0. gain0 (gmv locked) for neighoring dist, for det. calc. normalized to 16 as '1', default = 24

Table 8-1232 MCDI_REL_DET_GAIN_10 0x2f2b

Bit(s)	R/W	Default	Description
31-21	R/W		reserved.
20-16	R/W	0	reg_mcdi_reldetbsgain1. gain1 (lmv locked) for bs, for det. calc. normalized to 16 as '1', default = 0
15-14	R/W		reserved.
13-8	R/W	8	reg_mcdi_reldetbadwgain1. gain1 (lmv locked) for badw, for det. calc. normalized to 16 as '1', default = 8
7-5	R/W		reserved.
4-0	R/W	8	reg_mcdi_reldetsadgain1. gain1 (lmv locked) for qsad, for det. calc. normalized to 16 as '1', default = 8

Table 8-1233 MCDI_REL_DET_GAIN_11 0x2f2c

Bit(s)	R/W	Default	Description
31-14	R/W		reserved.
12-8	R/W	0	reg_mcdi_reldetchkedgegain1. gain1 (lmv locked) for chk_edge, for det. calc. normalized to 16 as '1', default = 0
7	R/W		reserved.
6-0	R/W	24	reg_mcdi_reldetnbrdstgain1. gain1 (lmv locked) for neighoring dist, for det. calc. normalized to 16 as '1', default = 24

Table 8-1234 MCDI_REL_DET_GAIN_20 0x2f2d

Bit(s)	R/W	Default	Description
31-21	R/W		reserved.
20-16	R/W	12	reg_mcdi_reldetbsgain2. gain2 (no locked) for bs, for det. calc. normalized to 16 as '1', default = 12
15-14	R/W		reserved.
13-8	R/W	32	reg_mcdi_reldetbadwgain2. gain2 (no locked) for badw, for det. calc. normalized to 16 as '1', default = 32
7-5	R/W		reserved.
4-0	R/W	16	reg_mcdi_reldetsadgain2. gain2 (no locked) for qsad, for det. calc. normalized to 16 as '1', default = 16

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Table 8-1235 MCDI_REL_DET_GAIN_21 0x2f2e

Bit(s)	R/W	Default	Description
31-26	R/W		reserved
25-16	R/W	0	reg_mcdi_reldetoffst. offset for rel calculation, for det. calc. -512~511, default = 0
15-14	R/W		reserved.
12-8	R/W	10	reg_mcdi_reldetchkedgegain2. gain2 (no locked) for chk_edge, for det. calc. normalized to 16 as '1', default = 10
7	R/W		reserved.
6-0	R/W	32	reg_mcdi_reldetnbrdstgain2. gain2 (no locked) for neighoring dist, for det. calc. normalized to 16 as '1', default = 32

Table 8-1236 MCDI_REL_DET_GMV_DIF_CHK 0x2f2f

Bit(s)	R/W	Default	Description
31-24	R/W		reserved.
23-16	R/W	0	reg_mcdi_reldetgmvfltthd. flat thd (>=) for gmv lock decision, default = 0
15	R/W		reserved.
14-12	R/W	3	reg_mcdi_reldetgmvdifthd. dif thd (>=) for current mv different from gmv for gmv dif check, actually used in Lmv lock check, default = 3
11	R/W		reserved.
10-8	R/W	1	reg_mcdi_reldetgmvdifmin. min mv dif for gmv dif check, default = 1, note: dif between reg_mcdi_rel_det_gmv_dif_max and reg_mcdi_rel_det_gmv_dif_min should be; 0,1,3,7, not work for others
7-4	R/W	4	reg_mcdi_reldetgmvdifmax. max mv dif for gmv dif check, default = 4
3-1	R/W		reserved
0	R/W	0	reg_mcdi_reldetgmvdifmvmode. mv mode used for gmv dif check, 0- use refmv, 1: use qmv, default = 0

Table 8-1237 MCDI_REL_DET_LMV_DIF_CHK 0x2f30

Bit(s)	R/W	Default	Description
31-24	R/W		reserved.
23-16	R/W	12	reg_mcdi_reldetlmvfltthd. flat thd (>=) for lmv lock decision, default = 12
15-14	R/W		reserved.
13-12	R/W	1	reg_mcdi_reldetlmvlockchkmode. lmv lock check mode, 0:cur Lmv, 1: cur & (last | next), 2: last & cur & next Lmv, default = 1
11	R/W		reserved.
10-8	R/W	1	reg_mcdi_reldetlmvdifmin. min mv dif for lmv dif check, default = 1, note: dif between reg_mcdi_rel_det_lmv_dif_max and reg_mcdi_rel_det_lmv_dif_min should be; 0,1,3,7, not work for others
7-4	R/W	4	reg_mcdi_reldetlmvdifmax. max mv dif for lmv dif check, default = 4

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Bit(s)	R/W	Default	Description
3-1	R/W		reserved
0	R/W	0	reg_mcdi_reldetlmvdifmvmode. mv mode used for lmv dif check, 0- use refmv, 1: use qmv, default = 0

Table 8-1238 MCDI_REL_DET_FRQ_CHK 0x2f31

Bit(s)	R/W	Default	Description
31-12	R/W		reserved.
11-8	R/W	10	reg_mcdi_reldetfrqgain. gain for frequency check, normalized to 4 as '1', default = 10
7-5	R/W		reserved
4-0	R/W	31	reg_mcdi_reldetfrqmax. max value for frequency check, default = 31

Table 8-1239 MCDI_REL_DET_PD22_CHK 0x2f32

Bit(s)	R/W	Default	Description
31-18	R/W		reserved.
30-21	R/W	512	reg_mcdi_reldetpd22chkoffst1. offset for pd22 check happened, default = 512
20-16	R/W	12	reg_mcdi_reldetpd22chkgain1. gain for pd22 check happened, normalized to 8 as '1', default = 12
14-5	R/W	512	reg_mcdi_reldetpd22chkoffst0. offset for pd22 check happened, default = 512
4-0	R/W	12	reg_mcdi_reldetpd22chkgain0. gain for pd22 check happened, normalized to 8 as '1', default = 12

Table 8-1240 MCDI_REL_DET_RPT_CHK_ROW 0x2f33

Bit(s)	R/W	Default	Description
31-27	R/W		reserved
26-16	R/W	2047	reg_mcdi_reldetrptchkendrow. end row (<) number for repeat check, default = 2047
15-11	R/W		reserved
10-0	R/W	0	reg_mcdi_reldetrptchkstartrow. start row (>=) number for repeat check, default = 0

Table 8-1241 MCDI_REL_DET_RPT_CHK_GAIN_QMV 0x2f34

Bit(s)	R/W	Default	Description
31-30	R/W		reserved
29-24	R/W	15	reg_mcdi_reldetrptchkqmvmax. max thd (<) of abs qmv for repeat check, default = 15, note that quarter mv's range is -63~63
23-22	R/W		reserved
21-16	R/W	10	reg_mcdi_reldetrptchkqmvmin. min thd (>=) of abs qmv for repeat check, default = 10, note that quarter mv's range is -63~63

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Bit(s)	R/W	Default	Description
15	R/W		reserved/
14-4	R/W	512	reg_mcdi_reldetrptchkoffst. offset for repeat check, default = 512
3-0	R/W	4	reg_mcdi_reldetrptchkgain. gain for repeat check, normalized to 8 as '1', default = 4

Table 8-1242 MCDI_REL_DET_RPT_CHK_THD_0 0x2f35

Bit(s)	R/W	Default	Description
31-24	R/W		reserved
23-16	R/W	255	reg_mcdi_reldetrptchkzerosadthd. zero sad thd (<) for repeat check, default = 255
15-14	R/W		reserved.
13-8	R/W	16	reg_mcdi_reldetrptchkzerobadwthd. zero badw thd (>=) for repeat check, default = 16
7-4	R/W		reserved
3-0	R/W	5	reg_mcdi_reldetrptchkfrqdifthd. frequency dif thd (<) for repeat check, 0~10, default = 5

Table 8-1243 MCDI_REL_DET_RPT_CHK_THD_1 0x2f36

Bit(s)	R/W	Default	Description
31-16	R/W		reserved
15-8	R/W	16	reg_mcdi_reldetrptchkvdifthd. vertical dif thd (<) for repeat check, default = 16
7-0	R/W	16	reg_mcdi_reldetrptchkhdifthd. horizontal dif thd (>=) for repeat check, default = 16

Table 8-1244 MCDI_REL_DET_LPF_DIF_THD 0x2f37

Bit(s)	R/W	Default	Description
31-24	R/W	9	reg_mcdi_reldetlpfdifthd3. hdif thd (<) for lpf selection of horizontal block, default = 9
23-16	R/W	48	reg_mcdi_reldetlpfdifthd2. vdif-hdif thd (>=) for lpf selection of horizontal block, default = 48
15-8	R/W	9	reg_mcdi_reldetlpfdifthd1. vdif thd (<) for lpf selection of vertical block, default = 9
7-0	R/W	48	reg_mcdi_reldetlpfdifthd0. hdif-vdif thd (>=) for lpf selection of vertical block, default = 48

Table 8-1245 MCDI_REL_DET_LPF_MSK_00_03 0x2f38

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-24	R/W	1	reg_mcdi_reldetlpfmsk03. det lpf mask03 for gmv/lmv locked mode, 0~16, default = 1

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Bit(s)	R/W	Default	Description
23-21	R/W		reserved
20-16	R/W	1	reg_mcdi_reldetlpfmsk02. det lpf mask02 for gmv/lmv locked mode, 0~16, default = 1
15-13	R/W		reserved
12-8	R/W	5	reg_mcdi_reldetlpfmsk01. det lpf mask01 for gmv/lmv locked mode, 0~16, default = 5
7-5	R/W		reserved
4-0	R/W	8	reg_mcdi_reldetlpfmsk00. det lpf mask00 for gmv/lmv locked mode, 0~16, default = 8

Table 8-1246 MCDI_REL_DET_LPF_MSK_04_12 0x2f39

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-24	R/W	0	reg_mcdi_reldetlpfmsk12. det lpf mask12 for vertical blocks, 0~16, default = 0
23-21	R/W		reserved
20-16	R/W	0	reg_mcdi_reldetlpfmsk11. det lpf mask11 for vertical blocks, 0~16, default = 0
15-13	R/W		reserved
12-8	R/W	16	reg_mcdi_reldetlpfmsk10. det lpf mask10 for vertical blocks, 0~16, default = 16
7-5	R/W		reserved
4-0	R/W	1	reg_mcdi_reldetlpfmsk04. det lpf mask04 for gmv/lmv locked mode, 0~16, default = 1

Table 8-1247 MCDI_REL_DET_LPF_MSK_13_21 0x2f3a

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-24	R/W	6	reg_mcdi_reldetlpfmsk21. det lpf mask21 for horizontal blocks, 0~16, default = 6
23-21	R/W		reserved
20-16	R/W	8	reg_mcdi_reldetlpfmsk20. det lpf mask20 for horizontal blocks, 0~16, default = 8
15-13	R/W		reserved
12-8	R/W	0	reg_mcdi_reldetlpfmsk14. det lpf mask14 for vertical blocks, 0~16, default = 0
7-5	R/W		reserved
4-0	R/W	0	reg_mcdi_reldetlpfmsk13. det lpf mask13 for vertical blocks, 0~16, default = 0

Table 8-1248 MCDI_REL_DET_LPF_MSK_22_30 0x2f3b

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-24	R/W	16	reg_mcdi_reldetlpfmsk30. det lpf mask30 for other blocks, 0~16, default = 16

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Bit(s)	R/W	Default	Description
23-21	R/W		reserved
20-16	R/W	1	reg_mcdi_reldetlpfmsk24. det lpf mask24 for horizontal blocks, 0~16, default = 1
15-13	R/W		reserved
12-8	R/W	0	reg_mcdi_reldetlpfmsk23. det lpf mask23 for horizontal blocks, 0~16, default = 0
7-5	R/W		reserved
4-0	R/W	1	reg_mcdi_reldetlpfmsk22. det lpf mask22 for horizontal blocks, 0~16, default = 1

Table 8-1249 MCDI_REL_DET_LPF_MSK_31_34 0x2f3c

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-24	R/W	0	reg_mcdi_reldetlpfmsk34. det lpf mask34 for other blocks, 0~16, default = 0
23-21	R/W		reserved
20-16	R/W	0	reg_mcdi_reldetlpfmsk33. det lpf mask33 for other blocks, 0~16, default = 0
15-13	R/W		reserved
12-8	R/W	0	reg_mcdi_reldetlpfmsk32. det lpf mask32 for other blocks, 0~16, default = 0
7-5	R/W		reserved
4-0	R/W	0	reg_mcdi_reldetlpfmsk31. det lpf mask31 for other blocks, 0~16, default = 0

Table 8-1250 MCDI_REL_DET_MIN 0x2f3d

Bit(s)	R/W	Default	Description
31-7	R/W		reserved
6-0	R/W	16	reg_mcdi_reldetmin. min of detected value, default = 16

Table 8-1251 MCDI_REL_DET_LUT_0_3 0x2f3e

Bit(s)	R/W	Default	Description
31-24	R/W	8	reg_mcdi_reldetmaplut3. default = 8
23-16	R/W	4	reg_mcdi_reldetmaplut2. default = 4
15-8	R/W	2	reg_mcdi_reldetmaplut1. default = 2
7-0	R/W	0	reg_mcdi_reldetmaplut0. default = 0

Table 8-1252 MCDI_REL_DET_LUT_4_7 0x2f3f

Bit(s)	R/W	Default	Description
31-24	R/W	64	reg_mcdi_reldetmaplut7. default = 64
23-16	R/W	48	reg_mcdi_reldetmaplut6. default = 48

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Bit(s)	R/W	Default	Description
15-8	R/W	32	reg_mcdi_reldetmaplut5. default = 32
7-0	R/W	16	reg_mcdi_reldetmaplut4. default = 16

Table 8-1253 MCDI_REL_DET_LUT_8_11 0x2f40

Bit(s)	R/W	Default	Description
31-24	R/W	160	reg_mcdi_reldetmaplut11. default = 160
23-16	R/W	128	reg_mcdi_reldetmaplut10. default = 128
15-8	R/W	96	reg_mcdi_reldetmaplut9. default = 96
7-0	R/W	80	reg_mcdi_reldetmaplut8. default = 80

Table 8-1254 MCDI_REL_DET_LUT_12_150x2f41

Bit(s)	R/W	Default	Description
31-24	R/W	255	reg_mcdi_reldetmaplut15. default = 255
23-16	R/W	240	reg_mcdi_reldetmaplut14. default = 240
15-8	R/W	224	reg_mcdi_reldetmaplut13. default = 224
7-0	R/W	192	reg_mcdi_reldetmaplut12. default = 192

Table 8-1255 MCDI_REL_DET_COL_CFD_THD 0x2f42

Bit(s)	R/W	Default	Description
31-24	R/W	5	reg_mcdi_reldetcolcfdfltthd. thd for flat smaller than (<) of column cofidence, default = 5
23-16	R/W	160	reg_mcdi_reldetcolcfdthd1. thd for rel larger than (>=) in rel calc. mode col confidence without gmv locking, default = 160
15-8	R/W	100	reg_mcdi_reldetcolcfdthd0. thd for rel larger than (>=) in rel calc. mode col confidence when gmv locked, default = 100
7-2	R/W	16	reg_mcdi_reldetcolcfdbadwthd. thd for badw larger than (>=) in qbadw calc. mode of column cofidence, default = 16
1	R/W		reserved
0	R/W	0	reg_mcdi_reldetcolcfdcalcmode. calc. mode for column cofidence, 0- use rel, 1: use qbadw, default = 0

Table 8-1256 MCDI_REL_DET_COL_CFD_AVG_LUMA 0x2f43

Bit(s)	R/W	Default	Description
31-24	R/W	235	reg_mcdi_reldetcolcfdavgmin1. avg luma min1 (>=) for column cofidence, valid between 16~235, default = 235
23-16	R/W	235	reg_mcdi_reldetcolcfdavgmax1. avg luma max1 (<) for column cofidence, valid between 16~235, default = 235

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Bit(s)	R/W	Default	Description
15-8	R/W	16	reg_mcdi_reldetcolcfdavgmin0. avg luma min0 (>=) for column cofidence, valid between 16~235, default = 16
7-0	R/W	21	reg_mcdi_reldetcolcfdavgmax0. avg luma max0 (<) for column cofidence, valid between 16~235, default = 21

Table 8-1257 MCDI_REL_DET_BAD_THD_00x2f44

Bit(s)	R/W	Default	Description
31-16	R/W		reserved
15-8	R/W	120	reg_mcdi_reldetbadsadthd. thd (>=) for bad sad, default = 120 (480/4)
7-6	R/W		reserved
5-0	R/W	12	reg_mcdi_reldetbadbadwthd. thd (>=) for bad badw, 0~42, default = 12

Table 8-1258 MCDI_REL_DET_BAD_THD_10x2f45

Bit(s)	R/W	Default	Description
31-24	R/W		reserved
23-16	R/W	4	reg_mcdi_reldetbadrelfltthd. thd (>=) of flat for bad rel detection, default = 4
15-8	R/W	160	reg_mcdi_reldetbadrelthd1. thd (>=) for bad rel without gmv/lmv locked, default = 160
7-0	R/W	120	reg_mcdi_reldetbadrelthd0. thd (>=) for bad rel with gmv/lmv locked, default = 120

Table 8-1259 MCDI_PD22_CHK_THD 0x2f46

Bit(s)	R/W	Default	Description
31-25	R/W		reserved
24-16	R/W	64	reg_mcdi_pd22chksaddifthd. sad dif thd (>=) for (pd22chksad - qsad) for pd22 check, default = 64
15-14	R/W		reserved
13-8	R/W	2	reg_mcdi_pd22chkqmvthd. thd (>=) of abs qmv for pd22 check, default = 2
7-0	R/W	4	reg_mcdi_pd22chkfltthd. thd (>=) of flat for pd22 check, default = 4

Table 8-1260 MCDI_PD22_CHK_GAIN_OFFST_0 0x2f47

Bit(s)	R/W	Default	Description
31-24	R/W	0	reg_mcdi_pd22chkedgeoffst0. offset0 of pd22chkedge from right film22 phase, -128~127, default = 0
23-21	R/W		reserved
20-16	R/W	16	reg_mcdi_pd22chkedgegain0. gain0 of pd22chkedge from right film22 phase, normalized to 16 as '1', default = 16
15-12	R/W		reserved

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Bit(s)	R/W	Default	Description
11-8	R/W	0	reg_mcdi_pd22chkbadwoffst0. offset0 of pd22chkbadw from right film22 phase, -8~7, default = 0
7-5	R/W		reserved
4-0	R/W	8	reg_mcdi_pd22chkbadwgain0. gain0 of pd22chkbadw from right film22 phase, normalized to 16 as '1', default = 8

Table 8-1261 MCDI_PD22_CHK_GAIN_OFFST_1 0x2f48

Bit(s)	R/W	Default	Description
31-24	R/W	0	reg_mcdi_pd22chkedgeoffst1. offset1 of pd22chkedge from right film22 phase, -128~127, default = 0
23-21	R/W		reserved
20-16	R/W	16	reg_mcdi_pd22chkedgegain1. gain1 of pd22chkedge from right film22 phase, normalized to 16 as '1', default = 16
15-12	R/W		reserved
11-8	R/W	0	reg_mcdi_pd22chkbadwoffst1. offset1 of pd22chkbadw from right film22 phase, -8~7, default = 0
7-5	R/W		reserved
4-0	R/W	12	reg_mcdi_pd22chkbadwgain1. gain1 of pd22chkbadw from right film22 phase, normalized to 16 as '1', default = 12

Table 8-1262 MCDI_LMV_LOCK_CNT_THD_GAIN 0x2f49

Bit(s)	R/W	Default	Description
31-20	R/W		reserved
19-16	R/W	6	reg_mcdi_lmvlockcntmax. max lmv lock count number, default = 6
15-12	R/W	0	reg_mcdi_lmvlockcntoffst. offset for lmv lock count, -8~7, default = 0
11-8	R/W	8	reg_mcdi_lmvlockcntgain. gain for lmv lock count, normalized 8 as '1', 15 is set to 16, default = 8
7-5	R/W		reserved
4-0	R/W	4	reg_mcdi_lmvlockcntthd. lmv count thd (>=) before be locked, 1~31, default = 4

Table 8-1263 MCDI_LMV_LOCK_ABS_DIF_THD 0x2f4a

Bit(s)	R/W	Default	Description
31-27	R/W		reserved
26-24	R/W	1	reg_mcdi_lmvlockdifthd2. lmv dif thd for third part, before locked, default = 1
23	R/W		reserved
22-20	R/W	1	reg_mcdi_lmvlockdifthd1. lmv dif thd for second part, before locked, default = 1
19	R/W		reserved
18-16	R/W	1	reg_mcdi_lmvlockdifthd0. lmv dif thd for first part, before locked, default = 1

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Bit(s)	R/W	Default	Description
15-13	R/W		reserved
12-8	R/W	24	reg_mcdi_lmvlockabsmax. max abs (<) of lmv to be locked, default = 24
7-5	R/W		reserved
4-0	R/W	1	reg_mcdi_lmvlockabsmin. min abs (>=) of lmv to be locked, default = 1

Table 8-1264 MCDI_LMV_LOCK_ROW 0x2f4b

Bit(s)	R/W	Default	Description
31-27	R/W		reserved
26-16	R/W	2047	reg_mcdi_lmvlockendrow. end row (<) for lmv lock, default = 2047
15-11	R/W		reserved
10-0	R/W	0	reg_mcdi_lmvlockstartrow. start row (>=) for lmv lock, default = 0

Table 8-1265 MCDI_LMV_LOCK_RT_MODE 0x2f4c

Bit(s)	R/W	Default	Description
31-27	R/W		reserved
26-24	R/W	2	reg_mcdi_lmvlockextmode. extend lines for lmv lock check, check how many lines for lmv locking, default = 2
23-16	R/W	32	reg_mcdi_lmvlockfltcntrt. ratio of flt cnt for lock check, normalized 256 as '1', 255 is set to 256, default = 32
15-8	R/W	48	reg_mcdi_lmvlocklmvcntrt1. ratio when use non-zero lmv for lock check, normalized 256 as '1', 255 is set to 256, default = 48
7-0	R/W	106	reg_mcdi_lmvlocklmvcntrt0. ratio when use max lmv for lock check, normalized 256 as '1', 255 is set to 256, default = 106

Table 8-1266 MCDI_GMV_LOCK_CNT_THD_GAIN 0x2f4d

Bit(s)	R/W	Default	Description
31-20	R/W		reserved
19-16	R/W	6	reg_mcdi_gmvlockcntmax. max gmv lock count number, default = 6
15-12	R/W	0	reg_mcdi_gmvlockcntoffst. offset for gmv lock count, -8~7, default = 0
11-8	R/W	8	reg_mcdi_gmvlockcntgain. gain for gmv lock count, normalized 8 as '1', 15 is set to 16, default = 8
7-5	R/W		reserved
4-0	R/W	4	reg_mcdi_gmvlockcntthd. gmv count thd (>=) before be locked, 1~31, default = 4

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Table 8-1267 MCDI_GMV_LOCK_ABS_DIF_THD 0x2f4e

Bit(s)	R/W	Default	Description
31-27	R/W		reserved
26-24	R/W	3	reg_mcdi_gmvlockdifthd2. gmv dif thd for third part, before locked, default = 3
23	R/W		reserved
22-20	R/W	2	reg_mcdi_gmvlockdifthd1. gmv dif thd for second part, before locked, default = 2
19	R/W		reserved
18-16	R/W	1	reg_mcdi_gmvlockdifthd0. gmv dif thd for first part, before locked, default = 1
15-13	R/W		reserved
12-8	R/W	15	reg_mcdi_gmvlockabsmax. max abs of gmv to be locked, default = 15
7-5	R/W		reserved
4-0	R/W	1	reg_mcdi_gmvlockabsmin. min abs of gmv to be locked, default = 1

Table 8-1268 MCDI_HIGH_VERT_FRQ_DIF_THD 0x2f4f

Bit(s)	R/W	Default	Description
31-0	R/W	103680	reg_mcdi_highvertfrqfldavgdifthd. high_vert_frq field average luma dif thd (>=), 3*Blk_Width*Blk_Height, set by software, default = 103680

Table 8-1269 MCDI_HIGH_VERT_FRQ_DIF_DIF_THD 0x2f50

Bit(s)	R/W	Default	Description
31-0	R/W	103680	reg_mcdi_highvertfrqfldavgdifdifthd. high_vert_frq field average luma dif's dif thd (<), 3*Blk_Width*Blk_Height, set by software, default = 103680

Table 8-1270 MCDI_HIGH_VERT_FRQ_RT_GAIN 0x2f51

Bit(s)	R/W	Default	Description
31-20	R/W		reserved
19-16	R/W	4	reg_mcdi_highvertfrqcntthd. high_vert_frq count thd (>=) before locked, 1~31, default = 4
15-8	R/W	24	reg_mcdi_highvertfrqbadsadrt. ratio for high_vert_frq bad sad count, normalized 256 as '1', 255 is set to 256, default = 24
7-0	R/W	130	reg_mcdi_highvertfrqbadbadwrt. ratio for high_vert_frq badw count, normalized 256 as '1', 255 is set to 256, default = 130

Table 8-1271 MCDI_MOTION_PARADOX_THD 0x2f52

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-24	R/W	4	reg_mcdi_motionparadoxcntthd. motion paradox count thd (>=) before locked, 1~31, default = 4
23-22	R/W		reserved

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Bit(s)	R/W	Default	Description
21-16	R/W	32	reg_mcdi_motionparadoxgmvthd. abs gmv thd (<) of motion paradox, 0~32, note that 32 means invalid gmv, be careful, default = 32
15-0	R/W		reserved

Table 8-1272 MCDI_MOTION_PARADOX_RT0x2f53

Bit(s)	R/W	Default	Description
31-24	R/W		reserved
23-16	R/W	24	reg_mcdi_motionparadoxbadsadrt. ratio for field bad sad count of motion paradox, normalized 256 as '1', 255 is set to 256, default = 24
15-8	R/W	120	reg_mcdi_motionparadoxbadrelrt. ratio for field bad reliabilty count of motion paradox, normalized 256 as '1', 255 is set to 256, default = 120
7-0	R/W	218	reg_mcdi_motionparadoxmtnrt. ratio for field motion count of motion paradox, normalized 256 as '1', 255 is set to 256, default = 218

Table 8-1273 MCDI_MOTION_REF_THD 0x2f54

Bit(s)	R/W	Default	Description
31-24	R/W		reserved
23-20	R/W	15	reg_mcdi_motionrefoffst. motion ref additive offset, default = 15
19-16	R/W	8	reg_mcdi_motionrefgain. motion ref gain, normalized 8 as '1', default = 8
15-13	R/W		reserved
12-8	R/W	1	reg_mcdi_motionrefrptmvthd. abs thd (>=) of rpt mv (0~31, 32 means invalid) for motion ref, default = 1
7-2	R/W	2	reg_mcdi_motionrefqmvthd. min thd (>=) of abs qmv for motion ref, note that quarter mv's range is -63~63, default = 2
1-0	R/W	1	reg_mcdi_motionreflpfmode. Mv and (8 x repeat flg) 's lpf mode of motion refinement, 0: no lpf, 1: [1 2 1], 2: [1 2 2 2 1], default = 1

Table 8-1274 MCDI_REL_COL_REF_RT 0x2f55

Bit(s)	R/W	Default	Description
31-8	R/W		reserved
7-0	R/W	135	reg_mcdi_relcolrefrt. ratio for column cofidence level against column number, for refinement, default = 135

Table 8-1275 MCDI_PD22_CHK_THD_RT 0x2f56

Bit(s)	R/W	Default	Description
31-27	R/W		reserved
26-16	R/W	1	reg_mcdi_pd22chkfltcntrt. ratio for flat count of field pulldown 22 check, normalized 2048 as '1', 2047 is set to 2048, default = 1
15-8	R/W	100	reg_mcdi_pd22chkcntrt. ratio of pulldown 22 check count, normalized 256 as '1', 255 is set to 256, default = 100

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Bit(s)	R/W	Default	Description
7-5	R/W		reserved
4-0	R/W	4	reg_mcdi_pd22chkcntthd. thd (>=) for pd22 count before locked, 1~31, default = 4

Table 8-1276 MCDI_CHAR_DET_DIF_THD 0x2f57

Bit(s)	R/W	Default	Description
31-24	R/W		reserved
23-16	R/W	64	reg_mcdi_chardetminmaxdifthd. thd (>=) for dif between min and max value, default = 64
15-8	R/W	17	reg_mcdi_chardetmaxdifthd. thd (<) for dif between max value, default = 17
7-0	R/W	17	reg_mcdi_chardetmindifthd. thd (<) for dif between min value, default = 17

Table 8-1277 MCDI_CHAR_DET_CNT_THD 0x2f58

Bit(s)	R/W	Default	Description
31-21	R/W		reserved
20-16	R/W	18	reg_mcdi_chardettotcntthd. thd (>=) for total count, 0~21, default = 18
15-13	R/W		reserved
12-8	R/W	1	reg_mcdi_chardetmaxcntthd. thd (>=) for max count, 0~21, default = 1
7-5	R/W		reserved
4-0	R/W	1	reg_mcdi_chardetmincntthd. thd (>=) for min count, 0~21, default = 1

Table 8-1278 MCDI_PD_22_CHK_WND0_X 0x2f59

Bit(s)	R/W	Default	Description
28-16	R/W	719	reg_mcdi_pd22chkwnd0_x1
12-0	R/W	0	reg_mcdi_pd22chkwnd0_x0

Table 8-1279 MCDI_PD_22_CHK_WND0_Y 0x2f5a

Bit(s)	R/W	Default	Description
28-16	R/W	39	reg_mcdi_pd22chkwnd0_y1
12-0	R/W	0	reg_mcdi_pd22chkwnd0_y0

Table 8-1280 MCDI_PD_22_CHK_WND1_X 0x2f5b

Bit(s)	R/W	Default	Description
28-16	R/W	719	reg_mcdi_pd22chkwnd1_x1
12-0	R/W	0	reg_mcdi_pd22chkwnd1_x0

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Table 8-1281 MCDI_PD_22_CHK_WND1_Y 0x2f5c

Bit(s)	R/W	Default	Description
28-16	R/W	199	reg_mcdi_pd22chkwnd1_y1
12-0	R/W	40	reg_mcdi_pd22chkwnd1_y0

Table 8-1282 MCDI_PD_22_CHK_FRC_LMV 0x2f5d

Bit(s)	R/W	Default	Description
10	R/W	1	reg_mcdi_pd22chklmvchk2
9	R/W	0	reg_mcdi_pd22chklmvchk1
8	R/W	0	reg_mcdi_pd22chklmvchk0
6	R/W	0	reg_mcdi_pd22chkfrcpd2
5	R/W	0	reg_mcdi_pd22chkfrcpd1
4	R/W	0	reg_mcdi_pd22chkfrcpd0
2	R/W	1	reg_mcdi_pd22chkfrcvof2
1	R/W	0	reg_mcdi_pd22chkfrcvof1
0	R/W	0	reg_mcdi_pd22chkfrcvof0

Table 8-1283 MCDI_PD_22_CHK_FRC_LMV 0x2f5e

Bit(s)	R/W	Default	Description
26	R/W	0	reg_mcdi_pd22chkflg2
25	R/W	0	reg_mcdi_pd22chkflg1
24	R/W	0	reg_mcdi_pd22chkflg
23-16	R/W	1	reg_mcdi_pd22chkcnt2
15-8	R/W	0	reg_mcdi_pd22chkcnt1
7-0	R/W	0	reg_mcdi_pd22chkcnt

Table 8-1284 MCDI_FIELD_MV 0x2f60

Bit(s)	R/W	Default	Description
31-24	R/W		reg_mcdi_pd22chkcnt
23-16	R/W		reg_mcdi_fieldgmvcnt
15	R/W		reg_mcdi_pd22chkflg
14	R/W		reg_mcdi_fieldgmvlock
13-8	R/W		reg_mcdi_fieldrptmv. last field rpt mv
7-6	R/W		reserved
5-0	R/W		reg_mcdi_fieldgmv. last field gmv

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Table 8-1285 MCDI_FIELD_HVF_PRDX_CNT 0x2f61

Bit(s)	R/W	Default	Description
31-24	R/W		reg_mcdi_motionparadoxcnt.
23-17	R/W		reserved
16	R/W		reg_mcdi_motionparadoxflg.
15-8	R/W		reg_mcdi_highvertfrqcnt.
7-4	R/W		reserved
3-2	R/W		reg_mcdi_highvertfrqphase.
1	R/W		reserved
0	R/W		reg_mcdi_highvertfrqflg.

Table 8-1286 MCDI_FIELD_LUMA_AVG_SUM_0 0x2f62

Bit(s)	R/W	Default	Description
31-0	R/W		reg_mcdi_fld_luma_avg_sum0.

Table 8-1287 MCDI_FIELD_LUMA_AVG_SUM_1 0x2f63

Bit(s)	R/W	Default	Description
31-0	R/W		reg_mcdi_fld_luma_avg_sum1.

Table 8-1288 MCDI_YCBCR_BLEND_CRTL 0x2f64

Bit(s)	R/W	Default	Description
31-16	R/W		reserved
15-8	R/W	0	reg_mcdi_ycbcrblendgain. ycbcr blending gain for cbcr in ycbcr. default = 0
7-2	R/W		reserved.
1-0	R/W	2	reg_mcdi_ycbcrblendmode. 0:y+cmb(cb,cr), 1:med(r,g,b), 2:max(r,g,b), default = 2

Table 8-1289 MCDI_MC_CRTL 0x2f70

Bit(s)	R/W	Default	Description
31-20	R/W	0	reserved
19	R/W	0	reg buf1 enable (if1)
18	R/W	0	reg buf2 enable (if2)
17	R/W	0	reg mv invert
16	R/W	0	mcvec force 0
15	R/W	0	buf2 always en
14-12	R/W	0	reg_mcdi_mcvec_offset: 0: disable 1: 1 pixel offset of mcvec 2: 2 pixel offset of mcvec

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Bit(s)	R/W	Default	Description
			3: 3 pixel offset of mcvec 4: 4 pixel offset of mcvec
11	R/w	0	reg_di_weave_both_side
10	R/W	0	reg_mcdi_mc_uv_en: mc for uv if needed, else use ma of uv
9-8	R/W	1	reg_mcdi_mcpreflg. flag to use previous field for MC, 0-forward field, 1: previous field,2-use forward & previous. default = 1
7	R/W	1	reg_mcdi_mcrelrefbycolcfden. enable rel refinement by column cofidence in mc blending, default = 1
6-5	R/W	0	reg_mcdi_mclpfen. enable mc pixles/rel lpf, 0:disable, 1: lpf rel, 2: lpf mc pxls, 3: lpf both rel and mc pxls, default = 0
4-2	R/W	0	reg_mcdi_mcdebugmode. enable mc debug mode, 0:disable, 1: split left/right, 2: split top/bottom, 3: debug mv, 4: debug rel, default = 0
1-0	R/W	1	reg_mcdi_mcen. mcdi enable mode, 0:disable, 1: blend with ma, 2: full mc, default = 1

Table 8-1290 MCDI_MC_LPF_MSK_0 0x2f71

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28	R		blend output ready
27	R		mcvec read input ready
26	R		mtn read input ready, same as bit24
25	R		ei dout ready (if0)
24	R		mtn read input ready
23	R		if2 input ready
22	R		if1 input ready
21	R		blend input ready
20-16	R/W	0	reg_mcdi_mclpfmsk02. mc lpf coef. 2 for pixel 0 of current block, normalized 16 as '1', default = 0
15-13	R/W		reserved
12-8	R/W	9	reg_mcdi_mclpfmsk01. mc lpf coef. 1 for pixel 0 of current block, normalized 16 as '1', default = 9
7-5	R/W		reserved
4-0	R/W	7	reg_mcdi_mclpfmsk00. mc lpf coef. 0 for pixel 0 of current block, normalized 16 as '1', default = 7

Table 8-1291 MCDI_MC_LPF_MSK_1 0x2f72

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-21	R		debug info

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Bit(s)	R/W	Default	Description
20-16	R/W	0	reg_mcdi_mclpfmsk12. mc lpf coef. 2 for pixel 1 of current block, 0~16, normalized 16 as '1', default = 0
15-13	R/W		reserved
12-8	R/W	11	reg_mcdi_mclpfmsk11. mc lpf coef. 1 for pixel 1 of current block, 0~16, normalized 16 as '1', default = 11
7-5	R/W		reserved
4-0	R/W	5	reg_mcdi_mclpfmsk10. mc lpf coef. 0 for pixel 1 of current block, 0~16, normalized 16 as '1', default = 5

Table 8-1292 MCDI_MC_LPF_MSK_2 0x2f73

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-21	R		debug info
20-16	R/W	1	reg_mcdi_mclpfmsk22. mc lpf coef. 2 for pixel 2 of current block, 0~16, normalized 16 as '1', default = 1
15-13	R/W		reserved
12-8	R/W	14	reg_mcdi_mclpfmsk21. mc lpf coef. 1 for pixel 2 of current block, 0~16, normalized 16 as '1', default = 14
7-5	R/W		reserved
4-0	R/W	1	reg_mcdi_mclpfmsk20. mc lpf coef. 0 for pixel 2 of current block, 0~16, normalized 16 as '1', default = 1

Table 8-1293 MCDI_MC_LPF_MSK_3 0x2f74

Bit(s)	R/W	Default	Description
31-29	R/W		reserved
28-21	R		debug info
20-16	R/W	5	reg_mcdi_mclpfmsk32. mc lpf coef. 2 for pixel 3 of current block, 0~16, normalized 16 as '1', default = 5
15-13	R/W		reserved
12-8	R/W	11	reg_mcdi_mclpfmsk31. mc lpf coef. 1 for pixel 3 of current block, 0~16, normalized 16 as '1', default = 11
7-5	R/W		reserved
4-0	R/W	0	reg_mcdi_mclpfmsk30. mc lpf coef. 0 for pixel 3 of current block, 0~16, normalized 16 as '1', default = 0

Table 8-1294 MCDI_MC_LPF_MSK_4 0x2f75

Bit(s)	R/W	Default	Description
31-21	R/W		reserved
20-16	R/W	7	reg_mcdi_mclpfmsk42. mc lpf coef. 2 for pixel 4 of current block, 0~16, normalized 16 as '1', default = 7

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Bit(s)	R/W	Default	Description
15-13	R/W		reserved
12-8	R/W	9	reg_mcdi_mclpfmsk41. mc lpf coef. 1 for pixel 4 of current block, 0~16, normalized 16 as '1', default = 9
7-5	R/W		reserved
4-0	R/W	0	reg_mcdi_mclpfmsk40. mc lpf coef. 0 for pixel 4 of current block, 0~16, normalized 16 as '1', default = 0

Table 8-1295 MCDI_MC_REL_GAIN_OFFST_0 0x2f76

Bit(s)	R/W	Default	Description
31-26	R/W		reserved
25	R/W	0	reg_mcdi_mcmotionparadoxflg. flag of motion paradox, initial with 0 and read from software, default = 0
24	R/W	0	reg_mcdi_mchighvertfrqflg. flag of high vert frq, initial with 0 and read from software, default = 0
23-16	R/W	128	reg_mcdi_mcmotionparadoxoffst. offset (r+ offset) for rel (MC blending coef.) refinement if motion paradox detected before MC blending before MC blending, default = 128
15-12	R/W		reserved
11-8	R/W	8	reg_mcdi_mcmotionparadoxgain. gain for rel (MC blending coef.) refinement if motion paradox detected before MC blending, normalized 8 as '1', set 15 to 16, default = 8
7-4	R/W	15	reg_mcdi_mchighvertfrqoffst. minus offset (alpha - offset) for motion (MA blending coef.) refinement if high vertical frequency detected before MA blending, default = 15
3-0	R/W	8	reg_mcdi_mchighvertfrqgain. gain for motion (MA blending coef.) refinement if high vertical frequency detected before MA blending, normalized 8 as '1', set 15 to 16, default = 8

Table 8-1296 MCDI_MC_REL_GAIN_OFFST_1 0x2f77

Bit(s)	R/W	Default	Description
31-24	R/W	255	reg_mcdi_mcoutofboundrayoffst. offset (rel + offset) for rel (MC blending coef.) refinement if MC pointed out of boundray before MC blending before MC blending, default = 255
23-20	R/W		reserved
19-16	R/W	8	reg_mcdi_mcoutofboundraygain. gain for rel (MC blending coef.) refinement if MC pointed out of boundray before MC blending, normalized 8 as '1', set 15 to 16, default = 8
15-8	R/W	255	reg_mcdi_mcrelrefbycolcfdoffst. offset (rel + offset) for rel (MC blending coef.) refinement if motion paradox detected before MC blending before MC blending, default = 255
7-4	R/W		reserved.
3-0	R/W	8	reg_mcdi_mcrelrefbycolcfdgain. gain for rel (MC blending coef.) refinement if column cofidence failed before MC blending, normalized 8 as '1', set 15 to 16, default = 8

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Table 8-1297 MCDI_MC_COL_CFD_0 0x2f78

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_0. column cofidence value 0 read from software. initial = 0

Table 8-1298 MCDI_MC_COL_CFD_1 0x2f79

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_1. column cofidence value 1 read from software. initial = 0

Table 8-1299 MCDI_MC_COL_CFD_2 0x2f7a

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_2. column cofidence value 2 read from software. initial = 0

Table 8-1300 MCDI_MC_COL_CFD_3 0x2f7b

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_3. column cofidence value 3 read from software. initial = 0

Table 8-1301 MCDI_MC_COL_CFD_4 0x2f7c

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 4 read from software. initial = 0

Table 8-1302 MCDI_MC_COL_CFD_5 0x2f7d

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 5 read from software. initial = 0

Table 8-1303 MCDI_MC_COL_CFD_6 0x2f7e

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 6 read from software. initial = 0

Table 8-1304 MCDI_MC_COL_CFD_7 0x2f7f

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 7 read from software. initial = 0

Table 8-1305 MCDI_MC_COL_CFD_8 0x2f80

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 8 read from software. initial = 0

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Table 8-1306 MCDI_MC_COL_CFD_9 0x2f81

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 9 read from software. initial = 0

Table 8-1307 MCDI_MC_COL_CFD_10 0x2f82

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 10 read from software. initial = 0

Table 8-1308 MCDI_MC_COL_CFD_11 0x2f83

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 11 read from software. initial = 0

Table 8-1309 MCDI_MC_COL_CFD_12 0x2f84

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 12 read from software. initial = 0

Table 8-1310 MCDI_MC_COL_CFD_13 0x2f85

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 13 read from software. initial = 0

Table 8-1311 MCDI_MC_COL_CFD_14 0x2f86

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 14 read from software. initial = 0

Table 8-1312 MCDI_MC_COL_CFD_15 0x2f87

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 15 read from software. initial = 0

Table 8-1313 MCDI_MC_COL_CFD_16 0x2f88

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 16 read from software. initial = 0

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Table 8-1314 MCDI_MC_COL_CFD_17 0x2f89

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 17 read from software. initial = 0

Table 8-1315 MCDI_MC_COL_CFD_18 0x2f8a

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 18 read from software. initial = 0

Table 8-1316 MCDI_MC_COL_CFD_19 0x2f8b

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 19 read from software. initial = 0

Table 8-1317 MCDI_MC_COL_CFD_20 0x2f8c

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 20 read from software. initial = 0

Table 8-1318 MCDI_MC_COL_CFD_21 0x2f8d

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 21 read from software. initial = 0

Table 8-1319 MCDI_MC_COL_CFD_22 0x2f8e

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 22 read from software. initial = 0

Table 8-1320 MCDI_MC_COL_CFD_23 0x2f8f

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 23 read from software. initial = 0

Table 8-1321 MCDI_MC_COL_CFD_24 0x2f90

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 24 read from software. initial = 0

Table 8-1322 MCDI_MC_COL_CFD_25 0x2f91

Bit(s)	R/W	Default	Description
31-0	R/W		mcdi_mc_col_cfd_4. column cofidence value 25 read from software. initial = 0

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Table 8-1323 MCDI_RO_FLD_LUMA_AVG_SUM 0x2fa0

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldlumaavgsum. block's luma avg sum of current filed (block based). initial = 0

Table 8-1324 MCDI_RO_GMV_VLD_CNT 0x2fa1

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_gmvvldcnt. valid gmv's count of pre one filed (block based). initial = 0

Table 8-1325 MCDI_RO_RPT_FLG_CNT 0x2fa2

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_rptflgcnt. repeat mv's count of pre one filed (block based). initial = 0

Table 8-1326 MCDI_RO_FLD_BAD_SAD_CNT 0x2fa3

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldbadsadcnt. bad sad count of whole pre one field (block based). initial = 0

Table 8-1327 MCDI_RO_FLD_BAD_BADW_CNT 0x2fa4

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldbadbadwcnt. bad badw count of whole pre one field (block based). initial = 0

Table 8-1328 MCDI_RO_FLD_BAD_REL_CNT 0x2fa5

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldbadrelcnt. bad rel count of whole pre one field (block based). initial = 0

Table 8-1329 MCDI_RO_FLD_MTN_CNT 0x2fa6

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldmtncnt. motion count of whole pre one field (pixel based). initial = 0

Table 8-1330 MCDI_RO_FLD_VLD_CNT 0x2fa7

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldvldcnt. valid motion count of whole pre one field (pixel based). initial = 0

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Table 8-1331 MCDI_RO_FLD_PD_22_PRE_CNT 0x2fa8

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldpd22precnt. prevoius pd22 check count of whole pre one field (block based). initial = 0

Table 8-1332 MCDI_RO_FLD_PD_22_FOR_CNT 0x2fa9

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldpd22forcnt. forward pd22 check count of whole pre one field (block based). initial = 0

Table 8-1333 MCDI_RO_FLD_PD_22_FLT_CNT 0x2faa

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_fldpd22fltcnt. flat count (for pd22 check) of whole pre one field (block based). initial = 0

Table 8-1334 MCDI_RO_HIGH_VERT_FRQ_FLG 0x2fab

Bit(s)	R/W	Default	Description
31-16	R		reserved.
15-8	R		ro_mcdi_highvertfrqcnt. high vertical frequency count till prevoius one field. initial = 0
7-3	R		reserved.
2-1	R		ro_mcdi_highvertfrqphase. high vertical frequency phase of prevoius one field. initial = 2
0	R		ro_mcdi_highvertfrqflg. high vertical frequency flag of prevoius one field. initial = 0

Table 8-1335 MCDI_RO_GMV_LOCK_FLG 0x2fac

Bit(s)	R/W	Default	Description
31-16	R		reserved.
15-8	R		ro_mcdi_gmvlckcnt. global mv lock count till prevoius one field. initial = 0
7-2	R		ro_mcdi_gmv. global mv of prevoius one field. -31~31, initial = 32 (invalid value)
1	R		ro_mcdi_zerogmvlckflg. zero global mv lock flag of prevoius one field. initial = 0
0	R		ro_mcdi_gmvlckflg. global mv lock flag of prevoius one field. initial = 0

Table 8-1336 MCDI_RO_RPT_MV 0x2fad

Bit(s)	R/W	Default	Description
5-0	R		ro_mcdi_rptmv. repeate mv of prevoius one field. -31~31, initial = 32 (invalid value)

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Table 8-1337 MCDI_RO_MOTION_PARADOX_FLG 0x2fae

Bit(s)	R/W	Default	Description
31-16	R		reserved.
15-8	R		ro_mcdi_motionparadoxcnt. motion paradox count till prevoius one field. initial = 0
7-1	R		reserved.
0	R		ro_mcdi_motionparadoxflg. motion paradox flag of prevoius one field. initial = 0

Table 8-1338 MCDI_RO_PD_22_FLG 0x2faf

Bit(s)	R/W	Default	Description
31-16	R		reserved.
26	R	0	ro_mcdi_pd22flg2. pull down 22 flag of prevoius one field. initial = 0
25	R	0	ro_mcdi_pd22flg1. pull down 22 flag of prevoius one field. initial = 0
24	R	0	ro_mcdi_pd22flg0. pull down 22 flag of prevoius one field. initial = 0
23-16	R	0	ro_mcdi_pd22cnt2. pull down 22 count till prevoius one field. initial = 0
15-8	R	0	ro_mcdi_pd22cnt1. pull down 22 count till prevoius one field. initial = 0
7-0	R	0	ro_mcdi_pd22cnt0. pull down 22 count till prevoius one field. initial = 0

Table 8-1339 MCDI_RO_COL_CFD_0 0x2fb0

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_0. column cofidence value 0. initial = 0

Table 8-1340 MCDI_RO_COL_CFD_1 0x2fb1

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_1. column cofidence value 1. initial = 0

Table 8-1341 MCDI_RO_COL_CFD_2 0x2fb2

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_2. column cofidence value 2. initial = 0

Table 8-1342 MCDI_RO_COL_CFD_3 0x2fb3

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_3. column cofidence value 3. initial = 0

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Table 8-1343 MCDI_RO_COL_CFD_4 0x2fb4

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_4. column cofidence value 4. initial = 0

Table 8-1344 MCDI_RO_COL_CFD_5 0x2fb5

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_5. column cofidence value 5. initial = 0

Table 8-1345 MCDI_RO_COL_CFD_6 0x2fb6

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_6. column cofidence value 6. initial = 0

Table 8-1346 MCDI_RO_COL_CFD_7 0x2fb7

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_7. column cofidence value 7. initial = 0

Table 8-1347 MCDI_RO_COL_CFD_8 0x2fb8

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_8. column cofidence value 8. initial = 0

Table 8-1348 MCDI_RO_COL_CFD_9 0x2fb9

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_9. column cofidence value 9. initial = 0

Table 8-1349 MCDI_RO_COL_CFD_10 0x2fba

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_10. column cofidence value 10. initial = 0

Table 8-1350 MCDI_RO_COL_CFD_11 0x2fbb

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_11. column cofidence value 11. initial = 0

Table 8-1351 MCDI_RO_COL_CFD_12 0x2fbc

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_12. column cofidence value 12. initial = 0

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Table 8-1352 MCDI_RO_COL_CFD_13 0x2fbd

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_13. column cofidence value 13. initial = 0

Table 8-1353 MCDI_RO_COL_CFD_14 0x2fbe

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_14. column cofidence value 14. initial = 0

Table 8-1354 MCDI_RO_COL_CFD_15 0x2fbf

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_15. column cofidence value 15. initial = 0

Table 8-1355 MCDI_RO_COL_CFD_16 0x2fc0

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_16. column cofidence value 16. initial = 0

Table 8-1356 MCDI_RO_COL_CFD_17 0x2fc1

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_17. column cofidence value 17. initial = 0

Table 8-1357 MCDI_RO_COL_CFD_18 0x2fc2

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_18. column cofidence value 18. initial = 0

Table 8-1358 MCDI_RO_COL_CFD_19 0x2fc3

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_19. column cofidence value 19. initial = 0

Table 8-1359 MCDI_RO_COL_CFD_20 0x2fc4

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_20. column cofidence value 20. initial = 0

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Table 8-1360 MCDI_RO_COL_CFD_21 0x2fc5

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_21. column cofidence value 21. initial = 0

Table 8-1361 MCDI_RO_COL_CFD_22 0x2fc6

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_22. column cofidence value 22. initial = 0

Table 8-1362 MCDI_RO_COL_CFD_23 0x2fc7

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_23. column cofidence value 23. initial = 0

Table 8-1363 MCDI_RO_COL_CFD_24 0x2fc8

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_24. column cofidence value 24. initial = 0

Table 8-1364 MCDI_RO_COL_CFD_25 0x2fc9

Bit(s)	R/W	Default	Description
31-0	R		ro_mcdi_col_cfd_25. column cofidence value 25. initial = 0

Table 8-1365 MCDI_RO_FLD_PD_22_PRE_CNT1 0x2fca

Bit(s)	R/W	Default	Description
31-0	R		previous pd22 check count of whole pre one field(block based). initial = 0

Table 8-1366 MCDI_RO_FLD_PD_22_POR_CNT1 0x2fcb

Bit(s)	R/W	Default	Description
31-0	R		forward pd22 check count of whole pre one field(block based). initial = 0

Table 8-1367 MCDI_RO_FLD_PD_22_FLT_CNT1 0x2fcc

Bit(s)	R/W	Default	Description
31-0	R		flat count(for pd22 check) of whole pre one field(block based). initial = 0

Table 8-1368 MCDI_RO_FLD_PD_22_PRE_CNT2 0x2fcd

Bit(s)	R/W	Default	Description
31-0	R		previous pd22 check count of whole pre one field(block based). initial = 0

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Table 8-1369 MCDI_RO_FLD_PD_22_POR_CNT2 0x2fce

Bit(s)	R/W	Default	Description
31-0	R		forward pd22 check count of whole pre one field(block based). initial = 0

Table 8-1370 MCDI_RO_FLD_PD_22_FLT_CNT2 0x2fcf

Bit(s)	R/W	Default	Description
31-0	R		flat count(for pd22 check) of whole pre one field(block based). initial = 0

8.2.3.13 PULLDOWN

Table 8-1371 DIPD_COMB_CTRL0 0x2fd0

Bit(s)	R/W	Default	Description
31-24	W		Cmb_v_dif_min
23-16	W		Cmb_v_dif_max
15-8	W		Cmb_crg_min
7-0	W		Cmb_crg_max

Table 8-1372 DIPD_COMB_CTRL1 0x2fd1

Bit(s)	R/W	Default	Description
31	W		Pd_check_en
29-24	W		Cmb_wv_min3
21-16	W		Cmb_wv_min2
13-8	W		Cmb_wv_min1
5-0	W		Cmb_wv_min0

Table 8-1373 DIPD_COMB_CTRL2 0x2fd2

Bit(s)	R/W	Default	Description
31-28	W		Cmb_wnd_cnt1
25-20	W		Ccnt_cmmin1
19-16	W		Ccnt_mtmin
13-8	W		Ccnt_cmmin
5-0	W		Cmb_wv_min4

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Table 8-1374 DIPD_COMB_CTRL3 0x2fd3

Bit(s)	R/W	Default	Description
31	W		Cmb32spcl
17-12	W		Cmb_wnd_mthd
11-4	W		Cmb_abs_nocmb
3-0	W		Cnt_minlen

Table 8-1375 DIPD_COMB_CTRL4 0x2fd4

Bit(s)	R/W	Default	Description
30	W		Flm_stamtn_en
29-28	W		In_horflt
27-20	W		Alpha
19-16	W		Rhtran_ctmtd
15-8	W		Htran_mnth1
7-0	W		Htran_mnth0

Table 8-1376 DIPD_COMB_CTRL5 0x2fd5

Bit(s)	R/W	Default	Description
31-24	W		Fld_mindif
23-16	W		Frm_mindif
13-8	W		Flm_smp_mtn_cnt
7-0	W		Flm_smp_mtn_thd

Table 8-1377 DIPD_RO_COMB_0 0x2fd6

Bit(s)	R/W	Default	Description
31-0	R		frmdif

Table 8-1378 DIPD_RO_COMB_1 0x2fd7

Bit(s)	R/W	Default	Description
31-0	R		Frmdif0

Table 8-1379 DIPD_RO_COMB_2 0x2fd8

Bit(s)	R/W	Default	Description
31-0	R		Frmdif1

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Table 8-1380 DIPD_RO_COMB_3 0x2fd9

Bit(s)	R/W	Default	Description
31-0	R		Frmdif2

Table 8-1381 DIPD_RO_COMB_4 0x2fda

Bit(s)	R/W	Default	Description
31-0	R		Frmdif3

Table 8-1382 DIPD_RO_COMB_5 0x2fdb

Bit(s)	R/W	Default	Description
31-0	R		Frmdif4

Table 8-1383 DIPD_RO_COMB_6 0x2fdc

Bit(s)	R/W	Default	Description
31-0	R		flddif

Table 8-1384 DIPD_RO_COMB_7 0x2fdd

Bit(s)	R/W	Default	Description
31-0	R		Flddif0

Table 8-1385 DIPD_RO_COMB_8 0x2fde

Bit(s)	R/W	Default	Description
31-0	R		Flddif1

Table 8-1386 DIPD_RO_COMB_9 0x2fdf

Bit(s)	R/W	Default	Description
31-0	R		Flddif2

Table 8-1387 DIPD_RO_COMB_10 0x2fe0

Bit(s)	R/W	Default	Description
31-0	R		Flddif3

Table 8-1388 DIPD_RO_COMB_11 0x2fe1

Bit(s)	R/W	Default	Description
31-0	R		Flddif4

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Table 8-1389 DIPD_RO_COMB_12 0x2fe2

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt0

Table 8-1390 DIPD_RO_COMB_13 0x2fe3

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt1

Table 8-1391 DIPD_RO_COMB_14 0x2fe4

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt2

Table 8-1392 DIPD_RO_COMB_15 0x2fe5

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt3

Table 8-1393 DIPD_RO_COMB_16 0x2fe6

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt4

Table 8-1394 DIPD_RO_COMB_17 0x2fe7

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt5

Table 8-1395 DIPD_RO_COMB_18 0x2fe8

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt6

Table 8-1396 DIPD_RO_COMB_19 0x2fe9

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt7

Table 8-1397 DIPD_RO_COMB_20 0x2fea

Bit(s)	R/W	Default	Description
31-0	R		Ro_rt8

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Table 8-1398 DIPD_COMB_CTRL5 0x2fd5

Bit(s)	R/W	Default	Description
31-24	W		Fld_mindif
23-16	W		Frm_mindif
13-8	W		Flm_smp_mtn_cnt
7-0	W		Flm_smp_mtn_thd

Table 8-1399 DIPD_COMB_CTRL6 0x2feb

Bit(s)	R/W	Default	Description
21	Rw	0	Reg_edit_sel
20	Rw	1	Reg_horflt_en
16-12	Rw	7	Reg_combseglen
9-4	Rw	6	Reg_trancombrat
3-0	Rw	4	Reg_combsegmin

8.2.3.14 DNR Registers

Table 8-1400 DNR_CTRL 0x2d00

Bit(s)	R/W	Default	Description
31:20	R/W		reserved
19	R/W		reg_dnr_bufctrl, deblock linebuf control, 0 : 3line 1: 5line, default =1
18	R/W		reg_dnr_dm_bufctrl, demosquito linebuf control, 0 : 3line 1: 5line, default =1
16	R/W		reg_dnr_en
15	R/W		reg_dnr_db_vdbstep , vdb step, 0: 4, 1: 8 . unsigned , default = 1
14	R/W		reg_dnr_db_vdbprten , vdb protectoin enable . unsigned , default = 1
13	R/W		reg_dnr_gbs_difen , enable dif (between LR and LL/RR) condition for gbs stat.. unsigned , default = 0
12	R/W		reg_dnr_luma_en , enable ycbcr2luma module . unsigned , default = 1
11:10	R/W		reg_dnr_db_mod , deblocking mode, 0: disable, 1: horizontal deblocking, 2: vertical deblocking, 3: horizontal & vertical deblocking. unsigned , default = 3
9	R/W		reg_dnr_db_chrmen , enable chroma deblocking . unsigned , default = 1
8	R/W		reg_dnr_hvdif_mod , 0: calc. difs by original Y, 1: by new luma. unsigned , default = 1
7	R/W		reserved
6: 4	R/W		reg_dnr_demo_lften , b0: Y b1:U b2:V . unsigned , default = 7

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Bit(s)	R/W	Default	Description
3	R/W		reserved
2: 0	R/W		reg_dnr_demo_rgten , b0: Y b1:U b2:V . unsigned , default = 7

Table 8-1401 DNR_HVSIZE 0x2d01

Bit(s)	R/W	Default	Description
31:29	R/W		reserved
28:16	R/W		reg_dnr_hsize , hsize . unsigned , default = 0
15:13	R/W		reserved
12: 0	R/W		reg_dnr_vsize , vsize . unsigned , default = 0

Table 8-1402 DNR_DBLK_BLANK_NUM 0x2d02

Bit(s)	R/W	Default	Description
31:16	R/W		reserved
15: 8	R/W		reg_dblk_hblank_num , deblock hor blank num . unsigned , default = 16
7: 0	R/W		reg_dblk_vblank_num , deblock ver blank num . unsigned , default = 45

Table 8-1403 DNR_BLK_OFFST 0x2d03

Bit(s)	R/W	Default	Description
31: 7	R/W		reserved
6: 4	R/W		reg_dnr_hbofst , horizontal block offset may provide by software calc.. unsigned , default = 0
3	R/W		reserved
2: 0	R/W		reg_dnr_vbofst , vertical block offset may provide by software calc.. unsigned , default = 0

Table 8-1404 DNR_GBS 0x2d04

Bit(s)	R/W	Default	Description
31: 2	R/W		reserved
1: 0	R/W		reg_dnr_gbs , global block strength may update by software calc.. unsigned , default = 0

Table 8-1405 DNR_HBOFFST_STAT 0x2d05

Bit(s)	R/W	Default	Description
31:24	R/W		reg_dnr_hbof_difthd , dif threshold (>=) between LR and LL/RR. unsigned , default = 2
23:16	R/W		reg_dnr_hbof_edgethd , edge threshold (<=) for LR . unsigned , default = 32
15: 8	R/W		reg_dnr_hbof_flatthd , flat threshold (>=) for LR . unsigned , default = 0

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Bit(s)	R/W	Default	Description
7	R/W		reserved
6: 4	R/W		reg_dnr_hbof_delta , delta for weighted bin accumulator. unsigned , default = 1
3	R/W		reserved
2: 0	R/W		reg_dnr_hbof_statmod , statistic mode for horizontal block offset, 0: count flags for 8-bin, 1: count LRs for 8-bin, 2: count difs for 8-bin, 3: count weighted flags for 8-bin, 4: count flags for first 32-bin, 5: count LRs for first 32-bin, 6 or 7: count difs for first 32-bin. unsigned , default = 2

Table 8-1406 DNR_VBOFFST_STAT 0x2d06

Bit(s)	R/W	Default	Description
31:24	R/W		reg_dnr_vbof_difthd , dif threshold (>=) between Up and Dw. unsigned , default = 1
23:16	R/W		reg_dnr_vbof_edgethd , edge threshold (<=) for Up/Dw. unsigned , default = 16
15: 8	R/W		reg_dnr_vbof_flatthd , flat threshold (>=) for Up/Dw. unsigned , default = 0
7	R/W		reserved
6: 4	R/W		reg_dnr_vbof_delta , delta for weighted bin accumulator. unsigned , default = 1
3	R/W		reserved
2: 0	R/W		reg_dnr_vbof_statmod , statistic mode for vertical block offset, 0: count flags for 8-bin, 1: count Ups for 8-bin, 2: count difs for 8-bin, 3: count weighted flags for 8-bin, 4: count flags for first 32-bin, 5: count Ups for first 32-bin, 6 or 7: count difs for first 32-bin. unsigned , default = 2

Table 8-1407 DNR_GBS_STAT 0x2d07

Bit(s)	R/W	Default	Description
31:24	R/W		reg_dnr_gbs_edgethd , edge threshold (<=) for LR . unsigned , default = 32
23:16	R/W		reg_dnr_gbs_flatthd , flat threshold (>=) for LR . unsigned , default = 0
15: 8	R/W		reg_dnr_gbs_varthd , variation threshold (<=) for Lvar/Rvar. unsigned , default = 16
7: 0	R/W		reg_dnr_gbs_difthd , dif threshold (>=) between LR and LL/RR. unsigned , default = 2

Table 8-1408 DNR_STAT_X_START_END 0x2d08

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29:16	R/W		reg_dnr_stat_xst . unsigned , default = 24
15:14	R/W		reserved
13: 0	R/W		reg_dnr_stat_xed . unsigned , default = HSIZE - 25

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Table 8-1409 DNR_STAT_Y_START_END 0x2d09

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29:16	R/W		reg_dnr_stat_yst . unsigned , default = 24
15:14	R/W		reserved
13: 0	R/W		reg_dnr_stat_yed . unsigned , default = VSIZE - 25

Table 8-1410 DNR_LUMA 0x2d0a

Bit(s)	R/W	Default	Description
31:27	R/W		reserved
28	R/W		Reg_dnr_luma_mode, ycbcr2luma mode selection, default =0
26:24	R/W		reg_dnr_luma_sqrtshft , left shift for fast squart of chroma, [0, 4]. unsigned , default = 2
23:21	R/W		reserved
20:16	R/W		reg_dnr_luma_sqrtoffst , offset for fast squart of chroma. signed , default = 0
15	R/W		reserved
14:12	R/W		reg_dnr_luma_wcmod , theta related to warm/cool segment line, 0: 0, 1: 45, 2: 90, 3: 135, 4: 180, 5: 225, 6: 270, 7: 315. . unsigned , default = 3
11: 8	R/W		reg_dnr_luma_cshft , shift for calc. delta part, 0~8, . unsigned , default = 8
7: 6	R/W		reserved
5: 0	R/W		reg_dnr_luma_cgain , final gain for delta part, 32 normalized to "1". unsigned , default = 4

Table 8-1411 DNR_DB_YEDGE_THD 0x2d0b

Bit(s)	R/W	Default	Description
31:24	R/W		reg_dnr_db_yedgethd0 , edge threshold0 for luma . unsigned , default = 12
23:16	R/W		reg_dnr_db_yedgethd1 , edge threshold1 for luma . unsigned , default = 15
15: 8	R/W		reg_dnr_db_yedgethd2 , edge threshold2 for luma . unsigned , default = 18
7: 0	R/W		reg_dnr_db_yedgethd3 , edge threshold3 for luma . unsigned , default = 25

Table 8-1412 DNR_DB_CEDGE_THD 0x2d0c

Bit(s)	R/W	Default	Description
31:24	R/W		reg_dnr_db_cedgethd0 , edge threshold0 for chroma . unsigned , default = 12
23:16	R/W		reg_dnr_db_cedgethd1 , edge threshold1 for chroma . unsigned , default = 15
15: 8	R/W		reg_dnr_db_cedgethd2 , edge threshold2 for chroma . unsigned , default = 18
7: 0	R/W		reg_dnr_db_cedgethd3 , edge threshold3 for chroma . unsigned , default = 25

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Table 8-1413 DNR_DB_HGAP 0x2d0d

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W		reg_dnr_db_hgapthd , horizontal gap thd (<=) for very sure blockiness . unsigned , default = 8
15: 8	R/W		reg_dnr_db_hgapdifthd , dif thd between hgap and lft/rgt hdifs. unsigned , default = 1
7: 1	R/W		reserved
0	R/W		reg_dnr_db_hgapmod , horizontal gap calc. mode, 0: just use current col x, 1: find max between (x-1, x, x+1) . unsigned , default = 0

Table 8-1414 DNR_DB_HBS 0x2d0e

Bit(s)	R/W	Default	Description
31: 6	R/W		reserved
5: 4	R/W		reg_dnr_db_hbsup , horizontal bs up value . unsigned , default = 1
3: 2	R/W		reg_dnr_db_hbsmax , max value of hbs for global control. unsigned , default = 3
1: 0	R/W		reg_dnr_db_hgbsthd , gbs thd (>=) for hbs calc. . unsigned , default = 1

Table 8-1415 DNR_DB_HACT 0x2d0f

Bit(s)	R/W	Default	Description
31:16	R/W		reserved
15: 8	R/W		reg_dnr_db_hactthd0 , thd0 of hact, for block classification. unsigned , default = 10
7: 0	R/W		reg_dnr_db_hactthd1 , thd1 of hact, for block classification. unsigned , default = 32

Table 8-1416 DNR_DB_YHDELTA_GAIN 0x2d10

Bit(s)	R/W	Default	Description
31:27	R/W		reserved
26:24	R/W		reg_dnr_db_yhdeltagain1 , (p1-q1) gain for Y's delta calc. when bs=1, normalized 8 as "1" . unsigned , default = 2
23	R/W		reserved
22:20	R/W		reg_dnr_db_yhdeltagain2 , (p1-q1) gain for Y's delta calc. when bs=2, normalized 8 as "1" . unsigned , default = 0
19	R/W		reserved
18:16	R/W		reg_dnr_db_yhdeltagain3 , (p1-q1) gain for Y's delta calc. when bs=3, normalized 8 as "1" . unsigned , default = 0
15	R/W		reserved
14: 8	R/W		reg_dnr_db_yhdeltaadjoffst , offset for adjust Y's hdelta (-64, 63). signed , default = 0

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Bit(s)	R/W	Default	Description
7: 6	R/W		reserved
5: 0	R/W		reg_dnr_db_yhdeltaadjgain , gain for adjust Y's hdelta, normalized 32 as "1" . unsigned , default = 32

Table 8-1417 DNR_DB_YHDELTA2_GAIN 0x2d11

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29:24	R/W		reg_dnr_db_yhdelta2gain2 , gain for bs=2's adjust Y's hdelta2, normalized 64 as "1" . unsigned , default = 8
23:21	R/W		reserved
20:16	R/W		reg_dnr_db_yhdelta2offst2 , offset for bs=2's adjust Y's hdelta2 (-16, 15). signed , default = 0
15:14	R/W		reserved
13: 8	R/W		reg_dnr_db_yhdelta2gain3 , gain for bs=3's adjust Y's hdelta2, normalized 64 as "1" . unsigned , default = 4
7: 5	R/W		reserved
4: 0	R/W		reg_dnr_db_yhdelta2offst3 , offset for bs=3's adjust Y's hdelta2 (-16, 15). signed , default = 0

Table 8-1418 DNR_DB_CHDELTA_GAIN 0x2d12

Bit(s)	R/W	Default	Description
31:27	R/W		reserved
26:24	R/W		reg_dnr_db_chdeltagain1 , (p1-q1) gain for UV's delta calc. when bs=1, normalized 8 as "1". unsigned , default = 2
23	R/W		reserved
22:20	R/W		reg_dnr_db_chdeltagain2 , (p1-q1) gain for UV's delta calc. when bs=2, normalized 8 as "1". unsigned , default = 0
19	R/W		reserved
18:16	R/W		reg_dnr_db_chdeltagain3 , (p1-q1) gain for UV's delta calc. when bs=3, normalized 8 as "1". unsigned , default = 0
15	R/W		reserved
14: 8	R/W		reg_dnr_db_chdeltaadjoffst , offset for adjust UV's hdelta (-64, 63). signed , default = 0
7: 6	R/W		reserved
5: 0	R/W		reg_dnr_db_chdeltaadjgain , gain for adjust UV's hdelta, normalized 32 as "1". unsigned , default = 32

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Table 8-1419 DNR_DB_CHDELTA2_GAIN 0x2d13

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29:24	R/W		reg_dnr_db_chdelta2gain2 , gain for bs=2's adjust UV's hdelta2, normalized 64 as "1" . unsigned , default = 8
23:21	R/W		reserved
20:16	R/W		reg_dnr_db_chdelta2offst2 , offset for bs=2's adjust UV's hdelta2 (-16, 15). signed , default = 0
15:14	R/W		reserved
13: 8	R/W		reg_dnr_db_chdelta2gain3 , gain for bs=2's adjust UV's hdelta2, normalized 64 as "1" . unsigned , default = 4
7: 5	R/W		reserved
4: 0	R/W		reg_dnr_db_chdelta2offst3 , offset for bs=2's adjust UV's hdelta2 (-16, 15). signed , default = 0

Table 8-1420 DNR_DB_YC_VEDGE_THD 0x2d14

Bit(s)	R/W	Default	Description
31:16	R/W		reserved
15: 8	R/W		reg_dnr_db_yvedgethd , special Y's edge thd for vdb. unsigned , default = 12
7: 0	R/W		reg_dnr_db_cvedgethd , special UV's edge thd for vdb. unsigned , default = 12

Table 8-1421 DNR_DB_VBS_MISC 0x2d15

Bit(s)	R/W	Default	Description
31:24	R/W		reg_dnr_db_vgapthd , vertical gap thd (<=) for very sure blockiness . unsigned , default = 8
23:16	R/W		reg_dnr_db_vactthd , thd of vact, for block classification . unsigned , default = 10
15: 8	R/W		reg_dnr_db_vgapdifthd , dif thd between vgap and vact. unsigned , default = 4
7: 4	R/W		reserved
3: 2	R/W		reg_dnr_db_vbsmax , max value of vbs for global control. unsigned , default = 2
1: 0	R/W		reg_dnr_db_vgbsthd , gbs thd (>=) for vbs calc. . unsigned , default = 1

Table 8-1422 DNR_DB_YVDELTA_GAIN 0x2d16

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29:24	R/W		reg_dnr_db_yvdeltaadjgain , gain for adjust Y's vdelta, normalized 32 as "1". unsigned , default = 32
23	R/W		reserved
22:16	R/W		reg_dnr_db_yvdeltaadjoffst , offset for adjust Y's vdelta (-64, 63). signed , default = 0

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Bit(s)	R/W	Default	Description
15:14	R/W		reserved
13: 8	R/W		reg_dnr_db_yvdelta2gain , gain for adjust Y's vdelta2, normalized 64 as "1". unsigned , default = 8
7: 5	R/W		reserved
4: 0	R/W		reg_dnr_db_yvdelta2offst , offset for adjust Y's vdelta2 (-16, 15). signed , default = 0

Table 8-1423 DNR_DB_CVDELTA_GAIN 0x2d17

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29:24	R/W		reg_dnr_db_cvdeltaadjgain , gain for adjust UV's vdelta, normalized 32 as "1". unsigned , default = 32
23	R/W		reserved
22:16	R/W		reg_dnr_db_cvdeltaadjoffst , offset for adjust UV's vdelta (-64, 63). signed , default = 0
15:14	R/W		reserved
13: 8	R/W		reg_dnr_db_cvdelta2gain , gain for adjust UV's vdelta2, normalized 64 as "1". unsigned , default = 8
7: 5	R/W		reserved
4: 0	R/W		reg_dnr_db_cvdelta2offst , offset for adjust UV's vdelta2 (-16, 15). signed , default = 0

Table 8-1424 DNR_RO_GBS_STAT_LR 0x2d19

Bit(s)	R/W	Default	Description
31: 0	R		ro_gbs_stat_lr . unsigned , default = 0

Table 8-1425 DNR_RO_GBS_STAT_LL 0x2d18

Bit(s)	R/W	Default	Description
31: 0	R		ro_gbs_stat_ll . unsigned , default = 0

Table 8-1426 DNR_RO_GBS_STAT_RR 0x2d1a

Bit(s)	R/W	Default	Description
31: 0	R		ro_gbs_stat_rr . unsigned , default = 0

Table 8-1427 DNR_RO_GBS_STAT_DIF 0x2d1b

Bit(s)	R/W	Default	Description
31: 0	R		ro_gbs_stat_dif . unsigned , default = 0

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Table 8-1428 DNR_RO_GBS_STAT_CNT 0x2d1c

Bit(s)	R/W	Default	Description
31: 0	R		ro_gbs_stat_cnt . unsigned , default = 0

Table 8-1429 DNR_RO_HBOF_STAT_CNT_0 0x2d1d

Bit(s)	R/W	Default	Description
31: 0	R		ro_hbof_stat_cnt0 . unsigned , default = 0

Table 8-1430 DNR_RO_HBOF_STAT_CNT_31 0x2d3c

Bit(s)	R/W	Default	Description
31: 0	R		ro_hbof_stat_cnt31 . unsigned , default = 0

Table 8-1431 DNR_RO_VBOF_STAT_CNT_0 0x2d3d

Bit(s)	R/W	Default	Description
31: 0	R		ro_vbof_stat_cnt0 . unsigned , default = 0

Table 8-1432 DNR_RO_VBOF_STAT_CNT_31 0x2d5c

Bit(s)	R/W	Default	Description
31: 0	R		ro_vbof_stat_cnt31 . unsigned , default = 0

Table 8-1433 DNR_DM_ADP_EN 0x2d5d

Bit(s)	R/W	Default	Description
31:8	R/W		reserved
7	R/W	1	Reg_dnr_scene_change_flag // scence change flag for dnr
6	R/W	1	Reg_dnr_dm_lpf_en // enable lpf fof demosquito filter
5	R/W	1	Reg_dnr_dm_adp_level_en // enable adptive demosquito level
4	R/W	1	Reg_dnr_dm_flag2bdif_en // enable edge flag to blkdif calc
3: 2	R/W	1	Reg_dnr_dm_edgeiir // edge flag iir mode, 0: cur, 1: max(pre,cur), 2,3: choice by org mtn
1	R/W	1	Reg_dnr_dm_dirdifmod // dif mode for direction calc , 0: abs(dif02), 1: (abs(dif01) +abs(dif21))/2
0	R/W	0	Reg_dnr_dm_sur_dir_mod // sure direction mode for cordif calc

Table 8-1434 DNR_DM_EDGE_DIR 0x2d5e

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	2	Reg_dnr_dm_dirdifcor // dif coring threshold for direction calc

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Bit(s)	R/W	Default	Description
15:8	R/W	160	Reg_dnr_dm_edgecordifthd // edge flag down when cordif is lager than theshold
7:0	R/W	60	Reg_dnr_dm_mtnrt // motion ration for mtn decision

Table 8-1435 DNR_DM_CTRL 0x2d60

Bit(s)	R/W	Default	Description
31:14	R/W		reserved
13	R/W	1	Reg_dnr_dm_fedgeflag_en, enable edge flag calc. Of each frame
12	R/W	1	Reg_dnr_dm_fedgeflag_cl, clear frame edge flag if needed
11:10	R/W	1	Reg_dnr_dm_fedgeflg_df, user defined edge when fedge_flg_en=0
9	R/W	1	Reg_dnr_dm_en, enable demosquito function
8	R/W	1	Reg_dnr_dm_chrmen, enable chroma proc. For demosquito
7:6	R/W	3	Reg_dnr_dm_level, demosquito level
5:4	R/W	1	Reg_dnr_dm_leveldw0, level down when gbs is small
3:2	R/W	1	Reg_dnr_dm_level_dw1, level down for no edge/flat blocks
1:0	R/W	0	Reg_dnr_dm_gbsthd, small/large threshold for gbs <=

Table 8-1436 DNR_DM_NR_BLND 0x2d61

Bit(s)	R/W	Default	Description
31:25	R/W		reserved
24	R/W	0	reg_dnr_dm_defalpen , 1 to enable user defined alpha for DM/NR blend
23:16	R/W	0	reg_dnr_dm_defalp , user defined alpha for DM/NR blend
15	R/W		reserved
14: 9	R/W	32	reg_dnr_dm_yalpgain , gain for DM/NR alpha, normalized 32 as 1
8: 0	R/W	0	reg_dnr_dm_yalpoffset , offset for DM/NR alpha, singed, (-255, 255)

Table 8-1437 DNR_DM_RNG_THD 0x2d62

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	2	reg_dnr_dm_rgnminthd
15: 8	R/W	64	reg_dnr_dm_rgnmaxthd
7: 0	R/W	4	reg_dnr_dm_rgndifthd

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Table 8-1438 DNR_DM_RNG_GAIN_OFST 0x2d63

Bit(s)	R/W	Default	Description
31:14	R/W		reserved
13: 8	R/W	16	reg_dnr_dm_rgngain , normalized 16 as 1
7: 0	R/W	0	reg_dnr_dm_rgnofst , unsigned

Table 8-1439 DNR_DM_DIR_MISC 0x2d64

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29	R/W	1	reg_dnr_dm_diralpen
28:24	R/W	0	reg_dnr_dm_diralpgain
23:22	R/W		reserved
21:16	R/W	0	reg_dnr_dm_diralpofst
15:13	R/W		reserved
12: 8	R/W	0	reg_dnr_dm_diralpmin
7:5	R/W		reserved
4: 0	R/W	31	reg_dnr_dm_diralpmax

Table 8-1440 DNR_DM_COR_DIF 0x2d65

Bit(s)	R/W	Default	Description
31:6	R/W		reserved
5	R/W	1	reg_dnr_dm_cordifprt , enable cordif proctection mode
4:2	R/W	3	reg_dnr_dm_cordifshft
1:0	R/W	3	reg_dnr_dm_cordifmod , 0: use max dir dif as cordif, 1: use max3x3-min3x3 as cordif 2: use min dif as cordif, 3: max's side dif.

Table 8-1441 DNR_DM_FLT_THD 0x2d66

Bit(s)	R/W	Default	Description
31:24	R/W	4	reg_dnr_dm_fltthd00 , block flat threshold0 for block average difference when gbs is small
23:16	R/W	6	reg_dnr_dm_fltthd01 , block flat threshold1 for block average difference when gbs is small
15: 8	R/W	9	reg_dnr_dm_fltthd10 , block flat threshold0 for block average difference when gbs is larger
7: 0	R/W	12	reg_dnr_dm_fltthd11 , block flat threshold1 for block average difference when gbs is larger

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Table 8-1442 DNR_DM_VAR_THD 0x2d67

Bit(s)	R/W	Default	Description
31:24	R/W	2	reg_dnr_dm_varthd00 , block variance threshold0 (>=) when gbs is small
23:16	R/W	15	reg_dnr_dm_varthd01 , block variance threshold1 (<=) when gbs is small
15: 8	R/W	3	reg_dnr_dm_varthd10 , block variance threshold0 (>=) when gbs is larger
7: 0	R/W	24	reg_dnr_dm_varthd11 , block variance threshold1 (<=) when gbs is larger

Table 8-1443 DNR_DM_EDGE_DIF_THD 0x2d68

Bit(s)	R/W	Default	Description
31:24	R/W	32	reg_dnr_dm_edgethd0 , block edge threshold (<=) when gbs is small
23:16	R/W	48	reg_dnr_dm_edgethd1 , block edge threshold (<=) when gbs is larger
15: 8	R/W	48	reg_dnr_dm_difthd0 , block dif threshold (<=) when gbs is small
7: 0	R/W	64	reg_dnr_dm_difthd1 , block dif threshold (<=) when gbs is larger

Table 8-1444 DNR_DM_AVG_THD 0x2d69

Bit(s)	R/W	Default	Description
31:16	R/W		reserved
15: 8	R/W	160	reg_dnr_dm_avgthd0 , block average threshold (>=) when gbs is small
7: 0	R/W	128	reg_dnr_dm_avgthd1 , block average threshold (<=) when gbs is larger

Table 8-1445 DNR_DM_AVG_VAR_DIF_THD 0x2d6a

Bit(s)	R/W	Default	Description
31:16	R/W		reserved
15: 8	R/W	12	reg_dnr_dm_avgdifthd , block average dif threshold(<) between cur and up block for flat block
7: 0	R/W	1	reg_dnr_dm_vardifthd , block variance dif threshold(>=) between cur and up block

Table 8-1446 DNR_DM_EDGE_DIF_THD2 0x2d6b

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	24	reg_dnr_dm_varthd2 , block variance threshold (>=) for edge block detect
15: 8	R/W	40	reg_dnr_dm_edgethd2 , block edge threshold (>=)
7: 0	R/W	80	reg_dnr_dm_difthd2 , block dif threshold (>=)

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Table 8-1447 DNR_DM_DIF_FLT_MISC 0x2d6c

Bit(s)	R/W	Default	Description
31:28	R/W	0	reg_dnr_dm_ldifoob, pre-defined large dif when pixel out of block
27:24	R/W	0	reg_dnr_dm_bdifoob, pre-defined block dif when pixel out of block
23:16	R/W	200	reg_dnr_dm_fltalp, pre-defined alpha for dm and nr blending when block is flat with mos
15:12	R/W		reserved
11:8	R/W	12	reg_dnr_dm_fltminbdif, pre-defined min block dif for dm filter when block is flat with mos
7	R/W		reserved
6:2	R/W	16	reg_dnr_dm_difnormgain, gain for pixel dif normalization for dm filter.
1	R/W	1	reg_dnr_dm_difnormen , enable pixel dif normalization for dm filter
0	R/W	0	reg_dnr_dm_difupden , enable block dif update using max of left,cur,right difs

Table 8-1448 DNR_DM_SDIF_LUT0_2 0x2d6d

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	16	reg_dnr_dm_sdiflut0, normally 0-16
15:13	R/W		reserved
12:8	R/W	14	reg_dnr_dm_sdiflut1
7:5	R/W		reserved
4:0	R/W	13	reg_dnr_dm_sdiflut2

Table 8-1449 DNR_DM_SDIF_LUT3_5 0x2d6e

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	10	reg_dnr_dm_sdiflut3
15:13	R/W		reserved
12:8	R/W	7	reg_dnr_dm_sdiflut4
7:5	R/W		reserved
4:0	R/W	5	reg_dnr_dm_sdiflut5

Table 8-1450 DNR_DM_SDIF_LUT6_8 0x2d6f

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	3	reg_dnr_dm_sdiflut6
15:13	R/W		reserved

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Bit(s)	R/W	Default	Description
12:8	R/W	1	reg_dnr_dm_sdiflut7
7:5	R/W		reserved
4:0	R/W	0	reg_dnr_dm_sdiflut8

Table 8-1451 DNR_DM_LDIF_LUT0_2 0x2d70

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	0	reg_dnr_dm_ldiflut0
15:13	R/W		reserved
12:8	R/W	4	reg_dnr_dm_ldiflut1
7:5	R/W		reserved
4:0	R/W	12	reg_dnr_dm_ldiflut2

Table 8-1452 DNR_DM_LDIF_LUT3_5 0x2d71

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	14	reg_dnr_dm_ldiflut3
15:13	R/W		reserved
12:8	R/W	15	reg_dnr_dm_ldiflut4
7:5	R/W		reserved
4:0	R/W	16	reg_dnr_dm_ldiflut5

Table 8-1453 DNR_DM_LDIF_LUT6_8 0x2d72

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	16	reg_dnr_dm_ldiflut6
15:13	R/W		reserved
12:8	R/W	16	reg_dnr_dm_ldiflut7
7:5	R/W		reserved
4:0	R/W	16	reg_dnr_dm_ldiflut8

Table 8-1454 DNR_DM_DIF2NORM_LUT0_2 0x2d73

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	16	reg_dnr_dm_dif2normlut0

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Bit(s)	R/W	Default	Description
15:13	R/W		reserved
12:8	R/W	5	reg_dnr_dm_dif2normlut1
7:5	R/W		reserved
4:0	R/W	3	reg_dnr_dm_dif2normlut2

Table 8-1455 DNR_DM_DIF2NORM_LUT3_5 0x2d74

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	2	reg_dnr_dm_dif2normlut3
15:13	R/W		reserved
12:8	R/W	2	reg_dnr_dm_dif2normlut4
7:5	R/W		reserved
4:0	R/W	1	reg_dnr_dm_dif2normlut5

Table 8-1456 DNR_DM_DIF2NORM_LUT6_8 0x2d75

Bit(s)	R/W	Default	Description
31:21	R/W		reserved
20:16	R/W	1	reg_dnr_dm_dif2normlut6
15:13	R/W		reserved
12:8	R/W	1	reg_dnr_dm_dif2normlut7
7:5	R/W		reserved
4:0	R/W	1	reg_dnr_dm_dif2normlut8

Table 8-1457 DNR_DM_GMS_THD 0x2d76

Bit(s)	R/W	Default	Description
31:16	R/W		reserved
15:8	R/W	0	reg_gms_stat_thd0
7:0	R/W	128	reg_gms_stat_thd1

Table 8-1458 DNR_RO_DM_GMS_STST_CNT 0x2d77

Bit(s)	R/W	Default	Description
31:0	RO		ro_dm_gms_stat_cnt

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Table 8-1459 DNR_RO_DM_GMS_STST_MS 0x2d78

Bit(s)	R/W	Default	Description
31:0	RO		ro_dm_gms_stat_ms

Table 8-1460 DNR_DM_EDGE_GAIN 0x2d79

Bit(s)	R/W	Default	Description
31:24	R/W	8	reg_dnr_dm_alpedgegain0 : // unsigned , default = 8 , edge based gain for alpha, normalized 16 as "1"
23:16	R/W	16	reg_dnr_dm_alpedgegain1 : // unsigned , default = 16 , edge based gain for alpha, normalized 16 as "1"
15: 8	R/W	24	reg_dnr_dm_alpedgegain2 : // unsigned , default = 24 , edge based gain for alpha, normalized 16 as "1"
7: 0	R/W	32	reg_dnr_dm_alpedgegain3 : // unsigned , default = 32 , edge based gain for alpha, normalized 16 as "1"

Table 8-1461 DNR_DM_FLG_BDIF 0x2d7a

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_dnr_dm_flg2bdif0 : // unsigned , default = 0 , (0-16), edge flg to blkdif calc.
23:16	R/W	96	reg_dnr_dm_flg2bdif1 : // unsigned , default = 96 , (0-16), edge flg to blkdif calc.
15: 8	R/W	160	reg_dnr_dm_flg2bdif2 : // unsigned , default = 160 , (0-16), edge flg to blkdif calc.
7: 0	R/W	192	reg_dnr_dm_flg2bdif3 : // unsigned , default = 192 , (0-16), edge flg to blkdif calc.

Table 8-1462 DNR_DM_GBS_RORM 0x2d7b

Bit(s)	R/W	Default	Description
15:12	R/W	0	reg_dnr_dm_gbs4difnorm0 : // unsigned , default = 0 , gbs=0 for dif norm calc.
11: 8	R/W	1	reg_dnr_dm_gbs4difnorm1 : // unsigned , default = 1 , gbs=1 for dif norm calc.
7: 4	R/W	4	reg_dnr_dm_gbs4difnorm2 : // unsigned , default = 4 , gbs=2 for dif norm calc.
3: 0	R/W	6	reg_dnr_dm_gbs4difnorm3 : // unsigned , default = 6 , gbs=3 for dif norm calc.

Table 8-1463 DNR_DM_FLG_LEV 0x2d7c

Bit(s)	R/W	Default	Description
13:12	R/W	1	reg_dnr_dm_flg2lev0 : // unsigned , default = 1 , edge flg to filter level calc.
9: 8	R/W	3	reg_dnr_dm_flg2lev1 : // unsigned , default = 3 , edge flg to filter level calc.
5: 4	R/W	3	reg_dnr_dm_flg2lev2 : // unsigned , default = 3 , edge flg to filter level calc.
1: 0	R/W	3	reg_dnr_dm_flg2lev3 : // unsigned , default = 3 , edge flg to filter level calc.

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Table 8-1464 DNR_DM_DIF_FLG_TH 0x2d7d

Bit(s)	R/W	Default	Description
15: 8	R/W	128	reg_dnr_dm_dif2flgthd1 : // unsigned , default = 128 , dif to edge flg threshold 1
7: 0	R/W	192	reg_dnr_dm_dif2flgthd2 : // unsigned , default = 192 , dif to edge flg threshold 2

Table 8-1465 DNR_DM_CALP_GAIN_OFST 0x2d7e

Bit(s)	R/W	Default	Description
21:16	R/W	16	reg_dnr_dm_calpgain : // unsigned , default = 16 , chroma gain for nr/dm alpha, normalized 32 as "1"
8: 0	R/W	-64	reg_dnr_dm_calpoffst : // signed , default = -64 , (-255, 255), chroma offset for nr/dm alpha

Table 8-1466 DECOMB_DET_VERT_CON0 0x2d80

Bit(s)	R/W	Default	Description
31:24	R/W	60	reg_di_dcmb_det_vcon_thd0 : default = 60 // u8
23:16	R/W	80	reg_di_dcmb_det_vcon_thd1 : default = 80 // u8
15: 8	R/W	63	reg_di_dcmb_det_valp_lmt0 : default = 63 // u8
7: 0	R/W	4	reg_di_dcmb_det_valp_lmt1 : default = 4 // u8

Table 8-1467 DECOMB_DET_VERT_CON1 0x2d81

Bit(s)	R/W	Default	Description
23:16	R/W	0	reg_di_dcmb_det_valp_lmt2 : default = 0 // u8
15: 8	R/W	32	reg_di_dcmb_det_vrate0 : default = 32 // u8
7: 0	R/W	4	reg_di_dcmb_det_vrate1 : default = 4 // u8

Table 8-1468 DECOMB_DET_EDGE_CON0 0x2d82

Bit(s)	R/W	Default	Description
31:24	R/W	60	reg_di_dcmb_det_econ_thd0 : default = 60 // u8
23:16	R/W	80	reg_di_dcmb_det_econ_thd1 : default = 80 // u8
15: 8	R/W	63	reg_di_dcmb_det_ealp_lmt0 : default = 63 // u8
7: 0	R/W	4	reg_di_dcmb_det_ealp_lmt1 : default = 4 // u8

Table 8-1469 DECOMB_DET_EDGE_CON1 0x2d83

Bit(s)	R/W	Default	Description
23:16	R/W	0	reg_di_dcmb_det_ealp_lmt2 : default = 0 // u8
15: 8	R/W	32	reg_di_dcmb_det_erate0 : default = 32 // u8
7: 0	R/W	4	reg_di_dcmb_det_erate1 : default = 4 // u8

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Table 8-1470 DECOMB_PARA 0x2d84

Bit(s)	R/W	Default	Description
29:28	R/W	1	reg_di_dcmb_cmb_lpf : default = 1 // u2, 0:no lpf, 1:[1 2 1], 2,3: [1 2 2 2 1]
27:26	R/W	0	reg_di_dcmb_vedge_chk : default = 0 // u2, vertical edge check, 0: no check, 1: vrt!=0, 2: vrt==3
25:24	R/W	0	reg_di_dcmb_nedge_chk : default = 0 // u2, no idea edge check, 0, no check, 1, check
23:20	R/W	0	reg_di_dcmb_edge_min : default = 0 // u4, min edge for edge cmb
19:16	R/W	15	reg_di_dcmb_edge_max : default = 15 // u4, min edge for edge cmb
15:8	R/W	255	reg_di_dcmb_bld_alp : default = 255 // u8, user defined alpha for di & decmb blend
7:0	R/W	40	reg_di_dcmb_bld_alp_beta : default = 40 // u8, beta for mtn & cmb blend, for bld alpha calc.

Table 8-1471 DECOMB_BLND_CON0 0x2d85

Bit(s)	R/W	Default	Description
31:24	R/W	100	reg_di_dcmb_bld_con_thd0 : default = 100 // u8
23:16	R/W	120	reg_di_dcmb_bld_con_thd1 : default = 120 // u8
15: 8	R/W	0	reg_di_dcmb_bld_alp_lmt0 : default = 0 // u8
7: 0	R/W	128	reg_di_dcmb_bld_alp_lmt1 : default = 128 // u8

Table 8-1472 DECOMB_BLND_CON1 0x2d86

Bit(s)	R/W	Default	Description
23:16	R/W	255	reg_di_dcmb_bld_alp_lmt2 : default = 255 // u8
15: 8	R/W	32	reg_di_dcmb_bld_rate0 : default = 32 // u8
7: 0	R/W	32	reg_di_dcmb_bld_rate1 : default = 32 // u8

Table 8-1473 DECOMB_YC_THRD 0x2d87

Bit(s)	R/W	Default	Description
15: 8	R/W	2	reg_di_dcmb_ythd : default = 2 // u8, default = 2
7: 0	R/W	2	reg_di_dcmb_cthd : default = 2 // u8, default = 2

Table 8-1474 DECOMB_MTN_GAIN_OFST 0x2d88

Bit(s)	R/W	Default	Description
21:16	R/W	16	reg_di_dcmb_mtn_alp_gain : default = 16 // u6, 16 is normalized to '1'
8:0	R/W	0	reg_di_dcmb_mtn_alp_ofst : default = 0 // s9, [-256, 255]

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Table 8-1475 DECOMB_CMB_SEL_GAIN_OFST 0x2d89

Bit(s)	R/W	Default	Description
21:16	R/W	48	reg_di_dcmb_cmb_sel_gain : default = 48 // u6, 16 is normalized to '1'
8:0	R/W	0	reg_di_dcmb_cmb_sel_ofst : default = 0 // s9, [-256, 255]

Table 8-1476 DECOMB_WIND00 0x2d8a

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_di_dcmb_wnd00 : default = 0 // u13, x0 for window 0, software control
12:0	R/W	719	reg_di_dcmb_wnd01 : default = 719 // u13, x1 for window 0, HSIZE-1, software control

Table 8-1477 DECOMB_WIND01 0x2d8b

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_di_dcmb_wnd02 : default = 0 // u13, y0 for window 0, software control
12:0	R/W	39	reg_di_dcmb_wnd03 : default = 39 // u13, y1 for window 0, software control

Table 8-1478 DECOMB_WIND10 0x2d8c

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_di_dcmb_wnd10 : default = 0 // u13, x0 for window 1, software control
12:0	R/W	719	reg_di_dcmb_wnd11 : default = 719 // u13, x1 for window 1, HSIZE-1, software control

Table 8-1479 DECOMB_WIND11 0x2d8d

Bit(s)	R/W	Default	Description
28:16	R/W	40	reg_di_dcmb_wnd12 : default = 40 // u13, y0 for window 1, software control
12:0	R/W	239	reg_di_dcmb_wnd13 : default = 239 // u13, y1 for window 1, VSIZE-1-40, software control

Table 8-1480 DECOMB_MODE 0x2d8e

Bit(s)	R/W	Default	Description
15	R/W	1	reg_di_dcmb_is_cmb_bef : default = 1 // u1, 1: decide is_cmb before cmbing refine, 0: decide is_cmb after cmbing refine
14	R/W	1	reg_di_dcmb_en0 : default = 1 // u1, enable decmobing for wind0
13	R/W	1	reg_di_dcmb_en1 : default = 1 // u1, enable decmobing for wind1
12	R/W	1	reg_di_dcmb_en2 : default = 1 // u1, enable decmobing for wind2
11:10	R/W	2	reg_di_dcmb_lpf_mod0 : default = 2 // u2, get combing free pixels of wind0 by: 0, vertical lpf, 1, edge lpf, 2,3, ei data
9:8	R/W	2	reg_di_dcmb_lpf_mod1 : default = 2 // u2, get combing free pixels of wind1 by: 0, vertical lpf, 1, edge lpf, 2,3, ei data

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Bit(s)	R/W	Default	Description
7:6	R/W	0	reg_di_dcmb_lpf_mod2 : default = 0 // u2, get combing free pixels of wind2 by: 0, vertical lpf, 1, edge lpf, 2,3, ei data
5	R/W	1	reg_di_dcmb_cmb_sel0 : default = 1 // u1, wind0 decmb based on: 0, vert cmb, 1, edge cmb
4	R/W	1	reg_di_dcmb_cmb_sel1 : default = 1 // u1, wind1 decmb based on: 0, vert cmb, 1, edge cmb
3	R/W	0	reg_di_dcmb_cmb_sel2 : default = 0 // u1, wind2 decmb based on: 0, vert cmb, 1, edge cmb
2	R/W	1	reg_di_dcmb_alp_mod0 : default = 1 // u1, wind0 decmb alpha based on: 0, user-defined, 1, motion adaptive
1	R/W	1	reg_di_dcmb_alp_mod1 : default = 1 // u1, wind1 decmb alpha based on: 0, user-defined, 1, motion adaptive
0	R/W	1	reg_di_dcmb_alp_mod2 : default = 1 // u1, wind2 decmb alpha based on: 0, user-defined, 1, motion adaptive

Table 8-1481 DECOMB_FRM_SIZE 0x2d8f

Bit(s)	R/W	Default	Description
28:16	R/W	1920	hsize_in : default = 1920 // u13, pic horz size in unit: pixel
12:0	R/W	1080	vsize_in : default = 1080 // u13, pic vert size in unit: pixel

Table 8-1482 DECOMB_HV_BLANK 0x2d90

Bit(s)	R/W	Default	Description
15:8	R/W	20	hblank_num : default = 20 // u8, hor blank time
7:0	R/W	50	vblank_num : default = 50 // u8, ver blank time

Table 8-1483 NR2_POLAR3_MODE 0x2d98

Bit(s)	R/W	Default	Description
19:18	R/W	3	reg_polar3_f02lpf_mod_0 : default = 3 //u2x2: low pass filter mode for field 0 and field2 before polar3 detection; 0 for no lpf, 1: [1 2 1]/4 vert lpf; 2: [1 2 1; 2 4 2; 1 2 1]/16 2d lpf, p1 no hlpf; 2: [1 2 1; 2 4 2; 1 2 1]/16 2d lpf, p1 [1 2 1]/4 hlpf
17:16	R/W	3	reg_polar3_f02lpf_mod_1 : default = 3 //u2x2: low pass filter mode for field 0 and field2 before polar3 detection; 0 for no lpf, 1: [1 2 1]/4 vert lpf; 2: [1 2 1; 2 4 2; 1 2 1]/16 2d lpf, p1 no hlpf; 2: [1 2 1; 2 4 2; 1 2 1]/16 2d lpf, p1 [1 2 1]/4 hlpf
15:8	R/W	5	reg_polar3_dif02_thrd_0 : default = 5 //u8x2: threshold of dif for polar3 detection except for 32 detection, only do polar3 detection on obvious motion, [0] for luma, 1[1] for chroma
7:0	R/W	5	reg_polar3_dif02_thrd_1 : default = 5 //u8x2: threshold of dif for polar3 detection except for 32 detection, only do polar3 detection on obvious motion, [0] for luma, 1[1] for chroma

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Table 8-1484 NR2_POLAR3_THRD 0x2d99

Bit(s)	R/W	Default	Description
31:24	R/W	30	reg_polar3_txtf02_thrd_0 : default = 30 //u8x2: threshold to vertical f0f2 texture, if texture larger than this threshold, will not do the polar3 decision.
23:16	R/W	30	reg_polar3_txtf02_thrd_1 : default = 30 //u8x2: threshold to vertical f0f2 texture, if texture larger than this threshold, will not do the polar3 decision.
15:8	R/W	20	reg_polar3_txtf1_thrd_0 : default = 20 //u8x2: threshold to vertical f0f2 texture, if texture larger than this threshold, will not do the polar3 decision.
7:0	R/W	20	reg_polar3_txtf1_thrd_1 : default = 20 //u8x2: threshold to vertical f0f2 texture, if texture larger than this threshold, will not do the polar3 decision.

Table 8-1485 NR2_POLAR3_PARA0 0x2d9a

Bit(s)	R/W	Default	Description
31:28	R/W	6	reg_polar3_rate0_0 : default = 6 //u4x2: delt = rate*dif02/32, e.g. f2<f0, if f1 within((f0+f2)/2 - delt), ((f0+f2)/2 + delt), then polar3_smoothmv++;
27:24	R/W	6	reg_polar3_rate0_1 : default = 6 //u4x2: delt = rate*dif02/32, e.g. f2<f0, if f1 within((f0+f2)/2 - delt), ((f0+f2)/2 + delt), then polar3_smoothmv++;
23:20	R/W	8	reg_polar3_rate1_0 : default = 8 //u4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< ((f0+f2)/2 - delt), then polar3_m1++; if f1>((f0+f2)/2 + delt), then polar3_p1++;
19:16	R/W	8	reg_polar3_rate1_1 : default = 8 //u4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< ((f0+f2)/2 - delt), then polar3_m1++; if f1>((f0+f2)/2 + delt), then polar3_p1++;
15:12	R/W	2	reg_polar3_rate2_0 : default = 2 //u4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< (f2 - delt- ofset2), then polar3_m2++; if f1>((f0 + delt+ ofset2), then polar3_p2++;
11:8	R/W	2	reg_polar3_rate2_1 : default = 2 //u4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< (f2 - delt- ofset2), then polar3_m2++; if f1>((f0 + delt+ ofset2), then polar3_p2++;
7:4	R/W	1	reg_polar3_ofst1_0 : default = 1 //s4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< (f2 - delt- ofset2), then polar3_m2++; if f1>((f0 + delt+ ofset2), then polar3_p2++;
3:0	R/W	1	reg_polar3_ofst1_1 : default = 1 //s4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< (f2 - delt- ofset2), then polar3_m2++; if f1>((f0 + delt+ ofset2), then polar3_p2++;

Table 8-1486 NR2_POLAR3_PARA1 0x2d9b

Bit(s)	R/W	Default	Description
31:24	R/W	48	reg_polar3_rate3_0 : default = 48 //u8x2: delt = rate*dif02/32, e.g. f2<f0, if f1<(f2 - delt- ofset3) or f1>((f0 + delt+ofst3), then polar3_32++;
23:16	R/W	48	reg_polar3_rate3_1 : default = 48 //u8x2: delt = rate*dif02/32, e.g. f2<f0, if f1<(f2 - delt- ofset3) or f1>((f0 + delt+ofst3), then polar3_32++;
15:12	R/W	2	reg_polar3_ofst3_0 : default = 2 //s4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< (f2 - delt-ofst3) or f1>((f0 + delt+ofst3), then polar3_32++;
11:8	R/W	2	reg_polar3_ofst3_1 : default = 2 //s4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< (f2 - delt-ofst3) or f1>((f0 + delt+ofst3), then polar3_32++;
7:4	R/W	2	reg_polar3_ofst2_0 : default = 2 //s4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< (f2 - delt- ofset2), then polar3_m2++; if f1>((f0 + delt+ ofset2), then polar3_p2++;
3:0	R/W	2	reg_polar3_ofst2_1 : default = 2 //s4x2: delt = rate*dif02/32, e.g. f2<f0, if f1< (f2 - delt- ofset2), then polar3_m2++; if f1>((f0 + delt+ ofset2), then polar3_p2++;

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Table 8-1487 NR2_POLAR3_CTRL 0x2d9c

Bit(s)	R/W	Default	Description
16	R.O	0	reg_polar3_ro_reset : default = 0 //u1: reset signal of the polar3 read only registers
15:8	R/W	10	reg_polar3_h_mute : default = 10 //u8: horizontally pixels to mute for left right sides for polar3 detection;
7:0	R/W	10	reg_polar3_v_mute : default = 10 //u8: horizontally pixels to mute for top and bottom sides for polar3 detection;

Table 8-1488 NR2_RO_POLAR3_NUMOFPIX 0x2d9d

Bit(s)	R/W	Default	Description
23:0	R.O	0	ro_polar3_numofpix : default = 0 //u24, number of pixels detected as polar3

Table 8-1489 NR2_RO_POLAR3_SMOOTHMV 0x2d9e

Bit(s)	R/W	Default	Description
23:0	R.O	0	ro_polar3_smoothmv : default = 0 //u24, number of pixels with smooth mv, F(t) is close between avg of f(t-1) and f(t+1);

Table 8-1490 NR2_RO_POLAR3_M1 0x2d9f

Bit(s)	R/W	Default	Description
23:0	R.O	0	ro_polar3_m1 : default = 0 //u24, number of pixels with F(t) is close to f(t-1) instead of f(t+1), but in between [f(t-1), f(t+1)];

Table 8-1491 NR2_RO_POLAR3_P1 0x2da0

Bit(s)	R/W	Default	Description
23:0	R.O	0	ro_polar3_p1 : default = 0 //u24, number of pixels with F(t) is close to f(t+1) instead of f(t-1), but in between [f(t-1), f(t+1)];

Table 8-1492 NR2_RO_POLAR3_M2 0x2da1

Bit(s)	R/W	Default	Description
23:0	R.O	0	ro_polar3_m2 : default = 0 //u24, number of pixels with F(t) is close to f(t-1) instead of f(t+1), but out side of (f(t-1), f(t+1));

Table 8-1493 NR2_RO_POLAR3_P2 0x2da2

Bit(s)	R/W	Default	Description
23:0	R.O	0	ro_polar3_p2 : default = 0 //u24, number of pixels with F(t) is close to f(t+1) instead of f(t-1), but out side of (f(t-1), f(t+1));

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Table 8-1494 NR2_RO_POLAR3_32 0x2da3

Bit(s)	R/W	Default	Description
23:0	R.O	0	ro_polar3_32 : default = 0 //u24, number of pixels with F(t) far from [f(t-1),f(t+1)] and f(t-1) is close to f(t+1);

Table 8-1495 NR4_DRT_CTRL 0x2da4

Bit(s)	R/W	Default	Description
31:24	R/W	16	reg_nr4_ydrt_3line_ssd_gain : // unsigned , default = 16 gain to max ssd normalized 16 as '1'
23:16	R/W	16	reg_nr4_ydrt_5line_ssd_gain : // unsigned , default = 16 gain to max ssd normalized 16 as '1'
14:13	R/W	1	reg_nr4_drt_yhsad_mode : // unsigned , default = 1 mode for luma horiztonal sad calc., 0: no vertical lpf, 1: vertical [1 2 1], 2 or 3: vertical [1 2 2 2 1] if 5 lines
12:11	R/W	1	reg_nr4_drt_chsad_mode : // unsigned , default = 1 mode for chroma horiztonal sad calc., 0: no vertical lpf, 1: vertical [1 2 1], 2 or 3: vertical [1 2 2 2 1] if 5 lines
10	R/W	1	reg_nr4_drt_yhsad_hlpf : // unsigned , default = 1 hlpf for luma hsad of drt calculation, 0: no lpf, 1: with [1 2 1] hlpf
9	R/W	1	reg_nr4_drt_yvsad_hlpf : // unsigned , default = 1 hlpf for luma vsad of drt calculation, 0: no lpf, 1: with [1 2 1] hlpf
8	R/W	1	reg_nr4_drt_ydsad_hlpf : // unsigned , default = 1 hlpf for luma dsad of drt calculation, 0: no lpf, 1: with [1 2 1] hlpf
7	R/W	1	reg_nr4_drt_chsad_hlpf : // unsigned , default = 1 hlpf for chrome hsad of drt calculation, 0: no lpf, 1: with [1 2 1] hlpf
6	R/W	1	reg_nr4_drt_cvsad_hlpf : // unsigned , default = 1 hlpf for chroma vsad of drt calculation, 0: no lpf, 1: with [1 2 1] hlpf
5	R/W	1	reg_nr4_drt_cdsad_hlpf : // unsigned , default = 1 hlpf for chroma dsad of drt calculation, 0: no lpf, 1: with [1 2 1] hlpf
4	R/W	1	reg_nr4_ydrt_dif_mode : // unsigned , default = 1 0:y_dif, 1: y_dif + (u_dif + v_dif)/2
3: 2	R/W	2	reg_nr4_cdrt_dif_mode : // unsigned , default = 2 0:(u_dif + v_dif), 1: y_dif/4 + (u_dif + v_dif)*3/4, 2:y_dif/2 + (u_dif + v_dif)/2, 3: y_dif (not recommended)
1:0	R/W		reserved

Table 8-1496 NR4_DRT_YSAD_GAIN 0x2da5

Bit(s)	R/W	Default	Description
31:24	R/W	16	reg_nr4_ysad_hrz_gain : // unsigned , default = 16 gain for horizontal sad, 16 normalized to "1"
23:16	R/W	20	reg_nr4_ysad_diag_gain : // unsigned , default = 20 gain for diagonal sad, 16 normalized to "1"
15: 8	R/W	16	reg_nr4_ysad_vrt_gain : // unsigned , default = 16 gain for vertical sad, 16 normalized to "1"
5: 0	R/W	6	reg_nr4_drt_ysad_core_rate : // unsigned , default = 6 rate of coring for sad(theta) - sad(theta+pi/2)*rate/64

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Table 8-1497 NR4_DRT_CSAD_GAIN 0x2da6

Bit(s)	R/W	Default	Description
31:24	R/W	16	reg_nr4_csad_hrz_gain : // unsigned , default = 16 gain for horizontal sad, 16 normalized to "1"
23:16	R/W	20	reg_nr4_csad_diag_gain : // unsigned , default = 20 gain for diagonal sad, 16 normalized to "1"
15: 8	R/W	16	reg_nr4_csad_vrt_gain : // unsigned , default = 16 gain for vertical sad, 16 normalized to "1"
5: 0	R/W	6	reg_nr4_drt_csad_core_rate : // unsigned , default = 6 rate of coring for sad(theta) - sad(theta+pi/2)*rate/64

Table 8-1498 NR4_DRT_SAD_ALP_CORE 0x2da7

Bit(s)	R/W	Default	Description
23:20	R/W	0	reg_nr4_ydrt_alp_core_rate : // unsigned , default = 0 luma ratio to min_err, alpha = (min_err - (max_err - min_err)*rate + ofst)/max_err * 64; dft = 0/32
19:16	R/W	0	reg_nr4_cdrt_alp_core_rate : // unsigned , default = 0 chroma ratio to min_err, alpha = (min_err - (max_err - min_err)*rate + ofst)/max_err * 64; dft = 0/32
13: 8	R/W	10	reg_nr4_ydrt_alp_core_ofst : // unsigned , default = 10 luma offset to min_err, alpha = (min_err - (max_err - min_err)*rate + ofst)/max_err * 64; dft = 10
5: 0	R/W	10	reg_nr4_cdrt_alp_core_ofst : // unsigned , default = 10 chroma offset to min_err, alpha = (min_err - (max_err - min_err)*rate + ofst)/max_err * 64; dft = 10

Table 8-1499 NR4_DRT_ALP_MINMAX 0x2da8

Bit(s)	R/W	Default	Description
29:24	R/W	0	reg_nr4_ydrt_alp_min : // unsigned , default = 0 luma min value of alpha, dft = 0
21:16	R/W	63	reg_nr4_ydrt_alp_max : // unsigned , default = 63 luma max value of alpha, dft = 63
13: 8	R/W	0	reg_nr4_cdrt_alp_min : // unsigned , default = 0 chroma min value of alpha, dft = 0
5: 0	R/W	63	reg_nr4_cdrt_alp_max : // unsigned , default = 63 chroma max value of alpha, dft = 63

Table 8-1500 NR4_SNR_CTRL_REG 0x2da9

Bit(s)	R/W	Default	Description
12	R/W	1	reg_nr4_bet2_sel : // unsigned , default = 1
11: 9	R/W	0	reg_nr4_snr2_sel_mode : // unsigned , default = 0 0: no filter, 1: adpgau, adp_drt_lpf blend; 2: adpgau, drt4_lpf blend; 3: adp_drt_lpf method, 4: drt4_lpf method, 5: adp_drt_ //original image blend, 6: drt4_lpf, original image blend, 7: adpgau method; dft=1
8	R/W	1	reg_nr4_snr2_gaulpf_mode : // unsigned , default = 1 0: 3*5 or 5*5 gaussian lpf; 1: 3*3 (window size) gaussian lpf; dft=1
7: 6	R/W	3	reg_nr4_snr2_alpha0_sad_mode : // unsigned , default = 3 0: max_sad*max_ssd; 1: max_sad*max_sad; 2: adp_max_sad*max_ssd; 3: adp_max_sad*adp_max_sad dft=3

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Bit(s)	R/W	Default	Description
5: 4	R/W	2	reg_nr4_snr2_alpha1_sad_mode : // unsigned , default = 2 0: max_sad; 1: cross_max_sad; 2 or 3: adp_sad dft=2
1: 0	R/W	3	reg_nr4_snr2_adp_drtlpf_mode : // unsigned , default = 3 0: adp_drtlpf [2 1 1]/4, 1: adp_drtlpf [4 2 1 1]/8; 2: adp_drtlpf [2 2 2 1 1]/8; 3: adp_drtlpf [7 7 7 6 5]/32; dft=3;

Table 8-1501 NR4_SNR_ALPHA0_MAX_MIN 0x2daa

Bit(s)	R/W	Default	Description
29:23	R/W	127	reg_nr4_snr2_alp0_ymin : // unsigned , default = 127 normalized to 128 as '1'
22:16	R/W	127	reg_nr4_snr2_alp0_ymax : // unsigned , default = 127 normalized to 128 as '1'
13: 7	R/W	127	reg_nr4_snr2_alp0_cmin : // unsigned , default = 127 normalized to 128 as '1'
6: 0	R/W	127	reg_nr4_snr2_alp0_cmax : // unsigned , default = 127 normalized to 128 as '1'

Table 8-1502 NR4_ALP0C_ERR2CURV_LIMIT0 0x2dab

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_snr2_alp0_minerr_cpar0 : // unsigned , default = 0 threshold0 of curve to map mierr to alp0 for chroma channel, this will be set value of flat region mierr that no need blur.
23:16	R/W	25	reg_nr4_snr2_alp0_minerr_cpar1 : // unsigned , default = 25 threshold1 of curve to map mierr to alp0 for chroma channel,this will be set value of texture region mierr that can not blur.
15: 8	R/W	40	reg_nr4_snr2_alp0_minerr_cpar5 : // unsigned , default = 40 rate0 (for mierr<th0) of curve to map mierr to alp0 for chroma channel. the larger of the value, the deep of the slope. 0~255.
7: 0	R/W	40	reg_nr4_snr2_alp0_minerr_cpar6 : // unsigned , default = 40 rate1 (for mierr>th1) of curve to map mierr to alp0 for chroma channel. the larger of the value, the deep of the slope. 0~255.

Table 8-1503 NR4_ALP0C_ERR2CURV_LIMIT1 0x2dac

Bit(s)	R/W	Default	Description
23:16	R/W	127	reg_nr4_snr2_alp0_minerr_cpar2 : // unsigned , default = 127 level limit(for mierr<th0) of curve to map mierr to alp0 for chroma channel, that we can do for flat region. 0~255.
15: 8	R/W	0	reg_nr4_snr2_alp0_minerr_cpar3 : // unsigned , default = 0 level limit(for th0<mierr<th1) of curve to map mierr to alp0 for chroma channel, that we can do for misc region. 0~255.
7: 0	R/W	127	reg_nr4_snr2_alp0_minerr_cpar4 : // unsigned , default = 127 level limit(for mierr>th1) of curve to map mierr to alp0 for chroma channel, that we can do for texture region. 0~255.

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Table 8-1504 NR4_ALP0Y_ERR2CURV_LIMIT0 0x2dad

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_snr2_alp0_minerr_ypar0 : // unsigned , default = 0 threshold0 of curve to map mierr to alp0 for luma channel, this will be set value of flat region mierr that no need blur. 0~255.
23:16	R/W	25	reg_nr4_snr2_alp0_minerr_ypar1 : // unsigned , default = 25 threshold1 of curve to map mierr to alp0 for luma channel,this will be set value of texture region mierr that can not blur.
15: 8	R/W	40	reg_nr4_snr2_alp0_minerr_ypar5 : // unsigned , default = 40 rate0 (for mierr<th0) of curve to map mierr to alp0 for luma channel. the larger of the value, the deep of the slope. 0~255.
7: 0	R/W	40	reg_nr4_snr2_alp0_minerr_ypar6 : // unsigned , default = 40 rate1 (for mierr>th1) of curve to map mierr to alp0 for luma channel. the larger of the value, the deep of the slope. 0~255.

Table 8-1505 NR4_ALP0Y_ERR2CURV_LIMIT1 0x2dae

Bit(s)	R/W	Default	Description
23:16	R/W	127	reg_nr4_snr2_alp0_minerr_ypar2 : // unsigned , default = 127 level limit(for mierr < th0) of curve to map mierr to alp0 for luma channel, set to alp0 that we can do for flat region. 0~255.
15: 8	R/W	0	reg_nr4_snr2_alp0_minerr_ypar3 : // unsigned , default = 0 level limit(for th0<mierr < th1) of curve to map mierr to alp0 for luma channel, alp0 that we can do for misc region. 0~255.
7: 0	R/W	127	reg_nr4_snr2_alp0_minerr_ypar4 : // unsigned , default = 127 level limit(for mierr > th1) of curve to map mierr to alp0 for luma channel, alp0 that we can do for texture region. 0~255.

Table 8-1506 NR4_SNR_ALPA1_RATE_AND_OFST 0x2daf

Bit(s)	R/W	Default	Description
23:18	R/W	0	reg_nr4_snr2_alp1_ycore_rate : // unsigned , default = 0 normalized 64 as "1"
17:12	R/W	0	reg_nr4_snr2_alp1_ccore_rate : // unsigned , default = 0 normalized 64 as "1"
11: 6	R/W	3	reg_nr4_snr2_alp1_ycore_ofst : // signed , default = 3 normalized 64 as "1"
5: 0	R/W	3	reg_nr4_snr2_alp1_ccore_ofst : // signed , default = 3 normalized 64 as "1"

Table 8-1507 NR4_SNR_ALPHA1_MAX_MIN 0x2db0

Bit(s)	R/W	Default	Description
23:18	R/W	0	reg_nr4_snr2_alp1_ymin : // unsigned , default = 0 normalized to 64 as '1'
17:12	R/W	63	reg_nr4_snr2_alp1_ymax : // unsigned , default = 63 normalized to 64 as '1'
11: 6	R/W	0	reg_nr4_snr2_alp1_cmin : // unsigned , default = 0 normalized to 64 as '1'
5: 0	R/W	63	reg_nr4_snr2_alp1_cmax : // unsigned , default = 63 normalized to 64 as '1'

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Table 8-1508 NR4_ALP1C_ERR2CURV_LIMIT0 0x2db1

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_snr2_alp1_minerr_cpar0 : // unsigned , default = 0 annel, this will be set value of flat region mierr that no need directional NR. 0~255.
23:16	R/W	24	reg_nr4_snr2_alp1_minerr_cpar1 : // unsigned , default = 24 hannel,this will be set value of texture region mierr that can not do directional NR. 0~255.
15: 8	R/W	0	reg_nr4_snr2_alp1_minerr_cpar5 : // unsigned , default = 0 a/chroma channel. the larger of the value, the deep of the slope.
7: 0	R/W	20	reg_nr4_snr2_alp1_minerr_cpar6 : // unsigned , default = 20 a/chroma channel. the larger of the value, the deep of the slope. 0~255

Table 8-1509 NR4_ALP1C_ERR2CURV_LIMIT1 0x2db2

Bit(s)	R/W	Default	Description
23:16	R/W	0	reg_nr4_snr2_alp1_minerr_cpar2 : // unsigned , default = 0 will be set to alp1 that we can do for flat region. 0~255.
15: 8	R/W	16	reg_nr4_snr2_alp1_minerr_cpar3 : // unsigned , default = 16 this will be set to alp1 that we can do for misc region. 0~255.
7: 0	R/W	63	reg_nr4_snr2_alp1_minerr_cpar4 : // unsigned , default = 63 will be set to alp1 that we can do for texture region. 0~255.255 before

Table 8-1510 NR4_ALP1Y_ERR2CURV_LIMIT0 0x2db3

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_snr2_alp1_minerr_ypar0 : // unsigned , default = 0 thra/chroma channel, this will be set value of flat region mierr that no need directional NR. 0~255.
23:16	R/W	24	reg_nr4_snr2_alp1_minerr_ypar1 : // unsigned , default = 24 thra/chroma channel,this will be set value of texture region mierr that can not do directional NR. 0~255.
15: 8	R/W	0	reg_nr4_snr2_alp1_minerr_ypar5 : // unsigned , default = 0 ratlp1 for luma/chroma channel. the larger of the value, the deep of the slope.
7: 0	R/W	20	reg_nr4_snr2_alp1_minerr_ypar6 : // unsigned , default = 20 ratlp1 for luma/chroma channel. the larger of the value, the deep of the slope. 0~255

Table 8-1511 NR4_ALP1Y_ERR2CURV_LIMIT1 0x2db4

Bit(s)	R/W	Default	Description
23:16	R/W	0	reg_nr4_snr2_alp1_minerr_ypar2 : // unsigned , default = 0 lev to alp1 for luma/chroma channel, this will be set to alp1 that we can do for flat region. 0~255.
15: 8	R/W	16	reg_nr4_snr2_alp1_minerr_ypar3 : // unsigned , default = 16 levierr to alp1 for luma/chroma channel, this will be set to alp1 that we can do for misc region. 0~255.
7: 0	R/W	63	reg_nr4_snr2_alp1_minerr_ypar4 : // unsigned , default = 63 lev to alp1 for luma/chroma channel, this will be set to alp1 that we can do for texture region. 0~255.255 before

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Table 8-1512 NR4_MTN_CTRL 0x2db5

Bit(s)	R/W	Default	Description
1	R/W	1	reg_nr4_mtn_ref_en : // unsigned , default = 1 enable motion refinement, dft = 1
0	R/W	0	reg_nr4_mtn_ref_bet_sel : // unsigned , default = 0 beta selection mode for motion refinement, 0: beta1, 1: beta2, dft = 0

Table 8-1513 NR4_MTN_REF_PAR0 0x2db6

Bit(s)	R/W	Default	Description
31:24	R/W	24	reg_nr4_mtn_ref_par0 : // unsigned , default = 24 par0 for beta to gain, dft =
23:16	R/W	60	reg_nr4_mtn_ref_par1 : // unsigned , default = 60 par1 for beta to gain, dft =
15: 8	R/W	4	reg_nr4_mtn_ref_par2 : // unsigned , default = 4 par2 for beta to gain, dft =
7: 0	R/W	32	reg_nr4_mtn_ref_par3 : // unsigned , default = 32 par3 for beta to gain, dft =

Table 8-1514 NR4_MTN_REF_PAR1 0x2db7

Bit(s)	R/W	Default	Description
23:16	R/W	128	reg_nr4_mtn_ref_par4 : // unsigned , default = 128 par4 for beta to gain, dft =
15: 8	R/W	40	reg_nr4_mtn_ref_par5 : // unsigned , default = 40 par5 for beta to gain, dft =
7: 0	R/W	20	reg_nr4_mtn_ref_par6 : // unsigned , default = 20 par6 for beta to gain, dft =

Table 8-1515 NR4_MCNR_LUMA_ENH_CTRL 0x2db8

Bit(s)	R/W	Default	Description
3	R/W	1	reg_nr4_luma_plus_en : // unsigned , default = 1 enable luma enhancement, dft = 1
2	R/W	1	reg_nr4_luma_plus_wt_mode : // unsigned , default = 1 luma weight calc mode, 0:sqrt(1+x^2), 1: 1+abs(x), dft = 0
1: 0	R/W	1	reg_nr4_luma_plus_orient_mode : // unsigned , default = 1 0: only use previous orient for pre and cur luma plus, 1: 0: only use current orient for pre and cur luma plus

Table 8-1516 NR4_MCNR_LUMA_STAT_LIMTX 0x2db9

Bit(s)	R/W	Default	Description
29:16	R/W	8	reg_nr4_luma_plus_xst : // unsigned , default = 8 start for luma plus statistic, dft = 8
13: 0	R/W	711	reg_nr4_luma_plus_xed : // unsigned , default = 711 end for luma plus statistic, dft = HSIZE-8-1;

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Table 8-1517 NR4_MCNR_LUMA_STAT_LIMTY 0x2dba

Bit(s)	R/W	Default	Description
29:16	R/W	8	reg_nr4_luma_plus_yst : // unsigned , default = 8 start for luma plus statistic, dft = 8
13: 0	R/W	231	reg_nr4_luma_plus_yed : // unsigned , default = 231 end for luma plus statistic, dft = VSIZE-8-1

Table 8-1518 NR4_MCNR_LUMA_DIF_CALC 0x2dbb

Bit(s)	R/W	Default	Description
29:24	R/W	8	reg_nr4_luma_plus_ugain : // unsigned , default = 8 U's gain for luma enhancement, 16 normalized as '1'
21:16	R/W	8	reg_nr4_luma_plus_vgain : // unsigned , default = 8 V's gain for luma enhancement, 16 normalized as '1'
15: 8	R/W	2	reg_nr4_luma_plus_ycor_thd : // unsigned , default = 2 Y coring threshold for difference calc., dft = 0
7: 0	R/W	0	reg_nr4_luma_plus_ccor_thd : // unsigned , default = 0 C coring threshold for difference calc., dft = 0

Table 8-1519 NR4_MCNR_LUMAPRE_CAL_PRAM 0x2dbc

Bit(s)	R/W	Default	Description
25:24	R/W	0	reg_nr4_pre_u_orient : // signed , default = 0 orientation of previous U, initial to 0, and will be updated by software
17:16	R/W	0	reg_nr4_pre_v_orient : // signed , default = 0 orientation of previous V, initial to 0, and will be updated by software
15: 8	R/W	0	reg_nr4_pre_u_mean : // unsigned , default = 0 mean of previous U, initial to 0, and will be updated by software
7: 0	R/W	0	reg_nr4_pre_v_mean : // unsigned , default = 0 mean of previousV, initial to 0, and will be updated by software

Table 8-1520 NR4_MCNR_LUMACUR_CAL_PRAM 0x2dbd

Bit(s)	R/W	Default	Description
25:24	R/W	0	reg_nr4_cur_u_orient : // signed , default = 0 orientation of current U, initial to 0, and will be updated by software
17:16	R/W	0	reg_nr4_cur_v_orient : // signed , default = 0 orientation of current V, initial to 0, and will be updated by software
15: 8	R/W	0	reg_nr4_cur_u_mean : // unsigned , default = 0 mean of current U, initial to 0, and will be updated by software
7: 0	R/W	0	reg_nr4_cur_v_mean : // unsigned , default = 0 mean of current, initial to 0, and will be updated by software

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Table 8-1521 NR4_MCNR_MV_CTRL_REG 0x2dbe

Bit(s)	R/W	Default	Description
13:12	R/W	2	reg_nr4_sad_bitw : // unsigned , default = 2 sad bit width (8 + x) before clip to u8, dft = 1
11: 4	R/W	64	reg_nr4_glb_gain : // unsigned , default = 64 global gain calc. by software, 64 is normalized as '1'
3: 0	R/W	8	reg_nr4_mv_err_rsft : // unsigned , default = 8 right shift for mv err calc., dft = 9

Table 8-1522 NR4_MCNR_MV_GAIN0 0x2dbf

Bit(s)	R/W	Default	Description
31:28	R/W	1	reg_nr4_lftmvx_gain : // unsigned , default = 1 left mvx gain for err calc., dft = 1
27:24	R/W	1	reg_nr4_lftmvy_gain : // unsigned , default = 1 left mvy gain for err calc., dft = 1
23:20	R/W	5	reg_nr4_zmvx_gain : // unsigned , default = 5 zero mvx gain for err calc., dft = 2
19:16	R/W	5	reg_nr4_zmvy_gain : // unsigned , default = 5 zero mvy gain for err calc., dft = 4
15:12	R/W	2	reg_nr4_lmvx0_gain : // unsigned , default = 2 line mvx0 gain for err calc., dft = 1
11: 8	R/W	2	reg_nr4_lmvx1_gain : // unsigned , default = 2 line mvx1 gain for err calc., dft = 1
7: 4	R/W	2	reg_nr4_lmvy0_gain : // unsigned , default = 2 line mvy0 gain for err calc., dft = 1
3: 0	R/W	2	reg_nr4_lmvy1_gain : // unsigned , default = 2 line mvy1 gain for err calc., dft = 1

Table 8-1523 NR4_MCNR_LMV_PARM 0x2dc0

Bit(s)	R/W	Default	Description
31:28	R/W	3	reg_nr4_lmv_rt0 : // unsigned , default = 3 ratio of max lmv
27:24	R/W	3	reg_nr4_lmv_rt1 : // unsigned , default = 3 ratio of second max lmv
21:16	R/W	16	reg_nr4_lmv_num_lmt0 : // unsigned , default = 16 lmv0 least/limit number of (total number - zero_bin)
13: 8	R/W	8	reg_nr4_lmv_num_lmt1 : // unsigned , default = 8 lmv1 least/limit number of (total number - zero_bin - max0)
1: 0	R/W	1	reg_nr4_max_sad_rng : // unsigned , default = 1 search range of max2 sad in small region, dft = 1

Table 8-1524 NR4_MCNR_ALP0_REG 0x2dc1

Bit(s)	R/W	Default	Description
25	R/W	1	reg_nr4_alp0_fail_chk : // unsigned , default = 1 enable check for alp0 fail status
24	R/W	1	reg_nr4_bet0_coef_ref_en : // unsigned , default = 1 bet1 refinement by coef_blt
23:16	R/W	255	reg_nr4_alp0_posad_gain : // unsigned , default = 255 the sad (norm) gain for pixel pointed by MV;
9: 8	R/W	0	reg_nr4_alp0_norm_mode : // unsigned , default = 0 alp0 select sad norm mode, 0: disable, 1: enable dc norm, 2: enable ac norm, 3: enable both (dc/ac) norm, dft = 3
5: 0	R/W	16	reg_nr4_alp0_norm_gain : // unsigned , default = 16 alp0 gain for sad norm, '32' as '1', dft = 1

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Table 8-1525 NR4_MCNR_ALP1_AND_BET0_REG 0x2dc2

Bit(s)	R/W	Default	Description
25:24	R/W	3	reg_nr4_alp1_norm_mode : // unsigned , default = 3 alp1 select sad norm mode, 0: disable, 1: enable dc norm, 2: enable ac norm, 3: enable both (dc/ac) norm, dft = 3
21:16	R/W	3	reg_nr4_alp1_norm_gain : // unsigned , default = 3 alp1 gain for sad norm, '32' as '1', dft = 1
9: 8	R/W	3	reg_nr4_bet0_norm_mode : // unsigned , default = 3 bet0 select sad norm mode, 0: disable, 1: enable dc norm, 2: enable ac norm, 3: enable both (dc/ac) norm, dft = 3
5: 0	R/W	8	reg_nr4_bet0_norm_gain : // unsigned , default = 8 bet0 gain for sad norm, '32' as '1', dft = 1

Table 8-1526 NR4_MCNR_BET1_AND_BET2_REG 0x2dc3

Bit(s)	R/W	Default	Description
25:24	R/W	3	reg_nr4_bet1_norm_mode : // unsigned , default = 3 bet1 select sad norm mode, 0: disable, 1: enable dc norm, 2: enable ac norm, 3: enable both (dc/ac) norm, dft = 3
21:16	R/W	8	reg_nr4_bet1_norm_gain : // unsigned , default = 8 bet1 gain for sad norm, '32' as '1', dft = 1
9: 8	R/W	0	reg_nr4_bet2_norm_mode : // unsigned , default = 0 bet2 select sad norm mode, 0: disable, 1: enable dc norm, 2: enable ac norm, 3: enable both (dc/ac) norm, dft = 3
5: 0	R/W	16	reg_nr4_bet2_norm_gain : // unsigned , default = 16 bet2 gain for sad norm, '32' as '1', dft = 1

Table 8-1527 NR4_MCNR_AC_DC_CRTL 0x2dc4

Bit(s)	R/W	Default	Description
11	R/W	1	reg_nr4_dc_mode : // unsigned , default = 1 mode for dc selection,0: Y_lpf, 1: Y_lpf + (U_Lpf+V_lpf)/2,
10	R/W	1	reg_nr4_ac_mode : // unsigned , default = 1 mode for ac selection, 0: Y_abs_dif, 1: Y_abs_dif + (U_abs_dif + V_abs_dif)/2
9	R/W	0	reg_nr4_dc_sel : // unsigned , default = 0 selection mode for dc value, 0: 3x5, 1: 5x5, dft = 1
8	R/W	0	reg_nr4_ac_sel : // unsigned , default = 0 selection mode for ac value, 0: 3x5, 1: 5x5, dft = 1
6: 4	R/W	2	reg_nr4_dc_shft : // unsigned , default = 2 right shift for dc value, dft = 2
2: 0	R/W	0	reg_nr4_ac_shft : // unsigned , default = 0 right shift for ac value, dft = 2

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Table 8-1528 NR4_MCNR_CM_CTRL0 0x2dc5

Bit(s)	R/W	Default	Description
28	R/W	0	reg_nr4_cm_skin_prc_bet0 : // unsigned , default = 0 enable skin tone processing for mcnr bet0 calc., dft = 1
27:26	R/W	1	reg_nr4_cm_chrm_sel : // unsigned , default = 1 chrome selection for color match, 0: 1x1, 1: 3X3LPF, 2: 3x5LPF, 3: 5x5LPF for 5lines, 3x5LPF for 3lines, dft = 3
25:24	R/W	1	reg_nr4_cm_luma_sel : // unsigned , default = 1 luma selection for color match, 0: 1x1, 1: 3X3LPF, 2: 3x5LPF, 3: 5x5LPF for 5lines, 3x5LPF for 3lines, dft = 3
23:21	R/W	3	reg_nr4_cm_skin_rshft_bet0 : // unsigned , default = 3 right shift for bet0's skin color gains, dft = 3
20	R/W	1	reg_nr4_cm_var_sel : // unsigned , default = 1 variation selection for color match, 0: 3x5, 1: 5x5 for 5lines, 3x5 for 3lines, dft = 1
19	R/W	1	reg_nr4_cm_green_prc_bet0 : // unsigned , default = 1 enable green processing for mcnr bet0 calc., dft = 1
18:16	R/W	4	reg_nr4_cm_green_rshft_bet0 : // unsigned , default = 4 right shift for bet0's green color gains, dft = 4
15:14	R/W	2	reg_nr4_preflt_mod : // unsigned , default = 2 pre filter mode in mcnr, 0: mv pointed pixel, 1: bilater filter
13:12	R/W	1	reg_nr4_alp1_mode : // unsigned , default = 1 mode for alpha1's sad selection, 0: max sad, 1: three min sads, 2: min sad, 3: co sad
9: 8	R/W	0	reg_nr4_bet0_mode : // unsigned , default = 0 mode for bet0's sad selection, 0: max sad, 1: three min sads, 2: min sad, 3: co sad, else: (co sad) - (min sad)
5: 4	R/W	2	reg_nr4_bet1_mode : // unsigned , default = 2 mode for bet1's sad selection, 0: max sad, 1: three min sads, 2: min sad, 3: co sad, else: (co sad) - (min sad)
1: 0	R/W	1	reg_nr4_bet2_mode : // unsigned , default = 1 mode for bet2's sad selection, 0: max sad, 1: three min sads, 2: min sad, 3: co sad, else: (co sad) - (min sad)

Table 8-1529 NR4_MCNR_CM_PRAM 0x2dc6

Bit(s)	R/W	Default	Description
29	R/W	1	reg_nr4_cm_blue_prc_alp0 : // unsigned , default = 1 enable blue processing for mcnr alpha0 calc., dft = 1
28	R/W	1	reg_nr4_cm_blue_prc_alp1 : // unsigned , default = 1 enable blue processing for mcnr alpha1 calc., dft = 1
27	R/W	1	reg_nr4_cm_skin_prc_alp0 : // unsigned , default = 1 enable skin tone processing for mcnr alpha0 calc., dft = 1
26	R/W	1	reg_nr4_cm_green_prc_alp0 : // unsigned , default = 1 enable green processing for mcnr alpha0 clac., dft = 1
25	R/W	1	reg_nr4_cm_skin_prc_alp1 : // unsigned , default = 1 enable skin tone processing for mcnr alpha0 calc., dft = 1
24	R/W	1	reg_nr4_cm_green_prc_alp1 : // unsigned , default = 1 enable green processing for mcnr alpha1 clac., dft = 1
23:20	R/W	13	reg_nr4_cm_blue_hue_st : // unsigned , default = 13 hue start of blue, dft =
19:16	R/W	15	reg_nr4_cm_blue_hue_ed : // unsigned , default = 15 hue end of blue, dft =
15:12	R/W	7	reg_nr4_cm_green_hue_st : // unsigned , default = 7 hue start of green, dft =
11: 8	R/W	10	reg_nr4_cm_green_hue_ed : // unsigned , default = 10 hue end of green, dft =

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Bit(s)	R/W	Default	Description
7: 4	R/W	5	reg_nr4_cm_skin_hue_st : // unsigned , default = 5 hue start of skin, dft =
3: 0	R/W	6	reg_nr4_cm_skin_hue_ed : // unsigned , default = 6 hue end of skin, dft =

Table 8-1530 NR4_MCNR_CM_RSHFT_ALP0 0x2dc7

Bit(s)	R/W	Default	Description
27:25	R/W	5	reg_nr4_cm_blue_rshft_bet0 : // unsigned , default = 5 right shift for bet0's blue color gains, dft = 5
24	R/W	1	reg_nr4_cm_blue_prc_bet0 : // unsigned , default = 1 enable blue processing for mcnr bet0 calc., dft = 1
22:20	R/W	5	reg_nr4_cm_blue_rshft_alp0 : // unsigned , default = 5 right shift for alpha0/1's blue color gains, dft = 5
18:16	R/W	5	reg_nr4_cm_blue_rshft_alp1 : // unsigned , default = 5 right shift for alpha0/1's blue color gains, dft = 5
14:12	R/W	4	reg_nr4_cm_green_rshft_alp0 : // unsigned , default = 4 right shift for alpha0/1's green color gains, dft = 4
10: 8	R/W	4	reg_nr4_cm_green_rshft_alp1 : // unsigned , default = 4 right shift for alpha0/1's green color gains, dft = 4
6: 4	R/W	3	reg_nr4_cm_skin_rshft_alp0 : // unsigned , default = 3 right shift for alpha0/1's skin color gains, dft = 3
2: 0	R/W	3	reg_nr4_cm_skin_rshft_alp1 : // unsigned , default = 3 right shift for alpha0/1's skin color gains, dft = 3

Table 8-1531 NR4_MCNR_BLUE_CENT 0x2dc8

Bit(s)	R/W	Default	Description
23:16	R/W	157	reg_nr4_cm_blue_centx : // unsigned , default = 157 x coordinate of center of blue, dft =
7: 0	R/W	110	reg_nr4_cm_blue_centy : // unsigned , default = 110 y coordinate of center of blue, dft =

Table 8-1532 NR4_MCNR_BLUE_GAIN_PAR0 0x2dc9

Bit(s)	R/W	Default	Description
31:24	R/W	32	reg_nr4_cm_blue_gain_par0 : // unsigned , default = 32 par0 for blue gain, dft =
23:16	R/W	255	reg_nr4_cm_blue_gain_par1 : // unsigned , default = 255 par1 for blue gain, dft =
15: 8	R/W	4	reg_nr4_cm_blue_gain_par2 : // unsigned , default = 4 par2 for blue gain, dft =
7: 0	R/W	32	reg_nr4_cm_blue_gain_par3 : // unsigned , default = 32 par3 for blue gain, dft =

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Table 8-1533 NR4_MCNR_BLUE_GAIN_PAR1 0x2dca

Bit(s)	R/W	Default	Description
23:16	R/W	32	reg_nr4_cm_blue_gain_par4 : // unsigned , default = 32 par4 for blue gain, dft =
15: 8	R/W	32	reg_nr4_cm_blue_gain_par5 : // unsigned , default = 32 par5 for blue gain, dft =
7: 0	R/W	0	reg_nr4_cm_blue_gain_par6 : // unsigned , default = 0 par6 for blue gain, dft =

Table 8-1534 NR4_MCNR_CM_BLUE_CLIP0 0x2dcb

Bit(s)	R/W	Default	Description
23:16	R/W	40	reg_nr4_cm_blue_luma_min : // unsigned , default = 40 luma min for blue color matching, dft =
7: 0	R/W	180	reg_nr4_cm_blue_luma_max : // unsigned , default = 180 luma max for blue color matching, dft =

Table 8-1535 NR4_MCNR_CM_BLUE_CLIP1 0x2dcc

Bit(s)	R/W	Default	Description
31:24	R/W	5	reg_nr4_cm_blue_sat_min : // unsigned , default = 5 saturation min for blue color matching, dft =
23:16	R/W	255	reg_nr4_cm_blue_sat_max : // unsigned , default = 255 saturation max for blue color matching, dft =
15: 8	R/W	0	reg_nr4_cm_blue_var_min : // unsigned , default = 0 variation min for blue color matching, dft =
7: 0	R/W	12	reg_nr4_cm_blue_var_max : // unsigned , default = 12 variation max for blue color matching, dft =

Table 8-1536 NR4_MCNR_GREEN_CENT 0x2dcd

Bit(s)	R/W	Default	Description
23:16	R/W	114	reg_nr4_cm_green_centx : // unsigned , default = 114 x coordinate of center of green, dft =
7: 0	R/W	126	reg_nr4_cm_green_centy : // unsigned , default = 126 y coordinate of center of green, dft =

Table 8-1537 NR4_MCNR_GREEN_GAIN_PAR0 0x2dce

Bit(s)	R/W	Default	Description
31:24	R/W	16	reg_nr4_cm_green_gain_par0 : // unsigned , default = 16 par0 for green gain, dft =
23:16	R/W	255	reg_nr4_cm_green_gain_par1 : // unsigned , default = 255 par1 for green gain, dft =
15: 8	R/W	255	reg_nr4_cm_green_gain_par2 : // unsigned , default = 255 par2 for green gain, dft =
7: 0	R/W	16	reg_nr4_cm_green_gain_par3 : // unsigned , default = 16 par3 for green gain, dft =

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Table 8-1538 NR4_MCNR_GREEN_GAIN_PAR1 0x2dcf

Bit(s)	R/W	Default	Description
23:16	R/W	16	reg_nr4_cm_green_gain_par4 : // unsigned , default = 16 par4 for green gain, dft =
15: 8	R/W	128	reg_nr4_cm_green_gain_par5 : // unsigned , default = 128 par5 for green gain, dft =
7: 0	R/W	0	reg_nr4_cm_green_gain_par6 : // unsigned , default = 0 par6 for green gain, dft =

Table 8-1539 NR4_MCNR_GREEN_CLIP0 0x2dd0

Bit(s)	R/W	Default	Description
23:16	R/W	40	reg_nr4_cm_green_luma_min : // unsigned , default = 40 luma min for green color matching, dft =
7: 0	R/W	160	reg_nr4_cm_green_luma_max : // unsigned , default = 160 luma max for green color matching, dft =

Table 8-1540 NR4_MCNR_GREEN_CLIP2 0x2dd1

Bit(s)	R/W	Default	Description
31:24	R/W	4	reg_nr4_cm_green_sat_min : // unsigned , default = 4 saturation min for green color matching, dft =
23:16	R/W	255	reg_nr4_cm_green_sat_max : // unsigned , default = 255 saturation max for green color matching, dft =
15: 8	R/W	0	reg_nr4_cm_green_var_min : // unsigned , default = 0 variation min for green color matching, dft =
7: 0	R/W	12	reg_nr4_cm_green_var_max : // unsigned , default = 12 variation max for green color matching, dft =

Table 8-1541 NR4_MCNR_SKIN_CENT 0x2dd2

Bit(s)	R/W	Default	Description
23:16	R/W	112	reg_nr4_cm_skin_centx : // unsigned , default = 112 x coordinate of center of skin tone, dft =
7: 0	R/W	149	reg_nr4_cm_skin_centy : // unsigned , default = 149 y coordinate of center of skin tone, dft =

Table 8-1542 NR4_MCNR_SKIN_GAIN_PAR0 0x2dd3

Bit(s)	R/W	Default	Description
31:24	R/W	20	reg_nr4_cm_skin_gain_par0 : // unsigned , default = 20 par0 for skin gain, dft =
23:16	R/W	255	reg_nr4_cm_skin_gain_par1 : // unsigned , default = 255 par1 for skin gain, dft =
15: 8	R/W	255	reg_nr4_cm_skin_gain_par2 : // unsigned , default = 255 par2 for skin gain, dft =
7: 0	R/W	8	reg_nr4_cm_skin_gain_par3 : // unsigned , default = 8 par3 for skin gain, dft =

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Table 8-1543 NR4_MCNR_SKIN_GAIN_PAR1 0x2dd4

Bit(s)	R/W	Default	Description
23:16	R/W	8	reg_nr4_cm_skin_gain_par4 : // unsigned , default = 8 par4 for skin gain, dft =
15: 8	R/W	128	reg_nr4_cm_skin_gain_par5 : // unsigned , default = 128 par5 for skin gain, dft =
7: 0	R/W	0	reg_nr4_cm_skin_gain_par6 : // unsigned , default = 0 par6 for skin gain, dft =

Table 8-1544 NR4_MCNR_SKIN_CLIP0 0x2dd5

Bit(s)	R/W	Default	Description
23:16	R/W	40	reg_nr4_cm_skin_luma_min : // unsigned , default = 40 luma min for skin color matching, dft =
7: 0	R/W	180	reg_nr4_cm_skin_luma_max : // unsigned , default = 180 luma max for skin color matching, dft =

Table 8-1545 NR4_MCNR_SKIN_CLIP1 0x2dd6

Bit(s)	R/W	Default	Description
31:24	R/W	5	reg_nr4_cm_skin_sat_min : // unsigned , default = 5 saturation min for skin color matching, dft =
23:16	R/W	255	reg_nr4_cm_skin_sat_max : // unsigned , default = 255 saturation max for skin color matching, dft =
15: 8	R/W	0	reg_nr4_cm_skin_var_min : // unsigned , default = 0 variation min for skin color matching, dft =
7: 0	R/W	12	reg_nr4_cm_skin_var_max : // unsigned , default = 12 variation max for skin color matching, dft =

Table 8-1546 NR4_MCNR_ALP1_GLB_CTRL 0x2dd7

Bit(s)	R/W	Default	Description
31	R/W	0	reg_nr4_alp1_glb_gain_en : // unsigned , default = 0 alp1 adjust by global gain, dft = 1
30:28	R/W	6	reg_nr4_alp1_glb_gain_lsft : // unsigned , default = 6 alp1 left shift before combine with global gain
27	R/W	1	reg_nr4_bet0_glb_gain_en : // unsigned , default = 1 bet0 adjust by global gain, dft = 1
26:24	R/W	6	reg_nr4_bet0_glb_gain_lsft : // unsigned , default = 6 bet1 left shift before combine with global gain
23	R/W	0	reg_nr4_bet1_glb_gain_en : // unsigned , default = 0 bet1 adjust by global gain, dft = 0
22:20	R/W	6	reg_nr4_bet1_glb_gain_lsft : // unsigned , default = 6 bet1 left shift before combine with global gain
19	R/W	1	reg_nr4_bet2_glb_gain_en : // unsigned , default = 1 bet2 adjust by global gain, dft = 1
18:16	R/W	6	reg_nr4_bet2_glb_gain_lsft : // unsigned , default = 6 bet2 left shift before combine with global gain
15	R/W	1	reg_nr4_alp1_ac_en : // unsigned , default = 1 alp1 adjust by ac, dft = 1

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Bit(s)	R/W	Default	Description
14:12	R/W	5	reg_nr4_alp1_ac_lsft : // unsigned , default = 5 alp1 left shift before combine with ac
11	R/W	0	reg_nr4_bet0_ac_en : // unsigned , default = 0 bet0 adjust by ac, dft = 1
10: 8	R/W	5	reg_nr4_bet0_ac_lsft : // unsigned , default = 5 bet0 left shift before combine with ac
7	R/W	0	reg_nr4_bet1_ac_en : // unsigned , default = 0 bet1 adjust by ac, dft = 1
6: 4	R/W	5	reg_nr4_bet1_ac_lsft : // unsigned , default = 5 bet1 left shift before combine with ac
3	R/W	0	reg_nr4_bet2_ac_en : // unsigned , default = 0 bet2 adjust by ac, dft = 1
2: 0	R/W	5	reg_nr4_bet2_ac_lsft : // unsigned , default = 5 bet2 left shift before combine with ac

Table 8-1547 NR4_MCNR_DC2NORM_LUT0 0x2dd8

Bit(s)	R/W	Default	Description
28:24	R/W	16	reg_nr4_dc2norm_lut0 : // unsigned , default = 16 normal 0~16, dc to norm for alpha adjust, dft =
20:16	R/W	16	reg_nr4_dc2norm_lut1 : // unsigned , default = 16 normal 0~16, dc to norm for alpha adjust, dft =
12: 8	R/W	16	reg_nr4_dc2norm_lut2 : // unsigned , default = 16 normal 0~16, dc to norm for alpha adjust, dft =
4: 0	R/W	16	reg_nr4_dc2norm_lut3 : // unsigned , default = 16 normal 0~16, dc to norm for alpha adjust, dft =

Table 8-1548 NR4_MCNR_DC2NORM_LUT1 0x2dd9

Bit(s)	R/W	Default	Description
28:24	R/W	16	reg_nr4_dc2norm_lut4 : // unsigned , default = 16 normal 0~16, dc to norm for alpha adjust, dft =
20:16	R/W	16	reg_nr4_dc2norm_lut5 : // unsigned , default = 16 normal 0~16, dc to norm for alpha adjust, dft =
12: 8	R/W	16	reg_nr4_dc2norm_lut6 : // unsigned , default = 16 normal 0~16, dc to norm for alpha adjust, dft =
4: 0	R/W	12	reg_nr4_dc2norm_lut7 : // unsigned , default = 12 normal 0~16, dc to norm for alpha adjust, dft =

Table 8-1549 NR4_MCNR_DC2NORM_LUT2 0x2dda

Bit(s)	R/W	Default	Description
4: 0	R/W	8	reg_nr4_dc2norm_lut8 : // unsigned , default = 8 normal 0~16, dc to norm for alpha adjust, dft =

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Table 8-1550 NR4_MCNR_AC2NORM_LUT0 0x2ddb

Bit(s)	R/W	Default	Description
28:24	R/W	2	reg_nr4_ac2norm_lut0 : // unsigned , default = 2 normal 0~16, ac to norm for alpha adjust, dft =
20:16	R/W	16	reg_nr4_ac2norm_lut1 : // unsigned , default = 16 normal 0~16, ac to norm for alpha adjust, dft =
12: 8	R/W	16	reg_nr4_ac2norm_lut2 : // unsigned , default = 16 normal 0~16, ac to norm for alpha adjust, dft =
4: 0	R/W	12	reg_nr4_ac2norm_lut3 : // unsigned , default = 12 normal 0~16, ac to norm for alpha adjust, dft =

Table 8-1551 NR4_MCNR_AC2NORM_LUT1 0x2ddc

Bit(s)	R/W	Default	Description
28:24	R/W	4	reg_nr4_ac2norm_lut4 : // unsigned , default = 4 normal 0~16, ac to norm for alpha adjust, dft =
20:16	R/W	2	reg_nr4_ac2norm_lut5 : // unsigned , default = 2 normal 0~16, ac to norm for alpha adjust, dft =
12: 8	R/W	1	reg_nr4_ac2norm_lut6 : // unsigned , default = 1 normal 0~16, ac to norm for alpha adjust, dft =
4: 0	R/W	1	reg_nr4_ac2norm_lut7 : // unsigned , default = 1 normal 0~16, ac to norm for alpha adjust, dft =

Table 8-1552 NR4_MCNR_AC2NORM_LUT2 0x2ddd

Bit(s)	R/W	Default	Description
4: 0	R/W	1	reg_nr4_ac2norm_lut8 : // unsigned , default = 1 normal 0~16, ac to norm for alpha adjust, dft =

Table 8-1553 NR4_MCNR_SAD2ALP0_LUT0 0x2dde

Bit(s)	R/W	Default	Description
31:24	R/W	255	reg_nr4_sad2alp0_lut0 : // unsigned , default = 255 sad to alpha0 for temporal pixel value, dft = 255
23:16	R/W	252	reg_nr4_sad2alp0_lut1 : // unsigned , default = 252 sad to alpha0 for temporal pixel value, dft = 252
15: 8	R/W	249	reg_nr4_sad2alp0_lut2 : // unsigned , default = 249 sad to alpha0 for temporal pixel value, dft = 249
7: 0	R/W	235	reg_nr4_sad2alp0_lut3 : // unsigned , default = 235 sad to alpha0 for temporal pixel value, dft = 70

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Table 8-1554 NR4_MCNR_SAD2ALP0_LUT1 0x2ddf

Bit(s)	R/W	Default	Description
31:24	R/W	185	reg_nr4_sad2alp0_lut4 : // unsigned , default = 185 sad to alpha0 for temporal pixel value, dft = 12
23:16	R/W	70	reg_nr4_sad2alp0_lut5 : // unsigned , default = 70 sad to alpha0 for temporal pixel value, dft = 1
15: 8	R/W	14	reg_nr4_sad2alp0_lut6 : // unsigned , default = 14 sad to alpha0 for temporal pixel value, dft = 0
7: 0	R/W	1	reg_nr4_sad2alp0_lut7 : // unsigned , default = 1 sad to alpha0 for temporal pixel value, dft = 0

Table 8-1555 NR4_MCNR_SAD2ALP0_LUT2 0x2de0

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_sad2alp0_lut8 : // unsigned , default = 0 sad to alpha0 for temporal pixel value, dft = 0
23:16	R/W	0	reg_nr4_sad2alp0_lut9 : // unsigned , default = 0 sad to alpha0 for temporal pixel value, dft = 0
15: 8	R/W	0	reg_nr4_sad2alp0_lut10 : // unsigned , default = 0 sad to alpha0 for temporal pixel value, dft = 0
7: 0	R/W	0	reg_nr4_sad2alp0_lut11 : // unsigned , default = 0 sad to alpha0 for temporal pixel value, dft = 0

Table 8-1556 NR4_MCNR_SAD2ALP0_LUT3 0x2de1

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_sad2alp0_lut12 : // unsigned , default = 0 sad to alpha0 for temporal pixel value, dft = 0
23:16	R/W	0	reg_nr4_sad2alp0_lut13 : // unsigned , default = 0 sad to alpha0 for temporal pixel value, dft = 0
15: 8	R/W	0	reg_nr4_sad2alp0_lut14 : // unsigned , default = 0 sad to alpha0 for temporal pixel value, dft = 0
7: 0	R/W	0	reg_nr4_sad2alp0_lut15 : // unsigned , default = 0 sad to alpha0 for temporal pixel value, dft = 0

Table 8-1557 NR4_MCNR_SAD2ALP1_LUT0 0x2de2

Bit(s)	R/W	Default	Description
31:24	R/W	192	reg_nr4_sad2alp1_lut0 : // unsigned , default = 192 sad to alpha1 for temporal blending, dft = 128
23:16	R/W	160	reg_nr4_sad2alp1_lut1 : // unsigned , default = 160 sad to alpha1 for temporal blending, dft = 128
15: 8	R/W	128	reg_nr4_sad2alp1_lut2 : // unsigned , default = 128 sad to alpha1 for temporal blending, dft = 128
7: 0	R/W	96	reg_nr4_sad2alp1_lut3 : // unsigned , default = 96 sad to alpha1 for temporal blending, dft = 64

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Table 8-1558 NR4_MCNR_SAD2ALP1_LUT1 0x2de3

Bit(s)	R/W	Default	Description
31:24	R/W	64	reg_nr4_sad2alp1_lut4 : // unsigned , default = 64 sad to alpha1 for temporal blending, dft = 64
23:16	R/W	32	reg_nr4_sad2alp1_lut5 : // unsigned , default = 32 sad to alpha1 for temporal blending, dft = 128
15: 8	R/W	16	reg_nr4_sad2alp1_lut6 : // unsigned , default = 16 sad to alpha1 for temporal blending, dft = 255
7: 0	R/W	8	reg_nr4_sad2alp1_lut7 : // unsigned , default = 8 sad to alpha1 for temporal blending, dft = 255

Table 8-1559 NR4_MCNR_SAD2ALP1_LUT2 0x2de4

Bit(s)	R/W	Default	Description
31:24	R/W	4	reg_nr4_sad2alp1_lut8 : // unsigned , default = 4 sad to alpha1 for temporal blending, dft = 255
23:16	R/W	0	reg_nr4_sad2alp1_lut9 : // unsigned , default = 0 sad to alpha1 for temporal blending, dft = 255
15: 8	R/W	16	reg_nr4_sad2alp1_lut10 : // unsigned , default = 16 sad to alpha1 for temporal blending, dft = 255
7: 0	R/W	64	reg_nr4_sad2alp1_lut11 : // unsigned , default = 64 sad to alpha1 for temporal blending, dft = 255

Table 8-1560 NR4_MCNR_SAD2ALP1_LUT3 0x2de5

Bit(s)	R/W	Default	Description
31:24	R/W	96	reg_nr4_sad2alp1_lut12 : // unsigned , default = 96 sad to alpha1 for temporal blending, dft = 255
23:16	R/W	224	reg_nr4_sad2alp1_lut13 : // unsigned , default = 224 sad to alpha1 for temporal blending, dft = 255
15: 8	R/W	255	reg_nr4_sad2alp1_lut14 : // unsigned , default = 255 sad to alpha1 for temporal blending, dft = 255
7: 0	R/W	255	reg_nr4_sad2alp1_lut15 : // unsigned , default = 255 sad to alpha1 for temporal blending, dft = 255

Table 8-1561 NR4_MCNR_SAD2BET0_LUT0 0x2de6

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_sad2bet0_lut0 : // unsigned , default = 0 sad to beta0 for tnr and mcnr blending, dft = 0
23:16	R/W	2	reg_nr4_sad2bet0_lut1 : // unsigned , default = 2 sad to beta0 for tnr and mcnr blending, dft = 2
15: 8	R/W	4	reg_nr4_sad2bet0_lut2 : // unsigned , default = 4 sad to beta0 for tnr and mcnr blending, dft = 4
7: 0	R/W	8	reg_nr4_sad2bet0_lut3 : // unsigned , default = 8 sad to beta0 for tnr and mcnr blending, dft = 8

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Table 8-1562 NR4_MCNR_SAD2BET0_LUT1 0x2de7

Bit(s)	R/W	Default	Description
31:24	R/W	16	reg_nr4_sad2bet0_lut4 : // unsigned , default = 16 sad to beta0 for tnr and mcnr blending, dft = 16
23:16	R/W	32	reg_nr4_sad2bet0_lut5 : // unsigned , default = 32 sad to beta0 for tnr and mcnr blending, dft = 32
15: 8	R/W	48	reg_nr4_sad2bet0_lut6 : // unsigned , default = 48 sad to beta0 for tnr and mcnr blending, dft = 48
7: 0	R/W	64	reg_nr4_sad2bet0_lut7 : // unsigned , default = 64 sad to beta0 for tnr and mcnr blending, dft = 64

Table 8-1563 NR4_MCNR_SAD2BET0_LUT2 0x2de8

Bit(s)	R/W	Default	Description
31:24	R/W	80	reg_nr4_sad2bet0_lut8 : // unsigned , default = 80 sad to beta0 for tnr and mcnr blending, dft = 80
23:16	R/W	96	reg_nr4_sad2bet0_lut9 : // unsigned , default = 96 sad to beta0 for tnr and mcnr blending, dft = 96
15: 8	R/W	112	reg_nr4_sad2bet0_lut10 : // unsigned , default = 112 sad to beta0 for tnr and mcnr blending, dft = 112
7: 0	R/W	128	reg_nr4_sad2bet0_lut11 : // unsigned , default = 128 sad to beta0 for tnr and mcnr blending, dft = 128

Table 8-1564 NR4_MCNR_SAD2BET0_LUT3 0x2de9

Bit(s)	R/W	Default	Description
31:24	R/W	196	reg_nr4_sad2bet0_lut12 : // unsigned , default = 196 sad to beta0 for tnr and mcnr blending, dft = 160
23:16	R/W	224	reg_nr4_sad2bet0_lut13 : // unsigned , default = 224 sad to beta0 for tnr and mcnr blending, dft = 192
15: 8	R/W	255	reg_nr4_sad2bet0_lut14 : // unsigned , default = 255 sad to beta0 for tnr and mcnr blending, dft = 224
7: 0	R/W	255	reg_nr4_sad2bet0_lut15 : // unsigned , default = 255 sad to beta0 for tnr and mcnr blending, dft = 255

Table 8-1565 NR4_MCNR_SAD2BET1_LUT0 0x2dea

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_sad2bet1_lut0 : // unsigned , default = 0 sad to beta1 for deghost blending, dft = 0
23:16	R/W	2	reg_nr4_sad2bet1_lut1 : // unsigned , default = 2 sad to beta1 for deghost blending, dft = 2
15: 8	R/W	4	reg_nr4_sad2bet1_lut2 : // unsigned , default = 4 sad to beta1 for deghost blending, dft = 4
7: 0	R/W	8	reg_nr4_sad2bet1_lut3 : // unsigned , default = 8 sad to beta1 for deghost blending, dft = 8

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Table 8-1566 NR4_MCNR_SAD2BET1_LUT1 0x2deb

Bit(s)	R/W	Default	Description
31:24	R/W	16	reg_nr4_sad2bet1_lut4 : // unsigned , default = 16 sad to beta1 for deghost blending, dft = 16
23:16	R/W	32	reg_nr4_sad2bet1_lut5 : // unsigned , default = 32 sad to beta1 for deghost blending, dft = 32
15: 8	R/W	48	reg_nr4_sad2bet1_lut6 : // unsigned , default = 48 sad to beta1 for deghost blending, dft = 48
7: 0	R/W	64	reg_nr4_sad2bet1_lut7 : // unsigned , default = 64 sad to beta1 for deghost blending, dft = 64

Table 8-1567 NR4_MCNR_SAD2BET1_LUT2 0x2dec

Bit(s)	R/W	Default	Description
31:24	R/W	80	reg_nr4_sad2bet1_lut8 : // unsigned , default = 80 sad to beta1 for deghost blending, dft = 80
23:16	R/W	96	reg_nr4_sad2bet1_lut9 : // unsigned , default = 96 sad to beta1 for deghost blending, dft = 96
15: 8	R/W	112	reg_nr4_sad2bet1_lut10 : // unsigned , default = 112 sad to beta1 for deghost blending, dft = 112
7: 0	R/W	128	reg_nr4_sad2bet1_lut11 : // unsigned , default = 128 sad to beta1 for deghost blending, dft = 128

Table 8-1568 NR4_MCNR_SAD2BET1_LUT3 0x2ded

Bit(s)	R/W	Default	Description
31:24	R/W	160	reg_nr4_sad2bet1_lut12 : // unsigned , default = 160 sad to beta1 for deghost blending, dft = 160
23:16	R/W	192	reg_nr4_sad2bet1_lut13 : // unsigned , default = 192 sad to beta1 for deghost blending, dft = 192
15: 8	R/W	224	reg_nr4_sad2bet1_lut14 : // unsigned , default = 224 sad to beta1 for deghost blending, dft = 224
7: 0	R/W	255	reg_nr4_sad2bet1_lut15 : // unsigned , default = 255 sad to beta1 for deghost blending, dft = 255

Table 8-1569 NR4_MCNR_SAD2BET2_LUT0 0x2dee

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_nr4_sad2bet2_lut0 : // unsigned , default = 0 sad to beta2 for snr and mcnr blending, dft = 0
23:16	R/W	1	reg_nr4_sad2bet2_lut1 : // unsigned , default = 1 sad to beta2 for snr and mcnr blending, dft = 2
15: 8	R/W	2	reg_nr4_sad2bet2_lut2 : // unsigned , default = 2 sad to beta2 for snr and mcnr blending, dft = 4
7: 0	R/W	4	reg_nr4_sad2bet2_lut3 : // unsigned , default = 4 sad to beta2 for snr and mcnr blending, dft = 8

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Table 8-1570 NR4_MCNR_SAD2BET2_LUT1 0x2def

Bit(s)	R/W	Default	Description
31:24	R/W	8	reg_nr4_sad2bet2_lut4 : // unsigned , default = 8 sad to beta2 for snr and mcnr blending, dft = 16
23:16	R/W	16	reg_nr4_sad2bet2_lut5 : // unsigned , default = 16 sad to beta2 for snr and mcnr blending, dft = 32
15: 8	R/W	32	reg_nr4_sad2bet2_lut6 : // unsigned , default = 32 sad to beta2 for snr and mcnr blending, dft = 48
7: 0	R/W	48	reg_nr4_sad2bet2_lut7 : // unsigned , default = 48 sad to beta2 for snr and mcnr blending, dft = 64

Table 8-1571 NR4_MCNR_SAD2BET2_LUT2 0x2df0

Bit(s)	R/W	Default	Description
31:24	R/W	64	reg_nr4_sad2bet2_lut8 : // unsigned , default = 64 sad to beta2 for snr and mcnr blending, dft = 80
23:16	R/W	80	reg_nr4_sad2bet2_lut9 : // unsigned , default = 80 sad to beta2 for snr and mcnr blending, dft = 96
15: 8	R/W	96	reg_nr4_sad2bet2_lut10 : // unsigned , default = 96 sad to beta2 for snr and mcnr blending, dft = 112
7: 0	R/W	112	reg_nr4_sad2bet2_lut11 : // unsigned , default = 112 sad to beta2 for snr and mcnr blending, dft = 128

Table 8-1572 NR4_MCNR_SAD2BET2_LUT3 0x2df1

Bit(s)	R/W	Default	Description
31:24	R/W	128	reg_nr4_sad2bet2_lut12 : // unsigned , default = 128 sad to beta2 for snr and mcnr blending, dft = 160
23:16	R/W	160	reg_nr4_sad2bet2_lut13 : // unsigned , default = 160 sad to beta2 for snr and mcnr blending, dft = 192
15: 8	R/W	224	reg_nr4_sad2bet2_lut14 : // unsigned , default = 224 sad to beta2 for snr and mcnr blending, dft = 224
7: 0	R/W	255	reg_nr4_sad2bet2_lut15 : // unsigned , default = 255 sad to beta2 for snr and mcnr blending, dft = 255

Table 8-1573 NR4_MCNR_RO_U_SUM 0x2df2

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_u_sum : // unsigned , default = 0 sum of U of current field/frame

Table 8-1574 NR4_MCNR_RO_V_SUM 0x2df3

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_v_sum : // unsigned , default = 0 sum of V of current field/frame

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Table 8-1575 NR4_MCNR_RO_GRDU_SUM 0x2df4

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_grdu_sum : // unsigned , default = 0 sum of gradient U of current field/ frame

Table 8-1576 NR4_MCNR_RO_GRDV_SUM 0x2df5

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_grdv_sum : // unsigned , default = 0 sum of gradient V of current field/ frame

Table 8-1577 NR4_TOP_CTRL 0x2dff

Bit(s)	R/W	Default	Description
31:20	R/W	0	reg_gclk_ctrl : // unsigned , default = 0
18	R/W	1	reg_nr4_mcnr_en : // unsigned , default = 1 ncnr enable or bypass, dft = 1
17	R/W	1	reg_nr2_en : // unsigned , default = 1 nr2 enable, dft = 1
16	R/W	1	reg_nr4_en : // unsigned , default = 1 nr4 enable, dft = 1
15	R/W	1	reg_nr2_proc_en : // unsigned , default = 1
14	R/W	1	reg_det3d_en : // unsigned , default = 1
13	R/W	1	di_polar_en : // unsigned , default = 1 do does not have in C
12	R/W	0	reg_cfr_enable : // unsigned , default = 0 0-disable; 1:enable
11: 9	R/W	7	reg_3dnr_enable_l : // unsigned , default = 7 b0: Y b1:U b2:V
8: 6	R/W	7	reg_3dnr_enable_r : // unsigned , default = 7 b0: Y b1:U b2:V
5	R/W	1	reg_nr4_lnbuf_ctrl : // unsigned , default = 1 line buf ctrl for nr4: 0, 3lines, 1, 5lines, dft = 1
4	R/W	0	reg_nr4_snr2_en : // unsigned , default = 0 snr2 enable, 0: use old snr, 1: use new snr2, dft = 1
3	R/W	1	reg_nr4_scene_change_en : // unsigned , default = 1 enable scene change proc. dft = 1
2	R/W	1	nr2_sw_en : // unsigned , default = 1 do does not have in C
0	R/W	0	reg_nr4_scene_change_flg : // unsigned , default = 0 flags for scene change, dft = 0

Table 8-1578 NR4_MCNR_SAD_GAIN 0x3700

Bit(s)	R/W	Default	Description
24	R/W	0	reg_nr4_bld12vs3_usemaxsad : // unsigned , default = 0 use minsad/maxsad instead of minsad/avgsad to decision if it was texture or flat region, 1: use minsad/ maxsad
23:16	R/W	64	reg_nr4_bld12vs3_rate_gain : // unsigned , default = 64 gain to minsad/maxsad or minsad/avgsad before LUT, 64 normalized as "1"

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Bit(s)	R/W	Default	Description
15: 8	R/W	32	reg_nr4_bld1vs2_rate_gain : // unsigned , default = 32 gain to minsad/maxsad or minsad/avgsad before the LUT, 64 normalized as"1"
7: 0	R/W	64	reg_nr4_coefblt_gain : // unsigned , default = 64 gain to final coefblt, normalized 64 as "1"

Table 8-1579 NR4_MCNR_LPF_CTRL 0x3701

Bit(s)	R/W	Default	Description
30:22	R/W	0	reg_nr4_preflt_alpofst : // signed , default = 0 pre filter alpha ofst
21:16	R/W	16	reg_nr4_preflt_alpgain : // unsigned , default = 16 pre filter alpha gain
15:14	R/W	3	reg_nr4_preflt_alpsel : // unsigned , default = 3 pre filter alpha selection for adaptive blending, 0: mv pointed sad, 1: weighted mv pointed sad, 2or3: coefblt
13: 8	R/W	8	reg_nr4_avgsad_gain : // unsigned , default = 8 gain for avg sad before luts
6	R/W	1	reg_nr4_maxsad_mod : // unsigned , default = 1 max sad select mode, 0: mx2_ sad, 1: max sad
5	R/W	1	reg_nr4_minsad_mod : // unsigned , default = 1 min sad select mode, 0: sad with min err, 1: min sad
4	R/W	1	reg_nr4_minmaxsad_lpf : // unsigned , default = 1 mode of lpf for minmaxsad, 0: no LPF, 1: [1 2 1]/4
3	R/W	1	reg_nr4_avgsad_lpf : // unsigned , default = 1 mode of lpf for avgsad, 0: no LPF, 1: [1 2 1]/4
2	R/W	1	reg_nr4_minavgsad_ratio_lpf : // unsigned , default = 1 mode of lpf for minsad/ avgsad and zmvsad/avgsad, 0: no LPF, 1: [1 2 1]/4
1	R/W	1	reg_nr4_bldvs_lut_lpf : // unsigned , default = 1 mode of lpf for bld12vs3 and bld1vs2 LUT results, 0: no LPF, 1: [1 2 1]/4
0	R/W	1	reg_nr4_final_coef_lpf : // unsigned , default = 1 mode of lpf for final coef_blt_ blend123, 0: no LPF, 1: [1 2 1]/4

Table 8-1580 NR4_MCNR_BLD_VS3LUT0 0x3702

Bit(s)	R/W	Default	Description
29:24	R/W	0	reg_nr4_bld12vs3_lut0 : // unsigned , default = 0
21:16	R/W	8	reg_nr4_bld12vs3_lut1 : // unsigned , default = 8
13: 8	R/W	10	reg_nr4_bld12vs3_lut2 : // unsigned , default = 10
5: 0	R/W	11	reg_nr4_bld12vs3_lut3 : // unsigned , default = 11

Table 8-1581 NR4_MCNR_BLD_VS3LUT1 0x3703

Bit(s)	R/W	Default	Description
29:24	R/W	12	reg_nr4_bld12vs3_lut4 : // unsigned , default = 12
21:16	R/W	14	reg_nr4_bld12vs3_lut5 : // unsigned , default = 14
13: 8	R/W	16	reg_nr4_bld12vs3_lut6 : // unsigned , default = 16
5: 0	R/W	24	reg_nr4_bld12vs3_lut7 : // unsigned , default = 24

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Table 8-1582 NR4_MCNR_BLD_VS3LUT2 0x3704

Bit(s)	R/W	Default	Description
29:24	R/W	50	reg_nr4_bld12vs3_lut8 : // unsigned , default = 50
21:16	R/W	58	reg_nr4_bld12vs3_lut9 : // unsigned , default = 58
13: 8	R/W	63	reg_nr4_bld12vs3_lut10 : // unsigned , default = 63
5: 0	R/W	63	reg_nr4_bld12vs3_lut11 : // unsigned , default = 63

Table 8-1583 NR4_MCNR_BLD_VS2LUT0 0x3705

Bit(s)	R/W	Default	Description
29:24	R/W	63	reg_nr4_bld1vs2_lut0 : // unsigned , default = 63
21:16	R/W	32	reg_nr4_bld1vs2_lut1 : // unsigned , default = 32
13: 8	R/W	16	reg_nr4_bld1vs2_lut2 : // unsigned , default = 16
5: 0	R/W	8	reg_nr4_bld1vs2_lut3 : // unsigned , default = 8

Table 8-1584 NR4_MCNR_BLD_VS2LUT1 0x3706

Bit(s)	R/W	Default	Description
29:24	R/W	4	reg_nr4_bld1vs2_lut4 : // unsigned , default = 4
21:16	R/W	2	reg_nr4_bld1vs2_lut5 : // unsigned , default = 2
13: 8	R/W	1	reg_nr4_bld1vs2_lut6 : // unsigned , default = 1
5: 0	R/W	0	reg_nr4_bld1vs2_lut7 : // unsigned , default = 0

Table 8-1585 NR4_COEFBLT_LUT10 0x3707

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_nr4_coefblt_lut10 : // signed , default = -128
23:16	R/W	0x0	reg_nr4_coefblt_lut11 : // signed , default = -128
15: 8	R/W	0x0	reg_nr4_coefblt_lut12 : // signed , default = -126
7: 0	R/W	0x0	reg_nr4_coefblt_lut13 : // signed , default = -124

Table 8-1586 NR4_COEFBLT_LUT11 0x3708

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_nr4_coefblt_lut14 : // signed , default = -120
23:16	R/W	0x0	reg_nr4_coefblt_lut15 : // signed , default = -110
15: 8	R/W	0x0	reg_nr4_coefblt_lut16 : // signed , default = -100
7: 0	R/W	0x0	reg_nr4_coefblt_lut17 : // signed , default = -90

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Table 8-1587 NR4_COEFBLT_LUT12 0x3709

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_nr4_coefblt_lut18 : // signed , default = -56
23:16	R/W	0x0	reg_nr4_coefblt_lut19 : // signed , default = -32
15: 8	R/W	0x0	reg_nr4_coefblt_lut110 : // signed , default = -64
7: 0	R/W	0x0	reg_nr4_coefblt_lut111 : // signed , default = -128

Table 8-1588 NR4_COEFBLT_LUT20 0x370a

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_nr4_coefblt_lut20 : // signed , default = -128
23:16	R/W	0x0	reg_nr4_coefblt_lut21 : // signed , default = -120
15: 8	R/W	0x0	reg_nr4_coefblt_lut22 : // signed , default = -112
7: 0	R/W	0x0	reg_nr4_coefblt_lut23 : // signed , default = -104

Table 8-1589 NR4_COEFBLT_LUT21 0x370b

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_nr4_coefblt_lut24 : // signed , default = -96
23:16	R/W	0x0	reg_nr4_coefblt_lut25 : // signed , default = -88
15: 8	R/W	0x0	reg_nr4_coefblt_lut26 : // signed , default = -76
7: 0	R/W	0x0	reg_nr4_coefblt_lut27 : // signed , default = -64

Table 8-1590 NR4_COEFBLT_LUT22 0x370c

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_nr4_coefblt_lut28 : // signed , default = -48
23:16	R/W	0x0	reg_nr4_coefblt_lut29 : // signed , default = -32
15: 8	R/W	0x0	reg_nr4_coefblt_lut210 : // signed , default = -64
7: 0	R/W	0x0	reg_nr4_coefblt_lut211 : // signed , default = -108

Table 8-1591 NR4_COEFBLT_LUT30 0x370d

Bit(s)	R/W	Default	Description
31:24	R/W	8	reg_nr4_coefblt_lut30 : // signed , default = 8
23:16	R/W	16	reg_nr4_coefblt_lut31 : // signed , default = 16
15: 8	R/W	24	reg_nr4_coefblt_lut32 : // signed , default = 24
7: 0	R/W	30	reg_nr4_coefblt_lut33 : // signed , default = 30

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Table 8-1592 NR4_COEFBLT_LUT31 0x370e

Bit(s)	R/W	Default	Description
31:24	R/W	36	reg_nr4_coefblt_lut34 : // signed , default = 36
23:16	R/W	48	reg_nr4_coefblt_lut35 : // signed , default = 48
15: 8	R/W	70	reg_nr4_coefblt_lut36 : // signed , default = 70
7: 0	R/W	96	reg_nr4_coefblt_lut37 : // signed , default = 96

Table 8-1593 NR4_COEFBLT_LUT32 0x370f

Bit(s)	R/W	Default	Description
31:24	R/W	120	reg_nr4_coefblt_lut38 : // signed , default = 120
23:16	R/W	64	reg_nr4_coefblt_lut39 : // signed , default = 64
15: 8	R/W	16	reg_nr4_coefblt_lut310 : // signed , default = 16
7: 0	R/W	0x0	reg_nr4_coefblt_lut311 : // signed , default = -8

Table 8-1594 NR4_COEFBLT_CONV 0x3710

Bit(s)	R/W	Default	Description
23:16	R/W	0	reg_nr4_coefblt_convmin : // unsigned , default = 0 minimum of coef. bilateral conversion
15: 8	R/W	255	reg_nr4_coefblt_convmax : // unsigned , default = 255 maximum of coef. bilateral conversion
7: 0	R/W	128	reg_nr4_coefblt_convmid : // unsigned , default = 128 value at midpoint of coef. bilateral conversion

Table 8-1595 NR4_DBGWIN_YX0 0x3711

Bit(s)	R/W	Default	Description
29:16	R/W	100	reg_nr4_dgbwin_yx0 : // unsigned , default = 100 ystart for debug window
13: 0	R/W	160	reg_nr4_dgbwin_yx1 : // unsigned , default = 160 yend for debug window

Table 8-1596 NR4_DBGWIN_YX1 0x3712

Bit(s)	R/W	Default	Description
29:16	R/W	200	reg_nr4_dgbwin_yx2 : // unsigned , default = 200 xstart for debug window
13: 0	R/W	300	reg_nr4_dgbwin_yx3 : // unsigned , default = 300 xend for debug window

Table 8-1597 NR4_NM_X_CFG 0x3713

Bit(s)	R/W	Default	Description
29:16	R/W	8	reg_nr4_nm_xst : // unsigned , default = 8 start for noise meter statistic, dft = 8
13: 0	R/W	711	reg_nr4_nm_xed : // unsigned , default = 711 end for noise meter statistic, dft = HSIZE-8-1;

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Table 8-1598 NR4_NM_Y_CFG 0x3714

Bit(s)	R/W	Default	Description
29:16	R/W	8	reg_nr4_nm_yst : // unsigned , default = 8 start for noise meter statistic, dft = 8;
13: 0	R/W	231	reg_nr4_nm_yed : // unsigned , default = 231 end for noise meter statistic, dft = VSIZE-8-1;

Table 8-1599 NR4_NM_SAD_THD 0x3715

Bit(s)	R/W	Default	Description
7: 0	R/W	255	reg_nr4_nm_sad_thd : // unsigned , default = 255 threshold for (flat region) sad count, dft = 4

Table 8-1600 NR4_MCNR_BANDSPLIT_PRAM 0x3716

Bit(s)	R/W	Default	Description
4	R/W	1	reg_nr4_mc_use_bandsplit : // unsigned , default = 1 separate lp and us for mc IIR filter, 0: no BS used; 1: use BS
3	R/W	1	reg_nr4_mc_apply_on_lp : // unsigned , default = 1 use mcnr only on lowpass portion;
2	R/W	1	reg_nr4_mc_apply_on_us : // unsigned , default = 1 use mcnr only on lp complimentary portion;
1: 0	R/W	1	reg_nr4_mc_zmvbs_use_adplpf : // unsigned , default = 1 use adapptive LPF for the zmv pointing data for MCNR, for abs(mvx)<th

Table 8-1601 NR4_MCNR_ALP1_SGN_COR 0x3717

Bit(s)	R/W	Default	Description
31:24	R/W	10	reg_nr4_mc_aph1_sgn_coring0 : // unsigned , default = 10 coring to cur-pre before do sgn decision
23:16	R/W	7	reg_nr4_mc_aph1_sgn_coring1 : // unsigned , default = 7 coring to cur-pre before do sgn decision
15: 8	R/W	90	reg_nr4_mc_aph1_sgn_core_max0 : // unsigned , default = 90 maximum of coring, default = 30/15
7: 0	R/W	15	reg_nr4_mc_aph1_sgn_core_max1 : // unsigned , default = 15 maximum of coring, default = 30/15

Table 8-1602 NR4_MCNR_ALP1_SGN_PRAM 0x3718

Bit(s)	R/W	Default	Description
10	R/W	1	reg_nr4_mc_alp1_sgn_half : // unsigned , default = 1 half block sgn sum mode enable, 0: only use 3x5 whole block sum of sgns; 1: use max(sgn_3x5, sqrt(sgn_left+sgn_righ))
9	R/W	1	reg_nr4_mc_alp1_sgn_frczmv : // unsigned , default = 1 force zmv to calculate the sign_sum;
8	R/W	1	reg_nr4_mc_alp1_sgnmvx_mode : // unsigned , default = 1 blend mode of sgnlut and mvxlut blend mode: 0: sgnlut+ mvxlut; 1: max(sgnlut, mvxlut), default =1

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Bit(s)	R/W	Default	Description
7: 4	R/W	4	reg_nr4_mc_aph1_sgn_crate0 : // unsigned , default = 4 rate to var, norm to 16 as 1, default = 2
3: 0	R/W	2	reg_nr4_mc_aph1_sgn_crate1 : // unsigned , default = 2 rate to var, norm to 16 as 1, default = 2

Table 8-1603 NR4_MCNR_ALP1_MVX_LUT1 0x3719

Bit(s)	R/W	Default	Description
31:28	R/W	14	reg_nr4_mc_alp1_mvx_luty3 : // unsigned , default = 14 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
27:24	R/W	14	reg_nr4_mc_alp1_mvx_lutc3 : // unsigned , default = 14 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)
23:20	R/W	12	reg_nr4_mc_alp1_mvx_luty2 : // unsigned , default = 12 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
19:16	R/W	12	reg_nr4_mc_alp1_mvx_lutc2 : // unsigned , default = 12 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)
15:12	R/W	5	reg_nr4_mc_alp1_mvx_luty1 : // unsigned , default = 5 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
11: 8	R/W	5	reg_nr4_mc_alp1_mvx_lutc1 : // unsigned , default = 5 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)
7: 4	R/W	3	reg_nr4_mc_alp1_mvx_luty0 : // unsigned , default = 3 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
3: 0	R/W	3	reg_nr4_mc_alp1_mvx_lutc0 : // unsigned , default = 3 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)

Table 8-1604 NR4_MCNR_ALP1_MVX_LUT2 0x371a

Bit(s)	R/W	Default	Description
31:28	R/W	15	reg_nr4_mc_alp1_mvx_luty7 : // unsigned , default = 15 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
27:24	R/W	15	reg_nr4_mc_alp1_mvx_lutc7 : // unsigned , default = 15 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)
23:20	R/W	15	reg_nr4_mc_alp1_mvx_luty6 : // unsigned , default = 15 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
19:16	R/W	15	reg_nr4_mc_alp1_mvx_lutc6 : // unsigned , default = 15 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)
15:12	R/W	15	reg_nr4_mc_alp1_mvx_luty5 : // unsigned , default = 15 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
11: 8	R/W	15	reg_nr4_mc_alp1_mvx_lutc5 : // unsigned , default = 15 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)
7: 4	R/W	15	reg_nr4_mc_alp1_mvx_luty4 : // unsigned , default = 15 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
3: 0	R/W	15	reg_nr4_mc_alp1_mvx_lutc4 : // unsigned , default = 15 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)

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Table 8-1605 NR4_MCNR_ALP1_MVX_LUT3 0x371b

Bit(s)	R/W	Default	Description
7: 4	R/W	6	reg_nr4_mc_alp1_mvx_luty8 : // unsigned , default = 6 alp1 of luma vas mvx (0~7), and alp1 vs mvy(0,1)
3: 0	R/W	6	reg_nr4_mc_alp1_mvx_lutc8 : // unsigned , default = 6 alp1 of chrm vas mvx (0~7), and alp1 vs mvy(0,1)

Table 8-1606 NR4_MCNR_ALP1_LP_PRAM 0x371c

Bit(s)	R/W	Default	Description
17:16	R/W	1	reg_nr4_mc_alp1_lp_sel : // unsigned , default = 1 mode for alp1_lp for lp portion IIR, 0: apha1, 1:dc_dif vs ac analysis; 2: gain/ofst of alp1; 3: max of 1/ 2 results
15: 8	R/W	64	reg_nr4_mc_alp1_lp_gain : // unsigned , default = 64 gain to alp1 to get the alp1_lp = alp1*gain/32 + ofset, default =64;
7: 0	R/W	0	reg_nr4_mc_alp1_lp_ofst : // signed , default = 0 offset to alp1 to get the alp1_lp = alp1*gain/32 + ofset, default =10;

Table 8-1607 NR4_MCNR_ALP1_SGN_LUT1 0x371d

Bit(s)	R/W	Default	Description
31:28	R/W	3	reg_nr4_mc_alp1_sgn_lut0 : // unsigned , default = 3 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
27:24	R/W	3	reg_nr4_mc_alp1_sgn_lut1 : // unsigned , default = 3 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
23:20	R/W	3	reg_nr4_mc_alp1_sgn_lut2 : // unsigned , default = 3 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
19:16	R/W	4	reg_nr4_mc_alp1_sgn_lut3 : // unsigned , default = 4 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
15:12	R/W	5	reg_nr4_mc_alp1_sgn_lut4 : // unsigned , default = 5 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
11: 8	R/W	6	reg_nr4_mc_alp1_sgn_lut5 : // unsigned , default = 6 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
7: 4	R/W	7	reg_nr4_mc_alp1_sgn_lut6 : // unsigned , default = 7 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
3: 0	R/W	8	reg_nr4_mc_alp1_sgn_lut7 : // unsigned , default = 8 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move

Table 8-1608 NR4_MCNR_ALP1_SGN_LUT2 0x371e

Bit(s)	R/W	Default	Description
31:28	R/W	9	reg_nr4_mc_alp1_sgn_lut8 : // unsigned , default = 9 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
27:24	R/W	10	reg_nr4_mc_alp1_sgn_lut9 : // unsigned , default = 10 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move
23:20	R/W	11	reg_nr4_mc_alp1_sgn_lut10 : // unsigned , default = 11 alp1 vs x=abs|sgn(cur-pre)|, if x is small, less possibility of flat region move

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Bit(s)	R/W	Default	Description
19:16	R/W	12	reg_nr4_mc_alp1_sgn_lut11 : // unsigned , default = 12 alp1 vs x=abs|sgn(curpre)|, if x is small, less possibility of flat region move
15:12	R/W	13	reg_nr4_mc_alp1_sgn_lut12 : // unsigned , default = 13 alp1 vs x=abs|sgn(curpre)|, if x is small, less possibility of flat region move
11: 8	R/W	14	reg_nr4_mc_alp1_sgn_lut13 : // unsigned , default = 14 alp1 vs x=abs|sgn(curpre)|, if x is small, less possibility of flat region move
7: 4	R/W	15	reg_nr4_mc_alp1_sgn_lut14 : // unsigned , default = 15 alp1 vs x=abs|sgn(curpre)|, if x is small, less possibility of flat region move
3: 0	R/W	15	reg_nr4_mc_alp1_sgn_lut15 : // unsigned , default = 15 alp1 vs x=abs|sgn(curpre)|, if x is small, less possibility of flat region move

Table 8-1609 NR4_RO_NM_SAD_SUM 0x371f

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_nm_sad_sum : // unsigned , default = 0 sum of sad, for scene change detectcion, in noise meter

Table 8-1610 NR4_RO_NM_SAD_CNT 0x3720

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_nm_sad_cnt : // unsigned , default = 0 cnt of sad, for scene change detectcion, in noise meter

Table 8-1611 NR4_RO_NM_VAR_SUM 0x3721

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_nm_var_sum : // unsigned , default = 0 sum of var, for noise level detection, in noise meter

Table 8-1612 NR4_RO_NM_VAR_SCNT 0x3722

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_nm_var_cnt : // unsigned , default = 0 cnt of var, for noise level detection, in noise meter

Table 8-1613 NR4_RO_NM_VAR_MIN_MAX 0x3723

Bit(s)	R/W	Default	Description
21:12	R.O	1023	ro_nr4_nm_min_var : // unsigned , default = 1023 min of var, for noise level detection, in noise meter
9: 0	R.O	0	ro_nr4_nm_max_var : // unsigned , default = 0 max of var, for noise level detection, in noise meter

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Table 8-1614 NR4_RO_NR4_DBGPIX_NUM 0x3724

Bit(s)	R/W	Default	Description
27: 0	R.O	0	ro_nr4_dbgpix_num : // unsigned , default = 0 number of pixels statistic invoved (removed?)

Table 8-1615 NR4_RO_NR4_BLDVS2_SUM 0x3725

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_bld1vs2_sum : // unsigned , default = 0 sum of blend_1vs2 with the debug window

Table 8-1616 NR4_BLDVS3_SUM 0x3726

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_bld12vs3_sum : // unsigned , default = 0 sum of blend_12vs3 with the debug window

Table 8-1617 NR4_COEF12_SUM 0x3727

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_coef12_sum : // signed , default = 0 sum of coef_blt_blend12 with the debug window, under 8 bits precision

Table 8-1618 NR4_COEF123_SUM 0x3728

Bit(s)	R/W	Default	Description
31: 0	R.O	0	ro_nr4_coef123_sum : // signed , default = 0 sum of coef_final with the debug window, under 8 bits precision

Table 8-1619 XLR_CTRL 0x3730

Bit(s)	R/W	Default	Description
22	R/W	1	reg_xlr_en : // unsigned , default = 1 enable bits for xlr function, 1: enable, 0: disable default = 1
21	R/W	1	reg_xlr_side_en : // unsigned , default = 1 enable to filter the above and below lines with xlr filter, default = 1
20	R/W	1	reg_xlr_3lines : // unsigned , default = 1 3 lines version enable default= 0
19:16	R/W	12	reg_xlr_simlp_gain : // unsigned , default = 12 gain to simlp to decide if need the XLR, default= 12
13: 8	R/W	6	reg_xlr_ooplp_gain : // unsigned , default = 6 gain to out-of-phase lp error to decide if need the XLR, birn to 16 as 1, default= 3
7: 0	R/W	64	reg_xlr_dislp_thrd : // unsigned , default = 64 threshold to lp error to discard XLR. default= 80

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Table 8-1620 XLR_THRD 0x3731

Bit(s)	R/W	Default	Description
27:24	R/W	3	reg_xlr_txt_core : // unsigned , default = 3 coring to texture default = 3, (3/32)
23:16	R/W	5	reg_xlr_err_thrd2 : // unsigned , default = 5 threshold to error to decide blending coef, 0, 1/4, 1/2, 1.0, defaut= {20, 10, 5}
15: 8	R/W	10	reg_xlr_err_thrd1 : // unsigned , default = 10 threshold to error to decide blending coef, 0, 1/4, 1/2, 1.0, defaut= {20, 10, 5}
7: 0	R/W	20	reg_xlr_err_thrd0 : // unsigned , default = 20 threshold to error to decide blending coef, 0, 1/4, 1/2, 1.0, defaut= {20, 10, 5}

Table 8-1621 XLR_HCT_THRD 0x3732

Bit(s)	R/W	Default	Description
27:26	R/W	1	reg_xlr_hct_step : // unsigned , default = 1 horizontal chroma diff steps, 0: abs[-1 (2) -1]; 1: abs[-1 0 (2) 0 -1], 2: abs[-1 0 0 (2) 0 0 -1]; 3:abs[-1 0 0 0 (2) 0 0 0 -1]
25:24	R/W	1	reg_xlr_hct_lpf : // unsigned , default = 1 horizontal chroma diff low-pass filter enable, 0: no HLPF; 1: [1 2 1]; 2: max[-1:1]; 3: max[-2:2]
23:16	R/W	20	reg_xlr_hct_thr : // unsigned , default = 20 horizontal chroma diff threshold for xlr enable, to save the horozontal no color tran mode
15: 8	R/W	8	reg_xlr_sat_thr : // unsigned , default = 8 saturation threshold for xlr enable, the smaller of the threshold , the more will do xlr;
7: 2	R/W	3	reg_xlr_hmargin : // unsigned , default = 3 left/right number of pixels without xlr;
0	R/W	0	reg_xlr_hpf_only : // unsigned , default = 0 enable for only do xlr filter on high pass portion of the data, instead full data

Table 8-1622 NR_DB_FLT_CTRL 0x3738

Bit(s)	R/W	Default	Description
26	R/W	0	reg_nrdeband_reset1 : // unsigned , default = 0 0 : no reset seed 1: reload chroma seed
25	R/W	0	reg_nrdeband_reset0 : // unsigned , default = 0 0 : no reset seed 1: reload luma seed
24	R/W	0	reg_nrdeband_rgb : // unsigned , default = 0 0 : yuv 1: RGB
23	R/W	1	reg_nrdeband_en11 : // unsigned , default = 1 debanding registers of side lines, [0] for luma, same for below
22	R/W	1	reg_nrdeband_en10 : // unsigned , default = 1 debanding registers of side lines, [1] for chroma, same for below
21	R/W	1	reg_nrdeband_siderand : // unsigned , default = 1 options to use side two lines use the rand, instead of use for the YUV three component of middle line, 0: seed [3]/bandrand[3] for middle line yuv; 1: seed[3]/bandrand[3] for nearby three lines Y;
20	R/W	0	reg_nrdeband_randmode : // unsigned , default = 0 mode of rand noise adding, 0: same noise strength for all difs; else: strenght of noise will not exceed the difs, MIN((pPKReg->reg_nrdeband_bandrand[m]), noise[m])
19:17	R/W	6	reg_nrdeband_bandrand2 : // unsigned , default = 6
15:13	R/W	6	reg_nrdeband_bandrand1 : // unsigned , default = 6
11: 9	R/W	6	reg_nrdeband_bandrand0 : // unsigned , default = 6

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Bit(s)	R/W	Default	Description
7	R/W	1	reg_nrdeband_hpxor1 : // unsigned , default = 1 debanding random hp portion xor, [0] for luma
6	R/W	1	reg_nrdeband_hpxor0 : // unsigned , default = 1 debanding random hp portion xor, [1] for chroma
5	R/W	1	reg_nrdeband_en1 : // unsigned , default = 1 debanding registers, for luma
4	R/W	1	reg_nrdeband_en0 : // unsigned , default = 1 debanding registers, for chroma
3: 2	R/W	2	reg_nrdeband_lpf_mode1 : // unsigned , default = 2 lpf mode, 0: 3x3, 1:3x5; 2: 5x5; 3:5x7
1: 0	R/W	2	reg_nrdeband_lpf_mode0 : // unsigned , default = 2 lpf mode, 0: 3x3, 1:3x5; 2: 5x5; 3:5x7

Table 8-1623 NR_DB_FLT_YC_THRD 0x3739

Bit(s)	R/W	Default	Description
31:28	R/W	9	reg_nrdeband_luma_th3 : // unsigned , default = 9 threshold to |Y-Ylpf|, if < th[0] use lpf
27:24	R/W	7	reg_nrdeband_luma_th2 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4
23:20	R/W	6	reg_nrdeband_luma_th1 : // unsigned , default = 6 elseif <th[1] use (lpf*3 + y)/4elseif < th[2] (lpf*1 + y)/2
19:16	R/W	5	reg_nrdeband_luma_th0 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/4elseif elseif < th[3] (lpf*1 + 3*y)/4; else
15:12	R/W	9	reg_nrdeband_chrm_th3 : // unsigned , default = 9 threshold to |Y-Ylpf|, if < th[0] use lpf
11: 8	R/W	7	reg_nrdeband_chrm_th2 : // unsigned , default = 7 elseif < th[1] use (lpf*3 + y)/4
7: 4	R/W	6	reg_nrdeband_chrm_th1 : // unsigned , default = 6 elseif < th[1] use (lpf*3 + y)/4elseif <th[2] (lpf*1 + y)/2
3: 0	R/W	5	reg_nrdeband_chrm_th0 : // unsigned , default = 5 elseif < th[1] use (lpf*3 + y)/4elseif elseif

Table 8-1624 NR_DB_FLT_RANDLUT 0x373a

Bit(s)	R/W	Default	Description
23:21	R/W	1	reg_nrdeband_randslut7 : // unsigned , default = 1 lut0
20:18	R/W	1	reg_nrdeband_randslut6 : // unsigned , default = 1 lut0
17:15	R/W	1	reg_nrdeband_randslut5 : // unsigned , default = 1 lut0
14:12	R/W	1	reg_nrdeband_randslut4 : // unsigned , default = 1 lut0
11: 9	R/W	1	reg_nrdeband_randslut3 : // unsigned , default = 1 lut0
8: 6	R/W	1	reg_nrdeband_randslut2 : // unsigned , default = 1 lut0
5: 3	R/W	1	reg_nrdeband_randslut1 : // unsigned , default = 1 lut0
2: 0	R/W	1	reg_nrdeband_randslut0 : // unsigned , default = 1 lut0

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Table 8-1625 NR_DB_FLT_PXI_THRD 0x373b

Bit(s)	R/W	Default	Description
25:16	R/W	0	reg_nrdeband_yc_th1 : // unsigned , default = 0 to luma/|u/v| for using the denoise
9: 0	R/W	0	reg_nrdeband_yc_th0 : // unsigned , default = 0 to luma/|u/v| for using the denoise

Table 8-1626 NR_DB_FLT_SEED_Y 0x373c

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8240	reg_nrdeband_seed0 : // unsigned , default = 1621438240 noise adding seed for Y. seed[0]= 0x60a52f20; as default

Table 8-1627 NR_DB_FLT_SEED_U 0x373d

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8247	reg_nrdeband_seed1 : // unsigned , default = 1621438247 noise adding seed for U. seed[0]= 0x60a52f27; as default

Table 8-1628 NR_DB_FLT_SEED_V 0x373e

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8242	reg_nrdeband_seed2 : // unsigned , default = 1621438242 noise adding seed for V. seed[0]= 0x60a52f22; as default

Table 8-1629 NR_DB_FLT_SEED3 0x373f

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8242	reg_nrdeband_seed3 : // unsigned , default = 1621438242 noise adding seed for V. seed[0]= 0x60a52f22; as default

8.2.3.15 VIUB Top-Level Registers

Table 8-1630 VIU_SW_RESET 0x1a01

Bit(s)	R/W	Default	Description
29	R/W	0	di_dsr1to2_reset : // unsigned , default = 0
28	R/W	0	vd2_lbuf_reset : // unsigned , default = 0
27	R/W	0	afbc_dec1_reset : // unsigned , default = 0
26	R/W	0	vd2_reset : // unsigned , default = 0
25	R/W	0	vd1_lbuf_reset : // unsigned , default = 0
24	R/W	0	afbc_dec0_reset : // unsigned , default = 0
23	R/W	0	vd1_reset : // unsigned , default = 0
22	R/W	0	osd1_afbcd_regs_reset : // unsigned , default = 0

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Bit(s)	R/W	Default	Description
21	R/W	0	osd1_afbcd_logic_reset : // unsigned , default = 0 Only active when MALI_AFBCD_TOP_CTRL[23] high
20	R/W	0	afbc_arb_reg_reset : // unsigned , default = 0
19	R/W	0	afbc_arb_reset : // unsigned , default = 0
17	R/W	0	osd4_reset : // unsigned , default = 0
16	R/W	0	osd3_reset : // unsigned , default = 0
15	R/W	0	osd2_reset : // unsigned , default = 0
14	R/W	0	osd1_reset : // unsigned , default = 0
12	R/W	0	vpp_axi_reset : // unsigned , default = 0
8	R/W	0	osd24bld_reset : // unsigned , default = 0
7	R/W	0	osd13bld_reset : // unsigned , default = 0
6	R/W	0	prime_reset : // unsigned , default = 0
5	R/W	0	hist_spl_reset : // unsigned , default = 0
4	R/W	0	ldim_stts_reset : // unsigned , default = 0
3	R/W	0	dolby1b_reset : // unsigned , default = 0
2	R/W	0	dolby1a_reset : // unsigned , default = 0
1	R/W	0	dolby0_reset : // unsigned , default = 0
0	R/W	0	vpp_reset : // unsigned , default = 0

Table 8-1631 VIU_SW_RESET0 0x1a02

Bit(s)	R/W	Default	Description
2	R/W	0	software : reset for mcinford_mif // unsigned , default = 0
1	R/W	0	software : reset for mcinfowr_mif // unsigned , default = 0
0	R/W	0	software : reset for mcvecrd_mif // unsigned , default = 0

Table 8-1632 VIU_SECURE_REG 0x1a04

Bit(s)	R/W	Default	Description
30-26	R/W	0	Dolby prime secure
25	R/W	0	Vpp probe function: 1:probe enable 0:disable
24	R/W	0	Dolby core3
23-16	R/W	0	Osd test data
9	R/W	0	Write back 1
8	R/W	0	Write back 0
7	R/W	0	Afbc dec1
6	R/W	0	Vd2 mif
5	R/W	0	Afbc dec0

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Bit(s)	R/W	Default	Description
4	R/W	0	Vd1 mif
2	R/W	0	Osd3 : // unsigned , default = 0
1	R/W	0	Osd2 : // unsigned , default = 0
0	R/W	0	osd1 : // unsigned , default = 0

Table 8-1633 DOLBY_INT_STAT 0x1a05

Bit(s)	R/W	Default	Description
3:0	R/W	0	dolby_int_state : // unsigned , default = 0

Table 8-1634 VIU_MISC_CTRL0 0x1a06

Bit(s)	R/W	Default	Description
8	R/W	0	vsync_int_ctrl : default = 0
0	R/W	0	scan_reg : default = 0

Table 8-1635 VIU_MISC_CTRL1 0x1a07

Bit(s)	R/W	Default	Description
31:0	R/W	0	viu_misc_ctrl1 : // unsigned , default = 0

Table 8-1636 VIU_SECURE_REG1 0x1a08

Bit(s)	R/W	Default	Description
31:0	R/W	0	viu_secure_reg1 : // unsigned , default = 0

Table 8-1637 VIU_SECURE_REG2 0x1a09

Bit(s)	R/W	Default	Description
31:0	R/W	0	viu_secure_reg2 : // unsigned , default = 0

Table 8-1638 VD1_AFBCD0_MISC_CTRL 0x1a0a

Bit(s)	R/W	Default	Description
22	R/W	0	Vd1_afbc0_mem_sel 0: vd_mif share afbc_memory 1: afbc_mif share afbc_ memory
21:20	R/W	0	Afbc_mif_go_field_sel : // unsigned , default = 0 0: go_file 1: go_field_post 2: go_ filed_pre
19:18	R/W	0	vd1_mif_go_field_sel : // unsigned , default = 0 0: go_file 1: go_field_post 2: go_ filed_pre
16	R/W	0	vd1_lbuf_ram_sel : // unsigned , default = 0 0: NO share ram 1: afbc0 share ext ram

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Bit(s)	R/W	Default	Description
15:14	R/W	0	vd1_in_mux_sel : // unsigned , default = 0 0: vd1 to dobly 2: vd1 to primel
13:12	R/W	0	vd1_axi_sel : // unsigned , default = 0 0: afbc0 mif to axi 1: vd1 mif to axi
11	R/W	0	afbc0_osd3_mux_vd1 : // unsigned , default = 0 0: afbc0 to vd1 1: osd3 to vd1
10	R/W	0	afbc_vd1_sel : // unsigned , default = 0 0: vd1_mif to vpp 1: afbc0_mif
9	R/W	0	afbc0_mux_vpp_mad : // unsigned , default = 0 0: afbc0 to vpp 1: afbc0 to di
8	R/W	0	di_mif0_en : // unsigned , default = 0 0: select mif to vpp 1: mif to di
7:0	R/W	0	afbc0_gclk_ctrl : // unsigned , default = 0

Table 8-1639 VD2_AFBCD1_MISC_CTRL 0x1a0b

Bit(s)	R/W	Default	Description
31:24	R/W	0	vd2_osd_mux_alpha : // unsigned , default = 0 config vd2_osd alpha value.
22	R/W	0	Vd2_afbc1_mem_sel 0: vd2_mif share afbc_memory 1: afbc1_mif share afbc_ memory
21:20	R/W	0	Vd2_afbc1_go_field_sel : // unsigned , default = 0 0: go_file 1: go_field_post 2: go_filed_pre
19:18	R/W	0	Vd2_mif_go_field_sel : // unsigned , default = 0 0: go_file 1: go_field_post 2: go_ filed_pre
16	R/W	0	vd2_lbuf_ram_sel : // unsigned , default = 0 0: NO share ram 1: afbc1 share ext ram
15:14	R/W	0	vd2_in_mux_sel : // unsigned , default = 0 0: afbc2_vd2 to vd2 2: osd4 to vd2
13:12	R/W	0	vd2_axi_sel : // unsigned , default = 0 0: afbc1 mif to axi 1: vd2 mif to axi
11	R/W	0	afbc1_osd4_mux_vd1 : // unsigned , default = 0 0: afbc1 to vd1 1: osd4 to vd2
10	R/W	0	afbc_vd2_sel : // unsigned , default = 0 0: vd2_mif to vpp 1: afbc1_mif
9	R/W	0	afbc1_mux_vpp_mad : // unsigned , default = 0 0: afbc1 to vpp 1: afbc1 to di
8	R/W	0	afbc_2mad_out_sel : // unsigned , default = 0 0: select vd1 to di 1: vd2 to di
8:0	R/W	0	Afbc1_gclk_ctrl : // unsigned , default = 0

Table 8-1640 DOLBY_PATH_CTRL 0x1a0c

Bit(s)	R/W	Default	Description
7	R/W	0	vpp_osd2_ext_mod : // unsigned , default = 0 // 0: 12bits 1: 10bits
6	R/W	0	vpp_osd1_ext_mod : // unsigned , default = 0 // 0: 12bits 1: 10bits
5	R/W	0	dolby1_vd2_ext_mod : // unsigned , default = 0 // 0: 12bits 1: 10bits
4	R/W	0	dolby0_vd1_ext_mod : // unsigned , default = 0 // 0: 12bits 1: 10bits
3	R/W	0	Osd2_dolby_bypass_en : // unsigned , default = 0
2	R/W	0	Osd1_dolby_bypass_en
1	R/W	0	Vd2_dolby_bypass_en
0	R/W	0	Vd1dolby_bypass_en

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Table 8-1641 WR_BACK_MISC_CTRL 0x1a0d

Bit(s)	R/W	Default	Description
1	R/W	0	wrbak_chan1_hsyn_en : // unsigned , default = 0 vd1 wrbak hsync enable
0	R/W	0	wrbak_chan0_hsyn_en : // unsigned , default = 0 vd0 wrbak hsync enable

Table 8-1642 OSD_PATH_MISC_CTRL 0x1a0e

Bit(s)	R/W	Default	Description
7:4	R/W	0	osd_axi_sel : // unsigned , default = 0 osd_axi_sel[0] 1:OSD1 get data from mali_afbcd 0: OSD1 get data from DDR osd_axi_sel[1] 1:OSD2 get data from mali_afbcd 0: OSD2 get data from DDR osd_axi_sel[2] 1:OSD3 get data from mali_afbcd 0: OSD3 get data from DDR osd_axi_sel[3] reserved
3	R/W	0	osd4_in_mux_sel : // unsigned , default = 0 0: osd4 to vpp 1: vd2_osd4 to vpp
2	R/W	0	osd4_mux_vpp_vd2 : // unsigned , default = 0 0: osd4 to vpp 1: osd4 to vd2
1	R/W	0	osd3_in_mux_sel : // unsigned , default = 0 0: osd3 to vpp 1: vd2_osd3 to vpp
0	R/W	0	osd3_mux_vpp_vd1 : // unsigned , default = 0 0: osd3 to vpp 1: osd3 to vd1

Table 8-1643 MALI_AFBCD_TOP_CTRL 0x1a0f

Bit(s)	R/W	Default	Description
31	R.O	0	ro_mali_dec_active : // unsigned , default = 0
23	R/W	0	Mali afbcd logic rst mode 1:logic rest controll by VIU_SW_RESET[21] 0:logic reset by vsync singal,don’t need software process
17:12	R/W	0	gclk_ctrl_osd1_afbcd : // unsigned , default = 0 mail afbc gate clock
8	R/W	1	axim_data_128b : // unsigned , default = 1 0: axi bus width=64 1: 128bit
6:4	R/W	7	axim_max_len : // unsigned , default = 7
2:0	R/W	5	axim_outstanding_trans : // unsigned , default =5

Table 8-1644 VIU_GCLK_CTRL 0x1a4f

Bit(s)	R/W	Default	Description
7:0	R/W	0	viu_gclk_ctrl : // unsigned , default =0

8.2.3.16 VPP_ARB_CTRL Registers

Table 8-1645 VPP_AFBC_RDARB_MODE 0x3970

Bit(s)	R/W	Default	Description
31-24	R/W	0	unused
23-20	R/W	0	Rdarb_sel

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Bit(s)	R/W	Default	Description
16	R/W	0	Rdarb_arb_mode
1-0	R/W	0	Rdarb_gate_clk_ctrl

Table 8-1646 VPP_AFBC_RDARB_REQEN_SLV 0x3971

Bit(s)	R/W	Default	Description
3-0	R/W	0	Rdarb_dc_req_en 1: Request is valid, 0: Invalid Bit0 : osd1 Bit1 : osd2 Bit2 : osd3 Bit3 : osd4

Table 8-1647 VPP_AFBC_RDARB_WEIGHT0_SLV 0x3972

Bit(s)	R/W	Default	Description
23-0	R/W	0	Rddc_weigh_sxn

Table 8-1648 VPP_AFBC_ARB_DBG_CTRL 0x3974

Bit(s)	R/W	Default	Description
31-0	RO	0	Det_cmd_ctrl

Table 8-1649 VPP_RDARB_MODE 0x3978

Bit(s)	R/W	Default	Description
27-20	R/W	0	Rdarb_sel 1: Request is valid, 0: Invalid Bit20: osd1 Bit21: osd2 Bit22: vd1 Bit23: vd2 Bit24: osd3 Bit25: osd4 Bit26: dolby0 Bit27:mali_afbc
17:16	R/W	0	Rdarb_arb_mode
3-0	R/W	0	Rdarb_gate_clk_ctrl

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Table 8-1650 VPP_RDARB_REQEN_SLV 0x3979

Bit(s)	R/W	Default	Description
15-0	R/W	0	Rdarb_dc_req_en 1: Request is valid, 0: Invalid Bit0: osd1 come from axi_port0 Bit1: osd2 come from axi_port0 Bit2: vd1 come from axi_port0 Bit3: vd2 come from axi_port0 Bit4: osd3 come from axi_port0 Bit5: osd4 come from axi_port0 Bit6: dolby0 come from axi_port0 Bit7:mali_afbc come from axi_port0 Bit8: osd1 come from axi_port1 Bit9: osd2 come from axi_port1 Bit10: vd1 come from axi_port1 Bit11: vd2 come from axi_port1 Bit12: osd3 come from axi_port1 Bit13: osd4 come from axi_port1 Bit14: dolby0 come from axi_port1 Bit15:mali_afbc come from axi_port1

Table 8-1651 VPP_RDARB_WEIGHT0_SLV 0x397a

Bit(s)	R/W	Default	Description
29-0	R/W	0	Rddc_weigh_sxn

Table 8-1652 VPP_RDARB_WEIGHT1_SLV 0x397b

Bit(s)	R/W	Default	Description
17-0	RO	0	Rddc_weigh_sxn

Table 8-1653 VPP_ARB_DBG_CTRL 0x397c

Bit(s)	R/W	Default	Description
31-0	RO	0	Det_cmd_ctrl

8.2.3.17 VD1 Path vd_rmem_if0 Registers

Table 8-1654 VD1 Path vd_rmem_if0 Registers VD1_IF0_GEN_REG 0x3200

Bit(s)	R/W	Default	Description
31	R/W	0	ENABLE_FREE_CLK. 0: Gated clock for power saving 1: Free-running clock to drive logic
30	R/W	0	SW_RESET: Write 1 to this bit to generate a pulse to reset everything except registers.
29	R/W	0	RESET_ON_GO_FIELD: Define whether to reset state machines on go_field pulse. 0: No reset on go_field 1: go_field reset everything except registers
28	R/W	0	URGENT_CHROMA: Set urgent level for chroma fifo request from DDR. 0: Non urgent

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Bit(s)	R/W	Default	Description
			1: Urgent
27	R/W	0	URGENT_LUMA: Set urgent level for luma fifo request from DDR. 0: Non urgent 1: Urgent
26	R/W	0	Chroma_end_at_last line: For chroma line, similar to luma_end_at_last_line, as below. Not used if data are stored together in one canvas.
25	R/W	0	Luma_end_at_last_line: Control whether continue outputting luma line past last line. 0: Repeat the last line or dummy pixels, after past the last line 1: Stop outputting data, once past the last line.
24-19	R/W	4	Hold_lines: After go_field, the number of lines to hold before the module is enabled.
18	R/W	0	LAST_LINE: This bit controls whether we simply repeat the last line or we push dummy pixels. ‘1’ tells the state-machines to repeat the last line using the dummy pixels defined in the register below. ‘0’ indicates that the state-machine should re-read the last line of real data.
17	R	0	Busy status of the state-machines. ‘1’ = busy, ‘0’ = idle
16	R/W	0	DEMUX_MODE: 0 = 4:2:2, 1 = RGB (24-bit). This value is used to control the de-muxing logic when the picture is stored together. When a picture is stored together, the data is read into a single FIFO (the Y FIFO) and must be demultiplexed into the “drain” outputs. In the case of 4:2:2 the data is assumed to be stored in memory in 16-bit chunks: <YCb><YCr><YCb><YCr>,… the Y, Cb and Cr 8-bit values are pulled from the single Y-FIFO and sent out in pairs. This value is only valid when the picture is stored together. If the picture is separated into different canvases, then this bit field is ignored.
15-14	R/W	0	BYTES_PER_PIXEL: This value is used to determine how many bytes are associated with each pixel. 0: This value should be used if the image is stored separately (e.g. RGB or Y, Cb, Cr). 1: This value should be used if the data is 4:2:2 data stored together. In this case each pixel , YCb or YCr, is 16-bits (two bytes). 2: This value should be used if the RGB (24-bit) data is stored together. 3: reserved for future use (alpha RGB).
13-12	R/W	0	DDR_BURST_SIZE_CR: This value is used to control the DDR burst request size for the Cr FIFO. 0: Maximum burst = 24 64-bit values 1: Maximum burst = 32 64-bit values 2: Maximum burst = 48 64-bit values 3: Maximum burst = 64 64-bit values
11-10	R/W	0	DDR_BURST_SIZE_CB: This value is used to control the DDR burst request size for the Cb FIFO. 0: Maximum burst = 24 64-bit values 1: Maximum burst = 32 64-bit values 2: Maximum burst = 48 64-bit values 3: Maximum burst = 64 64-bit values
9-8	R/W	0	DDR_BURST_SIZE_Y: This value is used to control the DDR burst request size for the Y FIFO. 0: Maximum burst = 24 64-bit values 1: Maximum burst = 32 64-bit values 2: Maximum burst = 48 64-bit values 3: Maximum burst = 64 64-bit values
7	R/W	0	MANUAL_START_FRAME: non-latching bit that can be used to simulate the go_ field signal for simulation.
6	R/W	0	CHRO_RPT_LASTL_CTRL: This bit controls whether to allow VPP’s chroma-repeat request.

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Bit(s)	R/W	Default	Description
			0: Chroma-repeat pulses from VPP are ignored 1: Chroma-repeat pulses from VPP are used.
5	R/W	0	Unused
4	R/W	0	LITTLE_ENDIAN: This bit defines the endianess of the memory data . 0: Pixel data are stored big-endian in memory 1: Pixel data are stored little-endian in memory
3	R/W	0	Chroma_hz_avg: For chroma line output control, similar to luma_hz_avg, as below. Not used if data are stored together in one canvas.
2	R/W	0	Luma_hz_avg: Enable output half amount of data per line to save bandwidth. 0: Output every pixel per line 1: Output half line, each data averaged between every 2 pixels Note: For 4:2:2 mode data stored together in one canvas, only do averaging over luma data.
1	R/W	0	SEPARATE_EN: Set this bit to 1 if the image is in separate canvas locations.
0	R/W	0	ENABLE: This bit is set to 1 to enable the FIFOs and other logic. This bit can be set to 0 to cleanup and put the logic into an IDLE state.

Table 8-1655 VD1_IF0_CANVAS0 – Picture 0 0x3201

Bit(s)	R/W	Default	Description
31-24	R/W	0	unused
23-16	R/W	0	CANVAS0_ADDR2: Canvas table address for picture 0 for component 2 (Cr FIFO). This value is ignored when the picture is stored together
15-8	R/W	0	CANVAS0_ADDR1: Canvas table address for picture 0 for component 1 (Cb FIFO). This value is ignored when the picture is stored together
7-0	R/W	0	CANVAS0_ADDR0: Canvas table address for picture 0 for component 0 (Y FIFO).

Table 8-1656 VD1_IF0_CANVAS1 – Picture 1 0x3202

Bit(s)	R/W	Default	Description
31-24	R/W	0	unused
23-16	R/W	0	CANVAS1_ADDR2: Canvas table address for picture 1 for component 2 (Cr FIFO). This value is ignored when the picture is stored together
15-8	R/W	0	CANVAS1_ADDR1: Canvas table address for picture 1 for component 1 (Cb FIFO). This value is ignored when the picture is stored together
7-0	R/W	0	CANVAS1_ADDR0: Canvas table address for picture 1 for component 0 (Y FIFO).

Table 8-1657 VD1_IF0_LUMA_X0 – Picture 0 0x3203

Bit(s)	R/W	Default	Description
31	R/W	0	Unused
30-16	R/W	0	LUMA_X_END0: Picture 0, luma X end value
15	R/W	0	Unused
14-0	R/W	0	LUMA_X_START0: Picture 0, luma X start value

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Table 8-1658 VD1_IF0_LUMA_Y0 – Picture 0 0x3204

Bit(s)	R/W	Default	Description
31-29	R/W	0	Unused
28-16	R/W	0	LUMA_Y_END0: Picture 0, luma Y end value
15-13	R/W	0	Unused
12-0	R/W	0	LUMA_Y_START0: Picture 0, luma Y start value

Table 8-1659 VD1_IF0_CHROMA_X0 – Picture 0 0x3205

Bit(s)	R/W	Default	Description
31	R/W	0	Unused
30-16	R/W	0	CHROMA_X_END0: Picture 0, chroma X end value. This value is only used when the picture is not stored together.
15	R/W	0	Unused
14-0	R/W	0	CHROMA _X_START0: Picture 0, chroma X start value. This value is only used when the picture is not stored together.

Table 8-1660 VD1_IF0_ CHROMA_Y0 – Picture 0 0x3206

Bit(s)	R/W	Default	Description
31-29	R/W	0	Unused
28-16	R/W	0	CHROMA _Y_END0: Picture 0, chroma Y end value. This value is only used when the picture is not stored together.
15-13	R/W	0	Unused
12-0	R/W	0	CHROMA _Y_START0: Picture 0, chroma Y start value. This value is only used when the picture is not stored together.

Table 8-1661 VD1_IF0_LUMA_X1 – Picture 1 0x3207

Bit(s)	R/W	Default	Description
31	R/W	0	Unused
30-16	R/W	0	LUMA_X_END1: Picture 1, luma X end value
15	R/W	0	Unused
14-0	R/W	0	LUMA_X_START1: Picture 1, luma X start value

Table 8-1662 VD1_IF0_LUMA_Y1 – Picture 1 0x3208

Bit(s)	R/W	Default	Description
31-29	R/W	0	Unused
28-16	R/W	0	LUMA_Y_END1: Picture 1, luma Y end value

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Bit(s)	R/W	Default	Description
15-13	R/W	0	Unused
12-0	R/W	0	LUMA_Y_START1: Picture 1, luma Y start value

Table 8-1663 VD1_IF0_CHROMA_X1 – Picture 1 0x3209

Bit(s)	R/W	Default	Description
31	R/W	0	Unused
30-16	R/W	0	CHROMA_X_END1: Picture 1, chroma X end value. This value is only used when the picture is not stored together.
15	R/W	0	Unused
14-0	R/W	0	CHROMA _X_START1: Picture 1, chroma X start value. This value is only used when the picture is not stored together.

Table 8-1664 VD1_IF0_ CHROMA_Y1 – Picture 1 0x320A

Bit(s)	R/W	Default	Description
31-29	R/W	0	Unused
28-16	R/W	0	CHROMA _Y_END1: Picture 1, chroma Y end value. This value is only used when the picture is not stored together.
15-13	R/W	0	Unused
12-0	R/W	0	CHROMA _Y_START1: Picture 1, chroma Y start value. This value is only used when the picture is not stored together.

Table 8-1665 VD1_IF0_REPEAT_LOOP – Pictures 0 and 1 0x320B

Bit(s)	R/W	Default	Description
31-24	R/W	0	CHROMA_RPT_LOOP1: Repeat loop for Picture 1. Bits[6:4] = start loop pointer, bits [2:0] = end loop pointer. Bits [7] and [3] are ignored.
23-16	R/W	0	LUMA_RPT_LOOP1: Repeat loop for Picture 1. Bits[6:4] = start loop pointer, bits [2:0] = end loop pointer. Bits [7] and [3] are ignored.
15-8	R/W	0	CHROMA_RPT_LOOP0: Repeat loop for Picture 0. Bits[6:4] = start loop pointer, bits [2:0] = end loop pointer. Bits [7] and [3] are ignored.
7-0	R/W	0	LUMA_RPT_LOOP0: Repeat loop for Picture 0. Bits[6:4] = start loop pointer, bits [2:0] = end loop pointer. Bits [7] and [3] are ignored.

Table 8-1666 VD1_IF0_LUMA0_RPT_PAT – Picture 0 LUMA repeat pattern 0x320C

Bit(s)	R/W	Default	Description
31-0	R/W	0	Luma repeat/skip pattern for picture 0

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Bits	Pattern Index	Pattern description
31-28	7	Repeat/skip pattern: Bit[3] = 0 indicates repeat. Bit[3] = 1 indicates either skip, or output this line and then skip. How to interpret this bit depends on the value of the previous pattern's Bit[3]. If previous Bit[3]=0, then skip; If previous Bit[3]=1, then output this line and then skip. Bits[2:0] indicate the skip / repeat count. Blow is an example of consecutive patterns, the start line is line 0: {0010} Repeat this line (line 0) two more times for a total of three line reads. Proceed to next line (line 1). {0000} Don't repeat this line (line 1). This line will be read just once. Proceed to next line (line 2). {1000} Skip one line (line 2) to get to the next line (line 3). The skip implies that the next line (line 3) should be read at least once. {1011} Read this line (line 3) once, and then skip the next four lines to get to the next line (line 8). The skip implies that the next line (line 8) should be read at least once. {0100} Repeat this line (line 8) four more times for a total of five line read. Proceed to next line (line 9). {1001} Skip two lines to get to the next line (line 11). The skip implies that the next line (line 11) should be read at least once.
27-24	6	See pattern definition above.
23-20	5	See pattern definition above.
19-16	4	See pattern definition above.
15-12	3	See pattern definition above.
11-8	2	See pattern definition above.
7-4	1	See pattern definition above.
3-0	0	See pattern definition above.

Table 8-1667 VD1_IF0_CHROMA0_RPT_PAT – Picture 0 CHROMA repeat pattern 0x320D

Bit(s)	R/W	Default	Description
31-0	R/W	0	Chroma repeat/skip pattern for picture 0. See picture 0 luma pattern for description. This value is only used when the picture is not stored together.

Table 8-1668 VD1_IF0_LUMA1_RPT_PAT – Picture 1 LUMA repeat pattern 0x320E

Bit(s)	R/W	Default	Description
31-0	R/W	0	Luma repeat/skip pattern for picture 1. See picture 0 luma pattern for description.

Table 8-1669 VD1_IF0_CHROMA1_RPT_PAT – Picture 1 CHROMA repeat pattern 0x320F

Bit(s)	R/W	Default	Description
31-0	R/W	0	Chroma repeat/skip pattern for picture 1. See picture 0 luma pattern for description. This value is only used when the picture is not stored together.

Table 8-1670 VD1_IF0_LUMA_PSEL – Picture 0 and 1's LUMA 0x3210

Bit(s)	R/W	Default	Description
31-28	R/W	0	unused
27-26	R/W	0	Luma_psel_mode: controls whether it's single-picture or two-picture mode. {00} Only picture 0 is used. Ignore settings defined in Luma_psel_last_line, Luma_psel_pattern and Luma_psel_loop.

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Bit(s)	R/W	Default	Description
			{01} Only picture 1 is used. Ignore settings defined in Luma_psel_last_line, Luma_psel_pattern and Luma_psel_loop. {1x} Two-picture mode.
25-24	R/W	0	Luma_psel_last_line: select which picture's last line to output, during repeat last line mode. Bit[0]=0, when picture 0 past the last line, use picture 0's last line during repeat last line mode; Bit[0]=1, when picture 0 past the last line, use picture 1's last line during repeat last line mode; Bit[1]=0, when picture 1 past the last line, use picture 0's last line during repeat last line mode; Bit[1]=1, when picture 1 past the last line, use picture 1's last line during repeat last line mode.
23-8	R/W	0	Luma_psel_pattern. If the value of the bit pointed by the loop pointer is 0, output picture 0's luma line, if the bit value is 1, output picture 1's luma line.
7-4	R/W	0	Luma_psel_loop start pointer.
3-0	R/W	0	Luma_psel_loop end pointer.

Table 8-1671 VD1_IF0_CHROMA_PSEL – Picture 0 and 1's CHROMA 0x3211

Bit(s)	R/W	Default	Description
31-28	R/W	0	unused
27-26	R/W	0	Chroma_psel_mode: see luma_psel_mode. This value is only used when the picture is not stored together.
25-24	R/W	0	Chroma_psel_last_line: See luma_psel_last_line. This value is only used when the picture is not stored together.
23-8	R/W	0	Chroma_psel_pattern. If the value of the bit pointed by the loop pointer is 0, output picture 0's chroma line, if the bit value is 1, output picture 1's chroma line. This value is only used when the picture is not stored together.
7-4	R/W	0	Chroma_psel_loop start pointer. This value is only used when the picture is not stored together.
3-0	R/W	0	Chroma_psel_loop end pointer. This value is only used when the picture is not stored together.

Table 8-1672 VD1_IF0_DUMMY_PIXEL 0x3212

Bit(s)	R/W	Default	Description
31-24	R/W	0x00	Y or R dummy pixel value ,8bit
23-16	R/W	0x80	Cb or G dummy pixel value ,8bit
15-8	R/W	0x80	Cr or B dummy pixel value ,8bit
7-0	R/W	0	unused

VD1_RANGE_MAP_Y 0x321A

VD1_RANGE_MAP_CB 0x321B

VD1_RANGE_MAP_CR 0x321C

Output data range conversion function:

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Y [n] = clip (Round (((Y [n] +DIN_OFFSET) * RANGE_MAP_COEF) / (1<<RANGE_MAP_SR)) +DOUT_OFFSET);

To perform VC-1 range reduction, set the following:

DIN_OFFSET = 0x180 = -128;

RANGE_MAP_COEF = RANGE_MAPY + 9

RANGE_MAP_SR = 3

DOUT_OFFSET = 0x080 = 128

To get the equivalent function:

Y[n] = clip((((Y[n]-128) * (RANGE_MAPY + 9) + 4) >> 3) +128);

Bit(s)	R/W	Default	Description
31-23	R/W	0	DIN_OFFSET
22-15	R/W	0	RANGE_MAP_COEF
14	R/W	0	unused
13-10	R/W	0	RANGE_MAP_SR
9-1	R/W	0	DOUT_OFFSET
0	R/W	0	RANGE_MAP_EN

Table 8-1673 VD1_IF0_GEN_REG2 0x321D

Bit(s)	R/W	Default	Description
31-2	R/W	0	unused
1-0	R/W	0	COLOR_MAP: Define color map for NV12 or NV21 mode. Only applicable when VD1_IF0_GEN_REG.SEPARATE_EN = 1. 0: NOT NV12 or NV21; 1: NV12 (CbCr); 2: NV21 (CrCb).

Table 8-1674 VD1_IF0_GEN_REG3 0x3216

Bit(s)	R/W	Default	Description
11-10	R/W	0	cntl_dbg_mode
9-8	R/W	0	cntl_bits_mode : 0->8bit 1->10bit 422 2->10bit 444
6-4	R/W	3	cntl_blk_len
2-1	R/W	1	cntl_burst_len
0	R/W	1	cntl_64bit_rev

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Table 8-1675 VIU_VD1_FMT_CTRL 0x3218

Bit(s)	R/W	Default	Description
31	R/W	0	gate_clk_en. 0=No clock gating, free-running; 1=Enable clock gating for power saving.
30	R/W	0	soft_rst. If true, reset formatters.
29	R/W	0	unused
28	R/W	0	if true, horizontal formatter use repeating to generete pixel, otherwise use bilinear interpolation
27-24	R/W	0	horizontal formatter initial phase
23	R/W	0	horizontal formatter repeat pixel 0 enable
22-21	R/W	0	horizontal Y/C ratio, 00: 1:1, 01: 2:1, 10: 4:1
20	R/W	0	horizontal formatter enable
19	R/W	0	if true, always use phase0 while vertical formater, meaning always repeat data, no interpolation
18	R/W	0	if true, disable vertical formatter chroma repeat last line
17	R/W	0	veritcal formatter dont need repeat line on phase0, 1: enable, 0: disable
16	R/W	0	veritcal formatter repeat line 0 enable
15-12	R/W	0	vertical formatter skip line num at the beginning
11-8	R/W	0	vertical formatter initial phase
7-1	R/W	0	vertical formatter phase step (3.4)
0	R/W	0	vertical formatter enable

Table 8-1676 VIU_VD1_FMT_W 0x3219

Bit(s)	R/W	Default	Description
27-16	R/W	0	horizontal formatter width
11-0	R/W	0	vertical formatter width

8.2.3.18 VD2 Path vd_rmem_if0 Registers

VD2_IF0_GEN_REG 0x3220

Same as VD1_IF0_GEN_REG

VD2_IF0_CANVAS0 0x3221

Same as VD1_IF0_CANVAS0

VD2_IF0_CANVAS1 0x3222

Same as VD1_IF0_CANVAS1

VD2_IF0_LUMA_X0 0x3223

Same as VD1_IF0_LUMA_X0

VD2_IF0_LUMA_Y0 0x3224

Same as VD1_IF0_LUMA_Y0

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VD2_IF0_CHROMA_X0 0x3225

Same as VD1_IF0_CHROMA_X0

VD2_IF0_CHROMA_Y0 0x3226

Same as VD1_IF0_CHROMA_Y0

VD2_IF0_LUMA_X1 0x3227

Same as VD1_IF0_LUMA_X1

VD2_IF0_LUMA_Y1 0x3228

Same as VD1_IF0_LUMA_Y1

VD2_IF0_CHROMA_X1 0x3229

Same as VD1_IF0_CHROMA_X1

VD2_IF0_CHROMA_Y1 0x327A

Same as VD1_IF0_CHROMA_Y1

VD2_IF0_RPT_LOOP 0x322B

Same as VD1_IF0_RPT_LOOP

VD2_IF0_LUMA0_RPT_PAT 0x322C

Same as VD1_IF0_LUMA0_RPT_PAT

VD2_IF0_CHROMA0_RPT_PAT 0x322D

Same as VD1_IF0_CHROMA0_RPT_PAT

VD2_IF0_LUMA1_RPT_PAT 0x322E

Same as VD1_IF0_LUMA1_RPT_PAT

VD2_IF0_CHROMA1_RPT_PAT 0x322F

Same as VD1_IF0_CHROMA1_RPT_PAT

VD2_IF0_LUMA_PSEL 0x3230

Same as VD1_IF0_LUMA_PSEL

VD2_IF0_CHROMA_PSEL 0x3231

Same as VD1_IF0_CHROMA_PSEL

VD2_IF0_DUMMY_PIXEL 0x3232

Same as VD1_IF0_DUMMY_PIXEL

VD2_RANGE_MAP_Y 0x323A

Same as VD1_RANGE_MAP_Y

VD2_RANGE_MAP_CB 0x323B

Same as VD1_RANGE_MAP_CB

VD2_RANGE_MAP_CR 0x323C

Same as VD1_RANGE_MAP_CR

VD2_IF0_GEN_REG2 0x323D

Same as VD1_IF0_GEN_REG2

VD2_IF0_GEN_REG3 0x3236

Same as VD1_IF0_GEN_REG3

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Table 8-1677 VIU_VD2_FMT_CTRL 0x3238

Bit(s)	R/W	Default	Description
31	R/W	0	gate_clk_en. 0=No clock gating, free-running; 1=Enable clock gating for power saving.
30	R/W	0	soft_rst. If true, reset formatters.
28	R/W	0	if true, horizontal formatter use repeating to generete pixel, otherwise use bilinear interpolation
27-24	R/W	0	horizontal formatter initial phase
23	R/W	0	horizontal formatter repeat pixel 0 enable
22-21	R/W	0	horizontal Y/C ratio, 00: 1:1, 01: 2:1, 10: 4:1
20	R/W	0	horizontal formatter enable
19	R/W	0	if true, always use phase0 while vertical formater, meaning always repeat data, no interpolation
18	R/W	0	if true, disable vertical formatter chroma repeat last line
17	R/W	0	veritcal formatter dont need repeat line on phase0, 1: enable, 0: disable
16	R/W	0	veritcal formatter repeat line 0 enable
15-12	R/W	0	vertical formatter skip line num at the beginning
11-8	R/W	0	vertical formatter initial phase
7-1	R/W	0	vertical formatter phase step (3.4)
0	R/W	0	vertical formatter enable

Table 8-1678 VIU_VD2_FMT_W 0x3239

Bit(s)	R/W	Default	Description
27-16	R/W	0	horizontal formatter width
11-0	R/W	0	vertical formatter width

8.2.3.19 Osd_blend Registers

Table 8-1679 VIU_OSD_BLEND_CTRL 0x39b0

Bit(s)	R/W	Default	Description
31:29	R/W	0x0	hold_lines : //unsigned , default = 3'h0, hold_lines(line) after go_field ,module active
28:27	R/W	0x3	blend2_premult_en : //unsigned , default = 2'h3, blend2 input premult label 1:pre-mult input 0:unpremult input
26	R/W	0x1	din0_byp_blend : //unsigned , default = 1'h1, blend_din0 bypass to dout0 1:by-pass 0:blend_din0 input to blend0
25	R/W	0x1	din2_osd_sel : //unsigned , default = 1'h1, blend1_dout bypass to blend2 0: blend1_dout to blend2 1:blend1_dout to dout1
24	R/W	0x1	din3_osd_sel : //unsigned , default = 1'h1, blend1_din3 bypass to dout1 1:bypass 0:blend_din3 input to blend1

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Bit(s)	R/W	Default	Description
23:20	R/W	0x5	blend_din_en : //unsigned , default = 4'h5, blend enable bits ,four bits for four input
19:16	R/W	0x0	din_premult_en : //unsigned , default = 4'h0, input premult label bits,four bits for four input
15:0	R/W	0x2341	din_reoder_sel : //unsigned , default = 16'h2341,osd_blend input reorder exp : din_reoder_sel[3:0] = 1 ,blend_din0 select osd1 din_reoder_sel [3:0] = 2 ,blend_ din0 select osd2 din_reoder_sel [3:0] = 3 ,blend_din0 select osd3 din_reoder_sel [3:0] = else,blend_din0 no src din_reoder_sel [7:4] fot blend_din1

Table 8-1680 VIU_OSD_BLEND_CTRL1 0x39c0

Bit(s)	R/W	Default	Description
17:16	R/W	0	reg_alp1_mapping_mode : //unsigned , default = 0 , osd_blend dout1 alpha divid- or mode 8bit alpha:set 0 9bit alpha:set 3
14:13	R/W	0	reg_div1_gclk_en : //unsigned , default = 0 , osd_blend dout1 alpha dividor gclk_ en
12	R/W	0	reg_div1_alpha_en : //unsigned , default = 0 , osd_blend dout1 alpha dividor gclk_en enable
9:8	R/W	0	osdbld_gclk_ctrl : //unsigned , default = 0 , osdbld_gclk_ctrl
5:4	R/W	0	reg_alp_mapping_mode : //unsigned , default = 0 , osd_blend dout0 alpha dividor mode 8bit alpha:set 0 9bit alpha:set 3
2:1	R/W	0	reg_div_gclk_en : //unsigned , default = 0 , osd_blend dout0 alpha dividor gclk_ en
0	R/W	0	reg_div_alpha_en : //unsigned , default = 0 , osd_blend dout0 alpha dividor gclk_ en enable

Table 8-1681 VIU_OSD_BLEND_DIN0_SCOPE_H 0x39b1

Bit(s)	R/W	Default	Description
28:16	R/W	0x2d0	bld_din0_h_end : ///unsigned , default = 13'h2d0,blend_din0 h_end
12:0	R/W	0x0	bld_din0_h_start : ///unsigned , default = 13'h0 ,blend_din0 h_start

Table 8-1682 VIU_OSD_BLEND_DIN0_SCOPE_V 0x39b2

Bit(s)	R/W	Default	Description
28:16	R/W	0x1e0	bld_din0_v_end : ///unsigned , default = 13'h1e0,blend_din0 v_end
12:0	R/W	0x0	bld_din0_v_start : ///unsigned , default = 13'h0,blend_din0 v_start

Table 8-1683 VIU_OSD_BLEND_DIN1_SCOPE_H 0x39b3

Bit(s)	R/W	Default	Description
28:16	R/W	0x2d0	bld_din1_h_end : ///unsigned , default = 13'h2d0
12:0	R/W	0x0	bld_din1_h_start : ///unsigned , default = 13'h0

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Table 8-1684 VIU_OSD_BLEND_DIN1_SCOPE_V 0x39b4

Bit(s)	R/W	Default	Description
28:16	R/W	0x1e0	bld_din1_v_end : ///unsigned , default = 13'h1e0
12:0	R/W	0x0	bld_din1_v_start : ///unsigned , default = 13'h0

Table 8-1685 VIU_OSD_BLEND_DIN2_SCOPE_H 0x39b5

Bit(s)	R/W	Default	Description
28:16	R/W	0x2d0	bld_din2_h_end : ///unsigned , default = 13'h2d0
12:0	R/W	0x0	bld_din2_h_start : ///unsigned , default = 13'h0

Table 8-1686 VIU_OSD_BLEND_DIN2_SCOPE_V 0x39b6

Bit(s)	R/W	Default	Description
28:16	R/W	0x1e0	bld_din2_v_end : ///unsigned , default = 13'h1e0
12:0	R/W	0x0	bld_din2_v_start : ///unsigned , default = 13'h0

Table 8-1687 VIU_OSD_BLEND_DIN3_SCOPE_H 0x39b7

Bit(s)	R/W	Default	Description
28:16	R/W	0x2d0	bld_din3_h_end : ///unsigned , default = 13'h2d0
12:0	R/W	0x0	bld_din3_h_start : ///unsigned , default = 13'h0

Table 8-1688 VIU_OSD_BLEND_DIN3_SCOPE_V 0x39b8

Bit(s)	R/W	Default	Description
28:16	R/W	0x1e0	bld_din3_v_end : ///unsigned , default = 13'h1e0
12:0	R/W	0x0	bld_din3_v_start : ///unsigned , default = 13'h0

Table 8-1689 VIU_OSD_BLEND_DUMMY_DATA0 0x39b9

Bit(s)	R/W	Default	Description
23:16	R/W	0x00	blend0_dummy_data_y : //unsigned , default = 8'h00
15:8	R/W	0x80	blend0_dummy_data_cb : //unsigned , default = 8'h80
7:0	R/W	0x80	blend0_dummy_data_cr : //unsigned , default = 8'h80

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Table 8-1690 VIU_OSD_BLEND_DUMMY_ALPHA 0x39ba

Bit(s)	R/W	Default	Description
28:20	R/W	0x0	blend0_dummy_alpha : //unsigned , default = 9'h0
19:11	R/W	0x0	blend1_dummy_alpha : //unsigned , default = 9'h0
8:0	R/W	0x0	blend2_dummy_alpha : //unsigned , default = 9'h0

Table 8-1691 VIU_OSD_BLEND_BLEND0_SIZE 0x39bb

Bit(s)	R/W	Default	Description
28:16	R/W	0x1e0	blend0_vsize : //unsigned , default = 13'h1e0,blend0_vsize
12:0	R/W	0x2d0	blend0_hsize : //unsigned , default = 13'h2d0,blend0_hsize

Table 8-1692 VIU_OSD_BLEND_BLEND1_SIZE 0x39bc

Bit(s)	R/W	Default	Description
28:16	R/W	0x1e0	blend1_vsize : //unsigned , default = 13'h1e0
12:0	R/W	0x2d0	blend1_hsize : ///unsigned , default = 13'h2d0

8.2.3.20 Pre/Post Blend Registers

Table 8-1693 VPP_PRE_BLEND_CTRL 0x3960

Bit(s)	R/W	Default	Description
27:20	R/W	0x4	hold_lines : //unsigned , default = 8'h4,blend work after hold_lines line after go_field
1:0	R/W	0x1	gclk_ctrl : //unsigned , default = 16'h1,gating ctrl

Table 8-1694 VPP_PRE_BLEND_BLEND_DUMMY_DATA 0x3961

Bit(s)	R/W	Default	Description
23:16	R/W	0x0	blend0_dummy_data_y : //unsigned , default = 8'h0,blend dummy data
15:8	R/W	0x80	blend0_dummy_data_cb : //unsigned , default = 8'h80,blend dummy data
7:0	R/W	0x80	blend0_dummy_data_cr : //unsigned , default = 8'h80 ,blend dummy data

Table 8-1695 VPP_PRE_BLEND_DUMMY_ALPHA 0x3962

Bit(s)	R/W	Default	Description
28:20	R/W	0x0	blend0_dummy_alpha : //unsigned , default = 9'h0,blend dummy alpha
19:11	R/W	0x0	blend1_dummy_alpha : //unsigned , default = 9'h0,blend dummy alpha
8:0	R/W	0x0	blend2_dummy_alpha : //unsigned , default = 9'h0,blend dummy alpha

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Table 8-1696 VPP_PRE_BLEND2_RO_CURRENT_XY 0x3963

Bit(s)	R/W	Default	Description
28:16	R.O	0x0	ro_current_x : //unsigned , default = 32'h0,blend out x point
12:0	R.O	0x0	ro_current_y : //unsigned , default = 32'h0,blend out x point

Table 8-1697 VPP_POST_PRE_BLEND_CTRL 0x3967

Bit(s)	R/W	Default	Description
27:20	R/W	0x4	hold_lines : //unsigned , default = 8'h4,blend work after hold_lines line after go_ field
1:0	R/W	0x1	gclk_ctrl : //unsigned , default = 16'h1,gating ctrl

Table 8-1698 VPP_POST_BLEND_BLEND_DUMMY_DATA 0x3968

Bit(s)	R/W	Default	Description
23:16	R/W	0x0	blend0_dummy_data_y : //unsigned , default = 8'h0,blend dummy data
15:8	R/W	0x80	blend0_dummy_data_cb : //unsigned , default = 8'h80,blend dummy data
7:0	R/W	0x80	blend0_dummy_data_cr : //unsigned , default = 8'h80 ,blend dummy data

Table 8-1699 VPP_POST-BLEND_DUMMY_ALPHA 0x3969

Bit(s)	R/W	Default	Description
28:20	R/W	0x0	blend0_dummy_alpha : //unsigned , default = 9'h0,blend dummy alpha
19:11	R/W	0x0	blend1_dummy_alpha : //unsigned , default = 9'h0,blend dummy alpha
8:0	R/W	0x0	blend2_dummy_alpha : //unsigned , default = 9'h0,blend dummy alpha

Table 8-1700 VPP_POST_BLEND2_RO_CURRENT_XY 0x396a

Bit(s)	R/W	Default	Description
28:16	R.O	0x0	ro_current_x : //unsigned , default = 32'h0,blend out x point
12:0	R.O	0x0	ro_current_y : //unsigned , default = 32'h0,blend out x point

8.2.3.21 VPU AFBC Registers

Table 8-1701 AFBC_ENABLE 0x1ae0

Bit(s)	R/W	Default	Description
22	R/W	0	reg_fmt_size_sw_mode : unsigned, default = 0, 0:hw mode 1:sw mode for format size
21	R/W	1	reg_addr_link_en : unsigned, default = 1, 1:enable
20	R/W	0	reg_fmt444_comb : unsigned, default = 0, 0: 444 8bit uncomb
19	R/W	0	reg_dos_uncomp_mode : unsigned , default = 0

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Bit(s)	R/W	Default	Description
18:16	R/W	4	soft_rst : unsigned , default = 4
13:12	R/W	1	ddr_blk_size : unsigned , default = 1
11:9	R/W	3	cmd_blk_size : unsigend , default = 3
8	R/W	0	dec_enable : unsigned , default = 0
1	R/W	1	head_len_sel : unsigned , default = 1
0	R/W	0	dec_frm_start : unsigned , default = 0

Table 8-1702 AFBC_MODE 0x1ae1

Bit(s)	R/W	Default	Description
29	R/W	0	ddr_sz_mode : uns, default = 0 , 0: fixed block ddr size 1 : unfixed block ddr size;
28	R/W	0	blk_mem_mode : uns, default = 0 , 0: fixed 16x128 size; 1 : fixed 12x128 size
27:26	R/W	0	rev_mode : uns, default = 0 , reverse mode
25:24	R/W	3	mif_urgent : uns, default = 3 , info mif and data mif urgent
22:16	R/W	4	hold_line_num : uns, default = 4 ,
15:14	R/W	2	burst_len : uns, default = 2, 0: burst1 1:burst2 2:burst4
13:8	R/W	0	compbits_yuv : uns, default = 0 , bit 1:0,: y component bitwidth : 00-8bit 01-9bit 10-10bit 11-12bit bit 3:2,: u component bitwidth : 00-8bit 01-9bit 10-10bit 11-12bit bit 5:4,: v component bitwidth : 00-8bit 01-9bit 10-10bit 11-12bit
7:6	R/W	0	vert_skip_y : uns, default = 0 , luma vert skip mode : 00-y0y1, 01-y0, 10-y1, 11-(y0+y1)/2
5:4	R/W	0	horz_skip_y : uns, default = 0 , luma horz skip mode : 00-y0y1, 01-y0, 10-y1, 11-(y0+y1)/2
3:2	R/W	0	vert_skip_uv : uns, default = 0 , chroma vert skip mode : 00-y0y1, 01-y0, 10-y1, 11-(y0+y1)/2
1:0	R/W	0	horz_skip_uv : uns, default = 0 , chroma horz skip mode : 00-y0y1, 01-y0, 10-y1, 11-(y0+y1)/2

Table 8-1703 AFBC_SIZE_IN 0x1ae2

Bit(s)	R/W	Default	Description
28:16	R/W	1920	hsize_in : uns, default = 1920 , pic horz size in unit: pixel
12:0	R/W	1080	vsize_in : uns, default = 1080 , pic vert size in unit: pixel

Table 8-1704 AFBC_DEC_DEF_COLOR 0x1ae3

Bit(s)	R/W	Default	Description
29:20	R/W	0	def_color_y : uns, default = 0, afbc dec y default setting value
19:10	R/W	0	def_color_u : uns, default = 0, afbc dec u default setting value
9: 0	R/W	0	def_color_v : uns, default = 0, afbc dec v default setting value

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Table 8-1705 AFBC_CONV_CTRL 0x1ae4

Bit(s)	R/W	Default	Description
13:12	R/W	2	fmt_mode : uns, default = 2, 0:yuv444 1:yuv422 2:yuv420
11: 0	R/W	256	conv_lbuf_len : uns, default = 256, unit=16 pixel need to set = 2^n

Table 8-1706 AFBC_LBUF_DEPTH 0x1ae5

Bit(s)	R/W	Default	Description
27:16	R/W	128	dec_lbuf_depth : uns, default = 128; // unit= 8 pixel
11:0	R/W	128	mif_lbuf_depth : uns, default = 128;

Table 8-1707 AFBC_HEAD_BADDR 0x1ae6

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	mif_info_baddr : uns, default = 32'h0;

Table 8-1708 AFBC_BODY_BADDR 0x1ae7

Bit(s)	R/W	Default	Description
31:0	R/W	0x0001_0000	mif_data_baddr : uns, default = 32'h0001_0000;

Table 8-1709 AFBC_SIZE_OUT 0x1ae8

Bit(s)	R/W	Default	Description
31:29	R/W	0	reserved
28:16	R/W	1920	hszie_out: uns, default = 1920 ; // unit: 1 pixel
15:13	R/W	0	reserved
12:0	R/W	1080	Vsize_out : uns, default = 1080 ; // unit: 1 pixel

Table 8-1710 AFBC_OUT_YSCOPE 0x1ae9

Bit(s)	R/W	Default	Description
28:16	R/W	0	out_vert_bgn : uns, default = 0 ; // unit: 1 pixel
12:0	R/W	1079	out_vert_end : uns, default = 1079 ; // unit: 1 pixel

Table 8-1711 AFBC_STAT 0x1aea

Bit(s)	R/W	Default	Description
31:1	RO	0	ro_dbg_top_info : uns, default = 0
0	R/W	0	frm_end_stat : uns, default = 0 frame end status

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Table 8-1712 AFBC_VD_CFMT_CTRL 0x1aeb

Bit(s)	R/W	Default	Description
31	R/W	0	cfmt_gclk_bit_dis : uns, default = 0 ; // it true, disable clock, otherwise enable clock
30	R/W	0	cfmt_soft_rst_bit : uns, default = 0 ; // soft rst bit
28	R/W	0	chfmt_rpt_pix : uns, default = 0 ; // if true, horizontal formatter use repeating to generete pixel, otherwise use bilinear interpolation
27:24	R/W	0	chfmt_ini_phase : uns, default = 0 ; // horizontal formatter initial phase
23	R/W	0	chfmt_rpt_p0_en : uns, default = 0 ; // horizontal formatter repeat pixel 0 enable
22:21	R/W	0	chfmt_yc_ratio : uns, default = 0 ; // horizontal Y/C ratio, 00: 1:1, 01: 2:1, 10: 4:1
20	R/W	0	chfmt_en : uns, default = 0 ; // horizontal formatter enable
19	R/W	0	cvfmt_phase0_always_en : uns, default = 0 ; //if true, always use phase0 while vertical formater, meaning always repeat data, no interpolation
18	R/W	0	cvfmt_rpt_last_dis : uns, default = 0 ; //if true, disable vertical formatter chroma repeat last line
17	R/W	0	cvfmt_phase0_nrpt_en : uns, default = 0 ; //veritcal formatter dont need repeat line on phase0, 1: enable, 0: disable
16	R/W	0	cvfmt_rpt_line0_en : uns, default = 0 ; //veritcal formatter repeat line 0 enable
15:12	R/W	0	cvfmt_skip_line_num : uns, default = 0 ; //vertical formatter skip line num at the beginning
11:8	R/W	0	cvfmt_ini_phase : uns, default = 0 ; //vertical formatter initial phase
7:1	R/W	0	cvfmt_phase_step : uns, default = 0 ; //vertical formatter phase step (3.4)
0	R/W	0	cvfmt_en : uns, default = 0 ; //vertical formatter enable

Table 8-1713 AFBC_VD_CFMT_W 0x1aec

Bit(s)	R/W	Default	Description
27:16	R/W	0x0	horizontal : formatter width
11:0	R/W	0x0	vertical : formatter width

Table 8-1714 AFBC_MIF_HOR_SCOPE 0x1aed

Bit(s)	R/W	Default	Description
25:16	R/W	0	mif_blk_bgn_h : uns, default = 0 ; // unit: 32 pixel/block hor
9: 0	R/W	59	mif_blk_end_h : uns, default = 59 ; // unit: 32 pixel/block hor

Table 8-1715 AFBC_MIF_VER_SCOPE 0x1aee

Bit(s)	R/W	Default	Description
27:16	R/W	0	mif_blk_bgn_v : uns, default = 0 ; // unit: 32 pixel/block ver
11: 0	R/W	269	mif_blk_end_v : uns, default = 269; // unit: 32 pixel/block ver

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Table 8-1716 AFBC_PIXEL_HOR_SCOPE 0x1aef

Bit(s)	R/W	Default	Description
28:16	R/W	0	dec_pixel_bgn_h : uns, default = 0 ; // unit: pixel
12: 0	R/W	1919	dec_pixel_end_h : uns, default = 1919 ; // unit: pixel

Table 8-1717 AFBC_PIXEL_VER_SCOPE 0x1af0

Bit(s)	R/W	Default	Description
28:16	R/W	0	dec_pixel_bgn_v : uns, default = 0 ; // unit: pixel
12: 0	R/W	1079	dec_pixel_end_v : uns, default = 1079 ; // unit: pixel

Table 8-1718 AFBC_VD_CFMT_H 0x1af1

Bit(s)	R/W	Default	Description
12:0	R/W	0x0	vertical : formatter height

Table 8-1719 AFBCDEC_IQUANT_ENABLE 0x1af2

Bit(s)	R/W	Default	Description
11	R/W	0x0	reg_quant_expand_en_1 : //unsigned, RW, enable for quantization value expansion
10	R/W	0x0	reg_quant_expand_en_0 : //unsigned, RW, enable for quantization value expansion
9: 8	R/W	0	reg_bcleav_ofst : //signed , RW, default = 0 bcleave ofset to get lower range, especially under lossy, for v1/v2, x=0 is equivalent, default = -1;
4	R/W	0	reg_quant_enable_1 : // unsigned , RW, default = 0 enable for quant to get some lossy
0	R/W	0	reg_quant_enable_0 : // unsigned , RW, default = 0 enable for quant to get some lossy

Table 8-1720 AFBCDEC_IQUANT_LUT_1 0x1af3

Bit(s)	R/W	Default	Description
30:28	R/W	0	reg_iquant_yclut_0_11 : // unsigned , RW, default = 0 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
26:24	R/W	1	reg_iquant_yclut_0_10 : // unsigned , RW, default = 1 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
22:20	R/W	2	reg_iquant_yclut_0_9 : // unsigned , RW, default = 2 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
18:16	R/W	3	reg_iquant_yclut_0_8 : // unsigned , RW, default = 3 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
14:12	R/W	4	reg_iquant_yclut_0_7 : // unsigned , RW, default = 4 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
10: 8	R/W	5	reg_iquant_yclut_0_6 : // unsigned , RW, default = 5 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

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Bit(s)	R/W	Default	Description
6: 4	R/W	5	reg_iquant_yclut_0_5 : // unsigned , RW, default = 5 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
2: 0	R/W	4	reg_iquant_yclut_0_4 : // unsigned , RW, default = 4 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

Table 8-1721 AFBCDEC_IQUANT_LUT_2 0x1af4

Bit(s)	R/W	Default	Description
14:12	R/W	3	reg_iquant_yclut_0_3 : // unsigned , RW, default = 3 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
10: 8	R/W	2	reg_iquant_yclut_0_2 : // unsigned , RW, default = 2 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
6: 4	R/W	1	reg_iquant_yclut_0_1 : // unsigned , RW, default = 1 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
2: 0	R/W	0	reg_iquant_yclut_0_0 : // unsigned , RW, default = 0 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

Table 8-1722 AFBCDEC_IQUANT_LUT_3 0x1af5

Bit(s)	R/W	Default	Description
30:28	R/W	0	reg_iquant_yclut_1_11 : // unsigned , RW, default = 0 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
26:24	R/W	1	reg_iquant_yclut_1_10 : // unsigned , RW, default = 1 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
22:20	R/W	2	reg_iquant_yclut_1_9 : // unsigned , RW, default = 2 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
18:16	R/W	3	reg_iquant_yclut_1_8 : // unsigned , RW, default = 3 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
14:12	R/W	4	reg_iquant_yclut_1_7 : // unsigned , RW, default = 4 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
10: 8	R/W	5	reg_iquant_yclut_1_6 : // unsigned , RW, default = 5 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
6: 4	R/W	5	reg_iquant_yclut_1_5 : // unsigned , RW, default = 5 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
2: 0	R/W	4	reg_iquant_yclut_1_4 : // unsigned , RW, default = 4 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

Table 8-1723 AFBCDEC_IQUANT_LUT_4 0x1af6

Bit(s)	R/W	Default	Description
14:12	R/W	3	reg_iquant_yclut_1_3 : // unsigned , RW, default = 3 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
10: 8	R/W	2	reg_iquant_yclut_1_2 : // unsigned , RW, default = 2 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

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Bit(s)	R/W	Default	Description
6: 4	R/W	1	reg_iquant_yclut_1_1 : // unsigned , RW, default = 1 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
2: 0	R/W	0	reg_iquant_yclut_1_0 : // unsigned , RW, default = 0 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

VD2_AFBC is the same as AFBC.BASE_addr is 0x3180

8.2.3.22 VPP Registers

Table 8-1724 VPP_DUMMY_DATA 0x1d00

Bit(s)	R/W	Default	Description
29:20	R/W	0	VD1_SC_Y : // unsigned , default = 0x10,dummy data used in the VD1 scaler,according VPP_DOLBY_CTRL[17] 1:set 8bit value 2:set 10bit value
19:10	R/W	0	VD1_SC_CB : // unsigned , default = 0x80,dummy data used in the VD1 scaler, according VPP_DOLBY_CTRL[17] 1:set 8bit value 2:set 10bit value
9 :0	R/W	0	VD1_SC_CR : // unsigned , default = 0x80,dummy data used in the VD1 scaler, according VPP_DOLBY_CTRL[17] 1:set 8bit value 2:set 10bit value

Table 8-1725 VPP_LINE_IN_LENGTH 0x1d01

Bit(s)	R/W	Default	Description
13:0	R/W	14	line_in_length : // unsigned , default = 14'd1920,VD1 scaler input hsize

Table 8-1726 VPP_PIC_IN_HEIGHT 0x1d02

Bit(s)	R/W	Default	Description
12:0	R/W	0x1fff	line_in_height : // unsigned , default = 13'h1fff,VD1 scaler input vsize

Table 8-1727 VPP_PREBLEND_VD1_H_START_END 0x1d1a

Bit(s)	R/W	Default	Description
28:16	R/W	0	start : //unsigned , default = 0x0000 ,preblend video1 horizontal start
12:0	R/W	0	end : //unsigned , default = 0x077f ,preblend video1 horizontal end

Table 8-1728 VPP_PREBLEND_VD1_V_START_END 0x1d1b

Bit(s)	R/W	Default	Description
28:16	R/W	0	start : //unsigned , default = 0x0000 ,preblend video1 vertical start
12:0	R/W	0	end : //unsigned , default = 0x0437 ,preblend video1 vertical end

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Table 8-1729 VPP_POSTBLEND_VD1_H_START_END 0x1d1c

Bit(s)	R/W	Default	Description
28:16	R/W	0	start : //unsigned , default = 0x0000 ,postblend video1 horizontal start
12:0	R/W	0	end : //unsigned , default = 0x077f ,postblend video1 horizontal end

Table 8-1730 VPP_POSTBLEND_VD1_V_START_END 0x1d1d

Bit(s)	R/W	Default	Description
31:29	/	/	/
28:16	R/W	0	//unsigned , default = 0x0000 ,postblend video1 vertical start Bit
15:13	/	/	/
12:0		0x077f	//unsigned , default = 0x077f ,postblend video1 vertical end

Table 8-1731 VPP_BLEND_VD2_H_START_END 0x1d1e

Bit(s)	R/W	Default	Description
28:16	R/W	0	start : //unsigned , default = 0x0000 ,preblend/postblend video2 horizontal start
12:0	R/W	0	end : //unsigned , default = 0x077f ,preblend/postblend video2 horizontal end

Table 8-1732 VPP_BLEND_VD2_V_START_END 0x1d1f

Bit(s)	R/W	Default	Description
28:16	R/W	0	start : //unsigned , default = 0x0000 ,preblend/postblend video2 vertical start
12:0	R/W	0	end : //unsigned , default = 0x077f ,preblend/postblend video2 vertical end

Table 8-1733 VPP_PREBLEND_H_SIZE 0x1d20

Bit(s)	R/W	Default	Description
29:16	R/W	14	prebld_v_size : //unsigned , default = 14'd1080 ,preblend output vsize
13:0	R/W	14	prebld_h_size : //unsigned , default = 14'd1920 ,preblend output hsize

Table 8-1734 VPP_POSTBLEND_H_SIZE 0x1d21

Bit(s)	R/W	Default	Description
29:16	R/W	14	postbld_v_size : //unsigned , default = 14'd1080 ,postblend output vsize
13:0	R/W	14	postbld_h_size : //unsigned , default = 14'd1920 ,postblend output hsize

Table 8-1735 VPP hold lines VPP_HOLD_LINES 0x1d22

Bit(s)	R/W	Default	Description
15:8	R/W	4	prebld_hold_lines : //unsigned , default = 4,preblend hold lines
7:0	R/W	4	postbld_hold_lines : //unsigned , default = 4,postblend hold lines

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Table 8-1736 VPP_MISC 0x1d26

Bit(s)	R/W	Default	Description
30	R/W	0	Color_manage_en //CM2 top enable
29	R/W	0	vpp_vks_en : //unsigned , default = 0 , vkstone enable
28	R/W	0	color_manage_en : //unsigned , default = 0 , color management enable
27	R/W	0	vd2_use_viu2_out_en : //unsigned , default = 0 , if true, vd2 use viu2 output as the input, otherwise use normal vd2 from memory
26:18	R/W	0	vd2 : alpha //unsigned , default = 0 , video 2 prebld/postbld alpha
7	R/W	1	postbld_en : //unsigned , default = 1 , postblend module enable ,din_reoder_sel of post blend can work when postbld_en high.
6	R/W	0	prebld_en : //unsigned , default = 0 , preblend module enable ,din_reoder_sel of post blend can work when postbld_en high.
3	R/W	0	sr4c1_path_sel : //unsigned , default = 0 , choose sr1 position 1:sr1 bettwen dnlp & CM 0:sr1 bettwen position after postblend
2	R/W	0	disable_rst_afifo : //unsigned , default = 0 , if true, disable resetting async fifo every vsync, otherwise every vsync the aync fifo will be reseted.
1	R/W	0	sr4c0_path_sel : //unsigned , default = 0 , choose sr0 position 1:sr0 bettwen prebld & vd1_scale 0:sr0 bettwen position after dnlp

Table 8-1737 VPP_OFIFO_SIZE 0x1d27

Bit(s)	R/W	Default	Description
31:20	R/W	0	ofifo_line_lenm1 : //unsigned , default = 0xfff , ofifo line length minus 1
19	R/W	0	vs_ctrl : //unsigned , default = 0 , if true invert input vs
18	R/W	0	hs_ctrl : //unsigned , default = 0 , if true invert input hs
17	R/W	0	force_top_bot_field_en : //unsigned , default = 0 , force top/bottom field, enable
16	R/W	0	fforce_top_bot_field : //unsigned , default = 0 , force top/bottom field, 0: top, 1: bottom
15	R/W	0	force_go_field : //unsigned , default = 0 , force one go_field, one pluse, write only
14	R/W	0	force_go_line : //unsigned , default = 0 , force one go_line, one pluse, write only
13:0	R/W	0	ofifo_size : //unsigned , default = 0x1000 , ofifo size (actually only bit 13:1 is valid), always even number ,max 4096

Table 8-1738 VPP_FIFO_STATUS 0x1d28

Bit(s)	R/W	Default	Description
29:25	R.O	0	ro_sco_ff_buf_count : //unsigned , default = 0,current scale out fifo counter
24:14	R.O	0	ro_afifo_count : //unsigned , default = 0,current enc afifo counter
13:1	R.O	0	ro_ofifo_buf_count : //unsigned , default = 0,current vpp line fifo ofifo counter

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Table 8-1739 VPP_MATRIX_PROBE_COLOR 0x1d5c

Bit(s)	R/W	Default	Description
31:24	R.O	0	ro_probe_y_l : //unsigned , default = 0,low 8 bits of component 0
23:12	R.O	0	ro_probe_cr : //unsigned , default = 0,component 1
11:0	R.O	0	ro_probe_cb : //unsigned , default = 0,component 2

Table 8-1740 VPP_MATRIX_PROBE_COLOR1 0x1dd7

Bit(s)	R/W	Default	Description
31	R.O	0x0	ro_probe_pix_v : //it means this probe is valid in the last field/frame
3:0	R.O	0	ro_probe_y_h : //unsigned , default = 0,high 4 bits of component 0

Table 8-1741 VPP_MATRIX_HL_COLOR 0x1d5d

Bit(s)	R/W	Default	Description
23:16	R/W	0	hl_y : //unsigned , default = 0,component 0
15:8	R/W	0	hl_cb : //unsigned , default = 0,component 1
7:0	R/W	0	hl_cr : //unsigned , default = 0,component 2

Table 8-1742 VPP_MATRIX_PROBE_POS 0x1d5e

Bit(s)	R/W	Default	Description
28:16	R/W	0	probe_x : //unsigned , default = 0,probe x, postion
12:0	R/W	0	probe_y : //unsigned , default = 0,probe y, position

Table 8-1743 VPP_MATRIX_CTRL 0x1d5f

Bit(s)	R/W	Default	Description
16	R/W	0	highlight_en : //unsigned , default = 0, highlight enable
15	R/W	0	probe_post : //unsigned , default = 0, if true, probe pixel data after matrix, otherwise probe pixel data before matrix
14:10	R/W	0	probel_sel : //unsigned , default = 0, active when VIU_SECURE_REG[9] high, probel sel : 5'b00001 :vadj1 5'b00010 :vadj2 5'b00100 :osd2 5'b01000 :postbld 5'b10000 :osd1

Table 8-1744 VPP_GAINOFF_CTRL0 0x1d6a

Bit(s)	R/W	Default	Description
31	R/W	0	enable : //unsigned , default = 0, gainoff module enable
30	R/W	0	enable_sel : //unsigned , default = 0, gainoff module enable sync sel

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Bit(s)	R/W	Default	Description
26:16	R/W	0	gain0 : //unsigned , default = 0, gainoff module gain0
10:0	R/W	0	gain1 : //unsigned , default = 0, gainoff module gain1

Table 8-1745 VPP_GAINOFF_CTRL1 0x1d6b

Bit(s)	R/W	Default	Description
26:16	R/W	0	gain2 : //unsigned , default = 0, gainoff module gain2
(DW +0):0	R/W	0	offset0 : //signed ,bitwidth is DW+1,DW is the chip path data width, default = 0, gainoff module offset0

Table 8-1746 VPP_GAINOFF_CTRL2 0x1d6c

Bit(s)	R/W	Default	Description
(DW +16):16	R/W	0	offset1 : //signed ,bitwidth is DW+1,DW is the chip path data width, default = 0, gainoff module offset1
(DW +0):0	R/W	0	offset2 : //signed , bitwidth is DW+1,DW is the chip path data width,default = 0, gainoff module offset2

Table 8-1747 VPP_GAINOFF_CTRL3 0x1d6d

Bit(s)	R/W	Default	Description
(DW +16):16	R/W	0	pre_offset0 : //signed , bitwidth is DW+1,DW is the chip path data width,default = 0, gainoff module pre_offset0
(DW +0):0	R/W	0	pre_offset1 : //signed , bitwidth is DW+1,DW is the chip path data width,default = 0, gainoff module pre_offset1

Table 8-1748 VPP_GAINOFF_CTRL4 0x1d6e

Bit(s)	R/W	Default	Description
(DW +0):0	R/W	0	pre_offset2 : //signed , bitwidth is DW+1,DW is the chip path data width,default = 0, gainoff module pre_offset2

Table 8-1749 VPP_GAINOFF_GCLK_CTRL 0x1d6f

Bit(s)	R/W	Default	Description
1:0	R/W	0	gainoff_gclk_ctrl : //unsigned , default = 0,gainoff_gclk_ctrl

Table 8-1750 VPP_GCLK_CTRL0 0x1d72

Bit(s)	R/W	Default	Description
5:4	R/W	0	ofifo_clk1 : //unsigned , default = 0,gating clock of out linefifo in vpp
3:2	R/W	0	clk0 : //unsigned , default = 0,clk swtich of all vpp module
1	R/W	0	reg_gclk : //unsigned , default = 0,registers gate clk of vpp module

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Table 8-1751 VPP_GCLK_CTRL1 0x1d73

Bit(s)	R/W	Default	Description
27:20	R/W	0	gclk_ctrl_wm : //unsigned , default = 0,gating clock of water_mark
18:15	R/W	0	dolby3_gclk_ctrl : //unsigned , default = 0,gating clock of dolby3
3:0	R/W	0	cm_gclk_ctrl : //unsigned , default = 0,gating clock of color manage

Table 8-1752 VPP_MISC1 0x1d76

Bit(s)	R/W	Default	Description
20:12	R/W	0	vd1_prebld_alpha : //unsigned , default = 0,VD1 alpha for preblend
8:0	R/W	0	vd1_postbld_alpha : //unsigned , default = 0,VD1 alpha for postblend

Table 8-1753 VPP_SRSCL_GCLK_CTRL 0x1d77

Bit(s)	R/W	Default	Description
15:8	R/W	0	gclk_ctrl_sr1 : //unsigned , default = 0 , gating clock of sr1
7:0	R/W	0	gclk_ctrl_sr0 : //unsigned , default = 0 , gating clock of sr0

Table 8-1754 VPP_BLACKEXT_CTRL 0x1d80

Bit(s)	R/W	Default	Description
31:24	R/W	0	blackext_start : //unsigned , default = 0 ,blackext_start
23:16	R/W	0	blackext_slope1 : //unsigned , default = 0 ,blackext_slope1
15:8	R/W	0	blackext_midpt : //unsigned , default = 0 ,blackext_midpt
7:0	R/W	0	blackext_slope2 : //unsigned , default = 0 ,blackext_slope2

Table 8-1755 VPP_DNLP_CTRL_00 0x1d81

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region03 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region02 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region01 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region00 output value

Table 8-1756 VPP_DNLP_CTRL_01 0x1d82

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region07 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region06 output value

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Bit(s)	R/W	Default	Description
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region05 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region04 output value

Table 8-1757 VPP_DNLP_CTRL_02 0x1d83

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region11 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region10 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region09 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region08 output value

Table 8-1758 VPP_DNLP_CTRL_03 0x1d84

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region15 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region14 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region13 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region12 output value

Table 8-1759 VPP_DNLP_CTRL_04 0x1d85

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region19 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region18 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region17 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region16 output value

Table 8-1760 VPP_DNLP_CTRL_05 0x1d86

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region23 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region22 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region21 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region20 output value

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Table 8-1761 VPP_DNLP_CTRL_06 0x1d87

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region27 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region26 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region25 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region24 output value

Table 8-1762 VPP_DNLP_CTRL_07 0x1d88

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region31 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region30 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region29 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region28 output value

Table 8-1763 VPP_DNLP_CTRL_08 0x1d89

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region35 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region34 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region33 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region32 output value

Table 8-1764 VPP_DNLP_CTRL_09 0x1d8a

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region39 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region38 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region37 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region36 output value

Table 8-1765 VPP_DNLP_CTRL_10 0x1d8b

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region43 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region42 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region41 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region40 output value

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Table 8-1766 VPP_DNLP_CTRL_11 0x1d8c

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region47 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region46 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region45 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region44 output value

Table 8-1767 VPP_DNLP_CTRL_12 0x1d8d

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region51 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region50 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region49 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region48 output value

Table 8-1768 VPP_DNLP_CTRL_13 0x1d8e

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region55 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region54 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region53 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region52 output value

Table 8-1769 VPP_DNLP_CTRL_14 0x1d8f

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region59 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region58 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region57 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region56 output value

Table 8-1770 VPP_DNLP_CTRL_15 0x1d90

Bit(s)	R/W	Default	Description
31:24	R/W	0	data0 : //unsigned , default = 0 ,bottom of region63 output value
23:16	R/W	0	data1 : //unsigned , default = 0 ,bottom of region62 output value
15:8	R/W	0	data2 : //unsigned , default = 0 ,bottom of region61 output value
7:0	R/W	0	data3 : //unsigned , default = 0 ,bottom of region60 output value

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Table 8-1771 VPP_SRSHARP0_CTRL 0x1d91

Bit(s)	R/W	Default	Description
28:16	R/W	0	srsharp_demo_split_sz : //unsigned , default = 0 ,srsharp demo top/bot left/right width
5:4	R/W	0	srsharp_demo_disp_post : //unsigned , default = 0 ,srsharp demo display postion
3	R/W	0	srsharp_demo_en : //unsigned , default = 0 ,srsharp demo enable
2	R/W	0	srsharp_c444to422_en : //unsigned , default = 0 ,srsharp format444 convert 422 enable
1	R/W	0	srsharp_buf_en : //unsigned , default = 0 ,srsharp buffer enable
0	R/W	0	srsharp_en : //unsigned , default = 0 ,srsharp enable

Table 8-1772 VPP_SRSHARP1_CTRL 0x1d92

Bit(s)	R/W	Default	Description
28:16	R/W	0	srsharp_demo_split_sz : //unsigned , default = 0 ,srsharp demo top/bot left/right width
5:4	R/W	0	srsharp_demo_disp_post : //unsigned , default = 0 ,srsharp demo display postion
3	R/W	0	srsharp_demo_en : //unsigned , default = 0 ,srsharp demo enable
2	R/W	0	srsharp_c444to422_en : //unsigned , default = 0 ,srsharp format444 convert 422 enable
1	R/W	0	srsharp_buf_en : //unsigned , default = 0 ,srsharp buffer enable
0	R/W	0	srsharp_en : //unsigned , default = 0 ,srsharp enable

Table 8-1773 VPP_DOLBY_CTRL 0x1d93

Bit(s)	R/W	Default	Description
17	R/W	0	pps_dummy_data_mode : //unsigned , default = 0 ,pps_dummy_data_mode 1: vd1_scale need setting 8 bits 0:vd1_scale need setting 10 bits
16	R/W	0	dolby3_path_sel : //unsigned , default = 0 ,1:dolby2->osd_mat->post_blend->dolby3->wm 0:dolby2->dolby3->osd_mat->post_blend->wm
10	R/W	0	vpp_clip_ext_mode2 : //unsigned , default = 0 ,Vpp out clip mode 1:10bit 0:12bit
9	R/W	0	vpp_clip_ext_mode1 : //unsigned , default = 0 ,Vpp Vd2 input clip mode 1:10bit 0:12bit
8	R/W	0	vpp_clip_ext_mode0 : //unsigned , default = 0 ,Vpp Vd1 input clip mode 1:10bit 0:12bit
3	R/W	0	vpp_dolby3_en : //unsigned , default = 0 ,dolby_core3 enable,active high
2	R/W	0	vpp_dpath_sel2 : //unsigned , default = 0 ,by_pass from post2_marix to outbuffer
1	R/W	0	vpp_dpath_sel1 : //unsigned , default = 0 ,by_pass from gainoff to vks
0	R/W	0	vpp_dpath_sel0 : //unsigned , default = 0 ,1: by_pass from preblend to VADJ1 0: unbypass

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Table 8-1774 VPP_SYNC_SEL0 0x1d96

Bit(s)	R/W	Default	Description
31:0	R/W	0	sync_sel_bits : //unsigned ,default = 0,sync_sel bits for VPP_DOLBY_CTRL

Table 8-1775 VPP_CCORING_CTRL 0x1da0

Bit(s)	R/W	Default	Description
27:16	R/W	0	reg_bypass_ccoring_ythd : // unsigned , default = 0 , bypass_ccoring_ythd
15:8	R/W	0	ccoring_th : // unsigned , default = 0 , Chroma coring threshold
3:0	R/W	0	ccoring_slope : // unsigned , default = 0 , Chroma coring slope

Table 8-1776 VPP_VE_ENABLE_CTRL 0x1da1

Bit(s)	R/W	Default	Description
29:28	R/W	0	dnlp_gclk_ctrl : // unsigned , default = 0 ,dnlp gclk ctrl
27:26	R/W	0	blackext_gclk_ctrl : // unsigned , default = 0 ,blackext gclk ctrl
25:24	R/W	0	ccoring_gclk_ctrl : // unsigned , default = 0 ,chroma coring gclk ctrl
20	R/W	0	demo_ccoring_enable : // unsigned , default = 0 ,demo chroma coring enable
19	R/W	0	demo_blackext_enable : // unsigned , default = 0 ,demo black enxtension enable
18	R/W	0	demo_dnlp_enable : // unsigned , default = 0 ,demo dynamic nonlinear luma processing enable
15:14	R/W	0	demo_disp_position : // unsigned , default = 0 ,2'b00: demo adjust on top, 2'b01: demo adjust on bottom, 2'b10: demo adjust on left, 2'b11: demo adjust on right
4	R/W	0	ccoring_en : // unsigned , default = 0 , chroma coring enable
3	R/W	0	blackext_en : // unsigned , default = 0 , black enxtension enable
2	R/W	0	dnlp_en : // unsigned , default = 0 , dynamic nonlinear luma processing enable

Table 8-1777 VPP_VE_DEMO_LEFT_TOP_SCREEN_WIDTH 0x1da2

Bit(s)	R/W	Default	Description
12:0	R/W	0	ve_demo_left_top_screen_width : // unsigned , default = 0 demo left or top screen width

Table 8-1778 VPP_VE_DEMO_CENTER_BAR 0x1da3

Bit(s)	R/W	Default	Description
31	R/W	0	ve_demo_center_bar : // unsigned , default = 0 center bar enable
27:24	R/W	0	ve_demo_center_bar : // unsigned , default = 0 center bar width (*2)
23:16	R/W	0	ve_demo_center_bar : // unsigned , default = 0 center bar Cr (*4)
15:8	R/W	0	ve_demo_center_bar : // unsigned , default = 0 center bar Cb (*4)
7:0	R/W	0	ve_demo_center_bar : // unsigned , default = 0 center bar y (*4)

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Table 8-1779 VPP_VE_H_V_SIZE 0x1da4

Bit(s)	R/W	Default	Description
28:16	R/W	780	ve_line_length : // unsigned , default = 780 ve_line_length
12:0	R/W	438	ve_pic_height : // unsigned , default = 438 ve_pic_height

Table 8-1780 VPP_OUT_H_V_SIZE 0x1da5

Bit(s)	R/W	Default	Description
28:16	R/W	780	vppout_line_length : / unsigned , default = 780 vd1_scale_out hsize
12:0	R/W	438	vppout_pic_height : // unsigned , default = 438 vd1_scale_out vsize

Table 8-1781 VPP_VDO_MEAS_CTRL 0x1da8

Bit(s)	R/W	Default	Description
10:0	R/W	0	vdo_meas_ctrl : // unsigned , default = 0 vdo_meas_ctrl

Table 8-1782 VPP_VDO_MEAS_VS_COUNT_HI 0x1da9

Bit(s)	R/W	Default	Description
19 :16	RO	0	ro_ind_meas_count_n // unsigned , default = 0 ind_meas_count_n, every number of sync_span vsyncs, this counter add 1
15 :0,	RO	0	ro_counter_h // unsigned , default = 0 high bit portion of counter

Table 8-1783 VPP_VDO_MEAS_VS_COUNT_LO 0x1daa

Bit(s)	R/W	Default	Description
31 :0	RO	0	ro_counter_l // unsigned , default = 0,low bit portion of counter

Table 8-1784 VPP_INPUT_CTRL 0x1dab

Bit(s)	R/W	Default	Description
11:9	R/W	0	vd2_sel : // unsigned , default = 0, 001: select vd1_din, 010: select vd2_din, 011: select d2d3_l_din, 100: d2d3_r_din, otherwise no selection
8:6	R/W	0	vd1_l_sel : // unsigned , default = 0, 001: select vd1_din, 010: select vd2_din, 011: select d2d3_l_din, 100: d2d3_r_din, otherwise no selection,vd1_l_sel selected cannot be used as the source of vd1_r_sel or vd2_sel
5:3	R/W	0	vd1_r_sel : // unsigned , default = 0, 001: select vd1_din, 010: select vd2_din, 011: select d2d3_l_din, 100: d2d3_r_din, otherwise no selection,useful only vd1_interleave_mode is not 00. And the source vd1_r_sel used can not used for the vd2_sel any more bit 2:0 vd1_interleave_mode // unsigned , default = 0, 000: no interleave, 001: pixel interleaving, 010: line interleaving, 011: 2 pixel interleaving,100: 2 line interleaving

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Table 8-1785 VPP_CTI_CTRL2 0x1dac

Bit(s)	R/W	Default	Description
25:24	R/W	0	cti_bpf_sel : // unsigned , default = 0
20:16	R/W	0	cti_blend_factor_gama : // unsigned , default = 0
12:8	R/W	0	cti_blend_factor_beta : // unsigned , default = 0
4:0	R/W	0	cti_blend_factor_alpha : // unsigned , default = 0

Table 8-1786 VPP_WRBAK_CTRL_SEC 0x1dad

Bit(s)	R/W	Default	Description
31	R/W	0	vpp_wrbak_sel // unsigned , default = 0, 1: VPP_WRBAK_CTRL regs set to vpp_wrbak_data_ini,Cbus can't access 0: VPP_WRBAK_CTRL reg can be writen bit
30:0	R/W	0	vpp_wrbak_data_ini // unsigned , default = 0,

Table 8-1787 VD1_BLEND_SRC_CTRL_SEC 0x1dae

Bit(s)	R/W	Default	Description
31	R/W	0	vd1_blend_src_sel : // unsigned , default = 0, 1: VD1_BLEND_SRC_CTRL regs set to vd1_blend_src_data_ini,Cbus can't access 0: VD1_BLEND_SRC_CTRL reg can be writen
30:0	R/W	0	vd1_blend_src_data_ini : // unsigned , default = 0,

Table 8-1788 VD2_BLEND_SRC_CTRL_SEC 0x1daf

Bit(s)	R/W	Default	Description
31	R/W	0	vd2_blend_src_sel : // unsigned , default = 0, 1: VD2_BLEND_SRC_CTRL regs set to vd1_blend_src_data_ini,Cbus can't access 0: VD2_BLEND_SRC_CTRL reg can be writen
30:0	R/W	0	vd2_blend_src_data_ini : // unsigned , default = 0,

Table 8-1789 OSD1_BLEND_SRC_CTRL_SEC 0x1db0

Bit(s)	R/W	Default	Description
31	R/W	0	osd1_blend_src_sel : // unsigned , default = 0, 1: OSD1_BLEND_SRC_CTRL regs set to osd1_blend_src_data_ini,Cbus can't access 0: OSD1_BLEND_SRC_CTRL reg can be writen
30:0	R/W	0	osd1_blend_src_data_ini : // unsigned , default = 0,

Table 8-1790 OSD2_BLEND_SRC_CTRL_SEC 0x1db1

Bit(s)	R/W	Default	Description
31	R/W	0	osd2_blend_src_sel : // unsigned , default = 0, 1: OSD2_BLEND_SRC_CTRL regs set to osd2_blend_src_data_ini,Cbus can't access 0: OSD2_BLEND_SRC_CTRL reg can be writen
30:0	R/W	0	osd2_blend_src_data_ini : // unsigned , default = 0,

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Table 8-1791 VPP_INT_LINE_NUM 0x1dce

Bit(s)	R/W	Default	Description
12:0	R/W	0x0	interrupt_line_num : //unsigned, default== 0x1fff,line number use to generate interrupt when line == this number

Table 8-1792 VPP_OFIFO_URG_CTRL 0x1dd8

Bit(s)	R/W	Default	Description
31	R/W	0x0	urgent_hold : //unsigned, default== 0, urgent fifo hold enable
28:12	R/W	0x0	urgent_fifo_th : //unsigned, default== 0, urgent fifo hold line threshold
15	R/W	0x0	urgent_ctrl_en : //unsigned, default== 0, urgent_ctrl_en
14	R/W	0x0	urgent_wr : //unsigned, default== 0, urgent_wr, if true for write buffer
13	R/W	0x0	out_inv_en : //unsigned, default== 0, out_inv_en
12	R/W	0x0	urgent_ini_value : //unsigned, default == 0, urgent_ini_value
11:6	R/W	0x0	up_th : //unsigned, default == 0, up_th up threshold
5:0	R/W	0x0	dn_th : //unsigned, default == 0, dn_th dn threshold

Table 8-1793 VPP_CLIP_MISC0 0x1dd9

Bit(s)	R/W	Default	Description
29:20	R/W	1023	r : // unsigned, default == 1023, final clip r channel top
19:10	R/W	1023	g : // unsigned, default == 1023, final clip g channel top
9: 0	R/W	1023	b : // unsigned, default == 1023, final clip b channel top

Table 8-1794 VPP_CLIP_MISC1 0x1dda

Bit(s)	R/W	Default	Description
29:20	R/W	0x0	r : // unsigned, default == 0, final clip r channel bottom
19:10	R/W	0x0	g : // unsigned, default == 0, final clip g channel bottom
9: 0	R/W	0x0	b : // unsigned, default == 0, final clip b channel bottom

Table 8-1795 VPP_VD1_CLIP_MISC0 0x1de1

Bit(s)	R/W	Default	Description
29:20	R/W	1023	r : //unsigned, default == 1023, vd1 clip r channel top
19:10	R/W	1023	g : //unsigned, default == 1023, vd1 clip g channel top
9: 0	R/W	1023	b : //unsigned, default == 1023, vd1 clip b channel top

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Table 8-1796 VPP_VD1_CLIP_MISC1 0x1de2

Bit(s)	R/W	Default	Description
29:20	R/W	0x0	r : //unsigned, default = 0, vd1 clip r channel bottom
19:10	R/W	0x0	g : //unsigned, default = 0, vd1 clip g channel bottom
9: 0	R/W	0x0	b : //unsigned, default = 0, vd1 clip b channel bottom

Table 8-1797 VPP_VD2_CLIP_MISC0 0x1de3

Bit(s)	R/W	Default	Description
29:20	R/W	1023	r : //unsigned, default = 1023, vd2 clip r channel top
19:10	R/W	1023	g : //unsigned, default = 1023, vd2 clip g channel top
9: 0	R/W	1023	b : //unsigned, default = 1023, vd2 clip b channel top

Table 8-1798 VPP_VD2_CLIP_MISC1 0x1de4

Bit(s)	R/W	Default	Description
29:20	R/W	0x0	r : // unsigned, default = 0, vd2 clip r channel bottom
19:10	R/W	0x0	g : // unsigned, default = 0, vd2 clip g channel bottom
9: 0	R/W	0x0	b : // unsigned, default = 0, vd2 clip b channel bottom

Table 8-1799 VPP_VD2_HDR_IN_SIZE 0x1df0

Bit(s)	R/W	Default	Description
18:16	R/W	0	vd2_in_v_size : // unsigned, default = 0x2d0, VPP VD2 input vsize
12:0	R/W	0	vd2_in_h_size : // unsigned, default = 0x1e0, VPP VD2 input hsize

Table 8-1800 VPP_OSD1_IN_SIZE 0x1df1

Bit(s)	R/W	Default	Description
18:16	R/W	0	osd1_in_v_size : // unsigned, default = 0x2d0, VPP osd1 input vsize
12:0	R/W	0	osd1_in_h_size : // unsigned, default = 0x1e0, VPP osd1 input hsize

Table 8-1801 VPP_GCLK_CTRL2 0x1df2

Bit(s)	R/W	Default	Description
13:13	R/W	0	vks_gclk_ctrl : // unsigned, default = 0 ,vks gating clock

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Table 8-1802 VD2_PPS_DUMMY_DATA 0x1df4

Bit(s)	R/W	Default	Description
23:16	R/W	0x0	Y : //unsigned, default = 0, vd2 scale dummy data
15:8	R/W	0x0	CB : //unsigned, default = 0, vd2 scale dummy data
7: 0	R/W	0x0	CR : //unsigned, default = 0, vd2 scale dummy data

Table 8-1803 VPP_OSD1_BLD_H_SCOPE 0x1df5

Bit(s)	R/W	Default	Description
28:16	R/W	0	blend_osd1_h_start : //unsigned, default = 0x0
12:0	R/W	0	blend_osd1_h_end : //unsigned, default = 0x2d0

Table 8-1804 VPP_OSD1_BLD_V_SCOPE 0x1df6

Bit(s)	R/W	Default	Description
28:16	R/W	0	blend_osd1_v_start : //unsigned, default = 0x0
12:0	R/W	0	blend_osd1_v_end : //unsigned, default = 0x1e0

Table 8-1805 VPP_OSD2_BLD_H_SCOPE 0x1df7

Bit(s)	R/W	Default	Description
28:16	R/W	0x0	blend_osd2_h_start : //unsigned, default = 0
12:0	R/W	0x0	blend_osd2_h_end : //unsigned, default = 0x2d0

Table 8-1806 VPP_OSD2_BLD_V_SCOPE 0x1df8

Bit(s)	R/W	Default	Description
28:16	R/W	0x0	blend_osd2_v_start : //unsigned, default = 0
12:0	R/W	0x0	blend_osd2_v_end : //unsigned, default = 0x1e0

Table 8-1807 VPP_WRBAK_CTRL 0x1df9

Bit(s)	R/W	Default	Description
27:16	R/W	0	wrbak_din_lnblank : //unsigned, default = 0,
11:8	R/W	0	wrbak_din_only_en : //unsigned, default = 0,
6:4	R/W	0	wrbak_chan1_sel : //unsigned, default = 0,1:vd1 2:vd2 3:osd1 4:osd2 5:posd_ blend
2:0	R/W	0	wrbak_chan0_sel : //unsigned, default = 0,1:vd1 2:vd2 3:osd1 4:osd2 5:posd_ blend

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Table 8-1808 VPP_SLEEP_CTRL 0x1dfa

Bit(s)	R/W	Default	Description
31	R/W	1	sleep_always_en : //unsigned, default = 1
30	R/W	0	sleep_always_dis : //unsigned, default = 0
29:16	R/W	0	sleep_line_len : //unsigned, default = 0
15:14	R/W	0	sleep_mode : //unsigned, default = 0
13:0	R/W	0	sleep_beg_line : //unsigned, default = 0

Table 8-1809 VD1_BLEND_SRC_CTRL 0x1dfb

Bit(s)	R/W	Default	Description
16	R/W	0	vd1_postbld_premult : //unsigned, default = 0
11:8	R/W	1	vd1_postbld_src : //unsigned, default = 1 , 0:close 1:vd1 2:vd2 3:osd1 4:osd2
4	R/W	0	vd1_prebld_premult : //unsigned, default = 0
3:0	R/W	1	vd1_prebld_src : //unsigned, default = 1 , 0:close 1:vd1 2:vd2 3:osd1 4:osd2

Table 8-1810 VD2_BLEND_SRC_CTRL 0x1dfc

Bit(s)	R/W	Default	Description
20	R/W	0	vd2_blend_path_sel : //unsigned, default = 0
16	R/W	0	vd2_postbld_premult : //unsigned, default = 0
11:8	R/W	2	vd2_postbld_src : //unsigned, default = 2 , 0:close 1:vd1 2:vd2 3:osd1 4:osd2
4	R/W	0	vd2_prebld_premult : //unsigned, default = 0
3:0	R/W	0	vd2_prebld_src : //unsigned, default = 0 ,0:close 1:vd1 2:vd2 3:osd1 4:osd2

Table 8-1811 OSD1_BLEND_SRC_CTRL 0x1dfd

Bit(s)	R/W	Default	Description
20	R/W	0	osd1_blend_path_sel : //unsigned, default = 0
16	R/W	0	osd1_postbld_premult : //unsigned, default = 0
11:8	R/W	3	osd1_postbld_src : //unsigned, default = 3 , 0:close 1:vd1 2:vd2 3:osd1 4:osd2
4	R/W	0	osd1_prebld_premult : //unsigned, default = 0
3:0	R/W	0	osd1_prebld_src : //unsigned, default = 0 ,0:close 1:vd1 2:vd2 3:osd1 4:osd2

Table 8-1812 OSD2_BLEND_SRC_CTRL 0x1dfe

Bit(s)	R/W	Default	Description
20	R/W	0	osd2_blend_path_sel : //unsigned, default = 0
16	R/W	0	osd2_postbld_premult : //unsigned, default = 0
11:8	R/W	4	osd2_postbld_src : //unsigned, default = 4 , 0:close 1:vd1 2:vd2 3:osd1 4:osd2

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Bit(s)	R/W	Default	Description
4	R/W	0	osd2_prebld_premult : //unsigned, default = 0
3:0	R/W	0	osd2_prebld_src : //unsigned, default = 0 , 0:close 1:vd1 2:vd2 3:osd1 4:osd2

Table 8-1813 VPP_CRC_CHK 0x1db3

Bit(s)	R/W	Default	Description
0	W	0	vpp crc check start (pulse)

Table 8-1814 VPP_RO_CRCSUM 0x1db2

Bit(s)	R/W	Default	Description
31-0	R	0	vpp crc result (lock by go_field)

8.2.3.23 CM Registers

Table 8-1815 VPP_CHROMA_ADDR_PORT 0x1d70

Bit(s)	R/W	Default	Description
31-0	R/W	0	Color management address port

Table 8-1816 VPP_CHROMA_DATA_PORT 0x1d71

Bit(s)	R/W	Default	Description
31-0	R/W	0	Color management data port

Color management internal registers is indirectly accessed by the registers VPP_CHROMA_ADDR_PORT and VPP_CHROMA_DATA_PORT.

Color management registers

The example to access the Color management registers is like this:

Wr(VPP_CHROMA_ADDR_PORT);

Wr(VPP_CHROMA_DATA_PORT):

Table 8-1817 REG_CHROMA_CONTROL 0x30

Bit(s)	R/W	Default	Description
31	R/W	0	reg_chroma_en. enable color manage function 1'b1: enable 1'b0: bypass
6	R/W	0	sat_sel. uv_max or u^2+v^2 selected as sat for reference 1'b1: uv_max(default) 1'b0: u^2+v^2
5	R/W	0	uv_adj_en. final uv_adjust enable 1'b1: enable 1'b0: bypass

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Bit(s)	R/W	Default	Description
2	R/W	0	hue_en. rgb to hue enable 1'b1: enable(default) 1'b0: bypass
1-0	R/W	0	csc_sel. define input YUV with different color type 2'b00: 601(16-235) 2'b01: 709(16-235) 2'b10: 601(0-255) 2'b11: 709(0-255)

Table 8-1818 SAT_BYYB_NODE0 0x200

Bit(s)	R/W	Default	Description
31-24	R/W	0	The 4th node, the same as below
23-16	R/W	0	The 3th node, the same as below
15-8	R/W	0	The 2th node, the same as below
7-0	R/W	0	Signed, The 1th node about saturation gain offset along Y coordinate, the gain normalized to 128 as “1”.

Table 8-1819 SAT_BYYB_NODE1 0x201

Bit(s)	R/W	Default	Description
31-24	R/W	0	The 8th node, the same as below
23-16	R/W	0	The 7th node, the same as below
15-8	R/W	0	The 6th node, the same as below
7-0	R/W	0	Signed, The 5th node about saturation gain offset along Y coordinate, the gain normalized to 128 as “1”.

Table 8-1820 SAT_BYYB_NODE2 0x202

Bit(s)	R/W	Default	Description
31-8	R/W	0	reserved
7-0	R/W	0	Signed, The 9th node about saturation gain offset along Y coordinate, the gain normalized to 128 as “1”.

Table 8-1821 SAT_SRC_NODE 0x203

Bit(s)	R/W	Default	Description
31-28	R/W	0	reserved
27-16	R/W	0x800	unsign, Threshold of input saturation for second & third piece. i.e. it is boundary for reg_CM2_Adj_Sat_via_HS[1][:]and reg_CM2_Adj_Sat_via_HS[2][:]
15-12	R/W	0	reserved
11-0	R/W	0x400	unsign, Threshold of input saturation for first and second piece.i.e. it is boundary for reg_CM2_Adj_Sat_via_HS[0][:] and reg_CM2_Adj_Sat_via_HS[1][:]

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Table 8-1822 CM_ENH_SFT_MODE 0x204

Bit(s)	R/W	Default	Description
31-11	R/W	0	reserved
10–8	R/W	0	Hue offset adjustments scale for Reg_CM2_Adj_Hue_via_H[:]& Reg_CM2_Adj_Hue_via_S[:]& Reg_CM2_Adj_Hue_via_Y[:]: 0: no scale up; 1: upscale by 2 - (-128,127)x2; 2: upscale by 4 - (-128,127)x4; 3: upscale by 8 - (-128,127)x8;
7–6	R/W	0	reserved
5-4	R/W	0	Luma offset adjustments scale for reg_CM2_Adj_Luma_via_Hue[i]: 0: no scale up; 1: upscale by2 - (-128,127)x2; 2: upscale by 4 - (-128,127)x4; 3: upscale by 8 - (-128,127)x8;
3-2	R/W	0	Saturation again adjustments scale for reg_CM2_Adj_Sat_via_Y[:][:] &Reg_CM2_Adj_SatGLBgain_via_Y[:]: 0: no scale up/down; 1: dnscale by 2 (-128,127)/2; 2: dnscale by 4 (-128,127)/4; 3: dnscale by 8 (-128,127)/8;
1-0	R/W	0	Saturation again adjustments scale for reg_CM2_Adj_Sat_via_HS[:][:]: 0: no scale up/down; 1: dnscale by 2 (-128,127)/2; 2: dnscale by 4 (-128,127)/4; 3: dnscale by 8 (-128,127)/8;

Table 8-1823 FRM_SIZE 0x205

Bit(s)	R/W	Default	Description
31-29	R/W	0	reserved
28-16	R/W	0x438	The frame height size
15-13	R/W	0	reserved
12-0	R/W	0x780	The frame width size

Table 8-1824 FITLER_CFG 0x206

Bit(s)	R/W	Default	Description
31-5	R/W	0	reserved
4	R/W	0	Horizontal Interleave filter (zero-padding) for 3D considerations: 0: using non-zero padding LPF 1: using zero-padding LPF
3-0	R/W	0	Apply CM on LP portion or original video pixels options: bits[1:0]: is for Luma path control; bits[3:2]: is for U/V path control; 0: no filter but still match the delay; 1: 5 taps LP filter 2: 9 taps LP filter 3: 13 taps LP filter

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Table 8-1825 CM_GLOBAL_GAIN 0x207

Bit(s)	R/W	Default	Description
31-28	R/W	0	reserved
27-16	R/W	0x200	Global Saturation Gain for general color adjustments (0~4095 <=> 0~8), 512 normalized to “1”.
15-12	R/W	0	reserved
11-0	R/W	0	Global Hue offsets for general color adjustments (0~4095 <=> 0~360 degree)

Table 8-1826 CM_ENH_CTL 0x208

Bit(s)	R/W	Default	Description
31-6	R/W	0	reserved
5	R/W	0	CM Bypass 1: Bypass 0: not bypass
4	R/W	0	Enable signal for CM2 Hue adjustments;
3	R/W	0	Enable signal for CM2 Saturation adjustments;
2	R/W	0	Enable signal for CM2 Luma adjustments;
1	R/W	0	cm2_filt_en :apply cm on lp portion enable
0	R/W	0	CM1 enable signal

Table 8-1827 ROI_X_SCOPE 0x209

Bit(s)	R/W	Default	Description
31-29	R/W	0	reserved
28-16	R/W	0	Ending col index of the Region of Interest (ROI)
15-12	R/W	0	reserved
12-0	R/W	0	Start col index of the Region of Interest (ROI)

Table 8-1828 ROI_Y_SCOPE 0x20a

Bit(s)	R/W	Default	Description
31-29	R/W	0	reserved
28-16	R/W	0	Ending col index of the Region of Interest (ROI)
15-13	R/W	0	reserved
12-0	R/W	0	Start col index of the Region of Interest (ROI)

Table 8-1829 POI_XY_DIR 0x20b

Bit(s)	R/W	Default	Description
31-29	R/W	0	reserved
28-16	R/W	0	Row index of the pixel(position) of Interest (POI)

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Bit(s)	R/W	Default	Description
15-13	R/W	0	reserved
12-0	R/W	0	Col index of the pixel(position) of Interest (POI)

Table 8-1830 COI_Y_SCOPE 0x20c

Bit(s)	R/W	Default	Description
31-16	R/W	0	reserved
15-8	R/W	0	Higher bound of luma value for color of interest (COI), 8bits precision
7-0	R/W	0	Lower bound of luma value for color of interest (COI), 8bits precision

Table 8-1831 COI_H_SCOPE 0x20d

Bit(s)	R/W	Default	Description
31-28	R/W	0	reserved
27-16	R/W	0	Higher bound of Hue value for color of interest (COI),12 8bits precision
15-12	R/W	0	reserved
11-0	R/W	0	Lower bound of Hue value for color of interest (COI), 12 bits precision

Table 8-1832 COI_S_SCOPE 0x20e

Bit(s)	R/W	Default	Description
31-28	R/W	0	reserved
27-16	R/W	0	Higher bound of Sat value for color of interest (COI),12 8bits precision
15-12	R/W	0	reserved
11-0	R/W	0	Lower bound of Sat value for color of interest (COI), 12 bits precision

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Table 8-1833 IFO_MODE 0x20f

Bit(s)	R/W	Default	Description
31-8	R/W	0	reserved
7-4	R/W	0	Mode control for COI replacement, bit[3:2] control COI pixels: 0: no replacement for COI pixels 1: disable CM2 enhance for COI pixels; 2: keep COI pixels Y but replace HS by [*HS]; 3: replace COI pixels to [*YHS] bit[1:0] controls non-COI pixels: 0: no replacement for non-COI pixels 1: disable CM2 enhance for non-COI pixels; 2: keep COI pixels Y but replace HS by [*HS]; 3: replace non- COI pixels to [*YHS]
3-0	R/W	0	Enhance mode control of pixels inside and outside Region of Interest (ROI) , bit [3:2] control ROI: 0: enable CM2 processing in ROI; 1: disable CM2 processing in ROI; 2: keep ROI pixels Y but replace HS by [*HS]; 3: ow ROI pixels to [*YHS] bit [1:0] control pixels other than ROI similarly. 0: enable CM2 processing in non-ROI; 1: disable CM2 processing in non-ROI; 2: keep ROI pixels Y but replace HS by [*HS]; 3: ow non-ROI pixels to [*YHS]

Table 8-1834 POI_RPL_MODE 0x210

Bit(s)	R/W	Default	Description
31-4	R/W	0	reserved
27-16	R/W	0	Reg_cm2_after_v_offset
15-4	R/W	0	Reg_cm2_after_u_offset
3-0	R/W	0	Pixel of interest (POI) replacement mode: 0: no replacements; 1: one pixel of POI position replaced to [*YHS] 2: 3X3 pixels centering POI position replaced to [*YHS] 3: 5X5 pixels centering POI position replaced to [*YHS] … 15: 29X29 pixels centering POI position replaced to [*YHS]

Table 8-1835 DEMO_OWR_YHS 0x211

Bit(s)	R/W	Default	Description
31-24	R/W	0	Saturation value overwriting to ROI/POI/COI; 12bits precision, equal to saturation precision.
23-12	R/W	0	Hue value overwriting to ROI/POI/COI; 12 bits precision, equal to 1/4 hue precision. E.g. { Reg_CM2Demo_OWR_H, 2'h0}
11-0	R/W	0	Luma value overwriting to ROI/POI/COI; 8bits precision, equal to 1/4 luma precision, e.g. { Reg_CM2Demo_OWR_Y, 2'h0}

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Table 8-1836 DEMO_POI_Y 0x212

Bit(s)	R/W	Default	Description
31-8	RO	0	Reserved
7-0	RO	0	Luma value for pixel of interest (POI), only get locked higher 8bits

Table 8-1837 DEMO_POI_H 0x213

Bit(s)	R/W	Default	Description
31-12	RO	0	Reserved
11-0	RO	0	Hue value for pixel of interest (POI), only get locked higher 12bits

Table 8-1838 DEMO_POI_S 0x214

Bit(s)	R/W	Default	Description
31-12	RO	0	Reserved
11-0	RO	0	Saturation value for pixel of interest (POI), only get locked higher 12bits

Table 8-1839 LUMA_ADJ_LIMT 0x215

Bit(s)	R/W	Default	Description
31-28	R/W	0	reserved
23-16	R/W	0	Slope to do the Luma adjust degrade speed based on Saturation. It was normalized to 16 as ‘1’.
15-12	R/W	0	reserved
11-0	R/W	0	Threshold to saturation to do Luma adjustment degrade. Only pixels’ saturation lower than this threshold will degrade the Luma adjustment.

Table 8-1840 SAT_ADJ_LIMT 0x216

Bit(s)	R/W	Default	Description
31-28	R/W	0	reserved
23-16	R/W	0	Slope to do the Sat adjust degrade speed based on Saturation. It was normalized to 16 as ‘1’.
15-12	R/W	0	reserved
11-0	R/W	0	Threshold to saturation to do Sat adjustment degrade. Only pixels’ saturation lower than this threshold will degrade the Luma adjustment.

Table 8-1841 HUE_ADJ_LIMT 0x217

Bit(s)	R/W	Default	Description
31-28	R/W	0	reserved
23-16	R/W	0	Slope to do the Hue adjust degrade speed based on Saturation. It was normalized to 16 as ‘1’.

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Bit(s)	R/W	Default	Description
15-12	R/W	0	reserved
11-0	R/W	0	Threshold to saturation to do Hue adjustment degrade. Only pixels' saturation lower than this threshold will degrade the Luma adjustment.

Table 8-1842 UVHS_OFST 0x218

Bit(s)	R/W	Default	Description
31-24	R/W	0	reserved
23-12	R/W	0	U offset after CM2, under s10 scale
11-8	R/W	0	reserved
7-0	R/W	0	U offset before CM2, under s10 scale.

Table 8-1843 HUE_CFG_PARA 0x219

Bit(s)	R/W	Default	Description
31-17	R/W	0	reserved
16	R/W	0	Options to protect HUE after CM2 adjustments. This will be added to avoid HUE distortion if Saturation is enhanced too much.
15-13	R/W	0	Hue adjustment via HS the Saturation division mode: 0: 1024/2048/3072, 4095; 1: 512, 1024, 1536, 2048; 2: 256, 512, 768, 1024; 3: 128, 256, 384, 512; 4: 512/1024/2048/4096; 5: 256/512/1024/2048; 6: 128/256/512/1024; 7: 64,128,256,512
12	R/W	0	Hue slice division mode: 0: 32 pieces, 360/32 degrees each slice; 1/up: first 20 slices with 360/64 degrees each slice, others 360/16 degrees each slices. Notes, this option provide options to get more precise Hue adjustments for FTC/Red and so on
11-0	R/W	0	Hue offset before CM2 adjustment, this will provide options to divide the Hue slices with a precise offset. But need to compensate back with the global Hue after CM2 adjusments

Table 8-1844 DEMO_SPLT_CFG 0x21a

Bit(s)	R/W	Default	Description
31-22	R/W	0	reserved
21-20	R/W	0	Demo split post
19-16	R/W	0	Demo split width
12-0	R/W	0	Demo split mode

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Table 8-1845 DEMO_SPLT_YHS 0x21b

Bit(s)	R/W	Default	Description
31-24	R/W	0	Luma value
23-12	R/W	0	Hue value
11-0	R/W	0	Sat value

Table 8-1846 XVYCC_YSCP_REG 0x21c

Bit(s)	R/W	Default	Description
27:16	R/W	0x3ff	xvycc_y_max
11:0	R/W	0x0	xvycc_y_min

Table 8-1847 XVYCC_USCP_REG 0x21d

Bit(s)	R/W	Default	Description
27:16	R/W	0x3ff	xvycc_u_max
11:0	R/W	0x0	xvycc_u_min

Table 8-1848 XVYCC_VSCP_REG 0x21e

Bit(s)	R/W	Default	Description
27:16	R/W	0x3ff	xvycc_v_max
11:0	R/W	0x0	xvycc_v_min

Table 8-1849 LUMA_ADJ0_REG 0x21f

Bit(s)	R/W	Default	Description
21:12	R/W	0x0	Reg_cm_luma_blacklevel
11:0	R/W	0x0	Reg_cm_luma_contrast

Table 8-1850 LUMA_ADJ1_REG 0x220

Bit(s)	R/W	Default	Description
28:16	R/W	0x0	Reg_sta_sat_hist_mode
12:0	R/W	0x0	Reg_cm_luma_brightness

Table 8-1851 STA_WIN_XYXY0_REG 0x221

Bit(s)	R/W	Default	Description
29:16	R/W	0x0	Reg_sta_win_xxyy1
13:0	R/W	0x0	Reg_sta_win_xxyy0

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Table 8-1852 STA_WIN_XYXY1_REG 0x222

Bit(s)	R/W	Default	Description
29:16	R/W	0x0	Reg_sta_win_xxyy3
13:0	R/W	0x0	Reg_sta_win_xxyy2

Table 8-1853 STA_CFG_REG 0x223

Bit(s)	R/W	Default	Description
31:30	R/W	0x1	Reg_sta_enable
29:24	R/W	0x1d	Reg_sta_hist_scale
23:16	R/W	0x20	Reg_sta_hue_hist_sat_thrd
15:8	R/W	0x1	Reg_sta_blk_thrd
7:0	R/W	0x254	Reg_sta_brt_thrd

Table 8-1854 STA_SAT_HIST0_REG 0x224

Bit(s)	R/W	Default	Description
24	R/W	0x0	Reg_cm_ro_frame
23:16	R/W	0x80	Reg_sta_sat_hist_rang_thrd2
15:8	R/W	0x40	Reg_sta_sat_hist_rang_thrd1
7:0	R/W	0x0	Reg_sta_sat_hist_rang_thrd0

Table 8-1855 STA_SAT_HIST1_REG 0x225

Bit(s)	R/W	Default	Description
23:16	R/W	0xff	Reg_sta_sat_hist_rang_thrd5
15:8	R/W	0xff	Reg_sta_sat_hist_rang_thrd4
7:0	R/W	0xc0	Reg_sta_sat_hist_rang_thrd3

Table 8-1856 RO_CM_HUE_HIST_BIN0□RO_CM_HUE_HIST_BIN31 0x226□0x245

Bit(s)	R/W	Default	Description
31:0	RO	0x0	RO_CM_HUE_HIST_BIN0~RO_CM_HUE_HIST_BIN31

Table 8-1857 RO_CM_SAT_HIST_BIN0□RO_CM_SAT_HIST_BIN31 0x246□0x265

Bit(s)	R/W	Default	Description
31:0	RO	0x0	RO_CM_SAT_HIST_BIN0~RO_CM_SAT_HIST_BIN31

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Table 8-1858 RO_CM_BLK_BIN 0x266

Bit(s)	R/W	Default	Description
31:0	RO	0x0	RO_CM_BLK_BIN

Table 8-1859 RO_CM_BRT_BIN 0x267

Bit(s)	R/W	Default	Description
31:0	RO	0x0	RO_CM_BRT_BIN

the main adjust parameter is saved according to 32 hue node order, one by one. The parameter of each hue node is same. All the parameter for each hue occupy 5 register-addr-space. For the addr-offset aligned, we allocate 8 addr-space to each node parameter, for example,

the parameter of 1th node uses the address space : 0x100, 0x101, 0x102, 0x103, 0x104,

and 2th node uses the address space : 0x108, 0x109, 0x10a, 0x10b, 0x10c,

and 2th node uses the address space : 0x110, 0x111, 0x112, 0x113, 0x114,

… … … …

Table 8-1860 CM2_ENH_COEF0_H00 0x100

Bit(s)	R/W	Default	Description
31-24	R/W	0	Same as last
23-16	R/W	0	Same as last
15-8	R/W	0	Signed, Saturation gain offset for three pieces saturation on Hue section (totally 32 sections) node 0; the gain normalized to 128 as “1”.
7-0	R/W	0	Signed, Luma offsets for Hue section (totally 32 sections) nodes 0 , range (-128,127)

Table 8-1861 CM2_ENH_COEF1_H00 0x101

Bit(s)	R/W	Default	Description
31-24	R/W	0	Signed, Hue offset for each four pieces Luma region on each Hue section (totally 32 sections) nodes, this is for y=2/4, hue node 0
23-16	R/W	0	Signed, Hue offset for each four pieces Luma region on each Hue section (totally 32 sections) nodes, this is for y=1/4, hue node 0
15-8	R/W	0	Signed, Hue offset for each four pieces Luma region on each Hue section (totally 32 sections) nodes, this is for y=0, hue node 0
7-0	R/W	0	Signed, Hue offset on Hue section (totally 32 sections) node 0

Table 8-1862 CM2_ENH_COEF2_H00 0x102

Bit(s)	R/W	Default	Description
31-24	R/W	0	Signed, Hue offset for each four pieces Saturation region on each Hue section (totally 32 sections) nodes; This is for sat = 1/4, hue node 0
23-16	R/W	0	Signed, Hue offset for each four pieces Saturation region on each Hue section (totally 32 sections) nodes; This is for sat = 0, hue node 0

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Bit(s)	R/W	Default	Description
15-8	R/W	0	Signed, Hue offset for each four pieces Luma region on each Hue section (totally 32 sections) nodes, this is for y=4/4, hue node 0
7-0	R/W	0	Signed, Hue offset for each four pieces Luma region on each Hue section (totally 32 sections) nodes, this is for y=3/4, hue node 0

Table 8-1863 CM2_ENH_COEF3_H00 0x103

Bit(s)	R/W	Default	Description
31-24	R/W	0	Saturation gain offsetfor four pieces Luma on each Hue section (totally 32 sections) nodes; the gain normalized to 128 as “1”. This is for sat = 0, hue node 0
23-16	R/W	0	Signed, Hue offset for each four pieces Saturation region on each Hue section (totally 32 sections) nodes; This is for sat = 4/4, hue node 0
15-8	R/W	0	Signed, Hue offset for each four pieces Saturation region on each Hue section (totally 32 sections) nodes; This is for sat = 3/4, hue node 0
7-0	R/W	0	Signed, Hue offset for each four pieces Saturation region on each Hue section (totally 32 sections) nodes; This is for sat = 2/4, hue node 0

Table 8-1864 CM2_ENH_COEF4_H00 0x104

Bit(s)	R/W	Default	Description
31-24	R/W	0	Saturation gain offsetfor four pieces Luma on each Hue section (totally 32 sections) nodes; the gain normalized to 128 as “1”. This is for sat = 4/4, hue node 0
23-16	R/W	0	Saturation gain offsetfor four pieces Luma on each Hue section (totally 32 sections) nodes; the gain normalized to 128 as “1”. This is for sat = 3/4, hue node 0
15-8	R/W	0	Saturation gain offsetfor four pieces Luma on each Hue section (totally 32 sections) nodes; the gain normalized to 128 as “1”. This is for sat = 2/4, hue node 0
7-0	R/W	0	Saturation gain offsetfor four pieces Luma on each Hue section (totally 32 sections) nodes; the gain normalized to 128 as “1”. This is for sat = 1/4, hue node 0

CM2_ENH_COEF0_H01 0x108

CM2_ENH_COEF1_H01 0x109

CM2_ENH_COEF2_H01 0x10a

CM2_ENH_COEF3_H01 0x10b

CM2_ENH_COEF4_H01 0x10c

CM2_ENH_COEF0_H02 0x110

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CM2_ENH_COEF1_H02 0x111

CM2_ENH_COEF2_H02 0x112

CM2_ENH_COEF3_H02 0x113

CM2_ENH_COEF4_H02 0x114

....

CM2_ENH_COEF0_H31 0x1f8

CM2_ENH_COEF1_H31 0x1f9

CM2_ENH_COEF2_H31 0x1fa

CM2_ENH_COEF3_H31 0x1fb

CM2_ENH_COEF4_H31 0x1fc

8.2.3.24 VPP OSD1 SCALER Registers

Table 8-1865 VPP_OSD_VSC_PHASE_STEP 0x1dc0

Bit(s)	R/W	Default	Description
27-0	R/W	0x0100-0000	4.24 format

Table 8-1866 VPP_OSD_VSC_INI_PHASE 0x1dc1

Bit(s)	R/W	Default	Description
31-16	R/W	0x0	botttom vertical scaler initial phase
15-0	R/W	0x0	top vertical scaler initial phase

Table 8-1867 VPP_OSD_VSC_CTRL0 0x1dc2

Bit(s)	R/W	Default	Description
25	R/W	0	Repeat last line en : 1:enable repeat last line 0:disable repeat last line
24	R/W	0x0	osd vertical Scaler enable
23	R/W	0x0	osd_prog_interlace 0: current field is progressive, 1: current field is interlace

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Bit(s)	R/W	Default	Description
22-21	R/W	0x0	osd_vsc_double_line_mode, bit1, double input width and half input height, bit0, change line buffer becomes 2 lines
20	R/W	0x0	osd_vsc_phase0_always_en
19	R/W	0x0	osd_vsc_nearest_en
17-16	R/W	0x0	osd_vsc_bot_rpt_l0_num
14-11	R/W	0x0	osd_vsc_bot_ini_rcv_num
9-8	R/W	0x0	osd_vsc_top_rpt_l0_num
6-3	R/W	0x0	osd_vsc_top_ini_rcv_num
2-0	R/W	0x0	osd_vsc_bank_length

Table 8-1868 VPP_OSD_HSC_PHASE_STEP 0x1dc3

Bit(s)	R/W	Default	Description
27-0	R/W	0x0100-0000	4.24 format

Table 8-1869 VPP_OSD_HSC_INI_PHASE 0x1dc4

Bit(s)	R/W	Default	Description
31-16	R/W	0x0	horizontal scaler initial phase1
15-0	R/W	0x0	horizontal scaler initial phase0

Table 8-1870 VPP_OSD_HSC_CTRL0 0x1dc5

Bit(s)	R/W	Default	Description
22	R/W	0x0	osd horizontal Scaler enable
21	R/W	0x0	osd_hsc_double_pix_mode
20	R/W	0x0	osd_hsc_phase0_always_en
19	R/W	0x0	osd_vsc_nearest_en
17-16	R/W	0x0	osd_hsc_rpt_p0_num1
14-11	R/W	0x0	osd_hsc_ini_rcv_num1
9-8	R/W	0x0	osd_hsc_rpt_p0_num0
6-3	R/W	0x0	osd_hsc_ini_rcv_num0
2-0	R/W	0x0	osd_hsc_bank_length

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Table 8-1871 VPP_OSD_HSC_INI_PAT_CTRL 0x1dc6

Bit(s)	R/W	Default	Description
15-8	R/W	0x0	for 3D quincunx sub-sampling. pattern, each patten 1 bit, from lsb -> msb
6-4	R/W	0x0	pattern start
2-0	R/W	0x0	pattern end

Table 8-1872 VPP_OSD_SC_DUMMY_DATA 0x1dc7

Bit(s)	R/W	Default	Description
31-24	R/W	0x0	componet 0 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
23-16	R/W	0x0	component 1 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
15-8	R/W	0x0	component 2 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
7-0	R/W	0x0	component 3 , alpha

Table 8-1873 VPP_OSD_SC_CTRL0 0x1dc8

Bit(s)	R/W	Default	Description
27:16	R/W	0	osd_sc_gclk_ctrl : //unsigned,default = 0,osd_sc_gclk_ctrl
13	R/W	0	osd_sc_din_osd_alpha_mode : //unsigned,default = 0,osc_sc_din_osd2_alpha_mode, 1: (alpha >= 128) ? alpha -1: alpha, 0: (alpha >=1) ? alpha - 1: alpha.
12	R/W	0	osd_sc_dout_alpha_mode : //unsigned,default = 0,osc_sc_alpha_mode, 1: (alpha >= 128) ? alpha + 1: alpha, 0: (alpha >=1) ? alpha + 1: alpha.
11:4	R/W	0	osd_sc_alpha : //unsigned,default = 0,default alpha for vd1 or vd2 if they are selected as the source
3	R/W	0	osd_sc_path_en : //unsigned,default = 0,osd scaler path enable
2	R/W	0	osd_sc_en : //unsigned,default = 0,osd scaler enable

Table 8-1874 VPP_OSD_SCI_WH_M1 0x1dc9

Bit(s)	R/W	Default	Description
28-16	R/W	0x0	OSD scaler input width minus 1
12-0	R/W	0x0	OSD scaler input height minus 1

Table 8-1875 VPP_OSD_SCO_H_START_END 0x1dca

Bit(s)	R/W	Default	Description
27-16	R/W	0x0	OSD scaler output horizontal start
11-0	R/W	0x0	OSD scaler output horizontal end

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Table 8-1876 VPP_OSD_SCO_V_START_END 0x1dcb

Bit(s)	R/W	Default	Description
27-16	R/W	0x0	OSD scaler output vertical start
11-0	R/W	0x0	OSD scaler output vertical end

Table 8-1877 VPP_OSD_SCALE_COEF_IDX 0x1dcc

Bit(s)	R/W	Default	Description

15	R/W	0x0	Because there are many coefficients used in the vertical filter and horizontal filters,
			//indirect access the coefficients of vertical filter and horizontal filter is used.
			//For vertical filter, there are 33x4 coefficients
			//For horizontal filter, there are 33x4 coefficients
			//Bit 15 index increment, if bit9 == 1 then (0: index increase 1, 1: index increase 2) else (index increase 2)

14	R/W	0x0	1: read coef through cbus enable, just for debug purpose in case when we wanna check the coef in ram in correct or not
9	R/W	0x0	if true, use 9bit resolution coef, other use 8bit resolution coef
8	R/W	0x0	type of index, 0: vertical coef, 1: horizontal coef
6-0	R/W	0x0	coef index

Table 8-1878 VPP_OSD_SCALE_COEF 0x1dcd

Bit(s)	R/W	Default	Description
31-0	R/W	0x0	

Table 8-1879 OSD_DB_FLT_CTRL 0x3140

Bit(s)	R/W	Default	Description
26	R/W	1	reg_nrdeband_reset1 : // unsigned , default = 0 0 : no reset seed 1: reload chroma seed
25	R/W	1	reg_nrdeband_reset0 : // unsigned , default = 0 0 : no reset seed 1: reload luma seed
24	R/W	0	reg_nrdeband_rgb : // unsigned , default = 0 0 : yuv 1: RGB
23	R/W	0	reg_nrdeband_en11 : // unsigned , default = 1 debanding registers of side lines, [0] for luma, same for below
22	R/W	0	reg_nrdeband_en10 : // unsigned , default = 1 debanding registers of side lines, [1] for chroma, same for below
21	R/W	1	reg_nrdeband_siderand : // unsigned , default = 1 options to use side two lines use the rand, instead of use for the YUV three component of middle line, 0: seed [3]/bandrand[3] for middle line yuv; 1: seed[3]/bandrand[3] for nearby three lines Y;
20	R/W	0	reg_nrdeband_randmode : // unsigned , default = 0 mode of rand noise adding, 0: same noise strength for all difs; else: strenght of noise will not exceed the difs, MIN((pPKReg->reg_nrdeband_bandrand[m]), noise[m])
19:17	R/W	6	reg_nrdeband_bandrand2 : // unsigned , default = 6
15:13	R/W	6	reg_nrdeband_bandrand1 : // unsigned , default = 6

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Bit(s)	R/W	Default	Description
11: 9	R/W	6	reg_nrdeband_bandrand0 : // unsigned , default = 6
7	R/W	1	reg_nrdeband_hpxor1 : // unsigned , default = 1 debanding random hp portion xor, [0] for luma
6	R/W	1	reg_nrdeband_hpxor0 : // unsigned , default = 1 debanding random hp portion xor, [1] for chroma
5	R/W	0	reg_nrdeband_en1 : // unsigned , default = 1 debanding registers, for luma
4	R/W	0	reg_nrdeband_en0 : // unsigned , default = 1 debanding registers, for chroma
3: 2	R/W	2	reg_nrdeband_lpf_mode1 : // unsigned , default = 2 lpf mode, 0: 3x3, 1:3x5; 2: 5x5; 3:5x7
1: 0	R/W	2	reg_nrdeband_lpf_mode0 : // unsigned , default = 2 lpf mode, 0: 3x3, 1:3x5; 2: 5x5; 3:5x7

Table 8-1880 OSD_DB_FLT_CTRL1 0x3141

Bit(s)	R/W	Default	Description
31:18			reserved
17:16	R/W	2	Reg_osddeband_noise_rs
15:12	R/W	8	Reg_osddeband_randgain
11			reserved
10:8	R/W	6	Reg_osddedband_bandrand5
7			reserved
6: 4	R/W	6	Reg_osddedband_bandrand4
3			reserved
2: 0	R/W	6	Reg_osddeband_bandrand3

Table 8-1881 OSD_DB_FLT_LUMA_THRD 0x3142

Bit(s)	R/W	Default	Description
31:30			
29:24	R/W	36	reg_nrdeband_luma_th3 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4
23:22			
21:16	R/W	28	reg_nrdeband_luma_th2 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/ 4elseif elseif <th[3] (lpf*1 + 3*y)/4; else
15:14			
13: 8	R/W	24	reg_nrdeband_luma_th1 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4
7: 6			
5: 0	R/W	20	reg_nrdeband_luma_th0 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/ 4elseif elseif

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Table 8-1882 OSD_DB_FLT_LUMA_THRD 0x3143

Bit(s)	R/W	Default	Description
31:30			
29:24	R/W	36	reg_nrdeband_chrm_th3 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4
23:22			
21:16	R/W	28	reg_nrdeband_chrm_th2 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/ 4elseif elseif <th[3] (lpf*1 + 3*y)/4; else
15:14			
13: 8	R/W	24	reg_nrdeband_chrm_th1 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4
7: 6			
5: 0	R/W	20	reg_nrdeband_chrm_th0 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/ 4elseif elseif

Table 8-1883 OSD_DB_FLT_RANDLUT 0x3144

Bit(s)	R/W	Default	Description
23:21	R/W	1	reg_nrdeband_randslut7 : // unsigned , default = 1 lut0
20:18	R/W	1	reg_nrdeband_randslut6 : // unsigned , default = 1 lut0
17:15	R/W	1	reg_nrdeband_randslut5 : // unsigned , default = 1 lut0
14:12	R/W	1	reg_nrdeband_randslut4 : // unsigned , default = 1 lut0
11: 9	R/W	1	reg_nrdeband_randslut3 : // unsigned , default = 1 lut0
8: 6	R/W	1	reg_nrdeband_randslut2 : // unsigned , default = 1 lut0
5: 3	R/W	1	reg_nrdeband_randslut1 : // unsigned , default = 1 lut0
2: 0	R/W	1	reg_nrdeband_randslut0 : // unsigned , default = 1 lut0

Table 8-1884 OSD_DB_FLT_PXI_THRD 0x3145

Bit(s)	R/W	Default	Description
25:16	R/W	0	reg_nrdeband_yc_th1 : // unsigned , default = 0 to luma/|u/v| for using the denoise
9: 0	R/W	0	reg_nrdeband_yc_th0 : // unsigned , default = 0 to luma/|u/v| for using the denoise

Table 8-1885 OSD_DB_FLT_SEED_Y 0x3146

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8240	reg_nrdeband_seed0 : // unsigned , default = 1621438240 noise adding seed for Y. seed[0]= 0x60a52f20; as default

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Table 8-1886 OSD_DB_FLT_SEED_U 0x3147

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8247	reg_nrdeband_seed1 : // unsigned , default = 1621438247 noise adding seed for U. seed[0]= 0x60a52f27; as default

Table 8-1887 OSD_DB_FLT_SEED_V 0x3148

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8242	reg_nrdeband_seed2 : // unsigned , default = 1621438242 noise adding seed for V. seed[0]= 0x60a52f22; as default

Table 8-1888 OSD_DB_FLT_SEED3 0x3149

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8242	reg_nrdeband_seed3 : // unsigned , default = 1621438242 noise adding seed for V. seed[0]= 0x60a52f22; as default

Table 8-1889 OSD_DB_FLT_SEED4 0x314a

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8242	reg_nrdeband_seed4 : // unsigned , default = 1621438242 noise adding seed for V. seed[0]= 0x60a52f22; as default

Table 8-1890 OSD_DB_FLT_SEED5 0x314b

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8242	reg_nrdeband_seed5 : // unsigned , default = 1621438242 noise adding seed for V. seed[0]= 0x60a52f22; as default

8.2.3.25 VPP OSD2 SCALER Registers

Table 8-1891 SCALER OSD2_VSC_PHASE_STEP 0x3d00

Bit(s)	R/W	Default	Description
27-0	R/W	0x0100-0000	4.24 format

Table 8-1892 OSD2_VSC_INI_PHASE 0x3d01

Bit(s)	R/W	Default	Description
31-16	R/W	0x0	botttom vertical scaler initial phase
15-0	R/W	0x0	top vertical scaler initial phase

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Table 8-1893 OSD2_VSC_CTRL0 0x3d02

Bit(s)	R/W	Default	Description
25	R/W	0	Repeat last line en : 1:enable repeat last line 0:disable repeat last line
24	R/W	0x0	osd vertical Scaler enable
23	R/W	0x0	osd_prog_interlace 0: current field is progressive, 1: current field is interlace
22-21	R/W	0x0	osd_vsc_double_line_mode, bit1, double input width and half input height, bit0, change line buffer becomes 2 lines
20	R/W	0x0	osd_vsc_phase0_always_en
19	R/W	0x0	osd_vsc_nearest_en
17-16	R/W	0x0	osd_vsc_bot_rpt_l0_num
14-11	R/W	0x0	osd_vsc_bot_ini_rcv_num
9-8	R/W	0x0	osd_vsc_top_rpt_l0_num
6-3	R/W	0x0	osd_vsc_top_ini_rcv_num
2-0	R/W	0x0	osd_vsc_bank_length

Table 8-1894 OSD2_HSC_PHASE_STEP 0x3d03

Bit(s)	R/W	Default	Description
27-0	R/W	0x0100-0000	4.24 format

Table 8-1895 OSD2_HSC_INI_PHASE 0x3d04

Bit(s)	R/W	Default	Description
31-16	R/W	0x0	horizontal scaler initial phase1
15-0	R/W	0x0	horizontal scaler initial phase0

Table 8-1896 OSD2_HSC_CTRL0 0x3d05

Bit(s)	R/W	Default	Description
22	R/W	0x0	osd horizontal Scaler enable
21	R/W	0x0	osd_hsc_double_pix_mode
20	R/W	0x0	osd_hsc_phase0_always_en
19	R/W	0x0	osd_vsc_nearest_en
17-16	R/W	0x0	osd_hsc_rpt_p0_num1
14-11	R/W	0x0	osd_hsc_ini_rcv_num1
9-8	R/W	0x0	osd_hsc_rpt_p0_num0
6-3	R/W	0x0	osd_hsc_ini_rcv_num0
2-0	R/W	0x0	osd_hsc_bank_length

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Table 8-1897 OSD2_HSC_INI_PAT_CTRL 0x3d06

Bit(s)	R/W	Default	Description
15-8	R/W	0x0	for 3D quincunx sub-sampling. pattern, each patten 1 bit, from lsb -> msb
6-4	R/W	0x0	pattern start
2-0	R/W	0x0	pattern end

Table 8-1898 OSD2_SC_DUMMY_DATA 0x3d07

Bit(s)	R/W	Default	Description
31-24	R/W	0x0	componet 0 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
23-16	R/W	0x0	component 1 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
15-8	R/W	0x0	component 2 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
7-0	R/W	0x0	component 3 , alpha

Table 8-1899 OSD2_SC_CTRL0 0x3d08

Bit(s)	R/W	Default	Description
27:16	R/W	0	osd_sc_gclk_ctrl : //unsigned,default = 0,osd_sc_gclk_ctrl
13	R/W	0	osd_sc_din_osd_alpha_mode : //unsigned,default = 0,osc_sc_din_osd2_alpha_mode, 1: (alpha >= 128) ? alpha -1: alpha, 0: (alpha >=1) ? alpha - 1: alpha.
12	R/W	0	osd_sc_dout_alpha_mode : //unsigned,default = 0,osc_sc_alpha_mode, 1: (alpha >= 128) ? alpha + 1: alpha, 0: (alpha >=1) ? alpha + 1: alpha.
11:4	R/W	0	osd_sc_alpha : //unsigned,default = 0,default alpha for vd1 or vd2 if they are selected as the source
3	R/W	0	osd_sc_path_en : //unsigned,default = 0,osd scaler path enable
2	R/W	0	osd_sc_en : //unsigned,default = 0,osd scaler enable

Table 8-1900 OSD2_SCI_WH_M1 0x3d09

Bit(s)	R/W	Default	Description
28-16	R/W	0x0	OSD scaler input width minus 1
12-0	R/W	0x0	OSD scaler input height minus 1

Table 8-1901 OSD2_SCO_H_START_END 0x3d0a

Bit(s)	R/W	Default	Description
27-16	R/W	0x0	OSD scaler output horizontal start
11-0	R/W	0x0	OSD scaler output horizontal end

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Table 8-1902 OSD2_SCO_V_START_END 0x3d0b

Bit(s)	R/W	Default	Description
27-16	R/W	0x0	OSD scaler output vertical start
11-0	R/W	0x0	OSD scaler output vertical end

Table 8-1903 OSD2_SCALE_COEF_IDX 0x3d18

Bit(s)	R/W	Default	Description

15	R/W	0x0	Because there are many coefficients used in the vertical filter and horizontal filters,
			//indirect access the coefficients of vertical filter and horizontal filter is used.
			//For vertical filter, there are 33x4 coefficients
			//For horizontal filter, there are 33x4 coefficients
			//Bit 15 index increment, if bit9 == 1 then (0: index increase 1, 1: index increase 2) else (index increase 2)

14	R/W	0x0	1: read coef through cbus enable, just for debug purpose in case when we wanna check the coef in ram in correct or not
9	R/W	0x0	if true, use 9bit resolution coef, other use 8bit resolution coef
8	R/W	0x0	type of index, 0: vertical coef, 1: horizontal coef
6-0	R/W	0x0	coef index

Table 8-1904 OSD2_SCALE_COEF 0x3d19

Bit(s)	R/W	Default	Description
31-0	R/W	0x0	

8.2.3.26 VPP OSD_BLD34 SCALER Registers

Table 8-1905 OSD34_VSC_PHASE_STEP 0x3d20

Bit(s)	R/W	Default	Description
27-0	R/W	0x0100-0000	4.24 format

Table 8-1906 OSD34_VSC_INI_PHASE 0x3d21

Bit(s)	R/W	Default	Description
31-16	R/W	0x0	botttom vertical scaler initial phase
15-0	R/W	0x0	top vertical scaler initial phase

Table 8-1907 OSD34_VSC_CTRL0 0x3d22

Bit(s)	R/W	Default	Description
25	R/W	0	Repeat last line en : 1:enable repeat last line 0:disable repeat last line
24	R/W	0x0	osd vertical Scaler enable

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Bit(s)	R/W	Default	Description
23	R/W	0x0	osd_prog_interlace 0: current field is progressive, 1: current field is interlace
22-21	R/W	0x0	osd_vsc_double_line_mode, bit1, double input width and half input height, bit0, change line buffer becomes 2 lines
20	R/W	0x0	osd_vsc_phase0_always_en
19	R/W	0x0	osd_vsc_nearest_en
17-16	R/W	0x0	osd_vsc_bot_rpt_l0_num
14-11	R/W	0x0	osd_vsc_bot_ini_rcv_num
9-8	R/W	0x0	osd_vsc_top_rpt_l0_num
6-3	R/W	0x0	osd_vsc_top_ini_rcv_num
2-0	R/W	0x0	osd_vsc_bank_length

Table 8-1908 OSD34_HSC_PHASE_STEP 0x3d23

Bit(s)	R/W	Default	Description
27-0	R/W	0x0100-0000	4.24 format

Table 8-1909 OSD34_HSC_INI_PHASE 0x3d24

Bit(s)	R/W	Default	Description
31-16	R/W	0x0	horizontal scaler initial phase1
15-0	R/W	0x0	horizontal scaler initial phase0

Table 8-1910 OSD34_HSC_CTRL0 0x3d25

Bit(s)	R/W	Default	Description
22	R/W	0x0	osd horizontal Scaler enable
21	R/W	0x0	osd_hsc_double_pix_mode
20	R/W	0x0	osd_hsc_phase0_always_en
19	R/W	0x0	osd_vsc_nearest_en
17-16	R/W	0x0	osd_hsc_rpt_p0_num1
14-11	R/W	0x0	osd_hsc_ini_rcv_num1
9-8	R/W	0x0	osd_hsc_rpt_p0_num0
6-3	R/W	0x0	osd_hsc_ini_rcv_num0
2-0	R/W	0x0	osd_hsc_bank_length

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Table 8-1911 OSD34_HSC_INI_PAT_CTRL 0x3d26

Bit(s)	R/W	Default	Description
15-8	R/W	0x0	for 3D quincunx sub-sampling. pattern, each patten 1 bit, from lsb -> msb
6-4	R/W	0x0	pattern start
2-0	R/W	0x0	pattern end

Table 8-1912 OSD34_SC_DUMMY_DATA 0x3d27

Bit(s)	R/W	Default	Description
31-24	R/W	0x0	componet 0 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
23-16	R/W	0x0	component 1 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
15-8	R/W	0x0	component 2 ,data = (dst_data >> 4) ,so 8 bit in 12bit mode, 6bit in 10bit mode
7-0	R/W	0x0	component 3 , alpha

Table 8-1913 OSD34_SC_CTRL0 0x3d28

Bit(s)	R/W	Default	Description
27:16	R/W	0	osd_sc_gclk_ctrl : //unsigned,default = 0,osd_sc_gclk_ctrl
13	R/W	0	osd_sc_din_osd_alpha_mode : //unsigned,default = 0,osc_sc_din_osd2_alpha_mode, 1: (alpha >= 128) ? alpha -1: alpha, 0: (alpha >=1) ? alpha - 1: alpha.
12	R/W	0	osd_sc_dout_alpha_mode : //unsigned,default = 0,osc_sc_alpha_mode, 1: (alpha >= 128) ? alpha + 1: alpha, 0: (alpha >=1) ? alpha + 1: alpha.
11:4	R/W	0	osd_sc_alpha : //unsigned,default = 0,default alpha for vd1 or vd2 if they are selected as the source
3	R/W	0	osd_sc_path_en : //unsigned,default = 0,osd scaler path enable
2	R/W	0	osd_sc_en : //unsigned,default = 0,osd scaler enable

Table 8-1914 OSD34_SCI_WH_M1 0x3d29

Bit(s)	R/W	Default	Description
28-16	R/W	0x0	OSD scaler input width minus 1
12-0	R/W	0x0	OSD scaler input height minus 1

Table 8-1915 OSD34_SCO_H_START_END 0x3d2a

Bit(s)	R/W	Default	Description
27-16	R/W	0x0	OSD scaler output horizontal start
11-0	R/W	0x0	OSD scaler output horizontal end

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Table 8-1916 OSD34_SCO_V_START_END 0x3d2b

Bit(s)	R/W	Default	Description
27-16	R/W	0x0	OSD scaler output vertical start
11-0	R/W	0x0	OSD scaler output vertical end

Table 8-1917 OSD34_SCALE_COEF_IDX 0x3d1e

Bit(s)	R/W	Default	Description

15	R/W	0x0	Because there are many coefficients used in the vertical filter and horizontal filters,
			//indirect access the coefficients of vertical filter and horizontal filter is used.
			//For vertical filter, there are 33x4 coefficients
			//For horizontal filter, there are 33x4 coefficients
			//Bit 15 index increment, if bit9 == 1 then (0: index increase 1, 1: index increase 2) else (index increase 2)

14	R/W	0x0	1: read coef through cbus enable, just for debug purpose in case when we wanna check the coef in ram in correct or not
9	R/W	0x0	if true, use 9bit resolution coef, other use 8bit resolution coef
8	R/W	0x0	type of index, 0: vertical coef, 1: horizontal coef
6-0	R/W	0x0	coef index

Table 8-1918 OSD34_SCALE_COEF 0x3d1f

Bit(s)	R/W	Default	Description
31-0	R/W	0x0	

8.2.3.27 VPP VD1 SCALER Registers

Because there are many coefficients used in the vertical filter and horizontal filters, indirect access the coefficients of vertical filter and horizontal filter is used. For vertical filter, there are 33x4 coefficients. For horizontal filter, there are 33x4 coefficients.

Table 8-1919 VPP_SCALE_COEF_IDX 0x1d03

Bit(s)	R/W	Default	Description
15	R/W	0	index_inc : // unsigned , default = 0x0 ,index increment, if bit9 == 1 then (0: index increase 1, 1: index increase 2) else (index increase 2)
14	R/W	0	rd_cbus_coef_en : // unsigned , default = 0x0 ,1: read coef through cbus enable, just for debug purpose in case when we wanna check the coef in ram in correct or not
13	R/W	0	vf_sep_coef_en : // unsigned , default = 0x0 ,if true, vertical separated coef enable
9	R/W	0	high_reso_en : // unsigned , default = 0x0 ,if true, use 9bit resolution coef, other use 8bit resolution coef
8:7	R/W	0	type_index : // unsigned , default = 0x0 ,type of index, 00: vertical coef, 01: vertical chroma coef: 10: horizontal coef, 11: resevered
6:0	R/W	0	coef_index : // unsigned , default = 0x0 ,coef index

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Table 8-1920 VPP_SCALE_COEF 0x1d04

Bit(s)	R/W	Default	Description
31:24	R/W	0	coef0 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
23:16	R/W	0	coef1 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
15:8	R/W	0	coef2 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
7 :0	R/W	0	coef3 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter

Table 8-1921 VPP_VSC_REGION12_STARTP 0x1d05

Bit(s)	R/W	Default	Description
28:16	R/W	0	region1_startp : //unsigned , default = 0 ,region1 startp
12:0	R/W	0	region2_startp : //unsigned , default = 0 ,region2 startp

Table 8-1922 VPP_VSC_REGION34_STARTP 0x1d06

Bit(s)	R/W	Default	Description
28:16	R/W	0	region3_startp : //unsigned , default = 0x0438,region3 startp
12:0	R/W	0	region4_startp : //unsigned , default = 0x0438,region4 startp

Table 8-1923 VPP_VSC_REGION4_ENDP 0x1d07

Bit(s)	R/W	Default	Description
12:0	R/W	13	region4_endp : //unsigned , default = 13'd1079 ,region4 endp

Table 8-1924 VPP_VSC_START_PHASE_STEP 0x1d08

Bit(s)	R/W	Default	Description
27:24	R/W	1	integer_part : //unsigned , default = 1,vertical start phase step, (source/dest)* (2^24),integer part of step
23:0	R/W	0	fraction_part : //unsigned , default = 0,vertical start phase step, (source/dest)* (2^24),fraction part of step

Table 8-1925 VPP_VSC_REGION0_PHASE_SLOPE 0x1d09

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,vertical scaler region0 phase slope, region0 phase slope

Table 8-1926 VPP_VSC_REGION1_PHASE_SLOPE 0x1d0a

Bit(s)	R/W	Default	Description
24:0	R/W	0	region1_phase_slope : //signed , default = 0,region1 phase slope

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Table 8-1927 VPP_VSC_REGION3_PHASE_SLOPE 0x1d0b

Bit(s)	R/W	Default	Description
24:0	R/W	0	region3_phase_slope : //signed , default = 0,region3 phase slope

Table 8-1928 VPP_VSC_REGION4_PHASE_SLOPE 0x1d0c

Bit(s)	R/W	Default	Description
24:0	R/W	0	region4_phase_slope : //signed , default = 0,region4 phase slope

Table 8-1929 VPP_VSC_PHASE_CTRL 0x1d0d

Bit(s)	R/W	Default	Description
18:17	R/W	0	vsc_double_line_mode : //unsigned , default = 0, double line mode, input/output line width of vscaler becomes 2X, so only 2 line buffer in this case, use for 3D line by line interleave scaling bit1 true, double the input width and half input height, bit0 true, change line buffer 2 lines instead of 4 lines
16	R.O	0	prog_interlace : //unsigned , default = 0,0: progressive output, 1: interlace output
15	R/W	0	vsc_bot_l0_out_en : //unsigned , default = 0,vertical scaler output line0 in advance or not for bottom field
14:13	R/W	1	vsc_bot_rpt_l0_num : //unsigned , default = 1,vertical scaler initial repeat line0 number for bottom field
11:8	R/W	4	vsc_bot_ini_rcv_num : //unsigned , default = 4,vertical scaler initial receiving number for bottom field
7	R/W	0	vsc_top_l0_out_en : //unsigned , default = 0,vertical scaler output line0 in advance or not for top field
6:5	R/W	1	vsc_top_rpt_l0_num : //unsigned , default = 1,vertical scaler initial repeat line0 number for top field
3:0	R/W	4	vsc_top_ini_rcv_num : //unsigned , default = 4,vertical scaler initial receiving number for top field

Table 8-1930 VPP_VSC_INI_PHASE 0x1d0e

Bit(s)	R/W	Default	Description
31:16	R/W	0,	//unsigned , default = 0,vertical scaler field initial phase for bottom field Bit 15:0 //unsigned , default = 0,vertical scaler field initial phase for top field

Table 8-1931 VPP_HSC_REGION12_STARTP 0x1d10

Bit(s)	R/W	Default	Description
28:16	R/W	0	region1_startp : //unsigned , default = 0,region1 startp
12:0	R/W	0	region2_startp : //unsigned , default = 0,region2 startp

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Table 8-1932 VPP_HSC_REGION34_STARTP 0x1d11

Bit(s)	R/W	Default	Description
28:16	R/W	0	region3 : startp //unsigned , default = 0x780,region3 startp
12:0	R/W	0	region4 : startp //unsigned , default = 0x780,region4 startp

Table 8-1933 VPP_HSC_REGION4_ENDP 0x1d12

Bit(s)	R/W	Default	Description
12:0	R/W	13	region4 : startp //unsigned , default = 13'd1919,region4 startp

horizontal start phase step, (source/dest)*(2^24)

Table 8-1934 VPP_HSC_START_PHASE_STEP 0x1d13

Bit(s)	R/W	Default	Description
27:24	R/W	1	integer_part : //unsigned , default = 1,integer part of step
23:0	R/W	0	fraction_part : //unsigned , default = 0,fraction part of step

Table 8-1935 VPP_HSC_REGION0_PHASE_SLOPE 0x1d14

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region0 phase slope

Table 8-1936 VPP_HSC_REGION1_PHASE_SLOPE 0x1d15

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region1 phase slope

Table 8-1937 VPP_HSC_REGION3_PHASE_SLOPE 0x1d16

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region3 phase slope

Table 8-1938 VPP_HSC_REGION4_PHASE_SLOPE 0x1d17

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region4 phase slope

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Table 8-1939 VPP_HSC_PHASE_CTRL 0x1d18

Bit(s)	R/W	Default	Description
22:21	R/W	1	hsc_rpt_p0_num0 : //unsigned , default = 1 ,horizontal scaler initial repeat pixel0 number0
19:16	R/W	4	hsc_ini_rcv_num0 : //unsigned , default = 4 ,horizontal scaler initial receiving number0
15:0	R/W	0	hsc_ini_phase0 : //unsigned , default = 0 ,horizontal scaler top field initial phase0

Table 8-1940 VPP_SC_MISC 0x1d19

Bit(s)	R/W	Default	Description
24	R/W	0	Repeat last line en : 1:enable repeat last line 0:disable repeat last line
22	R/W	0	hsc_len_div2_en : //unsigned , default = 0 ,if true, divide VSC line length 2 as the HSC input length, othwise VSC length length is the same as the VSC line length just for special usage, more flexibilityi
21	R/W	0	lbuf_mode : //unsigned , default = 0 ,if true, prevsc uses lin buffer, otherwise prevsc does not use line buffer, it should be same as prevsc_en
20	R/W	0	prehsc_en : //unsigned , default = 0 ,prehsc_en
19	R/W	0	prevsc_en : //unsigned , default = 0 ,prevsc_en
18	R/W	0	vsc_en : //unsigned , default = 0 ,vsc_en
17	R/W	0	hsc_en : //unsigned , default = 0 ,hsc_en
16	R/W	0	sc_top_en : //unsigned , default = 0 ,scale_top_en
15	R/W	0	sc_vd_en : //unsigned , default = 0 ,video1 scale out enable
12	R/W	1	hsc_nonlinear_4region_en : //unsigned , default = 1 ,if true, region0,region4 are nonlinear regions, otherwise they are not scaling regions, for horizontal scaler
10:8	R/W	0	hsc_bank_length : //unsigned , default = 0 ,horizontal scaler bank length
5	R/W	4	vsc_phase_field_mode : //unsigned , default = 4 ,vertical scaler phase field mode, if true, disable the opposite parity line output, more bandwith needed if output 1080i
4	R/W	0	vsc_nonlinear_4region_en : //unsigned , default = 0 ,if true, region0,region4 are nonlinear regions, otherwise they are not scaling regions, for vertical scaler
2:0	R/W	4	vsc_bank_length : //unsigned , default = 4 ,vertical scaler bank length

Table 8-1941 VPP_SCO_FIFO_CTRL 0x1d33

Bit(s)	R/W	Default	Description
27:16	R/W	0	sco_fifo_line_lenm1 : //unsigned , default = 0xfff, scale out fifo line length minus 1
12:0	R/W	0	sco_fifo_size : //unsigned , default = 0x200, scale out fifo size (actually only bit 11:1 is valid, 11:1, max 1024), always even number

for 3D quincunx sub-sampling and horizontal pixel by pixel 3D interleaving

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Table 8-1942 VPP_HSC_PHASE_CTRL1 0x1d34

Bit(s)	R/W	Default	Description
27:24	R/W	0	prehsc_mode : //unsigned , default = 0,prehsc_mode, bit 3:2, prehsc odd line interp mode, bit 1:0, prehsc even line interp mode, each 2bit, 00: pix0+pix1/2, average, 01: pix1, 10: pix0
23	R/W	0	hsc_double_pix_mode : //unsigned , default = 0,horizontal scaler double pixel mode
22:21	R/W	1	hsc_rpt_p0_num1 : //unsigned , default = 1,horizontal scaler initial repeat pixel0 number1
19:16	R/W	4	hsc_ini_rcv_num1 : //unsigned , default = 4,horizontal scaler initial receiving number1
15:0	R/W	0	hsc_ini_phase1 : //unsigned , default = 0,horizontal scaler top field initial phase1

Table 8-1943 for 3D quincunx sub-sampling VPP_HSC_INI_PAT_CTRL 0x1d35

Bit(s)	R/W	Default	Description
31:24	R/W	0	prehsc_pattern : //unsigned , default = 0, prehsc pattern, each patten 1 bit, from lsb -> msb
22:20	R/W	0	prehsc_pat_star : //unsigned , default = 0, prehsc pattern start
18:16	R/W	0	prehsc_pat_end : //unsigned , default = 0, prehsc pattern end
15:8	R/W	0	hsc_pattern : //unsigned , default = 0, hsc pattern, each patten 1 bit, from lsb -> msb
6:4	R/W	0	hsc_pat_start : //unsigned , default = 0, hsc pattern start
2:0	R/W	0	hsc_pat_end : //unsigned , default = 0, hsc pattern end

Table 8-1944 VPP_SC_GCLK_CTRL 0x1d35

Bit(s)	R/W	Default	Description
15:0	R/W	0	vpp_sc_gclk_ctrl : //unsigned , default = 0 ,scale clock gate

8.2.3.28 VPP VD2 SCALER Registers

Because there are many coefficients used in the vertical filter and horizontal filters, indirect access the coefficients of vertical filter and horizontal filter is used. For vertical filter, there are 33x4 coefficients. For horizontal filter, there are 33x4 coefficients□

Table 8-1945 VD2_SCALE_COEF_IDX 0x3943

Bit(s)	R/W	Default	Description
15	R/W	0	index_inc : // unsigned , default = 0x0 ,index increment, if bit9 == 1 then (0: index increase 1, 1: index increase 2) else (index increase 2)
14	R/W	0	rd_cbus_coef_en : // unsigned , default = 0x0 ,1: read coef through cbus enable, just for debug purpose in case when we wanna check the coef in ram in correct or not
13	R/W	0	vf_sep_coef_en : // unsigned , default = 0x0 ,if true, vertical separated coef enable

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Bit(s)	R/W	Default	Description
9	R/W	0	high_reso_en : // unsigned , default = 0x0 ,if true, use 9bit resolution coef, other use 8bit resolution coef
8:7	R/W	0	type_index : // unsigned , default = 0x0 ,type of index, 00: vertical coef, 01: vertical chroma coef: 10: horizontal coef, 11: resevered
6:0	R/W	0	coef_index : // unsigned , default = 0x0 ,coef index

Table 8-1946 VD2_SCALE_COEF 0x3944

Bit(s)	R/W	Default	Description
31:24	R/W	0	coef0 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
23:16	R/W	0	coef1 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
15:8	R/W	0	coef2 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter
7 :0	R/W	0	coef3 : //signed , default = 0x0 , coefficients for vertical filter and horizontal filter

Table 8-1947 VD2_VSC_REGION12_STARTP 0x3945

Bit(s)	R/W	Default	Description
28:16	R/W	0	region1_startp : //unsigned , default = 0 ,region1 startp
12:0	R/W	0	region2_startp : //unsigned , default = 0 ,region2 startp

Table 8-1948 VD2_VSC_REGION34_STARTP 0x3946

Bit(s)	R/W		Default
28:16		R/W	region3_startp : //unsigned , default = 0x0438,region3 startp
12:0		R/W	region4_startp : //unsigned , default = 0x0438,region4 startp

Table 8-1949 VD2_VSC_REGION4_ENDP 0x3947

Bit(s)	R/W	Default	Description
12:0	R/W	13	region4_endp : //unsigned , default = 13'd1079 ,region4 endp

Table 8-1950 VD2_VSC_START_PHASE_STEP 0x3948

Bit(s)	R/W	Default	Description
27:24	R/W	1	integer_part : //unsigned , default = 1,vertical start phase step, (source/dest)*(2^24),integer part of step
23:0	R/W	0	fraction_part : //unsigned , default = 0,vertical start phase step, (source/dest)*(2^24),fraction part of step

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Table 8-1951 VD2_VSC_REGION0_PHASE_SLOPE 0x3949

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,vertical scaler region0 phase slope, region0 phase slope

Table 8-1952 VD2_VSC_REGION1_PHASE_SLOPE 0x394a

Bit(s)	R/W	Default	Description
24:0	R/W	0	region1_phase_slope : //signed , default = 0,region1 phase slope

Table 8-1953 VD2_VSC_REGION3_PHASE_SLOPE 0x394b

Bit(s)	R/W	Default	Description
24:0	R/W	0	region3_phase_slope : //signed , default = 0,region3 phase slope

Table 8-1954 VD2_VSC_REGION4_PHASE_SLOPE 0x394c

Bit(s)	R/W	Default	Description
24:0	R/W	0	region4_phase_slope : //signed , default = 0,region4 phase slope

Table 8-1955 VD2_VSC_PHASE_CTRL 0x394d

Bit(s)	R/W	Default	Description
18:17	R/W	0	vsc_double_line_mode : //unsigned , default = 0, double line mode, input/output line width of vscaler becomes 2X, so only 2 line buffer in this case, use for 3D line by line interleave scaling bit1 true, double the input width and half input height, bit0 true, change line buffer 2 lines instead of 4 lines
16	R.O	0	prog_interlace : //unsigned , default = 0,0: progressive output, 1: interlace output
15	R/W	0	vsc_bot_l0_out_en : //unsigned , default = 0,vertical scaler output line0 in advance or not for bottom field
14:13	R/W	1	vsc_bot_rpt_l0_num : //unsigned , default = 1,vertical scaler initial repeat line0 number for bottom field
11:8	R/W	4	vsc_bot_ini_rcv_num : //unsigned , default = 4,vertical scaler initial receiving number for bottom field
7	R/W	0	vsc_top_l0_out_en : //unsigned , default = 0,vertical scaler output line0 in advance or not for top field
6:5	R/W	1	vsc_top_rpt_l0_num : //unsigned , default = 1,vertical scaler initial repeat line0 number for top field
3:0	R/W	4	vsc_top_ini_rcv_num : //unsigned , default = 4,vertical scaler initial receiving number for top field

Table 8-1956 VD2_VSC_INI_PHASE 0x394e

Bit(s)	R/W	Default	Description
31:16	R/W	0,	//unsigned , default = 0,vertical scaler field initial phase for bottom field Bit 15:0 //unsigned , default = 0,vertical scaler field initial phase for top field

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Table 8-1957 VD2_HSC_REGION12_STARTP 0x394f

Bit(s)	R/W	Default	Description
28:16	R/W	0	region1_startp : //unsigned , default = 0,region1 startp
12:0	R/W	0	region2_startp : //unsigned , default = 0,region2 startp

Table 8-1958 VD2_HSC_REGION34_STARTP 0x3950

Bit(s)	R/W	Default	Description
28:16	R/W	0	region3 : startp //unsigned , default = 0x780,region3 startp
12:0	R/W	0	region4 : startp //unsigned , default = 0x780,region4 startp

Table 8-1959 VD2_HSC_REGION4_ENDP 0x3951

Bit(s)	R/W	Default	Description
12:0	R/W	13	region4 : startp //unsigned , default = 13'd1919,region4 startp

horizontal start phase step, (source/dest)*(2^24)

Table 8-1960 VD2_HSC_START_PHASE_STEP 0x3952

Bit(s)	R/W	Default	Description
27:24	R/W	1	integer_part : //unsigned , default = 1,integer part of step
23:0	R/W	0	fraction_part : //unsigned , default = 0,fraction part of step

Table 8-1961 VD2_HSC_REGION0_PHASE_SLOPE 0x3953

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region0 phase slope

Table 8-1962 VD2_HSC_REGION1_PHASE_SLOPE 0x3954

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region1 phase slope

Table 8-1963 VD2_HSC_REGION3_PHASE_SLOPE 0x3955

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region3 phase slope

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Table 8-1964 VD2_HSC_REGION4_PHASE_SLOPE 0x3956

Bit(s)	R/W	Default	Description
24:0	R/W	0	region0_phase_slope : //signed , default = 0,region4 phase slope

Table 8-1965 VD2_HSC_PHASE_CTRL 0x3957

Bit(s)	R/W	Default	Description
22:21	R/W	1	hsc_rpt_p0_num0 : //unsigned , default = 1 ,horizontal scaler initial repeat pixel0 number0
19:16	R/W	4	hsc_ini_rcv_num0 : //unsigned , default = 4 ,horizontal scaler initial receiving number0
15:0	R/W	0	hsc_ini_phase0 : //unsigned , default = 0 ,horizontal scaler top field initial phase0

Table 8-1966 VD2_SC_MISC 0x3958

Bit(s)	R/W	Default	Description
24	R/W	0	Repeat last line en : 1:enable repeat last line 0:disable repeat last line
22	R/W	0	hsc_len_div2_en : //unsigned , default = 0 ,if true, divide VSC line length 2 as the HSC input length, othwise VSC length length is the same as the VSC line length just for special usage, more flexibilityi
21	R/W	0	lbuf_mode : //unsigned , default = 0 ,if true, prevsc uses lin buffer, otherwise prevsc does not use line buffer, it should be same as prevsc_en
20	R/W	0	prehsc_en : //unsigned , default = 0 ,prehsc_en
19	R/W	0	prevsc_en : //unsigned , default = 0 ,prevsc_en
18	R/W	0	vsc_en : //unsigned , default = 0 ,vsc_en
17	R/W	0	hsc_en : //unsigned , default = 0 ,hsc_en
16	R/W	0	sc_top_en : //unsigned , default = 0 ,scale_top_en
15	R/W	0	sc_vd_en : //unsigned , default = 0 ,video1 scale out enable
12	R/W	1	hsc_nonlinear_4region_en : //unsigned , default = 1 ,if true, region0,region4 are nonlinear regions, otherwise they are not scaling regions, for horizontal scaler
10:8	R/W	0	hsc_bank_length : //unsigned , default = 0 ,horizontal scaler bank length
5	R/W	4	vsc_phase_field_mode : //unsigned , default = 4 ,vertical scaler phase field mode, if true, disable the opposite parity line output, more bandwith needed if output 1080i
4	R/W	0	vsc_nonlinear_4region_en : //unsigned , default = 0 ,if true, region0,region4 are nonlinear regions, otherwise they are not scaling regions, for vertical scaler
2:0	R/W	4	vsc_bank_length : //unsigned , default = 4 ,vertical scaler bank length

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Table 8-1967 VD2_SCO_FIFO_CTRL 0x3959

Bit(s)	R/W	Default	Description
27:16	R/W	0	sco_fifo_line_lenm1 : //unsigned , default = 0xfff, scale out fifo line length minus 1
12:0	R/W	0	sco_fifo_size : //unsigned , default = 0x200, scale out fifo size (actually only bit 11:1 is valid, 11:1, max 1024), always even number

Table 8-1968 for 3D quincunx sub-sampling and horizontal pixel by pixel 3D interleaving VD2_HSC_PHASE_CTRL1 0x395a

Bit(s)	R/W	Default	Description
27:24	R/W	0	prehsc_mode : //unsigned , default = 0,prehsc_mode, bit 3:2, prehsc odd line interp mode, bit 1:0, prehsc even line interp mode, each 2bit, 00: pix0+pix1/2, average, 01: pix1, 10: pix0
23	R/W	0	hsc_double_pix_mode : //unsigned , default = 0,horizontal scaler double pixel mode
22:21	R/W	1	hsc_rpt_p0_num1 : //unsigned , default = 1,horizontal scaler initial repeat pixel0 number1
19:16	R/W	4	hsc_ini_rcv_num1 : //unsigned , default = 4,horizontal scaler initial receiving number1
15:0	R/W	0	hsc_ini_phase1 : //unsigned , default = 0,horizontal scaler top field initial phase1

Table 8-1969 for 3D quincunx sub-sampling VD2_HSC_INI_PAT_CTRL 0x395b

Bit(s)	R/W	Default	Description
31:24	R/W	0	prehsc_pattern : //unsigned , default = 0, prehsc pattern, each patten 1 bit, from lsb -> msb
22:20	R/W	0	prehsc_pat_star : //unsigned , default = 0, prehsc pattern start
18:16	R/W	0	prehsc_pat_end : //unsigned , default = 0, prehsc pattern end
15:8	R/W	0	hsc_pattern : //unsigned , default = 0, hsc pattern, each patten 1 bit, from lsb -> msb
6:4	R/W	0	hsc_pat_start : //unsigned , default = 0, hsc pattern start
2:0	R/W	0	hsc_pat_end : //unsigned , default = 0, hsc pattern end

Table 8-1970 VD2_SC_GCLK_CTRL 0x395c

Bit(s)	R/W	Default	Description
15:0	R/W	0	vpp_sc_gclk_ctrl : //unsigned , default = 0 ,scale clock gate

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8.2.3.29 Supper Scale/ Sharpness Registers

Table 8-1971 SRSHARP0_SHARP_HVSIZE 0x3e00

Bit(s)	R/W	Default	Description
28:16	R/W	0d1920	reg_pknr_hsize : . unsigned , default = 1920
12: 0	R/W	0d1080	reg_pknr_vsize : . unsigned , default = 1080

Table 8-1972 SRSHARP0_SHARP_HVBLANK_NUM 0x3e01

Bit(s)	R/W	Default	Description
23: 16	R/W	0d8	reg_deband_hblank : . unsigned , default = 8
15: 8	R/W	0d20	reg_pknr_hblank_num : . unsigned , default = 20
7: 0	R/W	0d60	reg_pknr_vblank_num : . unsigned , default = 60

Table 8-1973 SRSHARP0_NR_GAUSSIAN_MODE 0x3e02

Bit(s)	R/W	Default	Description
4	R/W	0d1	reg_nr_gau_ymode : : 0 3x3 filter; 1: 5x5 filter . unsigned , default = 1
0	R/W	0d1	reg_nr_gau_cmode : : 0 3x3 filter; 1: 5x5 filter . unsigned , default = 1

Table 8-1974 SRSHARP0_PK_CON_2CIRHPGAIN_TH_RATE 0x3e05

Bit(s)	R/W	Default	Description
31:24	R/W	0d25	reg_pk_cirhpcon2gain0 : : threshold0 of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 25
23:16	R/W	0d60	reg_pk_cirhpcon2gain1 : : threshold1 of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 60
15: 8	R/W	0d80	reg_pk_cirhpcon2gain5 : : rate0 (for hpcon<th0) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 80
7: 0	R/W	0d20	reg_pk_cirhpcon2gain6 : : rate1 (for hpcon>th1) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 20

Table 8-1975 SRSHARP0_PK_CON_2CIRHPGAIN_LIMIT 0x3e06

Bit(s)	R/W	Default	Description
31:24	R/W	0d96	reg_pk_cirhpcon2gain2 : : level limit(for hpcon<th0) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 96
23:16	R/W	0d96	reg_pk_cirhpcon2gain3 : : level limit(for th0<hpcon<th1) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 96
15: 8	R/W	0d5	reg_pk_cirhpcon2gain4 : : level limit(for hpcon>th1) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 5

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Table 8-1976 SRSHARP0_PK_CON_2CIRBPGAIN_TH_RATE 0x3e07

Bit(s)	R/W	Default	Description
31:24	R/W	0d20	reg_pk_cirbpcon2gain0 : : threshold0 of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 20
23:16	R/W	0d50	reg_pk_cirbpcon2gain1 : : threshold1 of curve to map bpcon to bpgain for circle bp filter (all 8 direction same).. unsigned , default = 50
15: 8	R/W	0d50	reg_pk_cirbpcon2gain5 : : rate0 (for bpcon<th0) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 50
7: 0	R/W	0d25	reg_pk_cirbpcon2gain6 : : rate1 (for bpcon>th1) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 25

Table 8-1977 SRSHARP0_PK_CON_2CIRBPGAIN_LIMIT 0x3e08

Bit(s)	R/W	Default	Description
31:24	R/W	0d40	reg_pk_cirbpcon2gain2 : : level limit(for bpcon<th0) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 40
23:16	R/W	0d40	reg_pk_cirbpcon2gain3 : : level limit(for th0<bpcon<th1) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 40
15: 8	R/W	0d5	reg_pk_cirbpcon2gain4 : : level limit(for bpcon>th1) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 5

Table 8-1978 SRSHARP0_PK_CON_2DRTHPGAIN_TH_RATE 0x3e09

Bit(s)	R/W	Default	Description
31:24	R/W	0d25	reg_pk_drthpcon2gain0 : : threshold0 of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 25
23:16	R/W	0d60	reg_pk_drthpcon2gain1 : : threshold1 of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 60
15: 8	R/W	0d80	reg_pk_drthpcon2gain5 : : rate0 (for hpcon<th0) of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 80
7: 0	R/W	0d20	reg_pk_drthpcon2gain6 : : rate1 (for hpcon>th1) of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 20

Table 8-1979 SRSHARP0_PK_CON_2DRTHPGAIN_LIMIT 0x3e0a

Bit(s)	R/W	Default	Description
31:24	R/W	0d90	reg_pk_drthpcon2gain2 : : level limit(for hpcon<th0) of curve to map hpcon to hpgain for directional hp filter (best direction).. unsigned , default = 90
23:16	R/W	0d96	reg_pk_drthpcon2gain3 : : level limit(for th0<hpcon<th1) of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 96
15: 8	R/W	0d5	reg_pk_drthpcon2gain4 : : level limit(for hpcon>th1) of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 5

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Table 8-1980 SRSHARP0_PK_CON_2DRTBPGAIN_TH_RATE 0x3e0b

Bit(s)	R/W	Default	Description
31:24	R/W	0d20	reg_pk_drtbpcon2gain0 : : threshold0 of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 20
23:16	R/W	0d50	reg_pk_drtbpcon2gain1 : : threshold1 of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 50
15: 8	R/W	0d50	reg_pk_drtbpcon2gain5 : : rate0 (for bpcon<th0) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 50
7: 0	R/W	0d25	reg_pk_drtbpcon2gain6 : : rate1 (for bpcon>th1) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 25

Table 8-1981 SRSHARP0_PK_CON_2DRTBPGAIN_LIMIT 0x3e0c

Bit(s)	R/W	Default	Description
31:24	R/W	0d40	reg_pk_drtbpcon2gain2 : : level limit(for bpcon<th0) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 40
23:16	R/W	0d40	reg_pk_drtbpcon2gain3 : : level limit(for th0<bpcon<th1) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 40
15: 8	R/W	0d5	reg_pk_drtbpcon2gain4 : : level limit(for bpcon>th1) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 5

Table 8-1982 SRSHARP0_PK_CIRFB_LPF_MODE 0x3e0d

Bit(s)	R/W	Default	Description
29:28	R/W	0d1	reg_cirhp_horz_mode : : no horz filter on HP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8. unsigned , default = 1
25:24	R/W	0d1	reg_cirhp_vert_mode : : no vert filter on HP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8. unsigned , default = 1
21:20	R/W	0d1	reg_cirhp_diag_mode : : filter on HP; 1: [1 2 1]/4; . unsigned , default = 1
13:12	R/W	0d1	reg_cirbp_horz_mode : : no horz filter on BP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8. unsigned , default = 1
9: 8	R/W	0d1	reg_cirbp_vert_mode : : no vert filter on BP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8. unsigned , default = 1
5: 4	R/W	0d1	reg_cirbp_diag_mode : : filter on BP; 1: [1 2 1]/4; . unsigned , default = 1

Table 8-1983 SRSHARP0_PK_DRTFB_LPF_MODE 0x3e0e

Bit(s)	R/W	Default	Description
29:28	R/W	0d1	reg_drthp_horz_mode : : no horz filter on HP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8 2 . unsigned , default = 1
25:24	R/W	0d1	reg_drthp_vert_mode : : no vert filter on HP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8 2 . unsigned , default = 1
21:20	R/W	0d1	reg_drthp_diag_mode : : filter on HP; 1: [1 2 1]/4; 1 . unsigned , default = 1
13:12	R/W	0d1	reg_drtbp_horz_mode : : no horz filter on BP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8 2 . unsigned , default = 1

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Bit(s)	R/W	Default	Description
9: 8	R/W	0d1	reg_drtbp_vert_mode : : no vert filter on BP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8 2 . unsigned , default = 1
5: 4	R/W	0d1	reg_drtbp_diag_mode : : filter on BP; 1: [1 2 1]/4; 1 . unsigned , default = 1

Table 8-1984 SRSHARP0_PK_CIRFB_HP_CORING 0x3e0f

Bit(s)	R/W	Default	Description
21:16	R/W	0d4	reg_cirhp_horz_core : : coring of HP for Horz. unsigned , default = 4
13: 8	R/W	0d4	reg_cirhp_vert_core : : coring of HP for Vert. unsigned , default = 4
5: 0	R/W	0d4	reg_cirhp_diag_core : : coring of HP for Diag. unsigned , default = 4

Table 8-1985 SRSHARP0_PK_CIRFB_BP_CORING 0x3e10

Bit(s)	R/W	Default	Description
21:16	R/W	0d4	reg_cirbp_horz_core : : coring of HP for Horz. unsigned , default = 4
13: 8	R/W	0d4	reg_cirbp_vert_core : : coring of HP for Vert. unsigned , default = 4
5: 0	R/W	0d4	reg_cirbp_diag_core : : coring of HP for Diag. unsigned , default = 4

Table 8-1986 SRSHARP0_PK_DRTFB_HP_CORING 0x3e11

Bit(s)	R/W	Default	Description
21:16	R/W	0d4	reg_drthp_horz_core : : coring of HP for Horz. unsigned , default = 4
13: 8	R/W	0d4	reg_drthp_vert_core : : coring of HP for Vert. unsigned , default = 4
5: 0	R/W	0d4	reg_drthp_diag_core : : coring of HP for Diag. unsigned , default = 4

Table 8-1987 SRSHARP0_PK_DRTFB_BP_CORING 0x3e12

Bit(s)	R/W	Default	Description
21:16	R/W	0d4	reg_drtbp_horz_core : : coring of HP for Horz. unsigned , default = 4
13: 8	R/W	0d4	reg_drtbp_vert_core : : coring of HP for Vert. unsigned , default = 4
5: 0	R/W	0d4	reg_drtbp_diag_core : : coring of HP for Diag. unsigned , default = 4

Table 8-1988 SRSHARP0_PK_CIRFB_BLEND_GAIN 0x3e13

Bit(s)	R/W	Default	Description
31:28	R/W	0d8	reg_hp_cir_hgain : : normalized 8 as '1' . unsigned , default = 8
27:24	R/W	0d8	reg_hp_cir_vgain : : normalized 8 as '1' . unsigned , default = 8
23:20	R/W	0d8	reg_hp_cir_dgain : : normalized 8 as '1' . unsigned , default = 8
15:12	R/W	0d8	reg_bp_cir_hgain : : normalized 8 as '1' . unsigned , default = 8

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Bit(s)	R/W	Default	Description
11: 8	R/W	0d8	reg_bp_cir_vgain : : normalized 8 as '1' . unsigned , default = 8
7: 4	R/W	0d8	reg_bp_cir_dgain : : normalized 8 as '1' . unsigned , default = 8

Table 8-1989 SRSHARP0_NR_ALPY_SSD_GAIN_OFST 0x3e14

Bit(s)	R/W	Default	Description
15: 8	R/W	0d16	reg_nr_alp0_ssd_gain : : gain to max ssd normalized 16 as '1'. unsigned , default = 16
5: 0	R/W	0x0	reg_nr_alp0_ssd_ofst : : offset to ssd before dividing to min_err . signed , default = -2

Table 8-1990 SRSHARP0_NR_ALP0Y_ERR2CURV_TH_RATE 0x3e15

Bit(s)	R/W	Default	Description
31:24	R/W	0d10	reg_nr_alp0_minerr_ypar0 : : threshold0 of curve to map mierr to alp0 for luma channel, this will be set value of flat region mierr that no need blur. 0~255.. unsigned , default = 10
23:16	R/W	0d25	reg_nr_alp0_minerr_ypar1 : : threshold1 of curve to map mierr to alp0 for luma channel,this will be set value of texture region mierr that can not blur.. unsigned , default = 25
15: 8	R/W	0d80	reg_nr_alp0_minerr_ypar5 : : rate0 (for mierr<th0) of curve to map mierr to alp0 for luma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 80
7: 0	R/W	0d64	reg_nr_alp0_minerr_ypar6 : : rate1 (for mierr>th1) of curve to map mierr to alp0 for luma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 64

Table 8-1991 SRSHARP0_NR_ALP0Y_ERR2CURV_LIMIT 0x3e16

Bit(s)	R/W	Default	Description
31:24	R/W	0d63	reg_nr_alp0_minerr_ypar2 : : level limit(for mierr<th0) of curve to map mierr to alp0 for luma channel, this will be set to alp0 that we can do for flat region. 0~255.. unsigned , default = 63
23:16	R/W	0d0	reg_nr_alp0_minerr_ypar3 : : level limit(for th0<mierr<th1) of curve to map mierr to alp0 for luma channel, this will be set to alp0 that we can do for misc region. 0~255.. unsigned , default = 0
15: 8	R/W	0d63	reg_nr_alp0_minerr_ypar4 : : level limit(for mierr>th1) of curve to map mierr to alp0 for luma channel, this will be set to alp0 that we can do for texture region. 0~255.. unsigned , default = 63

Table 8-1992 SRSHARP0_NR_ALP0C_ERR2CURV_TH_RATE 0x3e17

Bit(s)	R/W	Default	Description
31:24	R/W	0d10	reg_nr_alp0_minerr_cpar0 : : threshold0 of curve to map mierr to alp0 for chroma channel, this will be set value of flat region mierr that no need blur.. unsigned , default = 10
23:16	R/W	0d25	reg_nr_alp0_minerr_cpar1 : : threshold1 of curve to map mierr to alp0 for chroma channel,this will be set value of texture region mierr that can not blur.. unsigned , default = 25

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Bit(s)	R/W	Default	Description
15: 8	R/W	0d80	reg_nr_alp0_minerr_cpar5 : : rate0 (for mierr<th0) of curve to map mierr to alp0 for chroma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 80
7: 0	R/W	0d64	reg_nr_alp0_minerr_cpar6 : : rate1 (for mierr>th1) of curve to map mierr to alp0 for chroma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 64

Table 8-1993 SRSHARP0_NR_ALP0C_ERR2CURV_LIMIT 0x3e18

Bit(s)	R/W	Default	Description
31:24	R/W	0d63	reg_nr_alp0_minerr_cpar2 : : level limit(for mierr<th0) of curve to map mierr to alp0 for chroma channel, this will be set to alp0 that we can do for flat region. 0~255.. unsigned , default = 63
23:16	R/W	0d0	reg_nr_alp0_minerr_cpar3 : : level limit(for th0<mierr<th1) of curve to map mierr to alp0 for chroma channel, this will be set to alp0 that we can do for misc region. 0~255.. unsigned , default = 0
15: 8	R/W	0d63	reg_nr_alp0_minerr_cpar4 : : level limit(for mierr>th1) of curve to map mierr to alp0 for chroma channel, this will be set to alp0 that we can do for texture region. 0~255.. unsigned , default = 63

Table 8-1994 SRSHARP0_NR_ALP0_MIN_MAX 0x3e19

Bit(s)	R/W	Default	Description
29:24	R/W	0d2	reg_nr_alp0_ymin : : normalized to 64 as '1' . unsigned , default = 2
21:16	R/W	0d63	reg_nr_alp0_ymax : : normalized to 64 as '1' . unsigned , default = 63
13: 8	R/W	0d2	reg_nr_alp0_cmin : : normalized to 64 as '1' . unsigned , default = 2
5: 0	R/W	0d63	reg_nr_alp0_cmax : : normalized to 64 as '1' . unsigned , default = 63

Table 8-1995 SRSHARP0_NR_ALP1_MIERR_CORING 0x3e1a

Bit(s)	R/W	Default	Description
16	R/W	0d0	reg_nr_alp1_maxerr_mode : : 0 max err; 1: xerr . unsigned , default = 0
13: 8	R/W	0d0	reg_nr_alp1_core_rate : : normalized 64 as "1" . unsigned , default = 0
5: 0	R/W	0d3	reg_nr_alp1_core_ofst : : normalized 64 as "1". signed , default = 3

Table 8-1996 SRSHARP0_NR_ALP1_ERR2CURV_TH_RATE 0x3e1b

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_nr_alp1_minerr_par0 : : threshold0 of curve to map mierr to alp1 for luma/chroma channel, this will be set value of flat region mierr that no need directional NR. 0~255.. unsigned , default = 0
23:16	R/W	0d24	reg_nr_alp1_minerr_par1 : : threshold1 of curve to map mierr to alp1 for luma/chroma channel,this will be set value of texture region mierr that can not do directional NR. 0~255.. unsigned , default = 24

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Bit(s)	R/W	Default	Description
15: 8	R/W	0d0	reg_nr_alp1_minerr_par5 : : rate0 (for mierr<th0) of curve to map mierr to alp1 for luma/chroma channel. the larger of the value, the deep of the slope.. unsigned , default = 0
7: 0	R/W	0d20	reg_nr_alp1_minerr_par6 : : rate1 (for mierr>th1) of curve to map mierr to alp1 for luma/chroma channel. the larger of the value, the deep of the slope. 0~255. unsigned , default = 20

Table 8-1997 SRSHARP0_NR_ALP1_ERR2CURV_LIMIT 0x3e1c

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_nr_alp1_minerr_par2 : : level limit(for mierr<th0) of curve to map mierr to alp1 for luma/chroma channel, this will be set to alp1 that we can do for flat region. 0~255.. unsigned , default = 0
23:16	R/W	0d16	reg_nr_alp1_minerr_par3 : : level limit(for th0<mierr<th1) of curve to map mierr to alp1 for luma/chroma channel, this will be set to alp1 that we can do for misc region. 0~255.. unsigned , default = 16
15: 8	R/W	0d63	reg_nr_alp1_minerr_par4 : : level limit(for mierr>th1) of curve to map mierr to alp1 for luma/chroma channel, this will be set to alp1 that we can do for texture region. 0~255.255 before. unsigned , default = 63

Table 8-1998 SRSHARP0_NR_ALP1_MIN_MAX 0x3e1d

Bit(s)	R/W	Default	Description
29:24	R/W	0d0	reg_nr_alp1_ymin : : normalized to 64 as '1' . unsigned , default = 0
21:16	R/W	0d63	reg_nr_alp1_ymax : : normalized to 64 as '1' . unsigned , default = 63
13: 8	R/W	0d0	reg_nr_alp1_cmin : : normalized to 64 as '1' . unsigned , default = 0
5: 0	R/W	0d63	reg_nr_alp1_cmax : : normalized to 64 as '1' . unsigned , default = 63

Table 8-1999 SRSHARP0_PK_ALP2_MIERR_CORING 0x3e1e

Bit(s)	R/W	Default	Description
16	R/W	0d1	reg_pk_alp2_maxerr_mode : : 0 max err; 1: xerr . unsigned , default = 1
13: 8	R/W	0d13	reg_pk_alp2_core_rate : : normalized 64 as "1" . unsigned , default = 13
5: 0	R/W	0d1	reg_pk_alp2_core_ofst : : normalized 64 as "1" . signed , default = 1

Table 8-2000 SRSHARP0_PK_ALP2_ERR2CURV_TH_RATE 0x3e1f

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_pk_alp2_minerr_par0 : : threshold0 of curve to map mierr to alp2 for luma channel, this will be set value of flat region mierr that no need peaking.. unsigned , default = 0
23:16	R/W	0d24	reg_pk_alp2_minerr_par1 : : threshold1 of curve to map mierr to alp2 for luma channel,this will be set value of texture region mierr that can not do peaking. 0~255.. unsigned , default = 24

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Bit(s)	R/W	Default	Description
15: 8	R/W	0d0	reg_pk_alp2_minerr_par5 : : rate0 (for mierr<th0) of curve to map mierr to alp2 for luma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 0
7: 0	R/W	0d20	reg_pk_alp2_minerr_par6 : : rate1 (for mierr>th1) of curve to map mierr to alp2 for luma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 20

Table 8-2001 SRSHARP0_PK_ALP2_ERR2CURV_LIMIT 0x3e20

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_pk_alp2_minerr_par2 : : level limit(for mierr<th0) of curve to map mierr to alp2 for luma channel, this will be set to alp2 that we can do for flat region. 0~255.. unsigned , default = 0
23:16	R/W	0d16	reg_pk_alp2_minerr_par3 : : level limit(for th0<mierr<th1) of curve to map mierr to alp2 for luma channel, this will be set to alp2 that we can do for misc region. 0~255.. unsigned , default = 16
15: 8	R/W	0d63	reg_pk_alp2_minerr_par4 : : level limit(for mierr>th1) of curve to map mierr to alp2 for luma channel, this will be set to alp2 that we can do for texture region. 0~255. default = 63;. unsigned , default = 255

Table 8-2002 SRSHARP0_PK_ALP2_MIN_MAX 0x3e21

Bit(s)	R/W	Default	Description
13: 8	R/W	0d0	reg_pk_alp2_min : : normalized to 64 as '1' . unsigned , default = 0
5: 0	R/W	0d63	reg_pk_alp2_max : : normalized to 64 as '1' . unsigned , default = 63

Table 8-2003 SRSHARP0_PK_FINALGAIN_HP_BP 0x3e22

Bit(s)	R/W	Default	Description
17: 16	R/W	0d0	reg_final_gain_rs : : right shift bits for the gain normalization, 0 normal to 32 as 1; 1 normal to 64 as 1; -2 normal to 8 as 1; -1 normal to 16 as 1 . signed , default = 0
15: 8	R/W	0d40	reg_hp_final_gain : : gain to highpass boost result (including directional/circle blending), normalized 32 as '1', 0~255. 1.25 * 32. unsigned , default = 40
7: 0	R/W	0d30	reg_bp_final_gain : : gain to bandpass boost result (including directional/circle blending), normalized 32 as '1', 0~255. 1.25 * 32. unsigned , default = 30

Table 8-2004 SRSHARP0_PK_OS_HORZ_CORE_GAIN 0x3e23

Bit(s)	R/W	Default	Description
31:24	R/W	0d8	reg_pk_os_hsidecore : : side coring (not to current pixel) to adaptive overshoot margin in horizontal direction. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 8
23:16	R/W	0d20	reg_pk_os_hsidegain : : side gain (not to current pixel) to adaptive overshoot margin in horizontal direction. normalized to 32 as '1'. 0~255;. unsigned , default = 20

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Bit(s)	R/W	Default	Description
15: 8	R/W	0d2	reg_pk_os_hmidcore : : midd coring (to current pixel) to adaptive overshoot margin in horizontal direction. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 2
7: 0	R/W	0d20	reg_pk_os_hmidgain : : midd gain (to current pixel) to adaptive overshoot margin in horizontal direction. normalized to 32 as '1'. 0~255;. unsigned , default = 20

Table 8-2005 SRSHARP0_PK_OS_VERT_CORE_GAIN 0x3e24

Bit(s)	R/W	Default	Description
31:24	R/W	0d8	reg_pk_os_vsidecore : : side coring (not to current pixel) to adaptive overshoot margin in vertical direction. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 8
23:16	R/W	0d20	reg_pk_os_vsidegain : : side gain (not to current pixel) to adaptive overshoot margin in vertical direction. normalized to 32 as '1'. 0~255;. unsigned , default = 20
15: 8	R/W	0d2	reg_pk_os_vmidcore : : midd coring (to current pixel) to adaptive overshoot margin in vertical direction. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 2
7: 0	R/W	0d20	reg_pk_os_vmidgain : : midd gain (to current pixel) to adaptive overshoot margin in vertical direction. normalized to 32 as '1'. 0~255;. unsigned , default = 20

Table 8-2006 SRSHARP0_PK_OS_ADPT_MISC 0x3e25

Bit(s)	R/W	Default	Description
31:24	R/W	0d40	reg_pk_os_minerr_core : : coring to minerr for adaptive overshoot margin. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 40
23:16	R/W	0d6	reg_pk_os_minerr_gain : : gain to minerr based adaptive overshoot margin. normalized to 64 as '1'. 0~255;. unsigned , default = 6
15: 8	R/W	0d200	reg_pk_os_adpt_max : : maximum limit adaptive overshoot margin (4x). 0~255; . unsigned , default = 200
7: 0	R/W	0d20	reg_pk_os_adpt_min : : minimun limit adaptive overshoot margin (1x). 0~255; . unsigned , default = 20

Table 8-2007 SRSHARP0_PK_OS_STATIC 0x3e26

Bit(s)	R/W	Default	Description
29:28	R/W	0d2	reg_pk_osh_mode : : 0~3: (2x+1) window in H direction . unsigned , default = 2
25:24	R/W	0d2	reg_pk_osv_mode : : 0~3: (2x+1) window in V direction . unsigned , default = 2
21:12	R/W	0d200	reg_pk_os_down : : static negative overshoot margin. 0~1023; . unsigned , default = 200
9: 0	R/W	0d200	reg_pk_os_up : : static positive overshoot margin. 0~1023; . unsigned , default = 200

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Table 8-2008 SRSHARP0_PK_NR_ENABLE 0x3e27

Bit(s)	R/W	Default	Description
3: 2	R/W	0d0	reg_3d_mode : , 0: no 3D; 1: L/R; 2: T/B; 3: horizontal interleaved, dft = 0 //. unsigned , default = 0
1	R/W	0d1	reg_pk_en : . unsigned , default = 1
0	R/W	0d1	reg_nr_en : . unsigned , default = 1

Table 8-2009 SRSHARP0_PK_DRT_SAD_MISC 0x3e28

Bit(s)	R/W	Default	Description
31:24	R/W	0d24	reg_pk_sad_ver_gain : : gain to sad[4], 16 normalized to "1"; . unsigned , default = 24
23:16	R/W	0d24	reg_pk_sad_hor_gain : : gain to sad[0], 16 normalized to "1"; . unsigned , default = 24
10: 9	R/W	0d0	reg_pk_bias_diag : : bias towards diag . unsigned , default = 0
4: 0	R/W	0d24	reg_pk_drt_force : : force direction of drt peaking filter, h2b: 0:hp drt force, 1: bp drt force; 2: bp+hp drt force, 3: no force;. unsigned , default = 24

Table 8-2010 SRSHARP0_NR_TI_DNLP_BLEND 0x3e29

Bit(s)	R/W	Default	Description
10: 8	R/W	0d4	reg_dnlp_input_mode : : dnlp input options. 0: org_y; 1: gau_y; 2: gauadp_y; 3: edgadplpf_y; 4: nr_y;5: lti_y; 6: pk_y (before os);7: pk_y (after os). unsigned , default = 4
3: 2	R/W	0d1	reg_nr_cti_blend_mode : : blend mode of nr and lti result: 0: nr; 1:cti; 2: (nr+cti)/2; 3:cti + dlt_nr . unsigned , default = 1
1: 0	R/W	0d1	reg_nr_lti_blend_mode : : blend mode of nr and lti result: 0: nr; 1:lti; 2: (nr+lti)/2; 3:lti + dlt_nr . unsigned , default = 1

Table 8-2011 SRSHARP0_TI_DIR_CORE_ALPHA 0x3e2a

Bit(s)	R/W	Default	Description
29:24	R/W	0d10	reg_adp_lti_dir_alp_core_ofst : : ofst to min_err, alpha = (min_err - (max_err-min_err)*rate + ofst)/max_err*64; dft=10. unsigned , default = 10
19:16	R/W	0d0	reg_adp_lti_dir_alp_core_rate : : ofset to min_err, alpha = (min_err - (max_err-min_err)*rate + ofst)/max_err*64; dft=0/32. unsigned , default = 0
13: 8	R/W	0d0	reg_adp_lti_dir_alpmin : : min value of alpha, alpha = (min_err+x +ofst)/max_err*64; dft=10 . unsigned , default = 0
5: 0	R/W	0d63	reg_adp_lti_dir_alpmax : : max value of alpha, alpha = (min_err+x +ofst)/max_err*64; dft=63 . unsigned , default = 63

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Table 8-2012 SRSHARP0_CTI_DIR_ALPHA 0x3e2b

Bit(s)	R/W	Default	Description
29:24	R/W	0d5	reg_adp_cti_dir_alp_core_ofst : : ofst to min_err, alpha = (min_err - (max_err-min_err)*rate + ofst)/max_err*64; dft=10. unsigned , default = 5
19:16	R/W	0d0	reg_adp_cti_dir_alp_core_rate : : ofset to min_err, alpha = (min_err - (max_err-min_err)*rate + ofst)/max_err*64; dft=0/32. unsigned , default = 0
13: 8	R/W	0d0	reg_adp_cti_dir_alpmin : : min value of alpha, alpha = (min_err +x+ofst)/max_ err*64; dft=10 . unsigned , default = 0
5: 0	R/W	0d63	reg_adp_cti_dir_alpmax : : max value of alpha, alpha = (min_err +x+ofst)/max_ err*64; dft=63 . unsigned , default = 63

Table 8-2013 SRSHARP0_LTI_CTI_DF_GAIN 0x3e2c

Bit(s)	R/W	Default	Description
29:24	R/W	0d16	reg_adp_lti_hdf_gain : : 8 normalized to "1"; default = 16 . unsigned , default = 16
21:16	R/W	0d12	reg_adp_lti_vdf_gain : : 8 normalized to "1"; default = 12 . unsigned , default = 12
13: 8	R/W	0d16	reg_adp_cti_hdf_gain : : 8 normalized to "1"; default = 16 . unsigned , default = 16
5: 0	R/W	0d12	reg_adp_cti_vdf_gain : : 8 normalized to "1"; default = 12 . unsigned , default = 12

Table 8-2014 SRSHARP0_LTI_CTI_DIR_AC_DBG 0x3e2d

Bit(s)	R/W	Default	Description
30	R/W	0d1	reg_adp_lti_dir_lpf : : 0: no lpf; 1: [1 2 2 2 1]/8 lpf . unsigned , default = 1
28	R/W	0d0	reg_adp_lti_dir_difmode : : 0: y_dif; 1: y_dif + (u_dif+v_dif)/2; . unsigned , default = 0
26	R/W	0d1	reg_adp_cti_dir_lpf : : 0: no lpf; 1: [1 2 2 2 1]/8 lpf dft=1 . unsigned , default = 1
25:24	R/W	0d0	reg_adp_cti_dir_difmode : : 0: (u_dif+v_dif); 1: y_dif/2 + (u_dif+v_dif)*3/4; 2: y_ dif + (u_dif+v_dif)/2; 3: y_dif*2 (not recomended). unsigned , default = 0
23:22	R/W	0d3	reg_adp_hvlti_dcblend_mode : : 0: hlti_dc; 1:vlti_dc; 2: avg 3; blend on alpha . unsigned , default = 3
21:20	R/W	0d3	reg_adp_hvcti_dcblend_mode : : 0: hcti_dc; 1:vcti_dc; 2: avg 3; blend on alpha . unsigned , default = 3
19:18	R/W	0d3	reg_adp_hvlti_acblend_mode : : hlti_ac; 1:vlti_ac; 2: add 3;:adaptive to alpha . unsigned , default = 3
17:16	R/W	0d3	reg_adp_hvcti_acblend_mode : : hcti_ac; 1:vcti_ac; 2: add 3;: adaptive to alpha . unsigned , default = 3
14:12	R/W	0d0	reg_adp_hlti_debug : , for hlti debug, default = 0 . unsigned , default = 0
10: 8	R/W	0d0	reg_adp_vlti_debug : , for vlti debug, default = 0 . unsigned , default = 0
6: 4	R/W	0d0	reg_adp_hcti_debug : , for hcti debug, default = 0 . unsigned , default = 0
2: 0	R/W	0d0	reg_adp_vcti_debug : , for vcti debug, default = 0 . unsigned , default = 0

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Table 8-2015 SRSHARP0_HCTI_FLT_CLP_DC 0x3e2e

Bit(s)	R/W	Default	Description
28	R/W	0d1	reg_adp_hcti_en : , 0: no cti, 1: new cti, default = 1 . unsigned , default = 1
27:26	R/W	0d3	reg_adp_hcti_vdn_flt : , 0: no lpf; 1:[0,2,4,2,0], 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
25:24	R/W	0d2	reg_adp_hcti_hdn_flt : , 0: no lpf; 1:[0, 0, 0, 4, 8, 4, 0, 0, 0], 2:[0, 0, 2, 4, 4, 4, 2, 0, 0], 3: [1, 2, 2, 2, 2, 2, 2, 2, 1], default = 2. unsigned , default = 2
23:22	R/W	0d3	reg_adp_hcti_ddn_flt : , 0: no lpf; 1:[0,2,4,2,0], 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
21:20	R/W	0d2	reg_adp_hcti_lpf0_flt : , 0:no filter; 1:sigma=0.75, 2: sigma = 1.0, 3: sigma = 1.5, default = 2 . unsigned , default = 2
19:18	R/W	0d2	reg_adp_hcti_lpf1_flt : , 0:no filter; 1:sigma= 2.0, 2: sigma = 3.0, 3: sigma = 4.0, default = 2 . unsigned , default = 2
17:16	R/W	0d2	reg_adp_hcti_lpf2_flt : , 0:no filter; 1:sigma=5.0, 2: sigma = 9.0, 3: sigma = 13.0, default = 2 . unsigned , default = 2
15:12	R/W	0d7	reg_adp_hcti_hard_clp_win : , window size, 0~8, default = 7 . unsigned , default = 7
11: 8	R/W	0d3	reg_adp_hcti_hard_win_min : , window size, 0~8, default = 3 . unsigned , default = 3
4	R/W	0d1	reg_adp_hcti_clp_mode : , 0: hard clip, 1: adaptive clip, default = 1 . unsigned , default = 1
2: 0	R/W	0d0	reg_adp_hcti_dc_mode : , 0:dn, 1:lpf0, 2:lpf1, 3:lpf2, 4: lpf3: 5: vdn result; 6/7: org, default = 0 . unsigned , default = 0

Table 8-2016 SRSHARP0_HCTI_BST_GAIN 0x3e2f

Bit(s)	R/W	Default	Description
31:24	R/W	0d80	reg_adp_hcti_bst_gain0 : : gain of the bandpass 0 (lpf1-lpf2)- LBP, default = 80. unsigned , default = 80
23:16	R/W	0d96	reg_adp_hcti_bst_gain1 : : gain of the bandpass 1 (lpf0-lpf1)- BP, default = 96. unsigned , default = 96
15: 8	R/W	0d64	reg_adp_hcti_bst_gain2 : : gain of the bandpass 2 (hdn-lpf0)- HP, default = 64. unsigned , default = 64
7: 0	R/W	0d16	reg_adp_hcti_bst_gain3 : : gain of the unsharp band (yuvin-hdn) - US, default = 16. unsigned , default = 16

Table 8-2017 SRSHARP0_HCTI_BST_CORE 0x3e30

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_adp_hcti_bst_core0 : : core of the bandpass 0 (lpf1-lpf2)- LBP, default = 0 . unsigned , default = 0
23:16	R/W	0d0	reg_adp_hcti_bst_core1 : : core of the bandpass 1 (lpf0-lpf1)- BP, default = 0 . unsigned , default = 0
15: 8	R/W	0d0	reg_adp_hcti_bst_core2 : : core of the bandpass 2 (hdn-lpf0)- HP, default = 0 . unsigned , default = 0
7: 0	R/W	0d0	reg_adp_hcti_bst_core3 : : core of the unsharp band (yuvin-hdn) - US, default = 0. unsigned , default = 0

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Table 8-2018 SRSHARP0_HCTI_CON_2_GAIN_0 0x3e31

Bit(s)	R/W	Default	Description
31:29	R/W	0d2	reg_adp_hcti_con_mode : : con mode 0:[0, 0,-1, 1, 0, 0, 0]+[0, 0, 0, 1,-1, 0, 0], 1: [0, 0,-1, 0, 1, 0, 0], 2: [0,-1, 0, 0, 0, 1, 0], 3:[-1, 0, 0, 0, 0, 0, 1], 4: .... default = 2. unsigned , default = 2
28:26	R/W	0d3	reg_adp_hcti_dx_mode : : dx mode 0: [-1 1 0]; 1~7: [-1 (2x+1)"0" 1], default = 3 . unsigned , default = 3
25:24	R/W	0d1	reg_adp_hcti_con_lpf : : lpf mode of the con: 0: [1 2 1]/4; 1:[1 2 2 2 1]/8, default = 1 . unsigned , default = 1
23:16	R/W	0d25	reg_adp_hcti_con_2_gain0 : , default = 25. unsigned , default = 25
15: 8	R/W	0d60	reg_adp_hcti_con_2_gain1 : , default = 60. unsigned , default = 60
7: 0	R/W	0d0	reg_adp_hcti_con_2_gain2 : 0;, default = 0 . unsigned , default = 0

Table 8-2019 SRSHARP0_HCTI_CON_2_GAIN_1 0x3e32

Bit(s)	R/W	Default	Description
31:24	R/W	0d96	reg_adp_hcti_con_2_gain3 : 96;, default = 96 . unsigned , default = 96
23:16	R/W	0d5	reg_adp_hcti_con_2_gain4 : 5;, default = 5 . unsigned , default = 5
15: 8	R/W	0d80	reg_adp_hcti_con_2_gain5 : 80;, default = 80 . unsigned , default = 80
7: 0	R/W	0d20	reg_adp_hcti_con_2_gain6 : 20;, default = 20 . unsigned , default = 20

Table 8-2020 SRSHARP0_HCTI_OS_MARGIN 0x3e33

Bit(s)	R/W	Default	Description
7: 0	R/W	0d0	reg_adp_hcti_os_margin : : margin for hcti overshoot, default = 0 . unsigned , default = 0

Table 8-2021 SRSHARP0_HLTI_FLT_CLP_DC 0x3e34

Bit(s)	R/W	Default	Description
28	R/W	0d1	reg_adp_hlti_en : , 0: no cti, 1: new cti, default = 1 . unsigned , default = 1
27:26	R/W	0d2	reg_adp_hlti_vdn_flt : , 0: no lpf; 1:[0,2,4,2,0], 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 2 . unsigned , default = 2
25:24	R/W	0d2	reg_adp_hlti_hdn_flt : , 0: no lpf; 1:[0, 0, 0, 4, 8, 4, 0, 0, 0], 2:[0, 0, 2, 4, 4, 4, 2, 0, 0], 3: [1, 2, 2, 2, 2, 2, 2, 2, 1], default = 2. unsigned , default = 2
23:22	R/W	0d2	reg_adp_hlti_ddn_flt : , 0: no lpf; 1:[0,2,4,2,0], 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 2 . unsigned , default = 2
21:20	R/W	0d2	reg_adp_hlti_lpf0_flt : , 0:no filter; 1:sigma=0.75, 2: sigma = 1.0, 3: sigma = 1.5, default = 2 . unsigned , default = 2
19:18	R/W	0d2	reg_adp_hlti_lpf1_flt : , 0:no filter; 1:sigma= 2.0, 2: sigma = 3.0, 3: sigma = 4.0, default = 2 . unsigned , default = 2
17:16	R/W	0d2	reg_adp_hlti_lpf2_flt : , 0:no filter; 1:sigma=5.0, 2: sigma = 9.0, 3: sigma = 13.0, default = 2 . unsigned , default = 2

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Bit(s)	R/W	Default	Description
15:12	R/W	0d2	reg_adp_hlti_hard_clp_win : , window size, 0~8, default = 2 . unsigned , default = 2
11: 8	R/W	0d1	reg_adp_hlti_hard_win_min : , window size, 0~8, default = 1 . unsigned , default = 1
4	R/W	0d0	reg_adp_hlti_clp_mode : , 0: hard clip, 1: adaptive clip, default = 0 . unsigned , default = 0
2: 0	R/W	0d4	reg_adp_hlti_dc_mode : , 0:dn, 1:lpf0, 2:lpf1, 3:lpf2, 4: lpf3: 5: vdn result; 6/7:org, default = 4 . unsigned , default = 4

Table 8-2022 SRSHARP0_HLTI_BST_GAIN 0x3e35

Bit(s)	R/W	Default	Description
31:24	R/W	0d40	reg_adp_hlti_bst_gain0 : : gain of the bandpass 0 (lpf1-lpf2)- LBP, default = 40 . unsigned , default = 40
23:16	R/W	0d48	reg_adp_hlti_bst_gain1 : : gain of the bandpass 1 (lpf0-lpf1)- BP, default = 48 . unsigned , default = 48
15: 8	R/W	0d32	reg_adp_hlti_bst_gain2 : : gain of the bandpass 2 (hdn-lpf0)- HP, default = 32 . unsigned , default = 32
7: 0	R/W	0d16	reg_adp_hlti_bst_gain3 : : gain of the unsharp band (yuvin-hdn) - US, default = 16 . unsigned , default = 16

Table 8-2023 SRSHARP0_HLTI_BST_CORE 0x3e36

Bit(s)	R/W	Default	Description
31:24	R/W	0d5	reg_adp_hlti_bst_core0 : : core of the bandpass 0 (lpf1-lpf2)- LBP, default = 5 . unsigned , default = 5
23:16	R/W	0d5	reg_adp_hlti_bst_core1 : : core of the bandpass 1 (lpf0-lpf1)- BP, default = 5 . unsigned , default = 5
15: 8	R/W	0d5	reg_adp_hlti_bst_core2 : : core of the bandpass 2 (hdn-lpf0)- HP, default = 5 . unsigned , default = 5
7: 0	R/W	0d3	reg_adp_hlti_bst_core3 : : core of the unsharp band (yuvin-hdn) - US, default = 3. unsigned , default = 3

Table 8-2024 SRSHARP0_HLTI_CON_2_GAIN_0 0x3e37

Bit(s)	R/W	Default	Description
31:29	R/W	0d2	reg_adp_hlti_con_mode : : con mode 0:[0, 0,-1, 1, 0, 0, 0]+[0, 0, 0, 1,-1, 0, 0], 1: [0, 0,-1, 0, 1, 0, 0], 2: [0,-1, 0, 0, 0, 1, 0], 3:[-1, 0, 0, 0, 0, 0, 1], 4: ....., default = 2. unsigned , default = 2
28:26	R/W	0d3	reg_adp_hlti_dx_mode : : dx mode 0: [-1 1 0]; 1~7: [-1 (2x+1)"0" 1], default = 3 . unsigned , default = 3
25:24	R/W	0d1	reg_adp_hlti_con_lpf : : lpf mode of the con: 0: [1 2 1]/4; 1:[1 2 2 2 1]/8, default = 1 . unsigned , default = 1
23:16	R/W	0d25	reg_adp_hlti_con_2_gain0 : 25;, default = 25 . unsigned , default = 25
15: 8	R/W	0d60	reg_adp_hlti_con_2_gain1 : 60;, default = 60 . unsigned , default = 60
7: 0	R/W	0d90	reg_adp_hlti_con_2_gain2 : 0;, default = 90 . unsigned , default = 90

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Table 8-2025 SRSHARP0_HLTI_CON_2_GAIN_1 0x3e38

Bit(s)	R/W	Default	Description
31:24	R/W	0d96	reg_adp_hlti_con_2_gain3 : 96;, default = 96 . unsigned , default = 96
23:16	R/W	0d95	reg_adp_hlti_con_2_gain4 : 5;, default = 95 . unsigned , default = 95
15: 8	R/W	0d80	reg_adp_hlti_con_2_gain5 : 80;, default = 80 . unsigned , default = 80
7: 0	R/W	0d20	reg_adp_hlti_con_2_gain6 : 20;, default = 20 . unsigned , default = 20

Table 8-2026 SRSHARP0_HLTI_OS_MARGIN 0x3e39

Bit(s)	R/W	Default	Description
7: 0	R/W	0d0	reg_adp_hlti_os_margin : : margin for hlti overshoot, default = 0 . unsigned , default = 0

Table 8-2027 SRSHARP0_VLTI_FLT_CON_CLP 0x3e3a

Bit(s)	R/W	Default	Description
14	R/W	0d1	reg_adp_vlti_en : : enable bit of vlti, default = 1 . unsigned , default = 1
13:12	R/W	0d3	reg_adp_vlti_hxn_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
11:10	R/W	0d3	reg_adp_vlti_dxn_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
9: 8	R/W	0d3	reg_adp_vlti_han_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
7: 6	R/W	0d3	reg_adp_vlti_dan_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
5: 4	R/W	0d2	reg_adp_vlti_dx_mode : : 0:[-1 1] 1:[-1 0 -1]; 2/3: [-1 0 0 0 -1], default = 2 . unsigned , default = 2
2	R/W	0d1	reg_adp_vlti_con_lpf : : lpf mode of the con: 0: [1 2 1]/4; 1:[1 2 2 2 1]/8, default = 1 . unsigned , default = 1
0	R/W	0d1	reg_adp_vlti_hard_clp_win : : window size; 0: 1x3 window; 1: 1x5 window, default = 1. unsigned , default = 1

Table 8-2028 SRSHARP0_VLTI_BST_GAIN 0x3e3b

Bit(s)	R/W	Default	Description
23:16	R/W	0d32	reg_adp_vlti_bst_gain0 : : gain to boost filter [-1 2 -1];, default = 32. unsigned , default = 32
15: 8	R/W	0d32	reg_adp_vlti_bst_gain1 : : gain to boost filter [-1 0 2 0 -1];, default = 32 . unsigned , default = 32
7: 0	R/W	0d32	reg_adp_vlti_bst_gain2 : : gain to boost filter usf, default = 32 . unsigned , default = 32

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Table 8-2029 SRSHARP0_VLTI_BST_CORE 0x3e3c

Bit(s)	R/W	Default	Description
23:16	R/W	0d5	reg_adp_vlti_bst_core0 : : coring to boost filter [-1 2 -1];, default = 5 . unsigned , default = 5
15: 8	R/W	0d5	reg_adp_vlti_bst_core1 : : coring to boost filter [-1 0 2 0 -1];, default = 5 . unsigned , default = 5
7: 0	R/W	0d3	reg_adp_vlti_bst_core2 : : coring to boost filter usf, default = 3 . unsigned , default = 3

Table 8-2030 SRSHARP0_VLTI_CON_2_GAIN_0 0x3e3d

Bit(s)	R/W	Default	Description
31:24	R/W	0d25	reg_adp_vlti_con_2_gain0 : 25;, default = 25 . unsigned , default = 25
23:16	R/W	0d69	reg_adp_vlti_con_2_gain1 : 60;, default = 69 . unsigned , default = 60
15: 8	R/W	0d90	reg_adp_vlti_con_2_gain2 : 0;, default = 90 . unsigned , default = 90
7: 0	R/W	0d96	reg_adp_vlti_con_2_gain3 : 96;, default = 96 . unsigned , default = 96

Table 8-2031 SRSHARP0_VLTI_CON_2_GAIN_1 0x3e3e

Bit(s)	R/W	Default	Description
31:24	R/W	0d95	reg_adp_vlti_con_2_gain4 : 5;, default = 95 . unsigned , default = 95
23:16	R/W	0d80	reg_adp_vlti_con_2_gain5 : 80;, default = 80 . unsigned , default = 80
15: 8	R/W	0d20	reg_adp_vlti_con_2_gain6 : 20;, default = 20 . unsigned , default = 20
7: 0	R/W	0d0	reg_adp_vlti_os_margin : : margin for vlti overshoot, default = 0 . unsigned , default = 0

Table 8-2032 SRSHARP0_VCTI_FLT_CON_CLP 0x3e3f

Bit(s)	R/W	Default	Description
14	R/W	0d1	reg_adp_vcti_en : : enable bit of vlti, default = 1 . unsigned , default = 1
13:12	R/W	0d3	reg_adp_vcti_hxn_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
11:10	R/W	0d3	reg_adp_vcti_dxn_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
9: 8	R/W	0d3	reg_adp_vcti_han_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
7: 6	R/W	0d3	reg_adp_vcti_dan_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
5: 4	R/W	0d2	reg_adp_vcti_dx_mode : : 0:[-1 1] 1:[-1 0 -1]; 2/3: [-1 0 0 0 -1], default = 2 . unsigned , default = 2
2	R/W	0d1	reg_adp_vcti_con_lpf : : lpf mode of the con: 0: [1 2 1]/4; 1:[1 2 2 2 1]/8, default = 1 . unsigned , default = 1
0	R/W	0d1	reg_adp_vcti_hard_clp_win : : window size; 0: 1x3 window; 1: 1x5 window, default = 1. unsigned , default = 1

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Table 8-2033 SRSHARP0_VCTI_BST_GAIN 0x3e40

Bit(s)	R/W	Default	Description
23:16	R/W	0d0	reg_adp_vcti_bst_gain0 : : gain to boost filter [-1 2 -1];, default = 0 . unsigned , default = 0
15: 8	R/W	0d0	reg_adp_vcti_bst_gain1 : : gain to boost filter [-1 0 2 0 -1];, default = 0 . unsigned , default = 0
7: 0	R/W	0d0	reg_adp_vcti_bst_gain2 : : gain to boost filter usf, default = 0 . unsigned , default = 0

Table 8-2034 SRSHARP0_VCTI_BST_CORE 0x3e41

Bit(s)	R/W	Default	Description
23:16	R/W	0d0	reg_adp_vcti_bst_core0 : : coring to boost filter [-1 2 -1];, default = 0 . unsigned , default = 0
15: 8	R/W	0d0	reg_adp_vcti_bst_core1 : : coring to boost filter [-1 0 2 0 -1];, default = 0 . unsigned , default = 0
7: 0	R/W	0d0	reg_adp_vcti_bst_core2 : : coring to boost filter usf, default = 0 . unsigned , default = 0

Table 8-2035 SRSHARP0_VCTI_CON_2_GAIN_0 0x3e42

Bit(s)	R/W	Default	Description
31:24	R/W	0d25	reg_adp_vcti_con_2_gain0 : 25;, default = 25 . unsigned , default = 25
23:16	R/W	0d60	reg_adp_vcti_con_2_gain1 : 60;, default = 60 . unsigned , default = 60
15: 8	R/W	0d90	reg_adp_vcti_con_2_gain2 : 0;, default = 90 . unsigned , default = 90
7: 0	R/W	0d96	reg_adp_vcti_con_2_gain3 : 96;, default = 96 . unsigned , default = 96

Table 8-2036 SRSHARP0_VCTI_CON_2_GAIN_1 0x3e43

Bit(s)	R/W	Default	Description
31:24	R/W	0d95	reg_adp_vcti_con_2_gain4 : 5;, default = 95 . unsigned , default = 95
23:16	R/W	0d80	reg_adp_vcti_con_2_gain5 : 80;, default = 80 . unsigned , default = 80
15: 8	R/W	0d20	reg_adp_vcti_con_2_gain6 : 20;, default = 20 . unsigned , default = 20
7: 0	R/W	0d0	reg_adp_vcti_os_margin : : margin for vcti overshoot, default = 0 . unsigned , default = 0

Table 8-2037 SRSHARP0_SHARP_3DLIMIT 0x3e44

Bit(s)	R/W	Default	Description
28:16	R/W	0d0	reg_3d_mid_width : ,width of left part of 3d input, dft = half size of input width default = 0 . unsigned , default = 960
12: 0	R/W	0d0	reg_3d_mid_height : ,height of left part of 3d input, dft = half size of input height default = 0 . unsigned , default = 540

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Table 8-2038 SRSHARP0_DNLP_EN 0x3e45

Bit(s)	R/W	Default	Description
15:8	R/W	0d0	reg_dnlp_hblank: . unsigned , default = 8
0	R/W	0d1	reg_dnlp_en : . unsigned , default = 1

Table 8-2039 SRSHARP0_DEMO_CRTL 0x3e56

Bit(s)	R/W	Default	Description
18:17	R/W	0d2	demo_disp_position : . unsigned , default = 2
16	R/W	0d0	demo_hsvsharp_enable : . unsigned , default = 0
12: 0	R/W	0d360	demo_left_top_screen_width : : . unsigned , default = 360

Table 8-2040 SRSHARP0_SHARP_SR2_CTRL 0x3e57

Bit(s)	R/W	Default	Description
24	R/W	0	reg_sr2_bic_pknr_bypass : bypass peaking/TI/Cubic
23:22	R/W		reserved
21:16	R/W	24	sr2_pk_la_err_dis_rate, low angle and high angle error should not be no less than nearby_error* rate/64
15: 8	R/W	16	sr2_pk_sad_diag_gain, gain to sad[2] and sad[6], 16 normalized to 1
7	R/W	0	sr2_vert_outphs, vertical output pixel phase, 0: 0 phase; 1: 1/2 phase
6	R/W	0	sr2_horz_outphs, horizontal output pixel phase, 0: 0 phase; 1: 1/2 phase
5	R/W	0	sr2_vert_ratio , vertical scale ratio, 0-> 1:1; 1-> 1:2
4	R/W	0	sr2_horz_ratio , horizontal scale ratio, 0-> 1:1; 1-> 1:2
3	R/W	1	sr2_ bic_norm , normalization of bicubical: 0: 128; 1: 64
2	R/W	0	sr2_ enable , 1 to enable super scaler
1	R/W	0	sr2_ sharp_prc_lr_hbic,
0	R/W	0	sr2_ sharp_prc_lr, 1: LTI/CTI/NR/Peaking processing using LR grid. 0: on HR grid; 1:on LR grid, horizontally no upscale, but using simple bic.

Table 8-2041 SRSHARP0_SHARP_SR2_YBIC_HCOEF0 0x3e58

Bit(s)	R/W	Default	Description
31:24	R/W	0	sr2_y_bic_hcoeff03, signed
23:16	R/W	0	sr2_y_bic_hcoeff02, signed
15: 8	R/W	64	sr2_y_bic_hcoeff01, signed
7: 0	R/W	0	sr2_y_bic_hcoeff00, signed

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Table 8-2042 SRSHARP0_SHARP_SR2_YBIC_HCOEF1 0x3e59

Bit(s)	R/W	Default	Description
31:24	R/W	-4	sr2_y_bic_hcoeff13 , signed
23:16	R/W	36	sr2_y_bic_hcoeff12 , signed
15: 8	R/W	36	sr2_y_bic_hcoeff11 , signed
7: 0	R/W	-4	sr2_y_bic_hcoeff10 , signed

Table 8-2043 SRSHARP0_SHARP_SR2_CBIC_HCOEF0 0x3e5a

Bit(s)	R/W	Default	Description
31:24	R/W	0	sr2_c_bic_hcoeff03 , signed
23:16	R/W	21	sr2_c_bic_hcoeff02 , signed
15: 8	R/W	22	sr2_c_bic_hcoeff01 , signed
7: 0	R/W	21	sr2_c_bic_hcoeff00 , signed

Table 8-2044 SRSHARP0_SHARP_SR2_CBIC_HCOEF1 0x3e5b

Bit(s)	R/W	Default	Description
31:24	R/W	-4	sr2_c_bic_hcoeff13 , signed
23:16	R/W	36	sr2_c_bic_hcoeff12 , signed
15: 8	R/W	36	sr2_c_bic_hcoeff11 , signed
7: 0	R/W	-4	sr2_c_bic_hcoeff10 , signed

Table 8-2045 SHARP_SR2_YBIC_VCOEF0 0x3e5c

Bit(s)	R/W	Default	Description
31:24	R/W	0	sr2_y_bic_vcoeff03 , signed
23:16	R/W	0	sr2_y_bic_vcoeff02 , signed
15: 8	R/W	64	sr2_y_bic_vcoeff01 , signed
7: 0	R/W	0	sr2_y_bic_vcoeff00 , signed

Table 8-2046 SRSHARP0_SHARP_SR2_YBIC_VCOEF1 0x3e5d

Bit(s)	R/W	Default	Description
31:24	R/W	-4	sr2_y_bic_vcoeff13 , signed
23:16	R/W	36	sr2_y_bic_vcoeff12 , signed
15: 8	R/W	36	sr2_y_bic_vcoeff11 , signed
7: 0	R/W	-4	sr2_y_bic_vcoeff10 , signed

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Table 8-2047 SRSHARP0_SHARP_SR2_CBIC_VCOEF0 0x3e5e

Bit(s)	R/W	Default	Description
31:24	R/W	0	sr2_c_bic_vcoeff03 , signed
23:16	R/W	21	sr2_c_bic_vcoeff02 , signed
15: 8	R/W	22	sr2_c_bic_vcoeff01 , signed
7: 0	R/W	21	sr2_c_bic_vcoeff00 , signed

Table 8-2048 SRSHARP0_SHARP_SR2_CBIC_VCOEF1 0x3e5f

Bit(s)	R/W	Default	Description
31:24	R/W	-4	sr2_c_bic_vcoeff13 , signed
23:16	R/W	36	sr2_c_bic_vcoeff12 , signed
15: 8	R/W	36	sr2_c_bic_vcoeff11 , signed
7: 0	R/W	-4	sr2_c_bic_vcoeff10 , signed

Table 8-2049 SRSHARP0_SHARP_SR2_MISC 0x3e60

Bit(s)	R/W	Default	Description
31:2	R/W		reserved
1	R/W	0	sr2_cmpmux_bef , 0 : no swap for YUV/RGB; 1: swap for YUV/RGB, YUV/RGB->UVY/GBR
0	R/W	0	sr2_cmpmux_aft , 0 : no swap for YUV/RGB; 1: swap for YUV/RGB, UVY/GBR->YUV/RGB

Table 8-2050 SRSHARP0_SR3_SAD_CTRL 0x3e61

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29:24	R/W	0d6	reg_sr3_pk_sad_core_rate : rate of coring.
23:22	R/W		reserved
21:16	R/W	0d6	reg_sr3_lti_sad_core_rate : rate of coring.
15:14	R/W		reserved
13:8	R/W	0d6	reg_sr3_cti_sad_core_rate : rate of coring.
7	R/W	0d1	reg_sr3_lti_hsad_mode: mode for hsad of lti calculation, 0:block based; 1: other sharp
6	R/W	0d1	reg_sr3_cti_hsad_mode: mode for hsad of cti calculation, 0:block based; 1: other sharp
5	R/W	0d1	reg_sr3_lti_dsad_mode: mode for dsad of lti calculation, 0:block based; 1: other sharp
4	R/W	0d1	reg_sr3_cti_dsad_mode: mode for dsad of cti calculation, 0:block based; 1: other sharp
3	R/W	0d1	reg_sr3_lti_vsad_mode: mode for vsad of lti calculation, 0:block based; 1: other sharp

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Bit(s)	R/W	Default	Description
2	R/W	0d1	reg_sr3_cti_vsad_mode: mode for vsad of cti calculation, 0:block based; 1: other sharp
1	R/W	0d1	reg_sr3_lti_hsad_hlpf: hlpf for hsad of lti calculation, 0:no hlpf; 1: with [1 2 1] hlpf.
0	R/W	0d1	reg_sr3_cti_hsad_hlpf: hlpf for hsad of cti calculation, 0:no hlpf; 1: with [1 2 1] hlpf.

Table 8-2051 SRSHARP0_SR3_PK_CTRL0 0x3e62

Bit(s)	R/W	Default	Description
31:12	R/W		reserved
11	R/W	0d1	reg_sr3_pk_sad_mode: mode for sad of peaking calculation, 0: block based; 1: other sharp.
10	R/W	0d1	reg_sr3_pk_hsad_hlpf: hlpf for hsad for peaking calculation,0:no hlpf; 1: with [1 2 2 2 1] hlpf.
9	R/W	0d1	reg_sr3_pk_vsad_hlpf: hlpf for vsad for peaking calculation,0:no hlpf; 1: with [1 2 2 2 1] hlpf.
8	R/W	0d1	reg_sr3_pk_dsad_hlpf: hlpf for dsad for peaking calculation,0:no hlpf; 1: with [1 2 2 2 1] hlpf.
7:6	R/W	0d3	reg_sr3_pk_hpdrt_mode: mode for HPdrt filter
5:4	R/W	0d3	reg_sr3_pk_bpdrt_mode: mode for BPdrt filter
3:2	R/W	0d3	reg_pk_drtbld_range: range of the min2 and min direction distance
1	R/W		reserved
0	R/W	0d0	reg_sr3_pk_ti_blend_mode: blend mode of the TI and PK result

Table 8-2052 SRSHARP0_SR3_PK_CTRL1 0x3e63

Bit(s)	R/W	Default	Description
31	R/W		reserved
30:28	R/W	0d1	reg_sr3_pk_Hp_hvcon_replace8_maxsad: replace HP hvcon by maxsad
26:24	R/W	0d1	reg_sr3_pk_bp_hvcon_replace8_maxsad: replace BP hvcon by maxsad
23:16	R/W	0d32	reg_sr3_pk_hp_hvcon_replace8lv_gain: gain to local variant before calculating the hv gain for peaking.
15:8	R/W	0d32	reg_sr3_pk_bp_hvcon_replace8lv_gain: gain to local variant before calculating the hv gain for peaking.
7	R/W	0d1	reg_sr3_sad_intlev_mode: interleave detect xerr mode: 0 max; 1: sum
6	R/W	0d1	reg_sr3_sad_intlev_mode1: mode 1 of using diagonal protection: 1: with diagonal protection
5:0	R/W	0d12	reg_sr3_sad_intlev_gain: interleave detection for sad gain applied, normalized to 8 as 1

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Table 8-2053 SRSHARP0_DEJ_CTRL 0x3e64

Bit(s)	R/W	Default	Description
15:8	R/W	8	reg_sr3_dejaggy_hblank
7:4	R/W		reserved
3:2	R/W	0d3	reg_sr3_dejaggy_sameside_prtct: enable of sr3 dejaggy same side curve protect from filter, [0] for proc path; [1] for ctrl path.
1	R/W	0d1	reg_sr3_dejaggy_sameside_mode: mode of sameside flag decision
0	R/W	0d1	reg_sr3_dejaggy_enable: enable of sr3 dejaggy

Table 8-2054 SRSHARP0_DEJ_ALPHA 0x3e65

Bit(s)	R/W	Default	Description
31:28	R/W	0d0	reg_sr3_dejaggy_ctrlchrm_alpha_1 : alpha for LR video LPF
27:24	R/W	0d15	reg_sr3_dejaggy_ctrlchrm_alpha_0 : alpha for LR video LPF
23:20	R/W	0d0	reg_sr3_dejaggy_ctrlluma_alpha_1 : alpha for LR video LPF
19:16	R/W	0d15	reg_sr3_dejaggy_ctrlluma_alpha_0 : alpha for LR video LPF
15:12	R/W	0d4	reg_sr3_dejaggy_procchrm_alpha_1: alpha for LR video LPF
11:8	R/W	0d6	reg_sr3_dejaggy_procchrm_alpha_0: alpha for LR video LPF
7:4	R/W	0d4	reg_sr3_dejaggy_procluma_alpha_1: alpha for LR video LPF
3:0	R/W	0d6	reg_sr3_dejaggy_procluma_alpha_0: alpha for LR video LPF

Table 8-2055 SRSHARP0_SR3_DRTLPF_EN 0x3e66

Bit(s)	R/W	Default	Description
31:15	R/W		reserved
14:8	R/W	0d0	reg_pk_debug_edge
6:4	R/W	0d0	reg_sr3_drtlpf_theta_en
2:0	R/W	0d7	reg_sr3_drtlpf_enable: directional lpf on Y/U/V channels

Table 8-2056 SRSHARP0_SR3_DRTLPF_ALPHA_0 0x3e67

Bit(s)	R/W	Default	Description
31:28	R/W		reserved
27:24	R/W	0d9	reg_sr3_drtlpf_alpha_3
23:20	R/W		reserved
19:16	R/W	0d10	reg_sr3_drtlpf_alpha_2
15:12	R/W		reserved
11:8	R/W	0d11	reg_sr3_drtlpf_alpha_1

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Bit(s)	R/W	Default	Description
7:4	R/W		reserved
3:0	R/W	0d12	reg_sr3_drtlpf_alpha_0: directional lpf alpha coef for min_sad/max_sad compare

Table 8-2057 SRSHARP0_SR3_DRTLPF_ALPHA_1 0x3e68

Bit(s)	R/W	Default	Description
31:28	R/W		reserved
27:24	R/W	0d1	reg_sr3_drtlpf_alpha_7
23:20	R/W		reserved
19:16	R/W	0d4	reg_sr3_drtlpf_alpha_6
15:12	R/W		reserved
11:8	R/W	0d7	reg_sr3_drtlpf_alpha_5
7:4	R/W		reserved
3:0	R/W	0d8	reg_sr3_drtlpf_alpha_4: directional lpf alpha coef for min_sad/max_sad compare

Table 8-2058 SRSHARP0_SR3_DRTLPF_ALPHA_2 0x3e69

Bit(s)	R/W	Default	Description
31:28	R/W		reserved
27:24	R/W	0d0	reg_sr3_drtlpf_alpha_11
23:20	R/W		reserved
19:16	R/W	0d0	reg_sr3_drtlpf_alpha_10
15:12	R/W		reserved
11:8	R/W	0d0	reg_sr3_drtlpf_alpha_9
7:4	R/W		reserved
3:0	R/W	0d0	reg_sr3_drtlpf_alpha_8: directional lpf alpha coef for min_sad/max_sad compare

Table 8-2059 SRSHARP0_SR3_DRTLPF_ALPHA_OFST 0x3e6a

Bit(s)	R/W	Default	Description
31:28	R/W	0	reg_sr3_drtlpf_alpha_ofst7
27:24	R/W	0	reg_sr3_drtlpf_alpha_ofst6
23:20	R/W	0	reg_sr3_drtlpf_alpha_ofst5
19:16	R/W	-2	reg_sr3_drtlpf_alpha_ofst4
15:12	R/W	0	reg_sr3_drtlpf_alpha_ofst3
11:8	R/W	0	reg_sr3_drtlpf_alpha_ofst2

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Bit(s)	R/W	Default	Description
7:4	R/W	0	reg_sr3_drtlpf_alpha_ofst1
3:0	R/W	-2	reg_sr3_drtlpf_alpha_ofst0: directional lpf alpha coef offset of each direction.

Table 8-2060 SRSHARP0_SR3_DERING_CTRL 0x3e6b

Bit(s)	R/W	Default	Description
31	R/W		reserved
30:28	R/W	1	reg_sr3_dering_enable: dering enable
27	R/W		reserved
26:24	R/W	3	reg_sr3_dering_varlpf_mode: local variant LPF mode. 0: no filter; 1: errosion 3x3; 2: 3x3 lpf; 3: 3x3 errosion + lpf
23:20	R/W	9	reg_sr3_dering_maxrange: range of dering in LR resolution.
19:18	R/W		reserved
17:16	R/W	2	reg_sr3_dering_lcvar_blend_mode: mode for lcvar calculation. 0:HV blend; 1:di-ag blend; 2:HV blend + V; 3: HV blend+Diag blend
15:8	R/W	40	reg_sr3_dering_lcvar_gain: gain to local variant and normalized to 32 as 1
7:0	R/W	28	reg_sr3_dering_lcvar_nearby_maxsad_th: threshold to use near side maxsad if that side is larger than this threshold, otherwise use the max one.

Table 8-2061 SRSHARP0_SR3_DERING_LUMA2PKGAIN_0TO3 0x3e6c

Bit(s)	R/W	Default	Description
31:24	R/W	255	reg_sr3_dering_luma2pkgain3: level limit(for th0<bpcon<th1) of curve for dering pkgain base on LPF luma level
23:16	R/W	255	reg_sr3_dering_luma2pkgain2: level limit(for bpcon<th0) of curve for dering pkgain base on LPF luma level
15:8	R/W	200	reg_sr3_dering_luma2pkgain1: threshold 1 of curve for dering pkgain based on LPF luma level.
7:0	R/W	30	reg_sr3_dering_luma2pkgain0: threshold 0 of curve for dering pkgain based on LPF luma level.

Table 8-2062 SRSHARP0_SR3_DERING_LUMA2PKGAIN_4TO6 0x3e6d

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	24	reg_sr3_dering_luma2pkgain6: rate1 (for bpcon>th1) of curve for dering pkOS based on LPF luma level.
15:8	R/W	50	reg_sr3_dering_luma2pkgain5: rate0 (for bpcon<th0) of curve for dering pkOS based on LPF luma level.
7:0	R/W	255	reg_sr3_dering_luma2pkgain4: level limit(for bpcon>th1) of curve for dering pkgain base on LPF luma level

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Table 8-2063 SRSHARP0_SR3_DERING_LUMA2PKOS_0TO3 0x3e6e

Bit(s)	R/W	Default	Description
31:24	R/W	255	reg_sr3_dering_luma2pkos3: level limit(for th0<bpcon<th1) of curve for dering pkOS base on LPF luma level
23:16	R/W	255	reg_sr3_dering_luma2pkos2: level limit(for bpcon<th0) of curve for dering pkOS base on LPF luma level
15:8	R/W	200	reg_sr3_dering_luma2pkos1: threshold 1 of curve for dering pkOS based on LPF luma level.
7:0	R/W	30	reg_sr3_dering_luma2pkos0: threshold 0 of curve for dering pkOS based on LPF luma level.

Table 8-2064 SRSHARP0_SR3_DERING_LUMA2PKOS_4TO6 0x3e6f

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	24	reg_sr3_dering_luma2pkos6: rate1 (for bpcon>th1) of curve for dering pkOS based on LPF luma level.
15:8	R/W	50	reg_sr3_dering_luma2pkos5: rate0 (for bpcon<th0) of curve for dering pkOS based on LPF luma level.
7:0	R/W	255	reg_sr3_dering_luma2pkos4: level limit(for bpcon>th1) of curve for dering pkOS base on LPF luma level

Table 8-2065 SRSHARP0_SR3_DERING_GAINVS_MADSAD 0x3e70

Bit(s)	R/W	Default	Description
31:28	R/W	0	reg_sr3_dering_gainvs_maxsad7
27:24	R/W	0	reg_sr3_dering_gainvs_maxsad6
23:20	R/W	0	reg_sr3_dering_gainvs_maxsad5
19:16	R/W	0	reg_sr3_dering_gainvs_maxsad4
15:12	R/W	0	reg_sr3_dering_gainvs_maxsad3
11:8	R/W	0	reg_sr3_dering_gainvs_maxsad2
7:4	R/W	4	reg_sr3_dering_gainvs_maxsad1
3:0	R/W	8	reg_sr3_dering_gainvs_maxsad0: pkgain vs maxsad value, 8 node interpolations.

Table 8-2066 SRSHARP0_SR3_DERING_GAINVS_VR2MAX 0x3e71

Bit(s)	R/W	Default	Description
31:28	R/W	15	reg_sr3_dering_gainvs_vr2max7
27:24	R/W	15	reg_sr3_dering_gainvs_vr2max6
23:20	R/W	15	reg_sr3_dering_gainvs_vr2max5
19:16	R/W	15	reg_sr3_dering_gainvs_vr2max4
15:12	R/W	14	reg_sr3_dering_gainvs_vr2max3

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Bit(s)	R/W	Default	Description
11:8	R/W	12	reg_sr3_dering_gainvs_vr2max2
7:4	R/W	2	reg_sr3_dering_gainvs_vr2max1
3:0	R/W	0	reg_sr3_dering_gainvs_vr2max0: pkgain vs ratio

Table 8-2067 SRSHARP0_SR3_DERING_PARAM0 0x3e72

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	10	reg_sr3_dering_lcvar_floor
15:8	R/W	32	reg_sr3_dering_vr2max_gain: gain to max before feeding to LUT
7:6	R/W		reserved
5:0	R/W	16	reg_sr3_dering_vr2max_limt: limit of maxsad

Table 8-2068 SRSHARP0_SR3_DRTLPF_THETA 0x3e73

Bit(s)	R/W	Default	Description
31:0	R/W	0xfe-c96420	reg_sr3_drtlpf_theta: u4x8 directional lpf beta coef for min_sad/min2_sad compared to x=0:7 correspond to [1:8]/16; 0 means no drtLPF, 15: 100% alpha dependant drtLPF.

Table 8-2069 SRSHARP0_SATPRT_CTRL 0x3e74

Bit(s)	R/W	Default	Description
31:28	R/W		reserved
27:16	R/W	5	reg_satprt_sat_core: 4x will be coring to cor(irgb_max-irgb_min) to calculate the oy_delt, the smaller the more protection to color, the larger only the rich color will be protected.
15:8	R/W	64	reg_satprt_sat_rate: rate to cor(irgb_max-irgb_min) to calculate the oy_delt,the larger the more protection to color; norm 16 as 1
7:4	R/W		reserved
3:2	R/W	1	reg_satprt_csc_mode: CSC mode of current yuv input: 0:601; 1:709; 2:BT2020 NCL; 3 reserved
1	R/W	1	reg_satprt_is_lmt: flag telling the YUV is limited range data or full rang data; 1 is limited data
0	R/W	0	reg_satprt_enable: 1 to enable of saturation protection for dnlp adjustments

Table 8-2070 SRSHARP0_SATPRT_DIVM 0x3e75

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:0	R/W	{128,128,128}	reg_satprt_div_m: u8x3, 1/m, normalized to 128 as 1.

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Table 8-2071 SRSHARP0_DB_FLT_CTRL 0x3e77

Bit(s)	R/W	Default	Description
26	R/W	0	reg_nrdeband_reset1 : // unsigned , default = 0 0 : no reset seed 1: reload chroma seed
25	R/W	0	reg_nrdeband_reset0 : // unsigned , default = 0 0 : no reset seed 1: reload luma seed
24	R/W	0	reg_nrdeband_rgb : // unsigned , default = 0 0 : yuv 1: RGB
23	R/W	1	reg_nrdeband_en11 : // unsigned , default = 1 debanding registers of side lines, [0] for luma, same for below
22	R/W	1	reg_nrdeband_en10 : // unsigned , default = 1 debanding registers of side lines, [1] for chroma, same for below
21	R/W	1	reg_nrdeband_siderand : // unsigned , default = 1 options to use side two lines use the rand, instead of use for the YUV three component of middle line, 0: seed [3]/bandrand[3] for middle line yuv; 1: seed[3]/bandrand[3] for nearby three lines Y;
20	R/W	0	reg_nrdeband_randmode : // unsigned , default = 0 mode of rand noise adding, 0: same noise strength for all difs; else: strenght of noise will not exceed the difs, MIN((pPKReg->reg_nrdeband_bandrand[m]), noise[m])
19:17	R/W	6	reg_nrdeband_bandrand2 : // unsigned , default = 6
15:13	R/W	6	reg_nrdeband_bandrand1 : // unsigned , default = 6
11: 9	R/W	6	reg_nrdeband_bandrand0 : // unsigned , default = 6
7	R/W	1	reg_nrdeband_hpxor1 : // unsigned , default = 1 debanding random hp portion xor, [0] for luma
6	R/W	1	reg_nrdeband_hpxor0 : // unsigned , default = 1 debanding random hp portion xor, [1] for chroma
5	R/W	1	reg_nrdeband_en1 : // unsigned , default = 1 debanding registers, for luma
4	R/W	1	reg_nrdeband_en0 : // unsigned , default = 1 debanding registers, for chroma
3: 2	R/W	2	reg_nrdeband_lpf_mode1 : // unsigned , default = 2 lpf mode, 0: 3x3, 1:3x5; 2: 5x5; 3:5x7
1: 0	R/W	2	reg_nrdeband_lpf_mode0 : // unsigned , default = 2 lpf mode, 0: 3x3, 1:3x5; 2: 5x5; 3:5x7

Table 8-2072 SRSHARP0_DB_FLT_YC_THRD 0x3e78

Bit(s)	R/W	Default	Description
31:28	R/W	9	reg_nrdeband_luma_th3 : // unsigned , default = 9 threshold to |Y-Ylpf|, if < th[0] use lpf
27:24	R/W	7	reg_nrdeband_luma_th2 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4
23:20	R/W	6	reg_nrdeband_luma_th1 : // unsigned , default = 6 elseif <th[1] use (lpf*3 + y)/4elseif <th[2] (lpf*1 + y)/2
19:16	R/W	5	reg_nrdeband_luma_th0 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/4elseif elseif <th[3] (lpf*1 + 3*y)/4; else
15:12	R/W	9	reg_nrdeband_chrm_th3 : // unsigned , default = 9 threshold to |Y-Ylpf|, if < th[0] use lpf
11: 8	R/W	7	reg_nrdeband_chrm_th2 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4

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Bit(s)	R/W	Default	Description
7: 4	R/W	6	reg_nrdeband_chrm_th1 : // unsigned , default = 6 elseif <th[1] use (lpf*3 + y)/ 4elseif <th[2] (lpf*1 + y)/2
3: 0	R/W	5	reg_nrdeband_chrm_th0 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/ 4elseif elseif

Table 8-2073 SRSHARP0_DB_FLT_RANDLUT 0x3e79

Bit(s)	R/W	Default	Description
23:21	R/W	1	reg_nrdeband_randslut7 : // unsigned , default = 1 lut0
20:18	R/W	1	reg_nrdeband_randslut6 : // unsigned , default = 1 lut0
17:15	R/W	1	reg_nrdeband_randslut5 : // unsigned , default = 1 lut0
14:12	R/W	1	reg_nrdeband_randslut4 : // unsigned , default = 1 lut0
11: 9	R/W	1	reg_nrdeband_randslut3 : // unsigned , default = 1 lut0
8: 6	R/W	1	reg_nrdeband_randslut2 : // unsigned , default = 1 lut0
5: 3	R/W	1	reg_nrdeband_randslut1 : // unsigned , default = 1 lut0
2: 0	R/W	1	reg_nrdeband_randslut0 : // unsigned , default = 1 lut0

Table 8-2074 SRSHARP0_DB_FLT_PXI_THRD 0x3e7a

Bit(s)	R/W	Default	Description
25:16	R/W	0	reg_nrdeband_yc_th1 : // unsigned , default = 0 to luma/|u/v| for using the denoise
9: 0	R/W	0	reg_nrdeband_yc_th0 : // unsigned , default = 0 to luma/|u/v| for using the denoise

Table 8-2075 SRSHARP0_DB_FLT_SEED_Y 0x3e7b

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8240	reg_nrdeband_seed0 : // unsigned , default = 1621438240 noise adding seed for Y. seed[0]= 0x60a52f20; as default

Table 8-2076 SRSHARP0_DB_FLT_SEED_U 0x3e7c

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8247	reg_nrdeband_seed1 : // unsigned , default = 1621438247 noise adding seed for U. seed[0]= 0x60a52f27; as default

Table 8-2077 SRSHARP0_DB_FLT_SEED_V 0x3e7d

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8242	reg_nrdeband_seed2 : // unsigned , default = 1621438242 noise adding seed for V. seed[0]= 0x60a52f22; as default

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Table 8-2078 SRSHARP0_PKGAIN_VSLUMA_LUT_L 0x3e7e

Bit(s)	R/W	Default	Description
31:28	R/W	5	reg_pkgain_vsluma_lut7
27:24	R/W	6	reg_pkgain_vsluma_lut6
23:20	R/W	6	reg_pkgain_vsluma_lut5
19:16	R/W	6	reg_pkgain_vsluma_lut4
15:12	R/W	7	reg_pkgain_vsluma_lut3
11:8	R/W	10	reg_pkgain_vsluma_lut2
7:4	R/W	12	reg_pkgain_vsluma_lut1
3:0	R/W	8	reg_pkgain_vsluma_lut0

Table 8-2079 SRSHARP0_PKGAIN_VSLUMA_LUT_H 0x3e7f

Bit(s)	R/W	Default	Description
31:4	R/W		reserved
3:0	R/W	4	reg_pkgain_vsluma_lut8

Table 8-2080 SRSHARP0_PKOSHT_VSLUMA_LUT_L 0x3e80

Bit(s)	R/W	Default	Description
31:28	R/W	5	reg_pkosht_vsluma_lut7
27:24	R/W	6	reg_pkosht_vsluma_lut6
23:20	R/W	6	reg_pkosht_vsluma_lut5
19:16	R/W	6	reg_pkosht_vsluma_lut4
15:12	R/W	7	reg_pkosht_vsluma_lut3
11:8	R/W	10	reg_pkosht_vsluma_lut2
7:4	R/W	12	reg_pkosht_vsluma_lut1
3:0	R/W	8	reg_pkosht_vsluma_lut0

Table 8-2081 SRSHARP0_PKOSHT_VSLUMA_LUT_H 0x3e81

Bit(s)	R/W	Default	Description
31:4	R/W		reserved
3:0	R/W	4	reg_pkosht_vsluma_lut8

Table 8-2082 SRSHARP0_SATPRT_LMT_RGB1 0x3e82

Bit(s)	R/W	Default	Description
27:16	R/W	0d0	reg_satprt_lmt_g:
11: 0	R/W	0d0	reg_satprt_lmt_r: limit of RGB channel, for limited range RGB, 12bits

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Table 8-2083 SRSHARP0_SATPRT_LMT_RGB2 0x3e83

Bit(s)	R/W	Default	Description
31:16	R/W	0d0	reserved
11: 0	R/W	0d0	reg_satprt_lmt_b: limit of RGB channel, for limited range RGB

Table 8-2084 SRSHARP0_SHARP_GATE_CLK_CTRL_0 0x3e84

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Gate clock control [01:00]: sharp input control unit [03:02]: deband unit [05:04]: dejaggy unit [07:06]: dnlp unit [09:08]: demo control unit [11:10]: horiz interp unit

Table 8-2085 SRSHARP0_SHARP_GATE_CLK_CTRL_1 0x3e85

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Gate clock control [01:00]: sr_top “pipe_ctrl” [03:02]: drt [05:04]: ssd [07:06]: cubic [09:08]: edi [11:10]: pkgainsad [13:12]: bicomux [15:14]: bicin_fifo [17:16]: lpf4pkgain_fifo [19:18]: min2hvgain_fifo [21:20]: sad4pkgain_fifo [23:22]: dirminmax4xtl_fifo [25:24]: drtsad8_fifo [27:26]: ssdmax_fifo [29:28]: pkminmax_fifo [31:30]: cirdrtgain_fifo

Table 8-2086 SRSHARP0_SHARP_GATE_CLK_CTRL_2 0x3e86

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Gate clock control [01:00]: bufdiff_fifo [03:02]: osvar_fifo [05:04]: pkhvgain unit [07:06]: pkgain unit [09:08]: TI unit [11:10]: pk unit [13:12]: locvar unit [15:14]: hvconc unit

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Table 8-2087 SRSHARP0_SHARP_GATE_CLK_CTRL_3 0x3e87

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Gate clock control [01:00]: TI, htl Y [03:02]: TI, vti Y [05:04]: TI, htl U [07:06]: TI, vtl U [09:08]: TI, htl V [11:10]: TI, vtl V [13:12]: TI, lumaminmax_fifo [15:14]: TI, chrmminmax fifo

Table 8-2088 SRSHARP0_SHARP_DPS_CTRL 0x3e88

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Power saving control : hvcon : nrssd : os filter : cubic 5 lines to 9 lines : dering : locvar : drtlpf : hlti : hcti : vlti : vcti : lti blend : cti blend : htishort (no used) : nr Y filter : nr C filter : nr belnd : pkti blend : os ctrl : pk HP : pk BP [26:24] dejaggy power saving control [30:28] reserved

Table 8-2089 SRSHARP0_DNLP_00 0x3e90

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0010-0008	reg_dnlp_ygrid0 : : dnlp00 . unsigned , default = 32'h00100008

Table 8-2090 SRSHARP0_DNLP_01 0x3e91

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0020-0018	reg_dnlp_ygrid1 : : dnlp01 . unsigned , default = 32'h00200018

Table 8-2091 SRSHARP0_DNLP_02 0x3e92

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0030-0028	reg_dnlp_ygrid2 : : dnlp02 . unsigned , default = 32'h00300028

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Table 8-2092 SRSHARP0_DNLP_03 0x3e93

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0040-0038	reg_dnlp_ygrid3 : : dnlp03 . unsigned , default = 32'h00400038

Table 8-2093 SRSHARP0_DNLP_04 0x3e94

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0050-0048	reg_dnlp_ygrid4 : : dnlp04 . unsigned , default = 32'h00500048

Table 8-2094 SRSHARP0_DNLP_05 0x3e95

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0068-005c	reg_dnlp_ygrid5 : : dnlp05 . unsigned , default = 32'h0068005c

Table 8-2095 SRSHARP0_DNLP_06 0x3e96

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0080-0074	reg_dnlp_ygrid6 : : dnlp06 . unsigned , default = 32'h00800074

Table 8-2096 SRSHARP0_DNLP_07 0x3e97

Bit(s)	R/W	Default	Description
31: 0	R/W	0x00-a00090	reg_dnlp_ygrid7 : : dnlp07 . unsigned , default = 32'h00a00090

Table 8-2097 SRSHARP0_DNLP_08 0x3e98

Bit(s)	R/W	Default	Description
31: 0	R/W	0x00-c000b0	reg_dnlp_ygrid8 : : dnlp08 . unsigned , default = 32'h00c000b0

Table 8-2098 SRSHARP0_DNLP_09 0x3e99

Bit(s)	R/W	Default	Description
31: 0	R/W	0x00-e000d0	reg_dnlp_ygrid9 : : dnlp09 . unsigned , default = 32'h00e000d0

Table 8-2099 SRSHARP0_DNLP_10 0x3e9a

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0100-00f0	reg_dnlp_ygrid10 : : dnlp10 . unsigned , default = 32'h010000f0

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Table 8-2100 SRSHARP0_DNLP_11 0x3e9b

Bit(s)	R/W	Default	Description
31: 0	R/W	0x012-c0114	reg_dnlp_ygrid11 : : dnlp11 . unsigned , default = 32'h012c0114

Table 8-2101 SRSHARP0_DNLP_12 0x3e9c

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0154-0140	reg_dnlp_ygrid12 : : dnlp12 . unsigned , default = 32'h01540140

Table 8-2102 SRSHARP0_DNLP_13 0x3e9d

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0180-016c	reg_dnlp_ygrid13 : : dnlp13 . unsigned , default = 32'h0180016c

Table 8-2103 SRSHARP0_DNLP_14 0x3e9e

Bit(s)	R/W	Default	Description
31: 0	R/W	0x01-c001a0	reg_dnlp_ygrid14 : : dnlp14 . unsigned , default = 32'h01c001a0

Table 8-2104 SRSHARP0_DNLP_15 0x3e9f

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0200-01e0	reg_dnlp_ygrid15 : : dnlp15 . unsigned , default = 32'h020001e0

Table 8-2105 SRSHARP0_DNLP_16 0x3ea0

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0240-0220	reg_dnlp_ygrid16 : : dnlp16 . unsigned , default = 32'h02400220

Table 8-2106 SRSHARP0_DNLP_17 0x3ea1

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0280-0260	reg_dnlp_ygrid17 : : dnlp17 . unsigned , default = 32'h02800260

Table 8-2107 SRSHARP0_DNLP_18 0x3ea2

Bit(s)	R/W	Default	Description
31: 0	R/W	0x02-b00298	reg_dnlp_ygrid18 : : dnlp18 . unsigned , default = 32'h02b00298

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Table 8-2108 SRSHARP0_DNLP_19 0x3ea3

Bit(s)	R/W	Default	Description
31: 0	R/W	0x02-e002c8	reg_dnlp_ygrid19 : : dnlp19 . unsigned , default = 32'h02e002c8

Table 8-2109 SRSHARP0_DNLP_20 0x3ea4

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0300-02f0	reg_dnlp_ygrid20 : : dnlp20 . unsigned , default = 32'h030002f0

Table 8-2110 SRSHARP0_DNLP_21 0x3ea5

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0320-0310	reg_dnlp_ygrid21 : : dnlp21 . unsigned , default = 32'h03200310

Table 8-2111 SRSHARP0_DNLP_22 0x3ea6

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0338-032c	reg_dnlp_ygrid22 : : dnlp22 . unsigned , default = 32'h0338032c

Table 8-2112 SRSHARP0_DNLP_23 0x3ea7

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0350-0348	reg_dnlp_ygrid23 : : dnlp23 . unsigned , default = 32'h03500348

Table 8-2113 SRSHARP0_DNLP_24 0x3ea8

Bit(s)	R/W	Default	Description
31: 0	R/W	0x035-c0358	reg_dnlp_ygrid24 : : dnlp24 . unsigned , default = 32'h035c0358

Table 8-2114 SRSHARP0_DNLP_25 0x3ea9

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0368-0360	reg_dnlp_ygrid25 : : dnlp25 . unsigned , default = 32'h03680360

Table 8-2115 SRSHARP0_DNLP_26 0x3eaa

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0378-0370	reg_dnlp_ygrid26 : : dnlp26 . unsigned , default = 32'h03780370

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Table 8-2116 SRSHARP0_DNLP_27 0x3eab

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0388-0380	reg_dnlp_ygrid27 : : dnlp27 . unsigned , default = 32'h03880380

Table 8-2117 SRSHARP0_DNLP_28 0x3eac

Bit(s)	R/W	Default	Description
31: 0	R/W	0x03-a00390	reg_dnlp_ygrid28 : : dnlp28 . unsigned , default = 32'h03a00390

Table 8-2118 SRSHARP0_DNLP_29 0x3ead

Bit(s)	R/W	Default	Description
31: 0	R/W	0x03-c003b0	reg_dnlp_ygrid29 : : dnlp29 . unsigned , default = 32'h03c003b0

Table 8-2119 SRSHARP0_DNLP_30 0x3eae

Bit(s)	R/W	Default	Description
31: 0	R/W	0x03-e003d0	reg_dnlp_ygrid30 : : dnlp30 . unsigned , default = 32'h03e003d0

Table 8-2120 SRSHARP0_DNLP_31 0x3eaf

Bit(s)	R/W	Default	Description
31: 0	R/W	0x03f-c03f0	reg_dnlp_ygrid31 : : dnlp31 . unsigned , default = 32'h03fc03f0

Table 8-2121 SRSHARP0_SHARP_SYNC_CTRL 0x3eb0

Bit(s)	R/W	Default	Description
15:0	R/W	0d0	Shadow control for some setting: 0 to use VSYNC to shadow : SR all bypass : reg_sr2_bic_pknr_bypass [2] : reg_3d_mode [3] : reg_nrdeband_en11 [4]: reg_nrdeband_en10 [5]: reg_nrdeband_en1 [6]: reg_nrdeband_en0 [7]: reg_dejaggy_enable [8]: reg_sr2_sharp_prc_lr [9]: reg_sharp_pk_en [10]:reg_sharp_nr_en [11]:reg_adp_hlti_en [12]:reg_adp_hcti_en [13]:reg_adp_vlti_en [14]:reg_adp_vcti_en [15]:reg_dnlp_en

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Table 8-2122 SRSHARP1_SHARP_HVSIZE 0x3f00

Bit(s)	R/W	Default	Description
28:16	R/W	0d1920	reg_pknr_hsize : . unsigned , default = 1920
12: 0	R/W	0d1080	reg_pknr_vsize : . unsigned , default = 1080

Table 8-2123 SRSHARP1_SHARP_HVBLANK_NUM 0x3f01

Bit(s)	R/W	Default	Description
23: 16	R/W	0d8	reg_deband_hblank : . unsigned , default = 8
15: 8	R/W	0d20	reg_pknr_hblank_num : . unsigned , default = 20
7: 0	R/W	0d60	reg_pknr_vblank_num : . unsigned , default = 60

Table 8-2124 SRSHARP1_NR_GAUSSIAN_MODE 0x3f02

Bit(s)	R/W	Default	Description
4	R/W	0d1	reg_nr_gau_ymode : : 0 3x3 filter; 1: 5x5 filter . unsigned , default = 1
0	R/W	0d1	reg_nr_gau_cmode : : 0 3x3 filter; 1: 5x5 filter . unsigned , default = 1

Table 8-2125 SRSHARP1_PK_CON_2CIRHPGAIN_TH_RATE 0x3f05

Bit(s)	R/W	Default	Description
31:24	R/W	0d25	reg_pk_cirhpcon2gain0 : : threshold0 of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 25
23:16	R/W	0d60	reg_pk_cirhpcon2gain1 : : threshold1 of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 60
15: 8	R/W	0d80	reg_pk_cirhpcon2gain5 : : rate0 (for hpcon<th0) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 80
7: 0	R/W	0d20	reg_pk_cirhpcon2gain6 : : rate1 (for hpcon>th1) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 20

Table 8-2126 SRSHARP1_PK_CON_2CIRHPGAIN_LIMIT 0x3f06

Bit(s)	R/W	Default	Description
31:24	R/W	0d96	reg_pk_cirhpcon2gain2 : : level limit(for hpcon<th0) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 96
23:16	R/W	0d96	reg_pk_cirhpcon2gain3 : : level limit(for th0<hpcon<th1) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 96
15: 8	R/W	0d5	reg_pk_cirhpcon2gain4 : : level limit(for hpcon>th1) of curve to map hpcon to hpgain for circle hp filter (all 8 direction same). 0~255.. unsigned , default = 5

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Table 8-2127 SRSHARP1_PK_CON_2CIRBPGAIN_TH_RATE 0x3f07

Bit(s)	R/W	Default	Description
31:24	R/W	0d20	reg_pk_cirbpcon2gain0 : : threshold0 of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 20
23:16	R/W	0d50	reg_pk_cirbpcon2gain1 : : threshold1 of curve to map bpcon to bpgain for circle bp filter (all 8 direction same).. unsigned , default = 50
15: 8	R/W	0d50	reg_pk_cirbpcon2gain5 : : rate0 (for bpcon<th0) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 50
7: 0	R/W	0d25	reg_pk_cirbpcon2gain6 : : rate1 (for bpcon>th1) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 25

Table 8-2128 SRSHARP1_PK_CON_2CIRBPGAIN_LIMIT 0x3f08

Bit(s)	R/W	Default	Description
31:24	R/W	0d40	reg_pk_cirbpcon2gain2 : : level limit(for bpcon<th0) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 40
23:16	R/W	0d40	reg_pk_cirbpcon2gain3 : : level limit(for th0<bpcon<th1) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 40
15: 8	R/W	0d5	reg_pk_cirbpcon2gain4 : : level limit(for bpcon>th1) of curve to map bpcon to bpgain for circle bp filter (all 8 direction same). 0~255.. unsigned , default = 5

Table 8-2129 SRSHARP1_PK_CON_2DRTHPGAIN_TH_RATE 0x3f09

Bit(s)	R/W	Default	Description
31:24	R/W	0d25	reg_pk_drthpcon2gain0 : : threshold0 of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 25
23:16	R/W	0d60	reg_pk_drthpcon2gain1 : : threshold1 of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 60
15: 8	R/W	0d80	reg_pk_drthpcon2gain5 : : rate0 (for hpcon<th0) of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 80
7: 0	R/W	0d20	reg_pk_drthpcon2gain6 : : rate1 (for hpcon>th1) of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 20

Table 8-2130 SRSHARP1_PK_CON_2DRTHPGAIN_LIMIT 0x3f0a

Bit(s)	R/W	Default	Description
31:24	R/W	0d90	reg_pk_drthpcon2gain2 : : level limit(for hpcon<th0) of curve to map hpcon to hpgain for directional hp filter (best direction).. unsigned , default = 90
23:16	R/W	0d96	reg_pk_drthpcon2gain3 : : level limit(for th0<hpcon<th1) of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 96
15: 8	R/W	0d5	reg_pk_drthpcon2gain4 : : level limit(for hpcon>th1) of curve to map hpcon to hpgain for directional hp filter (best direction). 0~255.. unsigned , default = 5

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Table 8-2131 SRSHARP1_PK_CON_2DRTBPGAIN_TH_RATE 0x3f0b

Bit(s)	R/W	Default	Description
31:24	R/W	0d20	reg_pk_drtbpcon2gain0 : : threshold0 of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 20
23:16	R/W	0d50	reg_pk_drtbpcon2gain1 : : threshold1 of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 50
15: 8	R/W	0d50	reg_pk_drtbpcon2gain5 : : rate0 (for bpcon<th0) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 50
7: 0	R/W	0d25	reg_pk_drtbpcon2gain6 : : rate1 (for bpcon>th1) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 25

Table 8-2132 SRSHARP1_PK_CON_2DRTBPGAIN_LIMIT 0x3f0c

Bit(s)	R/W	Default	Description
31:24	R/W	0d40	reg_pk_drtbpcon2gain2 : : level limit(for bpcon<th0) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 40
23:16	R/W	0d40	reg_pk_drtbpcon2gain3 : : level limit(for th0<bpcon<th1) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 40
15: 8	R/W	0d5	reg_pk_drtbpcon2gain4 : : level limit(for bpcon>th1) of curve to map bpcon to bpgain for directional bp filter (best direction). 0~255.. unsigned , default = 5

Table 8-2133 SRSHARP1_PK_CIRFB_LPF_MODE 0x3f0d

Bit(s)	R/W	Default	Description
29:28	R/W	0d1	reg_cirhp_horz_mode : : no horz filter on HP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8. unsigned , default = 1
25:24	R/W	0d1	reg_cirhp_vert_mode : : no vert filter on HP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8. unsigned , default = 1
21:20	R/W	0d1	reg_cirhp_diag_mode : : filter on HP; 1: [1 2 1]/4; . unsigned , default = 1
13:12	R/W	0d1	reg_cirbp_horz_mode : : no horz filter on BP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8. unsigned , default = 1
9: 8	R/W	0d1	reg_cirbp_vert_mode : : no vert filter on BP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8. unsigned , default = 1
5: 4	R/W	0d1	reg_cirbp_diag_mode : : filter on BP; 1: [1 2 1]/4; . unsigned , default = 1

Table 8-2134 SRSHARP1_PK_DRTFB_LPF_MODE 0x3f0e

Bit(s)	R/W	Default	Description
29:28	R/W	0d1	reg_drthp_horz_mode : : no horz filter on HP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8 2 . unsigned , default = 1
25:24	R/W	0d1	reg_drthp_vert_mode : : no vert filter on HP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8 2 . unsigned , default = 1
21:20	R/W	0d1	reg_drthp_diag_mode : : filter on HP; 1: [1 2 1]/4; 1 . unsigned , default = 1
13:12	R/W	0d1	reg_drtbp_horz_mode : : no horz filter on BP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8 2 . unsigned , default = 1

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Bit(s)	R/W	Default	Description
9: 8	R/W	0d1	reg_drtbp_vert_mode : : no vert filter on BP; 1: [1 2 1]/4; 2/3: [1 2 2 2 1]/8 2 . unsigned , default = 1
5: 4	R/W	0d1	reg_drtbp_diag_mode : : filter on BP; 1: [1 2 1]/4; 1 . unsigned , default = 1

Table 8-2135 SRSHARP1_PK_CIRFB_HP_CORING 0x3f0f

Bit(s)	R/W	Default	Description
21:16	R/W	0d4	reg_cirhp_horz_core : : coring of HP for Horz. unsigned , default = 4
13: 8	R/W	0d4	reg_cirhp_vert_core : : coring of HP for Vert. unsigned , default = 4
5: 0	R/W	0d4	reg_cirhp_diag_core : : coring of HP for Diag. unsigned , default = 4

Table 8-2136 SRSHARP1_PK_CIRFB_BP_CORING 0x3f10

Bit(s)	R/W	Default	Description
21:16	R/W	0d4	reg_cirbp_horz_core : : coring of HP for Horz. unsigned , default = 4
13: 8	R/W	0d4	reg_cirbp_vert_core : : coring of HP for Vert. unsigned , default = 4
5: 0	R/W	0d4	reg_cirbp_diag_core : : coring of HP for Diag. unsigned , default = 4

Table 8-2137 SRSHARP1_PK_DRTFB_HP_CORING 0x3f11

Bit(s)	R/W	Default	Description
21:16	R/W	0d4	reg_drthp_horz_core : : coring of HP for Horz. unsigned , default = 4
13: 8	R/W	0d4	reg_drthp_vert_core : : coring of HP for Vert. unsigned , default = 4
5: 0	R/W	0d4	reg_drthp_diag_core : : coring of HP for Diag. unsigned , default = 4

Table 8-2138 SRSHARP1_PK_DRTFB_BP_CORING 0x3f12

Bit(s)	R/W	Default	Description
21:16	R/W	0d4	reg_drtbp_horz_core : : coring of HP for Horz. unsigned , default = 4
13: 8	R/W	0d4	reg_drtbp_vert_core : : coring of HP for Vert. unsigned , default = 4
5: 0	R/W	0d4	reg_drtbp_diag_core : : coring of HP for Diag. unsigned , default = 4

Table 8-2139 SRSHARP1_PK_CIRFB_BLEND_GAIN 0x3f13

Bit(s)	R/W	Default	Description
31:28	R/W	0d8	reg_hp_cir_hgain : : normalized 8 as '1' . unsigned , default = 8
27:24	R/W	0d8	reg_hp_cir_vgain : : normalized 8 as '1' . unsigned , default = 8
23:20	R/W	0d8	reg_hp_cir_dgain : : normalized 8 as '1' . unsigned , default = 8
15:12	R/W	0d8	reg_bp_cir_hgain : : normalized 8 as '1' . unsigned , default = 8

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Bit(s)	R/W	Default	Description
11: 8	R/W	0d8	reg_bp_cir_vgain : : normalized 8 as '1' . unsigned , default = 8
7: 4	R/W	0d8	reg_bp_cir_dgain : : normalized 8 as '1' . unsigned , default = 8

Table 8-2140 SRSHARP1_NR_ALPY_SSD_GAIN_OFST 0x3f14

Bit(s)	R/W	Default	Description
15: 8	R/W	0d16	reg_nr_alp0_ssd_gain : : gain to max ssd normalized 16 as '1'. unsigned , default = 16
5: 0	R/W	0x0	reg_nr_alp0_ssd_ofst : : offset to ssd before dividing to min_err . signed , default = -2

Table 8-2141 SRSHARP1_NR_ALP0Y_ERR2CURV_TH_RATE 0x3f15

Bit(s)	R/W	Default	Description
31:24	R/W	0d10	reg_nr_alp0_minerr_ypar0 : : threshold0 of curve to map mierr to alp0 for luma channel, this will be set value of flat region mierr that no need blur. 0~255.. unsigned , default = 10
23:16	R/W	0d25	reg_nr_alp0_minerr_ypar1 : : threshold1 of curve to map mierr to alp0 for luma channel,this will be set value of texture region mierr that can not blur.. unsigned , default = 25
15: 8	R/W	0d80	reg_nr_alp0_minerr_ypar5 : : rate0 (for mierr<th0) of curve to map mierr to alp0 for luma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 80
7: 0	R/W	0d64	reg_nr_alp0_minerr_ypar6 : : rate1 (for mierr>th1) of curve to map mierr to alp0 for luma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 64

Table 8-2142 SRSHARP1_NR_ALP0Y_ERR2CURV_LIMIT 0x3f16

Bit(s)	R/W	Default	Description
31:24	R/W	0d63	reg_nr_alp0_minerr_ypar2 : : level limit(for mierr<th0) of curve to map mierr to alp0 for luma channel, this will be set to alp0 that we can do for flat region. 0~255.. unsigned , default = 63
23:16	R/W	0d0	reg_nr_alp0_minerr_ypar3 : : level limit(for th0<mierr<th1) of curve to map mierr to alp0 for luma channel, this will be set to alp0 that we can do for misc region. 0~255.. unsigned , default = 0
15: 8	R/W	0d63	reg_nr_alp0_minerr_ypar4 : : level limit(for mierr>th1) of curve to map mierr to alp0 for luma channel, this will be set to alp0 that we can do for texture region. 0~255.. unsigned , default = 63

Table 8-2143 SRSHARP1_NR_ALP0C_ERR2CURV_TH_RATE 0x3f17

Bit(s)	R/W	Default	Description
31:24	R/W	0d10	reg_nr_alp0_minerr_cpar0 : : threshold0 of curve to map mierr to alp0 for chroma channel, this will be set value of flat region mierr that no need blur.. unsigned , default = 10
23:16	R/W	0d25	reg_nr_alp0_minerr_cpar1 : : threshold1 of curve to map mierr to alp0 for chroma channel,this will be set value of texture region mierr that can not blur.. unsigned , default = 25

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Bit(s)	R/W	Default	Description
15: 8	R/W	0d80	reg_nr_alp0_minerr_cpar5 : : rate0 (for mierr<th0) of curve to map mierr to alp0 for chroma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 80
7: 0	R/W	0d64	reg_nr_alp0_minerr_cpar6 : : rate1 (for mierr>th1) of curve to map mierr to alp0 for chroma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 64

Table 8-2144 SRSHARP1_NR_ALP0C_ERR2CURV_LIMIT 0x3f18

Bit(s)	R/W	Default	Description
31:24	R/W	0d63	reg_nr_alp0_minerr_cpar2 : : level limit(for mierr<th0) of curve to map mierr to alp0 for chroma channel, this will be set to alp0 that we can do for flat region. 0~255.. unsigned , default = 63
23:16	R/W	0d0	reg_nr_alp0_minerr_cpar3 : : level limit(for th0<mierr<th1) of curve to map mierr to alp0 for chroma channel, this will be set to alp0 that we can do for misc region. 0~255.. unsigned , default = 0
15: 8	R/W	0d63	reg_nr_alp0_minerr_cpar4 : : level limit(for mierr>th1) of curve to map mierr to alp0 for chroma channel, this will be set to alp0 that we can do for texture region. 0~255.. unsigned , default = 63

Table 8-2145 SRSHARP1_NR_ALP0_MIN_MAX 0x3f19

Bit(s)	R/W	Default	Description
29:24	R/W	0d2	reg_nr_alp0_ymin : : normalized to 64 as '1' . unsigned , default = 2
21:16	R/W	0d63	reg_nr_alp0_ymax : : normalized to 64 as '1' . unsigned , default = 63
13: 8	R/W	0d2	reg_nr_alp0_cmin : : normalized to 64 as '1' . unsigned , default = 2
5: 0	R/W	0d63	reg_nr_alp0_cmax : : normalized to 64 as '1' . unsigned , default = 63

Table 8-2146 SRSHARP1_NR_ALP1_MIERR_CORING 0x3f1a

Bit(s)	R/W	Default	Description
16	R/W	0d0	reg_nr_alp1_maxerr_mode : : 0 max err; 1: xerr . unsigned , default = 0
13: 8	R/W	0d0	reg_nr_alp1_core_rate : : normalized 64 as "1" . unsigned , default = 0
5: 0	R/W	0d3	reg_nr_alp1_core_ofst : : normalized 64 as "1". signed , default = 3

Table 8-2147 SRSHARP1_NR_ALP1_ERR2CURV_TH_RATE 0x3f1b

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_nr_alp1_minerr_par0 : : threshold0 of curve to map mierr to alp1 for luma/chroma channel, this will be set value of flat region mierr that no need directional NR. 0~255.. unsigned , default = 0
23:16	R/W	0d24	reg_nr_alp1_minerr_par1 : : threshold1 of curve to map mierr to alp1 for luma/chroma channel,this will be set value of texture region mierr that can not do directional NR. 0~255.. unsigned , default = 24

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Bit(s)	R/W	Default	Description
15: 8	R/W	0d0	reg_nr_alp1_minerr_par5 : : rate0 (for mierr<th0) of curve to map mierr to alp1 for luma/chroma channel. the larger of the value, the deep of the slope.. unsigned , default = 0
7: 0	R/W	0d20	reg_nr_alp1_minerr_par6 : : rate1 (for mierr>th1) of curve to map mierr to alp1 for luma/chroma channel. the larger of the value, the deep of the slope. 0~255. unsigned , default = 20

Table 8-2148 SRSHARP1_NR_ALP1_ERR2CURV_LIMIT 0x3f1c

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_nr_alp1_minerr_par2 : : level limit(for mierr<th0) of curve to map mierr to alp1 for luma/chroma channel, this will be set to alp1 that we can do for flat region. 0~255.. unsigned , default = 0
23:16	R/W	0d16	reg_nr_alp1_minerr_par3 : : level limit(for th0<mierr<th1) of curve to map mierr to alp1 for luma/chroma channel, this will be set to alp1 that we can do for misc region. 0~255.. unsigned , default = 16
15: 8	R/W	0d63	reg_nr_alp1_minerr_par4 : : level limit(for mierr>th1) of curve to map mierr to alp1 for luma/chroma channel, this will be set to alp1 that we can do for texture region. 0~255.255 before. unsigned , default = 63

Table 8-2149 SRSHARP1_NR_ALP1_MIN_MAX 0x3f1d

Bit(s)	R/W	Default	Description
29:24	R/W	0d0	reg_nr_alp1_ymin : : normalized to 64 as '1' . unsigned , default = 0
21:16	R/W	0d63	reg_nr_alp1_ymax : : normalized to 64 as '1' . unsigned , default = 63
13: 8	R/W	0d0	reg_nr_alp1_cmin : : normalized to 64 as '1' . unsigned , default = 0
5: 0	R/W	0d63	reg_nr_alp1_cmax : : normalized to 64 as '1' . unsigned , default = 63

Table 8-2150 SRSHARP1_PK_ALP2_MIERR_CORING 0x3f1e

Bit(s)	R/W	Default	Description
16	R/W	0d1	reg_pk_alp2_maxerr_mode : : 0 max err; 1: xerr . unsigned , default = 1
13: 8	R/W	0d13	reg_pk_alp2_core_rate : : normalized 64 as "1" . unsigned , default = 13
5: 0	R/W	0d1	reg_pk_alp2_core_ofst : : normalized 64 as "1" . signed , default = 1

Table 8-2151 SRSHARP1_PK_ALP2_ERR2CURV_TH_RATE 0x3f1f

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_pk_alp2_minerr_par0 : : threshold0 of curve to map mierr to alp2 for luma channel, this will be set value of flat region mierr that no need peaking.. unsigned , default = 0
23:16	R/W	0d24	reg_pk_alp2_minerr_par1 : : threshold1 of curve to map mierr to alp2 for luma channel,this will be set value of texture region mierr that can not do peaking. 0~255.. unsigned , default = 24

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Bit(s)	R/W	Default	Description
15: 8	R/W	0d0	reg_pk_alp2_minerr_par5 : : rate0 (for mierr<th0) of curve to map mierr to alp2 for luma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 0
7: 0	R/W	0d20	reg_pk_alp2_minerr_par6 : : rate1 (for mierr>th1) of curve to map mierr to alp2 for luma channel. the larger of the value, the deep of the slope. 0~255.. unsigned , default = 20

Table 8-2152 SRSHARP1_PK_ALP2_ERR2CURV_LIMIT 0x3f20

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_pk_alp2_minerr_par2 : : level limit(for mierr<th0) of curve to map mierr to alp2 for luma channel, this will be set to alp2 that we can do for flat region. 0~255.. unsigned , default = 0
23:16	R/W	0d16	reg_pk_alp2_minerr_par3 : : level limit(for th0<mierr<th1) of curve to map mierr to alp2 for luma channel, this will be set to alp2 that we can do for misc region. 0~255.. unsigned , default = 16
15: 8	R/W	0d63	reg_pk_alp2_minerr_par4 : : level limit(for mierr>th1) of curve to map mierr to alp2 for luma channel, this will be set to alp2 that we can do for texture region. 0~255. default = 63;. unsigned , default = 255

Table 8-2153 SRSHARP1_PK_ALP2_MIN_MAX 0x3f21

Bit(s)	R/W	Default	Description
13: 8	R/W	0d0	reg_pk_alp2_min : : normalized to 64 as '1' . unsigned , default = 0
5: 0	R/W	0d63	reg_pk_alp2_max : : normalized to 64 as '1' . unsigned , default = 63

Table 8-2154 SRSHARP1_PK_FINALGAIN_HP_BP 0x3f22

Bit(s)	R/W	Default	Description
17: 16	R/W	0d0	reg_final_gain_rs : : right shift bits for the gain normalization, 0 normal to 32 as 1; 1 normal to 64 as 1; -2 normal to 8 as 1; -1 normal to 16 as 1 . signed , default = 0
15: 8	R/W	0d40	reg_hp_final_gain : : gain to highpass boost result (including directional/circle blending), normalized 32 as '1', 0~255. 1.25 * 32. unsigned , default = 40
7: 0	R/W	0d30	reg_bp_final_gain : : gain to bandpass boost result (including directional/circle blending), normalized 32 as '1', 0~255. 1.25 * 32. unsigned , default = 30

Table 8-2155 SRSHARP1_PK_OS_HORZ_CORE_GAIN 0x3f23

Bit(s)	R/W	Default	Description
31:24	R/W	0d8	reg_pk_os_hsidecore : : side coring (not to current pixel) to adaptive overshoot margin in horizontal direction. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 8
23:16	R/W	0d20	reg_pk_os_hsidegain : : side gain (not to current pixel) to adaptive overshoot margin in horizontal direction. normalized to 32 as '1'. 0~255;. unsigned , default = 20

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Bit(s)	R/W	Default	Description
15: 8	R/W	0d2	reg_pk_os_hmidcore : : midd coring (to current pixel) to adaptive overshoot margin in horizontal direction. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 2
7: 0	R/W	0d20	reg_pk_os_hmidgain : : midd gain (to current pixel) to adaptive overshoot margin in horizontal direction. normalized to 32 as '1'. 0~255;. unsigned , default = 20

Table 8-2156 SRSHARP1_PK_OS_VERT_CORE_GAIN 0x3f24

Bit(s)	R/W	Default	Description
31:24	R/W	0d8	reg_pk_os_vsidecore : : side coring (not to current pixel) to adaptive overshoot margin in vertical direction. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 8
23:16	R/W	0d20	reg_pk_os_vsidegain : : side gain (not to current pixel) to adaptive overshoot margin in vertical direction. normalized to 32 as '1'. 0~255;. unsigned , default = 20
15: 8	R/W	0d2	reg_pk_os_vmidcore : : midd coring (to current pixel) to adaptive overshoot margin in vertical direction. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 2
7: 0	R/W	0d20	reg_pk_os_vmidgain : : midd gain (to current pixel) to adaptive overshoot margin in vertical direction. normalized to 32 as '1'. 0~255;. unsigned , default = 20

Table 8-2157 SRSHARP1_PK_OS_ADPT_MISC 0x3f25

Bit(s)	R/W	Default	Description
31:24	R/W	0d40	reg_pk_os_minerr_core : : coring to minerr for adaptive overshoot margin. the larger of this value, the less overshoot admitted 0~255;. unsigned , default = 40
23:16	R/W	0d6	reg_pk_os_minerr_gain : : gain to minerr based adaptive overshoot margin. normalized to 64 as '1'. 0~255;. unsigned , default = 6
15: 8	R/W	0d200	reg_pk_os_adpt_max : : maximum limit adaptive overshoot margin (4x). 0~255; . unsigned , default = 200
7: 0	R/W	0d20	reg_pk_os_adpt_min : : minimun limit adaptive overshoot margin (1x). 0~255; . unsigned , default = 20

Table 8-2158 SRSHARP1_PK_OS_STATIC 0x3f26

Bit(s)	R/W	Default	Description
29:28	R/W	0d2	reg_pk_osh_mode : : 0~3: (2x+1) window in H direction . unsigned , default = 2
25:24	R/W	0d2	reg_pk_osv_mode : : 0~3: (2x+1) window in V direction . unsigned , default = 2
21:12	R/W	0d200	reg_pk_os_down : : static negative overshoot margin. 0~1023; . unsigned , default = 200
9: 0	R/W	0d200	reg_pk_os_up : : static positive overshoot margin. 0~1023; . unsigned , default = 200

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Table 8-2159 SRSHARP1_PK_NR_ENABLE 0x3f27

Bit(s)	R/W	Default	Description
3: 2	R/W	0d0	reg_3d_mode : , 0: no 3D; 1: L/R; 2: T/B; 3: horizontal interleaved, dft = 0 //. unsigned , default = 0
1	R/W	0d1	reg_pk_en : . unsigned , default = 1
0	R/W	0d1	reg_nr_en : . unsigned , default = 1

Table 8-2160 SRSHARP1_PK_DRT_SAD_MISC 0x3f28

Bit(s)	R/W	Default	Description
31:24	R/W	0d24	reg_pk_sad_ver_gain : : gain to sad[4], 16 normalized to "1"; . unsigned , default = 24
23:16	R/W	0d24	reg_pk_sad_hor_gain : : gain to sad[0], 16 normalized to "1"; . unsigned , default = 24
10: 9	R/W	0d0	reg_pk_bias_diag : : bias towards diag . unsigned , default = 0
4: 0	R/W	0d24	reg_pk_drt_force : : force direction of drt peaking filter, h2b: 0:hp drt force, 1: bp drt force; 2: bp+hp drt force, 3: no force;. unsigned , default = 24

Table 8-2161 SRSHARP1_NR_TI_DNLP_BLEND 0x3f29

Bit(s)	R/W	Default	Description
10: 8	R/W	0d4	reg_dnlp_input_mode : : dnlp input options. 0: org_y; 1: gau_y; 2: gauadp_y; 3: edgadplpf_y; 4: nr_y;5: lti_y; 6: pk_y (before os);7: pk_y (after os). unsigned , default = 4
3: 2	R/W	0d1	reg_nr_cti_blend_mode : : blend mode of nr and lti result: 0: nr; 1:cti; 2: (nr+cti)/2; 3:cti + dlt_nr . unsigned , default = 1
1: 0	R/W	0d1	reg_nr_lti_blend_mode : : blend mode of nr and lti result: 0: nr; 1:lti; 2: (nr+lti)/2; 3:lti + dlt_nr . unsigned , default = 1

Table 8-2162 SRSHARP1_TI_DIR_CORE_ALPHA 0x3f2a

Bit(s)	R/W	Default	Description
29:24	R/W	0d10	reg_adp_lti_dir_alp_core_ofst : : ofst to min_err, alpha = (min_err - (max_err-min_err)*rate + ofst)/max_err*64; dft=10. unsigned , default = 10
19:16	R/W	0d0	reg_adp_lti_dir_alp_core_rate : : ofset to min_err, alpha = (min_err - (max_err-min_err)*rate + ofst)/max_err*64; dft=0/32. unsigned , default = 0
13: 8	R/W	0d0	reg_adp_lti_dir_alpmin : : min value of alpha, alpha = (min_err+x +ofst)/max_err*64; dft=10 . unsigned , default = 0
5: 0	R/W	0d63	reg_adp_lti_dir_alpmax : : max value of alpha, alpha = (min_err+x +ofst)/max_err*64; dft=63 . unsigned , default = 63

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Table 8-2163 SRSHARP1_CTI_DIR_ALPHA 0x3f2b

Bit(s)	R/W	Default	Description
29:24	R/W	0d5	reg_adp_cti_dir_alp_core_ofst : : ofst to min_err, alpha = (min_err - (max_err-min_err)*rate + ofst)/max_err*64; dft=10. unsigned , default = 5
19:16	R/W	0d0	reg_adp_cti_dir_alp_core_rate : : ofset to min_err, alpha = (min_err - (max_err-min_err)*rate + ofst)/max_err*64; dft=0/32. unsigned , default = 0
13: 8	R/W	0d0	reg_adp_cti_dir_alpmin : : min value of alpha, alpha = (min_err +x+ofst)/max_ err*64; dft=10 . unsigned , default = 0
5: 0	R/W	0d63	reg_adp_cti_dir_alpmax : : max value of alpha, alpha = (min_err +x+ofst)/max_ err*64; dft=63 . unsigned , default = 63

Table 8-2164 SRSHARP1_LTI_CTI_DF_GAIN 0x3f2c

Bit(s)	R/W	Default	Description
29:24	R/W	0d16	reg_adp_lti_hdf_gain : : 8 normalized to "1"; default = 16 . unsigned , default = 16
21:16	R/W	0d12	reg_adp_lti_vdf_gain : : 8 normalized to "1"; default = 12 . unsigned , default = 12
13: 8	R/W	0d16	reg_adp_cti_hdf_gain : : 8 normalized to "1"; default = 16 . unsigned , default = 16
5: 0	R/W	0d12	reg_adp_cti_vdf_gain : : 8 normalized to "1"; default = 12 . unsigned , default = 12

Table 8-2165 SRSHARP1_LTI_CTI_DIR_AC_DBG 0x3f2d

Bit(s)	R/W	Default	Description
30	R/W	0d1	reg_adp_lti_dir_lpf : : 0: no lpf; 1: [1 2 2 2 1]/8 lpf . unsigned , default = 1
28	R/W	0d0	reg_adp_lti_dir_difmode : : 0: y_dif; 1: y_dif + (u_dif+v_dif)/2; . unsigned , default = 0
26	R/W	0d1	reg_adp_cti_dir_lpf : : 0: no lpf; 1: [1 2 2 2 1]/8 lpf dft=1 . unsigned , default = 1
25:24	R/W	0d0	reg_adp_cti_dir_difmode : : 0: (u_dif+v_dif); 1: y_dif/2 + (u_dif+v_dif)*3/4; 2: y_ dif + (u_dif+v_dif)/2; 3: y_dif*2 (not recomended). unsigned , default = 0
23:22	R/W	0d3	reg_adp_hvlti_dcblend_mode : : 0: hlti_dc; 1:vlti_dc; 2: avg 3; blend on alpha . unsigned , default = 3
21:20	R/W	0d3	reg_adp_hvcti_dcblend_mode : : 0: hcti_dc; 1:vcti_dc; 2: avg 3; blend on alpha . unsigned , default = 3
19:18	R/W	0d3	reg_adp_hvlti_acblend_mode : : hlti_ac; 1:vlti_ac; 2: add 3;:adaptive to alpha . unsigned , default = 3
17:16	R/W	0d3	reg_adp_hvcti_acblend_mode : : hcti_ac; 1:vcti_ac; 2: add 3;: adaptive to alpha . unsigned , default = 3
14:12	R/W	0d0	reg_adp_hlti_debug : , for hlti debug, default = 0 . unsigned , default = 0
10: 8	R/W	0d0	reg_adp_vlti_debug : , for vlti debug, default = 0 . unsigned , default = 0
6: 4	R/W	0d0	reg_adp_hcti_debug : , for hcti debug, default = 0 . unsigned , default = 0
2: 0	R/W	0d0	reg_adp_vcti_debug : , for vcti debug, default = 0 . unsigned , default = 0

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Table 8-2166 SRSHARP1_HCTI_FLT_CLP_DC 0x3f2e

Bit(s)	R/W	Default	Description
28	R/W	0d1	reg_adp_hcti_en : , 0: no cti, 1: new cti, default = 1 . unsigned , default = 1
27:26	R/W	0d3	reg_adp_hcti_vdn_flt : , 0: no lpf; 1:[0,2,4,2,0], 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
25:24	R/W	0d2	reg_adp_hcti_hdn_flt : , 0: no lpf; 1:[0, 0, 0, 4, 8, 4, 0, 0, 0], 2:[0, 0, 2, 4, 4, 4, 2, 0, 0], 3: [1, 2, 2, 2, 2, 2, 2, 2, 1], default = 2. unsigned , default = 2
23:22	R/W	0d3	reg_adp_hcti_ddn_flt : , 0: no lpf; 1:[0,2,4,2,0], 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
21:20	R/W	0d2	reg_adp_hcti_lpf0_flt : , 0:no filter; 1:sigma=0.75, 2: sigma = 1.0, 3: sigma = 1.5, default = 2 . unsigned , default = 2
19:18	R/W	0d2	reg_adp_hcti_lpf1_flt : , 0:no filter; 1:sigma= 2.0, 2: sigma = 3.0, 3: sigma = 4.0, default = 2 . unsigned , default = 2
17:16	R/W	0d2	reg_adp_hcti_lpf2_flt : , 0:no filter; 1:sigma=5.0, 2: sigma = 9.0, 3: sigma = 13.0, default = 2 . unsigned , default = 2
15:12	R/W	0d7	reg_adp_hcti_hard_clp_win : , window size, 0~8, default = 7 . unsigned , default = 7
11: 8	R/W	0d3	reg_adp_hcti_hard_win_min : , window size, 0~8, default = 3 . unsigned , default = 3
4	R/W	0d1	reg_adp_hcti_clp_mode : , 0: hard clip, 1: adaptive clip, default = 1 . unsigned , default = 1
2: 0	R/W	0d0	reg_adp_hcti_dc_mode : , 0:dn, 1:lpf0, 2:lpf1, 3:lpf2, 4: lpf3: 5: vdn result; 6/7: org, default = 0 . unsigned , default = 0

Table 8-2167 SRSHARP1_HCTI_BST_GAIN 0x3f2f

Bit(s)	R/W	Default	Description
31:24	R/W	0d80	reg_adp_hcti_bst_gain0 : : gain of the bandpass 0 (lpf1-lpf2)- LBP, default = 80. unsigned , default = 80
23:16	R/W	0d96	reg_adp_hcti_bst_gain1 : : gain of the bandpass 1 (lpf0-lpf1)- BP, default = 96. unsigned , default = 96
15: 8	R/W	0d64	reg_adp_hcti_bst_gain2 : : gain of the bandpass 2 (hdn-lpf0)- HP, default = 64. unsigned , default = 64
7: 0	R/W	0d16	reg_adp_hcti_bst_gain3 : : gain of the unsharp band (yuvin-hdn) - US, default = 16. unsigned , default = 16

Table 8-2168 SRSHARP1_HCTI_BST_CORE 0x3f30

Bit(s)	R/W	Default	Description
31:24	R/W	0d0	reg_adp_hcti_bst_core0 : : core of the bandpass 0 (lpf1-lpf2)- LBP, default = 0 . unsigned , default = 0
23:16	R/W	0d0	reg_adp_hcti_bst_core1 : : core of the bandpass 1 (lpf0-lpf1)- BP, default = 0 . unsigned , default = 0
15: 8	R/W	0d0	reg_adp_hcti_bst_core2 : : core of the bandpass 2 (hdn-lpf0)- HP, default = 0 . unsigned , default = 0
7: 0	R/W	0d0	reg_adp_hcti_bst_core3 : : core of the unsharp band (yuvin-hdn) - US, default = 0. unsigned , default = 0

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Table 8-2169 SRSHARP1_HCTI_CON_2_GAIN_0 0x3f31

Bit(s)	R/W	Default	Description
31:29	R/W	0d2	reg_adp_hcti_con_mode : : con mode 0:[0, 0,-1, 1, 0, 0, 0]+[0, 0, 0, 1,-1, 0, 0], 1: [0, 0,-1, 0, 1, 0, 0], 2: [0,-1, 0, 0, 0, 1, 0], 3:[-1, 0, 0, 0, 0, 0, 1], 4: .... default = 2. unsigned , default = 2
28:26	R/W	0d3	reg_adp_hcti_dx_mode : : dx mode 0: [-1 1 0]; 1~7: [-1 (2x+1)"0" 1], default = 3 . unsigned , default = 3
25:24	R/W	0d1	reg_adp_hcti_con_lpf : : lpf mode of the con: 0: [1 2 1]/4; 1:[1 2 2 2 1]/8, default = 1 . unsigned , default = 1
23:16	R/W	0d25	reg_adp_hcti_con_2_gain0 : , default = 25. unsigned , default = 25
15: 8	R/W	0d60	reg_adp_hcti_con_2_gain1 : , default = 60. unsigned , default = 60
7: 0	R/W	0d0	reg_adp_hcti_con_2_gain2 : 0;, default = 0 . unsigned , default = 0

Table 8-2170 SRSHARP1_HCTI_CON_2_GAIN_1 0x3f32

Bit(s)	R/W	Default	Description
31:24	R/W	0d96	reg_adp_hcti_con_2_gain3 : 96;, default = 96 . unsigned , default = 96
23:16	R/W	0d5	reg_adp_hcti_con_2_gain4 : 5;, default = 5 . unsigned , default = 5
15: 8	R/W	0d80	reg_adp_hcti_con_2_gain5 : 80;, default = 80 . unsigned , default = 80
7: 0	R/W	0d20	reg_adp_hcti_con_2_gain6 : 20;, default = 20 . unsigned , default = 20

Table 8-2171 SRSHARP1_HCTI_OS_MARGIN 0x3f33

Bit(s)	R/W	Default	Description
7: 0	R/W	0d0	reg_adp_hcti_os_margin : : margin for hcti overshoot, default = 0 . unsigned , default = 0

Table 8-2172 SRSHARP1_HLTI_FLT_CLP_DC 0x3f34

Bit(s)	R/W	Default	Description
28	R/W	0d1	reg_adp_hlti_en : , 0: no cti, 1: new cti, default = 1 . unsigned , default = 1
27:26	R/W	0d2	reg_adp_hlti_vdn_flt : , 0: no lpf; 1:[0,2,4,2,0], 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 2 . unsigned , default = 2
25:24	R/W	0d2	reg_adp_hlti_hdn_flt : , 0: no lpf; 1:[0, 0, 0, 4, 8, 4, 0, 0, 0], 2:[0, 0, 2, 4, 4, 4, 2, 0, 0], 3: [1, 2, 2, 2, 2, 2, 2, 2, 1], default = 2. unsigned , default = 2
23:22	R/W	0d2	reg_adp_hlti_ddn_flt : , 0: no lpf; 1:[0,2,4,2,0], 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 2 . unsigned , default = 2
21:20	R/W	0d2	reg_adp_hlti_lpf0_flt : , 0:no filter; 1:sigma=0.75, 2: sigma = 1.0, 3: sigma = 1.5, default = 2 . unsigned , default = 2
19:18	R/W	0d2	reg_adp_hlti_lpf1_flt : , 0:no filter; 1:sigma= 2.0, 2: sigma = 3.0, 3: sigma = 4.0, default = 2 . unsigned , default = 2
17:16	R/W	0d2	reg_adp_hlti_lpf2_flt : , 0:no filter; 1:sigma=5.0, 2: sigma = 9.0, 3: sigma = 13.0, default = 2 . unsigned , default = 2

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Bit(s)	R/W	Default	Description
15:12	R/W	0d2	reg_adp_hlti_hard_clp_win : , window size, 0~8, default = 2 . unsigned , default = 2
11: 8	R/W	0d1	reg_adp_hlti_hard_win_min : , window size, 0~8, default = 1 . unsigned , default = 1
4	R/W	0d0	reg_adp_hlti_clp_mode : , 0: hard clip, 1: adaptive clip, default = 0 . unsigned , default = 0
2: 0	R/W	0d4	reg_adp_hlti_dc_mode : , 0:dn, 1:lpf0, 2:lpf1, 3:lpf2, 4: lpf3: 5: vdn result; 6/7:org, default = 4 . unsigned , default = 4

Table 8-2173 SRSHARP1_HLTI_BST_GAIN 0x3f35

Bit(s)	R/W	Default	Description
31:24	R/W	0d40	reg_adp_hlti_bst_gain0 : : gain of the bandpass 0 (lpf1-lpf2)- LBP, default = 40 . unsigned , default = 40
23:16	R/W	0d48	reg_adp_hlti_bst_gain1 : : gain of the bandpass 1 (lpf0-lpf1)- BP, default = 48 . unsigned , default = 48
15: 8	R/W	0d32	reg_adp_hlti_bst_gain2 : : gain of the bandpass 2 (hdn-lpf0)- HP, default = 32 . unsigned , default = 32
7: 0	R/W	0d16	reg_adp_hlti_bst_gain3 : : gain of the unsharp band (yuvin-hdn) - US, default = 16 . unsigned , default = 16

Table 8-2174 SRSHARP1_HLTI_BST_CORE 0x3f36

Bit(s)	R/W	Default	Description
31:24	R/W	0d5	reg_adp_hlti_bst_core0 : : core of the bandpass 0 (lpf1-lpf2)- LBP, default = 5 . unsigned , default = 5
23:16	R/W	0d5	reg_adp_hlti_bst_core1 : : core of the bandpass 1 (lpf0-lpf1)- BP, default = 5 . unsigned , default = 5
15: 8	R/W	0d5	reg_adp_hlti_bst_core2 : : core of the bandpass 2 (hdn-lpf0)- HP, default = 5 . unsigned , default = 5
7: 0	R/W	0d3	reg_adp_hlti_bst_core3 : : core of the unsharp band (yuvin-hdn) - US, default = 3. unsigned , default = 3

Table 8-2175 SRSHARP1_HLTI_CON_2_GAIN_0 0x3f37

Bit(s)	R/W	Default	Description
31:29	R/W	0d2	reg_adp_hlti_con_mode : : con mode 0:[0, 0,-1, 1, 0, 0, 0]+[0, 0, 0, 1,-1, 0, 0], 1: [0, 0,-1, 0, 1, 0, 0], 2: [0,-1, 0, 0, 0, 1, 0], 3:[-1, 0, 0, 0, 0, 0, 1], 4: ....., default = 2. unsigned , default = 2
28:26	R/W	0d3	reg_adp_hlti_dx_mode : : dx mode 0: [-1 1 0]; 1~7: [-1 (2x+1)"0" 1], default = 3 . unsigned , default = 3
25:24	R/W	0d1	reg_adp_hlti_con_lpf : : lpf mode of the con: 0: [1 2 1]/4; 1:[1 2 2 2 1]/8, default = 1 . unsigned , default = 1
23:16	R/W	0d25	reg_adp_hlti_con_2_gain0 : 25;, default = 25 . unsigned , default = 25
15: 8	R/W	0d60	reg_adp_hlti_con_2_gain1 : 60;, default = 60 . unsigned , default = 60
7: 0	R/W	0d90	reg_adp_hlti_con_2_gain2 : 0;, default = 90 . unsigned , default = 90

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Table 8-2176 SRSHARP1_HLTI_CON_2_GAIN_1 0x3f38

Bit(s)	R/W	Default	Description
31:24	R/W	0d96	reg_adp_hlti_con_2_gain3 : 96;, default = 96 . unsigned , default = 96
23:16	R/W	0d95	reg_adp_hlti_con_2_gain4 : 5;, default = 95 . unsigned , default = 95
15: 8	R/W	0d80	reg_adp_hlti_con_2_gain5 : 80;, default = 80 . unsigned , default = 80
7: 0	R/W	0d20	reg_adp_hlti_con_2_gain6 : 20;, default = 20 . unsigned , default = 20

Table 8-2177 SRSHARP1_HLTI_OS_MARGIN 0x3f39

Bit(s)	R/W	Default	Description
7: 0	R/W	0d0	reg_adp_hlti_os_margin : : margin for hlti overshoot, default = 0 . unsigned , default = 0

Table 8-2178 SRSHARP1_VLTI_FLT_CON_CLP 0x3f3a

Bit(s)	R/W	Default	Description
14	R/W	0d1	reg_adp_vlti_en : : enable bit of vlti, default = 1 . unsigned , default = 1
13:12	R/W	0d3	reg_adp_vlti_hxn_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
11:10	R/W	0d3	reg_adp_vlti_dxn_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
9: 8	R/W	0d3	reg_adp_vlti_han_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
7: 6	R/W	0d3	reg_adp_vlti_dan_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
5: 4	R/W	0d2	reg_adp_vlti_dx_mode : : 0:[-1 1] 1:[-1 0 -1]; 2/3: [-1 0 0 0 -1], default = 2 . unsigned , default = 2
2	R/W	0d1	reg_adp_vlti_con_lpf : : lpf mode of the con: 0: [1 2 1]/4; 1:[1 2 2 2 1]/8, default = 1 . unsigned , default = 1
0	R/W	0d1	reg_adp_vlti_hard_clp_win : : window size; 0: 1x3 window; 1: 1x5 window, default = 1. unsigned , default = 1

Table 8-2179 SRSHARP1_VLTI_BST_GAIN 0x3f3b

Bit(s)	R/W	Default	Description
23:16	R/W	0d32	reg_adp_vlti_bst_gain0 : : gain to boost filter [-1 2 -1];, default = 32. unsigned , default = 32
15: 8	R/W	0d32	reg_adp_vlti_bst_gain1 : : gain to boost filter [-1 0 2 0 -1];, default = 32 . unsigned , default = 32
7: 0	R/W	0d32	reg_adp_vlti_bst_gain2 : : gain to boost filter usf, default = 32 . unsigned , default = 32

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Table 8-2180 SRSHARP1_VLTI_BST_CORE 0x3f3c

Bit(s)	R/W	Default	Description
23:16	R/W	0d5	reg_adp_vlti_bst_core0 : : coring to boost filter [-1 2 -1];, default = 5 . unsigned , default = 5
15: 8	R/W	0d5	reg_adp_vlti_bst_core1 : : coring to boost filter [-1 0 2 0 -1];, default = 5 . unsigned , default = 5
7: 0	R/W	0d3	reg_adp_vlti_bst_core2 : : coring to boost filter usf, default = 3 . unsigned , default = 3

Table 8-2181 SRSHARP1_VLTI_CON_2_GAIN_0 0x3f3d

Bit(s)	R/W	Default	Description
31:24	R/W	0d25	reg_adp_vlti_con_2_gain0 : 25;, default = 25 . unsigned , default = 25
23:16	R/W	0d69	reg_adp_vlti_con_2_gain1 : 60;, default = 69 . unsigned , default = 60
15: 8	R/W	0d90	reg_adp_vlti_con_2_gain2 : 0;, default = 90 . unsigned , default = 90
7: 0	R/W	0d96	reg_adp_vlti_con_2_gain3 : 96;, default = 96 . unsigned , default = 96

Table 8-2182 SRSHARP1_VLTI_CON_2_GAIN_1 0x3f3e

Bit(s)	R/W	Default	Description
31:24	R/W	0d95	reg_adp_vlti_con_2_gain4 : 5;, default = 95 . unsigned , default = 95
23:16	R/W	0d80	reg_adp_vlti_con_2_gain5 : 80;, default = 80 . unsigned , default = 80
15: 8	R/W	0d20	reg_adp_vlti_con_2_gain6 : 20;, default = 20 . unsigned , default = 20
7: 0	R/W	0d0	reg_adp_vlti_os_margin : : margin for vlti overshoot, default = 0 . unsigned , default = 0

Table 8-2183 SRSHARP1_VCTI_FLT_CON_CLP 0x3f3f

Bit(s)	R/W	Default	Description
14	R/W	0d1	reg_adp_vcti_en : : enable bit of vlti, default = 1 . unsigned , default = 1
13:12	R/W	0d3	reg_adp_vcti_hxn_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
11:10	R/W	0d3	reg_adp_vcti_dxn_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
9: 8	R/W	0d3	reg_adp_vcti_han_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
7: 6	R/W	0d3	reg_adp_vcti_dan_flt : : 0: no dn; 1: [1 2 1]/4; 2 : [1 2 2 2 1]/8 3:[1 0 2 0 1]/4, default = 3 . unsigned , default = 3
5: 4	R/W	0d2	reg_adp_vcti_dx_mode : : 0:[-1 1] 1:[-1 0 -1]; 2/3: [-1 0 0 0 -1], default = 2 . unsigned , default = 2
2	R/W	0d1	reg_adp_vcti_con_lpf : : lpf mode of the con: 0: [1 2 1]/4; 1:[1 2 2 2 1]/8, default = 1 . unsigned , default = 1
0	R/W	0d1	reg_adp_vcti_hard_clp_win : : window size; 0: 1x3 window; 1: 1x5 window, default = 1. unsigned , default = 1

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Table 8-2184 SRSHARP1_VCTI_BST_GAIN 0x3f40

Bit(s)	R/W	Default	Description
23:16	R/W	0d0	reg_adp_vcti_bst_gain0 : : gain to boost filter [-1 2 -1];, default = 0 . unsigned , default = 0
15: 8	R/W	0d0	reg_adp_vcti_bst_gain1 : : gain to boost filter [-1 0 2 0 -1];, default = 0 . unsigned , default = 0
7: 0	R/W	0d0	reg_adp_vcti_bst_gain2 : : gain to boost filter usf, default = 0 . unsigned , default = 0

Table 8-2185 SRSHARP1_VCTI_BST_CORE 0x3f41

Bit(s)	R/W	Default	Description
23:16	R/W	0d0	reg_adp_vcti_bst_core0 : : coring to boost filter [-1 2 -1];, default = 0 . unsigned , default = 0
15: 8	R/W	0d0	reg_adp_vcti_bst_core1 : : coring to boost filter [-1 0 2 0 -1];, default = 0 . unsigned , default = 0
7: 0	R/W	0d0	reg_adp_vcti_bst_core2 : : coring to boost filter usf, default = 0 . unsigned , default = 0

Table 8-2186 SRSHARP1_VCTI_CON_2_GAIN_0 0x3f42

Bit(s)	R/W	Default	Description
31:24	R/W	0d25	reg_adp_vcti_con_2_gain0 : 25;, default = 25 . unsigned , default = 25
23:16	R/W	0d60	reg_adp_vcti_con_2_gain1 : 60;, default = 60 . unsigned , default = 60
15: 8	R/W	0d90	reg_adp_vcti_con_2_gain2 : 0;, default = 90 . unsigned , default = 90
7: 0	R/W	0d96	reg_adp_vcti_con_2_gain3 : 96;, default = 96 . unsigned , default = 96

Table 8-2187 SRSHARP1_VCTI_CON_2_GAIN_1 0x3f43

Bit(s)	R/W	Default	Description
31:24	R/W	0d95	reg_adp_vcti_con_2_gain4 : 5;, default = 95 . unsigned , default = 95
23:16	R/W	0d80	reg_adp_vcti_con_2_gain5 : 80;, default = 80 . unsigned , default = 80
15: 8	R/W	0d20	reg_adp_vcti_con_2_gain6 : 20;, default = 20 . unsigned , default = 20
7: 0	R/W	0d0	reg_adp_vcti_os_margin : : margin for vcti overshoot, default = 0 . unsigned , default = 0

Table 8-2188 SRSHARP1_SHARP_3DLIMIT 0x3f44

Bit(s)	R/W	Default	Description
28:16	R/W	0d0	reg_3d_mid_width : ,width of left part of 3d input, dft = half size of input width default = 0 . unsigned , default = 960
12: 0	R/W	0d0	reg_3d_mid_height : ,height of left part of 3d input, dft = half size of input height default = 0 . unsigned , default = 540

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Table 8-2189 SRSHARP1_DNLP_EN 0x3f45

Bit(s)	R/W	Default	Description
15:8	R/W	0d0	reg_dnlp_hblank: . unsigned , default = 8
0	R/W	0d1	reg_dnlp_en : . unsigned , default = 1

Table 8-2190 SRSHARP1_DEMO_CRTL 0x3f56

Bit(s)	R/W	Default	Description
18:17	R/W	0d2	demo_disp_position : . unsigned , default = 2
16	R/W	0d0	demo_hsvsharp_enable : . unsigned , default = 0
12: 0	R/W	0d360	demo_left_top_screen_width : : . unsigned , default = 360

Table 8-2191 SRSHARP1_SHARP_SR2_CTRL 0x3f57

Bit(s)	R/W	Default	Description
24	R/W	0	reg_sr2_bic_pknr_bypass : bypass peaking/TI/Cubic
23:22	R/W		reserved
21:16	R/W	24	sr2_pk_la_err_dis_rate, low angle and high angle error should not be no less than nearby_error* rate/64
15: 8	R/W	16	sr2_pk_sad_diag_gain, gain to sad[2] and sad[6], 16 normalized to 1
7	R/W	0	sr2_vert_outphs, vertical output pixel phase, 0: 0 phase; 1: 1/2 phase
6	R/W	0	sr2_horz_outphs, horizontal output pixel phase, 0: 0 phase; 1: 1/2 phase
5	R/W	0	sr2_vert_ratio , vertical scale ratio, 0-> 1:1; 1-> 1:2
4	R/W	0	sr2_horz_ratio , horizontal scale ratio, 0-> 1:1; 1-> 1:2
3	R/W	1	sr2_ bic_norm , normalization of bicubical: 0: 128; 1: 64
2	R/W	0	sr2_ enable , 1 to enable super scaler
1	R/W	0	sr2_ sharp_prc_lr_hbic,
0	R/W	0	sr2_ sharp_prc_lr, 1: LTI/CTI/NR/Peaking processing using LR grid. 0: on HR grid; 1:on LR grid, horizontally no upscale, but using simple bic.

Table 8-2192 SRSHARP1_SHARP_SR2_YBIC_HCOEF0 0x3f58

Bit(s)	R/W	Default	Description
31:24	R/W	0	sr2_y_bic_hcoeff03, signed
23:16	R/W	0	sr2_y_bic_hcoeff02, signed
15: 8	R/W	64	sr2_y_bic_hcoeff01, signed
7: 0	R/W	0	sr2_y_bic_hcoeff00, signed

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Table 8-2193 SRSHARP1_SHARP_SR2_YBIC_HCOEF1 0x3f59

Bit(s)	R/W	Default	Description
31:24	R/W	-4	sr2_y_bic_hcoeff13 , signed
23:16	R/W	36	sr2_y_bic_hcoeff12 , signed
15: 8	R/W	36	sr2_y_bic_hcoeff11 , signed
7: 0	R/W	-4	sr2_y_bic_hcoeff10 , signed

Table 8-2194 SRSHARP1_SHARP_SR2_CBIC_HCOEF0 0x3f5a

Bit(s)	R/W	Default	Description
31:24	R/W	0	sr2_c_bic_hcoeff03 , signed
23:16	R/W	21	sr2_c_bic_hcoeff02 , signed
15: 8	R/W	22	sr2_c_bic_hcoeff01 , signed
7: 0	R/W	21	sr2_c_bic_hcoeff00 , signed

Table 8-2195 SRSHARP1_SHARP_SR2_CBIC_HCOEF1 0x3f5b

Bit(s)	R/W	Default	Description
31:24	R/W	-4	sr2_c_bic_hcoeff13 , signed
23:16	R/W	36	sr2_c_bic_hcoeff12 , signed
15: 8	R/W	36	sr2_c_bic_hcoeff11 , signed
7: 0	R/W	-4	sr2_c_bic_hcoeff10 , signed

Table 8-2196 SHARP_SR2_YBIC_VCOEF0 0x3f5c

Bit(s)	R/W	Default	Description
31:24	R/W	0	sr2_y_bic_vcoeff03 , signed
23:16	R/W	0	sr2_y_bic_vcoeff02 , signed
15: 8	R/W	64	sr2_y_bic_vcoeff01 , signed
7: 0	R/W	0	sr2_y_bic_vcoeff00 , signed

Table 8-2197 SRSHARP1_SHARP_SR2_YBIC_VCOEF1 0x3f5d

Bit(s)	R/W	Default	Description
31:24	R/W	-4	sr2_y_bic_vcoeff13 , signed
23:16	R/W	36	sr2_y_bic_vcoeff12 , signed
15: 8	R/W	36	sr2_y_bic_vcoeff11 , signed
7: 0	R/W	-4	sr2_y_bic_vcoeff10 , signed

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Table 8-2198 SRSHARP1_SHARP_SR2_CBIC_VCOEF0 0x3f5e

Bit(s)	R/W	Default	Description
31:24	R/W	0	sr2_c_bic_vcoeff03 , signed
23:16	R/W	21	sr2_c_bic_vcoeff02 , signed
15: 8	R/W	22	sr2_c_bic_vcoeff01 , signed
7: 0	R/W	21	sr2_c_bic_vcoeff00 , signed

Table 8-2199 SRSHARP1_SHARP_SR2_CBIC_VCOEF1 0x3f5f

Bit(s)	R/W	Default	Description
31:24	R/W	-4	sr2_c_bic_vcoeff13 , signed
23:16	R/W	36	sr2_c_bic_vcoeff12 , signed
15: 8	R/W	36	sr2_c_bic_vcoeff11 , signed
7: 0	R/W	-4	sr2_c_bic_vcoeff10 , signed

Table 8-2200 SRSHARP1_SHARP_SR2_MISC 0x3f60

Bit(s)	R/W	Default	Description
31:2	R/W		reserved
1	R/W	0	sr2_cmpmux_bef , 0 : no swap for YUV/RGB; 1: swap for YUV/RGB, YUV/RGB->UVY/GBR
0	R/W	0	sr2_cmpmux_aft , 0 : no swap for YUV/RGB; 1: swap for YUV/RGB, UVY/GBR->YUV/RGB

Table 8-2201 SRSHARP1_SR3_SAD_CTRL 0x3f61

Bit(s)	R/W	Default	Description
31:30	R/W		reserved
29:24	R/W	0d6	reg_sr3_pk_sad_core_rate : rate of coring.
23:22	R/W		reserved
21:16	R/W	0d6	reg_sr3_lti_sad_core_rate : rate of coring.
15:14	R/W		reserved
13:8	R/W	0d6	reg_sr3_cti_sad_core_rate : rate of coring.
7	R/W	0d1	reg_sr3_lti_hsad_mode: mode for hsad of lti calculation, 0:block based; 1: other sharp
6	R/W	0d1	reg_sr3_cti_hsad_mode: mode for hsad of cti calculation, 0:block based; 1: other sharp
5	R/W	0d1	reg_sr3_lti_dsad_mode: mode for dsad of lti calculation, 0:block based; 1: other sharp
4	R/W	0d1	reg_sr3_cti_dsad_mode: mode for dsad of cti calculation, 0:block based; 1: other sharp
3	R/W	0d1	reg_sr3_lti_vsad_mode: mode for vsad of lti calculation, 0:block based; 1: other sharp

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Bit(s)	R/W	Default	Description
2	R/W	0d1	reg_sr3_cti_vsad_mode: mode for vsad of cti calculation, 0:block based; 1: other sharp
1	R/W	0d1	reg_sr3_lti_hsad_hlpf: hlpf for hsad of lti calculation, 0:no hlpf; 1: with [1 2 1] hlpf.
0	R/W	0d1	reg_sr3_cti_hsad_hlpf: hlpf for hsad of cti calculation, 0:no hlpf; 1: with [1 2 1] hlpf.

Table 8-2202 SRSHARP1_SR3_PK_CTRL0 0x3f62

Bit(s)	R/W	Default	Description
31:12	R/W		reserved
11	R/W	0d1	reg_sr3_pk_sad_mode: mode for sad of peaking calculation, 0: block based; 1: other sharp.
10	R/W	0d1	reg_sr3_pk_hsad_hlpf: hlpf for hsad for peaking calculation,0:no hlpf; 1: with [1 2 2 2 1] hlpf.
9	R/W	0d1	reg_sr3_pk_vsad_hlpf: hlpf for vsad for peaking calculation,0:no hlpf; 1: with [1 2 2 2 1] hlpf.
8	R/W	0d1	reg_sr3_pk_dsad_hlpf: hlpf for dsad for peaking calculation,0:no hlpf; 1: with [1 2 2 2 1] hlpf.
7:6	R/W	0d3	reg_sr3_pk_hpdrt_mode: mode for HPdrt filter
5:4	R/W	0d3	reg_sr3_pk_bpdrt_mode: mode for BPdrt filter
3:2	R/W	0d3	reg_pk_drtbld_range: range of the min2 and min direction distance
1	R/W		reserved
0	R/W	0d0	reg_sr3_pk_ti_blend_mode: blend mode of the TI and PK result

Table 8-2203 SRSHARP1_SR3_PK_CTRL1 0x3f63

Bit(s)	R/W	Default	Description
31	R/W		reserved
30:28	R/W	0d1	reg_sr3_pk_Hp_hvcon_replace8_maxsad: replace HP hvcon by maxsad
26:24	R/W	0d1	reg_sr3_pk_bp_hvcon_replace8_maxsad: replace BP hvcon by maxsad
23:16	R/W	0d32	reg_sr3_pk_hp_hvcon_replace8lv_gain: gain to local variant before calculating the hv gain for peaking.
15:8	R/W	0d32	reg_sr3_pk_bp_hvcon_replace8lv_gain: gain to local variant before calculating the hv gain for peaking.
7	R/W	0d1	reg_sr3_sad_intlev_mode: interleave detect xerr mode: 0 max; 1: sum
6	R/W	0d1	reg_sr3_sad_intlev_mode1: mode 1 of using diagonal protection: 1: with diagonal protection
5:0	R/W	0d12	reg_sr3_sad_intlev_gain: interleave detection for sad gain applied, normalized to 8 as 1

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Table 8-2204 SRSHARP1_DEJ_CTRL 0x3f64

Bit(s)	R/W	Default	Description
15:8	R/W	8	reg_sr3_dejaggy_hblank
7:4	R/W		reserved
3:2	R/W	0d3	reg_sr3_dejaggy_sameside_prtct: enable of sr3 dejaggy same side curve protect from filter, [0] for proc path; [1] for ctrl path.
1	R/W	0d1	reg_sr3_dejaggy_sameside_mode: mode of sameside flag decision
0	R/W	0d1	reg_sr3_dejaggy_enable: enable of sr3 dejaggy

Table 8-2205 SRSHARP1_DEJ_ALPHA 0x3f65

Bit(s)	R/W	Default	Description
31:28	R/W	0d0	reg_sr3_dejaggy_ctrlchrm_alpha_1 : alpha for LR video LPF
27:24	R/W	0d15	reg_sr3_dejaggy_ctrlchrm_alpha_0 : alpha for LR video LPF
23:20	R/W	0d0	reg_sr3_dejaggy_ctrlluma_alpha_1 : alpha for LR video LPF
19:16	R/W	0d15	reg_sr3_dejaggy_ctrlluma_alpha_0 : alpha for LR video LPF
15:12	R/W	0d4	reg_sr3_dejaggy_procchrm_alpha_1: alpha for LR video LPF
11:8	R/W	0d6	reg_sr3_dejaggy_procchrm_alpha_0: alpha for LR video LPF
7:4	R/W	0d4	reg_sr3_dejaggy_procluma_alpha_1: alpha for LR video LPF
3:0	R/W	0d6	reg_sr3_dejaggy_procluma_alpha_0: alpha for LR video LPF

Table 8-2206 SRSHARP1_SR3_DRTLPF_EN 0x3f66

Bit(s)	R/W	Default	Description
31:15	R/W		reserved
14:8	R/W	0d0	reg_pk_debug_edge
6:4	R/W	0d0	reg_sr3_drtlpf_theta_en
2:0	R/W	0d7	reg_sr3_drtlpf_enable: directional lpf on Y/U/V channels

Table 8-2207 SRSHARP1_SR3_DRTLPF_ALPHA_0 0x3f67

Bit(s)	R/W	Default	Description
31:28	R/W		reserved
27:24	R/W	0d9	reg_sr3_drtlpf_alpha_3
23:20	R/W		reserved
19:16	R/W	0d10	reg_sr3_drtlpf_alpha_2
15:12	R/W		reserved
11:8	R/W	0d11	reg_sr3_drtlpf_alpha_1

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Bit(s)	R/W	Default	Description
7:4	R/W		reserved
3:0	R/W	0d12	reg_sr3_drtlpf_alpha_0: directional lpf alpha coef for min_sad/max_sad compare

Table 8-2208 SRSHARP1_SR3_DRTLPF_ALPHA_1 0x3f68

Bit(s)	R/W	Default	Description
31:28	R/W		reserved
27:24	R/W	0d1	reg_sr3_drtlpf_alpha_7
23:20	R/W		reserved
19:16	R/W	0d4	reg_sr3_drtlpf_alpha_6
15:12	R/W		reserved
11:8	R/W	0d7	reg_sr3_drtlpf_alpha_5
7:4	R/W		reserved
3:0	R/W	0d8	reg_sr3_drtlpf_alpha_4: directional lpf alpha coef for min_sad/max_sad compare

Table 8-2209 SRSHARP1_SR3_DRTLPF_ALPHA_2 0x3f69

Bit(s)	R/W	Default	Description
31:28	R/W		reserved
27:24	R/W	0d0	reg_sr3_drtlpf_alpha_11
23:20	R/W		reserved
19:16	R/W	0d0	reg_sr3_drtlpf_alpha_10
15:12	R/W		reserved
11:8	R/W	0d0	reg_sr3_drtlpf_alpha_9
7:4	R/W		reserved
3:0	R/W	0d0	reg_sr3_drtlpf_alpha_8: directional lpf alpha coef for min_sad/max_sad compare

Table 8-2210 SRSHARP1_SR3_DRTLPF_ALPHA_OFST 0x3f6a

Bit(s)	R/W	Default	Description
31:28	R/W	0	reg_sr3_drtlpf_alpha_ofst7
27:24	R/W	0	reg_sr3_drtlpf_alpha_ofst6
23:20	R/W	0	reg_sr3_drtlpf_alpha_ofst5
19:16	R/W	-2	reg_sr3_drtlpf_alpha_ofst4
15:12	R/W	0	reg_sr3_drtlpf_alpha_ofst3
11:8	R/W	0	reg_sr3_drtlpf_alpha_ofst2

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Bit(s)	R/W	Default	Description
7:4	R/W	0	reg_sr3_drtlpf_alpha_ofst1
3:0	R/W	-2	reg_sr3_drtlpf_alpha_ofst0: directional lpf alpha coef offset of each direction.

Table 8-2211 SRSHARP1_SR3_DERING_CTRL 0x3f6b

Bit(s)	R/W	Default	Description
31	R/W		reserved
30:28	R/W	1	reg_sr3_dering_enable: dering enable
27	R/W		reserved
26:24	R/W	3	reg_sr3_dering_varlpf_mode: local variant LPF mode. 0: no filter; 1: errosion 3x3; 2: 3x3 lpf; 3: 3x3 errosion + lpf
23:20	R/W	9	reg_sr3_dering_maxrange: range of dering in LR resolution.
19:18	R/W		reserved
17:16	R/W	2	reg_sr3_dering_lcvar_blend_mode: mode for lcvar calculation. 0:HV blend; 1:di-ag blend; 2:HV blend + V; 3: HV blend+Diag blend
15:8	R/W	40	reg_sr3_dering_lcvar_gain: gain to local variant and normalized to 32 as 1
7:0	R/W	28	reg_sr3_dering_lcvar_nearby_maxsad_th: threshold to use near side maxsad if that side is larger than this threshold, otherwise use the max one.

Table 8-2212 SRSHARP1_SR3_DERING_LUMA2PKGAIN_0TO3 0x3f6c

Bit(s)	R/W	Default	Description
31:24	R/W	255	reg_sr3_dering_luma2pkgain3: level limit(for th0<bpcon<th1) of curve for dering pkgain base on LPF luma level
23:16	R/W	255	reg_sr3_dering_luma2pkgain2: level limit(for bpcon<th0) of curve for dering pkgain base on LPF luma level
15:8	R/W	200	reg_sr3_dering_luma2pkgain1: threshold 1 of curve for dering pkgain based on LPF luma level.
7:0	R/W	30	reg_sr3_dering_luma2pkgain0: threshold 0 of curve for dering pkgain based on LPF luma level.

Table 8-2213 SRSHARP1_SR3_DERING_LUMA2PKGAIN_4TO6 0x3f6d

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	24	reg_sr3_dering_luma2pkgain6: rate1 (for bpcon>th1) of curve for dering pkOS based on LPF luma level.
15:8	R/W	50	reg_sr3_dering_luma2pkgain5: rate0 (for bpcon<th0) of curve for dering pkOS based on LPF luma level.
7:0	R/W	255	reg_sr3_dering_luma2pkgain4: level limit(for bpcon>th1) of curve for dering pkgain base on LPF luma level

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Table 8-2214 SRSHARP1_SR3_DERING_LUMA2PKOS_0TO3 0x3f6e

Bit(s)	R/W	Default	Description
31:24	R/W	255	reg_sr3_dering_luma2pkos3: level limit(for th0<bpcon<th1) of curve for dering pkOS base on LPF luma level
23:16	R/W	255	reg_sr3_dering_luma2pkos2: level limit(for bpcon<th0) of curve for dering pkOS base on LPF luma level
15:8	R/W	200	reg_sr3_dering_luma2pkos1: threshold 1 of curve for dering pkOS based on LPF luma level.
7:0	R/W	30	reg_sr3_dering_luma2pkos0: threshold 0 of curve for dering pkOS based on LPF luma level.

Table 8-2215 SRSHARP1_SR3_DERING_LUMA2PKOS_4TO6 0x3f6f

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	24	reg_sr3_dering_luma2pkos6: rate1 (for bpcon>th1) of curve for dering pkOS based on LPF luma level.
15:8	R/W	50	reg_sr3_dering_luma2pkos5: rate0 (for bpcon<th0) of curve for dering pkOS based on LPF luma level.
7:0	R/W	255	reg_sr3_dering_luma2pkos4: level limit(for bpcon>th1) of curve for dering pkOS base on LPF luma level

Table 8-2216 SRSHARP1_SR3_DERING_GAINVS_MADSAD 0x3f70

Bit(s)	R/W	Default	Description
31:28	R/W	0	reg_sr3_dering_gainvs_maxsad7
27:24	R/W	0	reg_sr3_dering_gainvs_maxsad6
23:20	R/W	0	reg_sr3_dering_gainvs_maxsad5
19:16	R/W	0	reg_sr3_dering_gainvs_maxsad4
15:12	R/W	0	reg_sr3_dering_gainvs_maxsad3
11:8	R/W	0	reg_sr3_dering_gainvs_maxsad2
7:4	R/W	4	reg_sr3_dering_gainvs_maxsad1
3:0	R/W	8	reg_sr3_dering_gainvs_maxsad0: pkgain vs maxsad value, 8 node interpolations.

Table 8-2217 SRSHARP1_SR3_DERING_GAINVS_VR2MAX 0x3f71

Bit(s)	R/W	Default	Description
31:28	R/W	15	reg_sr3_dering_gainvs_vr2max7
27:24	R/W	15	reg_sr3_dering_gainvs_vr2max6
23:20	R/W	15	reg_sr3_dering_gainvs_vr2max5
19:16	R/W	15	reg_sr3_dering_gainvs_vr2max4
15:12	R/W	14	reg_sr3_dering_gainvs_vr2max3

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Bit(s)	R/W	Default	Description
11:8	R/W	12	reg_sr3_dering_gainvs_vr2max2
7:4	R/W	2	reg_sr3_dering_gainvs_vr2max1
3:0	R/W	0	reg_sr3_dering_gainvs_vr2max0: pkgain vs ratio

Table 8-2218 SRSHARP1_SR3_DERING_PARAM0 0x3f72

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:16	R/W	10	reg_sr3_dering_lcvar_floor
15:8	R/W	32	reg_sr3_dering_vr2max_gain: gain to max before feeding to LUT
7:6	R/W		reserved
5:0	R/W	16	reg_sr3_dering_vr2max_limt: limit of maxsad

Table 8-2219 SRSHARP1_SR3_DRTLPF_THETA 0x3f73

Bit(s)	R/W	Default	Description
31:0	R/W	0xfe-c96420	reg_sr3_drtlpf_theta: u4x8 directional lpf beta coef for min_sad/min2_sad compared to x=0:7 correspond to [1:8]/16; 0 means no drtLPF, 15: 100% alpha dependant drtLPF.

Table 8-2220 SRSHARP1_SATPRT_CTRL 0x3f74

Bit(s)	R/W	Default	Description
31:28	R/W		reserved
27:16	R/W	5	reg_satprt_sat_core: 4x will be coring to cor(irgb_max-irgb_min) to calculate the oy_delt, the smaller the more protection to color, the larger only the rich color will be protected.
15:8	R/W	64	reg_satprt_sat_rate: rate to cor(irgb_max-irgb_min) to calculate the oy_delt,the larger the more protection to color; norm 16 as 1
7:4	R/W		reserved
3:2	R/W	1	reg_satprt_csc_mode: CSC mode of current yuv input: 0:601; 1:709; 2:BT2020 NCL; 3 reserved
1	R/W	1	reg_satprt_is_lmt: flag telling the YUV is limited range data or full rang data; 1 is limited data
0	R/W	0	reg_satprt_enable: 1 to enable of saturation protection for dnlp adjustments

Table 8-2221 SRSHARP1_SATPRT_DIVM 0x3f75

Bit(s)	R/W	Default	Description
31:24	R/W		reserved
23:0	R/W	{128,128,128}	reg_satprt_div_m: u8x3, 1/m, normalized to 128 as 1.

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Table 8-2222 SRSHARP1_DB_FLT_CTRL 0x3f77

Bit(s)	R/W	Default	Description
26	R/W	0	reg_nrdeband_reset1 : // unsigned , default = 0 0 : no reset seed 1: reload chroma seed
25	R/W	0	reg_nrdeband_reset0 : // unsigned , default = 0 0 : no reset seed 1: reload luma seed
24	R/W	0	reg_nrdeband_rgb : // unsigned , default = 0 0 : yuv 1: RGB
23	R/W	1	reg_nrdeband_en11 : // unsigned , default = 1 debanding registers of side lines, [0] for luma, same for below
22	R/W	1	reg_nrdeband_en10 : // unsigned , default = 1 debanding registers of side lines, [1] for chroma, same for below
21	R/W	1	reg_nrdeband_siderand : // unsigned , default = 1 options to use side two lines use the rand, instead of use for the YUV three component of middle line, 0: seed [3]/bandrand[3] for middle line yuv; 1: seed[3]/bandrand[3] for nearby three lines Y;
20	R/W	0	reg_nrdeband_randmode : // unsigned , default = 0 mode of rand noise adding, 0: same noise strength for all difs; else: strenght of noise will not exceed the difs, MIN((pPKReg->reg_nrdeband_bandrand[m]), noise[m])
19:17	R/W	6	reg_nrdeband_bandrand2 : // unsigned , default = 6
15:13	R/W	6	reg_nrdeband_bandrand1 : // unsigned , default = 6
11: 9	R/W	6	reg_nrdeband_bandrand0 : // unsigned , default = 6
7	R/W	1	reg_nrdeband_hpxor1 : // unsigned , default = 1 debanding random hp portion xor, [0] for luma
6	R/W	1	reg_nrdeband_hpxor0 : // unsigned , default = 1 debanding random hp portion xor, [1] for chroma
5	R/W	1	reg_nrdeband_en1 : // unsigned , default = 1 debanding registers, for luma
4	R/W	1	reg_nrdeband_en0 : // unsigned , default = 1 debanding registers, for chroma
3: 2	R/W	2	reg_nrdeband_lpf_mode1 : // unsigned , default = 2 lpf mode, 0: 3x3, 1:3x5; 2: 5x5; 3:5x7
1: 0	R/W	2	reg_nrdeband_lpf_mode0 : // unsigned , default = 2 lpf mode, 0: 3x3, 1:3x5; 2: 5x5; 3:5x7

Table 8-2223 SRSHARP1_DB_FLT_YC_THRD 0x3f78

Bit(s)	R/W	Default	Description
31:28	R/W	9	reg_nrdeband_luma_th3 : // unsigned , default = 9 threshold to |Y-Ylpf|, if < th [0] use lpf
27:24	R/W	7	reg_nrdeband_luma_th2 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4
23:20	R/W	6	reg_nrdeband_luma_th1 : // unsigned , default = 6 elseif <th[1] use (lpf*3 + y)/4elseif <th[2] (lpf*1 + y)/2
19:16	R/W	5	reg_nrdeband_luma_th0 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/4elseif elseif <th[3] (lpf*1 + 3*y)/4; else
15:12	R/W	9	reg_nrdeband_chrm_th3 : // unsigned , default = 9 threshold to |Y-Ylpf|, if < th [0] use lpf
11: 8	R/W	7	reg_nrdeband_chrm_th2 : // unsigned , default = 7 elseif <th[1] use (lpf*3 + y)/4

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Bit(s)	R/W	Default	Description
7: 4	R/W	6	reg_nrdeband_chrm_th1 : // unsigned , default = 6 elseif <th[1] use (lpf*3 + y)/ 4elseif <th[2] (lpf*1 + y)/2
3: 0	R/W	5	reg_nrdeband_chrm_th0 : // unsigned , default = 5 elseif <th[1] use (lpf*3 + y)/ 4elseif elseif

Table 8-2224 SRSHARP1_DB_FLT_RANDLUT 0x3f79

Bit(s)	R/W	Default	Description
23:21	R/W	1	reg_nrdeband_randslut7 : // unsigned , default = 1 lut0
20:18	R/W	1	reg_nrdeband_randslut6 : // unsigned , default = 1 lut0
17:15	R/W	1	reg_nrdeband_randslut5 : // unsigned , default = 1 lut0
14:12	R/W	1	reg_nrdeband_randslut4 : // unsigned , default = 1 lut0
11: 9	R/W	1	reg_nrdeband_randslut3 : // unsigned , default = 1 lut0
8: 6	R/W	1	reg_nrdeband_randslut2 : // unsigned , default = 1 lut0
5: 3	R/W	1	reg_nrdeband_randslut1 : // unsigned , default = 1 lut0
2: 0	R/W	1	reg_nrdeband_randslut0 : // unsigned , default = 1 lut0

Table 8-2225 SRSHARP1_DB_FLT_PXI_THRD 0x3f7a

Bit(s)	R/W	Default	Description
25:16	R/W	0	reg_nrdeband_yc_th1 : // unsigned , default = 0 to luma/|u/v| for using the denoise
9: 0	R/W	0	reg_nrdeband_yc_th0 : // unsigned , default = 0 to luma/|u/v| for using the denoise

Table 8-2226 SRSHARP1_DB_FLT_SEED_Y 0x3f7b

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8240	reg_nrdeband_seed0 : // unsigned , default = 1621438240 noise adding seed for Y. seed[0]= 0x60a52f20; as default

Table 8-2227 SRSHARP1_DB_FLT_SEED_U 0x3f7c

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8247	reg_nrdeband_seed1 : // unsigned , default = 1621438247 noise adding seed for U. seed[0]= 0x60a52f27; as default

Table 8-2228 SRSHARP1_DB_FLT_SEED_V 0x3f7d

Bit(s)	R/W	Default	Description
31: 0	R/W	162143-8242	reg_nrdeband_seed2 : // unsigned , default = 1621438242 noise adding seed for V. seed[0]= 0x60a52f22; as default

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Table 8-2229 SRSHARP1_PKGAIN_VSLUMA_LUT_L 0x3f7e

Bit(s)	R/W	Default	Description
31:28	R/W	5	reg_pkgain_vsluma_lut7
27:24	R/W	6	reg_pkgain_vsluma_lut6
23:20	R/W	6	reg_pkgain_vsluma_lut5
19:16	R/W	6	reg_pkgain_vsluma_lut4
15:12	R/W	7	reg_pkgain_vsluma_lut3
11:8	R/W	10	reg_pkgain_vsluma_lut2
7:4	R/W	12	reg_pkgain_vsluma_lut1
3:0	R/W	8	reg_pkgain_vsluma_lut0

Table 8-2230 SRSHARP1_PKGAIN_VSLUMA_LUT_H 0x3f7f

Bit(s)	R/W	Default	Description
31:4	R/W		reserved
3:0	R/W	4	reg_pkgain_vsluma_lut8

Table 8-2231 SRSHARP1_PKOSHT_VSLUMA_LUT_L 0x3f80

Bit(s)	R/W	Default	Description
31:28	R/W	5	reg_pkosht_vsluma_lut7
27:24	R/W	6	reg_pkosht_vsluma_lut6
23:20	R/W	6	reg_pkosht_vsluma_lut5
19:16	R/W	6	reg_pkosht_vsluma_lut4
15:12	R/W	7	reg_pkosht_vsluma_lut3
11:8	R/W	10	reg_pkosht_vsluma_lut2
7:4	R/W	12	reg_pkosht_vsluma_lut1
3:0	R/W	8	reg_pkosht_vsluma_lut0

Table 8-2232 SRSHARP1_PKOSHT_VSLUMA_LUT_H 0x3f81

Bit(s)	R/W	Default	Description
31:4	R/W		reserved
3:0	R/W	4	reg_pkosht_vsluma_lut8

Table 8-2233 SRSHARP1_SATPRT_LMT_RGB1 0x3f82

Bit(s)	R/W	Default	Description
27:16	R/W	0d0	reg_satprt_lmt_g:
11: 0	R/W	0d0	reg_satprt_lmt_r: limit of RGB channel, for limited range RGB, 12bits

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Table 8-2234 SRSHARP1_SATPRT_LMT_RGB2 0x3f83

Bit(s)	R/W	Default	Description
31:16	R/W	0d0	reserved
11: 0	R/W	0d0	reg_satprt_lmt_b: limit of RGB channel, for limited range RGB

Table 8-2235 SRSHARP1_SHARP_GATE_CLK_CTRL_0 0x3f84

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Gate clock control [01:00]: sharp input control unit [03:02]: deband unit [05:04]: dejaggy unit [07:06]: dnlp unit [09:08]: demo control unit [11:10]: horiz interp unit

Table 8-2236 SRSHARP1_SHARP_GATE_CLK_CTRL_1 0x3f85

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Gate clock control [01:00]: sr_top “pipe_ctrl” [03:02]: drt [05:04]: ssd [07:06]: cubic [09:08]: edi [11:10]: pkgainsad [13:12]: bicomux [15:14]: bicin_fifo [17:16]: lpf4pkgain_fifo [19:18]: min2hvgain_fifo [21:20]: sad4pkgain_fifo [23:22]: dirminmax4xtl_fifo [25:24]: drtsad8_fifo [27:26]: ssdmax_fifo [29:28]: pkminmax_fifo [31:30]: cirdrtgain_fifo

Table 8-2237 SRSHARP1_SHARP_GATE_CLK_CTRL_2 0x3f86

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Gate clock control [01:00]: bufdiff_fifo [03:02]: osvar_fifo [05:04]: pkhvgain unit [07:06]: pkgain unit [09:08]: TI unit [11:10]: pk unit [13:12]: locvar unit [15:14]: hvconc unit

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Table 8-2238 SRSHARP1_SHARP_GATE_CLK_CTRL_3 0x3f87

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Gate clock control [01:00]: TI, htl Y [03:02]: TI, vti Y [05:04]: TI, htl U [07:06]: TI, vtl U [09:08]: TI, htl V [11:10]: TI, vtl V [13:12]: TI, lumaminmax_fifo [15:14]: TI, chrmminmax fifo

Table 8-2239 SRSHARP1_SHARP_DPS_CTRL 0x3f88

Bit(s)	R/W	Default	Description
31:0	R/W	0d0	Power saving control : hvcon : nrssd : os filter : cubic 5 lines to 9 lines : dering : locvar : drtlpf : hlti : hcti : vlti : vcti : lti blend : cti blend : htishort (no used) : nr Y filter : nr C filter : nr belnd : pkti blend : os ctrl : pk HP : pk BP [26:24] dejaggy power saving control [30:28] reserved

Table 8-2240 SRSHARP1_DNLP_00 0x3f90

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0010-0008	reg_dnlp_ygrid0 : : dnlp00 . unsigned , default = 32'h00100008

Table 8-2241 SRSHARP1_DNLP_01 0x3f91

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0020-0018	reg_dnlp_ygrid1 : : dnlp01 . unsigned , default = 32'h00200018

Table 8-2242 SRSHARP1_DNLP_02 0x3f92

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0030-0028	reg_dnlp_ygrid2 : : dnlp02 . unsigned , default = 32'h00300028

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Table 8-2243 SRSHARP1_DNLP_03 0x3f93

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0040-0038	reg_dnlp_ygrid3 : : dnlp03 . unsigned , default = 32'h00400038

Table 8-2244 SRSHARP1_DNLP_04 0x3f94

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0050-0048	reg_dnlp_ygrid4 : : dnlp04 . unsigned , default = 32'h00500048

Table 8-2245 SRSHARP1_DNLP_05 0x3f95

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0068-005c	reg_dnlp_ygrid5 : : dnlp05 . unsigned , default = 32'h0068005c

Table 8-2246 SRSHARP1_DNLP_06 0x3f96

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0080-0074	reg_dnlp_ygrid6 : : dnlp06 . unsigned , default = 32'h00800074

Table 8-2247 SRSHARP1_DNLP_07 0x3f97

Bit(s)	R/W	Default	Description
31: 0	R/W	0x00-a00090	reg_dnlp_ygrid7 : : dnlp07 . unsigned , default = 32'h00a00090

Table 8-2248 SRSHARP1_DNLP_08 0x3f98

Bit(s)	R/W	Default	Description
31: 0	R/W	0x00-c000b0	reg_dnlp_ygrid8 : : dnlp08 . unsigned , default = 32'h00c000b0

Table 8-2249 SRSHARP1_DNLP_09 0x3f99

Bit(s)	R/W	Default	Description
31: 0	R/W	0x00-e000d0	reg_dnlp_ygrid9 : : dnlp09 . unsigned , default = 32'h00e000d0

Table 8-2250 SRSHARP1_DNLP_10 0x3f9a

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0100-00f0	reg_dnlp_ygrid10 : : dnlp10 . unsigned , default = 32'h010000f0

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Table 8-2251 SRSHARP1_DNLP_11 0x3f9b

Bit(s)	R/W	Default	Description
31: 0	R/W	0x012-c0114	reg_dnlp_ygrid11 : : dnlp11 . unsigned , default = 32'h012c0114

Table 8-2252 SRSHARP1_DNLP_12 0x3f9c

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0154-0140	reg_dnlp_ygrid12 : : dnlp12 . unsigned , default = 32'h01540140

Table 8-2253 SRSHARP1_DNLP_13 0x3f9d

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0180-016c	reg_dnlp_ygrid13 : : dnlp13 . unsigned , default = 32'h0180016c

Table 8-2254 SRSHARP1_DNLP_14 0x3f9e

Bit(s)	R/W	Default	Description
31: 0	R/W	0x01-c001a0	reg_dnlp_ygrid14 : : dnlp14 . unsigned , default = 32'h01c001a0

Table 8-2255 SRSHARP1_DNLP_15 0x3f9f

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0200-01e0	reg_dnlp_ygrid15 : : dnlp15 . unsigned , default = 32'h020001e0

Table 8-2256 SRSHARP1_DNLP_16 0x3fa0

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0240-0220	reg_dnlp_ygrid16 : : dnlp16 . unsigned , default = 32'h02400220

Table 8-2257 SRSHARP1_DNLP_17 0x3fa1

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0280-0260	reg_dnlp_ygrid17 : : dnlp17 . unsigned , default = 32'h02800260

Table 8-2258 SRSHARP1_DNLP_18 0x3fa2

Bit(s)	R/W	Default	Description
31: 0	R/W	0x02-b00298	reg_dnlp_ygrid18 : : dnlp18 . unsigned , default = 32'h02b00298

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Table 8-2259 SRSHARP1_DNLP_19 0x3fa3

Bit(s)	R/W	Default	Description
31: 0	R/W	0x02-e002c8	reg_dnlp_ygrid19 : : dnlp19 . unsigned , default = 32'h02e002c8

Table 8-2260 SRSHARP1_DNLP_20 0x3fa4

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0300-02f0	reg_dnlp_ygrid20 : : dnlp20 . unsigned , default = 32'h030002f0

Table 8-2261 SRSHARP1_DNLP_21 0x3fa5

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0320-0310	reg_dnlp_ygrid21 : : dnlp21 . unsigned , default = 32'h03200310

Table 8-2262 SRSHARP1_DNLP_22 0x3fa6

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0338-032c	reg_dnlp_ygrid22 : : dnlp22 . unsigned , default = 32'h0338032c

Table 8-2263 SRSHARP1_DNLP_23 0x3fa7

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0350-0348	reg_dnlp_ygrid23 : : dnlp23 . unsigned , default = 32'h03500348

Table 8-2264 SRSHARP1_DNLP_24 0x3fa8

Bit(s)	R/W	Default	Description
31: 0	R/W	0x035-c0358	reg_dnlp_ygrid24 : : dnlp24 . unsigned , default = 32'h035c0358

Table 8-2265 SRSHARP1_DNLP_25 0x3fa9

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0368-0360	reg_dnlp_ygrid25 : : dnlp25 . unsigned , default = 32'h03680360

Table 8-2266 SRSHARP1_DNLP_26 0x3faa

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0378-0370	reg_dnlp_ygrid26 : : dnlp26 . unsigned , default = 32'h03780370

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Table 8-2267 SRSHARP1_DNLP_27 0x3fab

Bit(s)	R/W	Default	Description
31: 0	R/W	0x0388-0380	reg_dnlp_ygrid27 : : dnlp27 . unsigned , default = 32'h03880380

Table 8-2268 SRSHARP1_DNLP_28 0x3fac

Bit(s)	R/W	Default	Description
31: 0	R/W	0x03-a00390	reg_dnlp_ygrid28 : : dnlp28 . unsigned , default = 32'h03a00390

Table 8-2269 SRSHARP1_DNLP_29 0x3fad

Bit(s)	R/W	Default	Description
31: 0	R/W	0x03-c003b0	reg_dnlp_ygrid29 : : dnlp29 . unsigned , default = 32'h03c003b0

Table 8-2270 SRSHARP1_DNLP_30 0x3fae

Bit(s)	R/W	Default	Description
31: 0	R/W	0x03-e003d0	reg_dnlp_ygrid30 : : dnlp30 . unsigned , default = 32'h03e003d0

Table 8-2271 SRSHARP1_DNLP_31 0x3faf

Bit(s)	R/W	Default	Description
31: 0	R/W	0x03f-c03f0	reg_dnlp_ygrid31 : : dnlp31 . unsigned , default = 32'h03fc03f0

Table 8-2272 SRSHARP1_SHARP_SYNC_CTRL 0x3fb0

Bit(s)	R/W	Default	Description
15:0	R/W	0d0	Shadow control for some setting: 0 to use VSYNC to shadow : SR all bypass : reg_sr2_bic_pknr_bypass [2] : reg_3d_mode [3] : reg_nrdeband_en11 [4]: reg_nrdeband_en10 [5]: reg_nrdeband_en1 [6]: reg_nrdeband_en0 [7]: reg_dejaggy_enable [8]: reg_sr2_sharp_prc_lr [9]: reg_sharp_pk_en [10]:reg_sharp_nr_en [11]:reg_adp_hlti_en [12]:reg_adp_hcti_en [13]:reg_adp_vlti_en [14]:reg_adp_vcti_en [15]:reg_dnlp_en

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Table 8-2273 SRSHARP1_LC_INPUT_MUX 0x3fb1

Bit(s)	R/W	Default	Description
31:7	R/W	0	reserved
6:4	R/W	5	reg_lcinput_ysel: local contrast luma input options 0: org_y; 1: gau_y; 2: gauadp_y; 3: edgadplpf_y; 4: nr_y;5: lti_y; 6: pk_y (before os);7: pk_y (after os)
3	R/W	0	reserved
2:0	R/W	5	reg_lcinput_csel: local contrast chroma input options 0: org_c; 1: gau_c; 2: gauadp_c; 3: edgadplpf_c; 4: nr_c;5: cti_c; 6: pk_c

Table 8-2274 SRSHARP1_LC_TOP_CTRL 0x3fc0

Bit(s)	R/W	Default	Description
31:17	R/W	0	reserved
16	R/W	0	reg_lc_sync_ctrl: 0 to shadow reg_lc_enable.
15:8	R/W	8	reg_lc_hblank: blank between two line.
7:5	R/W	0	reserved
4	R/W	0	reg_lc_enable: enable signal for local contrast enhancement, 1-enable; 0 disable
3:1	R/W	0	reserved
0	R/W	0	reg_lc_blkblend_mode:use bilinear interpolation between blocks, 0: no interpolation 1: blender enabled

Table 8-2275 SRSHARP1_LC_HV_NUM 0x3fc1

Bit(s)	R/W	Default	Description
31:13	R/W	0	reserved
12:8	R/W	12	reg_lc_blk_hnum: lc processing region number of V, maximum to (STA_LEN_H-1) (0~12)
7:5	R/W	0	reserved
4:0	R/W	8	reg_lc_blk_vnum: lc processing region number of H, maximum to (STA_LEN_V-1) (0~8)

Table 8-2276 SRSHARP1_LC_SAT_LUT_0_1 0x3fc2

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	12	reg_lc_satur_lut_0: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	32	reg_lc_satur_lut_1: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

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Table 8-2277 SRSHARP1_LC_SAT_LUT_2_3 0x3fc3

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	56	reg_lc_satur_lut_2: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	84	reg_lc_satur_lut_3: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2278 SRSHARP1_LC_SAT_LUT_4_5 0x3fc4

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	115	reg_lc_satur_lut_4: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	149	reg_lc_satur_lut_5: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2279 SRSHARP1_LC_SAT_LUT_6_7 0x3fc5

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	185	reg_lc_satur_lut_6: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	223	reg_lc_satur_lut_7: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2280 SRSHARP1_LC_SAT_LUT_8_9 0x3fc6

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	263	reg_lc_satur_lut_8: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	305	reg_lc_satur_lut_9: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2281 SRSHARP1_LC_SAT_LUT_10_11 0x3fc7

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	348	reg_lc_satur_lut_10: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

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Bit(s)	R/W	Default	Description
15:12	R/W	0	reserved
11:0	R/W	393	reg_lc_satur_lut_11: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2282 SRSHARP1_LC_SAT_LUT_12_13 0x3fc8

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	440	reg_lc_satur_lut_12: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	488	reg_lc_satur_lut_13: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2283 SRSHARP1_LC_SAT_LUT_14_15 0x3fc9

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	537	reg_lc_satur_lut_14: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	588	reg_lc_satur_lut_15: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2284 SRSHARP1_LC_SAT_LUT_16_17 0x3fca

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	640	reg_lc_satur_lut_16: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	694	reg_lc_satur_lut_17: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2285 SRSHARP1_LC_SAT_LUT_18_19 0x3fcb

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	748	reg_lc_satur_lut_18: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	804	reg_lc_satur_lut_19: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

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Table 8-2286 SRSHARP1_LC_SAT_LUT_20_21 0x3fcc

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	861	reg_lc_satur_lut_20: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	919	reg_lc_satur_lut_21: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2287 SRSHARP1_LC_SAT_LUT_22_23 0x3fcd

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	978	reg_lc_satur_lut_22: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	1038	reg_lc_satur_lut_23: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2288 SRSHARP1_LC_SAT_LUT_24_25 0x3fce

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	1099	reg_lc_satur_lut_24: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	1161	reg_lc_satur_lut_25: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2289 SRSHARP1_LC_SAT_LUT_26_27 0x3fcf

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	1224	reg_lc_satur_lut_26: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	1287	reg_lc_satur_lut_27: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2290 SRSHARP1_LC_SAT_LUT_28_29 0x3fd0

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	1352	reg_lc_satur_lut_28: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

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Bit(s)	R/W	Default	Description
15:12	R/W	0	reserved
11:0	R/W	1418	reg_lc_satur_lut_29: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2291 SRSHARP1_LC_SAT_LUT_30_31 0x3fd1

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	1485	reg_lc_satur_lut_30: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	1552	reg_lc_satur_lut_31: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2292 SRSHARP1_LC_SAT_LUT_32_33 0x3fd2

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	1620	reg_lc_satur_lut_32: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	1690	reg_lc_satur_lut_33: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2293 SRSHARP1_LC_SAT_LUT_34_35 0x3fd3

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	1760	reg_lc_satur_lut_34: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	1830	reg_lc_satur_lut_35: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2294 SRSHARP1_LC_SAT_LUT_36_37 0x3fd4

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	1902	reg_lc_satur_lut_36: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	1974	reg_lc_satur_lut_37: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

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Table 8-2295 SRSHARP1_LC_SAT_LUT_38_39 0x3fd5

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	2047	reg_lc_satur_lut_38: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	2121	reg_lc_satur_lut_39: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2296 SRSHARP1_LC_SAT_LUT_40_41 0x3fd6

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	2196	reg_lc_satur_lut_40: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	2271	reg_lc_satur_lut_41: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2297 SRSHARP1_LC_SAT_LUT_42_43 0x3fd7

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	2347	reg_lc_satur_lut_42: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	2424	reg_lc_satur_lut_43: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2298 SRSHARP1_LC_SAT_LUT_44_45 0x3fd8

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	2502	reg_lc_satur_lut_44: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	2580	reg_lc_satur_lut_45: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2299 SRSHARP1_LC_SAT_LUT_46_47 0x3fd9

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	2659	reg_lc_satur_lut_46: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

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Bit(s)	R/W	Default	Description
15:12	R/W	0	reserved
11:0	R/W	2738	reg_lc_satur_lut_47: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2300 SRSHARP1_LC_SAT_LUT_48_49 0x3fda

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	2818	reg_lc_satur_lut_48: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	2899	reg_lc_satur_lut_49: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2301 SRSHARP1_LC_SAT_LUT_50_51 0x3fdb

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	2981	reg_lc_satur_lut_50: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	3063	reg_lc_satur_lut_51: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2302 SRSHARP1_LC_SAT_LUT_52_53 0x3fdc

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	3146	reg_lc_satur_lut_52: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	3229	reg_lc_satur_lut_53: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2303 SRSHARP1_LC_SAT_LUT_54_55 0x3fdd

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	3313	reg_lc_satur_lut_54: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	3398	reg_lc_satur_lut_55: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

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Table 8-2304 SRSHARP1_LC_SAT_LUT_56_57 0x3fde

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	3483	reg_lc_satur_lut_56: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	3569	reg_lc_satur_lut_57: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2305 SRSHARP1_LC_SAT_LUT_58_59 0x3fdf

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	3655	reg_lc_satur_lut_58: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	3742	reg_lc_satur_lut_59: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2306 SRSHARP1_LC_SAT_LUT_60_61 0x3fe0

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	3830	reg_lc_satur_lut_60: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2
15:12	R/W	0	reserved
11:0	R/W	3918	reg_lc_satur_lut_61: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2307 SRSHARP1_LC_SAT_LUT_62 0x3fe1

Bit(s)	R/W	Default	Description
31:12	R/W	0	reserved
11:0	R/W	4007	reg_lc_satur_lut_62: saturation protection curve, normalized to 4096 as 1, default = y=x^1.2

Table 8-2308 SRSHARP1_LC_CURVE_BLK_HIDX_0_1 0x3fe2

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	0	reg_lc_blk_hidx_0: block boundary x-index
15:14	R/W	0	reserved
13:0	R/W	320	reg_lc_blk_hidx_1: block boundary x-index

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Table 8-2309 SRSHARP1_LC_CURVE_BLK_HIDX_2_3 0x3fe3

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	640	reg_lc_blk_hidx_2: block boundary x-index
15:14	R/W	0	reserved
13:0	R/W	960	reg_lc_blk_hidx_3: block boundary x-index

Table 8-2310 SRSHARP1_LC_CURVE_BLK_HIDX_4_5 0x3fe4

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	1280	reg_lc_blk_hidx_4: block boundary x-index
15:14	R/W	0	reserved
13:0	R/W	1600	reg_lc_blk_hidx_5: block boundary x-index

Table 8-2311 SRSHARP1_LC_CURVE_BLK_HIDX_6_7 0x3fe5

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	1920	reg_lc_blk_hidx_6: block boundary x-index
15:14	R/W	0	reserved
13:0	R/W	2240	reg_lc_blk_hidx_7: block boundary x-index

Table 8-2312 SRSHARP1_LC_CURVE_BLK_HIDX_8_9 0x3fe6

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	2560	reg_lc_blk_hidx_8: block boundary x-index
15:14	R/W	0	reserved
13:0	R/W	2880	reg_lc_blk_hidx_9: block boundary x-index

Table 8-2313 SRSHARP1_LC_CURVE_BLK_HIDX_10_11 0x3fe7

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	3200	reg_lc_blk_hidx_10: block boundary x-index
15:14	R/W	0	reserved
13:0	R/W	3520	reg_lc_blk_hidx_11: block boundary x-index

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Table 8-2314 SRSHARP1_LC_CURVE_BLK_HIDX_12 0x3fe8

Bit(s)	R/W	Default	Description
31:14	R/W	0	reserved
13:0	R/W	3840	reg_lc_blk_hidx_1: block boundary x-index

Table 8-2315 SRSHARP1_LC_CURVE_BLK_VIDX_0_1 0x3fe9

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	0	reg_lc_blk_vidx_0: block boundary y-index
15:14	R/W	0	reserved
13:0	R/W	270	reg_lc_blk_vidx_1: block boundary y-index

Table 8-2316 SRSHARP1_LC_CURVE_BLK_VIDX_2_3 0x3fea

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	540	reg_lc_blk_vidx_2: block boundary y-index
15:14	R/W	0	reserved
13:0	R/W	810	reg_lc_blk_vidx_3: block boundary y-index

Table 8-2317 SRSHARP1_LC_CURVE_BLK_VIDX_4_5 0x3feb

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	1080	reg_lc_blk_vidx_4: block boundary y-index
15:14	R/W	0	reserved
13:0	R/W	1350	reg_lc_blk_vidx_5: block boundary y-index

Table 8-2318 SRSHARP1_LC_CURVE_BLK_VIDX_6_7 0x3fec

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:16	R/W	1620	reg_lc_blk_vidx_6: block boundary y-index
15:14	R/W	0	reserved
13:0	R/W	1890	reg_lc_blk_vidx_7: block boundary y-index

Table 8-2319 SRSHARP1_LC_CURVE_BLK_HIDX_8 0x3fed

Bit(s)	R/W	Default	Description
31:14	R/W	0	reserved
13:0	R/W	2160	reg_lc_blk_vidx_8: block boundary y-index

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Table 8-2320 SRSHARP1_LC_YUV2RGB_MAT_0_1 0x3fee

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	298	reg_lc_yuv2rgb_mat3x3_0: yuv2rgb 3x3 matrix
15:12	R/W	0	reserved
11:0	R/W	0	reg_lc_yuv2rgb_mat3x3_1: yuv2rgb 3x3 matrix

Table 8-2321 SRSHARP1_LC_YUV2RGB_MAT_2_3 0x3fef

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	408	reg_lc_yuv2rgb_mat3x3_2: yuv2rgb 3x3 matrix
15:12	R/W	0	reserved
11:0	R/W	298	reg_lc_yuv2rgb_mat3x3_3: yuv2rgb 3x3 matrix

Table 8-2322 SRSHARP1_LC_YUV2RGB_MAT_4_5 0x3ff0

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	-100	reg_lc_yuv2rgb_mat3x3_4: yuv2rgb 3x3 matrix
15:12	R/W	0	reserved
11:0	R/W	-208	reg_lc_yuv2rgb_mat3x3_5: yuv2rgb 3x3 matrix

Table 8-2323 SRSHARP1_LC_YUV2RGB_MAT_6_7 0x3ff1

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	298	reg_lc_yuv2rgb_mat3x3_6: yuv2rgb 3x3 matrix
15:12	R/W	0	reserved
11:0	R/W	516	reg_lc_yuv2rgb_mat3x3_7: yuv2rgb 3x3 matrix

Table 8-2324 SRSHARP1_LC_YUV2RGB_MAT_8 0x3ff2

Bit(s)	R/W	Default	Description
31:12	R/W	0	reserved
11:0	R/W	0	reg_lc_yuv2rgb_mat3x3_8: yuv2rgb 3x3 matrix

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Table 8-2325 SRSHARP1_LC_RGB2YUV_MAT_0_1 0x3ff3

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	263	reg_lc_rgb2yuv_mat3x3_0: rgb2yuv 3x3 matrix
15:12	R/W	0	reserved
11:0	R/W	516	reg_lc_rgb2yuv_mat3x3_1: rgb2yuv 3x3 matrix

Table 8-2326 SRSHARP1_LC_RGB2YUV_MAT_2_3 0x3ff4

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	100	reg_lc_rgb2yuv_mat3x3_2: rgb2yuv 3x3 matrix
15:12	R/W	0	reserved
11:0	R/W	-152	reg_lc_rgb2yuv_mat3x3_3: rgb2yuv 3x3 matrix

Table 8-2327 SRSHARP1_LC_RGB2YUV_MAT_4_5 0x3ff5

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	-298	reg_lc_rgb2yuv_mat3x3_4: rgb2yuv 3x3 matrix
15:12	R/W	0	reserved
11:0	R/W	450	reg_lc_rgb2yuv_mat3x3_5: rgb2yuv 3x3 matrix

Table 8-2328 SRSHARP1_LC_RGB2YUV_MAT_6_7 0x3ff6

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	450	reg_lc_rgb2yuv_mat3x3_6: rgb2yuv 3x3 matrix
15:12	R/W	0	reserved
11:0	R/W	-377	reg_lc_rgb2yuv_mat3x3_7: rgb2yuv 3x3 matrix

Table 8-2329 SRSHARP1_LC_RGB2YUV_MAT_8 0x3ff7

Bit(s)	R/W	Default	Description
31:12	R/W	0	reserved
11:0	R/W	-73	reg_lc_rgb2yuv_mat3x3_8: rgb2yuv 3x3 matrix

Table 8-2330 SRSHARP1_LC_YUV2RGB_OFST 0x3ff8

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	64	reg_lc_yuv2rgb_ofset_0: yuv2rgb pre-offset to yuv

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Bit(s)	R/W	Default	Description
15:12	R/W	0	reserved
11:0	R/W	512	reg_lc_yuv2rgb_ofset_1: yuv2rgb pre-offset to yuv

Table 8-2331 SRSHARP1_LC_YUV2RGB_CLIP 0x3ff9

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:28	R/W	0	reg_lc_yuv2rgb_rs: matrix normalization right shift extra bits, norm= (1<<(8+rs))
27:16	R/W	0	reg_lc_yuv2rgb_clip_0: yuv2rgb converted RGB clipping range
15:12	R/W	0	reserved
11:0	R/W	1023	reg_lc_yuv2rgb_clip_1: yuv2rgb converted RGB clipping range

Table 8-2332 SRSHARP1_LC_RGB2YUV_OFST 0x3ffa

Bit(s)	R/W	Default	Description
31:28	R/W	0	reserved
27:16	R/W	64	reg_lc_rgb2yuv_ofset_0: rgb2yuv post-offset to yuv
15:12	R/W	0	reserved
11:0	R/W	512	reg_lc_rgb2yuv_ofset_1: rgb2yuv post-offset to yuv

Table 8-2333 SRSHARP1_LC_RGB2YUV_CLIP 0x3ffb

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:28	R/W	0	reg_lc_rgb2yuv_rs: matrix normalization right shift extra bits, norm= (1<<(10+rs))
27:16	R/W	0	reg_lc_rgb2yuv_clip_0: rgb2yuv converted YUV clipping range
15:12	R/W	0	reserved
11:0	R/W	1023	reg_lc_rgb2yuv_clip_1: rgb2yuv converted YUV clipping range

Table 8-2334 SRSHARP1_LC_MAP_RAM_CTRL 0x3ffc

Bit(s)	R/W	Default	Description
31:1	R/W	0	reserved
0	R/W	0	reg_lc_cbus2ram_en: 1 to enable CBUS write/read data to/from LC MAP RAM

Table 8-2335 SRSHARP1_LC_MAP_RAM_ADDR 0x3ffd

Bit(s)	R/W	Default	Description
31	W	0	1 to pre-read from LC MAP RAM to CBUS
30:7	R/W	0	reserved

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Bit(s)	R/W	Default	Description
6:4	R/W	0	Vidx of LC MAP RAM
3:0	R/W	0	Hidx of LC MAP RA:

Table 8-2336 SRSHARP1_LC_MAP_RAM_DATA 0x3ffd

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:0	R/W	0	LC MAP RAM data for write/read

8.2.3.30 LC STTS / LC Curve Generator Registers

LC STTS / LC Curve Generator

Table 8-2337 LC_CURVE_CTRL 0x4000

Bit(s)	R/W	Default	Description
31	R/W	0	reg_lc_int_enable : 1 to enable LC_CURVE done interrupt
30: 14	R/W	0	reserved
13: 12	R/W	0	reg_lc_gclk_ctrl : gated clock control
11: 10	R/W	0	reserved
9: 8	R/W	1	reg_lc_hist_curve_nodes_hlpf: horizontal lpf of the ram_curve_nodes, 0: no LPF, 1= [1 2 1]; 2: [1 2 2 2 1]/8
7: 6	R/W	0	reserved
5: 4	R/W	1	reg_lc_hist_curve_nodes_vlpf: Vertical lpf of the ram_curve_nodes, 0: no LPF, 1= [1 2 1]; 2: [1 2 2 2 1]/8
3: 2	R/W	0	reserved
1	R/W	1	reg_lc_blackbar_mute_en: mute the black bar corresponding bin, 0: no mute, 1: mute enable
0	R/W	0	reg_lc_curve_en

Table 8-2338 LC_CURVE_HV_NUM 0x4001

Bit(s)	R/W	Default	Description
31: 13	R/W	0	reserved
12: 8	R/W	12	reg_lc_cur_blk_hnum: lc processing region number of H, maximum to 12
7: 5	R/W	0	reserved
4: 0	R/W	8	reg_lc_cur_blk_vnum: lc processing region number of V, maximum to 8

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Table 8-2339 LC_CURVE_LMT_RAT 0x4002

Bit(s)	R/W	Default	Description
31: 16	R/W	0	reserved
15: 8	R/W	6	reg_lmtrat_minmax: x/1024 of amount
7: 0	R/W	0x14	reg_lmtrat_valid: x/1024 of amount

Table 8-2340 LC_CURVE_CONTRAST_LH 0x4003

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x10	reg_lc_contrast_low: contrast gain to the lc for dark side, normalized 256 as "1"
15: 10	R/W	0	reserved
9: 0	R/W	0x18	reg_lc_contrast_hig: contrast gain to the lc for bright side, normalized 256 as "1"

Table 8-2341 LC_CURVE_CONTRAST__LMT_LH 0x4004

Bit(s)	R/W	Default	Description
31: 24	R/W	0xa	reg_lc_cntstlmt_low_0: limit for the contrast low
23: 16	R/W	0x14	reg_lc_cntstlmt_hig_0: limit for the contrast high
15: 8	R/W	0xff	reg_lc_cntstlmt_low_1:limit for the contrast low
7: 0	R/W	0xff	reg_lc_cntstlmt_hig_1:limit for the contrast high

Table 8-2342 LC_CURVE_CONTRAST_SCL_LH 0x4005

Bit(s)	R/W	Default	Description
31: 16	R/W	0	reserved
15: 8	R/W	0x40	reg_lc_cntstscl_low: scale for the contrast low, norm 8 as 1
7: 0	R/W	0x20	reg_lc_cntstscl_hig: scale for the contrast high,norm 8 as 1

Table 8-2343 LC_CURVE_CONTRAST_BVN_LH 0x4006

Bit(s)	R/W	Default	Description
31: 16	R/W	0	reserved
15: 8	R/W	32	reg_lc_cntstbvn_low: scale to num_m as limit of min_val to minBV distance, to protect mono-color.
7: 0	R/W	32	reg_lc_cntstscl_hig: scale for the contrast high,norm 8 as 1

Table 8-2344 LC_CURVE_MISC0 0x4007

Bit(s)	R/W	Default	Description
31: 20	R/W	0	reserved
19: 16	R/W	1	reg_lc_num_m_coring: coring to num_m, soft coring

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Bit(s)	R/W	Default	Description
15: 8	R/W	0x30	reg_lc_vbin_min: 4x is min width of valid histogram bin num
7: 0	R/W	0x38	reg_lc_slope_max_face: maximum slope for the pkBin-maxBV range curve to do face protection

Table 8-2345 LC_CURVE_YPKBV_RAT 0x4008

Bit(s)	R/W	Default	Description
31: 24	R/W	0x8c	reg_lc_ypkbv_ratio_0: x= ratio*(maxBv-minBv)+min_val as low bound of the ypkBV; normalized to 256 as 1
23: 16	R/W	0xa0	reg_lc_ypkbv_ratio_1
15: 8	R/W	0x78	reg_lc_ypkbv_ratio_2:
7: 0	R/W	0x60	reg_lc_ypkbv_ratio_3:

Table 8-2346 LC_CURVE_YPKBV_SLP_LMT 0x4009

Bit(s)	R/W	Default	Description
31: 16	R/W	0	reserved
15: 8	R/W	0xc	reg_lc_ypkbv_slope_lmt_0: min max slop for the curves to avoid artifacts, for min_slope
7: 0	R/W	0x60	reg_lc_ypkbv_slope_lmt_1: min max slop for the curves to avoid artifacts, for max_slope

Table 8-2347 LC_CURVE_YMINVAL_LMT_0_1 0x400a

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x30	reg_lc_yminval_lmt_0: lmt_val = lmt[minBV(64:64:768)], and yminV = MAX(yminV,lmt_val), for very dark region boost.
15: 10	R/W	0	reserved
9: 0	R/W	0x50	reg_lc_yminval_lmt_1

Table 8-2348 LC_CURVE_YMINVAL_LMT_2_3 0x400b

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x78	reg_lc_yminval_lmt_2
15: 10	R/W	0	reserved
9: 0	R/W	0x80	reg_lc_yminval_lmt_3

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Table 8-2349 LC_CURVE_YMINVAL_LMT_4_5 0x400c

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0xa0	reg_lc_yminval_lmt_4
15: 10	R/W	0	reserved
9: 0	R/W	0xc4	reg_lc_yminval_lmt_5

Table 8-2350 LC_CURVE_YMINVAL_LMT_6_7 0x400d

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0xe0	reg_lc_yminval_lmt_6
15: 10	R/W	0	reserved
9: 0	R/W	0x100	reg_lc_yminval_lmt_7

Table 8-2351 LC_CURVE_YMINVAL_LMT_8_9 0x400e

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x120	reg_lc_yminval_lmt_8
15: 10	R/W	0	reserved
9: 0	R/W	0x140	reg_lc_yminval_lmt_9

Table 8-2352 LC_CURVE_YMINVAL_LMT_10_11 0x400f

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x160	reg_lc_yminval_lmt_10
15: 10	R/W	0	reserved
9: 0	R/W	0x190	reg_lc_yminval_lmt_11

Table 8-2353 LC_CURVE_YPKBV_YMAXVAL_LMT_0_1 0x4010

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x78	reg_lc_YPKBV_ymaxval_lmt_0: lmt_val = 4*lmt[pkBV(64:64:768) or maxBV(64:64:768)], and ypkBV = MAX (ypkBV,lmt[pkBV]), ymaxV = MAX(ymaxV,lmt[maxBV])&& above min_slop for very dark region boost.
15: 10	R/W	0	reserved
9: 0	R/W	0x96	reg_lc_YPKBV_ymaxval_lmt_1

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Table 8-2354 LC_CURVE_YPKBV_YMAXVAL_LMT_2_3 0x4011

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0xe6	reg_lc_YPKBV_ymaxval_lmt_2
15: 10	R/W	0	reserved
9: 0	R/W	0x112	reg_lc_YPKBV_ymaxval_lmt_3

Table 8-2355 LC_CURVE_YPKBV_YMAXVAL_LMT_4_5 0x4012

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x14a	reg_lc_YPKBV_ymaxval_lmt_4
15: 10	R/W	0	reserved
9: 0	R/W	0x180	reg_lc_YPKBV_ymaxval_lmt_5

Table 8-2356 LC_CURVE_YPKBV_YMAXVAL_LMT_6_7 0x4013

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x1d4	reg_lc_YPKBV_ymaxval_lmt_6
15: 10	R/W	0	reserved
9: 0	R/W	0x200	reg_lc_YPKBV_ymaxval_lmt_7

Table 8-2357 LC_CURVE_YPKBV_YMAXVAL_LMT_8_9 0x4014

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x240	reg_lc_YPKBV_ymaxval_lmt_8
15: 10	R/W	0	reserved
9: 0	R/W	0x280	reg_lc_YPKBV_ymaxval_lmt_9

Table 8-2358 LC_CURVE_YPKBV_YMAXVAL_LMT_10_11 0x4015

Bit(s)	R/W	Default	Description
31: 26	R/W	0	reserved
25: 16	R/W	0x280	reg_lc_YPKBV_ymaxval_lmt_10
15: 10	R/W	0	reserved
9: 0	R/W	0x280	reg_lc_YPKBV_ymaxval_lmt_11

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Table 8-2359 LC_CURVE_HISTVLD_THRD 0x4016

Bit(s)	R/W	Default	Description
31: 12	R/W	0	reserved
11: 0	R/W	0x1fa	reg_lc_histvld_thrd: threshold to compare to bin to get number of valid bins

Table 8-2360 LC_CURVE_BB_MUTE_THRD 0x4017

Bit(s)	R/W	Default	Description
31: 24	R/W	0	reserved
23: 0	R/W	0x2a30	reg_lc_blackbar_mute_thrd: 1/8 of the region

Table 8-2361 LC_CURVE_INT_STATUS 0x4018

Bit(s)	R/W	Default	Description
31: 1	R/W	0	reserved
0	R/W	0	Status to indicate LC curve done, write 1 to clear.

Table 8-2362 LC_CURVE_RAM_CTRL 0x4020

Bit(s)	R/W	Default	Description
31:1	R/W	0	reserved
0	R/W	0	reg_lc_cbus2ram_en: 1 to enable CBUS read data from LC Curve RAM

Table 8-2363 LC_CURVE_RAM_ADDR 0x4021

Bit(s)	R/W	Default	Description
31:7	R/W	0	reserved
6:4	R/W	0	Vidx of LC CURVE RAM to read
3:0	R/W	0	Hidx of LC CURVE RAM to read

Table 8-2364 LC_CURVE_RAM_DATA 0x4022

Bit(s)	R/W	Default	Description
31:30	R/W	0	reserved
29:0	R/W	0	LC Curve RAM data for read

Table 8-2365 LC_STTS_GCLK_CTRL0 0x4028

Bit(s)	R/W	Default	Description
7:0	R/W	0	gate clock

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Table 8-2366 LC_STTS_CTRL0 0x4029

Bit(s)	R/W	Default	Description
31:24	R/W	0	reg_bit_sel : 12bit to 10bit for 8 input
23:22	R/W	0	no use
21	R/W	0	1: input format is RGB 0: input format is YUV
20	R/W	0	1: input format is RGB 0: input format is YUV
18:16	R/W	0	matrix_conv_rs
15:14	R/W	0	no use
12	R/W	0	no use
11	R/W	0	highlight enable
10	R/W	0	probe enable
9	R/W	0	probe post enable
8-6	R/W	0	input data component mux
5-3	R/W	0	input data sel
2	R/W	0	matrix enable
1	R/W	0	no use
0	R/W	0	soft reset

Table 8-2367 LC_STTS_WIDTHM1_HEIGHTM1 0x402a

Bit(s)	R/W	Default	Description
28:16	R/W	0	width m1
12:0	R/W	0	height m1

LC_STTS_MATRIX_COEF00_01 0x402b

See: VPP_VD1_MATRIX_COEF00_01

LC_STTS_MATRIX_COEF02_10 0x402c

See: VPP_VD1_MATRIX_COEF02_10

LC_STTS_MATRIX_COEF11_12 0x402d

See: VPP_VD1_MATRIX_COEF11_12

LC_STTS_MATRIX_COEF20_21 0x402e

See: VPP_VD1_MATRIX_COEF20_21

LC_STTS_MATRIX_COEF22 0x402f

See: VPP_VD1_MATRIX_COEF22

LC_STTS_MATRIX_OFFSET0_1 0x4030

See: VPP_VD1_MATRIX_OFFSET0_1

LC_STTS_MATRIX_OFFSET2 0x4031

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See: VPP_VD1_MATRIX_OFFSET2

LC_STTS_MATRIX_PRE_OFFSET0_1 0x4032

See: VPP_VD1_MATRIX_PRE_OFFSET0_1

LC_STTS_MATRIX_PRE_OFFSET2 0x4033

See: VPP_VD1_MATRIX_PRE_OFFSET2

Table 8-2368 LC_STTS_MATRIX_PROBE_COLOR 0x4036

Bit(s)	R/W	Default	Description
29-20	R	0	component 0
19-10	R	0	omponent 1
9-0	R	0	component 2

Table 8-2369 LC_STTS_MATRIX_HL_COLOR 0x4034

Bit(s)	R/W	Default	Description
23-16	R/W	0	component 0
15-8	R/W	0	component 1
7-0	R/W	0	component 2

Table 8-2370 LC_STTS_MATRIX_PROBE_POS 0x4035

Bit(s)	R/W	Default	Description
28-16	R/W	0	probe x, position
12-0	R/W	0	probe y, postion

Table 8-2371 LC_STTS_HIST_REGION_IDX 0x4037

Bit(s)	R/W	Default	Description
31	R	0	local contrast max statistic enable
28	R	0	eol enable
27-25	R	0	vertical line overlap number for max finding
24-22	R	0	horizontal pixel overlap number, 0: 17 pix, 1: 9 pix, 2: 5 pix, 3: 3 pix, 4: 0 pix
20	R	0	1,2,1 low pass filter enable before max/hist statistic
19-16	R	0	region H/V position index, refer to VDIN_LDIM_STTS_HIST_SET_REGION
15	R	0	1: region read index auto increase per read to VDIN_LDIM_STTS_HIST_READ_REGION
6-0	R	0	region read index

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Table 8-2372 LC_STTS_HIST_SET_REGION 0x4038

Bit(s)	R/W	Default	Description
28:16	R	0	if LC_STTS_HIST_REGION_IDX[19:16] == 5'h0: read/write hvstart0 if LC_STTS_HIST_REGION_IDX[19:16] == 5'h1: read/write hend01 if LC_STTS_HIST_REGION_IDX[19:16] == 5'h2: read/write vend01 if LC_STTS_HIST_REGION_IDX[19:16] == 5'h3: read/write hend23 if LC_STTS_HIST_REGION_IDX[19:16] == 5'h4: read/write vend23 if LC_STTS_HIST_REGION_IDX[19:16] == 5'h5: read/write hend45 if LC_STTS_HIST_REGION_IDX[19:16] == 5'h6: read/write vend45 if LC_STTS_HIST_REGION_IDX[19:16] == 5'd7: read/write hend67 if LC_STTS_HIST_REGION_IDX[19:16] == 5'h8: read/write vend67 if LC_STTS_HIST_REGION_IDX[19:16] == 5'h9: read/write hend89 if LC_STTS_HIST_REGION_IDX[19:16] == 5'ha: read/write vend89 //hvstart0, Bit 28:16 row0 vstart, Bit 12:0 col0 hstart //hend01, Bit 28:16 col1 hend, Bit 12:0 col0 hend //vend01, Bit 28:16 row1 vend, Bit 12:0 row0 vend //hend23, Bit 28:16 col3 hend, Bit 12:0 col2 hend //vend23, Bit 28:16 row3 vend, Bit 12:0 row2 vend //hend45, Bit 28:16 col5 hend, Bit 12:0 col4 hend //vend45, Bit 28:16 row5 vend, Bit 12:0 row4 vend //hend67, Bit 28:16 col7 hend, Bit 12:0 col6 hend //vend67, Bit 28:16 row7 vend, Bit 12:0 row6 vend //hend89, Bit 28:16 col9 hend, Bit 12:0 col8 hend //vend89, Bit 28:16 row9 vend, Bit 12:0 row8 vend
12:0	R	0	

Table 8-2373 LC_STTS_HIST_READ_REGION 0x4039

Bit(s)	R/W	Default	Description
29:20	R	0	Max_comp2
19:10	R	0	Max_comp1
9:0	R	0	Max_comp0

Table 8-2374 LC_STTS_HIST_START_RD_REGION 0x403a

Bit(s)	R/W	Default	Description
29:0	R	0	region data

Table 8-2375 LC_STTS_WHITE_INFO 0x403b

Bit(s)	R/W	Default	Description
31:8	R	0	white data count
7:0	R/W	0xf0	white data threshold

Table 8-2376 LC_STTS_BLACK_INFO 0x403c

Bit(s)	R/W	Default	Description
31:8	R	0	black data count
7:0	R/W	0x10	black data threshold

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8.2.3.31 VKSTONE Registers

Table 8-2377 VKS_CTRL 0x3100

Bit(s)	R/W	Default	Description
31	W	1	reg_vks_en : // unsigned , default = 1 enable signal of the vks function need set high in every frme
30	R/W	1	reg_vks_scl_mode0 : // unsigned , default = 1 : b0 mode of vks ofset mode, 0: offset= offset; 1: offset= offset*step= ofset/scale;
29	R/W	1	reg_vks_scl_mode1 : // unsigned , default = 1 : b0 mode of vks ofset mode, 0: offset= offset; 1: offset= offset*step= ofset/scale;
28	R/W	1	reg_vks_fill_mode : // unsigned , default = 1 mode of out-of-boundary fill, 0 extension, 1: fill with the fill_value
27:26	R/W	1	reg_vks_row_inp_mode : // unsigned , default = 1 , interpolation mode from 16pieces ofset/step to each line ofset and step; 0: linear interpolation; 1: cubic interpolation (using ccoef)
25	R/W	0	reg_vks_border_ext_mode0 : // unsigned , default = 0 , extend mode of the border data of luma and chroma, 0: copy the most border one; 1: extropolate the border one
24	R/W	0	reg_vks_border_ext_mode1 : // unsigned , default = 0 , extend mode of the border data of luma and chroma, 0: copy the most border one; 1: extropolate the border one
23	R/W	1	reg_vks_obuf_mode0 : // unsigned , default = 1 , mode of output buffer left/right side. 0: no precalculate active pixels during output fill region; 1: precaclc active pixels during output fill regions
22	R/W	1	reg_vks_obuf_mode1 : // unsigned , default = 1 , mode of output buffer left/right side. 0: no precalculate active pixels during output fill region; 1: precaclc active pixels during output fill regions
21:20	R/W	3	reg_vks_obuf_mrgn0 : // unsigned , default = 3 , margin pixels for left right most active pixel to the fill pixels to avoid jump
19:18	R/W	3	reg_vks_obuf_mrgn1 : // unsigned , default = 3 , margin pixels for left right most active pixel to the fill pixels to avoid jump
17:16	R/W	2	reg_vks_phs_qmode : // unsigned , default = 2 , interpolation mode of the phase, 0: floor to 1/64 phase; 1: round to 1/64 phase; 2/3 linear intp
15: 0	R/W	11651	reg_vks_row_scl : // unsigned , default = 11651 , scale of row to make it fit to the 16 pieces, scl = (2^23)/RowMax

Table 8-2378 VKS_OUT_WIN_SIZE 0x3101

Bit(s)	R/W	Default	Description
29:16	R/W	1280	reg_vks_ocolmax : // unsigned , default = 1280 output outer window col number, decided by the projector
13: 0	R/W	720	reg_vks_orowmax : // unsigned , default = 720 output outer window row number, decided by the projector

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Table 8-2379 VKS_PRELPF_YCOEF0 0x3102

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_vks_prelpf_ycoef0 : // signed , default = -128 coef of horizontal luma prelpf for Keystone, normalized 128 as '1'
23:16	R/W	0	reg_vks_prelpf_ycoef1 : // signed , default = 0 coef of horizontal luma prelpf for Keystone, normalized 128 as '1'
15: 8	R/W	0	reg_vks_prelpf_ycoef2 : // signed , default = 0 coef of horizontal luma prelpf for Keystone, normalized 128 as '1'
7: 0	R/W	0	reg_vks_prelpf_ycoef3 : // signed , default = 0 coef of horizontal luma prelpf for Keystone, normalized 128 as '1'

Table 8-2380 VKS_PRELPF_YCOEF1 0x3103

Bit(s)	R/W	Default	Description
15: 8	R/W	0	reg_vks_prelpf_ycoef4 : // signed , default = 0 coef of horizontal luma prelpf for Keystone, normalized 128 as '1'
7: 0	R/W	0	reg_vks_prelpf_ycoef5 : // signed , default = 0 coef of horizontal luma prelpf for Keystone, normalized 128 as '1'

Table 8-2381 VKS_PRELPF_CCOEF0 0x3104

Bit(s)	R/W	Default	Description
31:24	R/W	0x0	reg_vks_prelpf_ccoef0 : // signed , default = -128 mode of horizontal chroma prelpf for Keystone, normalized 128 as '1'
23:16	R/W	0	reg_vks_prelpf_ccoef1 : // signed , default = 0 mode of horizontal chroma prelpf for Keystone, normalized 128 as '1'
15: 8	R/W	0	reg_vks_prelpf_ccoef2 : // signed , default = 0 mode of horizontal chroma prelpf for Keystone, normalized 128 as '1'
7: 0	R/W	0	reg_vks_prelpf_ccoef3 : // signed , default = 0 mode of horizontal chroma prelpf for Keystone, normalized 128 as '1'

Table 8-2382 VKS_PRELPF_CCOEF1 0x3105

Bit(s)	R/W	Default	Description
15: 8	R/W	0	reg_vks_prelpf_ccoef4 : // signed , default = 0 mode of horizontal chroma prelpf for Keystone, normalized 128 as '1'
7: 0	R/W	0	reg_vks_prelpf_ccoef5 : // signed , default = 0 mode of horizontal chroma prelpf for Keystone, normalized 128 as '1'

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Table 8-2383 VKS_FILL_VAL 0x3106

Bit(s)	R/W	Default	Description
23:16	R/W	0	reg_vks_fill_value0 : // unsigned , default = 0 , border fill color define. yuv: [0 128 128]; rgb:[0 0 0] ,use 8 bits in 12bit path,6bits in 10bit path
15: 8	R/W	128	reg_vks_fill_value1 : // unsigned , default = 128 , border fill color define. yuv: [0 128 128]; rgb:[0 0 0] ,use 8 bits in 12bit path,6bits in 10bit path
7: 0	R/W	128	reg_vks_fill_value2 : // unsigned , default = 128 , border fill color define. yuv: [0 128 128]; rgb:[0 0 0] ,use 8 bits in 12bit path,6bits in 10bit path

Table 8-2384 VKS_IWIN_HSIZE 0x3107

Bit(s)	R/W	Default	Description
29:16	R/W	160	reg_vks_iwinx0 : // unsigned , default = 160 , input start-col and end-col;
13: 0	R/W	1279	reg_vks_iwinx1 : // unsigned , default = 1279 , input start-col and end-col;

Table 8-2385 VKS_IWIN_VSIZE 0x3108

Bit(s)	R/W	Default	Description
29:16	R/W	0	reg_vks_iwiny0 : // unsigned , default = 0 , input start-row and end-row;
13: 0	R/W	719	reg_vks_iwiny1 : // unsigned , default = 719 , input start-row and end-row;

Table 8-2386 VKS_TOP_MISC 0x3109

Bit(s)	R/W	Default	Description
18	R/W	1	reg_flt_en : // unsigned , default = 1
17	R/W	0	reg_frm_rst : // unsigned , default = 0
16	R/W	0	reg_ctrl_sync : // unsigned , default = 0
15: 8	R/W	4	blank_num : // unsigned , default = 4
7: 0	R/W	9	flt_blank_num : // unsigned , default = 9

Table 8-2387 VKS_START_CTRL 0x310a

Bit(s)	R/W	Default	Description
16	R/W	0	reg_vks_en_mode : // unsigned , default = 0
15: 0	R/W	5	reg_hold_phnum : // unsigned , default = 5

Table 8-2388 VKS_LBUF_SIZE 0x310b

Bit(s)	R/W	Default	Description
11: 0	R/W	1024	reg_lbuf_depth : // unsigned , default = 1024

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Table 8-2389 VKS_PARA_ADDR_PORT 0x310e

Bit(s)	R/W	Default	Description
31: 0	R/W	3ff	Access address for vkstone para_lut,start address must be 0x09

Table 8-2390 VKS_PARA_DATA_PORT 0x310f

Bit(s)	R/W	Default	Description
31: 0	R/W	/	Data access for vkstone para lut

Below are the detail registers of VKS_PARA_DATA_PORT and VKS_PARA_ADDR_PORT control:

Table 8-2391 VKS_SCL_OFFSET00 ~ VKS_SCL_OFFSET16 0X09~0X19

Bit(s)	R/W	Default	Description
31:20			reserved
19: 0	R/W	/	Left offset of the input pixel offset from the left 12bits pixel and 8 bits float phase

Table 8-2392 VKS_SCL_STEP00 ~ VKS_SCL_STEP16 0X1A~0X2A

Bit(s)	R/W	Default	Description
24:20			reserved
23: 0	R/W	/	Unsigned reg_scl_stepx for ratio of each line(defined piece),step:4.20 opxium = (ipxium << 20)/step scale:4.2 = 1/step

Table 8-2393 VKS_PPS_YCOEF00 ~ VKS_PPS_YCOEF32 0X2B~0X4B

Bit(s)	R/W	Default	Description
31:24	R/W	/	Signed reg_vks_ycoef0 Poly_phase scalar coef
23:16	R/W	/	Signed reg_vks_ycoef1 Poly_phase scalar coef
15:8	R/W	/	Signed reg_vks_ycoef2 Poly_phase scalar coef
7: 0	R/W	/	Signed reg_vks_ycoef3 Poly_phase scalar coef

Table 8-2394 VKS_PPS_CCOEF00 ~ VKS_PPS_CCOEF32 0X4C~0X6C

Bit(s)	R/W	Default	Description
31:24	R/W	/	Signed reg_vks_ccoef0 Poly_phase scalar coef
23:16	R/W	/	Signed reg_vks_ccoef1 Poly_phase scalar coef
15:8	R/W	/	Signed reg_vks_ccoef2 Poly_phase scalar coef
7: 0	R/W	/	Signed reg_vks_ccoef3 Poly_phase scalar coef

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8.2.3.32 OSD1 Registers

Table 8-2395 VIU_OSD1_CTRL_STAT 0x1A10

Bit(s)	R/W	Default	Description
31	R/W	0	osd_cfg_syn_en : // unsigned , default =0 1: module enable sync by go_field 0: normal
30	R/W	0	ENABLE_FREE_CLK. 1 = Use free-running clock; 0 = Use gated clock to save power.
29	R	0	TEST_RD_DSR: Applicable only when OSD debug mode is enabled. 1 = A new pixel is ready at register VIU_OSD1_TEST_RDDATA; 0 = No data ready.
/	/	/	/
27-24	R	0	OSD_BLK_MODE: the input pixel format of which the current OSD block is being processed. Mali src & normal src have different pixel format value for same format
23-22	R	0	OSD_BLK_PTR: The number of the current OSD block that is being processed.
21	R	0	OSD_ENABLE. 1 = OSD display is enabled; 0 = disabled.
20-12	R/W	0	GLOBAL_ALPHA: legal range 0 – 256. It is a 9-bit value that is multiplied to all output pixel's Alpha value, and then normalized, i.e.: Alpha_tmp = Alpha_internal + (Alpha_internal == 0 ? 0 : 1); Alpha_out = (Alpha_tmp * GLOBAL_ALPHA) / 256;
11	R/W	0	TEST_RD_EN: OSD debug mode enable. 1 = Output pixels are not routed to VPP, instead they are presented on registers VIU_OSD1_TEST_RDDATA, for CPU to read. 0 = Normal mode, pixels are output to VPP.
10-9	R/W	0	unused
8-5	R/W	0	CTRL_MTCH_Y: For OSD 444, 422 or 16-bit (COLOR_MATRIX = 0 or 1) mode, the input pixels contain no Alpha information, in order to associated the output pixel with an Alpha value, the following steps are taken: If TC_ALPHA_EN = 0, then all output pixels use a default Alpha value 0xFF; If TC_ALPHA_EN = 1, then the Alpha value is looked up by matching the pixel's Y/Cb/Cr against four Alpha registers' Y/Cb/Cr. If the pixel matches any one of the Alpha registers, then this register's Alpha value is used; If the pixel matches with more than one of the Alpha registers, then the lower Alpha register takes priority, e.g. use Alpha Reg0's value if the pixel match both Alpha Reg0 and Reg1; If no match, then use default Alpha value 0xFF. There are two ways of matching: one way is that the pixel has to compare all Y, Cb and Cr value with the Alpha registers; the other way is that the pixel only has to compare Y value with the Alpha registers. CTRL_MTCH_Y defines which way is used to determine a match. Bit[0] is for matching Alpha register 0: 1 = only need to compare Y; 0 = compare all Y, Cb and Cr. Bit[1] is for matching Alpha register 1: 1 = only need to compare Y; 0 = compare all Y, Cb and Cr. Bit[2] is for matching Alpha register 2: 1 = only need to compare Y; 0 = compare all Y, Cb and Cr. Bit[3] is for matching Alpha register 3: 1 = only need to compare Y; 0 = compare all Y, Cb and Cr.
4	R/W	0	CTRL_422TO444. 1 = Enable conversion of 422 format input to 444 format output; 0 = Disable 422 to 444 conversion.
/	/	/	/
2	R/W	0	osd_mem_mode : // unsigned , default =0 0:canvas_araddr 1: linear_araddr (mali src must use this mode)

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Bit(s)	R/W	Default	Description
1	R/W	0	premult_en : // unsigned , default =0
0	R/W	0	OSD_BLK_ENABLE: Each bit to enable display an OSD block

Table 8-2396 VIU_OSD1_CTRL_STAT2 0x1A2d

Bit(s)	R/W	Default	Description
31-16	R	0	unused
15	R/W	0	unused
14	R/W	0	Replaced_alpha_en
13-6	R/W	0	Replaced_alpha
5-4	R/W	0	Hold_fifo_lines[6:5]
3	R/W	0	RGBYUV_FULL_RANGE: Select coefficients for applicable output range. 1 = output full range 0-255; 0 = output range 16-235.
2	R/W	0	ALPHA_9B_MODE: Define how to expand 8-bit alpha value to 9-bit. 1 = The formula is (Alpha < 128) ? Alpha : Alpha + 1; 0 = The formula is (Alpha == 0) ? Alpha : Alpha + 1.
1	R/W	0	Pedding status cleanup
0	R/W	0	COLOR_EXPAND_MODE. 1 = Expand the color components to 8-bit by padding LSBs with MSBs. E.g. If the input is 5'b11000, the output is expanded to 8'b11000110; 0 = Expand the color components to 8-bit by padding LSBs with 0.

Table 8-2397 VIU_OSD1_COLOR_ADDR 0x1A11

Bit(s)	R/W	Default	Description
31-9	R	0	Unused
8	R/W	1	0 = Write LUT, 1 = Read LUT.
7-0	R/W	0	LUT_ADDR: For 2-bit, 4-bit and 8-bit color lookup mode. The initial read or write address of the look-up table.

Table 8-2398 VIU_OSD1_COLOR 0x1A12

Bit(s)	R/W	Default	Description
31-24	R/W	0	Current LUT entry's Y or R.
23-16	R/W	0	Current LUT entry's CB or G.
15-8	R/W	0	Current LUT entry's CR or B.
7-0	R/W	0	Current LUT entry's ALPHA.

VIU_OSD1_TCOLOR_AG0 0x1A17

VIU_OSD1_TCOLOR_AG1 0x1A18

VIU_OSD1_TCOLOR_AG2 0x1A19

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VIU_OSD1_TCOLOR_AG3 0x1A1a

Table 8-2399 Define Alpha register 0/1/2/3 values.

Bit(s)	R/W	Default	Description
31-24	R/W	0xFF	Y or R.
23-16	R/W	0xFF	CB or G.
15-8	R/W	0xFF	CR or B.
7-0	R/W	0xFF	ALPHA.

Table 8-2400 VIU_OSD1_BLK0_CFG_W0 0x1A1b Defines display block 0/1/2/3's property, word 0.

Bit(s)	R/W	Default	Description
31	R/W	0	Reserved
30	R/W	0	mali_src_en 1: read data from mali afbcd decoder 0: read data from DDR directly
29	R/W	0	y_rev: 0=normal read, 1=reverse read in Y direction
28	R/W	0	x_rev: 0=normal read, 1=reverse read in X direction
27-24	R/W	0	Reserved
23-16	R/W	0	TBL_ADDR. Virtual canvas LUT entry.
15	R/W	0	LITTLE_ENDIAN: define the of data stored in DDR. 1 = Data stored in DDR memory are of little endian; 0 = Data stored in DDR memory are of big endian.
14	R/W	0	RPT_Y: For reducing data size stored in DDR. 1 = For each line, OSD will display twice; 0 = No repeat, OSD display once per line.
13-12	R/W	0	INTERP_CTRL: If enabled, interpolate a data after each incoming pixels, in order to save DDR bandwidth. 0 = No interpolation; 1 = unused, no interpolation; 2 = Interpolate with preceding pixel value; 3 = Interpolate with the averaged value between the preceding pixel and the next pixel.
11-8	R/W	0	OSD_BLK_MODE: Define the OSD block's input pixel format according bit30. Bit30 == 0: 0 = 2-bit per pixel, totally 4 colors can be looked up from color palette LUT,Only OSD2 have; 1 = 4-bit per pixel, totally 16 colors can be looked up from color palette LUT,Only OSD2 have; 2 = 8-bit per pixel, totally 256 colors can be looked up from color palette LUT,Only OSD2 have; 3 = 4:2:2 mode. Input 32-bit data for 2 pixels. Bit[31:24] is Y0, bit [23:16] is Cb0, bit[15:8] is Y1, bit [7:0] is Cr0, for Alpha value refer to reg VIU_OSD1_CTRL_STAT.CTRL_MTCH_Y; 4 = 16-bit mode. Refer to COLOR_MATRIX; 5 = 32-bit mode. Refer to COLOR_MATRIX; 6 = unused; 7 = 24-bit mode. Refer to COLOR_MATRIX; 8-15 = unused; Bit30 == 1: 0:R8 1:8bit yuv422 2:RGB565 3:RGBA5551 4:RGBA4444 5: RGBA8888 7:RGB888 8:10bit yuv422 9:RGBA1010102 Other:unuse
6	R/W	0	TC_ALPHA_EN: refer to reg VIU_OSD1_CTRL_STAT.CTRL_MTCH_Y. 1 = Enable alpha register matching.

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Bit(s)	R/W	Default	Description
			0 = Disable.
5-2	R/W	0	COLOR_MATRIX: Applicable only to 16-bit color mode (OSD_BLK_MODE=4), 32-bit mode (OSD_BLK_MODE=5) and 24-bit mode (OSD_BLK_MODE=7), defines the bit-field allocation of the pixel data. For expanding the bit-fields to full 8-bit, refer to VIU_OSD1_CTRL_STAT2.color_expand_mode and VIU_OSD1_CTRL_STAT2.alpha_expand_mode. For 16-bit mode (OSD_BLK_MODE=4): 0 = 6:5:5 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:5] is Cb[7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3], for Alpha value refer to reg VIU_OSD1_CTRL_STAT.CTRL_MTCH_Y; 1 = 8:4:4 format. Bit[15:8] is Y or R, bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]], for Alpha value refer to reg VIU_OSD1_CTRL_STAT.CTRL_MTCH_Y; 2 = 6:4:4:2 format. Bit[15:10] is Y[7:2] or R[7:2], bit[9:6] is Cb[7:4] or G[7:4], bit[5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[7:6]; 3 = 4:4:4:4 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:8] is Cb[7:4] or G[7:4], bit[7:4] is Cr[7:4] or B[7:4], bit[3:0] is Alpha[7:4]; 4 = 5:6:5 format. Bit[15:11] is Y[7:3] or R[7:3], bit[10:5] is Cb[7:2] or G[7:2], bit[4:0] is Cr[7:3] or B[7:3], for Alpha value refer to reg VIU_OSD1_CTRL_STAT.CTRL_MTCH_Y; 5 = 4:4:4:4 format. Bit[15:12] is Alpha[7:4], bit[11:8] is Y[7:4] or R[7:4], bit[7:4] is Cb[7:4] or G[7:4], bit[3:0] is Cr[7:4] or B[7:4]; 6 = 1:5:5:5 format. Bit[15] is Alpha[7], bit[14:10] is Y[7:3] or R[7:3], bit[9:5] is Cb[7:3] or G[7:3], bit[4:0] is Cr[7:3] or B[7:3]; 7 = 4:6:4:2 format. Bit[15:12] is Y[7:4] or R[7:4], bit[11:6] is Cb[7:2] or G[7:2], bit[5:2] is Cr[7:4] or B[7:4], bit[1:0] is Alpha[1:0]. For 32-bit mode (OSD_BLK_MODE=5): 0 = RGBA 8:8:8:8 format. Bit[31:24] is Y or R, bit[23:16] is Cb or G; bit[15:8] is Cr or B; bit[7:0] is Alpha; 1 = ARGB 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Y or R; bit[15:8] is Cb or G; bit[7:0] is Cr or B; 2 = ABGR 8:8:8:8 format. Bit[31:24] is Alpha, bit[23:16] is Cr or B; bit[15:8] is Cb or G; bit[7:0] is Y or R; 3 = BGRA 8:8:8:8 format. Bit[31:24] is Cr or B, bit[23:16] is Cb or G; bit[15:8] is Y or R; bit[7:0] is Alpha. For 24-bit mode (OSD_BLK_MODE=7): 0 = RGB 8:8:8 mode. Bit[23:16] is Y or R, bit[15:8] is Cb or G, bit[7:0] is Cr or B, for Alpha value refer to reg VIU_OSD1_CTRL_STAT.CTRL_MTCH_Y; 1 = RGBA 5:6:5:8 mode. Bit[23:19] is Y[7:3] or R[7:3], bit[18:13] is Cb[7:2] or G[7:2], bit[12:8] is Cr[7:3] or B[7:3], bit[7:0] is Alpha; 2 = ARGB 8:5:6:5 mode. Bit[23:16] is Alpha, bit[15:11] is Y[7:3] or R[7:3], bit[10:5] is Cb[7:2] or G[7:2], bit[4:0] is Cr[7:3] or B[7:3]; 3 = RGBA 6:6:6:6 mode. Bit[23:18] is Y[7:2] or R[7:2], bit[17:12] is Cb[7:2] or G[7:2], bit[11:6] is Cr[7:2] or B[7:2], bit[5:0] is Alpha[7:2]; 4 = ARGB 6:6:6:6 mode. Bit[23:18] is Alpha[7:2];, bit[17:12] is Y[7:2] or R[7:2], bit[11:6] is Cb[7:2] or G[7:2], bit[5:0] is Cr[7:2] or B[7:2]; 5 = BGR 8:8:8 mode. Bit[23:16] is Cr or B, bit[15:8] is Cb or G, bit[7:0] is Y or R, for Alpha value refer to reg VIU_OSD1_CTRL_STAT.CTRL_MTCH_Y.
1	R/W	0	INTERLACE_EN. 1 = Enable interlace mode. 0 = Disable.
0	R/W	0	INTERLACE_SEL_ODD: Applicable only if INTERLACE_EN = 1. 1 = Only output odd lines; 0 = Only output even lines.

Table 8-2401 VIU_OSD1_BLK0_CFG_W1 0x1A1c Defines display block 0/1/2/3's property, word 1.

Bit(s)	R/W	Default	Description
31-29	R/W	0	Unused.
28-16	R/W	0	X_END. Virtual canvas co-ordinate.
15-13	R/W	0	Unused.
12-0	R/W	0	X_START. Virtual canvas co-ordinate.

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Table 8-2402 VIU_OSD1_BLK0_CFG_W2 0x1A1d Defines display block 0/1/2/3's property, word 2.

Bit(s)	R/W	Default	Description
31-29	R/W	0	Unused.
28-16	R/W	0	Y_END. Virtual canvas co-ordinate.
15-13	R/W	0	Unused.
12-0	R/W	0	Y_START. Virtual canvas co-ordinate.

Table 8-2403 VIU_OSD1_BLK0_CFG_W3 0x1A1e Defines display block 0/1/2/3's property, word 3.

Bit(s)	R/W	Default	Description
31-28	R/W	0	Unused.
27-16	R/W	0	H_END. Display horizontal co-ordinate.
15-12	R/W	0	Unused.
11-0	R/W	0	H_START. Display horizontal co-ordinate.

Table 8-2404 VIU_OSD1_BLK0_CFG_W4 0x1A13 Defines display block 0/1/2/3's property, word 4.

Bit(s)	R/W	Default	Description
31-28	R/W	0	Unused.
27-16	R/W	0	V_END. Display vertical co-ordinate.
15-12	R/W	0	Unused.
11-0	R/W	0	V_START. Display vertical co-ordinate.

Table 8-2405 VIU_OSD1_BLK1_CFG_W4 0x1a14

Bit(s)	R/W	Default	Description
31:0	R/W	0	Frame_addr: // unsigned , default =0 Frame_addr in linear_addr

Table 8-2406 VIU_OSD1_BLK2_CFG_W4 0x1a15

Bit(s)	R/W	Default	Description
31:0	R/W	0	Line_stride : // unsigned , default =0 Line_stride in linear_addr

Table 8-2407 VIU_OSD1_FIFO_CTRL_STAT 0x1A2b

Bit(s)	R/W	Default	Description
31	R/W	0	burst_len_sel[2] of [2:0]
30	R/W	0	BYTE_SWAP: In addition to endian control, further define whether to swap upper byte and lower byte within a 16-bit memory word. 1 = Swap, data[15:0] becomes {data[7:0], data[15:8]}; 0 = No swap, data[15:0] is still data[15:0].

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Bit(s)	R/W	Default	Description
29	R/W	0	Div_swap : swap the 2 64bits word in 128bits word
28-24	R/W	0	Fifo_lim : when osd fifo is small than the fifo_lim*16, closed the req port of osd_ rd_mif
23-22	R/W	0	Fifo_ctrl: 00 : for 1 word in 1 burst, 01 : for 2words in 1burst, 10 : for 4 words in 1burst, 11: reserved
21-20	R	0	FIFO_ST: State of the FIFO activity. 0 = Idle; 1 = FIFO requesting; 2 = FIFO request aborting.
19	R	0	FIFO_OVERFLOW.
18-12	R/W	32	FIFO_DEPTH_VAL: Define the depth of FIFO which stores 128-bit data from DDR to be FIFO_DEPTH_VAL * 8.
11-10	R/W	0	BURST_LEN_SEL[1:0] of [2:0]: Define DDR burst request length. 0 = up to 24 per burst; 1 = up to 32 per burst; 2 = up to 48 per burst;/ 3 = up to 64 per burst. 4 = up to 96 per burst, 5 = up to 128 per burst
9-5	R/W	4	HOLD_FIFO_LINES: The number of lines that OSD must wait after VSYNC, before it starts request data from DDR .
4	R/W	0	CLEAR_ERR: One pulse to clear error status.
3	R/W	0	FIFO_SYNC_RST: Set 1 to reset OSD FIFO.
2-1	R/W	0	ENDIAN: define the endianess of the 64-bit data stored in memory, and how to convert. 0 = No convertion; 1 = Convert to {din[31:0], din[63:32]}; 2 = Convert to {din[15:0], din[31:16], din[47:32], din[63:48]}; 3 = Convert to {din[47:32], din[63:48], din[15:0], din[31:16]};
0	R/W	0	URGENT. 1 = Set DDR request priority to be urgent; 0 = Set DDR request priority to be normal.

During OSD debug mode (VIU_OSD1_CTRL_STAT.TEST_RD_EN = 1), the output pixels will be presented at this register.

Table 8-2408 VIU_OSD1_TEST_RDDATA 0x1A2c

Bit(s)	R/W	Default	Description
31-24	R	0	Y or R.
23:16	R	0	Cb or G.
15-8	R	0	Cr or B.
7-0	R	0	Alpha[8:1].

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Table 8-2409 VIU_OSD1_PROT_CTRL 0x1a2e

Bit(s)	R/W	Default	Description
31:16	R/W	0	urgent_ctrl : // unsigned , default =0
15	R.O	0	prot_en : // unsigned , default =0; 1=Borrow PROT's FIFO storage, either for rotate or non-rotate.
12: 0	R.O	0	prot_fifo_size : // unsigned , default =0; effective FIFO size when prot_en=1.

Table 8-2410 VIU_OSD1_MALI_UNPACK_CTRL 0x1a2f

Bit(s)	R/W	Default	Description
31	R/W	0	mali_unpack_en 1: OSD will unpack mali_src 0: OSD will unpack normal src
28			Alpha_div_en: alpha dividor enable
27:26			Alpha_dividor gating clk
25:24			Alpa_mapping_mode In osd,this bit should be set 0 when Alpha_div_en active,means 8 bits alpha mode
17	/	/	/
16	R/W	0	afbcd_swap_64bit:
15: 12	R/W	1	afbcd_r_reorder,change osd output order when use mali src: 1: r_re = r ; 2: r_re = g; 3: r_re = b; 4: r_re = a; default: r_re = 0;
11: 8		2	afbcd_r_reorder,change osd output order when use mali src: 1: g_re = r ; 2: g_re = g; 3: g_re = b; 4: g_re = a; default: r_re = 0;
7: 4		3	afbcd_r_reorder,change osd output order when use mali src: 1: b_re = r ; 2: b_re = g; 3: b_re = b; 4: b_re = a; default: r_re = 0;
3: 0		4	afbcd_r_reorder,change osd output order when use mali src: 1: a_re = r ; 2: a_re = g; 3: a_re = b; 4: a_re = a; default: r_re = 0;

Table 8-2411 VIU_OSD1_DIMM_CTRL 0x1adf

Bit(s)	R/W	Default	Description
30	R/W	0	OSD dimm enable,osd out will be one color when this bit active
29:0	R/W	0	Osd_dim_rgb_out ,osd out will be this vaaule when bit30 active

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8.2.3.33 OSD2 Registers

VIU_OSD2_CTRL_STAT 0x1a30

See: VIU_OSD1_CTRL_STAT

VIU_OSD2_CTRL_STAT2 0x1a4d

See: VIU_OSD1_CTRL_STAT2

VIU_OSD2_COLOR_ADDR 0x1a31

See: VIU_OSD1_COLOR_ADDR

VIU_OSD2_COLOR 0x1a32

See: VIU_OSD1_COLOR

VIU_OSD2_TCOLOR_AG0 0x1a37

See: VIU_OSD1_TCOLOR_AG0

VIU_OSD2_TCOLOR_AG1 0x1a38

See: VIU_OSD1_TCOLOR_AG1

VIU_OSD2_TCOLOR_AG2 0x1a39

See: VIU_OSD1_TCOLOR_AG02

VIU_OSD2_TCOLOR_AG3 0x1a3a

See: VIU_OSD1_TCOLOR_AG3

VIU_OSD2_BLK0_CFG_W0 0x1a3b

See: VIU_OSD1_BLK0_CFG_W0

VIU_OSD2_BLK0_CFG_W1 0x1a3c

See: VIU_OSD1_BLK0_CFG_W1

VIU_OSD2_BLK0_CFG_W2 0x1a3d

See: VIU_OSD1_BLK0_CFG_W2

VIU_OSD2_BLK0_CFG_W3 0x1a3e

See: VIU_OSD1_BLK0_CFG_W3

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VIU_OSD2_BLK0_CFG_W4 0x1a64
See: VIU_OSD1_BLK0_CFG_W4

VIU_OSD2_BLK1_CFG_W4 0x1a65
See: VIU_OSD2_BLK1_CFG_W4

VIU_OSD2_BLK2_CFG_W4 0x1a66
See: VIU_OSD1_BLK2_CFG_W4

VIU_OSD2_FIFO_CTRL_STAT 0x1a4b
See: VIU_OSD1_FIFO_CTRL_STAT

VIU_OSD2_TEST_RDDATA 0x1a4c
See: VIU_OSD1_TEST_RDDATA

VIU_OSD2_PROT_CTRL 0x1a4e
See: VIU_OSD1_PROT_CTRL

VIU_OSD2_MALI_UNPACK_CTRL 0x1abd
See: VIU_OSD1_MALI_UNPACK_CTRL

VIU_OSD2_DIMM_CTRL 0x1acf
See: VIU_OSD1_DIMM_CTRL

8.2.3.34 OSD3 Registers

VIU_OSD3_CTRL_STAT 0x3d80
See: VIU_OSD1_CTRL_STAT

VIU_OSD3_CTRL_STAT2 0x3d81
See: VIU_OSD1_CTRL_STAT2

VIU_OSD3_COLOR_ADDR 0x3d82
See: VIU_OSD1_COLOR_ADDR

VIU_OSD3_COLOR 0x3d83
See: VIU_OSD1_COLOR

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VIU_OSD3_TCOLOR_AG0 0x3d84
See: VIU_OSD1_TCOLOR_AG0

VIU_OSD3_TCOLOR_AG1 0x3d85
See: VIU_OSD1_TCOLOR_AG1

VIU_OSD3_TCOLOR_AG2 0x3d86
See: VIU_OSD1_TCOLOR_AG02

VIU_OSD3_TCOLOR_AG3 0x3d87
See: VIU_OSD1_TCOLOR_AG3

VIU_OSD3_BLK0_CFG_W0 0x3d88
See: VIU_OSD1_BLK0_CFG_W0

VIU_OSD3_BLK0_CFG_W1 0x3d8c
See: VIU_OSD1_BLK0_CFG_W1

VIU_OSD3_BLK0_CFG_W2 0x3d90
See: VIU_OSD1_BLK0_CFG_W2

VIU_OSD3_BLK0_CFG_W3 0x3d94
See: VIU_OSD1_BLK0_CFG_W3

VIU_OSD3_BLK0_CFG_W4 0x3d98
See: VIU_OSD1_BLK0_CFG_W4

VIU_OSD3_BLK1_CFG_W4 0x3d99
See: VIU_OSD3_BLK1_CFG_W4

VIU_OSD3_BLK2_CFG_W4 0x3d9a
See: VIU_OSD1_BLK2_CFG_W4

VIU_OSD3_FIFO_CTRL_STAT 0x3d9c
See: VIU_OSD1_FIFO_CTRL_STAT

VIU_OSD3_TEST_RDDATA 0x3d9d
See: VIU_OSD1_TEST_RDDATA

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VIU_OSD3_PROT_CTRL 0x3d9e
See: VIU_OSD1_PROT_CTRL

VIU_OSD3_MALI_UNPACK_CTRL 0x3d9f
See: VIU_OSD1_MALI_UNPACK_CTRL

VIU_OSD3_DIMM_CTRL 0x3da0
See: VIU_OSD1_DIMM_CTRL

8.2.3.35 VPP_VD1_MATRIX Registers

Table 8-2412 VPP_VD1_MATRIX_COEF00_01 0x3290

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coefficient00, signed, 3.10
12-0	R/W	0	Coefficient01, signed, 3.10

Table 8-2413 VPP_VD1_MATRIX_COEF02_10 0x3291

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coefficient02, signed, 3.10
12-0	R/W	0	Coefficient10, signed, 3.10

Table 8-2414 VPP_VD1_MATRIX_COEF11_12 0x3292

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coefficient11, signed, 3.10
12-0	R/W	0	Coefficient12, signed, 3.10

Table 8-2415 VPP_VD1_MATRIX_COEF20_21 0x3293

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coefficient20, signed, 3.10
12-0	R/W	0	Coefficient21, signed, 3.10

Table 8-2416 VPP_VD1_MATRIX_COEF22 0x3294

Bit(s)	R/W	Default	Description
18-16	R/W	0	convrs
12-0	R/W	0	Coefficient22, signed, 3.10

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Table 8-2417 VPP_VD1_MATRIX_COEF13_14 0x3295

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coefficient13, signed, 3.10
12-0	R/W	0	Coefficient14, signed, 3.10

Table 8-2418 VPP_VD1_MATRIX_COEF23_24 0x3296

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coefficient23, signed, 3.10
12-0	R/W	0	Coefficient24, signed, 3.10

Table 8-2419 VPP_VD1_MATRIX_COEF15_25 0x3297

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coefficient15, signed, 3.10
12-0	R/W	0	Coefficient25, signed, 3.10

Table 8-2420 VPP_VD1_MATRIX_CLIP 0x3298

Bit(s)	R/W	Default	Description
31-8	R/W	0x0	reserved
7-5	R/W	0x1	Matrix rs
4-3	R/W	0x10	Matrix clmod 0: only 3x3, 1: pre_offseted_ch1,ch2> pre_offseted_ch0, use the added 2x3 coef 2: pre_offseted_ch1,ch2 > 0, use the added 2x3 coef 3: pre_offseted_ch1,ch2 > 512, use the added 2x3 coef
2-0	R/W	0x10	/

Table 8-2421 VPP_VD1_MATRIX_OFFSET0_1 0x3299

Bit(s)	R/W	Default	Description
27-16	R/W	0	Offset0, signed value
11-0	R/W	0	Offset1, signed value

Table 8-2422 VPP_VD1_MATRIX_OFFSET2 0x329a

Bit(s)	R/W	Default	Description
11-0	R/W	0	Offset2, signed value

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Table 8-2423 VPP_VD1_MATRIX_PRE_OFFSET0_1 0x329b

Bit(s)	R/W	Default	Description
27-16	R/W	0	pre_Offset0, signed value
11-0	R/W	0	Pre_Offset1, signed value

Table 8-2424 VPP_VD1_MATRIX_PRE_OFFSET2 0x329c

Bit(s)	R/W	Default	Description
11-0	R/W	0	Pre_Offset2, signed value

Table 8-2425 VPP_VD1_MATRIX_EN_CTRL 0x329d

Bit(s)	R/W	Default	Description
5:4	R/W	0	Gate clock ctrl
1	R/W	0	Enable_sync_sel
0	R/W	0	Conv_en_pre

8.2.3.36 VD2_HDR2_MATRIX Registers

VD2_HDR2_MATRIX_COEF00_01 0x3901

See: VPP_VD1_MATRIX_COEF00_01

VD2_HDR2_MATRIX_COEF02_10 0x3902

See: VPP_VD1_MATRIX_COEF02_10

VD2_HDR2_MATRIX_COEF11_12 0x3903

See: VPP_VD1_MATRIX_COEF11_12

VD2_HDR2_MATRIX_COEF20_21 0x3904

See: VPP_VD1_MATRIX_COEF20_21

VD2_HDR2_MATRIX_COEF22 0x3905

See: VPP_VD1_MATRIX_COEF22

VD2_HDR2_MATRIX_COEF13_14 0x3906

See: VPP_VD1_MATRIX_COEF13_14

VD2_HDR2_MATRIX_COEF23_24 0x3907

See: VPP_VD1_MATRIX_COEF23_24

VD2_HDR2_MATRIX_COEF15_25 0x3908

See: VPP_VD1_MATRIX_COEF15_25

VD2_HDR2_MATRIX_CLIP 0x3909

See: VPP_VD1_MATRIX_CLIP

VD2_HDR2_MATRIX_OFFSET0_1 0x390a

See: VPP_VD1_MATRIX_OFFSET0_1

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VD2_HDR2_MATRIX_OFFSET2 0x390b
See: VPP_VD1_MATRIX_OFFSET2

VD2_HDR2_MATRIX_PRE_OFFSET0_1 0x390c
See: VPP_VD1_MATRIX_PRE_OFFSET0_1

VD2_HDR2_MATRIX_PRE_OFFSET2 0x390d
See: VPP_VD1_MATRIX_PRE_OFFSET2

VD2_HDR2_MATRIX_EN_CTRL 0x390e
See: VPP_VD1_MATRIX_EN_CTRL

8.2.3.37 VPP_POST_MATRIX Registers

VPP_POST_MATRIX_COEF00_01 0x32b0
See: VPP_VD1_MATRIX_COEF00_01
VPP_POST_MATRIX_COEF02_10 0x32b1
See: VPP_VD1_MATRIX_COEF02_10
VPP_POST_MATRIX_COEF11_12 0x32b2
See: VPP_VD1_MATRIX_COEF11_12
VPP_POST_MATRIX_COEF20_21 0x32b3
See: VPP_VD1_MATRIX_COEF20_21
VPP_POST_MATRIX_COEF22 0x32b4
See: VPP_VD1_MATRIX_COEF22
VPP_POST_MATRIX_COEF13_14 0x32b5
See: VPP_VD1_MATRIX_COEF13_14
VPP_POST_MATRIX_COEF23_24 0x32b6
See: VPP_VD1_MATRIX_COEF23_24
VPP_POST_MATRIX_COEF15_25 0x32b7
See: VPP_VD1_MATRIX_COEF15_25
VPP_POST_MATRIX_CLIP 0x32b8
See: VPP_VD1_MATRIX_CLIP
VPP_POST_MATRIX_OFFSET0_1 0x32b9
See: VPP_VD1_MATRIX_OFFSET0_1

VPP_POST_MATRIX_OFFSET2 0x32ba
See: VPP_VD1_MATRIX_OFFSET2

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VPP_POST_MATRIX_PRE_OFFSET0_1 0x32bb

See: VPP_VD1_MATRIX_PRE_OFFSET0_1

VPP_POST_MATRIX_PRE_OFFSET2 0x32bc

See: VPP_VD1_MATRIX_PRE_OFFSET2

VPP_POST_MATRIX_EN_CTRL 0x32bd

See: VPP_VD1_MATRIX_EN_CTRL

Table 8-2426 VPP_POST_MATRIX_SAT 0x32c1

Bit(s)	R/W	Default	Description
2	R/W	0	sat_o_en:If true, the output result of post matrix output is saturated
1	R/W	0	sat_i_en:If true, the input result of post matrix output is saturated
0	R/W	0	misc_sat_en:same as sat_i_en

8.2.3.38 VPP_POST2_MATRIX Registers

VPP_POST2_MATRIX_COEF00_01 0x39a0

See: VPP_VD1_MATRIX_COEF00_01

VPP_POST2_MATRIX_COEF02_10 0x39a1

See: VPP_VD1_MATRIX_COEF02_10

VPP_POST2_MATRIX_COEF11_12 0x39a2

See: VPP_VD1_MATRIX_COEF11_12

VPP_POST2_MATRIX_COEF20_21 0x39a3

See: VPP_VD1_MATRIX_COEF20_21

VPP_POST2_MATRIX_COEF22 0x39a4

See: VPP_VD1_MATRIX_COEF22

VPP_POST2_MATRIX_COEF13_14 0x39a5

See: VPP_VD1_MATRIX_COEF13_14

VPP_POST2_MATRIX_COEF23_24 0x39a6

See: VPP_VD1_MATRIX_COEF23_24

VPP_POST2_MATRIX_COEF15_25 0x39a7

See: VPP_VD1_MATRIX_COEF15_25

VPP_POST2_MATRIX_CLIP 0x39a8

See: VPP_VD1_MATRIX_CLIP

VPP_POST2_MATRIX_OFFSET0_1 0x39a9

See: VPP_VD1_MATRIX_OFFSET0_1

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VPP_POST2_MATRIX_OFFSET2 0x39aa
See: VPP_VD1_MATRIX_OFFSET2

VPP_POST2_MATRIX_PRE_OFFSET0_1 0x39ab
See: VPP_VD1_MATRIX_PRE_OFFSET0_1

VPP_POST2_MATRIX_PRE_OFFSET2 0x39ac
See: VPP_VD1_MATRIX_PRE_OFFSET2

VPP_POST2_MATRIX_EN_CTRL 0x39ad
See: VPP_VD1_MATRIX_EN_CTRL

8.2.3.39 VPP_OSD2_MATRIX Registers

VPP_OSD2_MATRIX_COEF00_01 0x3920
See: VPP_VD1_MATRIX_COEF00_01

VPP_OSD2_MATRIX_COEF02_10 0x3921
See: VPP_VD1_MATRIX_COEF02_10

VPP_OSD2_MATRIX_COEF11_12 0x3922
See: VPP_VD1_MATRIX_COEF11_12

VPP_OSD2_MATRIX_COEF20_21 0x3923
See: VPP_VD1_MATRIX_COEF20_21

VPP_OSD2_MATRIX_COEF22 0x3924
See: VPP_VD1_MATRIX_COEF22

VPP_OSD2_MATRIX_COEF13_14 0x3925
See: VPP_VD1_MATRIX_COEF13_14

VPP_OSD2_MATRIX_COEF23_24 0x3926
See: VPP_VD1_MATRIX_COEF23_24

VPP_OSD2_MATRIX_COEF15_25 0x3927
See: VPP_VD1_MATRIX_COEF15_25

VPP_OSD2_MATRIX_CLIP 0x3928

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See: VPP_VD1_MATRIX_CLIP

VPP_OSD2_MATRIX_OFFSET0_1 0x3929
See: VPP_VD1_MATRIX_OFFSET0_1

VPP_OSD2_MATRIX_OFFSET2 0x392a
See: VPP_VD1_MATRIX_OFFSET2

VPP_OSD2_MATRIX_PRE_OFFSET0_1 0x392b
See: VPP_VD1_MATRIX_PRE_OFFSET0_1

VPP_OSD2_MATRIX_PRE_OFFSET2 0x392c
See: VPP_VD1_MATRIX_PRE_OFFSET2

VPP_OSD2_MATRIX_EN_CTRL 0x392d
See: VPP_VD1_MATRIX_EN_CTRL

8.2.3.40 VPP_WRAP_OSD1_MATRIX Registers

VPP_WRAP_OSD1_MATRIX_COEF00_01 0x3d60
See: VPP_VD1_MATRIX_COEF00_01

VPP_WRAP_OSD1_MATRIX_COEF02_10 0x3d61
See: VPP_VD1_MATRIX_COEF02_10

VPP_WRAP_OSD1_MATRIX_COEF11_12 0x3d62
See: VPP_VD1_MATRIX_COEF11_12

VPP_WRAP_OSD1_MATRIX_COEF20_21 0x3d63
See: VPP_VD1_MATRIX_COEF20_21

VPP_WRAP_OSD1_MATRIX_COEF22 0x3d64
See: VPP_VD1_MATRIX_COEF22

VPP_WRAP_OSD1_MATRIX_COEF13_14 0x3d65
See: VPP_VD1_MATRIX_COEF13_14

VPP_WRAP_OSD1_MATRIX_COEF23_24 0x3d66
See: VPP_VD1_MATRIX_COEF23_24

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VPP_WRAP_OSD1_MATRIX_COEF15_25 0x3d67
See: VPP_VD1_MATRIX_COEF15_25

VPP_WRAP_OSD1_MATRIX_CLIP 0x3d68
See: VPP_VD1_MATRIX_CLIP

VPP_WRAP_OSD1_MATRIX_OFFSET0_1 0x3d69
See: VPP_VD1_MATRIX_OFFSET0_1

VPP_WRAP_OSD1_MATRIX_OFFSET2 0x3d6a
See: VPP_VD1_MATRIX_OFFSET2

VPP_WRAP_OSD1_MATRIX_PRE_OFFSET0_1 0x3d6b
See: VPP_VD1_MATRIX_PRE_OFFSET0_1

VPP_WRAP_OSD1_MATRIX_PRE_OFFSET2 0x3d6c
See: VPP_VD1_MATRIX_PRE_OFFSET2

VPP_WRAP_OSD1_MATRIX_EN_CTRL 0x3d6d
See: VPP_VD1_MATRIX_EN_CTRL

8.2.3.41 VPP_WRAP_OSD2_MATRIX Registers

VPP_WRAP_OSD2_MATRIX_COEF00_01 0x3d70
See: VPP_VD1_MATRIX_COEF00_01

VPP_WRAP_OSD2_MATRIX_COEF02_10 0x3d71
See: VPP_VD1_MATRIX_COEF02_10

VPP_WRAP_OSD2_MATRIX_COEF11_12 0x3d72
See: VPP_VD1_MATRIX_COEF11_12

VPP_WRAP_OSD2_MATRIX_COEF20_21 0x3d73
See: VPP_VD1_MATRIX_COEF20_21

VPP_WRAP_OSD2_MATRIX_COEF22 0x3d74
See: VPP_VD1_MATRIX_COEF22

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VPP_WRAP_OSD2_MATRIX_COEF13_14 0x3d75
See: VPP_VD1_MATRIX_COEF13_14

VPP_WRAP_OSD2_MATRIX_COEF23_24 0x3d76
See: VPP_VD1_MATRIX_COEF23_24

VPP_WRAP_OSD2_MATRIX_COEF15_25 0x3d77
See: VPP_VD1_MATRIX_COEF15_25

VPP_WRAP_OSD2_MATRIX_CLIP 0x3d78
See: VPP_VD1_MATRIX_CLIP

VPP_WRAP_OSD2_MATRIX_OFFSET0_1 0x3d79
See: VPP_VD1_MATRIX_OFFSET0_1

VPP_WRAP_OSD2_MATRIX_OFFSET2 0x3d7a
See: VPP_VD1_MATRIX_OFFSET2

VPP_WRAP_OSD2_MATRIX_PRE_OFFSET0_1 0x3d7b
See: VPP_VD1_MATRIX_PRE_OFFSET0_1

VPP_WRAP_OSD2_MATRIX_PRE_OFFSET2 0x3d7c
See: VPP_VD1_MATRIX_PRE_OFFSET2

VPP_WRAP_OSD2_MATRIX_EN_CTRL 0x3d7d
See: VPP_VD1_MATRIX_EN_CTRL

8.2.3.42 VPP_WRAP_OSD3_MATRIX Registers

VPP_WRAP_OSD3_MATRIX_COEF00_01 0x3db0
See: VPP_VD1_MATRIX_COEF00_01

VPP_WRAP_OSD3_MATRIX_COEF02_10 0x3db1
See: VPP_VD1_MATRIX_COEF02_10

VPP_WRAP_OSD3_MATRIX_COEF11_12 0x3db2
See: VPP_VD1_MATRIX_COEF11_12

VPP_WRAP_OSD3_MATRIX_COEF20_21 0x3db3

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See: VPP_VD1_MATRIX_COEF20_21

VPP_WRAP_OSD3_MATRIX_COEF22 0x3db4
See: VPP_VD1_MATRIX_COEF22

VPP_WRAP_OSD3_MATRIX_COEF13_14 0x3db5
See: VPP_VD1_MATRIX_COEF13_14

VPP_WRAP_OSD3_MATRIX_COEF23_24 0x3db6
See: VPP_VD1_MATRIX_COEF23_24

VPP_WRAP_OSD3_MATRIX_COEF15_25 0x3db7
See: VPP_VD1_MATRIX_COEF15_25

VPP_WRAP_OSD3_MATRIX_CLIP 0x3db8
See: VPP_VD1_MATRIX_CLIP

VPP_WRAP_OSD3_MATRIX_OFFSET0_1 0x3db9
See: VPP_VD1_MATRIX_OFFSET0_1

VPP_WRAP_OSD3_MATRIX_OFFSET2 0x3dba
See: VPP_VD1_MATRIX_OFFSET2

VPP_WRAP_OSD3_MATRIX_PRE_OFFSET0_1 0x3dbb
See: VPP_VD1_MATRIX_PRE_OFFSET0_1

VPP_WRAP_OSD3_MATRIX_PRE_OFFSET2 0x3dbc
See: VPP_VD1_MATRIX_PRE_OFFSET2

VPP_WRAP_OSD3_MATRIX_EN_CTRL 0x3dbd
See: VPP_VD1_MATRIX_EN_CTRL

8.2.3.43 HDR Registers

Table 8-2427 HDR VDIN0_HDR2_CTRL 0x1280

Bit(s)	R/W	Default	Description
20:18	R/W	0	reg_din_swap : // unsigned , default = 0
17	R/W	0	reg_out_fmt : // unsigned , default = 0

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Bit(s)	R/W	Default	Description
16	R/W	0	reg_only_mat : // unsigned , default = 0 ,only use input matrix ,work when hdr disable
15	R/W	0	mat_o_en,//output matrix enable ,only work when hdr enable
14	R/W	0	mat_in_en //input matrix enable ,only work when hdr enable
13	R/W	0	reg_VDIN0_HDR2_top_en : // unsigned , default = 0,hdr enable singal
12	R/W	1	reg_cgain_mode : // unsigned , default = 1
7: 6	R/W	1	reg_gmut_mode : // unsigned , default = 1
5	R/W	0	reg_in_shift : // unsigned , default = 0
4	R/W	1	reg_in_fmt : // unsigned , default = 1
3	R/W	1	reg_eo_enable : // unsigned , default = 1
2	R/W	1	reg_oe_enable : // unsigned , default = 1
1	R/W	1	reg_ogain_enable : // unsigned , default = 1
0	R/W	1	reg_cgain_enable : // unsigned , default = 1

Table 8-2428 VDIN0_HDR2_CLK_GATE 0x1281

Bit(s)	R/W	Default	Description
31:30	R/W	0	clk_tm : gate clock ctrl (main clock) // unsigned , default = 0
29:28	R/W	0	output : matrix clock gate ctrl// unsigned , default = 0
25:24	R/W	0	input : matrix clock gate ctrl // unsigned , default = 0
23:22	R/W	0	hdr : top cbus clock gate ctrl // unsigned , default = 0
21:20	R/W	0	eotf : cbus clock gate ctrl // unsigned , default = 0
19:18	R/W	0	oetf : cbus clock gate ctrl // unsigned , default = 0
17:16	R/W	0	gamma : mult cbus clock gate ctrl // unsigned , default = 0
15:14	R/W	0	adaptive : cbus scaler clock gate ctrl // unsigned , default = 0
13:12	R/W	0	cgain : cbus clock gate ctrl// unsigned , default = 0
11:10	R/W	0	eotf : clock gate ctrl // unsigned , default = 0
9:8	R/W	0	oetf : clock gate ctrl // unsigned , default = 0
7:6	R/W	0	gamma : mult clock gate ctrl// unsigned , default = 0
5:4	R/W	0	adaptive : scaler clock gate ctrl // unsigned , default = 0
3:2	R/W	0	uv : gain clock gate ctrl // unsigned , default = 0
1:0	R/W	0	cgain : clock gate ctrl // unsigned , default = 0

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Table 8-2429 VDIN0_HDR2_MATRIXI_COEF00_01 0x1282

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2430 VDIN0_HDR2_MATRIXI_COEF02_10 0x1283

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-2431 VDIN0_HDR2_MATRIXI_COEF11_12 0x1284

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2432 VDIN0_HDR2_MATRIXI_COEF20_21 0x1285

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2433 VDIN0_HDR2_MATRIXI_COEF22 0x1286

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-2434 VDIN0_HDR2_MATRIXI_COEF30_31 0x1287

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2435 VDIN0_HDR2_MATRIXI_COEF32_40 0x1288

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

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Table 8-2436 VDIN0_HDR2_MATRIXI_COEF41_42 0x1289

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2437 VDIN0_HDR2_MATRIXI_OFFSET0_1 0x128A

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-2438 VDIN0_HDR2_MATRIXI_OFFSET2 0x128B

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2439 VDIN0_HDR2_MATRIXI_PRE_OFFSET0_1 0x128C

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2440 VDIN0_HDR2_MATRIXI_PRE_OFFSET2 0x128D

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-2441 VDIN0_HDR2_MATRIXO_COEF00_01 0x128E

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2442 VDIN0_HDR2_MATRIXO_COEF02_10 0x128F

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

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Table 8-2443 VDIN0_HDR2_MATRIXO_COEF11_12 0x1290

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2444 VDIN0_HDR2_MATRIXO_COEF20_21 0x1291

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2445 VDIN0_HDR2_MATRIXO_COEF22 0x1292

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-2446 VDIN0_HDR2_MATRIXO_COEF30_31 0x1293

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2447 VDIN0_HDR2_MATRIXO_COEF32_40 0x1294

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-2448 VDIN0_HDR2_MATRIXO_COEF41_42 0x1295

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2449 VDIN0_HDR2_MATRIXO_OFFSET0_1 0x1296

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

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Table 8-2450 VDIN0_HDR2_MATRIXO_OFFSET2 0x1297

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2451 VDIN0_HDR2_MATRIXO_PRE_OFFSET0_1 0x1298

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2452 VDIN0_HDR2_MATRIXO_PRE_OFFSET2 0x1299

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-2453 VDIN0_HDR2_MATRIXI_CLIP 0x129A

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2454 VDIN0_HDR2_MATRIXO_CLIP 0x129B

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2455 VDIN0_HDR2_CGAIN_OFFT 0x129C

Bit(s)	R/W	Default	Description
26:16	R/W	0	reg_cgain_oft2 : // signed , default = 0
10:0	R/W	0	reg_cgain_oft1 : // signed , default = 0

Table 8-2456 VDIN0_EOTF_LUT_ADDR_PORT 0x129E

Bit(s)	R/W	Default	Description
7:0	R/W	0	eotf_lut_addr : // unsigned , default = 0

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Table 8-2457 VDIN0_EOTF_LUT_DATA_PORT 0x129F

Bit(s)	R/W	Default	Description
19:0	R/W	0	eotf_lut_data : // unsigned , default = 0

Table 8-2458 VDIN0_OETF_LUT_ADDR_PORT 0x12A0

Bit(s)	R/W	Default	Description
7:0	R/W	0	oetf_lut_addr : // unsigned , default = 0

Table 8-2459 VDIN0_OETF_LUT_DATA_PORT 0x12A1

Bit(s)	R/W	Default	Description
11:0	R/W	0	oetf_lut_data : // unsigned , default = 0

Table 8-2460 VDIN0_CGAIN_LUT_ADDR_PORT 0x12A2

Bit(s)	R/W	Default	Description
7:0	R/W	0	cgain_lut_addr : // unsigned , default = 0

Table 8-2461 VDIN0_CGAIN_LUT_DATA_PORT 0x12A3

Bit(s)	R/W	Default	Description
11:0	R/W	0	cgain_lut_data : // unsigned , default = 0

Table 8-2462 VDIN0_HDR2_CGAIN_COEF0 0x12A4

Bit(s)	R/W	Default	Description
27:16	R/W	0	reg_cgain_coef1 : // unsigned , default = 0
11:0	R/W	0	reg_cgain_coef0 : //unsigned , default = 0

Table 8-2463 VDIN0_HDR2_CGAIN_COEF1 0x12A5

Bit(s)	R/W	Default	Description
11:0	R/W	0	reg_cgain_coef2 : // unsigned , default = 0

Table 8-2464 VDIN0_OGAIN_LUT_ADDR_PORT 0x12A6

Bit(s)	R/W	Default	Description
7:0	R/W	0	ogain_lut_addr : // unsigned , default = 0

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Table 8-2465 VDIN0_OGAIN_LUT_DATA_PORT 0x12A7

Bit(s)	R/W	Default	Description
11:0	R/W	0	ogain_lut_data : // unsigned , default = 0

Table 8-2466 VDIN0_HDR2_ADPS_CTRL 0x12A8

Bit(s)	R/W	Default	Description
6	R/W	1	reg_adpscl_bypass2 : // unsigned , default = 1
5	R/W	1	reg_adpscl_bypass1 : // unsigned , default = 1
4	R/W	1	reg_adpscl_bypass0 : // unsigned , default = 1
1:0	R/W	1	reg_adpscl_mode : // unsigned , default = 1

Table 8-2467 VDIN0_HDR2_ADPS_ALPHA0 0x12A9

Bit(s)	R/W	Default	Description
29:16	R/W	0x1000	reg_adpscl_alpha1 : // unsigned , default = 0x1000
13:0	R/W	0x1000	reg_adpscl_alpha0 : // unsigned , default = 0x1000

Table 8-2468 VDIN0_HDR2_ADPS_ALPHA1 0x12AA

Bit(s)	R/W	Default	Description
27:24	R/W	0xc	reg_adpscl_shift0 : // unsigned , default = 0xc
23:20	R/W	0xc	reg_adpscl_shift1 : // unsigned , default = 0xc
19:16	R/W	0xc	reg_adpscl_shift2 : // unsigned , default = 0xc
13:0	R/W	0x1000	reg_adpscl_alpha2 : // unsigned , default = 0x1000

Table 8-2469 VDIN0_HDR2_ADPS_BETA0 0x12AB

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta0_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta0 : // unsigned , default = 0xfc000

Table 8-2470 VDIN0_HDR2_ADPS_BETA1 0x12AC

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta1_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta1 : // unsigned , default = 0xfc000

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Table 8-2471 VDIN0_HDR2_ADPS_BETA2 0x12AD

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta2_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta2 : // unsigned , default = 0xfc000

Table 8-2472 VDIN0_HDR2_ADPS_COEF0 0x12AE

Bit(s)	R/W	Default	Description
27:16	R/W	460	reg_adpscl_ys_coef1 : // unsigned , default = 460
11:0	R/W	1188	reg_adpscl_ys_coef0 : // unsigned , default = 1188

Table 8-2473 VDIN0_HDR2_ADPS_COEF1 0x12AF

Bit(s)	R/W	Default	Description
11:0	R/W	104	reg_adpscl_ys_coef2 : // unsigned , default = 104

Table 8-2474 VDIN0_HDR2_GMUT_CTRL 0x12B0

Bit(s)	R/W	Default	Description
3:0	R/W	14	reg_gmut_shift : // unsigned , default = 14

Table 8-2475 VDIN0_HDR2_GMUT_COEF0 0x12B1

Bit(s)	R/W	Default	Description
31:16	R/W	674	reg_gmut_coef01 : // unsigned , default = 674
15:0	R/W	1285	reg_gmut_coef00 : // unsigned , default = 1285

Table 8-2476 VDIN0_HDR2_GMUT_COEF1 0x12B2

Bit(s)	R/W	Default	Description
31:16	R/W	142	reg_gmut_coef10 : // unsigned , default = 142
15:0	R/W	89	reg_gmut_coef02 : // unsigned , default = 89

Table 8-2477 VDIN0_HDR2_GMUT_COEF2 0x12B3

Bit(s)	R/W	Default	Description
31:16	R/W	23	reg_gmut_coef12 : // unsigned , default = 23
15:0	R/W	1883	reg_gmut_coef11 : // unsigned , default = 1883

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Table 8-2478 VDIN0_HDR2_GMUT_COEF3 0x12B4

Bit(s)	R/W	Default	Description
31:16	R/W	180	reg_gmut_coef21 : // unsigned , default = 180
15:0	R/W	34	reg_gmut_coef20 : // unsigned , default = 34

Table 8-2479 VDIN0_HDR2_GMUT_COEF4 0x12B5

Bit(s)	R/W	Default	Description
15:0	R/W	1834	reg_gmut_coef22 : // unsigned , default = 1834

Table 8-2480 VDIN0_HDR2_PIPE_CTRL1 0x12B6

Bit(s)	R/W	Default	Description
31:24	R/W	4	vblank_num_oetf : // unsigned , default = 4
23:16	R/W	4	hblank_num_oetf : // unsigned , default = 4
15:8	R/W	10	vblank_num_eotf : // unsigned , default = 10
7:0	R/W	10	hblank_num_eotf : // unsigned , default = 10

Table 8-2481 VDIN0_HDR2_PIPE_CTRL2 0x12B7

Bit(s)	R/W	Default	Description
31:24	R/W	10	vblank_num_cgain : // unsigned , default = 10
23:16	R/W	10	hblank_num_cgain : // unsigned , default = 10
15:8	R/W	11	vblank_num_gmut : // unsigned , default = 11
7:0	R/W	11	hblank_num_gmut : // unsigned , default = 11

Table 8-2482 VDIN0_HDR2_PIPE_CTRL3 0x12B8

Bit(s)	R/W	Default	Description
31:24	R/W	22	vblank_num_adps : // unsigned , default = 22
23:16	R/W	2	hblank_num_adps : // unsigned , default = 2
15:8	R/W	4	vblank_num_uv : // unsigned , default = 4
7:0	R/W	4	hblank_num_uv : // unsigned , default = 4

Table 8-2483 VDIN0_HDR2_PROC_WIN1 0x12B9

Bit(s)	R/W	Default	Description
28:16	R/W	0	proc_win_h_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_h_st : // unsigned , default = 0

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Table 8-2484 VDIN0_HDR2_PROC_WIN2 0x12BA

Bit(s)	R/W	Default	Description
31	R/W	0	proc_win_gmut_en : // unsigned , default = 0
30	R/W	0	proc_win_adps_en : // unsigned , default = 0
29	R/W	0	proc_win_cgain_en : // unsigned , default = 0
28:16	R/W	0	proc_win_v_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_v_st : // unsigned , default = 0

Table 8-2485 VDIN0_HDR2_MATRIXI_EN_CTRL 0x12BB

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

Table 8-2486 VDIN0_HDR2_MATRIXO_EN_CTRL 0x12BC

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

Table 8-2487 VDIN1_HDR2_CTRL 0x1380

Bit(s)	R/W	Default	Description
20:18	R/W	0	reg_din_swap : // unsigned , default = 0
17	R/W	0	reg_out_fmt : // unsigned , default = 0
16	R/W	0	reg_only_mat : // unsigned , default = 0 ,only use input matrix ,work when hdr disable
15	R/W	0	mat_o_en,//output matrix enable ,only work when hdr enable
14	R/W	0	mat_in_en //input matrix enable ,only work when hdr enable
13	R/W	0	reg_VDIN0_HDR2_top_en : // unsigned , default = 0,hdr enable singal
12	R/W	1	reg_cgain_mode : // unsigned , default = 1
7: 6	R/W	1	reg_gmut_mode : // unsigned , default = 1
5	R/W	0	reg_in_shift : // unsigned , default = 0
4	R/W	1	reg_in_fmt : // unsigned , default = 1
3	R/W	1	reg_eo_enable : // unsigned , default = 1
2	R/W	1	reg_oe_enable : // unsigned , default = 1

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Table 8-2488 VDIN1_HDR2_CLK_GATE 0x1381

Bit(s)	R/W	Default	Description
31:30	R/W	0	clk_tm : gate clock ctrl (main clock) // unsigned , default = 0
29:28	R/W	0	output : matrix clock gate ctrl// unsigned , default = 0
25:24	R/W	0	input : matrix clock gate ctrl // unsigned , default = 0
23:22	R/W	0	hdr : top cbus clock gate ctrl // unsigned , default = 0
21:20	R/W	0	eotf : cbus clock gate ctrl // unsigned , default = 0
19:18	R/W	0	oetf : cbus clock gate ctrl // unsigned , default = 0
17:16	R/W	0	gamma : mult cbus clock gate ctrl // unsigned , default = 0
15:14	R/W	0	adaptive : cbus scaler clock gate ctrl // unsigned , default = 0
13:12	R/W	0	cgain : cbus clock gate ctrl// unsigned , default = 0
11:10	R/W	0	eotf : clock gate ctrl // unsigned , default = 0
9:8	R/W	0	oetf : clock gate ctrl // unsigned , default = 0
7:6	R/W	0	gamma : mult clock gate ctrl// unsigned , default = 0
5:4	R/W	0	adaptive : scaler clock gate ctrl // unsigned , default = 0
3:2	R/W	0	uv : gain clock gate ctrl // unsigned , default = 0
1:0	R/W	0	cgain : clock gate ctrl // unsigned , default = 0

Table 8-2489 VDIN1_HDR2_MATRIXI_COEF00_01 0x1382

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2490 VDIN1_HDR2_MATRIXI_COEF02_10 0x1383

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-2491 VDIN1_HDR2_MATRIXI_COEF11_12 0x1384

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

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Table 8-2492 VDIN1_HDR2_MATRIXI_COEF20_21 0x1385

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2493 VDIN1_HDR2_MATRIXI_COEF22 0x1386

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-2494 VDIN1_HDR2_MATRIXI_COEF30_31 0x1387

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2495 VDIN1_HDR2_MATRIXI_COEF32_40 0x1388

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-2496 VDIN1_HDR2_MATRIXI_COEF41_42 0x1389

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2497 VDIN1_HDR2_MATRIXI_OFFSET0_1 0x138A

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-2498 VDIN1_HDR2_MATRIXI_OFFSET2 0x138B

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

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Table 8-2499 VDIN1_HDR2_MATRIXI_PRE_OFFSET0_1 0x138C

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2500 VDIN1_HDR2_MATRIXI_PRE_OFFSET2 0x138D

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-2501 VDIN1_HDR2_MATRIXO_COEF00_01 0x138E

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2502 VDIN1_HDR2_MATRIXO_COEF02_10 0x138F

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-2503 VDIN1_HDR2_MATRIXO_COEF11_12 0x1390

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2504 VDIN1_HDR2_MATRIXO_COEF20_21 0x1391

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2505 VDIN1_HDR2_MATRIXO_COEF22 0x1392

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

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Table 8-2506 VDIN1_HDR2_MATRIXO_COEF30_31 0x1393

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2507 VDIN1_HDR2_MATRIXO_COEF32_40 0x1394

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-2508 VDIN1_HDR2_MATRIXO_COEF41_42 0x1395

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2509 VDIN1_HDR2_MATRIXO_OFFSET0_1 0x1396

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-2510 VDIN1_HDR2_MATRIXO_OFFSET2 0x1397

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2511 VDIN1_HDR2_MATRIXO_PRE_OFFSET0_1 0x1398

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2512 VDIN1_HDR2_MATRIXO_PRE_OFFSET2 0x1399

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

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Table 8-2513 VDIN1_HDR2_MATRIXI_CLIP 0x139A

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2514 VDIN1_HDR2_MATRIXO_CLIP 0x139B

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2515 VDIN1_HDR2_CGAIN_OFFT 0x139C

Bit(s)	R/W	Default	Description
26:16	R/W	0	reg_cgain_oft2 : // signed , default = 0
10:0	R/W	0	reg_cgain_oft1 : // signed , default = 0

Table 8-2516 VDIN1_EOTF_LUT_ADDR_PORT 0x139E

Bit(s)	R/W	Default	Description
7:0	R/W	0	eotf_lut_addr : // unsigned , default = 0

Table 8-2517 VDIN1_EOTF_LUT_DATA_PORT 0x139F

Bit(s)	R/W	Default	Description
19:0	R/W	0	eotf_lut_data : // unsigned , default = 0

Table 8-2518 VDIN1_OETF_LUT_ADDR_PORT 0x13A0

Bit(s)	R/W	Default	Description
7:0	R/W	0	oetf_lut_addr : // unsigned , default = 0

Table 8-2519 VDIN1_OETF_LUT_DATA_PORT 0x13A1

Bit(s)	R/W	Default	Description
11:0	R/W	0	oetf_lut_data : // unsigned , default = 0

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Table 8-2520 VDIN1_CGAIN_LUT_ADDR_PORT 0x13A2

Bit(s)	R/W	Default	Description
7:0	R/W	0	cgain_lut_addr : // unsigned , default = 0

Table 8-2521 VDIN1_CGAIN_LUT_DATA_PORT 0x13A3

Bit(s)	R/W	Default	Description
11:0	R/W	0	cgain_lut_data : // unsigned , default = 0

Table 8-2522 VDIN1_HDR2_CGAIN_COEF0 0x13A4

Bit(s)	R/W	Default	Description
27:16	R/W	0	reg_cgain_coef1 : // unsigned , default = 0
11:0	R/W	0	reg_cgain_coef0 : //unsigned , default = 0

Table 8-2523 VDIN1_HDR2_CGAIN_COEF1 0x13A5

Bit(s)	R/W	Default	Description
11:0	R/W	0	reg_cgain_coef2 : // unsigned , default = 0

Table 8-2524 VDIN1_OGAIN_LUT_ADDR_PORT 0x13A6

Bit(s)	R/W	Default	Description
7:0	R/W	0	ogain_lut_addr : // unsigned , default = 0

Table 8-2525 VDIN1_OGAIN_LUT_DATA_PORT 0x13A7

Bit(s)	R/W	Default	Description
11:0	R/W	0	ogain_lut_data : // unsigned , default = 0

Table 8-2526 VDIN1_HDR2_ADPS_CTRL 0x13A8

Bit(s)	R/W	Default	Description
6	R/W	1	reg_adpscl_bypass2 : // unsigned , default = 1
5	R/W	1	reg_adpscl_bypass1 : // unsigned , default = 1
4	R/W	1	reg_adpscl_bypass0 : // unsigned , default = 1
1:0	R/W	1	reg_adpscl_mode : // unsigned , default = 1

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Table 8-2527 VDIN1_HDR2_ADPS_ALPHA0 0x13A9

Bit(s)	R/W	Default	Description
29:16	R/W	0x1000	reg_adpscl_alpha1 : // unsigned , default = 0x1000
13:0	R/W	0x1000	reg_adpscl_alpha0 : // unsigned , default = 0x1000

Table 8-2528 VDIN1_HDR2_ADPS_ALPHA1 0x13AA

Bit(s)	R/W	Default	Description
27:24	R/W	0xc	reg_adpscl_shift0 : // unsigned , default = 0xc
23:20	R/W	0xc	reg_adpscl_shift1 : // unsigned , default = 0xc
19:16	R/W	0xc	reg_adpscl_shift2 : // unsigned , default = 0xc
13:0	R/W	0x1000	reg_adpscl_alpha2 : // unsigned , default = 0x1000

Table 8-2529 VDIN1_HDR2_ADPS_BETA0 0x13AB

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta0_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta0 : // unsigned , default = 0xfc000

Table 8-2530 VDIN1_HDR2_ADPS_BETA1 0x13AC

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta1_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta1 : // unsigned , default = 0xfc000

Table 8-2531 VDIN1_HDR2_ADPS_BETA2 0x13AD

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta2_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta2 : // unsigned , default = 0xfc000

Table 8-2532 VDIN1_HDR2_ADPS_COEF0 0x13AE

Bit(s)	R/W	Default	Description
27:16	R/W	460	reg_adpscl_ys_coef1 : // unsigned , default = 460
11:0	R/W	1188	reg_adpscl_ys_coef0 : // unsigned , default = 1188

Table 8-2533 VDIN1_HDR2_ADPS_COEF1 0x13AF

Bit(s)	R/W	Default	Description
11:0	R/W	104	reg_adpscl_ys_coef2 : // unsigned , default = 104

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Table 8-2534 VDIN1_HDR2_GMUT_CTRL 0x13B0

Bit(s)	R/W	Default	Description
3:0	R/W	14	reg_gmut_shift : // unsigned , default = 14

Table 8-2535 VDIN1_HDR2_GMUT_COEF0 0x13B1

Bit(s)	R/W	Default	Description
31:16	R/W	674	reg_gmut_coef01 : // unsigned , default = 674
15:0	R/W	1285	reg_gmut_coef00 : // unsigned , default = 1285

Table 8-2536 VDIN1_HDR2_GMUT_COEF1 0x13B2

Bit(s)	R/W	Default	Description
31:16	R/W	142	reg_gmut_coef10 : // unsigned , default = 142
15:0	R/W	89	reg_gmut_coef02 : // unsigned , default = 89

Table 8-2537 VDIN1_HDR2_GMUT_COEF2 0x13B3

Bit(s)	R/W	Default	Description
31:16	R/W	23	reg_gmut_coef12 : // unsigned , default = 23
15:0	R/W	1883	reg_gmut_coef11 : // unsigned , default = 1883

Table 8-2538 VDIN1_HDR2_GMUT_COEF3 0x13B4

Bit(s)	R/W	Default	Description
31:16	R/W	180	reg_gmut_coef21 : // unsigned , default = 180
15:0	R/W	34	reg_gmut_coef20 : // unsigned , default = 34

Table 8-2539 VDIN1_HDR2_GMUT_COEF4 0x13B5

Bit(s)	R/W	Default	Description
15:0	R/W	1834	reg_gmut_coef22 : // unsigned , default = 1834

Table 8-2540 VDIN1_HDR2_PIPE_CTRL1 0x13B6

Bit(s)	R/W	Default	Description
31:24	R/W	4	vblank_num_oetf : // unsigned , default = 4
23:16	R/W	4	hblank_num_oetf : // unsigned , default = 4
15:8	R/W	10	vblank_num_eotf : // unsigned , default = 10
7:0	R/W	10	hblank_num_eotf : // unsigned , default = 10

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Table 8-2541 VDIN1_HDR2_PIPE_CTRL2 0x13B7

Bit(s)	R/W	Default	Description
31:24	R/W	10	vblank_num_cgain : // unsigned , default = 10
23:16	R/W	10	hblank_num_cgain : // unsigned , default = 10
15:8	R/W	11	vblank_num_gmut : // unsigned , default = 11
7:0	R/W	11	hblank_num_gmut : // unsigned , default = 11

Table 8-2542 VDIN1_HDR2_PIPE_CTRL3 0x13B8

Bit(s)	R/W	Default	Description
31:24	R/W	22	vblank_num_adps : // unsigned , default = 22
23:16	R/W	2	hblank_num_adps : // unsigned , default = 2
15:8	R/W	4	vblank_num_uv : // unsigned , default = 4
7:0	R/W	4	hblank_num_uv : // unsigned , default = 4

Table 8-2543 VDIN1_HDR2_PROC_WIN1 0x13B9

Bit(s)	R/W	Default	Description
28:16	R/W	0	proc_win_h_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_h_st : // unsigned , default = 0

Table 8-2544 VDIN1_HDR2_PROC_WIN2 0x13BA

Bit(s)	R/W	Default	Description
31	R/W	0	proc_win_gmut_en : // unsigned , default = 0
30	R/W	0	proc_win_adps_en : // unsigned , default = 0
29	R/W	0	proc_win_cgain_en : // unsigned , default = 0
28:16	R/W	0	proc_win_v_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_v_st : // unsigned , default = 0

Table 8-2545 VDIN1_HDR2_MATRIXI_EN_CTRL 0x13BB

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

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Table 8-2546 VDIN1_HDR2_MATRIXO_EN_CTRL 0x13BC

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

Table 8-2547 VD1_HDR2_CTRL 0x3800

Bit(s)	R/W	Default	Description
20:18	R/W	0	reg_din_swap : // unsigned , default = 0
17	R/W	0	reg_out_fmt : // unsigned , default = 0
16	R/W	0	reg_only_mat : // unsigned , default = 0 ,only use input matrix ,work when hdr disable
15	R/W	0	mat_o_en,//output matrix enable ,only work when hdr enable
14	R/W	0	mat_in_en //input matrix enable ,only work when hdr enable
13	R/W	0	reg_VDIN0_HDR2_top_en : // unsigned , default = 0,hdr enable singal
12	R/W	1	reg_cgain_mode : // unsigned , default = 1
7: 6	R/W	1	reg_gmut_mode : // unsigned , default = 1
5	R/W	0	reg_in_shift : // unsigned , default = 0
4	R/W	1	reg_in_fmt : // unsigned , default = 1
3	R/W	1	reg_eo_enable : // unsigned , default = 1
2	R/W	1	reg_oe_enable : // unsigned , default = 1

Table 8-2548 VD1_HDR2_CLK_GATE 0x3801

Bit(s)	R/W	Default	Description
31:30	R/W	0	clk_tm : gate clock ctrl (main clock) // unsigned , default = 0
29:28	R/W	0	output : matrix clock gate ctrl// unsigned , default = 0
25:24	R/W	0	input : matrix clock gate ctrl // unsigned , default = 0
23:22	R/W	0	hdr : top cbus clock gate ctrl // unsigned , default = 0
21:20	R/W	0	eotf : cbus clock gate ctrl // unsigned , default = 0
19:18	R/W	0	oetf : cbus clock gate ctrl // unsigned , default = 0
17:16	R/W	0	gamma : mult cbus clock gate ctrl // unsigned , default = 0
15:14	R/W	0	adaptive : cbus scaler clock gate ctrl // unsigned , default = 0
13:12	R/W	0	cgain : cbus clock gate ctrl// unsigned , default = 0
11:10	R/W	0	eotf : clock gate ctrl // unsigned , default = 0
9:8	R/W	0	oetf : clock gate ctrl // unsigned , default = 0
7:6	R/W	0	gamma : mult clock gate ctrl// unsigned , default = 0
5:4	R/W	0	adaptive : scaler clock gate ctrl // unsigned , default = 0

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Bit(s)	R/W	Default	Description
3:2	R/W	0	uv : gain clock gate ctrl // unsigned , default = 0
1:0	R/W	0	cgain : clock gate ctrl // unsigned , default = 0

Table 8-2549 VD1_HDR2_MATRIXI_COEF00_01 0x3802

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2550 VD1_HDR2_MATRIXI_COEF02_10 0x3803

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-2551 VD1_HDR2_MATRIXI_COEF11_12 0x3804

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2552 VD1_HDR2_MATRIXI_COEF20_21 0x3805

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2553 VD1_HDR2_MATRIXI_COEF22 0x3806

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-2554 VD1_HDR2_MATRIXI_COEF30_31 0x3807

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

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Table 8-2555 VD1_HDR2_MATRIXI_COEF32_40 0x3808

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-2556 VD1_HDR2_MATRIXI_COEF41_42 0x3809

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2557 VD1_HDR2_MATRIXI_OFFSET0_1 0x380A

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-2558 VD1_HDR2_MATRIXI_OFFSET2 0x380B

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2559 VD1_HDR2_MATRIXI_PRE_OFFSET0_1 0x380C

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2560 VD1_HDR2_MATRIXI_PRE_OFFSET2 0x380D

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-2561 VD1_HDR2_MATRIXO_COEF00_01 0x380E

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

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Table 8-2562 VD1_HDR2_MATRIXO_COEF02_10 0x380F

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-2563 VD1_HDR2_MATRIXO_COEF11_12 0x3810

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2564 VD1_HDR2_MATRIXO_COEF20_21 0x3811

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2565 VD1_HDR2_MATRIXO_COEF22 0x3812

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-2566 VD1_HDR2_MATRIXO_COEF30_31 0x3813

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2567 VD1_HDR2_MATRIXO_COEF32_40 0x3814

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-2568 VD1_HDR2_MATRIXO_COEF41_42 0x3815

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

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Table 8-2569 VD1_HDR2_MATRIXO_OFFSET0_1 0x3816

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-2570 VD1_HDR2_MATRIXO_OFFSET2 0x3817

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2571 VD1_HDR2_MATRIXO_PRE_OFFSET0_1 0x3818

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2572 VD1_HDR2_MATRIXO_PRE_OFFSET2 0x3819

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-2573 VD1_HDR2_MATRIXI_CLIP 0x381A

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2574 VD1_HDR2_MATRIXO_CLIP 0x381B

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2575 VD1_HDR2_CGAIN_OFFT 0x381C

Bit(s)	R/W	Default	Description
26:16	R/W	0	reg_cgain_oft2 : // signed , default = 0
10:0	R/W	0	reg_cgain_oft1 : // signed , default = 0

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Table 8-2576 VD1_EOTF_LUT_ADDR_PORT 0x381E

Bit(s)	R/W	Default	Description
7:0	R/W	0	eotf_lut_addr : // unsigned , default = 0

Table 8-2577 VD1_EOTF_LUT_DATA_PORT 0x381F

Bit(s)	R/W	Default	Description
19:0	R/W	0	eotf_lut_data : // unsigned , default = 0

Table 8-2578 VD1_OETF_LUT_ADDR_PORT 0x3820

Bit(s)	R/W	Default	Description
7:0	R/W	0	oetf_lut_addr : // unsigned , default = 0

Table 8-2579 VD1_OETF_LUT_DATA_PORT 0x3821

Bit(s)	R/W	Default	Description
11:0	R/W	0	oetf_lut_data : // unsigned , default = 0

Table 8-2580 VD1_CGAIN_LUT_ADDR_PORT 0x3822

Bit(s)	R/W	Default	Description
7:0	R/W	0	cgain_lut_addr : // unsigned , default = 0

Table 8-2581 VD1_CGAIN_LUT_DATA_PORT 0x3823

Bit(s)	R/W	Default	Description
11:0	R/W	0	cgain_lut_data : // unsigned , default = 0

Table 8-2582 VD1_HDR2_CGAIN_COEF0 0x3824

Bit(s)	R/W	Default	Description
27:16	R/W	0	reg_cgain_coef1 : // unsigned , default = 0
11:0	R/W	0	reg_cgain_coef0 : //unsigned , default = 0

Table 8-2583 VD1_HDR2_CGAIN_COEF1 0x3825

Bit(s)	R/W	Default	Description
11:0	R/W	0	reg_cgain_coef2 : // unsigned , default = 0

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Table 8-2584 VD1_OGAIN_LUT_ADDR_PORT 0x3826

Bit(s)	R/W	Default	Description
7:0	R/W	0	ogain_lut_addr : // unsigned , default = 0

Table 8-2585 VD1_OGAIN_LUT_DATA_PORT 0x3827

Bit(s)	R/W	Default	Description
11:0	R/W	0	ogain_lut_data : // unsigned , default = 0

Table 8-2586 VD1_HDR2_ADPS_CTRL 0x3828

Bit(s)	R/W	Default	Description
6	R/W	1	reg_adpscl_bypass2 : // unsigned , default = 1
5	R/W	1	reg_adpscl_bypass1 : // unsigned , default = 1
4	R/W	1	reg_adpscl_bypass0 : // unsigned , default = 1
1:0	R/W	1	reg_adpscl_mode : // unsigned , default = 1

Table 8-2587 VD1_HDR2_ADPS_ALPHA0 0x3829

Bit(s)	R/W	Default	Description
29:16	R/W	0x1000	reg_adpscl_alpha1 : // unsigned , default = 0x1000
13:0	R/W	0x1000	reg_adpscl_alpha0 : // unsigned , default = 0x1000

Table 8-2588 VD1_HDR2_ADPS_ALPHA1 0x382A

Bit(s)	R/W	Default	Description
27:24	R/W	0xc	reg_adpscl_shift0 : // unsigned , default = 0xc
23:20	R/W	0xc	reg_adpscl_shift1 : // unsigned , default = 0xc
19:16	R/W	0xc	reg_adpscl_shift2 : // unsigned , default = 0xc
13:0	R/W	0x1000	reg_adpscl_alpha2 : // unsigned , default = 0x1000

Table 8-2589 VD1_HDR2_ADPS_BETA0 0x382B

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta0_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta0 : // unsigned , default = 0xfc000

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Table 8-2590 VD1_HDR2_ADPS_BETA1 0x382C

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta1_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta1 : // unsigned , default = 0xfc000

Table 8-2591 VD1_HDR2_ADPS_BETA2 0x382D

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta2_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta2 : // unsigned , default = 0xfc000

Table 8-2592 VD1_HDR2_ADPS_COEF0 0x382E

Bit(s)	R/W	Default	Description
27:16	R/W	460	reg_adpscl_ys_coef1 : // unsigned , default = 460
11:0	R/W	1188	reg_adpscl_ys_coef0 : // unsigned , default = 1188

Table 8-2593 VD1_HDR2_ADPS_COEF1 0x382F

Bit(s)	R/W	Default	Description
11:0	R/W	104	reg_adpscl_ys_coef2 : // unsigned , default = 104

Table 8-2594 VD1_HDR2_GMUT_CTRL 0x3830

Bit(s)	R/W	Default	Description
3:0	R/W	14	reg_gmut_shift : // unsigned , default = 14

Table 8-2595 VD1_HDR2_GMUT_COEF0 0x3831

Bit(s)	R/W	Default	Description
31:16	R/W	674	reg_gmut_coef01 : // unsigned , default = 674
15:0	R/W	1285	reg_gmut_coef00 : // unsigned , default = 1285

Table 8-2596 VD1_HDR2_GMUT_COEF1 0x3832

Bit(s)	R/W	Default	Description
31:16	R/W	142	reg_gmut_coef10 : // unsigned , default = 142
15:0	R/W	89	reg_gmut_coef02 : // unsigned , default = 89

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Table 8-2597 VD1_HDR2_GMUT_COEF2 0x3833

Bit(s)	R/W	Default	Description
31:16	R/W	23	reg_gmut_coef12 : // unsigned , default = 23
15:0	R/W	1883	reg_gmut_coef11 : // unsigned , default = 1883

Table 8-2598 VD1_HDR2_GMUT_COEF3 0x3834

Bit(s)	R/W	Default	Description
31:16	R/W	180	reg_gmut_coef21 : // unsigned , default = 180
15:0	R/W	34	reg_gmut_coef20 : // unsigned , default = 34

Table 8-2599 VD1_HDR2_GMUT_COEF4 0x3835

Bit(s)	R/W	Default	Description
15:0	R/W	1834	reg_gmut_coef22 : // unsigned , default = 1834

Table 8-2600 VD1_HDR2_PIPE_CTRL1 0x3836

Bit(s)	R/W	Default	Description
31:24	R/W	4	vblank_num_oetf : // unsigned , default = 4
23:16	R/W	4	hblank_num_oetf : // unsigned , default = 4
15:8	R/W	10	vblank_num_eotf : // unsigned , default = 10
7:0	R/W	10	hblank_num_eotf : // unsigned , default = 10

Table 8-2601 VD1_HDR2_PIPE_CTRL2 0x3837

Bit(s)	R/W	Default	Description
31:24	R/W	10	vblank_num_cgain : // unsigned , default = 10
23:16	R/W	10	hblank_num_cgain : // unsigned , default = 10
15:8	R/W	11	vblank_num_gmut : // unsigned , default = 11
7:0	R/W	11	hblank_num_gmut : // unsigned , default = 11

Table 8-2602 VD1_HDR2_PIPE_CTRL3 0x3838

Bit(s)	R/W	Default	Description
31:24	R/W	22	vblank_num_adps : // unsigned , default = 22
23:16	R/W	2	hblank_num_adps : // unsigned , default = 2
15:8	R/W	4	vblank_num_uv : // unsigned , default = 4
7:0	R/W	4	hblank_num_uv : // unsigned , default = 4

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Table 8-2603 VD1_HDR2_PROC_WIN1 0x3839

Bit(s)	R/W	Default	Description
28:16	R/W	0	proc_win_h_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_h_st : // unsigned , default = 0

Table 8-2604 VD1_HDR2_PROC_WIN2 0x383A

Bit(s)	R/W	Default	Description
31	R/W	0	proc_win_gmut_en : // unsigned , default = 0
30	R/W	0	proc_win_adps_en : // unsigned , default = 0
29	R/W	0	proc_win_cgain_en : // unsigned , default = 0
28:16	R/W	0	proc_win_v_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_v_st : // unsigned , default = 0

Table 8-2605 VD1_HDR2_MATRIXI_EN_CTRL 0x383B

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

Table 8-2606 VD1_HDR2_MATRIXO_EN_CTRL 0x383C

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

Table 8-2607 VD1_HDR2_HIST_CTRL 0x383D

Bit(s)	R/W	Default	Description
23-16	R/W	0	hist result read index
15-8	R/W	0	gate clock ctrl
5	R/W	0	piecewise mode enable
4	R/W	0	hist window enable
3	R/W	0	data shfit enable
2-0	R/W	0	hist input select 0: e_rgb max 1: e_luma 2/3: e_noliear_sat 4/5: o_before_gamma 6/7 : after_gamma

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Table 8-2608 VD1_HDR2_HIST_H_START_END 0x383E

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist h start
12-0	R/W	0	hist h end

Table 8-2609 VD1_HDR2_HIST_V_START_END 0x383F

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist v start
12-0	R/W	0	hist v end

Table 8-2610 VD1_HDR2_HIST_RD 0x3840

Bit(s)	R/W	Default	Description
23-0	R	0	hist 128 bin result

Table 8-2611 VD2_HDR2_CTRL 0x3850

Bit(s)	R/W	Default	Description
20:18	R/W	0	reg_din_swap : // unsigned , default = 0
17	R/W	0	reg_out_fmt : // unsigned , default = 0
16	R/W	0	reg_only_mat : // unsigned , default = 0 ,only use input matrix ,work when hdr disable
15	R/W	0	mat_o_en,//output matrix enable ,only work when hdr enable
14	R/W	0	mat_in_en //input matrix enable ,only work when hdr enable
13	R/W	0	reg_VDIN0_HDR2_top_en : // unsigned , default = 0,hdr enable singal
12	R/W	1	reg_cgain_mode : // unsigned , default = 1
7: 6	R/W	1	reg_gmut_mode : // unsigned , default = 1
5	R/W	0	reg_in_shift : // unsigned , default = 0
4	R/W	1	reg_in_fmt : // unsigned , default = 1
3	R/W	1	reg_eo_enable : // unsigned , default = 1
2	R/W	1	reg_oe_enable : // unsigned , default = 1

Table 8-2612 VD2_HDR2_CLK_GATE 0x3851

Bit(s)	R/W	Default	Description
31:30	R/W	0	clk_tm : gate clock ctrl (main clock) // unsigned , default = 0
29:28	R/W	0	output : matrix clock gate ctrl// unsigned , default = 0
25:24	R/W	0	input : matrix clock gate ctrl // unsigned , default = 0
23:22	R/W	0	hdr : top cbus clock gate ctrl // unsigned , default = 0
21:20	R/W	0	eotf : cbus clock gate ctrl // unsigned , default = 0

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Bit(s)	R/W	Default	Description
19:18	R/W	0	oetf : cbus clock gate ctrl // unsigned , default = 0
17:16	R/W	0	gamma : mult cbus clock gate ctrl // unsigned , default = 0
15:14	R/W	0	adaptive : cbus scaler clock gate ctrl // unsigned , default = 0
13:12	R/W	0	cgain : cbus clock gate ctrl// unsigned , default = 0
11:10	R/W	0	eotf : clock gate ctrl // unsigned , default = 0
9:8	R/W	0	oetf : clock gate ctrl // unsigned , default = 0
7:6	R/W	0	gamma : mult clock gate ctrl// unsigned , default = 0
5:4	R/W	0	adaptive : scaler clock gate ctrl // unsigned , default = 0
3:2	R/W	0	uv : gain clock gate ctrl // unsigned , default = 0
1:0	R/W	0	cgain : clock gate ctrl // unsigned , default = 0

Table 8-2613 VD2_HDR2_MATRIXI_COEF00_01 0x3852

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2614 VD2_HDR2_MATRIXI_COEF02_10 0x3853

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-2615 VD2_HDR2_MATRIXI_COEF11_12 0x3854

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2616 VD2_HDR2_MATRIXI_COEF20_21 0x3855

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2617 VD2_HDR2_MATRIXI_COEF22 0x3856

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

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Table 8-2618 VD2_HDR2_MATRIXI_COEF30_31 0x3857

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2619 VD2_HDR2_MATRIXI_COEF32_40 0x3858

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-2620 VD2_HDR2_MATRIXI_COEF41_42 0x3859

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2621 VD2_HDR2_MATRIXI_OFFSET0_1 0x385A

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-2622 VD2_HDR2_MATRIXI_OFFSET2 0x385B

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2623 VD2_HDR2_MATRIXI_PRE_OFFSET0_1 0x385C

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2624 VD2_HDR2_MATRIXI_PRE_OFFSET2 0x385D

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

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Table 8-2625 VD2_HDR2_MATRIXO_COEF00_01 0x385E

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2626 VD2_HDR2_MATRIXO_COEF02_10 0x385F

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-2627 VD2_HDR2_MATRIXO_COEF11_12 0x3860

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2628 VD2_HDR2_MATRIXO_COEF20_21 0x3861

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2629 VD2_HDR2_MATRIXO_COEF22 0x3862

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-2630 VD2_HDR2_MATRIXO_COEF30_31 0x3863

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2631 VD2_HDR2_MATRIXO_COEF32_40 0x3864

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

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Table 8-2632 VD2_HDR2_MATRIXO_COEF41_42 0x3865

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2633 VD2_HDR2_MATRIXO_OFFSET0_1 0x3866

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-2634 VD2_HDR2_MATRIXO_OFFSET2 0x3867

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2635 VD2_HDR2_MATRIXO_PRE_OFFSET0_1 0x3868

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2636 VD2_HDR2_MATRIXO_PRE_OFFSET2 0x3869

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-2637 VD2_HDR2_MATRIXI_CLIP 0x386A

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2638 VD2_HDR2_MATRIXO_CLIP 0x386B

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

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Table 8-2639 VD2_HDR2_CGAIN_OFFT 0x386C

Bit(s)	R/W	Default	Description
26:16	R/W	0	reg_cgain_oft2 : // signed , default = 0
10:0	R/W	0	reg_cgain_oft1 : // signed , default = 0

Table 8-2640 VD2_EOTF_LUT_ADDR_PORT 0x386E

Bit(s)	R/W	Default	Description
7:0	R/W	0	eotf_lut_addr : // unsigned , default = 0

Table 8-2641 VD2_EOTF_LUT_DATA_PORT 0x386F

Bit(s)	R/W	Default	Description
19:0	R/W	0	eotf_lut_data : // unsigned , default = 0

Table 8-2642 VD2_OETF_LUT_ADDR_PORT 0x3870

Bit(s)	R/W	Default	Description
7:0	R/W	0	oetf_lut_addr : // unsigned , default = 0

Table 8-2643 VD2_OETF_LUT_DATA_PORT 0x3871

Bit(s)	R/W	Default	Description
11:0	R/W	0	oetf_lut_data : // unsigned , default = 0

Table 8-2644 VD2_CGAIN_LUT_ADDR_PORT 0x3872

Bit(s)	R/W	Default	Description
7:0	R/W	0	cgain_lut_addr : // unsigned , default = 0

Table 8-2645 VD2_CGAIN_LUT_DATA_PORT 0x3873

Bit(s)	R/W	Default	Description
11:0	R/W	0	cgain_lut_data : // unsigned , default = 0

Table 8-2646 VD2_HDR2_CGAIN_COEF0 0x3874

Bit(s)	R/W	Default	Description
27:16	R/W	0	reg_cgain_coef1 : // unsigned , default = 0
11:0	R/W	0	reg_cgain_coef0 : //unsigned , default = 0

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Table 8-2647 VD2_HDR2_CGAIN_COEF1 0x3875

Bit(s)	R/W	Default	Description
11:0	R/W	0	reg_cgain_coef2 : // unsigned , default = 0

Table 8-2648 VD2_OGAIN_LUT_ADDR_PORT 0x3876

Bit(s)	R/W	Default	Description
7:0	R/W	0	ogain_lut_addr : // unsigned , default = 0

Table 8-2649 VD2_OGAIN_LUT_DATA_PORT 0x3877

Bit(s)	R/W	Default	Description
11:0	R/W	0	ogain_lut_data : // unsigned , default = 0

Table 8-2650 VD2_HDR2_ADPS_CTRL 0x3878

Bit(s)	R/W	Default	Description
6	R/W	1	reg_adpscl_bypass2 : // unsigned , default = 1
5	R/W	1	reg_adpscl_bypass1 : // unsigned , default = 1
4	R/W	1	reg_adpscl_bypass0 : // unsigned , default = 1
1:0	R/W	1	reg_adpscl_mode : // unsigned , default = 1

Table 8-2651 VD2_HDR2_ADPS_ALPHA0 0x3879

Bit(s)	R/W	Default	Description
29:16	R/W	0x1000	reg_adpscl_alpha1 : // unsigned , default = 0x1000
13:0	R/W	0x1000	reg_adpscl_alpha0 : // unsigned , default = 0x1000

Table 8-2652 VD2_HDR2_ADPS_ALPHA1 0x387A

Bit(s)	R/W	Default	Description
27:24	R/W	0xc	reg_adpscl_shift0 : // unsigned , default = 0xc
23:20	R/W	0xc	reg_adpscl_shift1 : // unsigned , default = 0xc
19:16	R/W	0xc	reg_adpscl_shift2 : // unsigned , default = 0xc
13:0	R/W	0x1000	reg_adpscl_alpha2 : // unsigned , default = 0x1000

Table 8-2653 VD2_HDR2_ADPS_BETA0 0x387B

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta0_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta0 : // unsigned , default = 0xfc000

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Table 8-2654 VD2_HDR2_ADPS_BETA1 0x387C

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta1_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta1 : // unsigned , default = 0xfc000

Table 8-2655 VD2_HDR2_ADPS_BETA2 0x387D

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta2_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta2 : // unsigned , default = 0xfc000

Table 8-2656 VD2_HDR2_ADPS_COEF0 0x387E

Bit(s)	R/W	Default	Description
27:16	R/W	460	reg_adpscl_ys_coef1 : // unsigned , default = 460
11:0	R/W	1188	reg_adpscl_ys_coef0 : // unsigned , default = 1188

Table 8-2657 VD2_HDR2_ADPS_COEF1 0x387F

Bit(s)	R/W	Default	Description
11:0	R/W	104	reg_adpscl_ys_coef2 : // unsigned , default = 104

Table 8-2658 VD2_HDR2_GMUT_CTRL 0x3880

Bit(s)	R/W	Default	Description
3:0	R/W	14	reg_gmut_shift : // unsigned , default = 14

Table 8-2659 VD2_HDR2_GMUT_COEF0 0x3881

Bit(s)	R/W	Default	Description
31:16	R/W	674	reg_gmut_coef01 : // unsigned , default = 674
15:0	R/W	1285	reg_gmut_coef00 : // unsigned , default = 1285

Table 8-2660 VD2_HDR2_GMUT_COEF1 0x3882

Bit(s)	R/W	Default	Description
31:16	R/W	142	reg_gmut_coef10 : // unsigned , default = 142
15:0	R/W	89	reg_gmut_coef02 : // unsigned , default = 89

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Table 8-2661 VD2_HDR2_GMUT_COEF2 0x3883

Bit(s)	R/W	Default	Description
31:16	R/W	23	reg_gmut_coef12 : // unsigned , default = 23
15:0	R/W	1883	reg_gmut_coef11 : // unsigned , default = 1883

Table 8-2662 VD2_HDR2_GMUT_COEF3 0x3884

Bit(s)	R/W	Default	Description
31:16	R/W	180	reg_gmut_coef21 : // unsigned , default = 180
15:0	R/W	34	reg_gmut_coef20 : // unsigned , default = 34

Table 8-2663 VD2_HDR2_GMUT_COEF4 0x3885

Bit(s)	R/W	Default	Description
15:0	R/W	1834	reg_gmut_coef22 : // unsigned , default = 1834

Table 8-2664 VD2_HDR2_PIPE_CTRL1 0x3886

Bit(s)	R/W	Default	Description
31:24	R/W	4	vblank_num_oetf : // unsigned , default = 4
23:16	R/W	4	hblank_num_oetf : // unsigned , default = 4
15:8	R/W	10	vblank_num_eotf : // unsigned , default = 10
7:0	R/W	10	hblank_num_eotf : // unsigned , default = 10

Table 8-2665 VD2_HDR2_PIPE_CTRL2 0x3887

Bit(s)	R/W	Default	Description
31:24	R/W	10	vblank_num_cgain : // unsigned , default = 10
23:16	R/W	10	hblank_num_cgain : // unsigned , default = 10
15:8	R/W	11	vblank_num_gmut : // unsigned , default = 11
7:0	R/W	11	hblank_num_gmut : // unsigned , default = 11

Table 8-2666 VD2_HDR2_PIPE_CTRL3 0x3888

Bit(s)	R/W	Default	Description
31:24	R/W	22	vblank_num_adps : // unsigned , default = 22
23:16	R/W	2	hblank_num_adps : // unsigned , default = 2
15:8	R/W	4	vblank_num_uv : // unsigned , default = 4
7:0	R/W	4	hblank_num_uv : // unsigned , default = 4

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Table 8-2667 VD2_HDR2_PROC_WIN1 0x3889

Bit(s)	R/W	Default	Description
28:16	R/W	0	proc_win_h_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_h_st : // unsigned , default = 0

Table 8-2668 VD2_HDR2_PROC_WIN2 0x388A

Bit(s)	R/W	Default	Description
31	R/W	0	proc_win_gmut_en : // unsigned , default = 0
30	R/W	0	proc_win_adps_en : // unsigned , default = 0
29	R/W	0	proc_win_cgain_en : // unsigned , default = 0
28:16	R/W	0	proc_win_v_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_v_st : // unsigned , default = 0

Table 8-2669 VD2_HDR2_MATRIXI_EN_CTRL 0x388B

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

Table 8-2670 VD2_HDR2_MATRIXO_EN_CTRL 0x388C

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

Table 8-2671 OSD1_HDR2_CTRL 0x38A0

Bit(s)	R/W	Default	Description
20:18	R/W	0	reg_din_swap : // unsigned , default = 0
17	R/W	0	reg_out_fmt : // unsigned , default = 0
16	R/W	0	reg_only_mat : // unsigned , default = 0 ,only use input matrix ,work when hdr disable
15	R/W	0	mat_o_en,//output matrix enable ,only work when hdr enable
14	R/W	0	mat_in_en //input matrix enable ,only work when hdr enable
13	R/W	0	reg_VDIN0_HDR2_top_en : // unsigned , default = 0,hdr enable singal
12	R/W	1	reg_cgain_mode : // unsigned , default = 1
7: 6	R/W	1	reg_gmut_mode : // unsigned , default = 1

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Bit(s)	R/W	Default	Description
5	R/W	0	reg_in_shift : // unsigned , default = 0
4	R/W	1	reg_in_fmt : // unsigned , default = 1
3	R/W	1	reg_eo_enable : // unsigned , default = 1
2	R/W	1	reg_oe_enable : // unsigned , default = 1

Table 8-2672 OSD1_HDR2_CLK_GATE 0x38A1

Bit(s)	R/W	Default	Description
31:30	R/W	0	clk_tm : gate clock ctrl (main clock) // unsigned , default = 0
29:28	R/W	0	output : matrix clock gate ctrl// unsigned , default = 0
25:24	R/W	0	input : matrix clock gate ctrl // unsigned , default = 0
23:22	R/W	0	hdr : top cbus clock gate ctrl // unsigned , default = 0
21:20	R/W	0	eotf : cbus clock gate ctrl // unsigned , default = 0
19:18	R/W	0	oetf : cbus clock gate ctrl // unsigned , default = 0
17:16	R/W	0	gamma : mult cbus clock gate ctrl // unsigned , default = 0
15:14	R/W	0	adaptive : cbus scaler clock gate ctrl // unsigned , default = 0
13:12	R/W	0	cgain : cbus clock gate ctrl// unsigned , default = 0
11:10	R/W	0	eotf : clock gate ctrl // unsigned , default = 0
9:8	R/W	0	oetf : clock gate ctrl // unsigned , default = 0
7:6	R/W	0	gamma : mult clock gate ctrl// unsigned , default = 0
5:4	R/W	0	adaptive : scaler clock gate ctrl // unsigned , default = 0
3:2	R/W	0	uv : gain clock gate ctrl // unsigned , default = 0
1:0	R/W	0	cgain : clock gate ctrl // unsigned , default = 0

Table 8-2673 OSD1_HDR2_MATRIXI_COEF00_01 0x38A2

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2674 OSD1_HDR2_MATRIXI_COEF02_10 0x38A3

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

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Table 8-2675 OSD1_HDR2_MATRIXI_COEF11_12 0x38A4

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2676 OSD1_HDR2_MATRIXI_COEF20_21 0x38A5

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

Table 8-2677 OSD1_HDR2_MATRIXI_COEF22 0x38A6

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-2678 OSD1_HDR2_MATRIXI_COEF30_31 0x38A7

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2679 OSD1_HDR2_MATRIXI_COEF32_40 0x38A8

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-2680 OSD1_HDR2_MATRIXI_COEF41_42 0x38A9

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2681 OSD1_HDR2_MATRIXI_OFFSET0_1 0x38AA

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

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Table 8-2682 OSD1_HDR2_MATRIXI_OFFSET2 0x38AB

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2683 OSD1_HDR2_MATRIXI_PRE_OFFSET0_1 0x38AC

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

Table 8-2684 OSD1_HDR2_MATRIXI_PRE_OFFSET2 0x38AD

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-2685 OSD1_HDR2_MATRIXO_COEF00_01 0x38AE

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef00 : // signed , default = 0
12:0	R/W	0	coef01 : // signed , default = 0

Table 8-2686 OSD1_HDR2_MATRIXO_COEF02_10 0x38AF

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef02 : // signed , default = 0
12:0	R/W	0	coef10 : // signed , default = 0

Table 8-2687 OSD1_HDR2_MATRIXO_COEF11_12 0x38B0

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef11 : // signed , default = 0
12:0	R/W	0	coef12 : // signed , default = 0

Table 8-2688 OSD1_HDR2_MATRIXO_COEF20_21 0x38B1

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef20 : // signed , default = 0
12:0	R/W	0	coef21 : // signed , default = 0

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Table 8-2689 OSD1_HDR2_MATRIXO_COEF22 0x38B2

Bit(s)	R/W	Default	Description
12:0	R/W	0	coef22 : // signed , default = 0

Table 8-2690 OSD1_HDR2_MATRIXO_COEF30_31 0x38B3

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef13 : // signed , default = 0
12:0	R/W	0	coef14 : // signed , default = 0

Table 8-2691 OSD1_HDR2_MATRIXO_COEF32_40 0x38B4

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef23 : // signed , default = 0
12:0	R/W	0	coef24 : // signed , default = 0

Table 8-2692 OSD1_HDR2_MATRIXO_COEF41_42 0x38B5

Bit(s)	R/W	Default	Description
28:16	R/W	0	coef15 : // signed , default = 0
12:0	R/W	0	coef25 : // signed , default = 0

Table 8-2693 OSD1_HDR2_MATRIXO_OFFSET0_1 0x38B6

Bit(s)	R/W	Default	Description
26:16	R/W	0	offset0 : // signed , default = 0
10:0	R/W	0	offset1 : // signed , default = 0

Table 8-2694 OSD1_HDR2_MATRIXO_OFFSET2 0x38B7

Bit(s)	R/W	Default	Description
10:0	R/W	0	offset2 : // signed , default = 0

Table 8-2695 OSD1_HDR2_MATRIXO_PRE_OFFSET0_1 0x38B8

Bit(s)	R/W	Default	Description
26:16	R/W	0	pre_offset0 : // signed , default = 0
10:0	R/W	0	pre_offset1 : // signed , default = 0

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Table 8-2696 OSD1_HDR2_MATRIXO_PRE_OFFSET2 0x38B9

Bit(s)	R/W	Default	Description
10:0	R/W	0	pre_offset2 : // signed , default = 0

Table 8-2697 OSD1_HDR2_MATRIXI_CLIP 0x38BA

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2698 OSD1_HDR2_MATRIXO_CLIP 0x38BB

Bit(s)	R/W	Default	Description
19:8	R/W	0	comp_th : // unsigned , default = 0
7:5	R/W	0	conv_rs : // unsigned , default = 0
4:3	R/W	0	clmod : // unsigned , default = 0

Table 8-2699 OSD1_HDR2_CGAIN_OFFT 0x38BC

Bit(s)	R/W	Default	Description
26:16	R/W	0	reg_cgain_oft2 : // signed , default = 0
10:0	R/W	0	reg_cgain_oft1 : // signed , default = 0

Table 8-2700 OSD1_EOTF_LUT_ADDR_PORT 0x38be

Bit(s)	R/W	Default	Description
7:0	R/W	0	eotf_lut_addr : // unsigned , default = 0

Table 8-2701 OSD1_EOTF_LUT_DATA_PORT 0x38bf

Bit(s)	R/W	Default	Description
19:0	R/W	0	eotf_lut_data : // unsigned , default = 0

Table 8-2702 OSD1_OETF_LUT_ADDR_PORT 0x38c0

Bit(s)	R/W	Default	Description
7:0	R/W	0	oetf_lut_addr : // unsigned , default = 0

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Table 8-2703 OSD1_OETF_LUT_DATA_PORT 0x38c1

Bit(s)	R/W	Default	Description
11:0	R/W	0	oetf_lut_data : // unsigned , default = 0

Table 8-2704 OSD1_CGAIN_LUT_ADDR_PORT 0x38c2

Bit(s)	R/W	Default	Description
7:0	R/W	0	cgain_lut_addr : // unsigned , default = 0

Table 8-2705 OSD1_CGAIN_LUT_DATA_PORT 0x38c3

Bit(s)	R/W	Default	Description
11:0	R/W	0	cgain_lut_data : // unsigned , default = 0

Table 8-2706 OSD1_HDR2_CGAIN_COEF0 0x38c4

Bit(s)	R/W	Default	Description
27:16	R/W	0	reg_cgain_coef1 : // unsigned , default = 0
11:0	R/W	0	reg_cgain_coef0 : //unsigned , default = 0

Table 8-2707 OSD1_HDR2_CGAIN_COEF1 0x38c5

Bit(s)	R/W	Default	Description
11:0	R/W	0	reg_cgain_coef2 : // unsigned , default = 0

Table 8-2708 OSD1_OGAIN_LUT_ADDR_PORT 0x38c6

Bit(s)	R/W	Default	Description
7:0	R/W	0	ogain_lut_addr : // unsigned , default = 0

Table 8-2709 OSD1_OGAIN_LUT_DATA_PORT 0x38c7

Bit(s)	R/W	Default	Description
11:0	R/W	0	ogain_lut_data : // unsigned , default = 0

Table 8-2710 OSD1_HDR2_ADPS_CTRL 0x38c8

Bit(s)	R/W	Default	Description
6	R/W	1	reg_adpscl_bypass2 : // unsigned , default = 1
5	R/W	1	reg_adpscl_bypass1 : // unsigned , default = 1
4	R/W	1	reg_adpscl_bypass0 : // unsigned , default = 1
1:0	R/W	1	reg_adpscl_mode : // unsigned , default = 1

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Table 8-2711 OSD1_HDR2_ADPS_ALPHA0 0x38c9

Bit(s)	R/W	Default	Description
29:16	R/W	0x1000	reg_adpscl_alpha1 : // unsigned , default = 0x1000
13:0	R/W	0x1000	reg_adpscl_alpha0 : // unsigned , default = 0x1000

Table 8-2712 OSD1_HDR2_ADPS_ALPHA1 0x38ca

Bit(s)	R/W	Default	Description
27:24	R/W	0xc	reg_adpscl_shift0 : // unsigned , default = 0xc
23:20	R/W	0xc	reg_adpscl_shift1 : // unsigned , default = 0xc
19:16	R/W	0xc	reg_adpscl_shift2 : // unsigned , default = 0xc
13:0	R/W	0x1000	reg_adpscl_alpha2 : // unsigned , default = 0x1000

Table 8-2713 OSD1_HDR2_ADPS_BETA0 0x38cb

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta0_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta0 : // unsigned , default = 0xfc000

Table 8-2714 OSD1_HDR2_ADPS_BETA1 0x38cc

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta1_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta1 : // unsigned , default = 0xfc000

Table 8-2715 OSD1_HDR2_ADPS_BETA2 0x38cd

Bit(s)	R/W	Default	Description
20	R/W	0	reg_adpscl_beta2_s : // unsigned , default = 0
19:0	R/W	0xfc000	reg_adpscl_beta2 : // unsigned , default = 0xfc000

Table 8-2716 OSD1_HDR2_ADPS_COEF0 0x38ce

Bit(s)	R/W	Default	Description
27:16	R/W	460	reg_adpscl_ys_coef1 : // unsigned , default = 460
11:0	R/W	1188	reg_adpscl_ys_coef0 : // unsigned , default = 1188

Table 8-2717 OSD1_HDR2_ADPS_COEF1 0x38cf

Bit(s)	R/W	Default	Description
11:0	R/W	104	reg_adpscl_ys_coef2 : // unsigned , default = 104

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Table 8-2718 OSD1_HDR2_GMUT_CTRL 0x38d0

Bit(s)	R/W	Default	Description
3:0	R/W	14	reg_gmut_shift : // unsigned , default = 14

Table 8-2719 OSD1_HDR2_GMUT_COEF0 0x38d1

Bit(s)	R/W	Default	Description
31:16	R/W	674	reg_gmut_coef01 : // unsigned , default = 674
15:0	R/W	1285	reg_gmut_coef00 : // unsigned , default = 1285

Table 8-2720 OSD1_HDR2_GMUT_COEF1 0x38d2

Bit(s)	R/W	Default	Description
31:16	R/W	142	reg_gmut_coef10 : // unsigned , default = 142
15:0	R/W	89	reg_gmut_coef02 : // unsigned , default = 89

Table 8-2721 OSD1_HDR2_GMUT_COEF2 0x38d3

Bit(s)	R/W	Default	Description
31:16	R/W	23	reg_gmut_coef12 : // unsigned , default = 23
15:0	R/W	1883	reg_gmut_coef11 : // unsigned , default = 1883

Table 8-2722 OSD1_HDR2_GMUT_COEF3 0x38d4

Bit(s)	R/W	Default	Description
31:16	R/W	180	reg_gmut_coef21 : // unsigned , default = 180
15:0	R/W	34	reg_gmut_coef20 : // unsigned , default = 34

Table 8-2723 OSD1_HDR2_GMUT_COEF4 0x38d5

Bit(s)	R/W	Default	Description
15:0	R/W	1834	reg_gmut_coef22 : // unsigned , default = 1834

Table 8-2724 OSD1_HDR2_PIPE_CTRL1 0x38d6

Bit(s)	R/W	Default	Description
31:24	R/W	4	vblank_num_oetf : // unsigned , default = 4
23:16	R/W	4	hblank_num_oetf : // unsigned , default = 4
15:8	R/W	10	vblank_num_eotf : // unsigned , default = 10
7:0	R/W	10	hblank_num_eotf : // unsigned , default = 10

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Table 8-2725 OSD1_HDR2_PIPE_CTRL2 0x38d7

Bit(s)	R/W	Default	Description
31:24	R/W	10	vblank_num_cgain : // unsigned , default = 10
23:16	R/W	10	hblank_num_cgain : // unsigned , default = 10
15:8	R/W	11	vblank_num_gmut : // unsigned , default = 11
7:0	R/W	11	hblank_num_gmut : // unsigned , default = 11

Table 8-2726 OSD1_HDR2_PIPE_CTRL3 0x38d8

Bit(s)	R/W	Default	Description
31:24	R/W	22	vblank_num_adps : // unsigned , default = 22
23:16	R/W	2	hblank_num_adps : // unsigned , default = 2
15:8	R/W	4	vblank_num_uv : // unsigned , default = 4
7:0	R/W	4	hblank_num_uv : // unsigned , default = 4

Table 8-2727 OSD1_HDR2_PROC_WIN1 0x38d9

Bit(s)	R/W	Default	Description
28:16	R/W	0	proc_win_h_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_h_st : // unsigned , default = 0

Table 8-2728 OSD1_HDR2_PROC_WIN2 0x38da

Bit(s)	R/W	Default	Description
31	R/W	0	proc_win_gmut_en : // unsigned , default = 0
30	R/W	0	proc_win_adps_en : // unsigned , default = 0
29	R/W	0	proc_win_cgain_en : // unsigned , default = 0
28:16	R/W	0	proc_win_v_ed : // unsigned , default = 0
12:0	R/W	0	proc_win_v_st : // unsigned , default = 0

Table 8-2729 OSD1_HDR2_MATRIXI_EN_CTRL 0x38db

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

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Table 8-2730 OSD1_HDR2_MATRIXO_EN_CTRL 0x38dc

Bit(s)	R/W	Default	Description
5:4	R/W	0	matrix_gclk_ctrl : // unsigned , default = 0
1	R/W	0	enable_sync_sel : // unsigned , default = 0
0	R/W	0	matrix_conv_en : // unsigned , default = 0

Table 8-2731 OSD1_HDR2_HIST_H_START_END 0x38dd

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist h start
12-0	R/W	0	hist h end

Table 8-2732 OSD1_HDR2_HIST_V_START_END 0x38de

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist v start
12-0	R/W	0	hist v end

Table 8-2733 OSD1_HDR2_HIST_V_START_END 0x38df

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist v start
12-0	R/W	0	hist v end

Table 8-2734 OSD1_HDR2_HIST_RD 0x38e0

Bit(s)	R/W	Default	Description
23-0	R	0	hist 128 bin result

8.2.3.44 TCON Registers

Base address:□0xff660000

Table 8-2735 REG_TCON_TOP_CTRL

bit	Default	Name	Description
[0]		reg_bus_ctrl	bus control: 0 apb bus access, 1: i2c access??
[1]		reg_frm_chg	frame change signal, 1: tell the tcon incoming frame rate change, disable output
[3]		reg_ddrif_frm_rst_en	1: enable ddr interface frame auto reset
[4]		reg_tcon_out	1: select tcon MINI-LVDS/P2P data for analog PHY. 0: select Vx1/LVDS data for analog PHY
[5]		reg_tst_ddrif_en	1: use counter to test DDR interface

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bit	Default	Name	Description
[6]		reg_tst_clr_cmp_err	1: clear the error flag of ddrif testing module
[7]		reg_tcon_arugt	1: tcon read request urgent
[8]		reg_tcon_awugt	1: tcon write request urgent
[9]		reg_vs_inv	1: invert the vsync input polarity, TCON assumes positive vsync input
[10]		reg_hs_inv	1: invert the hsync input polarity, TCON assumes positive hsync input
[13:11]		reg_rev_msb_lsb	3'b00: reverse the LSB and MSB {b0,b1,b2,b3,b4,b5, b6,b7,b8,b9}. 3'b001: for 8bit output {b0,b1,b2,b3,b4, b5,b6,b7}. 3'b010: for 6bit output {b0,b1,b2,b3,b4,b5}. 3'b011: for 8bit cut to 6bit output {b2,b3,b4,b5,b6,b7}. 3'b100: for 10bit cut to 6bit output {b4, b5,b6,b7,b8,b9, b10}. else,bypass
[15:13]		reg_pwm_src_sel	input pwm source select: 0: pwmA, 1: pwmB, 2: pwmC, 3:pwmD, 4:pwmE, 5:pwmF, 6,7: pwm_VS
[16]		reg_tx2a_ckdi_en	1: tcon ip will control the analog TX CKDI
[17]		reg_tx2a_picode_en	1: tcon ip will control the analog TX PI CODE
[18]		reg_tx2a_pre_emp_en	1: tcon ip will control the analog TX PRE EMP
[19]		reg_tx2a_swing_en	1: tcon ip will control the analog TX SWING
[20]		reg_tx2a_tx_en	1: tcon ip will control the analog TX EN
[21]		reg_tx2a_p2s_rst_en	1: tcon ip will control the analog TX P2S RST
[22]		reg_tx2a_tx_rstn_en	1: tcon ip will control the analog TX RSTN
[15]		reg_p2p_lock_oen_ctrl_dout	1: patch for open drain IO. data is masked with OEN. 0: use 0 if lock/bcc is not from open drain IO.
[27:24]		reg_tx2a_dbg_sel	
[28]		reg_tcon_dbg_gpo_pe	
[29]		reg_tcon_dbg_gpo_pud	
[30]		reg_tcon_dbg_gpo_oen	
[31]		reg_tcon_dbg_en	

Table 8-2736 REG_TCON_RGB_IN_MUX

bit	Default	Name	Description
[1:0]		reg_rgb_in_mux, R	R channel, select input: 2'd2,2'd3: rgb_in[29:20], 2'd1: rgb_in[19:10], 2'd0:rgb_in[9:0]
[3:2]		reg_rgb_in_mux, G	G channel, select input: 2'd2,2'd3: rgb_in[29:20], 2'd1: rgb_in[19:10], 2'd0:rgb_in[9:0]
[5:4]		reg_rgb_in_mux, B	B channel, select input :2'd2,2'd3: rgb_in[29:20], 2'd1: rgb_in[19:10], 2'd0:rgb_in[9:0]

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Table 8-2737 REG_TCON_OUT_CH_SEL0

bit	Default	Name	Description
[3:0]		TX output channel 0 select	0: mlvds_out[7:0], 1: mlvds_out[15:8], 2: mlvds_out [23:16], 3: mlvds_out[31:24], 4: mlvds_out[39:32], 5: mlvds_out[47:40], 6: mlvds_out[55:48], 7: mlvds_out [63:56], else: 0
[7:4]		TX output channel 1 select	
[11:8]		TX output channel 2 select	
[15:12]		TX output channel 3 select	
[19:16]		TX output channel 4 select	
[23:20]		TX output channel 5 select	
[27:24]		TX output channel 6 select	
[31:28]		TX output channel 7 select	

Table 8-2738 REG_TCON_OUT_CH_SEL1

bit	Default	Name	Description
[3:0]		TX output channel 8 select	0: mlvds_out[7:0], 1: mlvds_out[15:8], 2: mlvds_out [23:16], 3: mlvds_out[31:24], 4: mlvds_out[39:32], 5: mlvds_out[47:40], 6: mlvds_out[55:48], 7: mlvds_out [63:56], else: 0
[7:4]		TX output channel 9 select	
[11:8]		TX output channel 10 select	
[15:12]		TX output channel 11 select	

Table 8-2739 REG_TCON_I2C_DEGLITCH_CNTL

bit	Default	Name	Description
[12:0]		scl filter tick gen period	generate a tick when the scl in_filter internal counter reaches this register
[15:13]		scl filter tick2 gen period	real filter happens at Ttick1 * Ttick2
[28:16]		sda filter tick gen period	generate a tick when the sda in_filter internal counter reaches this register
[31:29]		scl filter tick2 gen period	real filter happens at Ttick1 * Ttick2
0x8014			
[12:0]		aging signal filter tick gen period	generate a tick when the aging in_filter internal counter reaches this register
[15:13]		aging signal filter tick2 gen period	real filter happens at Ttick1 * Ttick2
[28:16]		flicker_en signal filter tick gen period	generate a tick when the flicker_en in_filter internal counter reaches this register
[31:29]		flicker_en filter tick2 gen period	real filter happens at Ttick1 * Ttick2
0x8018			

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bit	Default	Name	Description
[27:0]		pll lock signal filter HIGH parameter	applies this parameter when PLL_lock goes from low to high
0x801c			
[27:0]		pll lock signal filter LOW parameter	applies this parameter when PLL_lock goes from high to low

Table 8-2740 RO_TCON_STATUS0

bit	Default	Name	Description
[3:0]		RO_preline_no	
[4]		RO_reg_finish	
[5]		RO_obit_sel	output bitwidth: 1: 8bit, 0:6bit
[6]		RO_rd_urgent	
[7]		RO_wr_urgent	
[11:8]		RO_mpll_phase_selb	port1 clock phase
[15:12]		RO_mpll_phase_sela	port0 clock phase
[23:16]		RO_tx_clk_sel	this channel of 8 is clock channel if coresponding bit is 1
[31:24]		RO_tx_en	this channel of 8 is active if coresponding bit is 1

Table 8-2741 REG_TCON_PLLLOCK_CNTL

bit	Default	Name	Description
[0]		pll_lock	register configurable pll lock
[1]		lvds_lock	
[4:2]		reg_pll_lock_sel	3'd0: pll lock after deglitch. 3'd1: plllock signal directly from PLL. 3'd2,3'd3: PLL lock from register bit0. 3'd4: pll_lock after deglitch, sync to osc clock domain. 3'd5: plllock signal directly from PLL, sync to osc clock domain. 3'd6,7: PLL lock from register bit0, sync to osc clock domain
[5]		reg_system_ready	
[6]		reg_tcon-reset_lock_en	1: allow tcon ip to reset the pll lock signal automatically
[9:8]		reg_pll_lock_mon_sel	PLL lock signal for interrupt. 3'd0: pll lock after deglitch. 3'd1: plllock signal directly from PLL. 3'd2,3'd3: PLL lock from register bit0.

Table 8-2742 REG_TCON_PLLLCK_RST_CNT

bit	Default	Name	Description
[27:0]		reg_tcon_reset_lock_cnt	the low width of tcon ip auto reset PLL_LOCK

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Table 8-2743 REG_TCON_RST_CTRL

bit	Default	Name	Description
[0]		reg_tcon_rst_ctrl	1: reset all tcon modules
[1]			1: reset pix clock domain
[2]			1: reset p2s clock domain
[3]			1: reset osc clock domain
[4]			1: reset osc register clock domain
[5]			1: reset axi clock domain
[6]			1: reset interrupt clock domain
[7]			1: reset IP

Table 8-2744 REG_TCON_AXI_OFST0

bit	Default	Name	Description
[31:0]		reg_axi_addr_ofst0	Start Byte address for DDR interface0

Table 8-2745 REG_TCON_DDRIF_CTRL0

bit	Default	Name	Description
[11:0]		reg_ddrif_line_cmd_len	
[23:12]		reg_ddrif_line_end_push_th	
[26:24]		reg_ddrif_wr_req_th	
[30:28]		reg_ddrif_rd_req_th	

Table 8-2746 REG_TCON_CLK_CTRL

bit	Default	Name	Description
[0]		reg_tcon_clk_ctrl	1: enable pixel clock
[1]			1: enable p2s clock
[2]			1: enable axi clock
[3]			1: enable osc clock
[4]			1: enable interrupt clock

Table 8-2747 REG_TCON_DDRIF_CTRL1

bit	Default	Name	Description
[11:0]		reg_aw_time_out_th	
[27:16]		reg_ar_time_out_th	
[28]		reg_tst_frm_rst_en	
[29]		reg_ddrif_swrst_en	

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bit	Default	Name	Description
[30]		reg_fake_frm_rst	
[31]		reg_clr_ddrif_err	

Table 8-2748 RO_TCON_STATUS1

bit	Default	Name	Description
[0]		RO_pll_lock_mon	
[1]		RO_pll_lock_ip	
[2]		RO_sw_reset_ip	
[5:3]		RO_tx_mini_a_swing	
[8:6]		RO_tx_mini_b_swing	
[9]		RO_tst_ddrif_data_err	
[10]		RO_tst_ddrif_addr_err	
[11]		RO_aging_filtered	
[13:12]		RO_tx_pre_emp_swing_p0	
[15:14]		RO_tx_pre_emp_swing_p1	
[17:16]		RO_tx_pre_emp_swing_p2	
[19:18]		RO_tx_pre_emp_swing_p3	
[21:20]		RO_tx_pre_emp_swing_p4	
[23:22]		RO_tx_pre_emp_swing_p5	
[25:24]		RO_tx_pre_emp_swing_p6	
[27:26]		RO_tx_pre_emp_swing_p7	
[28]		RO_freq_p2s_clk	
[30:29]		RO_freq_mpll2x_clk	

Table 8-2749 REG_TCON_DDRIF_CTRL2

bit	Default	Name	Description
[7:0]		reg_ddrif_wff_depth	
[15:8]		reg_ddrif_rddata_ff_depth	

Table 8-2750 RO_STATUS2

bit	Default	Name	Description
[0]		ro_axi_towrite_cnt_err_p3	
[1]		ro_axi_toread_cnt_err_p3	
[2]		ro_axi_rd_addr_ff_cnt_err_p3	

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bit	Default	Name	Description
[3]		ro_axi_dwaff_cnt_err_p3	
[4]		ro_axi_towrite_cnt_err_p2	
[5]		ro_axi_toread_cnt_err_p2	
[6]		ro_axi_rd_addr_ff_cnt_err_p2	
[7]		ro_axi_dwaff_cnt_err_p2	
[8]		ro_axi_towrite_cnt_err_p1	
[9]		ro_axi_toread_cnt_err_p1	
[10]		ro_axi_rd_addr_ff_cnt_err_p1	
[11]		ro_axi_dwaff_cnt_err_p1	
[12]		ro_wr_ugt_p3	
[13]		ro_wr_ugt_p2	
[14]		ro_wr_ugt_p1	
[15]		ro_rd_ugt_p3	
[16]		ro_rd_ugt_p2	
[17]		ro_rd_ugt_p1	
[18]		ro_tst_addr_err_p3	
[19]		ro_tst_addr_err_p2	
[20]		ro_tst_addr_err_p1	
[21]		ro_tst_data_err_p3	
[22]		ro_tst_data_err_p2	
[23]		ro_tst_data_err_p1	

Table 8-2751 REG_TCON_AXI_OFST1

bit	Default	Name	Description
[31:0]		reg_axi_addr_ofst1	Start Byte address for DDR interface1

Table 8-2752 REG_TCON_AXI_OFST2

bit	Default	Name	Description
[31:0]		reg_axi_addr_ofst2	Start Byte address for DDR interface2

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Table 8-2753 REG_CNTL_INFILTER_LOCK

bit	Default	Name	Description
[18]			force lock value while [17:16]=2'b10.
[17:16]			2'b00: connect lock signal from GPIO directly 2'b01: connect lock filtered result to IP 2'b10: force lock value of bit 18 2'b11: resample lock filtered result to pix_clk domain, then sent to IP
[15:0]			

Table 8-2754 REG_TCON_INTR_MASKN

bit	Default	Name	Description
[0]		reg_intr_maskn	mask tcon vsync interrupt
[1]			mask tcon sw_reset interrupt
[2]			mask pll lock change interrupt
[3]			mask tcon p2s reset interrupt
[4]			mask tcon tx_en change interrupt
[5]			mask tcon aging change interrupt
[6]			mask ddr interface error interrupt

Table 8-2755 REG_TCON_GPO_CTRL0

bit	Default	Name	Description
[15:0]		tcon_gpo_ovrd_pe	override tcon gpo pull enable
[31:16]		tcon_gpo_ovrd_pud	override tcon gpo pull up/down

Table 8-2756 REG_TCON_GPO_CTRL1

bit	Default	Name	Description
[15:0]		tcon_gpo_ovrd_oen	override tcon gpo OEN
[31:16]		tcon_gpo_ovrd_en	enable overriding tcon gpo

Table 8-2757 REG_TCON_GPO_CTRL2

bit	Default	Name	Description
[15:0]		tcon_gpo_ovrd_gpo_o	override tcon gpo output value
[16]		tcon_gpo_ovrd_vs_latch	vsync latch the over ride setting

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Table 8-2758 RO_TCON_INTR

bit	Default	Name	Description
[0]		RO_intr	tcon vsync interrupt triggered
[1]			tcon swreset interrupt triggered
[2]			pll lock change interrupt triggered
[3]			tcon p2s reset interrupt triggered
[4]			tcon tx en change interrupt triggered
[5]			tcon aging signal change interrupt triggered
[6]			ddr interface error interrupt triggered

8.2.3.45 VDIN Registers

VDIN0_SCALE_COEF_IDX 0x1200

VDIN0_SCALE_COEF 0x1201

Table 8-2759 VDIN0_COM_CTRL0 0x1202

Bit(s)	R/W	Default	Description
31	R/W	0	mpeg_to_vdin_sel, 0: mpeg source to NR directly, 1: mpeg source pass through here
30	R/W	0	mpeg_field info which can be written by software
29	R/W	0	force go_field, pulse signal
28	R/W	0	force go_line, pulse signal
27	R/W	0	enable mpeg_go_field input signal
26-20	R/W	0	hold lines
19	R/W	0	delay go_field function enable
18-12	R/W	0	delay go_field line number
11-10	R/W	0	component2 output switch, 00: select component0 in, 01: select component1 in, 10: select component2 in
9-8	R/W	0	component1 output switch, 00: select component0 in, 01: select component1 in, 10: select component2 in
7-6	R/W	0	component0 output switch, 00: select component0 in, 01: select component1 in, 10: select component2 in
5	R/W	0	input window selection function enable
4	R/W	0	enable VDIN common data input, otherwise there will be no video data input
3-0	R/W	0	vdin selection, 1: mpeg_in from dram; 2: bt656 input; 3: Reserved (component input); 4: Reserved(tvdecoder input); 5: Reserved(hdmi rx input); 6: reserved(digital video input); 7: Wr_back 0; 8: reserved(MIPI CSI2) ; 9: Wr_back 1; 10: Reserved(second bt656 input); otherwise no input.

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Table 8-2760 VDIN0_ACTIVE_MAX_PIX_CNT_STATUS 0x1203

Bit(s)	R/W	Default	Description
28-16	R	0	active_max_pix_cnt, readonly
12-0	R	0	active_max_pix_cnt_shadow, readonly

Table 8-2761 VDIN0_LCNT_STATUS 0x1204

Bit(s)	R/W	Default	Description
28-16	R	0	go_line_cnt, readonly
12-0	R	0	active_line_cnt, readonly

Table 8-2762 VDIN0_COM_STATUS0 0x1205

Bit(s)	R/W	Default	Description
17	R	0	Vid_wr_pending_ddr_wrrsp
16	R	0	Curr_pic_sec
15	R	0	Curr_pic_sec_sav
14-3	R	0	lfifo_buf_cnt
2	R	0	vdin_direct_done status
1	R	0	vdin_nr_done status
0	R	0	field

Table 8-2763 VDIN0_COM_STATUS1 0x1206

Bit(s)	R/W	Default	Description
31	R	0	vdi4 fifo overflow
29-24	R	0	vdi3_asfifo_cnt
23	R	0	vdi3 fifo overflow
21-16	R	0	vdi3_asfifo_cnt
15	R	0	vdi2 fifo overflow
13-8	R	0	vdi2_asfifo_cnt
7	R	0	vdi1 fifo overflow
5-0	R	0	vdi1_asfifo_cnt

Table 8-2764 VDIN0_LCNT_SHADOW_STATUS 0x1207

Bit(s)	R/W	Default	Description
28-16	R	0	go_line_cnt_shadow, readonly
12-0	R	0	active_line_cnt_shadow, readonly

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Table 8-2765 VDIN0_ASFIFO_CTRL0 0x1208

Bit(s)	R/W	Default	Description
31	R/W	0	hv size select 0: write mif size 1:vdin hsc input size
30-28	R/W	0	v de start line number
27-20	R/W	0	v blank
19-12	R/W	0	h blank
10-9	R/W	0	bist pattern 0:horizontal gray scale 1: vertical gray scale 2/3:random data
8	R/W	0	Vdin bist enable : replace vdi6 (loop back path)
7	R/W	0	Vdi1 DE enable
6	R/W	0	Vdi1 go field enable
5	R/W	0	Vdi1 go line enable
4	R/W	0	Vdi1 if true, negative active input vsync
3	R/W	0	Vdi1 if true, negative active input hsync
2	R/W	0	Vdi1 vsync soft reset fifo enable
1	R/W	0	Vdi1 overflow status clear
0	R/W	0	Vdi1 asfifo soft reset, level signal

Table 8-2766 VDIN0_ASFIFO_CTRL1 0x1209

Bit(s)	R/W	Default	Description
23	R/W	0	Vdi4 DE enable
22	R/W	0	Vdi4 go field enable
21	R/W	0	Vdi4 go line enable
20	R/W	0	Vdi4 if true, negative active input vsync
19	R/W	0	Vdi4 if true, negative active input hsync
18	R/W	0	Vdi4 vsync soft reset fifo enable
17	R/W	0	Vdi4 overflow status clear
16	R/W	0	Vdi4 asfifo soft reset, level signal
7	R/W	0	Vdi3 DE enable
6	R/W	0	Vdi3 go field enable
5	R/W	0	Vdi3 go line enable
4	R/W	0	Vdi3 if true, negative active input vsync
3	R/W	0	Vdi3 if true, negative active input hsync
2	R/W	0	Vdi3 vsync soft reset fifo enable
1	R/W	0	Vdi3 overflow status clear
0	R/W	0	Vdi3 asfifo soft reset, level signal

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Table 8-2767 VDIN0_WIDTHM1I_WIDTHM1O 0x120a

Bit(s)	R/W	Default	Description
28-16	R/W	0	input width minus 1, after the window function
12-0	R/W	0	output width minus 1

Table 8-2768 VDIN0_SC_MISC_CTRL 0x120b

Bit(s)	R/W	Default	Description
14-8	R/W	0	hsc_ini_pixi_ptr, signed data, only useful when short_lineo_en is true
7	R/W	0	prehsc_en
6	R/W	0	hsc_en
5	R/W	0	hsc_short_lineo_en, short line output enable
4	R/W	0	hsc_nearest_en
3	R/W	0	Hsc_phase0_always_en
2-0	R/W	0	hsc_bank_length

Table 8-2769 VDIN0_HSC_PHASE_STEP 0x120c

Bit(s)	R/W	Default	Description
28-24	R/W	0	integer portion
23-0	R/W	0	fraction portion

Table 8-2770 VDIN0_HSC_INI_CTRL 0x120d

Bit(s)	R/W	Default	Description
30-29	R/W	0	hscale rpt_p0_num
28-24	R/W	0	hscale ini_rcv_num
23-0	R/W	0	hscale ini_phase

Table 8-2771 VDIN0_COM_STATUS2 0x120e

Bit(s)	R/W	Default	Description
23	R	0	Vdi7 fifo overflow
21-16	R	0	Vdi7_asfifo_cnt
15	R	0	Vdi6 fifo overflow
13-8	R	0	Vdi6_asfifo_cnt
7	R	0	vdi5 fifo overflow
5-0	R	0	vdi5_asfifo_cnt

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Table 8-2772 VDIN0_ASFIFO_CTRL2 0x120f

Bit(s)	R/W	Default	Description
25	R/W	0	if true, decimation counter sync with first valid DE in the field, //otherwise the decimation counter is not sync with external signal
24	R/W	0	decimation de enable
23-20	R/W	0	decimation phase, which counter value use to decimate,
19-16	R/W	0	decimation number, 0: not decimation, 1: decimation 2, 2: decimation 3 ....
7	R/W	0	Vdi5 DE enable
6	R/W	0	Vdi5 go field enable
5	R/W	0	Vdi5 go line enable
4	R/W	0	Vdi5 if true, negative active input vsync
3	R/W	0	Vdi5 if true, negative active input hsync
2	R/W	0	Vdi5 vsync soft reset fifo enable
1	R/W	0	Vdi5 overflow status clear
0	R/W	0	Vdi5 asfifo soft reset, level signal

Table 8-2773 VDIN0_MATRIX_CTRL 0x1210

Bit(s)	R/W	Default	Description
7	R/W	0	highlight_enable
6	R/W	0	probe_post, if true, probe pixel data after matrix, otherwise probe pixel data before matrix
5-4	R/W	0	probe_sel, 00: select matrix 0, 01: select matrix 1, otherwise select nothing
3-2	R/W	0	matrix coef idx selection, 00: select mat0, 01: select mat1, otherwise slect nothing
1	R/W	0	mat1 conversion matrix enable
0	R/W	0	Mat0 conversion matrix enable

Table 8-2774 VDIN0_MATRIX_COEF00_01 0x1211

Bit(s)	R/W	Default	Description
28-16	R/W	0	coef00
12-0	R/W	0	coef01

Table 8-2775 VDIN0_MATRIX_COEF02_10 0x1212

Bit(s)	R/W	Default	Description
28-16	R/W	0	coef02
12-0	R/W	0	Coef10

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Table 8-2776 VDIN0_MATRIX_COEF11_12 0x1213

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coef11
12-0	R/W	0	Coef12

Table 8-2777 VDIN0_MATRIX_COEF20_21 0x1214

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coef20
12-0	R/W	0	coef21

Table 8-2778 VDIN0_MATRIX_COEF22 0x1215

Bit(s)	R/W	Default	Description
18-16	R/W	0	convrs
7-0	R/W	0	Coef22

Table 8-2779 VDIN0_MATRIX_OFFSET0_1 0x1216

Bit(s)	R/W	Default	Description
26-16	R/W	0	offset0
10-0	R/W	0	Offset1

Table 8-2780 VDIN0_MATRIX_OFFSET2 0x1217

Bit(s)	R/W	Default	Description
10-0	R/W	0	Offset2

Table 8-2781 VDIN0_MATRIX_PRE_OFFSET0_1 0x1218

Bit(s)	R/W	Default	Description
26-16	R/W	0	Pre_offset0
10-0	R/W	0	Pre_Offset1

Table 8-2782 VDIN0_MATRIX_PRE_OFFSET2 0x1219

Bit(s)	R/W	Default	Description
10-0	R/W	0	Pre_Offset2

Table 8-2783 VDIN0_LFIFO_CTRL 0x121a

Bit(s)	R/W	Default	Description
11-0	R/W	0	lfifo_buf_size

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Table 8-2784 VDIN0_COM_GCLK_CTRL 0x121b

Bit(s)	R/W	Default	Description
15-14	R/W	0	Gate clock control for blackbar detector
13-12	R/W	0	Gate clock control for hist
11-10	R/W	0	Gate clock control for line fifo
9-8	R/W	0	Gate clock control for matrix
7-6	R/W	0	Gate clock control for horizontal scaler
5-4	R/W	0	Gate clock control for pre scaler
3-2	R/W	0	Gate clock control for vdin_com_proc
1-0	R/W	0	Gate clock control for the vdin reg

Table 8-2785 VDIN0_INTF_WIDTHM1 0x121c

Bit(s)	R/W	Default	Description
26	R/W	0	VDIN write mif bvalid_sel: 1. Bvalid_singnal from bus, 0: bytes_wr handshakes
25	R/W	0	VDIN write mif burst last sel: 1. All kind of burst last signal include ext_data_last. 0. Used the normal burst last signal
12-0	R/W	0	VDIN input interface width minus 1, before the window function, after the de decimation

Table 8-2786 VDIN0_WR_CTRL2 0x121f

Bit(s)	R/W	Default	Description
19	R/W	0	Vdin0 wr bit10 mode
18	R/W	0	Data_ext_en 1: send out data if req was interrupt by soft reset 0 : normal mode
17:16	R/W	1	Words lim[1:0] : it would not send out request before Words_lim *16 words were ready
15:12	R/W	1	Burst_ lim : 00 , 1 word in 1burst , 01 ,2 words in 1burst, 10, 4 words in 1burst , 11 reserved
10:9	R/W	0	Words_lim[3:2]
8	R/W	0	1: discard data before line fifo, 0: normal mode
7-0	R/W	0	Write chroma canvas address, for NV12/21 mode.

Table 8-2787 VDIN0_WR_CTRL 0x1220

Bit(s)	R/W	Default	Description
31:30	R/W	0	vdin0_wr_mif_hconv_mode. Applicable only to vdin_write_format=0 or 2. 0=Output every even pixel's CbCr; 1=Output every odd pixel's CbCr; 2=Output an average value per even&odd pair of pixels; 3=Output all CbCr. Only applys to vdin_write_format =2.
29	R/W	0	vdin0_wr_mif_no_clk_gate. If true, enable free-run clock.

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Bit(s)	R/W	Default	Description
28	R/W	0	clear write response counter in the vdin write memory interface
27	R/W	1	eol_sel, 1: use eol as the line end indication, 0: use width as line end indication in the vdin write memory interface
26	R/W	0	vcp_nr_en. Only used in VDIN0. NOT used in VDIN1
25	R/W	1	vcp_wr_en Only used in VDIN0. NOT used in VDIN1
24	R/W	1	vcp_in_en Only used in VDIN0. NOT used in VDIN1
23	R/W	1	vdin frame reset enble, if true, it will provide frame reset during go_field(vsync) to the modules after that
22	R/W	1	vdin line fifo soft reset enable, meaning, if true line fifo will reset during go_field (vsync)
21	R/W	0	vdin direct write done status clear bit
20	R/W	0	vdin NR write done status clear bit
19	R/W	0	Vdin0_wr words swap : swap the 2 64bits word in 128 words
18	R/W	0	vdin0_wr_mif_swap_cbcr. Applicable only to vdin_write_format =2. 0=Output CbCr (NV12); 1=Output CrCb (NV21);
17:16	R/W	0	vdin0_wr_mif_vconv_mode. Applicable only to vdin_write_format=2. 0=Output every even line's CbCr; 1=Output every odd line's CbCr; 2=Reserved; 3=Output all CbCr.
13-12	R/W	0	vdin_write_format, 0: 4:2:2 to one canvas; 1: 4:4:4 to one canvas; 2: Y to luma canvas, CbCr to chroma canvas, for NV12/21; 3: yuv422 full pack mode
11	R/W	0	vdin write canvas double buffer enable, means the canvas address will be latched by vsync before using
9	R/W	0	vdin write request urgent
8	R/W	0	vdin write request enable
7-0	R/W	0	Write canvas address (For NV12/21 mode, it's LUMA canvas)

Table 8-2788 VDIN0_WR_H_START_END 0x1221

Bit(s)	R/W	Default	Description
29	R/W	0	if true, horizontal reverse
28-16	R/W	0	start
12-0	R/W	0	end

Table 8-2789 VDIN0_WR_V_START_END 0x1222

Bit(s)	R/W	Default	Description
29	R/W	0	if true, vertical reverse
28-16	R/W	0	start
12-0	R/W	0	end

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Table 8-2790 VDIN0_VSC_PHASE_STEP 0x1223

Bit(s)	R/W	Default	Description
24-16	R/W	0	integer portion
19-0	R/W	0	fraction portion

Table 8-2791 VDIN0_VSC_INI_CTRL 0x1224

Bit(s)	R/W	Default	Description
23	R/W	0	vsc_en, vertical scaler enable
21	R/W	0	vsc_phase0_always_en, when scale up, you have to set it to 1
20-16	R/W	0	ini skip_line_num
15-0	R/W	0	vscaler ini_phase

Table 8-2792 VDIN0_SCIN_HEIGHTM1 0x1225

Bit(s)	R/W	Default	Description
12-0	R/W	0x437	scaler input height minus 1

Table 8-2793 VDIN0_DUMMY_DATA 0x1226

Bit(s)	R/W	Default	Description
23-16	R/W	0	dummy component 0
15-8	R/W	0x80	dummy component 1
7-0	R/W	0x80	dummy component 2

Table 8-2794 VDIN0_MATRIX_PROBE_COLOR 0x1228

Bit(s)	R/W	Default	Description
29-20	R/W	0	component 0
19-10	R/W	0	omponent 1
9-0	R/W	0	component 2

Table 8-2795 VDIN0_MATRIX_HL_COLOR 0x1229

Bit(s)	R/W	Default	Description
23-16	R/W	0	component 0
15-8	R/W	0	component 1
7-0	R/W	0	component 2

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Table 8-2796 VDIN0_MATRIX_PROBE_POS 0x122a

Bit(s)	R/W	Default	Description
28-16	R/W	0	probe x, position
12-0	R/W	0	probe y, postion

Table 8-2797 VDIN0_HIST_CTRL 0x1230

Bit(s)	R/W	Default	Description
31-24	R/W	0	No use
23-16	R/W	0	No use
11	R/W	0	Hist 34bin only mode
10-9	R/W	0	ldim_stts_din_sel, 00: from matrix0 dout, 01: from vsc_dout, 10: from matrix1 dout, 11: form matrix1 din
8	R/W	0	ldim_stts_en
6-5	R/W	0	hist_dnlp_low the real pixels in each bins got by VDIN_DNLP_HISTXX should multiple with 2^(dnlp_low+3)
3-2	R/W	0	hist_din_sel the source used for hist statistics. 2'b00: from MAT0_dout; 2'b01: from vsc_dout; 2'b10: from mat1_dout, 3: mat1_din
1	R/W	0	hist_win_en 1'b0: hist used for full picture; 1'b1: hist used for pixels within hist window
0	R/W	0	hist_spl_en 1'b0: disable hist readback; 1'b1: enable hist readback

Table 8-2798 VDIN0_HIST_H_START_END 0x1231

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist_hstart horizontal start value to define hist window
12-0	R/W	0	hist_hend horizontal end value to define hist window

Table 8-2799 VDIN0_HIST_V_START_END 0x1232

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist_vstart vertical start value to define hist window
12-0	R/W	0	hist_vend vertical end value to define hist window

Table 8-2800 VDIN0_HIST_MAX_MIN 0x1233

Bit(s)	R/W	Default	Description
15-8	R	0	hist_max maximum value
7-0	R	0	hist_min minimum value

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Table 8-2801 VDIN0_HIST_SPL_VAL 0x1234

Bit(s)	R/W	Default	Description
31-0	R	0	hist_spl_rd , counts for the total luma value

Table 8-2802 VDIN0_HIST_SPL_PIX_CNT 0x1235

Bit(s)	R/W	Default	Description
21-0	R	0	hist_spl_pixel_count, counts for the total calculated pixels

Table 8-2803 VDIN0_HIST_CHROMA_SUM 0x1236

Bit(s)	R/W	Default	Description
31-0	R	0	hist_chroma_sum , counts for the total chroma value

//0-255 are splited to 64 bins evenly, and VDIN_DNLP_HISTXX

//are the statistic number of pixels that within each bin.

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 2nd bin
15-0	R	0	counts for the 1st bin

Table 8-2804 VDIN0_DNLP_HIST01 0x1238

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 4th bin
15-0	R	0	counts for the 3rd bin

Table 8-2805 VDIN0_DNLP_HIST02 0x1239

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 6th bin
15-0	R	0	counts for the 5th bin

Table 8-2806 VDIN0_DNLP_HIST03 0x123a

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 8th bin
15-0	R	0	counts for the 7th bin

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Table 8-2807 VDIN0_DNLP_HIST04 0x123b

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 10th bin
15-0	R	0	counts for the 9th bin

Table 8-2808 VDIN0_DNLP_HIST05 0x123c

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 12th bin
15-0	R	0	counts for the 11th bin

Table 8-2809 VDIN0_DNLP_HIST06 0x123d

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 14th bin
15-0	R	0	counts for the 13th bin

Table 8-2810 VDIN0_DNLP_HIST07 0x123e

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 16th bin
15-0	R	0	counts for the 15th bin

Table 8-2811 VDIN0_DNLP_HIST08 0x123f

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 18th bin
15-0	R	0	counts for the 17th bin

Table 8-2812 VDIN0_DNLP_HIST09 0x1240

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 20th bin
15-0	R	0	counts for the 19th bin

Table 8-2813 VDIN0_DNLP_HIST10 0x1241

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 22nd bin
15-0	R	0	counts for the 21st bin

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Table 8-2814 VDIN0_DNLP_HIST11 0x1242

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 24th bin
15-0	R	0	counts for the 23rd bin

Table 8-2815 VDIN0_DNLP_HIST12 0x1243

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 26th bin
15-0	R	0	counts for the 25th bin

Table 8-2816 VDIN0_DNLP_HIST13 0x1244

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 28th bin
15-0	R	0	counts for the 27th bin

Table 8-2817 VDIN0_DNLP_HIST14 0x1245

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 30th bin
15-0	R	0	counts for the 29th bin

Table 8-2818 VDIN0_DNLP_HIST15 0x1246

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 32nd bin
15-0	R	0	counts for the 31st bin

Table 8-2819 VDIN0_DNLP_HIST16 0x1247

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 34th bin
15-0	R	0	counts for the 33rd bin

Table 8-2820 VDIN0_DNLP_HIST17 0x1248

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 36th bin
15-0	R	0	counts for the 35th bin

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Table 8-2821 VDIN0_DNLP_HIST18 0x1249

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 38th bin
15-0	R	0	counts for the 37th bin

Table 8-2822 VDIN0_DNLP_HIST19 0x124a

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 40th bin
15-0	R	0	counts for the 39th bin

Table 8-2823 VDIN0_DNLP_HIST20 0x124b

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 42nd bin
15-0	R	0	counts for the 41st bin

Table 8-2824 VDIN0_DNLP_HIST21 0x124c

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 44th bin
15-0	R	0	counts for the 43rd bin

Table 8-2825 VDIN0_DNLP_HIST22 0x124d

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 46th bin
15-0	R	0	counts for the 45th bin

Table 8-2826 VDIN0_DNLP_HIST23 0x124e

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 48th bin
15-0	R	0	counts for the 47th bin

Table 8-2827 VDIN0_DNLP_HIST24 0x124f

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 50th bin
15-0	R	0	counts for the 49th bin

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Table 8-2828 VDIN0_DNLP_HIST25 0x1250

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 52nd bin
15-0	R	0	counts for the 51st bin

Table 8-2829 VDIN0_DNLP_HIST26 0x1251

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 54th bin
15-0	R	0	counts for the 53rd bin

Table 8-2830 VDIN0_DNLP_HIST27 0x1252

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 56th bin
15-0	R	0	counts for the 55th bin

Table 8-2831 VDIN0_DNLP_HIST28 0x1253

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 58th bin
15-0	R	0	counts for the 57th bin

Table 8-2832 VDIN0_DNLP_HIST29 0x1254

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 60th bin
15-0	R	0	counts for the 59th bin

Table 8-2833 VDIN0_DNLP_HIST30 0x1255

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 62nd bin
15-0	R	0	counts for the 61st bin

Table 8-2834 VDIN0_DNLP_HIST31 0x1256

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 64th bin
15-0	R	0	counts for the 63rd bin

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Table 8-2835 VDIN0_LDIM_STTS_HIST_REGION_IDX 0x1257

Bit(s)	R/W	Default	Description
31	R	0	local dimming max statistic enable
28	R	0	eol enable
27-25	R	0	vertical line overlap number for max finding
24-22	R	0	horizontal pixel overlap number, 0: 17 pix, 1: 9 pix, 2: 5 pix, 3: 3 pix, 4: 0 pix
20	R	0	1,2,1 low pass filter enable before max/hist statistic
19-16	R	0	region H/V position index, refer to VDIN_LDIM_STTS_HIST_SET_REGION
15	R	0	1: region read index auto increase per read to VDIN_LDIM_STTS_HIST_READ_REGION
6-0	R	0	region read index

Table 8-2836 VDIN0_LDIM_STTS_HIST_SET_REGION 0x1258

Bit(s)	R/W	Default	Description
28:16	R	0	if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h0: read/write hvstart0 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h1: read/write hend01 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h2: read/write vend01 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h3: read/write hend23 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h4: read/write vend23 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h5: read/write hend45 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h6: read/write vend45 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'd7: read/write hend67 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h8: read/write vend67 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h9: read/write hend89 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'ha: read/write vend89 //hvstart0, Bit 28:16 row0 vstart, Bit 12:0 col0 hstart //hend01, Bit 28:16 col1 hend, Bit 12:0 col0 hend //vend01, Bit 28:16 row1 vend, Bit 12:0 row0 vend //hend23, Bit 28:16 col3 hend, Bit 12:0 col2 hend //vend23, Bit 28:16 row3 vend, Bit 12:0 row2 vend //hend45, Bit 28:16 col5 hend, Bit 12:0 col4 hend //vend45, Bit 28:16 row5 vend, Bit 12:0 row4 vend //hend67, Bit 28:16 col7 hend, Bit 12:0 col6 hend //vend67, Bit 28:16 row7 vend, Bit 12:0 row6 vend //hend89, Bit 28:16 col9 hend, Bit 12:0 col8 hend //vend89, Bit 28:16 row9 vend, Bit 12:0 row8 vend
12:0	R	0	

Table 8-2837 VDIN0_LDIM_STTS_HIST_READ_REGION 0x1259

Bit(s)	R/W	Default	Description
29:20	R	0	Max_comp2
19:10	R	0	Max_comp1
9:0	R	0	Max_comp0

Table 8-2838 VDIN0_MEAS_CTRL0 0x125a

Bit(s)	R/W	Default	Description
18	R/W	0	reset bit, high active
17	R/W	0	if true, widen hs/vs pulse
16	R/W	0	vsync total counter always accumulating enable

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Bit(s)	R/W	Default	Description
15-12	R/W	0	select hs/vs of video input channel to measure, 0: no selection, 1 vdi1, 2: vid2: 3: vid3, 4:vid4, 5:vdi5, 6:vdi6, 7:vdi7, 8:vdi8, 9:vdi9.
11-4	R/W	0	vsync_span, define how many vsync span need to measure
2-0	R/W	0	meas_hs_index, index to select which HS counter/range

Table 8-2839 VDIN0_MEAS_VS_COUNT_HI 0x125b

Bit(s)	R/W	Default	Description
19-16	R	0	meas_ind_total_count_n, every number of sync_span vsyncs, this count add 1
15-0	R	0	high bit portion of vsync total counter

Table 8-2840 VDIN0_MEAS_VS_COUNT_LO 0x125c

Bit(s)	R/W	Default	Description
31-0	R	0	low bit portion of vsync total counter

VDIN0_MEAS_HS_RANGE 0x125d

//according to the meas_hs_index in register VDIN_MEAS_CTRL0

//meas_hs_index == 0, first hs range

//meas_hs_index == 1, second hs range

//meas_hs_index == 2, third hs range

//meas_hs_index == 3, fourth hs range

Bit(s)	R/W	Default	Description
28-16	R	0	count_start
12-0	R	0	count_end

VDIN0_MEAS_HS_COUNT 0x125e

//according to the meas_hs_index in register VDIN_MEAS_CTRL0,

//meas_hs_index == 0, first range hs counter,

//meas_hs_index == 1, second range hs

//meas_hs_index == 2, third range hs

//meas_hs_index == 3, fourth range hs

Bit(s)	R/W	Default	Description
23-0	R	0	Hs counter

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Table 8-2841 VDIN0_BLKBAR_CTRL1 0x125f

Bit(s)	R/W	Default	Description
8	R/W	0	white_enable
7-0	R/W	0	blkbar_white_level

Table 8-2842 VDIN0_BLKBAR_CTRL0 0x1260

Bit(s)	R/W	Default	Description
31-24	R/W	0	blkbar_black_level threshold to judge a black point
20-8	R/W	0	blkbar_hwidth left and right region width
7-5	R/W	0	blkbar_comp_sel select yin or uin or vin to be the valid input
4	R/W	0	blkbar_sw_statistic_en enable software statistic of each block black points number
3	R/W	0	blkbar_det_en
2-1	R/W	0	blkbar_din_sel, 0:mat0_dout, 1:vsc_dout, 2:mat1_dout, 3:mat1_din
0	R/W	0	Blkbar_det_top_en

Table 8-2843 VDIN0_BLKBAR_H_START_END 0x1261

Bit(s)	R/W	Default	Description
28-16	R/W	0	blkbar_hstart. Left region start
12-0	R/W	0	blkbar_hend. Right region end

Table 8-2844 VDIN0_BLKBAR_V_START_END 0x1262

Bit(s)	R/W	Default	Description
28-16	R/W	0	blkbar_vstart.
12-0	R/W	0	blkbar_vend.

Table 8-2845 VDIN0_BLKBAR_CNT_THRESHOLD 0x1263

Bit(s)	R/W	Default	Description
19-0	R/W	0	blkbar_cnt_threshold. threshold to judge whether a block is totally black

Table 8-2846 VDIN0_BLKBAR_ROW_TH1_TH2 0x1264

Bit(s)	R/W	Default	Description
28-16	R/W	0	blkbar_row_th1. //threshold of the top blackbar
12-0	R/W	0	blkbar_row_th2 //threshold of the bottom blackbar

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Table 8-2847 VDIN0_BLKBAR_IND_LEFT_START_END 0x1265

Bit(s)	R/W	Default	Description
28-16	R	0	blkbar_ind_left_start. horizontal start of the left region in the current searching
12-0	R	0	blkbar_ind_left_end. horizontal end of the left region in the current searching

Table 8-2848 VDIN0_BLKBAR_IND_RIGHT_START_END 0x1266

Bit(s)	R/W	Default	Description
28-16	R	0	blkbar_ind_right_start.horizontal start of the right region in the current searching
12-0	R	0	blkbar_ind_right_end. horizontal end of the right region in the current searching

Table 8-2849 VDIN0_BLKBAR_IND_LEFT1_CNT 0x1267

Bit(s)	R/W	Default	Description
19-0	R	0	blkbar_ind_left1_cnt. Black pixel counter. left part of the left region

Table 8-2850 VDIN0_BLKBAR_IND_LEFT2_CNT 0x1268

Bit(s)	R/W	Default	Description
19-0	R	0	blkbar_ind_left2_cnt. Black pixel counter. right part of the left region

Table 8-2851 VDIN0_BLKBAR_IND_RIGHT1_CNT 0x1269

Bit(s)	R/W	Default	Description
19-0	R	0	blkbar_ind_right1_cnt. Black pixel counter. left part of the right region

Table 8-2852 VDIN0_BLKBAR_IND_RIGHT2_CNT 0x126a

Bit(s)	R/W	Default	Description
19-0	R	0	blkbar_ind_right2_cnt. Black pixel counter. right part of the right region

Table 8-2853 VDIN0_BLKBAR_STATUS0 0x126b

Bit(s)	R/W	Default	Description
29	R	0	blkbar_ind_black_det_done. LEFT/RIGHT Black detection done
28-16	R	0	blkbar_top_pos. Top black bar position
12-0	R	0	blkbar_bot_pos. Bottom black bar position

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Table 8-2854 VDIN0_BLKBAR_STATUS1 0x126c

Bit(s)	R/W	Default	Description
28-16	R	0	blkbar_left_pos. Left black bar posiont
12-0	R	0	blkbar_right_pos. Right black bar position

Table 8-2855 VDIN0_WIN_H_START_END 0x126d

Bit(s)	R/W	Default	Description
28-16	R/W	0	input window H start
12-0	R/W	0	input window H end

Table 8-2856 VDIN0_WIN_V_START_END 0x126e

Bit(s)	R/W	Default	Description
28-16	R/W	0	input window V start
12-0	R/W	0	input window V end

Table 8-2857 VDIN0_ASFIFO_CTRL3 0x126f

Bit(s)	R/W	Default	Description
31	R/W	0	Vdi9 DE enable
30	R/W	0	Vdi9 go field enable
29	R/W	0	Vdi9 go line enable
28	R/W	0	Vdi9 if true, negative active input vsync
27	R/W	0	Vdi9 if true, negative active input hsync
26	R/W	0	Vdi9 vsync soft reset fifo enable
25	R/W	0	Vdi9 overflow status clear
24	R/W	0	Vdi9 asfifo soft reset, level signal
15	R/W	0	Vdi7 DE enable
14	R/W	0	Vdi7 go field enable
13	R/W	0	Vdi7 go line enable
12	R/W	0	Vdi7 if true, negative active input vsync
11	R/W	0	Vdi7 if true, negative active input hsync
10	R/W	0	Vdi7 vsync soft reset fifo enable
9	R/W	0	Vdi7 overflow status clear
8	R/W	0	Vdi7 asfifo soft reset, level signal
7	R/W	0	Vdi6 DE enable
6	R/W	0	Vdi6 go field enable
5	R/W	0	Vdi6 go line enable

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Bit(s)	R/W	Default	Description
4	R/W	0	Vdi6 if true, negative active input vsync
3	R/W	0	Vdi6 if true, negative active input hsync
2	R/W	0	Vdi6 vsync soft reset fifo enable
1	R/W	0	Vdi6 overflow status clear
0	R/W	0	Vdi6 asfifo soft reset, level signal

Table 8-2858 VDIN0_DOLBY_DSC_CTRL0 0x1275

Bit(s)	R/W	Default	Description
31	R/W	0	Dolby check enable
30	R/W	0	Tunnel swap mode enable
29-24	R/W	0	Soft reset control, 29 : dsc, 28~27: crc check, 26~24, reserved
16	R/W	0	Little endian mode
15-0	R/W	0	Monitor metadata position

Table 8-2859 VDIN0_DOLBY_DSC_CTRL1 0x1276

Bit(s)	R/W	Default	Description
31-16	R/W	0	Metadata pixel start position
7-0	R/W	0	Crc check control

Table 8-2860 VDIN0_DOLBY_DSC_CTRL2 0x1277

Bit(s)	R/W	Default	Description
31	R/W	0	Metadata read enable
30	R/W	0	Metadata read address auto-increment
29-20	R/W	0	Metadata sum
17-0	R/W	0	Tunnel mode channel selection

Table 8-2861 VDIN0_DOLBY_DSC_CTRL3 0x1278

Bit(s)	R/W	Default	Description
15-0	R/W	0	Select metadata position

Table 8-2862 VDIN0_DOLBY_AXI_CTRL0 0x1279

Bit(s)	R/W	Default	Description
31	R/W	0	Memory read enable
30	R/W	0	AXI bus read enable
29	R/W	0	AXI write channel urgent

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Bit(s)	R/W	Default	Description
28	R/W	0	Pack mode little endian
27	R/W	0	AXI bus soft reset
26	R/W	0	Frame reset enable
25	R/W	0	Memory read protection enable
24-16	R/W	0	Memory read sum
15-8	R/W	0	AXI request hold line
7-6	R/W	0	AXI burst length
5	R/W	0	Frame buffer start
4	R/W	0	Buffer start
3-2	R/W	0	AXI status monitor control
1-0	R/W	0	awid

Table 8-2863 VDIN0_DOLBY_ AXI_CTRL1 0x127a

Bit(s)	R/W	Default	Description
31-0	R/W	0	Buffer start address

Table 8-2864 VDIN0_DOLBY_ AXI_CTRL2 0x127b

Bit(s)	R/W	Default	Description
31-16	R/W	0	Buffer size
15-0	R/W	0	Frame buffer size

Table 8-2865 VDIN0_DOLBY_ AXI_CTRL3 0x127c

Bit(s)	R/W	Default	Description
7-0	R/W	0	Hold cycle

Table 8-2866 VDIN0_WRARB_MODE 0x12c0

Bit(s)	R/W	Default	Description
23:16	R/W	0	wrarb_sel : uns, default = 0 , wrarb_sel [0]==0 slave dc0 connect master port0 wrarb_sel[0]==1 slave dc0 connect master port1 wrarb_sel [1]==0 slave dc1 connect master port0 wrarb_sel[1]==1 slave dc1 connect master port1 wrarb_sel [2]==0 slave dc2 connect master port0 wrarb_sel[2]==1 slave dc2 connect master port1 wrarb_sel [3]==0 slave dc3 connect master port0 wrarb_sel[3]==1 slave dc3 connect master port1 wrarb_sel [4]==0 slave dc4 connect master port0 wrarb_sel[4]==1 slave dc3 connect master port1 wrarb_sel [5]==0 slave dc5 connect master port0 wrarb_sel[5]==1 slave dc3 connect master port1
8	R/W	0	wrarb_arb_mode : uns, default = 0 , wrarb_arb_mode[0] master port0 arb way,
1:0	R/W	0	wrarb_gate_clk_ctrl : uns, default = 0 , wrarb_gate_clk_ctrl[1:0] master port0 clk gate control

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Table 8-2867 VDIN0_WRARB_REQEN_SLV 0x12c1

Bit(s)	R/W	Default	Description
7:0	R/W	0	wrarb_dc_req_en : unsigned , default = 0 wrarb_dc_req_en [0]: the slv0 req to mst port0 enable, wrarb_dc_req_en [1]: the slv1 req to mst port0 enable, wrarb_ dc_req_en [2]: the slv2 req to mst port0 enable, wrarb_dc_req_en [0]: the slv0 req to mst port1 enable, wrarb_dc_req_en [1]: the slv1 req to mst port1 enable, wrarb_dc_req_en [2]: the slv2 req to mst port1 enable,

Table 8-2868 VDIN0_WRARB_WEIGH0_SLV 0x12C2

Bit(s)	R/W	Default	Description
29:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 wrdc_weigh_sxn [0*6+:6]: the slv0 req weigh number wrdc_weigh_sxn [1*6+:6]: the slv1 req weigh number wrdc_ weigh_sxn [2*6+:6]: the slv2 req weigh number wrdc_weigh_sxn [3*6+:6]: the slv3 req weigh number wrdc_weigh_sxn [4*6+:6]: the slv4 req weigh number

Table 8-2869 VDIN0_WRARB_WEIGH1_SLV 0x12C3

Bit(s)	R/W	Default	Description
17:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 the slv5 req weigh number [0*6+:6],the slv6 req weighnumber [1*6+:6],the slv7 req weighnumber [2*6+:6],

Table 8-2870 VDIN0_RDWR_ARB_STATUS 0x12c4

Bit(s)	R/W	Default	Description
2	R.O	0	ro_wrarb_arb_busy : unsigned , default = 0
1	R/W	0x0	reserve :
0	R.O	0	ro_rdarb_arb_busy : unsigned , default = 0

Table 8-2871 VDIN0_ARB_DBG_CTRL 0x12c5

Bit(s)	R/W	Default	Description
31:0	R.O	0	ro_det_cmd_ctrl : unsigned , default = 0

Table 8-2872 VDIN0_ARB_DBG_STAT 0x12C6

Bit(s)	R/W	Default	Description
31:0	R.O	0	ro_det_dbg_stat : unsigned , default = 0

Table 8-2873 VDIN0_CRC_CHK 0x12c9

Bit(s)	R/W	Default	Description
0	W	0	vdin crc check start (pulse)

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Table 8-2874 VDIN0_RO_CRC 0x12ca

Bit(s)	R/W	Default	Description
31-0	R	0	vdin crc result (lock by go_field)

Table 8-2875 VDIN0_LINE_INT 0x12cb

Bit(s)	R/W	Default	Description
31	R	0	line interrupt flag
30	R	0	write done flag
17	R/W	0	clear line interrupt flag
16	R/W	0	clear write done interrupt flag
15	R/W	0	enable line interrupt
12-0	R/W	0	line interrupt number

VDIN1_SCALE_COEF_IDX 0x1300

VDIN1_SCALE_COEF 0x1301

Table 8-2876 VDIN1_COM_CTRL0 0x1302

Bit(s)	R/W	Default	Description
31	R/W	0	mpeg_to_vdin_sel, 0: mpeg source to NR directly, 1: mpeg source pass through here
30	R/W	0	mpeg_field info which can be written by software
29	R/W	0	force go_field, pulse signal
28	R/W	0	force go_line, pulse signal
27	R/W	0	enable mpeg_go_field input signal
26-20	R/W	0	hold lines
19	R/W	0	delay go_field function enable
18-12	R/W	0	delay go_field line number
11-10	R/W	0	component2 output switch, 00: select component0 in, 01: select component1 in, 10: select component2 in
9-8	R/W	0	component1 output switch, 00: select component0 in, 01: select component1 in, 10: select component2 in
7-6	R/W	0	component0 output switch, 00: select component0 in, 01: select component1 in, 10: select component2 in
5	R/W	0	input window selection function enable
4	R/W	0	enable VDIN common data input, otherwise there will be no video data input
3-0	R/W	0	vdin selection, 1: mpeg_in from dram; 2: bt656 input; 3: Reserved (component input); 4: Reserved(tvdecoder input); 5: Reserved(hdmi rx input); 6: reserved(digital video input); 7: Wr_back 0; 8: reserved(MIPI CSI2) ;9: Wr_back 1; 10: Reserved(second bt656 input); otherwise no input.

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Table 8-2877 VDIN1_ACTIVE_MAX_PIX_CNT_STATUS 0x1303

Bit(s)	R/W	Default	Description
28-16	R	0	active_max_pix_cnt, readonly
12-0	R	0	active_max_pix_cnt_shadow, readonly

Table 8-2878 VDIN1_LCNT_STATUS 0x1304

Bit(s)	R/W	Default	Description
28-16	R	0	go_line_cnt, readonly
12-0	R	0	active_line_cnt, readonly

Table 8-2879 VDIN1_COM_STATUS0 0x1305

Bit(s)	R/W	Default	Description
12-3	R	0	lfifo_buf_cnt
2	R	0	vdin_direct_done status
1	R	0	vdin_nr_done status
0	R	0	field

Table 8-2880 VDIN1_COM_STATUS1 0x1306

Bit(s)	R/W	Default	Description
31	R	0	vdi4 fifo overflow
29-24	R	0	vdi3_asfifo_cnt
23	R	0	vdi3 fifo overflow
21-16	R	0	vdi3_asfifo_cnt
15	R	0	vdi2 fifo overflow
13-8	R	0	vdi2_asfifo_cnt
7	R	0	vdi1 fifo overflow
5-0	R	0	vdi1_asfifo_cnt

Table 8-2881 VDIN1_LCNT_SHADOW_STATUS 0x1307

Bit(s)	R/W	Default	Description
28-16	R	0	go_line_cnt_shadow, readonly
12-0	R	0	active_line_cnt_shadow, readonly

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Table 8-2882 VDIN1_ASFIFO_CTRL0 0x1308

Bit(s)	R/W	Default	Description
31	R/W	0	hv size select 0: write mif size 1:vdin hsc input size
30-28	R/W	0	v de start line number
27-20	R/W	0	v blank
19-12	R/W	0	h blank
10-9	R/W	0	bist pattern 0:horizontal gray scale 1: vertical gray scale 2/3:random data
8	R/W	0	Vdin bist enable : replace vdi6 (loop back path)
7	R/W	0	Vdi1 DE enable
6	R/W	0	Vdi1 go field enable
5	R/W	0	Vdi1 go line enable
4	R/W	0	Vdi1 if true, negative active input vsync
3	R/W	0	Vdi1 if true, negative active input hsync
2	R/W	0	Vdi1 vsync soft reset fifo enable
1	R/W	0	Vdi1 overflow status clear
0	R/W	0	Vdi1 asfifo soft reset, level signal

Table 8-2883 VDIN1_ASFIFO_CTRL1 0x1309

Bit(s)	R/W	Default	Description
23	R/W	0	Vdi4 DE enable
22	R/W	0	Vdi4 go field enable
21	R/W	0	Vdi4 go line enable
20	R/W	0	Vdi4 if true, negative active input vsync
19	R/W	0	Vdi4 if true, negative active input hsync
18	R/W	0	Vdi4 vsync soft reset fifo enable
17	R/W	0	Vdi4 overflow status clear
16	R/W	0	Vdi4 asfifo soft reset, level signal
7	R/W	0	Vdi3 DE enable
6	R/W	0	Vdi3 go field enable
5	R/W	0	Vdi3 go line enable
4	R/W	0	Vdi3 if true, negative active input vsync
3	R/W	0	Vdi3 if true, negative active input hsync
2	R/W	0	Vdi3 vsync soft reset fifo enable
1	R/W	0	Vdi3 overflow status clear
0	R/W	0	Vdi3 asfifo soft reset, level signal

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Table 8-2884 VDIN1_WIDTHM1I_WIDTHM1O 0x130a

Bit(s)	R/W	Default	Description
28-16	R/W	0	input width minus 1, after the window function
12-0	R/W	0	output width minus 1

Table 8-2885 VDIN1_SC_MISC_CTRL 0x130b

Bit(s)	R/W	Default	Description
14-8	R/W	0	hsc_ini_pixi_ptr, signed data, only useful when short_lineo_en is true
7	R/W	0	prehsc_en
6	R/W	0	hsc_en
5	R/W	0	hsc_short_lineo_en, short line output enable
4	R/W	0	hsc_nearest_en
3	R/W	0	Hsc_phase0_always_en
3	R/W	0	phase0_always_en
2-0	R/W	0	hsc_bank_length

Table 8-2886 VDIN1_HSC_PHASE_STEP 0x130c

Bit(s)	R/W	Default	Description
28-24	R/W	0	integer portion
23-0	R/W	0	fraction portion

Table 8-2887 VDIN1_HSC_INI_CTRL 0x130d

Bit(s)	R/W	Default	Description
30-29	R/W	0	hscale rpt_p0_num
28-24	R/W	0	hscale ini_rcv_num
23-0	R/W	0	hscale ini_phase

Table 8-2888 VDIN1_COM_STATUS2 0x130e

Bit(s)	R/W	Default	Description
23	R	0	Vdi7 fifo overflow
21-16	R	0	Vdi7_asfifo_cnt
15	R	0	Vdi6 fifo overflow
13-8	R	0	Vdi6_asfifo_cnt
7	R	0	vdi5 fifo overflow
5-0	R	0	vdi5_asfifo_cnt

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Table 8-2889 VDIN1_ASFIFO_CTRL2 0x130f

Bit(s)	R/W	Default	Description
25	R/W	0	if true, decimation counter sync with first valid DE in the field, //otherwise the decimation counter is not sync with external signal
24	R/W	0	decimation de enable
23-20	R/W	0	decimation phase, which counter value use to decimate,
19-16	R/W	0	decimation number, 0: not decimation, 1: decimation 2, 2: decimation 3 ....
7	R/W	0	Vdi5 DE enable
6	R/W	0	Vdi5 go field enable
5	R/W	0	Vdi5 go line enable
4	R/W	0	Vdi5 if true, negative active input vsync
3	R/W	0	Vdi5 if true, negative active input hsync
2	R/W	0	Vdi5 vsync soft reset fifo enable
1	R/W	0	Vdi5 overflow status clear
0	R/W	0	Vdi5 asfifo soft reset, level signal

Table 8-2890 VDIN1_MATRIX_CTRL 0x1310

Bit(s)	R/W	Default	Description
0	R/W	0	post conversion matrix enable

Table 8-2891 VDIN1_MATRIX_COEF00_01 0x1311

Bit(s)	R/W	Default	Description
28-16	R/W	0	coef00
12-0	R/W	0	coef01

Table 8-2892 VDIN1_MATRIX_COEF02_10 0x1312

Bit(s)	R/W	Default	Description
28-16	R/W	0	coef02
12-0	R/W	0	Coef10

Table 8-2893 VDIN1_MATRIX_COEF11_12 0x1313

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coef11
12-0	R/W	0	Coef12

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Table 8-2894 VDIN1_MATRIX_COEF20_21 0x1314

Bit(s)	R/W	Default	Description
28-16	R/W	0	Coef20
12-0	R/W	0	coef21

Table 8-2895 VDIN1_ 0x1315

Bit(s)	R/W	Default	Description
18-16	R/W	0	convrs
7-0	R/W	0	Coef22

Table 8-2896 VDIN1_MATRIX_OFFSET0_1 0x1316

Bit(s)	R/W	Default	Description
26-16	R/W	0	offset0
10-0	R/W	0	Offset1

Table 8-2897 VDIN1_MATRIX_OFFSET2 0x1317

Bit(s)	R/W	Default	Description
10-0	R/W	0	Offset2

Table 8-2898 VDIN1_MATRIX_PRE_OFFSET0_1 0x1318

Bit(s)	R/W	Default	Description
26-16	R/W	0	Pre_offset0
10-0	R/W	0	Pre_Offset1

Table 8-2899 VDIN1_MATRIX_PRE_OFFSET2 0x1319

Bit(s)	R/W	Default	Description
10-0	R/W	0	Pre_Offset2

Table 8-2900 VDIN1_LFIFO_CTRL 0x131a

Bit(s)	R/W	Default	Description
11-0	R/W	0	lfifo_buf_size

Table 8-2901 VDIN1_COM_GCLK_CTRL 0x131b

Bit(s)	R/W	Default	Description
15-14	R/W	0	Gate clock control for blackbar detector
13-12	R/W	0	Gate clock control for hist

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Bit(s)	R/W	Default	Description
11-10	R/W	0	Gate clock control for line fifo
9-8	R/W	0	Gate clock control for matrix
7-6	R/W	0	Gate clock control for horizontal scaler
5-4	R/W	0	Gate clock control for pre scaler
3-2	R/W	0	Gate clock control for vdin_com_proc
1-0	R/W	0	Gate clock control for the vdin reg

Table 8-2902 VDIN1_INTF_WIDTHM1 0x131c

Bit(s)	R/W	Default	Description
26	R/W	0	VDIN write mif bvalid_sel: 1. Bvalid_singnal from bus, 0: bytes_wr handshakes
25	R/W	0	VDIN write mif burst last sel: 1. All kind of burst last signal include ext_data_last. 0. Used the normal burst last signal
12-0	R/W	0	VDIN input interface width minus 1, before the window function, after the de decimation

Table 8-2903 VDIN1_WR_CTRL2 0x131f

Bit(s)	R/W	Default	Description
19	R/W	0	Vdin1 wr bit10 mode
18	R/W	0	Data_ext_en 1: send out data if req was interrupt by soft reset 0 : normal mode
17:16	R/W	1	Words lim[1:0]: it would not send out request before Words_lim *16 words were ready
15:12	R/W	1	Burst_ lim : 00 , 1 word in 1burst , 01 ,2 words in 1burst, 10, 4 words in 1burst , 11 reserved
10:9	R/W	0	Words_lim[3:2]
8	R/W	0	1: discard data before line fifo, 0: normal mode
7-0	R/W	0	Write chroma canvas address, for NV12/21 mode.

Table 8-2904 VDIN1_WR_CTRL 0x1320

Bit(s)	R/W	Default	Description
31-30	R/W	0	vdin1_wr_mif_hconv_mode. Applicable only to vdin_write_format=0 or 2. 0=Output every even pixel's CbCr; 1=Output every odd pixel's CbCr; 2=Output an average value per even&odd pair of pixels; 3=Output all CbCr. Only applys to vdin_write_format =2.
29	R/W	0	vdin1_wr_mif_no_clk_gate. If true, enable free-run clock.
28	R/W	0	clear write response counter in the vdin write memory interface
27	R/W	1	eol_sel, 1: use eol as the line end indication, 0: use width as line end indication in the vdin write memory interface
23	R/W	1	vdin frame reset enble, if true, it will provide frame reset during go_field(vsync) to the modules after that

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Bit(s)	R/W	Default	Description
22	R/W	1	vdin line fifo soft reset enable, meaning, if true line fifo will reset during go_field (vsync)
21	R/W	0	vdin direct write done status clear bit
20	R/W	0	vdin NR write done status clear bit
19	R/W	0	Vdin0_wr words swap : swap the 2 64bits word in 128 words
18	R/W	0	vdin1_wr_mif_swap_cbcr. Applicable only to vdin_write_format =2. 0=Output CbCr (NV12); 1=Output CrCb (NV21);
17:16	R/W	0	vdin1_wr_mif_vconv_mode. Applicable only to vdin_write_format=2. 0=Output every even line's CbCr; 1=Output every odd line's CbCr; 2=Reserved; 3=Output all CbCr.
13-12	R/W	0	vdin_write_format, 0: 4:2:2 to one canvas; 1: 4:4:4 to one canvas; 2: Y to luma canvas, CbCr to chroma canvas, for NV12/21; 3: 422 10 bit full pack mode
11	R/W	0	vdin write canvas double buffer enable, means the canvas address will be latched by vsync before using
9	R/W	0	vdin write request urgent
8	R/W	0	vdin write request enable
7-0	R/W	0	Write canvas address (For NV12/21 mode, it's LUMA canvas)

Table 8-2905 VDIN1_WR_H_START_END 0x1321

Bit(s)	R/W	Default	Description
27-16	R/W	0	start
11-0	R/W	0	end

Table 8-2906 VDIN1_WR_V_START_END 0x1322

Bit(s)	R/W	Default	Description
27-16	R/W	0	start
11-0	R/W	0	end

Table 8-2907 VDIN1_VSC_PHASE_STEP 0x1323

Bit(s)	R/W	Default	Description
24-16	R/W	0	integer portion
19-0	R/W	0	fraction portion

Table 8-2908 VDIN1_VSC_INI_CTRL 0x1324

Bit(s)	R/W	Default	Description
23	R/W	0	vsc_en, vertical scaler enable
21	R/W	0	vsc_phase0_always_en, when scale up, you have to set it to 1

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Bit(s)	R/W	Default	Description
20-16	R/W	0	ini skip_line_num
15-0	R/W	0	vscaler ini_phase

Table 8-2909 VDIN1_SCIN_HEIGHTM1 0x1325

Bit(s)	R/W	Default	Description
12-0	R/W	0x437	scaler input height minus 1

Table 8-2910 VDIN1_DUMMY_DATA 0x1326

Bit(s)	R/W	Default	Description
23-16	R/W	0	dummy component 0
15-8	R/W	0x80	dummy component 1
7-0	R/W	0x80	dummy component 2

Table 8-2911 VDIN1_MATRIX_PROBE_COLOR 0x1328

Bit(s)	R/W	Default	Description
29-20	R/W	0	component 0
19-10	R/W	0	omponent 1
9-0	R/W	0	component 2

Table 8-2912 VDIN1_MATRIX_HL_COLOR 0x1329

Bit(s)	R/W	Default	Description
23-16	R/W	0	component 0
15-8	R/W	0	component 1
7-0	R/W	0	component 2

Table 8-2913 VDIN1_MATRIX_PROBE_POS 0x132a

Bit(s)	R/W	Default	Description
28-16	R/W	0	probe x, position
12-0	R/W	0	probe y, postion

Table 8-2914 VDIN1_HIST_CTRL 0x1330

Bit(s)	R/W	Default	Description
31-24	R/W	0	Hist pixel white threshold, larger than this will be counted as white pixel number
23-16	R/W	0	Hist pixel black threshold, less than this will be counted as black pixel number

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Bit(s)	R/W	Default	Description
11	R/W	0	Hist 32bin only mode
10-9	R/W	0	ldim_stts_din_sel, 00: from matrix0 dout, 01: from vsc_dout, 10: from matrix1 dout, 11: form matrix1 din
8	R/W	0	ldim_stts_en
6-5	R/W	0	hist_dnlp_low the real pixels in each bins got by VDIN_DNLP_HISTXX should multiple with 2^(dnlp_low+3)
3-2	R/W	0	hist_din_sel the source used for hist statistics. 2'b00: from MAT0_dout; 2'b01: from vsc_dout; 2'b10: from mat1_dout, 3: mat1_din
1	R/W	0	hist_win_en 1'b0: hist used for full picture; 1'b1: hist used for pixels within hist window
0	R/W	0	hist_spl_en 1'b0: disable hist readback; 1'b1: enable hist readback

Table 8-2915 VDIN1_HIST_H_START_END 0x1331

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist_hstart horizontal start value to define hist window
12-0	R/W	0	hist_hend horizontal end value to define hist window

Table 8-2916 VDIN1_HIST_V_START_END 0x1332

Bit(s)	R/W	Default	Description
28-16	R/W	0	hist_vstart vertical start value to define hist window
12-0	R/W	0	hist_vend vertical end value to define hist window

Table 8-2917 VDIN1_HIST_MAX_MIN 0x1333

Bit(s)	R/W	Default	Description
15-8	R	0	hist_max maximum value
7-0	R	0	hist_min minimum value

Table 8-2918 VDIN1_HIST_SPL_VAL 0x1334

Bit(s)	R/W	Default	Description
31-0	R	0	hist_spl_rd , counts for the total luma value

Table 8-2919 VDIN1_HIST_SPL_PIX_CNT 0x1335

Bit(s)	R/W	Default	Description
21-0	R	0	hist_spl_pixel_count, counts for the total calculated pixels

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Table 8-2920 VDIN1_HIST_CHROMA_SUM 0x1336

Bit(s)	R/W	Default	Description
31-0	R	0	hist_chroma_sum , counts for the total chroma value

//0-255 are splited to 64 bins evenly, and VDIN_DNLP_HISTXX

//are the statistic number of pixels that within each bin.

Table 8-2921 VDIN1_DNLP_HIST00 0x1337

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 2nd bin
15-0	R	0	counts for the 1st bin

Table 8-2922 VDIN1_DNLP_HIST01 0x1338

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 4th bin
15-0	R	0	counts for the 3rd bin

Table 8-2923 VDIN1_DNLP_HIST02 0x1339

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 6th bin
15-0	R	0	counts for the 5th bin

Table 8-2924 VDIN1_DNLP_HIST03 0x133a

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 8th bin
15-0	R	0	counts for the 7th bin

Table 8-2925 VDIN1_DNLP_HIST04 0x133b

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 10th bin
15-0	R	0	counts for the 9th bin

Table 8-2926 VDIN1_DNLP_HIST05 0x133c

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 12th bin
15-0	R	0	counts for the 11th bin

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Table 8-2927 VDIN1_DNLP_HIST06 0x133d

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 14th bin
15-0	R	0	counts for the 13th bin

Table 8-2928 VDIN1_DNLP_HIST07 0x133e

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 16th bin
15-0	R	0	counts for the 15th bin

Table 8-2929 VDIN1_DNLP_HIST08 0x133f

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 18th bin
15-0	R	0	counts for the 17th bin

Table 8-2930 VDIN1_DNLP_HIST09 0x1340

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 20th bin
15-0	R	0	counts for the 19th bin

Table 8-2931 VDIN1_DNLP_HIST10 0x1341

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 22nd bin
15-0	R	0	counts for the 21st bin

Table 8-2932 VDIN1_DNLP_HIST11 0x1342

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 24th bin
15-0	R	0	counts for the 23rd bin

Table 8-2933 VDIN1_DNLP_HIST12 0x1343

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 26th bin
15-0	R	0	counts for the 25th bin

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Table 8-2934 VDIN1_DNLP_HIST13 0x1344

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 28th bin
15-0	R	0	counts for the 27th bin

Table 8-2935 VDIN1_DNLP_HIST14 0x1345

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 30th bin
15-0	R	0	counts for the 29th bin

Table 8-2936 VDIN1_DNLP_HIST15 0x1346

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 32nd bin
15-0	R	0	counts for the 31st bin

Table 8-2937 VDIN1_DNLP_HIST16 0x1347

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 34th bin
15-0	R	0	counts for the 33rd bin

Table 8-2938 VDIN1_DNLP_HIST17 0x1348

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 36th bin
15-0	R	0	counts for the 35th bin

Table 8-2939 VDIN1_DNLP_HIST18 0x1349

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 38th bin
15-0	R	0	counts for the 37th bin

Table 8-2940 VDIN1_DNLP_HIST19 0x134a

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 40th bin
15-0	R	0	counts for the 39th bin

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Table 8-2941 VDIN1_DNLP_HIST20 0x134b

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 42nd bin
15-0	R	0	counts for the 41st bin

Table 8-2942 VDIN1_DNLP_HIST21 0x134c

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 44th bin
15-0	R	0	counts for the 43rd bin

Table 8-2943 VDIN1_DNLP_HIST22 0x134d

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 46th bin
15-0	R	0	counts for the 45th bin

Table 8-2944 VDIN1_DNLP_HIST23 0x134e

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 48th bin
15-0	R	0	counts for the 47th bin

Table 8-2945 VDIN1_DNLP_HIST24 0x134f

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 50th bin
15-0	R	0	counts for the 49th bin

Table 8-2946 VDIN1_DNLP_HIST25 0x1350

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 52nd bin
15-0	R	0	counts for the 51st bin

Table 8-2947 VDIN1_DNLP_HIST26 0x1351

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 54th bin
15-0	R	0	counts for the 53rd bin

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Table 8-2948 VDIN1_DNLP_HIST27 0x1352

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 56th bin
15-0	R	0	counts for the 55th bin

Table 8-2949 VDIN1_DNLP_HIST28 0x1353

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 58th bin
15-0	R	0	counts for the 57th bin

Table 8-2950 VDIN1_DNLP_HIST29 0x1354

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 60th bin
15-0	R	0	counts for the 59th bin

Table 8-2951 VDIN1_DNLP_HIST30 0x1355

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 62nd bin
15-0	R	0	counts for the 61st bin

Table 8-2952 VDIN1_DNLP_HIST31 0x1356

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 64th bin
15-0	R	0	counts for the 63rd bin

Table 8-2953 VDIN1_LDIM_STTS_HIST_REGION_IDX 0x1357

Bit(s)	R/W	Default	Description
31	R	0	local dimming max statistic enable
28	R	0	eol enable
27-25	R	0	vertical line overlap number for max finding
24-22	R	0	horizontal pixel overlap number, 0: 17 pix, 1: 9 pix, 2: 5 pix, 3: 3 pix, 4: 0 pix
20	R	0	1,2,1 low pass filter enable before max/hist statistic
19-16	R	0	region H/V position index, refer to VDIN_LDIM_STTS_HIST_SET_REGION
15	R	0	1: region read index auto increase per read to VDIN_LDIM_STTS_HIST_READ_REGION
6-0	R	0	region read index

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Table 8-2954 VDIN1_LDIM_STTS_HIST_SET_REGION 0x1358

Bit(s)	R/W	Default	Description
28:16	R	0	if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h0: read/write hvstart0 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h1: read/write hend01 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h2: read/write vend01 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h3: read/write hend23 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h4: read/write vend23 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h5: read/write hend45 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h6: read/write vend45 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'd7: read/write hend67 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h8: read/write vend67 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'h9: read/write hend89 if VDIN_LDIM_STTS_HIST_REGION_IDX[19:16] == 5'ha: read/write vend89 //hvstart0, Bit 28:16 row0 vstart, Bit 12:0 col0 hstart //hend01, Bit 28:16 col1 hend, Bit 12:0 col0 hend //vend01, Bit 28:16 row1 vend, Bit 12:0 row0 vend //hend23, Bit 28:16 col3 hend, Bit 12:0 col2 hend //vend23, Bit 28:16 row3 vend, Bit 12:0 row2 vend //hend45, Bit 28:16 col5 hend, Bit 12:0 col4 hend //vend45, Bit 28:16 row5 vend, Bit 12:0 row4 vend //hend67, Bit 28:16 col7 hend, Bit 12:0 col6 hend //vend67, Bit 28:16 row7 vend, Bit 12:0 row6 vend //hend89, Bit 28:16 col9 hend, Bit 12:0 col8 hend //vend89, Bit 28:16 row9 vend, Bit 12:0 row8 vend
12:0	R	0	

Table 8-2955 VDIN1_LDIM_STTS_HIST_READ_REGION 0x1359

Bit(s)	R/W	Default	Description
29:20	R	0	Max_comp2
19:10	R	0	Max_comp1
9:0	R	0	Max_comp0

Table 8-2956 VDIN1_MEAS_CTRL0 0x135a

Bit(s)	R/W	Default	Description
18	R/W	0	reset bit, high active
17	R/W	0	if true, widen hs/vs pulse
16	R/W	0	vsync total counter always accumulating enable
14-12	R/W	0	select hs/vs of video input channel to measure, 0: no selection, 1 vdi1, 2: vid2: 3: vid3, 4:vid4, 5:vdi5, 6:vdi6, 7:vdi7, 8:vdi8, 9:vdi9.
11-4	R/W	0	vsync_span, define how many vsync span need to measure
2-0	R/W	0	meas_hs_index, index to select which HS counter/range

Table 8-2957 VDIN1_MEAS_VS_COUNT_HI 0x135b

Bit(s)	R/W	Default	Description
19-16	R	0	meas_ind_total_count_n, every number of sync_span vsyncs, this count add 1
15-0	R	0	high bit portion of vsync total counter

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Table 8-2958 VDIN1_MEAS_VS_COUNT_LO 0x135c

Bit(s)	R/W	Default	Description
31-0	R	0	low bit portion of vsync total counter

VDIN1_MEAS_HS_RANGE 0x135d

//according to the meas_hs_index in register VDIN_MEAS_CTRL0

//meas_hs_index == 0, first hs range

//meas_hs_index == 1, second hs range

//meas_hs_index == 2, third hs range

//meas_hs_index == 3, fourth hs range

Bit(s)	R/W	Default	Description
28-16	R	0	count_start
12-0	R	0	count_end

VDIN1_MEAS_HS_COUNT 0x135e

//according to the meas_hs_index in register VDIN_MEAS_CTRL0,

//meas_hs_index == 0, first range hs counter,

//meas_hs_index == 1, second range hs

//meas_hs_index == 2, third range hs

//meas_hs_index == 3, fourth range hs

Bit(s)	R/W	Default	Description
23-0	R	0	Hs counter

Table 8-2959 VDIN1_BLKBAR_CTRL1 0x135f

Bit(s)	R/W	Default	Description
8	R/W	0	white_enable
7-0	R/W	0	blkbar_white_level

Table 8-2960 VDIN1_BLKBAR_CTRL0 0x1360

Bit(s)	R/W	Default	Description
31-24	R/W	0	blkbar_black_level threshold to judge a black point
20-8	R/W	0	blkbar_hwidth left and right region width
7-5	R/W	0	blkbar_comp_sel select yin or uin or vin to be the valid input
4	R/W	0	blkbar_sw_statistic_en enable software statistic of each block black points number
3	R/W	0	blkbar_det_en

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Bit(s)	R/W	Default	Description
2-1	R/W	0	blkbar_din_sel
0	R/W	0	Blkbar_det_top_en

Table 8-2961 VDIN1_BLKBAR_H_START_END 0x1361

Bit(s)	R/W	Default	Description
28-16	R/W	0	blkbar_hstart. Left region start
12-0	R/W	0	blkbar_hend. Right region end

Table 8-2962 VDIN1_BLKBAR_V_START_END 0x1362

Bit(s)	R/W	Default	Description
28-16	R/W	0	blkbar_vstart.
12-0	R/W	0	blkbar_vend.

Table 8-2963 VDIN1_BLKBAR_CNT_THRESHOLD 0x1363

Bit(s)	R/W	Default	Description
19-0	R/W	0	blkbar_cnt_threshold. threshold to judge whether a block is totally black

Table 8-2964 VDIN1_BLKBAR_ROW_TH1_TH2 0x1364

Bit(s)	R/W	Default	Description
28-16	R/W	0	blkbar_row_th1. //threshold of the top blackbar
12-0	R/W	0	blkbar_row_th2 //threshold of the bottom blackbar

Table 8-2965 VDIN1_BLKBAR_IND_LEFT_START_END 0x1365

Bit(s)	R/W	Default	Description
28-16	R	0	blkbar_ind_left_start. horizontal start of the left region in the current searching
12-0	R	0	blkbar_ind_left_end. horizontal end of the left region in the current searching

Table 8-2966 VDIN1_BLKBAR_IND_RIGHT_START_END 0x1366

Bit(s)	R/W	Default	Description
28-16	R	0	blkbar_ind_right_start.horizontal start of the right region in the current searching
12-0	R	0	blkbar_ind_right_end. horizontal end of the right region in the current searching

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Table 8-2967 VDIN1_BLKBAR_IND_LEFT1_CNT 0x1367

Bit(s)	R/W	Default	Description
19-0	R	0	blkbar_ind_left1_cnt. Black pixel counter. left part of the left region

Table 8-2968 VDIN1_BLKBAR_IND_LEFT2_CNT 0x1368

Bit(s)	R/W	Default	Description
19-0	R	0	blkbar_ind_left2_cnt. Black pixel counter. right part of the left region

Table 8-2969 VDIN1_BLKBAR_IND_RIGHT1_CNT 0x1369

Bit(s)	R/W	Default	Description
19-0	R	0	blkbar_ind_right1_cnt. Black pixel counter. left part of the right region

Table 8-2970 VDIN1_BLKBAR_IND_RIGHT2_CNT 0x136a

Bit(s)	R/W	Default	Description
19-0	R	0	blkbar_ind_right2_cnt. Black pixel counter. right part of the right region

Table 8-2971 VDIN1_BLKBAR_STATUS0 0x136b

Bit(s)	R/W	Default	Description
29	R	0	blkbar_ind_black_det_done. LEFT/RIGHT Black detection done
28-16	R	0	blkbar_top_pos. Top black bar position
12-0	R	0	blkbar_bot_pos. Bottom black bar position

Table 8-2972 VDIN1_BLKBAR_STATUS1 0x136c

Bit(s)	R/W	Default	Description
28-16	R	0	blkbar_left_pos. Left black bar posiont
12-0	R	0	blkbar_right_pos. Right black bar position

Table 8-2973 VDIN1_WIN_H_START_END 0x136d

Bit(s)	R/W	Default	Description
28-16	R/W	0	input window H start
12-0	R/W	0	input window H end

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Table 8-2974 VDIN1_WIN_V_START_END 0x136e

Bit(s)	R/W	Default	Description
28-16	R/W	0	input window V start
12-0	R/W	0	input window V end

Table 8-2975 VDIN1_ASFIFO_CTRL3 0x136f

Bit(s)	R/W	Default	Description
15	R/W	0	Vdi7 DE enable
14	R/W	0	Vdi7 go field enable
13	R/W	0	Vdi7 go line enable
12	R/W	0	Vdi7 if true, negative active input vsync
11	R/W	0	Vdi7 if true, negative active input hsync
10	R/W	0	Vdi7 vsync soft reset fifo enable
9	R/W	0	Vdi7 overflow status clear
8	R/W	0	Vdi7 asfifo soft reset, level signal
7	R/W	0	Vdi6 DE enable
6	R/W	0	Vdi6 go field enable
5	R/W	0	Vdi6 go line enable
4	R/W	0	Vdi6 if true, negative active input vsync
3	R/W	0	Vdi6 if true, negative active input hsync
2	R/W	0	Vdi6 vsync soft reset fifo enable
1	R/W	0	Vdi6 overflow status clear
0	R/W	0	Vdi6 asfifo soft reset, level signal

Table 8-2976 VDIN1_COM_GCLK_CTRL2 0x1370

Bit(s)	R/W	Default	Description
3-2	R/W	0	Vshrk_clk2 ctrl
1-0	R/W	0	Vshrk_clk1 ctrl

Table 8-2977 VDIN1_VSHRK_CTRL 0x1371

Bit(s)	R/W	Default	Description
27	R/W	0	Vshrk enable
26:25	R/W	0	Vshrk mode, 0: 1/2 shrink, 1: 1/4 shrink, 2: 1/8 shrink
24	R/W	0	Vshrink lpf mode, 1: 0.5,1.5,1.5,0.5 lpf for 1/4 shrink, 0.5,1.5,1.5...for 1/8 shrink
23:0	R/W	0	Vshrink padding dummy data

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Table 8-2978 VDIN1_HIST32 0x1372

Bit(s)	R/W	Default	Description
31:0	R	0	Hist 32 mode, [31:16] for white pixel number, 15:0 for black pixel number

Table 8-2979 VDIN1_COM_STATUS3 0x1373

Bit(s)	R/W	Default	Description
7	R	0	Vdi9 fifo overflow
5:0	R	0	Vdi9 asfifo cnt

Table 8-2980 VDIN1_CRC_CHK 0x13c9

Bit(s)	R/W	Default	Description
0	W	0	vdin crc check start (pulse)

Table 8-2981 VDIN1_RO_CRC 0x13ca

Bit(s)	R/W	Default	Description
31-0	R	0	vdin crc result (lock by go_field)

Table 8-2982 VDIN1_LINE_INT 0x13cb

Bit(s)	R/W	Default	Description
31	R	0	line interrupt flag
30	R	0	write done flag
17	R/W	0	clear line interrupt flag
16	R/W	0	clear write done interrupt flag
15	R/W	0	enable line interrupt
12-0	R/W	0	line interrupt number

Table 8-2983 VI_HIST_CTRL 0x2e00

Bit(s)	R/W	Default	Description
17-16	R/W	0	Spl_sft: the splt are right shift by spl_sft, 0: no shift, 1: right shift by 1. 2,3...
14	R/W	0	Hist_34bin_only, bin 32~63 are not valid, there are 34bins, bin0~bin31, and bin 64 for black pixel, bin 65 for white pixel
13-11	R/W	0	Hist_in_sel: 0: vpp_dout, 1: vpp_vd1_din, 2: vpp_vd2_din, 3: osd1, 4:osd2 5: di pre 6: vdin 7: post blend
10-8	R/W	0	Hist_din_comp_mux: mux of each component, din[9:0],[19:10],[29:20] switches
7-5	R/W	0	Hist_dnlp_low: hist number are shift by (hist_dnlp_low + 3). I.e. Dnlp_low =0, >> 3, dnlp_low=1, >> 4

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Bit(s)	R/W	Default	Description
1	R/W	0	Hist_win_en: hist statistic in a window
0	R/W	0	Luma_hist_spl_en, 1: enable the histogram statistic

Table 8-2984 VDIN1_WRARB_MODE 0x13c0

Bit(s)	R/W	Default	Description
23:16	R/W	0	wrarb_sel : uns, default = 0 , wrarb_sel [0]==0 slave dc0 connect master port0 wrarb_sel[0]==1 slave dc0 connect master port1 wrarb_sel [1]==0 slave dc1 connect master port0 wrarb_sel[1]==1 slave dc1 connect master port1 wrarb_sel [2]==0 slave dc2 connect master port0 wrarb_sel[2]==1 slave dc2 connect master port1 wrarb_sel [3]==0 slave dc3 connect master port0 wrarb_sel[3]==1 slave dc3 connect master port1 wrarb_sel [4]==0 slave dc4 connect master port0 wrarb_sel[4]==1 slave dc3 connect master port1 wrarb_sel [5]==0 slave dc5 connect master port0 wrarb_sel[5]==1 slave dc3 connect master port1
8	R/W	0	wrarb_arb_mode : uns, default = 0 , wrarb_arb_mode[0] master port0 arb way,
1:0	R/W	0	wrarb_gate_clk_ctrl : uns, default = 0 , wrarb_gate_clk_ctrl[1:0] master port0 clk gate control

Table 8-2985 VDIN1_WRARB_REQEN_SLV 0x13c1

Bit(s)	R/W	Default	Description
7:0	R/W	0	wrarb_dc_req_en : unsigned , default = 0 wrarb_dc_req_en [0]: the slv0 req to mst port0 enable, wrarb_dc_req_en [1]: the slv1 req to mst port0 enable, wrarb_dc_req_en [2]: the slv2 req to mst port0 enable, wrarb_dc_req_en [0]: the slv0 req to mst port1 enable, wrarb_dc_req_en [1]: the slv1 req to mst port1 enable, wrarb_dc_req_en [2]: the slv2 req to mst port1 enable,

Table 8-2986 VDIN1_WRARB_WEIGH0_SLV 0x13C2

Bit(s)	R/W	Default	Description
29:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 wrdc_weigh_sxn [0*6+:6]: the slv0 req weigh number wrdc_weigh_sxn [1*6+:6]: the slv1 req weigh number wrdc_weigh_sxn [2*6+:6]: the slv2 req weigh number wrdc_weigh_sxn [3*6+:6]: the slv3 req weigh number wrdc_weigh_sxn [4*6+:6]: the slv4 req weigh number

Table 8-2987 VDIN1_WRARB_WEIGH1_SLV 0x13C3

Bit(s)	R/W	Default	Description
17:0	R/W	0	wrdc_weigh_sxn : unsigned , default = 0 the slv5 req weigh number [0*6+:6],the slv6 req weighnumber [1*6+:6],the slv7 req weighnumber [2*6+:6],

Table 8-2988 VDIN1_RDWR_ARB_STATUS 0x13c4

Bit(s)	R/W	Default	Description
2	R.O	0	ro_wrarb_arb_busy : unsigned , default = 0
1	R/W	0x0	reserve :
0	R.O	0	ro_rdarb_arb_busy : unsigned , default = 0

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Table 8-2989 VDIN1_ARB_DBG_CTRL 0x13c5

Bit(s)	R/W	Default	Description
31:0	R.O	0	ro_det_cmd_ctrl : unsigned , default = 0

Table 8-2990 VDIN1_ARB_DBG_STAT 0x13C6

Bit(s)	R/W	Default	Description
31:0	R.O	0	ro_det_dbg_stat : unsigned , default = 0

Table 8-2991 VI_HIST_H_START_END 0x2e01

Bit(s)	R/W	Default	Description
28-16	R/W	0	Hist_hstart, refer to VI_HIST_CTRL[1]
12-0	R/W	0	Hist_hend

Table 8-2992 VI_HIST_V_START_END 0x2e02

Bit(s)	R/W	Default	Description
28-16	R/W	0	Hist_vstart, refer to VI_HIST_CTRL[1]
12-0	R/W	0	Hist_vend

Table 8-2993 VI_HIST_MAX_MIN 0x2e03

Bit(s)	R/W	Default	Description
15-8	R	0	hist_max maximum value
7-0	R	0	hist_min minimum value

Table 8-2994 VI_HIST_SPL_VAL 0x2e04

Bit(s)	R/W	Default	Description
31-0	R	0	hist_spl_rd , counts for the total luma value

Table 8-2995 VI_HIST_SPL_PIX_CNT 0x2e05

Bit(s)	R/W	Default	Description
21-0	R	0	hist_spl_pixel_count, counts for the total calculated pixels

Table 8-2996 VI_HIST_CHROMA_SUM 0x2e06

Bit(s)	R/W	Default	Description
31-0	R	0	hist_chroma_sum , counts for the total chroma value

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//0-255 are splited to 64 bins evenly, and VDIN_DNLP_HISTXX

//are the statistic number of pixels that within each bin.

Table 8-2997 VI_DNLP_HIST00 0x2e07

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 2nd bin
15-0	R	0	counts for the 1st bin

Table 8-2998 VI_DNLP_HIST01 0x2e08

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 4th bin
15-0	R	0	counts for the 3rd bin

Table 8-2999 VI_DNLP_HIST02 0x2e09

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 6th bin
15-0	R	0	counts for the 5th bin

Table 8-3000 VI_DNLP_HIST03 0x2e0a

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 8th bin
15-0	R	0	counts for the 7th bin

Table 8-3001 VI_DNLP_HIST04 0x2e0b

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 10th bin
15-0	R	0	counts for the 9th bin

Table 8-3002 VI_DNLP_HIST05 0x2e0c

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 12th bin
15-0	R	0	counts for the 11th bin

Table 8-3003 VI_DNLP_HIST06 0x2e0d

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 14th bin
15-0	R	0	counts for the 13th bin

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Table 8-3004 VI_DNLP_HIST07 0x2e0e

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 16th bin
15-0	R	0	counts for the 15th bin

Table 8-3005 VI_DNLP_HIST08 0x2e0f

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 18th bin
15-0	R	0	counts for the 17th bin

Table 8-3006 VI_DNLP_HIST09 0x2e10

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 20th bin
15-0	R	0	counts for the 19th bin

Table 8-3007 VI_DNLP_HIST10 0x2e11

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 22nd bin
15-0	R	0	counts for the 21st bin

Table 8-3008 VI_DNLP_HIST11 0x2e12

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 24th bin
15-0	R	0	counts for the 23rd bin

Table 8-3009 VI_DNLP_HIST12 0x2e13

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 26th bin
15-0	R	0	counts for the 25th bin

Table 8-3010 VI_DNLP_HIST13 0x2e14

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 28th bin
15-0	R	0	counts for the 27th bin

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Table 8-3011 VI_DNLP_HIST14 0x2e15

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 30th bin
15-0	R	0	counts for the 29th bin

Table 8-3012 VI_DNLP_HIST15 0x2e16

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 32nd bin
15-0	R	0	counts for the 31st bin

Table 8-3013 VI_DNLP_HIST16 0x2e17

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 34th bin
15-0	R	0	counts for the 33rd bin

Table 8-3014 VI_DNLP_HIST17 0x2e18

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 36th bin
15-0	R	0	counts for the 35th bin

Table 8-3015 VI_DNLP_HIST18 0x2e19

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 38th bin
15-0	R	0	counts for the 37th bin

Table 8-3016 VI_DNLP_HIST19 0x2e1a

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 40th bin
15-0	R	0	counts for the 39th bin

Table 8-3017 VI_DNLP_HIST20 0x2e1b

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 42nd bin
15-0	R	0	counts for the 41st bin

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Table 8-3018 VI_DNLP_HIST21 0x2e1c

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 44th bin
15-0	R	0	counts for the 43rd bin

Table 8-3019 VI_DNLP_HIST22 0x2e1d

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 46th bin
15-0	R	0	counts for the 45th bin

Table 8-3020 VI_DNLP_HIST23 0x2e1e

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 48th bin
15-0	R	0	counts for the 47th bin

Table 8-3021 VI_DNLP_HIST24 0x2e1f

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 50th bin
15-0	R	0	counts for the 49th bin

Table 8-3022 VI_DNLP_HIST25 0x2e20

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 52nd bin
15-0	R	0	counts for the 51st bin

Table 8-3023 VI_DNLP_HIST26 0x2e21

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 54th bin
15-0	R	0	counts for the 53rd bin

Table 8-3024 VI_DNLP_HIST27 0x2e22

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 56th bin
15-0	R	0	counts for the 55th bin

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Table 8-3025 VI_DNLP_HIST28 0x2e23

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 58th bin
15-0	R	0	counts for the 57th bin

Table 8-3026 VI_DNLP_HIST29 0x2e24

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 60th bin
15-0	R	0	counts for the 59th bin

Table 8-3027 VI_DNLP_HIST30 0x2e25

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 62nd bin
15-0	R	0	counts for the 61st bin

Table 8-3028 VI_DNLP_HIST31 0x2e26

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 64th bin
15-0	R	0	counts for the 63rd bin

Table 8-3029 VI_DNLP_HIST31 0x2e27

Bit(s)	R/W	Default	Description
31-16	R	0	counts for the 66th bin, for white pix
15-0	R	0	counts for the 65th bin, for black pix

Table 8-3030 VI_HIST_PIC_SIZE 0x2e28

Bit(s)	R/W	Default	Description
28-16	R/W	0	Hist_pic_height
12-0	R/W	0	Hist_pic_width

Table 8-3031 VI_HIST_GCLK_CTRL 0x2e2a

Bit(s)	R/W	Default	Description
5-4	R/W	0	Gated clock control of hist_clk
3-2	R/W	0	Gated clock control of clk0
1-0	R/W	0	Gated clock control of hist register clock

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8.2.3.46 Osd_mali_afbcd Registers

Table 8-3032 VPU_MAFBC_BLOCK_ID 0x3a00

Bit(s)	R/W	Default	Description
31:16	R.O	0x0	PRODUCT_ID : // Contains a product-specific value
15:12	R.O	0x0	VERSION_MAJOR : // Major release number of the AFBC decoder. This is the R part of an RnPn release number.
11:4	R.O	0x0	VERSION_MINOR : // Minor release number of the AFBC decoder. This is the P part of an RnPn release number.
3:0	R.O	0x0	VERSION_STATUS : // The version status of the AFBC decoder release. Starts at 0 and increases by one for each release.

Table 8-3033 VPU_MAFBC_IRQ_RAW_STATUS 0x3a01

Bit(s)	R/W	Default	Description
5	R.O	0x0	SECURE_ID_ERROR : // Activated when an invalid TrustZone Media Protection (TZMP) transaction is detected on the TZMP1 or TZMP2 data fields.
4	R.O	0x0	AXI_ERROR : // Activated when an AXI error is detected.
3	R.O	0x0	DETILING_ERROR : // Activated when a detiling error occurs.
2	R.O	0x0	DECODE_ERROR : // Activated when decoder core indicates a decoder error.
1	R.O	0x0	CONFIGURATION_SWAPPED : // Activated when configuration has been swapped from shadow registers to configuration registers.
0	R.O	0x0	SURFACES_COMPLETED : // Activated when all enabled surfaces have completed and have been fully read out. In continuous mode, this interrupt is triggered each time all surfaces have been read out.

Table 8-3034 VPU_MAFBC_IRQ_CLEAR 0x3a02

Bit(s)	R/W	Default	Description
5	R/W	0	SECURE_ID_ERROR : // unsigned , default = 0,Writing a 1 to this bit clears the corresponding bit in the IRQ_RAW_STATUS register.
4	R/W	0	AXI_ERROR : // unsigned , default = 0,Writing a 1 to this bit clears the corresponding bit in the IRQ_RAW_STATUS register.
3	R/W	0	DETILING_ERROR : // unsigned , default = 0,Writing a 1 to this bit clears the corresponding bit in the IRQ_RAW_STATUS register.
2	R/W	0	DECODE_ERROR : // unsigned , default = 0,Writing a 1 to this bit clears the corresponding bit in the IRQ_RAW_STATUS register.
1	R/W	0	CONFIGURATION_SWAPPED : // unsigned , default = 0,Writing a 1 to this bit clears the corresponding bit in the IRQ_RAW_STATUS register.
0	R/W	0	SURFACES_COMPLETED : // unsigned , default = 0,Writing a 1 to this bit clears the corresponding bit in the IRQ_RAW_STATUS register.

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Table 8-3035 VPU_MAFBC_IRQ_MASK 0x3a03

Bit(s)	R/W	Default	Description
5	R/W	0	SECURE_ID_ERROR : // unsigned , default = 0,When this is set to 1, the corresponding IRQ_RAW_STATUS interrupt is enabled. When this is set to zero, the corresponding IRQ_RAW_STATUS interrupt is disabled.
4	R/W	0	AXI_ERROR : // unsigned , default = 0,When this is set to 1, the corresponding IRQ_RAW_STATUS interrupt is enabled. When this is set to zero, the corresponding IRQ_RAW_STATUS interrupt is disabled.
3	R/W	0	DETILING_ERROR : // unsigned , default = 0,When this is set to 1, the corresponding IRQ_RAW_STATUS interrupt is enabled. When this is set to zero, the corresponding IRQ_RAW_STATUS interrupt is disabled.
2	R/W	0	DECODE_ERROR : // unsigned , default = 0, When this is set to 1, the corresponding IRQ_RAW_STATUS interrupt is enabled. When this is set to zero, the corresponding IRQ_RAW_STATUS interrupt is disabled.
1	R/W	0	CONFIGURATION_SWAPPED : // unsigned , default = 0, When this is set to 1, the corresponding IRQ_RAW_STATUS interrupt is enabled. When this is set to zero, the corresponding IRQ_RAW_STATUS interrupt is disabled.
0	R/W	0	SURFACES_COMPLETED : // unsigned , default = 0, When this is set to 1, the corresponding IRQ_RAW_STATUS interrupt is enabled. When this is set to zero, the corresponding IRQ_RAW_STATUS interrupt is disabled.

Table 8-3036 VPU_MAFBC_IRQ_STATUS 0x3a04

Bit(s)	R/W	Default	Description
5	R.O	0x0	SECURE_ID_ERROR : //When this is set to 1, it asserts the corresponding external interrupt. This signals an interruptrequest to the application processor.
4	R.O	0x0	AXI_ERROR : //When this is set to 1, it asserts the corresponding external interrupt. This signals an interruptrequest to the application processor.
3	R.O	0x0	DETILING_ERROR : //When this is set to 1, it asserts the corresponding external interrupt. This signals an interruptrequest to the application processor.
2	R.O	0x0	DECODE_ERROR : //When this is set to 1, it asserts the corresponding external interrupt. This signals an interruptrequest to the application processor.
1	R.O	0x0	CONFIGURATION_SWAPPED : //When this is set to 1, it asserts the corresponding external interrupt. This signals an interruptrequest to the application processor.
0	R.O	0x0	SURFACES_COMPLETED : //When this is set to 1, it asserts the corresponding external interrupt. This signals an interruptrequest to the application processor.

Table 8-3037 VPU_MAFBC_COMMAND 0x3a05

Bit(s)	R/W	Default	Description
1	R/W	0x0	PENDING_SWAP : //Performs a swap of shadow registers when the current decode operations are completed. After a swap,decoding of the new surfaces starts.
0	R/W	0x0	DIRECT_SWAP : //Performs a swap of shadow registers immediately.Current decode operations are terminated before completion . After termination, decoding of the new surfaces is started

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Table 8-3038 VPU_MAFBC_STATUS 0x3a06

Bit(s)	R/W	Default	Description
2	R.O	0x0	ERROR : //When this is 1, it indicates that the AFBC decoder is in an unrecoverable state and must be reset using RESETn to continue operation. This is flagged after a DETILING_ERROR.
1	R.O	0x0	SWAPPING : //When this is 1, the AFBC decoder is swapping surface configurations.
0	R.O	0x0	ACTIVE : //When this is 1, the AFBC decoder is decoding surfaces.

Table 8-3039 VPU_MAFBC_SURFACE_CFG 0x3a07

Bit(s)	R/W	Default	Description
16	R/W	0	CONTINUOUS_DECODING_ENABLE : //unsigned , default = 0,Enables continuous decoding of surfaces when it is set to 1. This bit describes what the decoder does when one set of surfaces has been completed. If this is 1, the decoder immediately begins decoding again. In this case, the configuration for the next surface depends on whether the software has performed a swap operation. If no swap is used, the same configuration is used. If this is 0, the decoder waits for a software input.
3	R/W	0	S3_ENABLE : //unsigned , default = 0,When this is 1, surface 3 enabled.
2	R/W	0	S2_ENABLE : //unsigned , default = 0,When this is 1, surface 2 enabled.
1	R/W	0	S1_ENABLE : //unsigned , default = 0,When this is 1, surface 1 enabled.
0	R/W	0	S0_ENABLE : //unsigned , default = 0,When this is 1, surface 0 enabled.

Table 8-3040 VPU_MAFBC_AXI_CFG 0x3a08

Bit(s)	R/W	Default	Description
7:4	R/W	0	CACHE : //unsigned , default = 0,This value is driven on the ARCACHEM signal.
3:0	R/W	0	QOS : //unsigned , default = 0,This value is driven on the ARQOSM signal.

Table 8-3041 VPU_MAFBC_HEADER_BUF_ADDR_LOW_S0 0x3a10

Bit(s)	R/W	Default	Description
31:0	R/W	0	HEADER_BUF_ADDR : [31:0] //unsigned , default = 0,Contains bits [31:0] of the header buffer address,Bits [5:0] must be set to 0 for alignment requirements.

Table 8-3042 VPU_MAFBC_HEADER_BUF_ADDR_HIGH_S0 0x3a11

Bit(s)	R/W	Default	Description
15:0	R/W	0	HEADER_BUF_ADDR : [47:32] //unsigned , default = 0,Contains the upper 16 bits of the header buffer address

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Table 8-3043 VPU_MAFBC_FORMAT_SPECIFIER_S0 0x3a12

Bit(s)	R/W	Default	Description
19	R/W	0	PAYLOAD_LIMIT_EN : // unsigned , default = 0,Enable payload read address check against the min and max payload address boundaries: 0 Disable 1 Enable. This means that out-of-bound reads output the color black.
18	R/W	0	TILED_HEADER_EN : // unsigned , default = 0,Enables tiled header mode:0 Disable tiled header 1 Enable tiled header
17:16	R/W	0	SUPER_BLOCK_ASPECT : // unsigned , default = 0,Selects superblock aspect ratio 00:16x16 pixels 01:32x8 pixels other:reserved
9	R/W	0	BLOCK_SPLIT : //unsigned , default = 0,Enables block split mode: 0 Block split mode off. 1 Block split mode on.
8	R/W	0	YUV_TRANSFORM : //unsigned , default = 0,Enables the internal YUV transform stage: 0 Internal YUV transform off. 1 Internal YUV transform on.
3:0	R/W	0	PIXEL_FORMAT : //unsigned , default = 0,Contains the pixel format configuration.

Table 8-3044 VPU_MAFBC_BUFFER_WIDTH_S0 0x3a13

Bit(s)	R/W	Default	Description
13:0	R/W	0	BUFFER_WIDTH : //unsigned , default = 0,AFBC buffer width in pixels

Table 8-3045 VPU_MAFBC_BUFFER_HEIGHT_S0 0x3a14

Bit(s)	R/W	Default	Description
13:0	R/W	0	BUFFER_HEIGHT : //unsigned , default = 0,AFBC buffer height in pixels

Table 8-3046 VPU_MAFBC_BOUNDING_BOX_X_START_S0 0x3a15

Bit(s)	R/W	Default	Description
12:0	R/W	0	BUFFER_X_START : //unsigned , default = 0,The AFBC buffer bounding box minimum x value. Given in pixels.

Table 8-3047 VPU_MAFBC_BOUNDING_BOX_X_END_S0 0x3a16

Bit(s)	R/W	Default	Description
12:0	R/W	0	BUFFER_X_START : //unsigned , default = 0,The AFBC buffer bounding box maximum x value. Given in pixels.

Table 8-3048 VPU_MAFBC_BOUNDING_BOX_Y_START_S0 0x3a17

Bit(s)	R/W	Default	Description
12:0	R/W	0	BUFFER_Y_START : //unsigned , default = 0,The AFBC buffer bounding box minimum y value. Given in pixels.

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Table 8-3049 VPU_MAFBC_BOUNDING_BOX_Y_END_S0 0x3a18

Bit(s)	R/W	Default	Description
12:0	R/W	0	BUFFER_Y_START : //unsigned , default = 0,The AFBC buffer bounding box maximum y value. Given in pixels.

Table 8-3050 VPU_MAFBC_OUTPUT_BUF_ADDR_LOW_S0 0x3a19

Bit(s)	R/W	Default	Description
31:0	R/W	0	OUTPUT_BUF_ADDR : [31:0] //unsigned , default = 0,Contains bits [31:0] of the output buffer address.Bits [6:0] must be set to 0 for alignment requirements. The address, frame size, and pixel format allocate area in the detiler SRAM for the current surface . You must ensure that the allocated area does not exceed the available SRAM, or overlap with other surfaces or planes that are defined in the output buffer.

Table 8-3051 VPU_MAFBC_OUTPUT_BUF_ADDR_HIGH_S0 0x3a1a

Bit(s)	R/W	Default	Description
15:0	R/W	0x0	OUTPUT_BUF_ADDR : [47:32] //Contains the upper 16 bits of the output buffer address

Table 8-3052 VPU_MAFBC_OUTPUT_BUF_STRIDE_S0 0x3a1b

Bit(s)	R/W	Default	Description
15:0	R/W	0	OUTPUT_BUF_STRIDE : [15:0] //unsigned , default = 0,Contains bits [15:0] of the output buffer stride.Bits [6:0] must be set to 0 for alignment requirements. The maximum permitted buffer stride is 8192 pixels wide. The byte size depends on the pixel format for the surface.

Table 8-3053 VPU_MAFBC_PREFETCH_CFG_S0 0x3a1c

Bit(s)	R/W	Default	Description
1	R/W	0	PREFETCH_READ_DIRECTION_Y : // unsigned , default = 0,Defines the pre-fetch read direction in Y: 0 Top to bottom 1 Bottom to top
0	R/W	0	PREFETCH_READ_DIRECTION_X : //unsigned , default = 0,Defines the pre-fetch read direction in X 0 Left to righ 1 Right to left

Table 8-3054 VPU_MAFBC_PAYLOAD_MIN_LOW_S0 0x3a1d

Bit(s)	R/W	Default	Description
31:0	R/W	0	PAYLOAD_MIN_LOW : //unsigned , default = 0, Indicates the lower 32 bits of the AFBC payload buffer minimum address.

Table 8-3055 VPU_MAFBC_PAYLOAD_MIN_HIGH_S0 0x3a1e

Bit(s)	R/W	Default	Description
15:0	R/W	0	PAYLOAD_MIN_HIGH : //unsigned , default = 0, Indicates the higher 16 bits of the AFBC payload buffer minimum address.

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Table 8-3056 VPU_MAFBC_PAYLOAD_MAX_LOW_S0 0x3a1f

Bit(s)	R/W	Default	Description
31:0	R/W	0x0	PAYLOAD_MIN_LOW : //unsigned , ndicates the lower 32 bits of the AFBC payload buffer maximum address.

Table 8-3057 VPU_MAFBC_PAYLOAD_MAX_HIGH_S0 0x3a20

Bit(s)	R/W	Default	Description
15:0	R/W	0	PAYLOAD_MIN_HIGH : //unsigned , default = 0, Indicates the higher 16 bits of the AFBC payload buffer maximum address.

VPU_MAFBC_HEADER_BUF_ADDR_LOW_S1 0x3a30

See VPU_MAFBC_HEADER_BUF_ADDR_LOW_S0

VPU_MAFBC_HEADER_BUF_ADDR_HIGH_S1 0x3a31

See VPU_MAFBC_HEADER_BUF_ADDR_HIGH_S0

VPU_MAFBC_FORMAT_SPECIFIER_S1 0x3a32

See VPU_MAFBC_FORMAT_SPECIFIER_S0

VPU_MAFBC_BUFFER_WIDTH_S1 0x3a33

See VPU_MAFBC_BUFFER_WIDTH_S0

VPU_MAFBC_BUFFER_HEIGHT_S1 0x3a34

See VPU_MAFBC_BUFFER_HEIGHT_S0

VPU_MAFBC_BOUNDING_BOX_X_START_S1 0x3a35

See VPU_MAFBC_BOUNDING_BOX_X_START_S0

VPU_MAFBC_BOUNDING_BOX_X_END_S1 0x3a36

See VPU_MAFBC_BOUNDING_BOX_X_END_S0

VPU_MAFBC_BOUNDING_BOX_Y_START_S1 0x3a37

See VPU_MAFBC_BOUNDING_BOX_Y_START_S0

VPU_MAFBC_BOUNDING_BOX_Y_END_S1 0x3a38

See VPU_MAFBC_BOUNDING_BOX_Y_END_S0

VPU_MAFBC_OUTPUT_BUF_ADDR_LOW_S1 0x3a39

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See VPU_MAFBC_OUTPUT_BUF_ADDR_LOW_S0

VPU_MAFBC_OUTPUT_BUF_ADDR_HIGH_S1 0x3a3a
See VPU_MAFBC_OUTPUT_BUF_ADDR_HIGH_S0

VPU_MAFBC_OUTPUT_BUF_STRIDE_S1 0x3a3b
See VPU_MAFBC_OUTPUT_BUF_STRIDE_S0

VPU_MAFBC_PREFETCH_CFG_S1 0x3a3c
See VPU_MAFBC_PREFETCH_CFG_S0

VPU_MAFBC_PAYLOAD_MIN_LOW_S1 0x3a3d
See VPU_MAFBC_PAYLOAD_MIN_LOW_S0

VPU_MAFBC_PAYLOAD_MIN_HIGH_S1 0x3a3e
See VPU_MAFBC_PAYLOAD_MIN_HIGH_S0

VPU_MAFBC_PAYLOAD_MAX_LOW_S1 0x3a3f
See VPU_MAFBC_PAYLOAD_MAX_LOW_S0

VPU_MAFBC_PAYLOAD_MAX_HIGH_S1 0x3a40
See VPU_MAFBC_PAYLOAD_MAX_HIGH_S0

VPU_MAFBC_HEADER_BUF_ADDR_LOW_S2 0x3a50
See VPU_MAFBC_HEADER_BUF_ADDR_LOW_S0

VPU_MAFBC_HEADER_BUF_ADDR_HIGH_S2 0x3a51
See VPU_MAFBC_HEADER_BUF_ADDR_HIGH_S0

VPU_MAFBC_FORMAT_SPECIFIER_S2 0x3a52
See VPU_MAFBC_FORMAT_SPECIFIER_S0

VPU_MAFBC_BUFFER_WIDTH_S2 0x3a53
See VPU_MAFBC_BUFFER_WIDTH_S0

VPU_MAFBC_BUFFER_HEIGHT_S2 0x3a54
See VPU_MAFBC_BUFFER_HEIGHT_S0

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VPU_MAFBC_BOUNDING_BOX_X_START_S2 0x3a55
See VPU_MAFBC_BOUNDING_BOX_X_START_S0

VPU_MAFBC_BOUNDING_BOX_X_END_S2 0x3a56
See VPU_MAFBC_BOUNDING_BOX_X_END_S0

VPU_MAFBC_BOUNDING_BOX_Y_START_S2 0x3a57
See VPU_MAFBC_BOUNDING_BOX_Y_START_S0

VPU_MAFBC_BOUNDING_BOX_Y_END_S2 0x3a58
See VPU_MAFBC_BOUNDING_BOX_Y_END_S0

VPU_MAFBC_OUTPUT_BUF_ADDR_LOW_S2 0x3a59
See VPU_MAFBC_OUTPUT_BUF_ADDR_LOW_S0

VPU_MAFBC_OUTPUT_BUF_ADDR_HIGH_S2 0x3a5a
See VPU_MAFBC_OUTPUT_BUF_ADDR_HIGH_S0

VPU_MAFBC_OUTPUT_BUF_STRIDE_S2 0x3a5b
See VPU_MAFBC_OUTPUT_BUF_STRIDE_S0

VPU_MAFBC_PREFETCH_CFG_S2 0x3a5c
See VPU_MAFBC_PREFETCH_CFG_S0

VPU_MAFBC_PAYLOAD_MIN_LOW_S2 0x3a5d
See VPU_MAFBC_PAYLOAD_MIN_LOW_S0

VPU_MAFBC_PAYLOAD_MIN_HIGH_S2 0x3a5e
See VPU_MAFBC_PAYLOAD_MIN_HIGH_S0

VPU_MAFBC_PAYLOAD_MAX_LOW_S2 0x3a5f
See VPU_MAFBC_PAYLOAD_MAX_LOW_S0

VPU_MAFBC_PAYLOAD_MAX_HIGH_S2 0x3a60
See VPU_MAFBC_PAYLOAD_MAX_HIGH_S0

VPU_MAFBC_HEADER_BUF_ADDR_LOW_S3 0x3a70
See VPU_MAFBC_HEADER_BUF_ADDR_LOW_S0

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VPU_MAFBC_HEADER_BUF_ADDR_HIGH_S3 0x3a71
See VPU_MAFBC_HEADER_BUF_ADDR_HIGH_S0

VPU_MAFBC_FORMAT_SPECIFIER_S3 0x3a72
See VPU_MAFBC_FORMAT_SPECIFIER_S0

VPU_MAFBC_BUFFER_WIDTH_S3 0x3a73
See VPU_MAFBC_BUFFER_WIDTH_S0

VPU_MAFBC_BUFFER_HEIGHT_S3 0x3a74
See VPU_MAFBC_BUFFER_HEIGHT_S0

VPU_MAFBC_BOUNDING_BOX_X_START_S3 0x3a75
See VPU_MAFBC_BOUNDING_BOX_X_START_S0

VPU_MAFBC_BOUNDING_BOX_X_END_S3 0x3a76
See VPU_MAFBC_BOUNDING_BOX_X_END_S0

VPU_MAFBC_BOUNDING_BOX_Y_START_S3 0x3a77
See VPU_MAFBC_BOUNDING_BOX_Y_START_S0

VPU_MAFBC_BOUNDING_BOX_Y_END_S3 0x3a78
See VPU_MAFBC_BOUNDING_BOX_Y_END_S0

VPU_MAFBC_OUTPUT_BUF_ADDR_LOW_S3 0x3a79
See VPU_MAFBC_OUTPUT_BUF_ADDR_LOW_S0

VPU_MAFBC_OUTPUT_BUF_ADDR_HIGH_S3 0x3a7a
See VPU_MAFBC_OUTPUT_BUF_ADDR_HIGH_S0

VPU_MAFBC_OUTPUT_BUF_STRIDE_S3 0x3a7b
See VPU_MAFBC_OUTPUT_BUF_STRIDE_S0

VPU_MAFBC_PREFETCH_CFG_S3 0x3a7c
See VPU_MAFBC_PREFETCH_CFG_S0

VPU_MAFBC_PAYLOAD_MIN_LOW_S3 0x3a7d

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See VPU_MAFBC_PAYLOAD_MIN_LOW_S0

VPU_MAFBC_PAYLOAD_MIN_HIGH_S3 0x3a7e
See VPU_MAFBC_PAYLOAD_MIN_HIGH_S0

VPU_MAFBC_PAYLOAD_MAX_LOW_S3 0x3a7f
See VPU_MAFBC_PAYLOAD_MAX_LOW_S0

VPU_MAFBC_PAYLOAD_MAX_HIGH_S3 0x3a80
See VPU_MAFBC_PAYLOAD_MAX_HIGH_S0

8.2.3.47 Primesl Registers

Table 8-3058 PRIMESL_LUTC_ADDR_PORT 0x3980

Bit(s)	R/W	Default	Description
8: 0	R/W	0	lutc_addr : // unsigned , default = 0 = 'h0,

Table 8-3059 PRIMESL_LUTC_DATA_PORT 0x3981

Bit(s)	R/W	Default	Description
10: 0	R/W	0	lutc_data : // unsigned , default = 0 = 'h0,

Table 8-3060 PRIMESL_LUTP_ADDR_PORT 0x3982

Bit(s)	R/W	Default	Description
8: 0	R/W	0	lutp_addr : // unsigned , default = 0 = 'h0,

Table 8-3061 PRIMESL_LUTP_DATA_PORT 0x3983

Bit(s)	R/W	Default	Description
13: 0	R/W	0	lutp_data : // unsigned , default = 0 = 'h0,

Table 8-3062 PRIMESL_LUTD_ADDR_PORT 0x3984

Bit(s)	R/W	Default	Description
8: 0	R/W	0	lutd_data : // unsigned , default = 0 = 'h0,

Table 8-3063 PRIMESL_LUTD_DATA_PORT 0x3985

Bit(s)	R/W	Default	Description
11: 0	R/W	0	lutd_data : // unsigned , default = 0 = 'h0,

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Table 8-3064 PRIMESL_CTRL0 0x3990

Bit(s)	R/W	Default	Description
29	R/W	0	legacy_mode_en : // unsigned , default = 0 = 'h0
28	R/W	1	clip_en : // unsigned , default = 1 = 'h0
26:16	R/W	0	inv_chroma_ratio : // unsigned , default = 0 = 'h0,
14: 4	R/W	0	inv_y_ratio : // unsigned , default = 0 = 'h0,
3	R/W	0	reg_gclk_ctrl : // unsigned , default = 0 = 'h0,
2: 1	R/W	0	gclk_ctrl : // unsigned , default = 0 = 'h0,
0	R/W	1	primesl_en : // unsigned , default = 1 = 'h0,

Table 8-3065 PRIMESL_CTRL1 0x3991

Bit(s)	R/W	Default	Description
25:16	R/W	0	l_headroom : // unsigned , default = 0 = 'h0,
9: 0	R/W	0	footroom : // unsigned , default = 0 = 'h0,

Table 8-3066 PRIMESL_CTRL2 0x3992

Bit(s)	R/W	Default	Description
9: 0	R/W	0	c_headroom : // unsigned , default = 0 = 'h0,

Table 8-3067 PRIMESL_CTRL3 0x3993

Bit(s)	R/W	Default	Description
29:16	R/W	0	mub : // unsigned , default = 0 = 'h0,
13: 0	R/W	0	mua : // unsigned , default = 0 = 'h0,

Table 8-3068 PRIMESL_CTRL4 0x3994

Bit(s)	R/W	Default	Description
25:16	R/W	0	oct_7_1 : // signed , default = 0 = 'h0,
9: 0	R/W	0	oct_7_0 : // signed , default = 0 = 'h0,

Table 8-3069 PRIMESL_CTRL5 0x3995

Bit(s)	R/W	Default	Description
25:16	R/W	0	oct_7_3 : // signed , default = 0 = 'h0,
9: 0	R/W	0	oct_7_2 : // signed , default = 0 = 'h0,

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Table 8-3070 PRIMESL_CTRL6 0x3996

Bit(s)	R/W	Default	Description
25:16	R/W		oct_7_5 : // signed , default = 0 = 'h0,
9: 0	R/W		oct_7_4 : // signed , default = 0 = 'h0,

Table 8-3071 PRIMESL_CTRL7 0x3997

Bit(s)	R/W	Default	Description
9: 0	R/W	0	oct_7_6 : // signed , default = 0 = 'h0,

Table 8-3072 PRIMESL_CTRL8 0x3998

Bit(s)	R/W	Default	Description
28:16	R/W	0	d_lut_threshold_3_1 : // unsigned , default = 0 = 'h0,
12: 0	R/W	0	d_lut_threshold_3_0 : // unsigned , default = 0 = 'h0,

Table 8-3073 PRIMESL_CTRL9 0x3999

Bit(s)	R/W	Default	Description
12: 0	R/W	0	d_lut_threshold_3_2 : // unsigned , default = 0 = 'h0,

Table 8-3074 PRIMESL_CTRL10 0x399a

Bit(s)	R/W	Default	Description
15:12	R/W	0	d_lut_step_4_3 : // unsigned , default = 0 = 'h0,
11: 8	R/W	0	d_lut_step_4_2 : // unsigned , default = 0 = 'h0,
7: 4	R/W	0	d_lut_step_4_1 : // unsigned , default = 0 = 'h0,
3: 0	R/W	0	d_lut_step_4_0 : // unsigned , default = 0 = 'h0,

Table 8-3075 PRIMESL_CTRL11 0x399b

Bit(s)	R/W	Default	Description
28:16	R/W	0	rgb2yuv_9_0 : // signed , default = 0 = 'h0,
12: 0	R/W	0	rgb2yuv_9_1 : // signed , default = 0 = 'h0,

Table 8-3076 PRIMESL_CTRL12 0x399c

Bit(s)	R/W	Default	Description
28:16	R/W	0	rgb2yuv_9_2 : // signed , default = 0 = 'h0,
12: 0	R/W	0	rgb2yuv_9_3 : // signed , default = 0 = 'h0,

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Table 8-3077 PRIMESL_CTRL13 0x399d

Bit(s)	R/W	Default	Description
28:16	R/W	0	rgb2yuv_9_4 : // signed , default = 0 = 'h0,
12: 0	R/W	0	rgb2yuv_9_5 : // signed , default = 0 = 'h0,

Table 8-3078 PRIMESL_CTRL14 0x399e

Bit(s)	R/W	Default	Description
28:16	R/W	0	rgb2yuv_9_6 : // signed , default = 0 = 'h0,
12: 0	R/W	0	rgb2yuv_9_7 : // signed , default = 0 = 'h0,

Table 8-3079 PRIMESL_CTRL15 0x399f

Bit(s)	R/W	Default	Description
12: 0	R/W	0	rgb2yuv_9_8 : // signed , default = 0 = 'h0,

Table 8-3080 PRIMESL_CTRL16 0x39e0

Bit(s)	R/W	Default	Description
31	R/W	0	reserved
30	R/W	0	byp_s_lut://unsigned
29	R/W	0	byp_d_lut://unsigned
28	R/W	0	byp_mat://unsigned
27	R/W	0	rgb_swap://unsigned
26:24	R/W	0	uv_shift://unsigned
23:22	R/W	0	rgb_shift://unsigned
21:20	R/W	3	rgb_clip://unsigned
19:18	R/W	/	reserved
17:16	R/W	0	rgb_s://unsigned
15:14	R/W	/	reserved
13:0	R/W	1024	reg_s://signed

Table 8-3081 PRIMESL_OMAT_OFFSET0 0x39e1

Bit(s)	R/W	Default	Description
31:28	R/W	/	reserved
27:16	R/W	512	pre_offset0// unsigned , default = 0 = 512,
15:12	R/W	/	reserved
11: 0	R/W	512	pre_offset1: // unsigned , default = 0 = 512,

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Table 8-3082 PRIMESL_OMAT_OFFSET1 0x39e2

Bit(s)	R/W	Default	Description
31:28	R/W	/	reserved
27:16	R/W	512	pre_offset2// unsigned , default = 0 = 512,
15:12	R/W	/	reserved
11: 0	R/W	256	Offset0: // unsigned , default = 0 = 512,

Table 8-3083 PRIMESL_OMAT_OFFSET2 0x39e3

Bit(s)	R/W	Default	Description
31:28	R/W	/	reserved
27:16	R/W	2048	offset1// unsigned , default = 0 = 512,
15:12	R/W	/	reserved
11: 0	R/W	2048	offset2: // unsigned , default = 0 = 512,

8.2.3.48 VADJ1/RGB_CONBRI Registers

Table 8-3084 VPP_VADJ1_MISC 0x3280

Bit(s)	R/W	Default	Description
5:4	R/W	0	Gate clk ctrl for vadj1
3	/	/	/
2	R/W	0	Sync enable 1:vadj1 regs work when go_field come 1:vadj1 regs directly
1	R/W	0	minus black level enable for vadj1
0	R/W	0	Module enable

Table 8-3085 VPP_VADJ1_BLACK_VAL 0X3281

Bit(s)	R/W	Default	Description
26:16	R/W	13'h200	Black_luma
10:0	R/W	13'h40	Black_chroma

Table 8-3086 VPP_VADJ1_Y 0x3282

Bit(s)	R/W	Default	Description
18-8	R/W	0	brightness, signed value
7-0	R/W	0x1d0	contrast, unsigned value, contrast from 0 <= contrast <2

cb' = cb*ma + cr*mb

cr' = cb*mc + cr*md

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Table 8-3087 VPP_VADJ1_MA_MB 0x3283

Bit(s)	R/W	Default	Description
25-16	R/W	0x100	MA, signed value, -2 < MA < 2
9-0	R/W	0	MB, signed value ,-2 < MB < 2

Table 8-3088 VPP_VADJ1_MC_MD 0x3284

Bit(s)	R/W	Default	Description
25-16	R/W	0	MC, signed value, -2 < MC < 2
9-0	R/W	0x100	MD, signed value ,-2 < MD < 2

Table 8-3089 VPP_VADJ1_CURV_0 0x3285

Bit(s)	R/W	Default	Description
31:24	R/W	0	vadj1_softcon_curv0_ci, u8
23:12	R/W	0	vadj1_softcon_curv0_b, u12
11:0	R/W	0	vadj1_softcon_curv0_a, s12

Table 8-3090 VPP_VADJ1_CURV_1 0x3286

Bit(s)	R/W	Default	Description
31:13			Reserved
12:4	R/W	0	vadj1_softcon_curv0_g, s9
3			Reserved
2:0	R/W	0	vadj1_softcon_curv0_cs, u3

Table 8-3091 VPP_VADJ1_CURV_2 0x3287

Bit(s)	R/W	Default	Description
31:24	R/W	0	vadj1_softcon_curv1_ci, u8
23:12	R/W	0	vadj1_softcon_curv1_b, u12
11:0	R/W	0	vadj1_softcon_curv1_a, s12

Table 8-3092 VPP_VADJ1_CURV_3 0x3288

Bit(s)	R/W	Default	Description
31:13			Reserved
12:4	R/W	0	vadj1_softcon_curv1_g, s9
3			Reserved
2:0	R/W	0	vadj1_softcon_curv1_cs, u3

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Table 8-3093 VPP_VD1_RGB_CTRST 0x3289

Bit(s)	R/W	Default	Description
31:28			Reserved
27:16	R/W	1024	Vd1_rgb_ctrst: contrast for rgb, normalized 1024 as '1.0'
15			Reserved
14:2	R/W	64	Vd1_rgb_ctrst_blklvl: contrast black level to be subtract before and add back after the contrast gain operation
1	R/W	0	Vd1_rgbbst_en: 1 to enable the RGB_BST
0	R/W	1	Vd1_rgb_ctrst_prt: enable signal to protect saturation in rgb (no clipping) during contrast adjustment

Table 8-3094 VPP_VD1_RGB_BRGHT 0x328A

Bit(s)	R/W	Default	Description
31:14			Reserved
13:2	R/W	0	Vd1_rgb_brght: brightness level in rgb domain
1	R/W	1	Vd1_rgb_brght_prt: enable signal to protect saturation in rgb (no clipping) during brightness adjustment
0	R/W	0	Vd1_rgb_ctrst_dlut_x2: Enable signal to do x2 to the dlut cells before subtracting from the normalized gain_max; 0:x1 1:x2

Table 8-3095 VPP_VD1_RGB_DLUT_0_3 0x328B

Bit(s)	R/W	Default	Description
31:24	R/W	255	Vd1_rgbbst_dlut0: Differential gain to normalized gain_max to customized protection curve. e.g. [255 205 171 147 128 113 102 93 85 78 73 68] for protection of not boost for pixels larger than 240
23:16	R/W	205	Vd1_rgbbst_dlut1: same as Vd1_rgbbst_dlut0
15:8	R/W	171	Vd1_rgbbst_dlut2: same as Vd1_rgbbst_dlut0
7:0	R/W	147	Vd1_rgbbst_dlut3: same as Vd1_rgbbst_dlut0

Table 8-3096 VPP_VD1_RGB_DLUT_4_7 0x328c

Bit(s)	R/W	Default	Description
31:24	R/W	128	Vd1_rgbbst_dlut4: same as Vd1_rgbbst_dlut0
23:16	R/W	113	Vd1_rgbbst_dlut5: same as Vd1_rgbbst_dlut0
15:8	R/W	102	Vd1_rgbbst_dlut6: same as Vd1_rgbbst_dlut0
7:0	R/W	93	Vd1_rgbbst_dlut7: same as Vd1_rgbbst_dlut0

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Table 8-3097 VPP_VD1_RGB_DLUT_8_11 0x328d

Bit(s)	R/W	Default	Description
31:24	R/W	85	Vd1_rgbbst_dlut8: same as Vd1_rgbbst_dlut0
23:16	R/W	78	Vd1_rgbbst_dlut9: same as Vd1_rgbbst_dlut0
15:8	R/W	73	Vd1_rgbbst_dlut10: same as Vd1_rgbbst_dlut0
7:0	R/W	68	Vd1_rgbbst_dlut11: same as Vd1_rgbbst_dlut0

Table 8-3098 VPP_VADJ2_MISC 0x32a0

Bit(s)	R/W	Default	Description
5:4	R/W	0	Gate clk ctrl for vadj2
3	/	/	/
2	R/W	0	Sync enable 1:vadj2 regs work when go_field come 1:vadj2 regs directly
1	R/W	0	minus black level enable for vadj2
0	R/W	0	Module enable

Table 8-3099 VPP_VADJ2_BLACK_VAL 0X32a1

Bit(s)	R/W	Default	Description
26:16	R/W	13'h200	Black_luma
10:0	R/W	13'h40	Black_chroma

Table 8-3100 VPP_VADJ2_Y 0x32a2

Bit(s)	R/W	Default	Description
18-8	R/W	0	brightness, signed value
7-0	R/W	0x1d0	contrast, unsigned value, contrast from 0 <= contrast <2

cb' = cb*ma + cr*mb

cr' = cb*mc + cr*md

Table 8-3101 VPP_VADJ2_MA_MB 0x32a3

Bit(s)	R/W	Default	Description
25-16	R/W	0x100	MA, signed value, -2 < MA < 2
9-0	R/W	0	MB, signed value ,-2 < MB < 2

Table 8-3102 VPP_VADJ2_MC_MD 0x32a4

Bit(s)	R/W	Default	Description
25-16	R/W	0	MC, signed value, -2 < MC < 2
9-0	R/W	0x100	MD, signed value ,-2 < MD < 2

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Table 8-3103 VPP_VADJ2_CURV_0 0x32a5

Bit(s)	R/W	Default	Description
31:24	R/W	0	Vadj2_softcon_curv0_ci, u8
23:12	R/W	0	Vadj2_softcon_curv0_b, u12
11:0	R/W	0	Vadj2_softcon_curv0_a, s12

Table 8-3104 VPP_VADJ2_CURV_1 0x32a6

Bit(s)	R/W	Default	Description
31:13			Reserved
12:4	R/W	0	Vadj2_softcon_curv0_g, s9
3			Reserved
2:0	R/W	0	Vadj2_softcon_curv0_cs, u3

Table 8-3105 VPP_VADJ2_CURV_2 0x32a7

Bit(s)	R/W	Default	Description
31:24	R/W	0	Vadj2_softcon_curv1_ci, u8
23:12	R/W	0	Vadj2_softcon_curv1_b, u12
11:0	R/W	0	Vadj2_softcon_curv1_a, s12

Table 8-3106 VPP_VADJ2_CURV_3 0x32a8

Bit(s)	R/W	Default	Description
31:13			Reserved
12:4	R/W	0	Vadj2_softcon_curv1_g, s9
3			Reserved
2:0	R/W	0	Vadj2_softcon_curv1_cs, u3

Table 8-3107 VPP_POST_RGB_CTRST 0x32a9

Bit(s)	R/W	Default	Description
31:28			Reserved
27:16	R/W	1024	Post_rgb_ctrst: contrast for rgb, normalized 1024 as ‘1.0’
15:12			Reserved
11:2	R/W	64	Post_rgb_ctrst_blklvl: contrast black level to be subtract before and add back after the contrast gain operation
1	R/W	0	Post_rgbbst_en: 1 to enable the RGB_BST
0	R/W	1	Post_rgb_ctrst_prt: enable signal to protect saturation in rgb (no clipping) during contrast adjustment

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Table 8-3108 VPP_POST_RGB_BRGHT 0x32aa

Bit(s)	R/W	Default	Description
31:14			Reserved
13:2	R/W	0	Post_rgb_brght: brightness level in rgb domain
1	R/W	1	Post_rgb_brght_prt: enable signal to protect saturation in rgb (no clipping) during brightness adjustment
0	R/W	0	Post_rgb_ctrst_dlut_x2: Enable signal to do x2 to the dlut cells before subtracting from the normalized gain_max; 0:x1 1:x2

Table 8-3109 VPP_POST_RGB_DLUT_0_3 0x32ab

Bit(s)	R/W	Default	Description
31:24	R/W	255	Post_rgbbst_dlut0: Differential gain to normalized gain_max to customized protection curve. e.g. [255 205 171 147 128 113 102 93 85 78 73 68] for protection of not boost for pixels larger than 240
23:16	R/W	205	Post_rgbbst_dlut1: same as Post_rgbbst_dlut0
15:8	R/W	171	Post_rgbbst_dlut2: same as Post_rgbbst_dlut0
7:0	R/W	147	Post_rgbbst_dlut3: same as Post_rgbbst_dlut0

Table 8-3110 VPP_POST_RGB_DLUT_4_7 0x32ac

Bit(s)	R/W	Default	Description
31:24	R/W	128	Post _rgbbst_dlut4: same as Post_rgbbst_dlut0
23:16	R/W	113	Post _rgbbst_dlut5: same as Post_rgbbst_dlut0
15:8	R/W	102	Post _rgbbst_dlut6: same as Post_rgbbst_dlut0
7:0	R/W	93	Post _rgbbst_dlut7: same as Post_rgbbst_dlut0

Table 8-3111 VPP_POST_RGB_DLUT_8_11 0x32ad

Bit(s)	R/W	Default	Description
31:24	R/W	85	Post_rgbbst_dlut8: same as Post_rgbbst_dlut0
23:16	R/W	78	Post_rgbbst_dlut9: same as Post_rgbbst_dlut0
15:8	R/W	73	Post_rgbbst_dlut10: same as Post_rgbbst_dlut0
7:0	R/W	68	Post_rgbbst_dlut11: same as Post_rgbbst_dlut0

Table 8-3112 VPP_POST_MATRIX_SAT 0x32c1

Bit(s)	R/W	Default	Description
2	R/W	85	Sato_en
1	R/W	78	Sati_en
0	R/W	73	Misc_sat_en

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8.2.3.49 3D LUT Registers

Table 8-3113 3D LUT VPP_LUT3D_CTRL 0x39d0

Bit(s)	R/W	Default	Description
9:8	R/W	0	Gated clock control
7	R/W		reserved
6:4	R/W		reg_lut3d_extnd_en: enable to set LUT value of 1023 to value 1024.
3	R/W		reserved
2	R/W	0	1 to shaddow the “reg_lut3d_enable” by VSYNC
1	R/W		reserved
0	R/W	0	reg_lut3d_enable: 1 to enable 3D LUT

Table 8-3114 VPP_LUT3D_CBUS2RAM_CTRL 0x39d1

Bit(s)	R/W	Default	Description
0	R/W	0	1 to enable CBUS to configure the LUT3D RAMs, 0 for in normal working status.

Table 8-3115 VPP_LUT3D_RAM_ADDR 0x39d2

Bit(s)	R/W	Default	Description
32	R/W	0	LUT3D RAMs Address port

Table 8-3116 VPP_LUT3D_RAM_DATA 0x39d3

Bit(s)	R/W	Default	Description
32	R/W	0	LUT3D RAMs data port

8.2.3.50 AFBCE Registers

Table 8-3117 AFBCE_ENABLE 0x41a0

Bit(s)	R/W	Default	Description
31:20	R/W	0	gclk_ctrl : unsigned , default = 0, gclk_ctrl of afbe submodule
19:16	R/W	0	afbce_sync_sel : unsigned , default = 0, shadow aome size/scope/mode registers in afnce
13	R/W	0	enc_rst_mode : unsigned , default = 0, 1: chose pulse rst mode ,need write a pule by
12	R/W	0	enc_en_mode : unsigned , default = 0, 1: chose pulse staert mode,need write a pule by AFBCE_ENABLE[0],0:automatic start
8	R/W	0	enc_enable : unsigned , default = 0, 1: open mode 0:close module

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Table 8-3118 AFBCE_MODE 0x41a1

Bit(s)	R/W	Default	Description
31:29	R/W	0	soft_rst : unsigned, default = 0 ,bit[0] async rst [1] sync rst [2] sync rst by vsync
27:26	R/W	0	rev_mode : unsigned, default = 0 , reverse mode
25:24	R/W	3	mif_urgent : unsigned, default = 3 , info mif and data mif urgent
22:16	R/W	4	hold_line_num : unsigned, default = 4, hold_line_num
15:14	R/W	1	burst_mode : unsigned, default = 1, 0: burst1 1:burst2 2:burst4
0	R/W	0	reg_fmt444_comb : unsigned, default = 0, 0: 444 8bit comb mode

Table 8-3119 AFBCE_SIZE_IN 0x41a2

Bit(s)	R/W	Default	Description
28:16	R/W	1920	hsize_in : unsigned, default = 1920 , pic size in unit: pixel
12:0	R/W	1080	vsize_in : unsigned, default = 1080 , pic size in unit: pixel

Table 8-3120 AFBCE_BLK_SIZE_IN 0x41a3

Bit(s)	R/W	Default	Description
28:16	R/W	60	hblk_size : unsigned, default = 60 , blk horz size out unit: 32*4 block
12:0	R/W	270	vblk_size : unsigned, default = 270, blk vert size out unit: 32*4 block

Table 8-3121 AFBCE_HEAD_BADDR 0x41a4

Bit(s)	R/W	Default	Description
31:0	R/W	0x00	head_baddr : unsigned, default = 32'h00,head_baddr

Table 8-3122 AFBCE_MIF_SIZE 0x41a5

Bit(s)	R/W	Default	Description
29:28	R/W	32	ddr_blk_size : unsigned, default = 32'd1,ddr_blk_size
26:24	R/W	32	cmd_blk_size : unsigned, default = 32'd3,cmd_blk_size
20:16	R/W	32	uncmp_size : unsigned, default = 32'd20 ,uncmp_size of a uncompressed 32*4 block
15:0	R/W	32	mmu_page_size : unsigned, default = 32'd4096,mmu_page_size 4096/8192

Table 8-3123 AFBCE_PIXEL_IN_HOR_SCOPE 0x41a6

Bit(s)	R/W	Default	Description
28:16	R/W	1919	enc_win_end_h : unsigned, default = 1919 ,pic scope in end,should be a integer multipe of 32 //
12:0	R/W	0	enc_win_bgn_h : unsigned, default = 0 ,pic scope in bgn,should be a integer multipe of 32//

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Table 8-3124 AFBCE_PIXEL_IN_VER_SCOPE 0x41a7

Bit(s)	R/W	Default	Description
28:16	R/W	1079	enc_win_end_v : unsigned, default = 1079 ,pic scope in end,should be a integer multipe of 4 //
12:0	R/W	0	enc_win_bgn_v : unsigned, default = 0 ,pic scope in bgn,should be a integer multipe of 4 //

Table 8-3125 AFBCE_CONV_CTRL 0x41a8

Bit(s)	R/W	Default	Description
28:16	R/W	2048	fmt_ybuf_depth : unsigned, default = 2048,fmt_ybuf_depth,fixed
11: 0	R/W	256	lbuf_depth : unsigned, default = 256, lbuf_depth,fixed

Table 8-3126 AFBCE_MIF_HOR_SCOPE 0x41a9

Bit(s)	R/W	Default	Description
25:16	R/W	0	blk_end_h : unsigned, default = 0 ,blk scope out end,should be a integer multipe of 4 //
9:0	R/W	59	blk_bgn_h : unsigned, default = 59 ,blk scope out bgn,should be a integer multipe of 4 //

Table 8-3127 AFBCE_MIF_VER_SCOPE 0x41aa

Bit(s)	R/W	Default	Description
27:16	R/W	0	blk_end_v : unsigned, default = 0 ,blk scope out end,should be a integer multipe of 4 //
11:0	R/W	269	blk_bgn_v : unsigned, default = 269 ,blk scope out bgn,should be a integer multipe of 4 //

Table 8-3128 AFBCE_STAT1 0x41ab

Bit(s)	R/W	Default	Description
31	R.O	0	ro_frm_end_pulse1 : unsigned, RO,default = 0 ;frame end status
30:0	R.O	0	ro_dbg_top_info1 : unsigned, RO,default = 0 ,ro_dbg_top_info1

Table 8-3129 AFBCE_STAT2 0x410c

Bit(s)	R/W	Default	Description
30:0	R.O	0	ro_dbg_top_info2 : unsigned, RO,default = 0 ,ro_dbg_top_info2

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Table 8-3130 AFBCE_FORMAT 0x41ad

Bit(s)	R/W	Default	Description
9: 8	R/W	2	reg_format_mode : // unsigned , RW, default = 2 data format;0 : YUV444, 1: YUV422, 2:YUV420, 3:RGB
7: 4	R/W	10	reg_compbits_c : // unsigned , RW, default = 10 chroma bitwidth 8 or 10
3: 0	R/W	10	reg_compbits_y : // unsigned , RW, default = 10 luma bitwidth 8 or 10

Table 8-3131 AFBCE_MODE_EN 0x41ae

Bit(s)	R/W	Default	Description
25	R/W	0	reg_adpt_interleave_ymode : // unsigned , RW, default = 0 force 0 to disable it: no HW implementation
24	R/W	0	reg_adpt_interleave_cmode : // unsigned , RW, default = 0 force 0 to disable it: not HW implementation
23	R/W	1	reg_adpt_yinterleave_luma_ride : // unsigned , RW, default = 1 vertical interleave piece luma reorder ride; 0: no reorder ride; 1: w/4 as ride
22	R/W	1	reg_adpt_yinterleave_chrm_ride : // unsigned , RW, default = 1 vertical interleave piece chroma reorder ride; 0: no reorder ride; 1: w/2 as ride
21	R/W	1	reg_adpt_xinterleave_luma_ride : // unsigned , RW, default = 1 vertical interleave piece luma reorder ride; 0: no reorder ride; 1: w/4 as ride
20	R/W	1	reg_adpt_xinterleave_chrm_ride : // unsigned , RW, default = 1 vertical interleave piece chroma reorder ride; 0: no reorder ride; 1: w/2 as ride
18	R/W	0	reg_disable_order_mode_i_6 : // unsigned , RW, default = 0 disable order mode0~6: each mode with one disable bit: 0: no disable, 1: diable
17	R/W	0	reg_disable_order_mode_i_5 : // unsigned , RW, default = 0 disable order mode0~6: each mode with one disable bit: 0: no disable, 1: diable
16	R/W	0	reg_disable_order_mode_i_4 : // unsigned , RW, default = 0 disable order mode0~6: each mode with one disable bit: 0: no disable, 1: diable
15	R/W	0	reg_disable_order_mode_i_3 : // unsigned , RW, default = 0 disable order mode0~6: each mode with one disable bit: 0: no disable, 1: diable
14	R/W	0	reg_disable_order_mode_i_2 : // unsigned , RW, default = 0 disable order mode0~6: each mode with one disable bit: 0: no disable, 1: diable
13	R/W	0	reg_disable_order_mode_i_1 : // unsigned , RW, default = 0 disable order mode0~6: each mode with one disable bit: 0: no disable, 1: diable
12	R/W	0	reg_disable_order_mode_i_0 : // unsigned , RW, default = 0 disable order mode0~6: each mode with one disable bit: 0: no disable, 1: diable
10	R/W	0	reg_minval_yenc_en : // unsigned , RW, default = 0 force disable, final decision to remove this ws 1% performance loss
9	R/W	0	reg_16x4block_enable : // unsigned , RW, default = 0 block as mission, but permit 16x4 block
8	R/W	0	reg_uncompress_split_mode : // unsigned , RW, default = 0 0: no split; 1: split
5	R/W	0	reg_input_padding_uv128 : // unsigned , RW, default = 0 input picture 32x4 block gap mode: 0: pad uv=0; 1: pad uv=128
4	R/W	0	reg_dwds_padding_uv128 : // unsigned , RW, default = 0 downsampled image for double write 32x gap mode: 0: pad uv=0; 1: pad uv=128

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Bit(s)	R/W	Default	Description
3: 1	R/W	0	reg_force_order_mode_value : // unsigned , RW, default = 0 force order mode 0~7
0	R/W	0	reg_force_order_mode_en : // unsigned , RW, default = 0 force order mode enable: 0: no force; 1: forced to force_value

Table 8-3132 AFBCE_DWSCALAR 0x41af

Bit(s)	R/W	Default	Description
7: 6	R/W	3	reg_dwscalar_w0 : // unsigned , RW, default = 3 horizontal 1st step scalar mode: 0: 1:1 no scalar; 1: 2:1 data drop (0,2,4, 6) pixel kept; 2: 2:1 data drop (1, 3, 5,7..) pixels kept; 3: avg
5: 4	R/W	0	reg_dwscalar_w1 : // unsigned , RW, default = 0 horizontal 2nd step scalar mode: 0: 1:1 no scalar; 1: 2:1 data drop (0,2,4, 6) pixel kept; 2: 2:1 data drop (1, 3, 5,7..) pixels kept; 3: avg
3: 2	R/W	2	reg_dwscalar_h0 : // unsigned , RW, default = 2 vertical 1st step scalar mode: 0: 1:1 no scalar; 1: 2:1 data drop (0,2,4, 6) pixel kept; 2: 2:1 data drop (1, 3, 5,7..) pixels kept; 3: avg
1: 0	R/W	3	reg_dwscalar_h1 : // unsigned , RW, default = 3 vertical 2nd step scalar mode: 0: 1:1 no scalar; 1: 2:1 data drop (0,2,4, 6) pixel kept; 2: 2:1 data drop (1, 3, 5,7..) pixels kept; 3: avg

Table 8-3133 AFBCE_DEFCOLOR_1 0x41b0

Bit(s)	R/W	Default	Description
23:12	R/W	4095	reg_enc_defalutcolor_3 : // unsigned , RW, default = 4095 Picture wise default color value in [Y Cb Cr]
11: 0	R/W	4095	reg_enc_defalutcolor_0 : // unsigned , RW, default = 4095 Picture wise default color value in [Y Cb Cr]

Table 8-3134 AFBCE_DEFCOLOR_2 0x41b1

Bit(s)	R/W	Default	Description
23:12	R/W	2048	reg_enc_defalutcolor_2 : // unsigned , RW, default = 2048 wise default color value in [Y Cb Cr]
11: 0	R/W	2048	reg_enc_defalutcolor_1 : // unsigned , RW, default = 2048 wise default color value in [Y Cb Cr]

Table 8-3135 AFBCE_QUANT_ENABLE 0x41b2

Bit(s)	R/W	Default	Description
11	R/W	0	reg_quant_expand_en_1 : // unsigned , RW, default = 0 enable for quantization value expansion
10	R/W	0	reg_quant_expand_en_0 : // unsigned , RW, default = 0 enable for quantization value expansion
9: 8	R/W	0	reg_bcleav_ofst : // signed , RW, default = 0 bcleave ofset to get lower range, especially under lossy, for v1/v2, x=0 is equivalent, default = -1;

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Bit(s)	R/W	Default	Description
4	R/W	0	reg_quant_enable_1 : // unsigned , RW, default = 0 enable for quant to get some lossy
0	R/W	0	reg_quant_enable_0 : // unsigned , RW, default = 0 enable for quant to get some lossy

Table 8-3136 AFBCE_IQUANT_LUT_1 0x41b3

Bit(s)	R/W	Default	Description
30:28	R/W	0	reg_iquant_yclut_0_11 : // unsigned , RW, default = 0 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
26:24	R/W	1	reg_iquant_yclut_0_10 : // unsigned , RW, default = 1 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
22:20	R/W	2	reg_iquant_yclut_0_9 : // unsigned , RW, default = 2 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
18:16	R/W	3	reg_iquant_yclut_0_8 : // unsigned , RW, default = 3 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
14:12	R/W	4	reg_iquant_yclut_0_7 : // unsigned , RW, default = 4 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
10: 8	R/W	5	reg_iquant_yclut_0_6 : // unsigned , RW, default = 5 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
6: 4	R/W	5	reg_iquant_yclut_0_5 : // unsigned , RW, default = 5 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
2: 0	R/W	4	reg_iquant_yclut_0_4 : // unsigned , RW, default = 4 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

Table 8-3137 AFBCE_IQUANT_LUT_2 0x41b4

Bit(s)	R/W	Default	Description
14:12	R/W	3	reg_iquant_yclut_0_3 : // unsigned , RW, default = 3 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
10: 8	R/W	2	reg_iquant_yclut_0_2 : // unsigned , RW, default = 2 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
6: 4	R/W	1	reg_iquant_yclut_0_1 : // unsigned , RW, default = 1 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
2: 0	R/W	0	reg_iquant_yclut_0_0 : // unsigned , RW, default = 0 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

Table 8-3138 AFBCE_IQUANT_LUT_3 0x41b5

Bit(s)	R/W	Default	Description
30:28	R/W	0	reg_iquant_yclut_1_11 : // unsigned , RW, default = 0 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
26:24	R/W	1	reg_iquant_yclut_1_10 : // unsigned , RW, default = 1 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
22:20	R/W	2	reg_iquant_yclut_1_9 : // unsigned , RW, default = 2 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

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Bit(s)	R/W	Default	Description
18:16	R/W	3	reg_iquant_yclut_1_8 : // unsigned , RW, default = 3 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
14:12	R/W	4	reg_iquant_yclut_1_7 : // unsigned , RW, default = 4 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
10: 8	R/W	5	reg_iquant_yclut_1_6 : // unsigned , RW, default = 5 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
6: 4	R/W	5	reg_iquant_yclut_1_5 : // unsigned , RW, default = 5 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
2: 0	R/W	4	reg_iquant_yclut_1_4 : // unsigned , RW, default = 4 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

Table 8-3139 AFBCE_IQUANT_LUT_4 0x41b6

Bit(s)	R/W	Default	Description
14:12	R/W	3	reg_iquant_yclut_1_3 : // unsigned , RW, default = 3 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
10: 8	R/W	2	reg_iquant_yclut_1_2 : // unsigned , RW, default = 2 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
6: 4	R/W	1	reg_iquant_yclut_1_1 : // unsigned , RW, default = 1 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)
2: 0	R/W	0	reg_iquant_yclut_1_0 : // unsigned , RW, default = 0 quantization lut for mintree leavs, iquant=2^lut(bc_leav_q+1)

Table 8-3140 AFBCE_RQUANT_LUT_1 0x41b7

Bit(s)	R/W	Default	Description
30:28	R/W	5	reg_rquant_yclut_0_11 : // unsigned , RW, default = 5 quantization lut for bctree leavs, quant=2^lut(bc_leav_r+1), can be calculated from iquant_yclut(fw_setting)
26:24	R/W	5	reg_rquant_yclut_0_10 : // unsigned , RW, default = 5
22:20	R/W	4	reg_rquant_yclut_0_9 : // unsigned , RW, default = 4
18:16	R/W	4	reg_rquant_yclut_0_8 : // unsigned , RW, default = 4
14:12	R/W	3	reg_rquant_yclut_0_7 : // unsigned , RW, default = 3
10: 8	R/W	3	reg_rquant_yclut_0_6 : // unsigned , RW, default = 3
6: 4	R/W	2	reg_rquant_yclut_0_5 : // unsigned , RW, default = 2
2: 0	R/W	2	reg_rquant_yclut_0_4 : // unsigned , RW, default = 2

Table 8-3141 AFBCE_RQUANT_LUT_2 0x41b8

Bit(s)	R/W	Default	Description
14:12	R/W	1	reg_rquant_yclut_0_3 : // unsigned , RW, default = 1
10: 8	R/W	1	reg_rquant_yclut_0_2 : // unsigned , RW, default = 1
6: 4	R/W	0	reg_rquant_yclut_0_1 : // unsigned , RW, default = 0
2: 0	R/W	0	reg_rquant_yclut_0_0 : // unsigned , RW, default = 0

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Table 8-3142 AFBCE_RQUANT_LUT_3 0x41b9

Bit(s)	R/W	Default	Description
30:28	R/W	5	reg_rquant_yclut_1_11 : // unsigned , RW, default = 5 quantization lut for bctree leavs, quant=2^lut(bc_leav_r+1), can be calculated from iquant_yclut(fw_setting)
26:24	R/W	5	reg_rquant_yclut_1_10 : // unsigned , RW, default = 5
22:20	R/W	4	reg_rquant_yclut_1_9 : // unsigned , RW, default = 4
18:16	R/W	4	reg_rquant_yclut_1_8 : // unsigned , RW, default = 4
14:12	R/W	3	reg_rquant_yclut_1_7 : // unsigned , RW, default = 3
10: 8	R/W	3	reg_rquant_yclut_1_6 : // unsigned , RW, default = 3
6: 4	R/W	2	reg_rquant_yclut_1_5 : // unsigned , RW, default = 2
2: 0	R/W	2	reg_rquant_yclut_1_4 : // unsigned , RW, default = 2

Table 8-3143 AFBCE_RQUANT_LUT_4 0x41ba

Bit(s)	R/W	Default	Description
14:12	R/W	1	reg_rquant_yclut_1_3 : // unsigned , RW, default = 1
10: 8	R/W	1	reg_rquant_yclut_1_2 : // unsigned , RW, default = 1
6: 4	R/W	0	reg_rquant_yclut_1_1 : // unsigned , RW, default = 0
2: 0	R/W	0	reg_rquant_yclut_1_0 : // unsigned , RW, default = 0

Table 8-3144 AFBCE_YUV_FORMAT_CONV_MODE 0x41bb

Bit(s)	R/W	Default	Description
6: 4	R/W	0	reg_444to422_mode : // unsigned , RW, default = 0
2: 0	R/W	0	reg_422to420_mode : // unsigned , RW, default = 0

Table 8-3145 AFBCE_DUMMY_DATA 0x41bc

Bit(s)	R/W	Default	Description
29: 0	R/W	0	reg_dummy_data : // unsigned , default = 0 ,bit[9:0] v bit[19:10] U bit[29:20] Y

Table 8-3146 AFBCE_CLR_FLAG 0x41bd

Bit(s)	R/W	Default	Description
1	R/W	0	enc_error_clr : // unsigned, default = 0 ;
0	R/W	0	frm_end_clr : // unsigned, default = 0 ;

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Table 8-3147 AFBCE_STA_FLAGT 0x41be

Bit(s)	R/W	Default	Description
31:26	R.O	0	reverse : // unsigned, RO,default = 0 ;
25:24	R.O	0	enc_st 0:idle 1:enc 2:hold 3:wait
23:21	R.O	0	emit_st 0:idle 1:tx_bcrot 2:tx_bc_lev 3:tx_zcd 4:tx_tree 5:tx_uncmp 6:tx_redu 7:tx_check
20:15	R.O	0	ro_cmd2bresp_cnt: // unsigned, RO,default = 0 ;
14:9	R.O	0	ro_enc2body_cnt: // unsigned, RO,default = 0 ;
8:3	R.O	0	ro_cmd2wd_cnt: // unsigned, RO,default = 0 ;
2	R.O	0	ro_wr_abort_flag: // unsigned, RO,default = 0 ;
1	R.O	0	ro_enc_error_flag: // unsigned, RO,default = 0 ;
0	R.O	0	ro_frm_end_flag: // unsigned, RO,default = 0 ;

Table 8-3148 AFBCE_MMU_NUM 0x41bf

Bit(s)	R/W	Default	Description
15: 0	R.O	0	ro_frm_mmu_num : // unsigned, RO,default = 0 ,mmu addr have been used in a frame

Table 8-3149 AFBCE_MMU_RMIF_CTRL1 0x41c0

Bit(s)	R/W	Default	Description
25:24	R/W	0	reg_sync_sel : // unsigned , default = 0, axi canvas id sync with frm rst
23:16	R/W	0	reg_canvas_id : // unsigned , default = 0, axi canvas id num
14:12	R/W	1	reg_cmd_intr_len : // unsigned , default = 1, interrupt send cmd when how many series axi cmd, 0 =12 1=16 2=24 3=32 4=40 5=48 6=56 7=64
11:10	R/W	1	reg_cmd_req_size : // unsigned , default = 1, how many room fifo have, then axi send series req, 0=16 1=32 2=24 3=64
9:8	R/W	2	reg_burst_len : // unsigned , default = 2, burst type: 0-single 1-bst2 2-bst4
7	R/W	0	reg_swap_64bit : // unsigned , default = 0, 64bits of 128bit swap enable
6	R/W	0	reg_little_endian : // unsigned , default = 0, big endian enable
5	R/W	0	reg_y_rev : // unsigned , default = 0, vertical reverse enable
4	R/W	0	reg_x_rev : // unsigned , default = 0, horizontal reverse enable
2:0	R/W	3	reg_pack_mode : // unsigned , default = 3, 0:4bit 1:8bit 2:16bit 3:32bit 4:64bit 5:128bit

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Table 8-3150 AFBCE_MMU_RMIF_CTRL2 0x41c1

Bit(s)	R/W	Default	Description
31:30	R/W	0	reg_sw_rst : // unsigned , default = 0,
23:18	R/W	0x0	reg_gclk_ctrl :
16:0	R/W	0	reg_urgent_ctrl : // unsigned , default = 0, urgent control reg : 16 reg_ugt_init : urgent initial value 15 reg_ugt_en : urgent enable 14 reg_ugt_type : 1= wrmif 0= rdmif 7 :4 reg_ugt_top_th: urgent top threshold 3 :0 reg_ugt_bot_th: urgent bottom threshold

Table 8-3151 AFBCE_MMU_RMIF_CTRL3 0x41c2

Bit(s)	R/W	Default	Description
16	R/W	1	reg_acc_mode : // unsigned , default = 1,
12:0	R/W	4096	reg_stride : // unsigned , default = 4096,

Table 8-3152 AFBCE_MMU_RMIF_CTRL4 0x41c3

Bit(s)	R/W	Default	Description
31:0	R/W	0	reg_baddr : // unsigned , default = 0, reg_mmu_baddr

Table 8-3153 AFBCE_MMU_RMIF_SCOPE_X 0x41c4

Bit(s)	R/W	Default	Description
28:16	R/W	4095	reg_x_end : // unsigned , default = 4095, the canvas hor end pixel position
12: 0	R/W	0	reg_x_start : // unsigned , default = 0, the canvas hor start pixel position

Table 8-3154 AFBCE_MMU_RMIF_SCOPE_Y 0x41c5

Bit(s)	R/W	Default	Description
28:16	R/W	0	reg_y_end : // unsigned , default = 0, the canvas ver end pixel position
12: 0	R/W	0	reg_y_start : // unsigned , default = 0, the canvas ver start pixel positio

Table 8-3155 AFBCE_MMU_RMIF_RO_STAT 0x41c6

Bit(s)	R/W	Default	Description
15:0	R/W	0x0	reg_status : // unsigned , reg_status

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8.2.3.51 LCD Registers

8.2.3.51.1 L_Gamma Control

Table 8-3156 L_Gamma Control L_GAMMA_CNTL_PORT (VCBUS: 0x1400)

Bit(s)	R/W	Default	Description
7	R/W	0	GAMMA_VCOM_POLReverse VCOM Polarity
6	R/W	0	GAMMA_RVS_OUTGAMMA DATA REVERSE OUTPUT FOLLOWING VCOM
5	R	0	ADR_RDYGAMMA ADDR PORT Is Ready for access
4	R	0	WR_RDY GAMMA DATA PORT IS RDY to Write
3	R	0	RD_RDY GAMMA DATA PORT IS RDY to Read
2	R/W	0	GAMMA_TRRGB10–>RGB8 using Trancate or Round off
1	R/W	0	GAMMA_SETGamma turn on syncing with Vsync
0	R/W	0	GAMMA_ENGamma Enable

Table 8-3157 L_GAMMA_DATA_PORT (VCBUS: 0x1401)

Bit(s)	R/W	Default	Description
15:0	-	0	GAMMA_DATA_PORTGAMMA DATA PORT ADDR, all gamma data read/write operation is via this port

Table 8-3158 L_GAMMA_ADDR_PORT (VCBUS: 0x1402)

Bit(s)	R/W	Default	Description
12	R/W	0	H-RDGAMMA is READY for Host to read/write
11	R/W	0	H-AUTO_INCBURST MODE *
10	R/W	0	H_SEL_RGAMMA R Selected
9	R/W	0	H_SEL_GGAMMA G Selected
8	R/W	0	H_SEL_BGAMMA B Selected
7:0	W	0	HADRGAMMA ADDR.

Note

↑When Programming the Gamma with Burst Mode, please turn off the IRQ service (Or make sure the IRQ service task have no Gamma programming operation)

Table 8-3159 L_GAMMA_VCOM_HSWITCH_ADDR (VCBUS: 0x1403)

Bit(s)	R/W	Default	Description
12:0	R/W	0	GAMMA_VCOM_HSWITCH_ADDRHorizontal Switch Point for VCOM with Gamma

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Table 8-3160 L_RGB_BASE_ADDR (VCBUS: 0x1405)

Bit(s)	R/W	Default	Description
13:10	R/W	0	rasp_ctrl bit3 : raspberry pi dpi en bit2:0 : raspberry pi dpi rgb mode
9:0	R/W	0	RGB_BASE_ADDR RGB Converter OFFSET

Table 8-3161 L_RGB_COEFF_ADDR (VCBUS: 0x1406)

Bit(s)	R/W	Default	Description
10:0	R/W	0	RGB_COEFF_ADDR RGB Converter SCALE

Table 8-3162 L_POL_CNTL_ADDR (VCBUS: 0x1407)

Bit(s)	R/W	Default	Description
15:14	R/W	0	DCLK_SELDCLK output
11	R/W	0	TCON_VSYNC_SEL_DVI VSYNC generated by TCON for RGB format DVI output (support Digital LCD interface)
10	R/W	0	TCON_HSYNC_SEL_DVI HSYNC generated by TCON for RGB format DVI output (support Digital LCD interface)
9	R/W	0	TCON_DE_SEL_DVI DE generated by TCON for RGB format DVI output (support Digital LCD interface)
8	R/W	0	CPH3_POL CPH3 Polarity Control
7	R/W	0	CPH2_POL CPH2 Polarity Control
6	R/W	0	CPH1_POL CPH1 Polarity Control
5	R/W	0	TCON_DE_SEL DE Generate by TCON
4	R/W	0	TCON_VS_SEL Vsync Generate by TCON
3	R/W	0	TCON_HS_SEL Hsync Generate by TCON
2	R/W	0	DE_POL De polarity control
1	R/W	0	VS_POL Vsync polarity control
0	R/W	0	HS_POL Hsync polarity control

Table 8-3163 L_DITH_CNTL_ADDR (VCBUS: 0x1408)

Bit(s)	R/W	Default	Description
13	R/W	0	DITH_R51=use the current dithering to dither input 10-bit R down to 5-bit. Applicable only if both dith10_en & dith8_en = 1, otherwise no effect
12	R/W	0	DITH_G51=use the current dithering to dither input 10-bit G down to 5-bit. Applicable only if both dith10_en & dith8_en = 1, otherwise no effect
11	R/W	0	DITH_B51=use the current dithering to dither input 10-bit B down to 5-bit. Applicable only if both dith10_en & dith8_en = 1, otherwise no effect
10	R/W	0	DITH10_EN 10-bits Dithering to 8 bits Enable
9	R/W	0	DITH8_EN 8-bits Dithering to 6 bits Enable

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Bit(s)	R/W	Default	Description
8	R/W	0	DITH_MD Dithering Mode select
7:4	R/W	0	DITH8_CNTL 8-bits Dithering control
3:0	R/W	0	DITH10_CNTL 10-bits Dithering control

Table 8-3164 L_GAMMA_PROBE_CTRL (VCBUS: 0x1409)

Bit(s)	R/W	Default	Description
1	R/W	0	HIGHLIGHT_EN
0	R/W	0	PROBE_EN

Table 8-3165 L_GAMMA_PROBE_COLOR_L (VCBUS: 0x140a)

Bit(s)	R/W	Default	Description
15:0	R	0	GAMMA_PROBE_COLOR_L Probe_pix[15:0]

Table 8-3166 L_GAMMA_PROBE_COLOR_H (VCBUS: 0x140b)

Bit(s)	R/W	Default	Description
15	R	0	PROBE_PIX_Vprobe_pix_v
14	R	0	reservereserve
13:0	R	0	GAMMA_PROBE_COLOR_H Probe_pix[29:16]

Table 8-3167 L_GAMMA_PROBE_HL_COLOR (VCBUS: 0x140c)

Bit(s)	R/W	Default	Description
15:0	R	0	GAMMA_PROBE_HL_COLOR 5:6:5 color

Table 8-3168 L_GAMMA_PROBE_POS_X (VCBUS: 0x140d)

Bit(s)	R/W	Default	Description
12:0	R/W	0	GAMMA_PROBE_POS_X Pos_x

Table 8-3169 L_GAMMA_PROBE_POS_Y (VCBUS: 0x140e)

Bit(s)	R/W	Default	Description
12:0	R/W	0	GAMMA_PROBE_POS_Y Pos_y

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8.2.3.51.2 L_TCON Control

Table 8-3170 L_STH1_HS_ADDR (VCBUS: 0x1410)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH1_HS_ADDR STH1 Horizontal Start

Table 8-3171 L_STH1_HE_ADDR (VCBUS: 0x1411)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH1_HE_ADDR STH1 Horizontal End

Table 8-3172 L_STH1_VS_ADDR (VCBUS: 0x1412)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH1_VS_ADDR STH1 Vertical Start

Table 8-3173 L_STH1_VE_ADDR (VCBUS: 0x1413)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH1_VE_ADDR STH1 Vertical End

Table 8-3174 L_STH2_HS_ADDR (VCBUS: 0x1414)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH2_HS_ADDR STH2 Horizontal Start

Table 8-3175 L_STH2_HE_ADDR (VCBUS: 0x1415)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH2_HE_ADDR STH2 Horizontal End

Table 8-3176 L_STH2_VS_ADDR (VCBUS: 0x1416)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH2_VS_ADDR STH2 Vertical Start

Table 8-3177 L_STH2_VE_ADDR (VCBUS: 0x1417)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH2_VE_ADDR STH2 Vertical End

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Table 8-3178 L_OEH_HS_ADDR (VCBUS: 0x1418)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEH_HS_ADDR OEH Horizontal Start

Table 8-3179 L_OEH_HE_ADDR (VCBUS: 0x1419)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEH_HE_ADDR OEH Horizontal End

Table 8-3180 L_OEH_VS_ADDR (VCBUS: 0x141a)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEH_VS_ADDR OEH Vertical Start

Table 8-3181 L_OEH_VE_ADDR (VCBUS: 0x141b)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEH_VE_ADDR OEH Vertical End

Table 8-3182 L_VCOM_HSWITCH_ADDR (VCBUS: 0x141c)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VCOM_HSWITCH_ADDR VCOM Horizontal Switch Point

Table 8-3183 L_VCOM_VS_ADDR (VCBUS: 0x141d)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VCOM_VS_ADDR VCOM Vertical Start

Table 8-3184 L_VCOM_VE_ADDR (VCBUS: 0x141e)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VCOM_VE_ADDR VCOM Vertical End

Table 8-3185 L_CPV1_HS_ADDR (VCBUS: 0x141f)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV1_HS_ADDR CPV1 Horizontal Start

Table 8-3186 L_CPV1_HE_ADDR (VCBUS: 0x1420)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV1_HE_ADDR CPV1 Horizontal End

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Table 8-3187 L_CPV1_VS_ADDR (VCBUS: 0x1421)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV1_VS_ADDR CPV1 Vertical Start

Table 8-3188 L_CPV1_VE_ADDR (VCBUS: 0x1422)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV1_VE_ADDR CPV1 Vertical End

Table 8-3189 L_CPV2_HS_ADDR (VCBUS: 0x1423)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV2_HS_ADDR CPV2 Horizontal Start

Table 8-3190 L_CPV2_HE_ADDR (VCBUS: 0x1424)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV2_HE_ADDR CPV2 Horizontal End

Table 8-3191 L_CPV2_VS_ADDR (VCBUS: 0x1425)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV2_VS_ADDR CPV2 Vertical Start

Table 8-3192 L_CPV2_VE_ADDR (VCBUS: 0x1426)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV2_VE_ADDR CPV2 Vertical End

Table 8-3193 L_STV1_HS_ADDR (VCBUS: 0x1427)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV1_HS_ADDR STV1 Horizontal Start

Table 8-3194 L_STV1_HE_ADDR (VCBUS: 0x1428)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV1_HE_ADDR STV1 Horizontal End

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Table 8-3195 L_STV1_VS_ADDR (VCBUS: 0x1429)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV1_VS_ADDR STV1 Vertical Start

Table 8-3196 L_STV1_VE_ADDR (VCBUS: 0x142a)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV1_VE_ADDR STV1 Vertical End

Table 8-3197 L_STV2_HS_ADDR (VCBUS: 0x142b)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV2_HS_ADDR STV2 Horizontal Start

Table 8-3198 L_STV2_HE_ADDR (VCBUS: 0x142c)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV2_HE_ADDR STV2 Horizontal End

Table 8-3199 L_STV2_VS_ADDR (VCBUS: 0x142d)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV2_VS_ADDR STV2 Vertical Start

Table 8-3200 L_STV2_VE_ADDR (VCBUS: 0x142e)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV2_VE_ADDR STV2 Vertical End

Table 8-3201 L_OEV1_HS_ADDR (VCBUS: 0x142f)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV1_HS_ADDR OEV1 Horizontal Start

Table 8-3202 L_OEV1_HE_ADDR (VCBUS: 0x1430)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV1_HE_ADDR OEV1 Horizontal End

Table 8-3203 L_OEV1_VS_ADDR (VCBUS: 0x1431)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV1_VS_ADDR OEV1 Vertical Start

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Table 8-3204 L_OEV1_VE_ADDR (VCBUS: 0x1432)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV1_VE_ADDR OEV1 Vertical End

Table 8-3205 L_OEV2_HS_ADDR (VCBUS: 0x1433)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV2_HS_ADDR OEV2 Horizontal Start

Table 8-3206 L_OEV2_HE_ADDR (VCBUS: 0x1434)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV2_HE_ADDR OEV2 Horizontal End

Table 8-3207 L_OEV2_VS_ADDR (VCBUS: 0x1435)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV2_VS_ADDR OEV2 Vertical Start

Table 8-3208 L_OEV2_VE_ADDR (VCBUS: 0x1436)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV2_VE_ADDR OEV2 Vertical End

Table 8-3209 L_OEV3_HS_ADDR (VCBUS: 0x1437)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV3_HS_ADDR OEV3 Horizontal Start

Table 8-3210 L_OEV3_HE_ADDR (VCBUS: 0x1438)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV3_HE_ADDR OEV3 Horizontal End

Table 8-3211 L_OEV3_VS_ADDR (VCBUS: 0x1439)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV3_VS_ADDR OEV3 Vertical Start

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Table 8-3212 L_OEV3_VE_ADDR (VCBUS: 0x143a)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV3_VE_ADDR OEV3 Vertical End

Table 8-3213 L_LCD_PWR_ADDR (VCBUS: 0x143b) (Unused)

Bit(s)	R/W	Default	Description
5	R/W	0	LCD_VDD LCD VDD control
4	R/W	0	LCD_VBL LCD VBL control
3:0	R/W	0	LCD_GPIO LCD GPIO control

Table 8-3214 L_LCD_PWM0_LO_ADDR (VCBUS: 0x143c) (Unused)

Bit(s)	R/W	Default	Description
12:0	R/W	0	LCD_PWM0_LO_ADDR LOW Count

Table 8-3215 L_LCD_PWM0_HI_ADDR (VCBUS: 0x143d) (Unused)

Bit(s)	R/W	Default	Description
12:0	R/W	0	LCD_PWM0_HI_ADDR High Count

Table 8-3216 L_LCD_PWM1_LO_ADDR (VCBUS: 0x143e) (Unused)

Bit(s)	R/W	Default	Description
12:0	R/W	0	LCD_PWM0_LO_ADDR LOW Count

Table 8-3217 L_LCD_PWM1_HI_ADDR (VCBUS: 0x143f) (Unused)

Bit(s)	R/W	Default	Description
12:0	R/W	0	LCD_PWM0_HI_ADDR High Count

Table 8-3218 L_INV_CNT_ADDR (VCBUS: 0x1440)

Bit(s)	R/W	Default	Description
4	R/W	0	INV_EN Invert data output Enable
3:0	R/W	0	INV_CNT Invert data count define

Table 8-3219 L_TCON_MISC_SEL_ADDR (VCBUS: 0x1441)

Bit(s)	R/W	Default	Description
12	R/W	0	STH2_SEL STH2 is Line signal
11	R/W	0	STH1_SEL STH1 is Line Signal

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Bit(s)	R/W	Default	Description
10	R/W	0	OEH_SEL OEH is Line Signal
9	R/W	0	VCOM_SEL VCOM is Line Signal
8	R/W	0	DB_LINE_SW VCOM switched double lines
7	R/W	0	CPV2_SEL CPV2 is Line Signal
6	R/W	0	CPV1_SEL CPV1 is Line Signal
5	R/W	1	STV2_SEL STV2 is frame Signal
4	R/W	1	STV1_SEL STV1 is frame Signal
3	R/W	1	OEV_UNITE OEV3/2/1 look like one signal
2	R/W	0	OEV3_SEL OEV3 is Line Signal
1	R/W	0	OEV2_SEL OEV2 is Line Signal
0	R/W	0	OEV1_SEL OEV1 is Line Signal

Table 8-3220 L_DUAL_PORT_CNTL_ADDR (VCBUS: 0x1442)

Bit(s)	R/W	Default	Description
15	R/W	0	OUTPUT_YUV
14:12	R.W	0	IDF
11:9	R/W	0	ISF
8	R/W	0	LCD_ANALOG_SEL_CPH3 CPH3/INVT pin select
7	R/W	0	LCD_ANALOG_3PHI_CLK_SEL CPH3/2/1 Output 3 phases
6	R/W	0	LCD_LVDS_SEL54
5	R/W	0	LCD_LVDS_SEL27
4	R/W	0	LCD_TTL_SEL
3	R/W	0	DUAL_LVDS_EN
2	R/W	0	PORT_SWP
1	R/W	0	RGB_SWP R/B Swapped in data path
0	R/W	0	BIT_SWP Bit Swap in data path(7:0à0:7)

Table 8-3221 MLVDS_CLK_CTL1_HI (VCBUS: 0x1443) (Unused in Athena)

Bit(s)	Default	Descrip-tion	R/W
15-0	0	Ask Chen Shi	R/W

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Table 8-3222 MLVDS_CLK_CTL1_LO (VCBUS: 0x1444) (Unused)

Bit(s)	R/W	Default	Description
31-30	R/W	0	Enable mlvds clocks
24	R/W	0	Mlvds_clk_half_delay
23:0	R/W	0	Mlvds_clk_pattern

Table 8-3223 L_TCON_DOUBLE_CTL (VCBUS: 0x1449)

Bit(s)	R/W	Default	Description
16	W	0	tcon_double_reset
15:8	R/W	0	tcon_double_ini
7:0	R/w	0	tcon_double_inv

Table 8-3224 L_TCON_PATTERN_HI (VCBUS: 0x144a)

Bit(s)	R/W	Default	Description
15:0	R/W	0	tcon_pattern[31:16]

Table 8-3225 L_TCON_PATTERN_LO (VCBUS: 0x144b)

Bit(s)	R/W	Default	Description
15:0	R/W	0	tcon_pattern[15:0]

Table 8-3226 LDIM_BL_ADDR_PORT (VCBUS: 0x144e)

Bit(s)	R/W	Default	Description
	R/W	0	LDIM_BL_ADDR_PORT N/A

Table 8-3227 LDIM_BL_DATA_PORT (VCBUS: 0x144f)

Bit(s)	R/W	Default	Description
	R/W	0	LDIM_BL_DATA_PORT N/A

Table 8-3228 L_DE_HS_ADDR (VCBUS: 0x1451)

Bit(s)	R/W	Default	Description
15:14	R/W	0	enable_tcon_double[7:6]enable_tcon_double[7:6]
12:0	R/W	0	de_hsDE Horizontal Start

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Table 8-3229 L_DE_HE_ADDR (VCBUS: 0x1452)

Bit(s)	R/W	Default	Description
15:14	R/W	0	enable_tcon_double[5:4]enable_tcon_double[5:4]
12:0	R/W	0	DE_HE_ADDR DE Horizontal End

Table 8-3230 L_DE_VS_ADDR (VCBUS: 0x1453)

Bit(s)	R/W	Default	Description
15:14	R/W	0	enable_tcon_double[3:2]enable_tcon_double[3:2]
12:0	R/W	0	DE_VS_ADDR DE Vertical Start

Table 8-3231 L_DE_VE_ADDR (VCBUS: 0x1454)

Bit(s)	R/W	Default	Description
15:14	R/W	0	enable_tcon_double[1:0]enable_tcon_double[1:0]
12:0	R/W	0	DE_VE_ADDR DE Vertical End

Table 8-3232 L_HSYNC_HS_ADDR (VCBUS: 0x1455)

Bit(s)	R/W	Default	Description
12:0	R/W	0	HSYNC_HS_ADDR HSYNC Horizontal Start

Table 8-3233 L_HSYNC_HE_ADDR (VCBUS: 0x1456)

Bit(s)	R/W	Default	Description
12:0	R/W	0	HSYNC_HE_ADDR HSYNC Horizontal End

Table 8-3234 L_HSYNC_VS_ADDR (VCBUS: 0x1457)

Bit(s)	R/W	Default	Description
12:0	R/W	0	HSYNC_VS_ADDR HSYNC Vertical Start

Table 8-3235 L_HSYNC_VE_ADDR (VCBUS: 0x1458)

Bit(s)	R/W	Default	Description
12:0	R/W	0	HSYNC_VE_ADDR HSYNC Vertical End

Table 8-3236 L_VSYNC_HS_ADDR (VCBUS: 0x1459)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VSYNC_HS_ADDR VSYNC Horizontal Start

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Table 8-3237 L_VSYNC_HE_ADDR (VCBUS: 0x145a)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VSYNC_HE_ADDR VSYNC Horizontal End

Table 8-3238 L_VSYNC_VS_ADDR (VCBUS: 0x145b)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VSYNC_VS_ADDR VSYNC Vertical Start

Table 8-3239 L_VSYNC_VE_ADDR (VCBUS: 0x145c)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VSYNC_VE_ADDR VSYNC Vertical End

Table 8-3240 L_LCD_MCU_CTL (VCBUS: 0x145d)

Bit(s)	R/W	Default	Description
11:10	R/W	0	MCU_SOURCEmcu_source
9	R/W	0	DISABLE_GAMMA_10BIf true, disable GAMMA 10bit input resolution(8bit input)
8	R/W	0	VFIFO_MCU_ENABLE Read data from viu vfifo output
7	R/W	0	HALT_VS_DE0: after vsync, halt the encp venc 1: after vsync and DE, halt encp venc
6	R/W	0	R8G8B8_FORMATIf true, RGB888
5	R/W	0	R6G6B6_FORMATIf true RGB666
4	R/W	0	R5G6B5_FORMATIf true RGB565
3	R/W	0	DAC_DITH_SELIf true, dither result is selected to VDAC
2	R	0	LCD_MCU_ENABLE_DERead only
1	R	0	LCD_MCU_ENABLE_VSYNCRead only
0	R/W	0	LCD_MCU_ENABLELCD MCU interface enable

8.2.3.51.3 Gamma Control

Table 8-3241 GAMMA_CNTL_PORT (VCBUS: 0x1480)

Bit(s)	R/W	Default	Description
7	R/W	0	GAMMA_VCOM_POLReverse VCOM Polarity
6	R/W	0	GAMMA_RVS_OUTGAMMA DATA REVERSE OUTPUT FOLLOWING VCOM
5	R	0	ADR_RDYGAMMA ADDR PORT Is Ready for access
4	R	0	WR_RDY GAMMA DATA PORT IS RDY to Write
3	R	0	RD_RDY GAMMA DATA PORT IS RDY to Read
2	R/W	0	GAMMA_TRRGB10–>RGB8 using Trancate or Round off

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Bit(s)	R/W	Default	Description
1	R/W	0	GAMMA_SETGamma turn on syncing with Vsync
0	R/W	0	GAMMA_ENGamma Enable

Table 8-3242 GAMMA_DATA_PORT (VCBUS: 0x1481)

Bit(s)	R/W	Default	Description
15:0	-	0	GAMMA_DATA_PORTGAMMA DATA PORT ADDR, all gamma data read/write operation is via this port

Table 8-3243 GAMMA_ADDR_PORT (VCBUS: 0x1482)

Bit(s)	R/W	Default	Description
12	R/W	0	H-RDGAMMA is READY for Host to read/write
11	R/W	0	H-AUTO_INCBURST MODE *
10	R/W	0	H_SEL_RGAMMA R Selected
9	R/W	0	H_SEL_GGAMMA G Selected
8	R/W	0	H_SEL_BGAMMA B Selected
7:0	W	0	HADRGAMMA ADDR.

Note

↑When Programming the Gamma with Burst Mode, please turn off the IRQ service (Or make sure the IRQ service task have no Gamma programming operation)

Table 8-3244 GAMMA_VCOM_HSWITCH_ADDR (VCBUS: 0x1483)

Bit(s)	R/W	Default	Description
12:0	R/W	0	GAMMA_VCOM_HSWITCH_ADDRHorizontal Switch Point for VCOM with Gamma

Table 8-3245 RGB_BASE_ADDR (VCBUS: 0x1485)

Bit(s)	R/W	Default	Description
13:10	R/W	0	rasp_ctrl bit3 : raspberry pi dpi en bit2:0 : raspberry pi dpi rgb mode
9:0	R/W	0	RGB_BASE_ADDR RGB Converter OFFSET

Table 8-3246 RGB_COEFF_ADDR (VCBUS: 0x1486)

Bit(s)	R/W	Default	Description
10:0	R/W	0	RGB_COEFF_ADDR RGB Converter SCALE

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Table 8-3247 POL_CNTL_ADDR (VCBUS: 0x1487)

Bit(s)	R/W	Default	Description
15:14	R/W	0	DCLK_SELDCLK output
11	R/W	0	TCON_VSYNC_SEL_DVI VSYNC generated by TCON for RGB format DVI output (support Digital LCD interface)
10	R/W	0	TCON_HSYNC_SEL_DVI HSYNC generated by TCON for RGB format DVI output (support Digital LCD interface)
9	R/W	0	TCON_DE_SEL_DVI DE generated by TCON for RGB format DVI output (support Digital LCD interface)
8	R/W	0	CPH3_POL CPH3 Polarity Control
7	R/W	0	CPH2_POL CPH2 Polarity Control
6	R/W	0	CPH1_POL CPH1 Polarity Control
5	R/W	0	TCON_DE_SEL DE Generate by TCON
4	R/W	0	TCON_VS_SEL Vsync Generate by TCON
3	R/W	0	TCON_HS_SEL Hsync Generate by TCON
2	R/W	0	DE_POL De polarity control
1	R/W	0	VS_POL Vsync polarity control
0	R/W	0	HS_POL Hsync polarity control

Table 8-3248 DITH_CNTL_ADDR (VCBUS: 0x1488)

Bit(s)	R/W	Default	Description
13	R/W	0	DITH_R51=use the current dithering to dither input 10-bit R down to 5-bit. Applicable only if both dith10_en & dith8_en = 1, otherwise no effect
12	R/W	0	DITH_G51=use the current dithering to dither input 10-bit G down to 5-bit. Applicable only if both dith10_en & dith8_en = 1, otherwise no effect
11	R/W	0	DITH_B51=use the current dithering to dither input 10-bit B down to 5-bit. Applicable only if both dith10_en & dith8_en = 1, otherwise no effect
10	R/W	0	DITH10_EN 10-bits Dithering to 8 bits Enable
9	R/W	0	DITH8_EN 8-bits Dithering to 6 bits Enable
8	R/W	0	DITH_MD Dithering Mode select
7:4	R/W	0	DITH8_CNTL 8-bits Dithering control
3:0	R/W	0	DITH10_CNTL 10-bits Dithering control

Table 8-3249 GAMMA_PROBE_CTRL (VCBUS: 0x1489)

Bit(s)	R/W	Default	Description
1	R/W	0	HIGHLIGHT_EN
0	R/W	0	PROBE_EN

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Table 8-3250 GAMMA_PROBE_COLOR_L (VCBUS: 0x148a)

Bit(s)	R/W	Default	Description
15:0	R	0	GAMMA_PROBE_COLOR_L Probe_pix[15:0]

Table 8-3251 GAMMA_PROBE_COLOR_H (VCBUS: 0x148b)

Bit(s)	R/W	Default	Description
15	R	0	PROBE_PIX_Vprobe_pix_v
14	R	0	reservereserve
13:0	R	0	GAMMA_PROBE_COLOR_H Probe_pix[29:16]

Table 8-3252 GAMMA_PROBE_HL_COLOR (VCBUS: 0x148c)

Bit(s)	R/W	Default	Description
15:0	R	0	GAMMA_PROBE_HL_COLOR 5:6:5 color

Table 8-3253 GAMMA_PROBE_POS_X (VCBUS: 0x148d)

Bit(s)	R/W	Default	Description
12:0	R/W	0	GAMMA_PROBE_POS_X Pos_x

Table 8-3254 GAMMA_PROBE_POS_Y (VCBUS: 0x148e)

Bit(s)	R/W	Default	Description
12:0	R/W	0	GAMMA_PROBE_POS_Y Pos_y

8.2.3.51.4 TCON Control

Table 8-3255 STH1_HS_ADDR (VCBUS: 0x1490)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH1_HS_ADDR STH1 Horizontal Start

Table 8-3256 STH1_HE_ADDR (VCBUS: 0x1491)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH1_HE_ADDR STH1 Horizontal End

Table 8-3257 STH1_VS_ADDR (VCBUS: 0x1492)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH1_VS_ADDR STH1 Vertical Start

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Table 8-3258 STH1_VE_ADDR (VCBUS: 0x1493)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH1_VE_ADDR STH1 Vertical End

Table 8-3259 STH2_HS_ADDR (VCBUS: 0x1494)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH2_HS_ADDR STH2 Horizontal Start

Table 8-3260 STH2_HE_ADDR (VCBUS: 0x1495)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH2_HE_ADDR STH2 Horizontal End

Table 8-3261 STH2_VS_ADDR (VCBUS: 0x1496)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH2_VS_ADDR STH2 Vertical Start

Table 8-3262 STH2_VE_ADDR (VCBUS: 0x1497)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STH2_VE_ADDR STH2 Vertical End

Table 8-3263 OEH_HS_ADDR (VCBUS: 0x1498)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEH_HS_ADDR OEH Horizontal Start

Table 8-3264 OEH_HE_ADDR (VCBUS: 0x1499)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEH_HE_ADDR OEH Horizontal End

Table 8-3265 OEH_VS_ADDR (VCBUS: 0x149a)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEH_VS_ADDR OEH Vertical Start

Table 8-3266 OEH_VE_ADDR (VCBUS: 0x149b)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEH_VE_ADDR OEH Vertical End

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Table 8-3267 VCOM_HSWITCH_ADDR (VCBUS: 0x149c)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VCOM_HSWITCH_ADDR VCOM Horizontal Switch Point

Table 8-3268 VCOM_VS_ADDR (VCBUS: 0x149d)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VCOM_VS_ADDR VCOM Vertical Start

Table 8-3269 VCOM_VE_ADDR (VCBUS: 0x149e)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VCOM_VE_ADDR VCOM Vertical End

Table 8-3270 CPV1_HS_ADDR (VCBUS: 0x149f)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV1_HS_ADDR CPV1 Horizontal Start

Table 8-3271 CPV1_HE_ADDR (VCBUS: 0x14a0)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV1_HE_ADDR CPV1 Horizontal End

Table 8-3272 CPV1_VS_ADDR (VCBUS: 0x14a1)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV1_VS_ADDR CPV1 Vertical Start

Table 8-3273 CPV1_VE_ADDR (VCBUS: 0x14a2)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV1_VE_ADDR CPV1 Vertical End

Table 8-3274 CPV2_HS_ADDR (VCBUS: 0x14a3)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV2_HS_ADDR CPV2 Horizontal Start

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Table 8-3275 CPV2_HE_ADDR (VCBUS: 0x14a4)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV2_HE_ADDR CPV2 Horizontal End

Table 8-3276 CPV2_VS_ADDR (VCBUS: 0x14a5)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV2_VS_ADDR CPV2 Vertical Start

Table 8-3277 CPV2_VE_ADDR (VCBUS: 0x14a6)

Bit(s)	R/W	Default	Description
12:0	R/W	0	CPV2_VE_ADDR CPV2 Vertical End

Table 8-3278 STV1_HS_ADDR (VCBUS: 0x14a7)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV1_HS_ADDR STV1 Horizontal Start

Table 8-3279 STV1_HE_ADDR (VCBUS: 0x14a8)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV1_HE_ADDR STV1 Horizontal End

Table 8-3280 STV1_VS_ADDR (VCBUS: 0x14a9)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV1_VS_ADDR STV1 Vertical Start

Table 8-3281 STV1_VE_ADDR (VCBUS: 0x14aa)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV1_VE_ADDR STV1 Vertical End

Table 8-3282 STV2_HS_ADDR (VCBUS: 0x14ab)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV2_HS_ADDR STV2 Horizontal Start

Table 8-3283 STV2_HE_ADDR (VCBUS: 0x14ac)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV2_HE_ADDR STV2 Horizontal End

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Table 8-3284 STV2_VS_ADDR (VCBUS: 0x14ad)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV2_VS_ADDR STV2 Vertical Start

Table 8-3285 STV2_VE_ADDR (VCBUS: 0x14ae)

Bit(s)	R/W	Default	Description
12:0	R/W	0	STV2_VE_ADDR STV2 Vertical End

Table 8-3286 OEV1_HS_ADDR (VCBUS: 0x14af)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV1_HS_ADDR OEV1 Horizontal Start

Table 8-3287 OEV1_HE_ADDR (VCBUS: 0x14b0)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV1_HE_ADDR OEV1 Horizontal End

Table 8-3288 OEV1_VS_ADDR (VCBUS: 0x14b1)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV1_VS_ADDR OEV1 Vertical Start

Table 8-3289 OEV1_VE_ADDR (VCBUS: 0x14b2)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV1_VE_ADDR OEV1 Vertical End

Table 8-3290 OEV2_HS_ADDR (VCBUS: 0x14b3)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV2_HS_ADDR OEV2 Horizontal Start

Table 8-3291 OEV2_HE_ADDR (VCBUS: 0x14b4)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV2_HE_ADDR OEV2 Horizontal End

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Table 8-3292 OEV2_VS_ADDR (VCBUS: 0x14b5)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV2_VS_ADDR OEV2 Vertical Start

Table 8-3293 OEV2_VE_ADDR (VCBUS: 0x14b6)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV2_VE_ADDR OEV2 Vertical End

Table 8-3294 OEV3_HS_ADDR (VCBUS: 0x14b7)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV3_HS_ADDR OEV3 Horizontal Start

Table 8-3295 OEV3_HE_ADDR (VCBUS: 0x14b8)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV3_HE_ADDR OEV3 Horizontal End

Table 8-3296 OEV3_VS_ADDR (VCBUS: 0x14b9)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV3_VS_ADDR OEV3 Vertical Start

Table 8-3297 OEV3_VE_ADDR (VCBUS: 0x14ba)

Bit(s)	R/W	Default	Description
12:0	R/W	0	OEV3_VE_ADDR OEV3 Vertical End

Table 8-3298 LCD_PWR_ADDR (VCBUS: 0x14bb) (Unused)

Bit(s)	R/W	Default	Description
5	R/W	0	LCD_VDD LCD VDD control
4	R/W	0	LCD_VBL LCD VBL control
3:0	R/W	0	LCD_GPIO LCD GPIO control

Table 8-3299 LCD_PWM0_LO_ADDR (VCBUS: 0x14bc) (Unused)

Bit(s)	R/W	Default	Description
12:0	R/W	0	LCD_PWM0_LO_ADDR LOW Count

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Table 8-3300 LCD_PWM0_HI_ADDR (VCBUS: 0x14bd) (Unused)

Bit(s)	R/W	Default	Description
12:0	R/W	0	LCD_PWM0_HI_ADDR High Count

Table 8-3301 LCD_PWM1_LO_ADDR (VCBUS: 0x14be) (Unused)

Bit(s)	R/W	Default	Description
12:0	R/W	0	LCD_PWM0_LO_ADDR LOW Count

Table 8-3302 LCD_PWM1_HI_ADDR (VCBUS: 0x14bf) (Unused)

Bit(s)	R/W	Default	Description
12:0	R/W	0	LCD_PWM0_HI_ADDR High Count

Table 8-3303 INV_CNT_ADDR (VCBUS: 0x14c0)

Bit(s)	R/W	Default	Description
4	R/W	0	INV_EN Invert data output Enable
3:0	R/W	0	INV_CNT Invert data count define

Table 8-3304 TCON_MISC_SEL_ADDR (VCBUS: 0x14c1)

Bit(s)	R/W	Default	Description
12	R/W	0	STH2_SEL STH2 is Line signal
11	R/W	0	STH1_SEL STH1 is Line Signal
10	R/W	0	OEH_SEL OEH is Line Signal
9	R/W	0	VCOM_SEL VCOM is Line Signal
8	R/W	0	DB_LINE_SW VCOM switched double lines
7	R/W	0	CPV2_SEL CPV2 is Line Signal
6	R/W	0	CPV1_SEL CPV1 is Line Signal
5	R/W	1	STV2_SEL STV2 is frame Signal
4	R/W	1	STV1_SEL STV1 is frame Signal
3	R/W	1	OEV_UNITE OEV3/2/1 look like one signal
2	R/W	0	OEV3_SEL OEV3 is Line Signal
1	R/W	0	OEV2_SEL OEV2 is Line Signal
0	R/W	0	OEV1_SEL OEV1 is Line Signal

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Table 8-3305 DUAL_PORT_CNTL_ADDR (VCBUS: 0x14c2)

Bit(s)	R/W	Default	Description
15	R/W	0	OUTPUT_YUV
14:12	R.W	0	IDF
11:9	R/W	0	ISF
8	R/W	0	LCD_ANALOG_SEL_CPH3 CPH3/INVT pin select
7	R/W	0	LCD_ANALOG_3PHI_CLK_SEL CPH3/2/1 Output 3 phases
6	R/W	0	LCD_LVDS_SEL54
5	R/W	0	LCD_LVDS_SEL27
4	R/W	0	LCD_TTL_SEL
3	R/W	0	DUAL_LVDS_EN
2	R/W	0	PORT_SWP
1	R/W	0	RGB_SWP R/B Swapped in data path
0	R/W	0	BIT_SWP Bit Swap in data path(7:0à0:7)

Table 8-3306 LVDS_PACK_CNTL_ADDR (VCBUS: 0x14d0)

Bit(s)	R/W	Default	Description
15:14	R/W	0	G_SELECT0:R, 1:G, 2:B, 3:0
13:12	R/W	0	G_SELECT0:R, 1:G, 2:B, 3:0
11:10	R/W	0	R_SELECT0:R, 1:G, 2:B, 3:0
9:8	R/W	0	BIT_SIZE0:10bits, 1:8bits, 2:6bits, 3:4bits
7	R/W	0	LVDS_USE_TCONhs, vs, de controlled by tcon
6	R/W	0	LVDS_DUALdual port mapping
5	R/W	0	PN_SWPpn_swp
4	R/W	0	LSB_FIRST0:MSB first, 1:LSB first
3	R/W	0	LVDS_RESVinvert hs and vs
2	R/W	0	ODD_EVEN_SWP0:normal, 1:swap
1:0	R/W	0	LVDS_rEPACK0:normal, 1,2:repack

Table 8-3307 DE_HS_ADDR (VCBUS: 0x14d1)

Bit(s)	R/W	Default	Description
12:0	R/W	0	DE_HS_ADDR DE Horizontal Start

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Table 8-3308 DE_HE_ADDR (VCBUS: 0x14d2)

Bit(s)	R/W	Default	Description
12:0	R/W	0	DE_HE_ADDR DE Horizontal End

Table 8-3309 DE_VS_ADDR (VCBUS: 0x14d3)

Bit(s)	R/W	Default	Description
12:0	R/W	0	DE_VS_ADDR DE Vertical Start

Table 8-3310 DE_VE_ADDR (VCBUS: 0x14d4)

Bit(s)	R/W	Default	Description
12:0	R/W	0	DE_VE_ADDR DE Vertical End

Table 8-3311 HSYNC_HS_ADDR (VCBUS: 0x14d5)

Bit(s)	R/W	Default	Description
12:0	R/W	0	HSYNC_HS_ADDR HSYNC Horizontal Start

Table 8-3312 HSYNC_HE_ADDR (VCBUS: 0x14d6)

Bit(s)	R/W	Default	Description
12:0	R/W	0	HSYNC_HE_ADDR HSYNC Horizontal End

Table 8-3313 HSYNC_VS_ADDR (VCBUS: 0x14d7)

Bit(s)	R/W	Default	Description
12:0	R/W	0	HSYNC_VS_ADDR HSYNC Vertical Start

Table 8-3314 HSYNC_VE_ADDR (VCBUS: 0x14d8)

Bit(s)	R/W	Default	Description
12:0	R/W	0	HSYNC_VE_ADDR HSYNC Vertical End

Table 8-3315 VSYNC_HS_ADDR (VCBUS: 0x14d9)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VSYNC_HS_ADDR VSYNC Horizontal Start

Table 8-3316 VSYNC_HE_ADDR (VCBUS: 0x14da)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VSYNC_HE_ADDR VSYNC Horizontal End

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Table 8-3317 VSYNC_VS_ADDR (VCBUS: 0x14db)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VSYNC_VS_ADDR VSYNC Vertical Start

Table 8-3318 VSYNC_VE_ADDR (VCBUS: 0x14dc)

Bit(s)	R/W	Default	Description
12:0	R/W	0	VSYNC_VE_ADDR VSYNC Vertical End

Table 8-3319 LCD_MCU_CTL (VCBUS: 0x14dd)

Bit(s)	R/W	Default	Description
11:10	R/W	0	MCU_SOURCEmcu_source
9	R/W	0	DISABLE_GAMMA_10BIf true, disable GAMMA 10bit input resolution(8bit input)
8	R/W	0	VFIFO_MCU_ENABLE Read data from viu vfifo output
7	R/W	0	HALT_VS_DE0: after vsync, halt the encp venc 1: after vsync and DE, halt encp venc
6	R/W	0	R8G8B8_FORMATIf true, RGB888
5	R/W	0	R6G6B6_FORMATIf true RGB666
4	R/W	0	R5G6B5_FORMATIf true RGB565
3	R/W	0	DAC_DITH_SELIf true, dither result is selected to VDAC
2	R	0	LCD_MCU_ENABLE_DERead only
1	R	0	LCD_MCU_ENABLE_VSYNCRead only
0	R/W	0	LCD_MCU_ENABLELCD MCU interface enable

Table 8-3320 LCD_MCU_DATA_0 (VCBUS: 0x14de)

Bit(s)	R/W	Default	Description
15-0	R	0	If RGB565 mode 15:11 R, 10:5 G, 4:0 B Else If (RGB666 or RGB888) 15:8 G, 7:0 R Else 14:10 G[4:0], 9:0 R

Table 8-3321 LCD_MCU_DATA_1 (VCBUS: 0x14de)

Bit(s)	R/W	Default	Description
15-0	R	0	If (RGB666 or RGB888) 15:8 G, 7:0 B Else 14:10 G[9:5], 9:0 B

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8.2.3.52 LVDS Registers

Table 8-3322 LVDS_GEN_CNTL (VCBUS: 0x14e0)

Bit(s)	R/W	Default	Description
15	R/W	0	Mini LVDS
14-11	R/W	0	unused
10:6	R/W	0	LVDS_PHY_MODE,unused
5-4	R/W	0	LVDS_FIFO_CLK_SEL: 0 = PHY clock divided by 6 1 = PHY clock divided by 7 2 = PHY clock divided by 8
3	R/W	0	FIFO_WR_EN: Set to 1 to enable the LVDS FIFO
2	R/W	0	FIFO_WR_BIST_GATE: This bit is used during production test to enable the FIFO writes at a particular time to ensure the data sent is consistent from run to run
1-0	R/W	0	FIFO_WR_MODE: 0 = FIFO disabled 1 = FIFO enabled 2 = FIFO write = DCLK / 2 3 = FIFO write = signal control (preferred)

Table 8-3323 LVDS_PHY_CNTL0 (VCBUS: 0x14e1) (Unused)

Bit(s)	R/W	Default	Description
15-11	R/W	0	unused
10	R/W	0	LVDS_CLKSEL_CHCK
9	R/W	0	LVDS_CLKSEL_CH5
8	R/W	0	LVDS_CLKSEL_CH4
7	R/W	0	LVDS_CLKSEL_CH3
6	R/W	0	LVDS_CLKSEL_CH2
5	R/W	0	LVDS_CLKSEL_CH1
4	R/W	0	LVDS_CLKSEL_CH0
3-0	R/W	0	LVDS_CTL_REG

Table 8-3324 LVDS_PHY_CNTL1 (VCBUS: 0x14e2) (Unused)

Bit(s)	R/W	Default	Description
15	R/W	0	unused
14	R/W	0	LVDS_PDN_CHCK
13	R/W	0	LVDS_PDN_CH5
12	R/W	0	LVDS_PDN_CH4
11	R/W	0	LVDS_PDN_CH3
10	R/W	0	LVDS_PDN_CH2

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Bit(s)	R/W	Default	Description
9	R/W	0	LVDS_PDN_CH1
8	R/W	0	LVDS_PDN_CH0
7	R/W	0	Unused
6	R/W	0	LVDS_DININV_CHCK
5	R/W	0	LVDS_DININV_CH5
4	R/W	0	LVDS_DININV_CH4
3	R/W	0	LVDS_DININV_CH3
2	R/W	0	LVDS_DININV_CH2
1	R/W	0	LVDS_DININV_CH1
0	R/W	0	LVDS_DININV_CH0

Table 8-3325 LVDS_PHY_CNTL2 (VCBUS: 0x14e3) (Unused)

Bit(s)	R/W	Default	Description
15-0	R/W	0	unused

Table 8-3326 LVDS_PHY_CNTL3 (VCBUS: 0x14e4) (Unused)

Bit(s)	R/W	Default	Description
15-12	R/W	0	unused
11-0	R/W	0	LVDS_COMMON

Table 8-3327 LVDS_PHY_CNTL4 (VCBUS: 0x14e5) (Unused)

Bit(s)	R/W	Default	Description
15-7	R/W	0	unused
6-0	R/W	0	LVDS_MDR_PU

Table 8-3328 LVDS_PHY_CNTL5 (VCBUS: 0x14e6) (Unused)

Bit(s)	R/W	Default	Description
15-11	R/W	0	LVDS_REFCTL
10-8	R/W	0	LVDS_VCM_CTL
7-4	R/W	0	LVDS_SWING_CTL
3-0	R/W	0	LVDS_PREM_CTL

Table 8-3329 LVDS_SRG_TEST (VCBUS: 0x14e8)) (Unused)

Bit(s)	R/W	Default	Description
12-0	R	0	unused

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Table 8-3330 LVDS_BIST_MUX0 (VCBUS: 0x14e9)

Bit(s)	R/W	Default	Description
15-14	R/W	0	B2 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED2 (see below)
13-12	R/W	0	B1 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED1 (see below)
11-10	R/W	0	B0 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED0 (see below)
9-8	R/W	0	A4 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED0 (see below)
7-6	R/W	0	A3 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED3 (see below)
5-4	R/W	0	A2 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED2 (see below)
3-2	R/W	0	A1 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED1 (see below)
1-0	R/W	0	A0 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED0 (see below)

Table 8-3331 LVDS_BIST_MUX1 (VCBUS: 0x14ea)

Bit(s)	R/W	Default	Description
15-8	R/W	0	unused
7-6	R/W	0	CLKB BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED3 (see below)
5-4	R/W	0	CLKA BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED2 (see below)

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Bit(s)	R/W	Default	Description
3-2	R/W	0	B4 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED0 (see below)
1-0	R/W	0	B3 BIST MUX 0 = pre-select random LFSR selection 1 = LFSR[8:1] 2 = LVDS_BIST_FIXED3 (see below)

Table 8-3332 LVDS_BIST_FIXED0 (VCBUS: 0x14eb)

Bit(s)	R/W	Default	Description
15-8	R/W	0	LVDS_BIST_FIXED1
7-0	R/W	0	LVDS_BIST_FIXED0

Table 8-3333 LVDS_BIST_FIXED1 (VCBUS: 0x14ec)

Bit(s)	R/W	Default	Description
15-8	R/W	0	LVDS_BIST_FIXED3
7-0	R/W	0	LVDS_BIST_FIXED2

Table 8-3334 LVDS_BIST_CNTL0 (VCBUS: 0x14ed)

Bit(s)	R/W	Default	Description
15-7	R	0	unused
6	R/W	0	LVDS_BIST_DUAL_PIXEL
5-1	R/W	0	LVDS_BIST_TAP_SEL: This selects the LFSR polynomial
0	R/W	0	BIST_EN

Table 8-3335 LVDS_CLKB_CLKA (VCBUS: 0x14ee)

Bit(s)	R/W	Default	Description
15	R/W	0	0
14-8	R/W	0x63	CLKB LVDS value shifted out serially
7	R/W	0	0
6-0	R/W	0x63	CLKA LVDS value shifted out serially

Table 8-3336 LVDS_PHY_CLK_CNTL (VCBUS: 0x14ef)

Bit(s)	R/W	Default	Description
15	R/W	0	LVDS_DIV_RESET_N
14	R/W	0	LVDS_DIV_SOFT_RESET
13-0	R/W	0	unused

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Table 8-3337 LVDS_SER_EN (VCBUS: 0x14f0)

Bit(s)	R/W	Default	Description
15-0	R/W	0	unused

Table 8-3338 LVDS_PHY_CNTL6 (VCBUS: 0x14f1)

Bit(s)	R/W	Default	Description
15-0	R/W	0	unused

Table 8-3339 LVDS_PHY_CNTL7 (VCBUS: 0x14f2)

Bit(s)	R/W	Default	Description
15-0	R/W	0	Unused

Table 8-3340 LVDS_PHY_CNTL8 (VCBUS: 0x14f3)

Bit(s)	R/W	Default	Description
15-0	R/W	0	unused

Table 8-3341 MLVDS_CLK_CTL_HI (VCBUS: 0x14f4)

Bit(s)	R/W	Default	Description
15-0	R/W	0	

Table 8-3342 MLVDS_CLK_CTL_LO (VCBUS: 0x14f5)

Bit(s)	R/W	Default	Description
15-0	R/W	0	

Table 8-3343 MLVDS_DUAL_GATE_WR_START(VCBUS: 0x14f6)

Bit(s)	R/W	Default	Description
15-0	R/W	0	

Table 8-3344 MLVDS_DUAL_GATE_WR_END(VCBUS: 0x14f7)

Bit(s)	R/W	Default	Description
15-0	R/W	0	

Table 8-3345 MLVDS_DUAL_GATE_RD_START(VCBUS: 0x14f8)

Bit(s)	R/W	Default	Description
15-0	R/W	0	

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Table 8-3346 MLVDS_DUAL_GATE_RD_END(VCBUS: 0x14f9)

Bit(s)	R/W	Default	Description
15-0	R/W	0	

Table 8-3347 MLVDS_SECOND_RESET_CTL(VCBUS: 0x14fa)

Bit(s)	R/W	Default	Description
15-0	R/W	0	

Table 8-3348 MLVDS_GATE_CTL_HI (VCBUS: 0x14fb)

Bit(s)	R/W	Default	Description
15-0	R/W	0	mlvds_dual_gate_ctl[31:16]

Table 8-3349 MLVDS_GATE_CTL_LO (VCBUS: 0x14fc)

Bit(s)	R/W	Default	Description
15-0	R/W	0	mlvds_dual_gate_ctl[15:0]

8.2.3.53 Video Encoder Registers

Table 8-3350 ENCI_VIDEO_MODE(VCBUS: 0x1b00)

Bit(s)	R/W	Default	Description
7	R/W	0	Enci_video_mode_pal_ln309_avonFix the bug pal video is not on some fields of ln309
6	R/W	0	ENCI_VIDEO_MODE_SQPX Square pixel mode
5-4	R/W	0	ENCI_VIDEO_MODE_FSCSEL 0: Fc=3.5795M, 1:Fsc=4.4336M, 2: Fsc= 3.5756M, 3:Fsc=3.582M.
3	R/W	0	ENCI_VIDEO_MODE_PED 0: No pedestal, 1: Pedestal enable
2	R/W	0	ENCI_VIDEO_MODE_PDRST 0: Phase reset every field, 1: Phase not reset
1	R/W	0	ENCI_VIDEO_MODE_PHALT 0: Phase alternate disable, 1: Phase alternate enable
0	R/W	0	ENCI_VIDEO_MODE_LNFMT 0: 525 lines, 1:625 lines

Table 8-3351 ENCI_VIDEO_MODE_ADV(VCBUS: 0x1b01)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_VIDEO_MODE_EN_CSYNC_MACVEnable macrovision signal on CSYNC
14	R/W	0	CFG_LUMA_FOR_CVBS_HIGHLuma for CVBS high bandwidth
13	R/W	0	CFG_CHROMA_MOD_HIGHChroma modulation high bandwidth

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Bit(s)	R/W	Default	Description
12	R/W	0	CFG_LUMA_FOR_RGB_HIGHUsing high bandwidth luma for RGB
11	R/W	0	ENCI_VIDEO_MODE_EN_CMPN_HUE Enable component HUE matrix
10	R/W	0	ENCI_VIDEO_MODE_EN_HY_SYLPF Enable low pass filter on sync of high bandwidth Luma signal
11-10	R/W	0	ENCI_VIDEO_MODE_ADV_HY_SYLPF_SEL Low pass filter selection on sync of high bandwidth Luma signal. 0=Bypass LPF; 1=Use 3-tap LPF; 2=Use 5-tap LPF.
9	R/W	0	ENCI_VIDEO_MODE_EN_PMACV Enable macrovision on progressive mode of ENCI
8	R/W	0	ENCI_VIDEO_MODE_EN_PROG Enable progressive mode on itnerlace TV encoder
9 -8	R/W	0	ENCI_VIDEO_MODE_ADV_CMPT_BURST_WIN_SEL
7 -6	R/W	0	ENCI_VIDEO_MODE_ADV_CBW Chroma Filter bandwidth, 0:Low, 1:Medium, 2:Higher
5 -4	R/W	0	ENCI_VIDEO_MODE_ADV_YBW Luma Filter bandwidth , 0:Medium, 1:Low, 2: High
2	R/W	1	ENCI_VIDEO_MODE_ADV_VBICTL 0: Blank line end at line6 1: Blank line end at line17/22
1 -0	R/W	2'b10	ENCI_VIDEO_MODE_ADV_DMXMD Video input demux shifting mode, adjust this value When input video stream is shifted.

Table 8-3352 ENCI_VIDEO_FSC_ADJ(VCBUS: 0x1b02)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'd0	ENCI_VIDEO_FSC_ADJ FSC adjust , 16 bit value that can adjust FSC frequency (Add with setting value)

Table 8-3353 ENCI_VIDEO_BRIGHT(VCBUS: 0x1b03)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd0	ENCI_VIDEO_BRIGHT Brightness

Table 8-3354 ENCI_VIDEO_CONT(VCBUS: 0x1b04)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd0	ENCI_VIDEO_CONT Contrast

Table 8-3355 ENCI_VIDEO_SAT(VCBUS: 0x1b05)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd0	ENCI_VIDEO_SAT Saturation

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Table 8-3356 ENCI_VIDEO_HUE(VCBUS: 0x1b06)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd0	ENCI_VIDEO_HUE Hue

Table 8-3357 ENCI_VIDEO_SCH(VCBUS: 0x1b07)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd0	ENCI_VIDEO_SCH Sch adjust, adjust the phase of the FSC

Table 8-3358 ENCI_SYNC_MODE(VCBUS: 0x1b08)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd7	ENCI_SYNC_MODE Video input Synchronization mode Define Field sync/Vertical Sync/Horizontal Sync source, 0: FFF All from fsi (Not used in T25) 1: FVH From FSI, VSI, HSI respectively 2: FVV From FSI, VSI, VSI respectively 3: 0VV No FSI, Vsync/Hsync from VSI, HSI respectively 4: FVH_EAV From CCIR601 stream directly 5: FFF_EAV From CCIR601 stream, but use Field sync only 6: FMVH field rst when vs is near around line start. 7: MASTER Master mode, free run, send HSO/VSO out

Note

↑Suggest use Master mode only

Table 8-3359 ENCI_SYNC_HSO_BEGIN(VCBUS: 0x1b09)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd3	ENCI_SYNC_HSO_BEGIN HSO begin position

Table 8-3360 ENCI_SYNC_HSO_END(VCBUS: 0x1b0a)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd5	ENCI_SYNC_HSO_END HSO end postion

Table 8-3361 ENCI_SYNC_VSO_EVNLN(VCBUS: 0x1b0b)

Bit(s)	R/W	Default	Description
15 -8	R/W	8	ENCI_SYNC_VSO_EVN_STRTLN VSO output start line in even field
7 -0	R/W	8	ENCI_SYNC_VSO_EVN_ENDLN VSO output end line in even field

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Table 8-3362 ENCI_SYNC_VSO_ODDLN(VCBUS: 0x1b0e)

Bit(s)	R/W	Default	Description
15 -8	R/W	9	ENCI_SYNC_VSO_ODD_STRTLN VSO output start line in odd field
7 -0	R/W	9	ENCI_SYNC_VSO_ODD_ENDLN VSO output end line in odd field

Table 8-3363 ENCI_SYNC_CTRL(VCBUS: 0x1b0f)

Bit(s)	R/W	Default	Description
8	R/W	0	ENCI_SYNC_DE_V
7	R/W	0	ENCI_SYNC_VSO_INACTIVE_MODE
6	R/W	0	ENCI_SYNC_CTRL_VSOMD VSO position in odd field, 0:Half line after VSO in Even field, 1:Same as Even Field
5	R/W	0	ENCI_SYNC_CTRL_FSOP FSO polarity
4	R/W	0	ENCI_SYNC_CTRL_VSOP VSO polarity
3	R/W	0	ENCI_SYNC_CTRL_HSOP HSO polarity
2	R/W	1	ENCI_SYNC_CTRL_FSIP FSI polarity
1	R/W	0	ENCI_SYNC_CTRL_VSIP VSI polarity
0	R/W	0	ENCI_SYNC_CTRL_HSIP HSI polarity

Table 8-3364 ENCI_SYNC_HOFFST(VCBUS: 0x1b10)

Bit(s)	R/W	Default	Description
10 -0	R/W	2	ENCI_SYNC_HOFFST Horizontal offset after HSI in slave mode

Table 8-3365 ENCI_SYNC_VOFFST(VCBUS: 0x1b11)

Bit(s)	R/W	Default	Description
8 -0	R/W	0	ENCI_SYNC_VOFFST Horizontal offset aftet VSI in slave mode

Table 8-3366 ENCI_SYNC_ADJ(VCBUS: 0x1b12)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_SYNC_ADJ_EN Analog Synchronization and color burst value adjust enable
14 -10	R/W	0	ENCI_SYNC_ADJ_BU_V Analog V burst adjustment, Signed value
9 -5	R/W	0	ENCI_SYNC_ADJ_BU_U Analog U burst adjustment, Signed value
4 -0	R/W	0	ENCI_SYNC_ADJ_SYNC Analog sync adjustment, Signed value

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Table 8-3367 ENCI_RGB_SETTING(VCBUS: 0x1b13)

Bit(s)	R/W	Default	Description
12 -3	R/W	0	Reserved Resevered
2 -0	R/W	0	ENCI_RGB_SYNCAnalog RGB sync disable (1:Enable Sync, 0:Disable) , Bit2 for Blue, Bit1 for Green. Bit0 for Red

Table 8-3368 ENCI_CMPN_MATRIX_CB(VCBUS: 0x1b14)

Bit(s)	R/W	Default	Description
15:8	R/W	0	ENCP_VIDEO_MATRIX_maCbCr matrix parameter Ma (signed value)
7:0	R/W	0	ENCP_VIDEO_MATRIX_mBCbCr matrix parameter Mb (signed value)

Note

↑These 2 register is for HUE adjustment. The formula is : Cb' = Cb*Ma + Cr*Mb, Cr' = Cb*Mc + Cr*Md

(effective only when Enable HUE matrix by bit ENCI_VIDEO_MODE_ADV[11])

Suggest software prepare a table for MATRIX_CB/CR, The code to generate the table is illustrate here:

```
int tbl_matrix_cb[360];
int tbl_matrix_cr[360];

For(i=0;i<360;i++)
{
double cita = 2*3.14159*i/360;
int cosm = (int)(cos(cita)*128);
int sinm = (int)(sin(cita)*128);
tbl_matrix_cb[i] = ((cosm << 8)&0xff00) + (sinm &0xff);
tbl_matrix_cr[i] = (((-sinm)<<8)&0xff00) + (cosm & 0xff);
}
```

Table 8-3369 ENCI_CMPN_MATRIX_CR(VCBUS: 0x1b15)

Bit(s)	R/W	Default	Description
15:8	R/W	0	ENCP_VIDEO_MATRIX_mCCbCr matrix parameter Mc (signed value)
7:0	R/W	0	ENCP_VIDEO_MATRIX_mDCbCr matrix parameter Md (signed value)

Note

↑See above note.

Table 8-3370 ENCI_VBI_SETTING(VCBUS: 0x1b20)

Bit(s)	R/W	Default	Description
13	R/W	0	ENCI_VBI_SETTING_CCCSTM Close caption phase adjustment (Value is from TTXDT0)
12	R/W	0	ENCI_VBI_SETTING_TTX_ODD_EN Teletext odd field enable
11	R/W	0	ENCI_VBI_SETTING_TTX_EVN_EN Teletext even field enable

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Bit(s)	R/W	Default	Description
10 -8	R/W	0	ENCI_VBI_SETTING_TTX_FREQ Teletext frequency selection
5	R/W	0	ENCI_VBI_SETTING_CGMS_ODD_EN CGMS odd field enable
4	R/W	0	ENCI_VBI_SETTING_CGMS_EVN_EN CGMS even field enable
3	R/W	0	ENCI_VBI_SETTING_WSS_ODD_EN WSS odd field enable
2	R/W	0	ENCI_VBI_SETTING_WSS_EVN_EN WSS Even Field enable
1	R/W	0	ENCI_VBI_SETTING_CC_ODD_EN Close Caption Odd Field enable
0	R/W	0	ENCI_VBI_SETTING_CC_EVN_EN Close Caption Even Field enable

Table 8-3371 ENCI_VBI_CCDT_EVN(VCBUS: 0x1b21)

Bit(s)	R/W	Default	Description
15 -0	R/W	0	ENCI_VBI_CCDT_EVN Close caption even field data

Table 8-3372 ENCI_VBI_CCDT_ODD(VCBUS: 0x1b22)

Bit(s)	R/W	Default	Description
15 -0	R/W	0	ENCI_VBI_CCDT_ODD Close caption even odd data

Table 8-3373 ENCI_VBI_CC525_LN(VCBUS: 0x1b23)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h1211	ENCI_VBI_CC525_LN Close caption line in 525 format

Table 8-3374 ENCI_VBI_CC625_LN(VCBUS: 0x1b24)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h1615	ENCI_VBI_CC625_LN Close caption line in 625 format

Table 8-3375 ENCI_VBI_WSSDT(VCBUS: 0x1b25)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h0	ENCI_VBI_WSSDT WSS data

Table 8-3376 ENCI_VBI_WSS_LN(VCBUS: 0x1b26)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'd22	ENCI_VBI_WSS_LN WSS line

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Table 8-3377 ENCI_VBI_CGMSDT_L(VCBUS: 0x1b27)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h0	ENCI_VBI_CGMSDT_L CGMS data low 16 bits

Table 8-3378 ENCI_VBI_CGMSDT_H(VCBUS: 0x1b28)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'h0	ENCI_VBI_CGMSDT_H CGMS data high 8 bits

Table 8-3379 ENCI_VBI_CGMS_LN(VCBUS: 0x1b29)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h1110	ENCI_VBI_CGMS_LN CGMS line

Table 8-3380 ENCI_VBI_TTX_HTIME(VCBUS: 0x1b2a)

Bit(s)	R/W	Default	Description
15 -10	R/W	6'h2d	ENCI_VBI_TTX_HTIME_TOTAL_BYTES Teletext total bytes
9 -0	R/W	11'h135	ENCI_VBI_TTX_HTIME_STRT_POS Teletext start postion

Table 8-3381 ENCI_VBI_TTX_LN(VCBUS: 0x1b2b)

Bit(s)	R/W	Default	Description
15 -9	R/W	16'h1415	ENCI_VBI_TTX_STRT_LN Teletext Start line
8 -0	R/W	16'h1415	ENCI_VBI_TTX_END_LN Teletext End Line

Table 8-3382 ENCI_VBI_TTXDT0(VCBUS: 0x1b2c)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h0	ENCI_VBI_TTXDT0 Teletext data0

Table 8-3383 ENCI_VBI_TTXDT1(VCBUS: 0x1b2d)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h0	ENCI_VBI_TTXDT1 Teletext data1

Table 8-3384 ENCI_VBI_TTXDT2(VCBUS: 0x1b2e)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h0	ENCI_VBI_TTXDT2 Teletext data2

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Table 8-3385 ENCI_VBI_TTXDT3(VCBUS: 0x1b2f)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h0	ENCI_VBI_TTXDT3 Teletext data3

Table 8-3386 ENCI_MACV_N0(VCBUS: 0x1b30)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'h00	ENCI_MACV_N0 Macrovision register N0

Table 8-3387 ENCI_MACV_N1(VCBUS: 0x1b31)

Bit(s)	R/W	Default	Description
5 -0	R/W	6'h17	ENCI_MACV_N1 Macrovision register N1

Table 8-3388 ENCI_MACV_N2(VCBUS: 0x1b32)

Bit(s)	R/W	Default	Description
5 -0	R/W	6'h15	ENCI_MACV_N2 Macrovision register N2

Table 8-3389 ENCI_MACV_N3(VCBUS: 0x1b33)

Bit(s)	R/W	Default	Description
5 -0	R/W	6'h21	ENCI_MACV_N3 Macrovision register N3

Table 8-3390 ENCI_MACV_N4(VCBUS: 0x1b34)

Bit(s)	R/W	Default	Description
5 -0	R/W	6'h15	ENCI_MACV_N4 Macrovision register N4

Table 8-3391 ENCI_MACV_N5(VCBUS: 0x1b35)

Bit(s)	R/W	Default	Description
2 -0	R/W	3'h5	ENCI_MACV_N5 Macrovision register N5

Table 8-3392 ENCI_MACV_N6(VCBUS: 0x1b36)

Bit(s)	R/W	Default	Description
2 -0	R/W	3'h5	ENCI_MACV_N6 Macrovision register N6

Table 8-3393 ENCI_MACV_N7(VCBUS: 0x1b37)

Bit(s)	R/W	Default	Description
1 -0	R/W	2'h2	ENCI_MACV_N7 Macrovision register N7

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Table 8-3394 ENCI_MACV_N8(VCBUS: 0x1b38)

Bit(s)	R/W	Default	Description
5 -0	R/W	6'h1b	ENCI_MACV_N8 Macrovision register N8

Table 8-3395 ENCI_MACV_N9(VCBUS: 0x1b39)

Bit(s)	R/W	Default	Description
5 -0	R/W	6'h1b	ENCI_MACV_N9 Macrovision register N9

Table 8-3396 ENCI_MACV_N10(VCBUS: 0x1b3a)

Bit(s)	R/W	Default	Description
5 -0	R/W	6'h24	ENCI_MACV_N10 Macrovision register N10

Table 8-3397 ENCI_MACV_N11(VCBUS: 0x1b3b)

Bit(s)	R/W	Default	Description
14 -0	R/W	15'h07f8	ENCI_MACV_N11 Macrovision register N11

Table 8-3398 ENCI_MACV_N12(VCBUS: 0x1b3c)

Bit(s)	R/W	Default	Description
14 -0	R/W	15'h0	ENCI_MACV_N12 Macrovision register N12

Table 8-3399 ENCI_MACV_N13(VCBUS: 0x1b3d)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'h0f	ENCI_MACV_N13 Macrovision register N13

Table 8-3400 ENCI_MACV_N14(VCBUS: 0x1b3e)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'h0f	ENCI_MACV_N14 Macrovision register N14

Table 8-3401 ENCI_MACV_N15(VCBUS: 0x1b3f)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'h60	ENCI_MACV_N15 Macrovision register N15

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Table 8-3402 ENCI_MACV_N16(VCBUS: 0x1b40)

Bit(s)	R/W	Default	Description
0 -0	R/W	1'h1	ENCI_MACV_N16 Macrovision register N16

Table 8-3403 ENCI_MACV_N17(VCBUS: 0x1b41)

Bit(s)	R/W	Default	Description
3 -0	R/W	4'ha	ENCI_MACV_N17 Macrovision register N17

Table 8-3404 ENCI_MACV_N18(VCBUS: 0x1b42)

Bit(s)	R/W	Default	Description
3 -0	R/W	4'h0	ENCI_MACV_N18 Macrovision register N18

Table 8-3405 ENCI_MACV_N19(VCBUS: 0x1b43)

Bit(s)	R/W	Default	Description
3 -0	R/W	4'h5	ENCI_MACV_N19 Macrovision register N19

Table 8-3406 ENCI_MACV_N20(VCBUS: 0x1b44)

Bit(s)	R/W	Default	Description
2 -0	R/W	3'h4	ENCI_MACV_N20 Macrovision register N20

Table 8-3407 ENCI_MACV_N21(VCBUS: 0x1b45)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h3ff	ENCI_MACV_N21 Macrovision register N21

Table 8-3408 ENCI_MACV_N22(VCBUS: 0x1b46)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h0	ENCI_MACV_N22 Macrovision register N22

Table 8-3409 ENCI_DBG_PX_RST(VCBUS: 0x1b48)

Bit(s)	R/W	Default	Description
15	R/W	1	ENCI_DBG_PX_RST_EN 1:Reset enable (Interlaced TV encoder is disabled), 0: Normal
10 -0	R/W	0	ENCI_DBG_PX_RST_VAL Pixel value after reset

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Table 8-3410 ENCI_DBG_FLDLN_RST(VCBUS: 0x1b49)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_DBG_LN_RST_EN Line reset enable
14	R/W	0	ENCI_DBG_FLD_RST_EN Field reset enable
11 -9	R/W	0	ENCI_DBG_FLD_RST_VAL Field reset value
8 -0	R/W	0	ENCI_DBG_LN_RST_VAL Line reset value

Table 8-3411 ENCI_DBG_PX_INT(VCBUS: 0x1b4a)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_DBG_PX_INT_EN Pixel Interrupt enable
10 -0	R/W	0	ENCI_DBG_PX_INT_VAL Pixel value that trig the interrupt

Table 8-3412 ENCI_DBG_FLDLN_INT(VCBUS: 0x1b4b)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_DBG_FLD_INT_EN Field interrupt enable
11	R/W	0	ENCI_DBG_LN_INT_EN Line interrupt enable
14 -12	R/W	0	ENCI_DBG_FLD_INT_VAL Field value that trig the interrupt
8 -0	R/W	0	ENCI_DBG_LN_INT_VAL Line value that trig the interrupt

Table 8-3413 ENCI_DBG_MAXPX(VCBUS: 0x1b4c)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_DBG_MAXPX_EN Debug mode, change the max pixel.
10 -0	R/W	0	ENCI_DBG_MAXPX_CHGVAL The value of max pixel

Table 8-3414 ENCI_DBG_MAXLN(VCBUS: 0x1b4d)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_DBG_MAXLN_EN Debug mode, change the max line counter
8 -0	R/W	0	ENCI_DBG_MAXLN_CHGVAL The value want to change

Table 8-3415 ENCI_MACV_MAX_AMP(VCBUS: 0x1b50)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_MACV_MAX_AMP_EN Macrovision max amplitude change enable
10 -0	R/W	0	ENCI_MACV_MAX_AMP_VAL The value want to change

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Table 8-3416 ENCI_MACV_PULSE_LO(VCBUS: 0x1b51)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_MACV_PULSE_LO_EN Macrovision low pulse change enable.
10 -0	R/W	0	ENCI_MACV_PULSE_LO_VAL The value want to change

Table 8-3417 ENCI_MACV_PULSE_HI(VCBUS: 0x1b52)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_MACV_PULSE_HI_EN Macrovision high pulse change enable.
10 -0	R/W	0	ENCI_MACV_PULSE_HI_VAL The value want to change

Table 8-3418 ENCI_MACV_BKP_MAX(VCBUS: 0x1b53)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCI_MACV_BKP_MAX_EN Macrovision back porch max value change enable.
9 -0	R/W	0	ENCI_MACV_BKP_MAX_VAL The value want to change

Table 8-3419 ENCI_CFILT_CTRL(VCBUS: 0x1b54)

Bit(s)	R/W	Default	Description
12	R/W	1	ENCI_CFILT_CMPT_SMOOTH_FLTComponent Cb/Cr smooth filter
11-10	R/W	0	ENCI_CFILT_CMPT_SELFilter component bandwith Cb/Cr sel 0: High, 1: Medium 2: Low
9-8	R/W	0	ENCI_CFILT_CVBS_CB_DLYFilter CVBS Cb delay
7-6	R/W	0	ENCI_CFILT_CVBS_CR_DLYFilter CVBS Cr delay
5-4	R/W	0	ENCI_CFILT_CMPT_CB_DLYFilter component Cb delay
3-2	R/W	0	ENCI_CFILT_CMPT_CR_DLY Filter component Cr delay
1	R/W	1	ENCI_CFILT_CMPT_SELFilter component CbCr bandwith sel (1:high, 0:low)
0	R/W	0	ENCI_CFILT_CVBS_SELFilter CVBS CbCr bandwith sel (1:high, 0:low)

Table 8-3420 ENCI_CFILT7(VCBUS: 0x1b55)

Bit(s)	R/W	Default	Description
14-0	R/W	15'h4d53	ENCI_CFILT7Filter7 parameters

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Table 8-3421 ENCI_YC_DELAY(VCBUS: 0x1b56)

Bit(s)	R/W	Default	Description
9-8	R/W	0	ENCI_COMPY_DELAYComponent Y delay
7-4	R/W	4'h2	ENCI_Y_DELAYInterlace Y delay
3-0	R/W	4'h0	ENCI_C_DELAYInterlace C delay

Table 8-3422 ENCI_VIDEO_EN(VCBUS: 0x1b57)

Bit(s)	R/W	Default	Description
0	R/W	0	ENCI_VIDEO_ENInterlace video enable

Table 8-3423 ENCP_VFIFO2VD_CTL (VCBUS: 0x1b58)

Bit(s)	R/W	Default	Description
15-8	R/W	0	VFIFO2VD_VD_SELvfifo2vd_vd_sel
7	R/W	0	VFIFO2VD_DROP vfifo2vd_drop
6-1	R/W	0	VFIFO2VD_DELAY vfifo2vd_delay
0	R/W	0	VFIFO2VD_EN vfifo2vd_en

Table 8-3424 ENCP_VFIFO2VD_PIXEL_START(VCBUS: 0x1b59)

Bit(s)	R/W	Default	Description
12-0	R/W	0	vFIFO2VD_PIXEL_STARTPixel start

Table 8-3425 ENCP_VFIFO2VD_PIXEL_END(VCBUS: 0x1b5a)

Bit(s)	R/W	Default	Description
15	R/W	0	VFIFO2VD_YC_USE_FIRST_VFIFO_REQ0=Keep as previous version; 1=On the very first pixel since VFIFO is enabled, YC pipeline values take from default setting, rather than from values stored in the pipeline.
12-0	R/W	0	VFIFO2VD_PIXEL_ENDPixel end

Table 8-3426 ENCP_VFIFO2VD_LINE_TOP_START(VCBUS: 0x1b5b)

Bit(s)	R/W	Default	Description
10-0	R/W	0	VFIFO2VD_LINE_TOP_STARTTop field line end

Table 8-3427 ENCP_VFIFO2VD_LINE_TOP_END(VCBUS: 0x1b5c)

Bit(s)	R/W	Default	Description
10-0	R/W	0	VFIFO2VD_LINE_TOP_ENDTop field line end

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Table 8-3428 ENCP_VFIFO2VD_LINE_BOT_START(VCBUS: 0x1b5d)

Bit(s)	R/W	Default	Description
10-0	R/W	0	VFIFO2VD_LINE_BOT_STARTBottom field line end

Table 8-3429 ENCP_VFIFO2VD_LINE_BOT_END(VCBUS: 0x1b5e)

Bit(s)	R/W	Default	Description
10-0	R/W	0	VFIFO2VD_LINE_BOT_ENDBottom field line end

Table 8-3430 VENC_SYNC_ROUTE(VCBUS: 0x1b60)

Bit(s)	R/W	Default	Description
2	R/W	0	VENC_SYNC_ROUTE Internal Vencoder hsync/vsync source (0-VIU, 1-External Venc)
1	R/W	0	VIU_SYNC_ROUTE VIU hsync/vsync input source (0-Venc, 1-External Venc)
0	R/W	1	VPINS_SYNC_ROUTE External Vencoder hsync/vsync source (0-Internal Venc, 1-VIU)

Table 8-3431 VENC_VIDEO_EXSRC(VCBUS: 0x1b61)

Bit(s)	R/W	Default	Description
0	R/W	1	VENC_VIDEO_EXSRC External Video Source enable, get video data from external pins.

Note

↑This is not supported by T25, clear this when init.

Table 8-3432 VENC_DVI_SETTING(VCBUS: 0x1b62)

Bit(s)	R/W	Default	Description
15	R/W	0	VENC_DVI_SEL_DVI1=use programmable vs/hs/de whose only connection is to DVI/HDMI.
14	R/W	0	VENC_DVI_SEL_INTL_DEFor interlace mode only. 1=use intl_de for DE.
13	R/W	0	DVI_SEL_DELAYED_VSDVI sync use delayed vsync from analog output
13	R/W	0	VENC_DVI_GAMMA_EN0=Send to DVI/HDMI interface with data NOT from Gamma table. 1=Send to DVI/HDMI interface with data from Gamma table.
12	R/W	0	VENC_DVI_DDR_DESELFor external DVI device only.
11	R/W	0	VENC_DVI_DDR_CKSELFor external DVI device only.
10	R/W	0	VENC_DVI_DDR_CKPHIFor external DVI device only.
9	R/W	0	VENC_DVI_DDR_SELFor external DVI device only.
8	R/W	0	VENC_DVI_INV_CLKInvert DVI clock

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Bit(s)	R/W	Default	Description
7	R/W	0	VENC_DVI_SEL_INTERNAL_HDMI0=Select the external DVI device which is compatible to chips predate M1 1=Select the internal HDMI tx which is in chips starting from meson.
56:4	R/W	0	VENC_DVI_CLK_SELDVI Clock selection: 0: clk27 1: clk54 2: ½ of ENCP pixel_clk 3: clk108 4: ½ of ENCI pixel_clk 5: cts_vclk1 6-7: Reserved (Note : here the clk27/54/108 is signal name, not the real frequency)
3	R/W	0	VENC_DVI_INV_VSYNC1=Invert DVI/HDMI Vsync output polarity
2	R/W	0	VENC_DVI_INV_HSYNC1=Invert DVI/HDMI Hsync output polarity
1	R/W	0	SYNC_SEL_OUT_SYNCDVI/601 sync use output sync (not viu sync)
0	R/W	1	VENC_DVI_SRC_SELDVI source select (0: interlace, 1:progressive)

Table 8-3433 VENC_C656_CTRL (VCBUS: 0x1b63)

Bit(s)	R/W	Default	Description
3	R/W	0	VENC_C656_clkpClock out Polarity
2	R/W	0	VENC_C656_FIDPField Polarity
1	R/W	0	VENC_C656_VSoPVsyn Polarity
0	R/W	0	VENC_C656_hsoPHsyn Polarity

Upsamping is only used for 480i/576i, 480p/576p, not for 720p/1080i/1080p mode,

For 480i/576i upsampling, video signal is upsampled from 27Mhz to 108Mhz,

For 480p/576p upsampling, video signal is upsampled from 54Mz to 108Mhz

set VENC_UPSAMPLE_CTRL0 = 0x9061

set VENC_UPSAMPLE_CTRL1 = 0xa061

set VENC_UPSAMPLE_CTRL2 = 0x1b061

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Table 8-3434 VENC_UPSAMPLE_CTRL0 (VCBUS: 0x1b64)

Bit(s)	R/W	Default	Description
15-12	R/W	0	upsAMPLE_SELUpsample0 din selection 0: Interlace High bandwidth Luma 1: CVBS 2: S-Video luma 3: S-Video Chroma 4: Interlace Pb 5: Interlace Pr 6: Interlace R 7: Interlace G 8: Interlace B 9: Progressive Y a: Progressive Pb b: Progressive Pr c: Progressive R d: Progressive G e: Progressive B f: VDAC test value (VENC_VDAC_TST_VAL)
11-9	R	0	UnusedUnused
8	R/W	0	bYPASSIf true, bypass
7	R/W	0	F1_DIV_INVIf true, invert div
6	R/W	0	F1_UPSAMPLE_ENFilter1 upsample enable
5	R/W	0	F1_ENFilter1 filtering enable
4	R/W	0	F1_CLK_RATIOIf true, Filter1 clk ratio is 2, input data sample every 2 clocks, otherwise 1
3	R/W	0	F0_DIV_INVIf true, invert div
2	R/W	0	F0_UPSAMPLE_ENFilter0 upsample enable
1	R/W	0	F0_ENFilter0 filtering enable
0	R/W	0	F0_CLK_RATIOIf true, Filter0 clk ratio is 2, input data sample every 2 clocks, otherwise 1

Table 8-3435 VENC_UPSAMPLE_CTRL1 (VCBUS: 0x1b65)

Bit(s)	R/W	Default	Description
15-12	R/W	0	upsAMPLE_SELUpsample1 din selection 0: Interlace High bandwidth Luma 1: CVBS 2: S-Video luma 3: S-Video Chroma 4: Interlace Pb 5: Interlace Pr 6: Interlace R 7: Interlace G 8: Interlace B 9: Progressive Y a: Progressive Pb b: Progressive Pr c: Progressive R d: Progressive G e: Progressive B f: VDAC test value (VENC_VDAC_TST_VAL)
11-9	R	0	UnusedUnused
8	R/W	0	bYPASSIf true, bypass

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Bit(s)	R/W	Default	Description
7	R/W	0	F1_DIV_INVIf true, invert div
6	R/W	0	F1_UPSAMPLE_ENFilter1 upsample enable
5	R/W	0	F1_ENFilter1 filtering enable
4	R/W	0	F1_CLK_RATIOIf true, Filter1 clk ratio is 2, input data sample every 2 clocks, otherwise 1
3	R/W	0	F0_DIV_INVIf true, invert div
2	R/W	0	F0_UPSAMPLE_ENFilter0 upsample enable
1	R/W	0	F0_ENFilter0 filtering enable
0	R/W	0	F0_CLK_RATIOIf true, Filter0 clk ratio is 2, input data sample every 2 clocks, otherwise 1

Table 8-3436 VENC_UPSAMPLE_CTRL2 (VCBUS: 0x1b66)

Bit(s)	R/W	Default	Description
15-12	R/W	0	upsAMPLE_SELUpsample2 din selection 0: Interlace High bandwidth Luma 1: CVBS 2: S-Video luma 3: S-Video Chroma 4: Interlace Pb 5: Interlace Pr 6: Interlace R 7: Interlace G 8: Interlace B 9: Progressive Y a: Progressive Pb b: Progressive Pr c: Progressive R d: Progressive G e: Progressive B f: VDAC test value (VENC_VDAC_TST_VAL)
11-9	R	0	UnusedUnused
8	R/W	0	bYPASSIf true, bypass
7	R/W	0	F1_DIV_INVIf true, invert div
6	R/W	0	F1_UPSAMPLE_ENFilter1 upsample enable
5	R/W	0	F1_ENFilter1 filtering enable
4	R/W	0	F1_CLK_RATIOIf true, Filter1 clk ratio is 2, input data sample every 2 clocks, otherwise 1
3	R/W	0	F0_DIV_INVIf true, invert div
2	R/W	0	F0_UPSAMPLE_ENFilter0 upsample enable
1	R/W	0	F0_ENFilter0 filtering enable
0	R/W	0	F0_CLK_RATIOIf true, Filter0 clk ratio is 2, input data sample every 2 clocks, otherwise 1

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Table 8-3437 TCON_INVERT_CTL (VCBUS: 0x1b67)

Bit(s)	R/W	Default	Description
15	R/W	1'b0	vsync invert
14	R/W	1'b0	hsync invert
13	R/W	1'b0	Oev3 invert
12	R/W	1'b0	Oev2 invert
11	R/W	1'b0	Cpv2 invert
10	R/W	1'b0	Cph3 invert
9	R/W	1'b0	Cph2 invert
8	R/W	1'b0	Cph1 invert
7	R/W	1'b0	oeh invert
6	R/W	1'b0	vcom invert
5	R/W	1'b0	Stv2 invert
4	R/W	1'b0	Stv1 invert
3	R/W	1'b0	Cpv1 invert
2	R/W	1'b0	Oev1 invert
1	R/W	1'b0	Sth1 invert
0	R/W	1'b0	Sth2 invert

Table 8-3438 VENC_VIDEO_PROG_MODE (VCBUS: 0x1b68)

Bit(s)	R/W	Default	Description
9	R/W	1'b0	VENC_DAC_CLK_SEL_CLK108Select clk108 as DAC clock
8	R/W	1'b0	VENC_BIST_FIELD_SELVenc bist and venc_field output selection. 0:Interlace, 1:Progressive.
7	R/W	1'b0	VENC_VIU_VSIN_SEL VIU Vsync input select, 0:Progressive, 1:Interlace
6	R/W	1'b0	VENC_VIU_HSIN_SEL VIU Hsync input select, 0:Progressive, 1:Interlace
5	R/W	1'b0	VENC_VPIN_VSIN_SEL Pin Vsync input select, 0:Progressive, 1:Interlace
4	R/W	1'b0	VENC_VPIN_HSIN_SEL Pin Hsync input select, 0:Progressive, 1:Interlace
3	R/W	1'b0	VENC_DAC1_CLK_SEL DAC1 clock select, 0:clk54, 1:clk27
2	R/W	1'b0	VENC_DAC0_CLK_SEL DAC0 clock select, 0:clk54, 1:clk27

Table 8-3439 VENC_ENCI_LINE (VCBUS: 0x1b69)

Bit(s)	R/W	Default	Description
9 -0	R	-	VENC_ENCI_LINECurrent interlace encoder line

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Table 8-3440 VENC_ENCI_PIXEL (VCBUS: 0x1b6a)

Bit(s)	R/W	Default	Description
9 -0	R	-	VENC_ENCI_LINECurrent interlace encoder line

Table 8-3441 VENC_ENCP_LINE (VCBUS: 0x1b6b)

Bit(s)	R/W	Default	Description
9 -0	R	-	VENC_ENCP_LINECurrent interlace encoder line

Table 8-3442 VENC_ENCP_PIXEL (VCBUS: 0x1b6c)

Bit(s)	R/W	Default	Description
9 -0	R	-	VENC_ENCP_LINECurrent interlace encoder line

Table 8-3443 VENC_STATA (VCBUS: 0x1b6d)

Bit(s)	R/W	Default	Description
2 -0	R	-	VENC_FILED Current Venc Field

Table 8-3444 VENC_INTCTRL (VCBUS: 0x1b6e)

Bit(s)	R/W	Default	Description
13	R/W	0	ENCP_TIMER_INT_EN
12	R/W	0	ENCP_FLDINT_EN
11	R/W	0	ENCP_PXRST_EN
10	R/W	0	ENCP_TIMER_INT_EN
9	R /W	0	ENCP_LNRST_INT_EN Progressive encoder filed change interrupt enable
8	R /W	0	ENCP_PXRST_INT_EN Progressive encoder line change interrupt enable
7	R/W	0	ENCL_TIMER_INT_EN
6	R/W	0	ENCL_FLDINT_EN
5	R/W	0	ENCL_PXRST_EN
4	R/W	0	ENCI_TIMER_INT_EN
3	R /W	0	ENCI_TTXLOAD_INT_EN Interlace encoder teletext data interrupt enable
2	R /W	0	ENCI_FDRST_INT_EN Interlace encoder field reset interrupt enable
1	R /W	0	ENCI_LNRST_INT_EN Interlace encoder filed change interrupt enable
0	R /W	0	ENCI_PXRST_INT_EN Interlace encoder line change interrupt enable

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Table 8-3445 VENC_INTFLAG (VCBUS: 0x1b6f)

Bit(s)	R/W	Default	Description
9	R /W	0	ENCP_LNRST_INT_F Progressive encoder filed change interrupt flag
8	R /W	0	ENCP_PXRST_INT_F Progressive encoder line change interrupt flag
3	R /W	0	ENCI_TTXLOAD_INT_F Interlace encoder teletext data load interrupt flag
2	R /W	0	ENCI_FDRST_INT_F Interlace encoder field reset interrupt flag
1	R /W	0	ENCI_LNRST_INT_F Interlace encoder filed change interrupt flag
0	R /W	0	ENCI_PXRST_INT_FInterlace encoder line change interrupt flag

Table 8-3446 VENC_VIDEO_TST_EN (VCBUS: 0x1b70)

Bit(s)	R/W	Default	Description
0	R/W	0	VENC_VIDEO_TST_EN BIST enable

Table 8-3447 VENC_VIDEO_TST_MDSEL (VCBUS: 0x1b71)

Bit(s)	R/W	Default	Description
7-0	R/W	8'h01	VENC_VIDEO_TST_MDSEL BIST Mode 0: TST_MODE_FIXVAL—— Fix Value on Y/Cb/Cr 1: TST_MODE_COLORBAR —— 100/75 color bar 2: TST_MODE_THINLINE —— Thin horizonl/vertical lines on screen 3: TST_MODE_DOTGRID —— Dot grid on screen

Note

↑In none color bar mode, value is from register TST_Y/TST_CB/TST_CR.

Table 8-3448 VENC_VIDEO_TST_Y (VCBUS: 0x1b72)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd512	VENC_VIDEO_TST_Y BIST fix value Y

Note

↑Value is 10 bists,X4 if convert from 8 bits CCIR601 value

Table 8-3449 VENC_VIDEO_TST_CB (VCBUS: 0x1b73)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd512	VENC_VIDEO_TST_CBBIST fix value Cb

Note

↑Value is 10 bists,X4 if convert from 8 bits CCIR601 value

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Table 8-3450 VENC_VIDEO_TST_CR (VCBUS: 0x1b74)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd512	VENC_VIDEO_TST_CR BIST fix value Cr

Note

↑Value is 10 bists,X4 if convert from 8 bits CCIR601 value

Table 8-3451 VENC_VIDEO_TST_CLRBAR_STRT(VCBUS: 0x1b75)

Bit(s)	R/W	Default	Description
12-0	R/W	11'd275	VENC_VIDEO_TST_CLRBAR_STRTColorbar BIST start position

Note

↑VENC_VIDEO_TST_CLRBAR_STRT Interlace mode : 235, Progressive mode : 274

Table 8-3452 VENC_VIDEO_TST_CLRBAR_WIDTH(VCBUS: 0x1b76)

Bit(s)	R/W	Default	Description
12 -0	R/W	11'd360	VENC_VIDEO_TST_CLRBAR_WIDTHColorbar BIST Width

Note

↑VENC_VIDEO_TST_CLRBAR_WIDTH Interlace mode : 360, Progressive mode: 180

Table 8-3453 VENC_VIDEO_TST_VDCNT_STSET(VCBUS: 0x1b77)

Bit(s)	R/W	Default	Description
1-0	R/W	0	VENC_VIDEO_TST_VDCNT_STSET BIST video data shifting setting, adjust this value if the data sequence generated by BIST is incorrect.

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Table 8-3454 VENC_VDAC_DACSEL0(VCBUS: 0x1b78)

Bit(s)	R/W	Default	Description
15-12	R/W	0	VENC_VDAC_DLYVDAC delay (0-15 clocks)
5	R/W	0	VENC_VDAC_SEL_ATV_DMD1: select VDAC0 source from ATV demod
4-0	R/W	4'h2	VENC_VDAC_DACSEL0 Video Dac0 selection 0: Interlace Y 1: CVBS 2: S-Video luma 3: S-Video Chroma 4: Interlace Pb 5: Interlace Pr 6: Interlace R 7: Interlace G 8: Interlace B 9: Progressive Y a: Progressive Pb b: Progressive Pr c: Progressive R d: Progressive G e: Progressive B f: VDAC test value (VENC_VDAC_TST_VAL) 0x10: upsampled data0 0x11: upsampled data1 0x12 upsampled data2

Table 8-3455 VENC_VDAC_DACSEL1(VCBUS: 0x1b79)

Bit(s)	R/W	Default	Description
		16’h4	Same as above

Table 8-3456 VENC_VDAC_DACSEL2(VCBUS: 0x1b7a)

Bit(s)	R/W	Default	Description
		16’h5	Same as above

Table 8-3457 VENC_VDAC_DACSEL3(VCBUS: 0x1b7b)

Bit(s)	R/W	Default	Description
		16’h1	Same as above

Table 8-3458 VENC_VDAC_DACSEL4(VCBUS: 0x1b7c)

Bit(s)	R/W	Default	Description
		16’h2	Same as above

Table 8-3459 VENC_VDAC_DACSEL5(VCBUS: 0x1b7d)

Bit(s)	R/W	Default	Description
		16’h3	Same as above

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Table 8-3460 VENC_VDAC_SETTING(VCBUS: 0x1b7e)

Bit(s)	R/W	Default	Description
0	R/W	0	VENC_VDAC0_PWDN Powers down video DAC 0
1	R/W	0	VENC_VDAC1_PWDN Powers down video DAC 1
2	R/W	0	VENC_VDAC2_PWDN Powers down video DAC 2
3	R/W	0	VENC_VDAC3_PWDN Powers down video DAC 3
4	R/W	0	VENC_VDAC4_PWDN Powers down video DAC 4
5	R/W	0	VENC_VDAC5_PWDN Powers down video DAC 5
6	R/W	0	Unused
7	R/W	0	VENC_VDAC_all_PWDN Powers down all video DACs
8	R/W	0	VENC_VDAC_SOGVDAC SOG signal (Sync on green, not used)
9	R/W	0	VENC_VDAC_irENVDAC IREN signal (7.5 IRE setup, not used)
12	R/W	0	VENC_VDAC_SYNCVDAC sync signal when SYBL_en is 0
13	R/W	0	VENC_VDAC_BLNKVDAC blank signal when SYBL_EN is 0
15	R/W	0	VENC_VDAC_SYBL_ENEnable tv encoder to control VDAC sync and blank

Table 8-3461 VENC_VDAC_TST_VAL(VCBUS: 0x1b7f)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h200	VENC_VDAC_TST_VAL Video DAC test value

Table 8-3462 ENCP_VIDEO_EN(VCBUS: 0x1b80)

Bit(s)	R/W	Default	Description
0	R/W	0	ENCP_VIDEO_EN Progressive encoder enable

Table 8-3463 ENCP_VIDEO_SYNC_MODE(VCBUS: 0x1b81)

Bit(s)	R/W	Default	Description
15:0	R/W	8'h0	DVI_VSYNC_DELAYDelay lines from normal analog Vsync, can be used for DVI output
7 -0	R/W	8'h7	ENCP_VIDEO_SYNC_MODE Video input Synchronization mode (4:Slave mode, 7:Master mode)

Table 8-3464 ENCP_MACV_EN(VCBUS: 0x1b82)

Bit(s)	R/W	Default	Description
0	R/W	0	ENCP_MACV_EN Macrovision enable

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Table 8-3465 ENCP_VIDEO_Y_SCL(VCBUS: 0x1b83)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd81	ENCP_VIDEO_Y_SCL Y scale

Table 8-3466 ENCP_VIDEO_PB_SCL(VCBUS: 0x1b84)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd79	ENCP_VIDEO_PB_SCL Cb scale

Table 8-3467 ENCP_VIDEO_PR_SCL(VCBUS: 0x1b85)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd79	ENCP_VIDEO_PR_SCL Cr scale

Table 8-3468 ENCP_VIDEO_SYNC_SCL(VCBUS: 0x1b86)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd128	ENCP_VIDEO_SYNC_SCL Analog Sync value scale

Table 8-3469 ENCP_VIDEO_MACV_SCL(VCBUS: 0x1b87)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd128	ENCP_VIDEO_MACV_SCL Macrovision value scale

Table 8-3470 ENCP_VIDEO_Y_OFFST(VCBUS: 0x1b88)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h3c0	ENCP_VIDEO_Y_OFFST Y offset

Table 8-3471 ENCP_VIDEO_PB_OFFST(VCBUS: 0x1b89)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h0	ENCP_VIDEO_PB_OFFST Pb offset

Table 8-3472 ENCP_VIDEO_PR_OFFST(VCBUS: 0x1b8a)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h0	ENCP_VIDEO_PR_OFFST Pr offset

Table 8-3473 ENCP_VIDEO_SYNC_OFFST(VCBUS: 0x1b8b)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h0	ENCP_VIDEO_SYNC_OFFST Sync pulse offset

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Table 8-3474 ENCP_VIDEO_MACV_OFFST(VCBUS: 0x1b8c)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h0	ENCP_VIDEO_MACV_OFFST Macrovision offset

Table 8-3475 ENCP_VIDEO_MODE(VCBUS: 0x1b8d)

Bit(s)	R/W	Dflt	Description
15	R/W	0	ENCP_PX_LN_CNT_SHADOW_ENpixel count and line count shadow enable
14	R/W	0	ENCP_VIDEO_mode_DE_V1=DE signal's polarity is active high. 0=DE signal's polarity is active low.
13	R/W	0	ENCP_VIDEO_mode_en_VSO_OFLD_FIX_EN1=Extend ENCP_VIDEO_MAX_PXCNT by 1 pixel. 0=Treat ENCP_VIDEO_MAX_PXCNT as is.
12	R/W	0	ENCP_VIDEO_mode_en_field_oddEnable odd field (for 1080i)
11	R/W	0	ENCP_VIDEO_mode_en_vso_ofldEnable VSO odd field
10	R/W	0	ENCP_VIDEO _mode_en_vavon_ofldEnable Vertical Active On for odd field
9	R/W	0	ENCP_VIDEO _mode_en_equ_ofldEnable Equalization pulse for odd field
8	R/W	0	ENCP_VIDEO _mode_en_equEnable Equalization pulse
6	R/W	0	ENCP_VIDEO _mode_en_hseq_switchEnable Hsync and equalization pulse switch in center
5	R/W	0	ENCP_VIDEO _mode_en_vp_ofldEnable Vertical Pulse for odd field
4	R/W	0	ENCP_VIDEO _mode_en_vp2Enable 2nd vertical pulse in a line (1080i)
3	R/W	0	ENCP_VIDEO _mode_en_vpe_half_ofldVertical pulse on end line of odd field is half
2	R/W	0	ENCP_VIDEO _mode_en_vpb_half_ofldVertical pulse on begin line of odd field is half
1	R/W	0	ENCP_VIDEO _mode_en_vpe_halfVertical pulse on end line of even field is half
0	R/W	0	ENCP_VIDEO _mode_en_vpb_halfVertical pulse on begin line of even field is half

Note

↑This register should be set as 0x140 for 720p , and 0x1ffc for 1080i, and 0x0 for 480p/576p

Table 8-3476 ENCP_VIDEO_MODE_ADV(VCBUS: 0x1b8e)

Bit(s)	R/W	Default	Description
15-14	R/W	0	ENCP_SP_TIMING_CTRL
13	R/W	0	ENCP_CR_BAPASS_LIM
12	R/W	0	ENCP_CB_BAPASS_LIM
11	R/W	0	ENCP_Y_BAPASS_LIM
10	R/W	0	ENCP_SEL_GAMMA_RGB_IN

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Bit(s)	R/W	Default	Description
9-8	R/W	0	Unused
7	R/W	0	ENCP_VIDEO_MODE_ADV_EN_HUE_MATRIXEnable HUE matrix
6	R/W	0	ENCP_VIDEO_MODE_ADV_SWAP_PBPRSwap PB PR
5	R/W	0	ENCP_VIDEO_MODE_ADV_EN_HS_PBPREnable sync pulse on PB PR
4	R/W	0	ENCP_VIDEO_MODE_ADV_GAIN_HDTVYPBPR gain as HDTV type
3	R/W	0	ENCP_VIDEO_MODE_ADV_VFIFO_ENData input from VFIFO
2 :0	R/W	1	ENCP_VIDEO_MODE_ADV_VFIFO_UPMODESampling rate: 0: 1 1: ½ 2: ¼ 3: 1/8

Table 8-3477 ENCP_DBG_PX_RST(VCBUS: 0x1b90)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCP_DBG_PX_RST_EN Debug mode pixel reset enable
12 -0	R/W	0	ENCP_DBG_PX_RST_VAL Debug mode pixel reset value

Table 8-3478 ENCP_DBG_LN_RST(VCBUS: 0x1b91)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCP_DBG_LN_RST_EN Debug mode line reset enable
10 -0	R/W	0	ENCP_DBG_LN_RST_VAL Debug mode line reset value

Table 8-3479 ENCP_DBG_PX_INT(VCBUS: 0x1b92)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCP_DBG_PX_INT_EN Pixel Interrupt enable
12 -0	R/W	0	ENCP_DBG_PX_INT_VAL Pixel value that trig the interrupt

Table 8-3480 ENCP_DBG_LN_INT(VCBUS: 0x1b93)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCP_DBG_LN_INT_EN Line Interrupt enable
10 -0	R/W	0	ENCP_DBG_LN_INT_VAL Line value that trig the interrupt

Table 8-3481 ENCP_VIDEO_YFP1_HTIME(VCBUS: 0x1b94)

Bit(s)	R/W	Default	Description
12 -0	R/W	240	ENCP_VIDEO_YFP1_HTIME Filter switch start point

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Table 8-3482 ENCP_VIDEO_YFP2_HTIME(VCBUS: 0x1b95)

Bit(s)	R/W	Default	Description
12 -0	R/W	1657	ENCP_VIDEO_YFP2_HTIME Filter switch end point

Table 8-3483 ENCP_VIDEO_YC_DLY(VCBUS: 0x1b96)

Bit(s)	R/W	Default	Description
5 –4	R/W	16'h0	ENCP_VIDEO_Y_DLY Y delay
3 –2	R/W	16'h0	ENCP_VIDEO_Cb_DLY Cb delay
1 –0	R/W	16'h0	ENCP_VIDEO_Cr_DLY Cr delay

Table 8-3484 ENCP_VIDEO_MAX_PXCNT(VCBUS: 0x1b97)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd1715	ENCP_VIDEO_MAX_PXCNT Max pixel counter

Table 8-3485 ENCP_VIDEO_HSPULS_BEGIN(VCBUS: 0x1b98)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd1715	ENCP_VIDEO_HSPULS_BEGIN Analog Horizontal Sync Begin

Table 8-3486 ENCP_VIDEO_HSPULS_END(VCBUS: 0x1b99)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd125	ENCP_VIDEO_HSPULS_END Analog Horizontal Sync End

Table 8-3487 ENCP_VIDEO_HSPULS_SWITCH(VCBUS: 0x1b9a)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd88	ENCP_VIDEO_HSPULS_switch Analog Horizontal switch point (for HDTV)

Table 8-3488 ENCP_VIDEO_VSPULS_BEGIN(VCBUS: 0x1b9b)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd0	ENCP_VIDEO_VSPULS_BEGIN Analog Veritcal Sync begin point

Table 8-3489 ENCP_VIDEO_VSPULS_END(VCBUS: 0x1b9c)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd1589	ENCP_VIDEO_VSPULS_END Analog Veritcal Sync end point

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Table 8-3490 ENCP_VIDEO_VSPULS_BLINE(VCBUS: 0x1b9d)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd5	ENCP_VIDEO_VSPULs_BLINEVso begin line

Table 8-3491 ENCP_VIDEO_VSPULS_ELINE(VCBUS: 0x1b9e)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd10	ENCP_VIDEO_VSPULS_ELINEVso end line

Table 8-3492 ENCP_VIDEO_EQPULS_BEGIN(VCBUS: 0x1b9f)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd0	ENCP_VIDEO_EQPULS_BEGIN Analog Equalization pulse begin point

Table 8-3493 ENCP_VIDEO_EQPULS_END(VCBUS: 0x1ba0)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd1589	ENCP_VIDEO_EQPULS_END Analog Equalization pulse end point

Table 8-3494 ENCP_VIDEO_EQPULS_BLINE(VCBUS: 0x1ba1)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd5	ENCP_VIDEO_EQPULs_BLINEEqualization pulse begin line

Table 8-3495 ENCP_VIDEO_EQPULS_ELINE(VCBUS: 0x1ba2)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd10	ENCP_VIDEO_EQPULS_ELINEEqualization pulse end line

Table 8-3496 ENCP_VIDEO_HAVON_END(VCBUS: 0x1ba3)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd1656	ENCP_VIDEO_havon_END Vertical active video end point

Table 8-3497 ENCP_VIDEO_HAVON_BEGIN(VCBUS: 0x1ba4)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd217	ENCP_VIDEO_havon_BEGINVertical active video start point

Table 8-3498 ENCP_VIDEO_VAVON_ELINE(VCBUS: 0x1baf)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd519	ENCP_VIDEO_vavon_ElineVertical active video on end line

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Table 8-3499 ENCP_VIDEO_VAVON_BLINE(VCBUS: 0x1ba6)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd42	ENCP_VIDEO_vavon_BlineVertical active video on start line

Table 8-3500 ENCP_VIDEO_HSO_BEGIN(VCBUS: 0x1ba7)

Bit(s)	R/W	Default	Description
12 -0	R/W	16	ENCP_VIDEO_hSO_begin Digital Hsync out start point

Table 8-3501 ENCP_VIDEO_HSO_END(VCBUS: 0x1ba8)

Bit(s)	R/W	Default	Description
12 -0	R/W	32	ENCP_VIDEO_hSO_endDigital Hsync out end point

Table 8-3502 ENCP_VIDEO_VSO_BEGIN(VCBUS: 0x1ba9)

Bit(s)	R/W	Default	Description
12 -0	R/W	16	ENCP_VIDEO_VSO_begin Digital Vsync out start point

Table 8-3503 ENCP_VIDEO_VSO_END(VCBUS: 0x1baa)

Bit(s)	R/W	Default	Description
12 -0	R/W	32	ENCP_VIDEO_VSO_endDigital Vsync out end point

Table 8-3504 ENCP_VIDEO_VSO_BLINE(VCBUS: 0x1bab)

Bit(s)	R/W	Default	Description
10 -0	R/W	37	ENCP_VIDEO_vSO_blineDigital Vsync out start line

Table 8-3505 ENCP_VIDEO_VSO_ELINE(VCBUS: 0x1bac)

Bit(s)	R/W	Default	Description
10 -0	R/W	39	ENCP_VIDEO_vSO_elineDigital Vsync out end line

Table 8-3506 ENCP_VIDEO_SYNC_WAVE_CURVE(VCBUS: 0x1bad)

Bit(s)	R/W	Default	Description
0	R/W	1'b1	ENCP_VIDEO_SYNC_WAVE_CURVE Enable curve wave on analog sync edge

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Table 8-3507 ENCP_VIDEO_MAX_LNCNT(VCBUS: 0x1bae)

Bit(s)	R/W	Default	Description
10 -0	R/W	10'd524	ENCP_VIDEO_MAX_LNCNT Max line counter

Table 8-3508 ENCP_VIDEO_SY_VAL(VCBUS: 0x1bb0)

Bit(s)	R/W	Default	Description
9 -0	R/W	0	ENCP_VIDEO_SY_VAL Analog Sync Pulse value 1

Table 8-3509 ENCP_VIDEO_SY2_VAL(VCBUS: 0x1bb1)

Bit(s)	R/W	Default	Description
9 -0	R/W	480	ENCP_VIDEO_SY2_VAL Analog Sync Pulse value 2

Table 8-3510 ENCP_VIDEO_BLANKY_VAL(VCBUS: 0x1bb2)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd240	ENCP_VIDEO_BLANKY_VAL Blank Y value

Table 8-3511 ENCP_VIDEO_BLANKPB_VAL(VCBUS: 0x1bb3)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd512	ENCP_VIDEO_BLANKPB_VAL Blank Pb value

Table 8-3512 ENCP_VIDEO_BLANKPR_VAL(VCBUS: 0x1bb4)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd512	ENCP_VIDEO_BLANKPR_VAL Blank Pr value

Table 8-3513 ENCP_VIDEO_HOFFST(VCBUS: 0x1bb5)

Bit(s)	R/W	Default	Description
13 -0	R/W	2	ENCP_VIDEO_HOFFST Horizontal offset after HSI

Table 8-3514 ENCP_VIDEO_VOFFST(VCBUS: 0x1bb6)

Bit(s)	R/W	Default	Description
10 -0	R/W	0	ENCP_VIDEO_VOFFST Vertical offset after VSI

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Table 8-3515 ENCP_VIDEO_RGB_CTRL(VCBUS: 0x1bb7)

Bit(s)	R/W	Default	Description
12-8	R/W	4	CFG_BLANK_DLY
6	R/W	0	Cfg_macv_R_en
5	R/W	0	Cfg_macv_G_en
4	R/W	0	Cfg_macv_b_en
3	R/W	0	CFG_RGB_SEPERATE_BLANK
2	R/W	0	Cfg_sync_on_r
1	R/W	1	Cfg_sync_on_g
0	R/W	0	Cfg_sync_on_b

Table 8-3516 ENCP_VIDEO_FILT_CTRL(VCBUS: 0x1bb8)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCP_VIDEO_FILT_YF_NO_RSTDisable auto reset Y filter each line
14	R/W	0	ENCP_VIDEO_FILT_CF_NO_RSTDisable auto reset C filter each line
14	R/W	0	CFILT_CHROMA444_ENIf true, chroma input is 444 instead of 422
12	R/W	0	ENCP_VIDEO_FILT_BYPASS_TOPBy pass all ENCP filter
11	R/W	0	ENCP_VIDEO_FILT_CHROMA_SWAPSwap chroma Cb/Cr
10:8	R/W	0	ENCP_VIDEO_FILT_CF_BYPASSChroma filter by pass control
7:4	R/W	0	ENCP_VIDEO_FILT_YF_CFG14 bits Y filter parameter 1, will be effect after ENCP_VIDEO_YFP1_HTIME bit 3 : Edge mode bit 2:0 : By pass ctrl
3:0	R/W	0	ENCP_VIDEO_FILT_YF_CFG24 bits Y filter parameter 1, will be effect after ENCP_VIDEO_YFP1_HTIME bit 3 : Edge mode bit 2:0 : By pass ctrl

Table 8-3517 ENCP_ VIDEO_OFLD_VPEQ_OFST(VCBUS: 0x1bb9)

Bit(s)	R/W	Default	Description
15:12	R/W	0	ENCP_VIDEO_OFLD_VPULS_OFST_BEGINOdd field Vsync pulse offset begin, 4 bits signed.
11:8	R/W	1	ENCP_VIDEO_OFLD_VPULS_OFST_ENDOdd field Vsync pulse offset end, 4 bits signed.
7:4	R/W	0	ENCP_VIDEO_OFLD_EQPULS_OFST_BEGINOdd field EQU pulse offset begin, 4 bits signed.
3:0	R/W	0	ENCP_VIDEO_OFLD_EQPULS_OFST_ENDOdd field EQU pulse offset end, 4 bits signed.

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Table 8-3518 ENCP_ VIDEO_OFLD_VOAV_OFST(VCBUS: 0x1bba)

Bit(s)	R/W	Default	Description
15:12	R/W	0	ENCP_VIDEO_OFLD_VSO_OFST_BEGINOdd field VSO offset begin, 4 bits signed.
11:8	R/W	1	ENCP_VIDEO_OFLD_VSO_OFST_ENDOdd field VSO offset end, 4 bits signed.
7:4	R/W	1	ENCP_VIDEO_OFLD_VAVON_OFST_BEGINOdd field VAVON offset begin, 4 bits signed.
3:0	R/W	1	ENCP_VIDEO_OFLD_VAVON_OFST_ENDOdd field VAVON offset end, 4 bits signed.

Table 8-3519 ENCP_ VIDEO_MATRIX_CB(VCBUS: 0x1bbb)

Bit(s)	R/W	Default	Description
15:8	R/W	0	ENCP_VIDEO_MATRIX_maCbCr matrix parameter Ma (signed value)
7:0	R/W	0	ENCP_VIDEO_MATRIX_mBCbCr matrix parameter Mb (signed value)

Note

These 2 register is for HUE adjustment. The formula is : Cb' = Cb*Ma + Cr*Mb, Cr' = Cb*Mc + Cr*Md

(effective only when Enable HUE matrix by bit ENCP_VIDEO_MODE_ADV[7])

Note

See the descritption of register ENCI_CMPN_MATRIX_CB.

Table 8-3520 ENCP_ VIDEO_MATRIX_CR(VCBUS: 0x1bbc)

Bit(s)	R/W	Default	Description
15:8	R/W	0	ENCP_VIDEO_MATRIX_mCCbCr matrix parameter Mc (signed value)
7:0	R/W	0	ENCP_VIDEO_MATRIX_mDCbCr matrix parameter Md (signed value)

Table 8-3521 ENCP_ VIDEO_RGBIN_CTRL(VCBUS: 0x1bbd)

Bit(s)	R/W	Default	Description
1	R/W	0	USE RGB data from VIU
0	R/W	0	VIU RGB IN BLANK ZERO

Table 8-3522 ENCP_MACV_BLANKY_VAL(VCBUS: 0x1bc0)

Bit(s)	R/W	Default	Description
9 -0	R/W	0	ENCP_MACV_BLANKY_VAL Macrovision Blank Y value

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Table 8-3523 ENCP_MACV_MAXY_VAL(VCBUS: 0x1bc1)

Bit(s)	R/W	Default	Description
9 -0	R/W	590	ENCP_MACV_MAXY_VAL Macrovision max Y value

Table 8-3524 ENCP_MACV_1ST_PSSYNC_STRT(VCBUS: 0x1bc2)

Bit(s)	R/W	Default	Description
10 -0	R/W	238	ENCP_MACV_1ST_PSSYNC_STRT Macrovision first pseudo sync start

Table 8-3525 ENCP_MACV_PSSYNC_STRT(VCBUS: 0x1bc3)

Bit(s)	R/W	Default	Description
7 -0	R/W	1	ENCP_MACV_PSSYNC_STRT Macrovision pseudo sync start

Table 8-3526 ENCP_MACV_AGC_STRT(VCBUS: 0x1bc4)

Bit(s)	R/W	Default	Description
7 -0	R/W	61	ENCP_MACV_AGC_STRT Macrovision AGC pulse start

Table 8-3527 ENCP_MACV_AGC_END(VCBUS: 0x1bc5)

Bit(s)	R/W	Default	Description
7 -0	R/W	141	ENCP_MACV_AGC_END Macrovision AGC pulse end

Table 8-3528 ENCP_MACV_WAVE_END(VCBUS: 0x1bc6)

Bit(s)	R/W	Default	Description
7 -0	R/W	175	ENCP_MACV_WAVE_END Macrovision wave end

Table 8-3529 ENCP_MACV_STRTLINE(VCBUS: 0x1bc7)

Bit(s)	R/W	Default	Description
9 -0	R/W	11	ENCP_MACV_STRTLINE Macrovision start line

Table 8-3530 ENCP_MACV_ENDLINE(VCBUS: 0x1bc8)

Bit(s)	R/W	Default	Description
9 -0	R/W	19	ENCP_MACV_ENDLINE Macrovision end line

Table 8-3531 ENCP_MACV_TS_CNT_MAX_L(VCBUS: 0x1bb9)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'h7dcd	ENCP_MACV_TS_CNT_MAX_L Macrovistion Cyclic Variation Time step Counter, Low 16bits

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Table 8-3532 ENCP_MACV_TS_CNT_MAX_H(VCBUS: 0x1bca)

Bit(s)	R/W	Default	Description
3 -0	R/W	4'h3	ENCP_MACV_TS_CNT_MAX_H Macrovistion Cyclic Variation Time step Counter, High 4bits

Table 8-3533 ENCP_MACV_TIME_DOWN(VCBUS: 0x1bcb)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'd3068	ENCP_MACV_TIME_DOWN Macrovistion Cyclic Variation Time start going down

Table 8-3534 ENCP_MACV_TIME_LO(VCBUS: 0x1bbc)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'd3658	ENCP_MACV_TIME_LO Macrovistion Cyclic Variation Time start maitain low

Table 8-3535 ENCP_MACV_TIME_UP(VCBUS: 0x1bcd)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'd4366	ENCP_MACV_TIME_UP Macrovistion Cyclic Variation Time start going up

Table 8-3536 ENCP_MACV_TIME_RST(VCBUS: 0x1bce)

Bit(s)	R/W	Default	Description
15 -0	R/W	16'd4956	ENCP_MACV_TIME_RST Macrovistion Cyclic Variation reset

Table 8-3537 ENCP_VBI_CTRL(VCBUS: 0x1bd0)

Bit(s)	R/W	Default	Description
1	R/W	0	ENCP_VBI_EN Enable VBI
0	R/W	0	ENCP_VBI_NO_CURVE VBI data no curve at edge

Table 8-3538 ENCP_VBI_SETTING(VCBUS: 0x1bd1)

Bit(s)	R/W	Default	Description
1	R/W	0x16	ENCP_VBI_LNVBI line number
0	R/W	0x28	ENCP_VBI_CNTVBI data bit count

Table 8-3539 ENCP_VBI_BEGIN(VCBUS: 0x1bd2)

Bit(s)	R/W	Default	Description
12-0	R/W	0	ENCP_VBI_BEGIN VBI pulse begin point

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Table 8-3540 ENCP_VBI_WIDTH(VCBUS: 0x1bd3)

Bit(s)	R/W	Default	Description
12-0	R/W	0	ENCP_VBI_WIDTH VBI data bit width

Table 8-3541 ENCP_VBI_HVAL(VCBUS: 0x1bd4)

Bit(s)	R/W	Default	Description
9-0	R/W	0	ENCP_VBI_HVAL VBI pulse value when data bit is high

Table 8-3542 ENCP_VBI_DATA0(VCBUS: 0x1bd5)

Bit(s)	R/W	Default	Description
15-0	R/W	0	ENCP_VBI_data0 VBI data 0 (LSB first)

Table 8-3543 ENCP_VBI_DATA1(VCBUS: 0x1bd6)

Bit(s)	R/W	Default	Description
15-0	R/W	0	ENCP_VBI_data1 VBI data 1 (LSB first)

Table 8-3544 VENC_VDAC_DAC0_GAINCTRL(VCBUS: 0x1bf0)

Bit(s)	R/W	Default	Description
15	R/W	0	DAC_GAIN_BPgain bypass, 1:bypass
14-8	R/W	0	Unused
7-0	R/W	0	DAC_GAINgain setting 8-bit

Table 8-3545 VENC_VDAC_DAC0_OFFSET (VCBUS: 0x1bf1)

Bit(s)	R/W	Default	Description
10-0	R/W	0	DAC_OFFSET_VALoffset setting, 11 bits

Table 8-3546 VENC_VDAC_DAC1_GAINCTRL(VCBUS: 0x1bf2)

Bit(s)	R/W	Default	Description
15	R/W	0	DAC_GAIN_BPgain bypass, 1:bypass
14-8	R/W	0	Unused
7-0	R/W	0	DAC_GAINgain setting 8-bit

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Table 8-3547 VENC_VDAC_DAC1_OFFSET (VCBUS: 0x1bf3)

Bit(s)	R/W	Default	Description
10-0	R/W	0	DAC_OFFSET_VALoffset setting, 11 bits

Table 8-3548 VENC_VDAC_DAC2_GAINCTRL(VCBUS: 0x1bf4)

Bit(s)	R/W	Default	Description
15	R/W	0	DAC_GAIN_BPgain bypass, 1:bypass
14-8	R/W	0	Unused
7-0	R/W	0	DAC_GAINgain setting 8-bit

Table 8-3549 VENC_VDAC_DAC2_OFFSET (VCBUS: 0x1bf5)

Bit(s)	R/W	Default	Description
10-0	R/W	0	DAC_OFFSET_VALoffset setting, 11 bits

Table 8-3550 VENC_VDAC_DAC3_GAINCTRL(VCBUS: 0x1bf6)

Bit(s)	R/W	Default	Description
15	R/W	0	DAC_GAIN_BPgain bypass, 1:bypass
14-8	R/W	0	Unused
7-0	R/W	0	DAC_GAINgain setting 8-bit

Table 8-3551 VENC_VDAC_DAC3_OFFSET (VCBUS: 0x1bf7)

Bit(s)	R/W	Default	Description
10-0	R/W	0	DAC_OFFSET_VALoffset setting, 11 bits

Table 8-3552 VENC_VDAC_DAC4_GAINCTRL(VCBUS: 0x1bf8)

Bit(s)	R/W	Default	Description
15	R/W	0	DAC_GAIN_BPgain bypass, 1:bypass
14-8	R/W	0	Unused
7-0	R/W	0	DAC_GAINgain setting 8-bit

Table 8-3553 VENC_VDAC_DAC4_OFFSET (VCBUS: 0x1bf9)

Bit(s)	R/W	Default	Description
10-0	R/W	0	DAC_OFFSET_VALoffset setting, 11 bits

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Table 8-3554 VENC_VDAC_DAC5_GAINCTRL(VCBUS: 0x1bfa)

Bit(s)	R/W	Default	Description
15	R/W	0	DAC_GAIN_BPgain bypass, 1:bypass
14-8	R/W	0	Unused
7-0	R/W	0	DAC_GAINgain setting 8-bit

Table 8-3555 VENC_VDAC_DAC5_OFFSET (VCBUS: 0x1bfb)

Bit(s)	R/W	Default	Description
10-0	R/W	0	DAC_OFFSET_VALoffset setting, 11 bits

Table 8-3556 VENC_VDAC_FIFO_CTRL (VCBUS: 0x1bfc)

Bit(s)	R/W	Default	Description
14	R/W	0	FIFO_EN_ENCT
13	R/W	0	FIFO_EN_ENCI
12	R/W	0	FIFO_EN_CNCP
11-6	R/W	0	DAC_CLOCK_2X
5-0	R/W	0	DAC_CLOCK_4X

Table 8-3557 ENCL_TCON_INVERT_CTL (VCBUS: 0x1bfd)

Bit(s)	R/W	Default	Description
15	R/W	1'b0	vsync invert
14	R/W	1'b0	hsync invert
13	R/W	1'b0	Oev3 invert
12	R/W	1'b0	Oev2 invert
11	R/W	1'b0	Cpv2 invert
10	R/W	1'b0	Cph3 invert
9	R/W	1'b0	Cph2 invert
8	R/W	1'b0	Cph1 invert
7	R/W	1'b0	oeh invert
6	R/W	1'b0	vcom invert
5	R/W	1'b0	Stv2 invert
4	R/W	1'b0	Stv1 invert
3	R/W	1'b0	Cpv1 invert
2	R/W	1'b0	Oev1 invert
1	R/W	1'b0	Sth1 invert
0	R/W	1'b0	Sth2 invert

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Table 8-3558 ENCI_DVI_HSO_BEGIN(VCBUS: 0x1c00)

Bit(s)	R/W	Default	Description
10-0	R/W	1713	ENCI_DVI_HSO_begINDVI/HDMI Interlace HSO begin position.

Table 8-3559 ENCI_DVI_HSO_END(VCBUS: 0x1c01)

Bit(s)	R/W	Default	Description
10-0	R/W	121	ENCI_DVI_HSO_ENDDVI/HDMI Interlace HSO end position.

Table 8-3560 ENCI_DVI_VSO_BLINE_EVN(VCBUS: 0x1c02)

Bit(s)	R/W	Default	Description
8-0	R/W	261	ENCI_DVI_VSO_bLINE_eVNDVI/HDMI Interlace VSO start line for even field.

Table 8-3561 ENCI_DVI_VSO_BLINE_ODD(VCBUS: 0x1c03)

Bit(s)	R/W	Default	Description
8-0	R/W	261	ENCI_DVI_VSO_bLINE_ODDDVI/HDMI Interlace VSO start line for odd field.

Table 8-3562 ENCI_DVI_VSO_ELINE_EVN(VCBUS: 0x1c04)

Bit(s)	R/W	Default	Description
8-0	R/W	1	ENCI_DVI_VSO_ELINE_eVNDVI/HDMI Interlace VSO end line for even field.

Table 8-3563 ENCI_DVI_VSO_ELINE_ODD(VCBUS: 0x1c05)

Bit(s)	R/W	Default	Description
8-0	R/W	2	ENCI_DVI_VSO_ELINE_ODDDVI/HDMI Interlace VSO end line for odd field.

Table 8-3564 ENCI_DVI_VSO_BEGIN_EVN(VCBUS: 0x1c06)

Bit(s)	R/W	Default	Description
10-0	R/W	855	ENCI_DVI_VSO_BEGIN_eVNDVI/HDMI Interlace VSO begin position for even field.

Table 8-3565 ENCI_DVI_VSO_BEGIN_ODD(VCBUS: 0x1c07)

Bit(s)	R/W	Default	Description
10-0	R/W	1713	ENCI_DVI_VSO_BEGIN_ODDDVI/HDMI Interlace VSO begin position for odd field.

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Table 8-3566 ENCI_DVI_VSO_END_EVN(VCBUS: 0x1c08)

Bit(s)	R/W	Default	Description
10-0	R/W	1713	ENCI_DVI_VSO_END_eVNDVI/HDMI Interlace VSO end position for even field.

Table 8-3567 ENCI_DVI_VSO_END_ODD(VCBUS: 0x1c09)

Bit(s)	R/W	Default	Description
10-0	R/W	855	ENCI_DVI_VSO_END_ODDDVI/HDMI Interlace VSO end position for odd field.

Table 8-3568 ENCI_CFILT_CTRL2(VCBUS: 0x1c0a)

Bit(s)	R/W	Default	Description
15-12	R/W	0	ENCI_CFILT_CVBS_CB_DLYFilter CVBS Cb delay. Delay same as below.
11-8	R/W	0	ENCI_CFILT_CVBS_CR_DLYFilter CVBS Cr delay. Delay same as below.
7-4	R/W	0	ENCI_CFILT_CMPT_CB_DLYFilter component Cb delay. Delay same as below.
3-0	R/W	0	ENCI_CFILT_CMPT_CR_DLYFilter component Cr delay. 0=No delay; 1=Delay by 1 cycle; … 6=Delay by 6 cycles; 7-16=Reserved.

Table 8-3569 ENCI_DACSEL0(VCBUS: 0x1c0b)

Bit(s)	R/W	Default	Description
15-12	R/W	0	VENC_I_DACSEL_0dac3
11-8	R/W	3	VENC_I_DACSEL_0dac2
7-4	R/W	2	VENC_I_DACSEL_0dac1
3-0	R/W	1	VENC_I_DACSEL_0dac0

Table 8-3570 ENCI_DACSEL1(VCBUS: 0x1c0c)

Bit(s)	R/W	Default	Description
9-8	R/W	0	DVI_SYNC_SELDvi_sync_sel
7-4	R/W	5	VENC_I_DACSEL_1dac5
3-0	R/W	4	VENC_I_DACSEL_1dac4

Table 8-3571 ENCP_DACSEL0(VCBUS: 0x1c0d)

Bit(s)	R/W	Default	Description
15-12	R/W	0	VENC_P_DACSEL_0dac3
11-8	R/W	3	VENC_P_DACSEL_0dac2

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Bit(s)	R/W	Default	Description
7-4	R/W	2	VENC_P_DACSEL_0dac1
3-0	R/W	1	VENC_P_DACSEL_0dac0

Table 8-3572 ENCP_DACSEL1(VCBUS: 0x1c0e)

Bit(s)	R/W	Default	Description
9-8	R/W	0	DVI_SYNC_SELDvi_sync_sel
7-4	R/W	5	VENC_P_DACSEL_1dac5
3-0	R/W	4	VENC_P_DACSEL_1dac4

Table 8-3573 ENCP_MAX_LINE_SWITCH_POINT(VCBUS: 0x1c0f)

Bit(s)	R/W	Default	Description
12-0	R/W	13'h1fff	ENCP_DE_V_END_ODDmax_pxcnt = (line >= cfg_max_line_switch_point) ? max_pxcnt_tmp - 1: max_pxcnt_tmp;

Table 8-3574 ENCI_TST_EN(VCBUS: 0x1c10)

Bit(s)	R/W	Default	Description
0	R/W	0	ENCI_TST_ENEnci_tst_en

Table 8-3575 ENCI_TST_MDSEL(VCBUS: 0x1c11)

Bit(s)	R/W	Default	Description
1-0	R/W	1	ENCI_TST_MDSELVideo test mode selectl: 0: enci_tst_mode_fixval 1: enci_tst_mode_colorbar 2: enci_tst_mode_thinline 3: enci_tst_mode_dotgrid

Table 8-3576 ENCI_TST_Y(VCBUS: 0x1c12)

Bit(s)	R/W	Default	Description
9-0	R/W	512	ENCI_TST_YDefault value of Y in test mode

Table 8-3577 ENCI_TST_CB(VCBUS: 0x1c13)

Bit(s)	R/W	Default	Description
9-0	R/W	512	ENCI_TST_CBDefault value of Pb in test mode

Table 8-3578 ENCI_TST_CR(VCBUS: 0x1c14)

Bit(s)	R/W	Default	Description
9-0	R/W	512	ENCI_TST_CRDefault value of Pr in test mode

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Table 8-3579 ENCI_TST_CLRBAR_STRT (VCBUS: 0x1c15)

Bit(s)	R/W	Default	Description
10-0	R/W	275	ENCI_TST_CLRBAR_STRTColor bar start position

Table 8-3580 ENCI_TST_CLRBAR_WIDTH (VCBUS: 0x1c16)

Bit(s)	R/W	Default	Description
10-0	R/W	360	ENCI_TST_CLRBAR_WIDTHColor bar width

Table 8-3581 ENCI_TST_VDCNT_STSEL (VCBUS: 0x1c17)

Bit(s)	R/W	Default	Description
1-0	R/W	0	ENCI_TST_VDCNT_STSELFor interlace, vdata count set when color bar start

Table 8-3582 ENCI_VFIFO2VD_CTL (VCBUS: 0x1c18)

Bit(s)	R/W	Default	Description
15-8	R/W	‘h1b	Vfifo2vd_vd_selVfifo2vd_vd_sel: 00_01_10_11 -> Y_Cb_Y_Cr
7	R/W	0	Vfifo2vd_dropVfifo2vd_drop
6-1	R/W	0	Vfifo2vd_delayVfifo2vd_delay
0	R/W	0	Vfifo2vd_enVfifo2vd_en

Table 8-3583 ENCI_VFIFO2VD_PIXEL_START (VCBUS: 0x1c19)

Bit(s)	R/W	Default	Description
12-0	R/W	0	Enci_vfifo2vd_pixel_startVfifo2vd_pixel_start

Table 8-3584 ENCI_VFIFO2VD_PIXEL_END (VCBUS: 0x1c1a)

Bit(s)	R/W	Default	Description
15	R/W	0	VDATA_YC_USE_FIRST_VFIFO_REGVdata_yc_use_vfifo_req
14-13			RESERVED
12-0	R/W	0	Enci_vfifo2vd_pixel_ENDVfifo2vd_pixel_end

Table 8-3585 ENCI_VFIFO2VD_LINE_TOP_START (VCBUS: 0x1c1b)

Bit(s)	R/W	Default	Description
10-0	R/W	0	Enci_vfifo2vd_LINE_TOP_STARTenci_vfifo2vd_line_top_start

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Table 8-3586 ENCI_VFIFO2VD_LINE_TOP_END (VCBUS: 0x1c1c)

Bit(s)	R/W	Default	Description
10-0	R/W	0	Enci_vfifo2vd_LINE_TOP_ENDenci_vfifo2vd_line_top_end

Table 8-3587 ENCI_VFIFO2VD_LINE_BOT_START (VCBUS: 0x1c1d)

Bit(s)	R/W	Default	Description
10-0	R/W	0	Enci_vfifo2vd_LINE_BOT_STARTenci_vfifo2vd_line_bot_start

Table 8-3588 ENCI_VFIFO2VD_LINE_BOT_END (VCBUS: 0x1c1e)

Bit(s)	R/W	Default	Description
10-0	R/W	0	Enci_vfifo2vd_LINE_BOT_ENDenci_vfifo2vd_line_bot_end

Table 8-3589 ENCI_VFIFO2VD_CTL2 (VCBUS: 0x1c1f)

Bit(s)	R/W	Default	Description
1	R/W	1	CFG_VFIFO2VD_OUT_SCALER_BYPASScfg_vfifo2vd_out_scaler_bypass
0	R/W	0	CFG_VFIFO_DIN_FULL_RANGEcfg_vfifo_din_full_range

Table 8-3590 ENCP_DVI_HSO_BEGIN(VCBUS: 0x1c30)

Bit(s)	R/W	Default	Description
12-0	R/W	16	ENCP_DVI_HSO_BEGIN DVI/HDMI Hsync out start point

Table 8-3591 ENCP_DVI_HSO_END(VCBUS: 0x1c31)

Bit(s)	R/W	Default	Description
12-0	R/W	32	ENCP_DVI_HSO_ENdDVI/HDMI Hsync out end point

Table 8-3592 ENCP_DVI_VSO_BLINE_EVN(VCBUS: 0x1c32)

Bit(s)	R/W	Default	Description
10-0	R/W	40	ENCP_DVI_VSO_BLINE_EVNDVI/HDMI Vsync out start line for even field.

Table 8-3593 ENCP_DVI_VSO_BLINE_ODD(VCBUS: 0x1c33)

Bit(s)	R/W	Default	Description
10-0	R/W	40	ENCP_DVI_VSO_BLINE_ODDDVI/HDMI Vsync out start line for odd field.

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Table 8-3594 ENCP_DVI_VSO_ELINE_EVN(VCBUS: 0x1c34)

Bit(s)	R/W	Default	Description
10-0	R/W	42	ENCP_DVI_VSO_ELINE_EVNDVI/HDMI Vsync out end line for even field.

Table 8-3595 ENCP_DVI_VSO_ELINE_ODD(VCBUS: 0x1c35)

Bit(s)	R/W	Default	Description
10-0	R/W	42	ENCP_DVI_VSO_ELINE_ODDDVI/HDMI Vsync out end line for odd field.

Table 8-3596 ENCP_DVI_VSO_BEGIN_EVN(VCBUS: 0x1c36)

Bit(s)	R/W	Default	Description
12-0	R/W	16	ENCP_DVI_VSO_BEGIN_EVNDVI/HDMI Vsync out start point for even field.

Table 8-3597 ENCP_DVI_VSO_BEGIN_ODD(VCBUS: 0x1c37)

Bit(s)	R/W	Default	Description
12-0	R/W	16	ENCP_DVI_VSO_BEGIN_ODDDVI/HDMI Vsync out start point for odd field.

Table 8-3598 ENCP_DVI_VSO_END_EVN(VCBUS: 0x1c38)

Bit(s)	R/W	Default	Description
12-0	R/W	32	ENCP_DVI_VSO_END_EVNDVI/HDMI Vsync out end point for even field.

Table 8-3599 ENCP_DVI_VSO_END_ODD(VCBUS: 0x1c39)

Bit(s)	R/W	Default	Description
12-0	R/W	32	ENCP_DVI_VSO_END_ODDDVI/HDMI Vsync out end point for odd field.

Table 8-3600 ENCP_DE_H_BEGIN(VCBUS: 0x1c3a)

Bit(s)	R/W	Default	Description
12-0	R/W	217	ENCP_DE_H_BEGINDVI/HDMI horizontal active video start point.

Table 8-3601 ENCP_DE_H_END(VCBUS: 0x1c3b)

Bit(s)	R/W	Default	Description
12-0	R/W	1657	ENCP_DE_H_ENDDVI/HDMI horizontal active video end point.

Table 8-3602 ENCP_DE_V_BEGIN_EVEN(VCBUS: 0x1c3c)

Bit(s)	R/W	Default	Description
10-0	R/W	42	ENCP_DE_V_BEGIN_EVENDVI/HDMI vertical active video start line for even field.

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Table 8-3603 ENCP_DE_V_END_EVEN(VCBUS: 0x1c3d)

Bit(s)	R/W	Default	Description
10-0	R/W	519	ENCP_DE_V_END_EVENDVI/HDMI vertical active video end line for even field.

Table 8-3604 ENCP_DE_V_BEGIN_ODD(VCBUS: 0x1c3e)

Bit(s)	R/W	Default	Description
10-0	R/W	42	ENCP_DE_V_BEGIN_ODDDVI/HDMI vertical active video start line for odd field.

Table 8-3605 ENCP_DE_V_END_ODD(VCBUS: 0x1c3f)

Bit(s)	R/W	Default	Description
10-0	R/W	519	ENCP_DE_V_END_ODDDVI/HDMI vertical active video end line for odd field.

Table 8-3606 ENCI_SYNC_LINE_LENGTH(VCBUS: 0x1c40)

Bit(s)	R/W	Default	Description
15-11	R/W	0	SYNC_PULSE_LENGTHSync_pulse_length
10-0	R/W	0	SYNC_PULSE_START_LINESync_pulse_start_line

Table 8-3607 ENCI_SYNC_PIXEL_EN(VCBUS: 0x1c41)

Bit(s)	R/W	Default	Description
15	R/W	0	SYNC_PULSE_ENSync_pulse_enable
14	R/W	0	RESERVED
13	R/W	0	Short_fussy_syncShort_fussy_sync
12-0	R/W	0	SYNC_PULSE_START_PIXELSync_pulse_start_pixel

Table 8-3608 ENCI_SYNC_TO_LINE_EN(VCBUS: 0x1c42)

Bit(s)	R/W	Default	Description
15	R/W	0	Enci_sync_enalbeEnci_sync_enable
14	R/W	0	Encp_sync_enalbeEncp_sync_enable
13	R/W	0	Enct_sync_enalbeEnct_sync_enable
12	R/W	0	Encl_sync_enalbeEncl_sync_enable
11	R/W	0	Fussy_sync_enableFussy_sync_enable
10-0	R/W	0	Sync_pulse_target_lineSync_pulse_target_line

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Table 8-3609 ENCI_SYNC_TO_PIXEL(VCBUS: 0x1c43)

Bit(s)	R/W	Default	Description
12-0	R/W	0	SYNC_PULSE_target_pixelSync_pulse_target_pixel

Table 8-3610 ENCP_SYNC_LINE_LENGTH(VCBUS: 0x1c44)

Bit(s)	R/W	Default	Description
15-11	R/W	0	SYNC_PULSE_LENGTHSync_pulse_length
10-0	R/W	0	SYNC_PULSE_START_LINESync_pulse_start_line

Table 8-3611 ENCP_SYNC_PIXEL_EN(VCBUS: 0x1c45)

Bit(s)	R/W	Default	Description
15	R/W	0	SYNC_PULSE_ENSync_pulse_enable
14	R/W	0	RESERVED
13	R/W	0	Short_fussy_syncShort_fussy_sync
12-0	R/W	0	SYNC_PULSE_START_PIXELSync_pulse_start_pixel

Table 8-3612 ENCP_SYNC_TO_LINE_EN(VCBUS: 0x1c46)

Bit(s)	R/W	Default	Description
15	R/W	0	Enci_sync_enalbeEnci_sync_enable
14	R/W	0	Encp_sync_enalbeEncp_sync_enable
13	R/W	0	Enct_sync_enalbeEnct_sync_enable
12	R/W	0	Encl_sync_enalbeEncl_sync_enable
11	R/W	0	Fussy_sync_enableFussy_sync_enable
10-0	R/W	0	Sync_pulse_target_lineSync_pulse_target_line

Table 8-3613 ENCP_SYNC_TO_PIXEL(VCBUS: 0x1c47)

Bit(s)	R/W	Default	Description
12-0	R/W	0	SYNC_PULSE_target_pixelSync_pulse_target_pixel

Table 8-3614 ENCT_SYNC_LINE_LENGTH(VCBUS: 0x1c48)

Bit(s)	R/W	Default	Description
15-11	R/W	0	SYNC_PULSE_LENGTHSync_pulse_length
10-0	R/W	0	SYNC_PULSE_START_LINESync_pulse_start_line

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Table 8-3615 ENCT_SYNC_PIXEL_EN(VCBUS: 0x1c49)

Bit(s)	R/W	Default	Description
15	R/W	0	SYNC_PULSE_ENSync_pulse_enable
14	R/W	0	RESERVED
13	R/W	0	Short_fussy_syncShort_fussy_sync
12-0	R/W	0	SYNC_PULSE_START_PIXELSync_pulse_start_pixel

Table 8-3616 ENCT_SYNC_TO_LINE_EN(VCBUS: 0x1c4a)

Bit(s)	R/W	Default	Description
15	R/W	0	Enci_sync_enalbeEnci_sync_enable
14	R/W	0	Encp_sync_enalbeEncp_sync_enable
13	R/W	0	Enct_sync_enalbeEnct_sync_enable
12	R/W	0	Encl_sync_enalbeEncl_sync_enable
11	R/W	0	Fussy_sync_enableFussy_sync_enable
10-0	R/W	0	Sync_pulse_target_lineSync_pulse_target_line

Table 8-3617 ENCT_SYNC_TO_PIXEL(VCBUS: 0x1c4b)

Bit(s)	R/W	Default	Description
12-0	R/W	0	SYNC_PULSE_target_pixelSync_pulse_target_pixel

Table 8-3618 ENCL_SYNC_LINE_LENGTH(VCBUS: 0x1c4c)

Bit(s)	R/W	Default	Description
15-11	R/W	0	SYNC_PULSE_LENGTHSync_pulse_length
10-0	R/W	0	SYNC_PULSE_START_LINESync_pulse_start_line

Table 8-3619 ENCL_SYNC_PIXEL_EN(VCBUS: 0x1c4d)

Bit(s)	R/W	Default	Description
15	R/W	0	SYNC_PULSE_ENSync_pulse_enable
14	R/W	0	RESERVED
13	R/W	0	Short_fussy_syncShort_fussy_sync
12-0	R/W	0	SYNC_PULSE_START_PIXELSync_pulse_start_pixel

Table 8-3620 ENCL_SYNC_TO_LINE_EN(VCBUS: 0x1c4e)

Bit(s)	R/W	Default	Description
15	R/W	0	Enci_sync_enalbeEnci_sync_enable
14	R/W	0	Encp_sync_enalbeEncp_sync_enable

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Bit(s)	R/W	Default	Description
13	R/W	0	Enct_sync_enalbeEnct_sync_enable
12	R/W	0	Encl_sync_enalbeEncl_sync_enable
11	R/W	0	Fussy_sync_enableFussy_sync_enable
10-0	R/W	0	Sync_pulse_target_lineSync_pulse_target_line

Table 8-3621 ENCL_SYNC_TO_PIXEL(VCBUS: 0x1c4f)

Bit(s)	R/W	Default	Description
12-0	R/W	0	SYNC_PULSE_target_pixelSync_pulse_target_pixel

Table 8-3622 ENCP_VFIFO2VD_CTL2(VCBUS: 0x1c50)

Bit(s)	R/W	Default	Description
3	R/W	1	VFIFO2VD_encp_lcd_scaler_bypass0=Scale LCD input data to y [16*4,235*4], c [16*4,240*4] 1=Do not scale LCD input data
2	R/W	1	VFIFO2VD_encp_vadj_scaler_bypass0=Scale enc480p_vadj input data to y [16*4,235*4], c [16*4,240*4] 1=Do not scale data to enc480p_vadj
1	R/W	1	VFIFO2VD_out_scaler_bypass0=Scale vfifo2vd's output vdata to y[16,235], c [16,240] 1=Do not scale vfifo2vd's output vdata
0	R/W	0	VFIFO_DIN_FULL_RANGE0=Data from viu fifo is y[16*4,235*4], c[16*4,240*4] 1=Data from viu fifo is full range [0,1023]

Table 8-3623 VENC_DVI_SETTING_MORE(VCBUS: 0x1c51)

Bit(s)	R/W	Default	Description
0	R/W	0	VENC_DVI_SEL_RGB Applicable for using on-chip hdmi tx module only. This bit controls correct bit-mapping from Venc to hdmi_tx depending on whether YCbCr or RGB mode. 0=Map data bit from Venc to hdmi_tx for YCbCr mode; 1=Map data bit from Venc to hdmi_tx for RGB mode.

Table 8-3624 VENC_VDAC_DAC0_FILT_CTRL0(VCBUS: 0x1c58)

Bit(s)	R/W	Default	Description
0	R/W	0	VENC_VDAC_DAC0_FILT_EN 0=Bypass filter; 1=Enable filter.

Note

↓dout = ((din + din_d2) * coef1 + (din_d1 * coef0) + 32) >> 6

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Table 8-3625 VENC_VDAC_DAC0_FILT_CTRL1(VCBUS: 0x1c59)

Bit(s)	R/W	Default	Description
15-8	R/W	8'h00	VENC_VDAC_DAC0_FILT_COEF1Filter coef1
7-0	R/W	8'h40	VENC_VDAC_DAC0_FILT_COEF0Filter coef0

Table 8-3626 VENC_VDAC_DAC1_FILT_CTRL0(VCBUS: 0x1c5a)

Bit(s)	R/W	Default	Description
0	R/W	0	VENC_VDAC_DAC1_FILT_EN 0=Bypass filter; 1=Enable filter.

Note

↓dout = ((din + din_d2) * coef1 + (din_d1 * coef0) + 32) >> 6

Table 8-3627 VENC_VDAC_DAC1_FILT_CTRL1(VCBUS: 0x1c5b)

Bit(s)	R/W	Default	Description
15-8	R/W	8'h00	VENC_VDAC_DAC1_FILT_COEF1Filter coef1
7-0	R/W	8'h40	VENC_VDAC_DAC1_FILT_COEF0Filter coef0

Table 8-3628 VENC_VDAC_DAC2_FILT_CTRL0(VCBUS: 0x1c5c)

Bit(s)	R/W	Default	Description
0	R/W	0	VENC_VDAC_DAC2_FILT_EN 0=Bypass filter; 1=Enable filter.

Note

↓dout = ((din + din_d2) * coef1 + (din_d1 * coef0) + 32) >> 6

Table 8-3629 VENC_VDAC_DAC2_FILT_CTRL1(VCBUS: 0x1c5d)

Bit(s)	R/W	Default	Description
15-8	R/W	8'h00	VENC_VDAC_DAC2_FILT_COEF1Filter coef1
7-0	R/W	8'h40	VENC_VDAC_DAC2_FILT_COEF0Filter coef0

Table 8-3630 VENC_VDAC_DAC3_FILT_CTRL0(VCBUS: 0x1c5e)

Bit(s)	R/W	Default	Description
0	R/W	0	VENC_VDAC_DAC3_FILT_EN 0=Bypass filter; 1=Enable filter.

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Note

↓dout = ((din + din_d2) * coef1 + (din_d1 * coef0) + 32) >> 6

Table 8-3631 VENC_VDAC_DAC3_FILT_CTRL1(VCBUS: 0x1c5f)

Bit(s)	R/W	Default	Description
15-8	R/W	8'h00	VENC_VDAC_DAC3_FILT_COEF1Filter coef1
7-0	R/W	8'h40	VENC_VDAC_DAC3_FILT_COEF0Filter coef0

Table 8-3632 ENCL_VFIFO2VD_CTL (VCBUS: 0x1c90)

Bit(s)	R/W	Default	Description
15-8	R/W	'h1b	Vfifo2vd_vd_selVfifo2vd_vd_sel: 00_01_10_11 -> Y_Cb_Y_Cr
7	R/W	0	Vfifo2vd_dropVfifo2vd_drop
6-1	R/W	0	Vfifo2vd_delayVfifo2vd_delay
0	R/W	0	Vfifo2vd_enVfifo2vd_en

Table 8-3633 ENCL_VFIFO2VD_PIXEL_START (VCBUS: 0x1c91)

Bit(s)	R/W	Default	Description
12-0	R/W	0	ENCL_vfifo2vd_pixel_startVfifo2vd_pixel_start

Table 8-3634 ENCL_VFIFO2VD_PIXEL_END (VCBUS: 0x1c92)

Bit(s)	R/W	Default	Description
15	R/W	0	VDATA_YC_USE_FIRST_VFIFO_REGVdata_yc_use_vfifo_req
14-13			RESERVED
12-0	R/W	0	ENCL_vfifo2vd_pixel_ENDVfifo2vd_pixel_end

Table 8-3635 ENCL_VFIFO2VD_LINE_TOP_START (VCBUS: 0x1c93)

Bit(s)	R/W	Default	Description
10-0	R/W	0	ENCL_vfifo2vd_LINE_TOP_STARTENCL_vfifo2vd_line_top_start

Table 8-3636 ENCL_VFIFO2VD_LINE_TOP_END (VCBUS: 0x1c94)

Bit(s)	R/W	Default	Description
10-0	R/W	0	ENCL_vfifo2vd_LINE_TOP_ENDENCL_vfifo2vd_line_top_end

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Table 8-3637 ENCL_VFIFO2VD_LINE_BOT_START (VCBUS: 0x1c95)

Bit(s)	R/W	Default	Description
10-0	R/W	0	ENCL_vfifo2vd_LINE_BOT_STARTENCL_vfifo2vd_line_bot_start

Table 8-3638 ENCL_VFIFO2VD_LINE_BOT_END (VCBUS: 0x1c96)

Bit(s)	R/W	Default	Description
10-0	R/W	0	ENCL_vfifo2vd_LINE_BOT_ENDENCL_vfifo2vd_line_bot_end

Table 8-3639 ENCL_VFIFO2VD_CTL2 (VCBUS: 0x1c97)

Bit(s)	R/W	Default	Description
3	R/W	1	CFG_ENCL_LCD_SCALER_BYPASScfg_encl_lcd_scaler_bypass
2	R/W	1	CFG_ENCL_VADJ_SCALER_BYPASScfg_encl_vadj_scaler_bypass
1	R/W	1	CFG_VFIFO2VD_OUT_SCALER_BYPASScfg_vfifo2vd_out_scaler_bypass
0	R/W	0	CFG_VFIFO_DIN_FULL_RANGEcfg_vfifo_din_full_range

Table 8-3640 ENCL_TST_EN(VCBUS: 0x1c98)

Bit(s)	R/W	Default	Description
0	R/W	0	ENCL_TST_ENENCL_tst_en

Table 8-3641 ENCL_TST_MDSEL(VCBUS: 0x1c99)

Bit(s)	R/W	Default	Description
1-0	R/W	1	ENCL_TST_MDSELVideo test mode selectl: 0: ENCL_tst_mode_fixval 1: ENCL_tst_mode_colorbar 2: ENCL_tst_mode_thinline 3: ENCL_tst_mode_dotgrid

Table 8-3642 ENCL_TST_Y(VCBUS: 0x1c9a)

Bit(s)	R/W	Default	Description
9-0	R/W	512	ENCL_TST_YDefault value of Y in test mode

Table 8-3643 ENCL_TST_CB(VCBUS: 0x1c9b)

Bit(s)	R/W	Default	Description
9-0	R/W	512	ENCL_TST_CBDefault value of Pb in test mode

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Table 8-3644 ENCL_TST_CR(VCBUS: 0x1c9c)

Bit(s)	R/W	Default	Description
9-0	R/W	512	ENCL_TST_CRDefault value of Pr in test mode

Table 8-3645 ENCL_TST_CLRBAR_STRT (VCBUS: 0x1c9d)

Bit(s)	R/W	Default	Description
10-0	R/W	275	ENCL_TST_CLRBAR_STRTColor bar start position

Table 8-3646 ENCL_TST_CLRBAR_WIDTH (VCBUS: 0x1c9e)

Bit(s)	R/W	Default	Description
10-0	R/W	360	ENCL_TST_CLRBAR_WIDTHColor bar width

Table 8-3647 ENCL_TST_VDCNT_STSEL (VCBUS: 0x1c9f)

Bit(s)	R/W	Default	Description
1-0	R/W	0	ENCL_TST_VDCNT_STSELFor interlace, vdata count set when color bar start

Table 8-3648 ENCL_VIDEO_EN(VCBUS: 0x1ca0)

Bit(s)	R/W	Default	Description
0	R/W	0	ENCL_VIDEO_EN Progressive encoder enable

Table 8-3649 ENCL_VIDEO_Y_SCL(VCBUS: 0x1ca1)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd81	ENCL_VIDEO_Y_SCL Y scale

Table 8-3650 ENCL_VIDEO_PB_SCL(VCBUS: 0x1ca2)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd79	ENCL_VIDEO_PB_SCL Cb scale

Table 8-3651 ENCL_VIDEO_PR_SCL(VCBUS: 0x1ca3)

Bit(s)	R/W	Default	Description
7 -0	R/W	8'd79	ENCL_VIDEO_PR_SCL Cr scale

Table 8-3652 ENCL_VIDEO_Y_OFFST(VCBUS: 0x1ca4)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h3c0	ENCL_VIDEO_Y_OFFST Y offset

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Table 8-3653 ENCL_VIDEO_PB_OFFST(VCBUS: 0x1ca5)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h0	ENCL_VIDEO_PB_OFFST Pb offset

Table 8-3654 ENCL_VIDEO_PR_OFFST(VCBUS: 0x1ca6)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'h0	ENCL_VIDEO_PR_OFFST Pr offset

Table 8-3655 ENCL_VIDEO_MODE(VCBUS: 0x1ca7)

Bit(s)	R/W	Dflt	Description
15	R/W	0	ENCL_PX_LN_CNT_SHADOW_ENpixel count and line count shadow enable
14	R/W	0	ENCL_VIDEO_mode_DE_V1=DE signal's polarity is active high. 0=DE signal's polarity is active low.
13	R/W	0	ENCL_VIDEO_mode_en_VSO_OFLD_FIX_EN1=Extend ENCL_VIDEO_MAX_PXCNT by 1 pixel. 0=Treat ENCL_VIDEO_MAX_PXCNT as is.
12	R/W	0	ENCL_VIDEO_mode_en_field_oddEnable odd field (for 1080i)
11	R/W	0	ENCL_VIDEO_mode_en_vso_ofldEnable VSO odd field
10	R/W	0	ENCL_VIDEO _mode_en_vavon_ofldEnable Vertical Active On for odd field
9	R/W	0	ENCL_VIDEO _mode_en_equ_ofldEnable Equalization pulse for odd field
8	R/W	0	ENCL_VIDEO _mode_en_equEnable Equalization pulse
6	R/W	0	ENCL_VIDEO _mode_en_hseq_switchEnable Hsync and equalization pulse switch in center
5	R/W	0	ENCL_VIDEO _mode_en_vp_ofldEnable Vertical Pulse for odd field
4	R/W	0	ENCL_VIDEO _mode_en_vp2Enable 2nd vertical pulse in a line (1080i)
3	R/W	0	ENCL_VIDEO _mode_en_vpe_half_ofldVertical pulse on end line of odd field is half
2	R/W	0	ENCL_VIDEO _mode_en_vpb_half_ofldVertical pulse on begin line of odd field is half
1	R/W	0	ENCL_VIDEO _mode_en_vpe_halfVertical pulse on end line of even field is half
0	R/W	0	ENCL_VIDEO _mode_en_vpb_halfVertical pulse on begin line of even field is half

Note

↑This register should be set as 0x140 for 720p , and 0x1ffc for 1080i, and 0x0 for 480p/576p

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Table 8-3656 ENCL_VIDEO_MODE_ADV(VCBUS: 0x1ca8)

Bit(s)	R/W	Default	Description
15-14	R/W	0	ENCL_SP_TIMING_CTRL
13-11	R/W	0	RESERVED
10	R/W	0	ENCL_SEL_GAMMA_RGB_IN
9-8	R/W	0	Unused
7	R/W	0	ENCL_VIDEO_MODE_ADV_EN_HUE_MATRIXEnable HUE matrix
6	R/W	0	ENCL_VIDEO_MODE_ADV_SWAP_PBPRSwap PB PR
5	R/W	0	ENCL_VIDEO_MODE_ADV_EN_HS_PBPREnable sync pulse on PB PR
4	R/W	0	ENCL_VIDEO_MODE_ADV_GAIN_HDTVYPBPR gain as HDTV type
3	R/W	0	ENCL_VIDEO_MODE_ADV_VFIFO_ENData input from VFIFO
2 :0	R/W	1	ENCL_VIDEO_MODE_ADV_VFIFO_UPMODESampling rate: 0: 1 1: ½ 2: ¼ 3: 1/8

Table 8-3657 ENCL_DBG_PX_RST(VCBUS: 0x1ca9)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCL_DBG_PX_RST_EN Debug mode pixel reset enable
12 -0	R/W	0	ENCL_DBG_PX_RST_VAL Debug mode pixel reset value

Table 8-3658 ENCL_DBG_LN_RST(VCBUS: 0x1caa)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCL_DBG_LN_RST_EN Debug mode line reset enable
10 -0	R/W	0	ENCL_DBG_LN_RST_VAL Debug mode line reset value

Table 8-3659 ENCL_DBG_PX_INT(VCBUS: 0x1cab)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCL_DBG_PX_INT_EN Pixel Interrupt enable
12 -0	R/W	0	ENCL_DBG_PX_INT_VAL Pixel value that trig the interrupt

Table 8-3660 ENCL_DBG_LN_INT(VCBUS: 0x1cac)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCL_DBG_LN_INT_EN Line Interrupt enable
10 -0	R/W	0	ENCL_DBG_LN_INT_VAL Line value that trig the interrupt

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Table 8-3661 ENCL_VIDEO_YFP1_HTIME(VCBUS: 0x1cad)

Bit(s)	R/W	Default	Description
12 -0	R/W	240	ENCL_VIDEO_YFP1_HTIME Filter switch start point

Table 8-3662 ENCL_VIDEO_YFP2_HTIME(VCBUS: 0x1cae)

Bit(s)	R/W	Default	Description
12 -0	R/W	1657	ENCL_VIDEO_YFP2_HTIME Filter switch end point

Table 8-3663 ENCL_VIDEO_YC_DLY(VCBUS: 0x1caf)

Bit(s)	R/W	Default	Description
10	R/W	0	ENCL_VIDEO_Y_DLY Y delay[2]
8	R/W	0	ENCL_VIDEO_CB_DLY Cb delay[2]
6	R/W	0	ENCL_VIDEO_CR_DLY Cr delay[2]
5 –4	R/W	0	ENCL_VIDEO_Y_DLY Y delay[1:0]
3 –2	R/W	0	ENCL_VIDEO_Cb_DLY Cb delay[1:0]
1 –0	R/W	0	ENCL_VIDEO_Cr_DLY Cr delay[1:0]

Table 8-3664 ENCL_VIDEO_MAX_PXCNT(VCBUS: 0x1cb0)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd1715	ENCL_VIDEO_MAX_PXCNT Max pixel counter

Table 8-3665 ENCL_VIDEO_HAVON_END(VCBUS: 0x1cb1)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd1656	ENCL_VIDEO_havon_END Vertical active video end point

Table 8-3666 ENCL_VIDEO_HAVON_BEGIN(VCBUS: 0x1cb2)

Bit(s)	R/W	Default	Description
12 -0	R/W	13'd217	ENCL_VIDEO_havon_BEGINVertical active video start point

Table 8-3667 ENCL_VIDEO_VAVON_ELINE(VCBUS: 0x1cb3)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd519	ENCL_VIDEO_vavon_ElineVertical active video on end line

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Table 8-3668 ENCL_VIDEO_VAVON_BLINE(VCBUS: 0x1cb4)

Bit(s)	R/W	Default	Description
10 -0	R/W	11'd42	ENCL_VIDEO_vavon_BlineVertical active video on start line

Table 8-3669 ENCL_VIDEO_HSO_BEGIN(VCBUS: 0x1cb5)

Bit(s)	R/W	Default	Description
12 -0	R/W	16	ENCL_VIDEO_hSO_begin Digital Hsync out start point

Table 8-3670 ENCL_VIDEO_HSO_END(VCBUS: 0x1cb6)

Bit(s)	R/W	Default	Description
12 -0	R/W	32	ENCL_VIDEO_hSO_endDigital Hsync out end point

Table 8-3671 ENCL_VIDEO_VSO_BEGIN(VCBUS: 0x1cb7)

Bit(s)	R/W	Default	Description
12 -0	R/W	16	ENCL_VIDEO_VSO_begin Digital Vsync out start point

Table 8-3672 ENCL_VIDEO_VSO_END(VCBUS: 0x1cb8)

Bit(s)	R/W	Default	Description
12 -0	R/W	32	ENCL_VIDEO_VSO_endDigital Vsync out end point

Table 8-3673 ENCL_VIDEO_VSO_BLINE(VCBUS: 0x1cb9)

Bit(s)	R/W	Default	Description
10 -0	R/W	37	ENCL_VIDEO_vSO_blineDigital Vsync out start line

Table 8-3674 ENCL_VIDEO_VSO_ELINE(VCBUS: 0x1cba)

Bit(s)	R/W	Default	Description
10 -0	R/W	39	ENCL_VIDEO_vSO_elineDigital Vsync out end line

Table 8-3675 ENCL_VIDEO_MAX_LNCNT(VCBUS: 0x1cbb)

Bit(s)	R/W	Default	Description
10 -0	R/W	10'd524	ENCL_VIDEO_MAX_LNCNT Max line counter

Table 8-3676 ENCL_VIDEO_BLANKY_VAL(VCBUS: 0x1cbc)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd240	ENCL_VIDEO_BLANKY_VAL Blank Y value

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Table 8-3677 ENCL_VIDEO_BLANKPB_VAL(VCBUS: 0x1cbd)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd512	ENCL_VIDEO_BLANKPB_VAL Blank Pb value

Table 8-3678 ENCL_VIDEO_BLANKPR_VAL(VCBUS: 0x1cbe)

Bit(s)	R/W	Default	Description
9 -0	R/W	10'd512	ENCL_VIDEO_BLANKPR_VAL Blank Pr value

Table 8-3679 ENCL_VIDEO_HOFFST(VCBUS: 0x1cbf)

Bit(s)	R/W	Default	Description
13 -0	R/W	2	ENCL_VIDEO_HOFFST Horizontal offset after HSI

Table 8-3680 ENCL_VIDEO_VOFFST(VCBUS: 0x1cc0)

Bit(s)	R/W	Default	Description
10 -0	R/W	0	ENCL_VIDEO_VOFFST Vertical offset after VSI

Table 8-3681 ENCL_VIDEO_RGB_CTRL(VCBUS: 0x1cc1)

Bit(s)	R/W	Default	Description
12-8	R/W	4	CFG_BLANK_DLY
6	R/W	0	Cfg_macv_R_en
5	R/W	0	Cfg_macv_G_en
4	R/W	0	Cfg_macv_b_en
3	R/W	0	CFG_RGB_SEPERATE_BLANK
2	R/W	0	Cfg_sync_on_r
1	R/W	1	Cfg_sync_on_g
0	R/W	0	Cfg_sync_on_b

Table 8-3682 ENCL_VIDEO_FILT_CTRL(VCBUS: 0x1cc2)

Bit(s)	R/W	Default	Description
15	R/W	0	ENCL_VIDEO_FILT_YF_NO_RSTDisable auto reset Y filter each line
14	R/W	0	ENCL_VIDEO_FILT_CF_NO_RSTDisable auto reset C filter each line
14	R/W	0	CFILT_CHROMA444_ENIf true, chroma input is 444 instead of 422
12	R/W	0	ENCL_VIDEO_FILT_BYPASS_TOPBy pass all ENCL filter
11	R/W	0	ENCL_VIDEO_FILT_CHROMA_SWAPSwap chroma Cb/Cr

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Bit(s)	R/W	Default	Description
10:8	R/W	0	ENCL_VIDEO_FILT_CF_BYPASSChroma filter by pass control
7:4	R/W	0	ENCL_VIDEO_FILT_YF_CFG14 bits Y filter parameter 1, will be effect after ENCL_VIDEO_YFP1_HTIME bit 3 : Edge mode bit 2:0 : By pass ctrl
3:0	R/W	0	ENCL_VIDEO_FILT_YF_CFG24 bits Y filter parameter 1, will be effect after ENCL_VIDEO_YFP1_HTIME bit 3 : Edge mode bit 2:0 : By pass ctrl

Table 8-3683 ENCL_ VIDEO_OFLD_VPEQ_OFST(VCBUS: 0x1cc3)

Bit(s)	R/W	Default	Description
15:12	R/W	0	ENCL_VIDEO_OFLD_VPULS_OFST_BEGINOdd field Vsync pulse offset begin, 4 bits signed.
11:8	R/W	1	ENCL_VIDEO_OFLD_VPULS_OFST_ENDOdd field Vsync pulse offset end, 4 bits signed.
7:4	R/W	0	ENCL_VIDEO_OFLD_EQPULS_OFST_BEGINOdd field EQU pulse offset begin, 4 bits signed.
3:0	R/W	0	ENCL_VIDEO_OFLD_EQPULS_OFST_ENDOdd field EQU pulse offset end, 4 bits signed.

Table 8-3684 ENCL_VIDEO_OFLD_VOAV_OFST(VCBUS: 0x1cc4)

Bit(s)	R/W	Default	Description
15:12	R/W	0	ENCL_VIDEO_OFLD_VSO_OFST_BEGINOdd field VSO offset begin, 4 bits signed.
11:8	R/W	1	ENCL_VIDEO_OFLD_VSO_OFST_ENDOdd field VSO offset end, 4 bits signed.
7:4	R/W	1	ENCL_VIDEO_OFLD_VAVON_OFST_BEGINOdd field VAVON offset begin, 4 bits signed.
3:0	R/W	1	ENCL_VIDEO_OFLD_VAVON_OFST_ENDOdd field VAVON offset end, 4 bits signed.

Table 8-3685 ENCL_VIDEO_MATRIX_CB(VCBUS: 0x1cc5)

Bit(s)	R/W	Default	Description
15:8	R/W	0	ENCL_VIDEO_MATRIX_maCbCr matrix parameter Ma (signed value)
7:0	R/W	0	ENCL_VIDEO_MATRIX_mBCbCr matrix parameter Mb (signed value)

Note

↑These 2 register is for HUE adjustment. The formula is : Cb' = Cb*Ma + Cr*Mb, Cr' = Cb*Mc + Cr*Md

(effective only when Enable HUE matrix by bit ENCL_VIDEO_MODE_ADV[7])

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Note

See the descritption of register ENCI_CMPN_MATRIX_CB.

Table 8-3686 ENCL_VIDEO_MATRIX_CR(VCBUS: 0x1cc6)

Bit(s)	R/W	Default	Description
15:8	R/W	0	ENCL_VIDEO_MATRIX_mCCbCr matrix parameter Mc (signed value)
7:0	R/W	0	ENCL_VIDEO_MATRIX_mDCbCr matrix parameter Md (signed value)

Table 8-3687 ENCL_VIDEO_RGBIN_CTRL(VCBUS: 0x1cc7)

Bit(s)	R/W	Default	Description
1	R/W	0	USE RGB data from VIU
0	R/W	0	CFG_VIDEO_RGBIN_ZBLKVIU RGB IN BLANK ZERO

Table 8-3688 ENCL_MAX_LINE_SWITH_POINT(VCBUS: 0x1cc8)

Bit(s)	R/W	Default	Description
13	R/W	0	CFG_VID_LOCK_ADJ_EN
12-0	R/W	‘h1fff	CFG_MAX_LINE_SWITH_POINT

Table 8-3689 ENCL_DACSEL0(VCBUS: 0x1cc9)

Bit(s)	R/W	Default	Description
15-12	R/W	0	VENC_L_DACSEL_0dac3
11-8	R/W	3	VENC_L_DACSEL_0dac2
7-4	R/W	2	VENC_L_DACSEL_0dac1
3-0	R/W	1	VENC_L_DACSEL_0dac0

Table 8-3690 ENCL_DACSEL1(VCBUS: 0x1cca)

Bit(s)	R/W	Default	Description
7-4	R/W	5	VENC_L_DACSEL_1dac5
3-0	R/W	4	VENC_L_DACSEL_1dac4

Table 8-3691 ENCL_INBUF_CNTL0(VCBUS: 0x1cd3)

Bit(s)	R/W	Default	Description
13-0	R/W	512	VENC_L_INBUF_SIZENew ENCL buffer size, always 512 for TL1

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Table 8-3692 ENCL_INBUF_CNTL1(VCBUS: 0x1cd4)

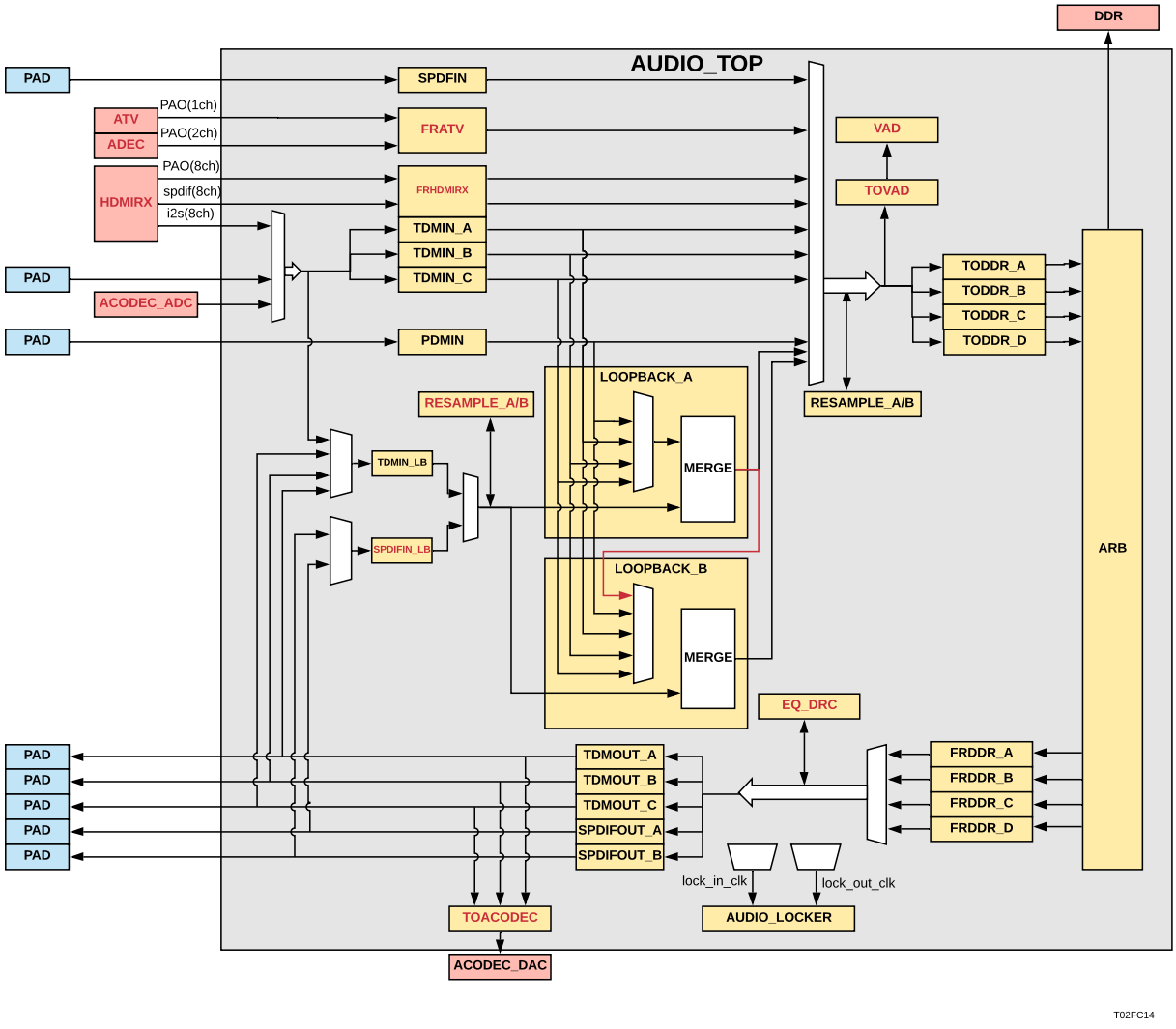
Bit(s)	R/W	Default	Description
15	R/W	0	VENC_L_INBUF_RSTFor INBUF reset rising edge reset, this value should be 0 normally, set to 1 to reset. For INBUF reset falling edge reset, this value should be 1 normally, set to 0 to reset.
14	R/W	0	VENC_L_INBUF_SOFT_RST_EDGE1: reset VENCL INBUF by VS rising edge, 0: reset VENCL INBUF by VS falling edge. Please make sure this edge is the same as the VENC_L_INT reset edge which generates go_field. Refer to VPP_ OFIFO_SIEZE[19].
12-0	R/W	3839	VENC_L_INBUF_LINE_LENM1H active length minus 1

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9 Audio Path

The chip integrates 3 TDM input/output interface, 1 SPDIF input, 2 SPDIF output interface, 1 PDM interface upto 8 channels, 4 TODDR (FIFO) for transfer input data to DDR, 4 FRDDR (FIFO) for transfer data from DDR to output, 1 TDM LB and 1 SPDIF LB and 2 Loopback for AEC, 2 HW resample for clock synchronization, 1 VAD voice wake up, and 1 clock locker detect difference of two clock. Below is the diagram for the audio path.

Figure 9-1 Audio Path



9.1 Audio Input

9.1.1 Overview

This section describes TDM input interface, SPDIF input interface and PDM input interface.

9.1.2 TDM Input Interface

TDM input interface works in the following way:

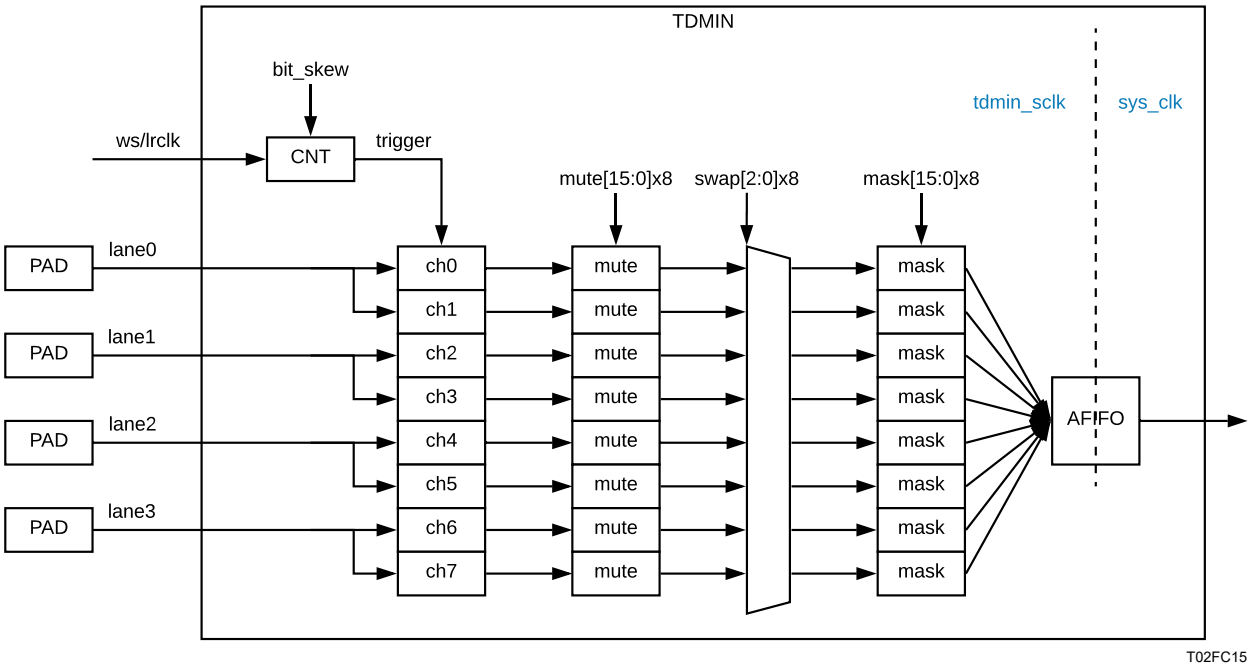
- All worked at tdm_sclk before AFIFO,

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- Output is worked at sysclk;
- Add skew control between lrclk and data;
- Detect rise or fall edge of lrclk and clear bit_cnt/slot_cnt;
- Shift 4 serial data in to 8 parallel register (max 32bits) ;
- Swap 8 parallel register to 8 sample data;
- Send 8 sample data out in serials by mask/mute configure;

Below is the diagram of TDM input interface.

Figure 9-2 TDM Input



9.1.3 SPDIF Input Interface

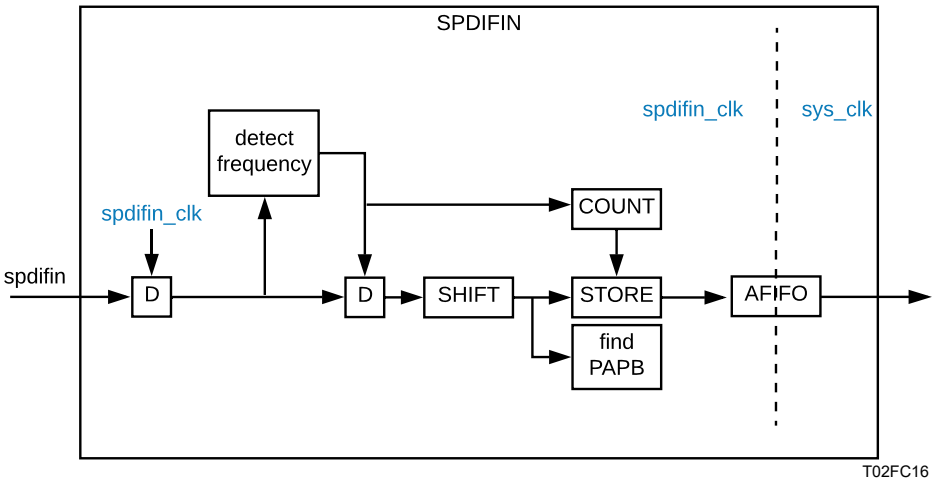
SPDIF input interface works in the following way:

- All worked at spdifin_clk before AFIFO;
- After AFIFO worked at sysclk;
- Detect frequency by rise or fall edge;
- Capture data to shift reg;
- Store data to sample L channel or R channel;
- Detect PaPb if it's IEC60937;

Below is the diagram of SPDIF input interface.

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Figure 9-3 SPDIF Input



9.1.4 PDM

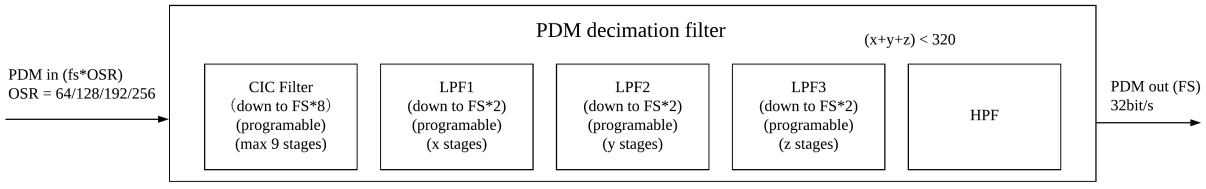
PDM input decimation filter is a highly programmable multistage decimation filter. It supports 4 inputs or 8 channel PDM digtal micphone interface. Each channel contains one 9 stage CIC filter, 3 low power filters and a high pass filter. The PDM input bit rate is calculate with fs*OSR. OSR can be 64, 128, 192 or 256.

CIC filter may have 3 to 9 CIC stages and the downsample rate also can be programable. There is a multiplier and shifter to adjust the result to match the accuarcy and CIC bit width.

There are 3 low pass filters which shared a 336x24 Coeffient memory and a 336x28 data memory. That means these 3 filters can be configed upto 336 taps together. The coeffient memory need to be programmed through APB bus. For example, if the filter 1 contains 140 taps, filter 2 contains 32 taps, filter 3 contain 146 steps, the filter 1 coefficient will use 0~139 of the coeffient memory address; the filter 2 coefficient will use 140~171 of coefficient memory address and the filter 3 coefficient will use 172~317 of the coeffient memory address. The fillter stage controller will based on the downsample rate, filter taps and the rounding mode of each filter to arrange the filter.

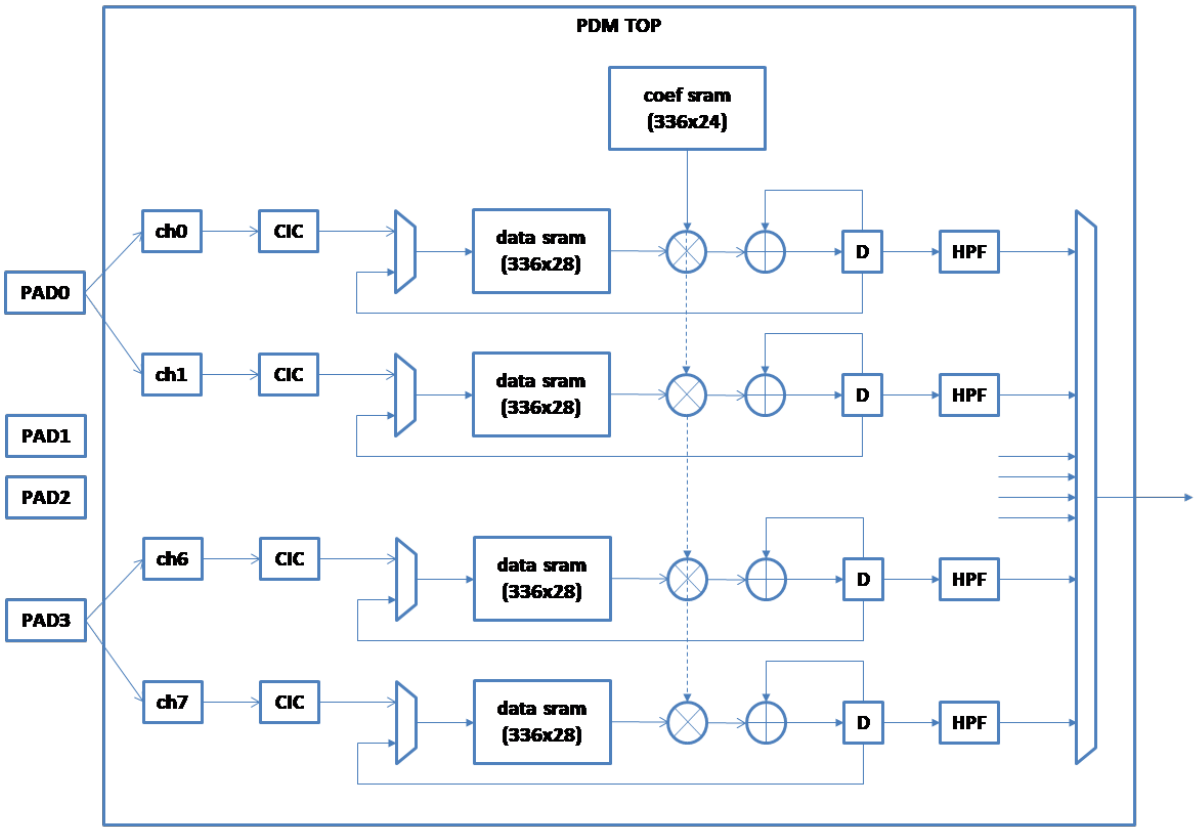
The final high pass filter is used to filter the DC curent.

Figure 9-4 PDM Decimation Filter



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Figure 9-5 PDM Structure



9.1.5 FRHDMIRX

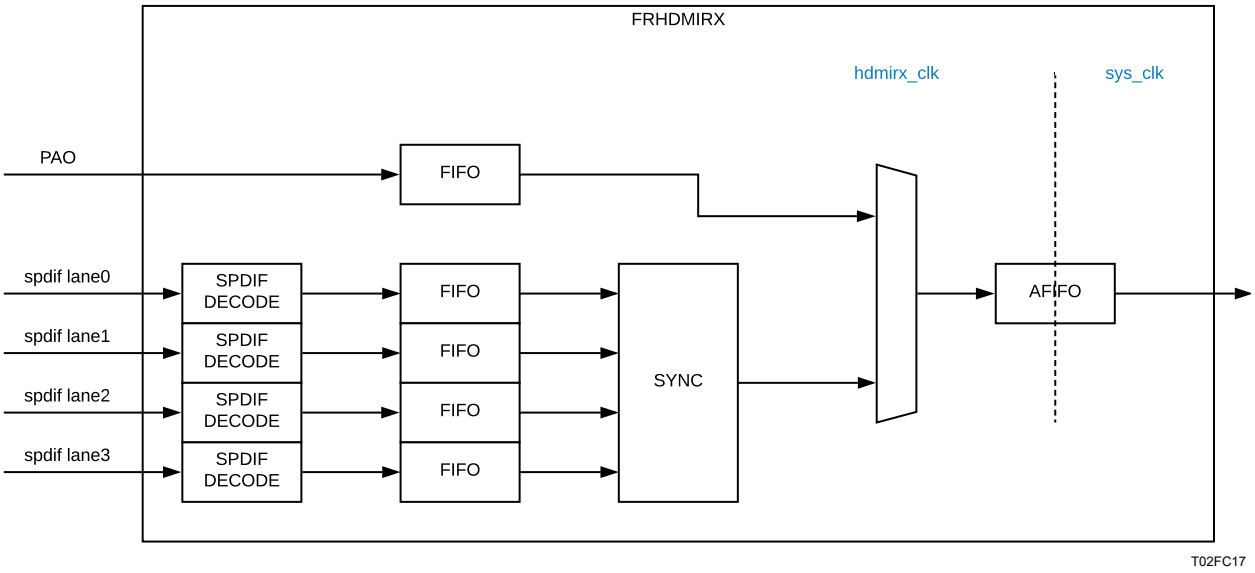
HDMIRX can send 3 audio signal:

- PAO, send sample value;
- I2S, connect to TDMIN;
- SPDIF, include 4 lanes;

FRHDMIRX will recevie PAO or SPDIF data, and send to TODDR.

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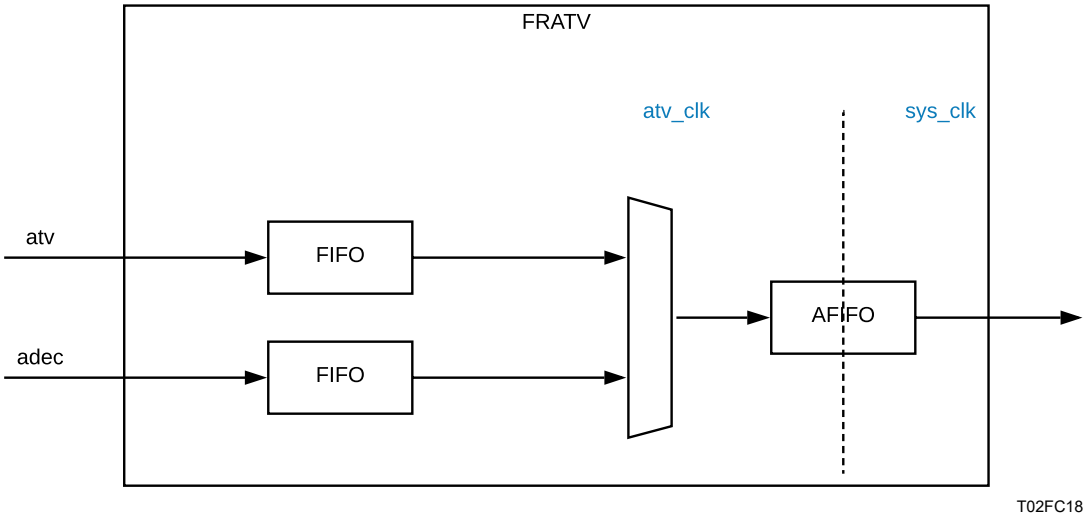
Figure 9-6 FRHDMIRX



9.1.6 FRATV

FRATV has two kind of sources: ATV mono or ADEC stero, as the figure shown below.

Figure 9-7 FRATV



9.2 Audio Output

9.2.1 Overview

This section describes TDM output interface, SPDIF output interface and audio REQ_FRDDR submodule.

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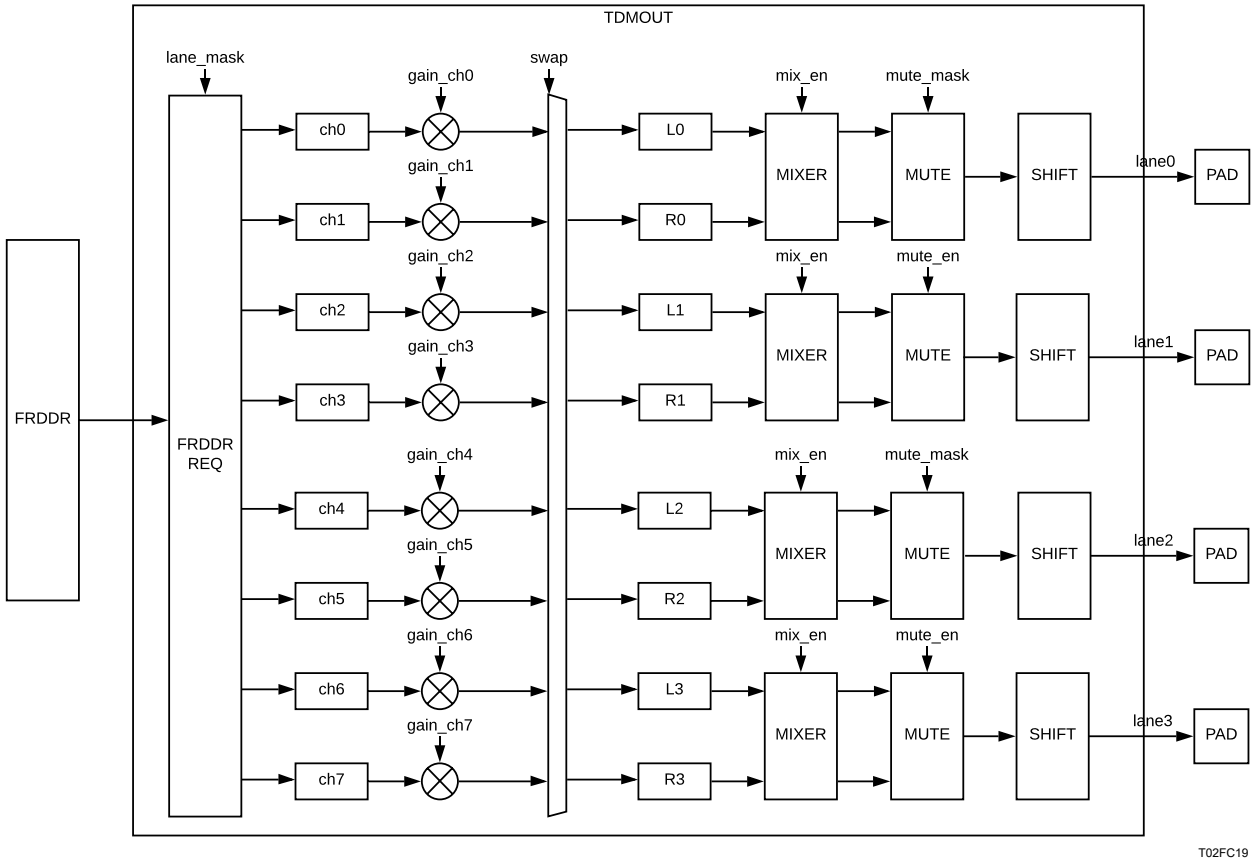
9.2.2 TDM Output Interface

TDM output interface works in the following way:

- All worked at tdm_sclk;
- Detect sample valid by tdm_lrclk rise edge and clear bit_cnt/slot_cnt;
- Request data from FRDDR and store to 8 sample register;
- Swap 8 sample register;
- Shift send out data;

Below is the digram of TDM output interface.

Figure 9-8 TDM Output



9.2.3 SPDIF Output Interface

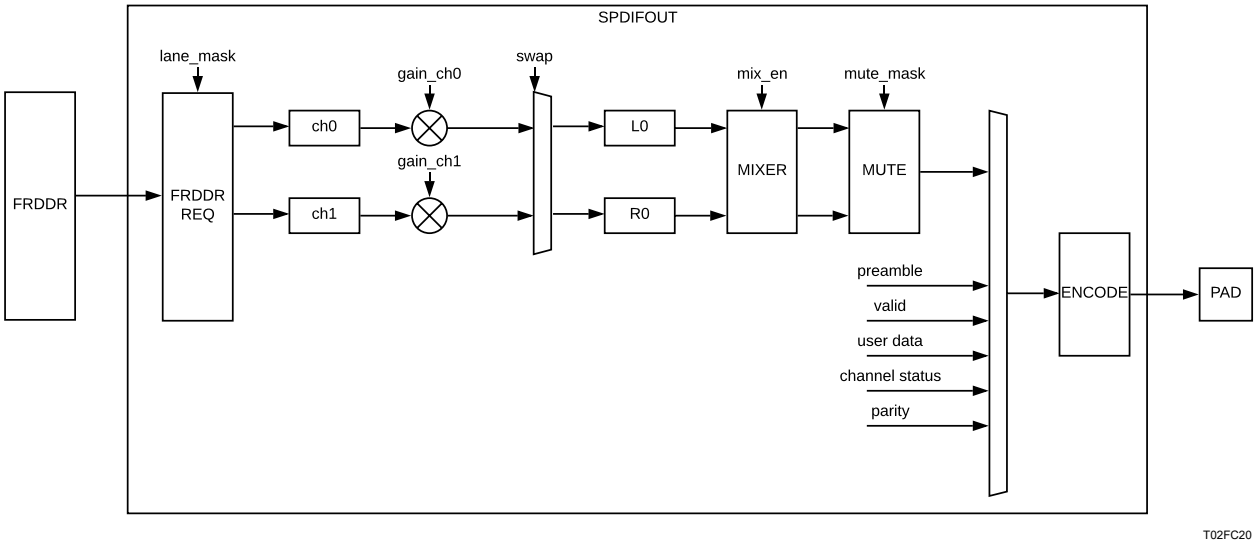
The SPDIF output interface works in the following way:

- All worked at spdifout clk;
- Request data from FRDDR and store to sample L0/R0
- Select data by bit_cnt/slot_cnt and send out;

Below is the digram of SPDIF output interface.

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Figure 9-9 SPDIF Output Interface



The SPDIF (encode) add same source select. The SPDIF out can be the same with i2s 0/1/2/3, and can select before or after eq/drc.

9.2.4 Audio REQ_FRDDR

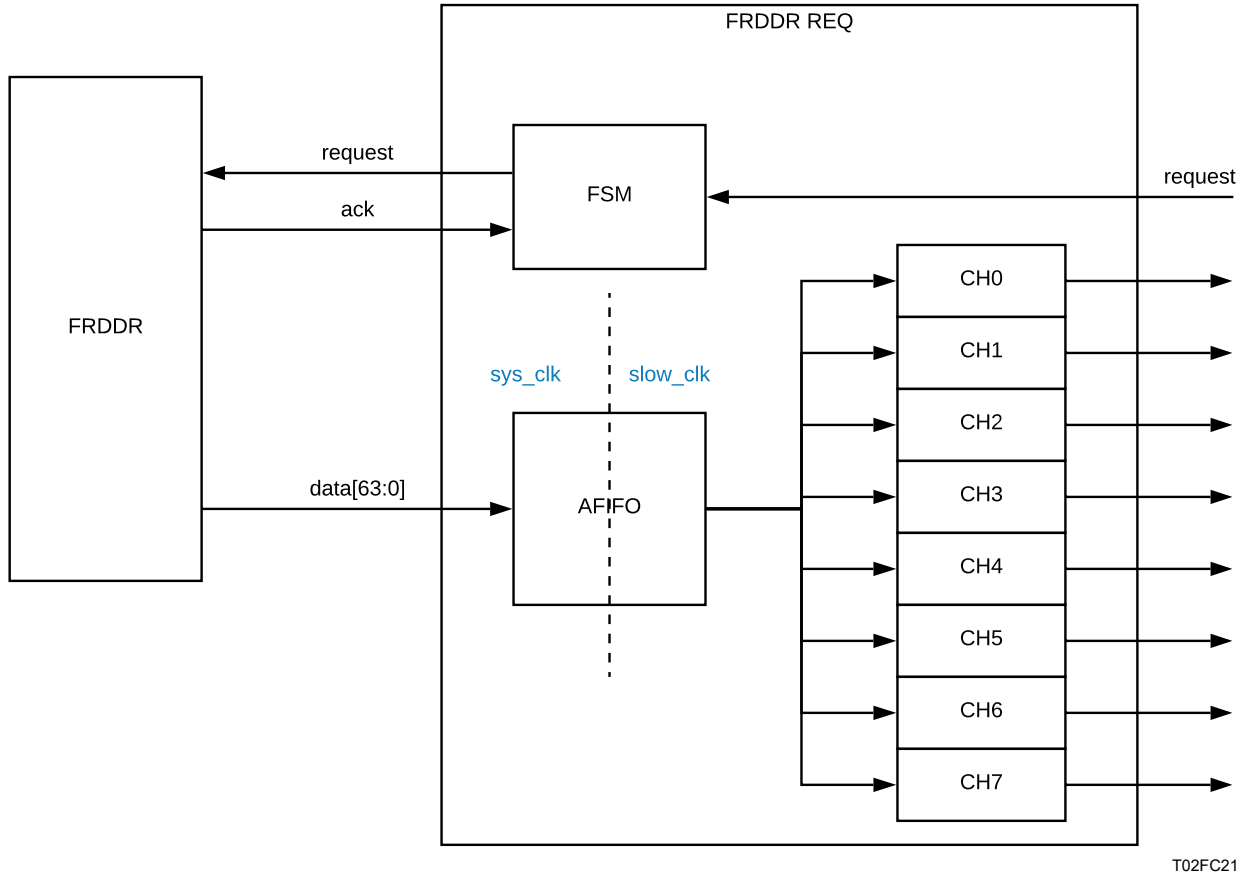
Audio Req_Frddr transfers data from sys clock to out clock and change format in the following way;

- Wait for OUT enable and FRDDR initial done;
- Request first time and fill 8 sample;
- When received update data, it will send request to FRDDR and update 8 sample;

Below is a diagram of Audio REQ_FRDDR.

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Figure 9-10 Audio REQ_FRDDR



9.3 DDR Datapath

9.3.1 Overview

This part describes the datapath between audio module and DDR.

9.3.2 Audio TODDR

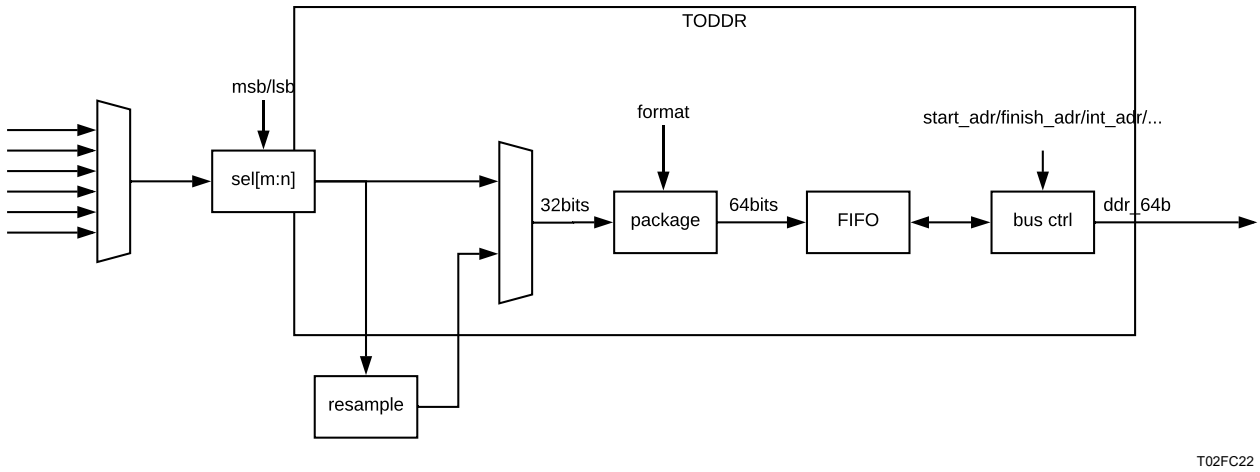
The chip has 4 TODDR(FIFO), TODDR_A's FIFO depth is 4096 x 64; B/C/D are 128 x 64. TODDR module works in the following way:

- All TODDR work at sysclk;
- Resample if need;
- Change format and package to 64 bits data by configuration;
- Write to fifo;
- Read data from fifo and send to DDR automatically by configuration;

Below is the Diagram of Audio TODDR.

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Figure 9-11 Audio TODDR



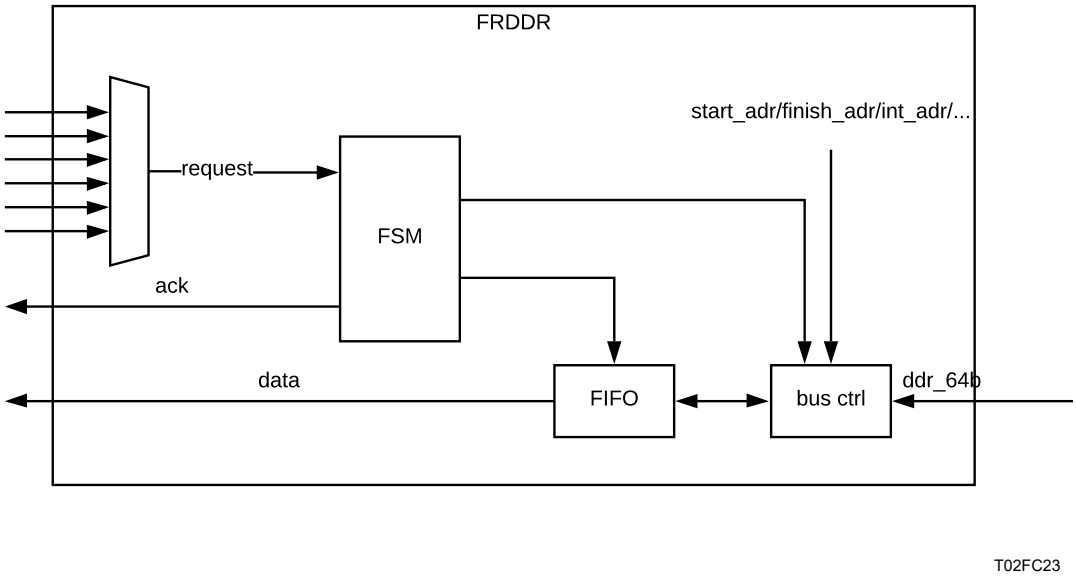
9.3.3 Audio FRDDR

The chip has 4 FRDDR(FIFO), FRDDR_A's FIFO depth is 256 x 64, B/C/D are 128 x 64;

- All FRDDR work at sysclk;
- When enable FRDDR, it will fill FIFO from DDR first;
- When FRDDR receive request, it will read data from fifo and send out;
- FRDDR will fill FIFO automatically by configure;

Below is the Diagram of Audio FRDDR.

Figure 9-12 Audio FRDDR



9.4 Audio Loopback

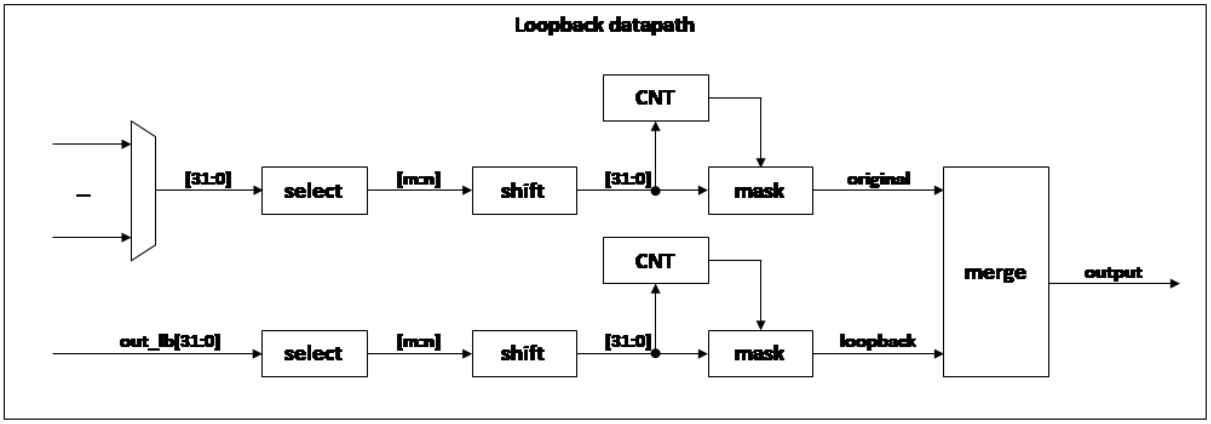
TDMIN_LB can receive one TDMOUT or SPDIFOUT, it can merge with one TDMIN/SPDIFIN/PDMIN in the following way:

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- Store one source to temp register;
- When another source arrived, send out direct;
- When finished, send out temp register;

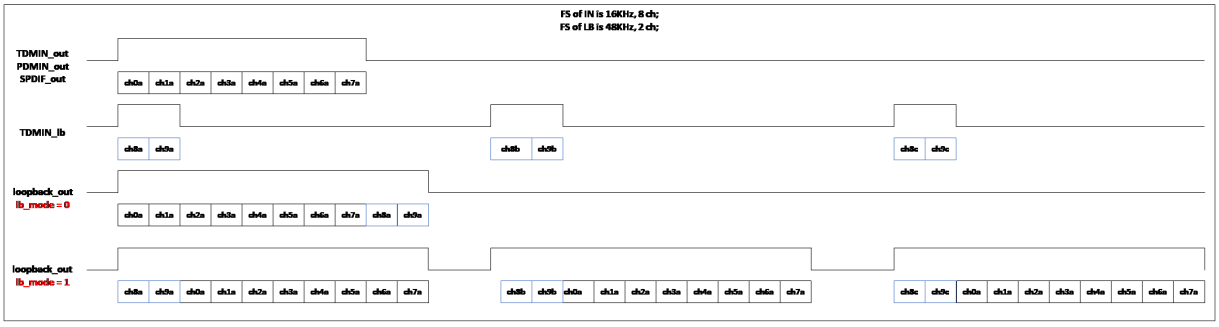
Below is the diagram of audio loopback datapath.

Figure 9-13 Audio Loopback Datapath



Audio loopback wave form are shown below.

Figure 9-14 Audio Loopback Waveform



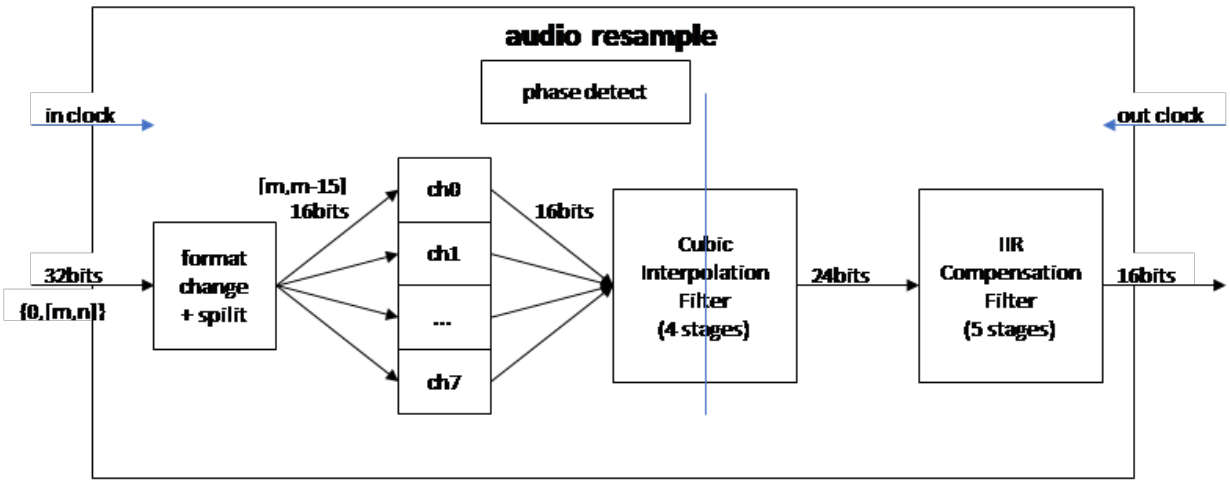
9.5 Audio Resample

Audio resample changes data from one symbol rate to another symbol rate in the following steps:

1. Input is from Toddr and selected by toddr's m/n register. Example A: toddr_m = 27, toddr_n=4, resample_in[31:0] = {8'd0, toddr_in[27:4]};
2. Cut or extend input from 32 bits to 16 bits. By example A, set resample_M = 23, then filter_in = toddr_in[27:12];
3. Split to each channel (max is 8ch);
4. Interpolation by a cubic filter;
5. Use an IIR filter compensate;

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Figure 9-15 Audio Resample



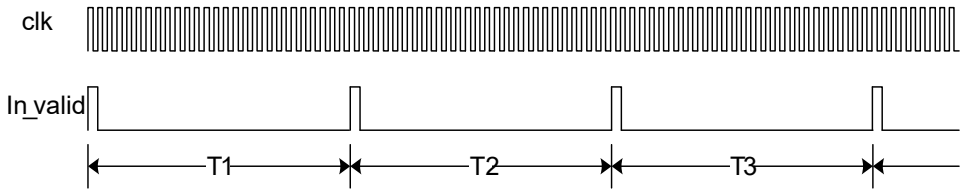
Note

In the diagram, the module phase_det is used to calculate the interpolation phase, the module interp is used to implement the interpolation processing. There are two methods which can be implemented to calculate the interpolation phase. One is front feedback method, the other is accumulation method. They can be switched by register.

(A) method 0

The method 0 is the direct feedback method. According to the out_valid, it interpolates the mapping sample. The operation flow is given below:

1. calculate the input data rate(average the count every input valid signal). For hdmi_rx audio data, the input rate is basically even.



T02TM02

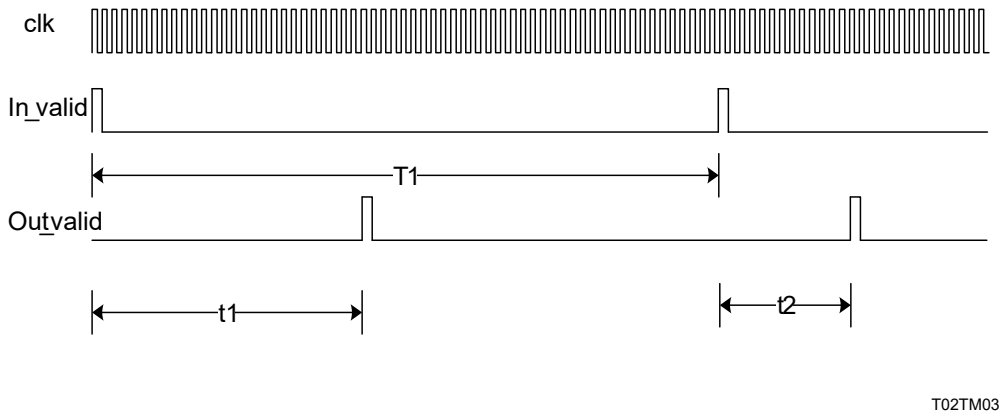
Count the clock number in T1,T2, and so on. get CNT then do average:

T = 0.999*T + 0.001*CNT

For 8 channels signal, the CNT should be count between 8 in_valid signal.

2. In the output clock domain, generate the output valid signal. Because the output clock should be integral times of output frequency, the valid signal can easily generated by counting the output clock. Then change the valid signal to main clock domain.
3. Calculate the interpolation phase mu.

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Count the clock number in t1 get cnt, then mu can be calculated:

Mu = cnt/T

The bit width of mu is 9 bits.

4. Do the interpolation (4 points cubic interpolation)

   Basic function is:

   Y = ((a*mu + b) * mu + c)*mu + d, mu means interpolation phase.

   and

   a = 16*x0 - 48*x1 + 48*x2 - 16*x3

   b = 31*x0 - 31*x1 - 31*x2 + 31*x3

   c = 23*x0 + 59*x1 - 59*x2 - 23*x3

   d = 6*x0 + 58*x1 + 58*x2 + 6*x3

   (x0, x1, x2 and x3 are the delay line of input data.)

Performance Analysis

The advantage of this method is that the output data rate is generated by output clock and match the acquirement of software absolutely.

The disadvantage is that the interpolation phase has bias. The bias relies on the main clock. More higher the main clock frequency is, more precise the interpolation result is.

Actually the precision of interpolation phase is 9 bits, so the frequency of main clock should be more than 512 times of input data rate.

(A) method 1

The method 1 is accumulation method. It assume the output frequency is accurate and stable. It sets a fixed phase step. The phase step is the ratio of the input frequency to output frequency. Every step, It outputs a sample. About the bias of input frequency, there is a register to report the statistics value. It can be used to adjust the initial phase step by firmware.

Performance Analysis

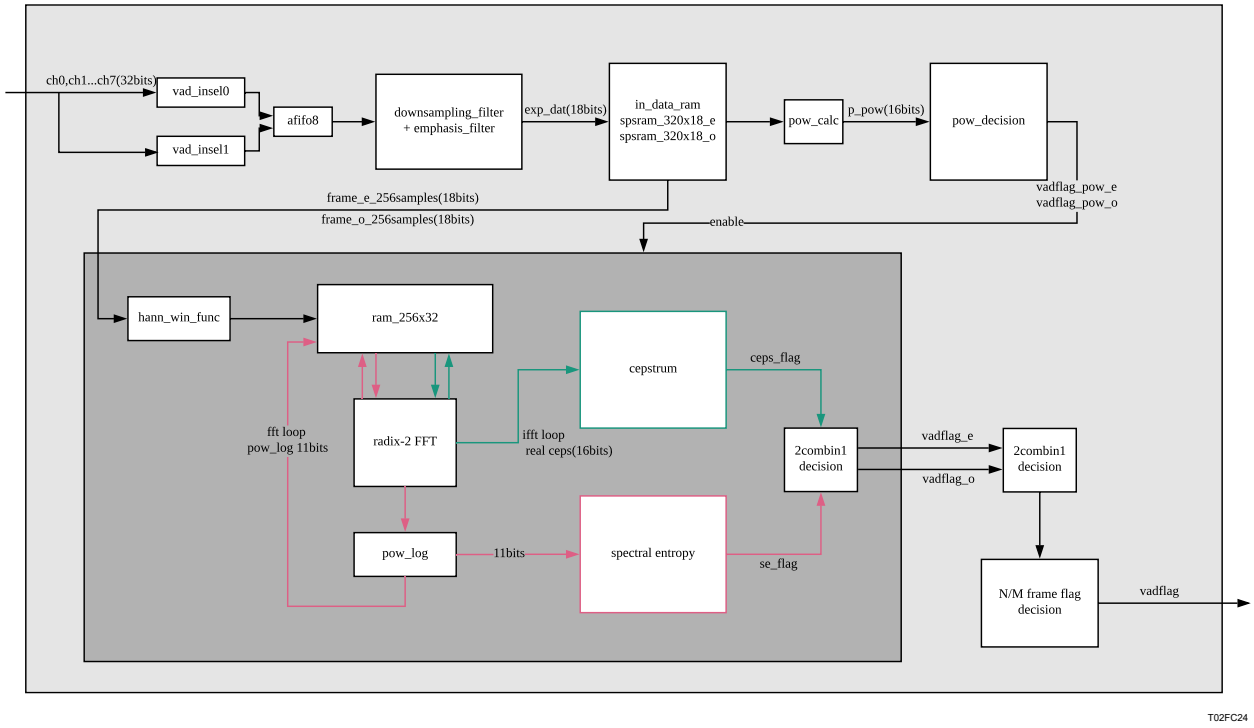
This method requires the audio output frequency precision, because there is no information statistics of output frequency. When the audio output frequency is not stable, it is recommended it not be used.

9.6 Audio VAD

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VAD will wake up DSP when it determine if voice is present in a particular audio signal. sIt support 2 channels audio streams searching which combined from PDM filter output audio stream. The audio stream will be down-sampled to 8K/s rate and de-noised before detection processing. VAD can provide power detection, cepstral detection and spectral entropy detection. It will do the detection per 10ms. It contains one 256p FFT function for cepstral and spectral entropy calculation.

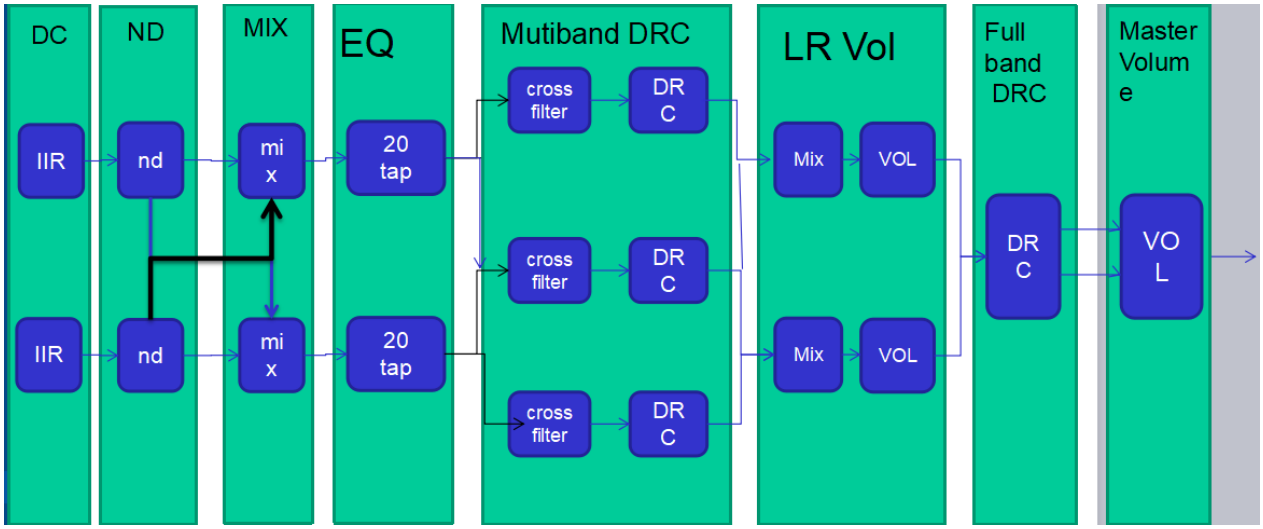
Figure 9-16 Audio VAD



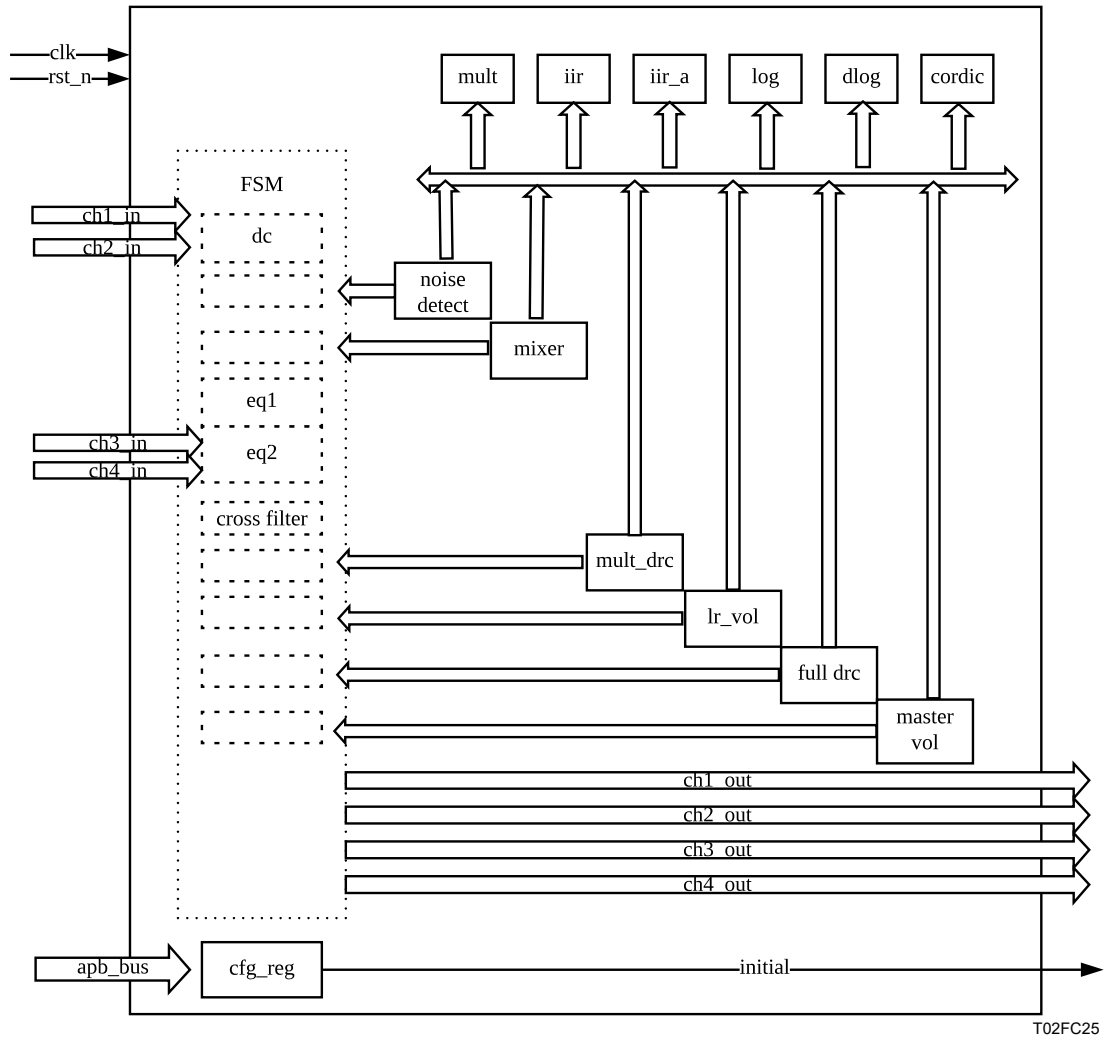
9.7 Audio EQDRC

It support 4 channels audio streams. Two channels do eq and drc, the others only do eq.

The basic structure diagram is shown below.



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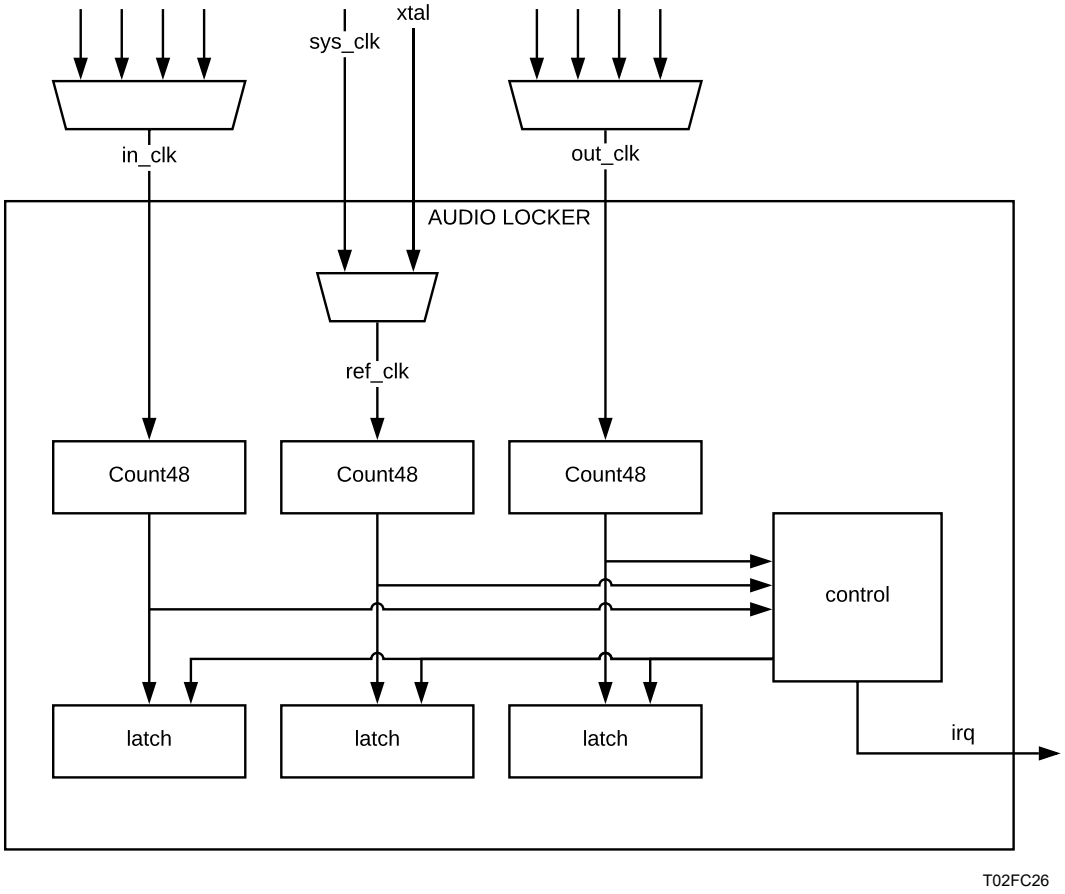
9.8 Audio Locker

The audio input clock and output clock are in different power domain and may have little difference, the difference leads to FIFO problem, the chip uses Audio Locker module to fix this problem, the basic method is to measure the difference between different clocks and adjust (software dynamically) the clock frequency to make them match.

The diagram of Audio Locker module is shown below.

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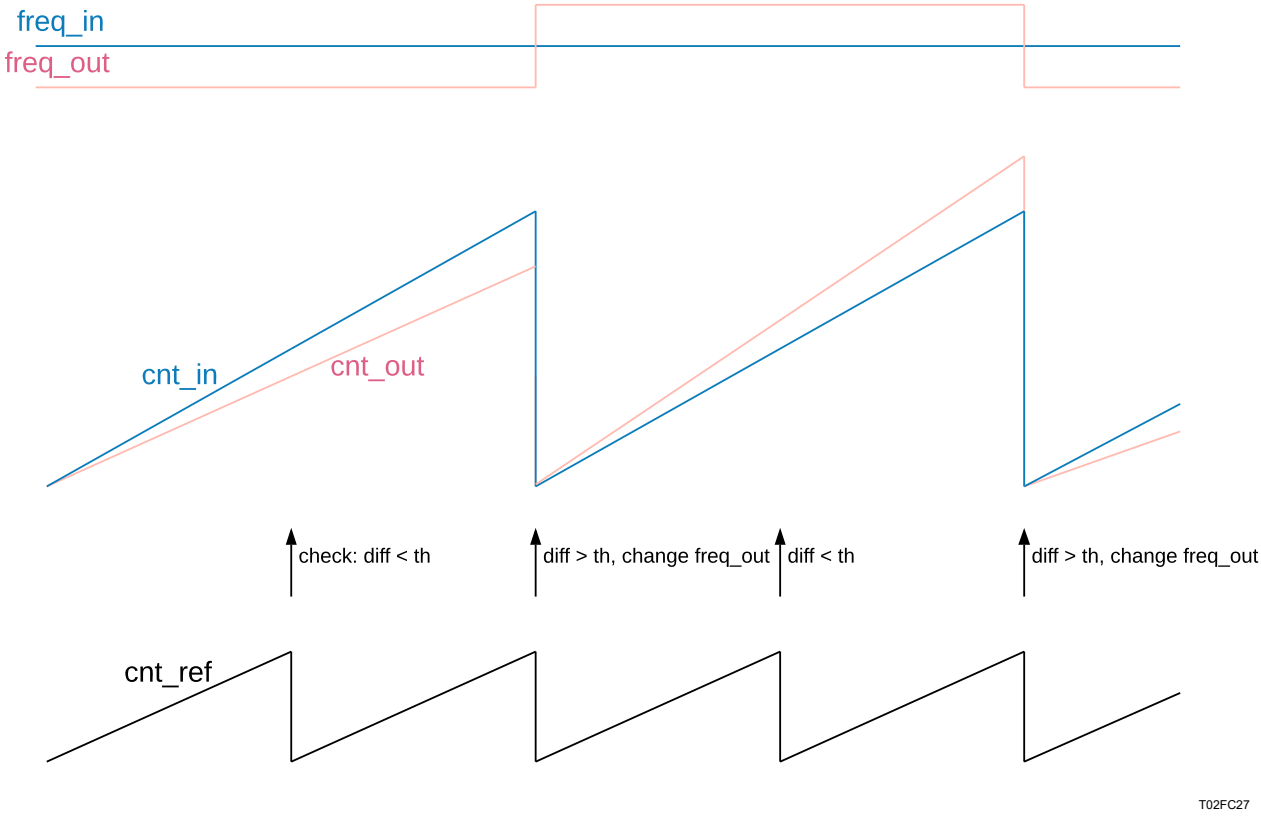
Figure 9-17 Audio Locker



For example: when the diff of two freq creater than threshold is detected, change frequency of clock_ out by SW, will keep clock_out close clock_in.

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Figure 9-18 Audio Locker Example



9.9 Audio Timing Specification

There are two modes for the audio I2S/TDM interface: master mode and slave mode, as shown below.

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Figure 9-19 I2S/TDM Timing Diagram, Master Mode

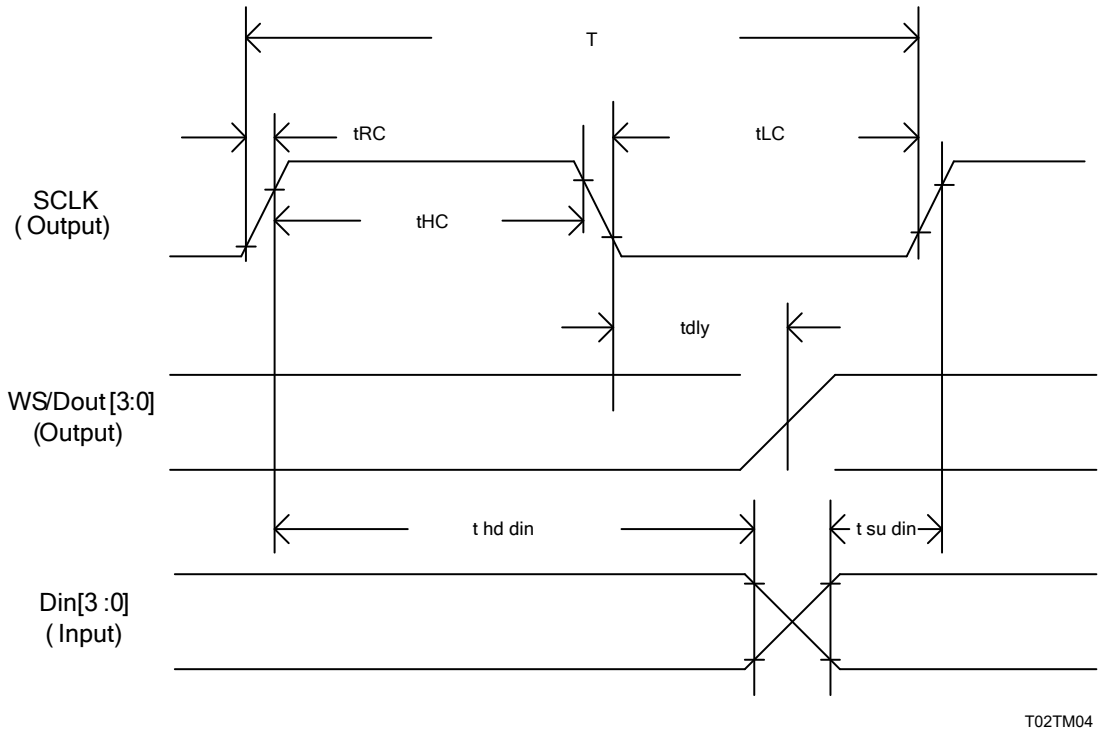


Table 9-1 Audio I2S/TDM Timing Specification, Transmitter, Master Mode

Symbol	Parameter	Min	Typ	Max	Unit
T	Clock period	10			ns
tHC	High level of SCLK	0.4			T
tLC	Low level of SCLK	0.4			T
tRC	Edge time of SCLK			0.8	ns
tdly	Delay from SCLK to WS	-2	3	5	
tsuin	Setup time of WS/Din	4			ns
thdin	Hold time of WS/Din	4			ns

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Figure 9-20 I2S/TDM Timing Diagram, Slave Mode

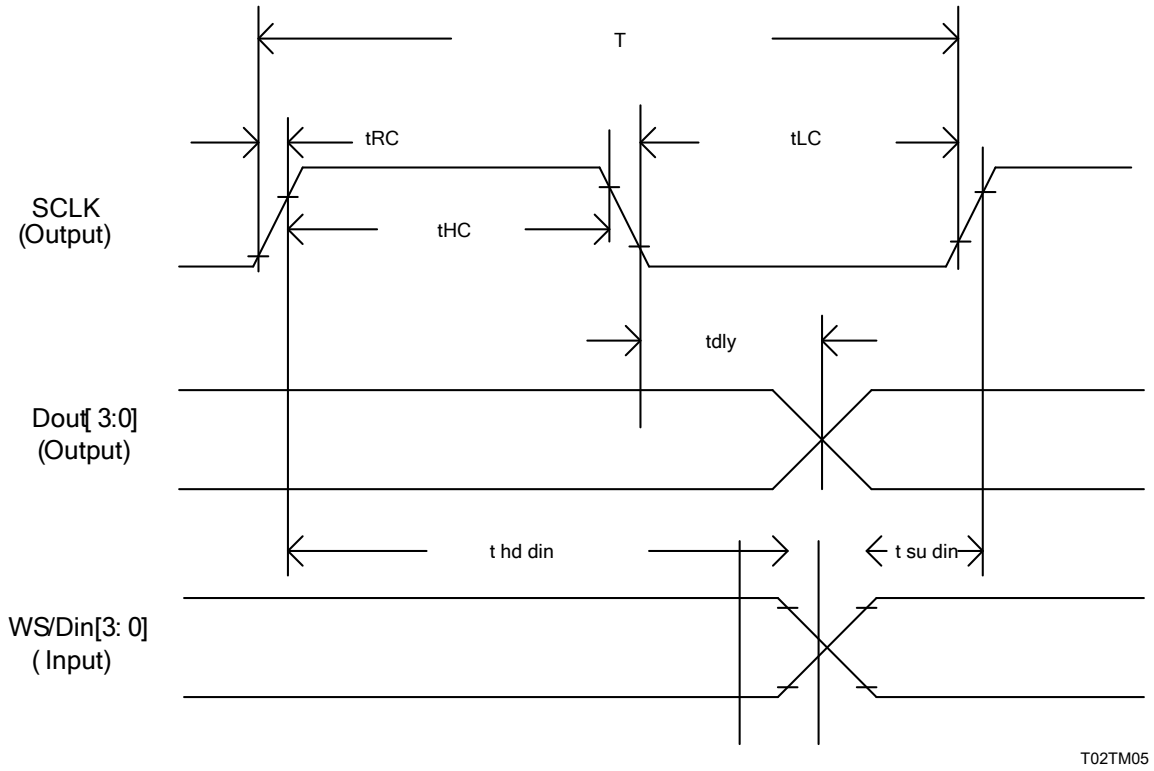


Table 9-2 Audio I2S/TDM Timing Specification, Transmitter, Slave Mode

Symbol	Parameter	Min	Typ	Max	Unit
T(out)	Clock period	40			ns
T(in)	Clock period	10			ns
tHC	High level of SCLK	0.4			T
tLC	Low level of SCLK	0.4			T
tRC	Edge time of SCLK			0.8	ns
tsu in	Setup time of WS/Din	4			ns
thd in	Hold time of WS/Din	4			ns
tdly	Delay between SCLK and Dout	7	12	15	ns

9.10 Register Description

Sub-module	Base Address	Range
AUDIO_TOP	0xFF60_0000	4KB
PDM	0xFF60_1000	1KB
LOCKER	0xFF60_1400	1KB
VAD	0xFF60_1800	1KB

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Sub-module	Base Address	Range
RESAMPLE_A	0xFF60_1C00	1KB
EQDRC	0xFF60_2000	4KB
EARCTX	0xFF60_3000	2KB
EARCRX	0xFF60_3800	2KB
RESAMPLE_B	0xFF60_4000	1KB
LOCKER_B	0xFF60_4400	1KB

Each register final address = module base address+ address * 4

9.10.1 CLK/RESET Registers

Table 9-3 EE_AUDIO_CLK_GATE_EN0 0x00

Bits	R/W	Default	Description
31	R/W	0x0000-0000	frhdmirx, 0:disable; 1: enable;
30	R/W		fratv, 0:disable; 1: enable;
29	R/W		Spdifin_lb, 0:disable; 1: enable;
28	R/W		Audio locker, 0:disable; 1: enable;
27	R/W		tovad, 0:disable; 1: enable;
26	R/W		resampleB, 0:disable; 1: enable;
25	R/W		reserved
24	R/W		reserved
23	R/W		reserved
22	R/W		eqdrc, 0:disable; 1: enable;
21	R/W		spdifoutB, 0:disable; 1: enable;
18	R/W		resampleA, 0:disable; 1: enable;
17	R/W		spdifout, 0:disable; 1: enable;
16	R/W		spdifin, 0:disable; 1: enable;
15	R/W		loopbackA, 0:disable; 1: enable;
14	R/W		toddrc, 0:disable; 1: enable;
13	R/W		toddrb, 0:disable; 1: enable;
12	R/W		toddra, 0:disable; 1: enable;
11	R/W		frddrc, 0:disable; 1: enable;
10	R/W		frddrb, 0:disable; 1: enable;
9	R/W		frddra, 0:disable; 1: enable;
8	R/W		tdmoutc, 0:disable; 1: enable;
7	R/W		tdmoutb, 0:disable; 1: enable;

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Bits	R/W	Default	Description
6	R/W		tmdouta, 0:disable; 1: enable;
5	R/W		tdminlb, 0:disable; 1: enable;
4	R/W		tdminc, 0:disable; 1: enable;
3	R/W		tdminb, 0:disable; 1: enable;
2	R/W		tdmina, 0:disable; 1: enable;
1	R/W		pdm, 0:disable; 1: enable;
0	R/W		ddr_arb, 0:disable; 1: enable;

Table 9-4 EE_AUDIO_CLK_GATE_EN1 0x01

Bits	R/W	Default	Description
31:3	R/W	0x0000-0000	reserved
2	R/W		loopbackB, 0:disable; 1: enable;
1	R/W		toddrd, 0:disable; 1: enable;
0	R/W		frddrd, 0:disable; 1: enable;

Table 9-5 EE_AUDIO_MCLK_A_CTRL 0x02

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
26:24	R/W		clk_sel, 0:mp0_pll; 1:mp1_pll; 2:mp2_pll; 3:mp3_pll; 4:hifi_pll; 5:fclk_div3(666M); 6:fclk_div4(500M); 7:cts_oscin(24M);
15:0	R/W		clk_div, the frequency of mclk = pll/clk_div;

EE_AUDIO_MCLK_B_CTRL 0x03

Same as EE_AUDIO_MCLK_A_CTRL.

EE_AUDIO_MCLK_C_CTRL 0x04

Same as EE_AUDIO_MCLK_A_CTRL.

EE_AUDIO_MCLK_D_CTRL 0x05

Same as EE_AUDIO_MCLK_A_CTRL.

EE_AUDIO_MCLK_E_CTRL 0x06

Same as EE_AUDIO_MCLK_A_CTRL.

EE_AUDIO_MCLK_F_CTRL 0x07

Same as EE_AUDIO_MCLK_A_CTRL.

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Table 9-6 EE_AUDIO_MST_PAD_CTRL0 0x08

Bits	R/W	Default	Description
6:4	R/W		Mclk_pad_1_sel: 0: mclk_a; 1: mclk_b; 2: mclk_c; 3: mclk_d; 4: mclk_e; 5: mclk_f;
2:0	R/W		Mclk_pad_0_sel: 0: mclk_a; 1: mclk_b; 2: mclk_c; 3: mclk_d; 4: mclk_e; 5: mclk_f;

Table 9-7 EE_AUDIO_MST_PAD_CTRL1 0x09

Bits	R/W	Default	Description
26:24	R/W		lrclk_pad_2_sel: 0: lrclk_a; 1: lrclk_b; 2: lrclk_c; 3: lrclk_d; 4: lrclk_e; 5: lrclk_f;
22:20	R/W		lrclk_pad_1_sel: 0: lrclk_a; 1: lrclk_b; 2: lrclk_c; 3: lrclk_d; 4: lrclk_e; 5: lrclk_f;
18:16	R/W		lrclk_pad_0_sel: 0: lrclk_a; 1: lrclk_b; 2: lrclk_c; 3: lrclk_d; 4: lrclk_e; 5: lrclk_f;
10:8	R/W		sclk_pad_2_sel: 0: sclk_a; 1: sclk_b; 2: sclk_c; 3: sclk_d; 4: sclk_e; 5: sclk_f;

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Bits	R/W	Default	Description
6:4	R/W		sclk_pad_1_sel: 0: sclk_a; 1: sclk_b; 2: sclk_c; 3: sclk_d; 4: sclk_e; 5: sclk_f;
2:0	R/W		sclk_pad_0_sel: 0: sclk_a; 1: sclk_b; 2: sclk_c; 3: sclk_d; 4: sclk_e; 5: sclk_f;

Table 9-8 EE_AUDIO_SW_RESET0 0x0A

Bits	R/W	Default	Description
31	R/W	0	Frhdmirx
30	R/W	0	Fratv
29	R/W	0	Spdifin_lb
28	R/W	0	Locker
27	R/W	0	Tovad
26	R/W	0	resampleB
25	R/W	0	clk tree
24	R/W	0	-
23	R/W	0	toacodec
20	R/W	0	ddrarb
19	R/W	0	resample
18	R/W	0	eqdrc
17	R/W	0	spdifin
16	R/W	0	spdifoutB
15	R/W	0	spdifout
14	R/W	0	tdmoutc
13	R/W	0	tdmoutb
12	R/W	0	tdmouta
11	R/W	0	frddrc
10	R/W	0	frddrb
9	R/W	0	frddra
8	R/W	0	toddrc
7	R/W	0	toddrb
6	R/W	0	toddra
5	R/W	0	loopback

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Bits	R/W	Default	Description
4	R/W	0	tdmin_lb
3	R/W	0	tdminc
2	R/W	0	tdminb
1	R/W	0	tdmina
0	R/W	0	pdm

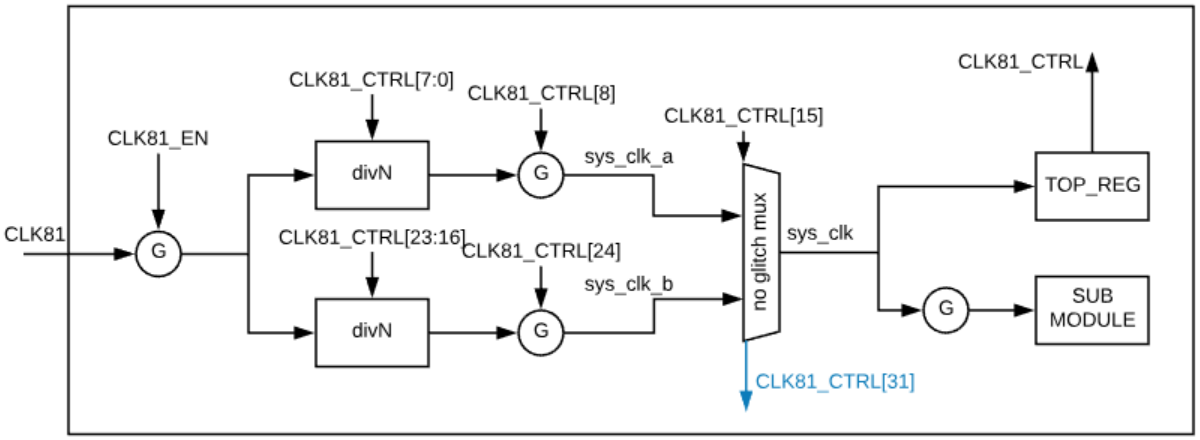
Table 9-9 EE_AUDIO_SW_RESET1 0x0B

Bits	R/W	Default	Description
2	R/W	0	loopbackB
1	R/W	0	toddrd
0	R/W	0	frddrd

Table 9-10 EE_AUDIO_CLK81_CTRL 0x0C

Bits	R/W	Default	Description
31	R	0	Select out
24	R/W	0	Sys_clk_B enable
23:16	R/W	0	Sys_clk_B div
15	R/W	0	0: Select sys_clk_A; 1: Select sys_clk_B;
8	R/W	1	Sys_clk_A enable
7:0	R/W	0	Sys_clk_A div

Figure 9-21 EE_AUDIO_CLK81_CTRL



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Table 9-11 EE_AUDIO_CLK81_EN 0x0D

Bits	W	Default	Description
31	R/W	1	Write 0 will off all clk81(sys_clk) in audio top. And will make error if access to all audio register if EN = 0.

Table 9-12 EE_AUDIO_MST_A_SCLK_CTRL0 0x10

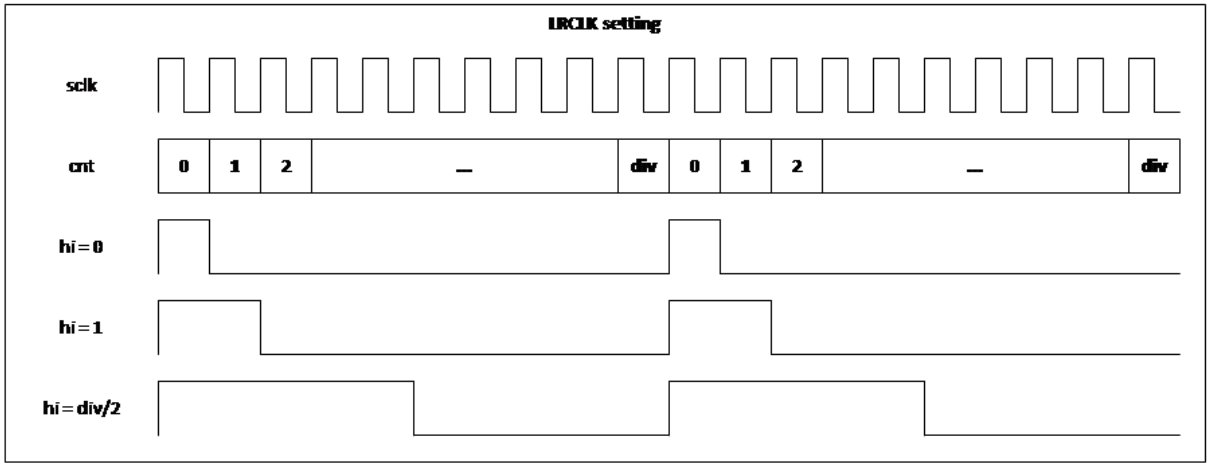
Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_in_en, 0:disable; 1: enable;
30	R/W		clk_out_en, 0:disable; 1: enable;
29:20	R/W		sclk_div, the frequency of sclk = mclk/sclk_div;
19:10	R/W		lrclk_hi, duty cycle of LRCLK, less than lrclk_div; example 0: lrclk_hi = 1, LRCLK will only keep one cycle; example 1: lrclk_hi = lrclk_div/2, LRCLK will be 50/50 duty cycle;
9:0	R/W		lrclk_div, the frequency of lrclk = sclk/lrclk_div;

Table 9-13 EE_AUDIO_MST_A_SCLK_CTRL1 0x11

Bits	R/W	Default	Description
31:28	R/W	0x0000-0000	sclk_ph0_sel, select from sclk_delay_line(depth is 16)
27:24	R/W		lrclk_ph0_sel, select from lrclk_delay_line(depth is 16)
23:20	R/W		sclk_ph1_sel, select from sclk_delay_line(depth is 16)
19:16	R/W		lrclk_ph1_sel, select from lrclk_delay_line(depth is 16)
15:12	R/W		sclk_ph2_sel, select from sclk_delay_line(depth is 16)
11:8	R/W		lrclk_ph2_sel, select from lrclk_delay_line(depth is 16)
5:0	R/W		clk_inv, invert clk; [5]: lrclk_ph2; [4]: sclk_ph2; [3]: lrclk_ph1; [2]: sclk_ph1; [1]: lrclk_ph0; [0]: sclk_ph0;

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Figure 9-22 LRCLK Setting



EE_AUDIO_MST_B_SCLK_CTRL0 0x12

Same as EE_AUDIO_MST_A_SCLK_CTRL0

EE_AUDIO_MST_B_SCLK_CTRL1 0x13

Same as EE_AUDIO_MST_A_SCLK_CTRL1

EE_AUDIO_MST_C_SCLK_CTRL0 0x14

Same as EE_AUDIO_MST_A_SCLK_CTRL0

EE_AUDIO_MST_C_SCLK_CTRL1 0x15

Same as EE_AUDIO_MST_A_SCLK_CTRL1

EE_AUDIO_MST_D_SCLK_CTRL0 0x16

Same as EE_AUDIO_MST_A_SCLK_CTRL0

EE_AUDIO_MST_D_SCLK_CTRL1 0x17

Same as EE_AUDIO_MST_A_SCLK_CTRL1

EE_AUDIO_MST_E_SCLK_CTRL0 0x18

Same as EE_AUDIO_MST_A_SCLK_CTRL0

EE_AUDIO_MST_E_SCLK_CTRL1 0x19

Same as EE_AUDIO_MST_A_SCLK_CTRL1

EE_AUDIO_MST_F_SCLK_CTRL0 0x1a

Same as EE_AUDIO_MST_A_SCLK_CTRL0

EE_AUDIO_MST_F_SCLK_CTRL1 0x1b

Same as EE_AUDIO_MST_A_SCLK_CTRL1

Table 9-14 EE_AUDIO_CLK_TDMIN_A_CTRL 0x20

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_in_en, 0:disable; 1: enable;
30	R/W		clk_out_en, 0:disable; 1: enable;

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Bits	R/W	Default	Description
29	R/W		sclk_inv, 0:not revert; 1: revert clock;
27:24	R/W		sclk_sel, 0:mst_a_sclk_ph1; 1:mst_b_sclk_ph1; 2:mst_c_sclk_ph1; 3:mst_d_ sclk_ph1; 4:mst_e_sclk_ph1; 5:mst_f_sclk_ph1; 6:i_slv_sclk_a 7:i_slv_sclk_b 8:i_slv_sclk_c 9::i_slv_sclk_d ; 10:i_slv_sclk_e ; 11:i_slv_sclk_f ; 12:i_slv_sclk_g ; 13:i_slv_sclk_h ; 14:i_slv_sclk_i ; 15:i_slv_sclk_j ;
23:20	R/W		lrclk_sel, 0:mst_a_lrclk_ph1; 1:mst_b_lrclk_ph1; 2:mst_c_lrclk_ph1; 3:mst_d_ lrclk_ph1; 4:mst_e_lrclk_ph1; 5:mst_f_lrclk_ph1; 6:i_slv_lrclk_a 7:i_slv_lrclk_b 8:i_slv_lrclk_c 9::i_slv_lrclk_d ; 10:i_slv_lrclk_e ; 11:i_slv_lrclk_f ; 12:i_slv_lrclk_ g ; 13:i_slv_lrclk_h ; 14:i_slv_lrclk_i ; 15:i_slv_lrclk_j ;

EE_AUDIO_CLK_TDMIN_B_CTRL 0x21

Same as EE_AUDIO_CLK_TDMIN_A_CTRL

EE_AUDIO_CLK_TDMIN_C_CTRL 0x22

Same as EE_AUDIO_CLK_TDMIN_A_CTRL

EE_AUDIO_CLK_TDMIN_LB_CTRL 0x23

Same as EE_AUDIO_CLK_TDMIN_A_CTRL

Table 9-15 EE_AUDIO_CLK_TDMOUT_A_CTRL 0x24

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_in_en, 0:disable; 1: enable;
30	R/W		clk_out_en, 0:disable; 1: enable;
29	R/W		sclk_inv, 0:not revert; 1: revert clock;
28	R/W		Sclk_ws_inv, for the capture ws sclk; 0: not revert; 1: revert clock;

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Bits	R/W	Default	Description
27:24	R/W		sclk_sel, 0:mst_a_sclk_ph1; 1:mst_b_sclk_ph1; 2:mst_c_sclk_ph1; 3:mst_d_sclk_ph1; 4:mst_e_sclk_ph1; 5:mst_f_sclk_ph1; 6:i_slv_sclk_a ; 7:i_slv_sclk_b ; 8:i_slv_sclk_c ; 9::i_slv_sclk_d ; 10:i_slv_sclk_e ; 11:i_slv_sclk_f ; 12:i_slv_sclk_g ; 13:i_slv_sclk_h ; 14:i_slv_sclk_i ; 15:i_slv_sclk_j ;
23:20	R/W		lrclk_sel, 0:mst_a_lrclk_ph1; 1:mst_b_lrclk_ph1; 2:mst_c_lrclk_ph1; 3:mst_d_lrclk_ph1; 4:mst_e_lrclk_ph1; 5:mst_f_lrclk_ph1; 6:i_slv_lrclk_a ; 7:i_slv_lrclk_b ; 8:i_slv_lrclk_c ; 9::i_slv_lrclk_d ; 10:i_slv_lrclk_e ; 11:i_slv_lrclk_f ; 12:i_slv_lrclk_g ; 13:i_slv_lrclk_h ; 14:i_slv_lrclk_i ; 15:i_slv_lrclk_j ;

EE_AUDIO_CLK_TDMOUT_B_CTRL 0x25

Same as EE_AUDIO_CLK_TDMOUT_A_CTRL

EE_AUDIO_CLK_TDMOUT_C_CTRL 0x26

Same as EE_AUDIO_CLK_TDMOUT_A_CTRL

Table 9-16 EE_AUDIO_CLK_SPDIFIN_CTRL 0x27

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
26:24	R/W		clk_sel, 0:mp0_pll; 1:mp1_pll; 2:mp2_pll; 3:mp3_pll; 4:hifi_pll; 5:fclk_div3(666M); 6:fclk_div4(500M); 7:fclk_div5(400M);
7:0	R/W		clk_div, the frequency of mclk = pll/clk_div;

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Table 9-17 EE_AUDIO_CLK_SPDIFOUT_CTRL 0x28

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
26:24	R/W		clk_sel, 0:mp0_pll; 1:mp1_pll; 2:mp2_pll; 3:mp3_pll; 4:hifi_pll; 5:fclk_div3(666M); 6:fclk_div4(500M); 7:fclk_div5(400M)
9:0	R/W		clk_div, the frequency of mclk = pll/clk_div;

Table 9-18 EE_AUDIO_CLK_RESAMPLEA_CTRL 0x29

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_ en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
27:24	R/W		sclk_sel, 0:mst_a_sclk_ph1; 1:mst_b_sclk_ph1; 2:mst_c_sclk_ph1; 3:mst_d_sclk_ph1; 4:mst_e_sclk_ph1; 5:mst_f_sclk_ph1; 6:i_slv_sclk_a ; 7:i_slv_sclk_b ; 8:i_slv_sclk_c ; 9::i_slv_sclk_d ; 10:i_slv_sclk_e ; 11:i_slv_sclk_f ; 12:i_slv_sclk_g ; 13:i_slv_sclk_h ; 14:i_slv_sclk_i ; 15:i_slv_sclk_j ;
7:0	R/W		Clk_div

Table 9-19 EE_AUDIO_CLK_LOCKER_CTRL 0x2a

Bits	R/W	Default	Description
31	R/W	0x0000-0000	lock_out_clk; 0:disable; 1:enable;
30	R/W		Force_oscin for lock_out_clk, 0:disable; 1: force clock source as oscin(24M);
27:24	R/W		clk_sel, 0:mst_a_sclk_ph1; 1:mst_b_sclk_ph1; 2:mst_c_sclk_ph1; 3:mst_d_sclk_ph1; 4:mst_e_sclk_ph1; 5:mst_f_sclk_ph1; 6:i_slv_sclk_a ; 7:i_slv_sclk_b ;

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Bits	R/W	Default	Description
			8:i_slv_sclk_c ; 9::i_slv_sclk_d ; 10:i_slv_sclk_e ; 11:i_slv_sclk_f ; 12:i_slv_sclk_g ; 13:i_slv_sclk_h ; 14:i_slv_sclk_i ; 15:i_slv_sclk_j ;
23:16	R/W		clk_div, lock_out_clk; out = in/clk_div;
15	R/W		clk_en, lock_in_clk; 0:disable; 1:enable;
14	R/W		Force_oscin for lock_in_clk, 0:disable; 1: force clock source as oscin(24M);
11:8	R/W		clk_sel, lock_in_clk; 0:mst_a_mclk; 1:mst_b_mclk; 2:mst_c_mclk; 3:mst_d_mclk; 4:mst_e_mclk; 5:mst_f_mclk; 6:i_slv_sclk_a ; 7:i_slv_sclk_b ; 8:i_slv_sclk_c ; 9::i_slv_sclk_d ; 10:i_slv_sclk_e ; 11:i_slv_sclk_f ; 12:i_slv_sclk_g ; 13:i_slv_sclk_h ; 14:i_slv_sclk_i ; 15:i_slv_sclk_j ;
7:0	R/W		clk_div, lock_in_clk; out = in/clk_div;

Table 9-20 EE_AUDIO_CLK_PDMIN_CTRL0 0x2b

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
26:24	R/W		clk_sel, 0:mp0_pll; 1:mp1_pll; 2:mp2_pll; 3:mp3_pll; 4:hifi_pll; 5:fclk_div3(666M); 6:fclk_div4(500M); 7:fclk_div5(400M);
15:0	R/W		the frequency of pdm_dclk = pll/clk_div;

Table 9-21 EE_AUDIO_CLK_PDMIN_CTRL1 0x2c

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);

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Bits	R/W	Default	Description
26:24	R/W		clk_sel, 0:mp0_pll; 1:mp1_pll; 2:mp2_pll; 3:mp3_pll; 4:hifi_pll; 5:fclk_div3(666M); 6:fclk_div4(500M); 7:fclk_div5(400M);
15:0	R/W		the frequency of pdm_sysclk = pll/clk_div;

Table 9-22 EE_AUDIO_CLK_SPDIFOUT_B_CTRL 0x2d

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
26:24	R/W		clk_sel, 0:mp0_pll; 1:mp1_pll; 2:mp2_pll; 3:mp3_pll; 4:hifi_pll; 5:fclk_div3(666M); 6:fclk_div4(500M); 7:fclk_div5(400M);
9:0	R/W		clk_div, the frequency of mclk = pll/clk_div;

Table 9-23 EE_AUDIO_CLK_RESAMPLEB_CTRL 0x2E

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_ en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
27:24	R/W		sclk_sel, 0:mst_a_sclk_ph1; 1:mst_b_sclk_ph1; 2:mst_c_sclk_ph1; 3:mst_d_sclk_ph1; 4:mst_e_sclk_ph1; 5:mst_f_sclk_ph1; 6:i_slv_sclk_a ; 7:i_slv_sclk_b ; 8:i_slv_sclk_c ; 9::i_slv_sclk_d ; 10:i_slv_sclk_e ; 11:i_slv_sclk_f ; 12:i_slv_sclk_g ; 13:i_slv_sclk_h ; 14:i_slv_sclk_i ; 15:i_slv_sclk_j ;
7:0	R/W		Clk_div

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Table 9-24 EE_AUDIO_CLK_SPDIFIN_LB_CTRL 0x2F

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_ en, 0:disable; 1: enable;
30	R/W		Clk_sel: 0: spdif_out_a clk; 1: spdif_out_b clk;
29	R/W		Clk_inv

Table 9-25 EE_AUDIO_CLK_EQDRC_CTRL 0x30

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_ en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
26:24	R/W		0:mp0_pll; 1:mp1_pll; 2:mp2_pll; 3:mp3_pll; 4:hifi_pll; 5:fclk_div3(666M); 6:fclk_div4(500M); 7:fclk_div5(400M)
15:0	R/W		Clk_div

Table 9-26 EE_AUDIO_CLK_VAD_CTRL 0x31

Bits	R/W	Default	Description
31	R/W	0x0000-0000	clk_ en, 0:disable; 1: enable;
30	R/W		Force_oscin, 0:disable; 1: force clock source as oscin(24M);
26:24	R/W		0:mp0_pll; 1:mp1_pll; 2:mp2_pll; 3:mp3_pll; 4:hifi_pll; 5:fclk_div3(666M); 6:fclk_div4(500M); 7:fclk_div5(400M)
15:0	R/W		Clk_div

9.10.2 TODDR Registers

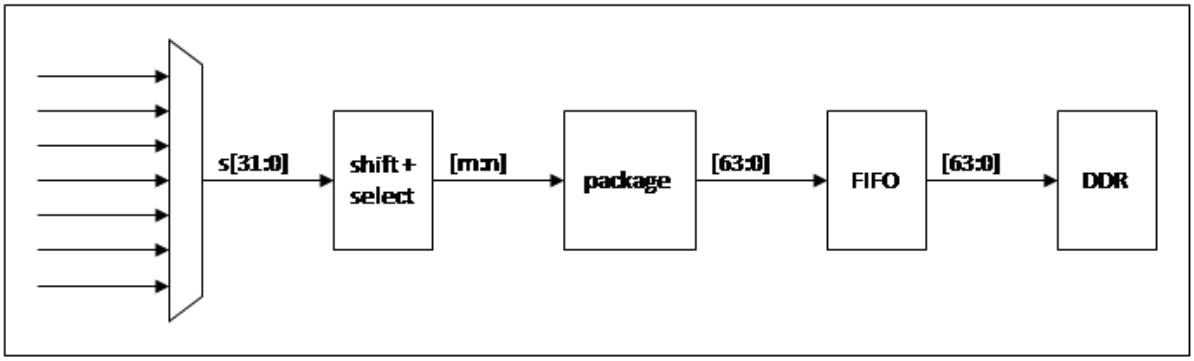
Table 9-27 EE_AUDIO_TODDR_A_CTRL0 0x40

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_toddr_en, 0: disable; 1: enable;
29	R/W		reg_ext_signed, 0: select write to only one buff (start_addr,finish_addr); 1: select write to two buff (start_addr,finish_addr) (start_addrb, finish_addrb);

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Bits	R/W	Default	Description
28	R/W		pp_mode: if enable, will jump addr between addr and addrB
27	R/W		Enable_sync_chnum; 1: start store data when first ch ;
26:24	R/W		reg_toddr_endian: transfer data endian of ddr bus (8bytes)
23:16	R/W		[23] : reserved; [22] : reserved; [21] : fifo overflow, write when fifocnt = depth; [20] : fifo overflow, read when fifocnt = 0; [19] : when write to ddr "int_addr" data (only once); [18] : when write to ddr "int_addr" data (repeat); [17] : when write to ddr address match "int_addr"; [16]: when write to ddr address match "finish_addr";
15:13	R/W		reg_toddr_sel, 0: combined data[m:n] without gap; like S0[m:n],S1[m:n],S2[m:n], … 1: combined data[m:n] as 16bits; like {S0[11:0],4'd0},{S1[11:0],4'd0}… 2: combined data[m:n] as 16bits; like {4'd0,S0[11:0]},{4'd0,{S1[11:0]}… 3: combined data[m:n] as 32bits; like {S0[27:4],8'd0},{S1[27:4],8'd0}… 4: combined data[m:n] as 32bits; like {8'd0,S0[27:4]},{8'd0,{S1[27:4]}…
12:8	R/W		reg_toddr_m_sel, the msb positioin in data
7:3	R/W		reg_toddr_n_sel, the lsb position in data
2	R/W		Clear adr/cnt value which captured by vad frame sync
1	R/W		Clear adr/cnt value which captured by vadflag
0	R/W		Ddr bus ugt bit

We can change format before write to DDR: TODDR-format



For example:

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m = 28; n = 9;
toddr_sel = 0;

	63			0
ADR = 0	s3[12:9]	s2[28:9]	s1[28:9]	s0[28:9]
ADR = 8	s6[16:9]	s5[28:9]	s4[28:9]	s3[28:13]
ADR = 16	s9[20:9]	s8[28:9]	s7[28:9]	s6[28:17]

m = 8; n = 1;
toddr_sel = 1;

	63			0
ADR = 0	s3[8:1],8'd0	s2[8:1],8'd0	s1[8:1],8'd0	s0[8:1],8'd0
ADR = 8	s7[8:1],8'd0	s6[8:1],8'd0	s5[8:1],8'd0	s4[8:1],8'd0

m = 22; n = 12;
toddr_sel = 2;

	63			0
ADR = 0	5'd0,s3[22:12]	5'd0,s2[22:12]	5'd0,s1[22:12]	5'd0,s0[22:12]
ADR = 8	5'd0,s7[22:12]	5'd0,s6[22:12]	5'd0,s5[22:12]	5'd0,s4[22:12]

m = 18; n = 4;
toddr_sel = 3;

	63	0
ADR = 0	s1[18:4],17'd0	s0[18:4],17'd0
ADR = 8	s3[18:4],17'd0	s2[18:4],17'd0

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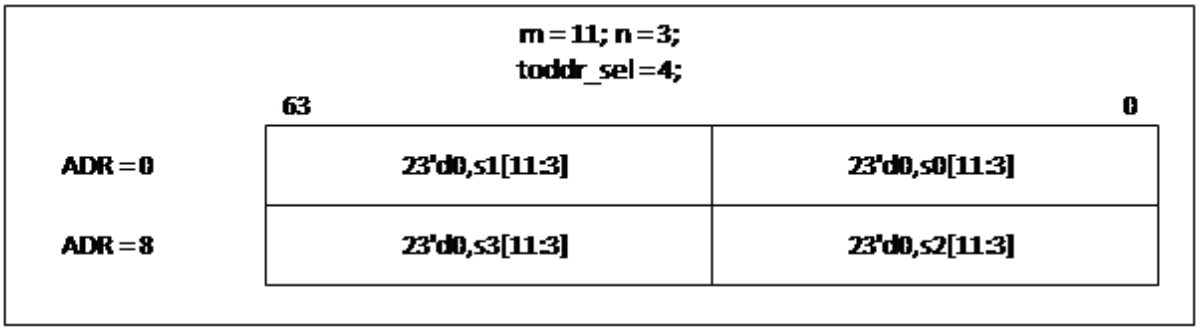


Table 9-28 EE_AUDIO_TODDR_A_CTRL1 0x41

Bits	R/W	Default	Description
31:28	R/W	0x0000-0000	Src select: 0: tdmin_a; 1: tdmin_b; 2: tdmin_c; 3: spdifin; 4: pdmin; 5: fratv; 6: tdmin_lb; 7: loopback_a; 8: frhdmirx; 9: loopback_b; 10: spdifin_lb; 15: vad;
27	R/W		Select from resample_A
26	R/W		Select from resample_B
25	R/W		force_finish; the value from 0-> 1: force finish by current address and jump to start_address;
24	R/W		Insert_chnum; 0: disable; 1: insert chnum[9:0] to data[9:0]
23:12	R/W		reg_fifo_start_rd_th, each time, when fifo_cnt greater than this register, control will start read data from fifo and write to DDR; write length is "reg_fifo_start_rd_th + 1" * 8bytes;
11:8	R/W		reg_status_sel, control status2 source;
7:0	R/W		reg_int_status_clr,clear each bits of int_status register

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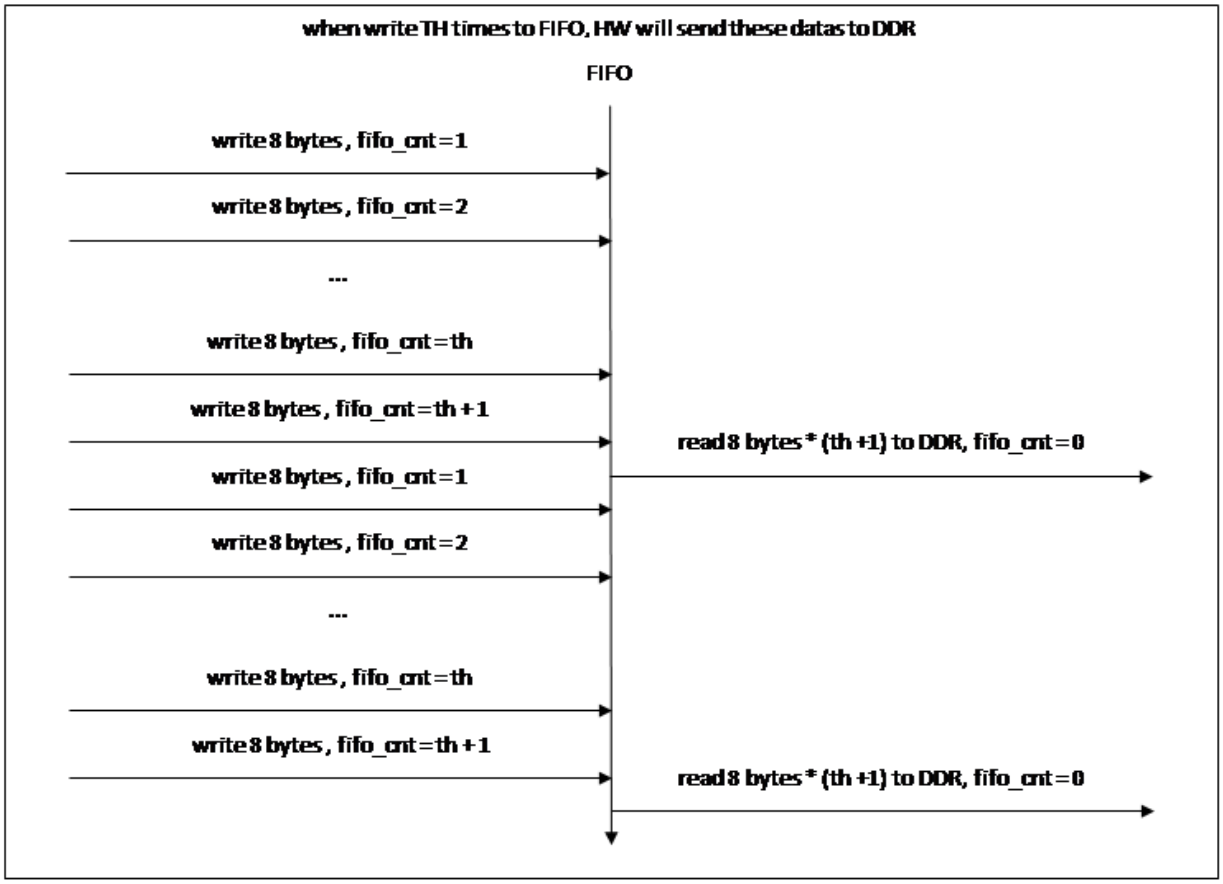


Table 9-29 EE_AUDIO_TODDR_A_START_ADDR 0x42

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	start_addr, buff_A start address, ignore [2:0]

Table 9-30 EE_AUDIO_TODDR_A_FINISH_ADDR 0x43

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	finish_addr, buff_A finish address, ignore [2:0]

Table 9-31 EE_AUDIO_TODDR_A_INT_ADDR 0x44

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	int_addr, usage A : as an address of interrupt; usage B : as a count of interrupt;

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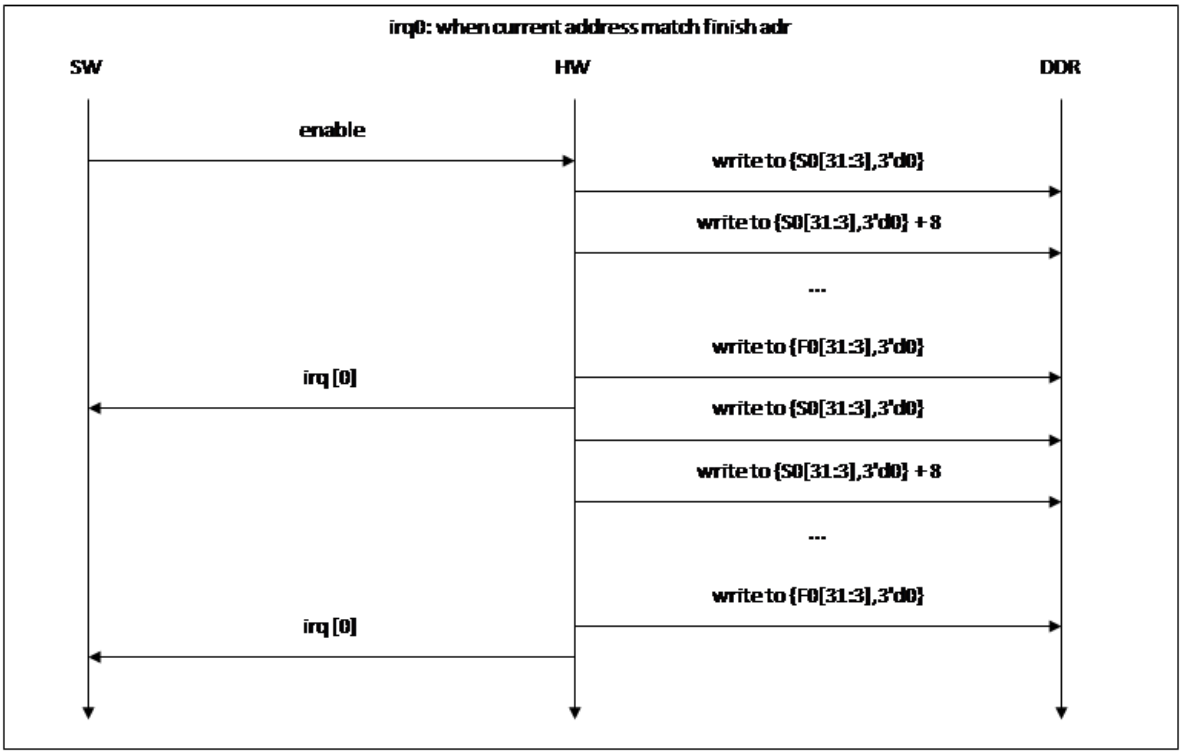
Table 9-32 EE_AUDIO_TODDR_A_STATUS1 0x45

Bits	R/W	Default	Description
21	R/W	0x0000-0000	sel_b_true
20	R/W		sel_b
19:8	R/W		fifo count, the num in fifo
7:0	R/W		int_status, when irq generate, related bit will changed to 1 and can only clear by reg_int_status_clr

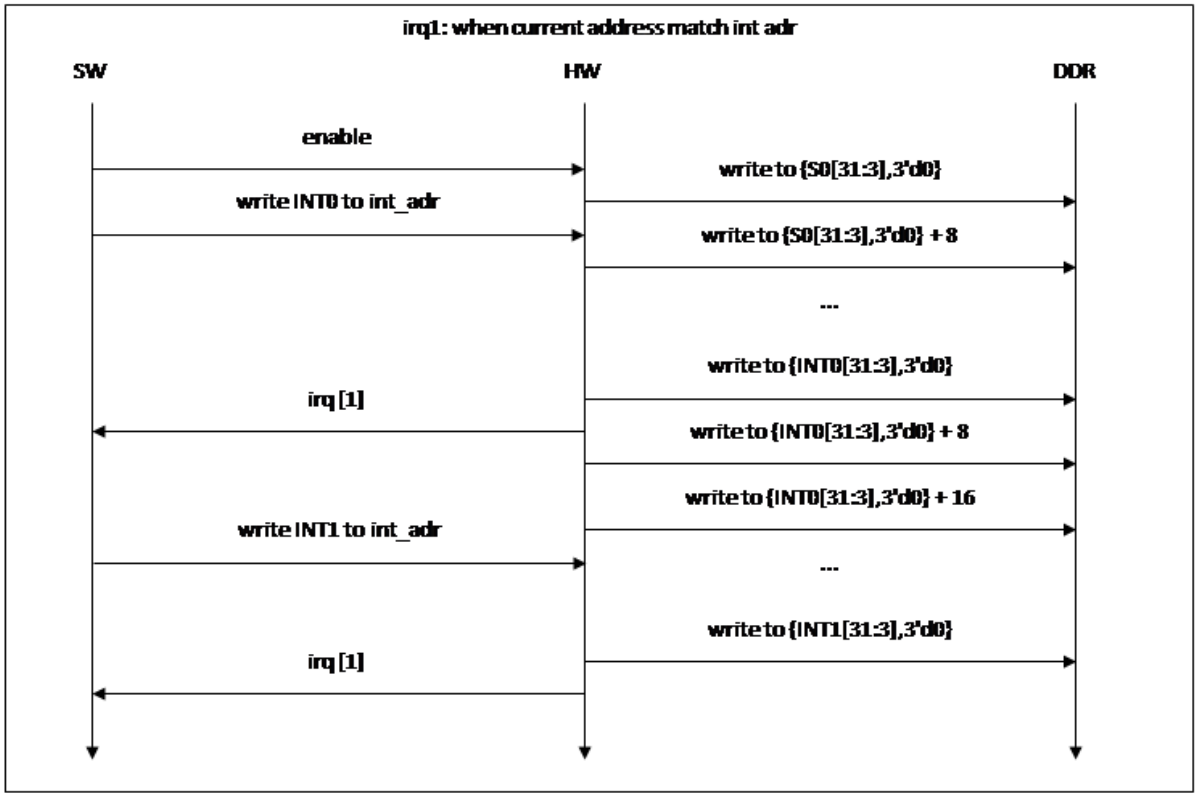
Table 9-33 EE_AUDIO_TODDR_A_STATUS2 0x46

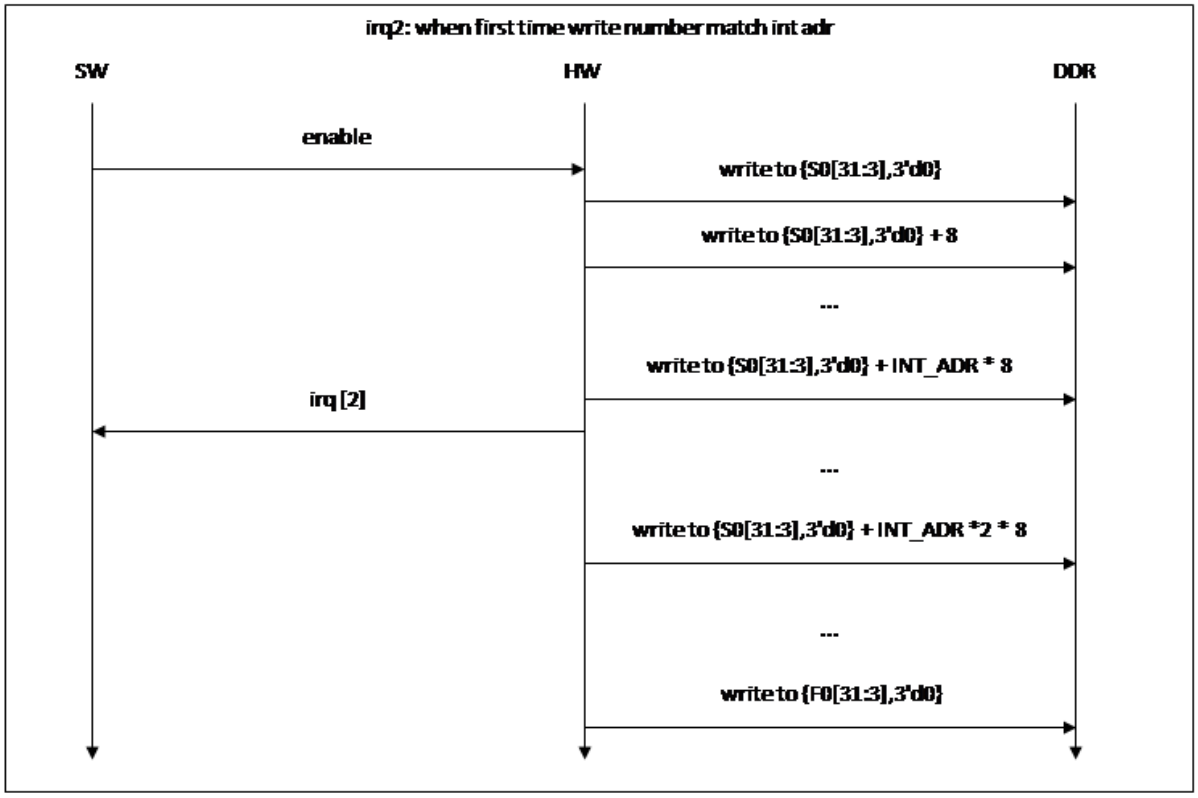
Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	status2, by reg_status_sel: 0: current ddr write address; 1: next finish address; 2: count by ddr reply, current ddr write address; 3: count by ddr reply, next finish address; 4: ddr address captured by vad flag; 5: ddr address captured by vad frame sync; 6: [31:16]: fifo_cnt captured by vad frame sync; [15:0]: fifo_cnt captured by vad flag;

We can generate 8 irq and add them together to CPU. Then can read int_status to know which irq it is:



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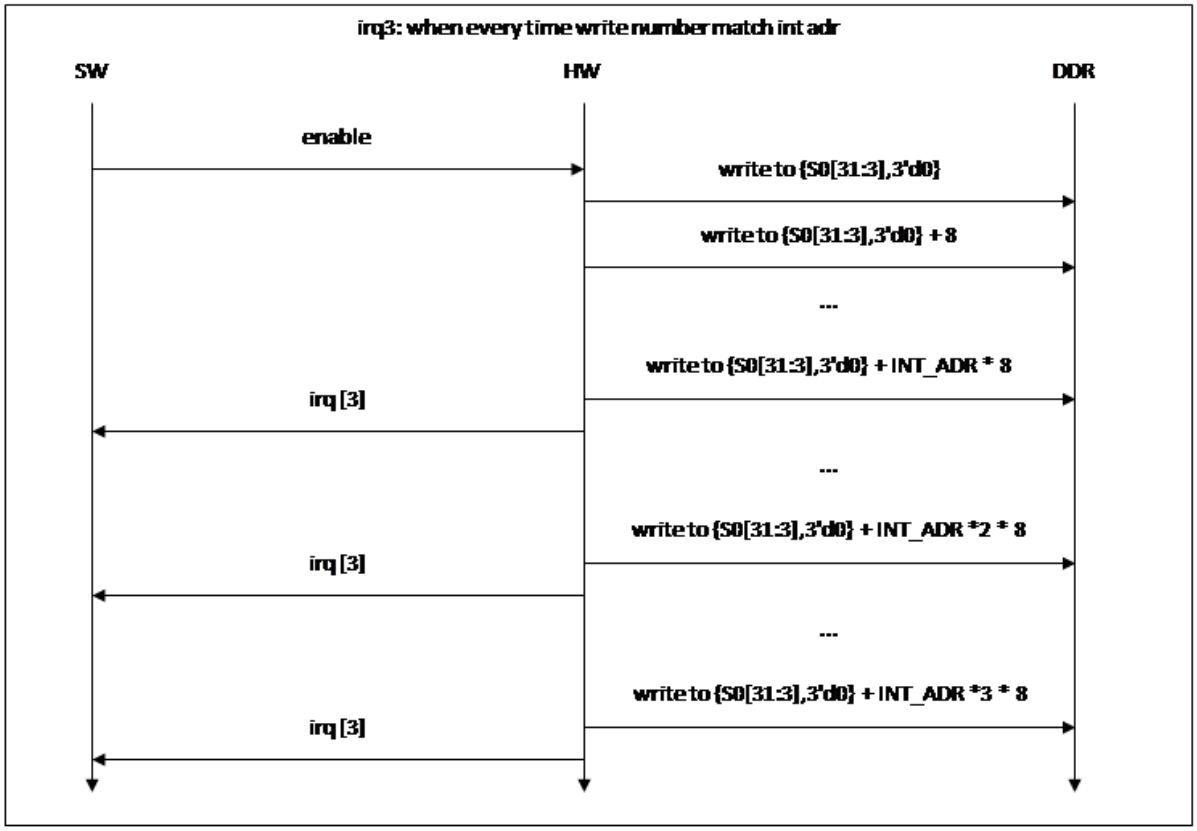


Table 9-34 EE_AUDIO_TODDR_A_START_ADDRB 0x47

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	istart_addrb, buff_B start address, ignore [2:0]

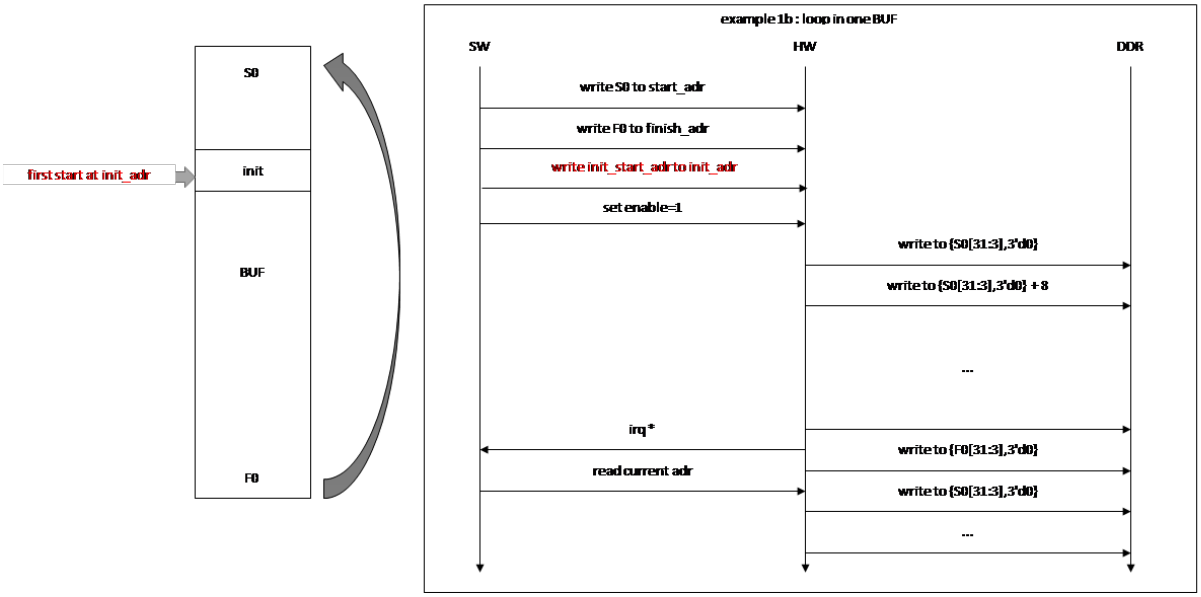
Table 9-35 EE_AUDIO_TODDR_A_FINISH_ADDRB 0x48

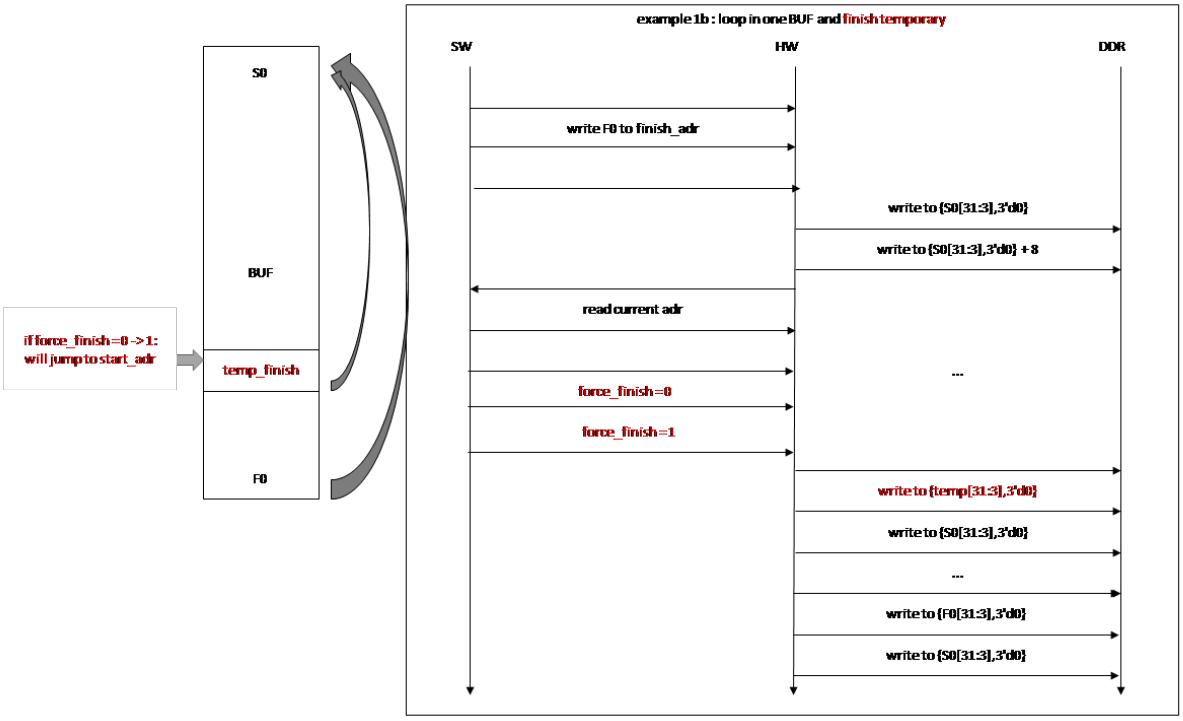
Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	finish_addrb, buff_B finish address, ignore [2:0]

Table 9-36 EE_AUDIO_TODDR_A_INIT_ADDR 0x49

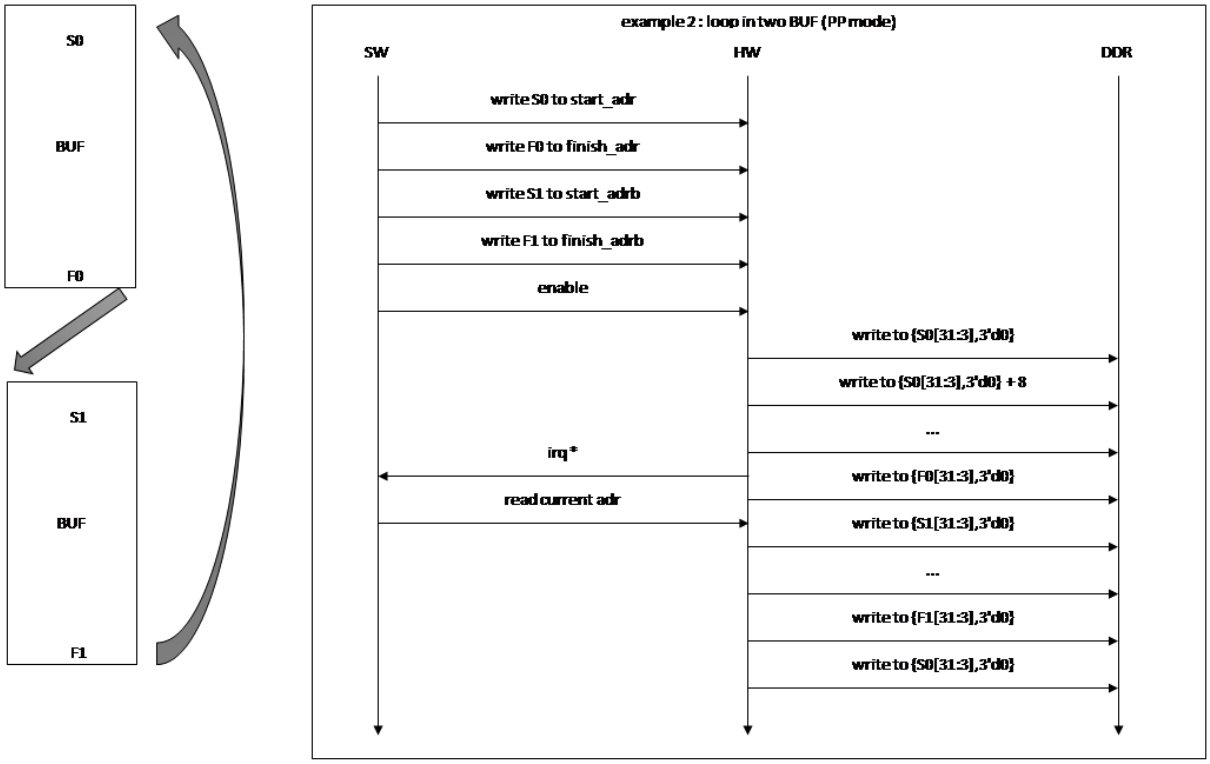
Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	initial address, the first ddr address after enable

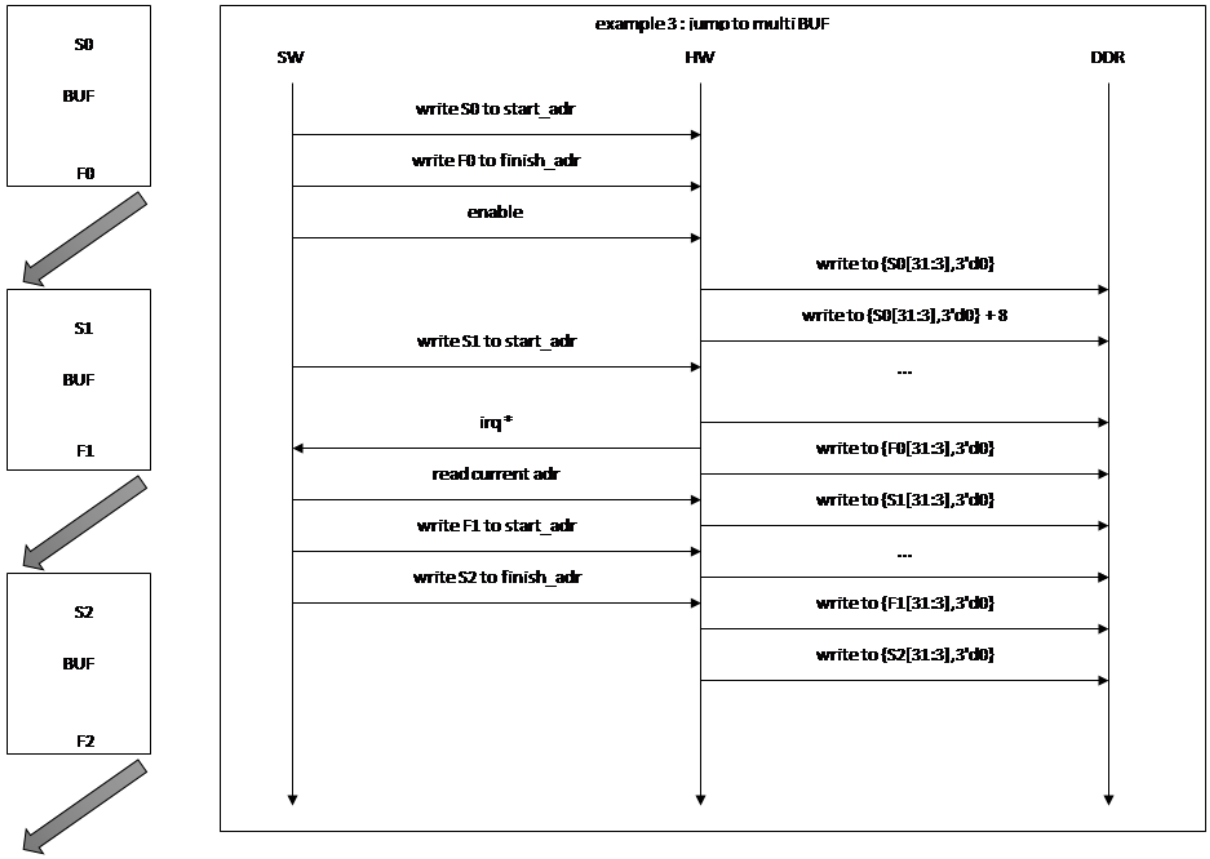
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Table 9-37 EE_AUDIO_TODDR_A_CTRL2 0x4a

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Reserved for hold read.
11:0	R/W		Reserved for hold read start offset.

Table 9-38 EE_AUDIO_TODDR_A_CHNUM_ID0 0x300

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ID of ch3
23:16	R/W		ID of ch2
15:8	R/W		ID of ch1
7:0	R/W		ID of ch0

Table 9-39 EE_AUDIO_TODDR_A_CHNUM_ID1 0x301

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ID of ch7
23:16	R/W		ID of ch6
15:8	R/W		ID of ch5
7:0	R/W		ID of ch4

Table 9-40 EE_AUDIO_TODDR_A_CHNUM_ID2 0x302

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ID of ch11
23:16	R/W		ID of ch10
15:8	R/W		ID of ch9
7:0	R/W		ID of ch8

Table 9-41 EE_AUDIO_TODDR_A_CHNUM_ID3 0x303

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ID of ch15
23:16	R/W		ID of ch14
15:8	R/W		ID of ch13
7:0	R/W		ID of ch12

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Table 9-42 EE_AUDIO_TODDR_A_CHNUM_ID4 0x304

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ID of ch19
23:16	R/W		ID of ch18
15:8	R/W		ID of ch17
7:0	R/W		ID of ch16

Table 9-43 EE_AUDIO_TODDR_A_CHNUM_ID5 0x305

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ID of ch23
23:16	R/W		ID of ch22
15:8	R/W		ID of ch21
7:0	R/W		ID of ch20

Table 9-44 EE_AUDIO_TODDR_A_CHNUM_ID6 0x306

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ID of ch27
23:16	R/W		ID of ch26
15:8	R/W		ID of ch25
7:0	R/W		ID of ch24

Table 9-45 EE_AUDIO_TODDR_A_CHNUM_ID7 0x307

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ID of ch31
23:16	R/W		ID of ch30
15:8	R/W		ID of ch29
7:0	R/W		ID of ch28

Table 9-46 EE_AUDIO_TODDR_A_CHSYNC_CTRL 0x30F

Bits	R/W	Default	Description
31	R/W	0x0000-0000	bypass_n; 0: out=in; 1: sync data by chnum info;
30	R/W		sw_reset; 1: reset chnum_sync module
29	R/W		mode; 0: work as BUFF_MODE; 1: work as DROP_MODE;
28	R/W		id_se; 0: ID = input valid count; 1: ID= reg_id_***
27	R/W		fix_stable; 1: ignore stable info of input; 0: compensate dat by stable info;

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Bits	R/W	Default	Description
26	R/W		debug_en0, for BUF_MODE only; 1: send chnum info from input; 0: generate new chnum info;
7:0	R/W		chnum_max; if input is 8ch, set this to 7;

EE_AUDIO_TODDR_B_CTRL0 0x50
EE_AUDIO_TODDR_B_CTRL1 0x51
EE_AUDIO_TODDR_B_START_ADDR 0x52
EE_AUDIO_TODDR_B_FINISH_ADDR 0x53
EE_AUDIO_TODDR_B_INT_ADDR 0x54
EE_AUDIO_TODDR_B_STATUS1 0x55
EE_AUDIO_TODDR_B_STATUS2 0x56
EE_AUDIO_TODDR_B_START_ADDRB 0x57
EE_AUDIO_TODDR_B_FINISH_ADDR 0x58
EE_AUDIO_TODDR_B_INIT_ADDR 0x59
EE_AUDIO_TODDR_B_CTRL2 0x5a

EE_AUDIO_TODDR_C_CTRL0 0x60
EE_AUDIO_TODDR_C_CTRL1 0x61
EE_AUDIO_TODDR_C_START_ADDR 0x62
EE_AUDIO_TODDR_C_FINISH_ADDR 0x63
EE_AUDIO_TODDR_C_INT_ADDR 0x64
EE_AUDIO_TODDR_C_STATUS1 0x65
EE_AUDIO_TODDR_C_STATUS2 0x66
EE_AUDIO_TODDR_C_START_ADDRB 0x67
EE_AUDIO_TODDR_C_FINISH_ADDR 0x68
EE_AUDIO_TODDR_C_INIT_ADDR 0x69
EE_AUDIO_TODDR_C_CTRL2 0x6a

EE_AUDIO_TODDR_D_CTRL0 0x210
EE_AUDIO_TODDR_D_CTRL1 0x211
EE_AUDIO_TODDR_D_START_ADDR 0x212
EE_AUDIO_TODDR_D_FINISH_ADDR 0x213
EE_AUDIO_TODDR_D_INT_ADDR 0x214
EE_AUDIO_TODDR_D_STATUS1 0x215
EE_AUDIO_TODDR_D_STATUS2 0x216
EE_AUDIO_TODDR_D_START_ADDRB 0x217
EE_AUDIO_TODDR_D_FINISH_ADDR 0x218

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EE_AUDIO_TODDR_D_INIT_ADDR 0x219

EE_AUDIO_TODDR_D_CTRL2 0x21a

9.10.3 FRDDR Registers

Table 9-47 EE_AUDIO_FRDDR_A_CTRL0 0x70

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_frddr_en, 0: disable; 1: enable;
30	R/W		reg_frddr_pp_mode, 0: select write to only one buff (start_addr,finish_addr); 1: select write to two buff (start_addr,finish_addr) (start_addrb, finish_addrb);
26:24	R/W		reg_frdd_endian
23:16	R/W		reg_frddr_int_en, [23] : reserved; [22] : reserved; [21] : fifo overflow, write when fifocnt = depth; [20] : fifo overflow, read when fifocnt = 0; [19] : first time when read from ddr "int_addr" data(only once); [18] : once time when read from ddr "int_addr" data(repeat); [17] : when read from ddr address match "int_addr"; [16]: when read from ddr address match "finish_addr";
15:12	R/W		reg_frddr_ack_dly, add delay to frddr ack
0	R/W		Ddr bus ugt setting.

Table 9-48 EE_AUDIO_FRDDR_A_CTRL1 0x71

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	reg_fifo_depth, the max depth of fifo; for high bit rates like 384k*32bits*8ch, set this register higher; for low bit rates like 48k*32bits*2ch, set this register lower;
23:16			reg_fifo_start_wr_th, when the fifo cnt less than "reg_fifo_depth - reg_fifo_start_wr_th", start reqeuest and read data from DDR; each time request "reg_fifo_start_wr_th" * 8 bytes data;
12			force finish; when the value changed from 0 to 1; will finished by current address and jump to start address;
11:8			reg_status_sel, control status2 source;
7:0			reg_int_status_clr, clear each bits of int_status register

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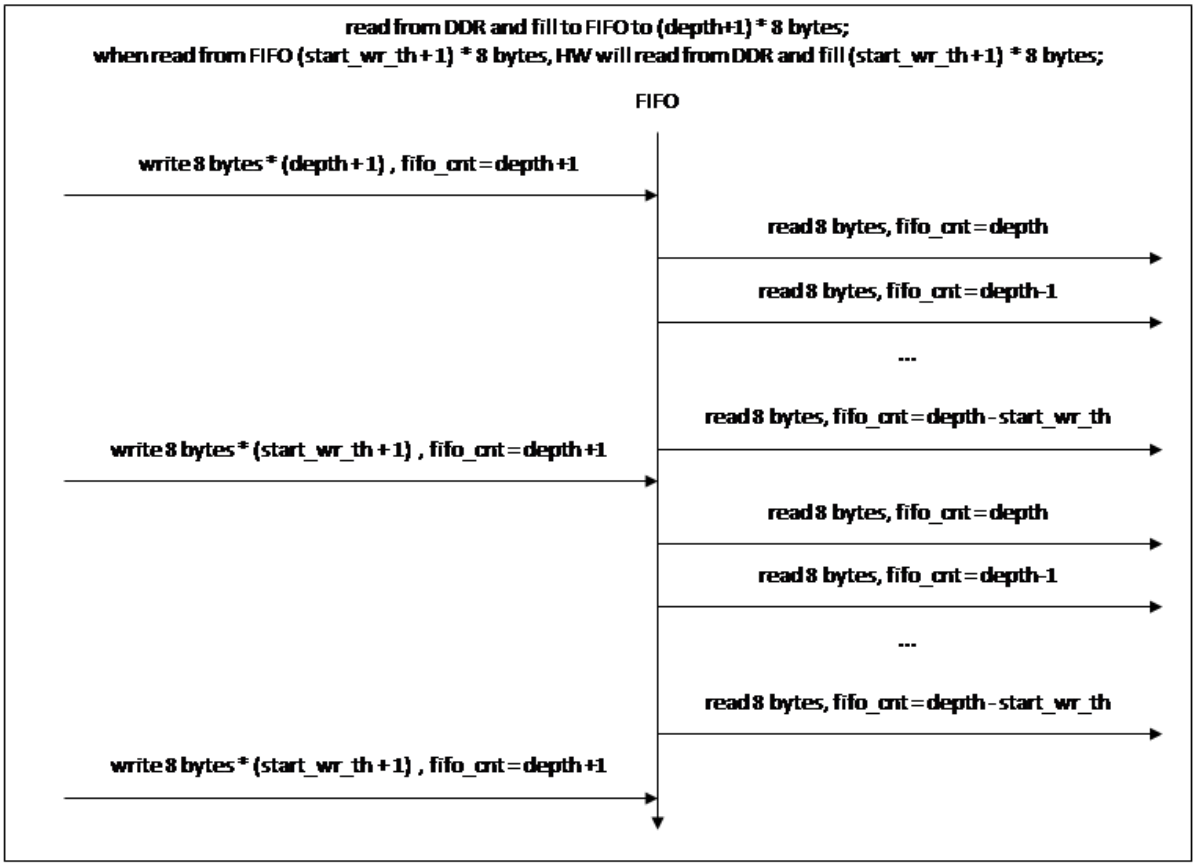


Table 9-49 EE_AUDIO_FRDDR_A_START_ADDR 0x72

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	start_addr,buff_B start address

Table 9-50 EE_AUDIO_FRDDR_A_FINISH_ADDR 0x73

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	finish_addr,buff_B finish address

Table 9-51 EE_AUDIO_FRDDR_A_INT_ADDR 0x74

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	int_addr,usage A : as an address of interrupt; usage B : as a count of interrupt;

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Table 9-52 EE_AUDIO_FRDDR_A_STATUS1 0x75

Bits	R/W	Default	Description
19	R/W	0x0000-0000	r_selb_true
18	R/W		r_selb
17:8	R/W		fifo count, the num in fifo
7:0	R/W		int_status, when irq generate, related bit will changed to 1 and can only clear by reg_int_status_clr

Table 9-53 EE_AUDIO_FRDDR_A_STATUS2 0x76

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	status2, by reg_status_sel: 0: current ddr write address; 1: next finish address; 2: count by ddr reply, current ddr write address; 3: count by ddr reply, next finish address;

The same design as TODDR:

We can generate 8 irq and add them together to CPU.

Then can read int_status to know it's which irq.

EE_AUDIO_FRDDR_A_START_ADDRB 0x77

EE_AUDIO_FRDDR_A_FINISH_ADDRB 0x78

The same design as TODDR:

We have an internal register to store reg_start_adr and reg_finish_adr.

We call them as r_start_adr and r_finish_adr;

Each time, when curr_adr match r_finish_adr, curr_adr will jump to r_start_adr , then update r_start_adr and r_finish_adr;

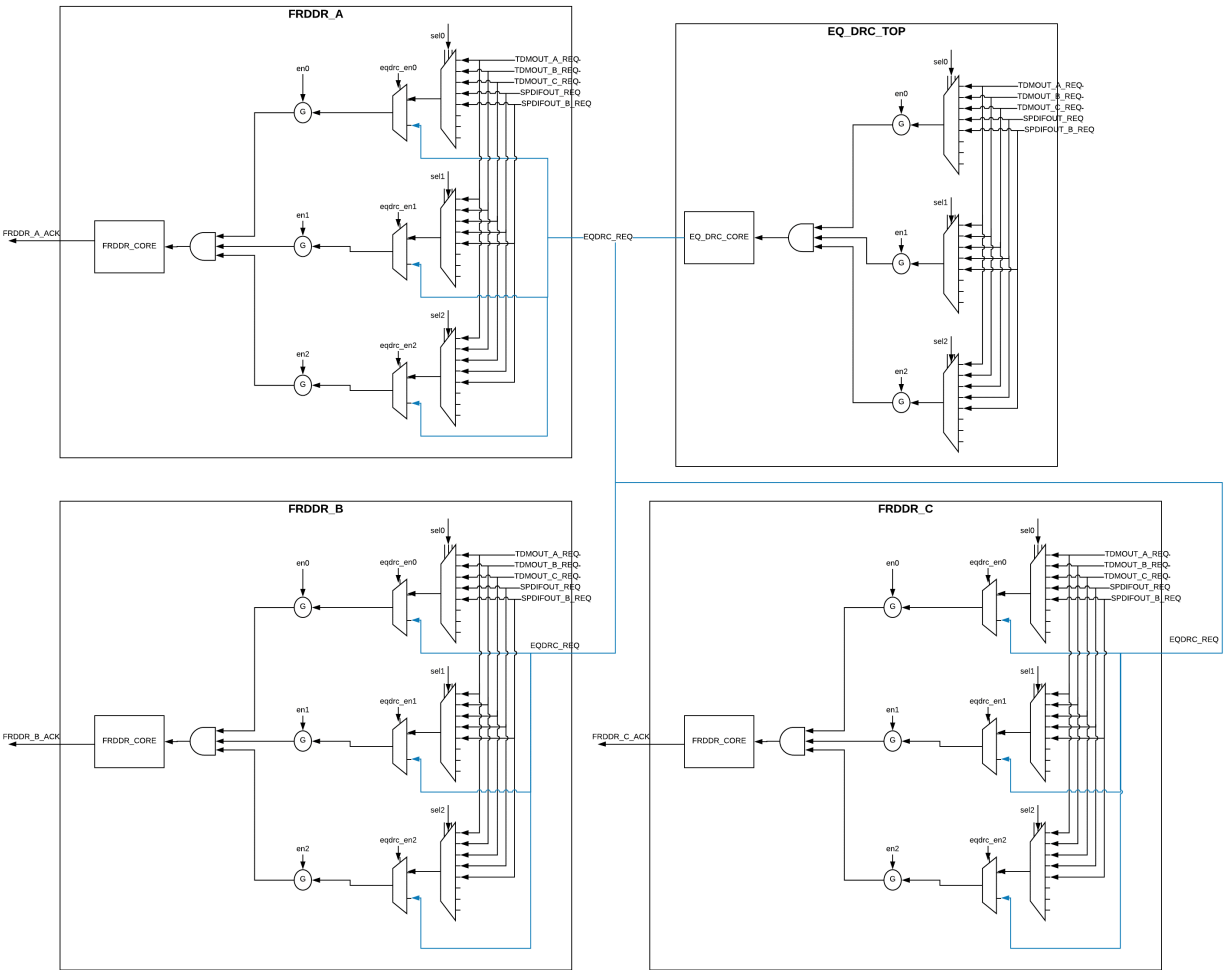
That's mean SW can write new start adr and finish adr before curr_adr match old finish adr .

Table 9-54 EE_AUDIO_FRDDR_A_CTRL2 0x7a

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	ch num for share buffer feature; if needn't share buffer, set it as 0; if need share buffer , set it as TDMOUT/SPDIFOUT ch number;
20	R/W		Src_sel2_en
19	R/W		Src_sel2_eq; 1: select request from EQDRC; 0: select request from TDMOUT/SPDIFOUT;
18:16	R/W		Src_sel2; 0: tdmout_a; 1: tdmout_b; 2: tdmout_c; 3: spdifout; 4: spdifout_b; 5: reserved; 6: reserved; 7: reserved;

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Bits	R/W	Default	Description
12	R/W		Src_sel1_en
11	R/W		Src_sel1_eq; 1: select request from EQDRC; 0: select request from TDMOUT/ SPDIFOUT;
10:8	R/W		Src_sel1; 0: tdmout_a; 1: tdmout_b; 2: tdmout_c; 3: spdifout; 4: spdifout_b; 5: reserved; 6: reserved; 7: reserved;
4	R/W		Src_sel0_en
3	R/W		Src_sel0_eq; 1: select request from EQDRC; 0: select request from TDMOUT/ SPDIFOUT;
2:0	R/W		Src_sel0; 0: tdmout_a; 1: tdmout_b; 2: tdmout_c; 3: spdifout; 4: spdifout_b; 5: reserved; 6: reserved; 7: reserved;



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EE_AUDIO_FRDDR_B_CTRL0 0x80
EE_AUDIO_FRDDR_B_CTRL1 0x81
EE_AUDIO_FRDDR_B_START_ADDR 0x82
EE_AUDIO_FRDDR_B_FINISH_ADDR 0x83
EE_AUDIO_FRDDR_B_INT_ADDR 0x84
EE_AUDIO_FRDDR_B_STATUS1 0x85
EE_AUDIO_FRDDR_B_STATUS2 0x86
EE_AUDIO_FRDDR_B_START_ADDRB 0x87
EE_AUDIO_FRDDR_B_FINISH_ADDR 0x88
EE_AUDIO_FRDDR_B_INIT_ADDR 0x89
EE_AUDIO_FRDDR_B_CTRL2 0x8a

EE_AUDIO_FRDDR_C_CTRL0 0x90
EE_AUDIO_FRDDR_C_CTRL1 0x91
EE_AUDIO_FRDDR_C_START_ADDR 0x92
EE_AUDIO_FRDDR_C_FINISH_ADDR 0x93
EE_AUDIO_FRDDR_C_INT_ADDR 0x94
EE_AUDIO_FRDDR_C_STATUS1 0x95
EE_AUDIO_FRDDR_C_STATUS2 0x96
EE_AUDIO_FRDDR_C_START_ADDRB 0x97
EE_AUDIO_FRDDR_C_FINISH_ADDR 0x98
EE_AUDIO_FRDDR_C_INIT_ADDR 0x99
EE_AUDIO_FRDDR_C_CTRL2 0x9a

EE_AUDIO_FRDDR_D_CTRL0 0x220
EE_AUDIO_FRDDR_D_CTRL1 0x221
EE_AUDIO_FRDDR_D_START_ADDR 0x222
EE_AUDIO_FRDDR_D_FINISH_ADDR 0x223
EE_AUDIO_FRDDR_D_INT_ADDR 0x224
EE_AUDIO_FRDDR_D_STATUS1 0x225
EE_AUDIO_FRDDR_D_STATUS2 0x226
EE_AUDIO_FRDDR_D_START_ADDRB 0x227
EE_AUDIO_FRDDR_D_FINISH_ADDR 0x228
EE_AUDIO_FRDDR_D_INIT_ADDR 0x229
EE_AUDIO_FRDDR_D_CTRL2 0x22a

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9.10.4 DDR ARB Registers

Table 9-55 EE_AUDIO_ARB_CTRL 0xa0

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_arb_en, 0:disable; 1: enable;
7:0	R/W		reg_arb_mask, [7]: frddr_d; [6]: frddr_c; [5]: frddr_b; [4]: frddr_a; [3]: toddr_d; [2]: toddr_c; [1]: toddr_b; [0]: toddr_a;

9.10.5 LoopBack Registers

Table 9-56 EE_AUDIO_LB_A_CTRL0 0xb0

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_lb_en, 0:disable; 1: enable;
30	R/W		reg_lb_mode, 0: out rate = in data rate; 1: out rate = loopback data rate;
29	R/W		reg_ext_signed, 0: extend bits as "0"; 1: extend bits as "msb";
28	R/W		Enable_sync_chnum; 1: start store data when ch_num can match ID;
27	R/W		chnum_en; 1: start send new ch num out;
17:16	R/W		Reg_sts_sel, refer to REG_LB_STATUS.
15:13	R/W		reg_dat_sel, shift [m:n] to [31:0]; 0: right justified, out = { ext,[m:n]}; 1: left justified, out = {[m:n],all0}; 2: right justified, out = { ext,[m:n]}; 3: left justified, out = {[m:n],all0}; 4: right justified, out = { ext,[m:n]};
12:8	R/W		reg_dat_m_sel, the msb positioin in data
7:3	R/W		reg_dat_n_sel, the lsb position in data
2:0	R/W		reg_dat_src_sel, [7]: reserved; [6]: reserved; [5]: for loopbackB, will select loopbackA output; [4]: pdmin; [3]: spdifin; [2]: tdmin_c; [1]: tdmin_b; [0]: tdmin_a;

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Table 9-57 EE_AUDIO_LB_A_CTRL1 0xb1

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Lb_sel_resampleA;1: select from resample A
30	R/W		Lb_sel_resampleB;1: select from resample B; need set Lb_sel_resampleA = 0;
29	R/W		reg_lb_ext_signed, 0: extend bits as "0"; 1: extend bits as "msb";
28	R/W		Enable_sync_chnum; 1: start store data when ch_num can match ID;
27	R/W		hold_insertion 1: hold insert data store(update) and insert 0 after original data;
15:13			reg_lb_sel, shift [m:n] to [31:0]; 0: right justified, out = {ext,[m:n]}; 1: left justified, out = {[m:n],all0}; 2: right justified, out = { ext,[m:n]}; 3: left justified, out = {[m:n], all0}; 4: right justified, out = { ext,[m:n]};
12:8			reg_lb_m_sel, the msb positioin loopback data
7:3			reg_lb_n_sel, the lsb position loopback data
0			Lb src sel; 1: spdifin_lb; 0: tdmin_lb;

Table 9-58 EE_AUDIO_LB_A_CTRL2 0xb2

Bits	R/W	Default	Description
19:16	R/W	0x0000-0000	Dat_ch_num, max channel number of data source; max is 15 (equal 16 ch)
15:0	R/W		Dat_ch_mask, 16bits match 16 ch, set 1 will sending out, set0 will drop off this ch;

Table 9-59 EE_AUDIO_LB_A_CTRL3 0xb3

Bits	R/W	Default	Description
19:16	R/W	0x0000-0000	Lb_ch_num, max channel number of data source; max is 15 (equal 16 ch)
15:0	R/W		Lb_ch_mask, 16bits match 16 ch, set 1 will sending out, set0 will drop off this ch;

Table 9-60 EE_AUDIO_ LB_A_DAT_ID0 0xb4

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	data_ch7_id
23:16	R/W		data_ch6_id
15:8	R/W		data_ch5_id
7:0	R/W		data_ch4_id

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Table 9-61 EE_AUDIO_LB_A_DAT_ID1 0xb5

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	data_ch3_id
23:16	R/W		data_ch2_id
15:8	R/W		data_ch1_id
7:0	R/W		data_ch0_id

Table 9-62 EE_AUDIO_LB_A_DAT_ID2 0xb6

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	data_ch11_id
23:16	R/W		data_ch10_id
15:8	R/W		data_ch9_id
7:0	R/W		data_ch8_id

Table 9-63 EE_AUDIO_LB_A_DAT_ID3 0xb7

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	data_ch15_id
23:16	R/W		data_ch14_id
15:8	R/W		data_ch13_id
7:0	R/W		data_ch12_id

Table 9-64 EE_AUDIO_LB_A_ID0 0xb8

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	lb_data_ch7_id
23:16	R/W		lb_data_ch6_id
15:8	R/W		lb_data_ch5_id
7:0	R/W		lb_data_ch4_id

Table 9-65 EE_AUDIO_LB_A _ID1 0xb9

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	lb_data_ch3_id
23:16	R/W		lb_data_ch2_id

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Bits	R/W	Default	Description
15:8	R/W		lb_data_ch1_id
7:0	R/W		lb_data_ch0_id

Table 9-66 EE_AUDIO_LB_A _ID2 0xba

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	lb_data_ch11_id
23:16	R/W		lb_data_ch10_id
15:8	R/W		lb_data_ch9_id
7:0	R/W		lb_data_ch8_id

Table 9-67 EE_AUDIO_LB_A _ID3 0xbb

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	lb_data_ch15_id
23:16	R/W		lb_data_ch14_id
15:8	R/W		lb_data_ch13_id
7:0	R/W		lb_data_ch12_id

Table 9-68 EE_AUDIO_LB_A _STS 0xbc

Bits	R/W	Default	Description
31:0	R	0x0000-0000	reg_sts_sel = 0: [31:0] = 0; reg_sts_sel = 1: [31]:pipe_cnt_err2; mean pipe cnt decr when it's 0; [30]:pipe_cnt_err1; mean pipe cnt incr when it's 3; [29:28]:pipe_cnt; current pipe cnt value; [27]:output_en_err; when output, received a new start; [26]:insert_storing_err2; it it’s receving, but received first ch; [25]:insert_storing_err1, if it’s not receving, but received data [24]:orig_storing_err2; it it’s receving, but received first ch; [23]:orig_storing_err1, if it’s not receving, but received data [22]:1st_send_done, mean finished first time send insert data; [21]:insert_storing, mean it’s receiving insert data now; [20]:orig_storing, mean it’s receiving insert data now; [19]:insert_store_cnt_err, when first_ch, if store_cnt is not 0, mean error, missed ch or another reason; [9]:orig_store_cnt_err, when first_ch, if store_cnt is not 0, mean error, missed ch or another reason; reg_sts_sel = 2: [31:0]:insert_store_cnt_debug, when start sending data, stored insert dat ch number; reg_sts_sel = 3: [31:0]:orig_store_cnt_debug, when start sending data, stored orig dat ch number;

EE_AUDIO_LB_B_CTRL0 0x230

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EE_AUDIO_LB_B_CTRL1 0x231

EE_AUDIO_LB_B_CTRL2 0x232

EE_AUDIO_LB_B_CTRL3 0x233

EE_AUDIO_ LB_B_DAT_ID0 0x234

EE_AUDIO_LB_B_DAT_ID1 0x235

EE_AUDIO_LB_B_DAT_ID2 0x236

EE_AUDIO_LB_B_DAT_ID3 0x237

EE_AUDIO_LB_B_ID0 0x238

EE_AUDIO_LB_B_ID1 0x239

EE_AUDIO_LB_B_ID2 0x23a

EE_AUDIO_LB_B_ID3 0x23b

EE_AUDIO_LB_B_STS 0x23c

9.10.6 TDM Registers

TDMin_B/C are the same as TDMin_A.

TDMout_B/C are the same as TDMout_A.

TDMin_LB is the same as TDMin_A.

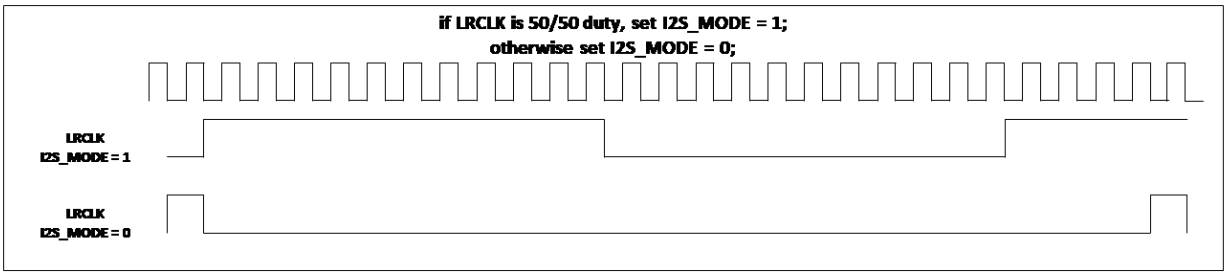
Table 9-69 EE_AUDIO_TDMIN_A_CTRL 0xc0

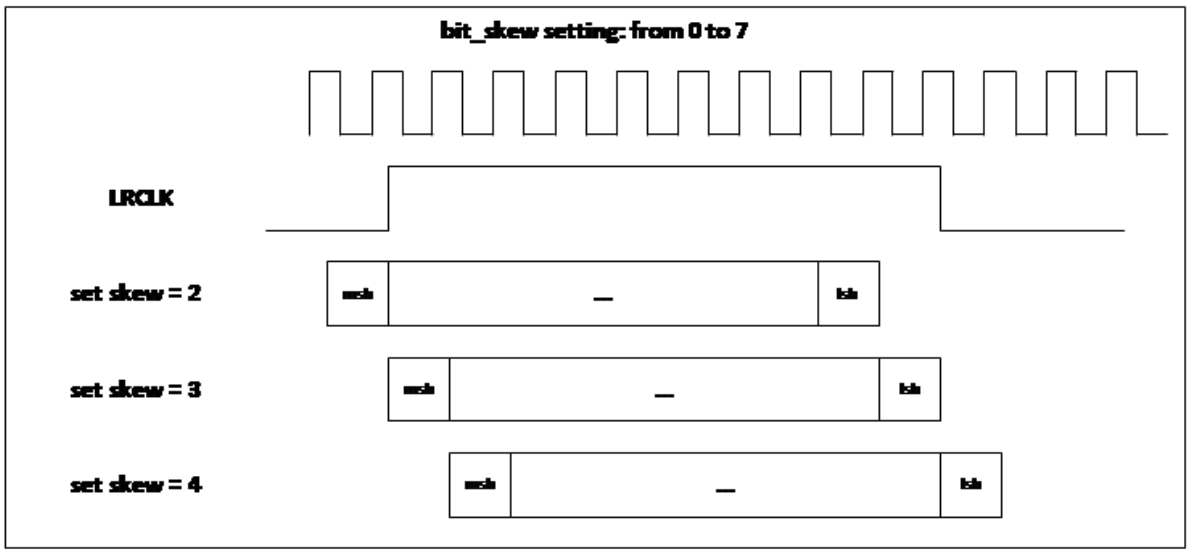
Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_work_enable 0:disable; 1:enable;
30	R/W		reg_i2s_mode,0:tdm mode; 1: i2s mode;
29	R/W		reg_rst_afifo_out_n, reset afifo out side; need set to 1 before reg_rst_afifo_in_n;
28	R/W		reg_rst_afifo_in_n, reset afifo in side; need set 1 after set reg_rst_afifo_out_n to 1;
27	R/W		reg_tdmin_in_debug_en, 0: disable debug mode; 1: enable;
26	R/W		reg_tdmin_in_auto_en,0: disalbe; 1: enable detect and store max/min of bit_cnt ;
25	R/W		reg_tdmin_in_rev_ws, revert ws(lrclk); 0 :disable; 1: enable;
24	R/W		reg_tdmin_in_rev_dat, revert data; 0:disable; 1:enable;

23:20	R/W		select tdmin src;
			0: PAD_TDMINA_DIN*;//for tdmin_lb, it's tdmoutA
			1: PAD_TDMINB_DIN*; //for tdmin_lb, it's tdmoutB
			2: PAD_TDMINC_DIN*; //for tdmin_lb, it's tdmoutC
			3: PAD_TDMINA_D*;//for tdmin_lb, it's PAD_TDMINA_DIN*
			4: PAD_TDMINB_D*; //for tdmin_lb, it's PAD_TDMINB_DIN*
			5: PAD_TDMINC_D*; //for tdmin_lb, it's PAD_TDMINC_DIN*
			6:HDMIRX_I2S;//for tdmin_lb, it's PAD_TDMINA_D*
			7:ACODEC_ADC; //for tdmin_lb, it's PAD_TDMINB_D*
			8:fix0;//for tdmin_lb, it's PAD_TDMINC_D*
			9:fix0; //for tdmin_lb, it's HDMIRX_I2S
			10:fix0; //for tdmin_lb, it's ACODEC_ADC;
			11:fix0;
			12:fix0;
			13: TDMOUTA; //for tdmin_lb, it's 0
			14: TDMOUTB; //for tdmin_lb, it's 0
			15: TDMOUTC; //for tdmin_lb, it's 0

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Bits	R/W	Default	Description
18:16	R/W		reg_tdmin_in_bit_skew, add delay to ws or data for skew modification;
6	R/W		chnum_en; 1: add ch cnt to chnum;
5	R/W		reg_lsb_first, 0: store first bit received to data_store[0]; 1: store first bit received to data_store[31];
4:0	R/W		reg_tdmin_bit_num, bitwidth of each slot, if slot is 16bits, set this register to 15;





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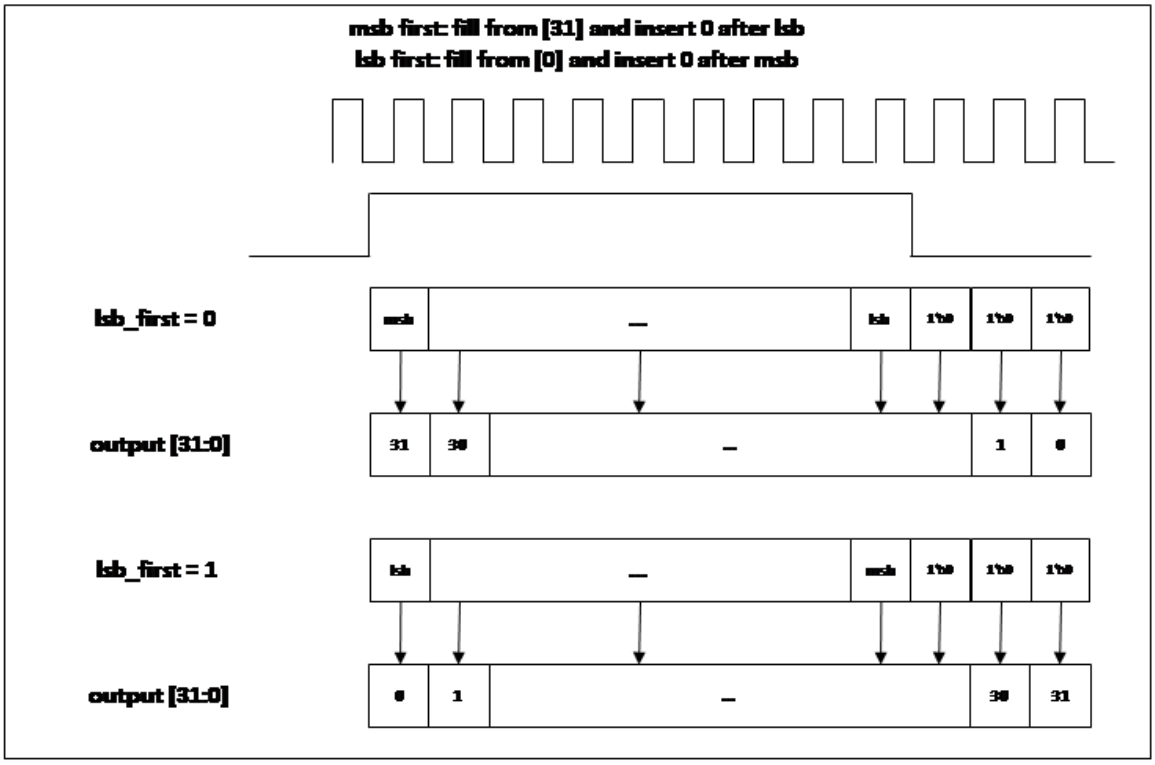


Table 9-70 EE_AUDIO_TDMIN_A_SWAP 0xc1

Bits	R/W	Default	Description
30:28	R/W	0x0000-0000	ch7_sel, 0: lane0 left channel;
26:24	R/W		ch6_sel, 1: lane0 right channel;
22:20	R/W		ch5_sel, 2: lane1 left channel;
18:16	R/W		ch4_sel, 3: lane1 right channel;
14:12	R/W		ch3_sel, 4: lane2 left channel;
10:8	R/W		ch2_sel, 5: lane2 right channel;
6:4	R/W		ch1_sel, 6: lane3 left channel;
2:0	R/W		ch0_sel, 7: lane3 right channel;

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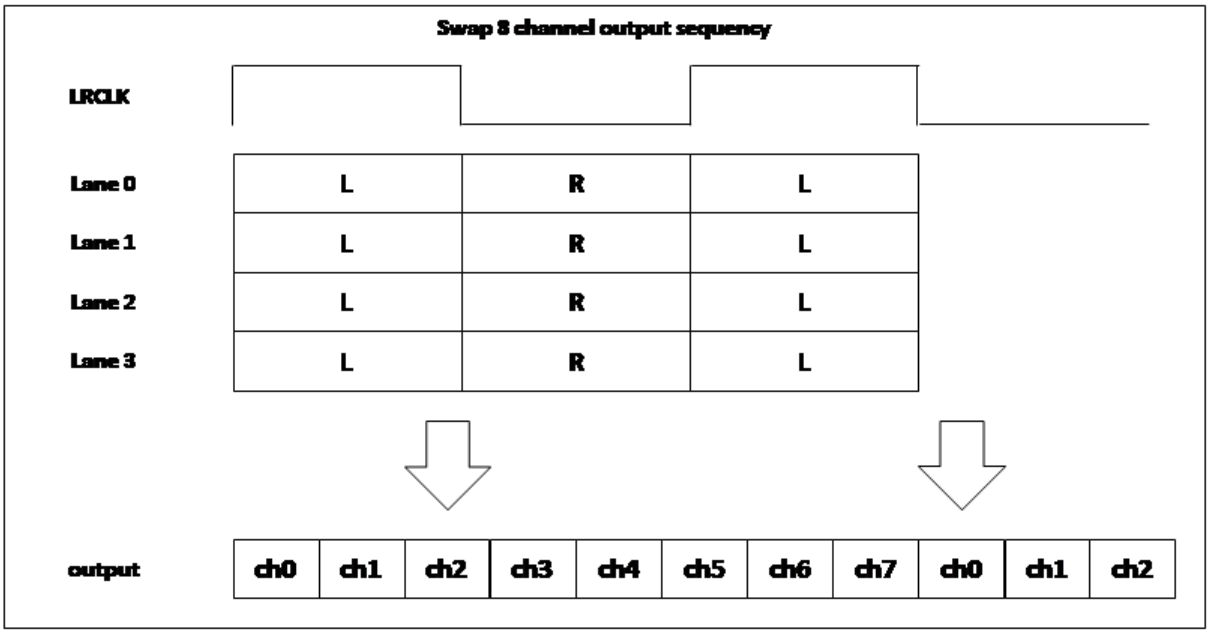


Table 9-71 EE_AUDIO_TDMIN_A_MASK0 0xc2

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane0_mask, mask each channel in lane0, max is 32 ch;

Table 9-72 EE_AUDIO_TDMIN_A_MASK1 0xc3

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane1_mask, mask each channel in lane1, max is 32 ch;

Table 9-73 EE_AUDIO_TDMIN_A_MASK2 0xc4

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane2_mask, mask each channel in lane2, max is 32 ch;

Table 9-74 EE_AUDIO_TDMIN_A_MASK3 0xc5

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane3_mask, mask each channel in lane3, max is 32 ch;

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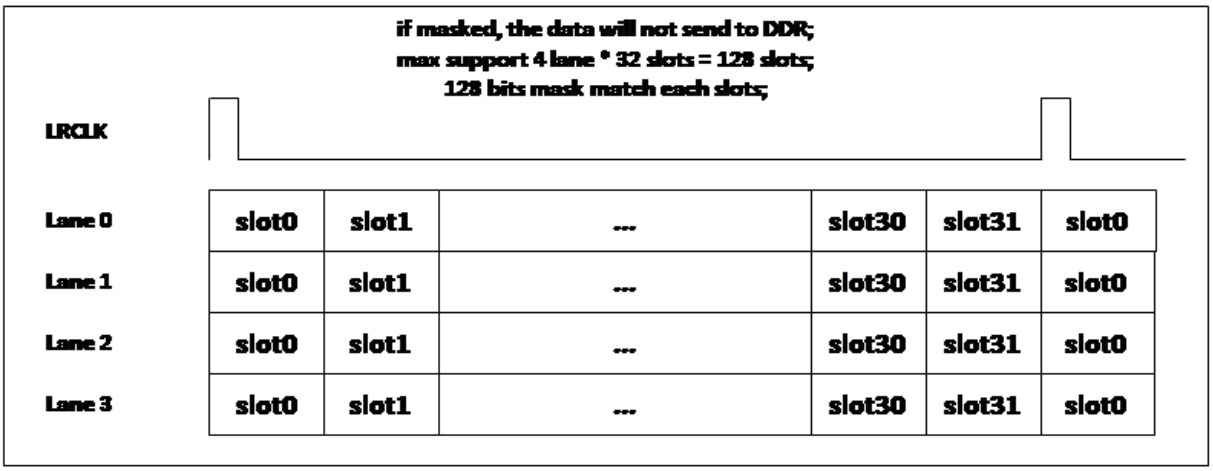


Table 9-75 EE_AUDIO_TDMIN_A_STAT 0xc6

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	r_input_slot_cnt_max, the maxnum of slot_cnt
23:16	R/W		r_input_slot_cnt_min, the minnum of slot_cnt
14	R/W		overflow_flag, overflow flag of afifo
13:10	R/W		max_fifo_cnt, the maxnum of afifo_cnt
9:5	R/W		r_input_bit_cnt_max, the maxnum of bit_cnt
4:0	R/W		r_input_bit_cnt_min, the minnum of bit_cnt

Table 9-76 EE_AUDIO_TDMIN_A_MUTE_VAL 0xc7

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_tmdin_a_mute_val, when mute , the channel value

Table 9-77 EE_AUDIO_TDMIN_A_MUTE0 0xc8

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane0_mute, mute each channel in lane0, max is 32 ch;

Table 9-78 EE_AUDIO_TDMIN_A_MUTE1 0xc9

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane1_mute, mute each channel in lane1, max is 32 ch;

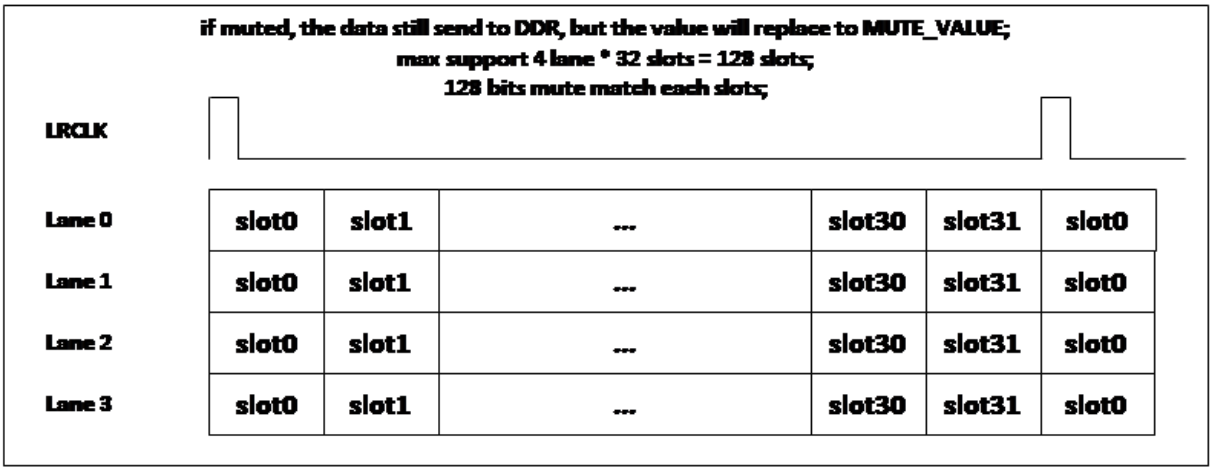
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Table 9-79 EE_AUDIO_TDMIN_A_MUTE2 0xca

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane2_mute, mute each channel in lane2, max is 32 ch;

Table 9-80 EE_AUDIO_TDMIN_A_MUTE3 0xcb

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane3_mute, mute each channel in lane3, max is 32 ch;



EE_AUDIO_TDMIN_B_CTRL 0xd0
EE_AUDIO_TDMIN_B_SWAP 0xd1
EE_AUDIO_TDMIN_B_MASK0 0xd2
EE_AUDIO_TDMIN_B_MASK1 0xd3
EE_AUDIO_TDMIN_B_MASK2 0xd4
EE_AUDIO_TDMIN_B_MASK3 0xd5
EE_AUDIO_TDMIN_B_STAT 0xd6
EE_AUDIO_TDMIN_B_MUTE_VAL 0xd7
EE_AUDIO_TDMIN_B_MUTE0 0xd8
EE_AUDIO_TDMIN_B_MUTE1 0xd9
EE_AUDIO_TDMIN_B_MUTE2 0xda
EE_AUDIO_TDMIN_B_MUTE3 0xdb

EE_AUDIO_TDMIN_C_CTRL 0xe0
EE_AUDIO_TDMIN_C_SWAP 0xe1
EE_AUDIO_TDMIN_C_MASK0 0xe2
EE_AUDIO_TDMIN_C_MASK1 0xe3

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EE_AUDIO_TDMIN_C_MASK2 0xe4
EE_AUDIO_TDMIN_C_MASK3 0xe5
EE_AUDIO_TDMIN_C_STAT 0xe6
EE_AUDIO_TDMIN_C_MUTE_VAL 0xe7
EE_AUDIO_TDMIN_C_MUTE0 0xe8
EE_AUDIO_TDMIN_C_MUTE1 0xe9
EE_AUDIO_TDMIN_C_MUTE2 0xea
EE_AUDIO_TDMIN_C_MUTE3 0xeb

EE_AUDIO_TDMIN_LB_CTRL 0xf0
EE_AUDIO_TDMIN_LB_SWAP 0xf1
EE_AUDIO_TDMIN_LB_MASK0 0xf2
EE_AUDIO_TDMIN_LB_MASK1 0xf3
EE_AUDIO_TDMIN_LB_MASK2 0xf4
EE_AUDIO_TDMIN_LB_MASK3 0xf5
EE_AUDIO_TDMIN_LB_STAT 0xf6
EE_AUDIO_TDMIN_LB_MUTE_VAL 0xf7
EE_AUDIO_TDMIN_LB_MUTE0 0xf8
EE_AUDIO_TDMIN_LB_MUTE1 0xf9
EE_AUDIO_TDMIN_LB_MUTE2 0xfa
EE_AUDIO_TDMIN_LB_MUTE3 0xfb

Table 9-81 EE_AUDIO_TDMOUT_A_CTRL0 0x140

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_work_enable, 0:disable; 1:enable;
29	R/W		reg_rst_afifo_out_n, reset afifo out side;
28	R/W		reg_rst_afifo_in_n, reset afifo in side; need set 1 after set reg_rst_afifo_out_n to 1;
27:24	R/W		force_oe[3:0]; 1: oe = force_val; 0: oe = mask_val;
23:20	R/W		reg_tdm_lr_mix, [23]: mix l3 and r3;[22]: mix l2 and r2;[21]: mix l1 and r1;[20]: mix l0 and r0;
19:15	R/W		reg_tdm_init_bitnum, initial count value of bitcnt
14:10	R/W		reg_tdm_init_slotnum, initial count value of slotcnt
9:5	R/W		reg_tdmout_slot_num , max value of slotcnt; if each frame has 8 slots, set it to 7;
4:0	R/W		reg_tdmout_bit_num ,max value of bitcnt; if each slot has 16bits, set it to 15;

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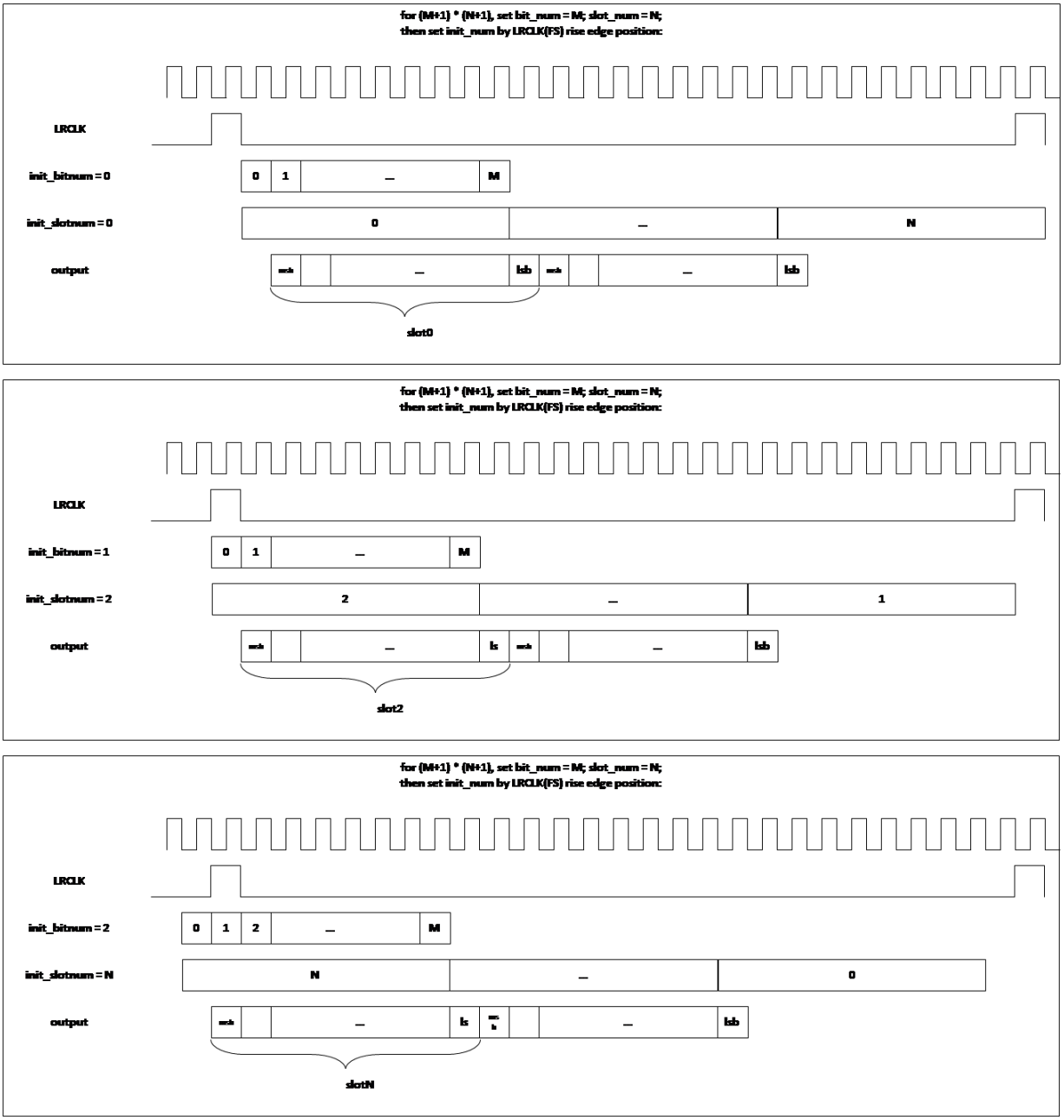


Table 9-82 EE_AUDIO_TDMOUT_A_CTRL1 0x141

Bits	R/W	Default	Description
31	R/W	0x0000-0000	eq_drc_sel; 1: select eq_drc output;
30	R/W		reg_debug_en,0:disable debug feature; 1: enable;
29	R/W		reg_out_lsb_first, 0: msb first; 1: lsb first;
28	R/W		reg_rev_ws_in, revert ws; 0: disable; 1: enable;
27	R/W		reg_rev_dat, revert data; 0: disable; 1: enable;
26	R/W		reg_gain_en, 0:disable; 1: enable data * gain;
25:24	R/W		reg_frddr_sel, 0:frddr_A;1:frddr_B;2:frddr_C;

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Bits	R/W	Default	Description
23:16	R/W		reg_wait_cnt, wait some time when enable set to 1;then start request data from frddr;
15:14	R/W		reg_gain_shift; 0: data * 1; 1: data * 2; 2: data * 4; 3: data * 8;
12:8	R/W		reg_frddr_msb, msb position of data
6:4	R/W		reg_frddr_type, 0: split 64bits ddr data to 8 sample, each sample need 8 bits; if bitwidth < 8, left-justified; 1: split 64bits ddr data to 4 sample, each sample need 16 bits; if bitwidth < 16, left-justified ; 2: split 64bits ddr data to 4 sample, each sample need 16 bits; if bitwidth < 16, right-justified ; 3: split 64bits ddr data to 2 sample, each sample need 32 bits; if bitwidth < 32, left-justified; 4: split 64bits ddr data to 2 sample, each sample need 32 bits; if bitwidth < 32, right-justified;
3:0	R/W		force_oe_val[3:0];

For position of sample in DDR:

	ddr_data [7:0] (byte)							
type	byte7	byte6	byte5	byte4	byte3	byte2	byte1	byte0
0	s7	s6	s5	s4	s3	s2	s1	s0
	s15	s14	s13	s12	s11	s10	s9	s8
1/2	s3		s2		s1		s0	
	s7		s6		s5		s4	
3/4	s1				s0			
	s3				s2			

For encoder, it will start output[31] first, then output[30], finished at output[32-bit_num]; If lsb_first, will start s[0], then s[1], finished at s[bit_num+1];

	output [31:0] (bit)																														
type	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	s[7]	s[6]	s[5]	s[4]	s[3]	s[2]	s[1]	s[0]	0																							
1	s[15]	s[14]	s[13]	s[12]	s[11]	s[10]	s[9]	s[8]	s[7]	s[6]	s[5]	s[4]	s[3]	s[2]	s[1]	s[0]	0															
2	s[msb]	s[msb-1]	...			s[1]	s[0]	0																								
3	s[31]	s[30]	s[29]	s[28]	s[27]	s[26]	s[25]	s[24]	s[23]	s[22]	s[21]	s[20]	s[19]	s[18]	s[17]	s[16]	s[15]	s[14]	s[13]	s[12]	s[11]	s[10]	s[9]	s[8]	s[7]	s[6]	s[5]	s[4]	s[3]	s[2]	s[1]	s[0]
4	s[msb]	s[msb-1]	...																			s[1]	s[0]	0								

Table 9-83 EE_AUDIO_TDMOUT_A_SWAP 0x142

Bits	R/W	Default	Description
30:28	R/W	0x0000-0000	lane3 right channel, 0: ch0;
26:24	R/W		lane3 left channel, 1: ch1;
22:20	R/W		lane2 right channel, 2: ch2;
18:16	R/W		lane2 left channel, 3: ch3;
14:12	R/W		lane1 right channel, 4: ch4;
10:8	R/W		lane1 left channel, 5: ch5;

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Bits	R/W	Default	Description
6:4	R/W		lane0 right channel, 6: ch6;
2:0	R/W		lane0 left channel, 7: ch7;

Table 9-84 EE_AUDIO_TDMOUT_A_MASK0 0x143

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane0_mask, mask each channel in lane0, max is 32 ch;

Table 9-85 EE_AUDIO_TDMOUT_A_MASK1 0x144

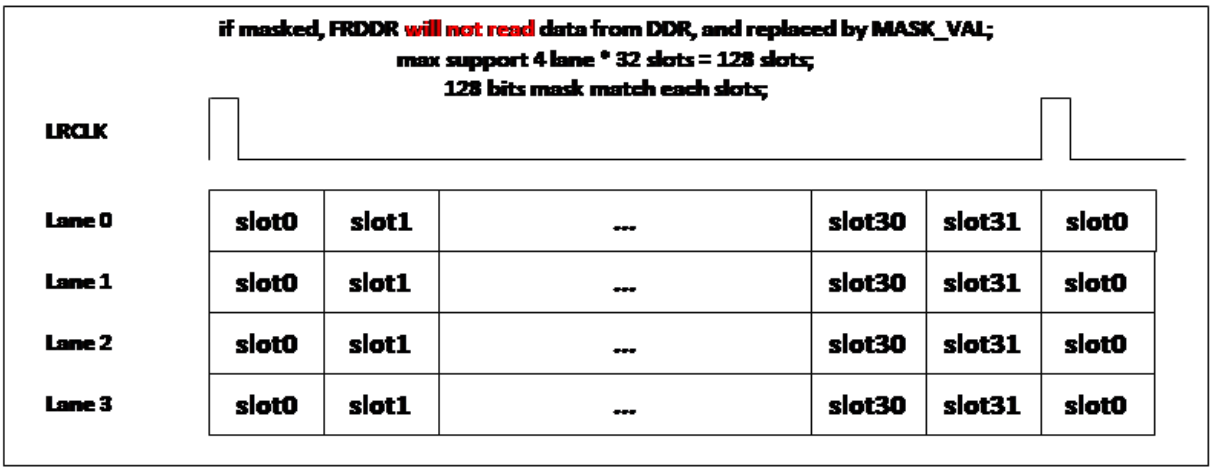
Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane1_mask, mask each channel in lane1, max is 32 ch;

Table 9-86 EE_AUDIO_TDMOUT_A_MASK2 0x145

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane2_mask, mask each channel in lane2, max is 32 ch;

Table 9-87 EE_AUDIO_TDMOUT_A_MASK3 0x146

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane3_mask, mask each channel in lane3, max is 32 ch;



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Table 9-88 EE_AUDIO_TDMOUT_A_STAT 0x147

Bits	R/W	Default	Description
31:29	R/W	0x0000-0000	fifo_cnt, afifo cnt
28	R/W		up_error, change to 1 if overflow
27:24	R/W		req_frdd_fsm_stat
21	R/W		r_slot_cnt_err
20	R/W		r_bit_cnt_err
16:12	R/W		r_max_slot_cnt
8:4	R/W		r_max_bit_cnt
3	R/W		r_out_en
2	R/W		r_out_en_pre
1	R/W		r_first_fs
0	R/W		c_frdd_init_finish

Table 9-89 EE_AUDIO_TDMOUT_A_GAIN0 0x148

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	gain_ch3
23:16	R/W	0x0000-0000	gain_ch2
15:8	R/W	0x0000-0000	gain_ch1
7:0	R/W	0x0000-0000	gain_ch0

Table 9-90 EE_AUDIO_TDMOUT_A_GAIN1 0x149

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	gain_ch7
23:16	R/W	0x0000-0000	gain_ch6
15:8	R/W	0x0000-0000	gain_ch5
7:0	R/W	0x0000-0000	gain_ch4

Table 9-91 EE_AUDIO_TDMOUT_A_MUTE_VAL 0x14a

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_tmdin_a_mute_val, when mute , the channel value

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Table 9-92 EE_AUDIO_TDMOUT_A_MUTE0 0x14b

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane0_mute, mute each channel in lane0, max is 32 ch;

Table 9-93 EE_AUDIO_TDMOUT_A_MUTE1 0x14c

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane1_mute, mute each channel in lane1, max is 32 ch;

Table 9-94 EE_AUDIO_TDMOUT_A_MUTE2 0x14d

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane2_mute, mute each channel in lane2 max is 32 ch;

Table 9-95 EE_AUDIO_TDMOUT_A_MUTE3 0x14e

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_lane3_mute, mute each channel in lane3, max is 32 ch;

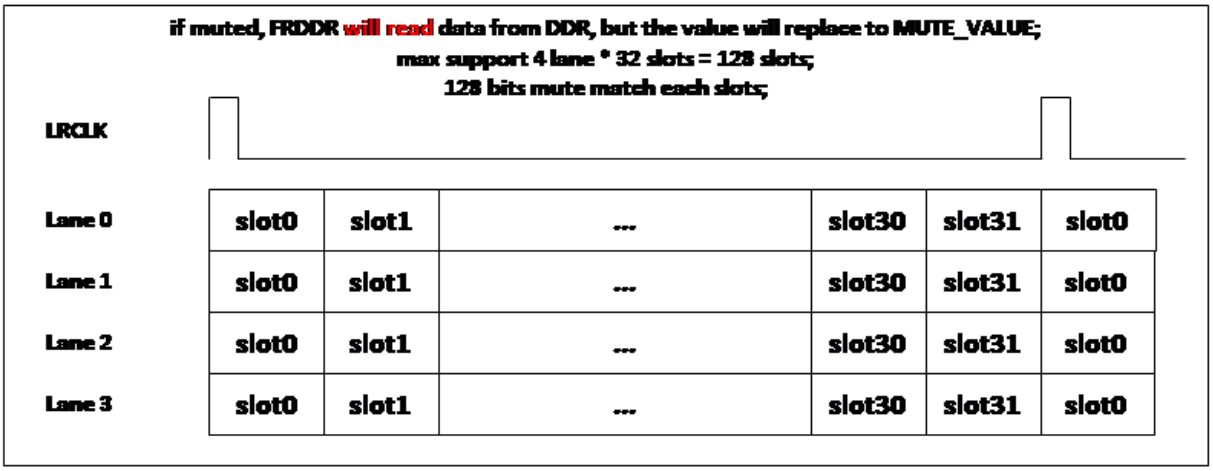


Table 9-96 EE_AUDIO_TDMOUT_A_MASK_VAL 0x14f

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_tdmout_mask_val, when masked, the channel value

EE_AUDIO_TDMOUT_B_CTRL0 0x150

EE_AUDIO_TDMOUT_B_CTRL1 0x151

EE_AUDIO_TDMOUT_B_SWAP 0x152

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EE_AUDIO_TDMOUT_B_MASK0 0x153
EE_AUDIO_TDMOUT_B_MASK1 0x154
EE_AUDIO_TDMOUT_B_MASK2 0x155
EE_AUDIO_TDMOUT_B_MASK3 0x156
EE_AUDIO_TDMOUT_B_STAT 0x157
EE_AUDIO_TDMOUT_B_GAIN0 0x158
EE_AUDIO_TDMOUT_B_GAIN1 0x159
EE_AUDIO_TDMOUT_B_MUTE_VAL 0x15a
EE_AUDIO_TDMOUT_B_MUTE0 0x15b
EE_AUDIO_TDMOUT_B_MUTE1 0x15c
EE_AUDIO_TDMOUT_B_MUTE2 0x15d
EE_AUDIO_TDMOUT_B_MUTE3 0x15e
EE_AUDIO_TDMOUT_B_MASK_VAL 0x15f

EE_AUDIO_TDMOUT_C_CTRL0 0x160
EE_AUDIO_TDMOUT_C_CTRL1 0x161
EE_AUDIO_TDMOUT_C_SWAP 0x162
EE_AUDIO_TDMOUT_C_MASK0 0x163
EE_AUDIO_TDMOUT_C_MASK1 0x164
EE_AUDIO_TDMOUT_C_MASK2 0x165
EE_AUDIO_TDMOUT_C_MASK3 0x166
EE_AUDIO_TDMOUT_C_STAT 0x167
EE_AUDIO_TDMOUT_C_GAIN0 0x168
EE_AUDIO_TDMOUT_C_GAIN1 0x169
EE_AUDIO_TDMOUT_C_MUTE_VAL 0x16a
EE_AUDIO_TDMOUT_C_MUTE0 0x16b
EE_AUDIO_TDMOUT_C_MUTE1 0x16c
EE_AUDIO_TDMOUT_C_MUTE2 0x16d
EE_AUDIO_TDMOUT_C_MUTE3 0x16e
EE_AUDIO_TDMOUT_C_MASK_VAL 0x16f

9.10.7 SPDIF Registers

Table 9-97 EE_AUDIO_SPDIFIN_CTRL0 0x100

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_work_enable, 0: disable; 1:enable;
30	R/W		reg_chnum_sel, 0: ch_num = 0~383(include frame cnt); 1: ch_num = 0~1(only L/R);

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Bits	R/W	Default	Description
29	R/W		reg_rst_afifo_out_n, reset afifo out side;
28	R/W		reg_rst_afifo_in_n, reset afifo in side; need set 1 after set reg_rst_afifo_out_n to 1;
27	R/W		reg_debug_en, 0:disable debug; 1: enable;
26	R/W		reg_chunm_en; 1: start add ch_cnt to ch_num;
25	R/W		reg_findpapb_en, 0: disable NonPCM mode; 1: enable;
24	R/W		reg_width_sel, 0: detect sample mode by max_width;1: detect sample mode by min_width;
23:12	R/W		reg_nonpcm2pcm_th, when detected NonPcm mode;if long time (z_cnt >= th) didn't detect PaPb again, will generate interrupt to SW: now changed to PCM mode;
11:8	R/W		reg_ch_status_sel, For EE_AUDIO_SPDIFIN_STAT1
7	R/W		reg_mute_l, 0: disable ; 1: mute channel L;
6	R/W		reg_mute_r, 0: disable ; 1: mute channel R;
5:4	R/W		reg_spdifin_src_sel, 0: PAD of spdifin;1: spdifout;
3	R/W		reg_check_valid, 0: disable valid check ; 1: enable;
2	R/W		reg_check_parity, 0: disable parity check ; 1: enable;
1	R/W		reg_invert_data, 0: disalbe; 1: invert [27:4] to [4:27];
0	R/W		reg_spdifin_phase, 0: disable invert; 1: enable;

Table 9-98 EE_AUDIO_SPDIFIN_CTRL1 0x101

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_force_sample_mode, 0: auto detect sample mode; 1: force a fixed sample mode;
30:28	R/W		reg_sample_mode
27:20	R/W		reg_interrupt_mask, mask each interrupt;
19:0	R/W		reg_base_timer, define a base timer to detect sample mode changed;

Table 9-99 EE_AUDIO_SPDIFIN_CTRL2 0x102

Bits	R/W	Default	Description
29:20	R/W	0x0000-0000	reg_sample_mode0_timer_th
19:10	R/W		reg_sample_mode1_timer_th
9:0	R/W		reg_sample_mode2_timer_th

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Table 9-100 EE_AUDIO_SPDIFIN_CTRL3 0x103

Bits	R/W	Default	Description
29:20	R/W	0x0000-0000	reg_sample_mode3_timer_th
19:10	R/W		reg_sample_mode4_timer_th
9:0	R/W		reg_sample_mode5_timer_th

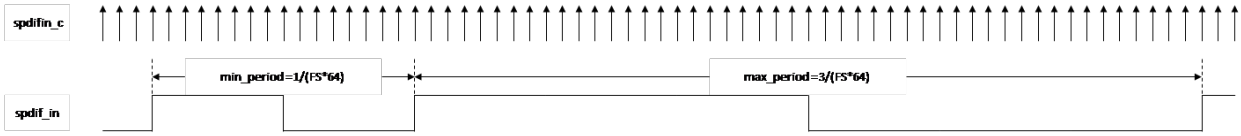
Table 9-101 EE_AUDIO_SPDIFIN_CTRL4 0x104

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	reg_sample_mode0_timer
23:16	R/W		reg_sample_mode1_timer
15:8	R/W		reg_sample_mode2_timer
7:0	R/W		reg_sample_mode3_timer

Table 9-102 EE_AUDIO_SPDIFIN_CTRL5 0x105

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	reg_sample_mode4_timer
23:16	R/W		reg_sample_mode5_timer
15:8	R/W		reg_sample_mode6_timer

We need automatically detect the frequency of sample rate (FS). It will done by search min_period or max_period of SPDIF IN and compared to thresholds. We can support 7 mode settings.



For example: If spdifin_clk is 166MHz, and we want support 48KHz and 96KHz.

The min_period of 48KHz is 1/(48000*64) and the counter number by 166MHz is 166000000/(48000*64) = 54;

96KHz counter number is 166000000/(96000*64) = 27;

Then set:

Width_sel = 1;

Mode0_th = (54+27)/2 = 41;

Mode1_th = 0;

Mode0_timer = 27;

Mode1_timer = 13;

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Or:

Width_sel = 0;

Mode0_th = (162+81)/2 = 121;

Mode1_th = 0;

Mode0_timer = 27;

Mode1_timer = 13;

More example:

spdifin_clk = 166MHz								
			width_sel = 1			width_sel = 0		
	FS(Hz)	timer	min_ period			max_ period		
mode0	24000	54	108.0729-167	th0	95	324.2187-5	th0	283
mode1	32000	40	81.05468-75	th1	70	243.1640-625	th1	209
mode2	44100	29	58.81519-274	th2	58	176.4455-782	th2	172
mode3	46000	28	56.38586-957	th3	55	169.1576-087	th3	165
mode4	48000	27	54.03645-833	th4	41	162.1093-75	th4	121
mode5	96000	13	27.01822-917	th5	20	81.05468-75	th5	60
mode6	192000	6	13.50911-458			40.52734-375		

spdifin_clk = 250MHz								
			width_sel = 1			width_sel = 0		
	FS(Hz)	timer	min_ period			max_ period		
mode0	24000	81	162.7604-167	th0	142	488.2812-5	th0	427
mode1	32000	61	122.0703-125	th1	105	366.2109-375	th1	315
mode2	44100	44	88.57709-751	th2	87	265.7312-925	th2	260
mode3	46000	42	84.91847-826	th3	83	254.7554-348	th3	249
mode4	48000	40	81.38020-833	th4	61	244.1406-25	th4	183
mode5	96000	20	40.69010-417	th5	31	122.0703-125	th5	91
mode6	192000	10	20.34505-208			61.03515-625		

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spdifin_clk = 400MHz								
			width_sel = 1			width_sel = 0		
	FS(Hz)	timer	min_ period			max_ period		
mode0	24000	130	260.4166-667	th0	228	781.25	th0	683
mode1	32000	97	195.3125	th1	169	585.9375	th1	505
mode2	44100	70	141.7233-56	th2	139	425.1700-68	th2	416
mode3	46000	67	135.8695-652	th3	133	407.6086-957	th3	399
mode4	48000	65	130.2083-333	th4	98	390.625	th4	292
mode5	96000	32	65.10416-667	th5	49	195.3125	th5	146
mode6	192000	16	32.55208-333			97.65625		

Table 9-103 EE_AUDIO_SPDIFIN_CTRL6 0x106

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	Reg_clr_internal_sts[7:0]; 7: clear valid bit status; 6: clear parity bit status; 5: clear ch status; 4: clear z_cnt status; 3: clear find_nonpcm status; 2: clear pd_data status; 1: clear pc_data status; 0: clear find_papb status;
23:16	R/W		Reg_clr_interrupt[7:0] for each bit of irq_status[7:0];
8	R/W		reg_papb_ext_sync, 1: add ext "0" sync check for papb; 0: disable;
7:0	R/W		reg_papb_ext_mask, mask 8 channel "0" sync

Table 9-104 EE_AUDIO_SPDIFIN_STAT0 0x107

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Valid bit
30:28	R/W		r_sample_mode, current sample mode;
27:18	R/W		r_width_min, the min width of two edge;

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Bits	R/W	Default	Description
17:8	R/W		r_width_max, the max width of two edge;
7:0	R/W		r_interrupt_status, interrupt status, need clear by reg_clk_interrupt; [7]: find PaPb; [6]: valid changed; [5]: find nonpcm to pcm (reg_nonpcm2pcm_th) ; [4]: find Pc or Pd changed; [3]: find CH status changed; [2]: find sample mode changed; [1]: find parity error; [0]: find overflow;

Table 9-105 EE_AUDIO_SPDIFIN_STAT1 0x108

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel status, by reg_ch_status_sel; reg_ch_status_sel[3]: 0: channel A; 1: channel B; reg_ch_status_sel[2:0]: 0: ch_status[31:0]; 1: ch_status[63:32]; 2: ch_status[95:64]; 3: ch_status[127:96]; 4: ch_status[159:128]; 5: ch_status[191:160]; 6: pc[15:0],pd[15:0];

Table 9-106 EE_AUDIO_SPDIFIN_STAT2 0x109

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	debug status, by reg_ch_status_sel; 0: r_z_width, the width of two preamble Z; 1: {16'd0, frame_cnt_min[7:0], frame_cnt_max[7:0]}; 2: {6'd0,width_min[9:0],6'd0, width_max[9:0]};

Table 9-107 EE_AUDIO_SPDIFIN_MUTE_VAL 0x10a

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_spdifin_mute_val, when muted, the channel value

Table 9-108 EE_AUDIO_SPDIFOUT_STAT 0x120

Bits	R/W	Default	Description
7:5	R/W	0x0000-0000	fifo_cnt, afifo cnt
4	R/W		up_error, change to 1 if overflow
3:0	R/W		req_frdd_fsm_stat

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Table 9-109 EE_AUDIO_SPDIFOUT_GAIN0 0x121

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	gain_ch3
23:16	R/W	0x0000-0000	gain_ch2
15:8	R/W	0x0000-0000	gain_ch1
7:0	R/W	0x0000-0000	gain_ch0

Table 9-110 EE_AUDIO_SPDIFOUT_GAIN1 0x122

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	gain_ch7
23:16	R/W	0x0000-0000	gain_ch6
15:8	R/W	0x0000-0000	gain_ch5
7:0	R/W	0x0000-0000	gain_ch4

Table 9-111 EE_AUDIO_SPDIFOUT_CTRL0 0x123

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_work_enable, 0: disable ; 1: enable;
29	R/W		reg_rst_afifo_out_n, reset afifo out side;
28	R/W		reg_rst_afifo_in_n, reset afifo in side; need set 1 after set reg_rst_afifo_out_n to 1;
27	R/W		reg_hold_start_en; 1: add delay to match TDM out when share buff;
26	R/W		reg_userdata_sel, 0: "user data" = reg_userdata_set; 1: "user data " = data [29];
25	R/W		reg_userdata_set
24	R/W		reg_chdata_sel, 0: "ch status" = reg_chsts0~B ;1: "ch status" = data [30];
23	R/W		reg_mix_lr, 0: disable; 1: L = (L+R)/2; R = (L+R)/2;
22	R/W		reg_mute_l, 0: disable; 1: ch_l_data = reg_mute_val;
21	R/W		reg_mute_r, 0: disable; 1: ch_r_data = reg_mute_val;
20	R/W		reg_data_sel, 0: insert data from 31bits;1: insert data from 27bits;
19	R/W		reg_out_msb_first, 0: lsb first; 1: msb first;
18	R/W		reg_valid_sel, 0: "valid flag" = reg_valid_set; 1: "valid flag" = data [28];
17	R/W		reg_valid_set

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Bits	R/W	Default	Description
11:4	R/W		reg_mask, [11:10]: mask lane3 L/R; [9:8]: mask lane2 L/R; [7:6]: mask lane1 L/R; [5:4]: mask lane0 L/R;
3:0	R/W		reg_parity_mask, [0]: initial parity value;

Table 9-112 EE_AUDIO_SPDIFOUT_CTRL1 0x124

Bits	R/W	Default	Description
21	R/W	0x0000-0000	eq_drc_sel; 1: select eq_drc data;
26	R/W		reg_gain_en, 0:disable; 1: enable data * gain;
25:24	R/W		reg_frddr_sel, 0:frddr_A; 1:frddr_B; 2:frddr_C;
23:16	R/W		reg_wait_cnt, wait some time when enable set to 1; then start request data from frddr;
15:14	R/W		reg_gain_shift; 0: data * 1; 1: data * 2; 2: data * 4; 3: data * 8;
12:8	R/W		reg_frddr_msb, msb position of data
7	R/W		Reg_force_start; 1: needn't ack from FRDDR; for only transmit "ch status" usage.
6:4	R/W		reg_frddr_type, 0: split 64bits ddr data to 8 sample, each sample need 8 bits; if bitwidth < 8, left-justified; 1: split 64bits ddr data to 4 sample, each sample need 16 bits; if bitwidth < 16, left-justified ; 2: split 64bits ddr data to 4 sample, each sample need 16 bits; if bitwidth < 16, right-justified ; 3: split 64bits ddr data to 2 sample, each sample need 32 bits; if bitwidth < 32, left-justified; 4: split 64bits ddr data to 2 sample, each sample need 32 bits; if bitwidth < 32, right-justified;

Table 9-113 EE_AUDIO_SPDIFOUT_PREAMB 0x125

Bits	R/W	Default	Description
31	R/W	0x0000-0000	set preamble Z, 0: preamble Z = 8'b11101000; 1: reg[7:0];
20	R/W		set preamble Y, 0: preamble Z = 8'b11100100; 1: reg[15:8];
29	R/W		set preamble X, 0: preamble Z = 8'b11100010; 1: reg[23:16];
23:16	R/W		preamble X
15:8	R/W		preamble Y
7:0	R/W		preamble Z

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Table 9-114 EE_AUDIO_SPDIFOUT_SWAP 0x126

Bits	R/W	Default	Description
31:16	R/W	0x0000-0000	hold start cnt, work when CTRL0[27] = 1;
6:4	R/W		lane0 right ch sel, 0: ch0; 1: ch1; 2: ch2; 3: ch3; 4: ch4; 5: ch5; 6: ch6; 7: ch7;
2:0	R/W		lane0 left ch sel 0: ch0; 1: ch1; 2: ch2; 3: ch3; 4: ch4; 5: ch5; 6: ch6; 7: ch7;

Table 9-115 EE_AUDIO_SPDIFOUT_CHSTS0 0x127

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel A status[31:0]

Table 9-116 EE_AUDIO_SPDIFOUT_CHSTS1 0x128

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel A status[63:32]

Table 9-117 EE_AUDIO_SPDIFOUT_CHSTS2 0x129

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel A status[95:64]

Table 9-118 EE_AUDIO_SPDIFOUT_CHSTS3 0x12a

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel A status[127:96]

Table 9-119 EE_AUDIO_SPDIFOUT_CHSTS4 0x12b

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel A status[159:128]

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Table 9-120 EE_AUDIO_SPDIFOUT_CHSTS5 0x12cv

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel A status[191:160]

Table 9-121 EE_AUDIO_SPDIFOUT_CHSTS6 0x12d

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel B status[31:0]

Table 9-122 EE_AUDIO_SPDIFOUT_CHSTS7 0x12e

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel B status[63:32]

Table 9-123 EE_AUDIO_SPDIFOUT_CHSTS8 0x12f

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel B status[95:64]

Table 9-124 EE_AUDIO_SPDIFOUT_CHSTS9 0x130

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel B status[127:96]

Table 9-125 EE_AUDIO_SPDIFOUT_CHSTSA 0x131

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel B status[159:128]

Table 9-126 EE_AUDIO_SPDIFOUT_CHSTSB 0x132

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel B status[191:160]

EE_AUDIO_SPDIFOUT_MUTE_VAL 0x133

EE_AUDIO_SPDIFOUT_B_STAT 0x1a0

EE_AUDIO_SPDIFOUT_B _GAIN0 0x1a1

EE_AUDIO_SPDIFOUT_B _GAIN1 0x1a2

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EE_AUDIO_SPDIFOUT_B _CTRL0 0x1a3
EE_AUDIO_SPDIFOUT_B _CTRL1 0x1a4
EE_AUDIO_SPDIFOUT_B _PREAMB 0x1a5
EE_AUDIO_SPDIFOUT_B _SWAP 0x1a6
EE_AUDIO_SPDIFOUT_B _CHSTS0 0x1a7
EE_AUDIO_SPDIFOUT_B _CHSTS1 0x1a8
EE_AUDIO_SPDIFOUT_B _CHSTS2 0x1a9
EE_AUDIO_SPDIFOUT_B _CHSTS3 0x1aa
EE_AUDIO_SPDIFOUT_B _CHSTS4 0x1ab
EE_AUDIO_SPDIFOUT_B _CHSTS5 0x1ac
EE_AUDIO_SPDIFOUT_B _CHSTS6 0x1ad
EE_AUDIO_SPDIFOUT_B _CHSTS7 0x1ae
EE_AUDIO_SPDIFOUT_B _CHSTS8 0x1af
EE_AUDIO_SPDIFOUT_B _CHSTS9 0x1b0
EE_AUDIO_SPDIFOUT_B _CHSTSA 0x1b1
EE_AUDIO_SPDIFOUT_B _CHSTSB 0x1b2
EE_AUDIO_SPDIFOUT_B _MUTE_VAL 0x1b3

Table 9-127 EE_AUDIO_SPDIFIN_ LB_CTRL0 0x1f0

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_work_enable, 0: disable; 1:enable;
30	R/W		reg_chnum_sel, 0: ch_num = 0~383(include frame cnt); 1: ch_num = 0~1(only L/R);
29	R/W		reg_rst_afifo_out_n, reset afifo out side;
28	R/W		reg_rst_afifo_in_n, reset afifo in side; need set 1 after set reg_rst_afifo_out_n to 1;
26	R/W		reg_chunm_en; 1: start add ch_cnt to ch_num;
25	R/W		reg_findpapb_en, 0: disable NonPCM mode; 1: enable;
23:12	R/W		reg_nonpcm2pcm_th, when detected NonPcm mode;if long time (z_cnt >= th) didn't detect PaPb again, will generate interrupt to SW: now changed to PCM mode;
11:8	R/W		reg_ch_status_sel, For EE_AUDIO_SPDIFIN_STAT1
7	R/W		reg_mute_l, 0: disable ; 1: mute channel L;
6	R/W		reg_mute_r, 0: disable ; 1: mute channel R;
5:4	R/W		reg_spdifin_src_sel, 0: PAD of spdifin;1: spdifout;
3	R/W		reg_check_valid, 0: disable valid check ; 1: enable;
2	R/W		reg_check_parity, 0: disable parity check ; 1: enable;
1	R/W		reg_invert_data, 0: disalbe; 1: invert [27:4] to [4:27];
0	R/W		reg_spdifin_phase, 0: disable invert; 1: enable;

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Table 9-128 EE_AUDIO_SPDIFIN_ LB_CTRL1 0x1f1

Bits	R/W	Default	Description
27:20	R/W		reg_interrupt_mask, mask each interrupt;

Table 9-129 EE_AUDIO_SPDIFIN_ LB_CTRL6 0x1f6

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	Reg_clr_internal_sts[7:0]; 7: clear valid bit status; 6: clear parity bit status; 5: clear ch status; 4: clear z_cnt status; 3: clear find_nonpcm status; 2: clear pd_data status; 1: clear pc_data status; 0: clear find_papb status;
23:16	R/W		Reg_clr_interrupt[7:0] for each bit of irq_status[7:0];
8	R/W		reg_papb_ext_sync, 1: add ext "0" sync check for papb; 0: disable;
7:0	R/W		reg_papb_ext_mask, mask 8 channel "0" sync

Table 9-130 EE_AUDIO_SPDIFIN_ LB_STAT0 0x1f7

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Valid bit
7:0	R/W		r_interrupt_status, interrupt status, need clear by reg_clk_interrupt; [7]: find PaPb; [6]: valid changed; [5]: find nonpcm to pcm (reg_nonpcm2pcm_th) ; [4]: find Pc or Pd changed; [3]: find CH status changed; [1]: find parity error; [0]: find overflow;

Table 9-131 EE_AUDIO_SPDIFIN_ LB_STAT1 0x1f8

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel status, by reg_ch_status_sel; reg_ch_status_sel[3]: 0: channel A; 1: channel B; reg_ch_status_sel[2:0]: 0: ch_status[31:0]; 1: ch_status[63:32]; 2: ch_status[95:64]; 3: ch_status[127:96]; 4: ch_status[159:128]; 5: ch_status[191:160]; 6: pc[15:0],pd[15:0];

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Table 9-132 EE_AUDIO_SPDIFIN_ LB_MUTE_VAL 0x1fa

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	reg_spdifin_mute_val, when muted, the channel value

9.10.8 Resample Registers

Table 9-133 EE_AUDIO_RESAMPLEA_CTRL0 0x110

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Soft_reset
28	R/W		Enable,
27:26	R/W		Method_sel,
25:16	R/W		outrdy_Cnt_ctrl[9:0],
15:0	R/W		Avg_cnt_init,

Table 9-134 EE_AUDIO_RESAMPLEA_CTRL1 0x111

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	Phase_step,It is used to set the phase step of the accumulator. It is equal to fs_in/fs_out*(1<<28).

Table 9-135 EE_AUDIO_RESAMPLEA_CTRL2 0x112

Bits	R/W	Default	Description
30:28	R/W	0x0000-0000	outrdy_Cnt_ctrl[12:10],
27	R/W		resample_start_mode;
25	R/W		IIR filter enable
24	R/W		Pause_en
23:0	R/W		Pause_cnt_thd

Table 9-136 EE_AUDIO_RESAMPLEA_CTRL3 0x113

Bits	R/W	Default	Description
18:16	R/W	0x0000-0000	Source sel; 0: toddr_A; 1: toddr_B; 2: toddr_C; 3: toddr_D; 4: loopback_A; 5: loopback_B;
14:12	R/W		resample_st_cnt;

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Bits	R/W	Default	Description
11:8	R/W		Reg_ch_num_sel, The channel number of input
4:0	R/W		Reg_in_msb, the msb of input[31:0]

Table 9-137 EE_AUDIO_RESAMPLEA_COEF0 0x114

Bits	R/W	Default	Description
25:0	R/W	0x0000-0000	IIR filter coef0

Table 9-138 EE_AUDIO_RESAMPLEA_COEF1 0x115

Bits	R/W	Default	Description
25:0	R/W	0x0000-0000	IIR filter coef1

Table 9-139 EE_AUDIO_RESAMPLEA_COEF2 0x116

Bits	R/W	Default	Description
25:0	R/W	0x0000-0000	IIR filter coef2

Table 9-140 EE_AUDIO_RESAMPLEA_COEF3 0x117

Bits	R/W	Default	Description
25:0	R/W	0x0000-0000	IIR filter coef3

Table 9-141 EE_AUDIO_RESAMPLEA_COEF4 0x118

Bits	R/W	Default	Description
25:0	R/W	0x0000-0000	IIR filter coef4

Table 9-142 EE_AUDIO_RESAMPLEA_STATUS1 0x119

Bits	R/W	Default	Description
24	R/W	0x0000-0000	The pause status
21:0	R/W		The real frequency of input data

EE_AUDIO_RESAMPLEB_CTRL0 0x1e0

EE_AUDIO_RESAMPLEB_CTRL1 0x1e1

EE_AUDIO_RESAMPLEB_CTRL2 0x1e2

EE_AUDIO_RESAMPLEB_CTRL3 0x1e3

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EE_AUDIO_RESAMPLEB_COEF0 0x1e4

EE_AUDIO_RESAMPLEB_COEF1 0x1e5

EE_AUDIO_RESAMPLEB_COEF2 0x1e6

EE_AUDIO_RESAMPLEB_COEF3 0x1e7

EE_AUDIO_RESAMPLEB_COEF4 0x1e8

EE_AUDIO_RESAMPLEB_STATUS1 0x1e9

9.10.9 TOACODEC Registers

Table 9-143 EE_AUDIO_TOACODEC_CTRL0 0x1d0

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_toacodec_en_dat, 0: disable; 1: enable;
23:20	R/W		Dat1_sel: 0: tdmout_a_dat[0]; 1: tdmout_a_dat[1]; 2: tdmout_a_dat[2]; 3: tdmout_a_dat[3]; 4: tdmout_b_dat[0]; 5: tdmout_b_dat[1]; 6: tdmout_b_dat[2]; 7: tdmout_b_dat[3]; 8: tdmout_c_dat[0]; 9: tdmout_c_dat[1]; 10: tdmout_c_dat[2]; 11: tdmout_c_dat[3];
19:16	R/W		Dat0_sel: 0: tdmout_a_dat[0]; 1: tdmout_a_dat[1]; 2: tdmout_a_dat[2]; 3: tdmout_a_dat[3]; 4: tdmout_b_dat[0]; 5: tdmout_b_dat[1]; 6: tdmout_b_dat[2]; 7: tdmout_b_dat[3]; 8: tdmout_c_dat[0]; 9: tdmout_c_dat[1]; 10: tdmout_c_dat[2]; 11: tdmout_c_dat[3];
14:12	R/W		lrclk_sel: 0: tdmout_a_lrclk; 1: tdmout_b_lrclk; 2: tdmout_c_lrclk; 4: tdmin_a_lrclk; 5: tdmin_b_lrclk; 6: tdmin_c_lrclk;
10	R/W		Lrclk_inv; 1: invert lrclk;
9	R/W		Bclk_cap_inv: The dat_o and lrclk_o will captured for timing balance after select; If this bit set to 1, will use invert bclk to capture;
8	R/W		Bclk_o_inv: if set 1, the final bclk connect to acodec will invert;

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Bits	R/W	Default	Description
6:4	R/W		Bclk_sel: 0: tdmout_a_bclk; 1: tdmout_b_bclk; 2: tdmout_c_bclk; 4: tdmin_a_bclk; 5: tdmin_b_bclk; 6: tdmin_c_bclk;
2:0	R/W		Mclk_sel: 0:mst_mclk_a; 1:mst_mclk_b; 2:mst_mclk_c; 3:mst_mclk_d; 4:mst_mclk_e; 5:mst_mclk_f;

9.10.10 TOVAD Registers

Table 9-144 EE_AUDIO_TOVAD_CTRL0 0x1d2

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_tovad_en, 0: disable; 1: enable;
30	R/W		reg_tovad_v_sel, 0: level; 1:pulse;
14:12	R/W		Data_sel: 0: tdmin_a; 1: tdmin_b; 2: tdmin_c; 3: spdifin; 4: pdmin; 5: loopback_b; 6: tdmin_lb; 7: loopback_a;

9.10.11 FRATV Registers

Table 9-145 EE_AUDIO_FRATV_CTRL0 0x1d3

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_work_enable 0:disable; 1:enable;
29	R/W		reg_rst_afifo_out_n, reset afifo out side; need set to 1 before reg_rst_afifo_in_n;
28	R/W		reg_rst_afifo_in_n, reset afifo in side; need set 1 after set reg_rst_afifo_out_n to 1;
21	R/W		Reg_atv_1ch; 0: copy atv 1ch to 2ch; 1: keep atv 1ch only;
20	R/W		Reg_in_sel; 0: select from ATV; 1: select from ADEC;

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Bits	R/W	Default	Description
19	R/W		Reg_adec_r_fisrt; 0: store adec_left ch first; 1: store adec_right ch first;
5	R/W		Reg_clk_inv; 0: capture input by rise edge of clock; 1: capture input by fall edge of clock;

9.10.12 FRHDMIRX Registers

Table 9-146 EE_AUDIO_FRHDMIRX_CTRL0 0x200

Bits	R/W	Default	Description
31	R/W	0x0000-0000	reg_work_enable, 0: disable; 1:enable;
30	R/W		reg_chnum_sel, 0: ch_num = 0~383(include frame cnt); 1: ch_num = 0~1(only L/R);
29	R/W		reg_rst_afifo_out_n, reset afifo out side;
28	R/W		reg_rst_afifo_in_n, reset afifo in side; need set 1 after set reg_rst_afifo_out_n to 1;
27:24	R/W		Reg_spdif_en[3:0]; HDMIRX will send 4 spdif lane, this reg will enable each spdif lane;
23	R/W		Reg_in_sel; 0: select spdif lane; 1: select PAO mode;
22	R/W		R eg_clk_inv; 0: capture input by rise edge of clock; 1: capture input by fall edge of clock;
21	R/W		Reg_spdifin_phase; 1: invert spdifin;
20	R/W		Reg_no_clock_gate; reserved for stop clock auto gating;
12:11	R/W		Reg_req_sel; Sync 4 spdifin by one of them; 0: sync by spdif lane0; 1: sync by spdif lane1; 2: sync by spdif lane2; 3: sync by spdif lane3;
10	R/W		Reg_chnum_debu1; reserved for debug; 1: send frame cnt + L/R
9	R/W		Reg_chnum_debu0; reserved for debug; 1: send frame cnt + CH_CNT
8	R/W		Reg_findpapb_en; 1: start detect PAPB;
7	R/W		Reg_chnum_en; reserved for debug; 1: start sending ch num info out;
6:0	R/W		Reg_status_sel; refer to reg_stat;

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Table 9-147 EE_AUDIO_ FRHDMIRX _CTRL1 0x201

Bits	R/W	Default	Description
31:20	R/W	0x0000-0000	reg_nonpcm2pcm_th, when detected NonPcm mode; if long time (z_cnt >= th) didn't detect PaPb again, will generate interrupt to SW: now changed to PCM mode;
19:16	R/W		reg_mute_l, 0: disable ; 1: mute channel L;
15:12	R/W		reg_mute_r, 0: disable ; 1: mute channel R;
11	R/W		reg_check_valid, 0: disable valid check ; 1: enable;
10	R/W		reg_check_parity, 0: disable parity check ; 1: enable;
9	R/W		reg_invert_data, 0: disalbe; 1: invert [27:4] to [4:27];
8	R/W		reg_papb_ext_sync, 1: add ext "0" sync check for papb; 0: disable;
7:0	R/W		reg_papb_ext_mask, mask 8 channel "0" sync

Table 9-148 EE_AUDIO_ FRHDMIRX _CTRL2 0x202

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Enable interrupt by spdif lane3: find papb;
30	R/W		Enable interrupt by spdif lane3: valid changed;
29	R/W		Enable interrupt by spdif lane3: find nonpcm to pcm;
28	R/W		Enable interrupt by spdif lane3: find pcpd changed;
27	R/W		Enable interrupt by spdif lane3: find ch status changed;
26	R/W		Reserved;
25	R/W		Enable interrupt by spdif lane3: find parity error;
24	R/W		Enable interrupt by PAO data: find papb;
23	R/W		Enable interrupt by spdif lane2: find papb;
22	R/W		Enable interrupt by spdif lane2: valid changed;
21	R/W		Enable interrupt by spdif lane2: find nonpcm to pcm;
20	R/W		Enable interrupt by spdif lane2: find pcpd changed;
19	R/W		Enable interrupt by spdif lane2: find ch status changed;
18	R/W		Reserved;
17	R/W		Enable interrupt by spdif lane2: find parity error;
16	R/W		Enable interrupt by PAO data: find pcpd changed;
15	R/W		Enable interrupt by spdif lane1: find papb;
14	R/W		Enable interrupt by spdif lane1: valid changed;
13	R/W		Enable interrupt by spdif lane1: find nonpcm to pcm;
12	R/W		Enable interrupt by spdif lane1: find pcpd changed;
11	R/W		Enable interrupt by spdif lane1: find ch status changed;

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Bits	R/W	Default	Description
10	R/W		Reserved;
9	R/W		Enable interrupt by spdif lane1: find parity error;
8	R/W		Reserved;
7	R/W		Enable interrupt by spdif lane0: find papb;
6	R/W		Enable interrupt by spdif lane0: valid changed;
5	R/W		Enable interrupt by spdif lane0: find nonpcm to pcm;
4	R/W		Enable interrupt by spdif lane0: find pcpd changed;
3	R/W		Enable interrupt by spdif lane0: find ch status changed;
2	R/W		Reserved;
1	R/W		Enable interrupt by spdif lane0: find parity error;
0	R/W		Reserved;

Table 9-149 EE_AUDIO_ FRHDMIRX _CTRL3 0x203

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Clear status in spdif lane3: valid_bit;
30	R/W		Clear status in spdif lane3: parity error;
29	R/W		Clear status in spdif lane3: ch status;
28	R/W		Clear status in spdif lane3: z_cnt;
27	R/W		Clear status in spdif lane3: find_nonpcm;
26	R/W		Clear status in spdif lane3: pd_data
25	R/W		Clear status in spdif lane3: pc_data
24	R/W		Clear status in spdif lane3: find_papb;
23	R/W		Clear status in spdif lane2: valid_bit;
22	R/W		Clear status in spdif lane2: parity error;
21	R/W		Clear status in spdif lane2: ch status;
20	R/W		Clear status in spdif lane2: z_cnt;
19	R/W		Clear status in spdif lane2: find_nonpcm;
18	R/W		Clear status in spdif lane2: pd_data
17	R/W		Clear status in spdif lane2: pc_data
16	R/W		Clear status in spdif lane2: find_papb;
15	R/W		Clear status in spdif lane1: valid_bit;
14	R/W		Clear status in spdif lane1: parity error;
13	R/W		Clear status in spdif lane1: ch status;
12	R/W		Clear status in spdif lane1: z_cnt;

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Bits	R/W	Default	Description
11	R/W		Clear status in spdif lane1: find_nonpcm;
10	R/W		Clear status in spdif lane1: pd_data
9	R/W		Clear status in spdif lane1: pc_data
8	R/W		Clear status in spdif lane1: find_papb;
7	R/W		Clear status in spdif lane0: valid_bit;
6	R/W		Clear status in spdif lane0: parity error;
5	R/W		Clear status in spdif lane0: ch status;
4	R/W		Clear status in spdif lane0: z_cnt;
3	R/W		Clear status in spdif lane0: find_nonpcm;
2	R/W		Clear status in spdif lane0: pd_data
1	R/W		Clear status in spdif lane0: pc_data
0	R/W		Clear status in spdif lane0: find_papb;

Table 9-150 EE_AUDIO_FRHDMIRX_CTRL4 0x204

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Clear interrupt status in spdif lane3: find papb;
30	R/W		Clear interrupt status in spdif lane3: valid changed;
29	R/W		Clear interrupt status in spdif lane3: find nonpcm to pcm;
28	R/W		Clear interrupt status in spdif lane3: find pcpd changed;
27	R/W		Clear interrupt status in spdif lane3: find ch status changed;
26	R/W		Reserved;
25	R/W		Clear interrupt status in spdif lane3: find parity error;
24	R/W		Clear interrupt status in PAO data: find papb;
23	R/W		Clear interrupt status in spdif lane2: find papb;
22	R/W		Clear interrupt status in spdif lane2: valid changed;
21	R/W		Clear interrupt status in spdif lane2: find nonpcm to pcm;
20	R/W		Clear interrupt status in spdif lane2: find pcpd changed;
19	R/W		Clear interrupt status in spdif lane2: find ch status changed;
18	R/W		Reserved;
17	R/W		Clear interrupt status in spdif lane2: find parity error;
16	R/W		Clear interrupt status in PAO data: find pcpd changed;
15	R/W		Clear interrupt status in spdif lane1: find papb;
14	R/W		Clear interrupt status in spdif lane1: valid changed;
13	R/W		Clear interrupt status in spdif lane1: find nonpcm to pcm;

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Bits	R/W	Default	Description
12	R/W		Clear interrupt status in spdif lane1: find pcpd changed;
11	R/W		Clear interrupt status in spdif lane1: find ch status changed;
10	R/W		Reserved;
9	R/W		Clear interrupt status in spdif lane1: find parity error;
8	R/W		Reserved;
7	R/W		Clear interrupt status in spdif lane0: find papb;
6	R/W		Clear interrupt status in spdif lane0: valid changed;
5	R/W		Clear interrupt status in spdif lane0: find nonpcm to pcm;
4	R/W		Clear interrupt status in spdif lane0: find pcpd changed;
3	R/W		Clear interrupt status in spdif lane0: find ch status changed;
2	R/W		Reserved;
1	R/W		Clear interrupt status in spdif lane0: find parity error;
0	R/W		Reserved;

Table 9-151 EE_AUDIO_ FRHDMIRX _CTRL5 0x205

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	Reg_spdifin_mute_val;

Table 9-152 EE_AUDIO_ FRHDMIRX _STAT0 0x20a

Bits	R/W	Default	Description
31	R/W	0x0000-0000	Interrupt status in spdif lane3: find papb;
30	R/W		Interrupt status in spdif lane3: valid changed;
29	R/W		Interrupt status in spdif lane3: find nonpcm to pcm;
28	R/W		Interrupt status in spdif lane3: find pcpd changed;
27	R/W		Interrupt status in spdif lane3: find ch status changed;
26	R/W		Reserved;
25	R/W		Interrupt status in spdif lane3: find parity error;
24	R/W		Interrupt status in PAO data: find papb;
23	R/W		Interrupt status in spdif lane2: find papb;
22	R/W		Interrupt status in spdif lane2: valid changed;
21	R/W		Interrupt status in spdif lane2: find nonpcm to pcm;
20	R/W		Interrupt status in spdif lane2: find pcpd changed;
19	R/W		Interrupt status in spdif lane2: find ch status changed;
18	R/W		Reserved;

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Bits	R/W	Default	Description
17	R/W		Interrupt status in spdif lane2: find parity error;
16	R/W		Interrupt status in PAO data: find pcpd changed;
15	R/W		Interrupt status in spdif lane1: find papb;
14	R/W		Interrupt status in spdif lane1: valid changed;
13	R/W		Interrupt status in spdif lane1: find nonpcm to pcm;
12	R/W		Interrupt status in spdif lane1: find pcpd changed;
11	R/W		Interrupt status in spdif lane1: find ch status changed;
10	R/W		Reserved;
9	R/W		Interrupt status in spdif lane1: find parity error;
8	R/W		Reserved;
7	R/W		Interrupt status in spdif lane0: find papb;
6	R/W		Interrupt status in spdif lane0: valid changed;
5	R/W		Interrupt status in spdif lane0: find nonpcm to pcm;
4	R/W		Interrupt status in spdif lane0: find pcpd changed;
3	R/W		Interrupt status in spdif lane0: find ch status changed;
2	R/W		Reserved;
1	R/W		Interrupt status in spdif lane0: find parity error;
0	R/W		Reserved;

Table 9-153 EE_AUDIO_ FRHDMIRX _STAT1 0x20b

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	channel status, by reg_status_sel; reg_ status_sel[6:4]; 0: spdif lane0; 1: spdif lane1; 2: spdif lane2; 3: spdif lane3; 4: pao pc/pd value; 5: valid bits; + afifo info; for spdif lane0/1/2/3: reg_status_sel[3]: 0: channel A; 1: channel B; reg_status_sel[2:0]: 0: ch_status[31:0]; 1: ch_status[63:32]; 2: ch_status[95:64]; 3: ch_status[127:96]; 4: ch_status[159:128]; 5: ch_status[191:160]; 6: pc[15:0],pd[15:0]; For valid bits:+ afifo info Bit4: sync fifo overflow; Bit3: spdif lane3 valid bit; Bit2: spdif lane2 valid bit; Bit1: spdif lane1 valid bit; Bit0: spdif lane0 valid bit;

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reg_status_sel			register info	reg_status_sel			register info	reg_status_sel			register info
[6:-4]	[3]	[2:-0]		[6:-4]	[3]	[2:-0]		[6:-4]	[-3]	[-2-:-0]	
0	0	0	lane0 ch_sts L [31:0]	2	0	0	lane2 ch_sts L [31:0]	4	X	X	PAO mode PC [15:0]PD[15:0]
0	0	1	lane0 ch_sts L [63:32]	2	0	1	lane2 ch_sts L [63:32]	5	X	X	bit4: overflow_flag
0	0	2	lane0 ch_sts L [95:64]	2	0	2	lane2 ch_sts L [95:64]				bit3: lane3 valid bit
0	0	3	lane0 ch_sts L [127:96]	2	0	3	lane2 ch_sts L [127:96]				bit2: lane2 valid bit
0	0	4	lane0 ch_sts L [159:128]	2	0	4	lane2 ch_sts L [159:128]				bit1: lane1 valid bit
0	0	5	lane0 ch_sts L [191:160]	2	0	5	lane2 ch_sts L [191:160]				bit0: lane0 valid bit
0	0	6	lane0 PC [15:0],PD [15:0]	2	0	6	lane2 PC[15:0], PD[15:0]				
0	1	0	lane0 ch_sts R [31:0]	2	1	0	lane2 ch_sts R [31:0]				
0	1	1	lane0 ch_sts R [63:32]	2	1	1	lane2 ch_sts R [63:32]				
0	1	2	lane0 ch_sts R [95:64]	2	1	2	lane2 ch_sts R [95:64]				
0	1	3	lane0 ch_sts R [127:96]	2	1	3	lane2 ch_sts R [127:96]				
0	1	4	lane0 ch_sts R [159:128]	2	1	4	lane2 ch_sts R [159:128]				
0	1	5	lane0 ch_sts R [191:160]	2	1	5	lane2 ch_sts R [191:160]				
0	1	6	lane0 PC [15:0],PD [15:0]	2	1	6	lane3 PC[15:0], PD[15:0]				
1	0	0	lane1 ch_sts L [31:0]	3	0	0	lane3 ch_sts L [31:0]				
1	0	1	lane1 ch_sts L [63:32]	3	0	1	lane3 ch_sts L [63:32]				
1	0	2	lane1 ch_sts L [95:64]	3	0	2	lane3 ch_sts L [95:64]				
1	0	3	lane1 ch_sts L [127:96]	3	0	3	lane3 ch_sts L [127:96]				

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reg_status_sel			register info	reg_status_sel			register info	reg_status_sel			register info
[6:-4]	[3]	[2:-0]		[6:-4]	[3]	[2:-0]		[6:-4]	[-3]	[-2-:-0]	
1	0	4	lane1 ch_sts L [159:128]	3	0	4	lane3 ch_sts L [159:128]				
1	0	5	lane1 ch_sts L [191:160]	3	0	5	lane3 ch_sts L [191:160]				
1	0	6	lane1 PC [15:0],PD [15:0]	3	0	6	lane3 PC[15:0], PD[15:0]				
1	1	0	lane1 ch_sts R [31:0]	3	1	0	lane3 ch_sts R [31:0]				
1	1	1	lane1 ch_sts R [63:32]	3	1	1	lane3 ch_sts R [63:32]				
1	1	2	lane1 ch_sts R [95:64]	3	1	2	lane3 ch_sts R [95:64]				
1	1	3	lane1 ch_sts R [127:96]	3	1	3	lane3 ch_sts R [127:96]				
1	1	4	lane1 ch_sts R [159:128]	3	1	4	lane3 ch_sts R [159:128]				
1	1	5	lane1 ch_sts R [191:160]	3	1	5	lane3 ch_sts R [191:160]				
1	1	6	lane1 PC [15:0],PD [15:0]	3	1	6	lane3 PC[15:0], PD[15:0]				

9.10.13 PDM Registers

Base Address: 0xFF601000

Each register final address = module base address+ address * 4

Table 9-154 PDM_CTRL 0x00

Bits	R/W	Default	Description
31	R/W	0x0000-0000	PDM enable
30	R/W		invert the PDM_DCLK
29	R/W		output mode: 1: 24bits. 0: 32 bits
28	R/W		bypass mode. 1: bypass all filter. directly output the PDM input to DDR. 0: normal mode.
27:20	R/W		pdm_mute_mask[7:0]; [7] mute pdm ch7; … [0] mute pdm ch0;

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Bits	R/W	Default	Description
17	R/W		chnum_en. 1: start send ch_cnt out.
16	R/W		PDM Asynchronous FIFO soft reset. write 1 to soft reset AFIFO
15:8	R/W		PDM channel reset. 0: to reset each PDM channel. 1: normal mode
7:0	R/W		PDM channel enable. each bit for one channel

Table 9-155 PDM_HCIC_CTRL1 0x01

Bits	R/W	Default	Description
31	R/W	0x0000-0000	hcic filter enable. 1 use sinc filter. 0 bypass input to output
29:24	R/W		hcic final gain shift parameter
23:16	R/W		hcic final gain multiplier
8:4	R/W		hcic down sample rate
3:0	R/W		hcic stage number. must be between 3 to 9

PDM_HCIC_CTRL2 0x02

Table 9-156 PDM_F1_CTRL 0x03

Bits	R/W	Default	Description
31	R/W	0x0000-0000	filter 1 enable
17:16	R/W		f1 round mode. 2'b00 : sign bit at bit 49. 28bits output [49:22] round at bit 21. 32bits output [49:18]. 24bits output [49:26] 2'b01 : sign bit at bit 50. 28bits output [50:23] round at bit 22. 32bits output [49:18]. 24bits output [49:26] 2'b10 : sign bit at bit 51. 28bits output [51:24] round at bit 23 32bits output [49:18]. 24bits output [49:26].
15:12	R/W		filter 1 down sample rate
8:0	R/W		filter 1 stage number

Table 9-157 PDM_F2_CTRL 0x04

Bits	R/W	Default	Description
31	R/W	0x0000-0000	filter 2 enable
17:16	R/W		F2 round mode. 2'b00 : round at bit 21. 2'b01 : round at bit 22.2'b10 : round at bit 23
15:12	R/W		filter 2 down sample rate
8:0	R/W		filter 2 stage number

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Table 9-158 PDM_F3_CTRL 0x05

Bits	R/W	Default	Description
31	R/W	0x0000-0000	filter 3 enable
17:16	R/W		F3 round mode. 2'b00 : round at bit 21. 2'b01 : round at bit 22.2'b10 : round at bit 23
15:12	R/W		filter 3 down sample rate
8:0	R/W		filter 3 stage number

Table 9-159 PDM_HPF_CTRL 0x06

Bits	R/W	Default	Description
31	R/W	0x0000-0000	High pass filter enable
20:16	R/W		high pass filter shift steps. 6~19 steps
15:0	R/W		high pass filter output factor

Table 9-160 PDM_CHAN_CTRL 0x07

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	Chan3 data sample pointer vs edge of the PDM_DCLK
23:16	R/W		Chan2 data sample pointer vs edge of the PDM_DCLK
15:8	R/W		Chan1 data sample pointer vs edge of the PDM_DCLK
7:0	R/W		Chan0 data sample pointer vs edge of the PDM_DCLK

Table 9-161 PDM_CHAN_CTRL1 0x08

Bits	R/W	Default	Description
31:24	R/W	0x0000-0000	Chan7 data sample pointer vs edge of the PDM_DCLK
23:16	R/W		Chan6 data sample pointer vs edge of the PDM_DCLK
15:8	R/W		Chan5 data sample pointer vs edge of the PDM_DCLK
7:0	R/W		Chan4 data sample pointer vs edge of the PDM_DCLK

Table 9-162 PDM_COEFF_ADDR 0x09

Bits	R/W	Default	Description
8:0	R/W	0x0000-0000	address of the write/read of coeff data

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Table 9-163 PDM_COEFF_DATA 0x0A

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	write/read data to coeff memory

Table 9-164 PDM_CLKG_CTRL 0x0B

Bits	R/W	Default	Description
6	R/W	0x0000-0000	filt_ctrl module auto clock gating control
5	R/W		sinc fifo module auto clock gating control
4	R/W		filter module auto clock gating control
3	R/W		apb module auto clock gating control
2	R/W		coeff memory module auto clock gating control
1	R/W		each channel module auto clock gating control
0	R/W		cts_pdm_clk auto clock gating control

Table 9-165 PDM_STS 0x0C

Bits	R/W	Default	Description
1	R/W	0x0000-0000	HPF filter output overflow. means the PCLK is too slow
0	R/W		HCIC filter output overflow. means the CTS_PDM_CLK is too slow. can't finished the filter function.

Table 9-166 PDM_MUTE_VALUE 0x0D

Bits	R/W	Default	Description
31:0	R/W	0x0000-0000	mute value if mute_mask = 1;

Table 9-167 PDM_MASK_NUM 0x0E

Bits	R/W	Default	Description
15:0	R/W	0x0000-0000	When PDM power on, there are some invalid data because filter initialize. So set this register can mask these invalid data. If PDM out frequency is 48K, set MASK_NUM = 49 will mask 50/48k = 1.04ms.

9.10.14 LOCKER Registers

Base Address: 0xFF601400

Each register final address = module base address+ address * 4

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Table 9-168 AUD_LOCK_EN 0X00

Bit(s)	R/W	Default	Description
31:1			reserved
0	R/W	0x0	Audio_lock_en: 0 = disable; 1= enable.

Table 9-169 AUD_LOCK_SW_RESET 0X01

Bit(s)	R/W	Default	Description
31:1			reserved
0	R/W	0x0	Audio_lock_soft_reset: 1=generate soft reset pulse

Table 9-170 AUD_LOCK_SW_LATCH 0X02

Bit(s)	R/W	Default	Description
31:4			reserved
3	R/W	0x0	omclk2ref software latch: 1= generate the latch pulse
2	R/W	0x0	Ref2omclk software latch: 1= generate the latch pulse
1	R/W	0x0	Ref2imclk software latch: 1= generate the latch pulse
0	R/W	0x0	Imclk2ref software latch: 1= generate the latch pulse

Table 9-171 AUD_LOCK_HW_LATCH 0X03

Bit(s)	R/W	Default	Description
31:4			reserved
3	R/W	0x0	omclk2ref hardware latch enble: 1= enable□0= disable
2	R/W	0x0	Ref2omclk hardware latch enble: 1= enable□0= disable
1	R/W	0x0	Ref2imclk hardware latch enble: 1= enable□0= disable
0	R/W	0x0	Imclk2ref hardware latch enble: 1= enable□0= disable

Table 9-172 AUD_LOCK_REFCLK_SRC 0X04

Bit(s)	R/W	Default	Description
31:2			reserved
1:0	R/W	0x0	Ref clk source sel: 0= pclk; 1=oscinclk; 2,3,reserved

Table 9-173 AUD_LOCK_REFCLK_LAT_INT 0X05

Bit(s)	R/W	Default	Description
31:0			U32 number of reference clock cycles to latch the imclk and omclk

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Table 9-174 AUD_LOCK_IMCLK_LAT_INT 0X06

Bit(s)	R/W	Default	Description
31:0			U32 number of imclk clock cycles to latch the reference clock

Table 9-175 AUD_LOCK_OMCLK_LAT_INT 0X07

Bit(s)	R/W	Default	Description
31:0			U32 number of omclk clock cycles to latch the reference clock

Table 9-176 AUD_LOCK_REFCLK_DS_INT 0X08

Bit(s)	R/W	Default	Description
31:10			reserved
9:0	R/W	0x0	U10 downsample step of reference clock for the couter48ds to be measured, module = x+1

Table 9-177 AUD_LOCK_IMCLK_DS_INT 0X09

Bit(s)	R/W	Default	Description
31:10			reserved
9:0	R/W	0x0	U10 downsample step of imclk for the couter48ds to be measured, module = x+1

Table 9-178 AUD_LOCK_OMCLK_DS_INT 0X0a

Bit(s)	R/W	Default	Description
31:10			reserved
9:0	R/W	0x0	U10 downsample step of omclk for the couter48ds to be measured, module = x +1

Table 9-179 AUD_LOCK_INT_CLR 0X0b

Bit(s)	R/W	Default	Description
31:2			reserved
1	R/W	0x0	It is used to generate pulse to clear the interrupt status
0	R/W	0x0	It is used to generate pulse to clear the interrupt

Table 9-180 AUD_LOCK_GCLK_CTRL 0X0c

Bit(s)	R/W	Default	Description
31:4			reserved
3:2	R/W	0x0	It is used to gate the module clock
1	R/W	0x0	It is used to gate the register clock
0			reserved

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Table 9-181 AUD_LOCK_INT_CTRL 0X0d

Bit(s)	R/W	Default	Description
31:4			reserved
3	R/W	0x0	It is used to mask interrupt for omclk_state
2	R/W	0x0	It is used to mask interrupt for imclk_state
1	R/W	0x0	It is used to mask interrupt for refclk_state1
0	R/W	0x0	It is used to mask interrupt for refclk_state0

Table 9-182 RO_REF2IMCLK_CNT_L 0X10

Bit(s)	R/W	Default	Description
31:0	R	0x0	U48 latched imclk counter48ds of each reg_refclk_latch_interval; Will start from 0 reaching 2^48

Table 9-183 RO_REF2IMCLK_CNT_H 0X11

Bit(s)	R/W	Default	Description
31:16			reserved
15:0	R	0x0	U48 latched imclk counter48ds of each reg_refclk_latch_interval; Will start from 0 reaching 2^48

Table 9-184 RO_REF2OMCLK_CNT_L 0X12

Bit(s)	R/W	Default	Description
31:0	R	0x0	U48 latched omclk counter48ds of each reg_refclk_latch_interval; Will start from 0 reaching 2^48

Table 9-185 RO_REF2OMCLK_CNT_H 0X13

Bit(s)	R/W	Default	Description
31:16			reserved
15:0	R	0x0	U48 latched omclk counter48ds of each reg_refclk_latch_interval; Will start from 0 reaching 2^48

Table 9-186 RO_IMCLK2REF_CNT_L 0X14

Bit(s)	R/W	Default	Description
31:0	R	0x0	U48 latched reference clock counter48ds of each reg_imclk_latch_interval; Will start from 0 reaching 2^48

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Table 9-187 RO_IMCLK2REF_CNT_H 0X15

Bit(s)	R/W	Default	Description
31:16			reserved
15:0	R	0x0	U48 latched reference clock counter48ds of each reg_imclk_latch_interval; Will start from 0 reaching 2^48

Table 9-188 RO_OMCLK2REF_CNT_L 0X16

Bit(s)	R/W	Default	Description
31:0	R	0x0	U48 latched reference clock counter48ds of each reg_omclk_latch_interval; Will start from 0 reaching 2^48

Table 9-189 RO_OMCLK2REF_CNT_H 0X17

Bit(s)	R/W	Default	Description
31:16			reserved
15:0	R	0x0	U48 latched reference clock counter48ds of each reg_omclk_latch_interval; Will start from 0 reaching 2^48

Table 9-190 RO_REFCLK_PKG_CNT 0X18

Bit(s)	R/W	Default	Description
31:16	R	0x0	U16,number of reference clk latch interval period conter for imclk latch
15:0	R	0x0	U16,number of reference clk latch interval period conter for imclk latch

Table 9-191 RO_IMCLK_PKG_CNT 0X19

Bit(s)	R/W	Default	Description
31:16			reserved
15:0	R	0x0	U16,number of imclk latch interval period counter

Table 9-192 RO_OMCLK_PKG_CNT 0X1a

Bit(s)	R/W	Default	Description
31:16			reserved
15:0	R	0x0	U16,number of omclk latch interval period counter

Table 9-193 RO_AUD_LOCK_INT_STATUS 0X1b

Bit(s)	R/W	Default	Description
31:4			reserved
3	R	0x0	It is used to report interrupt status for omclk_state
2	R	0x0	It is used to report interrupt status for imclk_state

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Bit(s)	R/W	Default	Description
1	R	0x0	It is used to report interrupt status for refclk_state1
0	R	0x0	It is used to report interrupt status for refclk_state0

9.10.15 VAD Registers

Base Address: 0xFF601800

Each register final address = module base address+ address * 4

Table 9-194 EE_VAD_TOP_CTRL0 0x0

Bits	R/W	Default	Description
31	R/W	0	vad_en
30	R/W	0	dec_fir_en
29	R/W	0	pre_emp_en
28	R/W	0	pre_ram_en
27	R/W	0	frame_his_en
26	R/W	0	frame_his_save_en
23	R/W	0	ceps_ceps_en
22	R/W	0	ceps_spec_en
21	R/W	0	post_dec_en
20	R/W	0	two_channel_en
10	R/W	0	soft_rst
9	R/W	0	dec_fir_soft_rst
8	R/W	0	pre_emp_soft_rst
7	R/W	0	proc_soft_rst
6	R/W	0	frame_his_soft_rst
5	R/W	0	frame_his_save_soft_rst
4	R/W	0	ceps_win_soft_rst
3	R/W	0	ceps_fft_soft_rst
2	R/W	0	ceps_ceps_soft_rst
1	R/W	0	ceps_spec_soft_rst
0	R/W	0	post_dec_soft_rst

Table 9-195 EE_VAD_TOP_CTRL1 0x1

Bits	R/W	Default	Description
31:30	R/W	0	vad_clk_gate
29:28	R/W	0	prepare_clk_gate

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Bits	R/W	Default	Description
27:26	R/W	0	proc_clk_gate
25:24	R/W	0	frame_his_clk_gate
23:22	R/W	0	ceps_ceps_clk_gate
21:20	R/W	0	ceps_spec_clk_gate
19:18	R/W	0	ch_sel1_clk_gate
17:16	R/W	0	ch_sel0_clk_gate
7	R/W	0	vad_sw_reset
6	R/W	0	prepare_sw_reset
5	R/W	0	proc_sw_reset
4	R/W	0	frame_his_sw_reset
3	R/W	0	ceps_win_sw_reset
2	R/W	0	ceps_ceps_sw_reset
1	R/W	0	ceps_spec_sw_reset
0	R/W	0	post_dec_sw_reset

Table 9-196 EE_VAD_TOP_CTRL2 0x2

Bits	R/W	Default	Description
25:24	R/W	0	ch1_shift :0:0, 1:1, 2:2, 3:3
9:8	R/W	0	ch0_shift:0:0, 1:1, 2:2, 3:3

Table 9-197 VAD_FIR_CTRL 0x3

Bits	R/W	Default	Description
29:24	R/W	0	dec_filter_order
15:8	R/W	0	dec_filter_depthm1 ;>= filter pipe+ order
3:0	R/W	0	dec_rate;decimation_rate-1(0~11)

Table 9-198 VAD_FIR_EMP 0x4

Bits	R/W	Default	Description
9:0	R/W	0	pre_emp_coef :float unsigned

Table 9-199 VAD_FIR_COEF 0 0x5

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_0 : 8.1.1.7
15:0	R/W	0	dec_coef_1 : 8.1.1.7

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Table 9-200 VAD_FIR_COEF 1 0x6

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_2 : 8.1.1.7
15:0	R/W	0	dec_coef_3 : 8.1.1.7

Table 9-201 VAD_FIR_COEF 2 0x7

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_4 : 8.1.1.7
15:0	R/W	0	dec_coef_5 : 8.1.1.7

Table 9-202 VAD_FIR_COEF 3 0x8

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_6 : 8.1.1.7
15:0	R/W	0	dec_coef_7 : 8.1.1.7

Table 9-203 VAD_FIR_COEF 4 0x9

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_8 : 8.1.1.7
15:0	R/W	0	dec_coef_9 : 8.1.1.7

Table 9-204 VAD_FIR_COEF 5 0xa

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_10 : 8.1.1.7
15:0	R/W	0	dec_coef_11 : 8.1.1.7

Table 9-205 VAD_FIR_COEF 6 0xb

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_12 : 8.1.1.7
15:0	R/W	0	dec_coef_13 : 8.1.1.7

Table 9-206 VAD_FIR_COEF 7 0xc

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_14 : 8.1.1.7
15:0	R/W	0	dec_coef_15 : 8.1.1.7

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Table 9-207 VAD_FIR_COEF 8 0xd

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_16 : 8.1.1.7
15:0	R/W	0	dec_coef_17 : 8.1.1.7

Table 9-208 VAD_FIR_COEF 9 0xe

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_18 : 8.1.1.7
15:0	R/W	0	dec_coef_19 : 8.1.1.7

Table 9-209 VAD_FIR_COEF 10 0xf

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_20 : 8.1.1.7
15:0	R/W	0	dec_coef_21 : 8.1.1.7

Table 9-210 VAD_FIR_COEF 11 0x10

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_22 : 8.1.1.7
15:0	R/W	0	dec_coef_23 : 8.1.1.7

Table 9-211 VAD_FIR_COEF 12 0x11

Bits	R/W	Default	Description
31:16	R/W	0	dec_coef_24 : 8.1.1.7

Table 9-212 VAD_FRAME_CTRL0 0x12

Bits	R/W	Default	Description
31:30	R/W	0	frame_len 0: 64, 1: 128, 2: 256
26:24	R/W	0	pow_diff_dist:0:4, 1:8, 3:16, 5:24
17:16	R/W	0	pow_old_avglen:0:4, 1:8,3:16

Table 9-213 VAD_FRAME_CTRL1 0x13

Bits	R/W	Default	Description
11:0	R/W	0	pow_rampup_thr: 8.16.0.13, threshold for power ramp up detect

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Table 9-214 VAD_FRAME_CTRL2 0x14

Bits	R/W	Default	Description
31:16	R/W	0	pow_cur_thr: 10.0.0.52
15:0	R/W	0	pow_cur_thr_high: 10.0.0.52

Table 9-215 VAD_CEP_CTRL0 0x15

Bits	R/W	Default	Description
29:28	R/W	0	ceps_decision_sel: 0 : ||, 1: &&, 2:naxm 3:maxmin
25:24	R/W	0	decision_sel_each:0 : ||, 1: &&, 2:spectral_entropy 3:ceps
20:16	R/W	0	login_min: 2^(-logn_min),5.5.0, [0~20]
10:0	R/W	0	ceps_log_blk: 11.6.1

Table 9-216 VAD_CEP_CTRL1 0x16

Bits	R/W	Default	Description
8:0	R/W	0	ceps_weight_step : 9,-3,0

Table 9-217 VAD_CEP_CTRL2 0x17

Bits	R/W	Default	Description
31:16	R/W	0	ceps_max_thr :16.9.1
15:0	R/W	0	ceps_maxmin_thr :16.9.1

Table 9-218 VAD_CEP_CTRL3 0x18

Bits	R/W	Default	Description
31:24	R/W	0	ceps_det_start: [0~128]
23:16	R/W	0	ceps_det_end: [0~128], end>start
14:8	R/W	0	ceps_freq_blk_low: 0: no blk, 1blk[0], 2 blank [255 0 1]…
6:0	R/W	0	ceps_freq_blk_high: 0:no blk,1:blk[128],2 blk[127:129]…

Table 9-219 VAD_CEP_CTRL4 0x19

Bits	R/W	Default	Description
30:24	R/W	0	spectral_entropy_start: [0~127]
22:16	R/W	0	spectral_entropy_end: [0~127]

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Table 9-220 VAD_CEP_CTRL5 0x1a

Bits	R/W	Default	Description
27:16	R/W	0	spectral_entropy_thr_max: 12.7.1
11:0	R/W	0	spectral_entropy_thr_min: 12.7.1

Table 9-221 VAD_DEC_CTRL 0x1b

Bits	R/W	Default	Description
29:28	R/W	0	decision_sel_combine: 0 is &, 1is |,2 is +
20:16	R/W	0	vadflag_confirm_m: <32
5:0	R/W	0	vadflag_confirm_n

Table 9-222 VAD_TOP_STS0 0x1c

Bits	R/W	Default	Description
4	R	0	Afifo_err
3:0	R	0	lafifo_cnt

Table 9-223 VAD_TOP_STS1 0x1d

Bits	R/W	Default	Description
20	R	0	Pre_overflow
19	R	0	Pre_fst
18	R	0	Pre_val
17:0	R	0	Pre_dat

Table 9-224 VAD_TOP_STS2 0x1e

Bits	R/W	Default	Description
18	R	0	Ceps_val
17:0	R	0	Ceps_dat

Table 9-225 VAD_FIR_STS0 0x1f

Bits	R/W	Default	Description
31:0	R	0	reserved

Table 9-226 VAD_FIR_STS1 0x20

Bits	R/W	Default	Description
31:-	R	0	reserved

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Table 9-227 VAD_POW_STS0 0x21

Bits	R/W	Default	Description
31:16	R	0	reserved
15:0	R	0	Pow_o

Table 9-228 VAD_POW_STS1 0x22

Bits	R/W	Default	Description
31:0	R	0	reserved

Table 9-229 VAD_POW_STS2 0x23

Bits	R/W	Default	Description
31:0	R	0	reserved

Table 9-230 VAD_FFT_STS0 0x24

Bits	R/W	Default	Description
31	R	0	Ceps_done_latch
30	R	0	Spec_entropy_done
29	R	0	Cepstrum_done

Table 9-231 VAD_FFT_STS1 0x25

Bits	R/W	Default	Description
11:0	R	0	Se_o

Table 9-232 VAD_SPE_STS0 0x26

Bits	R/W	Default	Description
31:16	R	0	Cepsw_o_h
15:0	R	0	Cepsw_o_l

Table 9-233 VAD_SPE_STS1 0x27

Bits	R/W	Default	Description
31:16	R	0	Cur_pow

Table 9-234 VAD_SPE_STS2 0x28

Bits	R/W	Default	Description
31:0	R	0	Calc_q_pre

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Table 9-235 VAD_SPE_STS3 0x29

Bits	R/W	Default	Description
31:0	R	0	Reserved

Table 9-236 VAD_DEC_STS0 0x2a

Bits	R/W	Default	Description
31:0	R	0	Reserved

Table 9-237 VAD_DEC_STS1 0x2b

Bits	R/W	Default	Description
31:0	R	0	Reserved

Table 9-238 VAD_LUT_CTRL 0x2c

Bits	R/W	Default	Description
31	R/W	0	win_lut_w
30	R/W	0	win_lut_r
6:0	R/W	0	win_lut_addr

Table 9-239 VAD_LUT_WR 0x2d

Bits	R/W	Default	Description
31:0	R/W	0	Win_lut_wr:12.0.0.7

Table 9-240 VAD_LUT_RD 0x2e

Bits	R/W	Default	Description
31:0	R	0	Win_lut_rd:12.0.0.7

Table 9-241 VAD_IN_SEL0 0x2f

Bits	R/W	Default	Description
31:0	R/W	0	ch0_sel: reg_vad.gen_vadin_coeff, bit[n] for ch_n enable

Table 9-242 VAD_IN_SEL1 0x30

Bits	R/W	Default	Description
31:0	R/W	0	Ch1_sel: reg_vad.gen_vadin_coeff, bit[n] for ch_n enable

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Table 9-243 VAD_TO_DDR 0x31

Bits	R/W	Default	Description
31	R/W	0	vad_to_ddr_en
30	R/W	0	vad_to_ddr_rst
29	R/W	0	vad_to_ddr_eo
27:16	R/W	0	vad_to_ddr_cur_cnt
11:0	R/W	0	vad_to_ddr_max_cnt

9.10.16 EQDRC Registers

EQDRC Registers Base Address: 0xFF602000

Each register final address = module base address+ address * 4

Table 9-244 EE_AED_COEF_RAM_CNTL 0x0

Bits	R/W	Default	Description
31-10	R/W	0x0	reserved;
9-2	R/W	0x0	addr: read or write the addr of coef ram
1	R/W	0x0	rd/wr: 0 read from coef ram; 1 write to coef ram
0	R/W	0x0	valid: 0 the command had been finish; 1 the command wait to be done

Table 9-245 EE_AED_COEF_RAM_DATA 0x1

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	data: read from coef ram or write to coef ram

Table 9-246 EE_AED_EQ_EN 0x2

Bits	R/W	Default	Description
31-2	R/W	0x0	reserved;
1	R/W	0x0	eq2_en: It is used to enable EQ for ch3 and ch4. 1= enable, 0 = bypass.
0	R/W	0x0	eq1_en: It is used to enable EQ for ch1 and ch2. 1= enable, 0 = bypass.

Table 9-247 EE_AED_TAP_CNTL 0x3

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
4-0	R/W	0xa	eq2_tap: It is used to set the tap for EQ2.
4-0	R/W	0xa	eq1_tap: It is used to set the tap for EQ1.

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Table 9-248 EE_AED_EQ_VOLUME 0x4

Bits	R/W	Default	Description
31-30	R/W	0x2	volume_step: It is used to set volume adjust step.
29-26	R/W	0x0	reserved;
25-16	R/W	0x3ff	volume_master: It is used to set master volume.
15-8	R/W	0x30	volume_ch2: It is used to set channel2 volume.
7-0	R/W	0x30	volume_ch1: It is used to set channel1 volume.

Table 9-249 EE_AED_EQ_VOLUME_SLEW_CNT 0x5

Bits	R/W	Default	Description
31-12	R/W	0x0	reserved;
11-0	R/W	0x0	volume_slew_cnt: It is used to control volume change and mute ramp rate.

Table 9-250 EE_AED_MUTE 0x6

Bits	R/W	Default	Description
31	R/W	0x0	mute_master: It is used to mute all.
30-2	R/W	0x0	reserved;
1	R/W	0x0	mute_ch2: It is used to mute channel1.
0	R/W	0x0	mute_ch1: It is used to mute channel1.

Table 9-251 EE_AED_DRC_CNTL 0x7

Bits	R/W	Default	Description
31-6	R/W	0x0	reserved;
5-3	R/W	0x5	drc_tap: It is used to set the power tap for DRC.
2	R/W	0x0	drc_pow_sel: It is used to select the input of gain alpha filter. 0 = dB domain, 1 = decimal domain
1	R/W	0x0	drc_rms_mode: It is used to select the mode of pow calculation. 0 = RMS, 1= peaking
0	R/W	0x0	drc_en: It is used to enable the post DRC. 1= enable, 0 = bypass.

Table 9-252 EE_AED_DRC_RMS_COEF0 0x8

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x34ebb	It is used to set the coefficient for power calculation filter.

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Table 9-253 EE_AED_DRC_RMS_COEF1 0x9

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7c-b145	It is used to set the coefficient for power calculation filter.

Table 9-254 EE_AED_DRC_THD0 0x0a

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold0 for DRC.

Table 9-255 EE_AED_DRC_THD1 0x0b

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold1 for DRC.

Table 9-256 EE_AED_DRC_THD2 0x0c

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold2 for DRC.

Table 9-257 EE_AED_DRC_THD3 0x0d

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold3 for DRC.

Table 9-258 EE_AED_DRC_THD4 0x0e

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold4 for DRC.

Table 9-259 EE_AED_DRC_K0 0x0f

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the k-slope of gain for tap 0.

Table 9-260 EE_AED_DRC_K1 0x10

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the k-slope of gain for tap 1.

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Table 9-261 EE_AED_DRC_K2 0x11

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the k-slope of gain for tap 2.

Table 9-262 EE_AED_DRC_K3 0x12

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the k-slope of gain for tap 3.

Table 9-263 EE_AED_DRC_K4 0x13

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the k-slope of gain for tap 4.

Table 9-264 EE_AED_DRC_K5 0x14

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the k-slope of gain for tap 5.

Table 9-265 EE_AED_DRC_THD_OUT0 0x15

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold_out0 for DRC.

Table 9-266 EE_AED_DRC_THD_OUT1 0x16

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold_out1 for DRC.

Table 9-267 EE_AED_DRC_THD_OUT2 0x17

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold_out2 for DRC.

Table 9-268 EE_AED_DRC_THD_OUT3 0x18

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold_out3 for DRC.

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Table 9-269 EE_AED_DRC_OFFSET 0x19

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power offset of DRC.

Table 9-270 EE_AED_DRC_RELEASE_COEF00 0x1a

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-271 EE_AED_DRC_RELEASE_COEF01 0x1b

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

Table 9-272 EE_AED_DRC_RELEASE_COEF10 0x1c

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-273 EE_AED_DRC_RELEASE_COEF11 0x1d

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

Table 9-274 EE_AED_DRC_RELEASE_COEF20 0x1e

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-275 EE_AED_DRC_RELEASE_COEF21 0x1f

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

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Table 9-276 EE_AED_DRC_RELEASE_COEF30 0x20

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-277 EE_AED_DRC_RELEASE_COEF31 0x21

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

Table 9-278 EE_AED_DRC_RELEASE_COEF40 0x22

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-279 EE_AED_DRC_RELEASE_COEF41 0x23

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

Table 9-280 EE_AED_DRC_RELEASE_COEF50 0x24

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-281 EE_AED_DRC_RELEASE_COEF51 0x25

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

Table 9-282 EE_AED_DRC_ATTACK_COEF00 0x26

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

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Table 9-283 EE_AED_DRC_ATTACK_COEF01 0x27

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

Table 9-284 EE_AED_DRC_ATTACK_COEF10 0x28

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

Table 9-285 EE_AED_DRC_ATTACK_COEF11 0x29

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

Table 9-286 EE_AED_DRC_ATTACK_COEF20 0x2a

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

Table 9-287 EE_AED_DRC_ATTACK_COEF21 0x2b

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

Table 9-288 EE_AED_DRC_ATTACK_COEF30 0x2c

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

Table 9-289 EE_AED_DRC_ATTACK_COEF31 0x2d

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

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Table 9-290 EE_AED_DRC_ATTACK_COEF40 0x2e

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

Table 9-291 EE_AED_DRC_ATTACK_COEF41 0x2f

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

Table 9-292 EE_AED_DRC_ATTACK_COEF50 0x30

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

Table 9-293 EE_AED_DRC_ATTACK_COEF51 0x31

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

Table 9-294 EE_AED_DRC_LOOPBACK_CNTL 0x32

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
17	R/W	0x0	dis_fifo_rst: 0 if drc_loopback_cnt change, then reset fifo; 1 don`t reset
16	R/W	0x0	drc_loopback_bypass: It is used to set bypass look-ahead mode. 1= bypass.
15-9	R/W	0x0	reserved;
8-0	R/W	0x90	drc_loopback_cnt: It is used to set the look-ahead count.

Table 9-295 EE_AED_MDRC_CNTL 0x33

Bits	R/W	Default	Description
31-17	R/W	0x0	reserved;
16	R/W	0x0	mdrc_pow_sel: It is used to select the input of gain alpha filter. 0 = dB domain, 1 = decimal domain
15-9	R/W	0x0	reserved;

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Bits	R/W	Default	Description
8	R/W	0x0	mdrc_all_en: It is used to enable the entire multi-band DRC. 1= enable, 0 = bypass.
7-6	R/W	0x0	reserved;
5-3	R/W	0x0	[3] mdrc_rms_mode: It is used to select the band0 mode of pow calculation. 0 = RMS, 1= peaking It is used to select the band1 mode of pow calculation. 0 = RMS, 1= peaking It is used to select the band2 mode of pow calculation. 0 = RMS, 1= peaking
2-0	R/W	0x0	[0] mdrc_en: It is used to enable the band 0 of MDRC. 1= enable, 0 = bypass. It is used to enable the band 1 of MDRC. 1= enable, 0 = bypass. It is used to enable the band 2 of MDRC. 1= enable, 0 = bypass.

Table 9-296 EE_AED_MDRC_RMS_COEF00 0x34

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x34ebb	It is used to set the coefficient for power calculation filter.

Table 9-297 EE_AED_MDRC_RMS_COEF01 0x35

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7c-b145	It is used to set the coefficient for power calculation filter.

Table 9-298 EE_AED_MDRC_RELEASE_COEF00 0x36

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-299 EE_AED_MDRC_RELEASE_COEF01 0x37

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

Table 9-300 EE_AED_MDRC_ATTACK_COEF00 0x38

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

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Table 9-301 EE_AED_MDRC_ATTACK_COEF01 0x39

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

Table 9-302 EE_AED_MDRC_THD0 0x3a

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold for DRC.

Table 9-303 EE_AED_MDRC_K0 0x3b

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the k-slope of gain.

Table 9-304 EE_AED_MDRC_LOW_GAIN 0x3c

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the gain for DRC band 0.

Table 9-305 EE_AED_MDRC_OFFSET0 0x3d

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power offset of DRC.

Table 9-306 EE_AED_MDRC_RMS_COEF10 0x3e

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x34ebb	It is used to set the coefficient for power calculation filter.

Table 9-307 EE_AED_MDRC_RMS_COEF11 0x3f

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7c-b145	It is used to set the coefficient for power calculation filter.

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Table 9-308 EE_AED_MDRC_RELEASE_COEF10 0x40

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-309 EE_AED_MDRC_RELEASE_COEF11 0x41

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

Table 9-310 EE_AED_MDRC_ATTACK_COEF10 0x42

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

Table 9-311 EE_AED_MDRC_ATTACK_COEF11 0x43

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

Table 9-312 EE_AED_MDRC_THD1 0x44

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold for DRC.

Table 9-313 EE_AED_MDRC_K1 0x45

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the k-slope of gain.

Table 9-314 EE_AED_MDRC_OFFSET1 0x46

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power offset of DRC.

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Table 9-315 EE_AED_MDRC_MID_GAIN 0x47

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the gain for DRC band 1.

Table 9-316 EE_AED_MDRC_RMS_COEF20 0x48

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x34ebb	It is used to set the coefficient for power calculation filter.

Table 9-317 EE_AED_MDRC_RMS_COEF21 0x49

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7c-b145	It is used to set the coefficient for power calculation filter.

Table 9-318 EE_AED_MDRC_RELEASE_COEF20 0x4a

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x5188	It is used to set the coefficient for gain release filter.

Table 9-319 EE_AED_MDRC_RELEASE_COEF21 0x4b

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7fae78	It is used to set the coefficient for gain release filter.

Table 9-320 EE_AED_MDRC_ATTACK_COEF20 0x4c

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x3263a	It is used to set the coefficient for gain attack filter.

Table 9-321 EE_AED_MDRC_ATTACK_COEF21 0x4d

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x7cd9-c6	It is used to set the coefficient for gain attack filter.

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Table 9-322 EE_AED_MDRC_THD2 0x4e

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power threshold for DRC.

Table 9-323 EE_AED_MDRC_K2 0x4f

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the k-slope of gain.

Table 9-324 EE_AED_MDRC_OFFSET2 0x50

Bits	R/W	Default	Description
31-0	R/W	0x0	It is used to set the power offset of DRC.

Table 9-325 EE_AED_MDRC_HIGH_GAIN 0x51

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the gain for DRC band 1.

Table 9-326 EE_AED_ED_CNTL 0x52

Bits	R/W	Default	Description
31-2	R/W	0x0	reserved;
1	R/W	0x0	ed_sign:
0	R/W	0x0	ed_int: soft reset

Table 9-327 EE_AED_DC_EN 0x53

Bits	R/W	Default	Description
31-1	R/W	0x0	reserved;
0	R/W	0x0	It is used to enable the noise detection. 1= enable, 0 = bypass.

Table 9-328 EE_AED_ND_LOW_THD 0x54

Bits	R/W	Default	Description
31-24	R/W	0x0	reserved;
23-0	R/W	0x100	It is used to set the low threshold for detection.

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Table 9-329 EE_AED_ND_HIGH_THD 0x55

Bits	R/W	Default	Description
31-24	R/W	0x0	reserved;
23-0	R/W	0x200	It is used to set the high threshold for detection. If the signal amplitude is bigger than it, the signal is not noise.

Table 9-330 EE_AED_ND_CNT_THD 0x56

Bits	R/W	Default	Description
31-16	R/W	0x0	reserved;
15-0	R/W	0x100	It is used to set the counter threshold for amplitude detection. If the counter for which the signal is smaller than reg_nd_low_thd continuously is bigger than it, the signal is noise.

Table 9-331 EE_AED_ND_SUM_NUM 0x57

Bits	R/W	Default	Description
31-16	R/W	0x0	reserved;
15-0	R/W	0x200	It is used to set the statistical number for sum of signal amplitude.

Table 9-332 EE_AED_ND_CZ_SUM 0x58

Bits	R/W	Default	Description
31-16	R/W	0x0	reserved;
15-0	R/W	0x800	It is used to set the statistical number for zero-crossing.

Table 9-333 EE_AED_ND_SUM_THD0 0x59

Bits	R/W	Default	Description
31-0	R/W	0x20000	It is used to set the low threshold for sum of amplitude.

Table 9-334 EE_AED_ND_SUM_THD1 0x5a

Bits	R/W	Default	Description
31-0	R/W	0x30000	It is used to set the high threshold for sum of amplitude.

Table 9-335 EE_AED_ND_CZ_THD0 0x5b

Bits	R/W	Default	Description
31-16	R/W	0x0	reserved;
15-0	R/W	0x200	It is used to set the low threshold for zero-crossing number.

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Table 9-336 EE_AED_ND_CZ_THD1 0x5c

Bits	R/W	Default	Description
31-16	R/W	0x0	reserved;
15-0	R/W	0x100	It is used to set the high threshold for zero-crossing number.

Table 9-337 EE_AED_ND_COND_CNTL 0x5d

Bits	R/W	Default	Description
31-6	R/W	0x0	reserved;
5	R/W	0x0	Reg_nd_audio_cond2_en: It is used to enable the audio detection for amplitude.
4	R/W	0x0	Reg_nd_audio_cond1_en: It is used to enable the audio detection for amplitude.
3	R/W	0x0	Reg_nd_audio_cond0_en: It is used to enable the audio detection for amplitude.
2	R/W	0x0	Reg_nd_noise_cond2_en: It is used to enable the noise detection for amplitude.
1	R/W	0x0	Reg_nd_noise_cond1_en: It is used to enable the noise detection for amplitude.
0	R/W	0x0	Reg_nd_noise_cond0_en: It is used to enable the noise detection for amplitude.

Table 9-338 EE_AED_ND_RELEASE_COEF0 0x5e

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the coefficients for release filter when gain adjusting.

Table 9-339 EE_AED_ND_RELEASE_COEF1 0x5f

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the coefficients for release filter when gain adjusting.

Table 9-340 EE_AED_ND_ATTACK_COEF0 0x60

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the coefficients for attack filter when gain adjusting.

Table 9-341 EE_AED_ND_ATTACK_COEF1 0x61

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the coefficients for attack filter when gain adjusting.

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Table 9-342 EE_AED_ND_ATTACK_COEF1 0x61

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x0	It is used to set the coefficients for attack filter when gain adjusting.

Table 9-343 EE_AED_ND_CNTL 0x62

Bits	R/W	Default	Description
31-3	R/W	0x0	reserved;
2-1	R/W	0x0	nd_gain_sel: It is used to set the gain when noise is detected. 0 = 1/2 1 = 1/4 2 = 1/8 3 = 0
0	R/W	0x0	It is used to enable the dc cut module. 1= enable, 0 = bypass.

Table 9-344 EE_AED_MIX0_LL 0x63

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the ch1 gain to ch1 mixer.

Table 9-345 EE_AED_MIX0_RL 0x64

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the ch2 gain to ch1 mixer.

Table 9-346 EE_AED_MIX0_LR 0x65

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the ch1 gain to ch2 mixer.

Table 9-347 EE_AED_MIX0_RR 0x66

Bits	R/W	Default	Description
31-26	R/W	0x0	reserved;
25-0	R/W	0x40000	It is used to set the ch2 gain to ch2 mixer.

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Table 9-348 EE_AED_CLIP_THD 0x67

Bits	R/W	Default	Description
31-23	R/W	0x0	reserved;
22-0	R/W	0x7fffff	It is used to set the clipping threshold.

Table 9-349 EE_AED_CH1_ND_SUM_OUT 0x68

Bits	R/W	Default	Description
31-0	R	0x0	It is used to report the sum of amplitude of ch1.

Table 9-350 EE_AED_CH2_ND_SUM_OUT 0x69

Bits	R/W	Default	Description
31-0	R	0x0	It is used to report the sum of amplitude of ch2.

Table 9-351 EE_AED_CH1_ND_CZ_OUT 0x6a

Bits	R/W	Default	Description
31-16	R	0x0	reserved;
15-0	R	0x0	It is used to report the sum of zero-crossing of ch1.

Table 9-352 EE_AED_CH2_ND_CZ_OUT 0x6b

Bits	R/W	Default	Description
31-16	R	0x0	reserved;
15-0	R	0x0	It is used to report the sum of zero-crossing of ch2.

Table 9-353 EE_AED_NOISE_STATUS 0x6c

Bits	R/W	Default	Description
31-7	R	0x0	reserved;
6	R	0x0	It is used to report the noise flag.
5-3	R	0x0	It is used to report the audio state.
2-0	R	0x0	It is used to report the noise state.

Table 9-354 EE_AED_POW_CURRENT_S0 0x6d

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the mdrc0 current power.

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Table 9-355 EE_AED_POW_CURRENT_S1 0x6e

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the mdrc1 current power.

Table 9-356 EE_AED_POW_CURRENT_S2 0x6f

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the mdrc2 current power.

Table 9-357 EE_AED_POW_OUT0 0x70

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the mdrc0 destination power.

Table 9-358 EE_AED_POW_OUT1 0x71

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the mdrc1 destination power.

Table 9-359 EE_AED_POW_OUT2 0x72

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the mdrc2 destination power.

Table 9-360 EE_AED_ADJ_INDEX0 0x73

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R/W	0x0	It is used to report the mdrc0 adjusting power.

Table 9-361 EE_AED_ADJ_INDEX1 0x74

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the mdrc1 adjusting power.

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Table 9-362 EE_AED_ADJ_INDEX2 0x75

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the mdrc2 adjusting power.

Table 9-363 EE_AED_DRC_GAIN_INDEX0 0x76

Bits	R/W	Default	Description
31-26	R	0x0	reserved;
25-0	R	0x0	It is used to report mdrc0 the gain.

Table 9-364 EE_AED_DRC_GAIN_INDEX1 0x77

Bits	R/W	Default	Description
31-26	R	0x0	reserved;
25-0	R	0x0	It is used to report mdrc0 the gain.

Table 9-365 EE_AED_DRC_GAIN_INDEX2 0x78

Bits	R/W	Default	Description
31-26	R	0x0	reserved;
25-0	R	0x0	It is used to report mdrc0 the gain.

Table 9-366 EE_AED_CH1_VOLUME_STATE 0x79

Bits	R/W	Default	Description
31-11	R	0x0	reserved;
10-0	R	0x0	It is used to report the volume state of ch1.

Table 9-367 EE_AED_CH2_VOLUME_STATE 0x7a

Bits	R/W	Default	Description
31-11	R	0x0	reserved;
10-0	R	0x0	It is used to report the volume state of ch2.

Table 9-368 EE_AED_CH1_VOLUME_GAIN 0x7b

Bits	R/W	Default	Description
31-25	R	0x0	reserved;
25-0	R	0x0	It is used to report the volume gain of ch1.

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Table 9-369 EE_AED_CH2_VOLUME_GAIN 0x7c

Bits	R/W	Default	Description
31-25	R	0x0	reserved;
25-0	R	0x0	It is used to report the volume gain of ch2.

Table 9-370 EE_AED_FULL_POW_CURRENT 0x7d

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the current power.

Table 9-371 EE_AED_FULL_POW_OUT 0x7e

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the destination power.

Table 9-372 EE_AED_FULL_POW_ADJ 0x7f

Bits	R/W	Default	Description
-	R	0x0	reserved;
31-0	R	0x0	It is used to report the adjusting power.

Table 9-373 EE_AED_FULL_DRC_GAIN 0x80

Bits	R/W	Default	Description
31-26	R	0x0	reserved;
25-0	R	0x0	It is used to report the gain.

Table 9-374 EE_AED_MASTER_VOLUME_STATE 0x81

Bits	R/W	Default	Description
31-11	R	0x0	reserved;
10-0	R	0x0	It is used to report the volume state of master.

Table 9-375 EE_AED_MASTER_VOLUME_GAIN 0x82

Bits	R/W	Default	Description
31-26	R	0x0	reserved;
25-0	R	0x0	It is used to report the volume gain of master.

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Table 9-376 EE_AED_AED_TOP_CTL 0x83

Bits	R/W	Default	Description
31	R/W	0x0	reg_start_en:write 1 to start eqdrc function, will auto clear to 0
29-26	R/W	0x0	reg_lane_eq_ch34: eqdrc channel 34 select,can only set one bit valid 0000: audio data to channel 34 will be zero 0001:audio data from lane0 0010:audio data from lane1 0100:audio data from lane2 1000:audio data from lane3
25-18	R/W	0x0	reg_channel_valid: audio channel valid for each bit
17-14	R/W	0x0	reg_lane_eq_ch12: eqdrc channel 12 select,can only set one bit valid 0000: audio data to channel 12 will be zero 0001:audio data from lane0 0010:audio data from lane1 0100:audio data from lane2 1000:audio data from lane3
13-11	R/W	0x0	reg_data_type:0: split 64bits ddr data to 8 sample, each sample need 8 bits; if bit-width < 8, left-justified; 1: split 64bits ddr data to 4 sample, each sample need 16 bits; if bitwidth < 16, left-justified ; 2: split 64bits ddr data to 4 sample, each sample need 16 bits; if bitwidth < 16, right-justified ; 3: split 64bits ddr data to 2 sample, each sample need 32 bits; if bitwidth < 32, left-justified; 4: split 64bits ddr data to 2 sample, each sample need 32 bits; if bitwidth < 32, right-justified;
10-6	R/W	0x0	reg_data_msb:msb position of data
5-4	R/W	0x0	reg_frddr_source: 0:frddr_A; 1:frddr_B; 2:frddr_C;
2	R/W	0x0	reg_afifo_in_rst: reset afifo in side;
1	R/W	0x0	reg_afifo_out_rst:reset afifo out side;
0	R/W	0x0	req_eqdrc_en:eqdrc module enable

Table 9-377 EE_AED_AED_TOP_CTL 0x84

Bits	R/W	Default	Description
31-20	R/W	0x0	reserved;
19-12	R/W	0x0	reg_ack_num: if needn't share buffer, set it as 0; if need share buffer , set it as TDMOUT/SPDIFOUT ch number;
11	R/W	0x0	reg_req_sel2_en
10-8	R/W	0x0	reg_req_sel2: 0: tdmout_a; 1: tdmout_b; 2: tdmout_c; 3: spdifout; 4: spdifout_b; 5: reserved; 6: reserved; 7: reserved;
7	R/W	0x0	reg_req_sel1_en

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Bits	R/W	Default	Description
6-4	R/W	0x0	reg_req_sel1: 0: tdmout_a; 1: tdmout_b; 2: tdmout_c; 3: spdifout; 4: spdifout_b; 5: reserved; 6: reserved; 7: reserved;
3	R/W	0x0	reg_req_sel0_en
2-0	R/W	0x0	req_req_sel0: 0: tdmout_a; 1: tdmout_b; 2: tdmout_c; 3: spdifout; 4: spdifout_b; 5: reserved; 6: reserved; 7: reserved;

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10 AFIFO

10.1 Register Description

Base address: 0xffd05000

Table 10-1 AIU_AIFIFO_CTRL 0x00

Bit(s)	R/W	Default	Description
13	R/W	0	Aififo request to dcu status
12	R/W	0	Dcu select status
11:5	R/W	0	Aififo word counter number
4:0	R/W	0	How many bits left in the first pop register

Table 10-2 AIU_AIFIFO_STATUS 0x01

Bit(s)	R/W	Default	Description
4:0	R	0x0	How many Bits left in the first pop register

Table 10-3 AIU_AIFIFO_GBIT 0x02

Bit(s)	R/W	Default	Description
15:0	R	0x0	The gb data

Table 10-4 AIU_AIFIFO_CLB 0x03

Bit(s)	R/W	Default	Description
15:0	R	0x0	The gb data

Table 10-5 AIU_MEM_AIFIFO_START_PTR 0x04

Bit(s)	R/W	Default	Description
31:0	RW	0x0	The start address from DDR

Table 10-6 AIU_MEM_AIFIFO_CURR_PTR 0x05

Bit(s)	R/W	Default	Description
31:0	R	0x0	The current address from DDR

Table 10-7 AIU_MEM_AIFIFO_END_PTR 0x06

Bit(s)	R/W	Default	Description
31:0	RW	0x0	The end address from DDR

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Table 10-8 AIU_MEM_AIFIFO_BYTES_AVAIL 0x07

Bit(s)	R/W	Default	Description
15:11	R/W	0	Unused.
10	R/W	0	Set this bit to 1 to enable filling of the FIFO controlled by the buffer level control. If this bit is 0, then use bit[1] to control the enabling of filling
9	R/W	0	This bit is set when data can be popped
8	R/W	0	This bit will be high when we're fetching data from the DDR memory To reset this module, set cntl_enable = 0, and then wait for busy = 0. After that you can pulse cntl_init to start over
7	R/W	0	Just in case endian. last minute byte swap of the data out of the FIFO to getbit
6	R/W	0	Unused.
5:3	R/W	0	
2	R/W	0	Set to 1 to enable reading the DDR memory FIFO and filling the pipeline to get-bit Set cntl_empty_en = cntl_fill_en = 0 when pulsing cntl_init
1	R/W	0	Set to 1 to enable reading data from DDR memory
0	R/W	0	After setting the read pointers, sizes, channel masks and read masks, set this bit to 1 and then to 0 NOTE: You don't need to pulse cntl_init if only the start address is being changed

Table 10-9 AIU_MEM_AIFIFO_CONTROL 0x08

Bit(s)	R/W	Default	Description
15:11		0x0	Unused
10	RW	0x0	Use_level Set This bit to 1 to enable filling of the FIFO controlled by the buffer level control. If This bit is 0, then use Bit[1] to control the enabling of filling
9	RW	0x0	Data Ready. This bit is set when data can be popped
8	RW	0x0	Fill busy This bit will be high when we're fetching data from the DDR memory
7	RW	0x0	Cntl_endian_jic Just in case endian. Last minute byte swap of the data out of the FIFO to get bit
6	RW	0x0	Unused
5:3	RW	0x0	Endian
2	RW	0x0	Cntl_empty_en Set to 1 to enable reading the DDR memory FIFO and filling the pipeline to get-bit
1	RW	0x0	Cntl_fill_en Set to 1 to enable reading data from DDR memory
0	RW	0x0	Cntl_init

Table 10-10 AIU_MEM_AIFIFO_MAN_WP 0x09

Bit(s)	R/W	Default	Description
31:0	RW	0x0	Manual write ptr.

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Table 10-11 AIU_MEM_AIFIFO_MAN_RP 0x0a

Bit(s)	R/W	Default	Description
31:0	RW	0x0	Manual read ptr.

Table 10-12 AIU_MEM_AIFIFO_LEVEL 0x0b

Bit(s)	R/W	Default	Description
1	R/W	0	Set to 1 for manual write pointer mode
0	R/W	0	Set high then low after everything has been initialized

Table 10-13 AIU_MEM_AIFIFO_BUF_CNTL 0x0c

Bit(s)	R/W	Default	Description
1	RW	0x0	manual mode Set to 1 for manual write pointer mode
0	RW	0x0	Init Set high then low after everything has been initialized

Table 10-14 AIU_MEM_AIFIFO_BUF_WRAP_COUNT 0x0d

Bit(s)	R/W	Default	Description
29:24	R/W	0	A_brst_num
21:16	R/W	0	A_id
15:0	R/W	0	level_hold

Table 10-15 AIU_MEM_AIFIFO_MEM_CTL 0x0f

Bit(s)	R/W	Default	Description
31:16	R/W	0	drop_bytes
15:14	R/W	0	drop_status (Read-Only)
13:12	R/W	0	sync_match_position (Read-Only)
11:6	R/W	0	reserved
5:4	R/W	0	TIME_STAMP_NUMBER, 0-32bits, 1-64bits, 2-96bits, 3-128bits
3	R/W	0	stamp_soft_reset
2	R/W	0	TIME_STAMP_length_enable
1	R/W	0	TIME_STAMP_sync64_enable
0	R/W	0	TIME_STAMP_enable

Table 10-16 AIFIFO_TIME_STAMP_CNTL 0x10

Bit(s)	R/W	Default	Description
31:0	R/W	0	TIME_STAMP_SYNC_CODE_0

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Table 10-17 AIFIFO_TIME_STAMP_SYNC_0 0x11

Bit(s)	R/W	Default	Description
31:0	R/W	0	TIME_STAMP_SYNC_CODE_1

Table 10-18 AIFIFO_TIME_STAMP_SYNC_1 0x12

Bit(s)	R/W	Default	Description
31:0	R/W	0	TIME_STAMP_0

Table 10-19 AIFIFO_TIME_STAMP_0 0x13

Bit(s)	R/W	Default	Description
31:0	R/W	0	TIME_STAMP_1

Table 10-20 AIFIFO_TIME_STAMP_1 0x14

Bit(s)	R/W	Default	Description
31:0	R/W	0	TIME_STAMP_2

Table 10-21 AIFIFO_TIME_STAMP_2 0x15

Bit(s)	R/W	Default	Description
31:0	R/W	0	TIME_STAMP_3

Table 10-22 AIFIFO_TIME_STAMP_3 0x16

Bit(s)	R/W	Default	Description
31:0	R/W	0	TIME_STAMP_LENGTH

Table 10-23 AIFIFO_TIME_STAMP_LENGTH 0x17

Bit(s)	R/W	Default	Description
31:0	RW	0x0	TIME_STAMP_LENGTH

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11 Memory Interface

This chapter describes the chip's memory interfaces from the following aspects:

- DDR
- NAND
- EMMC/SDIO/SD
- SPICC
- SPIFC

11.1 DDR

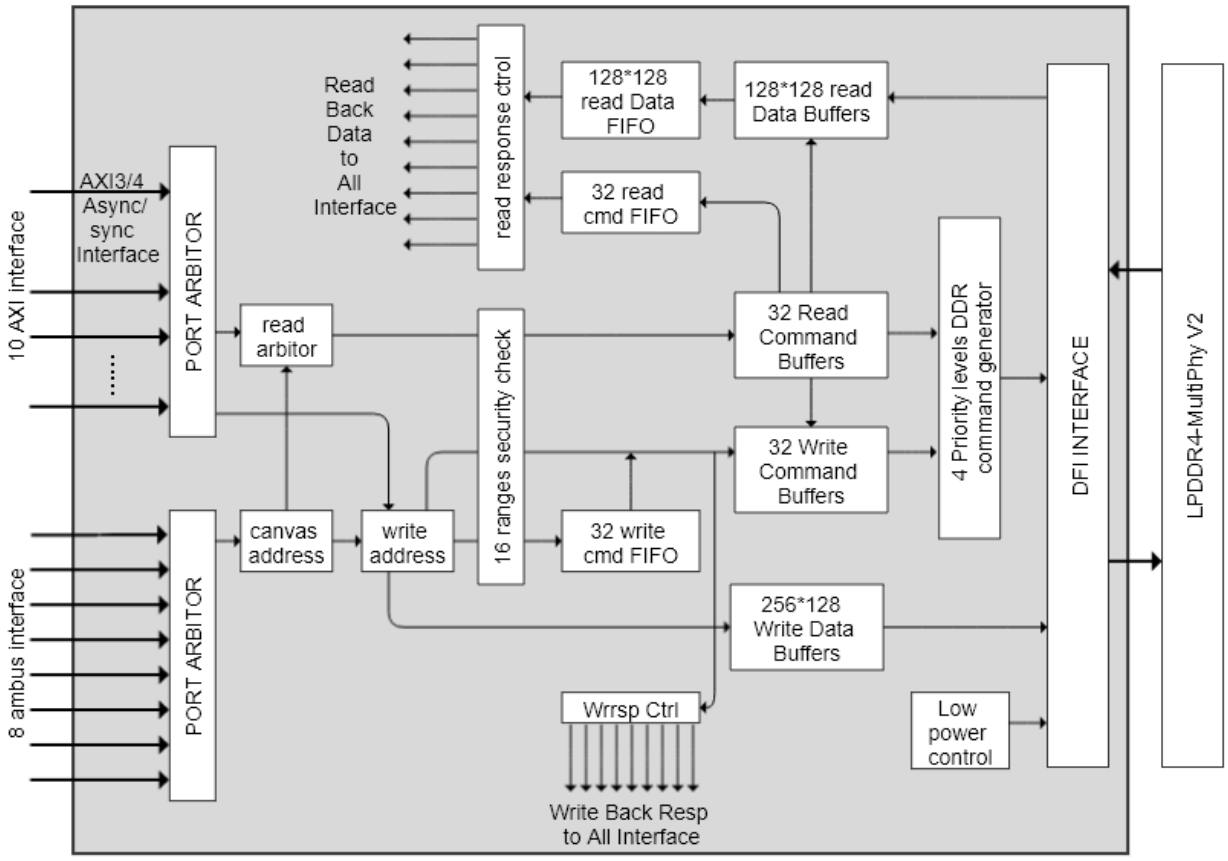
11.1.1 Overview

DDR consists of the 2 parts: DDR memory controller (DMC) and DDR PHY controller. The main features of this module are listed below:

- Support DDR3/4 SDRAM, LPDDR3, LPDDR4 SDRAM
- Support 32bits and 16bits data mode.
- Unbalanced 16bits DDR3/4 SDRAM size in 32 bits data bus mode.
- Each DDR command buffer contains 64bytes data.
- 32 write command buffer and 32 read command buffer for DDR command generation.
- READ command reordering regardless ID.
- Write command reordering regardless IDs.
- Write and read data coherence.
- Optimized DDR command reordering based on:
  - Priority
  - Rank
  - Bank
  - Bank group interleaving(DDR4)
  - Read/write
- 4 level priority control to reduce the latency for the urgent request.
- Unblocking urgent control through Asynchronous FIFO and pipelines.
- Additional 128 depth write data FIFO to balance DMC and AXI master write throughputs.
- Additional 128*128 read data FIFO to balance DMC and AXI master read throughputs.
- Per Port Per ID Security Control through the all security ranges.
- Up to 16 security range can be defined.
- Support up to 16 AXI3/4 and 8 AMBUS(Amlogic) interfaces.
- Optimized AMBUS to support 0~3 burst to improve DDR efficiency and reduce
- Low power control with LPDDR4 Multi-PHY V2.
  - Adding auto Self-Refresh mode for super low bandwidth application.
  - Auto power down mode for most of application.
  - Hardware Fast Frequency change less than 10uS.

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Figure 11-1 DDR Interface



11.1.2 Register Description

DMC unsecure register. Base address 0xFF638000. Each register takes 4 byte address.

Each register's final address = 0xFF638C00+ offset * 4.

Table 11-1 AM_DDR_PLL_CNTL 0 0x0000

Bit(s)	R/W	Default	Description
29			dpll_reset.
28			dpll_en.
27:26			dpll_clk_en
20:19			od1
18:16			od
14:10			dpll_ref_div_n
8:0			dpll_int_num

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Table 11-2 AM_DDR_PLL_CNTL1 0x0001

Bit(s)	R/W	Default	Description
18:0			ddr_dpll_frac

Table 11-3 AM_DDR_PLL_CNTL2 0x0002

Bit(s)	R/W	Default	Description
22:20			fref_sel
17:16			os_ssc
15:12			ssc_str_m
8			ssc_en
7:4			ssc_dep_sel
1:0			dpll ss_mode

Table 11-4 AM_DDR_PLL_CNTL3 0x0003

Bit(s)	R/W	Default	Description
31			afc bypass
30			afc clk sel
29			code new
28			dco_m_en
27			dco_sdm_en
26			div2
25			div mode
24			fast_lock mode
23			fb_pre_div
22			filter_mode
21			fix_en
20			freq_shift_en
19			load
18			load_en
17			lock_f
16			pulse_width_en
15			sdmnc_en
14			sdmnc_mode
13			sdmnc_range
12			tdc_en

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Bit(s)	R/W	Default	Description
11			tdc_mode_sel
10			wait_en

Table 11-5 AM_DDR_PLL_CNTL4 0x0004

Bit(s)	R/W	Default	Description
1:0			pfd_gain
7:4			filter_pvt1
11:8			filter pvt2
13:12			acq_gain
18:16			lambda0
22:20			lambda1
26:24			rou
30:28			alpha

Table 11-6 AM_DDR_PLL_CNTL5 0x0005

Bit(s)	R/W	Default	Description
15:0			reve
21:16			lm_s
27:24			lm_w
30:28			adj_vco_ldo

Table 11-7 AM_DDR_PLL_CNTL6 0x0006

Bit(s)	R/W	Default	Description
15:0			reve
21:16			lm_s
27:24			lm_w
30:28			adj_vco_ldo

Table 11-8 AM_DDR_PLL_STS 0x0007

Bit(s)	R/W	Default	Description
31			DDR_PLL_LOCK
30:19			not used
18			DDR_AFC_DONE
17			DDR_PLL_LOCK

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Bit(s)	R/W	Default	Description
16:7			DDR_DPLL_OUT_RSV
6:0			DDR_SDMNC_MONITOR

Table 11-9 DDR_CLK_CNTL 0x0008

Bit(s)	R/W	Default	Description
31			ddr_pll_clk enable. enable the clock from DDR_PLL to clock generateion.
30			ddr_pll_prod_test_en. enable the clock to clock/32 which to clock frequency measurement and production test pin.
29			not used.
28			clock generation logic soft reset. 0 = reset.
27			phy_4xclk phase inverter
26			pll_freq divide/2. 1: use pll div/2 clock as the n_clk. 0: use pll clock as n_clk. this setting is used for the DDR PHY PLL fast lock mode.
2			enable dmc_clk.
1			enable LPDDR4-PHY DfiClk.
0			enable LPDDR4-PHY DfiCtlClk.

Table 11-10 DDR_PHY_CTRL 0x0009

Bit(s)	R/W	Default	Description
4			DDR PHY PwrOkIn pin.
1			DDR PHY APB soft reset_n.
0			phy_reset_n.

Table 11-11 AM_DDR_PLL_FREQ1_OD 0x000c

Bit(s)	R/W	Default	Description
8			currunt FREQ selection. it can forced to change to select which freqency to select, or it can auto changed by FREQ change hardware.
5:4			OD1.
2:0			OD.

The following registers' base address is 0xff638000. Each register takes 4 byte address.

Each register's final address = 0xff638000 + offset * 4.

Table 11-12 DMC_REQ_CTRL 0x0000

Bit(s)	R/W	Default	Description
23			enable dmc request of ambus chan 7. Reserved for GE2D interface. Async interface.
22			enable dmc request of ambus chan 6. DOS HCODEC interface Sync interface.

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Bit(s)	R/W	Default	Description
21			enable dmc request of ambus chan 5. DOS VDEC interface Sync interface.
20			enable dmc request of ambus chan 4. VPU write interface 1 Sync interface.
19			enable dmc request of ambus chan 3. VPU write interface 0 Sync interface.
18			enable dmc request of ambus chan 2. VPU read interface 2. Sync interface.
17			enable dmc request of ambus chan 1. VPU read interface 1. Sync interface.
16			enable dmc request of ambus chan 0. VPU read interface 0. Sync interface.
9			enable dmc request of axibus chan 9. wave async interface.
8			enable dmc request of axibus chan 8 hevc_b async interface.
7			enable dmc request of axibus chan 7. DEVICE. Async interface.
6			enable dmc request of axibus chan 6. USB Async interface.
5			enable dmc request of axibus chan 5. reserved for dmc_test.
4			enable dmc request of axibus chan 4. hevc front Async interface.
3			enable dmc request of axibus chan 3. HDCP/HDMI Async interface.
2			-
1			enable dmc request of axibus chan 1. Dvalin . async interface.
0			enable dmc request of axibus chan 0. CPU/A53 async interface.

Table 11-13 DMC_SOFT_RST 0x0001

Bit(s)	R/W	Default	Description
31:24			Reserved.
23:16			8 AMBUS input interface n_clk domain reset_n signal. 0 : reset. 1: normal working mode.
15:0			16 AXI BUS input intefaces n_clk domain reset_n signal. 0: reset. 1: normal working mode. each bit for one interface.

Table 11-14 DMC_SOFT_RST1 0x0002

Bit(s)	R/W	Default	Description
31:24			Not used.
23.16			8 am bus interfaces master clock domain reset_n signal. 0 : reset : 1 normal working mode.
15:0.			16 AXI bus interfaces master clock domain reset_n signal. 0 : reset : 1 normal working mode.

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Table 11-15 DMC_SOFT_RST2 0x0003

Bit(s)	R/W	Default	Description
31~11			Reserved.
10			DMC DFI cmd soft reset_n
9			DMC DFI MISC soft reset_n
8			DMC DFI data soft reset_n
7			DMC DFI dcu soft reset_n
6			DMC siu soft reset_n
5			DMC test soft reset_n. 0 : reset. 1 : normal working mode.
4			DMC low power control moudle soft reset_n. 0 : reset. 1 : normal working mode.
3			DMC QOS monitor module soft reset_n. 0 : reset. 1 : normal working mode.
2			DMC register modle soft reset_n.0 : reset. 1 : normal working mode.
1			DMC canvas transfer module soft reset_n. 0 : reset. 1 : normal working mode.
0			DMC command buffers and command generation modules soft reset. 0 = reset. 1:

Table 11-16 DMC_RST_STS1 0x0004

Bit(s)	R/W	Default	Description
31:24			Not used.
23.0			Read only. The DMC_SOFT_RST1 signal in n_clk domain. When one of the 2 clocks is too slow or too fast, we can read this register to make sure another clock domain reset is done.

DMC_VERSION 0x0005

DMC version number.

DC_CAV_LUT_DATAL 0x0012

low 32 bits of canvas data which need to be configured to canvas memory.

Table 11-17 DC_CAV_LUT_DATAH 0x0013

Bit(s)	R/W	Default	Description
60:58			Canvas block mode. 2: 64x32, 1: 32x32; 0: linear mode.
57:56			canvas Y direction wrap control. 1: wrap back in y. 0: not wrap back.
55			canvas X direction wrap control. 1: wrap back in X. 0: not wrap back.
54			canvas Hight.
53:41			canvas Width, unit: 8bytes. must in 32bytes boundary. that means last 2 bits must be 0.
40:29			cavnas start address. unit. 8 bytes. must be in 32bytes boundary. that means last 2bits must be 0.
28:0			Canvas block mode. 2 : 64x32, 1: 32x32; 0 : linear mode.

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Table 11-18 DC_CAV_LUT_ADDR 0x0014

Bit(s)	R/W	Default	Description
9:8			Write 9:8 2'b10. the canvas data will saved in canvas memory with addres 7:0.
7:0			256 canvas Look up table address.

DC_CAV_LUT_RDATAL 0x0015

CBUS low 32bytes canvas read back data from LUT.

DC_CAV_LUT_RDATAH 0x0016

CBUS low 32bytes canvas read back data from LUT.

Table 11-19 DC_CAV_BLK_CTRL0 0x0018

Bit(s)	R/W	Default	Description
31:0			blkmode. 1 : 32x32. 0 : others.

Table 11-20 DC_CAV_BLK_CTRL1 0x0019

Bit(s)	R/W	Default	Description
63:32			blkmode. 1 : 32x32. 0 : others.

Table 11-21 DC_CAV_BLK_CTRL2 0x001a

Bit(s)	R/W	Default	Description
95:64			blkmode. 1 : 32x32. 0 : others.

Table 11-22 DC_CAV_BLK_CTRL3 0x001b

Bit(s)	R/W	Default	Description
127:96			blkmode. 1 : 32x32. 0 : others.

Table 11-23 DC_CAV_BLK_CTRL4 0x001c

Bit(s)	R/W	Default	Description
159:128			blkmode. 1 : 32x32. 0 : others.

Table 11-24 DC_CAV_BLK_CTRL5 0x001d

Bit(s)	R/W	Default	Description
191:160			blkmode. 1 : 32x32. 0 : others.

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Table 11-25 DC_CAV_BLK_CTRL6 0x001e

Bit(s)	R/W	Default	Description
223:192			blkmode. 1 : 32x32. 0 : others.

Table 11-26 DC_CAV_BLK_CTRL7 0x001f

Bit(s)	R/W	Default	Description
255:224			blkmode. 1 : 32x32. 0 : others.

Table 11-27 DMC_MON_CTRL0 0x0020

Bit(s)	R/W	Default	Description
31			qos_mon_en. write 1 to trigger the enable. polling this bit 0, means finished. or use interrupt to check finish.
30			qos_mon interrupt clear. clear the qos monitor result. read 1 = qos mon finish interrupt.
3			qos monitor 3 enable.
2			qos monitor 2 enable.
1			qos monitor 1 enable.
0			qos monitor 0 enable.

Table 11-28 DMC_MON_CTRL1 0x0021

Bit(s)	R/W	Default	Description
23:0			qos monitor 0 channel select. 8 ambus port and 16 AXI port selection. 1 bit for one port.

Table 11-29 DMC_MON_CTRL2 0x0022

Bit(s)	R/W	Default	Description
15:0			port select for the selected channel.

Table 11-30 DMC_MON_CTRL3 0x0023

Bit(s)	R/W	Default	Description
23:0			qos monitor 0 channel select. 8 ambus port and 16 AXI port selection. 1 bit for one port.

Table 11-31 DMC_MON_CTRL4 0x0024

Bit(s)	R/W	Default	Description
15:0			port select for the selected channel.

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Table 11-32 DMC_MON_CTRL5 0x0025

Bit(s)	R/W	Default	Description
23:0			qos monitor 0 channel select. 8 ambus port and 16 AXI port selection. 1 bit for one port.

Table 11-33 DMC_MON_CTRL6 0x0026

Bit(s)	R/W	Default	Description
15:0			port select for the selected channel.

Table 11-34 DMC_MON_CTRL7 0x0027

Bit(s)	R/W	Default	Description
23:0			port select for the selected channel.

Table 11-35 DMC_MON_CTRL8 0x0028

Bit(s)	R/W	Default	Description
15:0			port select for the selected channel.

DMC_MON_ALL_REQ_CNT 0x0029

At the test period, the whole MMC request time.

DMC_MON_ALL_GRANT_CNT 0x002a

At the test period, the whole MMC granted data cycles. 64bits unit.

DMC_MON_ONE_GRANT_CNT 0x002b

At the test period, the granted data cycles for the selected channel and ports.

DMC_MON_SEC_GRANT_CNT 0x002c

At the test period, the granted data cycles for the selected channel and ports.

DMC_MON_THD_GRANT_CNT 0x002d

At the test period, the granted data cycles for the selected channel and ports.

DMC_MON_FOR_GRANT_CNT 0x002e

At the test period, the granted data cycles for the selected channel and ports.

DMC_MON_TIMER 0x002f

Timer for the monitor period.

Table 11-36 DMC_CLKG_CTRL0 0x0030

Bit(s)	R/W	Default	Description
23:16			Enable the 8 ambus interfaces both main and n_clk auto clock gating function. each 1 bit for one interface.
15:0			Enable the 16 axi interfaces both main and n_clk auto clock gating function. each 1 bit for one interface.

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Table 11-37 DMC_CLKG_CTRL1 0x0031

Bit(s)	R/W	Default	Description
23:16			Force to disable the 8 ambus interfaces both main and n_clk. each 1 bit for one interface.
15:0			Force to disable the 16 axi interfaces both main and n_clk. each 1 bit for one interface.

Table 11-38 DMC_CLKG_CTRL2 0x0032

Bit(s)	R/W	Default	Description
7			Force to disalbe the clock of write rsp generation.
6			Force to disalbe the clock of read rsp generation.
5			Force to disalbe the clock of command filter.
4			Force to disalbe the clock of write reorder buffer.
3			Force to disalbe the clock of write data buffer.
2			Force to disalbe the clock of read reorder buffer.
1			Force to disalbe the clock of read canvas.
0			Force to disalbe the clock of write canvas.

Table 11-39 DMC_CLKG_CTRL3 0x0033

Bit(s)	R/W	Default	Description
7			Enalbe auto clock gating for write rsp generation.
6			Enalbe auto clock gating for read rsp generation.
5			Enalbe auto clock gating for ddr0 command filter.
4			Enalbe auto clock gating for ddr0 write reorder buffer.
3			Enalbe auto clock gating for ddr0 write data buffer.
2			Enalbe auto clock gating for ddr0 read reorder buffer.
1			Enalbe auto clock gating for read canvas.
0			Enalbe auto clock gating for write canvas.

Table 11-40 DMC_CHAN_STS 0x0036

Bit(s)	R/W	Default	Description
27	RO		always 1
26	RO		ddr0 write data buffer idle. 1 : idle 0: busy.
25	RO		always 1.
24	RO		ddr0 wbuf idle. 1 : idle 0: busy.
23:16	RO		ambus channel idle. 1 : idle 0: busy.
15:0	RO		axibus channel idle. 1 : idle 0: busy.

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Table 11-41 DMC_2ARB_CTRL 0x0038

Bit(s)	R/W	Default	Description
32:24			Not used.
22.			always 1
21:12			ddr0 write data buffer idle. 1 : idle 0: busy.
11:6.			always 1.
5:0.			ddr0 wbuf idle. 1 : idle 0: busy.

Table 11-42 DMC_CMD_FILTER_CTRL1 0x0040

Bit(s)	R/W	Default	Description
30			1 : use DDR4 special filter.

DMC_CMD_FILTER_CTRL2 0x0041

Not used.

Table 11-43 DMC_CMD_FILTER_CTRL3 0x0042

Bit(s)	R/W	Default	Description
31.			force wbuf empty.
30:26			wbuf high level number
25:21			wbuf mid level number
20:16			wbuf low level number
14:10			rbuf high level number
9:5			rbuf middle level number
4:0			rbuf low level number

Table 11-44 DMC_CMD_FILTER_CTRL4 0x0043

Bit(s)	R/W	Default	Description
29:25			tITW.long
24:20			tITW. short
19:12			tAP auto precharge the bank not used if idle that long time.
11:6			write to read accesses if there write hit request.
5:0			read to write accesses if there write hit request.

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Table 11-45 DMC_CMD_FILTER_CTRL5 0x0044

Bit(s)	R/W	Default	Description
31:24			Once ddr data bus switch to read, the maxmum read command number to give up the bus when there's write request pending for write buffer.
23:16			Once ddr data bus switch to write, the maxmum write command number to give up the bus when there's read request pending too long.
15:8.			Once ddr data bus switch to read, the minimum read command number to transfer back to write stage if there's still pending read request.
7:0.			Once ddr data bus switch to write, the minimum write command number to transfer back to read stage if there's still pending write request.

Table 11-46 DMC_CMD_BUFFER_CTRL 0x0045

Bit(s)	R/W	Default	Description
31:26			Total write buffer number. Default 32.
25:20			Total read buffer number. Default 32.
19:8			Reserved
7:0			aw_pending_inc_num. incease write ugent level 1 when write command waiting to in write buffer that long.

Table 11-47 DMC_CMD_BUFFER_CTRL1 0x0046

Bit(s)	R/W	Default	Description
31:26			Read buffer number in non-urgent request.
25:20			Read buffer bank miss watch dog threshold.
19:8			Read buffer urgent level 3 counter inc weight.
7:0			Read buffer urgent level 2 counter inc weight.
3:0			Read buffer urgent level 2 counter inc weight.

Table 11-48 DMC_CMD_FILTER_CTRL6 0x0047

Bit(s)	R/W	Default	Description
31:24			write urgent 3 request pending hold num.
23:16			write urgent 2 request pending hold num.
15:8			write urgent 1 request pending hold num.
7:0			write urgent 0 request pending hold num.

DMC_RDDBUF_CTRL 0x0048

Not used.

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Table 11-49 DMC_CMD_FILTER_CTRL7 0x0049

Bit(s)	R/W	Default	Description
15:8			Write to read waiting cycles if there write hit request.
7:0			Read to write waiting cycles if there write hit request.

Table 11-50 DMC_AM0_CHAN_CTRL 0x0060

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non-urgent request.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-51 DMC_AM0_HOLD_CTRL 0x0061

Bit(s)	R/W	Default	Description
15:8			Read hold num. Max outstanding request number.
7:0			Read hold release num. If the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-52 DMC_AM0_CHAN_CTRL1 0x0062

Bit(s)	R/W	Default	Description
31			Side band signal used as block other request.
30			Side band urgent increase enable.
29			Side band urgent decrease urgent enable.
23:16			When bit 31 enabled, block the ambus related bits read request.
15:0			When bit 31 enabled, block the axi bus related bits read request.

Table 11-53 DMC_AM0_CHAN_CTRL2 0x0063

Bit(s)	R/W	Default	Description
31:24			Not used.
23:16			When side band signal used as block other request, and side bank signal is high, block the ambus related bits write request.
15:0			When side band signal used as block other request, and side bank signal is high, block the axi bus related bits write request.

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Table 11-54 DMC_AM1_CHAN_CTRL 0x0064

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non urgent request.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-55 DMC_AM1_HOLD_CTRL 0x0065

Bit(s)	R/W	Default	Description
15:8			Read hold num. Max outstanding request number.
7:0			Read hold release num. If the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-56 DMC_AM1_CHAN_CTRL1 0x0066

Bit(s)	R/W	Default	Description
31:			Side band signal used as block other request.
30 :			Side band urgent increase enable.
29 :			Side band urgent decrease urgent enable.
23:16			When bit 31 enabled, block the ambus related bits read request.
15:0			When bit 31 enabled, block the axi bus related bits read request.

Table 11-57 DMC_AM1_CHAN_CTRL2 0x0067

Bit(s)	R/W	Default	Description
31:24			Not used.
23:16			When side band signal used as block other request, and side bank signal is high, block the ambus related bits write request.
15:0			When side band signal used as block other request, and side bank signal is high, block the axi bus related bits write request.

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Table 11-58 DMC_AM2_CHAN_CTRL 0x0068

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non urgent request.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-59 DMC_AM2_HOLD_CTRL 0x0069

Bit(s)	R/W	Default	Description
15:8			Read hold num. Max outstanding request number.
7:0			Read hold release num. If the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-60 DMC_AM2_CHAN_CTRL1 0x006a

Bit(s)	R/W	Default	Description
31:			Side band signal used as block other request.
30 :			Side band urgent increase enable.
29 :			Side band urgent decrease urgent enable.
23:16			When bit 31 enabled, block the ambus related bits read request.
15:0			When bit 31 enabled, block the axi bus related bits read request.

Table 11-61 DMC_AM2_CHAN_CTRL1 0x006b

Bit(s)	R/W	Default	Description
31:24			Not used.
23:16			When side band signal used as block other request, and side bank signal is high, block the ambus related bits write request.
15:0			When side band signal used as block other request, and side bank signal is high, block the axi bus related bits write request.

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Table 11-62 DMC_AM3_CHAN_CTRL 0x006c

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non urgent request.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-63 DMC_AM3_HOLD_CTRL 0x006d

Bit(s)	R/W	Default	Description
31:24			Write hold num. Max outstanding request number.
23:16			Write hold release num. If the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-64 DMC_AM3_CHAN_CTRL1 0x006e

Bit(s)	R/W	Default	Description
31			Side band signal used as block other request.
30			Side band urgent increase enable.
29			Side band urgent decrease urgent enable.
23:16			When bit 31 enabled, block the ambus related bits read request.
15:0			When bit 31 enabled, block the axi bus related bits read request.

Table 11-65 DMC_AM3_CHAN_CTRL2 0x006f

Bit(s)	R/W	Default	Description
31:24			Not used.
23:16			When side band signal used as block other request, and side bank signal is high, block the ambus related bits write request.
15:0			When side band signal used as block other request, and side bank signal is high, block the axi bus related bits write request.

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Table 11-66 DMC_AM4_CHAN_CTRL 0x0070

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non urgent request.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-67 DMC_AM4_HOLD_CTRL 0x0071

Bit(s)	R/W	Default	Description
31:24			Write hold num. Max outstanding request number.
23:16			Write hold release num. If the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-68 DMC_AM4_CHAN_CTRL1 0x0072

Bit(s)	R/W	Default	Description
31:			Side band signal used as block other request.
30 :			Side band urgent increase enable.
29 :			Side band urgent decrease urgent enable.
23:16			When bit 31 enabled, block the ambus related bits read request.
15:0			When bit 31 enabled, block the axi bus related bits read request.

Table 11-69 DMC_AM4_CHAN_CTRL2 0x0073

Bit(s)	R/W	Default	Description
31:24			Not used.
23:16			When bit 31 enabled, block the ambus related bits write request.
15:0			When bit 31 enabled, block the axi bus related bits write request.

Table 11-70 DMC_AM5_CHAN_CTRL 0x0074

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.

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Bit(s)	R/W	Default	Description
29:20			Write request pending cycle number to inc urgent level if not granted.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non urgent request.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-71 DMC_AM5_HOLD_CTRL 0x0075

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-72 DMC_AM6_CHAN_CTRL 0x0078

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
18			Force this channel all request to be super urgent request.
17			Force this channel all request to be urgent request.
16			Force this channel all request to be non urgent request.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-73 DMC_AM6_HOLD_CTRL 0x0079

Bit(s)	R/W	Default	Description
31:24			Enable to incr 2 urgent levels if the pending cycles is doubled.
23:16			Enable to incr 3 urgent levels.
15:8			Write request pending cycle number to inc urgent level if not granted.
7:0			Force this channel all request to be super urgent request.

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Table 11-74 DMC_AM7_CHAN_CTRL 0x007c

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non urgent request.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-75 DMC_AM7_HOLD_CTRL 0x007d

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-76 DMC_AXI0_CHAN_CTRL 0x0080

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19			Axi0 default urgent control : 1 use AWUGT/ARUGT pins in the port. 0 : use bit [15:14] of this register.
18			Force this channel all request to be super urgent request.
17			Force this channel all request to be urgent request.
16			Force this channel all request to be non urgent request.
15:14			Axi0 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Arbiter weight

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Table 11-77 DMC_AXI0_HOLD_CTRL 0x0081

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-78 DMC_AXI0_CHAN_CTRL1 0x0082

Bit(s)	R/W	Default	Description
31:28			FIQ status
27:24			IRQ status.
22			use ACTIVE input as clock gating control.
21:20			chan0 QOS mode.
19:16			AXI0 QOS high limit.
15:12			AXI0 QOS mit limit.
11			ARM FIQ controlled super urgent enable.
10			ARM FIQ controlled urgent enable.
9			ARM IRQ controlled super urgent enable.
8			ARM IRQ controlled urgent enable.
7			IRQ/FIQ controll enable.
6:5			not used.
4			enable AXI0 auto urgent enable. When there's no other request, treat the AXI0 as super urgent request. other wise, use the bit3:0 to set the urgent.
3:2			A9 urgent if there's VIU request.
1:0			A9 urgent if there's request other than VIU

Table 11-79 DMC_AXI1_CHAN_CTRL 0x0084

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19.			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.

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Bit(s)	R/W	Default	Description
16.			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-80 DMC_AXI1_HOLD_CTRL 0x0085

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-81 DMC_AXI1_CHAN_CTRL1 0x0086

Bit(s)	R/W	Default	Description
31:28			FIQ status
27:24			IRQ status.
21:20			Mali QOS mode.
19:16			mail QOS high limit.
15:12			mail QOS mit limit.
11			ARM FIQ controlled super urgent enable.
10			ARM FIQ controlled urgent enable.
9			ARM IRQ controlled super urgent enable.
8			ARM IRQ controlled urgent enable.
7			IRQ/FIQ controll enable.
6:0			not used.

Table 11-82 DMC_AXI2_CHAN_CTRL 0x0088

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19.			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.

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Bit(s)	R/W	Default	Description
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4.			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-83 DMC_AXI2_HOLD_CTRL 0x0089

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-84 DMC_AXI3_CHAN_CTRL 0x008c

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19.			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-85 DMC_AXI3_HOLD_CTRL 0x008d

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

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Bit(s)	R/W	Default	Description
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-86 DMC_AXI4_CHAN_CTRL 0x0090

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19.			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18.			Force this channel all request to be super urgent request.
17.			Force this channel all request to be urgent request.
16.			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-87 DMC_AXI4_HOLD_CTRL 0x0091

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

DMC_AXI5_CHAN_CTRL 0x0094

Not used.

DMC_AXI5_HOLD_CTRL 0x0095

Table 11-88 DMC_AXI6_CHAN_CTRL 0x0096

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.

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Bit(s)	R/W	Default	Description
19			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18			Force this channel all request to be super urgent request.
17			Force this channel all request to be urgent request.
16			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-89 DMC_AXI6_HOLD_CTRL 0x0097

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-90 DMC_AXI7_CHAN_CTRL 0x009c

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18			Force this channel all request to be super urgent request.
17			Force this channel all request to be urgent request.
16			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

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Table 11-91 DMC_AXI7_HOLD_CTRL 0x009d

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-92 DMC_AXI8_CHAN_CTRL 0x00a0

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18			Force this channel all request to be super urgent request.
17			Force this channel all request to be urgent request.
16			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-93 DMC_AXI8_HOLD_CTRL 0x00a1

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-94 DMC_AXI9_CHAN_CTRL 0x00a4

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.

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Bit(s)	R/W	Default	Description
29:20			Write request pending cycle number to inc urgent level if not granted.
19			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18			Force this channel all request to be super urgent request.
17			Force this channel all request to be urgent request.
16			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

Table 11-95 DMC_AXI9_HOLD_CTRL 0x00a5

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-96 DMC_AXI10_CHAN_CTRL 0x00a8

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18			Force this channel all request to be super urgent request.
17			Force this channel all request to be urgent request.
16			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

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Table 11-97 DMC_AXI10_HOLD_CTRL 0x00a9

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-98 DMC_AXI10_CHAN_CTRL1 0x00aa

Bit(s)	R/W	Default	Description
11			Mali/NNA channel FIQ controlled super urgent enable.
10			Mali/NNA channel FIQ controlled urgent enable.
9			Mali/NNA channel IRQ controlled super urgent enable.
8			Mali/NNA channel IRQ controlled urgent enable.
7			IRQ/FIQ controll enable.
6:0			Not used.

Table 11-99 DMC_AXI11_CHAN_CTRL 0x00ac

Bit(s)	R/W	Default	Description
31			Enable to incr 2 urgent levels if the pending cycles is doubled.
30			Enable to incr 3 urgent levels.
29:20			Write request pending cycle number to inc urgent level if not granted.
19			Axi0 default urgent control: 1 use AWUGT/ARUGT pins in the port. 0: use bit [15:14] of this register.
18			Force this channel all request to be super urgent request.
17			Force this channel all request to be urgent request.
16			Force this channel all request to be non-urgent request.
15:14			Axi1 default urgent level.
13:4			Read request pending cycle number to inc urgent level if not granted.
3:0			Canvas arbiter arbiter weight

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Table 11-100 DMC_AXI11_HOLD_CTRL 0x00ad

Bit(s)	R/W	Default	Description
31:24			Write hold num. max outstanding request number.
23:16			Write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.
15:8			Read hold num. max outstanding request number.
7:0			Read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number below the hold release number, then continue to request.

Table 11-101 DMC_AXI12_CHAN_CTRL 0x00ae

Bit(s)	R/W	Default	Description
31			enable to incr 2 urgent levels if the pending cycles is doubled.
30			enable to incr 3 urgent levels.
29:20			write request pending cycle number to inc urgent level if not granted.
19			axi0 default urgent control : 1 use AWUGT/ARUGT pins in the port. 0 : use bit [15:14] of this register.
18			force this channel all request to be super urgent request.
17			force this channel all request to be urgent request.
16			force this channel all request to be non urgent request.
15:14			axi1 default urgent level.
13:4			read request pending cycle number to inc urgent level if not granted.
3:0			canvas arbiter arbiter weight

Table 11-102 DMC_AXI12_HOLD_CTRL 0x00af

Bit(s)	R/W	Default	Description
31:24			write hold num. max outstanding request number.
23:16			write hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number bellow the hold release number, then continue to request.
15:8			read hold num. max outstanding request number.
7:0			read hold release num. if the outstanding request == hold num, then hold this request unless the outstanding request number bellow the hold release number, then continue to request.

The following registers' base address is 0xff639000. Each register takes 4 byte address.

Each register's final address = 0xff639000 + offset * 4.

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Table 11-103 DMC_SEC_RANGE_CTRL 0x0000

Bit(s)	R/W	Default	Description
31:16			16 range security level. each
15:0			16 range enable. each bit for one range to indentify the range is enabled or not.

Table 11-104 DMC_SEC_RANGE0_CTRL 0x0001

Bit(s)	R/W	Default	Description
31:16			range 0 end address higher 16bits.
15:0			range 0 start address higher 16bits.

DMC_SEC_RANGE1_CTRL 0x0002

DMC_SEC_RANGE2_CTRL 0x0003

DMC_SEC_RANGE3_CTRL 0x0004

DMC_SEC_RANGE4_CTRL 0x0005

DMC_SEC_RANGE5_CTRL 0x0006

DMC_SEC_RANGE6_CTRL 0x0007

DMC_SEC_RANGE7_CTRL 0x0008

DMC_SEC_RANGE8_CTRL 0x0009

DMC_SEC_RANGE9_CTRL 0x000a

DMC_SEC_RANGE10_CTRL 0x000b

DMC_SEC_RANGE11_CTRL 0x000c

DMC_SEC_RANGE12_CTRL 0x000d

DMC_SEC_RANGE13_CTRL 0x000e

DMC_SEC_RANGE14_CTRL 0x000f

Table 11-105 DMC_SEC_RANGE0_RID_CTRL0 0x0010

Bit(s)	R/W	Default	Description
31:0			range 0 end address higher 16bits.

Table 11-106 DMC_SEC_RANGE0_RID_CTRL1 0x0011

Bit(s)	R/W	Default	Description
63:32			range 0 end address higher 16bits.

Table 11-107 DMC_SEC_RANGE0_RID_CTRL2 0x0012

Bit(s)	R/W	Default	Description
95:64			range 0 end address higher 16bits.

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Table 11-108 DMC_SEC_RANGE0_RID_CTRL3 0x0013

Bit(s)	R/W	Default	Description
127:96			range 0 end address higher 16bits.

DMC_SEC_RANGE1_RID_CTRL0 0x0014
DMC_SEC_RANGE1_RID_CTRL1 0x0015
DMC_SEC_RANGE1_RID_CTRL2 0x0016
DMC_SEC_RANGE1_RID_CTRL3 0x0017

DMC_SEC_RANGE2_RID_CTRL0 0x0018
DMC_SEC_RANGE2_RID_CTRL1 0x0019
DMC_SEC_RANGE2_RID_CTRL2 0x001a
DMC_SEC_RANGE2_RID_CTRL3 0x001b

DMC_SEC_RANGE3_RID_CTRL0 0x001c
DMC_SEC_RANGE3_RID_CTRL1 0x001d
DMC_SEC_RANGE3_RID_CTRL2 0x001e
DMC_SEC_RANGE3_RID_CTRL3 0x001f

DMC_SEC_RANGE4_RID_CTRL0 0x0020
DMC_SEC_RANGE4_RID_CTRL1 0x0021
DMC_SEC_RANGE4_RID_CTRL2 0x0022
DMC_SEC_RANGE4_RID_CTRL3 0x0023

DMC_SEC_RANGE5_RID_CTRL0 0x0024
DMC_SEC_RANGE5_RID_CTRL1 0x0025
DMC_SEC_RANGE5_RID_CTRL2 0x0026
DMC_SEC_RANGE5_RID_CTRL3 0x0027

DMC_SEC_RANGE6_RID_CTRL0 0x0028
DMC_SEC_RANGE6_RID_CTRL1 0x0029
DMC_SEC_RANGE6_RID_CTRL2 0x002a
DMC_SEC_RANGE6_RID_CTRL3 0x002b

DMC_SEC_RANGE7_RID_CTRL0 0x002c
DMC_SEC_RANGE7_RID_CTRL1 0x002d
DMC_SEC_RANGE7_RID_CTRL2 0x002e
DMC_SEC_RANGE7_RID_CTRL3 0x002f

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DMC_SEC_RANGE8_RID_CTRL0 0x0030
DMC_SEC_RANGE8_RID_CTRL1 0x0031
DMC_SEC_RANGE8_RID_CTRL2 0x0032
DMC_SEC_RANGE8_RID_CTRL3 0x0033

DMC_SEC_RANGE9_RID_CTRL0 0x0034
DMC_SEC_RANGE9_RID_CTRL1 0x0035
DMC_SEC_RANGE9_RID_CTRL2 0x0036
DMC_SEC_RANGE9_RID_CTRL3 0x0037

DMC_SEC_RANGE10_RID_CTRL0 0x0038
DMC_SEC_RANGE10_RID_CTRL1 0x0039
DMC_SEC_RANGE10_RID_CTRL2 0x003a
DMC_SEC_RANGE10_RID_CTRL3 0x003b

DMC_SEC_RANGE11_RID_CTRL0 0x003c
DMC_SEC_RANGE11_RID_CTRL1 0x003d
DMC_SEC_RANGE11_RID_CTRL2 0x003e
DMC_SEC_RANGE11_RID_CTRL3 0x003f

DMC_SEC_RANGE12_RID_CTRL0 0x0040
DMC_SEC_RANGE12_RID_CTRL1 0x0041
DMC_SEC_RANGE12_RID_CTRL2 0x0042
DMC_SEC_RANGE12_RID_CTRL3 0x0043

DMC_SEC_RANGE13_RID_CTRL0 0x0044
DMC_SEC_RANGE13_RID_CTRL1 0x0045
DMC_SEC_RANGE13_RID_CTRL2 0x0046
DMC_SEC_RANGE13_RID_CTRL3 0x0047

DMC_SEC_RANGE14_RID_CTRL0 0x0048
DMC_SEC_RANGE14_RID_CTRL1 0x0049
DMC_SEC_RANGE14_RID_CTRL2 0x004a
DMC_SEC_RANGE14_RID_CTRL3 0x004b

DMC_SEC_RANGE15_RID_CTRL0 0x004c
DMC_SEC_RANGE15_RID_CTRL1 0x004d

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DMC_SEC_RANGE15_RID_CTRL2 0x004e
DMC_SEC_RANGE15_RID_CTRL3 0x004f

DMC_SEC_RANGE0_WID_CTRL0 0x0050
DMC_SEC_RANGE0_WID_CTRL1 0x0051
DMC_SEC_RANGE0_WID_CTRL2 0x0052
DMC_SEC_RANGE0_WID_CTRL3 0x0053

DMC_SEC_RANGE1_WID_CTRL0 0x0054
DMC_SEC_RANGE1_WID_CTRL1 0x0055
DMC_SEC_RANGE1_WID_CTRL2 0x0056
DMC_SEC_RANGE1_WID_CTRL3 0x0057

DMC_SEC_RANGE2_WID_CTRL0 0x0058
DMC_SEC_RANGE2_WID_CTRL1 0x0059
DMC_SEC_RANGE2_WID_CTRL2 0x005a
DMC_SEC_RANGE2_WID_CTRL3 0x005b

DMC_SEC_RANGE3_WID_CTRL0 0x005c
DMC_SEC_RANGE3_WID_CTRL1 0x005d
DMC_SEC_RANGE3_WID_CTRL2 0x005e
DMC_SEC_RANGE3_WID_CTRL3 0x005f

DMC_SEC_RANGE4_WID_CTRL0 0x0060
DMC_SEC_RANGE4_WID_CTRL1 0x0061
DMC_SEC_RANGE4_WID_CTRL2 0x0062
DMC_SEC_RANGE4_WID_CTRL3 0x0063

DMC_SEC_RANGE5_WID_CTRL0 0x0064
DMC_SEC_RANGE5_WID_CTRL1 0x0065
DMC_SEC_RANGE5_WID_CTRL2 0x0066
DMC_SEC_RANGE5_WID_CTRL3 0x0067

DMC_SEC_RANGE6_WID_CTRL0 0x0068
DMC_SEC_RANGE6_WID_CTRL1 0x0069
DMC_SEC_RANGE6_WID_CTRL2 0x006a
DMC_SEC_RANGE6_WID_CTRL3 0x006b

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DMC_SEC_RANGE7_WID_CTRL0 0x006c
DMC_SEC_RANGE7_WID_CTRL1 0x006d
DMC_SEC_RANGE7_WID_CTRL2 0x006e
DMC_SEC_RANGE7_WID_CTRL3 0x006f

DMC_SEC_RANGE8_WID_CTRL0 0x0070
DMC_SEC_RANGE8_WID_CTRL1 0x0071
DMC_SEC_RANGE8_WID_CTRL2 0x0072
DMC_SEC_RANGE8_WID_CTRL3 0x0073

DMC_SEC_RANGE9_WID_CTRL0 0x0074
DMC_SEC_RANGE9_WID_CTRL1 0x0075
DMC_SEC_RANGE9_WID_CTRL2 0x0076
DMC_SEC_RANGE9_WID_CTRL3 0x0077

DMC_SEC_RANGE10_WID_CTRL0 0x0078
DMC_SEC_RANGE10_WID_CTRL1 0x0079
DMC_SEC_RANGE10_WID_CTRL2 0x007a
DMC_SEC_RANGE10_WID_CTRL3 0x007b

DMC_SEC_RANGE11_WID_CTRL0 0x007c
DMC_SEC_RANGE11_WID_CTRL1 0x007d
DMC_SEC_RANGE11_WID_CTRL2 0x007e
DMC_SEC_RANGE11_WID_CTRL3 0x007f

DMC_SEC_RANGE12_WID_CTRL0 0x0080
DMC_SEC_RANGE12_WID_CTRL1 0x0081
DMC_SEC_RANGE12_WID_CTRL2 0x0082
DMC_SEC_RANGE12_WID_CTRL3 0x0083

DMC_SEC_RANGE13_WID_CTRL0 0x0084
DMC_SEC_RANGE13_WID_CTRL1 0x0085
DMC_SEC_RANGE13_WID_CTRL2 0x0086
DMC_SEC_RANGE13_WID_CTRL3 0x0087

DMC_SEC_RANGE14_WID_CTRL0 0x0088
DMC_SEC_RANGE14_WID_CTRL1 0x0089
DMC_SEC_RANGE14_WID_CTRL2 0x008a

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DMC_SEC_RANGE14_WID_CTRL3 0x008b

DMC_SEC_RANGE15_WID_CTRL0 0x008c

DMC_SEC_RANGE15_WID_CTRL1 0x008d

DMC_SEC_RANGE15_WID_CTRL2 0x008e

DMC_SEC_RANGE15_WID_CTRL3 0x008f

Table 11-109 DMC_DES_CTRL 0x0095

Bit(s)	R/W	Default	Description
1			DES enable. 1: DES enable. 0: DES disable. default is 1.
0			DES register mask. if write 1 only. after write 1, DES_CTRL, DES_KEY, DES_ padding, and CFG_CA_REMAP regsiter can't be write and read.

Table 11-110 DMC_CA_REMAP_L 0x009b

Bit(s)	R/W	Default	Description
63:60			New address for index 15
59:56			New address for index 14
55:52			New address for index 13
51:48			New address for index 12
47:44			New address for index 11
43:40			New address for index 10
39:36			New address for index 9
35:32			New address for index 8
31:28			New address for index 7
27:24			New address for index 6
23:20			New address for index 5
19:16			New address for index 4
15:12			New address for index 3
11:8			New address for index 2
7:4			New address for index 1
3:0			New address for index 0

Table 11-111 DMC_CA_REMAP_H 0x009c

Bit(s)	R/W	Default	Description
63:60			New address for index 15
59:56			New address for index 14
55:52			New address for index 13
51:48			New address for index 12

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Bit(s)	R/W	Default	Description
47:44			New address for index 11
43:40			New address for index 10
39:36			New address for index 9
35:32			New address for index 8
31:28			New address for index 7
27:24			New address for index 6
23:20			New address for index 5
19:16			New address for index 4
15:12			New address for index 3
11:8			New address for index 2
7:4			New address for index 1
3:0			New address for index 0

Table 11-112 DMC_PROT0_RANGE 0x00a0

Bit(s)	R/W	Default	Description
31:16			Range end address.
15:0			Range end address.

Table 11-113 DMC_PROT0_CTRL 0x00a1

Bit(s)	R/W	Default	Description
26			protection 0 read access protection enable.
25			Protection 0 write access block function. If enabled, the access wouldn't write to the ddr sdram. If not enabled only generate a interrupt, but the access still wrote to ddr.
24			Protection range 0 write access protection enable.
23:16			Each bit to enable one of the 8 ambus channel for the protection function.
15:0			Each bit to enable one of the 15 channel input for the protection function.

Table 11-114 DMC_PROT1_RANGE 0x00a2

Bit(s)	R/W	Default	Description
31:16			Range end address.
15:0			Range end address.

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Table 11-115 DMC_PROT1_CTRL 0x00a3

Bit(s)	R/W	Default	Description
26			Protection range 1 read access protection enable bit.
25			Protection 1 write access block function. If enabled, the access wouldn't write to the ddr sdram. If not enabled only generate a interrupt, but the access still wrote to ddr.
24			Protection range 1 write access protection enable bit.
23:16			Each bit to enable one of the 8 ambus channel for the protection function.
15:0			Each bit to enable one of the 15 channel input for the protection function.

DMC_WTCH0_D0 0x00a4

DMC_WTCH0_D1 0x00a5

DMC_WTCH0_D2 0x00a6

DMC_WTCH0_D3 0x00a7

Table 11-116 DMC_WTCH0_RANGE 0x00a8

Bit(s)	R/W	Default	Description
31:16			Start address high 16
15:0			Start address high 16

Table 11-117 DMC_WTCH0_CTRL 0x00a9

Bit(s)	R/W	Default	Description
23:0			8 ambus and 16 axibus input channels select.

Table 11-118 DMC_WTCH0_CTRL1 0x00aa

Bit(s)	R/W	Default	Description
31:16			Start address high 16.
2			Watch point 0 enable.
1:0			Watch point0 type. 2'b00: double bytes. Only watchpoint data 15:0 and data strb 1:0 is valid. 2'b01: 4 bytes. 2'b10: 8 bytes. 2'b11, all 16bytes.

DMC_WTCH1_D0 0x00ab

DMC_WTCH1_D1 0x00ac

DMC_WTCH1_D2 0x00ad

DMC_WTCH1_D3 0x00ae

Table 11-119 DMC_WTCH1_RANGE 0x00af

Bit(s)	R/W	Default	Description
15:0			Start address high 16

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Table 11-120 DMC_WTCH1_CTRL 0x00b0

Bit(s)	R/W	Default	Description
23:0			8 ambus and 16 axibus input channels select.

Table 11-121 DMC_WTCH1_CTRL1 0x00b1

Bit(s)	R/W	Default	Description
31:16			Start address high 16.
2			Watch point 0 enable.
1:0			Watch point0 type. 2'b00: double bytes. Only watchpoint data 15:0 and data strb 1:0 is valid. 2'b01: 4 bytes. 2'b10: 8 bytes. 2'b11, all 16bytes.

Table 11-122 DMC_TRAP0_RANGE 0x00b2

Bit(s)	R/W	Default	Description
31:16			Trap0 end address
2			Start0 address.

Table 11-123 DMC_TRAP0_CTRL 0x00b3

Bit(s)	R/W	Default	Description
31			Trap0 port ID 3 enable.
30			Trap0 port ID 2 enable.
29			Trap0 port ID 1 enable.
28			Trap0 port ID 0 enable.
27			Trap0 port ID 3 subid enable.
26			Trap0 port ID 2 subid enable.
25			Trap0 port ID 1 subid enable.
24			Trap0 port ID 0 subid enable.
16:20			Trap0 port ID 1 ID number.
14:11			Trap0 port ID 1 subid ID number.
8:4			Trap0 port ID 0 ID number.
3:0			Trap0 port ID 0 subid ID number.

Table 11-124 DMC_TRAP0_CTRL2 0x00b4

Bit(s)	R/W	Default	Description
31:17			Not used.
16:20			Trap0 port ID 3 ID number.
14:11			Trap0 port ID 3 subid ID number.

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Bit(s)	R/W	Default	Description
8:4.			Trap0 port ID 2 ID number.
3:0			Trap0 port ID 2 subid ID number.

Table 11-125 DMC_TRAP1_RANGE 0x00b5

Bit(s)	R/W	Default	Description
31:16			Trap end address
15:0			Start address

Table 11-126 DMC_TRAP1_CTRL 0x00b6

Bit(s)	R/W	Default	Description
31			Trap0 port ID 3 enable.
30			Trap0 port ID 2 enable.
29			Trap0 port ID 1 enable.
28			Trap0 port ID 0 enable.
27			Trap0 port ID 3 subid enable.
26			Trap0 port ID 2 subid enable.
25			Trap0 port ID 1 subid enable.
24			Trap0 port ID 0 subid enable.
16:20			Trap0 port ID 1 ID number.
14:11			Trap0 port ID 1 subid ID number.
8:4			Trap0 port ID 0 ID number.
3:0			Trap0 port ID 0 subid ID number.

Table 11-127 DMC_TRAP1_CTRL2 0x00b7

Bit(s)	R/W	Default	Description
31:17			Not used.
16:20			Trap1 port ID 1 ID number.
14:11			Trap1 port ID 1 subid ID number.
8:4.			Trap1 port ID 0 ID number.
3:0.			Trap1 port ID 0 subid ID number.

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Table 11-128 DMC_SEC_STATUS 0x00b8

Bit(s)	R/W	Default	Description
31:2			Not used.
1			Write security violation
0			Read security violation.

DMC_VIO_ADDR0 0x00b9

Table 11-129 DMC_VIO_ADDR1 0x00ba

Bit(s)	R/W	Default	Description
31:24			Not used. Always 0.
24			ddr0 secure check violation.
23			ddr0 protection 1 vilation.
22			ddr0 protection 0 vilation.
21			ddr0 watch 1 catch
20			ddr0 watch 0 catch.
19			ddr0 write address overflow. write out of DDR size.
18:16			ddr0 write violation AWPROT bits.
15:00			ddr0_write violation ID.bit 15:11 is for port ID. bit 9:6 for subport ID.

DMC_VIO_ADDR2 0x00bb

Table 11-130 DMC_VIO_ADDR3 0x00bc

Bit(s)	R/W	Default	Description
24			ddr0 read secure check violation.
23			ddr0 read protection 1 violation.
22			ddr0 read protection 0 violation.
21			ddr0 read trap1 violation
20			ddr0 read trap0 violation
19			ddr 0 read address overflow. write out of DDR size.
18:16			ddr 0 read violation ARPROT bits.
15:0			ddr 0 read violation ID. bit 15:11 is for port ID. bit 9:6 for subport ID.

Table 11-131 DDR0_ADDRMAP_0 0x00d0

Bit(s)	R/W	Default	Description
29:25			ca8
24:20			ca7
19:15			ca6

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Bit(s)	R/W	Default	Description
14:10			ca5
9:5			ca4
4:0			ca3

Table 11-132 DDR0_ADDRMAP_1 0x00d1

Bit(s)	R/W	Default	Description
29:25			ra2
24:20			ra1
19:15			ra0
14:10			ca11
9:5			ca10
4:0			ca9

Table 11-133 DDR0_ADDRMAP_2 0x00d2

Bit(s)	R/W	Default	Description
29:25			ra8
24:20			ra7
19:15			ra6
14:10			ra5
9:5			ra4
4:0			ra3

Table 11-134 DDR0_ADDRMAP_3 0x00d3

Bit(s)	R/W	Default	Description
29:25			ra14
24:20			ra13
19:15			ra12
14:10			ra11
9:5			ra10
4:0			ra9

Table 11-135 DDR0_ADDRMAP_4 0x00d4

Bit(s)	R/W	Default	Description
29:25			ra16 for DDR4 SDRAM
24:20			bg1 for DDR4 SDRAM.

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Bit(s)	R/W	Default	Description
19:15			ba2. or bg0 for DDR4.
14:10			ba1.
9:5			ba0.
4:0			ra15.

Table 11-136 DDR1_ADDRMAP_0 0x00d5

Bit(s)	R/W	Default	Description
29:25			ca8
24:20			ca7
19:15			ca6
14:10			ca5
9:5			ca4
4:0			ca3

Table 11-137 DDR1_ADDRMAP_1 0x00d6

Bit(s)	R/W	Default	Description
29:25			ra2
24:20			ra1
19:15			ra0
14:10			ca11
9:5			ca10
4:0			ca9

Table 11-138 DDR1_ADDRMAP_2 0x00d7

Bit(s)	R/W	Default	Description
29:25			ra8
24:20			ra7
19:15			ra6
14:10			ra5
9:5			ra4
4:0			ra3

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Table 11-139 DDR1_ADDRMAP_3 0x00d8

Bit(s)	R/W	Default	Description
29:25			ra14
24:20			ra13
19:15			ra12
14:10			ra11
9:5			ra10
4:0			ra9

Table 11-140 DDR1_ADDRMAP_4 0x00d9

Bit(s)	R/W	Default	Description
29:25			ra16 for DDR4 SDRAM
24:20			bg1 for DDR4 SDRAM.
19:15			ba2. or bg0 for DDR4 SDRAM.
14:10			ba1.
9:5			ba0.
4:0			ra15.

Table 11-141 DMC_DDR_CTRL 0x00da

Bit(s)	R/W	Default	Description
31:28			16bit selection for DDR3/4 not balanced mode. 4'b1000: 3Gbyte mode. low 2Gbyte is in 32bits mode. 2G~3G is 16bits mode. 4'b0100: 1.5Gbyte mode. low 1Gbyte is in 32bits mode. 1G~1.5G is in 16bits mode. 4'b0010: 768Mbyte mode. low 512Mbyte is in 32bits mode. 512M~768M is in 16bits mode. 4'b0001: 384Mbyte mode. low 25Mbyte is in 32bits mode. 256M~384M is in 16bits m0doe. 4'b0000: balance mode. ether 32bits mode or 16bits mode depends on bit
27			0 : cavnas use 64bytes boundary 1 : canvas use 32bytes boundary.
24:22			3'b000 : ddr3 mode. 3'b001 : ddr4 mode. 3'b010 : lpddr3 mode. 3'b011 : lpddr4 mode.
21			rank1 enable bit. if 1, rank1 used the address map is as bit 5:3 defined.
20			DDR4 BG1 enable bit.
18			always 0.
16			1 only use 16bits data in a 32bits phy data interface. 0 : normal data interface.

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Bit(s)	R/W	Default	Description
7:4			DDR rank 1 size. 4'b0000 : DDR rank 1 is 128Mbyte. 4'b0001 : DDR rank 1 is 256Mbyte. 4'b0010 : DDR rank 1 is 512Mbyte. 4'b0011 : DDR rank 1 is 1Gbyte. 4'b0100 : DDR rank 1 is 2Gbyte. 4'b1000 : DDR rank 1 is 192Mbyte. 4'b1001 : DDR rank 1 is 374Mbyte. 4'b1010 : DDR rank 1 is 768Mbyte. 4'b1011 : DDR rank 1 is 1.5Gbyte. 4'b1100 : DDR rank 1 is 3Gbyte. others : reserved.
3:0			DDR rank 0 size. 4'b0000 : DDR rank 0 is 128Mbyte. 4'b0001 : DDR rank 0 is 256Mbyte. 4'b0010 : DDR rank 0 is 512Mbyte. 4'b0011 : DDR rank 0 is 1Gbyte. 4'b0100 : DDR rank 0 is 2Gbyte. 4'b1000 : DDR rank 0 is 192Mbyte. 4'b1001 : DDR rank 0 is 374Mbyte. 4'b1010 : DDR rank 0 is 768Mbyte. 4'b1011 : DDR rank 0 is 1.5Gbyte. 4'b1100 : DDR rank 0 is 3Gbyte. others : reserved.

Table 11-142 DDR_APB_SEC_CTRL 0x00db

Bit(s)	R/W	Default	Description
19:16			DMC normal APB register secure control.
19:			1: all can write those register. 0: the APB_PROT[0] must match the bit 0 to write those regsiter.
15:12			DMC sticky APB regsiter secure control.
15.			1: All APB bus can write those registers. 0: The APB_PROT[0] must match the bit 12 to write those register.
11:8.			DMC DDR PLL clock related APB regsiter secure control.
11.			1: All APB bus can write those registers. 0: The APB_PROT[0] must match the bit 8 to write those register.
7:4.			DMC DDR SDRAM protocal contorl register contorl
11.			1: All APB bus can write those registers. 0: The APB_PROT[0] must match the bit 4 to write those register.
3:0.			LPDDR4 PHY APB regsiter secure control.
3			1: All APB bus can write those registers. 0: The APB_PROT[0] must match the bit 0 to write those register.

Table 11-143 DDR_MEM_PD_CTRL 0x00e0

Bit(s)	R/W	Default	Description
31:14			Not used.
13:12			wc_fifo
11:10			wd_mem
9:8			rd_mem
7:6			dfifo
5:4			cfifo

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Bit(s)	R/W	Default	Description
3:2			cav_ram
1:0			sticky ram

DMC_TEST_WRCMD_ADDR 0x0000

DMC_TEST_RDRSP_ADDR 0x0001

DMC_TEST_RDCMD_ADDR 0x0002

Table 11-144 DMC_TEST_WDG 0x0003

Bit(s)	R/W	Default	Description
31:16			Write response watch dog.
15:0			Read response watch dog.

DMC_TEST_STA 0x0004

DMC_TEST_EDA 0x0005

Table 11-145 DMC_TEST_CTRL 0x0006

Bit(s)	R/W	Default	Description
31			enable test.
30			when enable test, enable the write to DDR function.
29			when enable test, enable the read from DDR function.
28			when enable test, enable the sha calculation function must be same as read enable but without write function.
27			enabe to compare data. when do the read enable to enable the error comparaion. suppose the read data should be same as the data in the write buffer.
26			reserved.
25			address generation type.
24			0 : use the DMC_TEST_NUM register as the counter of test numbers.
23			1 : the first write is {WD3, WD2,WD1,WD0}, then the latter is the previous data plus a pattern.({ + WD7, + WD6, + WD5, + WD4}).
23			reserved.
22:20			read repeat times. for non-sha function, we can define multi times of the read. the test module would repeat the same adddress repeat times.
19			limit write.
18			limit read.
17:16			sha mode for sha function enabled.
15:8.			write outstanding commands limit.
7:0			read outstanding commands limit.

DMC_TEST_NUM 0x0007

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DMC_TEST_WD0 0x0010
DMC_TEST_WD1 0x0011
DMC_TEST_WD2 0x0012
DMC_TEST_WD3 0x0013
DMC_TEST_WD4 0x0014
DMC_TEST_WD5 0x0015
DMC_TEST_WD6 0x0016
DMC_TEST_WD7 0x0017
DMC_TEST_WD8 0x0018
DMC_TEST_WD9 0x0019
DMC_TEST_WD10 0x001a
DMC_TEST_WD11 0x001b
DMC_TEST_WD12 0x001c
DMC_TEST_WD13 0x001d
DMC_TEST_WD14 0x001e
DMC_TEST_WD15 0x001f
DMC_TEST_RD0 0x0020
DMC_TEST_RD1 0x0021
DMC_TEST_RD2 0x0022
DMC_TEST_RD3 0x0023
DMC_TEST_RD4 0x0024
DMC_TEST_RD5 0x0025
DMC_TEST_RD6 0x0026
DMC_TEST_RD7 0x0027
DMC_TEST_RD8 0x0028
DMC_TEST_RD9 0x0029
DMC_TEST_RD10 0x002a
DMC_TEST_RD11 0x002b
DMC_TEST_RD12 0x002c
DMC_TEST_RD13 0x002d
DMC_TEST_RD14 0x002e
DMC_TEST_RD15 0x002f
DMC_TEST_ERR_ADDR 0x0030
DMC_TEST_ERR_CNT 0x0031

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Table 11-146 DMC_TEST_STS 0x0032

Bit(s)	R/W	Default	Description
31			test done bit. write 1 to clean.
30			indicate address err
29~7			not used.
6			read data resp error(caused by security or rd latency).
5			test MRR/MPR rd latency error. write 1 clear
4			sha done. write 1 to clean.
3			write done. write 1 to clean.
2			read done. write 1 to clean
1			write watchdog triggered. write 1 to clean
0			read watchdog triggered. write 1 to clean.

DMC_TEST_COMP_MASK 0x0033

1 : to MASK this bit. 0: compare this bit.

Table 11-147 DMC_TEST_RDBI0 0x0034

Bit(s)	R/W	Default	Description
31:16			the second cycle.
15:0			the first cycle.

Table 11-148 DMC_TEST_RDBI1 0x0035

Bit(s)	R/W	Default	Description
31:16			Forth cycle.
15:0			third cycle.

Table 11-149 DMC_TEST_WSTRB0 0x0036

Bit(s)	R/W	Default	Description
31:16			the second cycle.
15:0			the first cycle.

Table 11-150 DMC_TEST_WSTRB1 0x0037

Bit(s)	R/W	Default	Description
31:16			Forth cycle.
15:0			third cycle.

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Table 11-151 DMC_TEST_DRAM_CMD 0x0038

Bit(s)	R/W	Default	Description
31			cmd done. write 0 to clean.
30			data done. write 0 to clean.
4:0			only one bit can be 1. read data stored in DMC_TEST_RD* write data from DMC_TEST_W
4			LPDDR4 MPC write data command(MPC WR FIFO).
3			LPDDR4 MPC read data command (MPC RD Calibration and RD FIFO).
2			LPDDR4 MPC-1 command (NOP, Start DQS interval ....)
1			mrr comand.
0			mrw command.

Table 11-152 DMC_TEST_DRAM_CMD_CODE 0x0039

Bit(s)	R/W	Default	Description
27:26			128bits data cycles . 0: 1 clock cycles; 1: 2 clock cycles; 2: 3 clock cycles; 3:4 clock cycles.
25			MRW/MRR/MPC command rank 1 select. 1: select. 0: not select.
24.			MRW/MRR/MPC command rank 0 select. 1: select. 0: not select.
23:16			MR addr. DDR4 case : 18:16 ba[2:0]. 20:19 BG[1:0].
15:0			opcode.

Table 11-153 DMC_TEST_DRAM_CMD_TIME 0x003a

Bit(s)	R/W	Default	Description
31:16			PRE CMD timer. delay how many cycle to start the command.
15:0			POST CMD timer delay how many cycle after the command execute.

DMC_TEST_SHA_MSG0 0x0040

DMC_TEST_SHA_MSG1 0x0041

DMC_TEST_SHA_MSG2 0x0042

DMC_TEST_SHA_MSG3 0x0043

DMC_TEST_SHA_MSG4 0x0044

DMC_TEST_SHA_MSG5 0x0045

DMC_TEST_SHA_MSG6 0x0046

DMC_TEST_SHA_MSG7 0x0047

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Table 11-154 DMC_DRAM_TMRD 0x0000

Bit(s)	R/W	Default	Description
3:0			tMRD.

Table 11-155 DMC_DRAM_TRFC 0x0001

Bit(s)	R/W	Default	Description
9:0			tRFC

Table 11-156 DMC_DRAM_TRP 0x0002

Bit(s)	R/W	Default	Description
21:16			tRP for precharge all banks.
5:0			tRP for precharge one bank.

Table 11-157 DMC_DRAM_TRTW 0x0003

Bit(s)	R/W	Default	Description
5:0			tRTW

Table 11-158 DMC_DRAM_TCL 0x0004

Bit(s)	R/W	Default	Description
5:0			CL/tRL. read latency.

Table 11-159 DMC_DRAM_TCWL 0x0005

Bit(s)	R/W	Default	Description
5:0			CWL: write latency.

Table 11-160 DMC_DRAM_TRAS 0x0006

Bit(s)	R/W	Default	Description
7:0			tRAS. minimum active to precharge time for same bank.

Table 11-161 DMC_DRAM_TRC 0x0007

Bit(s)	R/W	Default	Description
7:0			tRC. minimum active to active time for same bank.

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Table 11-162 DMC_DRAM_TRCD 0x0008

Bit(s)	R/W	Default	Description
7:0			tRCD active to read/write timing for same bank.

Table 11-163 DMC_DRAM_TRRD 0x0009

Bit(s)	R/W	Default	Description
21:16			tRRD_l active bank A to active B in same band group for DDR4.
5:0			tRRD/tRRD_s active bank A to active bank b time.

Table 11-164 DMC_DRAM_TFAW 0x000a

Bit(s)	R/W	Default	Description
8:0			tFAW. four active command windows

Table 11-165 DMC_DRAM_TRTP 0x000b

Bit(s)	R/W	Default	Description
5:0			tRTP.

Table 11-166 DMC_DRAM_TWR 0x000c

Bit(s)	R/W	Default	Description
5:0			tWR.

Table 11-167 DMC_DRAM_TWTR 0x000d

Bit(s)	R/W	Default	Description
5:0			tWTR.

Table 11-168 DMC_DRAM_TCCD 0x000e

Bit(s)	R/W	Default	Description
19:16			tCCD/tCCD_l.
3:0			tCCD/tCCD_s read to read command time or write to write command time.

Table 11-169 DMC_DRAM_TEXSR 0x000f

Bit(s)	R/W	Default	Description
9:0			tEXSR. EXIT SELF-REFRESH to read/write command.

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Table 11-170 DMC_DRAM_TXS 0x0010

Bit(s)	R/W	Default	Description
9:0			tEXSR. EXIT SELF-REFRESH to read/write command.

Table 11-171 DMC_DRAM_TXP 0x0011

Bit(s)	R/W	Default	Description
3:0			tXP. EXIT power down to other command time

Table 11-172 DMC_DRAM_TXPDLL 0x0012

Bit(s)	R/W	Default	Description
9:0			tXPDLL, EXIT power down to read/write command time(need to relock PLL).

Table 11-173 DMC_DRAM_TZQCS 0x0013

Bit(s)	R/W	Default	Description
11:0			ZQCS command to other command time.

Table 11-174 DMC_DRAM_TCKSRE 0x0014

Bit(s)	R/W	Default	Description
4:0			enter self-refresh to disable clock time.

Table 11-175 DMC_DRAM_TCKSRX 0x0015

Bit(s)	R/W	Default	Description
4:0			enable clock to exit self-refresh time.

Table 11-176 DMC_DRAM_TCKE 0x0016

Bit(s)	R/W	Default	Description
4:0			CKE high or low minimum time.

Table 11-177 DMC_DRAM_TMOD 0x0017

Bit(s)	R/W	Default	Description
4:0			tMOD. MRR/MRW to other command time.

Table 11-178 DMC_DRAM_TDQS 0x0018

Bit(s)	R/W	Default	Description
3:0			tDQS. the delay to access different rank.

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DMC_DRAM_TRSTL 0x0019

Not used.

Table 11-179 DMC_DRAM_TZQLAT 0x001a

Bit(s)	R/W	Default	Description
5:0			ZQ LATCH command to other comand timing in LPDDR4 mode.

Table 11-180 DMC_DRAM_TMRR 0x001b

Bit(s)	R/W	Default	Description
7:0			tMRR not used in DMC. not support MR READ.

Table 11-181 DMC_DRAM_TCKESR 0x001c

Bit(s)	R/W	Default	Description
9:0			tCKESR. CKE low minimum pulse in self refresh mode.

DMC_DRAM_TDPD 0x001d

Not support.

Table 11-182 DMC_DRAM_DFITCTRLDELAY 0x001e

Bit(s)	R/W	Default	Description
3:0			DFI_t_ctrldealy

Table 11-183 DMC_DRAM_DFITPHYWRDATA 0x001f

Bit(s)	R/W	Default	Description
5:0			dfi_t_phy_wrdata.

Table 11-184 DMC_DRAM_DFITPHYWRLAT 0x0020

Bit(s)	R/W	Default	Description
5:0			dfi_t_phy_wrlat. in DDR3/4/LPDDR3 mode: WL -5. in LPDDR4 mode: WL -5 + 2.

Table 11-185 DMC_DRAM_DFITRDDATAEN 0x0021

Bit(s)	R/W	Default	Description
5:0			dfi_t_rddata_en. in DDR3/4/LPDDR3 mode: RL -5. in LPDDR4 mode : RL -5 + 1.

Table 11-186 DMC_DRAM_DFITPHYRDLAT 0x0022

Bit(s)	R/W	Default	Description
5:0			dfi_t_rdlat.

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Table 11-187 DMC_DRAM_DFITCTRLUPDMIN 0x0023

Bit(s)	R/W	Default	Description
7:0			CTRLUPD_MIN minimux clock cycle to maintain CTRLUPD_REQ.

Table 11-188 DMC_DRAM_DFITCTRLUPDMAX 0x0024

Bit(s)	R/W	Default	Description
7:0			CTRLUPD_MAX. maxmum clock cycle to maintain CTRLUPD_REQ if no CTRLUPD_ACK response.

DMC_DRAM_DFITMSTRRESP 0x0025

Not used.

DMC_DRAM_DFITREFMSKI 0x0026

Not used.

DMC_DRAM_DFITCTRLUPDI 0x0027

Not used.

Table 11-189 DMC_DRAM_DFITDRAMCLK 0x0028

Bit(s)	R/W	Default	Description
17			dram clk1 enable.
16			dram clk0 enable.
15:8			DRAM CLK disable waiting time
7:0			DRAM CLK enable enable timer

Table 11-190 DMC_DRAM_DFITLPRESP 0x002a

Bit(s)	R/W	Default	Description
3:0			dfi_lp_ctrl_req response time. after dfi_lp_ctrl_req asserted, and after response time if there's still no dfi_lp_ack respone, then drop the dfi_lp_ctrl_req.

Table 11-191 DMC_DRAM_DFITPHYMSTR 0x002b

Bit(s)	R/W	Default	Description
15			1: enable DFIPHYMASTER INTERFACE 0 disable DFIPHYMSTR en response.

Table 11-192 DMC_DRAM_TCKECK 0x002c

Bit(s)	R/W	Default	Description
4:0			tCKECK from CKE low to assert dfi_dram_clk_disable time. this time + dfi_t_ctrl_delay

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Table 11-193 DMC_DRAM_TREFI 0x002d

Bit(s)	R/W	Default	Description
31:24			tZQCI dmc send zqci period. unit is how much auto refresh period.
23:16			pvti dmc send dfi_ctrlupd_req period. unit is one auto refresh period.
15:8			tREFI.dmc send auto refresh command period. unit is 100ns.
7:0			t100ns period. unit is dmc clock cycles

Table 11-194 DMC_DRAM_TSR 0x002e

Bit(s)	R/W	Default	Description
5:0			tSR. self resfresh enter to exit time.

Table 11-195 DMC_DRAM_TCCDMW 0x002f

Bit(s)	R/W	Default	Description
5:0			4*tCCD in LPDDR4 mask write.

Table 11-196 DMC_DRAM_TESCKE 0x0030

Bit(s)	R/W	Default	Description
5:0			tESCKE. enter self refresh to power time for LPDDR4.

Table 11-197 DMC_DRAM_TREFI_DDR3 0x0031

Bit(s)	R/W	Default	Description
7:0			8*DDR3 SDRAM tREFI time . the unit is t100ns. use this to check in 8*tREFI time, the DMC should sent more than 16 auto REFRESH command.

Table 11-198 DMC_DRAM_TZQCAL 0x0032

Bit(s)	R/W	Default	Description
7:0			ZQCAL for LPDDR4.

DMC_DRAM_T10US 0x0033

10us clock cycle number used for LP2 mode.

Table 11-199 DMC_DRAM_TMRRI 0x0034

Bit(s)	R/W	Default	Description
7:0			tMRRI for MRR

DMC_NFQ_TMRD 0x0040

DMC_NFQ_TRFC 0x0041

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DMC_NFQ_TRP 0x0042

DMC_NFQ_TRTW 0x0043

DMC_NFQ_TCL 0x0044

DMC_NFQ_TCWL 0x0045

DMC_NFQ_TRAS 0x0046

DMC_NFQ_TRC 0x0047

DMC_NFQ_TRCD 0x0048

DMC_NFQ_TRRD 0x0049

DMC_NFQ_TFAW 0x004a

DMC_NFQ_TRTP 0x004b

DMC_NFQ_TWR 0x004c

DMC_NFQ_TWTR 0x004d

DMC_NFQ_TCCD 0x004e

DMC_NFQ_TEXSR 0x004f

DMC_NFQ_TXS 0x0050

DMC_NFQ_TXP 0x0051

DMC_NFQ_TXPDLL 0x0052

DMC_NFQ_TZQCS 0x0053

DMC_NFQ_TCKSRE 0x0054

DMC_NFQ_TCKSRX 0x0055

DMC_NFQ_TCKE 0x0056

DMC_NFQ_TMOD 0x0057

DMC_NFQ_TDQS 0x0058

DMC_NFQ_TRSTL 0x0059

DMC_NFQ_TZQLAT 0x005a

DMC_NFQ_TMRR 0x005b

DMC_NFQ_TCKESR 0x005c

DMC_NFQ_TDPD 0x005d

DMC_NFQ_DFITCTRLDELAY 0x005e

DMC_NFQ_DFITPHYWRDATA 0x005f

DMC_NFQ_DFITPHYWRLAT 0x0060

DMC_NFQ_DFITRDDATAEN 0x0061

DMC_NFQ_DFITPHYRDLAT 0x0062

DMC_NFQ_DFITCTRLUPDMIN 0x0063

DMC_NFQ_DFITCTRLUPDMAX 0x0064

DMC_NFQ_DFITMSTRRESP 0x0065

DMC_NFQ_DFITREFMSKI 0x0066

DMC_NFQ_DFITCTRLUPDI 0x0067

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DMC_NFQ_DFITDRAMCLK 0x0068

DMC_NFQ_DFITLPRESP 0x006a

DMC_NFQ_DFITPHYMSTR 0x006b

DMC_NFQ_TCKECK 0x006c

DMC_NFQ_TREFI 0x006d

DMC_NFQ_TSR 0x006e

DMC_NFQ_TCCDMW 0x006f

DMC_NFQ_TESCKE 0x0070

DMC_NFQ_TREFI_DDR3 0x0071

DMC_NFQ_TZQCAL 0x0072

DMC_NFQ_T10US 0x0073

DMC_NFQ_TMRRI 0x0074

DMC_DRAM_DFITPHYUPDTYPE0 0x0080

DMC_DRAM_DFITPHYUPDTYPE1 0x0081

DMC_DRAM_DFITPHYUPDTYPE2 0x0082

DMC_DRAM_DFITPHYUPDTYPE3 0x0083

Table 11-200 DMC_DRAM_DFIODTCFG 0x0084

Bit(s)	R/W	Default	Description
12			rank1 ODT default. default vulue for ODT[1] pins if theres no read/write activity.
11			rank1 ODT write sel. enable ODT[1] if there's write occur in rank1.
10			rank1 ODT write nsel. enable ODT[1] if theres's write occur in rank0.
9			rank1 odt read sel. enable ODT[1] if there's read occur in rank1.
8			rank1 odt read nsel. enable ODT[1] if there's read occure in rank0.
4			rank0 ODT default. default vulue for ODT[0] pins if theres no read/write activity.
3			rank0 ODT write sel. enable ODT[0] if there's write occur in rank0.
2			rank0 ODT write nsel. enable ODT[0] if theres's write occur in rank1.
1			rank0 odt read sel. enable ODT[0] if there's read occur in rank0.
0			rank0 odt read nsel. enable ODT[0] if there's read occure in rank1.

Table 11-201 DMC_DRAM_DFIODTCFG1 0x0085

Bit(s)	R/W	Default	Description
27:24			ODT length for BL8 read transfer.
19:16			ODT length for BL8 write transfer.
12:8.			ODT latency for reads. suppose to be 0.
4:0.			ODT latency for writes. suppose to be 0

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Table 11-202 DMC_DRAM_MCFG 0x0086

Bit(s)	R/W	Default	Description
12			1: dbi inversion. 0: dbi high inversion.
11			1: dbi read enable. 0: dbi not enabled.
10			1: enable staggered chip select for 2 ranks DRAM.
9			1: enable send auto refresh command to DDR SDRAM when PCTL is in CFG/ STOP state.
8			send auto refr cmd before enter register triggered self refresh
4			send auto refr cmd after exit regsiter triggered self refresh mode.
3			disable dram clock after enter register triggered self refresh.
2			send DFI_LP_REQ to PHY after enter register triggered elf refresh mode.
1			send DRAM to power down mode after enter self refresh. ONLY for LPDDR4.
0			send DFI_CTRLUPD_REQ after exit register triggered self refresh.

Table 11-203 DMC_DRAM_DFI_CTRL 0x0089

Bit(s)	R/W	Default	Description
20			siu_dfi1_phymstr_ack_en
19			siu_dfi_phymstr_req_and
18			siu_dfi_phymstr_req_or
17			siu_dfi_phymstr_type_sel
16			siu_dfi_phymstr_cs_sel
15			siu_dfi1_lp_en
14			siu_dfi_lp_ack_and
13			siu_dfi_lp_ack_or
12			siu_dfi1_init_start_en
11			siu_dfi_init_com_and
10			siu_dfi_init_com_or
9			siu_dfi1_freq_en
8			siu_dfi1_dram_clk_dis_en
7			siu_dfi_phyupd_type_sel
6			siu_dfi1_phyupd_ack_en
5			siu_dfi_phyupd_req_and
4			siu_dfi_phyupd_req_or
3			siu_dfi_ctrlupd_ack_and
2			siu_dfi_ctrlupd_ack_or

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Bit(s)	R/W	Default	Description
1			siu_dfi1_ctrlupd_req_en
0			siu_dfi1_cmd_en

Table 11-204 DMC_DRAM_DFIINITCFG 0x008a

Bit(s)	R/W	Default	Description
31.			dfi_init_complete status. read only.
15:14			Frequency set 1 dfi_freq_ratio value.
12:8			Frequency set 1 dfi_freq value.
7:6			Frequency set 0 dfi_freq_ratio value.
5:1			Frequency set 0 dfi_freq value.
0.			dfi_init_start value can be use manually config dfi_init_start signal.

Table 11-205 DMC_DRAM_ZQ_CTRL 0x008b

Bit(s)	R/W	Default	Description
2			send ZQCS command to RANK0 then send comand to RANK1.
1			send ZQCS command to both RANK0 and RANK1 together.
0			send ZQCS command to only rank0.

Table 11-206 DMC_DRAM_APD_CTRL 0x008c

Bit(s)	R/W	Default	Description
19:16			DFI_LP_WAKEUP value in APD DFI_LP_REQ mode
12			1: exit power down slow mode(waiting PLL LOCK). 0 : fast mode.
11			enable DFI_LP_REQ when enter Auto power down mode.
10			disable DFI_clk_disable when enter auto power down mode.

Table 11-207 DMC_DRAM_ASR_CTRL 0x008d

Bit(s)	R/W	Default	Description
23:20			DFI_LP_WAKEUP value in self refresh DFI_LP_REQ mode.
17			send REFRESH command after exit from auto self refersh mode(ASR).
16			send REFERSH command before enter to Auto self refresh mode(ASR).
15			send ZQCS command after exit from Auto self refresh mode(ASR).
14			send dfi_ctrl_upd after exit from ASR mode
13			send power down command when enter ASR mode. //for LPDDR4 only.
12			set the PHY enter LP2 mode after enter ASR mode.

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Bit(s)	R/W	Default	Description
11			send DFI_LP_REQ after enter ASR mode.
10			set DFI_CLK_DISABLE after enter ASR mode.
9:0			0 disable auto ASR mode.

DMC_DRAM_PHYMSTR_CTRL 0x0090

Not used.

DMC_DRAM_DFIODTRANKMAP 0x0091

Not used.

Table 11-208 DMC_DRAM_REFR_CTRL 0x0092

Bit(s)	R/W	Default	Description
17:8			auto refresh request pending cnt if there's page hit request.
6			Disabled auto refresh command if over 16 auto refresh command sent in 2 TREFI_DDR3 period
5			enable dmc send ZQCS command .
4.			enable dmc send DFI_CTRUPD_REQ.
3:1			how many refresh command send for one period. = this number + 1
0			enable dmc send auto refresh command.

Table 11-209 DMC_DRAM_FREQ_CTRL 0x0093

Bit(s)	R/W	Default	Description
31 .			Wiret 1 to change freqency read 0: finished.
30:9			Not used.
9.			1 : FREQ MRW done. Let FREQ change machine continue.
8			Freq wait. 1 when freq change finishes, state machine stop at self refresh state in case there's something need to handle.
7			When change PLL setting, disable dmc clock
6			When change PLL setting, disable PHY dficlk and dfictlclk.
5			Check vpu_sleep_en ==1 when do FREQ change. If vpu_sleep_en == 0, just wait.
4			Nxt frequency selection. 1 = freq1. 0 = freq0.
3:1.			Not used.
0.			Current frequency selection.

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Table 11-210 DMC_DRAM_SCFG 0x0094

Bit(s)	R/W	Default	Description
2:0			only one bit can be high at same time.
2			1 : to ask PCTL enter ACCESS STATE. 0 : deassert the request.
1			1 : to ask PCTL enter SELF REFRESH STATE. 0 : deassert the request.
0			1 : to ask PCTL enter STOP/CONFIG STATE . 0 : deassert the request.

Table 11-211 DMC_DRAM_STAT 0x0095

Bit(s)	R/W	Default	Description
31			rd latency error. 1: means after dfiphytrdlat cycles, the read data still not back.
28:24			dram_sr_state
23:20			stop_st
19:15			sleep_st
14:12			ACCESS STATUS 0 : ACCESS is in normal working mode. 1 : ACCESS sending precharege command. 2 : ACCESS sending AUTO REFESH command. 3 : ACCESS sending DIF_CTRLUPD_REQ command. 4 : ACCESS sending ZQCS command to DDR DRAM(ZQCAL for LPDDR4). 5 : ACCESS sending ZQLATCH command to LPDDR4 only.
11:8			APD STATUS: 0 : APD_IDLE 1 : APD sending PRECHARGE command 2 : APD sending CKE low command 3 : APD sending DISABLE DRAM CLOCK command 4 : APD sending DFI_LP_CTRL_REQ 5 : APD in Auto Power down mode. 6 : APD deassert DFI_LP_CTRL_REQ 7 : APD sending enable DRAM CLOCK command 8 : APD sending out CKE high command.
7:4			DRAM_STATUS: 0 : DRAM IDLE 1 : DRAM_STOP/DRAM_CFG 2 : DRAM_ACCESS 3 : DRAM_SLEEP 4 : DRAM APD(AUTO POWER DOWN). 5 : IDLE -> STOP/CONFIG 6 : STOP -> SLEEP 7 : STOP -> ACCESS 8 : ACCESS -> SLEEP. 9 : ACCESS -> STOP A : ACCESS -> APD B : SLEEP -> STOP C : SLEEP -> ACCESS D : APD -> ACCESS
3			Reserved.
2			1 : DRAM enter normal working state.
1			1 : DRAM enter sleep state. self refresh state.
0			1 : dram enter cfg state.

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Table 11-212 DMC_DRAM_STAT1 0x0096

Bit(s)	R/W	Default	Description
11:8			freq_st.
7:5			train_st
4:0			dram_phy_st

Table 11-213 DMC_PHY_RETRAINING_CTRL 0x0097

Bit(s)	R/W	Default	Description
31			phy_retraining enable.
30			check vpu sleep_en.
25:24			retraining dfi_freq[4:3], the [2:0] bit still use the dfi_freq bits to keep the frequency.
23:0			retraining period unit : 100ns.

Table 11-214 DMC_DFI_ERR_STAT 0x0098

Bit(s)	R/W	Default	Description
31:20			not used.
9			ddr0_dfi_error
8:5			ddr0_dfi_error_info.
4			ddr1_dfi_error.
3:0			ddr1_dfi_error_info.

Table 11-215 DMC_LP2_TIMER 0x009a

Bit(s)	R/W	Default	Description
15:0			timer setting to measure how long the chip is entered LP2 mode.

Table 11-216 DMC_DRAM_DFI_SWAP_0 0x00a0

Bit(s)	R/W	Default	Description
5:0			dfi_act_n function select

Table 11-217 DMC_DRAM_DFI_SWAP_1 0x00a1

Bit(s)	R/W	Default	Description
5:0			dfi_ras_n function select

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Table 11-218 DMC_DRAM_DFI_SWAP_2 0x00a2

Bit(s)	R/W	Default	Description
5:0			dfi_cas_n function select

Table 11-219 DMC_DRAM_DFI_SWAP_3 0x00a3

Bit(s)	R/W	Default	Description
5:0			dfi_we_n function select

Table 11-220 DMC_DRAM_DFI_SWAP_4 0x00a4

Bit(s)	R/W	Default	Description
5:0			dfi_bg0 function select

Table 11-221 DMC_DRAM_DFI_SWAP_5 0x00a5

Bit(s)	R/W	Default	Description
5:0			dfi_bg[1] function select

Table 11-222 DMC_DRAM_DFI_SWAP_6 0x00a6

Bit(s)	R/W	Default	Description
5:0			dfi_ba[0] function select

Table 11-223 DMC_DRAM_DFI_SWAP_7 0x00a7

Bit(s)	R/W	Default	Description
5:0			dfi_ba[1] function select

Table 11-224 DMC_DRAM_DFI_SWAP_8 0x00a8

Bit(s)	R/W	Default	Description
5:0			dfi_ba[2] function select

Table 11-225 DMC_DRAM_DFI_SWAP_9 0x00a9

Bit(s)	R/W	Default	Description
5:0			dfi_a[0] function select

Table 11-226 DMC_DRAM_DFI_SWAP_10 0x00aa

Bit(s)	R/W	Default	Description
5:0			dfi_a[1] function select

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Table 11-227 DMC_DRAM_DFI_SWAP_11 0x00ab

Bit(s)	R/W	Default	Description
5:0			dfi_a[2] function select

Table 11-228 DMC_DRAM_DFI_SWAP_12 0x00ac

Bit(s)	R/W	Default	Description
5:0			dfi_a[3] function select

Table 11-229 DMC_DRAM_DFI_SWAP_13 0x00ad

Bit(s)	R/W	Default	Description
5:0			dfi_a[4] function select

Table 11-230 DMC_DRAM_DFI_SWAP_14 0x00ae

Bit(s)	R/W	Default	Description
5:0			dfi_a[5] function select

Table 11-231 DMC_DRAM_DFI_SWAP_15 0x00af

Bit(s)	R/W	Default	Description
5:0			dfi_a[6] function select

Table 11-232 DMC_DRAM_DFI_SWAP_16 0x00b0

Bit(s)	R/W	Default	Description
5:0			dfi_a[7] function select

Table 11-233 DMC_DRAM_DFI_SWAP_17 0x00b1

Bit(s)	R/W	Default	Description
5:0			dfi_a[8] function select

Table 11-234 DMC_DRAM_DFI_SWAP_18 0x00b2

Bit(s)	R/W	Default	Description
5:0			dfi_a[9] function select

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Table 11-235 DMC_DRAM_DFI_SWAP_19 0x00b3

Bit(s)	R/W	Default	Description
5:0			dfi_a[10] function select

Table 11-236 DMC_DRAM_DFI_SWAP_20 0x00b4

Bit(s)	R/W	Default	Description
5:0			dfi_a[11] function select

Table 11-237 DMC_DRAM_DFI_SWAP_21 0x00b5

Bit(s)	R/W	Default	Description
5:0			dfi_a[12] function select

Table 11-238 DMC_DRAM_DFI_SWAP_22 0x00b6

Bit(s)	R/W	Default	Description
5:0			dfi_a[13] function select

Table 11-239 DMC_DRAM_DFI_SWAP_23 0x00b7

Bit(s)	R/W	Default	Description
5:0			dfi_a[14] function select

Table 11-240 DMC_DRAM_DFI_SWAP_24 0x00b8

Bit(s)	R/W	Default	Description
5:0			dfi_a[15] function select

Table 11-241 DMC_DRAM_DFI_SWAP_25 0x00b9

Bit(s)	R/W	Default	Description
5:0			dfi_a[16] function select

Table 11-242 DMC_DRAM_DFI_SWAP_26 0x00bb

Bit(s)	R/W	Default	Description
5:0			dfi_a[17] function select

The following registers' base address is 0xff638800. Each register takes 4 byte address.

Each register's final address = 0xff638800 + offset * 4.

DMC_STICKY_0 0x0000

DMC_STICKY_1 0x0001

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DMC_STICKY_2 0x0002
DMC_STICKY_3 0x0003
DMC_STICKY_4 0x0004
DMC_STICKY_5 0x0005
DMC_STICKY_6 0x0006
DMC_STICKY_7 0x0007
DMC_STICKY_8 0x0008
DMC_STICKY_9 0x0009
DMC_STICKY_10 0x000a
DMC_STICKY_11 0x000b
DMC_STICKY_12 0x000c
DMC_STICKY_13 0x000d
DMC_STICKY_14 0x000e
DMC_STICKY_15 0x000f
DMC_STICKY_16 0x0010
DMC_STICKY_17 0x0011
DMC_STICKY_18 0x0012
DMC_STICKY_19 0x0013
DMC_STICKY_20 0x0014
DMC_STICKY_21 0x0015
DMC_STICKY_22 0x0016
DMC_STICKY_23 0x0017
DMC_STICKY_24 0x0018
DMC_STICKY_25 0x0019
DMC_STICKY_26 0x001a
DMC_STICKY_27 0x001b
DMC_STICKY_28 0x001c
DMC_STICKY_29 0x001d
DMC_STICKY_30 0x001e
DMC_STICKY_31 0x001f
DMC_STICKY_32 0x0020
DMC_STICKY_33 0x0021
DMC_STICKY_34 0x0022
DMC_STICKY_35 0x0023
DMC_STICKY_36 0x0024
DMC_STICKY_37 0x0025
DMC_STICKY_38 0x0026
DMC_STICKY_39 0x0027

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DMC_STICKY_40 0x0028

DMC_STICKY_41 0x0029

DMC_STICKY_42 0x002a

DMC_STICKY_43 0x002b

DMC_STICKY_44 0x002c

DMC_STICKY_45 0x002d

DMC_STICKY_46 0x002e

DMC_STICKY_47 0x002f

DMC_STICKY_48 0x0030

DMC_STICKY_49 0x0031

DMC_STICKY_50 0x0032

DMC_STICKY_51 0x0033

DMC_STICKY_52 0x0034

DMC_STICKY_53 0x0035

DMC_STICKY_54 0x0036

DMC_STICKY_55 0x0037

DMC_STICKY_56 0x0038

DMC_STICKY_57 0x0039

DMC_STICKY_58 0x003a

DMC_STICKY_59 0x003b

DMC_STICKY_60 0x003c

DMC_STICKY_61 0x003d

DMC_STICKY_62 0x003e

DMC_STICKY_63 0x003f

11.2 NAND

11.2.1 Overview

The chip supports SLC NAND Flash with 8-bit ECC.

SPI NAND is an SLC NAND Flash memory device with Serial peripheral interface.

It offers superior write performance and cost per bit over SPI NOR and low pin-count over ONFI NAND.

11.2.2 Features

The features of SPI NAND are as follows.

- Industry-standard serial peripheral interface.
- Internal ECC parity(BCH-8/BCH-4).
- Density: 1Gbit/2GBits/4Gbits
- 2x/4x bus mode supported

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11.2.3 Descriptor Commands

Command “Standby”

All CEs are high, all other signals are “don’t care”, the bits [9:0] specify how many extra NAND cycles the controller stays in “standby”, if the number is “0”, only one NAND cycle, if the number is 10, the controller will stay in “standby” for 11 NAND cycles.

Command “Idle”

CE is low, ALE and CLE is low, WE and RE is high, the NAND bus is taken over by controller, but the bus is let to idle for one NAND cycle, if extra NAND cycles number is set, then extra NAND cycles is idled, useful when NAND controller needs time to switch to another mode, or wait for RB.

Command “Command”

NAND command is sent, CE is low, ALE is low, CLE is high, WE is low and RE is high, usually only one NAND CE is low, all the other 3 should be high, otherwise the NAND command is accepted by multiple Dies and may cause conflict problem, but it is fine when used to reset the NAND, the command itself is sent by bit 7 to bit 0, one command sends one NAND command.

Command “Address”

NAND address cycle, the CE is low, ALE is high, CLE is low, WE is low and RE is high. Usually only one NAND CE is low, all the other 3 CEs should be high. The address is sent in the bit 7 to bit 0, it lasts one NAND cycle.

Command “Data to NAND”

Write NAND bus directly by this command, CE is low, ALE and CLE are low, WE is low and RE is high, the low bit7 to bit 0 is write to NAND flash when this command is done, it lasts one NAND command cycle. Usually used to program NAND features, it is very low efficient to program NAND by this command, could be used to debug NAND flash software.

Command “NAND to Data”

Read NAND bus and save the data to registers, CE is low, ALE and CLE are low, WE is high and RE is low, the NAND output is locked in registers at perfect timing, the extra NAND cycles will make this command repeats extra NAND cycle, if it is “0”, only one byte is locked in registers, since the registers used to lock the data is only 32 bits, extra NAND cycles number larger than 3 will cause the data overflow. This command is used to read status and features from NAND, not for read NAND data, it is very low efficient to read NAND by this command, it could be used to debug NAND software.

Command “Sync Read”

Same as “NAND to Data”, used in synchronous mode, each NAND cycle will read in two bytes in synchronous mode.

Command “RB pin”

Hardware ready/busy detect with timer and interrupt, the timer is bit 4 to bit 0, it is power of 2 NAND cycle, for example, if timer is set to 10, then 2^10 or 1024 NAND cycles is set to timer out the RB command. The maximum setting is 31, that is 2^31 NAND cycles. Or 42 seconds when NAND cycle is 20 ns. When the RB waiting is timed out, the hardware is still in “waiting”, an interrupt is sent to CPU, the “RB” command can be programmed to send out interrupt when RB is high when the correspond INT bits is set to high, once RB is high, the command queue advanced to next command, no matter interrupt or not, interrupt setting is “one time setting”, it means one RB command can be programmed with interrupt, the next can be programmed without interrupt. There are few bits in the RB command for software debugging purpose. it can be set to different numbers to tell which RB command is interrupted. CPU can read out the NAND status registers to tell which RB causes interrupt.

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Command “RB IO”

Assume the NAND controller is used without ready/busy pin, the ready/busy is checked through status register, This command should be used with NAND read status command, like the old RB command, this command is designed with option to check which IO bits as ready/busy, “IO6”, “IO5” and “IO4”, and “INT6”, “INT5” and “INT4” to issue “IRQ” if “Ready”, time out logic is the same as old RB, in case of “Read” NAND, an extra “0x00” command is needed to switch NAND flash back to “Read” mode.

The RB IO command is associated with CE state, there is no CE option in RB IO command, valid CE from previous command is used.

NAND status is checked every 16 NAND cycles, if not ready when time out, an “IRQ” is issued and timer is reset.

Change to read status mode.

Set proper timer out parameter.

Monitor which IO bits, IO6, IO5 or IO4.

Set IRQ option, set NAND_CFG cmd_irq_en bit.

Return back to read mode if needed.

Command “MEM to NAND”

DMA command to send data from memory to NAND, this command consist of ECC mode, the data is read from DDR memory by NAND controller at 64 bits/system cycle, BCH encoded by hardware and sent to NAND in the fly. The NAND data address can be programmed by register or command “set address”.

The data size can be any number, if ECC is off, the exact number of data bytes are read from DDR memory and sent to NAND without ECC encode, if ECC is on, and final page is not 512 bytes, then “0xff” is appended to the data to make it 512 bytes and send to BCH encoded and send to NAND.

When ECC is off, no spare data bytes are read from DDR, the data is sent exactly the same as DDR. When ECC is on, 2/16/0 bytes of spare bytes are read from DDR and appended to 512/1024 main data, encoded and parity bytes all sent to NAND.

The final BCH code word is rounded to integer number of bytes, see BCH table.

If ECC is OFF, the command writes [13:0] number of bytes to NAND.

If ECC is ON, the command writes “pages” of bytes to NAND.

If “short” is 0, the page size is BCH code data size, 512 or 1024.

If “short” is 1, shortened BCH mode, the page size is “page size”*8 bytes.

The number of pages is up to 63.

Data randomization option, if cmd[19] is set to “1” and the random seed is not equal to “0”, the data and parity written to NAND is randomized with PRBS random number, the same seed must be used to read the page back.

Random seed is set by command “Seed” cmd[14:0], suggest to use NAND physical address as “seed”.

Command “NAND to MEM”

DMA command to read data from NAND and save to DDR, this command consist of ECC mode, and data size, this command will correct the data by hardware and save the decode result information to DDR too. The data address in DDR is programmed by “set address” command, if not programmed the address increases to next available data space.

When ECC is off, the data is save to DDR without change, the information bytes is also saved in DDR with ECC mode bits off, no spare bytes, when ECC is on, the data is BCH decoded and corrected by

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hardware, if the data size is less than 512/1024, more data will be read out from NAND, but only the required number of bytes are sent to DDR.

If ECC is OFF, the command reads [13:0] number of bytes from NAND.

If ECC is ON, the command reads “pages” of bytes to NAND.

If “short” is 0, the page size is BCH code data size, 512 or 1024.

If “short” is 1, shortened BCH mode, the page size is “page size”*8 bytes.

The number of pages is up to 63.

Data randomization option is the same as “MEM to NAND”.

Command “Set Address”

DDR address setup command, used to change the memory address when read or write NAND, the D/S bits is for data or spare, “0” for data, “1” for spare, L/H is for low and high 16 bits of address, “0” for low 16 bits, “1” for high 16 bits, the DDR address is 32 bits.

This command has nothing to do with NAND bus, but it will take one NAND command to execute, one NAND idle command is used in time.

Mode	Cmd[18:17]	Description
Data	00	Set Data Address
Info	01	Set Info Address
Status	10	Set Status Address

Command “STS”

Read status and save to DDR memory, the status registers is 32 bits, with

Field	Name	Description
15:0	Sts	Status
23:16	Sts counter	Sequence number
31	Done	This status is valid.

STS 1: read status once, without IRQ.

STS 2: read status twice, with IRQ.

If NAND_CFG[20] Sts_irq_en is set to “1”, the STS 2 command will issue an IRQ to system, if used after NAND read command “N2M”, it guarantees the DMA is done. Because the STS itself is a DMA command, it will wait till the previous DMA command done before its DMA.

STS 2 command resets the sequence number, the sequence number in itself is the number of STS 1’s before IRQ, wrap around to 8 bits.

Command “Set Seed”

Random seed for data randomizer, any number except 0 starts random number generator, if “N2M” or “M2N” random bit is set, the random number is XORed with data.

Set none zero random seed, in cmd[14:0].

Set cmd[19] of “N2M” or “M2N” to 1, enable scrambler.

Use same seed to program/read the same NAND page.

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11.2.4 Register Description

The base address of NAND registers is 0xffe07000, and the final address of each register is listed below.

Table 11-243 NAND Register List

Register Name	Description	Address	R/W
P_NAND_CMD	Write Command and Read Status	Base + 0x00	R/W
P_NAND_CFG	Configuration	Base + 0x04	R/W
P_NAND_DADR	Data Address	Base + 0x08	R/W
P_NAND_IADR	Information Address	Base + 0x0c	R/W
P_NAND_BUF	Read Data Buffer	Base + 0x10	R
P_NAND_INFO	Information	Base + 0x14	R
P_NAND_DC	DDR interface	Base + 0x18	R
P_NAND_ADR	DDR Address	Base + 0x1c	R
P_NAND_DL	DDR Low 32 Bits Data	Base + 0x20	R/W
P_NAND_DH	DDR High 32 Bits Data	Base + 0x24	R/W
P_NAND_CADR	Command Queus Address	Base + 0x28	R/W
P_NAND_SADR	Status Address	Base + 0x2c	R/W
P_NAND_PINS	CS2: SDRAM/NAND pin sharing	Base + 0x30	R/W
P_NAND_VER	Version number	Base + 0x38	R

P_NAND_CMD

Write : Send NAND command to controller, the command format is specified in previous section.

Bit(s)	Name	Description
21:0	Cmd	NAND command sent to NAND queue buffer
30	Cmd_go	When 1, and NAND bus is in waiting Rb mode, due to time out or longer than expected Rb waiting, the command queue will move on by disable RB waiting in current command.
31	Cmd_reset	When 1 the NAND command queue buffer is reset to zero.

Read : Read NAND controller status

Bit(s)	Name	Description
19:0	Cmd_curr	NAND command current on NAND bus, still going, not finished.
24:20	Cmd_cnt	Number of NAND commands still in NAND command queue buffer, the buffer size is 32.
25	Timer out	When 1 wait Rb command timed out.
26	Rb0	Current Rb0 status, 1: ready, 0: busy.
27	Rb1	Current Rb1 status, 1: ready, 0: busy.

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Bit(s)	Name	Description
28	Rb2	Current Rb2 status, 1: ready, 0: busy.
29	Rb3	Current Rb3 status, 1: ready, 0: busy.
30	Mem_rdy	When 1, DDR interface is idle and ready to accept memory movement request.
31	Ecc_rdy	When 1, ECC BCH encoder/decoder is idle and read to accept encode or decode activity.

P_NAND_CFG

Bit(s)	Name	Description
4:0	Bus_cyc	The number of system clock cycles in one NAND cycle – 1, for example, if the bus_cyc is 3, then the NAND cycle is 4 system clock cycles, the minimum setting is 3, the maximum setting is 31. program this register according NAND timing mode.
9:5	Bus_tim	The timing to lock the NAND data when read NAND data or status, please refer to “Timing Calculator” for details.
11:10	Sync	00: Async mode 01: Micron Sync mode 10: Toshiba-Samsung toggle mode
12	Cmd_start	When set to “1”, if the NAND controller internal 32 command buffer has less than 16 commands, the command DMA starts reading commands from DDR and saves them to internal buffer, the DMA keeps watching the internal buffer, reads whenever there are less than 16 commands left, if an all “zero” command is met, This bit is cleared and current command DMA is done.
13	Cmd_auto	When set to ‘1’, the command DMA will check the previous command queue end location, whether it is changed from all “zero” back to valid command, the auto check period is 1 ms.
14	Apb_mode	Special NAND mode for ROM boot or debug, when 1, DDR interface is redirected to APB register, all the read/write activities are through APB registers. When used in ROM boot and DDR is not ready.
15	Spare_only	When 1, the NAND controller read NAND with/without ECC, but only save the information bytes into DDR memory, the main data is discarded, designed for software to survey the NAND flash spare bytes and prepare NAND programming.
16	Sync_adj	Used to adjust data timing in sync or toggle mode, 0: default timing, 1: delay 1 system clock cycle.
20	Sts_irq_en	Enable STS IRQ.
21	Cmd_irq_en	Enable RB pin or RB IO IRQ.
26	Oob_on	Set to 1 oob_mode 16/0, Set to 0 no oob bytes.
27	Oob_mode	New in M8 v2, Set to 1 enable new oob mode. First page 16 bytes, all other pages 0 byte.
28	Dc_ugt	Set NAND controller DDR interface to Urgent mode.
29	Nand_wpn	When 1, the NAND wpn pin is set to low, the NAND is in write protection mode, default to 0, the NAND is not protected.

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Bit(s)	Name	Description
30	Core_power	When 1, internal NAND controller core clock gating is override to always on. The clock gating is disabled.
31	Top_power	When 1, internal NAND top clock gating is override to always on, the clock gating for top is disabled.

P_NAND_DADR

Set DDR data address by registers, the address is 32 Bits, since the DDR data address can also be set by NAND commands, when both happens at the same time, register setting is ignored, the NAND command setting takes effect. Software should avoid conflict address setting.

P_NAND_IADR

Set DDR information (spare bytes) address by registers, the address is 32 Bits, since the DDR information address can also be set by NAND commands, when both happens at the same time, register setting is ignored, the NAND command setting takes effect. Software should avoid conflict address setting.

P_NAND_BUF

When read NAND status, features or data, the results are buffer in this register, the register is 32 Bits, it can only hold 4 bytes, if the host not doesn't read out the results, it will be over written by the following "read".

P_NAND_INFO

One 32 Bits information per each 512 bytes in ECC mode.

Bit(s)	Name	Description
7:0	Info 0	Information (spare) byte 0, errors already corrected by BCH.
15:8	Info 1	Information (spare) byte 1, errors already corrected by BCH
21:16	Pages	Count down page number in current DMA read, starts from the total page size, count down to 1.
28:24	Errcnt	Number of errors corrected by BCH in current page, 0 means no error in current page, 0x1f means this page is uncorrectable.
29	Unc	When 1, this page is uncorrectable by BCH, this page is bad.
30	Ecc	When 1, current NAND read is with ECC on.
31	Done	When 1, the information content and data read from NAND are valid, otherwise the "read" is not done.

P_NAND_DC

Used for apb_mode, internal NAND controller still uses DDR interface, only the DDR request and DDR grant are redirected from DDR to apb registers, this enables the host to read NAND without DDR, for ROM boot and debug.

Bit(s)	Name	Description
7:0	Dc_wr_dm	DDR write data mask, Each Bit masks one byte.
8	Dc_wr	When 1, write data from NAND to DDR. When 0, read data from DDR to NAND.
9	Dc_lbrst	When 1, the 64 Bits data is the last in current DDR burst, used with Dc_req.

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Bit(s)	Name	Description
10	Dc_ugt	When 1, current DDR request of read/write is urgent, in NAND controller, the Dc_ugt is set to 0, none-urgent.
11	Dc_req	When 1, the NAND controller send request to DDR to read or write data, in case of apb_mode, when dc_req is “1”, the host is responsible for send to or receive from NAND controller. Note: this is the only signal the host needs to check when in apb_mode.

P_NAND_ADR

32 Bits DDR address when NAND controller read or write to DDR memory, any address space within the DDR memory installed in the system is valid.

P_NAND_DL

The DDR interface uses 64 Bits width bus, this register is for low 32 Bits, [31:0].

P_NAND_DH

The DDR interface uses 64 Bits width bus, this register is for high 32 Bits, [63:32].

Read and write to this register will generate “grant” and “dc_wr_avail” or “dc_rd_avail”.

Always read or write low 32 Bits first, then read or write to high 32 Bits and NAND controller hardware will generate “grant” and “dc_wr_avail” or “dc_rd_avail”, combined with the data, the DDR address advances to next address.

NAND controller programs NAND flash in apb_mode:

- Check P_NAND_DC till “dc_req” is high.
- Write low 32 Bits to P_NAND_DL.
- Write high 32 Bits to P_NAND_DH.
- Go back to beginning till all the data is written.

NAND controller reads data from NAND flash in apb_mode:

- Check P_NAND_DC till “dc_req” is high.
- Read low 32 Bits from P_NAND_DL.
- Read high 32 Bits from P_NAND_DH.
- Go back to beginning till all the data is read.

Since the DDR interface in NAND controller process the data in group of 16 double words (64 Bits), the software can only check “dc_req” at the beginning of each 16 double words.

P_NAND_CADR

Set command queue memory address, 32 Bits, any memory location.

This address can only be programmed by APB bus.

P_NAND_SADR

Set status memory location, 32 Bits, any memory location.

This address can also be programmed through command queue.

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P_NAND_PINS

Bit(s)	Name	Description
13:0	Pins_len	When pins are acquired by NAND, it will use Pins_len number of NAND bus cycles before releasing pins. Defult is 8.
27:14	Pins_off	When pins are released by NAND, it will wait Pins_off number of NAND bus cycles before sending next request. Default is 2.
31	Not shared	When 1 the pins is not shared, default is 0, pins are shared.

11.3 eMMC/SD

11.3.1 Overview

The chip has the following features of eMMC/SD.

- Supports SDSC/SDHC/SDXC card
- Supports eMMC and MMC card specification version 5.0 up to HS200 with data content TDES crypto
- 1 bit, 4 Bits, 8 Bits date lines supported (8 Bits only for MMC)
- Descriptor chain architecture, timing tuning and adjustment
- Supports descriptor-based internal DMA controller

This module uses eMMC/SD CONTROLLERs to connect varied SD/MMC Card, or eMMC protocol compatible memory with high throughput.

11.3.2 Pin Description

Table 11-244 Pin Description of eMMC/SD/SDIO Module

Name	Type	Description	Speed (MHz)
CLK	Output	SD eMMC clock, 0~200MHz	200
DS	DS Data strobe for eMMC HS400 mode	-	-
DAT[7:0]	SD Card 4 Bits, eMMC 8 Bits Input/Output/Push-Pull Internal pull-up for pins not used	Data, 1,4,8 mode	200
CMD	Input/Output/Push-Pull/Open-Drain Open-drain for initialization Push-pull for fast command transfer ROD is connected when in open-drain mode.	Command Response	200
Rst_n	eMMC required	Hardware reset	Low
IRQ	SD Card or eMMC not used.	Device interrupt can be replaced by DAT[1]	Low

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11.3.3 eMMC/SD Mode

eMMC Mode

Table 11-245 eMMC Mode

Mode Name	Data Rate	IO Voltage	Bus Width	Frequency	Max Data Transfer
Legacy MMC card	Single	3/1.8V	1, 4, 8	0-26MHz	26MB/s
High Speed SDR	Single	3/1.8V	1,4, 8	0-52MHz	52MB/s
High Speed DDR	Dual	3/1.8V	4, 8	0-52MHz	104MB/s
HS200	Single	1.8V	4, 8	0-200MHz	200MB/s

The HS200 mode offers the following features:

- SDR Data sampling method
- CLK frequency up to 200MHz Data rate – up to 200MB/s
- 4 or 8-bits bus width supported
- Single ended signaling with 4 Drive Strengths
- Signaling levels of 1.8V
- Tuning concept for Read Operations

SD Mode

Table 11-246 SD Mode

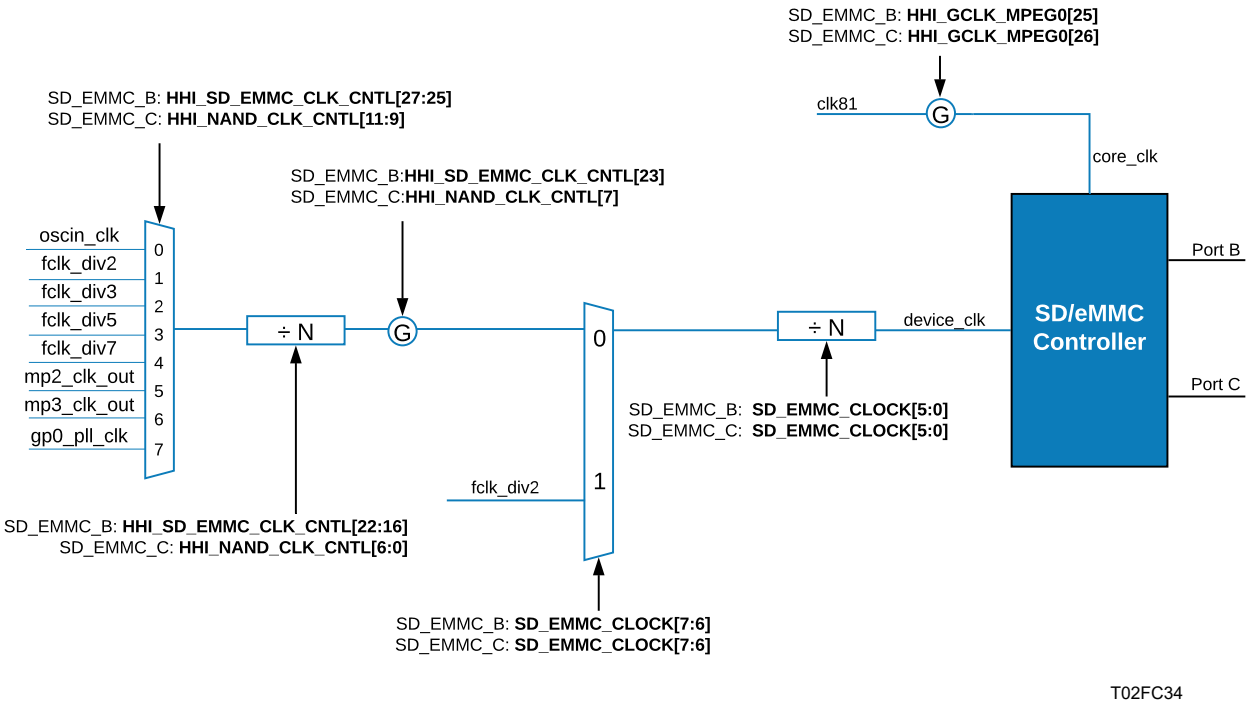
Mode Name	Data Rate	IO Voltage	Bus Width	Frequency	Max Data Transfer
Default Speed	Single	3.3V	1, 4	0-25MHz	12.5MB/s
High Speed	Single	3.3V	1, 4	0-50MHz	25MB/s
SDR12	Single	1.8V	1,4	0-25MHz	12.5MB/s
SDR25	Single	1.8V	1,4	0-50MHz	25MB/s
SDR50	Single	1.8V	1,4	0-100MHz	50MB/s
SDR104 (highest)	Single	1.8V	1,4	0-208MHz	104MB/s
DDR50	Dual	1.8V	4	0-50MHz	50MB/s

11.3.4 Clock

The clock source of EMMC is shown in the figure below.

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Figure 11-2 EMMC Clock Source



11.3.5 Descriptor

Structure

The descriptor has a size of 4x32 Bits.

Table 11-247 Descriptor Structure

byte	7	6	5	4	3	2	1	0
0	length[7:0]							
1	Timeout 4 Bits				End of chain	R1b	block mode	length[8]
2	data num	resp num	resp 128	resp nocrc	Data wr	Data io	No cmd	No resp
3	owner	error	cmd index 6 Bits					
4	cmd argument 32 Bits							
5								
6								
7								
8	data address 32 Bits or data 0-4 bytes [1]Big Endian, [0]SRAM							
9								
10								
11								
12	response address 32 Bits or response irq en [0]SRAM							
13								

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byte	7	6	5	4	3	2	1	0
14								
15								

Definition

Name	Bits	Description
Length	Cmd_cfg[8:0]	same as spec, copy the content from command argument into this field, different byte size and 512 bytes, different number of blocks and infinite blocks. If the command is operating on bytes, block mode = 0, this field contains the number of bytes to read or write, A value of 0 shall cause 512 bytes to be read to written, if the command is operating on blocks, block mode = 1, this field contains the number of blocks, a value of 0 is infinite number of blocks.
Block_mode	Cmd_cfg[9]	1: the read or write shall be performed on block basis. The block size is from SD/eMMC device, and saved in APB3 register in module. 0: the read or write is byte based.
R1b	Cmd_cfg[10]	1: check the DAT0 busy after received response R1 0: do not check the DAT0 busy state.
End_of_chain	Cmd_cfg[11]	1: it is the end of descriptor chain, the host stops and issues IRQ after this descriptor is done. 0: the host reads next descriptor and continues. The command chain execution is started by write an APB3 register and stopped by the “end of chain” or clear a APB3 start register, or found one descriptor with owner is set to 0.
Timeout	Cmd_cfg[15:12]	2timeout ms when timeout != 0, max timeout 32.768s, when over the time-out limit, error bit is set, IRQ is issued. When timeout is 0, no time limit.
No_resp	Cmd_cfg[16]	1: this command doesn’t have response, used with command doesn’t have response. 0: there is a response. The module waits for response, the response time-out setting is in APB3 register.
No_cmd	Cmd_cfg[17]	1: this descriptor doesn’t have command in it, it does data DMA only, used with command to read or write SD/eMMC with data from multiple locations.
Data_io	Cmd_cfg[18]	1: there is data action in this descriptor, used with command have data process. 0: there is no data read/write action.
Data_wr	Cmd_cfg[19]	1: host writes data to SD/eMMC 0: host read data from SD/eMMC
Resp_nocrc	Cmd_cfg[20]	1: R3 response doesn’t have CRC. 0: host does CRC check.
Resp_128	Cmd_cfg[21]	1: R3 response with 128 Bits information. 0: 32 Bits responses.
Resp_num	Cmd_cfg[22]	1: the resp_addr is the IRQ enable Bits, used to check the response error status, when there is an error, IRQ[14] is issued, the first 4 bytes of response is saved into resp_addr. 0: save response into SRAM or DDR location.
Data_num	Cmd_cfg[23]	1: save 4 bytes of data back into descriptor itself at bytes 8~11.
Cmd_index	Cmd_cfg[29:24]	The SD/eMMC command index. Desc REG wr: 4 reg44, 12 reg4c.
Error	Cmd_cfg[30]	Write back by host.

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Name	Bits	Description
		The combined error from command, response, data, includes CRC error and timeout. When it is set the descriptor execution is stopped and an IRQ is issued. The CPU can read SD_EMMC_STATUS register to get detail information.
Owner	Cmd_cfg[31]	Programmed by CPU to 1, cleared by host to 0. 1: the descriptor is valid and owned by host, after it is done, even it has error, the owner bit is cleared, the descriptor is owned by CPU. In case of descriptor chain execution when host found a descriptor with “0” owner bit, it will stop.
Cmd_arg	Desc 4~7 bytes	32 Bits. The actual command argument some of the previous fields are copied from this command argument, the software need to make sure they are consistent. Desc REG wr: new value Data_addr: write mask, 1: change, 0: no change.
Data_addr	Desc 8~11 bytes	32 Bits. If the data_num is 0, the content is data address. If the data_num is 1, ths content is 4 data bytes. When it is an address: Data_addr[0]: 1: SRAM address, 0: DDR address. If the data_addr[31:12] matches with SD_EMMC_BASE, it is SRAM address. Data_addr[1]: 1: 4 bytes big endian, 0: little endian(default).
Resp_addr	Desc 12~15 bytes	32 Bits If the resp_num is 0, the content is resp address. If the resp_num is 1, before execution, it is the response IRQ enable Bits, after execution, it is the first 4 response bytes. When it is an address: Resp_addr[0]: 1: SRAM address, 0: DDR address. If the resp_addr[31:12] matches with SD_EMMC_BASE, it is SRAM address.

11.3.6 Register Description

Each register final address = module base address+ address * 4

Where module address addresses are 0xFFE07000 for portB (eMMC), 0xFFE05000 for port C (eMMC).

Table 11-248 SD_EMMC_CLOCK 0x0

Bit(s)	R/W	Default	Description
30	R/W	0	Cfg_irq_sdio_sleep_ds : 1: select DS as IRQ source during sleep.
29	R/W	0	Cfg_irq_sdio_sleep: 1: enable IRQ sdio when in sleep mode. When DAT1 IRQ, the controller uses PCLK to detect DAT1 level and starts core clock, the core initials IRQ_period and detect DAT1 IRQ.
28	R/W	0	Cfg_always_on: 1: Keep clock always on 0: Clock on/off controlled by activities. Any APB3 access or descriptor execution will turn clock on. Recommended value: 0
27:22	R/W	0	Cfg_rx_delay: RX clock delay line, 6 bits 0: no delay, n: delay n*50ps Maximum delay 3150ps.

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Bit(s)	R/W	Default	Description
21:16	R/W	0	Cfg_tx_delay: TX clock delay line, 6 bits 0: no delay, n: delay n*50ps Maximum delay 63*50ps =3150ps.
15:14	R/W	0	Cfg_sram_pd: Sram power down
13:12			Cfg_rx_phase: RX clock phase 0: 0 phase, 1: 90 phase, 2: 180 phase, 3: 270 phase. Recommended value: 0
11:10	R/W	0	Cfg_tx_phase: TX clock phase 0: 0 phase, 1: 90 phase, 2: 180 phase, 3: 270 phase. Recommended value: 2
9:8	R/W	0	Cfg_co_phase: Core clock phase 0: 0 phase, 1: 90 phase, 2: 180 phase, 3: 270 phase. Recommended value: 2
7:6	R/W	0	Cfg_src: Clock source 0: Crystal 24MHz or other frequencies selected by clock reset test control register. 1: Fix PLL, 1000MHz Recommended value: 1
5:0	R/W	0	Cfg_div: Clock divider Frequency = clock source/cfg_div Clock off: cfg_div==0, the clock is disabled Divider bypass: cfg_div==1, clock source is used as core clock without divider Maximum divider 63.

Table 11-249 SD_EMMC_DELAY1 0x4

Bit(s)	R/W	Default	Description
31:30	R/W	0	Reserved.
29:24	R/W	0	Dly[4]: Data 4 delay line
23:18	R/W	0	Dly[3]: Data 3 delay line
17:12	R/W	0	Dly[2]: Data 2 delay line
11:6	R/W	0	Dly[1]: Data 1 delay line
5:0	R/W	0	Dly[0]: Data 0 delay line Total delay = 50ps * Dly When Dly == 0, no delay. When Dly ==63, 3150ps delay. NOTE: the 50ps is typical delay, actually delay may vary from chip to chip, from different temperature.

Table 11-250 SD_EMMC_DELAY2 0x8

Bit(s)	R/W	Default	Description
31:30			Reserved.
29:24			Dly[9]: Data 9 delay line
23:18			Dly[8]: Data 8 delay line
17:12			Dly[7]: Data 7 delay line
11:6			Dly[6]: Data 6 delay line
5:0			Dly[5]: Data 5 delay line Total delay = 50ps * Dly When Dly == 0, no delay. When Dly ==63, 3150ps delay. NOTE: the 50ps is typical delay, actually delay may vary from chip to chip, from different temperature.

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Table 11-251 SD_EMMC_ADJUST 0xc

Bit(s)	R/W	Default	Description
31:23			Unused
22	R/W	0	Adj_auto 1: Use cali_dut's first falling edge to adjust the timing, set cali_enable to 1 to use this function, simulation shows it can tracking 2.5ns range with 800ppm. 0: disable Working for HS200 mode, set Cali_enable to 1. Enhanced after gxlx project. Use RESP and DAT0 as reference, Separate RESP and DAT0, adjust the timing whenever there is a transition, insert a sample when there is no transition.
21:16	R/W	0	Adj_delay: Resample the input signals when clock index==adj_delay
15	R/W	0	Reserved
14	R/W	0	Cali_rise: 1: test the rising edge, recording rising edge location only. 0: test the falling edge
13	R/W	0	Adj_fixed: Adjust interface timing by resampling the input signals
12	R/W	0	Cali_enable: 1: Enable calibration 0: shut off to save power.
11:8	R/W	0	Cali_sel: Select one signal to be tested Signals are labeled from 0 to 9 the same as delay lines. Only one signal is tested at anytime. For example: Cali_sel == 9, test CMD line.

Table 11-252 SD_EMMC_CALOUT 0x10

Bit(s)	R/W	Default	Description
31:16			Unused
15:8	R		Cali_setup
7	R		Cali_vld: The reading is valid When there is no rising edge or falling edge event, the valid is low, this reading is not valid.
5:0	R		Cali_idx: The event happens at this index, The index starts from rising edge of core clock from 0, 1, 2, …

Table 11-253 SD_EMMC_ADJ_IDX_LOG 0x20

Bit(s)	R/W	Default	Description
31:30			Current, Auto_adj mode, Current 6 bits adj_idx
29:24			Previous 1
23:18			Previous 2
17:12			Previous 3
11:6			Previous 4
5:0			Previous 5: Last two bits of previous 5

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Table 11-254 SD_EMMC_CLKTEST_LOG 0x24

Bit(s)	R/W	Default	Description
31			Clktest_done, Test done
30:0			Clktest_times, Test clock core for 2^Clktest_exp

Table 11-255 SD_EMMC_CLKTEST_OUT 0x28

Bit(s)	R/W	Default	Description
31:0			Clktest_out, All 2^clktest_exp test results add up. Note: divided by 2^clktest_exp and get average clock core period length measured by 50ps delay cells.

Table 11-256 SD_EMMC_EYETEST_LOG 0x2c

Bit(s)	R/W	Default	Description
31			eyetest_done, Test done
30:0			eyetest_times, Test eye for 2^eyetest_exp

Table 11-257 SD_EMMC_EYETEST_OUT0 0x30

Bit(s)	R/W	Default	Description
31:0			All 2^eyetest_exp test results “OR” together.

Table 11-258 SD_EMMC_EYETEST_OUT1 0x34

Bit(s)	R/W	Default	Description
31:0			eyetest_out1, All 2^eyetest_exp test results “OR” together. Total eyeout [62:0] = {Eyetest_out1[30:0], eyetest_out0[31:0]} EYEtest output changed to 64 bits after TXLX and A113 {eyetest_out1, eyetest_out0}

Table 11-259 SD_EMMC_INTF3 0x38

Bit(s)	R/W	Default	Description
26			Eyetest_sel, 0 : select core clock as eyetest clock. 1 : select DS after delay line as eyetest clock. Eyetest point is DS after delay line.
25:23			NAND_EDO, NAND Async interface EDO position after RE rising edge. [not for SD_eMMC]
22			Sd_intf3, Using SD interface 3
21:18			Ds_sht_exp, 0: DS shift setting never expires, always using the DS_sht_m as shift length. None-zero: 2^ds_sht_exp after 2^ds_sht_exp “ms”, the setting expired The internal FSM will automatically change the Ds shift setting.
17:12			Ds_sht_m, Shift DS by number of 50ps delay cells. If using auto FSM mode, it is the initial value.
11			Eyetest_on, 1: Turn on eye test After eyetest_done, set this bit to 0 to reset eyetest internal registers.

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Bit(s)	R/W	Default	Description
10:6			Eyetest_exp, Repeat the eye test for 2^eyetest_exp times, Or the test results together, report the final results.
5			Clktest_on_m, Manual turn on clock test
4:0			Clktest_exp, Repeat the clock test for 2^clktest_exp times. Add the clock length together, report the sum.

Table 11-260 SD_EMMC_START 0x40

Bit(s)	R/W	Default	Description
31:2	R/W	0	Desc_addr[31:2]: Descriptor address, the last 2 Bits are 0, SRAM: 4 bytes aligned, the valid address range is from 0x200~0x3ff DDR: 8 bytes aligned the valid address is anywhere in DDR, the length of chain is unlimited. Desc_addr = ADDR>>2.
1	R/W	0	Desc_busy: Start/Stop 1: Start command chain execution process. 0: Stop Write 1 to this register starts execution. Write 0 to this register stops execution.
0	R/W	0	Desc_int: SRAM/DDR 1: Read descriptor from internal SRAM, limited to 32 descriptors. 0: Read descriptor from external DDR

Table 11-261 SD_EMMC_CFG 0x44

Bit(s)	R/W	Default	Description
31:28	R/W	0	Cfg_ip_txd_adj: Data 1 interrupt, when in TXD mode, the data 1 irq is a input signal, the round trip delay is uncertain factor, change this cfg to compensate the delay.
27	R/W	0	Cfg_err_abort: 1: abort current read/write and issue IRQ 0: continue on current read/write blocks.
26	R/W	0	Cfg_irq_ds: 1: Use DS pin as SDIO IRQ input, 0: Use DAT1 pin as SDIO IRQ input.
25	R/W	0	Cfg_txd_retry: When TXD CRC error, host sends the block again. The total number of retries of one descriptor is limited to 15, after 15 retries, the TXD_err is set to high.
24	R/W	0	Cfg_txd_add_err: TXD add error test. Test feature, should not be used in normal condition. It will inverted the first CRC Bits of the 3rd block. Block index starts from 0, 1, 2, …
23	R/W	0	Cfg_auto_clk: SD/eMMC Clock Control 1: when BUS is idle and no descriptor is available, automatically turn off clock, to save power. 0:.whenever core clock is on the SD/eMMC clock is ON, it is still on/off during read data from SD/eMMC.
22	R/W	0	Cfg_stop_clk: SD/eMMC Clock Control 1: no clock for external SD/eMMC, used in voltage switch. 0: normal clock, the clock is automatically on/off during reading mode to back off reading in case of DDR slow response.
21	R/W	0	Cfg_cmd_low: Hold CMD as output Low eMMC boot mode.
20	R/W	0	Reserved
19	R/W	0	Cfg_ignore_owner: Use this descriptor even if its owner bit is “0”.

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Bit(s)	R/W	Default	Description
18	R/W	0	Cfg_sdclk_always_on: 1: SD/eMMC clock is always ON 0: SD/eMMC clock is controlled by host. WARNING: Set SD/eMMC clock to always ON, host may lose data when DDR is slow.
17	R/W	0	Cfg_blk_gap_ip: 1: Enable SDIO data block gap interrupt period 0: Disabled.
16	R/W	0	Cfg_out_fall: DDR mode only The command and TXD start from rising edge. Set 1 to start from falling edge.
15:12	R/W	0	Cfg_rc_cc: Wait response-command, command-command gap before next command, 2cfg_rc_cc core clock cycles.
11:8	R/W	0	Cfg_resp_timeout: Wait response till 2cfg_resp_timeout core clock cycles. Maximum 32768 core cycles.
7:4	R/W	0	Cfg_bl_len: Block length 2cfg_bl_len, because internal buffer size is limited to 512 bytes, the cfg_bl_len <=9.
3	R/W	0	Cfg_dc_ugt: 1: DDR access urgent 0: DDR access normal
2	R/W	0	Cfg_ddr: 1: DDR mode 0: SDR mode
1:0	R/W	0	Cfg_bus_width: 0: 1 bit 1: 4 Bits 2: 8 Bits 3: 2 Bits (not supported)

Table 11-262 SD_EMMC_STATUS 0x48

Bit(s)	R/W	Default	Description
31	R		Core_busy: 1: core is busy, desc_busy or sd_emmc_irq or bus_fsm is not idle. 0: core is idle.
30	R		Desc_busy: 1: Desc input process is busy, more descriptors in chain. 0: no more descriptor in chain or desc_err.
29:26	R		Bus_fsm: BUS fsm
25	R		DS: Input data strobe
24	R		CMD_i: Input response signal
23:16	R		DAT_i: Input data signals
15	R/W		IRQ_sdio: SDIO device uses DAT[1] to request IRQ
14	R/W		Resp_status: When resp_num is set to 1, the resp_addr is the response status IRQ enable Bits, if there is an error.
13	R/W		End_of_Chain: End of Chain IRQ, Normal IRQ
12	R/W		Desc_timeout: Descriptor execution time over time limit. The timeout limit is set by descriptor itself. Consider the multiple block read/write, set the proper timeout limits.
11	R/W		Resp_timeout: No response received before time limit. The timeout limit is set by cfg_resp_timeout.
10	R/W		Resp_err: Response CRC error
9	R/W		Desc_err: SD/eMMC controller doesn't own descriptor. The owner bit is "0", set cfg_ignore_owner to ignore this error.

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Bit(s)	R/W	Default	Description
8	R/W		Txd_err: TX data CRC error, For multiple block write, any one of blocks CRC error.
7:0	R/W		Rxd_err: RX data CRC error per wire, for multiple block read, the CRC errors are Ored together.

Table 11-263 SD_EMMC_IRQ_EN 0x4c

Bit(s)	R/W	Default	Description
31:18			unused
17			Cfg_cmd_setup, 1: improve CMD setup time by half SD_CLK cycle. 0: CMD is two cycle aligned with DATA.
16	R/W	0	Cfg_secure: Data read/write with crypto DES
15	R/W	0	en_IRQ_sdio: Enable sdio interrupt.
14	R/W	0	En_resp_status: Response status error.
13	R/W	0	en_End_of_Chain: End of Chain IRQ
12	R/W	0	en_Desc_timeout: Descriptor execution time over time limit.
11	R/W	0	en_Resp_timeout: No response received before time limit.
10	R/W	0	en_Resp_err: Response CRC error
9	R/W	0	en_Desc_err: SD/eMMC controller doesn't own descriptor.
8	R/W	0	En_txd_err: TX data CRC error
7:0	R/W	0	en_Rxd_err: RX data CRC error per wire.

Table 11-264 Descriptor_REG0 0x50

Bit(s)	R/W	Default	Description
31:0	R/W		SD_EMMC_CMD_CFG APB read wait Same as descriptor first word, resp_num = 1, response saved back into descriptor only. Read from this APB will hold APB bus

Table 11-265 Descriptor_REG1 0x54

Bit(s)	R/W	Default	Description
31:0	R/W		SD_EMMC_CMD_ARG APB write start Same as descriptor second word. Write to this APB address starts execution. If the current desc is busy, it will be executed after current descriptor is done.

Table 11-266 Descriptor_REG2 0x58

Bit(s)	R/W	Default	Description
31:0	R/W		SD_EMMC_CMD_DAT : Same as descriptor third word, 32 Bits data.

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Table 11-267 Descriptor_REG3 0x5c

Bit(s)	R/W	Default	Description
31:0	R/W		SD_EMMC_CMD_RSP: Write: response status IRQ enable Bits. Read: Response Bit 31:0

Table 11-268 Descriptor_REG4 0x60

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_CMD_RSP1: Response bit 63:32

Table 11-269 Descriptor_REG5 0x64

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_CMD_RSP2: Response bit 95:64

Table 11-270 Descriptor_REG6 0x68

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_CMD_RSP3: Response bit 127:96

Table 11-271 Descriptor_REG7 0x6c

Bit(s)	R/W	Default	Description
31:0			Reserved

Table 11-272 Current_Next_Descriptor_REG0 0x70

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_CURR_CFG: Current descriptor under execution.

Table 11-273 Current_Next_Descriptor_REG1 0x74

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_CURR_ARG

Table 11-274 Current_Next_Descriptor_REG2 0x78

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_CURR_DAT

Table 11-275 Current_Next_Descriptor_REG3 0x7c

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_CURR_RSP

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Table 11-276 Current_Next_Descriptor_REG4 0x80

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_NEXT_CFG: Next descriptor waiting for execution, already read out from SRAM or DDR, can't be changed.

Table 11-277 Current_Next_Descriptor_REG5 0x84

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_NEXT_ARG

Table 11-278 Current_Next_Descriptor_REG6 0x88

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_NEXT_DAT

Table 11-279 Current_Next_Descriptor_REG7 0x8c

Bit(s)	R/W	Default	Description
31:0	R		SD_EMMC_NEXT_RSP

Table 11-280 SD_EMMC_RXD 0x90

Bit(s)	R/W	Default	Description
31:25	R		Unused
24:16	R		Data_blk: Rxd Blocks received from BUS Txd blocks received from DDR.
15:10			unused
9:0	R		Data_cnt: Rxd words received from BUS. Txd words received from DDR.

Table 11-281 SD_EMMC_TXD 0x94

Bit(s)	R/W	Default	Description
31:25			Unused
24:16	R		Txd_blk: Txd BUS block counter
15			unused
14:0	R		Txd_cnt: Txd BUS cycle counter

11.4 Serial Peripheral Interface Communication Controller

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11.4.1 Overview

SPI Communication Controller is designed for connecting general SPI protocol compatible module. This controller allows rapid data communication with less software interrupts than conventional serial communications.

11.4.2 Features

- Full-duplex synchronous serial interface
- Master/Slave configurable
- Four chip selects to support multiple peripherals
- Transfer continuation function allows unlimited length data transfers
- 64-bit wide by 16-entry FIFO for both transmit and receive data
- Polarity and phase of the Chip Select (SS) and SPI Clock (SCLK) are configurable
- Both PIO(Programming In/Out interface) and DMA(Direct Memory Access interface) supported

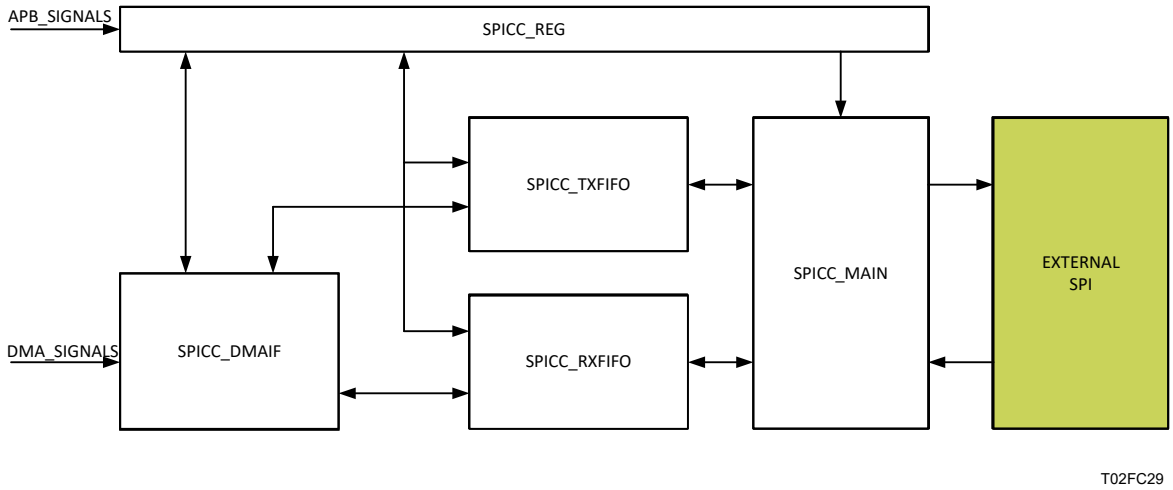
11.4.3 Function

The following are two SPICC modes of operation:

- **Master Mode**—When the SPICC module is configured as a master, it uses a serial link to transfer data between the SPICC and an external device. A chip-enable signal and a clock signal are used to transfer data between these two devices. If the external device is a transmit-only device, the SPICC master's output port can be ignored and used for other purposes. To use the internal TXFIFO and RXFIFO, two auxiliary output signals, SS and SPI_RDY, are used for data transfer rate control. The user can also program the sample period control register to a fixed data transfer rate.
- **Slave Mode**–When the SPICC module is configured as a slave, the user can configure the SPICC Control register to match the external SPI master's timing. In this configuration, SS becomes an input signal, and is used to control data transfers through the Shift register, as well as to load/store the data FIFO.

There are 5 sub-modules in spi communication controller, i. e. spicc_reg, spicc_dmaif, spicc_txfifo, spicc_rxfifo, and spicc_main. Transmitting and receiving are using different channel, that means they have different buffer.

Figure 11-3 SPICC



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Note

- spicc_reg is driven by host cpu, and spicc_reg is responsible for configuring other modules.
- spicc_dmaif is responsible for dealing with DMA operatoins.
- spicc_txfifo contains a transmission FIFO.
- spicc_rxfifo contains a receiving FIFO.
- spicc_main is responsible for main control of basic spi operation.

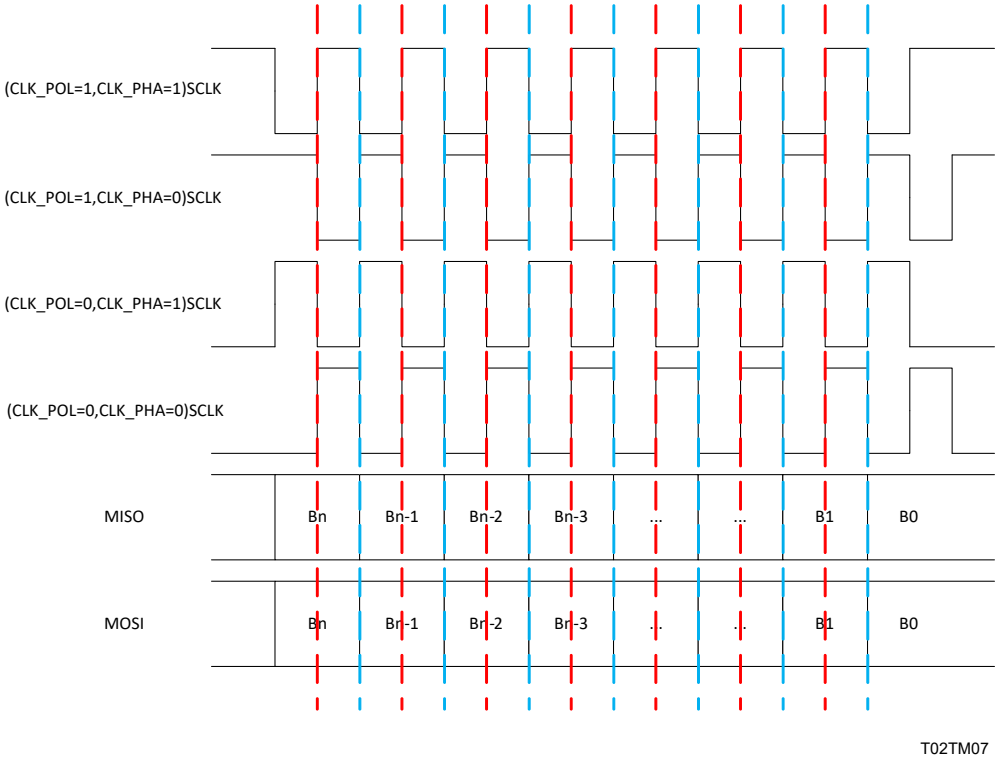
Here are SPI External Signals:

Table 11-282 SPI External Signals

Signal Name	I/O	Description
spicc_sclk	IO	SCLK, SPI Clock
spicc_miso	IO	MISO, Master Input Slave Output
spicc_mosi	IO	MOSI, Master Output Slave In
spicc_ss[3:0]	IO	SS, SPI chip Select, Supports up to 4 slaves.

And here is SPI Generic Timing:

Figure 11-4 SPI Generic Timing



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11.4.4 Register Description

Table 11-283 RXDATA 0xffd13000

Bit(s)	R/W	Default	Description
31:0	R	0	Rx Data

Note

When PIO mode, programmer can get data from this register.

Table 11-284 TXDATA 0xffd13004

Bit(s)	R/W	Default	Description
31:0	W	0	Tx Data

Note

When PIO mode, programmer need send data to this register.

Table 11-285 CONREG 0xffd13008

Bit(s)	R/W	Default	Description
31:19	RW	0	[13]burst_length ([5:0]bit number of one word/package, [12:6]burst length-1)
18:16	RW	0	[3]data_rate (sclk will be divided by system clock with equation: 2^(data_rate+2), Example: if system clock = 128MHz and data_rate=2, sclk's frequency equals 8MHz)
15:14			Reserved
13:12	RW	0	[2]chip_select (00:select ss_0, 01:select ss_1, 10:select ss_2, 11:select ss_3,)
11:10			Reserved
9:8	RW	0	[2]drctl (0:ignore RDY input, 1:Data ready using pin rdy_i's failing edge, 2:Data ready using pin rdy_i's low level, 3:reserved)
7	RW	0	[1]sspol (0:SS polarity Low active,1:High active)
6	RW	0	[1]ssctl (see details in Note1)
5	RW	0	[1]pha (clock/data phase control, see section 2.2)
4	RW	0	[1]pol (clock polarity control, see section 2.2)
3	RW	0	[1]smc (start mode control, see Note2)
2	RW	0	[1]xch(exchange bit, ATTN:will automatically cleared when burst finished, see Note3)

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Bit(s)	R/W	Default	Description
1	RW	0	[1]mode (0:slave,1:master)
0	RW	0	[1]en (0:spicc disable,1:enable)

Note

- In one burst of master mode, if ssctl ==1, ss will output 1 between each spi transition. And if ssctl ==0, ss will output 0.
- Smc is for start mode control. If smc ==0, burst will start when xch is set to 1'b1; if smc==1, burst will start when txfifo is not empty.
- Setting xch will issue a burst when smc==0, and This bit will be self-cleared after burst is finished.

Table 11-286 INTREG 0xffd1300c

Bit(s)	R/W	Default	Description
31:8			Reserved
7	RW	0	[1]tcen(transfer completed interrrupt enable)
6	RW	0	Reserved
5	RW	0	[1]rfen(rxfifo full interrrupt enable)
4	RW	0	Reserved
3	RW	0	[1]rren(rxfifo ready interrrupt enable)
2	RW	0	[1]tfen(txfifo full interrrupt enable)
1	RW	0	Reserved
0	RW	0	[1]teen(txfifo empty interrrupt enable)

Note

Interrupt Status presents in STATREG.

Table 11-287 DMAREG 0xffd13010

Bit(s)	R/W	Default	Description
31:26	RW	0	[6]DMA Burst Number
25:20	RW	0	[6]DMA Thread ID
19	RW	0	[1]DMA Urgent
18:15	RW	0x7	[4]Number in one Write request burst(0:1,1:2...)
14:11	RW	0x7	[4]Number in one Read request burst(0:1,1:2...)
10:6	RW	0x8	[5]RxFIFO threshold(RxFIFO's count>=thres, will request write)
5:1	RW	0	[5]TxFIFO threshold(TxFIFO's count<=thres, will request read)
0	RW	0	[1]DMA Enable

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Table 11-288 STATREG 0xffd13014

Bits	R/W	Defaults	Description
31:8			Reserved
7	RW	0	[1]tc(transfer completed, w1c, see Note1)
6	R	0	Reserved
5	R	0	[1]rf(rxfifo full)
4	R	0	Reserved
3	R	0	[1]rr(rxfifo ready)
2	R	0	[1]tf(txfifo full)
1	R	0	Reserved
0	R	0	[1]te(txfifo empty)

Note

Tc is the status bit which indicates a burst transfer is completed. And a burst transfer should be started by writing xch 1'b1. This bit supports w1c(Write 1 clear).

Table 11-289 PERIODREG 0xffd13018

Bit(s)	R/W	Default	Description
31:15			Reserved
14:0	RW	0	[15]period(wait cycles, see Note1)

Note

Programmer can add wait cycles through this register if transmission rate need to be controlled.

Table 11-290 TESTREG 0xffd1301c

Bit(s)	R/W	Default	Description
31:23	RW	0	Reserved
23:22	RW	0	[2]fiforst(fifo soft reset)
21:16	RW	0x15	[6]dlyctl(delay control)
15	RW	0	[1]swap(data swap for reading rxfifo)
14	RW	0	[1]lbc(loop back control)
12:10	R	0	[3]smstatus(internal state machine status)
9:5	R	0	[5]rxcnt(internal RxFIFO counter)
4:0	R	0	[5]txcnt(internal TxFIFO counter)

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Note

Programmer can only use the TESTREG[9:0], rxcnt(internal RxFIFO counter) and txcnt(internal TxFIFO counter) , and other Bits just for test.

Table 11-291 DRADDR 0xffd13020

Bit(s)	R/W	Default	Description
31:0	RW	0	Read Address of DMA

Table 11-292 DWADDR 0xffd13024

Bit(s)	R/W	Default	Description
31:0	RW	0	Write Address of DMA

Table 11-293 LD_CNTL0 0xffd13028

Bits	R/W	Defaults	Description
31:9	RW	0	Reserved
8	RW	0	dma raddr/waddr load by dma_enable signal
7	RW	0	dma waddr load by vsync irq
6	RW	0	dma raddr load by vsync irq
5	RW	0	dma write counter enable
4	RW	0	dma read counter enable
3	RW	0	xch enable set by vsync irq
2	RW	0	dma enable set by vsync irq
1	RW	0	Reserved
0	RW	0	Vsync irq source select

Table 11-294 LD_CNTL1 0xffd1302c

Bits	R/W	Defaults	Description
31:16	RW	0	dma write counter
15:0	RW	0	dma read counter

Table 11-295 LD_RADDR 0xffd13030

Bits	R/W	Defaults	Description
31:0	RW	0	shadow dma read address for load

Table 11-296 LD_WADDR 0xffd13034

Bits	R/W	Defaults	Description
31:0	RW	0	shadow dma write address for load

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Table 11-297 ENHANCE_CNTL 0xffd13038

Bits	R/W	Defaults	Description
31:30	R	0	Reserved
29	R/W	0	main clock always on
28	R/W	0	clk-cs delay enable
27	R/W	0	cs_oen enhance enable
26	R/W	0	clk_oen enhance enable
25	R/W	0	mosi_oen enhance enable
24	R/W	0	spi clk select 0: controlled by data_rate in CONREG 1: controlled by enhance_clk_div in ENHANCE_CNTL
23:16	R/W	0	enhance_clk_div
15:0	R/W	0	clk-cs delay value

Table 11-298 ENHANCE_CNTL1 0xffd1303c

Bits	R/W	Defaults	Description
31:29	R/W	0	enhance_fclk_mosi_oen_dlyctl: mosi_oen delay control in fclk
28	R/W	0	enhance_fclk_mosi_oen_dlyctl_en: enable dlyctl
27:25	R/W	0	enhance_fclk_mosi_o_dlyctl: mosi_o delay control in fclk
24	R/W	0	enhance_fclk_mosi_o_dlyctl_en: enable dlyctl
23:21	R/W	0	enhance_fclk_miso_i_dlyctl: miso_i delay control in fclk
20	R/W	0	enhance_fclk_miso_i_dlyctl_en: enable dlyctl
19:17	R/W	0	enhance_fclk_mosi_i_dlyctl: mosi_i delay control in fclk
16	R/W	0	enhance_fclk_mosi_i_dlyctl_en: enable dlyctl
15	R/W	0	enhance_fclk_en: fclk gate enable
14	R/W	0	enhance_mosi_i_capture_en: enable, 1=select enhance capture function for mosi_i (slave mode)
9:1	R/W	0	enhance_clk_tcnt: adjust capturing timing for miso_i data (master mode). when clk_cnt=enhance_clk_tcnt, capture input data. the value of enhance_clk_tcnt must be less than the most value of clk_cnt. sclk will be divided by system clock with clk_cnt.
0	R/W	0	enhance_miso_i_capture_en: enable, 1=select enhance capture function for miso_i (master mode)

Table 11-299 ENHANCE_CNTL2 0xffd13040

Bits	R/W	Defaults	Description
31	R/W	0	clk_cs_tt delay enable
30:16	R/W	0	clk_cs_tt delay value

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Bits	R/W	Defaults	Description
15	R/W	0	clk_cs_ti delay enable
14:0	R/W	0	clk_cs_ti delay value

SPICC1: See the registers for SPICC0

11.5 Serial Peripheral Interface Flash Controller

11.5.1 Overview

SPI Flash Controller is designed for connecting varied SPI Flash memory.

11.5.2 Features

- Support three operation modes, NOR Flash mode, Master mode, and Slave mode.
- Support read/write buffer up to 64bytes.
- Support no clock toggling during DUMMY state.
- Support hold by an external pin during a transition.
- AHB read support byte and halfword.
- Support bit-number rather than byte-number for each stage.
- Support 2/4 wire writing like fast reading
- Support both rising-edge and falling-edge for SPI slave sampling and SPI master sampling.
- Support 1 wire for SPI_D and SPI_Q.
- Support SPI_CK setup and hold time by cycles
- Support 8 bit clock divider, so SPI_CK can be low as 1/256 HCLK
- Support byte-order in a word
- Support no command state, so the command is sent/received in address state by 2/4 wires.
- Support both data input and data output in a transition. SPI_DOUT->(SPI_DUMMY)->SPI_DIN

11.5.3 Register Description

Note

If the bit2 “Backward Compatible” in “SPI User Register” is 1, the registers and functions of this SPI controller are as same as Apollo SPI controller.

To use SPI Flash commands, please set the bit2 “Backward Compatible” in “SPI User Register” as “1”.

Table 11-300 SPI FLASH Command register 0xffd14000

Bit(s)	R/W	Default	Description
31	R/W	0	READ command. 1 = read. When it becames 0, the read command is finished. The READ command could be (0xEB, 0x6B, 0xBB, 0x3B, 0x0B, 0x03). By default is 0x0B. One read command will read erilog 32x8bits data. And saved in data cache.
30	R/W	0	WREN command. (0x06)

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Bit(s)	R/W	Default	Description
29	R/W	0	WRDI command. (0x04).
28	R/W	0	RDID command. (0x9f).
27	R/W	0	RDSR command. (0x05).
26	R/W	0	WRSR command. (0x01).
25	R/W	0	Page program command. (0xAD or 0x02).
24	R/W	0	SE command (0x20).
23	R/W	0	BE command (0xD8).
22	R/W	0	CE command.(0xC7).
21	R/W	0	Deep Power Down command(0xB9).
20	R/W	0	RES command. (0xAB).
19	R/W	0	HPM command.(0xA3). (Just For winbond SPI flash).
18	R/W	0	USER defined command.
17:0	R/W	0	Reserved for future.

If the bit2 “Backward Compatible” in “SPI User Register” is 1, the bit15:0 of this registers are defined as same as Apollo SPI controller.

Bit(s)	R/W	Default	Description
15	R/W	0	USER command address bit. 1 = user command includes address. 0 = no address.
14	R/W	0	USER command dummy bit. 1= user command includes Dummy bytes.
13	R/W	0	USER command DIN bit. 1 = use command includes data in. 0 = no data in.
12	R/W	0	USER command DO bit. 1 -= use command includes data output. 0 = no data output. Only DIN or DO support for one User command. Not both.
11:10	R/W	0	User command dummy byte number. How many dummy bytes for user command.
9:8	R/W	0	Reserved for future.
7:0	R/W	0	command value for User command bit[7:0].

Table 11-301 SPI address register 0xffd14004

Bit(s)	R/W	Default	Description
31:0	R/W	0	The address[31:0] of the user command

If the bit2 “Backward Compatible” in “SPI User Register” is 1, the bit15:0 of this registers are defined as same as Apollo SPI controller.

31:30	R/W	0	Reserved.
29:24	R/W	0	DIN/DO data bytes number.
23:0	R/W	0	24 bits address.

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Table 11-302 SPI control register 0xffd14008

Bit(s)	R/W	Default	Description
31:27	R/W	0	Reserved.
26	R/W	0	Write bit order. 1 = 0, 1, 2, 3, 4, 5, 6, 7. 0 = 7, 6, 5, 4, 3, 2, 1, 0.
25	R/W	0	Read bit order. . 1 = 0, 1, 2, 3, 4, 5, 6, 7. 0 = 7, 6, 5, 4, 3, 2, 1, 0.
24	R/W	0	Fast read QIO mode.
23	R/W	0	Fast read DIO mode.
22	R/W	0	Write 2 bytes status mode. For some of winbond SPI flash, the status register is 16bits.
21	R/W	1	SPI flash WP pin value if use SPI flash WP pin as write protection.
20	R/W	0	Fast read QOUT mode.
19	R/W	1	1 = SPI share pins with SDRAM. 0 = doesn't share.
18	R/W	0	SPI hold mode. 1=SPI controller would use SPI hold function. 0 = SPI controller won't use hold function. The SPI flash hold pin can be tie high on the board. Or SPI controller can use hold pin as QIO/QOUT mode.
17	R/W	1	1 = enable AHB request. 0 = disable AHB request when you reconfigure SPI controller or running APB bus commands.
16	R/W	0	1 =enable SST SPI Flash aai command. The APB bus PP command will send AAI command.
15	R/W	1	1 = release from Deep Power-Down command is with read electronic signature.
14	R/W	0	Fast read DOUT mode.
13	R/W	1	Fast read mode. AHB bus read requirement and APB bus read command use the command 0x0Bh.
12:0	R/W	0	Reserved for future.

If the bit2 "Backward Compatible" in "SPI User Register" is 1, the bit12:0 of this registers are defined as same as Apollo SPI controller.

Bit(s)	R/W	Default	Description
12	R/W	1	1=SPI clock frequency is same as system clock. 0 = SPI clock frequency will use clock divider.
11:8	R/W	0	Clock counter for clock divider.
7:4	R/W	0	Clock high counter.
3:0	R/W	0	Clock low counter. If the SPI clock frequcncy = sys_clock_frequency / n. Then the clock divider counter = n – 1; the clock high counter = n /2 – 1; the clock low counter = n– 1; For example, if you want to SPI clock erilog is divided by 2 of the system clock. The clock divider counter = 1, clock high counter = 0, clock low counter = 1. For SPI clock frequency = system clock / 4. The clock divider counter = 3, clock high counter = 1, clock low counter = 3.

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Table 11-303 SPI control register 1 0xffd1400c

Bit(s)	R/W	Default	Description
31:28	R/W	5	SPI Clock cycles for SPI flash timing requirement tCSH.
27:16	R/W	0xfff	SPI Clock cycles for SPI flash timing requirement tRES.
15:0	R/W	0x0120	System clock cycles for SPI bus timer. In SPI share bus and SPI hold function mode. SPI bus timer used , if SPI use the bus for a limit time, SPI controller will diassert SPI hold pin to halt the SPI Flash, and give the bus control to SDRAM.

Table 11-304 SPI status register

Bit(s)	R/W	Default	Description
31:24	R/W	0	Reserved.
23:16	R/W	0	For winbond SPI flash, this 8 bits used for DIOmode M7~M0,
15:0	R/W	0	SPI status register value. WRSR command will write this value to SPI flash status. RDSR or RES command will save the read result to this register.

When SPI controller in the slave mode, this register are the status for the SPI master to read out.

Bit(s)	R/W	Default	Description
31:0	R/W	0	In SPI Slave mode, the read status of the user command

Table 11-305 SPI control register 2 0xffd14014

Bit(s)	R/W	Default	Description
31:28	R/W	0	Delay cycle number of SPI_CS input in SPI slave mode or SPI_CS output in SPI master mode 0= not delay, 1 = delayed by 1 cycle of system clock, 2 = delayed by 2 cycles of system clock , …
27:26	R/W	0	delay mode of SPI_CS input in SPI slave mode or SPI_CS output in SPI master mode 0= not latched by the edges of SPI_CK 1= latched by the falling edges of SPI_CK 2 = latched by the rising edges of SPI_CK
25:23	R/W	0	Delay cycle number of SPI Data from SPI Master to SPI Slave In SPI master mode, it is for data outputs; in SPI slave mode, it is for data inputs. 0= not delay, 1 = delayed by 1 cycle of system clock, 2 = delayed by 2 cycles of system clock , …
22:21	R/W	0	Delay mode of SPI Data from SPI Master to SPI Slave In SPI master mode, it is for data outputs; in SPI slave mode, it is for data inputs. 0= not latched by the edges of SPI_CK 1= latched by the falling edges of SPI_CK 2 = latched by the rising edges of SPI_CK
20:18	R/W	0	Delay cycle number of SPI Data from SPI Slave to SPI Master. In SPI master mode, it is for data inputs; in SPI slave mode, it is for data outputs. 0= not delay, 1 = delayed by 1 cycle of system clock, 2 = delayed by 2 cycles of system clock , …
17:16	R/W	0	Delay mode of SPI Data from SPI Slave to SPI Master. In SPI master mode, it is for data inputs; in SPI slave mode, it is for data outputs. 0= not latched by the edges of SPI_CK

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Bit(s)	R/W	Default	Description
			1= latched by the falling edges of SPI_CK 2 = latched by the rising edges of SPI_CK
15:12	R/W	0	In SPI master mode, SPI_CK rising edge mode 4'b1000 = later by 1/4 cycle of SPI_CK 4'b1001 = later by 1/8 cycle of SPI_CK 4'b1010 = later by 1/16 cycle of SPI_CK 4'b1011 = later by 1/32 cycle of SPI_CK 4'b1100 = earlier by 1/4 cycle of SPI_CK 4'b1101 = earlier by 1/8 cycle of SPI_CK 4'b1110 = earlier by 1/16 cycle of SPI_CK 4'b1111 = earlier by 1/32 cycle of SPI_CK Others = Normal
11:8	R/W	0	In SPI master mode, SPI_CK falling edge mode 4'b1000 = later by 1/4 cycle of SPI_CK 4'b1001 = later by 1/8 cycle of SPI_CK 4'b1010 = later by 1/16 cycle of SPI_CK 4'b1011 = later by 1/32 cycle of SPI_CK 4'b1100 = earlier by 1/4 cycle of SPI_CK 4'b1101 = earlier by 1/8 cycle of SPI_CK 4'b1110 = earlier by 1/16 cycle of SPI_CK 4'b1111 = earlier by 1/32 cycle of SPI_CK Others = Normal
7:4	R/W	1	In master mode, SPI clock cycles for SPI hold timing.
3:0	R/W	1	In master mode, SPI clock sycles for SPI setup timing. SPI setup time and SPI hold time is used to configure how soon the controller can enable spi_cs_n after the controller get the bus and how long the controller still keep the bus after the spi_cs_n become to be high.

Table 11-306 SPI Clock register 0xffd14018

Bit(s)	R/W	Default	Description
31	R/W	1	1=SPI clock frequency is same as system clock. 0 = SPI clock frequency will use clock divider.
30:18	R/W	0	Clock counter for Pre-scale divider: 0= not pre-scale divider, 1= pre-scale divided by 2, 2= pre-scale divided by 3, ….,
17:12	R/W	0	Clock counter for clock divider.
11:6	R/W	0	Clock high counter in SPI master mode. In SPI slave mode, it is for the delay counter for the rising edges of spi_ck_i
5:0	R/W	0	Clock low counter, in SPI master mode. In SPI slave mode, it is for the delay counter for the falling edges of spi_ck_i If the SPI clock frequcncy = sys_clock_frequency / n. Then the clock divider counter = n – 1; the clock high counter = n /2 – 1; the clock low counter = n– 1; For example, if you want to SPI clock erilog is divided by 2 of the system clock. The clock divider counter = 1, clock high counter = 0, clock low counter = 1. For SPI clock frequency = system clock / 4. The clock divider counter = 3, clock high counter = 1, clock low counter = 3.

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Table 11-307 SPI User register 0xffd1401c

Bit(s)	R/W	Default	Description
31	R/W	1	USER command COMMAND bit. 1 = user command includes command. 0 = no command. If some SPI slaves may support 2/4 IO at the first cycle, clear this bit.
30	R/W	0	USER command ADDRESS bit. 1 = user command includes address. 0 = no address.
29	R/W	0	USER command DUMMY bit. 1= user command includes Dummy bytes.
28	R/W	0	USER command DIN bit. 1 = user command includes data in. 0 = no data in.
27	R/W	0	USER command DO bit. 1 = user command includes data output. 0 = no data output. If both DIN and DO are valid, SPI master is firstly in data output state and then in data input state. If all of DUMMY, DO and DIN are valid, SPI master is firstly in data output state and then in dummy state, finally in data input state.
26	R/W	0	USER command dummy idle bit. 1= no SPI clock toggling in dummy state. 0= normal
25	R/W	0	USER command highpart bit for SPI_DOUT stage. It is for data-output in spi master mode and for data-input in spi slave mode. 1 = only high half part of buffer are used. 0 = low half part or the whole 64bytes are used.
24	R/W	0	USER command highpart bit for SPI_DIN stage. It is for data-input in spi master mode and for data-output in spi slave mode. 1 = only high half part of buffer are used. 0 = low half part or the whole 64bytes are used.
23	R/W	0	User command external hold bit for prep. 1 = in prep state, SPI master controller can be hold by the external pin SPI_HOLD
22	R/W	0	User command external hold bit for command. 1 = in command state, SPI master controller can be hold by the external pin SPI_HOLD
21	R/W	0	User command external hold bit for address. 1 = in address state, SPI master controller can be hold by the external pin SPI_HOLD
20	R/W	0	User command external hold bit for dummy. 1 = in dummy state, SPI master controller can be hold by the external pin SPI_HOLD
19	R/W	0	User command external hold bit for data input. 1 = in data input state, SPI master controller can be hold by the external pin SPI_HOLD
18	R/W	0	User command external hold bit for data output. 1 = in data output state, SPI master controller can be hold by the external pin SPI_HOLD
17	R/W	1	User command external hold polarity bit. 1 = high is valid for hold, 0 = low is valid for hold.
16	R/W	0	Single DIO mode: Data output and input apply only 1 wire.
15	R/W	0	Fast write QIO mode.
14	R/W	0	Fast write DIO mode.
13	R/W	0	Fast write QOUT mode.
12	R/W	0	Fast write DOUT mode.
11	R/W	0	Write byte order. 0 = d[7:0], d[15:8], d[23:16], d[31:24]. 1 = d[31:24], d[23:16], d[15:8], d[7:0]
10	R/W	0	Read byte order. . 0 = d[7:0], d[15:8], d[23:16], d[31:24]. 1 = d[31:24], d[23:16], d[15:8], d[7:0]
9:8	R/W	0	AHB endian mode: 0= little-endian; 1= big-endian; 2~3 reserved

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Bit(s)	R/W	Default	Description
7	R/W	0	In SPI master mode, the clock output edge bit: 0 = SPI_CK is inverted, 1 = SPI_ CK is not inverted
6	R/W	1	In SPI slave mode, the clock input edge bit: 0 = SPI_CK_I is inverted, 1 = SPI_ CK_I is not inverted
5	R/W	0	SPI CS setup bit: 1 = valid in prep state
4	R/W	0	SPI CS hold bit: 1 = valid in done state
3	R/W	0	AHB-read apply the configurations of user-command, such as command value, bit-length,…
2	R/W	1	Backward Compatible: 1 = compatible to Apollo SPI This bit affect the three registers: “SPI Flash Commmand Register”, “SPI Address Register” and “SPI Control Register”
1	R/W	0	AHB-read support 4byte address, when AHB-read apply the configurations of user-command. 1 = 4byte address, 0 = 3byte address
0	R/W	0	In SPI master mode, Enable bit for Data input during SPI_DOUT stage. 1 = enable; 0 = disable This bit shall not be used in 2/4wire or SIO. When this bit is 1, during SPI_DOUT stage, data input are stored into cacheline/ buffer from address 0, i.e., Bit24 is not controlling the start address. The data output can be specified by bit25

Table 11-308 SPI User register 1 0xffd14020

Bit(s)	R/W	Default	Description
31:26	R/W	0	USER command bit number for address state 0 = 1 bit, 1= 2 bits, …
25:17	R/W	0	USER command bit number for data output state 0 = 1 bit, 1= 2 bits, …
16:8	R/W	0	USER command bit number for data input state 0 = 1 bit, 1= 2 bits, …
7:0	R/W	0	USER command cycle number for dummy state 0 = 1 cycle, 1= 2 cycles, …

Table 11-309 SPI User register 2 0xffd14024

Bit(s)	R/W	Default	Description
31:28	R/W	0	USER command bit number for command state 0 = 1 bit, 1= 2 bits, …
27:16	R/W	0	Reserved
15:0	R/W	0	The command content of the user command

Table 11-310 SPI User register 3 0xffd14028

Bit(s)	R/W	Default	Description
31:0	R/W	0	In SPI Master mode, the address[63:32] of the user command In SPI Slave mode, the write status of the user command

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Table 11-311 SPI PIN register 0xffd1402c

Bit(s)	R/W	Default	Description
31	R/W	0	Pin swap when it is in DIN stage and SPI data input are 4wire. This feature is for Ubec Zigbee chips. 1 = swap between {spi_q, spi_d_i} and {spi_hold_i, spi_wp_i} 0 = normal
30	R/W	0	In SPI Master mode, CS keep active after a transition. 1 = enable; 0 = disable
29	R/W	0	Idle edge of SPI_CK 0 = low when it is idle 1 = high when it is idle
28:24	R/W	0	Reserved
23	R/W	0	In the SPI slave mode, spi_cs_i polarity: 1= high voltage is active 0= low voltage is active
22:21	R/W	0	In the SPI slave mode, spi_ck_i and spi_cs_i source pins 0=SPI_CK and SPI_CS pins, respectively 1=SPI_CS2 and SPI_CS1 pins, respectively 2=SPI_HOLD and SPI_WP pins, respectively
20	R/W	0	SPI_CS2 and SPI_CS1 pin function MUX 0= spi_ck and spi_cs in the SPI master mode 1= input pads for spi_ck_i and spi_cs_i, respectively, in the SPI slave mode
19	R/W	0	SPI_CK and SPI_CS pin function MUX 0= spi_ck and spi_cs in the SPI master mode 1= input pads for spi_ck_i and spi_cs_i, respectively, in the SPI slave mode
18:17	R/W	0	SPI_HOLD and SPI_WP pin function MUX 0= normal 1= spi_q and spi_d, respectively 2= spi_cs3 and spi_cs2, repectively 3= input pads for spi_ck_i and spi_cs_i, respectively, in the SPI slave mode
16	R/W	0	SPI_D and SPI_Q switch 1= SPI_D and SPI_Q pin-functions are swapped 0= normal
15:11	R/W	0	In Master mode, these are spi_ck MUX bit[4:0] for spi_cs4 (SPI_HOLD pin), spi_cs3 (SPI_WP pin), spi_cs2, spi_cs1 and spi_cs, respectively 1= this pin is spi_ck, if this pin is not idle 0= this pin is spi_cs, if this pin is not idle
10:6	R/W	0	In Master mode, these are polarity bit[4:0] for spi_cs4 (SPI_HOLD pin), spi_cs3 (SPI_WP pin), spi_cs2, spi_cs1 and spi_cs, respectively 1= high voltage is active 0= low voltage is active
5:0	R/W	0x1E	In Master mode, these are idle bit[5:0] for SPI_CK, for spi_cs4 (SPI_HOLD pin), spi_cs3 (SPI_WP pin), spi_cs2, spi_cs1 and spi_cs, respectively 1= idle, i.e., the spi_ck signal is 0 or the spi_cs is at the inactive level 0= active if SPI controller is working

Table 11-312 SPI Slave register 0xffd14030

Bit(s)	R/W	Default	Description
31	R/W	0	SPI controller SW reset: 1 = reset, 0 = none
30	R/W	0	SPI slave mode: 1 = slave, 0 = master
29	R/W	0	In SPI slave mode, the enable bit for the command of write-buffer-and-read-buffer

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Bit(s)	R/W	Default	Description
			0= disable, 1=enable
28	R/W	0	In SPI slave mode, the enable bit for the command of write-status-and-read-status 0= disable, 1=enable
27	R/W	0	SPI slave command define enable 0=Apply the last 3bits of Flash commands and two extra command. 1=Apply the user defined command in SPI Slave register 3
26:4	R/W	0	Reserved
11:10	R/W	0	spi_cs_i recovery mode: 0= and 1= or 2= normal 3= delayed
9:5	R/W	0	Interrupt enable for bit4:0 1= enable, 0=disable
4	R/W	0	A SPI transition is done. (whatever it is in SPI master mode or SPI slave mode)
3	R/W	0	In SPI slave mode, a status write is done
2	R/W	0	In SPI slave mode, a status read is done
1	R/W	0	In SPI slave mode, a buffer write is done
0	R/W	0	In SPI slave mode, a buffer read is done

Table 11-313 SPI Slave register 1 0xffd14034

Bit(s)	R/W	Default	Description
31:27	R/W	0	In SPI slave mode, status bit number 0 = 1 bit, 1= 2 bits, …
26	R/W	0	In SPI slave mode, status fast read/write enable bit: 1 = enable, 0 = disable
25	R/W	0	In SPI slave mode, status read back enable bit: 1 = reading status is written status in SPI User register 3, 0 = reading status is in SPI Status register.
24:16	R/W	0	In SPI slave mode, buffer bit number 0 = 1 bit, 1= 2 bits, …
15:10	R/W	0	In SPI slave mode, address bit number for reading buffer 0 = 1 bit, 1= 2 bits, …
9:4	R/W	0	In SPI slave mode, address bit number for writing buffer 0 = 1 bit, 1= 2 bits, …
3	R/W	0	In SPI slave mode, dummy enable bit for writing status 1=enable, 0=disable
2	R/W	0	In SPI slave mode, Dummy enable bit for reading status 1=enable, 0=disable
1	R/W	0	In SPI slave mode, Dummy enable bit for writing buffer 1=enable, 0=disable
0	R/W	0	In SPI slave mode, Dummy enable bit for reading buffer 1=enable, 0=disable

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Table 11-314 SPI Slave register 2 0xffd14038

Bit(s)	R/W	Default	Description
31:24	R/W	0	In SPI slave mode, Dummy cycle number for writing buffer 0 = 1 cycle, 1= 2 cycles, …
23:16	R/W	0	In SPI slave mode, Dummy cycle number for reading buffer 0 = 1 cycle, 1= 2 cycles, …
15:8	R/W	0	In SPI slave mode, Dummy cycle number for writing status 0 = 1 cycle, 1= 2 cycles, …
7:0	R/W	0	In SPI slave mode, Dummy cycle number for reading status 0 = 1 cycle, 1= 2 cycles, …

Table 11-315 SPI Slave register 3 0xffd1403C

Bit(s)	R/W	Default	Description
31:24	R/W	0	In SPI slave mode, Command value for writing status, when bit27 “SPI slave command define enable” in SPI Slave register is 1
23:16	R/W	0	In SPI slave mode, Command value for reading status, when bit27 “SPI slave command define enable” in SPI Slave register is 1
15:8	R/W	0	In SPI slave mode, Command value for writing buffer, when bit27 “SPI slave command define enable” in SPI Slave register is 1
7:0	R/W	0	In SPI slave mode, Command value for reading buffer, when bit27 “SPI slave command define enable” in SPI Slave register is 1

Table 11-316 SPI controller cache 0~7 0xffd14040~0xffd1405c

Bit(s)	R/W	Default	Description
31:0	R/W	0	Cache line Word 0~7. Cache is used to read data both for AHB or APB read command. Cache is also used for APB page programming etc.

Table 11-317 SPI controller buffer 8~15 0xffd14060~0xffd1407c

Bit(s)	R/W	Default	Description
31:0	R/W	0	Buffer Word 8. Buffer is used to read/write data only for APB read/write user commands.

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12 I/O Interface

12.1 Universal Serial Bus

12.1.1 Overview

- Triple USB 2.0 high-speed USB I/O, two USB Hosts and one USB OTG

12.1.2 Features

The USB2.0 OTG controller features

- Support for the following speeds: High-Speed (HS, 480-Mbps),Full-Speed (FS, 12-Mbps) and Low-Speed (LS, 1.5-Mbps) modes
- Multiple DMA/non DMA mode access support on the application side
- Supports up to 16 bidirectional endpoints, including control endpoint 0.
- Supports Session Request Protocol (SRP) and Host Negotiation Protocol (HNP)
- Supports up to 16 host channels.
- Supports ACA ID detector, refer to Amlogic USB2.0 OTGID Detector Specification.pdf for detail

The USB2.0 Host controller features:

- Support for the following speeds: High-Speed (HS, 480-Mbps),Full-Speed (FS, 12-Mbps) and Low-Speed (LS, 1.5-Mbps) modes
- Multiple DMA/non DMA mode access support on the application side
- Supports up to 16 host channels.

The USB2.0 PHY features:

- Support for the following speeds: High-Speed (HS, 480-Mbps),Full-Speed (FS, 12-Mbps) and Low-Speed (LS, 1.5-Mbps) modes

The USB3.0 Host controller features:

- Support for the following speed: Super-Speed, High-Speed, Full-Speed and Low-Speed
- Compliant with the xHCI specification

The USB3.0 SS PHY features:

- 5-Gbps SuperSpeed data transmission rate over 3-m USB 3 cable
- Integrated PHY includes transmitter, receiver, spread spectrum clock (SSC) generation, PLL, digital core, and electrostatic discharge (ESD) protection circuits
- Supports legacy Half-rate mode for power-saving

12.1.3 Register Description

Base address:

PHY 20: 0xff63_6000.

PHY 21: 0xff63_A000.

PHY 22: 0xff65_8000.

For the following registers, each register’s final address= base address + offset *4.

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Table 12-1 reg32_00 0x00

Bit(s)	R/W	Default	Description
31	R/W	0	PLL_Fine_Tuning_2
30	R/W	1	PLL_Fine_Tuning_1
29	R/W	0	PLL_Fine_Tuning_0
28	R/W	0	PLL_Bypass_Enable
27	R/W	0	PLL_Lock_over_ride
26	R/W	0	pll_clock_divide_5
25	R/W	0	pll_clock_divide_4
24	R/W	0	pll_clock_divide_3
23	R/W	0	pll_clock_divide_2
22	R/W	0	pll_clock_divide_1
21	R/W	0	pll_clock_divide_0
20	R/W	0	refclk_multiplier_5
19	R/W	0	refclk_multiplier_4
18	R/W	1	refclk_multiplier_3
17	R/W	0	refclk_multiplier_2
16	R/W	1	refclk_multiplier_1
15	R/W	0	refclk_multiplier_0
14	R/W	1	reset_FS_LS_Clock_Divider
13	R/W	1	reset_HS_CDR
12	R/W	1	reset_FS_LS_CDR
11:10	R/W	0	reg32_00_11_10_reserved
9	R/W	0	BIAS_Power_Down
8	R/W	0	BGR_Power_Down
7	R/W	0	Calibration_Power_Down
6	R/W	0	HS_Disconnect_Power_Down
5	R/W	0	HS_Squelch_Power_Down
4	R/W	0	FS_LS_RX_Power_Down
3	R/W	0	HS_RX_Power_Down
2	R/W	0	FS_LS_Driver_Power_Down
1	R/W	0	HS_TX_Driver_Power_Down
0	R/W	0	PLL_Power_Down

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Table 12-2 reg32_01 0x04

Bit(s)	R/W	Default	Description
31:24	R/W	0	TBD_3
23:16	R/W	0	TBD_2
15:10	R/W	0	TBD_1
9	R/W	0	hs_en_mode
8	R/W	0	spare
7:4	R/W	0	TBD_0
3:2	R/W	0	bypass_en
1:0	R/W	0x3	slew_control

Table 12-3 reg32_02 0x08

Bit(s)	R/W	Default	Description
31	RO	0	Phy_status
30	RO	0	cal_en_flag
29:27	RO	0	reg32_02_29_27_reserved
26	RO	0	HS_Disconnect_Status
25	RO	0	HS_Squelch_Status
24	RO	0	PRBS_Sync_Out
23:16	RO	0	Calibration_code_Value_23_16
15:8	RO	0	Calibration_code_Value_15_8
7:0	RO	0	Calibration_code_Value_7_0

Table 12-4 reg32_03 0x0c

Bit(s)	R/W	Default	Description
31:24	R/W	0	TBD_7
23:16	R/W	0	TBD_6
15:8	R/W	0	TBD_5
7:4	R/W	0x2	TBD_4
4:2	R/W	0x1	hsdic_ref
2:0	R/W	0	squelch_ref

Table 12-5 reg32_04 0x10

Bit(s)	R/W	Default	Description
31:30	R/W	0	i_c2l_bias_trim_3_2
29:28	R/W	0	i_c2l_bias_trim_1_0

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Bit(s)	R/W	Default	Description
27	R/W	0	TEST_Bypass_mode_enable
26	RO	0	i_c2l_cal_done
25	R/W	0	i_c2l_cal_reset_n
24	R/W	0	i_c2l_cal_en
23:16	R/W	0	Calibration_code_Value_23_16
15:8	R/W	0	Calibration_code_Value_15_8
7:0	R/W	0	Calibration_code_Value_7_0

Table 12-6 reg32_05 0x14

Bit(s)	R/W	Default	Description
31:24	R/W	0	i_c2l_obs_7_0
23:22	R/W	0	reg32_05_23_22_reserved
21:20	R/W	0	i_c2l_pll_perfcfg_21_20
19	R/W	0x1	i_c2l_pll_perfcfg_19
18:16	R/W	0	i_c2l_pll_perfcfg_18_16
15:12	R/W	0	i_c2l_pll_perfcfg_15_12
11:10	R/W	0	i_c2l_pll_perfcfg_11_10
9	R/W	0x1	i_c2l_pll_perfcfg_9
8	R/W	0x1	i_c2l_pll_perfcfg_8
7:4	R/W	0x7	i_c2l_pll_perfcfg_7_4
3:2	R/W	0x3	i_c2l_pll_perfcfg_3_2
1:0	R/W	0	i_c2l_pll_perfcfg_1_0

Table 12-7 reg32_06 0x18

Bit(s)	R/W	Default	Description
31	R/W	0x1	hub_extra_bit_cntr
30:24	R/W	0	cntr_timeout
23	R/W	0	Internal_loopback
22	R/W	0	reg32_06_22_reserved
21	R/W	0	PCS_Reset_Transmit_State_machine
20	R/W	0	PCS_Reset_Receive_State_machine
19:16	R/W	0xf	fsls_farend_device_disconnect_micro_second_count_11_8
15:12	R/W	0xa	reg32_06_15_12_reserved
11:8	R/W	0	bypass_disc_cntr_3_0
7:0	R/W	0x17	PCS_microsecond_timer_done_count_value_7_0

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Table 12-8 reg32_07 0x1c

Bit(s)	R/W	Default	Description
31:24	RO	0	Prbs_Error_count
23:21	R/W	0	reg32_07_23_21_reserved
20:17	R/W	0xf	RX_ERROR_Turn_Around_Timer_Count
16	R/W	0	acceptable_bit_drops
15	R/W	0	host_tristate
14:12	R/W	0x4	fs_ls_minimum_count
11:8	R/W	0xf	cntr_done_value_7_4
7:4	R/W	0xf	cntr_done_value_3_0
3:0	R/W	0	HS_CDR_internal_tap_select

Table 12-9 reg32_08 0x20

Bit(s)	R/W	Default	Description
31:24	R/W	0	Custom_Pattern_2
23:16	R/W	0	Custom_Pattern_1
15:8	R/W	0x4	Custom_Pattern_0
7	R/W	0	Enable_RX_ERROR_Timeout_Mode
6	R/W	0	reset_us_timer
5	R/W	0	PRBS_ERROR_Insert
4	R/W	0	PRBS_comparison_enable
3	R/W	0	PRBS_Enable
2:0	R/W	0	pattern

Table 12-10 reg32_09 0x24

Bit(s)	R/W	Default	Description
31:24	R/W	0	Custom_Pattern_6
23:16	R/W	0	Custom_Pattern_5
15:8	R/W	0	Custom_Pattern_4
7:0	R/W	0	Custom_Pattern_3

Table 12-11 reg32_10 0x28

Bit(s)	R/W	Default	Description
31:24	R/W	0	Custom_Pattern_10
23:16	R/W	0	Custom_Pattern_9

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Bit(s)	R/W	Default	Description
15:8	R/W	0	Custom_Pattern_8
7:0	R/W	0	Custom_Pattern_7

Table 12-12 reg32_11 0x2c

Bit(s)	R/W	Default	Description
31:24	R/W	0	Custom_Pattern_14
23:16	R/W	0	Custom_Pattern_13
15:8	R/W	0	Custom_Pattern_12
7:0	R/W	0	Custom_Pattern_11

Table 12-13 reg32_12 0x30

Bit(s)	R/W	Default	Description
31:24	R/W	0	Custom_Pattern_18
23:16	R/W	0	Custom_Pattern_17
15:8	R/W	0	Custom_Pattern_16
7:0	R/W	0	Custom_Pattern_15

Table 12-14 reg32_13 0x34

Bit(s)	R/W	Default	Description
31	R/W	0	reg32_13_31_reserved
30	R/W	0	bypass_reg[6]:i_c2l_fsls_rx_en
29	R/W	0	bypass_reg[5]:i_c2l_hs_rx_en
28	R/W	0	bypass_reg[4]:i_c2l_fs_oe
27	R/W	0	bypass_reg[3]:i_c2l_hs_oe
26	R/W	0	bypass_reg[2]:i_c2l_ls_en
25	R/W	0	bypass_reg[1]:i_c2l_fs_en
24	R/W	0	bypass_reg[0]:i_c2l_hs_en
23	R/W	0	Bypass_Host_Disconnect_Enable
22	R/W	0	Bypass_Host_Disconnect_Value
21	R/W	0	Clear_Hold_HS_disconnect
20:16	R/W	0x8	minimum_count_for_sync_detection
15	R/W	0	Update_PMA_signals
14	R/W	0	load_stat
13:8	R/W	0	reg32_13_13_8_reserved
7:0	R/W	0	Custom_Pattern_19

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Table 12-15 reg32_14 0x38

Bit(s)	R/W	Default	Description
31:24	R/W	0	Bypass_ctrl_15_8 bypass_ctrl[8]: bypass i_rpd_en bypass_ctrl[9]: bypass i_rpu_sw2_en bypass_ctrl[10]: i_rpu_sw1_en
23:16	R/W	0	Bypass_ctrl_7_0 bypass_ctrl[0]: hs bypass_ctrl[1]: fs bypass_ctrl[2]: ls bypass_ctrl[3]:hs_out_en bypass_ctrl[4]:fsls_out_en bypass_ctrl[5]:hs_rx_en bypass_ctrl[6]:hls_rx_en
15:8	R/W	0	reg32_14_7_reserved
7	R/W	0	i_c2l_assert_single_enable_zero
6	R/W	0	i_c2l_data_16_8
5	R/W	0	pg_rstn
4	R/W	0	bypass_reg[11:10]:i_rpu_sw1_en
3:2	R/W	0	bypass_reg[9]:i_rpu_sw2_en
1	R/W	0	bypass_reg[8]: i_rpd_en
0	R/W	0	i_rpd_en

Table 12-16 reg32_15 0x3c

Bit(s)	R/W	Default	Description
31:29	R/W	0	reg32_15_31_29_reserved
28:16	R/W	0xfa0	ms_4_cntr
15:8	R/W	0x3c	non_se0_cntr
7:0	R/W	0x3c	se0_cntr

Table 12-17 reg32_16 0x40

Bit(s)	R/W	Default	Description
31	RO	0	usb2_mppll_lock_dig
30	RO	0	usb2_mppll_lock
29	R/W	0	usb2_mppll_reset
28	R/W	0	usb2_mppll_en
27	R/W	0x1	usb2_mppll_fast_lock
26	R/W	0	usb2_mppll_lock_f
25:24	R/W	0x1	usb2_mppll_lock_long
23	R/W	0	usb2_mppll_dco_sdm_en
22	R/W	0x1	usb2_mppll_load
21	R/W	0	usb2_mppll_sdm_en
20	R/W	0	usb2_mppll_tdc_mode

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Bit(s)	R/W	Default	Description
19:15	R/W	0	reg32_16_19_15_reserved
14:10	R/W	0x1	usb2_mppll_n
9	R/W	0	reg32_16_9_reserved
8:0	R/W	0x14	usb2_mppll_m

Table 12-18 reg32_17 0x44

Bit(s)	R/W	Default	Description
31:28	R/W	0x9	usb2_mppll_filter_pvt1
27:24	R/W	0x2	usb2_mppll_filter_pvt2
23	R/W	0	usb2_mppll_filter_mode
22:20	R/W	0x7	usb2_mppll_lambda0
19:17	R/W	0x7	usb2_mppll_lambda1
16	R/W	0	usb2_mppll_fix_en
15:14	R/W	0	reg32_17_15_14_reserved
13:0	R/W	0	usb2_mppll_frac_in

Table 12-19 reg32_18 0x48

Bit(s)	R/W	Default	Description
31	R/W	0	usb2_mppll_acg_range
30:29	R/W	0x3	usb2_mppll_adj_ldo
28:26	R/W	0x3	usb2_mppll_alpha
25:24	R/W	0x1	usb2_mppll_bb_mode
23:22	R/W	0x1	usb2_mppll_bias_adj
21:19	R/W	0x3	usb2_mppll_data_sel
18:16	R/W	0x3	usb2_mppll_rou
15:14	R/W	0	usb2_mppll_pfd_gain
13	R/W	0x1	usb2_mppll_dco_clk_sel
12	R/W	0	usb2_mppll_dco_m_en
11:6	R/W	0x27	usb2_mppll_lk_s
5:2	R/W	0x9	usb2_mppll_lk_w
1:0	R/W	0x1	usb2_mppll_lkw_sel

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Table 12-20 reg32_19 0x4c

Bit(s)	R/W	Default	Description
31	RO	0	usb2_mppll_lock_dig
30	RO	0	usb2_mppll_lock
29:10	RO	0	reg32_19_29_10_reserved
9:0	RO	0	usb2_mppll_reg_out

Table 12-21 reg32_20 0x50

Bit(s)	R/W	Default	Description
31	R/W	0	bypass_cal_done_r5
30:29	R/W	0	usb2_bgr_dbg_1_0
28:24	R/W	0	usb2_bgr_vref_4_0
23:22	R/W	0	squelch_sel: 01: debounce 1; 10: debounce 2; 00/11: no debounce
21	R/W	0	usb2_bgr_start
20:16	R/W	0	usb2_bgr_adj_4_0
15:14	R/W	0	usb2_edgedrv_trim_1_0
13	R/W	0	usb2_edgedrv_en
12:9	R/W	0xf	usb2_dmon_sel_3_0
8	R/W	0	usb2_dmon_en
7	R/W	0	bypass_otg_det
6	R/W	0	usb2_cal_code_r5
5	R/W	0	usb2_amon_en
4	R/W	0x1	usb2_otg_vbusdet_en
3:1	R/W	0x4	usb2_otg_vbus_trim_2_0
0	R/W	0	usb2_otg_iddet_en

Table 12-22 reg32_21 0x54

Bit(s)	R/W	Default	Description
31:26	R/W	0	reg32_21_31_26_reserved
25:20	R/W	0	bypass_utmi_reg [21:20]: xcvr_select ctrl reg [22]: term_select ctrl reg [23]: suspend ctrl reg [25:24]: opmode ctrl reg
19:16	R/W	0	bypass_utmi_cntr [16]: bypass xcvr_select [17]: bypass term_select [18]: bypass suspend [19]: bypass opmode
15:8	R/W	0	hs cdr ctrl
7	R/W		hs cdr sel
6	R/W		reg32_21_6_reserved
5:4	R/W	0x2	usb2_otg_aca_trim_1_0

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Bit(s)	R/W	Default	Description
3	R/W	0	usb2_tx_strg_pd
2	R/W	0x1	usb2_otg_aca_en
1	R/W	0x1	usb2_cal_ack_en
0	R/W	0	usb2_bgr_force

Table 12-23 reg32_22 0x58

Bit(s)	R/W	Default	Description
31:14	RO	0	reg32_22_31_14_reserved
[13:6]			hs cdr state
5:3	RO	0	usb2_otg_aca_iddig
2	RO	0	usb2_otg_vbus_vld
1	RO	0	usb2_otg_sess_vld
0	RO	0	usb2_otg_id_dig

Table 12-24 reg32_23 0x5c

Bit(s)	R/W	Default	Description
31:16	RO	0	test_bus_data_int_15_0
15:13		0	ldo_trim
12		1	ldo_en
11		0	new_hs_disc_ctrl[1]
10		0	new_hs_disc_ctrl[0]
9		0	sel_cdr
8		0	pcs_sel
7	R/W	0	orw_test_bus_en
6:1	R/W	0	orw_test_bus_sel_5_0
0	R/W	0	orw_usb2_bgr_en

12.2 Ethernet

12.2.1 Overview

The Ethernet MAC controller provides a complete ethernet interface from the chip to a Reduced Gigabit Media Independent Interface(RGMII) compliant Ethernet PHY.

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12.2.2 Ethernet MAC

12.2.2.1 Features

Ethernet MAC has the following features:

- 10/100/1000 MAC 5.10a
- RGMII/RMII
- AHB 32 Bits internal bus
- RX FIFO 8KB, TX FIFO 8KB
- 2 MAC addresses
- EEE
- Power Management

12.2.3 Ethernet PHY

12.2.3.1 Features

Ethernet PHY has the following features:

- Integrated IEEE 802.3/802.3u compliant 10/100Mbps Ethernet PHY
- Supporting both full and half-duplex for either 10 or 100 Mb/s data rate
- Auto MDIX capable
- Supports wake-on-LAN
- 100 Base-T support
- MII/RMII/SMII interface
- Supports auto-negotiation
- Full set of power down modes
- Interface available to 100Base-FX Fiber-PMD
- Serial Management Interface (SMI)
- Fix configurations for LED status indicators
- Supporting military temperature range -20℃ to 80℃
- Perfect mix of analog and digital lends itself to robustness, portability, and performance
- Multiple input clock options
- Stand-alone core

12.2.3.2 Register Description

Base Address: 0xFF64C000

Table 12-25 ETH_PHY_DBG_CTL0 0x0

Bit(s)	R/W	Default	Description
7:0	R/W	0	ETH digital debug registers

Table 12-26 ETH_PHY_DBG_CTL1 0x4

Bit(s)	R/W	Default	Description
23:0	R/W	0	ETH digital debug registers

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Table 12-27 ETH_PHY_DBG_CFG0 0x8

Bit(s)	R/W	Default	Description
31:0	R/W	0	ETH digital debug registers

Table 12-28 ETH_PHY_DBG_CFG1 0xc

Bit(s)	R/W	Default	Description
31:0	R/W	0	ETH digital debug registers

Table 12-29 ETH_PHY_DBG_CFG2 0x10

Bit(s)	R/W	Default	Description
31:0	R/W	0	ETH digital debug registers

Table 12-30 ETH_PHY_DBG_CFG3 0x14

Bit(s)	R/W	Default	Description
31:0	R/W	0	ETH digital debug registers

Table 12-31 ETH_PHY_DBG_CFG4 0x18

Bit(s)	R/W	Default	Description
31:0	R/W	0	ETH digital debug registers

Table 12-32 ETH_PLL_STS 0x40

Bit(s)	R/W	Default	Description
31	R		eth_mppll_lock
30	R		eth_mppll_lock_dig
29:10	R		reserved
9:0	R		eth_mppll_reg_out

Table 12-33 ETH_PLL_CTL0 0x44

Bit(s)	R/W	Default	Description
31	R		eth_mppll_lock
30	R		eth_mppll_lock_dig
29	R/W		eth_mppll_reset
28	R/W		eth_mppll_en
27	R/W		eth_mppll_fast_lock
26	R/W		eth_mppll_lock_f

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Bit(s)	R/W	Default	Description
25:24	R/W		eth_mppll_lock_long
23	R/W		eth_mppll_sel_ref
22	R/W		eth_mppll_load
21	R/W		eth_mppll_sdm_en
20	R/W		eth_mppll_tdc_mode
19:15	R/W		reserved
14:10	R/W		eth_mppll_n
9	R/W		reserved
8:0	R/W		eth_mppll_m

Table 12-34 ETH_PLL_CTL1 0x48

Bit(s)	R/W	Default	Description
31:28	R/W		eth_mppll_filter_pvt1
27:24	R/W		eth_mppll_filter_pvt2
23	R/W		eth_mppll_filter_mode
22:20	R/W		eth_mppll_lambda0
19:17	R/W		eth_mppll_lambda1
16	R/W		eth_mppll_fix_en
15:14	R/W		reserved
13:0	R/W		eth_mppll_frac_in

Table 12-35 ETH_PLL_CTL2 0x4C

Bit(s)	R/W	Default	Description
31	R/W		eth_mppll_acq_range
30:29	R/W		eth_mppll_adj_ldo
28:26	R/W		eth_mppll_alpha
25:24	R/W		eth_mppll_bb_mode
23:22	R/W		eth_mppll_bias_adj
21:19	R/W		eth_mppll_data_sel
18:16	R/W		eth_mppll_rou
15:14	R/W		eth_mppll_pfd_gain
13	R/W		eth_mppll_dco_clk_sel
12	R/W		eth_mppll_dco_m_en
11:6	R/W		eth_mppll_lk_s

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Bit(s)	R/W	Default	Description
5:2	R/W		eth_mppll_lk_w
1:0	R/W		eth_mppll_lkw_sel

Table 12-36 ETH_PLL_CTL3 0x50

Bit(s)	R/W	Default	Description
31:1	R		reserved
0	R/W		eth_mppll_dco_sdm_en

Table 12-37 ETH_PLL_CTL4 0x54

Bit(s)	R/W	Default	Description
31:0	R/W		reserved

Table 12-38 ETH_PLL_CTL5 0x58

Bit(s)	R/W	Default	Description
31:16	R/W	0	ETH_PHY_RXADC0
15:3	R/W	0	ETH_PHY_CTLIO
2:0	R/W	0	reserved

Table 12-39 ETH_PLL_CTL6 0x5C

Bit(s)	R/W	Default	Description
31:16	R/W	0	reserved
15:0	R/W	0	ETH_PHY_RXSQLD

Table 12-40 ETH_PLL_CTL7 0x60

Bit(s)	R/W	Default	Description
31:16	R/W	0	reserved
15:0	R/W	0	ETH_PHY_RXADC1

Table 12-41 ETH_PHY_CNTL0 0x80

Bit(s)	R/W	Default	Description
31:16	R/W	0	co_reg3_oui_in, SMI register 3 default value
15:0	R/W	0	co_reg2_oui_in, SMI register 2 default value

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Table 12-42 ETH_PHY_CNTL1 0x84

Bit(s)	R/W	Default	Description
31:24	R/W	0	led_cfg, led_cfg7:5: co_activity_led output select
23	R/W	0	reserved
22	R/W	0	co_pwruprst_byp
21	R/W	0	co_clk_ext
20	R/W	0	co_st_scan
19	R/W	0	co_rxclk_inv
18	R/W	0	co_phy_enb
17	R/W	0	co_clkfreq
16	R/W	0	eth_clk_enable
15:14	R/W	0	co_st_miimode1:0
13	R/W	0	co_smii_source_sync
12	R/W	0	co_st_pllbp
11	R/W	0	co_st_adcbp
10	R/W	0	co_st_fxmode
9	R/W	0	co_en_high
8	R/W	0	co_automdix_en
7:3	R/W	0	co_st_phyadd4:0
2:0	R/W	0	co_st_mode2:0

Table 12-43 ETH_PHY_CNTL2 0x88

Bit(s)	R/W	Default	Description
31	R/W	0	reserved
30	R/W	0	eth_phy_clk25min_en:
29:28	R/W	0	reserved
27:24	R/W	0	debug output bus select
23:13	R/W	0	reserved
12	R/W	0	analog production test mode enable
11	R/W	0	mdi source select
10	R/W	0	reserved
9	R/W	0	source select for rx_clk to mac
8	R/W	0	source select for tx_clk output to gpio
7	R/W	0	rx_dv/col switch for mac
6	R/W	0	ephy smi source select
5	R/W	0	use internal phy
4	R/W	0	ephy smi source select

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Bit(s)	R/W	Default	Description
3	R/W	0	co_clkin source select
2	R/W	0	co_mdclk source select
1	R/W	0	enable reset in test mode
0	R/W	0	ephy loopback mode enable, in/out through debug in/out gpio

Table 12-44 ETH_PHY_STS0 0x94

Bit(s)	R/W	Default	Description
31:28	R		reserved
27:19	R		co_int_vec8:0
19	R		reserved
18:0	R		ETH_PLL_STS18:0

Table 12-45 ETH_PHY_STS1 0x98

Bit(s)	R/W	Default	Description
31:0	R		eth_phy_dbg_prb

Table 12-46 ETH_PHY_STS2 0x9C

Bit(s)	R/W	Default	Description
31:26	R		eth_phy_rxda5:0
25	R		eth_phy_rxadcoflw
24	R		eth_phy_rxadcuflw
23	R		eth_phy_rxprepga
22:20	R		eth_phy_rxgain2:0
19:0	R		eth_test_status

Table 12-47 ETH_PHY_DBG_REG 0xA0

Bit(s)	R/W	Default	Description
31:16	R		eth_phy_dbg_reg
15:8			reserved
7:0	R/W	0	eth_phy_dbg_reg_mux

12.3 Inter-Integrated Circuit (I2C)

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12.3.1 Overview

Inter-Integrated Circuit (IIC or I2C) is a multi-slave serial communication bus between ICs. The chip integrates the I2C interface and signals allowing communications with other I2C peripheral devices.

12.3.2 Features

The I2C Master Module has the following features:

- Support for 7-bit and 10-bit addressable devices
- Programmable bus speed including standard speed (100 KBits/s) and fast speed (400 KBits/s)
- Error transfer detection
- “Transfer complete” indication by polling or interrupt (Interrupts handled by the ISA module).
- Internal buffer holding up to 8 bytes for transfer (in either direction)
- Flexible architecture allowing the software to dictate the format of the I2C bit streams
- Manual setting of the I2C bus to accommodate a software only mode

12.3.3 Register Description

For I2C module in EE domain, each register final address = 0x FF805000 + offset * 4 for master mode, final address = 0x FF806000 + offset * 4 for slave mode.

Table 12-48 I2C_M_0_CONTROL_REG 0x7c00

Bit(s)	R/W	Default	Description
31	R/W	0	CNTL_JIC: There is internal logic to dynamically enable the gated clocks. If this gated clock logic doesn’t work, you can set This bit to always enable the clock. Setting This bit wastes power.
30	R	0	Unused
29-28	R/W	0	QTR_CLK_EXT: These two Bits extend the clock divider to 12 Bits: QTR_CLK = {[29:28],[21:12]}
27	R	0	unused
26	R	0	Read back level of the SDA line
25	R	0	Read back level of the SCL line
24	R/W	0	Sets the level of the SDA line if manual mode is enabled. If This bit is ‘0’, then the SDA line is pulled low. If This bit is ‘1’ then the SDA line is tri-stated.
23	R/W	0	Sets the level of the SCL line if manual mode is enabled. If This bit is ‘0’, then the SCL line is pulled low. If This bit is ‘1’ then the SCL line is tri-stated.
22	R/W	0	This bit is used to enable manual mode. Manual I2C mode is controlled by Bits 12,13,14 and 15 above.
21:12	R/W	0x142	QTR_CLK_DLY. This value corresponds to period of the SCL clock divided by 4 Quarter Clock Delay = * System Clock Frequency For example, if the system clock is 133Mhz, and the I2C clock period is 10uS (100khz), then Quarter Clock Delay = * 133 Mhz = 332
11:8	R	-	READ_DATA_COUNT: This value corresponds to the number of bytes READ over the I2C bus. If this value is zero, then no data has been read. If this value is 1, then Bits [7:0] in TOKEN_RDATA_REG0 contains valid data. The software can read this register after an I2C transaction to get the number of bytes to read from the I2C device.

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Bit(s)	R/W	Default	Description
7:4	R	-	CURRENT_TOKEN: This value reflects the current token being processed. In the event of an error, the software can use this value to determine the error location.
3	R	-	ERROR: This read only Bit is set if the I2C device generates a NACK during writing. This bit is cleared at on the clock cycle after the START Bit is set to 1 indicating the start of list processing. Errors can be ignored by setting the ACK_ IGNORE Bit(s) below. Errors will be generated on Writes to devices that return NACK instead of ACK. A NACK is returned by a device if it is unable to accept any more data (for example because it is processing some other real-time function). In the event of an ERROR, the I2C module will automatically generate a STOP condition on the bus.
2	R	-	STATUS: This bit reflects the status of the List processor: 0: IDLE 1: Running. The list processor will enter this state on the clock cycle after the START Bit is set. The software can poll the status register to determine when processing is complete.
1	R/W	0	ACK_IGNORE: Set to 1 to disable I2C ACK detection. The I2C bus uses an ACK signal after every byte transfer to detect problems during the transfer. Current Software implementations of the I2C bus ignore this ACK. This bit is for compatibility with the current Amlogic software. This bit should be set to 0 to allow NACK operations to abort I2C bus transactions. If a NACK occurs, the ERROR bit above will be set.
0	R/W	0	START: Set to 1 to start list processing. Setting This bit to 0 while the list processor is operating causes the list processor to abort the current I2C operation and generate an I2C STOP command on the I2C bus. Normally This bit is set to 1 and left high until processing is complete. To re-start the list processor with a new list (after a previous list has been exhausted), simply set This bit to zero then to one.

Table 12-49 I2C_M_0_SLAVE ADDRESS 0x7c01

Bit(s)	R/W	Default	Description
31:29	R	0	Reserved
28	R/W	0	USE_CNTL_SCL_LOW: If This bit is set to 1, then Bits[27:16] control the SCL low time.
27:16	R/W	0	SCL Low delay.
15:14	R	0	Unused
13-11	R/W	0	SCL_FILTER: A filter was added in the SCL input path to allow for filtering of slow rise times. 0 = no filtering, 7 = max filtering
10:8	R/W	0	SDA FILTER: A filter was added in the SDA input path to allow for filtering of slow rise times. 0 = no filtering, 7 = max filtering
7:0	R/W	0x00	SLAVE_ADDRESS. This is a 7-bit value for a 7-bit I2C device, or (0xF0 | {A9, A8}) for a 10-bit I2C device. By convention, the slave address is typically stored in by first left shifting it so that it's MSB is D7 (The I2C bus assumes the 7-bit address is left shifted one). Additionally, since the SLAVE address is always an 7-bit value, D0 is always 0. NOTE: The I2C always transfers 8-bits even for address. The I2C hardware will use D0 to dictate the direction of the bus. Therefore, D0 should always be '0' when this register is set.

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Table 12-50 I2C_M_0_TOKEN_LIST_REG0 0x7c02 The register below describes the first 8 tokens in the token list.

Bit(s)	R/W	Default	Description
31:28	R/W	0x00	8th token in the list to process
27:24	R/W	0x00	7th token in the list to process
23:20	R/W	0x00	6th token in the list to process
19:16	R/W	0x00	5th token in the list to process
15:12	R/W	0x00	4th token in the list to process
11:8	R/W	0x00	3rd token in the list to process
7:4	R/W	0x00	2nd token in the list to process
3:0	R/W	0x00	1st token in the list to process (See the table below for token definitions)

Table 12-51 Token Definitions

Command Token	Value	Data	Description
END	0x0	N/A	Used to tell the I2C module that this is the end of the Token list. This token is not associated with the I2C bus, but rather with the state-machine that drives the token list processor.
START	0x1	N/A	The START Token is used to tell an I2C device that this is the beginning of an I2C transfer
SLAVE_ ADDR-- WRITE	0x2	7-bits	This bit-sequence is used to address a device and tell the device it is being WRITTEN
SLAVE_ ADDR-- READ	0x3	7-bits	This bit sequence is used to address a device and tell the device it is being READ.
DATA	0x4	8-bits	This 8-bit byte sequence is a byte transfer (READ or WRITE). The DATA token corresponds to a WRITE if it follows a SLAVE_ADDR-WRITE token. The DATA token corresponds to a READ if it follows a SLAVE_ADDR-READ token.
DATA-LAST	0x5	8-bits	Used to indicate the last 8-bit byte transfer is a byte transfer of a READ.
STOP	0x6	N/A	This tells the I2C device it is no longer being addressed

Write data associated with the DATA token should be placed into the I2C_TOKEN_WDATA_REG0 or I2C_TOKEN_WDATA_REG1 registers. Read data associated with the DATA or DATA-LAST token can be read from the I2C_TOKEN_RDATA_REG0 or I2C_TOKEN_RDATA_REG1 registers.

Table 12-52 I2C_M_0_TOKEN_LIST_REG1 0x7c03

Bit(s)	R/W	Default	Description
31:28	R/W	0x00	16th token in the list to process
27:24	R/W	0x00	15th token in the list to process
23:20	R/W	0x00	14th token in the list to process
19:16	R/W	0x00	13th token in the list to process
15:12	R/W	0x00	12th token in the list to process

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Bit(s)	R/W	Default	Description
11:8	R/W	0x00	11th token in the list to process
7:4	R/W	0x00	10th token in the list to process
3:0	R/W	0x00	9th token in the list to process

Table 12-53 I2C_M_0_TOKEN_WDATA_REG0 0x7c04

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	4th data byte written for a DATA (write) token.
23:16	R/W	0x00	3rd data byte written for a DATA (write) token.
15:8	R/W	0x00	2nd data byte written for a DATA (write) token.
7:0	R/W	0x00	1st data byte written for a DATA (write) token.

Table 12-54 I2C_M_0_TOKEN_WDATA_REG1 0x7c05

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	8th data byte written for a DATA (write) token.
23:16	R/W	0x00	7th data byte written for a DATA (write) token.
15:8	R/W	0x00	6th data byte written for a DATA (write) token.
7:0	R/W	0x00	5th data byte written for a DATA (write) token.

Table 12-55 I2C_M_0_TOKEN_RDATA_REG0 0x7c06

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	4th data byte read for a DATA or DATA-LAST (READ) token.
23:16	R/W	0x00	3rd data byte read for a DATA or DATA-LAST (READ) token.
15:8	R/W	0x00	2nd data byte read for a DATA or DATA-LAST (READ) token.
7:0	R/W	0x00	1st data byte read for a DATA or DATA-LAST (READ) token.

Table 12-56 I2C_M_0_TOKEN_RDATA_REG1 0x7c07

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	8th data byte read for a DATA or DATA-LAST (READ) token.
23:16	R/W	0x00	7th data byte read for a DATA or DATA-LAST (READ) token.
15:8	R/W	0x00	6th data byte read for a DATA or DATA-LAST (READ) token.
7:0	R/W	0x00	5th data byte read for a DATA or DATA-LAST (READ) token.

I2C_M_1_ 0x7800~0x7807

See I2C_M_0

I2C_M_2_ 0x7400~0x7407

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See I2C_M_0

I2C_M_3_ 0x7000~0x7007

See I2C_M_0

For I2C module in AO domain, each register final address = 0xFF805000 + offset * 4 for master mode, final address = 0x FF806000 + offset * 4 for slave mode.

Table 12-57 AO_I2C_M_0_CONTROL_REG 0x0

Bit(s)	R/W	Default	Description
31	R/W	0	CNTL_JIC: There is internal logic to dynamically enable the gated clocks. If this gated clock logic doesn't work, you can set this bit to always enable the clock. Setting this bit wastes power.
30	R	0	Unused
29-28	R/W	0	QTR_CLK_EXT: These two bits extend the clock divider to 12 bits: QTR_CLK = {[29:28],[21:12]}
27	R	0	unused
26	R	0	Read back level of the SDA line
25	R	0	Read back level of the SCL line
24	R/W	0	Sets the level of the SDA line if manual mode is enabled. If this bit is '0', then the SDA line is pulled low. If this bit is '1' then the SDA line is tri-stated.
23	R/W	0	Sets the level of the SCL line if manual mode is enabled. If this bit is '0', then the SCL line is pulled low. If this bit is '1' then the SCL line is tri-stated.
22	R/W	0	This bit is used to enable manual mode. Manual I2C mode is controlled by bits 12,13,14 and 15 above.
21:12	R/W	0x142	QTR_CLK_DLY. This value corresponds to period of the SCL clock divided by 4 Quarter Clock Delay = * System Clock Frequency For example, if the system clock is 133Mhz, and the I2C clock period is 10uS (100khz), then Quarter Clock Delay = * 133 Mhz = 332
11:8	R	-	READ_DATA_COUNT: This value corresponds to the number of bytes READ over the I2C bus. If this value is zero, then no data has been read. If this value is 1, then bits [7:0] in TOKEN_RDATA_REG0 contains valid data. The software can read this register after an I2C transaction to get the number of bytes to read from the I2C device.
7:4	R	-	CURRENT_TOKEN: This value reflects the current token being processed. In the event of an error, the software can use this value to determine the error location.
3	R	-	ERROR: This read only bit is set if the I2C device generates a NACK during writing. This bit is cleared at on the clock cycle after the START bit is set to 1 indicating the start of list processing. Errors can be ignored by setting the ACK_ IGNORE bit below. Errors will be generated on Writes to devices that return NACK instead of ACK. A NACK is returned by a device if it is unable to accept any more data (for example because it is processing some other real-time function). In the event of an ERROR, the I2C module will automatically generate a STOP condition on the bus.
2	R	-	STATUS: This bit reflects the status of the List processor: 0: IDLE 1: Running. The list processor will enter this state on the clock cycle after the START bit is set. The software can poll the status register to determine when processing is complete.

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Bit(s)	R/W	Default	Description
1	R/W	0	ACK_IGNORE: Set to 1 to disable I2C ACK detection. The I2C bus uses an ACK signal after every byte transfer to detect problems during the transfer. Current Software implementations of the I2C bus ignore this ACK. This bit is for compatibility with the current Amlogic software. This bit should be set to 0 to allow NACK operations to abort I2C bus transactions. If a NACK occurs, the ERROR bit above will be set.
0	R/W	0	START: Set to 1 to start list processing. Setting this bit to 0 while the list processor is operating causes the list processor to abort the current I2C operation and generate an I2C STOP command on the I2C bus. Normally this bit is set to 1 and left high until processing is complete. To re-start the list processor with a new list (after a previous list has been exhausted), simply set this bit to zero then to one.

Table 12-58 AO_I2C_M_0_SLAVE_ADDR 0x1

Bit(s)	R/W	Default	Description
31:29	R	0	Unused
28	R/W	0	USE_CNTL_SCL_LOW: If this bit is set to 1, then bits[27:16] control the SCL low time.
27:16	R/W	0	SCL Low delay. This is a new feature in M8baby. In the previous M8baby design, the SCL low time was controlled by bits[21:12] of the register above. In this design, the SCL delay is controlled independently by these bits.
15:14	R	0	Unused
13:11	R/W	0	SCL_FILTER: A filter was added in the SCL input path to allow for filtering of slow rise times. 0 = no filtering, 7 = max filtering
10:8	R/W	0	SDA FILTER: A filter was added in the SDA input path to allow for filtering of slow rise times. 0 = no filtering, 7 = max filtering
7:0	R/W	0x00	SLAVE_ADDRESS. This is a 7-bit value for a 7-bit I2C device, or (0xF0 | {A9, A8}) for a 10-bit I2C device. By convention, the slave address is typically stored in by first left shifting it so that it's MSB is D7 (The I2C bus assumes the 7-bit address is left shifted one). Additionally, since the SLAVE address is always an 7-bit value, D0 is always 0. NOTE: The I2C always transfers 8-bits even for address. The I2C hardware will use D0 to dictate the direction of the bus. Therefore, D0 should always be '0' when this register is set.

The register below describes the first 8 tokens in the token list.

Table 12-59 AO_I2C_M_0_TOKEN_LIST0 0x2

Bit(s)	R/W	Default	Description
31:28	R/W	0x00	8th token in the list to process
27:24	R/W	0x00	7th token in the list to process
23:20	R/W	0x00	6th token in the list to process
19:16	R/W	0x00	5th token in the list to process
15:12	R/W	0x00	4th token in the list to process
11:8	R/W	0x00	3rd token in the list to process
7:4	R/W	0x00	2nd token in the list to process
3:0	R/W	0x00	1st token in the list to process (See the table below for token definitions)

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Table 12-60 Token Definitions

Com-mand Token	Value	Data	Descriptiaon
END	0x0	N/A	Used to tell the I2C module that this is the end of the Token list. This token is not associated with the I2C bus, but rather with the state-machine that drives the to-ken list processor.
START	0x1	N/A	The START Token is used to tell an I2C device that this is the beginning of an I2C transfer
SLAVE_ ADDR--WRITE	0x2	7-bits	This bit-sequence is used to address a device and tell the device it is being WRITTEN
SLAVE_ ADDR--READ	0x3	7-bits	This bit sequence is used to address a device and tell the device it is being READ.
DATA	0x4	8-bits	This 8-bit byte sequence is a byte transfer (READ or WRITE). The DATA token corresponds to a WRITE if it follows a SLAVE_ADDR-WRITE token. The DATA token corresponds to a READ if it follows a SLAVE_ADDR-READ token.
DATA-LAST	0x5	8-bits	Used to indicate the last 8-bit byte transfer is a byte transfer of a READ.
STOP	0x6	N/A	This tells the I2C device it is no longer being addressed

Write data associated with the DATA token should be placed into the I2C_TOKEN_WDATA_REG0 or I2C_TOKEN_WDATA_REG1 registers. Read data associated with the DATA or DATA-LAST token can be read from the I2C_TOKEN_RDATA_REG0 or I2C_TOKEN_RDATA_REG1 registers.

Table 12-61 AO_I2C_M_0_TOKEN_LIST1 0x3

Bit(s)	R/W	Default	Description
31:28	R/W	0x00	16th token in the list to process
27:24	R/W	0x00	15th token in the list to process
23:20	R/W	0x00	14th token in the list to process
19:16	R/W	0x00	13th token in the list to process
15:12	R/W	0x00	12th token in the list to process
11:8	R/W	0x00	11th token in the list to process
7:4	R/W	0x00	10th token in the list to process
3:0	R/W	0x00	9th token in the list to process

Table 12-62 AO_I2C_M_0_WDATA_REG0 0x4

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	4th data byte written for a DATA (write) token.
23:16	R/W	0x00	3rd data byte written for a DATA (write) token.
15:8	R/W	0x00	2nd data byte written for a DATA (write) token.
7:0	R/W	0x00	1st data byte written for a DATA (write) token.

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Table 12-63 AO_I2C_M_0_WDATA_REG1 0x5

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	8th data byte written for a DATA (write) token.
23:16	R/W	0x00	7th data byte written for a DATA (write) token.
15:8	R/W	0x00	6th data byte written for a DATA (write) token.
7:0	R/W	0x00	5th data byte written for a DATA (write) token.

Table 12-64 AO_I2C_M_0_RDATA_REG0 0x6

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	4th data byte read for a DATA or DATA-LAST (READ) token.
23:16	R/W	0x00	3rd data byte read for a DATA or DATA-LAST (READ) token.
15:8	R/W	0x00	2nd data byte read for a DATA or DATA-LAST (READ) token.
7:0	R/W	0x00	1st data byte read for a DATA or DATA-LAST (READ) token.

Table 12-65 AO_I2C_M_0_RDATA_REG1 0x7

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	8th data byte read for a DATA or DATA-LAST (READ) token.
23:16	R/W	0x00	7th data byte read for a DATA or DATA-LAST (READ) token.
15:8	R/W	0x00	6th data byte read for a DATA or DATA-LAST (READ) token.
7:0	R/W	0x00	5th data byte read for a DATA or DATA-LAST (READ) token.

Table 12-66 AO_I2C_M_0_TIMEOUT_TH 0x8

Bit(s)	R/W	Default	Description
31:24	R/W	0x00	8th data byte read for a DATA or DATA-LAST (READ) token.
23:16	R/W	0x00	7th data byte read for a DATA or DATA-LAST (READ) token.
15:8	R/W	0x00	6th data byte read for a DATA or DATA-LAST (READ) token.
7:0	R/W	0x00	5th data byte read for a DATA or DATA-LAST (READ) token.

Table 12-67 AO_I2C_S_CONTROL_REG 0x0

Bit(s)	R/W	Default	Description
31-29	R	0	REG POINTER: There are 5 internal registers inside the I2C slave module. The I2C Master sets this value using the byte that follows the address byte in the I2C data stream. Register 4 (numbered 0,1,…4) is the status register.
28	R/W	0	SEND READY: This bit is set to ‘1’ by the ARC to indicate to the slave machine that the I2C slave module is ready to send data. This bit is cleared by the I2C module when it has sent 4 bytes to the I2C master. This bit is also available in the status register that can be read by the I2C master. The I2C master can read the status register to see when the I2C slave module has data to send.

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Bit(s)	R/W	Default	Description
27	R/W	0	RECEIVE READY: This bit is set to '1' by the ARC to indicate to the slave machine that the I2C slave module is ready to receive data. This bit is cleared by the I2C module when it has received 4 bytes from the I2C master. This bit is also available in the status register that can be read by the I2C master. The I2C master can read the status register to see when the I2C slave module is ready to receive data.
26	R	0	BUSY: Read only status bit. '1' indicates that the I2C slave module is sending or receiving data.
25	R/W	0	IRQ_EN: If this bit is set, then an interrupt will be sent to the ARC whenever 4 bytes have been read or 4 bytes have been written to the I2C slave module.
24	R/W	0	ACK Always: Typically the ACK of a slave I2C device is dependent upon the availability of data (if reading) and room to store data (when we are being written). Our I2C module has a status register that can be read continuously. This bit can be set if the I2C master wants to continually read the status register.
23-16	R/W	0	Slave Address: Bits [7:1] are used to identify the device. Bit [0] is ignored since this corresponds to the R/W bit.
15-8	R/W	0x27	HOLD TIME: Data hold time after the falling edge of SCL. This hold time is computed as Hold time = (MPEG system clock period) * (value + 1).
7	R/W	0	Enable: A '1' enables the I2C slave state machine.
6-0	R/W	0x06	Sampling rate. Defined as MPEG system clock / (value + 1). The SDA and SCL inputs into the slave module are sampled as a way of filtering the inputs. A rising or falling edge is determined by when 3 successive samples are either high or low respectively.

Table 12-68 AO_I2C_S_SEND_REG: Send Data 0x1

Bit(s)	R/W	Default	Description
31-0	R/W	0	The I2C slave module can send up to 4 bytes of data starting with the byte located at bits [7:0]

Table 12-69 AO_I2C_S_RECV_REG: Received Data 0x2

Bit(s)	R/W	Default	Description
31-0	R	0	This read only register corresponds to the 4 bytes of data written to the I2C slave module by an external I2C master. Bits [7:0] correspond to the first byte written by the I2C master.

Table 12-70 AO_I2C_S_CNTL1_REG 0x3

Bit(s)	R/W	Default	Description
31-6	R	0	Unused
5-3	R/W	0	SCL_FILTER: A filter was added in the SCL input path to allow for filtering of slow rise times. 0 = no filtering, 7 = max filtering
2-0	R/W	0x00	SDA FILTER: A filter was added in the SDA input path to allow for filtering of slow rise times. 0 = no filtering, 7 = max filtering

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12.4 Universal Asynchronous Receiver And Transmitter

12.4.1 Overview

There are a number of UARTs in the chip that offer 2-wire (RX/TX) and 4-wire (RX/TX, CTS/RTS) connections at the digital I/O pins. Each UART contains one transmit FIFO and a receive FIFO (see depths below). The FIFO's are filled by the CPU and read by the CPU. In some cases, the receive FIFO can be configured to be pushed directly to DDR memory without CPU intervention.

Table 12-71 UART List

UART	RX/TX FIFO depths	RX FIFO DMA to DDR	Comment
UART0	128 bytes	Yes	Located in the EE domain
UART1	64 bytes	Yes	Located in the EE domain
UART2	64 bytes	Yes	Located in the EE domain
UART0-AO	64 bytes	No	Located in the Always On domain
UART2-AO	64 bytes	No	Located in the Always On domain

12.4.2 Features

Input filters

The CTS (clear to send) and RX (receive) input paths have input filters to deal with slow rise times. The filters are configurable to use a 125ns or 1us sampling mechanism. There is an implied 3 system clock cycle delay (15ns for a typical system clock of 200MHz) that is used to synchronize and detect the rising/falling edge of the RXD signal. The RXD signal may be passed through an optional filter to deglitch the external signal in noisy conditions. The deglitch filter has two settings which add to the "detection delay" of the RXD signal by the internal logic:

- Filter setting 1 (125ns strobe): 375ns ~ 2.6us
- Filter setting 2 (1us strobe): 3us ~ 21us

The filter is described in the register specification. If the filter is disabled, the shortest RXD low time and high time is 12 system clock cycles (60ns for a system clock of 200MHz).

Clear to Send

CTS is a signal sent from the receiver UART back to the transmitting UART to tell the transmitting UART to stop sending data. The CTS signal must be received before the next START symbol is sent. The transmitting UART is allowed to send one more byte after the CTS signal is recognized. The CTS signal coming into the chip goes through some synchronization and detection which adds an additional 5 system clocks (typically 25ns for a 200Mhz system clock). This setup time for CTS detection is called CTSstop. The CTS input also has an optional filter can be used to deglitch the incoming CTS signal. If the filter is disabled, the CTS signal must be de-asserted 5 system clock cycles before the start of the next BYTE transfer. If the CTS filter is enabled, then additional time must be added to the 25ns requirement. There are two programmable filter settings that effectively delay CTS being seen by the internal logic:

- Filter setting 1 (125ns strobe): 375ns ~ 2.6us
- Filter setting 2 (1us strobe): 3us ~ 21us

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Interrupts

The UARTs can generate interrupts if the receive FIFO exceeds a pre-programmed threshold. An interrupt can also be generated if there is a frame or parity error.

Clock Independent Operation

Because the system clock can be altered to accommodate dynamic frequency scaling, the UARTs have an option in which they use the 24Mhz crystal clock as the source for the UART.

12.4.3 Functional Description

The UART requires that a Baud Rate be established. The UART supports rates as slow as 1Hz up to rates as high as 8M bits/s. Once the baud rate has been established, bytes are transmitted as they are written to the transmit-FIFO by the CPU. A large transmit-FIFO exists to allow the CPU to pre-load a transmit package because the CPU can often write faster than the UART can transmit the data.

Data this automatically received by the UART is placed into the receive FIFO one byte at a time. The receive-FIFO decouples the UART from the CPU allowing the CPU to read the UART byte data at a rate not dictated by the UART.

Figure 12-1 UART Timing Diagram

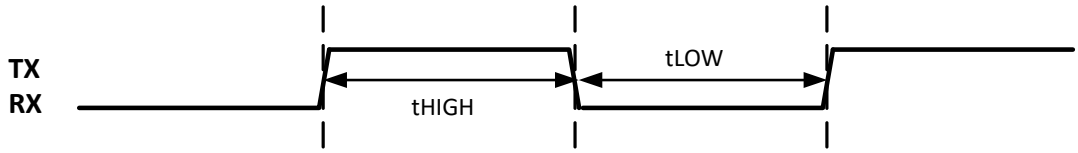


Table 12-72 UART Timing Specification

Symbol	Description	Min.	Max.	Unit	Notes
tHIGH	TX/RX high time	60ns	167ms		
tLOW	TX/RX low time	60ns	167ms		

12.4.4 Register Description

The following register description is uniformly applied to all UART instantiations in the chip.

Base_adr:0xffd00000

Final_adr = base_adr + offset *4

UART0 = 128x8 FIFOs

UART1 = 64x8 FIFOs

UART2 = 64x8 FIFOs

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Table 12-73 UARTx_WFIFO: Write data 0x9000

Bit(s)	R/W	Default	Description
31-8	R	0	unused
7-0	R/W	-	Write FIFO data. The Write FIFO holds 128 or 64 bytes. The Write FIFO can be written as long as it is not full.

Table 12-74 UARTx_RFIFO: Read Data 0x9001

Bit(s)	R/W	Default	Description
31-8	R	0	unused
7-0	R/W	-	Read FIFO data. The Read FIFO holds 128 or 64 bytes. The empty flag can be used to determine if data is available

Table 12-75 UARTx_CONTROL: UART Mode 0x9002

Bit(s)	R/W	Default	Description
31	R/W	0	Invert the RTS signal
30	R/W	0	Mask Error: Set to 1 to mask errors
29	R/W	0	Invert the CTS signal
28	R/W	0	Transmit byte Interrupt: Set to 1 to enable the generation an interrupt whenever a byte is read from the transmit FIFO
27	R/W	0	Receive byte Interrupt: Set to 1 to enable the generation an interrupt whenever a byte is written to the receive FIFO
26	R/W	0	Set to 1 to invert the TX pin
25	R/W	0	Set to 1 to invert the RX pin
24	R/W	0	Clear Error
23	R/W	0	Reset the receive state machine
22	R/W	0	Reset the transmit state machine
21-20	R/W	0	Character length: 00 = 8 Bits, 01 = 7 Bits, 10 = 6 Bits, 11 = 5 Bits
19	R/W	1	Parity Enable: Set to 1 to enable parity
18	R/W	0	Parity type: 0 = even, 1 = odd
17-16	R/W	0	Stop bit length: 00 = 1 bit, 01 = 2 Bits
15	R/W	0	Two Wire mode:
14	R/W	0	Unused
13	R/W	0	Receive Enable. Set to 1 to enable the UART receive function
12	R/W	0	Transmit Enable. Set to 1 to enable the UART transmit function
11-0	R/W	0x120	Old Baud rate

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Table 12-76 UARTx_STATUS: UART Status 0x9003

Bit(s)	R/W	Default	Description
31-27	R	0	Unused
26	R	0	UART_RECV_BUSY: This bit will be 1 if the uart receive state machine is busy
25	R	0	UART_XMIT_BUSY: This bit will be 1 if the uart transmit state machine is busy
24	R	0	RECV_FIFO_OVERFLOW:
23	R	0	CTS Level
22	R	0	Transmit FIFO Empty
21	R	0	Transmit FIFO Full
20	R	0	Receive FIFO empty
19	R	0	Receive FIFO full
18	R	0	This bit is set if the FIFO is written when it is full. To clear This bit, wriTE bit 24 of register 0x2132
17	R	0	Frame error. To clear This bit, wriTE bit 24 of register UART0_CONTROL
16	R	0	Parity error. To clear This bit, wriTE bit 24 of register UART0_CONTROL
15	R	0	Unused
14-8	R	0	Transmit FIFO count. Number of bytes in the transmit FIFO
7	R	0	Unused
6-0	R	0	Receive FIFO count. Number of bytes in the receive FIFO

Table 12-77 UARTx_MISC: UART IRQ CONTROL 0x9004

Bit(s)	R/W	Default	Description
31	R/W	0	Added a “just in case” bit that can be set to 1 to enable clocks always. The default is 0 meaning the auto-clock gating logic is enabled.
30	R/W	0	USE old Rx Baud: There was a bug in the RX baud rate generator. The Rx baud rate generator was re-designed to compute a baud rate correctly. If you want to use the old (stupid) logic, you can set This bit to 1.
29	R/W	0	ASYNC_FIFO_PURGE: This bit can be set to 1 after all UART bytes have been received in order to purge the data into the Async FIFO. This bit is needed because the UART receives 8-bit data, but the ASYNC FIFO can only be written with 16-bit data. In this case there might be a residual byte if the UART is not receiving an even number of bytes.
28	R/W	0	ASYNC_FIFO_EN: If This bit is set to 1, then the UART received data is automatically sent to the Async FIFO module which will in turn automatically send the data to DDR memory
27	R/W	0	CTS: Filter Timebase select: 1 = 1uS, 0 = 111nS timebase. A filter was added to the CTS input to allow for a little digital filtering. The amount of filtering is controlled by this timebase (longer = more filtering) and the value in Bits FILTER_ SEL below.
26-24	R/W	0	CTS: FILTER_SEL: 0 = no filter, 7 = max filtering
23-20	R/W	0	old BAUD_RATE_EXT: These 4 Bits extend the baud rate divider to 16-bits: Baud Rate = {Reg4[23:20],Reg2[11:0]}
19	R/W	0	RX: Filter Timebase select: 1 = 1uS, 0 = 111nS timebase. A filter was added to the RX input to allow for a little digital filtering. The amount of filtering is controlled

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Bit(s)	R/W	Default	Description
			by this timebase (longer = more filtering) and the value in Bits FILTER_SEL below.
18-16	R/W	0	RX: FILTER_SEL: 0 = no filter, 7 = max filtering
15-8	R/W	32	XMIT_IRQ_CNT: The UART can be configured to generate an interrupt if the number of bytes in the transmit FIFO drops below this value.
7:0	R/W	15	RECV_IRQ_CNT: The UART can be configured to generate an interrupt after a certain number of bytes have been received by the UART.

Table 12-78 UARTx_REG5 0x9005

Bit(s)	R/W	Default	Description
31-28	R/W	0	unused
27	R/W	0	Xtal2_clk_sel: 0: see Xtal_clk_sel 1: xtal_div2(12M)
26	R/W	0	Xtal_clk_sel: 0: xtal_div3(8M); 1: xtal(24M);(need set xtal_tick_en =1 first);
24	R/W	0	USE_XTAL_CLK: If this bit is set, then the clock for generating the UART Baud rate comes from the crystal pad. This allows the UART to operate independent of clk81.
23	R/W	0	USE New Baud rate. Over the years, the baud rate has been extended by concatenating bits from different registers. To take advantage of the full 23-bit baud rate generate (extended to 23 bits to accommodate very low baud rates), you must set this bit. If this bit is set, then the baud rate is configured using bits [22:0] below
22:0	R/W	15	NEW_BAUD_RATE: If bit[23] = 1 above, then the baud rate for the UART is computed using these bits. This was added in M6 to accommodate lower baud rates.

UART1_WFIFO: Write data 0x8c00

See UART0 bit descriptions

UART1_RFIFO: Read Data 0x8c01

See UART0 bit descriptions

UART1_CONTROL: UART Mode 0x8c02

See UART0 bit descriptions

UART1_STATUS: UART Status 0x8c03

See UART0 bit descriptions

UART1_MISC: UART IRQ CONTROL 0x8c04

See UART0 bit descriptions

UART1_REG5 0x8c05

See UART0 bit descriptions

UART2_YFIFO: write data 0x8800

See UART0 bit descriptions

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UART2_RFIFO: Read Data 0x8801

See UART0 bit descriptions

UART2_CONTROL: UART Mode 0x8802

See UART0 bit descriptions

UART2_STATUS: UART Status 0x8803

See UART0 bit descriptions

UART2_MISC: UART IRQ CONTROL 0x8804

See UART0 bit descriptions

UART2_REG5 0x8805

See UART0 bit descriptions

UART3_DF_REG_A73 0x8400

See UART3_SLIP bit descriptions

UART3_DF_REG_A74 0x8401

See UART3_SLIP bit descriptions

UART3_DF_REG_A75 0x8402

See UART3_SLIP bit descriptions

UART3_DF_REG_A76 0x8403

See UART3_SLIP bit descriptions

UART3_DF_REG_A77 0x8404

See UART3_SLIP bit descriptions

UART3_DF_REG_A78 0x8405

See UART3_SLIP bit descriptions

UART3_DF_REG_A79 0x8406

See UART3_SLIP bit descriptions

UART3_DF_REG_A80 0x8407

See UART3_SLIP bit descriptions

UART3_DF_REG_A81 0x8408

See UART3_SLIP bit descriptions

UART3_DF_REG_A82 0x8409

See UART3_SLIP bit descriptions

UART3_DF_REG_A83 0x8410

See UART3_SLIP bit descriptions

UART3_DF_REG_A84 0x8411

See UART3_SLIP bit descriptions

UART3_DF_REG_A85 0x8412

See UART3_SLIP bit descriptions

UART3_DF_REG_A86 0x8413

See UART3_SLIP bit descriptions

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UART3_DF_REG_A87 0x8414

See UART3_SLIP bit descriptions

UART3_DF_REG_A88 0x8415

See UART3_SLIP bit descriptions

UART3_DF_REG_A89 0x8416

See UART3_SLIP bit descriptions

UART3_DF_REG_A96 0x8417

See UART3_SLIP bit descriptions

UART3_DF_REG_A97 0x8418

See UART3_SLIP bit descriptions

UART3_DF_REG_A128 0x8420

See UART3_SLIP bit descriptions

UART3_DF_REG_A129 0x8421

See UART3_SLIP bit descriptions

UART3_DF_REG_A130 0x8422

See UART3_SLIP bit descriptions

UART3_DF_REG_A131 0x8423

See UART3_SLIP bit descriptions

UART3_DF_REG_A132 0x8424

See UART3_SLIP bit descriptions

UART3_DF_REG_A133 0x8425

See UART3_SLIP bit descriptions

UART3_DF_REG_A134 0x8426

See UART3_SLIP bit descriptions

UART3_DF_REG_A135 0x8427

See UART3_SLIP bit descriptions

UART3_DF_REG_A136 0x8428

See UART3_SLIP bit descriptions

12.5 Infrared Remote

12.5.1 Overview

IR module includes 3 sub modules: IR demodulation, Legacy IR remote control and Multi-format IR remote control.

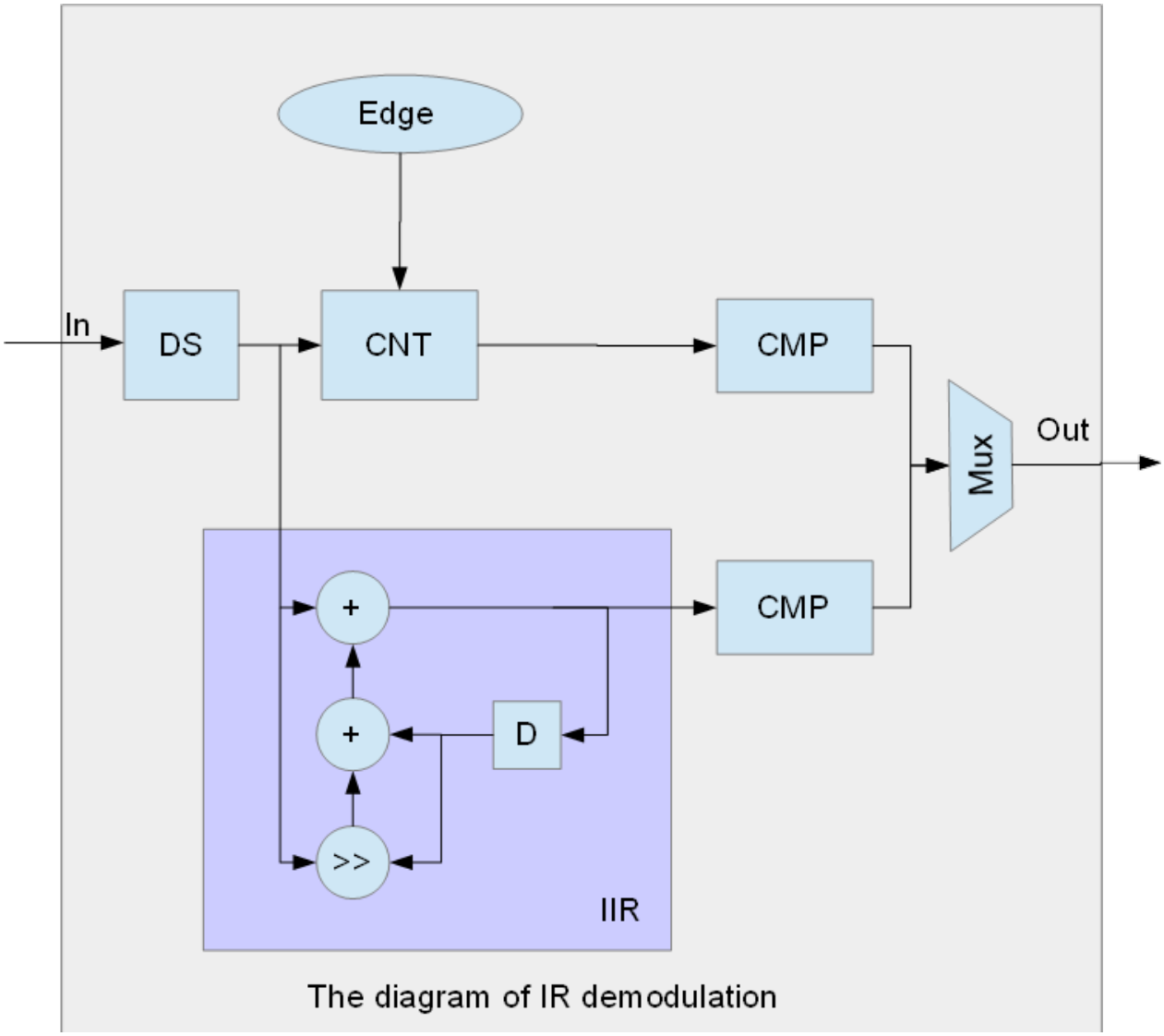
12.5.2 IR Demodulation

IR demolation module demodulate the modulate signal to get the envelop signal.

IR demulation diagram is shown below.

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Figure 12-2 IR Demulation



It implements two methods for demodulation. The input data is an one-bit stream.

The input data are down-sampled firstly. The rate of downsample are configured by register, ranges from 0 to 255.

One method is implemented in the upper path.

The CNT is used to counte the time for ‘0’ and ‘1’. It is cleared when input data changes (from 1 to 0 or from 0 to 1).

When (CNT>REG_IR_PROCT1 && in==1) , it outputs ‘1’. REG_IR_PROCT1 is used to canceled the glitch. Actually, the operation of downsample also have some utility to cancel the glitch.

When (CNT>REG_IR_DECT0 && in==0) , it outputs ‘0’. REG_IR_DECT0 is used to confirm the ‘0’. The value should be bigger than one carrier period.

The other method is implemented in the lower path.

It is a one tap IIR filter. The input data control the integrating factor and the factor can be configured from registers. When 1 inputs, the accumulator can grow quickly. When 0 input, the accumulator decays slowly, and when the number of 0 is bigger enough, it decays soon.

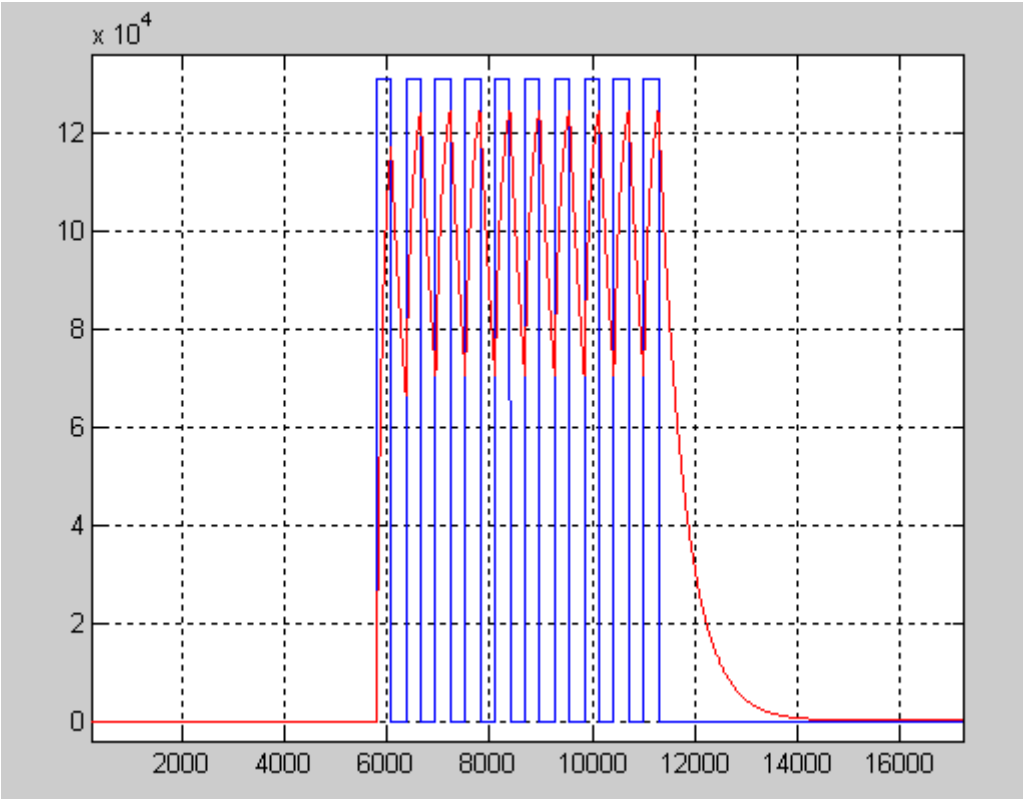
When (accum> REG_IR_IIR_THD1), it outputs ‘1’.

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When (accum< REG_IR_IIR_THD0), it outputs ‘0’.

The diagram shows the simple simulation result. The blue is the input data, the red is the filter out.

Figure 12-3 Simulation Result of IR Demulation

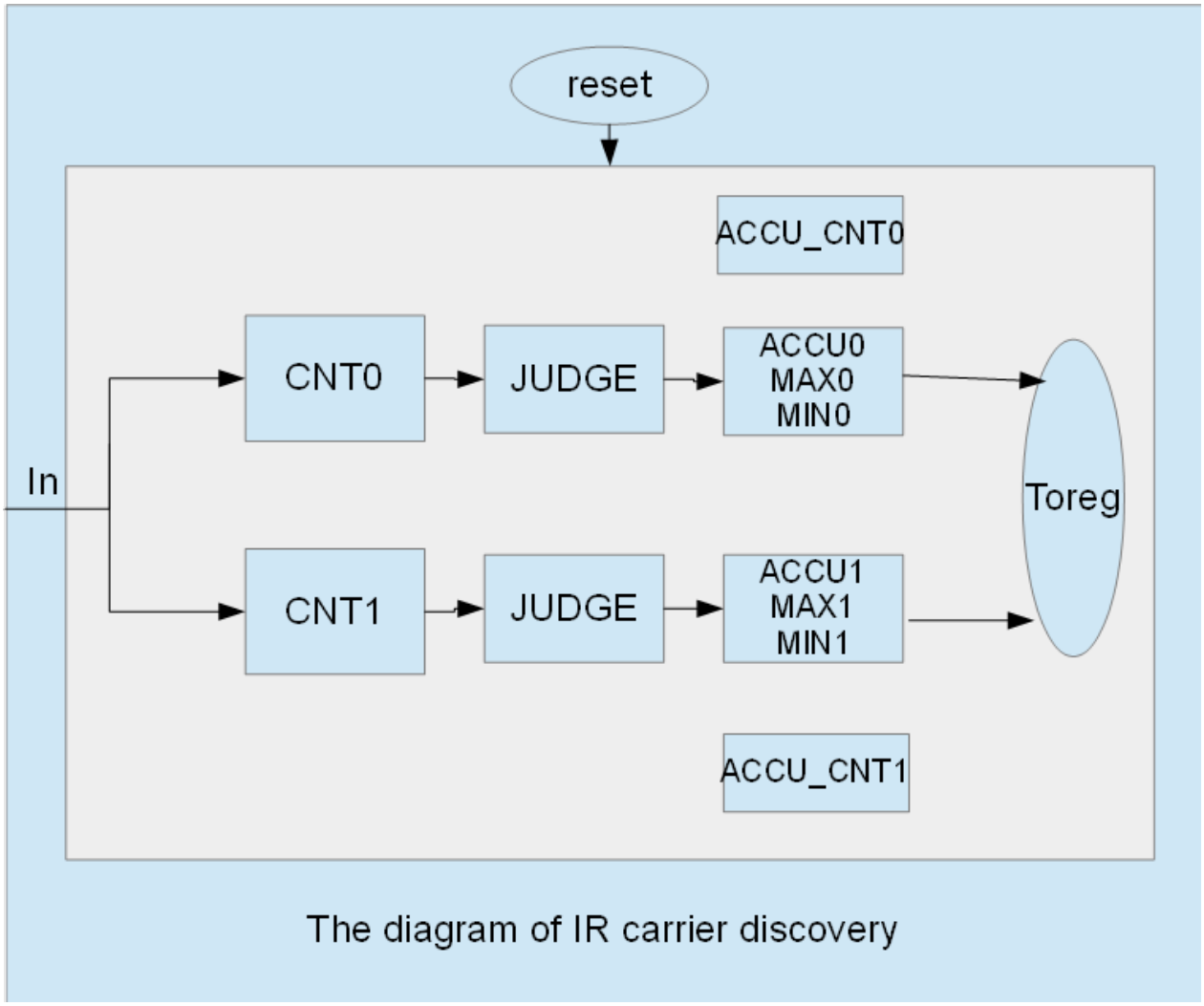


Because the input carrier signal is a rectangular wave signal, the carrier can be estimated easily by counting the wave period. For estimating precisely, the counter for the long “0” and start period and glitch should be removed.

The simple diagram is shown below:

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Figure 12-4 IR Carrier Discovery



CNT is the counter for time of “0” and “1”.

JUDGE is used to remove the long “0” and glitch.

ACC_CNT is the counter for number of accumulating. If ACC_CNT== REG_IR_ST_CNT_THD, then write the value to registers.

For example:

Set REG_IR_ST_CNT_THD to 256, then accumulates 256 times.

The carrier can be calculated:

Fc = Fsys/(RO_IR_SUM_CNT0 +RO_IR_SUM_CNT1)*256;

DUT = RO_IR_SUM_CNT1/(RO_IR_SUM_CNT0 +RO_IR_SUM_CNT1);

Where Fc is the carrier frequency, Fsys is the system clock frequency, DUT is the carrier duty ratio.

12.5.3 Legacy IR Control

The Legacy IR Remote control module has two modes of operation:

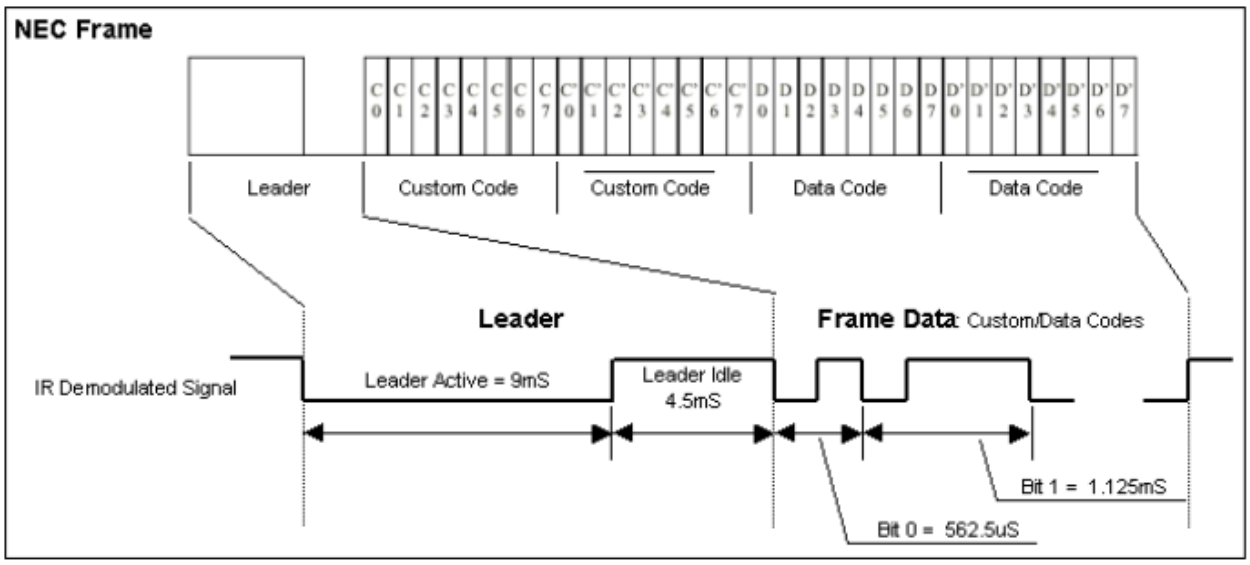
- NEC Frame decoder mode
- General Time Measurement mode

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Note

NEC Frame Decoder: The NEC Frame Decoder mode operates by analyzing the waveform of a TV remote.

Figure 12-5 NEC Frame Decoder



The waveform has a number of components, each of which must fit within a time window to be considered valid. If the entire waveform meets the specifications described by the registers below, then the TV remote codes are captured and an interrupt is generated.

Note

General Time Measurement: Some remotes don't follow the standard NEC format, so additional registers provide the ability to measure the time between rising and/or falling edges of the IR signal. Since the time measurement is done in hardware, the software only needs to read a "width" measurement from a register for every rising and/or falling edge event.

12.5.4 Multi Format IR Control

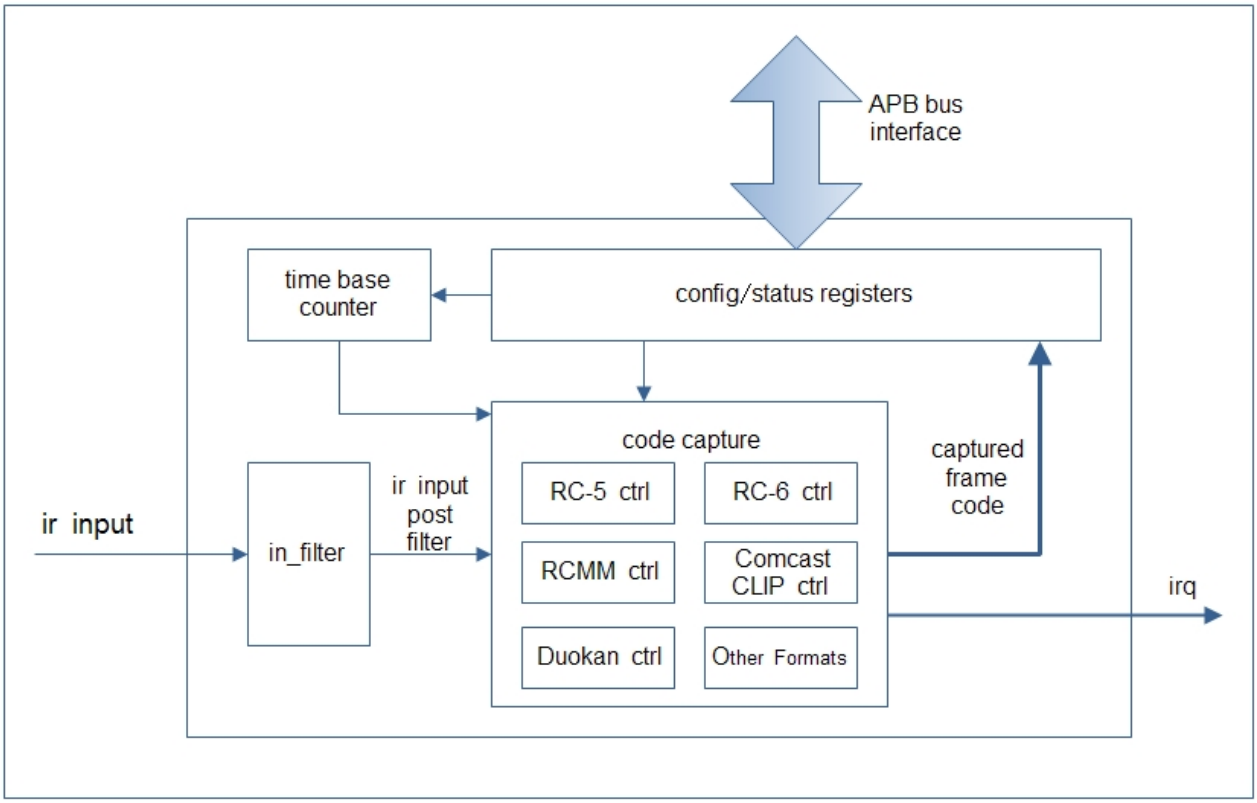
The decoder mainly consisted of two blocks:

- Decoder with input filter
- A set of registers including control & clock, data and tuning

The function diagram of IR decoder is illustrated in the figure below.

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Figure 12-6 IR Decoder Function Block



IR Decoder decodes the IR remote control input signal. 13 operation modes are supported:

- Hardware Decode IR transmission protocol compatible frame decoder mode (NEC MITSUBISHI Thomson Toshiba Sony SIRC RC5 RC6 RCMM Duokan Comcast Sanyo Modes)
- General programmable time measurement frame decoder mode (General Mode)

In Hardware Decode Mode, the Decoder uses signal pattern search mechanism to decode data frame. It can detect logical “0”, “1” , “00”, “01”, “10” and “11”, as well as data frame start and end. Whenever Decoder detects and decodes the data frame, the data are kept in data register.

In General Mode, the Decoder uses edge detection mechanism to decode data frame. It can detect each input signal edge and record the time between two edges. The time measurement result is kept in control register.

The user should set proper operation mode corresponding to the selection of remote controller.

There is a simple time-based signal Filter between the signal input and the Decoder. The Filter is programmable and helps to improve signal integrity.

12.5.5 Register Description

For the following registers:

Base_adr: 0xFF800000

Final_adr = base_adr + offset *4

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Table 12-79 AO_IR_BLASTER_ADDR0 0x53

Bit(s)	R/W	Default	Description
31-27	R	0	unused
26	R	-	BUSY: If this bit is 1, then the IR Blaster module is busy.
25	R	-	This output is 1 when the FIFO is Full
24	R	-	This output is 1 when the FIFO is Empty
23-16	R	-	FIFO Level
15-14	R/W	0	Unused
13-12	R/W	0	MODULATOR_TB: This input controls the clock used to create the modulator output. The modulator is typically run between 32khz and 56khz. The modulator output will equal a divided value of the following: 00: system clock “clk” 01: mpeg_xtal3_tick 10: mpeg_1uS_tick 11: mpeg_10uS_tick
11-4	R/W	0	SLOW_CLOCK_DIV: This is a divider value used to divide down the input “clk”. The divider is N+1 so a value of 0 equals divide by 1.
3	R/W	0	SLOW_CLOCK_MODE: Set this signal high to use a special mode in which the “clk” input is driven by a slow clock less than 1Mhz. This is used for low power cases where we want to run the IR Blaster between 32khz and 1Mhz
2	R/W	0	INIT_LOW: Setting this bit to 1 initializes the output to be high. Please set this bit back to 0 when done
1	R/W	0	INIT_LOW: Setting this bit to 1 initializes the output to be low. Please set this bit back to 0 when done
0	R/W	0	ENABLE: 1 = Enable. If this bit is set to 0, then the IR blaster module is reset and put into an IDLE state.

Table 12-80 AO_IR_BLASTER_ADDR1 0x54

Bit(s)	R/W	Default	Description
31-28	R/W	0	unused
27-16	R/W	0	This value is used with “modulator_tb[1:0]” above to create a low pulse. The time is computed as (mod_lo_count+1) x modulator_tb. The purpose for having a low/high count is the modulator output might not be 50% duty cycle. Hi/Lo counters allow us to modulate using a non-50% duty cycle waveform.
15-12	R/W	0	Unused
11-0	R/W	0	This value is used with “modulator_tb[1:0]” above to create a high pulse. The time is computed as (mod_hi_count+1) x modulator_tb

Table 12-81 AO_IR_BLASTER_ADDR2 0x55

Bit(s)	R/W	Default	Description
31-17	R	0	unused
16	W	0	Set this bit to 1 to write the data below to the FIFO

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Bit(s)	R/W	Default	Description
15-12	R	0	Unused
11-0	R/W	0	FIFO data to be written: Bit[12] output level (or modulation enable/disable: 1 = enable) Bit[11:10] Timebase: 00 = 1uS 01 = 10uS 10 = 100uS 11 = Modulator clock Bit[9:0] Count of timebase units to delay

Table 12-82 AO_IR_BLASTER_ADDR3 0x56

Bit(s)	R/W	Default	Description
31-17	R	0	unused
16	R	0	fifo thd pending
8	R/W	0	FIFO irq enable
7:0	R/W	0	FIFO irq threshold.

For the following registers:

Base_adr: 0xff808000

Final_adr = base_adr + offset *4

Table 12-83 AO_IR_DEC_DEMOD_CNTL0 0x30

Bits	R/W	Default	Description
31			Ir demodulor soft_reset:write 1 to reset ir_demodulor,will auto clr to 0
30			Reg_ir_fd_reset :It is used to reset the carrier detection module.write 1 to reset , will auto clr to 0
29			Reg_ir_demod_mode:It is uset to set the demod mode. 0 = count mode; 1 = iir mode.default is 0
28			Ir demodulor clk gate bypass:write 1 will bypass clk gate,default is 0
27:16			Reg_ir_st_cnt_thd:It is used to set the statistics number for carrier detection.default is 0x40
15:8			Reg_ir_ds_rate:It is uset to set the downsample rate.default is 0x8
7:4			Reg_ir_fsft_1:It is used to set the shift value for input data “1”.default is 0x7
3:0			Reg_ir_fsft_0:It is used to set the shift value for input data “0”.default is 0x9

Table 12-84 AO_IR_DEC_DEMOD_CNTL1 0x31

Bits	R/W	Default	Description
31			Reg_ir_demod_en:ir demod enable ,default is 0
30			Reg_ir_invt:ir input invert, invert input at 1.default is 0

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Bits	R/W	Default	Description
29:16			Reg_ir_proctect1:It is used to set the protection threshold for 1. It is used to filter the glitch.default is 0xa
13:0			Reg_ir_detect0:It is used to set the detection threshold for signal “0”.default is 0x3e8

Table 12-85 AO_IR_DEC_DEMOD_IIR_THD 0x32

Bits	R/W	Default	Description
31:16			Reg_ir_iir_thd1:It is used to set the detection threshold for “1” in iir mode.default is 0xdac
15:0			Reg_ir_iir_thd0:It is used to set the detection threshold for “0” in iir mode.default is 0xdac

Table 12-86 AO_IR_DEC_DEMOD_THD0 0x33

Bits	R/W	Default	Description
29:16			Reg_ir_thd0_low:It is used to set the low threshold for “0” when statistics.default is 0x12c
13:0			Reg_ir_thd0_high:It is used to set the high threshold for “0” when statistics.default is 0x1770

Table 12-87 AO_IR_DEC_DEMOD_THD1 0x34

Bits	R/W	Default	Description
29:16			Reg_ir_thd1_low:It is used to set the low threshold for “1” when statistics.default is 0x12c
13:0			Reg_ir_thd1_high:It is used to set the high threshold for “1” when statistics.default is 0x1770

Table 12-88 AO_IR_DEC_DEMOD_SUM_CNT0 0x35

Bits	R/W	Default	Description
25:0			Ro_ir_sum_cnt0:It is used to report the sum value for the statistics data “0”. READ ONLY

Table 12-89 AO_IR_DEC_DEMOD_SUM_CNT1 0x36

Bits	R/W	Default	Description
25:0			Ro_ir_sum_cnt1:It is used to report the sum value for the statistics data “1”. READ ONLY

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Table 12-90 AO_IR_DEC_DEMOD_CNT0 0x37

Bits	R/W	Default	Description
29:16			Ro_ir_max_cnt0:It is used to report the maximum value for the statistics data “0”. READ ONLY
13:0			Ro_ir_min_cnt0:It is used to report the minimum value for the statistics data “0”. READ ONLY

Table 12-91 AO_IR_DEC_DEMOD_CNT1 0x38

Bits	R/W	Default	Description
29:16			Ro_ir_max_cnt1:It is used to report the maximum value for the statistics data “1”. READ ONLY
13:0			Ro_ir_min_cnt1:It is used to report the minimum value for the statistics data “1”. READ ONLY

Table 12-92 AO_IR_DEC_LDR_ACTIVE: Leader Active Time 0x00

Bit(s)	R/W	Default	Description
31-29	R	0	unused
28-16	R/W	0x 1d8	Max Leader ACTIVE time: 9.44mS assuming base rate = 20uS
15-13	R	0	unused
12-0	R/W	0x1ac	Min Leader ACTIVE time: 8.56mS assuming base rate = 20uS

This register controls the min/max leader active time window. For most TV remote controls, the leader active time is about 9mS. The values in this register correspond to counts of the base rate programmed by register 0x2124

Table 12-93 AO_IR_DEC_LDR_IDLE: Leader Idle Time 0x01

Bit(s)	R/W	Default	Description
31-26	R	0	unused
28-16	R/W	0xf8	Max Leader IDLE time: 4.96mS assuming base rate = 20uS
15-13	R	0	unused
12-0	R/W	0xca	Min Leader IDLE time: 4.04mS assuming base rate = 20uS

This register controls the min/max leader IDLE time window. For most TV remote controls, the leader idle time is about 4.5mS. The values in this register correspond to counts of the base rate programmed by register: 0x2124

Table 12-94 AO_IR_DEC_LDR_REPEAT: Repeat Leader Idle Time 0x02

Bit(s)	R/W	Default	Description
31-26	R	0	Unused
25-16	R/W	0x7a	Max REPEAT Leader IDLE time: 2.44mS assuming base rate = 20uS
15-10	R	0	unused
9-0	R/W	0x66	Min REPEAT Leader IDLE time: 2.04mS assuming base rate = 20uS

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This register controls the repeat leader IDLE time window. The repeat key uses the standard leader active time (9ms) but a shorter leader idle time.

Table 12-95 AO_IR_DEC_BIT_0: BIT 0 Identification Time 0x03

Bit(s)	R/W	Default	Description
31-26	R	0	Unused
25-16	R/W	0x42	Max BIT 0 time: 1.32mS assuming base rate = 20uS
15-10	R	0	Unused
9-0	R/W	0x2e	Min BIT 0 time: 0.92mS assuming base rate = 20uS

This register controls the min/max BIT 0 time window. For most TV remote controls, the bit 0 time is about 1.125mS. The values in this register correspond to counts of the base rate programmed by register: 0x2124

Table 12-96 AO_IR_DEC_REG0: Base Rate Generator 0x04

Bit(s)	R/W	Default	Description
31	R	0	Just in case bit. Normally this bit is set to 0 so that the auto-clock gating is enabled. If there is a problem related to the auto-clock gating, then this bit can be set to one to disable the auto-clock gating.
30-28	R/W	0	FILTER_COUNT: This is a new feature to Nike. The IR remote input now has a simple filter that accommodates slow rise times by providing a little hysteresis. The logic works as follows. If the input is low, then an input signal will only be considered HIGH if it remains high for (FILTER_COUNT * 111nS). Similarly, if the input is currently high, it will only be considered LOW if the input signal remains low for (FILTER_COUNT * 111nS).
27-25	R	0	Unused
24-12	R/w	0xFA0	Max Frame Time. 80mS assuming base rate = 20uS This value is used to determine if a code is a repeat code (e.g. leader followed by no data for this amount of time). This value can also be used to catch slow remote codes (i.e. code sequences that are longer than expected).
11-0	R/W	0x13	This value dictates the base rate time for all measurements associated with the IR decoder. In the past, the base rate was divided from the system clock. In the current design, the base rate is divided from a fixed 1uS timer. This 1uS timer is constant and doesn't change (even when the system clock does) Base rate = (count + 1) * 1uS

This register controls the master rate generator for all width measurements made by the IR decoder module.

Table 12-97 AO_IR_DEC_FRAME: Frame Data 0x05

Bit(s)	R/W	Default	Description
31-0	R	0	Frame Data. Format: {custom code, ~custom code, data, ~data}

Note

New keys will be ignored until this register is read if the hold first key bit is set in the decode control register. Reading this register resets an internal hold first flag.

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Table 12-98 AO_IR_DEC_STATUS: Frame Status 0x06

Bit(s)	R/W	Default	Description
31	R/W	0	Sim faster: Reserved
30	R/W	0	BIT_1_MATCH_EN: Set this bit to 1 to enable qualification of bit 1 times. In the previous IR decoder module, frame detection only looked at the BIT 0 time to identify a zero bit. If a zero bit time wasn't found, then it was assumed that the bit was a 1. In the updated IR decoder module, the module will look at the BIT 0 time to find zero bits, and the BIT 1 time to find 1 bits. If the width of a pulse doesn't match the zero or the one bit width time, then the frame is considered invalid.
29-20	R/W	0x89	Max BIT 1 time: 2.74mS assuming base rate = 20uS
19-10	R/W	0x57	Min BIT 1 time: 1.74mS assuming base rate = 20uS
9	R	-	IRQ Status. 1 if there is an interrupt
8	R	-	IR Decoder input. This is the level of the digital signal coming into the IR module for decoding. This is the same as reading the I/O pad level.
7	R	-	BUSY. 1 if the decoder isbusy
6-4	R	0	Decoder Status: For debug only 000: OK 001: last frame timed out 010: leader time error (invalid IR signal) 011: repeat error (repeat leader, but other IR transitions found). 100: Invalid bit
3-0	R	0	Frame Status Bit 3: Frame data valid Bit 2: data code error (data != ~data in IR bit stream) Bit 1: custom code error (custom_code != ~custom_code in IR bit stream) Bit 0: 1 = repeat key, 0 = standard key

Table 12-99 AO_IR_DEC_REG1: Decode / Interrupt Control 0x07

Bit(s)	R/W	Default	Description
31	R	0	Unused
30	R/W	0	CNTL_1uS_EQ_CLK: This bit should be set to 1 if the clk81 (system clock) is less than 50 Mhz.
29	R/W	0	CNTL_111nS_EQ_CLK: This bit should be set to 1 if the clk81 (system clock) is less than 50 Mhz.
28-16	R	0	Time measurement since the last time the internal time counter was reset by the rising and/or falling edge of the IR signal. The selection of reset on rising and/or falling edge is determined by the selection of the IRQ (Bits 3-2 below)
15	R/W	1	ENABLE: If this bit is 1, then the state-machines are enabled. If this bit is zero, then the state-machines cleanup and immediately return to idle.
14	R/W	0	USE SYSTEM CLOCK: This is a new feature. 1 = use the system clock, 0 = use the 1uS timebase tick. During normal operation, the module is setup to create a 20uS tick from the 1uS internal timebase of the chip. If the chip is configured to operate using the 32khz RTC oscillator, the 1uS timebase is invalid and therefore the 20uS timebase is invalid. In order to measure time correctly, the IR remote circuit can use the system clock (which in this case is the 32khz oscillator clock) as the master timebase.
13-9	R/W	1f	Number of bits in the IR frame (N-1)
8-7	R/W	0	Decoder mode 00: NEC Frames: Decode Leader and 32 bits 01: Only accumulate bits (skip the leader)

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Bit(s)	R/W	Default	Description
			10: Measure Mode: The internal width measuring counter is reset on the rising and/or falling edge of the IR remote signal based on the settings of IRQ Selection below. Just before being reset, the measured width is captured and stored so that it can be read in bits [28:16] of this register. 11: NEC Frames: Decode Leader and 32 bits
6	R/W	1	Hold First Key. If this bit is set true, then the frame data register (0x2125) will only be updated if hasn't already been updated. Once updated, the frame data register will not be updated again until it has been read. This bit can be used to guarantee the first TV remote code captured will not be overwritten by subsequent transmissions from a TV remote. NOTE: You must read the frame data register to clear an internal hold first flag if this bit is set.
5-4	R/W	11	Frame mask. These bits are used to qualify frames for capture. 00: Capture all frames good or bad 01: Capture only frames where data=~data. Ignore custom codes 10: Capture only frames where custom_code = ~custom_code. Ignore data codes 11: Capture only frames where (data=~data) and (custom_code = ~custom_ code)
3-2	R/W	0	IRQ Selection and width measurement reset: 00: IR Decoder done 01: IR input rising or falling edge detected 10: IR input falling edge detected 11: IR rising edge detected
1	R/W	0	IR Polarity. Polarity of the input signal (VD[0])
0	R/W	0	Set to 1 to reset the IR decoder. This is useful because the IR remote state machine thinks in terms of milliseconds and may take tens of milliseconds to return to idle by itself.

AO_MF_IR_DEC_LDR_ACTIVE: Leader Active control 0x10 This register controls the min/max Leader Active time window. For example, for NEC format, the Leader Active time is about 9mS. To identify a Leader Active time between 8.60 mS and 9.40 mS (assuming base resolution = 20uS), user can set Max duration = 0x1d6 ('d470) to represent 9.40 mS, and set Min duration = 0x1ae ('d430) to represent 8.60 mS.

Bit(s)	R/W	Default	Description
31-29	R	0	Unused
28-16	R/W	0	Max duration of Leader's active part
15-13	R	0	Unused
12-0	R/W	0	Min duration of Leader's active part

Table 12-100 AO_MF_IR_DEC_LDR_IDLE: Leader Idle control 0x11

Bit(s)	R/W	Default	Description
31-29	R	0	Unused
28-16	R/W	0	Max duration of Leader's idle part
15-13	R	0	Unused
12-0	R/W	0	Min duration of Leader's idle part

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Table 12-101 AO_MF_IR_DEC_LDR_REPEAT: Repeat Leader Idle Time 0x12

Bit(s)	R/W	Default	Description
31-26	R	0	Unused
25-16	R/W	0	Max duration of Repeat Code's Leader. In NECformat, it defines for the repeat leader's idle part. In Toshiba format, it defines for the repeat leader's second idle part (In Toshiba format, the repeat leader's first idle part has the same duration time as the normal leader idle part.)
15-10	R	0	Unused
9-0	R/W	0	Min duration of Repeat Code's Leader

Table 12-102 AO_MF_IR_DEC_BIT_0 0x13

Bit(s)	R/W	Default	Description
31-26	R	0	Unused
25-16	R/W	0	Max duration of Duration Setting Regiser 0. It defines max tining duration for: Logic"0" for NEC/Toshiba/Sony/Thomas format or Half trailer bit for RC6 format (RC6's half trailer bit typically 888.89us) or time of Duokan/RCMM/4ppm format's Logic "00"
15-10	R	0	Unused
9-0	R/W	0	Min duration of Duration Setting Regiser 0.

Table 12-103 AO_MF_IR_DEC_REG0 0x14

Bit(s)	R/W	Default	Description
31	R/W	0	Clock gating control just in case. Set 1 can force clock gating disabled.
30-28	R/W	0	Filter ctrl. Set the monitor timing for input filter, bigger value means longer monitor time. Value 0 = no filtering.
27-25	R	0	Unused
24-12	R/W	0	Max frame time. Max duration of one whole frame.
11-0	R/W	0	Base time parameter. Used to generate the timing resolution. Resolution = (base_time_paramter + 1) * (1/ Freq_sys_clk). For example, if Frequency of sys_clk is 1Mhz, and base_time_parameter=19, Then resolution = (19+1)*(1uS) = 20uS.

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Table 12-104 AO_MF_IR_DEC_STATUS 0x16

Bit(s)	R/W	Default	Description
31	R/W	0	Frame data valid 1. (This bit is set to 1 when a captured frame is updated/stored into “FrameBody_1” register. A read of “FrameBody_1” register will clear This bit. “FrameBody_1” register is used to store the over 32bit MSBs of the formats whose length is more than 32 bit)
30	R/W	0	bit_1_match_en. Set to 1 to enable the check of whether logic”1” bit matches timing configure during the frame input process.
29-20	R/W	0	Max Duration 1. Max duration of Duration Setting Regiser 1. It defines max duration for: Logic”1” for NEC/Toshiba/Sony format or Whole trailer bit for RC6 format (RC6’s whole trailer bit typically 1777.78us) or time of Duokan/RCMM/4ppm format’s Logic “01”
19-10	R/W	0	Min Duration 1. Min duration of Duration Setting Regiser 1.
9	R	0	irq_status. Appear as 1 if there is an interrupt.
8	R	0	ir_i_sync. IR remote serial input after synchronization. This is the level of the digital signal coming into the IR module for decoding. This is the same as reading the I/O pad level.
7	R	0	Busy. When =1, means state machine is active.
6-4	R	0	Decoder_status (for debug only). 000: OK 001: last frame timed out 010: leader time error (invalid IR signal) 011: repeat error (repeat leader, but other IR transitions found). 100: Invalid bit
3-0	R	0	Frame status. bit 3: Frame data valid (This bit is set to 1 when a captured frame is updated/stored into “FrameBody” register. A read of “FrameBody” register will clear This bit. If store and read occurs at the same time, This bit is set to 1 in common, But if “Hold first” is set to true and this valid Bit is already 1, a read clear takes precedance and This bit is clear to 0.) bit 2: data code error (data != ~data in IR bit stream) bit 1: custom code error (custom_code != ~custom_code in IR bit stream) bit 0: 1 = received frame is repeat key, 0 = received frame is normal key

Table 12-105 AO_MF_IR_DEC_REG1 0x17

Bit(s)	R/W	Default	Description
31	R/W	0	Set to 1 to use faster timebase. .
30	R/W	0	cntl_1us_eq_clk. Just use sys_clk to relace 1uS tick.
29	R/W	0	cntl_xtal3_eq_clk. Just use sys_clk to relace 111ns tick.
28-16	R	0	Pulse Width Counter. It stores the internal counter of pulse width duration. Commonly used as time measurement when decode_mode is set to measure width mode (software decode). Time measurement starts at the last time the internal time counter was reset by the rising and/or falling edge of the IR signal. The selection of reset on rising and/or falling edge is determined by the IRQ Selection field (Bits 3-2 below)
15	R/W	0	Enable. 1 = enable the state machine of IR decoder.

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Bit(s)	R/W	Default	Description
			0 = disable the state machine of IR decoder.
14	R/W	0	cntl_use_sys_clk. Use sys_clk for the timebase. It's useful when sys_clk at low frequency (such as 32Khz) and cannot create 1uS timebase tick. 1 = use the system clock as timebase. 0 = use the 1uS timebase tick as timebase.
13-8	R/W	0	bit_length minus 1. (N-1). Used to set the value of frame body's bit length (frame body commonly includes address and data code part). If a format has 24 bit frame body, this value shall be set to 23.
7	R/W	0	Record_at_error. 1= record the frame body and status forcibly, even if data/custom code error check enabled by frame_mask and relative error occurs. 0 = if data/custom code error check enabled by frame_mask and relative error occurs, not record the frame body and status forcibly
6	R/W	0	Hold_first Used to hold the first captured frame data. If This bit is set to 1, then the "FrameBody/FrameBody_1" register will only be updated if hasn't already been updated. Once updated, the "FrameBody/FrameBody_1" register will not be updated again until it has been read. This bit can be used to guarantee the first TV remote code captured will not be overwritten by subsequent transmissions from a TV remote. NOTE: Read the"FrameBody" register can clear the internal "Frame data valid" flag, and read the "FrameBody_1" register can clear the "Frame data valid 1" flag.
5-4	R/W	0	Frame_mask. Some formats' body include bit-inversed data or custom/address code for error check. 00 = ignore error check from either data or custom/address code 01= check if data code matches its inverse values, ignore error check from custom/address code 10= check if custom/address code matches its inverse values, ignore error check from data code 11= check if data and custom codes match their inverse values
3-2	R/W	0	Irq_sel. IRQ Selection and width measurement reset: 00: IR Decoder done 01: IR input rising or falling edge detected 10: IR input falling edge detected 11: IR rising edge detected
1	R/W	0	IR input polarity selection. Used to adjust/invert the polarity of IR input waveform.
0	R/W	0	Decoder Reset. Set to 1 to reset the IR decoder. This is useful because the IR remote state machine thinks in terms of milliseconds and may take tens of milliseconds to return to idle by itself.

Table 12-106 AO_MF_IR_DEC_REG2 0x18

Bit(s)	R/W	Default	Description
31-27	R	0	Unused
26	R/W	0	Width_low_enable. Enable counter record of low pulse width duration. 0 = do not force enable of width low counter record 1 = force enable of width low counter record Some IR formats' decoding need to use internal width low counter record. By default, the width low counter record is enabled automatically for related formats. This bit is used for enable forcibly just in case. Besides, if "leader plus stop bit" method is enabled for repeat detection, This bit is also need to be enabled.
25	R/W	0	Width_high_enable. Enable counter record of high pulse width duration. 0 = do not force enable of width high counter record

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Bit(s)	R/W	Default	Description
			1 = force enable of width high counter record Some IR formats' decoding need to use internal width high counter record. By default, the width high counter record is enabled automatically for related formats. This bit is used for enable forcibly just in case.
24	R/W	0	Enable "leader plus stop bit" method for repeat detection. 0 = "leader plus stop bit" method disabled 1 = "leader plus stop bit" method enabled Some IR formats use one normal frame's leader followed by a stop bit to represet repeat. There is no frame data in this kind repeat frame. To use this method, width_low_enable (Bit 26 of 0x20 offset register) shall be set to 1, and max_duration_3 and min_duration_3 in 0x28 offset register shall be set to appropriate value for stop bit's timing duration.
23-22	R	0	Unused
21-16	R/W	0	Repeat_Bit_index. These Bits are used for compare bit method to set the index of the bit that is used as repeat flag. The index value can be 0 to 63. Compare bit method is one of the methods for repeat detection . Some IR formats use one bit in frame to represet whether the frame is repeat.
15	R/W	0	Running_count_tick_mode. This bit is only valid when use_clock_to_counter Bit is 0. 0 = use 100uS as increasing time unit of frame-to-frame counter 1 = use 10uS as increasing time unit of frame-to-frame counter
14	R/W	0	Use_clock_to_counter. If This bit is set to 1, the running_count_tick_mode Bit is ignored. 0 = do not use system clock as increasing time unit of frame-to-frame counter 1 = use system clock as increasing time unit of frame-to-frame counter
13	R/W	0	Enable frame-to-frame time counter (running-counter). 0 = frame-to-frame time counter disabled 1 = frame-to-frame time counter enabled If enabled, the frame-to-frame counter increases every 100uS or 10uS until it reaches its max value(all Bits are 1) or it is reset. When it reaches its max value, it keeps the value until it is reset. When it is reset, it becomes zero and then begin increasing again. The counter can be reset even when it has not reached its max value. The increasing time unit can be 100uS or 10uS or system clock frequency which is set by running_count_tick_mode and use_clock_to_counter settings. When a frame's data are capured and stored into FrameBody/FrameBody_1 register, frame-to-frame counter is reset to zero. After reset to zero, the frame-to-frame counter will begin increasing again, until it reaches its max value or it is reset. For repeat frame detection, users can use hardware detection by enabling compare frame or compare bit method, or users can read frame-to-frame counter to let software to make the decision.
12	R/W	0	Enable repeat time check for repeat detection. This bit is valid only when compare frame method or compare Bit method is enabled. 0 = repeat time check disabled 1 = repeat time check enabled When repeat frame detection is enabled by enabling compare frame or compare Bit method, the frame time interval may need to be checked in order to decide whether the frames are repeat (key pressed without release) or not. You can configure the repeat_time_max value by setting 0x38 offset register. If frame interval is smaller than the "repeat time max", it may considered as repeat. If frame interval is bigger than the "repeat time max", it is considered as not repeat.
11	R/W	0	Enable compare frame method for repeat detection. 0 = compare frame method disabled 1 = compare frame method enabled Some IR formats transer the same data frame as repeat frame when the key is kept pressed without release. For repeat detection, compare frame method can be used.

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Bit(s)	R/W	Default	Description
			If a new frame and the old received frame are the same and the repeat time is under the limit(frame-to-frame time counter value is smaller than the repeat_time_max), the status register's frame_status0 is set to 1 automatically as repeat detected flag. You can configure the repeat_time_max value by setting 0x38 offset register.
10	R/W	0	Enable compare Bit method for repeat detection. 0 = compare Bit method disabled 1 = compare Bit method enabled Some IR formats use only one bit to represet whether the frame is repeat. You can compare only one bit instead of compare the whole frame for repeat detection. If compare frame method is enabled, then This bit is ignored.
9	R/W	0	Disable read-clear of FrameBody/FrameBody_1. 0 = read-clear enabled 1 = read-clear disabled FrameBody/FrameBody_1 registers are read-cleared in default. When these register are read, they are cleared to zero. This bit is used to disable this read-clear feature. (FrameBody/FrameBody_1 registers are used to store captured frame data).
8	R/W	0	input stream bit order. 0 = LSB first mode (first bit in input stream is considered as LSB) 1= MSB first mode (first bit in input stream is considered as MSB) Note: Commonly the following formats shall set 1 to enable MSB first mode (unless you insist on LSBfirst mode for your specified use): RC5, RC5 extend, RC6, RCMM, Duokan, Comcast
7:4	R	0	Unused
3:0	R/W	0	Decode_mode.(format selection) 0x0 =NEC 0x1= skip leader (just Bits, without leader) 0x2=General time measurement (measure width, software decode) 0x3=MITSUBISHI 0x4=Thomson 0x5=Toshiba 0x6=Sony SIRC 0x7=RC5 0x8=Reserved 0x9=RC6 0xA=RCMM 0xB=Duokan 0xC=Reserved 0xD=Reserved 0xE=Comcast 0xF=Sanyo

Table 12-107 AO_MF_IR_DEC_DURATN2 0x19

Bit(s)	R/W	Default	Description
31-26	R	0	Unused
25-16	R/W	0	Max duration of Duration Setting Regiser 2. It defines max duration for: Half bit for RC5/6 format (RC5 typically 888.89us for half bit, RC6 typically 444.44us) or time of Duokan/RCMM/4ppm format’s Logic “10” or time of Comcast/16ppm’s base duration

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Bit(s)	R/W	Default	Description
15-10	R	0	Unused
9-0	R/W	0	Min duration of Duration Setting Regiser 2.

Table 12-108 AO_MF_IR_DEC_DURATN3 0x1a

Bit(s)	R/W	Default	Description
31-26	R	0	Unused
25-16	R/W	0	Max duration of Duration Setting Regiser 3. It defines max duration for: Whole bit for RC5/6 format (RC5 typically 1777.78us for whole bit, RC6 typically 888.89us) or time of Duokan/RCMM/4ppm format's Logic "11" or time of Comcast/16ppm's offset duration
15-10	R	0	Unused
9-0	R/W	0	Min duration of Duration Setting Regiser 3.

Table 12-109 AO_MF_IR_DEC_FRAME: Frame Body (Frame Data, LSB 32Bit) 0x1b

Bit(s)	R/W	Default	Description
31-0	R	0	32 bit Read-Only register stores frame body (LSB 32 bit) captured from IR remote data flow, commonly includes custom/address code and data code.

Note

New keys will be ignored until **FrameBody** register is read if the hold first key Bit is set in the decode control register. Reading this register resets an internal frame data valid flag.

Table 12-110 AO_MF_IR_DEC_FRAME: Frame Body 1 (Frame Data, MSB 32Bit) 0x1b

Bit(s)	R/W	Default	Description
31-0	R	0	Stores frame body excess 32 bit range. (MSB 32 bit)

Note

New keys will be ignored until **FrameBody** register is read if the hold first key Bit is set in the decode control register. Reading this register resets an internal frame data valid flag.

Table 12-111 AO_MF_IR_DEC_STATUS_1 0x1c

Bit(s)	R/W	Default	Description
31-20	R	0	Unused
19-0	R	0	Stores the last frame-to-frame counter value before the last counter reset caused by the last frame data record/update.

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Table 12-112 AO_MF_IR_DEC_STATUS_2 0x1d

Bit(s)	R/W	Default	Description
31-20	R	0	Unused
19-0	R	0	Stores the value of the frame-to-frame counter which is running currently.

12.6 Pulse-Width Modulation

12.6.1 Overview

The chip has 5 PWM modules that can be connected to various digital I/O pins, among which 3 are in EE domain and 2 is in AO domain. Each PWM is driven by a programmable divider driven by a 4:1 clock selector. The PWM signal is generated using two 16-bit counters. One is the High and Low counter, which is individually programmable with values between 1 and 65535. Using a combination of the divided clock (divide by N) and the HIGH and LOW counters, a wide number of PWM configurations are possible. The other is delta-sigma couter, generate 18-bit sigma, the PWM-out is the highest sigma. The PWM outputs vs counters are also illustrate below.

Figure 12-7 PWM Block Diagram

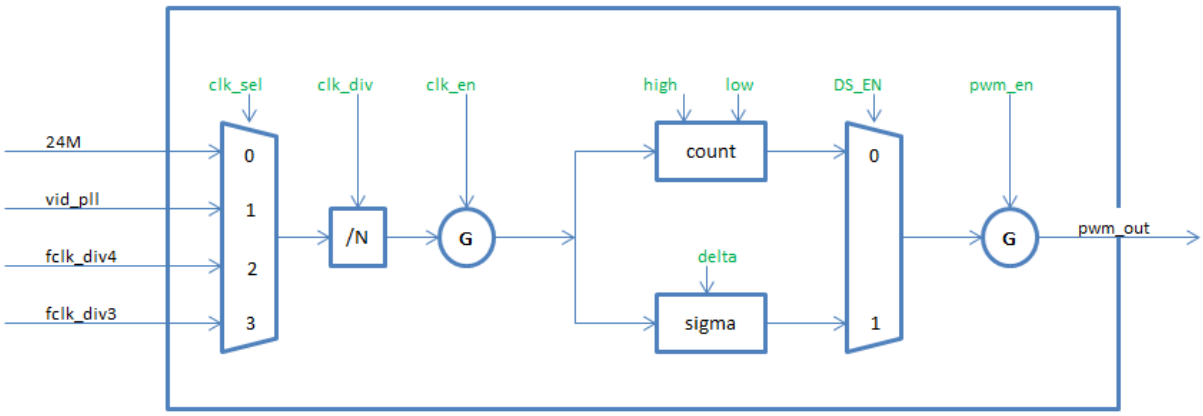
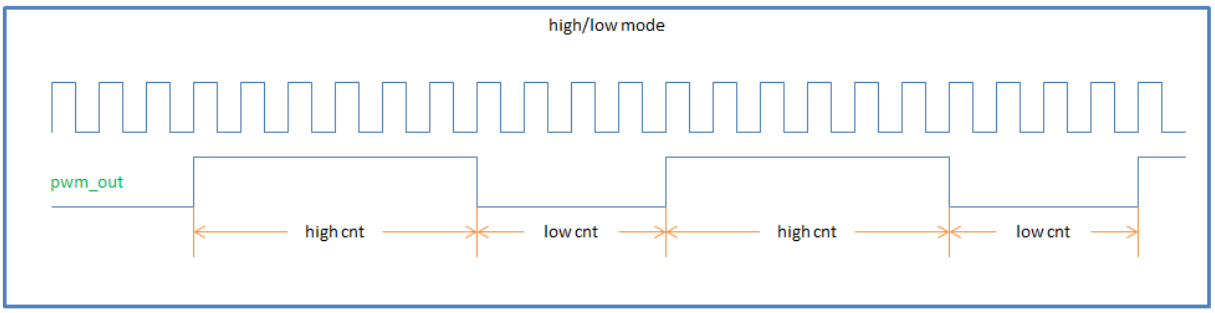
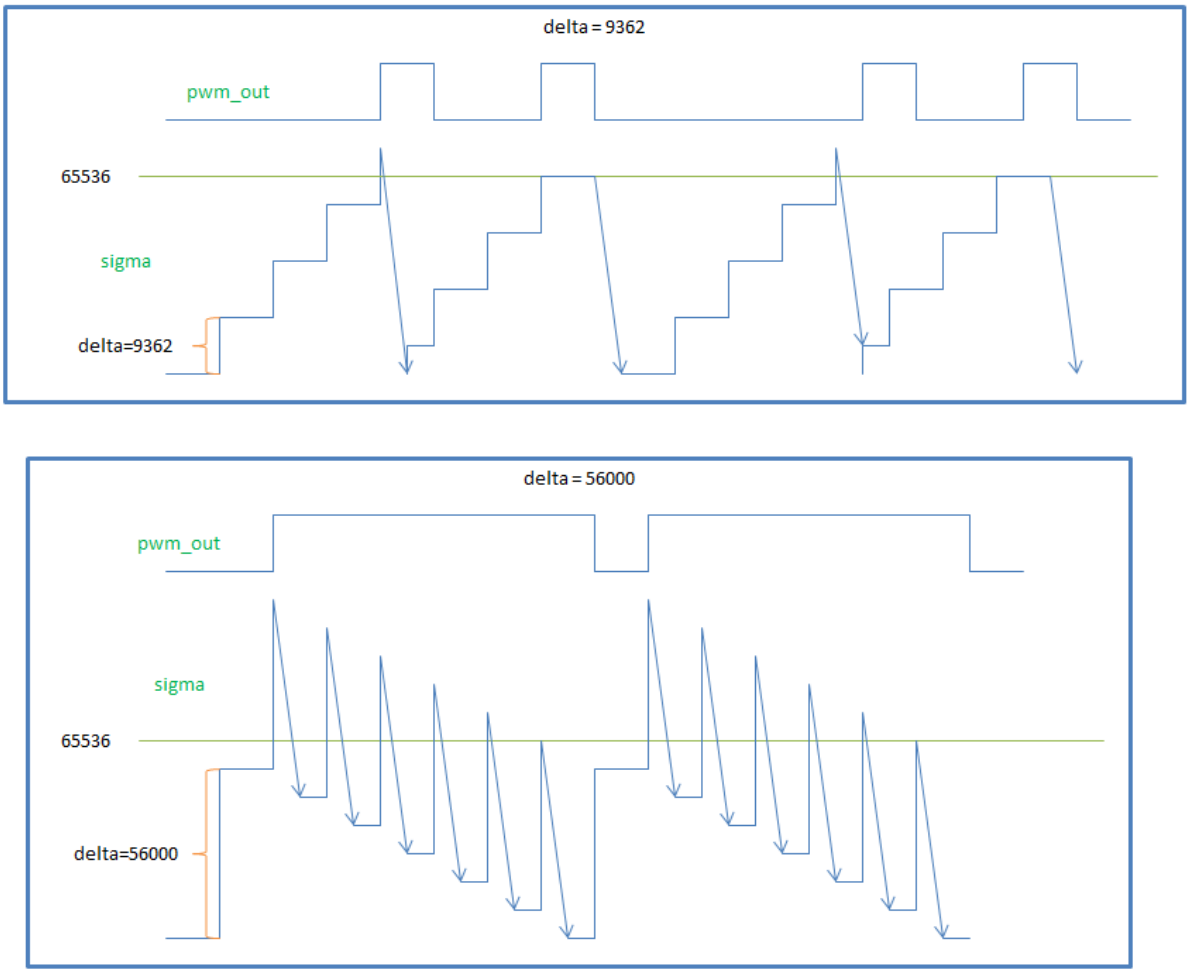


Figure 12-8 High/Low Counter



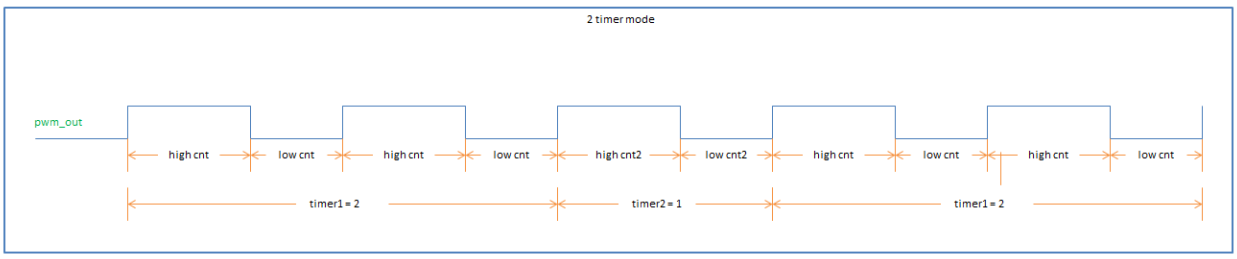
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Figure 12-9 Delta-sigma Conter



PWM 2 timer mode is illustrated as following:

Figure 12-10 2 Timer Mode



12.6.2 Register Description

Each PWM module contains two PWM generators call A and B and controlled by the following registers. For PWM modules in EE domain, the each register's final address =0xffd00000 + offset*4

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Table 12-113 PWM_PWM_A 0x6c00

Bit(s)	R/W	Default	Description
31-15	R/W	0x86	PWM_A_HIGH: This sets the high time (in clock counts) for the PWM_A generator output
15-0	R/W	0x86	PWM_A_LOW: This sets the high time (in clock counts) for the PWM_A generator output

Table 12-114 PWM_PWM_B 0x6c01

Bit(s)	R/W	Default	Description
31-15	R/W	0x50	PWM_B_HIGH: This sets the high time (in clock counts) for the PWM_B generator output
15-0	R/W	0x8	PWM_B_LOW: This sets the high time (in clock counts) for the PWM_B generator output

Table 12-115 PWM_MISC_REG_AB 0x6c02

Bit(s)	R/W	Default	Description
31	R/W	0	pwm_B_hiz when hiz mode, pwm_o will connect to pad_oe, then pad_o will equal this bit
30	R/W	0	pwm_A_hiz when hiz mode, pwm_o will connect to pad_oe, then pad_o will equal this bit
29	R/W	0	pwm_B_constant_en
28	R/W	0	pwm_A_constant_en
27	R/W	0	pwm_B_inv_en
26	R/W	0	pwm_A_inv_en
25	R/W	0	cntl_pwm_a2_en
24	R/W	0	cntl_pwm_b2_en
23	R/W	0	PWM_B_CLK_EN: Set this bit to 1 to enable PWM B clock
22-16	R/W	0	PWM_B_CLK_DIV: Selects the divider (N+1) for the PWM B clock. See the clock tress document
15	R/W	0	PWM_A_CLK_EN: Set this bit to 1 to enable PWM A clock
14-8	R/W	0	PWM_A_CLK_DIV: Selects the divider (N+1) for the PWM A clock. See the clock tress document
7-6	R/W	0	PWM_B_CLK_SEL: Select the clock for the PWM B. See the clock tress document
5-4	R/W	0	PWM_A_CLK_SEL: Select the clock for the PWM A. See the clock tress document
3	R/W	0	DS_B_EN: This bit is only valid if PWM_B_EN is 0: if this bit is set to 1, then the PWM_B output is configured to generate a delta sigma output based on the settings in the register below. If this bit is set to 0, then the PWM_B output is set low.
2	R/W	0	DS_A_EN: This bit is only valid if PWM_A_EN is 0: if this bit is set to 1, then the PWM_A output is configured to generate a delta sigma output based on the settings in the register below. If this bit is set to 0, then the PWM_A output is set low.

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Bit(s)	R/W	Default	Description
1	R/W	0	PWM_B_EN: If this bit is set to 1, then the PWM_B output is configured to generate a PWM output based on the register above. If this bit is 0, then the PWM_B output is controlled by DS_B_EN above.
0	R/W	0	PWM_A_EN: If this bit is set to 1, then the PWM_A output is configured to generate a PWM output based on the register above. If this bit is 0, then the PWM_A output is controlled by DS_A_EN above.

Table 12-116 DS_A_B 0x6c03

Bit(s)	R/W	Default	Description
31-15	R/W	0x8000	DS_B_VAL: This value represents the delta sigma setting for channel B (PWM_B)
15-0	R/W	0x8000	DS_A_VAL: This value represents the delta sigma setting for channel A (PWM_A)

Table 12-117 PWM_TIME_AB 0x6c04

Bit(s)	R/W	Default	Description
31-24	R/W	0xa	A_timer1
23:16	R/W	0xa	A_timer2
15:8	R/W	0xa	B_timer1
7:0	R/W	0xa	B_timer2

Table 12-118 PWM_A2 0x6c05

Bit(s)	R/W	Default	Description
31-15	R/W	0x86	PWM_A2_HIGH: This sets the high time (in clock counts) for the PWM_A2 generator output
15-0	R/W	0x86	PWM_A2_LOW: This sets the high time (in clock counts) for the PWM_A2 generator output

Table 12-119 PWM_B2 0x6c06

Bit(s)	R/W	Default	Description
31-15	R/W	0x50	PWM_B2_HIGH: This sets the high time (in clock counts) for the PWM_B2 generator output
15-0	R/W	0x8	PWM_B2_LOW: This sets the high time (in clock counts) for the PWM_B2 generator output

Table 12-120 PWM_BLINK_AB 0x6c07

Bit(s)	R/W	Default	Description
31-10	R	0	Reserved
9	R/W	0	blink enable for pwm B

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Bit(s)	R/W	Default	Description
8	R/W	0	blink enable for pwm A
7-4	R/W	0	blink times for pwm B
3-0	R/W	0	blink times for pwm A

PWM_PWM_C_D: 0x6800~0x6807

See the registers for PWM A/B

PWM_PWM_E_F: 0x6400~0x6407

See the registers for PWM A/B

AO PWM' clock sources are xtal, clk81, fclk_div3, fclk_div4.

For the following register, each register's final address =0xFF807000 + offset*4

Table 12-121 AO_PWM_PWM_A: PWM_A_DUTY_CYCLE 0x0

Bit(s)	R/W	Default	Description
31-15	R/W	0	PWM_A_HIGH: This sets the high time (in clock counts) for the PWM_A generator output
15-0	R/W	0	PWM_A_LOW: This sets the high time (in clock counts) for the PWM_A generator output

Table 12-122 AO_PWM_PWM_B: PWM_B_DUTY_CYCLE 0x1

Bit(s)	R/W	Default	Description
31-15	R/W	0	PWM_B_HIGH: This sets the high time (in clock counts) for the PWM_B generator output
15-0	R/W	0	PWM_B_LOW: This sets the high time (in clock counts) for the PWM_B generator output

Table 12-123 AO_PWM_MISC_REG_AB: 0x2

Bit(s)	R/W	Default	Description
31	R/W	0	pwm_B_hiz when hiz mode, pwm_o will connect to pad_oe, then pad_o will equal this bit
30	R/W	0	pwm_A_hiz when hiz mode, pwm_o will connect to pad_oe, then pad_o will equal this bit
29	R/W	0	pwm_B_constant_en set this bit to 1, then pwm can support 0%(100%) duty output
28	R/W	0	pwm_A_constant_en set this bit to 1, then pwm can support 0%(100%) duty output
27	R/W	0	pwm_B_inv_en set this bit to 1, pwm output is inverted
26	R/W	0	pwm_A_inv_en set this bit to 1, pwm output is inverted
25	R/W	0	Pwm_a2_en
24	R/W	0	Pwm_b2_en

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Bit(s)	R/W	Default	Description
23	R/W	0	PWM_B_CLK_EN: Set this bit to 1 to enable PWM B clock
22-16	R/W	0	PWM_B_CLK_DIV: Selects the divider (N+1) for the PWM B clock. See the clock tress document
15	R/W	0	PWM_A_CLK_EN: Set this bit to 1 to enable PWM A clock
14-8	R/W	0	PWM_A_CLK_DIV: Selects the divider (N+1) for the PWM A clock. See the clock tress document
7-6	R/W	0	PWM_B_CLK_SEL: Select the clock for the PWM B. See the clock tress document
5-4	R/W	0	PWM_A_CLK_SEL: Select the clock for the PWM A. See the clock tress document
3	R/W	0	DS_B_EN: This bit is only valid if PWM_B_EN is 0: if this bit is set to 1, then the PWM_B output is configured to generate a delta sigma output based on the settings in the register below. If this bit is set to 0, then the PWM_B output is set low.
2	R/W	0	DS_A_EN: This bit is only valid if PWM_A_EN is 0: if this bit is set to 1, then the PWM_A output is configured to generate a delta sigma output based on the settings in the register below. If this bit is set to 0, then the PWM_A output is set low.
1	R/W	0	PWM_B_EN: If this bit is set to 1, then the PWM_B output is configured to generate a PWM output based on the register above. If this bit is 0, then the PWM_B output is controlled by DS_B_EN above.
0	R/W	0	PWM_A_EN: If this bit is set to 1, then the PWM_A output is configured to generate a PWM output based on the register above. If this bit is 0, then the PWM_A output is controlled by DS_A_EN above.

Table 12-124 AO_PWM_DELTA_SIGMA_AB 0x3

Bit(s)	R/W	Default	Description
31-15	R/W	0	DS_B_VAL: This value represents the delta sigma setting for channel B (PWM_B)
15-0	R/W	0	DS_A_VAL: This value represents the delta sigma setting for channel A (PWM_A)

Table 12-125 AO_PWM_TIME_AB 0x4

Bit(s)	R/W	Default	Description
31-24	R/W	0	A1_timer
23:16	R/W	0	A2_timer
15:8	R/W	0	B1_timer
7:0	R/W	0	B2_timer

Table 12-126 AO_PWM_A2 0x5

Bit(s)	R/W	Default	Description
31-15	R/W	0	PWM_A2_HIGH: This sets the high time (in clock counts) for the PWM_A2 generator output
15-0	R/W	0	PWM_A2_LOW: This sets the high time (in clock counts) for the PWM_A2 generator output

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Table 12-127 AO_PWM_B2 0x6

Bit(s)	R/W	Default	Description
31-15	R/W	0	PWM_B2_HIGH: This sets the high time (in clock counts) for the PWM_B 2 generator output
15-0	R/W	0	PWM_B2_LOW: This sets the high time (in clock counts) for the PWM_B 2 generator output

Table 12-128 AO_PWM_BLINK_AB 0x7

Bit(s)	R/W	Default	Description
31-10	R	0	Reserved
9	R/W	0	blink enable for pwm B
8	R/W	0	blink enable for pwm A
7-4	R/W	0	blink times for pwm B
3-0	R/W	0	blink times for pwm A

For the following register, each register's final address= 0xFF802000 + offset * 4

Table 12-129 AO_PWM_PWM_C: PWM_C_DUTY_CYCLE 0x0

Bit(s)	R/W	Default	Description
31-15	R/W	0	PWM_C_HIGH: This sets the high time (in clock counts) for the PWM_C generator output
15-0	R/W	0	PWM_C_LOW: This sets the high time (in clock counts) for the PWM_C generator output

Note

This is a new module to Nike. It replaces the older delta sigma (PWM like) generator in the HIU. This module allows the software to select either a PWM or delta-sigma output using the same module. There are two outputs: PWM_A and PWM_B. Either of these can be programmed to be PWM outputs or delta sigma outputs.

Table 12-130 AO_PWM_PWM_D: PWM_D_DUTY_CYCLE 0x1

Bit(s)	R/W	Default	Description
31-15	R/W	0	PWM_D_HIGH: This sets the high time (in clock counts) for the PWM_D generator output
15-0	R/W	0	PWM_D_LOW: This sets the high time (in clock counts) for the PWM_D generator output

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Table 12-131 AO_PWM_MISC_REG_CD: 0x2

Bit(s)	R/W	Default	Description
31	R/W	0	pwm_D_hiz when hiz mode, pwm_o will connect to pad_oe, then pad_o will equal this bit
30	R/W	0	pwm_C_hiz when hiz mode, pwm_o will connect to pad_oe, then pad_o will equal this bit
29	R/W	0	pwm_D_constant_en set this bit to 1, then pwm can support 0%(100%) duty output
28	R/W	0	pwm_C_constant_en set this bit to 1, then pwm can support 0%(100%) duty output
27	R/W	0	pwm_D_inv_en set this bit to 1, pwm output is inverted
26	R/W	0	pwm_C_inv_en set this bit to 1, pwm output is inverted
25	R/W	0	Pwm_C2_en
24	R/W	0	Pwm_D2_en
23	R/W	0	PWM_D_CLK_EN: Set this bit to 1 to enable PWM D clock
22-16	R/W	0	PWM_D_CLK_DIV: Selects the divider (N+1) for the PWM D clock. See the clock tress document
15	R/W	0	PWM_C_CLK_EN: Set this bit to 1 to enable PWM C clock
14-8	R/W	0	PWM_C_CLK_DIV: Selects the divider (N+1) for the PWM C clock. See the clock tress document
7-6	R/W	0	PWM_D_CLK_SEL: Select the clock for the PWM D. See the clock tress document
5-4	R/W	0	PWM_C_CLK_SEL: Select the clock for the PWM C. See the clock tress document
3	R/W	0	DS_D_EN: This bit is only valid if PWM_D_EN is 0: if this bit is set to 1, then the PWM_D output is configured to generate a delta sigma output based on the settings in the register below. If this bit is set to 0, then the PWM_D output is set low.
2	R/W	0	DS_C_EN: This bit is only valid if PWM_C_EN is 0: if this bit is set to 1, then the PWM_C output is configured to generate a delta sigma output based on the settings in the register below. If this bit is set to 0, then the PWM_C output is set low.
1	R/W	0	PWM_D_EN: If this bit is set to 1, then the PWM_D output is configured to generate a PWM output based on the register above. If this bit is 0, then the PWM_D output is controlled by DS_D_EN above.
0	R/W	0	PWM_C_EN: If this bit is set to 1, then the PWM_C output is configured to generate a PWM output based on the register above. If this bit is 0, then the PWM_C output is controlled by DS_C_EN above.

Table 12-132 AO_PWM_DELTA_SIGMA_CD 0x3

Bit(s)	R/W	Default	Description
31-15	R/W	0	DS_D_VAL: This value represents the delta sigma setting for channel D (PWM_D)
15-0	R/W	0	DS_C_VAL: This value represents the delta sigma setting for channel C (PWM_C)

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Table 12-133 AO_PWM_TIME_CD 0x4

Bit(s)	R/W	Default	Description
31-24	R/W	0	C1_timer
23:16	R/W	0	C2_timer
15:8	R/W	0	D1_timer
7:0	R/W	0	D2_timer

Table 12-134 AO_PWM_C2 0x5

Bit(s)	R/W	Default	Description
31-15	R/W	0	PWM_C2_HIGH: This sets the high time (in clock counts) for the PWM_C generator output
15-0	R/W	0	PWM_C2_LOW: This sets the high time (in clock counts) for the PWM_C generator output

Table 12-135 AO_PWM_D2 0x6

Bit(s)	R/W	Default	Description
31-15	R/W	0	PWM_D2_HIGH: This sets the high time (in clock counts) for the PWM_D generator output
15-0	R/W	0	PWM_D2_LOW: This sets the high time (in clock counts) for the PWM_D generator output

Table 12-136 AO_PWM_BLINK_CD 0x7

Bit(s)	R/W	Default	Description
31-10	R	0	Reserved
9	R/W	0	blink enable for pwm D
8	R/W	0	blink enable for pwm C
7-4	R/W	0	blink times for pwm D
3-0	R/W	0	blink times for pwm C

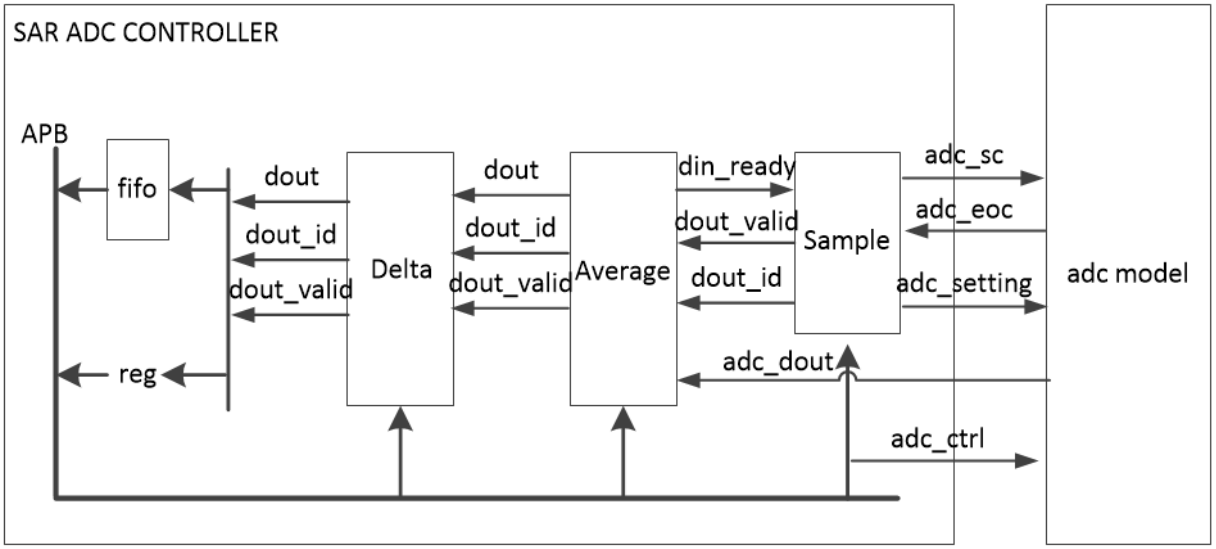
12.7 SAR ADC

12.7.1 Overview

The SAR ADC is a general purpose ADC for measuring analog signals. The module can make RAW ADC measurements or average a number of measurements to introduce filtering. The SAR ADC is a single block so an analog mux is placed in front to allow multiple different measurements to be made sequentially. Timing of the samples, and delays between muxing are all programmable as is the averaging to be applied to the SAR ADC.

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Figure 12-11 SAR ADC Controller



Manual Sample Mode

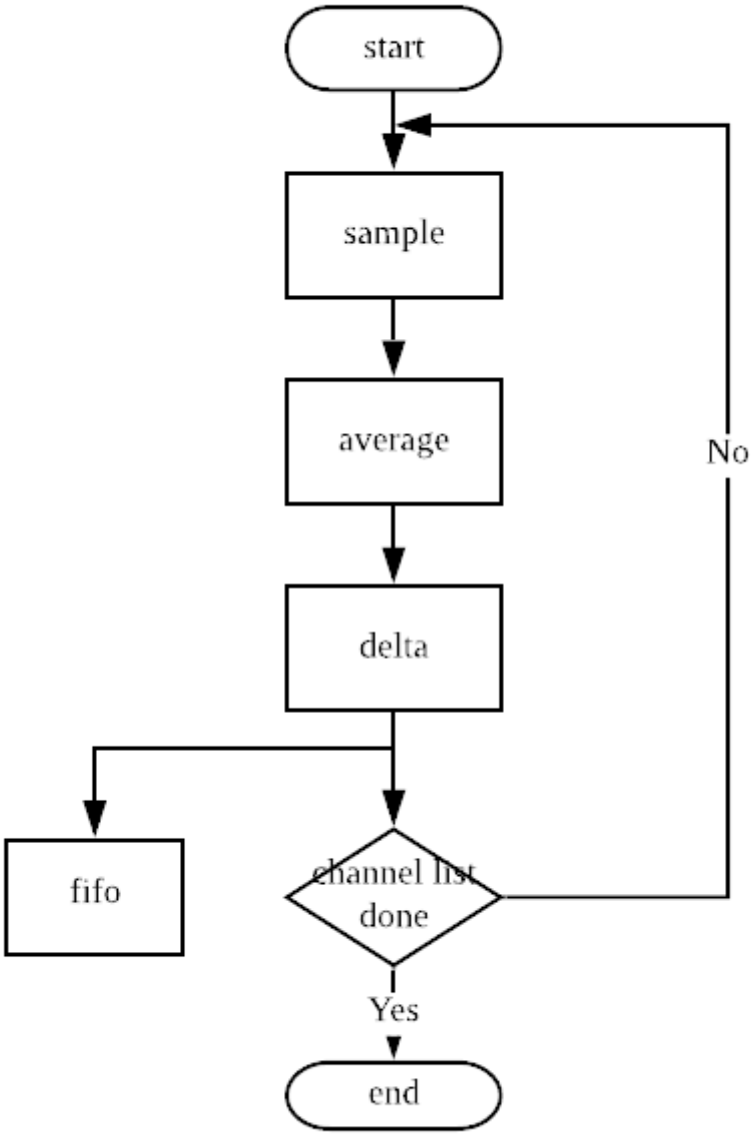
In this mode, the control signals (adc_sc, adc_setting, adc_ctrl, adc_clk) are configured by registers directly. The sample data will be registered directly and cannot be processed by average and delta.

Batch Sample Mode

The sar adc controller will process the sample data according to channel list which is configurable, and write the sample data into FIFO. The depth of the FIFO is 16. There are 8 channels in the channel list.

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Figure 12-12 SAR ADC Controller

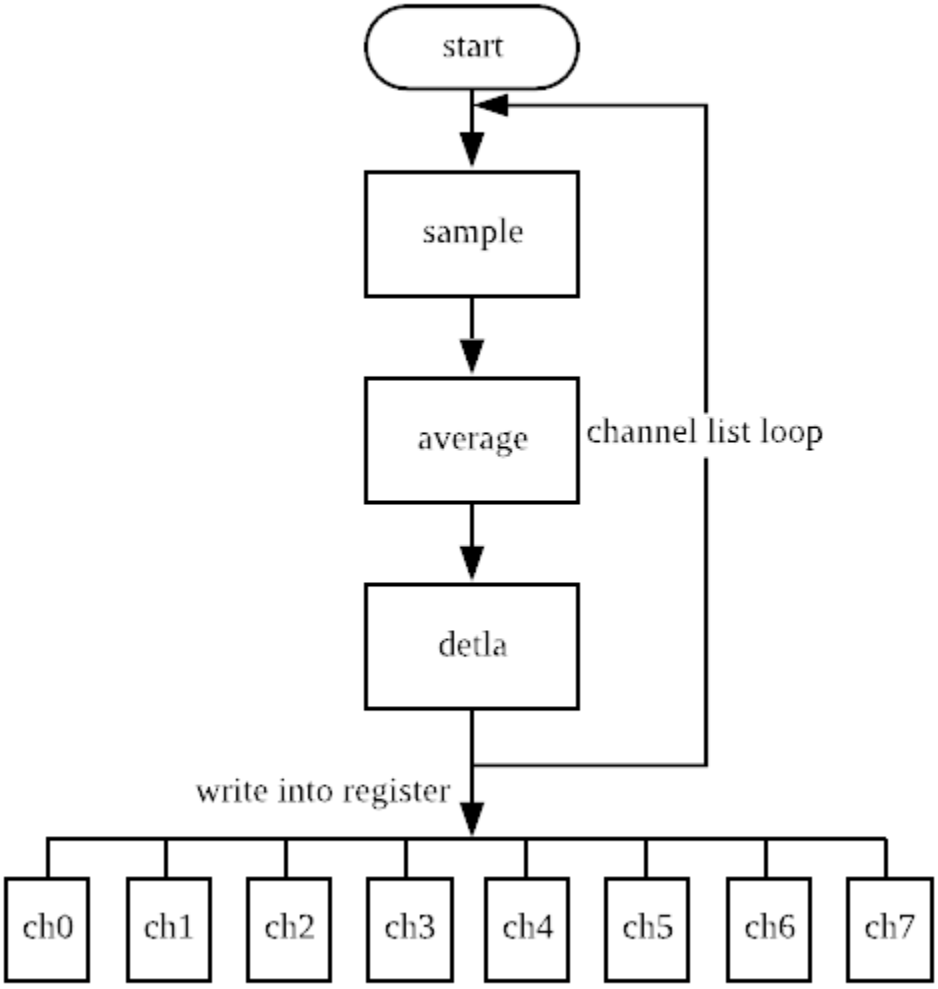


Periodic Sample Mode

The sar adc controller will process the sample data according to channel list periodically, and write the sample data into each register of the corresponding channel. Only the latest sample data is present in the register, previous samples are overridden.

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Figure 12-13 SAR ADC Controller



12.7.2 Register Description

Each register final address = 0xff809000 + offset * 4

Table 12-137 AO_SAR_ADC_REG0: Control Register #0 0x0

Bit(s)	R/W	Default	Description
31	R	0	PANEL DETECT level.
30	R	0	DELTA_BUSY: If This bit is 1, then it indicates the delta processing engine is busy
29	R	0	AVG_BUSY: If This bit is 1, then it indicates the averaging engine is busy
28	R	0	SAMPLE_BUSY: If This bit is 1, then it indicates the sampling engine is busy
27	R	0	FIFO_FULL:
26	R	0	FIFO_EMPTY:

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Bit(s)	R/W	Default	Description
25-21	R/W	4	FIFO_COUNT: Current count of samples in the acquisition FIFO
20-19	R/W	0	ADC_BIAS_CTRL
18-16	R/W	0	CURR_CHAN_ID: These Bits represent the current channel (0..7) that is being sampled.
15	R	0	Unused
14	R/W	0	SAMPLING_STOP: This bit can be used to cleanly stop the sampling process in the event that continuous sampling is enabled. To stop sampling, simply set This bit and wait for all processing modules to no longer indicate that they are busy.
13-12	R/W	0	CHAN_DELTA_EN: There are two Bits corresponding to Channels 0 and 1. Channel 0 and channel 1 can be individually enabled to take advantage of the delta processing module.
11	R/W	0	Unused
10	R/W	0	DETECT_IRQ_POL: This bit sets the polarity of the detect signal. The detect signal is used during X/Y panel applications to detect if the panel is touched
9	R/W	0	DETECT_IRQ_EN: If This bit is set to 1, then an interrupt will be generated if the DETECT signal is low/high. The polarity is set in the bit above.
8-4	R/W	0	FIFO_CNT_IRQ: When the FIFO contains N samples, then generate an interrupt (if bit 3 is set below).
3	R/W	0	FIFO_IRQ_EN: Set This bit to 1 to enable an IRQ when the acquisition FIFO reaches a certain level.
2	W	0	SAMPLE_START: This bit should be written to 1 to start sampling.
1	R/W	0	CONTINUOUS_EN: If This bit is set to 1, then the channel list will be continually processed
0	R/W	0	SAMPLING_ENABLE: Setting This bit to ‘1’ enables the touch panel controller sampling engine, averaging module, XY processing engine and the FIFO.

Table 12-138 AO_SAR_ADC_CHAN_LIST:Channel List 0x1

Bit(s)	R/W	Default	Description
31-27	R/W	0	unused
26-24	R/W	2	Length of the list of channels to process. If this value is 2, then only channels in Bits [8:0] below are processed.
23-21	R/W	7	8th channel
20-18	R/W	6	7th channel
17-15	R/W	5	6th channel
14-12	R/W	4	5th channel
11-9	R/W	3	4th channel
8-6	R/W	2	3rd channel
5-3	R/W	1	2nd channel
2-0	R/W	0	First channel in the list of channels to process

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Table 12-139 AO_SAR_ADC_AVG_CNTL:Sampling/Averaging Modes 0x2

Bit(s)	R/W	Default	Description
31-30	R/W	0	Channel 7: Averaging mode: 0 = no averaging, 1 = simple averaging, 2 = median averaging.
29-28	R/W	0	Channel 6: Averaging mode: 0 = no averaging, 1 = simple averaging, 2 = median averaging.
27-26	R/W	0	Channel 5: Averaging mode: 0 = no averaging, 1 = simple averaging, 2 = median averaging.
25-24	R/W	0	Channel 4: Averaging mode: 0 = no averaging, 1 = simple averaging, 2 = median averaging.
23-22	R/W	0	Channel 3: Averaging mode: 0 = no averaging, 1 = simple averaging, 2 = median averaging.
21-20	R/W	0	Channel 2: Averaging mode: 0 = no averaging, 1 = simple averaging, 2 = median averaging.
19-18	R/W	0	Channel 1: Averaging mode: 0 = no averaging, 1 = simple averaging, 2 = median averaging.
17-16	R/W	0	Channel 0: Averaging mode: 0 = no averaging. 1 = simple averaging of the number of samples acquired (1,2,4 or 8). 2 = median averaging. NOTE: If these Bits are set to 2, then you must set the number of samples to acquire below to 8.
15-13	R/W	0	Channel 7: Number of samples to acquire 2N:
13-12	R/W	0	Channel 6: Number of samples to acquire 2N:
11-10	R/W	0	Channel 5: Number of samples to acquire 2N:
9-8	R/W	0	Channel 4: Number of samples to acquire 2N:
7-6	R/W	0	Channel 3: Number of samples to acquire 2N:
5-4	R/W	0	Channel 2: Number of samples to acquire 2N:
3-2	R/W	0	Channel 1: Number of samples to acquire 2N:
1-0	R/W	0	Channel 0: Number of samples to acquire 2N: 0 = 1, 1 = 2, 2 = 4, 4 = 8.

Each channel listed in the CHANNEL_LIST is given independent control of the number of samples to acquire and averaging mode.

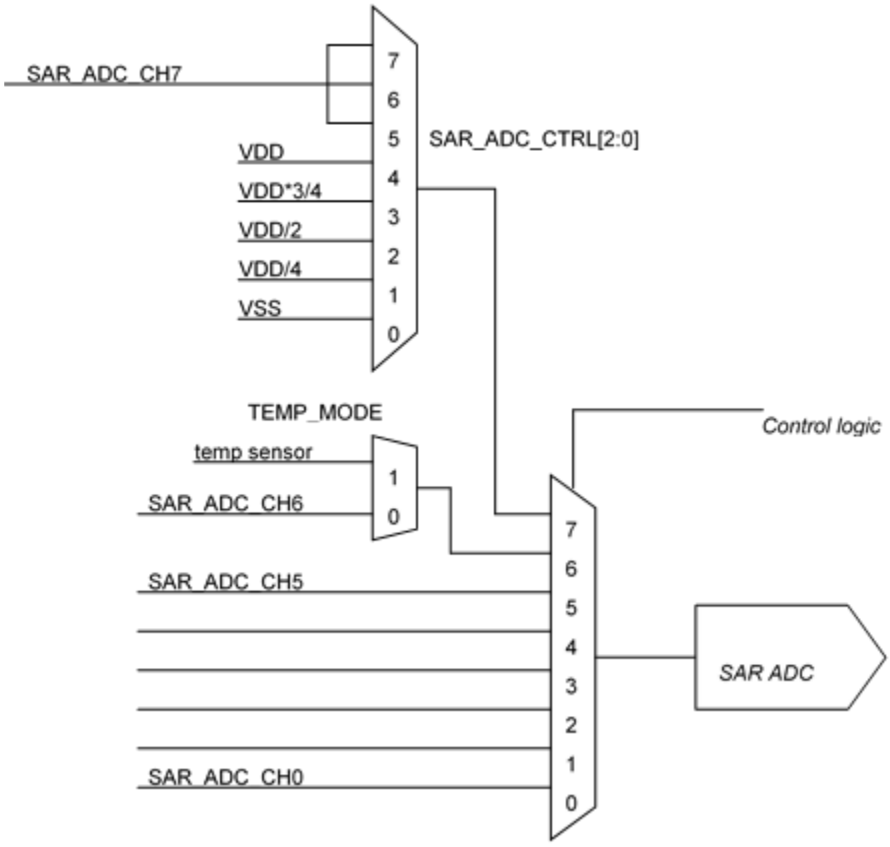
Table 12-140 AO_SAR_ADC_REG3: Control Register #3 0x3

Bit(s)	R/W	Default	Description
31	R/W	0	CNTL_USE_SC_DLY: hold time delay was added to the start conversion clock. Unfortunately, it appears that the analog ADC design requires that we use the inverted clock so This bit is meaningless.
30	R/W	0	SAR ADC_CLK_EN: 1 = enable the SAR ADC clock
29	R/W	0	reserved
28	R/W	0	reserved
27	R/W	0	SARADC_CTRL[4]: is used to control the internal ring counter. 1 = enable the continuous ring counter. 0 = disable
26	R/W	0	SARADC_CTRL[3]: used to select the internal sampling clock phase
25~23	R/W	0	SARADC_CTRL[2:0]: 000 ssa 001 vdda/4

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Bit(s)	R/W	Default	Description
			010 vdda/2 011 vdda*3/4 100 vdda 101, 110, 111 unused
22	R/W	0	DETECT_EN: This bit controls the analog switch that connects a 50k resistor to the X+ signal. Setting This bit to 1 closes the analog switch
21	R/W	0	ADC_EN: Set This bit to 1 to enable the ADC
20-18	R/W	2	PANEL_DETECT_COUNT: Increasing this value increases the filtering on the panel detect signal using the timebase settings in Bits [17:16] below.
17-16	R/W	0	PANEL_DETECT_FILTER_TB: 0 = count 1uS ticks, 1 = count 10uS ticks, 2 = count 100uS ticks. 3 = count 1mS ticks
15-10	R/W	20	ADC_CLK_DIV: The ADC clock is derived by dividing the 27Mhz crystal by N+1. This value divides the 27Mhz clock to generate an ADC clock. A value of 20 for example divides the 27Mhz clock by 21 to generate an equivalent 1.28Mhz clock.
9-8	R/W	1	BLOCK_DLY_SEL: 0 = count 1uS ticks, 1 = count 10uS ticks, 2 = count 100uS ticks. 3 = count 1mS ticks
7-0	R/W	10	BLOCK_DLY: After all channels in the CHANNEL_LIST have been processed, the sampling engine will delay for an amount of time before re-processing the CHANNEL_LIST again. Combined with Bits [9:8] above, this value is used to generate a delay between processing blocks of channels.

Figure 12-14 AO_SAR_ADC_REG3



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As the CHANNEL_LIST is process, the input switches are set according to the requirements of the channel. After setting the switches there is a programmable delay before sampling begins. Additionally, each channel specifies the number of samples for that particular channel. The sampling rate is programmed below.

Table 12-141 AO_SAR_ADC_DELAY:INPUT / SAMPLING DELAY 0x4

Bit(s)	R/W	Default	Description
31-29	R	0	unused
28	R/W	0	CNTL_EOC_BY_CNT: ADC dout valid controlled by counter
27	R/W	0	CNTL_USE_LATCHED_DATA: ADC dout be latched first, then be sampled
26	R	0	unused
25-24	R/W	0	INPUT_DLY_SEL: 0 = 111nS ticks, 1 = count 1uS ticks, 2 = count 10uS ticks, 3 = count 100uS ticks
23-16	R/W	3	INPUT_DLY_CNY: For channels that acquire 2,4 or 8 samples, the delay between two samples is controlled by this count (N+1) combined with the delay selection in the two bits above.
15-10	R/W	14	CNTL_EOC_DLY_CNT: the delay between SC and ADC output data ready for latch/sample
9-8	R/W	0	SAMPLE_DLY_SEL: 0 = count 1uS ticks, 1 = count 10uS ticks, 2 = count 100uS ticks. 3 = count 1mS ticks
7-0	R/W	9	SAMPLE_DLY_CNY: For channels that acquire 2,4 or 8 samples, the delay between two samples is controlled by this count (N+1) combined with the delay selection in the two bits above.

For channel 0 and channel 1, (the special X/Y channels) the last sample pushed into the FIFO for each channel is saved in a register. This allows the software to see the last sample for channel 0 and channel 1 even when the FIFO overflows. For example, if we are sampling quickly and there is a gesture on the screen, we can use the contents of the FIFO to see the direction of the gesture and use the last sample values to see where the pen finally came to rest.

Table 12-142 AO_SAR_ADC_LAST_RD: Last Sample 0x5

Bit(s)	R/W	Default	Description
31-24	R	0	unused
23-16	R	0	LAST_CHANNEL1
15-10	R	0	unused
9-0	R	0	LAST_CHANNEL0

Table 12-143 AO_SAR_ADC_FIFO_RD: Control Register #6 (FIFO RD) 0x6

Bit(s)	R/W	Default	Description
31-16	R	0	Unused
15	R	0	Unused
14-12	R	0	Channel ID. This value identifies the channel associated with the data in Bits [9:0] below
11-0	R	0	Sample value: 12-bit raw or averaged ADC sample written to the FIFO.

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Channels 2 ~ 7 can program the ADC input mux to any selection between 0 and 7. This register allows the software to associate a mux selection with a particular channel. In addition to the ADC mux, there are a number of switches that can be set in any particular state. Channels 2 ~ 7 share a common switch setting. Channels 0 and 1 on the other hand have programmable switch settings (see other registers below).

Table 12-144 AO_SAR_ADC_AUX_SW:Channel 2~7 ADC MUX, Switch Controls 0x7

Bit(s)	R/W	Default	Description
31	R	0	unused
30	R/W	0	CNTL_MANUAL_SC
29	R/W	0	CNTL_MANUAL_CLK
28	R/W	0	CNTL_MANUAL_MODE
25-23	R/W	7	Channel 7 ADC_MUX setting when channel 7 is being measured.
22-20	R/W	7	Channel 6 ADC_MUX setting when channel 6 is being measured.
19-17	R/W	7	Channel 5 ADC_MUX setting when channel 5 is being measured.
16-14	R/W	6	Channel 4 ADC_MUX setting when channel 4 is being measured.
13-11	R/W	0	Channel 3 ADC_MUX setting when channel 3 is being measured.
10-8	R/W	1	Channel 2 ADC_MUX setting when channel 2 is being measured.
7	R	0	unused
6	R/W	0	VREF_P_MUX setting when channel 2,3..7 is being measured
5	R/W	0	VREF_N_MUX setting when channel 2,3..7 is being measured
4	R/W	0	MODE_SEL setting when channel 2,3..7 is being measured
3	R/W	1	YP_DRIVE_SW setting when channel 2,3..7 is being measured
2	R/W	1	XP_DRIVE_SW setting when channel 2,3..7 is being measured
1	R/W	0	YN_DRIVE_SW setting when channel 2,3..7 is being measured
0	R/W	0	XN_DRIVE_SW setting when channel 2,3..7 is being measured

Channels 0 and 1 have independent programmable switch settings when either/both of these channels are being measured.

Table 12-145 AO_SAR_ADC_CHAN_10_SW:Channel 0, 1 ADC MUX, Switch Controls 0x8

Bit(s)	R/W	Default	Description
31-26	R	0	unused
25-23	R/W	2	Channel 1 ADC MUX setting
22	R/W	0	Channel 1 VREF_P_MUX
21	R/W	0	Channel 1 VREF_N_MUX
20	R/W	0	Channel 1 MODE_SEL
19	R/W	1	Channel 1 YP_DRIVE_SW setting: 0: TADC_CH6N = 3.3v 1: TADC_CH6N = floating
18	R/W	1	Channel 1 XP_DRIVE_SW setting: 0: TADC_CH6N = 3.3v

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Bit(s)	R/W	Default	Description
			1: TADC_CH6N = floating
17	R/W	0	Channel 1 YN_DRIVE_SW setting: 0: TADC_CH4N = floating 1: TADC_CH4N = GND
16	R/W	0	Channel 1 XN_DRIVE_SW setting: 0: TADC_CH4N = floating 1: TADC_CH4N = GND
15-10	R		unused
9-7	R/W	3	Channel 0 ADC MUX setting
6	R/W	0	Channel 0 VREF_P_MUX
5	R/W	0	Channel 0 VREF_N_MUX
4	R/W	0	Channel 0 MODE_SEL
3	R/W	1	Channel 0 YP_DRIVE_SW setting: 0: TADC_CH6N = 3.3v 1: TADC_CH6N = floating
2	R/W	1	Channel 0 XP_DRIVE_SW setting: 0: TADC_CH6N = 3.3v 1: TADC_CH6N = floating
1	R/W	0	Channel 0 YN_DRIVE_SW setting: 0: TADC_CH4N = floating 1: TADC_CH4N = GND
0	R/W	0	Channel 0 XN_DRIVE_SW setting: 0: TADC_CH4N = floating 1: TADC_CH4N = GND

IDLE MODE: When nothing is being measured, the switches should be put into a safe state. This safe state is accomplished using Bits [9:0] below.

DETECT MODE: When bit [26] is set, the input muxes / switches are configured according to the Bits below. Typically the software configures the switches below to correspond to the detect touch mode. That is, Y- internal MOSFET is closed so that the Y plane of the touch screen is connected to Ground. Additionally, the DETECT_EN bit (different register) set to 1 so that the 50k resistor to VDD is connected to X+. In this configuration, the detect comparator connected to the 50k resistor will be weakly pulled up to VDD through the 50k resistor. If the user touches the screen, the X and Y planes of the touch screen will contact causing the X+ signal to be pulled to ground.

Table 12-146 AO_SAR_ADC_DETECT_IDLE_SW:DETECT / IDLE Mode switches 0x9

Bit(s)	R/W	Default	Description
31-27	R	0	unused
26	R/W	0	DETECT_SW_EN: If This bit is set, then Bits [25:16] below are applied to the analog muxes/switches of the touch panel controller.
25-23	R/W	5	DETECT MODE ADC MUX setting
22	R/W	0	DETECT MODE VREF_P_MUX setting
21	R/W	0	DETECT MODE VREF_N_MUX setting
20	R/W	0	DETECT MODE MODE_SEL setting
19	R/W	1	DETECT MODE YP_DRIVE_SW setting: 0: TADC_CH6N = 3.3v

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Bit(s)	R/W	Default	Description
			1: TADC_CH6N = floating
18	R/W	1	DETECT MODE XP_DRIVE_SW setting: 0: TADC_CH6N = 3.3v 1: TADC_CH6N = floating
17	R/W	0	DETECT MODE YM_DRIVE_SW setting: 0: TADC_CH4N = floating 1: TADC_CH4N = GND
16	R/W	0	DETECT MODE YM_DRIVE_SW setting: 0: TADC_CH4N = floating 1: TADC_CH4N = GND
15-10	R		Unused
9-7	R/W	5	IDLE MODE ADC MUX setting
6	R/W	0	IDLE MODE VREF_P_MUX setting
5	R/W	0	IDLE MODE VREF_N_MUX setting
4	R/W	0	IDLE MODE MODE_SEL setting
3	R/W	1	IDLE MODE YP_DRIVE_SW setting: 0: TADC_CH6N = 3.3v 1: TADC_CH6N = floating
2	R/W	1	IDLE MODE XP_DRIVE_SW setting: 0: TADC_CH6N = 3.3v 1: TADC_CH6N = floating
1	R/W	0	IDLE MODE YN_DRIVE_SW setting: 0: TADC_CH4N = floating 1: TADC_CH4N = GND
0	R/W	0	IDLE MODE XN_DRIVE_SW setting: 0: TADC_CH4N = floating 1: TADC_CH4N = GND

Table 12-147 AO_SAR_ADC_DELTA_10:Delta Mode Deltas 0xa

Bit(s)	R/W	Default	Description
31-28	R	0	unused
27-16	R/W	0	Channel 1 delta value when delta processing for channel 1 is enabled.
15-12	R/W		unused
11-0	R/W	0	Channel 0 delta value when delta processing for channel 0 is enabled.

Table 12-148 AO_SAR_ADC_REG11: 0xb

Bit(s)	R/W	Default	Description
31	R/W	0	FIFO_DISABLE
30	R/W	0	PERIOD_SAMPLE_EN
29-13	R/W	0	ts_cntl_int
12-0	R/W	0	sar_bg_cntl

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Table 12-149 AO_SAR_ADC_REG12: 0xc

Bit(s)	R/W	Default	Description
31-0	R/W	0	reserved

Table 12-150 AO_SAR_ADC_REG13: 0xd

Bit(s)	R/W	Default	Description
15-8	R/W	0	SARADC_RSV2
7-0	R/W	0	reserved

Table 12-151 AO_SAR_ADC_CHNL01: 0xe

Bit(s)	R/W	Default	Description
31	R/W	0	dout_chnl1_valid
30-28	R/W	0	dout_chnl1_id
27-16	R/W	0	dout_chnl1
15	R/W	0	dout_chnl0_valid
14-12	R/W	0	dout_chnl0_id
11-0	R/W	0	dout_chnl0

Table 12-152 AO_SAR_ADC_CHNL23: 0xf

Bit(s)	R/W	Default	Description
31	R/W	0	dout_chnl3_valid
30-28	R/W	0	dout_chnl3_id
27-16	R/W	0	dout_chnl3
15	R/W	0	dout_chnl2_valid
14-12	R/W	0	dout_chnl2_id
11-0	R/W	0	dout_chnl2

Table 12-153 AO_SAR_ADC_CHNL45: 0x10

Bit(s)	R/W	Default	Description
31	R/W	0	dout_chnl5_valid
30-28	R/W	0	dout_chnl5_id
27-16	R/W	0	dout_chnl5
15	R/W	0	dout_chnl4_valid
14-12	R/W	0	dout_chnl4_id
11-0	R/W	0	dout_chnl4

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Table 12-154 AO_SAR_ADC_CHNL67: 0x11

Bit(s)	R/W	Default	Description
31	R/W	0	dout_chnl7_valid
30-28	R/W	0	dout_chnl7_id
27-16	R/W	0	dout_chnl7
15	R/W	0	dout_chnl6_valid
14-12	R/W	0	dout_chnl6_id
11-0	R/W	0	dout_chnl6

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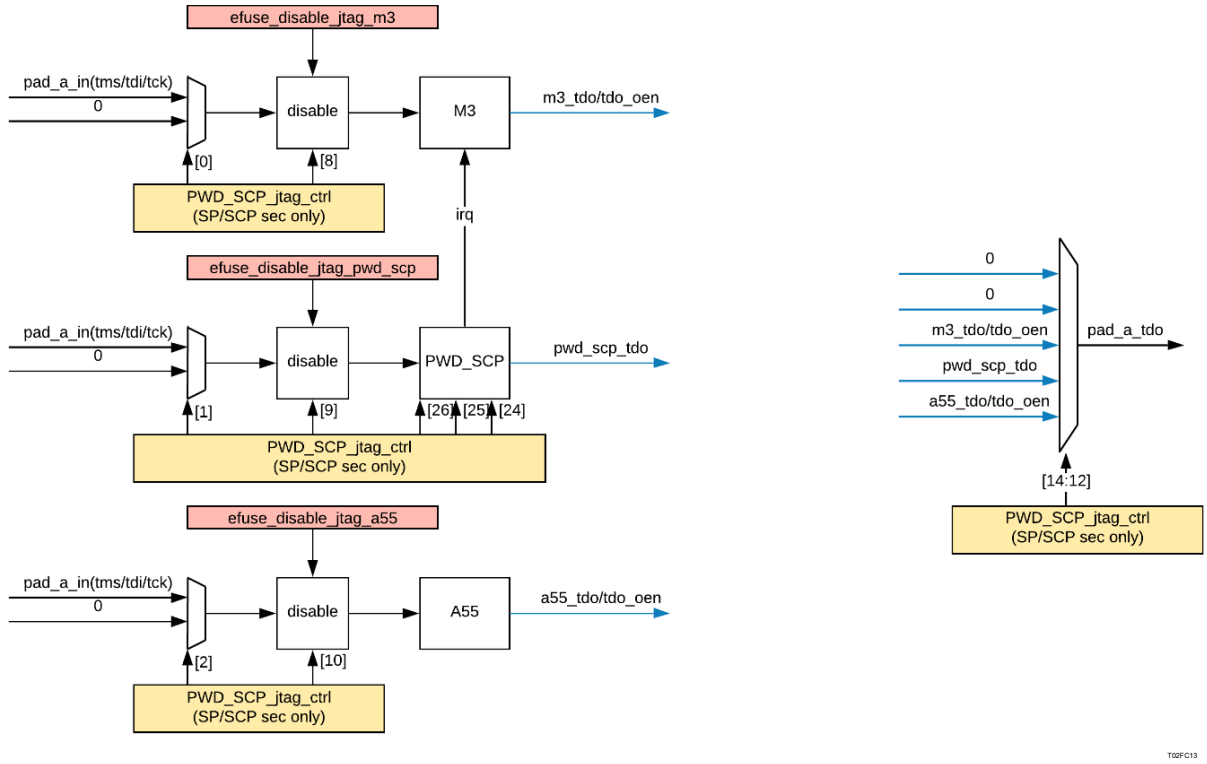
13 System Interface

13.1 JTAG

13.1.1 Overview

JTAG is an interface for internal test. The structure of the JTAG module is shown in the following diagram.

Figure 13-1 JTAG Structure



13.1.2 Register Description

Base address: 0xFF800000

Register address: 0xFF800000 + offset * 4

reg name	address	write sec id	read sec id
AO_SEC_JTAG_SP_CTRL	0xac	111	111
AO_SEC_JTAG_PWD_SP_0	0xad	no	111
AO_SEC_JTAG_PWD_SP_1	0xae	no	111
AO_SEC_JTAG_PWD_SP_2	0xaf	no	111
AO_SEC_JTAG_PWD_SP_3	0xb0	no	111
AO_SEC_JTAG_PWD_SP_CNTL	0xb1	111	111

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reg name	address	write sec id	read sec id
AO_SEC_JTAG_PWD_SP_ADDR0	0xb2	111	111
AO_SEC_JTAG_PWD_SP_ADDR1	0xb3	111	111
AO_SEC_JTAG_PWD_SP_ADDR2	0xb4	111	111
AO_SEC_JTAG_PWD_SP_ADDR3	0xb5	111	111
AO_SEC_JTAG_SCP_CTRL	0xb6	x11	x11
AO_SEC_JTAG_PWD_SCP_0	0xb7	no	x11
AO_SEC_JTAG_PWD_SCP_1	0xb8	no	x11
AO_SEC_JTAG_PWD_SCP_2	0xb9	no	x11
AO_SEC_JTAG_PWD_SCP_3	0xba	no	x11
AO_SEC_JTAG_PWD_SCP_CNTL	0xbb	x11	x11
AO_SEC_JTAG_PWD_SCP_ADDR0	0xbc	x11	x11
AO_SEC_JTAG_PWD_SCP_ADDR1	0xbd	x11	x11
AO_SEC_JTAG_PWD_SCP_ADDR2	0xbe	x11	x11
AO_SEC_JTAG_PWD_SCP_ADDR3	0xbf	x11	x11

Table 13-1 AO_SEC_JTAG_SP_CNTL 0xAC

Bit(s)	R/W	Default	Description
31	R	0	jrag_pwd_sp_data_valid; 1: finished write jtag_pwd_addr3;
30	R	0	jrag_pwd_sp_data_timeout; 1: jtag_pwd_sp_wd count greater threshold (time out);
26	R/W	0	jtag_pwd_sp_clr; 1: clear valid and irq status;
25	R/W	0	jtag_pwd_sp_wd_resetn; 0: reset pwd_sp_wd, then no irq;
24	R/W	0	jtag_pwd_sp_en; 0: disable jtag_pwd_sp in/out; 1: enable jtag_pwd_sp in/out;
9	R/W	0	jtag_disable_pwd_sp_in; 0: enable; 1: disable pwd_sp_jtag_in;
8	R/W	0	jtag_disable_m4_in; 0: enable; 1: disable m4_jtag_in;
4	R/W	0	jtag_sel_force_m4_tdo_en_low; 0: tdo_en = m4_jtag_tdo_en; 1: tdo_en = 1'b0 (always output);
1	R/W	0	jtag_sel_pwd_sp_in; 0: pad_a; 1: pad_b;
0	R/W	0	jtag_sel_m4_in; 0: pad_a; 1: pad_b;

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Table 13-2 AO_SEC_JTAG_PWD_SP_0 0xAD

Bit(s)	R/W	Default	Description
31-0	R	0	jtag_pwd_sp received pwd[31:0]

Table 13-3 AO_SEC_JTAG_PWD_SP_1 0xAE

Bit(s)	R/W	Default	Description
31-0	R	0	jtag_pwd_sp received pwd[63:32]

Table 13-4 AO_SEC_JTAG_PWD_SP_2 0xAF

Bit(s)	R/W	Default	Description
31-0	R	0	jtag_pwd_sp received pwd[95:64]

Table 13-5 AO_SEC_JTAG_PWD_SP_3 0xB0

Bit(s)	R/W	Default	Description
31-0	R	0	jtag_pwd_sp received pwd[127:96]

Table 13-6 AO_SEC_JTAG_PWD_SP_CNTL 0xB1

Bit(s)	R/W	Default	Description
31:5	R/W	0	timeout threshold;
4	R/W	0	activity by tdo_en; 1: detect tdo activity and use it reset watchdog count;
3	R/W	0	activity by tck_en; 1: detect tck activity and use it reset watchdog count;
2	R/W	0	activity by tms_en; 1: detect tms activity and use it reset watchdog count;
1	R/W	0	activity by tdi_en; 1: detect tdi activity and use it reset watchdog count;
0	R/W	0	wd_timeout_en; 0: disable watchdog timeout; 1: enable watchdog timeout;

Table 13-7 AO_SEC_JTAG_PWD_SP_ADDR0 0xB2

Bit(s)	R/W	Default	Description
31-0	R/W	0	addr0

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Table 13-8 AO_SEC_JTAG_PWD_SP_ADDR1 0xB3

Bit(s)	R/W	Default	Description
31-0	R/W	0	addr1

Table 13-9 AO_SEC_JTAG_PWD_SP_ADDR2 0xB4

Bit(s)	R/W	Default	Description
31-0	R/W	0	addr2

Table 13-10 AO_SEC_JTAG_PWD_SP_ADDR3 0xB5

Bit(s)	R/W	Default	Description
31-0	R/W	0	addr3

Table 13-11 AO_SEC_JTAG_SCP_CNTL 0xB6

Bit(s)	R/W	Default	Description
31	R	0	jrag_pwd_scp_data_valid; 1: finished write jtag_pwd_addr3;
30	R	0	jrag_pwd_scp_data_timeout; 1: jtag_pwd_sp_wd count greater threshold (time out);
26	R/W	0	jtag_pwd_scp_clr; 1: clear valid and irq status;
25	R/W	0	jtag_pwd_scp_wd_resetn; 0: reset pwd_scp_wd, then no irq;
24	R/W	0	jtag_pwd_scp_en; 0: disable jtag_pwd_scp in/out; 1: enable jtag_pwd_scp in/out;
18:16	R/W	0	jtag_pad_b_out sel: 0: none; 1:m4; 2:jtag_pwd_sp; 3:m3; 4:jtag_pwd_scp; 5:A55
14:12	R/W	0	jtag_pad_a_out_sel: 0: none; 1:m4; 2:jtag_pwd_sp; 3:m3; 4:jtag_pwd_scp; 5:A55
10	R/W	0	jtag_disable_A55_in; 0: enable; 1: disable A55_jtag_in;
9	R/W	0	jtag_disable_pwd_scp_in; 0: enable; 1: disable pwd_scp_jtag_in;
8	R/W	0	jtag_disable_m3_in; 0: enable; 1: disable m3_jtag_in;

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Bit(s)	R/W	Default	Description
5	R/W	0	jtag_sel_force_A55_tdo_en_low; 0: tdo_en = A55_jtag_tdo_en; 1: tdo_en = 1'b0 (always output);
4	R/W	0	jtag_sel_force_m3_tdo_en_low; 0: tdo_en = m3_jtag_tdo_en; 1: tdo_en = 1'b0 (always output);
2	R/W	0	jtag_sel_A55_in; 0: pad_a; 1: pad_b;
1	R/W	0	jtag_sel_pwd_scp_in; 0: pad_a; 1: pad_b;
0	R/W	0	jtag_sel_m3_in; 0: pad_a; 1: pad_b;

Table 13-12 AO_SEC_JTAG_PWD_SCP_0 0xB7

Bit(s)	R/W	Default	Description
31-0	R	0	jtag_pwd_scp received pwd[31:0]

Table 13-13 AO_SEC_JTAG_PWD_SCP_1 0xB8

Bit(s)	R/W	Default	Description
31-0	R	0	jtag_pwd_scp received pwd[63:32]

Table 13-14 AO_SEC_JTAG_PWD_SCP_2 0xB9

Bit(s)	R/W	Default	Description
31-0	R	0	jtag_pwd_scp received pwd[95:64]

Table 13-15 AO_SEC_JTAG_PWD_SCP_3 0xBA

Bit(s)	R/W	Default	Description
31-0	R	0	jtag_pwd_scp received pwd[127:96]

Table 13-16 AO_SEC_JTAG_PWD_SCP_CNTL 0xBB

Bit(s)	R/W	Default	Description
31:5	R/W	0	timeout threshold;
4	R/W	0	activity by tdo_en; 1: detect tdo activity and use it reset watchdog count;
3	R/W	0	activity by tck_en; 1: detect tck activity and use it reset watchdog count;
2	R/W	0	activity by tms_en; 1: detect tms activity and use it reset watchdog count;

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Bit(s)	R/W	Default	Description
1	R/W	0	activity by tdi_en; 1: detect tdi activity and use it reset watchdog count;
0	R/W	0	wd_timeout_en; 0: disable watchdog timeout; 1: enable watchdog timeout;

Table 13-17 AO_SEC_JTAG_PWD_SCP_ADDR0 0xBC

Bit(s)	R/W	Default	Description
31-0	R/W	0	addr0

Table 13-18 AO_SEC_JTAG_PWD_SCP_ADDR1 0xBD

Bit(s)	R/W	Default	Description
31-0	R/W	0	addr1

Table 13-19 AO_SEC_JTAG_PWD_SCP_ADDR2 0xBE

Bit(s)	R/W	Default	Description
31-0	R/W	0	addr2

Table 13-20 AO_SEC_JTAG_PWD_SCP_ADDR3 0xBF

Bit(s)	R/W	Default	Description
31-0	R/W	0	addr3

13.2 Temp Sensor

13.2.1 Overview

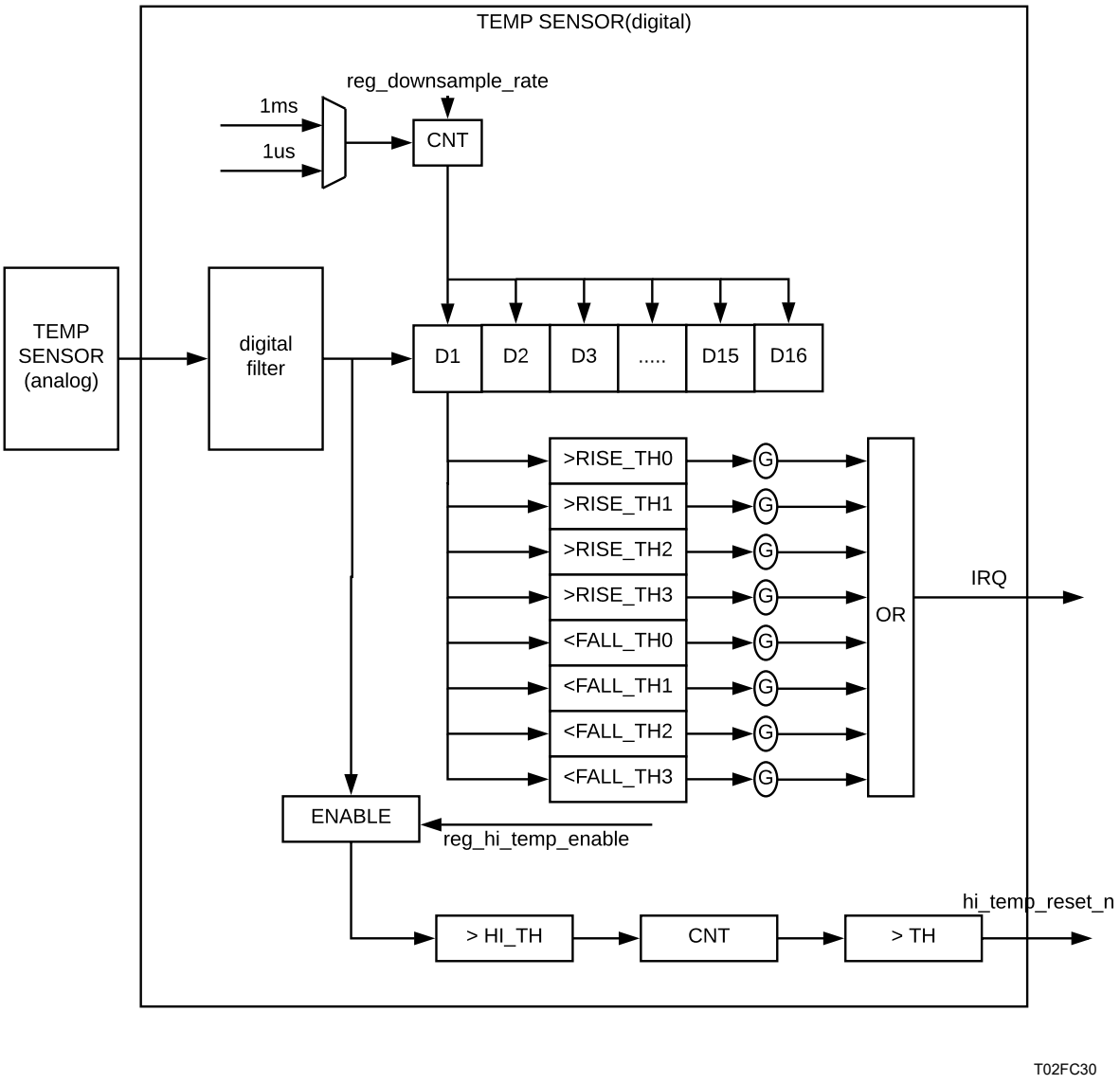
The chip integrates 3 Temp Sensor, one is close to DDR, one is close to PLL (between CPU and GPU),and one is close to SAR ADC.

Each Temp Sensor are the same design.

- Capture temperature by programable tick;
- Store 16 temperature value;
- Can reset all chip if detect high temperature;
- Can generate IRQ by 8 threshold;

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Figure 13-2 Temp Sensor



13.2.2 Register Description

Temp Sensor PLL Registers

Base Address: 0xFF634800

Each register final address = module base address+ address * 4

Temp Sensor DDR Registers

Base Address: 0xFF634C00

Each register final address = module base address+ address * 4

Temp Sensor SAR ADC Registers

Base Address: 0xFF635000

Each register final address = module base address+ address * 4

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Table 13-21 TS_CFG_REG1 0x001

Bits	R/W	Default	Description
31	R/W	0x0000-0000	fall_th3_irq_en
30	R/W		fall_th2_irq_en
29	R/W		fall_th1_irq_en
28	R/W		fall_th0_irq_en
27	R/W		rise_th3_irq_en
26	R/W		rise_th2_irq_en
25	R/W		rise_th1_irq_en
24	R/W		rise_th0_irq_en
23	R/W		fall_th3_irq_stat_clr
22	R/W		fall_th2_irq_stat_clr
21	R/W		fall_th1_irq_stat_clr
20	R/W		fall_th0_irq_stat_clr
19	R/W		rise_th3_irq_stat_clr
18	R/W		rise_th2_irq_stat_clr
17	R/W		rise_th1_irq_stat_clr
16	R/W		rise_th0_irq_stat_clr
15	R/W		1: enable IRQ related function.
14	R/W		fast_mode: 0 : downsample unit = 1ms; 1: downsample unit = 1us;
13	R/W		clr_hi_temp_stat
12	R/W		ts_ana_rset_vbg reset vbg(set 0, if have error place set it plus 01000..)
11	R/W		ts_ana_rst_sd reset adc(set 0, if have error place set it plus 01000..)
10	R/W		ts_ana_en_vcm enable vcm (disable:0; enable:1)
9	R/W		ts_ana_en_vbg enable vbg (disable:0; enable: 1)
8:7	R/W		filter hcic mode 0: downsample rate = 128; 1: downsample rate = 256; 2/3: downsample rate = 512;
6	R/W		filter ts_out_ctrl; 1: add more delay for filter lock;
5	R/W		filter en(disable:0; enable:1)
4	R/W		ts_ana_en_iptat, useless.

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Bits	R/W	Default	Description
3	R/W		Temp Sensor DEM enable. (disable:0; enable:1)
2:0	R/W		Bipolar bias current input control. recommend value : 3. ts_ana_ch_sel; 0: 8'b00000001; 1: 8'b00000011; 2: 8'b00000111; 3: 8'b00001111; 4: 8'b00011111; 5: 8'b00111111; 6: 8'b01111111; 7: 8'b11111111;

Table 13-22 TS_CFG_REG2 0x002

Bits	R/W	Default	Description
31	R/W	0x0000-0000	hi_temp_enable
30	R/W		reset_en, if = 0, will not reset all chip;
27:16	R/W		high temperature times, if continuous detect high temperature, then will reset all chip.
15:0	R/W		high temperature threshold, if temperature value > this th , mean detected once high temperature

Table 13-23 TS_CFG_REG3 0x03

Bits	R/W	Default	Description
31:16	R/W	0x0000-0000	
15:0	R/W		down_sample rate

Table 13-24 TS_CFG_REG4 0x04

Bits	R/W	Default	Description
23:12	R/W	0x0000-0000	rise_th0
11:0	R/W		rise_th1

Table 13-25 TS_CFG_REG5 0x05

Bits	R/W	Default	Description
23:12	R/W	0x0000-0000	rise_th2
11:0	R/W		rise_th3

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Table 13-26 TS_CFG_REG6 0x06

Bits	R/W	Default	Description
23:12	R/W	0x0000-0000	fall_th0
11:0	R/W		fall_th1

Table 13-27 TS_CFG_REG7 0x07

Bits	R/W	Default	Description
23:12	R/W	0x0000-0000	fall_th2
11:0	R/W		fall_th3

Table 13-28 TS_STAT0 0x10

Bits	R/W	Default	Description
31:18	R	0x0000-0000	detect_hi_temp_cnt
17	R		detected_hi_temp_stat
16	R		filter lock
15:0	R		filter out

Table 13-29 TS_STAT1 0x11

Bits	R/W	Default	Description
31:9	R	0x0000-0000	
8	R		hi_temp_stat
7	R		fall_th3_irq
6	R		fall_th2_irq
5	R		fall_th1_irq
4	R		fall_th0_irq
3	R		rise_th3_irq
2	R		rise_th2_irq
1	R		rise_th1_irq
0	R		rise_th0_irq

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Table 13-30 TS_STAT2 0x12

Bits	R/W	Default	Description
31:16	R	0x0000-0000	temperature value D2
15:0	R		temperature value D1

Table 13-31 TS_STAT3 0x13

Bits	R/W	Default	Description
31:16	R	0x0000-0000	temperature value D4
15:0	R		temperature value D3

Table 13-32 TS_STAT4 0x14

Bits	R/W	Default	Description
31:16	R	0x0000-0000	temperature value D6
15:0	R		temperature value D5

Table 13-33 TS_STAT5 0x15

Bits	R/W	Default	Description
31:16	R	0x0000-0000	temperature value D8
15:0	R		temperature value D7

Table 13-34 TS_STAT6 0x16

Bits	R/W	Default	Description
31:16	R	0x0000-0000	temperature value D10
15:0	R		temperature value D9

Table 13-35 TS_STAT7 0x17

Bits	R/W	Default	Description
31:16	R	0x0000-0000	temperature value D12
15:0	R		temperature value D11

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Table 13-36 TS_STAT8 0x18

Bits	R/W	Default	Description
31:16	R	0x0000-0000	temperature value D14
15:0	R		temperature value D13

Table 13-37 TS_STAT9 0x19

Bits	R/W	Default	Description
31:16	R	0x0000-0000	temperature value D16
15:0	R		temperature value D15

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